Integrated Disposal Facility

Change Control Log

Change Control Logs ensure that changes to this unit are performed in a methodical, controlled, coordinated and transparent manner. Each unit addendum will have a "**Last Modification Date**" which represents the last date the portion of the unit has been modified. The "**Modification Number**" represents Ecology's method for tracking the different versions of the permit. This log will serve as an up to date record of modifications and version history of the unit.

Last modification to Integrated Disposal Facility **September 30, 2015**

Chapters	Last Modification Date	Modification Number
Conditions	09/30/2015	8c.2015.Q3
1.0 Part A Form	10/01/2008	
2.0 Topographic Map Description	09/30/2014	
3.0 Waste Analysis Plan	06/30/2013	
4.0 Process Information	12/31/2008	
4A1 Phase I Critical Systems Design Report	03/31/2008	
4A2 Critical Systems Tables & Data Sheets	03/31/2008	
4A3 Critical Systems Design Drawings	03/31/2008	
4B Detailed Design Cell 1 Construction Quality Assurance Plan	04/09/2006	
4C Facility Response Action Plan	04/09/2006	
4D Construction Specifications (C-1)	12/31/2006	
5.0 Groundwater Monitoring	06/30/2010	
6.0 Procedures to Prevent Hazards	06/20/2013	
7.0 Reserved		
8.0 Personnel Training	09/30/2014	
9.0 Reserved		
10.0 Reserved		
11.0 Closure Plan	09/30/2014	
12.0 Reserved		
13.0 Other Federal and State Laws	04/09/2006	
Addendum J.1 Pre-Active Life Contingency Plan (OUO)	06/30/2012	
Addendum J.2 Active Life Contingency Plan (OUO)	06/30/2012	

PART III, OPERATING UNIT 11 UNIT-SPECIFIC CONDITIONS
INTEGRATED DISPOSAL FACILITY

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PART III, OPERATING UNIT 11 UNIT-SPECIFIC CONDITIONS
INTEGRATED DISPOSAL FACILITY

This document sets forth the operating conditions for the Integrated Disposal Facility (IDF).

III.11.A COMPLIANCE WITH APPROVED PERMIT

The Permittees shall comply with all requirements set forth in the Integrated Disposal Facility (IDF) Permit conditions, the Chapters and Appendices specified in Permit Condition III.11.A and the Amendments specified in Permit Conditions III.11.B through III.11.I. All subsections, figures, and tables included in these portions are enforceable unless stated otherwise:

OPERATING UNIT 11:

Chapter 1.0	Part A Form, dated October 1, 2008
Chapter 2.0	Topographic Map Description, dated September 30, 2014
Chapter 3.0	Waste Analysis Plan, dated June 30, 2013
Chapter 4.0	Process Information, dated December 31, 2008
Appendix 4A	Design Report (as applicable to critical systems), dated March 31, 2008
Appendix 4B	Construction Quality Assurance Plan, dated April 9, 2006
Appendix 4C	Response Action Plan, dated April 9, 2006
Appendix 4D	Construction Specifications (RPP-18489, Rev. 1), dated December 31, 2006
Chapter 5.0	Ground Water Monitoring, dated June 30, 2010
Chapter 6.0	Procedure to Prevent Hazards, dated June 20, 2013
Addendum J.1	Contingency Plan – Pre-Active Life, dated June 30, 2012
Addendum J.2	Contingency Plan – Active Life, dated June 30, 2012
Chapter 8.0	Personnel Training, dated September 30, 2014
Chapter 11.0	Closure, dated September 30, 2014
Chapter 13.0	Other Federal and State Laws, dated April 9, 2006

General and Standard Hanford Facility RCRA Permit, WA7 89000 8967 (Permit) conditions (Part I and Part II Conditions) applicable to the IDF are identified in Permit Attachment 9 (Permit Applicability Matrix).

III.11.B AMENDMENTS TO THE APPROVED PERMIT

III.11.B.1 Portions of Permit Attachment 4, *Hanford Emergency Management Plan* that are not made enforceable by inclusion in the applicability matrix for that document, are not made enforceable by reference in this document.

III.11.B.2 Permittees must comply with all applicable portions of the Permit. The facility and unit-specific recordkeeping requirements are distinguished in the General Information Portion of the Permit, and are tied to the Permit conditions.

III.11.B.3 The scope of this Permit is restricted to the landfill construction and operation as necessary to dispose of: 1) immobilized low activity waste from the WTP, and 2) the Demonstration Bulk Vitrification System and IDF operational waste as identified in Chapter 4.0. Future expansion of the RCRA trench, or disposal of other wastes not specified in this Permit, is prohibited unless authorized via modification of this Permit.

III.11.B.4 In accordance with WAC 173-303-806(11)(d), this Permit shall be reviewed every five (5) years after the effective date and modified, as necessary, in accordance with WAC 173-303-830(3).

III.11.B.5 Inspection Requirements – Pre-Active Life Period and Active Life Period

III.11.B.5.a The Permittees will conduct inspections of the IDF according to the following requirements:

III.11.B.5.a.i Prior to the start of the active life of the IDF as defined in WAC 173-303-040, according to Chapter 6.0, Table 6.2.

III.11.B.5.a.ii Following the start of the active life of the IDF as defined in WAC 173-303-040, according to Chapter 6.0, Table 6.2A.

III.11.B.5.b The Permittees will remedy any problems revealed by inspections conducted pursuant to permit condition III.11.B.5.a on a schedule, which prevents hazards to the public health and the environment and as agreed to in writing, by Ecology. Where a hazard is imminent or has already occurred, remedial action must be taken immediately.

III.11.B.5.c Reserved

III.11.B.5.d Rainwater Management

III.11.B.5.e Prior to the start of the active life of the IDF, the Permittees will manage the discharge of such water in accordance with the pollution prevention and best management practices required by State Waste Discharge Permit Number ST 4511.

III.11.B.5.e.i Management of Liquids Collected in the Leachate Collection and Removal System (LCRS), Leak Detection System (LDS), and Secondary Leak Detection System (SLDS) prior to the start of the active life of the IDF.

III.11.B.5.e.ii Permittees shall manage the liquid in the LCRS system in a manner that does not allow the fluid head to exceed 30.5 cm above the flat 50-foot by 50-foot LCRS sump HDPE bottom liner, and the LCRS sump trough, except for storms that exceed the 25-year, 24-hour storm event [(WAC 173-303-665(2)(h)(ii)(B). Liquid with a depth greater than 30.5 cm above the LCRS liner will be removed at the earliest practicable time after detection (not to exceed 5 working days).

III.11.B.5.e.iii Accumulated liquid of pumpable quantities in the LDS and SLDS will be managed in a manner that does not allow the fluid head to exceed 30.5 cm above the LDS liner or SLDS liner [WAC 173-303-665(2)(h)(i)(C)(iii)]. Liquid with a depth greater than 30.5 cm above a liner will be removed at the earliest practicable time after detection (not to exceed 5 working days).

III.11.B.5.e.iv The Permittees will use a flow meter to check if the amount of actual liquid pumped corresponds to the amount accumulated in the leachate collection tank to verify the proper function of the leachate collection and removal sump pumps with each use. The Permittees will document in the IDF portion of the facility operating record appropriate quality assurance/quality control requirements for selection and operation of the flow meter based on the required verification. In addition, the Permittees will evaluate the leachate transfer lines for freeze and thaw damage when ambient conditions may cause such damage to occur. The Permittees will document the methods and criteria used for purposes of this evaluation, along with an appropriate justification.

III.11.B.5.e.v The Permittee will inspect for liquids after significant rainfall events.

III.11.B.5.e.vi The Permittee will annually verify monitoring gauges and instruments are in current calibration; calibration will be performed annually or more frequently at intervals suggested by the manufacturer (refer to Chapter 4.0, §4.3.7.4)

III.11.B.5.f The Permittees will monitor liquids in the Leachate Collection and Removal System and Leak Detection System to ensure the action leakage rate (Chapter 4.0, Appendix 4A) is not exceeded.

III.11.B.5.g Soil Stabilization

Prior to the first placement of waste in the IDF, the Permittee will apply soil stabilization materials as needed to prevent soil erosion in and around the landfill.

III.11.C Design Requirements

III.11.C.1 IDF is designed in accordance with WAC 173-303-665 and WAC 173-303-640 as described in Chapter 4.0. Design changes impacting IDF critical systems shall be performed in accordance with Permit Conditions III.11.D.1.d.i and III.11.D.1.d.ii.

III.11.C.1.a IDF Critical Systems include the following: The leachate collection and removal system (LCRS), leachate collection tank (LCT), leak detection system (LDS), liner system (LS), and closure cap. H-2 Drawings for the LCRS, LCT, LDS, and LS are identified in Appendix 4A, Section 3 of this Permit. Drawings for the closure cap will be provided pursuant to Permit Condition III.11.C.1.c.

The Permittees shall construct and operate the IDF in accordance with all specifications contained in RPP-18489 Rev 0. Critical systems, as defined in the definitions section of the Site-Wide RCRA Permit, are identified in Appendix 4A, Section 1 of this Permit.

III.11.C.1.b Landfill Cap

At final closure of the landfill, the Permittees shall cover the landfill with a final cover (closure cap) designed and constructed [WAC 173-303-665(6), WAC 173-303-806(4)(h)] to: Provide long-term minimization of migration of liquids through the closed landfill; Function with minimum maintenance; Promote drainage and minimize erosion or abrasion of the cover; Accommodate settling and subsidence so that the cover's integrity is maintained; and have a permeability less than or equal to the permeability of any bottom liner system or natural sub soils present.

III.11.C.1.c Compliance Schedule

Proposed conceptualized final cover design is presented in Chapter 11, Closure Requirements. Six months prior to start of construction of IDF landfill final cover (but no later than 6 months prior to acceptance of the last shipment of waste at the IDF), the Permittees shall submit IDF landfill final cover design, specifications and Construction Quality Assurance (CQA) plan to Ecology for review and approval. No construction of the final cover may proceed until Ecology approval of the final design is given, through a permit modification.

III.11.C.1.d The Permittees shall notify Ecology at least sixty (60) calendar days prior to the date it expects to begin closure of the IDF landfill in accordance with WAC 173-303-610(c).

III.11.C.2 Design Reports

III.11.C.2.a New Tank Design Assessment Report

Permittees shall generate a written report in accordance with WAC 173-303-640(3)(a), providing the results of the leachate collection tank system design assessment. The report shall be reviewed and certified by an Independent Qualified Registered Professional Engineer (IQRPE)[1] in accordance with WAC-173-303-810(13)(a).

[1] "Independent qualified registered professional engineer," as used here and elsewhere with respect to Operating Unit 11, means a person who is licensed by the state of Washington, or a state which has reciprocity with the state of Washington as defined in

III.11.C.2.b Compliance Schedule

Permittees shall submit the leachate collection tank design assessment report to Ecology along with the IQRPE certification, prior to construction of any part of the tank system including ancillary equipment.

III.11.D CONSTRUCTION REQUIREMENTS

III.11.D.1 Construction Quality Assurance

III.11.D.1.a Ecology shall provide field oversight during construction of critical systems. In cases where an Engineering Change Notice (ECN) and/or Non Conformance Report (NCR) are required, Ecology and the Permittees shall follow steps for processing changes to the approved design per Permit Conditions III.11.D.1.d.i and III.11.D.1.d.ii.

III.11.D.1.b Permittees shall implement the CQA plan (Appendix 4B of the permit) during construction of IDF.

III.11.D.1.b.i The Permittees will not receive waste in the IDF until the owner or operator has submitted to Ecology by certified mail or hand delivery a certification signed by the CQA officer that the approved CQA plan has been successfully carried out and that the unit meets the requirements of WAC 173-303-665(2)(h) or (j); and the procedure in WAC 173-303-810(14)(a) has been completed. Documentation supporting the CQA officer's certification shall be furnished to Ecology upon request.

III.11.D.1.c Construction inspection reports

Permittees shall submit a report documenting the results of the leachate tank installation inspection. This report must be prepared by an independent, qualified installation inspector or a professional independent, qualified, registered, professional engineer either of whom is trained and experienced in the proper installation of tank systems or components. The Permittees will remedy all discrepancies before the tank system is placed in use. This report shall be submitted to Ecology 90 days prior to IDF operation and be included in the IDF Operating Record. [WAC 173-303-640(3)(h)].

III.11.D.1.d ECN/NCR Process for Critical Systems

Portions of the following conditions for processing engineering change notices and non-conformance reporting were extracted from and supersede Site Wide General Permit Condition II.L.

III.11.D.1.d.i Engineering Change Notice for Critical Systems

During construction of the IDF, the Permittees shall formally document changes to the approved designs, plans, and specifications, identified in Appendices 4A, 4B, 4C, and 4D of this permit, with an Engineering Change Notice (ECN).

The Permittees shall maintain all ECNs in the IDF unit-specific Operating Record and shall make them available to Ecology upon request or during the course of an inspection. The Permittees shall provide to Ecology copies of proposed ECNs affecting any critical system within five (5) working days of initiating the ECN. Identification of critical systems is included in Permit Condition III.11.C.1 and Appendix 4A of this permit. Within five (5) working days, Ecology will review a proposed ECN modifying a critical system and inform the Permittees whether the proposed ECN, when issued, will require a Class 1, 2, or 3 Permit modification.

RCW 18.43.100, and who is not an employee of the owner or operator of the facility for which construction or modification certification is required. A qualified professional engineer is an engineer with expertise in the specific area for which a certification is given.

III.11.D.1.d.ii Non-conformance Reporting for Critical Systems

III.11.D.1.d.ii.a During construction of the IDF, the Permittees shall formally document with a Nonconformance Report (NCR), any work completed which does not meet or exceed the standards of the approved design, plans and specifications, identified in Appendices 4A, 4B, 4C and 4D of this Permit. The Permittees shall maintain all NCRs in the IDF unit-specific Operating Record and shall make them available to Ecology upon request, or during the course of an inspection.

III.11.D.1.d.ii.b The Permittees shall provide copies of NCRs affecting any critical or regulated system to Ecology within five (5) working days after identification of the nonconformance. Identification of critical systems is included in Permit Condition III.11.C.1 and Appendix 4A of this permit. Ecology will review a NCR affecting a critical system and notify the Permittees within five (5) working days, in writing, whether a Permit modification is required for any nonconformance, and whether prior approval is required from Ecology before work proceeds, which affects the nonconforming item.

III.11.D.1.d.ii.c As-Built Drawings

Upon completing construction of IDF, the Permittees shall produce as-built drawings of the project, which incorporate the design and construction modifications resulting from all project ECNs and NCRs, as well as modifications made pursuant to WAC 173-303-830. The Permittees shall place the drawings into the Operating Record within twelve (12) months of completing construction.

III.11.D.2 The Permittees shall not reduce the minimum frequency of destructive testing less than one test per 500 feet of seam, without prior approval in writing from Ecology

III.11.E GROUND WATER AND GROUND WATER MONITORING

Ground water shall be monitored in accordance with WAC 173-303 and the provisions contained in the Ecology-approved facility ground water monitoring plan (Chapter 5.0). All wells used to monitor the ground water beneath the unit shall be constructed in accordance with the provisions of WAC 173-160.

III.11.E.1 Ground Water Monitoring Program

III.11.E.1.a Prior to initial waste placement in the IDF landfill, the Permittees shall sample all ground water monitoring wells in the IDF network twice quarterly for one first year to determine baseline conditions. For the first sampling event (and only the first), samples for each well will include all constituents in 40 CFR 264 Appendix IX. Thereafter, sampling will include only those constituents as specified in Chapter 5.0, Table 5-2: chromium (filtered and unfiltered the first year to compare results), specific conductance, TOC, TOX, and pH. Other constituents to be monitored but not statistically compared include alkalinity, anions, ICP metals, and turbidity. These will provide important information on hydrogeologic characteristics of the aquifer and may provide indications of encroaching contaminants from other facilities not associated with IDF.

III.11.E.1.b After the baseline monitoring is completed, and data is analyzed, the Permittees and Ecology shall assess revisions to Chapter 5.0, Table 5-2. Subsequent samples will be collected annually and will include constituents listed in Table 5-2 as approved by Ecology. All data analysis will employ Ecology approved statistical methods pursuant to WAC 173-303-645. Changes to Chapter 5.0 will be subject to the permit modification procedures under WAC 173-303-830.

III.11.E.1.c All constituents used as tracers to assess performance of the facility through computer modeling should be sampled at least annually to validate modeling results.

Groundwater monitoring data and analytes to be monitored will be reviewed periodically as defined in Chapter 5.0 of this Permit.

III.11.E.1.d Upon Ecology approval of the leachate monitoring plan, leachate monitoring and groundwater monitoring activities should be coordinated as approved by Ecology to form an effective and efficient means of monitoring the performance of the IDF facility.

III.11.E.1.e Groundwater monitoring data shall be reported to Ecology annually by July 31. The annual report shall include monitoring results for the 12-month period from January 1 through December 31.

III.11.F LEACHATE COLLECTION COMPONENT MANAGEMENT

Permittees shall design, construct, and operate all leachate collection systems to minimize clogging during the active life and post closure period

III.11.F.1 Leachate Collection and Removal System (LCRS)

III.11.F.1.a At least 120 days prior to initial waste placement in the IDF, the Permittees shall submit a Leachate monitoring plan to Ecology for review, approval, and incorporation into the permit. Upon approval by Ecology, this plan will be incorporated into the Permit as a class 1' modification. The Permittees shall not accept waste into the IDF until the requirements of the leachate monitoring plan have been incorporated into this permit.

III.11.F.1.b Leachate in the LCRS (primary sump) shall be sampled and analyzed monthly for the first year of operation of the facility and quarterly thereafter (pursuant to WAC 173-303-200). Additionally, leachate shall be sampled and analyzed to meet waste acceptance criteria at the receiving treatment storage and disposal facility.

III.11.F.1.c Permittees shall manage the leachate in the LCRS system in a manner that does not allow the fluid head to exceed 30.5 cm above the flat 50-foot by 50-foot LCRS sump HDPE bottom liner except for rare storm events as discussed in Chapter 4.0, §4.3.6.1 and the LCRS sump trough (WAC 173-303-665(2)(h)(ii)(B). Liquid with a depth greater than 30.5 cm above the SLDS liner will be removed at the earliest practicable time after detection (not to exceed 5 working days).

III.11.F.1.d After initial waste placement, Permittees shall manage all leachate from the permitted cell as dangerous waste (designated with Dangerous Waste Number F039) in accordance with WAC 173-303.

III.11.F.2 Monitoring and Management of Leak Detection System (LDS/secondary sump)

III.11.F.2.a Permittees shall manage the leachate in the LDS system in a manner that does not allow the fluid head to exceed 30.5 cm above the LDS liner (WAC 173-303-665(2)(h)(ii)(B).

III.11.F.2.b Permittees shall monitor and record leachate removal for comparison to the Action Leakage Rate (ALR) as described in Appendix 4C, Response Action Plan. If the leachate flow rate in the LDS exceeds the ALR, the Permittees shall implement the Ecology approved response action plan (Appendix 4C).

III.11.F.2.c Leachate from the LDS (secondary sump) shall be sampled semi-annually if a pumpable quantity of leachate is available for sampling.

III.11.F.2.d Accumulated liquid of pumpable quantities in the LDS will be managed in a manner that does not allow the fluid head to exceed 30.5 cm above the LDS liner [WAC 173-303-665(2)(h)(i)(C)(iii)].

Liquid with a depth greater than 30.5 cm above the LDS liner will be removed at the earliest practicable time after detection (not to exceed 5 working days).

III.11.F.3 Monitoring and Management of the Secondary Leak Detection System (SLDS)

III.11.F.3.a At least 180 days prior to initial waste placement, the, the Permittees shall submit to Ecology for approval a sub-surface liquids monitoring and operations plan (SLMOP) for the SLDS to include the following: monitoring frequency, pressure transducer configuration, liquid collection and storage processes, sampling and analysis and response actions. The SLMOP shall be approved by Ecology prior to placement of waste in the IDF, and incorporated into the Permit as a Class 1' modification.

III.11.F.3.b Permittees shall monitor and manage the SLDS (tertiary sump) pursuant to the approved sub-surface liquids monitoring and operations plan.

III.11.F.3.c Accumulated liquid of pumpable quantities in the SLDS will be managed in a manner that does not allow the fluid head to exceed 30.5 cm above the SLDS liner [WAC 173-303-665(2)(h)(i)(C)(iii)]. Liquid with a depth greater than 30.5 cm above the SLDS liner will be removed at the earliest practicable time after detection (not to exceed 5 working days).

III.11.F.3.d After initial waste placement, Permittees shall manage all leachate from the permitted cell as dangerous waste in accordance with WAC 173-303.

III.11.G CONSTRUCTION WATER MANAGEMENT

III.11.G.1 During construction, it is anticipated that liquids will accumulate on top of all liners and sumps. Permittees shall manage the construction wastewater in accordance with State Waste Discharge Permit ST 4511.

III.11.G.2 Liquid accumulation within the LCRS, LDS, and SLDS prior to initial waste placement will be considered construction wastewater (i.e., not leachate).

III.11.H LANDFILL LINER INTEGRITY MANAGEMENT & LANDFILL OPERATIONS

III.11.H.1 Permittees shall design, construct, and operate the landfill in a manner to protect the liners from becoming damaged. Temperature: Waste packages with elevated temperatures shall be evaluated and managed in a manner to maintain the primary (upper) liner below the design basis temperature for the liner (e.g.,160 F). Weight: Waste, fill material and closure cover shall be placed in a manner that does not exceed the allowable load bearing capacity of the liner (weight per area 13,000 lb/ft^2). Puncture: At least 3 feet of clean backfill material shall be placed as an operations layer over the leachate collection and removal system to protect the system from puncture damage.

III.11.H.1.a All equipment used for construction and operations inside of the IDF shall meet the weight limitation as specified in Permit Condition III.11.H.1. Only equipment that can be adequately supported by the operations layer as specified in Permit Condition III.11.H.1 (e.g., will not have the potential to puncture the liner) shall be used inside of the IDF. All equipment used for construction and operations outside of the IDF shall not damage the berms. Changes to any equipment will follow the process established by condition II.R of the site wide permit. Within 120 days from the effective date for the permit, a process for demonstrating compliance with this condition shall be submitted for review by Ecology. This process will be incorporated into appropriate IDF operating procedures prior to IDF operations.

III.11.H.2 The Permittees shall construct berms and ditches to prevent run-on and run-off in accordance with the requirements of Chapter 4, Section 4.3.8 of the IDF portion of this permit.

Before the first placement of waste in the IDF, the Permittees shall submit to Ecology a final grading and topographical map on a scale sufficient to identify berms and ditches used to control run-on and run-off. Upon approval, Ecology will incorporate these maps into the permit as a Class 1' modification.

III.11.H.3 The Permittees shall operate the Resource Conservation and Recovery Act (RCRA) IDF Cell (Cell1) in accordance with WAC 173-303-665(2) and the operating practices described in Chapters 3.0, 4.0, 6.0, 8.0, Addendum J.1, Addendum J.2, and Appendix 4A, §1, subsection 7, except as otherwise specified in this Permit.

III.11.H.4 The Permittees shall maintain a permanent and accurate record of the three-dimensional location of each waste type, based on grid coordinates, within the RCRA IDF Cell (Cell1) in accordance with WAC 173-303-665(5).

III.11.I WASTE ACCEPTANCE CRITERIA

The only acceptable waste form approved for disposal at the RCRA cell of IDF are IDF operational waste, Immobilized Low Activity Waste (ILAW) in glass form from the Waste Treatment Plant (WTP) Low Activity Waste (LAW) Vitrification facility and ILAW from the Bulk Vitrification Research Demonstration and Development facility (up to 50 boxes). Specifics about waste acceptance criteria for each of these wastes are detailed below.

No other waste forms may be disposed at the RCRA cell of IDF unless authorized via a Final Permit modification decision. Requests for Permit modifications must be accompanied by an analysis adequate for Ecology to comply with SEPA, as well as by a risk assessment and groundwater modeling to show the environmental impact. Permit Condition III.11.I.5 outlines the process by which waste sources in the IDF are modeled in an ongoing risk budget and a ground water impact analysis.

III.11.I.1 Six months prior to IDF operations Permittees shall submit to Ecology for review, approval, and incorporation into the permit, all waste acceptance criteria to address, at a minimum, the following: physical/chemical criteria, liquids and liquid containing waste, land disposal restriction treatment standards and prohibitions, compatibility of waste with liner, gas generation, packaging, handling of packages, minimization of subsidence.

III.11.I.1.a All containers/packages shall meet void space requirements pursuant to WAC 173-303-665(12).

III.11.I.1.b Compliance Schedule

III.11.I.1.b.i Six months prior to IDF operations, the Permittees shall submit to Ecology for review, approval, and incorporation into the permit any necessary modifications to the IDF Waste Analysis Plan (Chapters 3.0 of the IDF portion of this permit).

III.11.I.2 ILAW Waste Acceptance Criteria

The only ILAW forms acceptable for disposal at IDF are: (1) approved glass canisters that are produced in accordance with the terms, conditions, and requirements of the WTP portion of the Permit, and (2) the 50 bulk vitrification test boxes as specified in the Demonstration Bulk Vitrification System (DBVS) test plans.

To assure protection of human health and the environment, it is necessary that the appropriate quality of glass be disposed at IDF. The Land Disposal Restrictions (LDR) Treatment Standard for eight metals (arsenic, barium, cadmium, chromium, lead, mercury, selenium and silver), when associated with High Level Waste, is HLVIT (40 CFR 268). Because these metals are constituents in the Hanford Tanks Waste, the LDR standard for ILAW disposed to IDF is HLVIT.

For any ILAW glass form(s) that DOE intends to dispose of in IDF, DOE will provide to Ecology for review, an ILAW Waste Form Technical Requirements Document (IWTRD). The IWTRD will contain:

III.11.I.2.a WTP ILAW Waste Acceptance Criteria

III.11.I.2.a.i A description of each specific glass formulation that DOE intends to use including a basis for why each specific formulation is proposed for use, which specific tank wastes the glass formulation is proposed for use with, the characteristics of the glass that are key to satisfactory performance (e.g., VHT, PCT, and TCLP and/or other approved performance testing methodologies that the parties agree are appropriate and necessary), the range in key characteristics anticipated if the specific glass formulation is produced on a production basis with tank waste, and the factors that DOE must protect against in producing the glass to ensure the intended glass characteristics will exist in the actual ILAW.

III.11.I.2.a.ii A performance assessment that provides a reasonable basis for assurance that each glass formulation will, once disposed of in IDF in combination with the other waste volumes and waste forms planned for disposal at the entire Integrated Disposal Facility, be adequately protective of human health and the environment; and will not violate or be projected to violate all applicable state and federal laws, regulations and environmental standards.

Within 60 days of a request by Ecology, the Permittees shall provide a separate model run using Ecology's assumptions and model input.

III.11.I.2.a.iii A description of production processes including management controls and quality assurance/quality control requirements that assure that glass produced for each formulation will perform in a reasonably similar manner to the waste form assumed in the performance assessment for that formulation.

The Permittees shall update the IWTRD consistent with the above requirements for review by Ecology consistent with their respective roles and authority as provided under the TPA. Ecology comments shall be dispositioned through the Review Comment Record (RCR) process and will be reflected in further modeling to modify the IDF ILAW Chapter 3.0, Waste Analysis Plan as appropriate.

The initial IWTRD contained glass formulation data as required by Permit Condition III.11.I.2.a.i, and was submitted on December 18, 2006 (AR Accession # 0906020182). The performance assessment required by Permit Condition III.11.I.2.a.ii, and the quality assurance/quality control requirements process required by Permit Condition III.11.I.2.a.iii shall be submitted for Ecology review as soon as possible after issuance of the Final Tank Closure and Waste Management Environment Impact Statement (EIS) and receipt of underlying codes and data packages, and at least 180 days prior to the date DOE expects to receive waste at IDF. At a minimum, the Permittees shall submit updates to the IWTRD to Ecology every five years or more frequently with the next one due June 30, 2015, if any of the following conditions exist:

- The Permittees submits a permit modification request allowing additional waste forms to be disposed of at IDF.
- The WTP or other vitrification facility change their glass formulations from those previously included in the IWTRD.
- An unanticipated event or condition occurs that Ecology determines would warrant an update to the IWTRD.

III.11.I.2.a.iv The Permittees shall not dispose of any WTP ILAW not described and evaluated in the IWTRD.

III.11.I.3 ILAW Waste Acceptance Criteria Verification

III.11.I.3.a Six months prior to disposing of ILAW in the IDF, the Permittees will submit an ILAW verification plan to Ecology for review and approval. This plan will be coordinated with WTP, Ecology, and the Permittees personnel. This plan will outline the specifics of verifying ILAW waste acceptance through WTP operating parameters, and/or glass sampling. The Plan will include physical sampling requirements for batches, glass formulations, and/or feed envelopes.

III.11.I.4 Demonstration Bulk Vitrification System (DBVS) Bulk Vitrification Waste Acceptance Criteria

III.11.I.4.a Bulk Vitrification waste forms that are acceptable to be disposed of at IDF are up to 50 boxes of vitrified glass produced pursuant to the DBVS Research, Development, and Demonstration (RD&D) Permit from processing Hanford Tank S-109 tank waste.

III.11.I.4.b If Bulk Vitrification is selected as a technology to supplement the Waste Treatment Plant, the IDF portion of the Permit will need to be modified to accept Bulk Vitrification Full Scale production waste forms. This modification will need to be accompanied by appropriate Tri-Party Agreement (TPA) changes (per M-062 requirements) and adequate risk assessment information sufficient for the Department of Ecology to meet its State Environmental Policy Act (SEPA) obligations.

III.11.I.4.c DBVS Waste Acceptance Verification will occur on 100% of the waste packages. Pursuant to the DBVS RD&D Permit, a detailed campaign test report will be produced and submitted to Ecology detailing results of all testing performed on each waste package that is produced. IDF personnel shall review these reports to verify that the waste packages meet IDF Waste Acceptance Criteria.

III.11.I.4.d The Permittees shall not dispose of any waste forms that do not comply with all appropriate and applicable treatment standards, including all applicable LDR.

III.11.I.5 Modeling – Risk Budget Tool

III.11.I.5.a The Permittees must create and maintain a modeling - risk budget tool, which models the future impacts of the planned IDF waste forms (including input from analyses performed as specified in Permit Conditions III.11.I.2.a through III.11.I.2.a.ii) and their impact to underlying vadose and ground water. This software tool will be submitted for Ecology review as soon as possible after issuance of Final Tank Closure and Waste Management EIS and receipt of underlying codes and data packages, and at least 180 days prior to the date DOE expects to receive waste at IDF. The risk budget tool shall be updated at least every 5 years. The model will be updated more frequently if needed, to support permit modifications or SEPA Threshold Determinations whenever a new waste stream or significant expansion is being proposed for the IDF. This risk budget tool shall be conducted in manner that is consistent with state and federal requirements, and represents a risk analysis of all waste previously disposed of in the entire IDF (both cell 1 and cell 2) and those wastes expected to be disposed of in the future for the entire IDF to determine cumulative impacts. The groundwater impact should be modeled to evaluate fate and transport in the groundwater aquifer(s) and should be compared against various performance standards including but not limited to drinking water standards (40 CFR 141 and 40 CFR 143). Ecology will review modeling assumptions, input parameters, and results and will provide comments to the Permittees. Ecology comments shall be dispositioned through the Review Comment Record (RCR) process and will be reflected in further modeling to modify the IDF ILAW waste acceptance criteria as appropriate.

III.11.I.5.a.i The modeling-risk budget tool will include a sensitivity analysis reflecting parameters and changes to parameters as requested by Ecology.

III.11.I.5.a.ii If these modeling efforts indicate results within 75% of a performance standard [including but not limited to federal drinking water standards (40 CFR 141 and 40 CFR 143)], Ecology and the Permittees will meet to discuss mitigation measures or modified waste acceptance criteria for specific waste forms.

III.11.I.5.a.iii When considering all the waste forms to be disposed of in IDF, the Permittees shall not dispose of any waste that will result (through forward looking modeling or in real groundwater concentrations data) in a violation of any state or federal regulatory limit, specifically including but not limited to drinking water standards for any constituent as defined in 40 CFR 141 and 40 CFR 143.

III.11.I.6 The Permittees shall not dispose of any waste that is not in compliance with state and federal requirements as identified in Chapter 13.0.

III.11.I.6.a In accordance with DOE's authority under the Atomic Energy Act of 1954, as amended and other applicable law, prior to disposing of any mixed immobilized low-activity waste (ILAW) in the IDF, DOE will certify to the State of Washington that it has determined that such ILAW is not high-level waste and meets the criteria and requirements outlined in DOE's consultation with the U.S. Nuclear Regulatory Commission beginning in 1993 (Letter from R.M. Bernero, USNRC to J. Lytle, USDOE, dated March 2, 1993; Letter from J. Kinzer, USDOE, to C. J. Paperiello, USNRC, Classification of Hanford Low-Activity Tank Waste Fraction, dated March 7, 1996; and Letter from C.J. Paperiello, USNRC, to J. Kinzer, USDOE, Classification of Hanford Low-Activity Tank Waste Fraction, dated June 9, 1997). While the requirement to provide such certification is an enforceable obligation of this permit, the provision of such certification does not convey, or purport to convey, authority to Ecology to regulate the radioactive hazards of the waste under this permit.

III.11.I.7 IDF Operational Waste Acceptance Criteria.

III.11.I.7.a IDF operational activities (including decontamination, cleanup, and maintenance) will generate a small amount of waste. Waste that can meet IDF waste acceptance without treatment will be disposed of at the IDF. All other IDF operational waste will be managed pursuant to WAC 173-303-200.

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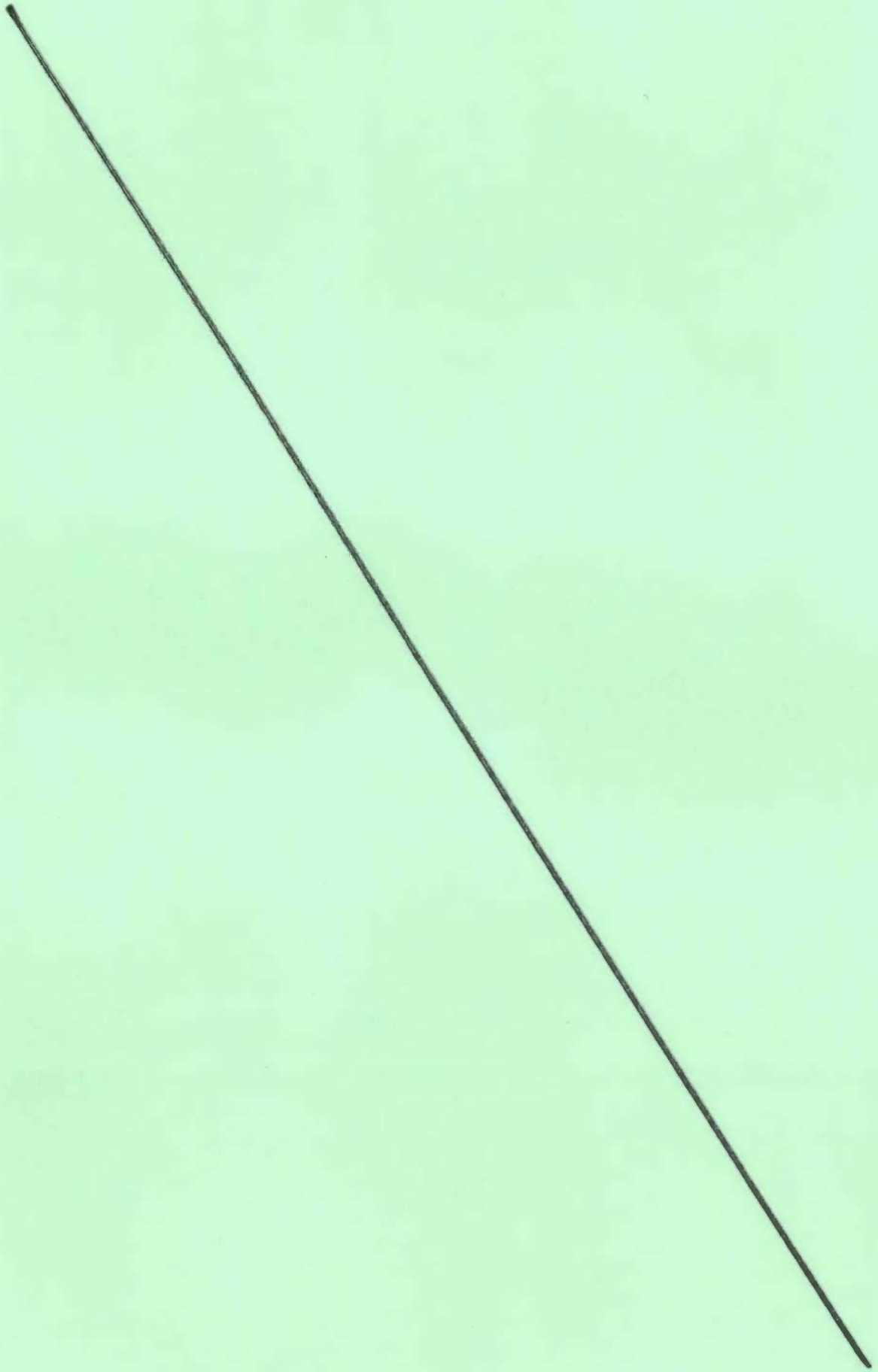

WASHINGTON STATE DEPARTMENT OF ECOLOGY

Dangerous Waste Permit Application Part A Form

Date Received	Reviewed by:	Date:
Month Day Year	Approved by:	Date:
09 19 2008		09 22 2008

I. This form is submitted to: (place an "X" in the appropriate box)

- [X] Request modification to a final status permit (commonly called a "Part B" permit)
- [ ] Request a change under interim status
- [ ] Apply for a final status permit. This includes the application for the initial final status permit for a site or for a permit renewal (i.e., a new permit to replace an expiring permit).
- [ ] Establish interim status because of the wastes newly regulated on: (Date)
  List waste codes:

II. EPA/State ID Number

WA 7890008967

III. Name of Facility

US Department of Energy - Hanford Facility

IV. Facility Location (Physical address not P.O. Box or Route Number)

A. Street

825 Jadwin

City or Town	State	ZIP Code
Richland	WA	99352

County Code (if known)	County Name
005	Benton

B. Land Type	C. Geographic Location Latitude (degrees, mins, secs)	Longitude (degrees, mins, secs)	D. Facility Existence Date Month Day Year
F	Refer to TOPO Map (Section XV.)		03 02 1943

V. Facility Mailing Address

Street or P.O. Box

P.O. Box 550

City or Town	State	ZIP Code
Richland	WA	99352

VI. Facility contact (Person to be contacted regarding waste activities at facility)		
Name (last)	**(first)**	
Brockman	David	
Job Title	**Phone Number (area code and number)**	
Manager	(509) 376-7395	
Contact Address		
Street or P.O. Box		
P.O. Box 550		
City or Town	**State**	**ZIP Code**
Richland	WA	99352

VII. Facility Operator Information	
A. Name	**Phone Number**
Department of Energy Owner/Operator CH2M HILL Plateau Remediation Company Co-Operator for Integrated Disposal Facility*	(509) 376-7395 (509) 376-0556*
Street or P.O. Box	
P.O. Box 550 P.O. Box 1600 *	

City or Town	State	ZIP Code
Richland	WA	99352

B. Operator Type F

C. Does the name in VII.A reflect a proposed change in operator? ☐ Yes ☒ No Co-Operator* change

If yes, provide the scheduled date for the change:

Month		Day		Year					
1	0		0	1		2	0	0	8

D. Is the name listed in VII.A. also the owner? If yes, skip to Section VIII.C. ☐ Yes ☒ No

VIII. Facility Owner Information	
A. Name	**Phone Number (area code and number)**
David A. Brockman, Operator/Facility-Property Owner	(509) 376-7395
Street or P.O. Box	
P.O. Box 550	

City or Town	State	ZIP Code
Richland	WA	99352

B. Owner Type F

C. Does the name in VIII.A reflect a proposed change in owner? ☐ Yes ☒ No

If yes, provide the scheduled date for the change:

Month		Day		Year					

IX. NAICS Codes (5/6 digit codes)

A. First		B. Second	
562211	Waste Treatment & Disposal	924110	Administration of Air & Water Resource & Solid Waste Management Programs
C. Third		**D. Fourth**	
541713	Research & Development in the Physical, Engineering, & Life Sciences		

X. Other Environmental Permits (see instructions)		
A. Permit Type	B. Permit Number	C. Description

XI. Nature of Business (provide a brief description that includes both dangerous waste and non-dangerous waste areas and activities)

The Integrated Disposal Facility (IDF) is an expandable lined landfill located in the 200 East Area of the Hanford Facility. The landfill is divided lengthwise into distinct east and west cells, one for disposal of low-level radioactive waste (the east cell) and the other for disposal of mixed waste (the west cell). The cell for disposal of low-level radioactive waste is outside the scope of this permit.

Mixed waste disposed at the IDF is currently limited to vitrified low-activity waste (LAW) from the Waste Treatment Plant (WTP) and Demonstration Bulk Vitrification System (DBVS). Additionally, mixed waste generated by IDF operations will be disposed of in IDF. (The IDF Permit requires modification if other waste streams are proposed for disposal.) The vitrified waste form generated by both the WTP and the DBVS facilities is known as Immobilized Low Activity Waste (ILAW) . The amount shown in Section XII of 8.2-hectare meters (82,000 cubic meters) is the waste capacity of the initial construction. The amount will be revised as required for future expansion to accommodate the entire waste volume through an approved permit modification.

D80

Mixed radioactive high level wastes stored in the Double-Shell and Single-Shell Tank System carry the characteristic dangerous waste numbers D002, and D004 through D011 . The specified technology based treatment standard for high-level radioactive waste as described in 40 CFR 268.40 (vitrification) will be used to produce the waste form that will be placed in steel canisters or steel boxes. Tank waste will meet this vitrification standard as the waste exits at the Waste Treatment Plant or Demonstration Bulk Vitrification System Facility. (Permit conditions for the WTP and DBVS require that the D001 and D003 waste codes be removed prior to the waste stream entering these facilities.)

IDF operational activities (including decontamination, cleanup, and maintenance) will generate a small amount of waste. Waste that can meet IDF waste acceptance without treatment will be buried at the IDF. All other IDF operational waste will be managed pursuant to WAC 173-303-200 and either sent to a 90-day accumulation area or directly to another permitted TSD for treatment. Treated IDF operational waste will either be buried at IDF or sent to another permitted Hanford TSD for final disposition.

S01

Process Code S01 (container storage) has been included within this Part A Form in the event that storage is required before final disposal (e.g., to support the staging and confirmation process of the waste or cooling of vitrified waste if required).

EXAMPLE FOR COMPLETING ITEMS XII and XIII (shown in lines numbered X-1, X-2, and X-3 below): A facility has two storage tanks that hold 1200 gallons and 400 gallons respectively. There is also treatment in tanks at 20 gallons/hr. Finally, a one-quarter acre area that is two meters deep will undergo *in situ vitrification*.

Section XII. Process Codes and Design Capacities								Section XIII. Other Process Codes							
Line Number		A. Process Codes (enter code)			B. Process Design Capacity 1. Amount	B. Process Design Capacity 2. Unit of Measure (enter code)	C. Process Total Number of Units	Line Number		A. Process Codes (enter code)			B Process Design Capacity 1. Amount	B Process Design Capacity 2. Unit of Measure (enter code)	C. Process Total Number of Units	D. Process Description
X	1	S	0	2	1,600	G	002	X	1	T	0	4	700	C	001	In situ vitrification
X	2	T	0	3	20	E	001									
X	3	T	0	4	700	C	001									
	1	D	8	0	8.2	F	1		1							
	2	S	0	1	*	*	1		2							
	3								3							
	4								4							
	5								5							
	6								6							
	7								7							
	8								8							
	9								9							
1	0							1	0							
1	1							1	1							
1	2							1	2							
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1	6							1	6							
1	7							1	7							
1	8							1	8							
1	9							1	9							
2	0							2	0							
2	1							2	1							
2	2							2	2							
2	3							2	3							
2	4							2	4							
2	5							2	5							

XIV. Description of Dangerous Wastes

Example for completing this section: A facility will receive three non-listed wastes, then store and treat them on-site. Two wastes are corrosive only, with the facility receiving and storing the wastes in containers. There will be about 200 pounds per year of each of these two wastes, which will be neutralized in a tank. The other waste is corrosive and ignitable and will be neutralized then blended into hazardous waste fuel. There will be about 100 pounds per year of that waste, which will be received in bulk and put into tanks.

Line Number	A. Dangerous Waste No.				B. Estimated Annual Quantity of Waste	C. Unit of Measure	D. Processes (1) Process Codes									(2) Process Description [If a code is not entered in D (1)]
X 1	D	0	0	2	400	P	S	0	1	T	0	1				
X 2	D	0	0	1	100	P	S	0	2	T	0	1				
X 3	D	0	0	2												Included with above
1	D	0	0	2	20,000,000	K	D	8	0							Includes Debris
2	D	0	0	4		K	D	8	0							Includes Debris
3	D	0	0	5		K	D	8	0							Includes Debris
4	D	0	0	6		K	D	8	0							Includes Debris
5	D	0	0	7		K	D	8	0							Includes Debris
6	D	0	0	8		K	D	8	0							Includes Debris
7	D	0	0	9		K	D	8	0							Includes Debris
8	D	0	1	0		K	D	8	0							Includes Debris
9	D	0	1	1		K	D	8	0							Includes Debris
10	D	0	1	8		K	D	8	0							Includes Debris
11	D	0	1	9		K	D	8	0							Includes Debris
12	D	0	2	2		K	D	8	0							Includes Debris
13	D	0	2	8		K	D	8	0							Includes Debris
14	D	0	2	9		K	D	8	0							Includes Debris
15	D	0	3	0		K	D	8	0							Includes Debris
16	D	0	3	3		K	D	8	0							Includes Debris
17	D	0	3	4		K	D	8	0							Includes Debris
18	D	0	3	5		K	D	8	0							Includes Debris
19	D	0	3	6		K	D	8	0							Includes Debris
20	D	0	3	8		K	D	8	0							Includes Debris
21	D	0	3	9		K	D	8	0							Includes Debris
22	D	0	4	0		K	D	8	0							Includes Debris
23	D	0	4	1		K	D	8	0							Includes Debris
24	D	0	4	3		K	D	8	0							Includes Debris
25	W	T	0	1		K	D	8	0							Includes Debris

EPA/State ID Number	W	A	7	8	9	0	0	0	8	9	6	7

Continuation of Section XIV. Description of Dangerous Waste

Line Number	A. Dangerous Waste No. (enter code)	B. Estimated Annual Quantity of Waste	C. Unit of Measure (enter code)	D. Process (1) Process Codes (enter)	D. Process (2) Process Description [If a code is not entered in D (1)]
26	WT02		K	D80	Includes Debris
27	WP01		K	D80	Includes Debris
28	WP02		K	D80	Includes Debris
29	F001		K	D80	Includes Debris
30	F002		K	D80	Includes Debris
31	F003		K	D80	Includes Debris
32	F004		K	D80	Includes Debris
33	F005		K	D80	Includes Debris
34	F039		K	D80	Includes Debris
35	D001	600,000*	K	S01*	Includes Debris
36	D002		K	S01*	Includes Debris
37	D003		K	S01*	Includes Debris
38	D004		K	S01*	Includes Debris
39	D005		K	S01*	Includes Debris
40	D006		K	S01*	Includes Debris
41	D007		K	S01*	Includes Debris
42	D008		K	S01*	Includes Debris
43	D009		K	S01*	Includes Debris
44	D010		K	S01*	Includes Debris
45	D011		K	S01*	Includes Debris
46	D018		K	S01*	Includes Debris
47	D019		K	S01*	Includes Debris
48	D022		K	S01*	Includes Debris
49	D028		K	S01*	Includes Debris
50	D029		K	S01*	Includes Debris
51	D030		K	S01*	Includes Debris
52	D033		K	S01*	Includes Debris
53	D034		K	S01*	Includes Debris
54	D035		K	S01*	Includes Debris
55	D036		K	S01*	Includes Debris
56	D038		K	S01*	Includes Debris
57	D039		K	S01*	Includes Debris
58	D040		K	S01*	Includes Debris

EPA/State ID Number	W	A	7	8	9	0	0	0	8	9	6	7

Continuation of Section XIV. Description of Dangerous Waste

Line Number	A. Dangerous Waste No. (enter code)				B. Estimated Annual Quantity of Waste	C. Unit of Measure (enter code)	D. Process									(2) Process Description [If a code is not entered in D (1)]
							(1) Process Codes (enter)									
59	D	0	4	1		K	S	0	1*							Includes Debris
60	D	0	4	3		K	S	0	1*							Includes Debris
61	W	T	0	1		K	S	0	1*							Includes Debris
62	W	T	0	2		K	S	0	1*							Includes Debris
63	W	P	0	1		K	S	0	1*							Includes Debris
64	W	P	0	2		K	S	0	1*							Includes Debris
65	F	0	0	1		K	S	0	1*							Includes Debris
66	F	0	0	2		K	S	0	1*							Includes Debris
67	F	0	0	3		K	S	0	1*							Includes Debris
68	F	0	0	4		K	S	0	1*							Includes Debris
69	F	0	0	5		K	S	0	1*							Includes Debris
70	F	0	3	9		K	S	0	1*							Includes Debris
71																
72																
73																
74																
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XV. Map

Attach to this application a topographic map of the area extending to at least one (1) mile beyond property boundaries. The map must show the outline of the facility; the location of each of its existing and proposed intake and discharge structures; each of its dangerous waste treatment, storage, recycling, or disposal units; and each well where fluids are injected underground. Include all springs, rivers, and other surface water bodies in this map area, plus drinking water wells listed in public records or otherwise known to the applicant within ¼ mile of the facility property boundary. The instructions provide additional information on meeting these requirements.

Topographic map is located in the Ecology Library

XVI. Facility Drawing

All existing facilities must include a scale drawing of the facility (refer to instructions for more detail).

XVII. Photographs

All existing facilities must include photographs (aerial or ground-level) that clearly delineate all existing structures; existing storage, treatment, recycling, and disposal areas; and sites of future storage, treatment, recycling, or disposal areas (refer to instructions for more detail).

XVIII. Certifications

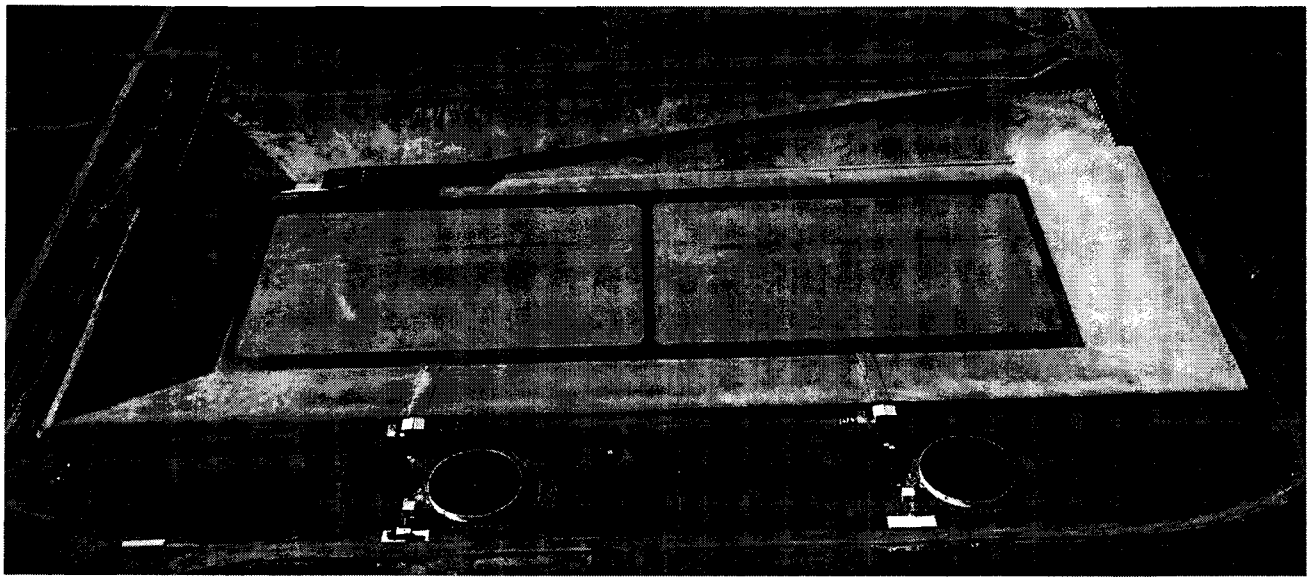
I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

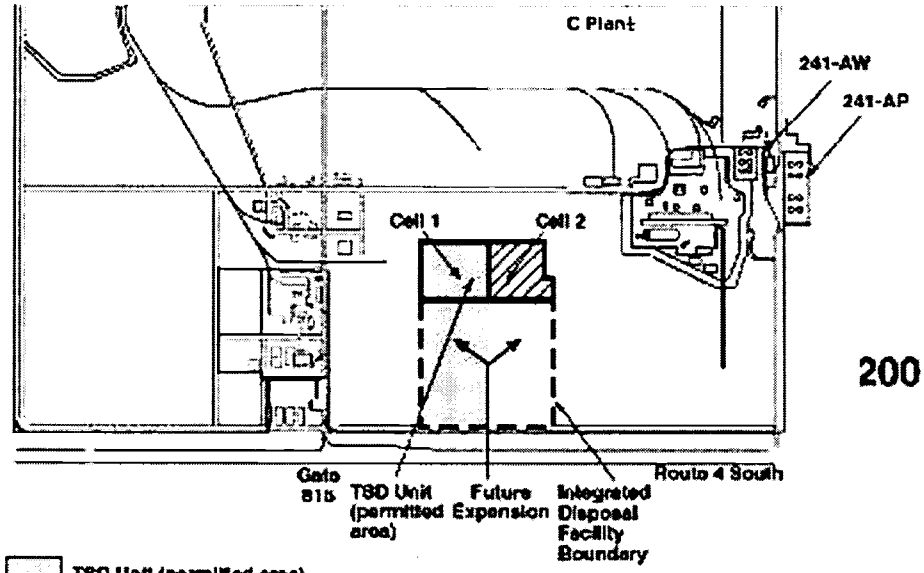
Operator Name and Official Title (type or print) David A. Brockman, Manager U.S. Department of Energy Richland Operations Office	**Signature** Doug A. Sloop for	**Date Signed** 9/19/08
Co-Operator* Name and Official Title (type or print) John G. Lehew, III President and Chief Executive Officer CH2M HILL Plateau Remediation Company	**Signature**	**Date Signed** 9/2/08
Co-Operator – Address and Telephone Number* P.O. Box 1600 Richland, WA 99352 (509) 376-0556		
Facility-Property Owner Name and Official Title (type or print) David A. Brockman, Manager U.S. Department of Energy Richland Operations Office	**Signature** Doug A. Sloop for	**Date Signed** 9/19/08

Comments

In Section VII. Facility Operator Information, there is no change to DOE as the Facility Owner/Operator; only a change in Co-Operator*. The change in Co-Operator* will be effective October 1, 2008.

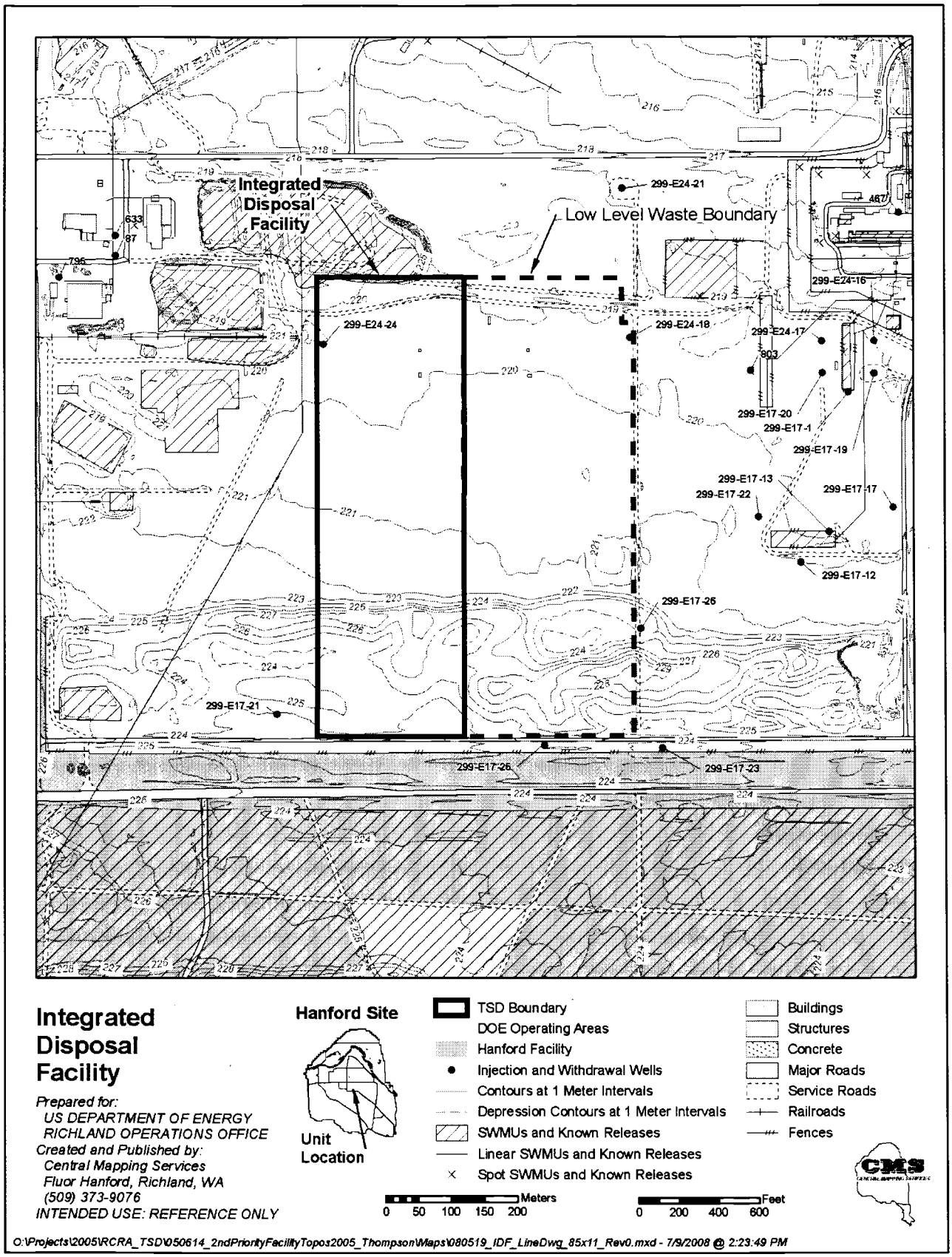
Integrated Disposal Facility





200 East Area

IDF 200 East Area Locational References


Integrated Disposal Facility
Low Level Waste Boundary
299-E24-21
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299-E17-1
299-E17-19
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299-E17-17
299-E17-12
299-E17-25
299-E17-21
299-E17-26
299-E17-23
633
87
796
487
803
Integrated Disposal Facility
Prepared for:
US DEPARTMENT OF ENERGY
RICHLAND OPERATIONS OFFICE
Created and Published by:
Central Mapping Services
Fluor Hanford, Richland, WA
(509) 373-9076
INTENDED USE: REFERENCE ONLY
Hanford Site
Unit Location
TSD Boundary
DOE Operating Areas
Hanford Facility
Injection and Withdrawal Wells
Contours at 1 Meter Intervals
Depression Contours at 1 Meter Intervals
SWMUs and Known Releases
Linear SWMUs and Known Releases
Spot SWMUs and Known Releases
Buildings
Structures
Concrete
Major Roads
Service Roads
Railroads
Fences
Meters
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Feet
0 200 400 600
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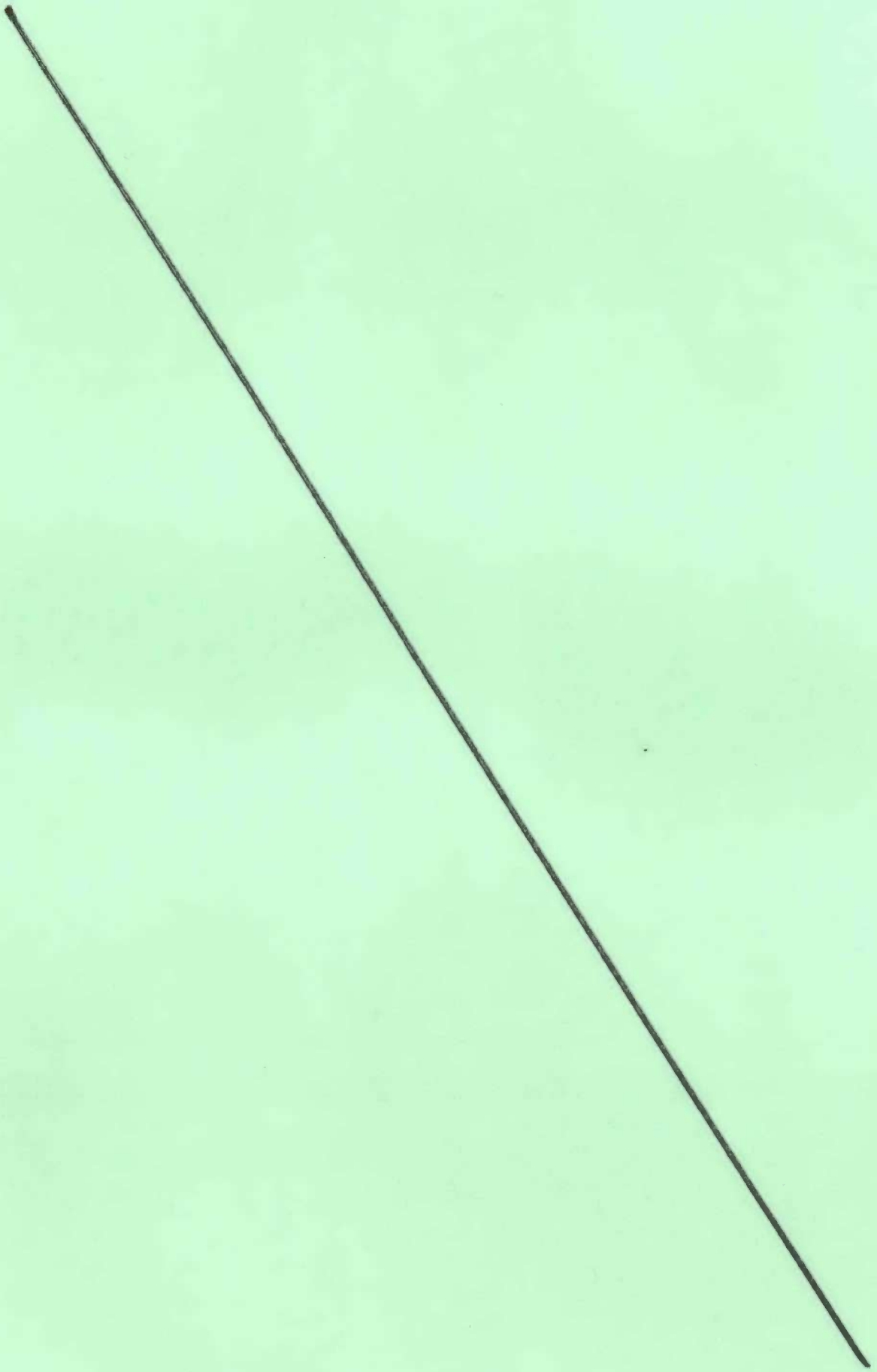

CHAPTER 2.0
TOPOGRAPHIC MAP DESCRIPTION

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CHAPTER 2.0
TOPOGRAPHIC MAP DESCRIPTION

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2.0 TOPOGRAPHIC MAP DESCRIPTION

2.1 Introduction

A topographic map is located in the Department of Ecology library (3100 Port of Benton Boulevard, Richland, WA 99354) reflecting general topographic requirements and the area set aside for IDF. The actual dimensions and waste volume capacity of the Resource Conservation and Recovery Act (RCRA) trench that is being permitted are described in the Part A and Appendix 4A of the IDF portion of this permit. The IDF is located on the Hanford Facility, which limits the use of surrounding land to Department of Energy activities. There are no surface waters in the area defined on the topographical map. Chapter 5.0 includes figures that reflect additional requirements for topographic maps. For the point of compliance and proposed groundwater wells see Figure 5-8, and for the aquifer location see Figure 5-4 and Section 5.3 for the identification of the aquifer.

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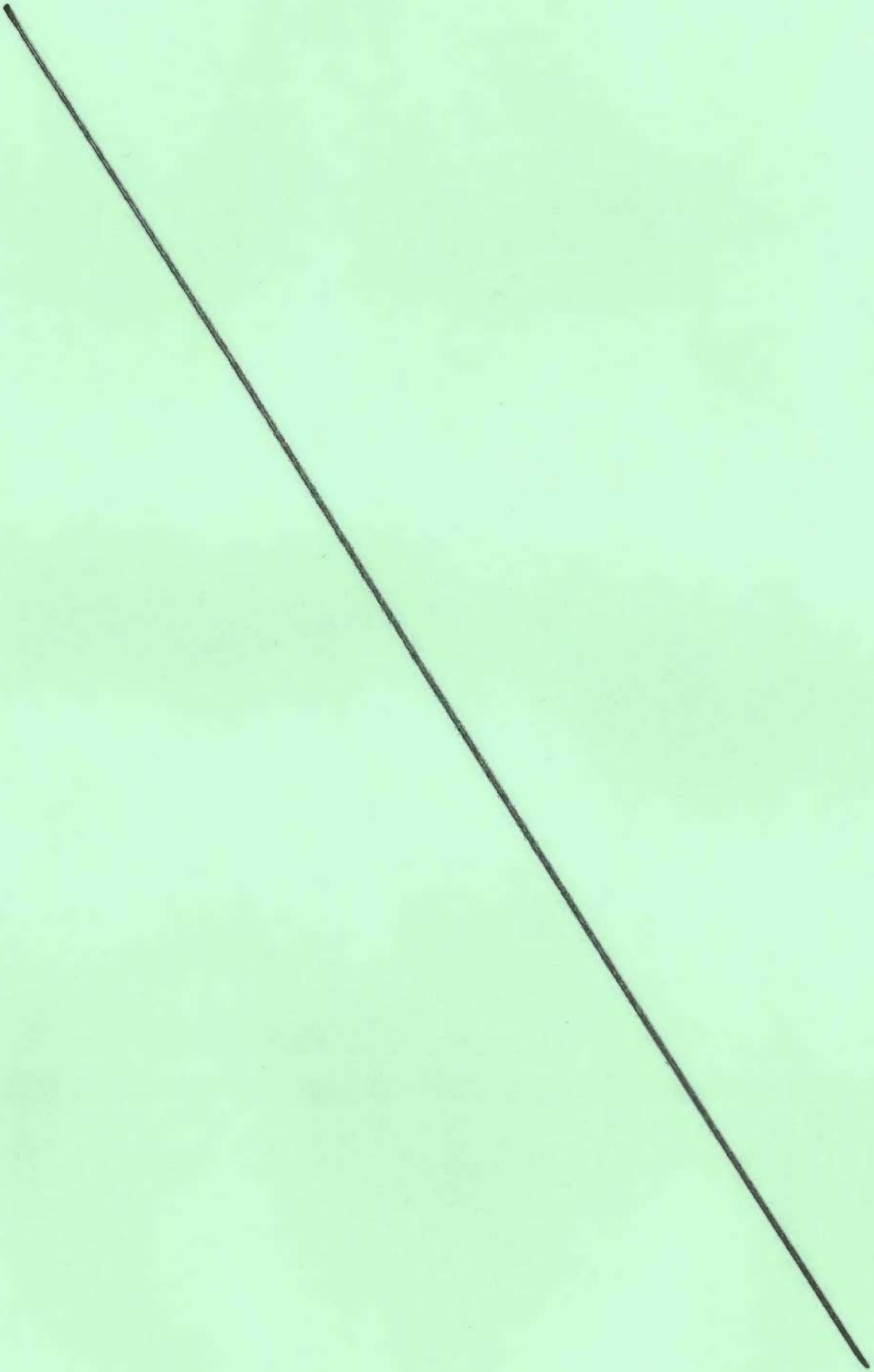

CHAPTER 3.0
WASTE ANALYSIS PLAN

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CHAPTER 3.0
WASTE ANALYSIS PLAN

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WASTE ANALYSIS [C]

This chapter provides information on the chemical, biological, and physical characteristics of the waste treated for disposal. The information includes descriptions required by WAC 173-303-300(5) contained in the *Waste Analysis Plan for the Integrated Disposal Facility*.

CHEMICAL, BIOLOGICAL, AND PHYSICAL ANALYSIS [C-1]

The primary mission of the IDF will be to dispose of vitrified waste generated on the Hanford Site. This includes vitrified LAW from the RPP-WTP and DBVS, and low-level radioactive waste. Additionally, waste generated through IDF operations will be disposed of in IDF. Waste to be disposed of in IDF is assigned dangerous waste numbers found in Chapter 1.0.

WASTE ANALYSIS PLAN [C-2]

The *Waste Analysis Plan for the Integrated Disposal Facility* summarizes waste acceptance processes and contains the following information: unit description, confirmation process, selection of waste analysis parameters, selection of sampling procedures, selection of a laboratory, laboratory testing, and analytical methods, selection of waste re-evaluation frequencies, special procedural requirements, and recordkeeping requirements.

GLOSSARY

AEA	Atomic Energy Act of 1954
BVW	bulk vitrification waste
CAP	corrective action plan
CFR	Code of Federal Regulations
COLIWASA	composite liquid waste sampler
°C	degree Celsius
DOE-ORP	U.S. Department of Energy, Office of River Protection
DOE-RL	U.S. Department of Energy, Richland Operations Office
DBVS	Demonstration Bulk Vitrification System
DST	double-shell tank
Ecology	Washington State Department of Ecology
IDF	Integrated Disposal Facility
ILAW	immobilized low-activity waste
LDR	land disposal restriction
NDE	nondestructive examination
PPE	personal protective equipment
QA	quality assurance
QC	quality control
RCRA	*Resource Conservation and Recovery Act of 1976*
RCW	Revised Code of Washington
RPP-WTP	River Protection Project-Waste Treatment Plant
SWITS	Solid Waste Information Tracking System
TRU	transuranic
TSCA	*Toxic Substances Control Act of 1976*
TSD	treatment, storage, and/or disposal
WAC	Washington Administrative Code
WAP	waste analysis plan

METRIC CONVERSION CHART

Into metric units			Out of metric units		
If you know	Multiply by	To get	If you know	Multiply by	To get
Length			**Length**		
inches	25.40	millimeters	millimeters	0.03937	inches
inches	2.54	centimeters	centimeters	0.393701	inches
feet	0.3048	meters	meters	3.28084	feet
yards	0.9144	meters	meters	1.0936	yards
miles (statute)	1.60934	kilometers	kilometers	0.62137	miles (statute)
Area			**Area**		
square inches	6.4516	square centimeters	square centimeters	0.155	square inches
square feet	0.09290304	square meters	square meters	10.7639	square feet
square yards	0.8361274	square meters	square meters	1.19599	square yards
square miles	2.59	square kilometers	square kilometers	0.386102	square miles
acres	0.404687	hectares	hectares	2.47104	acres
Mass (weight)			**Mass (weight)**		
ounces (avoir)	28.34952	grams	grams	0.035274	ounces (avoir)
pounds	0.45359237	kilograms	kilograms	2.204623	pounds (avoir)
tons (short)	0.9071847	tons (metric)	Tons (metric)	1.1023	tons (short)
Volume			**Volume**		
ounces (U.S., liquid)	29.57353	milliliters	milliliters	0.033814	ounces (U.S., liquid)
quarts (U.S., liquid)	0.9463529	liters	liters	1.0567	quarts (U.S., liquid)
gallons (U.S., liquid)	3.7854	liters	liters	0.26417	gallons (U.S., liquid)
cubic feet	0.02831685	cubic meters	cubic meters	35.3147	cubic feet
cubic yards	0.7645549	cubic meters	cubic meters	1.308	cubic yards
Temperature			**Temperature**		
Fahrenheit	subtract 32 then multiply by 5/9ths	Celsius	Celsius	multiply by 9/5ths, then add 32	Fahrenheit
Energy			**Energy**		
kilowatt hour	3,412	British thermal unit	British thermal unit	0.000293	kilowatt hour
kilowatt	0.94782	British thermal unit per second	British thermal unit per second	1.055	kilowatt
Force/Pressure			**Force/Pressure**		
pounds (force) per square inch	6.894757	kilopascals	kilopascals	0.14504	pounds per square inch

06/2001

Source: *Engineering Unit Conversions*, M. R. Lindeburg, PE., Third Ed., 1993, Professional Publications, Inc., Belmont, California.

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3.0 INTEGRATED DISPOSAL FACILITY WASTE ANALYSIS PLAN

Pursuant to WAC 173-303-300(5) this waste analysis plan (WAP) documents the waste acceptance process, sampling methodologies, analytical techniques, and overall processes that will be undertaken for mixed waste accepted for disposal at the Integrated Disposal Facility (IDF). . Mixed waste disposed at the IDF will be limited to vitrified low-activity waste (LAW) from the RPP-WTP and DBVS and mixed waste generated by IDF operations. (see Chapter 1, Part A Form). Vitrified LAW generated by RPP-WTP is known as Immobilized Low Activity Waste (ILAW) and generated by DBVS is known as Bulk Vitrified Waste (BVW). The IDF will be located in the 200 East Area of the Hanford Facility.

The IDF also will receive low-level waste for disposal. Mixed waste will not be placed in the low-level waste portion of the IDF. The requirements of this WAP are applicable to mixed waste and are not applicable to the low-level radioactive waste. The term 'treatment, storage, and/or disposal (TSD) unit' is used throughout this WAP to refer to the IDF. Activities will be performed by the IDF operating organization, waste acceptance organization, or its delegated representative.

Although the treatment and disposal of radioactive waste (i.e., source, special nuclear, and by-product materials as defined by the *Atomic Energy Act of 1954*) are not within the scope of *Resource Conservation and Recovery Act* (RCRA) *of 1976* or WAC 173-303, information is provided for general knowledge.

3.1 Description of Unit Processes and Activities

The IDF will be a single, expandable disposal facility constructed to RCRA Subtitle C standards, half of which is for disposal of mixed waste the other half will be for disposal of low-level waste. Initial capacity for mixed waste disposal is 82,000 cubic meters of waste with an ultimate capacity of up to 450,000 cubic meters of waste. Disposal capacity beyond the initial 82,000 cubic meters will require a modification to the Part B Permit. The mixed waste types to be disposed in the IDF include vitrified LAW from the RPP-WTP and DBVS. Additionally, mixed waste generated by IDF operations will be disposed of in IDF.

The mission of the RCRA portion of the IDF is to provide an approved disposal facility for the permanent, environmentally safe disposition of mixed waste and RCRA waste.

For ILAW, and BVW the container packaging and handling will be designed to maintain containment of each waste type, limit intrusion, and limit human exposure at the IDF. ILAW containers will be transported from the RPP-WTP to the IDF using a tractor-trailer system. BVW will be transported from the DBVS staging area to IDF using a similar system. Transport of the ILAW and BVW to the landfill will occur along a pre-determined route.

The lined landfill will have a leachate collection and removal system. The leachate collection tanks will be operated in accordance with the generator provisions of WAC 173-303-200 and are not subject to this WAP.

Additional information is located in Chapter 1.0 (IDF Part A) and Chapter 4.0 (Process Information).

3.2 Identification and Classification of Waste

The ILAW, BVW, and newly generated mixed waste will be accepted for disposal. The mixed waste disposed of at the IDF is received from waste generated within IDF, and two other Hanford Facility TSD units (RPP-WTP and DBVS). The following waste will not be accepted for disposal at this TSD unit:

- Waste is not accepted for disposal when the waste contains free-standing liquid unless all free-standing liquid:
  - Has been removed by decanting or other methods.
  - Has been mixed with sorbent or stabilized (solidified) so that free-standing liquid is no longer observed.
  - Otherwise has been eliminated.

- Container is very small, such as an ampoule.
- Container is a labpack and is disposed in accordance with WAC 173-303-161 or 40 CFR 264.316.
- Container is designed to hold free liquids for use other than storage, such as a battery or capacitor.

There could be cases in which small amounts of residual liquids are present in mixed waste containers because condensate has formed following packaging or free liquids remain in debris items (e.g., pumps, tubing) even after draining. When it is not practical to remove this residual liquid, the free liquid must be eliminated to the extent possible by adding a quantity of sorbent sufficient to sorb all residual liquids.

Free liquid is determined by SW-846, *Test Methods for Evaluating Solid Waste: Physical/Chemical Method*, Method 9095 (Paint Filter Liquids Test) [WAC 173-303-140(4)(b) and 40 CFR 264.314(d)] only for waste that has the potential for free liquid formation.

- Gaseous waste not accepted for disposal if the is waste packaged at a pressure in excess of 1.5 atmospheres at 20°C.
- Pyrophoric waste is not accepted for disposal. Waste containing less that 1 weight percent pyrophoric material partially or completely dispersed in each package is not considered pyrophoric for the purposes of this requirement.
- Solid acid waste is not accepted for disposal [WAC 173-303-140(4)(c)].
- Extremely hazardous waste that does not meet WAC 173-303-140(4)(d) is not accepted for disposal. Extremely hazardous waste that has been treated could be disposed in accordance with Revised Code of Washington (RCW) 70.105.050(2), "Hazardous Waste Management."
- Organic/carbonaceous waste that does not meet WAC 173-303-140(4)(d) is not accepted for disposal.
- Waste not meeting the LDR treatment standards is not accepted for disposal [40 CFR 268 and WAC 173-303-140(4)].
- Waste streams will be evaluated during pre-shipment review to ensure that the waste streams do not contain constituents incompatible with the liner system in concentration sufficient to degrade the liner. Table 1 provides a list of chemicals shown to be incompatible with the liner material at 100% concentrations (WHC-SD-WM-TI-714). In general, mixed waste that meets federal and state treatment standards would be compatible with the TSD unit liner system. Waste accepted at the IDF will be compatible with the liner. Constituents in Table 1 will not be accepted for disposal (refer to Section 3.4.9 for waste stream compatibility).

3.3 Management of Waste

The ILAW, BVW, and newly generated wastes (see Section 3.3.1) generated during normal operations of this TSD unit are accepted at this TSD unit for disposal. The two onsite TSD units (RPP-WTP and DBVS) transferring/shipping waste to this TSD unit hereafter are referred to as the 'generator' unless otherwise denoted in this WAP. The waste acceptance process for transfers from the generator is identified in Figure 1.

Written waste tracking procedure(s) are implemented to ensure waste received at the TSD unit matches the manifest or transfer papers, to ensure that the waste is tracked though the TSD unit to final disposition, and to maintain the information required in WAC 173-303-380. The waste tracking process provides a mechanism to track waste through a uniquely identified container. The unique identifier is a barcode (or equivalent) that is recorded in the Solid Waste Information Tracking System (SWITS). This mechanism encompasses the waste acceptance process, the movement of waste, the processing of waste, and management of the waste.

The container identification number provides traceability between the TSD unit and the hard copy of records that are maintained as part of the operating record to ensure information relative to the location, quantity, and physical and chemical characteristics of the waste are available.

The following sections describe the process for waste acceptance and the different types of information and knowledge reviewed/required during the acceptance process. The process for management of waste is described in Section 3.4.

3.3.1 Newly Generated Waste within the IDF

This TSD unit generates mixed waste as a result of operational (e.g., chemical, radiological) activities. These activities include, transfer functions along with inspection, decontamination, cleanup, maintenance tasks and leachate collection. The IDF generated operational waste will be maintained in accordance with generator provisions of WAC 173-303-200 and WAC 173-303-600(3)(d). Any newly generated waste (except leachate) not meeting IDF waste acceptance criteria will be designated and sent to another permitted TSD or to a 90 day accumulation area. IDF leachate will be managed in accordance with WAC 173-303-200 and transferred to the Liquid Effluent Retention Facility/Effluent Treatment Facility (LERF/ETF) (or other permitted TSD) for treatment. Solids or residuals resulting from IDF leachate treatment may be designated/packaged and sent back to the IDF for burial or to another permitted TSD.

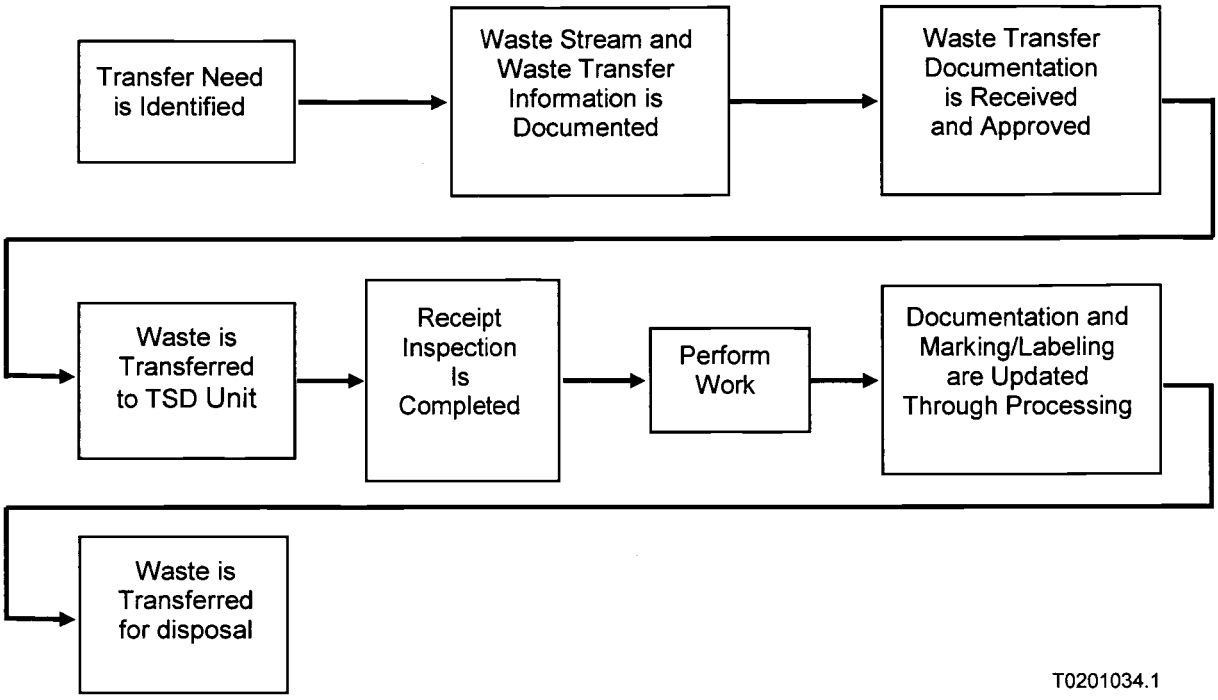


Figure 1. Waste Transfers and Analysis Plan Onsite TSD Units Flow Diagram.

Table 1. Chemicals Incompatible with the High Density Polyethylene Liner (in concentrated form)*

Chemical	CAS Number
Amyl chloride	543-59-9
Aqua regia	8007-56-5
Bromic acid	15541-45-4
Bromobenzene	108-86-1
Bromoform	75-25-2
Calcium bisulfite	13780-03-5
Calcium sulfide	20548-54-3
Diethyl benzene	25340-17-4
Diethyl ether	60-29-7
Bromine	7726-95-6
Chlorine	7782-50-5
Fluorine	7782-41-4
Ethyl chloride	75-00-3
Ethylene trichloride	79-01-6
Nitrobenzene	98-95-3
Perchlorobenzene	118-74-1
Propylene dichloride	78-87-5
Sulfur trioxide	7446-11-9
Sulfuric acid (fuming)	8014-95-7
Thionyl chloride	7719-09-7
Vinylidene chloride.	75-35-4

CAS = Chemical Abstract Service.
* WHC-SD-WM-TI-714

3.4 Confirmation Process

WAC 173-303-300(1) requires confirmation on mixed waste before acceptance of waste into a waste management unit. The confirmation process consists of two parts, pre-shipment review, and verification. Confirmation activities are performed in accordance with TSD unit-specific governing documentation. The confirmation process is detailed in Figure 2 for ILAW and BVW.

3.4.1 Pre-Shipment Review

Pre-shipment review takes place before waste can be scheduled for transfer or shipment to this TSD unit. The review focuses on whether the waste stream is defined accurately and meets the TSD unit waste acceptance criteria and whether the LDR status is determined correctly. Only waste determined to be acceptable for storage (see Section 3.4.1.2.2) and/or disposal is scheduled. This determination is based on the information provided by the generator. The pre-shipment review consists of waste stream approval and the waste shipment approval process. The following sections discuss the pre-shipment review process. The information obtained during the pre-shipment review, at a minimum, includes all information necessary to safely dispose of the waste. The pre-shipment review ensures the waste is characterized and the data provided qualify as 'acceptable knowledge' (Section 3.4.1.4).

3.4.1.1 Pre-Shipment Review of Wastes

Pre-shipment review for ILAW and BVW waste containers will take place at RPP-WTP and the DBVS staging area respectively before either type of containers can be scheduled for transfer to the IDF. The review will focus on whether the waste stream is defined accurately, meets the waste acceptance criteria, and the land disposal restrictions (LDR) status was determined correctly. Only waste determined to be acceptable for storage (see Section 3.4.1.2.2) and/or disposal will be scheduled. This determination will be based on the information provided by the generator. The pre-transfer review will consist of the waste profile documentation and waste transfer approval process. The following sections discuss the pre-transfer review process. ILAW and BVW containers received for land disposal will be at least 90% full. The information obtained from the generator, at a minimum, will contain five elements:

(1) Documentation to ensure waste can be managed pursuant to the Part A, Form 3.

(2) Documentation to ensure the waste is not a prohibited waste in accordance with Section 3.2,

(3) A determination if the waste is an ignitable, reactive, or incompatible waste as defined in WAC 173-303-040.

(4) Documentation that waste meets LDR requirements of 40 CFR 268 and WAC 173-303-140.

(5) Operational restrictions on acceptance of waste.

During the waste profile documentation process for ILAW and BVW containers, the generator will have the responsibility to provide relevant information pertaining to the proper management of the waste. Characterization information pertaining to the treatment of ILAW and BVW will be obtained during the waste profile documentation process.

3.4.1.2 Waste Stream Approval Process for Wastes

The waste stream approval process consists of reviewing stream information supplied on a waste stream profile and supporting documentation to allow receipt of the waste into the IDF. Waste stream compatibility (i.e., compatibility between individual waste streams and compatibility between waste streams and landfill design and construction parameters) will be assessed on a case-by-case basis. Criteria for assessing and determining compatibility will be identified in either the facility Waste Acceptance Criteria, Waste Analysis Plan, or other protocol or procedure as appropriate.

3.4.1.2.1 Waste Stream Approval for ILAW and BVW

During the waste profile documentation process, the IDF waste acceptance organization will obtain the following information:

- Description of waste generating process
- Characterization data
- Dangerous waste numbers
- LDR data (as specified in Section 3.4.10)
- Composition of ILAW and BVW including regulated constituents of concern (refer to Chapter 1.0 of the permit application -Part A Form)

The waste profile documentation process will be as follows.

1. Appropriate generator fills out waste profile documentation.
2. The IDF designated waste acceptance organization reviews the waste profile information against the waste acceptance criteria for each ILAW or BVW transfer.
3. If discrepancies are noted, the IDF designated waste acceptance organization requests additional information from the generator to address discrepancies for either: (1) inconsistent information and (2) information not constituting acceptable knowledge (refer to Section 3.4.1.4.1).

Information (waste profile documentation) is resubmitted by the generator addressing concerns in Item 3.

- If concerns are addressed, waste profile documentation is approved.
- If concerns are not addressed and met, waste profile documentation is not approved until concerns are corrected.

3.4.1.2.2 Waste Stream Approval for Newly Generated Mixed Waste

The waste stream approval process for wastes generated during IDF operations (except for leachate) consists of reviewing stream information supplied on a waste stream profile and supporting documentation. The waste stream profile requires the following supporting documentation:

- Generator information (e.g., name, address, point-of-contact, telephone number)
- Waste stream name
- Waste generating process description
- Waste numbers
- Chemical characterization information (e.g., characterization method(s), chemicals present, concentration ranges)
- Designation information
- LDR information including identification of underlying hazardous constituents if applicable
- Waste type information (e.g., physical state, adsorbents used, inert materials, stabilizing agents used)
- Packaging information (e.g., container type, maximum weight, size).

Attachments could consist of container drawings, process flow information, analytical data, etc.

In some cases, such as variable waste streams, the waste stream profile information could be general in nature. In these cases, more detailed information is gathered during the waste shipment approval process on a per shipment basis. This information is reviewed against the TSD unit waste acceptance criteria to ensure the waste is acceptable for receipt. If conformance issues are found during this review, additional information is requested that could include analytical data or a sample to be analyzed.

If the waste cannot be received, the TSD unit pursues acceptance of the waste at an alternate TSD unit. Once the waste meets the waste acceptance criteria, the TSD unit assigns the profile to a waste specification record and establishes a waste verification frequency based on the requirements found in Section 3.4.2. Profile information is re-evaluated as discussed in Section 3.4.7.

3.4.1.3 Waste Transfer/Shipment Approval Process

After the appropriate generator has received the waste profile documentation approval from IDF (refer to Section 3.4.1.2.1), the generator waste transfer will be subjected to the waste transfer approval process. Only those ILAW and BVW containers approved under the waste profile documentation as part of the waste transfer approval process will be transferred to the IDF. During the waste transfer approval process, the IDF designated waste acceptance organization will obtain the following information.

For each ILAW or BVW container transfer that is a candidate for disposal in the TSD unit, the generator will provide the following information:

- Container identification number
- Profile number
- Waste description
- Generator information (e.g., name, address, point-of-contact, telephone number)
- Container information (e.g., type, size, weight)
- Waste numbers
- LDR certification
- Packaging materials and quantities

The ILAW and BVW container transfer approval process will be as follows.

1. The generator obtains information from existing database, operating record, or generator records on each ILAW container to be transferred under the approved waste profile documentation.
2. Information is submitted to the TSD unit designated waste acceptance organization by the generator and is reviewed for the following:
   - Consistency with approved waste profile documentation.
   - Consistency with waste acceptance criteria within the IDF.
3. If discrepancies are identified, the TSD unit designated waste acceptance organization will request additional information from the generator to address any discrepancies.
4. Information (waste package documentation) is resubmitted by the generator addressing concerns in Item 3.
5. If discrepancies are addressed, this information is forwarded to the TSD waste acceptance organization.
6. If discrepancies are not addressed, transfer is not approved until discrepancies are corrected.

3.4.1.4 Acceptable Knowledge Requirements

The TSD unit ensures that all information used to make waste management decisions is based on adequate characterization data as described in the following sections. The TSD unit evaluates the data to ensure that the data are adequate acceptable knowledge for management of the waste.

3.4.1.4.1 General Acceptable Knowledge Requirements

One or more of the following types of information could be considered, provided that the information is of sufficient quality to demonstrate compliance with applicable waste acceptance criteria:

- Mass balance from a controlled process that has a specified output for a specified input.
- Material safety data sheet on chemical products.
- Test data from a surrogate sample.
- Analytical data on the waste or a waste from a similar process.

In addition, acceptable knowledge requirements can be met using a combination of analytical data or screening results and one and/or more of the following information:

- Interview information
- Logbooks
- Procurement records
- Qualified analytical data
- Radiation work package
- Procedures and/or methods
- Process flow charts
- Inventory sheets
- Vendor information
- Mass balance from an uncontrolled process (e.g., spill cleanup)
- Mass balance from a process with variable inputs and outputs (e.g., washing/cleaning methods)

If the information is sufficient to quantify the constituents of regulatory concern and to determine waste characteristics as required by the regulations and TSD unit waste acceptance criteria, the information is considered acceptable. Adequate acceptable knowledge includes (1) general waste knowledge requirements and/or (2) LDR waste knowledge requirements.

(1) General waste knowledge requirements. At a minimum, the generator supplies enough information for the waste to be managed at this TSD unit (refer to Section 3.4.1.1). The minimum level of acceptable knowledge consists of designation data where the constituents causing a waste number to be assigned are quantified and that data address any TSD unit operational parameters necessary for proper management of the waste.

When process knowledge indicates that constituents, which if present in the waste might cause the waste to be regulated, are input to a process, but not expected to be in the waste, sampling and analysis must be performed to ensure the constituents do not appear in the waste above applicable regulatory levels. This requirement can be met through chemical screening. This sampling and analysis are required only for initial characterization of the waste stream.

When the available information does not qualify as acceptable knowledge or is not sufficient to characterize a waste for management, the sampling and testing methods outlined in WAC 173-303-110 are used to determine whether a waste designates as ignitable, corrosive, reactive, and/or toxic and whether the waste contains free liquids as applicable. If the analysis is performed to complete characterization after acceptance of the waste by the TSD unit, this WAP governs the sampling and testing requirements.

(2) LDR waste knowledge. The TSD unit operating record contains all information required to document that the appropriate treatment standards have been met or will be met after the waste is treated unless otherwise excepted in this section.

- Both ILAW and BVW will be LDR compliant waste streams prior to acceptance at the IDF. Vitrification at the WTP and DBVS will facilitate LDR compliance for the majority of the mixed waste disposed of at IDF. IDF operational waste will be treated as needed to meet LDR at another TSD other than WTP or the DBVS
- This TSD unit may use analytical data as necessary to ensure that the applicable requirements found in 40 CFR 268.7 and WAC 173-303-140(4) are met.

3.4.1.4.2 Methodology to Ensure Compliance with LDR Requirements

The generators are subject to LDR requirements and are required to submit all information notifications and certifications described in WAC 173-303-380(1), (j), (k), (n), and (o). Mixed waste not meeting the treatment standards cannot be disposed at this TSD unit.
The following are general requirements for certification or information notification.

- The waste is subject to LDR and the waste has been treated. The generator supplies the appropriate LDR certification information (40 CFR 268).
- The waste is subject to LDR and the generator has determined that the waste meets the LDR as generated. The generator develops the certification based on process knowledge and/or analytical data and supplies the appropriate LDR certification information necessary to demonstrate compliance with the LDR treatment standards of 40 CFR 268 and WAC 173-303-140. State-only LDRs do not require this type of certification.

When demonstrating that a concentration-based LDR treatment standard has been met, a representative sample of the waste must be submitted for analysis. This sample could be taken by the treatment facility or the generator and is required to comply with the LDR treatment standards contained in 40 CFR 268.40 and 268.48 for underlining hazardous constituents.

3.4.2 Verification

Verification is an assessment performed by this TSD unit to substantiate that the waste received is the same as represented by the analysis supplied by the generator for the pre-shipment review. Verification for ILAW and BVW containers will contain one element, a 100% container receipt inspection. Physical/chemical screening will not be performed on the ILAW or BVW containers. Waste is not accepted by the TSD unit for disposal until the required elements of verification have been completed, including evaluation of any data obtained from verification activities. All conformance issues identified during the verification process are resolved in accordance with Section 3.4.3. Verification activity results will be documented by the IDF designated waste acceptance organization.

Sampling and analysis for non-vitrification mixed waste (e.g., treatment residues from treatment of IDF leachate that are returned to IDF for disposal) will not occur at the IDF but will occur at another permitted TSD.

3.4.2.1 Container Receipt Inspection

Container receipt inspection is a mandatory element of the confirmation process.

3.4.2.1.1 Container Receipt Inspection for ILAW and BVW

The ILAW and BVW container receipt inspection will be performed by IDF designated waste acceptance organization. The following criteria will be evaluated during container receipt inspection:

- Number of containers
- Size of containers
- Labels
- Container integrity

Discrepancies identified during the container receipt inspection will be communicated to generator. Discrepancies will be resolved before the containers are unloaded. Once the discrepancies are resolved, the ILAW containers will be unloaded and disposed. Should discrepancies remain unresolved after 30 days, Ecology will be notified and daily walk around inspections conducted.

3.4.2.2 Physical Screening Process

The ILAW and BVW containers are not required to be physically screened because the generator verifies the waste meet the waste acceptance criteria for IDF.

3.4.2.3 Chemical Screening Process

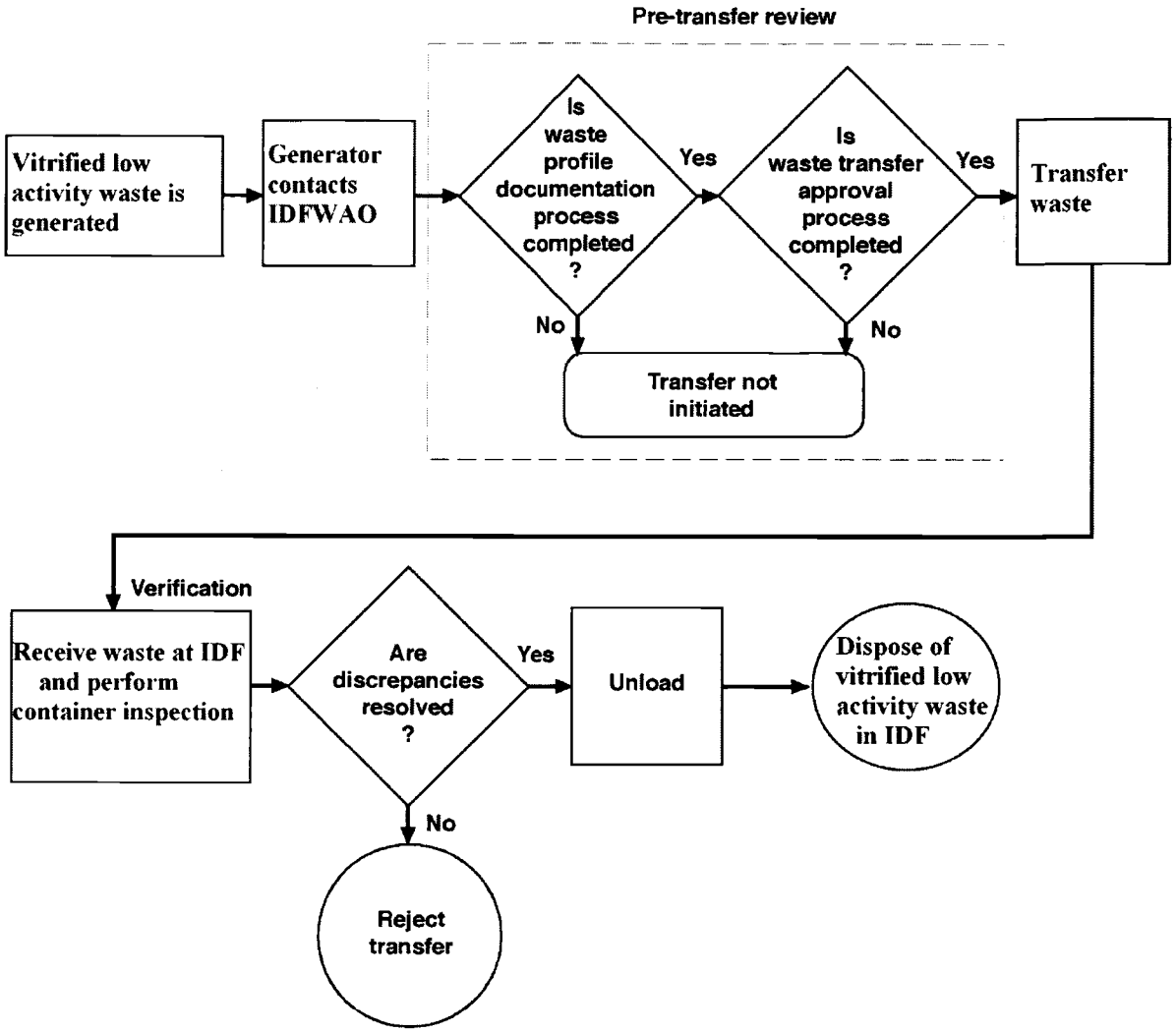
Chemical screening is a verification element for containerized mixed waste. The ILAW and BVW containers are not required to be chemically screened because the generator verifies the waste meet the waste acceptance criteria for IDF.

3.4.3 Waste Acceptance

Initial acceptance of waste occurs only after the confirmation process described in Section 3.4 is complete. Conformance issues identified during the confirmation process are documented and managed in accordance with Section 3.4.1.2.2. Conformance issues that must be corrected before waste acceptance include the following:

- Waste that is not identified in the Part A, Form 3 (Chapter 1.0).
- Waste does not match approved profile documentation.
- Designation, physical, and/or chemical characterization discrepancy.
- Incorrect LDR paperwork.
- Packaging discrepancy.
- Manifest discrepancies as described in WAC 173-303-370(4).

For waste shipments with unresolved conformance issue(s) that exceed 90 days, this TSD will notify Ecology at least once per calendar quarter.



IDF = Integrated Disposal Facility
IDFWAO = IDF Waste Acceptance Organization

Figure 2. Vitrification or Alternative Method Transfer and Waste Analysis Plan Process Flow Diagram

3.4.4 Selecting Waste Analysis Parameters

The ILAW and BVW containers will be managed without the need to perform sampling and analysis at the TSD. No parameters will be required to be identified.

Table 2. Parameters and Rationale for Physical Screening

Parameter	Method*	Rationale for selection
Nondestructive examination	Field method	Confirm consistency between waste and shipping documentation.

*Procedures based on manufacturer's recommended methodology unless otherwise noted. When regulations require a specific method, the method is followed.

SW-846, *Test Methods for Evaluating Solid Waste*, latest edition, U.S. Environmental Protection Agency, Washington, D.C.

WAC 173-303, "Dangerous Waste Regulations

3.4.5 Selecting Sampling Procedures

Any required sampling and analysis of the ILAW and BVW containers will be performed at the generator before the containers are closed. Sampling and analysis for IDF operational mixed waste will not occur at the IDF but at another Hanford TSD.

3.4.6 Selecting A Laboratory, Laboratory Testing, and Analytical Methods

Any required sampling and analysis of the ILAW and BVW containers will be performed before the containers are closed at the RPP-WTP and DBVS respectively. No Laboratory, laboratory testing or analytical methods will be required to be identified.

3.4.7 Selecting Waste Re-Evaluation Frequencies

The re-evaluation (repeat and review) frequency for ILAW to review a waste generating process and associated waste profile documentation is every two years, or more often if conditions in WAC 173-303-300(4)(a) arise. Since BVW will be generated over a shorter time period, frequency for review will be every six months.

When a waste generating process and associated waste profile documentation is re-evaluated, IDF personnel or designated waste acceptance organization could request the generator to do one or more of the following:

- Verify the current waste profile documentation is accurate.
- Supply new waste profile documentation.

When a waste profile is re-evaluated, the TSD unit could request the organization generating the waste to do one of the following:

- Verify the current waste profile is accurate.
- Supply a new waste profile.
- Submit a sample for parameter analysis.

3.4.8 Special Waste Analysis Procedural Requirements

Special procedural requirements for the IDF will include procedures for ignitable, reactive, and incompatible waste, and provisions for complying with federal and state LDR requirements. This section discusses any special process requirements for receiving mixed waste at this TSD unit.

3.4.9 Procedures for Ignitable, Reactive, and Incompatible Waste

Waste stream compatibility (i.e., compatibility between individual waste streams and compatibility between waste streams and landfill design and construction parameters) and waste stream ignitability will be assessed on a case-by-case basis. Criteria for assessing and determining compatibility and ignitability will be identified in either the facility Waste Acceptance Criteria, Waste Analysis Plan, or other protocol or procedure as appropriate. Should these wastes be accepted, appropriate administrative and engineering controls will be implemented as necessary.

This TSD unit does not accept reactive waste (refer to Section 3.2 and Section 3.4.1.1). The TSD unit ensures that reactive waste is not accepted at this TSD unit in the following manner.

- Pre-shipment review will identify whether the waste is reactive based on the definition contained in WAC 173-303-040.
- If analysis of the characterization information leads to a conclusion that the waste is a reactive waste, the containers, or waste will not be accepted.

The types of prohibited waste not accepted at this TSD unit as listed in Section 3.2.

3.4.10 Provisions for Complying With Federal and State Land Disposal Restriction Requirements

State-only and federal LDR requirements restrict the land disposal of certain types of waste subject to RCRA and RCW 70.105, "Hazardous Waste Management", as amended. Waste managed on the Hanford Facility falls within the purview of these LDRs per 40 CFR 268 and WAC 173-303-140. The treatment standards for mixed waste disposed at IDF are based on the dangerous waste numbers accepted as documented on the IDF Part A as well as additional information necessary for identifying treatability groups etc.

The IDF will not perform sampling and analysis to determine compliance with treatment standards contained in 40 CFR 268. Any sampling and analysis results required to demonstrate compliance with concentration-based treatment standards contained in 40 CFR 268.40 will be obtained by IDF waste acceptance organization from the generator, during the waste profile documentation process to meet the requirements of 40 CFR 268.7(c)(2). Sampling and analysis results will be placed into the unit-specific portion of the Hanford Facility operating record. Other LDR records are identified in WAC 173-303-380(1)(m) and will be obtained from the generator, by IDF personnel as part of either the waste profile documentation process or the waste transfer approval process. The treated waste must meet all applicable LDRs to be accepted for disposal at the IDF. IDF will obtain the LDR certification from the treatment unit.

Mixed waste constituents that are subject to LDRs are identified in 40 CFR 268.40 by reference in WAC 173-303-140(2), the extremely hazardous waste disposal requirements for DOE facilities contained in RCW 70.105.050(2), and the state-only LDRs contained in WAC 173-303-140(4)(b)-(d). The mixed waste must meet certain treatment standards, as specified in 40 CFR 268.40, RCW 70.105.050(2), and WAC 173-303-140(4)(b)-(d), if the waste is to be land disposed. Any waste requiring LDR treatment must be treated prior to acceptance into the IDF.

State-only LDRs for mixed waste will be met in the following manner:

- Extremely hazardous waste disposal requirements in RCW 70.105.050(2) concerning "all reasonable methods" will be met by the treatment performed to meet 40 CFR 268, WAC 173-303-140(4)(b)-(d), and DOE requirements for disposal. If no treatment is required to meet 40 CFR 268, WAC 173-303-140(4)(b)-(d), or DOE requirements, no treatment is required to dispose of extremely hazardous waste at the IDF.
- Special requirements for bulk and containerized liquids in WAC 173-303-140(4)(b) are identical to the landfill requirements contained in 40 CFR 264.314.

For mixed waste, including the provisions when to perform the paint filter test, these requirements are described in Section 3.2 of the WAP.

- Solid acid waste requirements in WAC 173-303-140(4)(c) can be met through knowledge of the treatment process. Sampling and analysis following treatment is not required to meet this state-only LDR. Disposal of treated solid acid waste still displaying the WSC2 characteristic can occur only when the waste is treated to reduce the harmful properties or characteristics of the waste.
- Organic/Carbonaceous waste prohibition requirements in WAC 173-303-140(4)(d) do not apply to the Hanford Facility because the Hanford Facility is operating under WAC 173-303-140(4)(d)(iii), in accordance with a sitewide 1,609 kilometers (1,000-mile) inapplicability certification. Sampling and analysis is not required to determine the organic/carbonaceous content of a mixed waste.
- Ecology allows treatment of Organic/Carbonaceous waste in lieu of meeting the inapplicability certification requirements WAC-173-303-140(4)(d)(iii) through macro-encapsulation for hazardous debris only.

3.4.11 Off-Specification Waste

Off-Specification ILAW or BVW is waste not meeting the waste acceptance criteria as described in Section 3.4, Confirmation Process. ILAW or BVW streams determined to be off-specification may be temporarily stored in the RCRA lined portion of the IDF pending resolution of discrepancy or return to generating TSD as long as these wastes meet LDR. ILAW and BVW may be temporarily stored in the RCRA lined portion of the IDF, provided the temperature administrative control limit is not exceeded, until sufficiently cool for disposal.

3.5 Waste Tracking

The IDF will monitor and record the placement of waste packages. At the time of final placement of each package, the position and serial number of the package will be logged.

3.6 Recordkeeping

Recordkeeping requirements that will be applicable to this WAP are as follows:

- Confirmation records described in Section 3.4 will be maintained in accordance with Condition II.I.1.b of the Hanford Facility RCRA Permit, Dangerous Waste Portion (Ecology 2001).
- Waste profile documentation described in Section 3.4.1.2.1 will be maintained in accordance with Condition II.I.1.j of the Hanford Facility RCRA Permit, Dangerous Waste Portion.
- LDR records described in Section 3.4.10 will be maintained in accordance with WAC 173-303-380(1)(m) in the IDF unit-specific portion of the Hanford Facility operating record.

3.7 References

Ecology, 2001, "Hanford Facility RCRA Permit, Dangerous Waste Portion", Washington State Department of Ecology, Olympia, Washington as amended.

SW-846, *Test Methods for Evaluating Solid Waste: Physical/Chemical Method*, latest edition, Office of Solid Waste, U.S. Environmental Protection Agency, Washington, D.C.

WHC-SD-WM-TI-714, *High-Density Polyethylene Liner Chemical Compatibility for Radioactive Mixed Waste Trenches*, 1995, Westinghouse Hanford Company, Richland, Washington.

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CHAPTER 4.0
PROCESS INFORMATION

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PROCESS INFORMATION

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4.0 PROCESS INFORMATION

This chapter discusses the processes that will be used to dispose waste in the IDF and includes a discussion of the design and function of the following:

- Container
- Disposal landfill
- Leak detection system
- Leachate collection and removal system
- Secondary leak detection system

Note that the SLDS is not a design requirement of WAC 173-303-665, however DOE is adding the design feature pursuant to its authority under the Atomic Energy Act of 1954 (AEA) and not for the purposes of compliance with the dangerous waste regulations. Therefore, information regarding the design, construction, and operation of the secondary leak detection system is provided in this application as information only. Pursuant to AEA, DOE has sole and exclusive responsibility and authority to regulate the source, special nuclear and by-product material component of radioactive mixed waste at DOE-owned nuclear facilities. Source, special nuclear and by-product materials, as defined by AEA, are not subject to regulation under Resource Conservation and Recovery Act (RCRA) or the Hazardous Waste Management Act, by the State of Washington and are not be subject to State dangerous waste permit, orders, or any other enforceable instrument issued there under. DOE recognizes that radionuclide data may be useful in the development and confirmation of geohydrologic conceptual models. Radionuclide data contained herein is therefore provided as a matter of comity so the information may be used for such purposes.

Waste stream compatibility (i.e., compatibility between individual waste streams and compatibility between waste streams and landfill design and construction parameters) will be assessed on a case-by-case basis. Criteria for assessing and determining compatibility is identified in the Waste Acceptance Criteria, Waste Analysis Plan, or other protocol or procedure as appropriate (Chapter 3.0, for further discussion of waste stream compatibility).

Process Code S01 (container storage) has been included within this permit, in the event that storage is required before final disposal (e.g., to support the confirmation process of the waste or cooling of vitrified waste if required). Waste failing the confirmation process (Chapter 3.0) will be identified as off-specification and may require storage prior to disposal. Only off-specification waste or vitrified waste requiring cooling (due to process heat) may be stored in the lined portion of the IDF pending disposition. To maintain operational flexibility, off-specification containers and vitrified waste requiring cooling could be left on the transport vehicles at the IDF until disposal can occur but may be off-loaded into the lined portion of the IDF pending final disposal provided the temperature administrative control limit is not exceeded. Off-specification waste and vitrified waste requiring cooling will be separated from other waste via tape, ropes, chains, or other cordon mechanism.

4.1 Containers

All mixed waste accepted for disposal at the IDF will be packaged in standard containers [U.S. Department of Transportation (DOT) and/or DOE], unless alternate packages are dictated by the size, shape, or form of waste (49 CFR 173) (e.g., metal boxes), and self-contained bulk waste.

4.1.1 Description of Containers

Mixed waste disposed at the IDF is limited to vitrified low-activity waste (LAW) from the RPP-WTP and DBVS. Additionally, mixed waste generated by IDF operations will be disposed of in IDF.

The RPP-WTP and DBVS containers are designed specifically for the vitrified low activity waste form. Nominal RPP-WTP container dimensions will be 122 centimeters base outside dimension, 107 centimeters top by 230 centimeters in length, with a wall thickness of 0.357 centimeter with a container volume of 2.55 cubic meters.

The DBVS container dimensions are approximately 2.4 meters wide by 3.1 meters tall and 7.3 meters long and a container volume of 54 cubic meters. The vitrified low activity waste will be compatible with the containers, stainless steel for RPP-WTP and carbon steel for DBVS. Before receipt at the IDF, containers will be closed by the generator.

Due to the radioactivity and remote handling of the RPP-WTP immobilized waste containers, conventional labeling of the vitrified immobilized waste containers will not be feasible and an alternative to the standard labeling requirements will be used. This alternative labeling approach will use a unique alphanumeric identifier that will be welded onto each immobilized glass waste container. The welded "identifier" will ensure that the number is always legible, will not be removed or damaged during container decontamination, will not be damaged by heat or radiation, and will not degrade over time.

The identifier will be welded onto the shoulder and sidewall of each immobilized glass container at two locations 180 degrees apart. Characters will be approximately 2 in. high by 1.5 in. wide. The identifier will be formed by welding on stainless steel filler material at the time of container construction. This identifier will be used to track the container from receipt at the RPP-WTP, throughout its subsequent path of shipment and disposal at the IDF.

Each identifier will be composed of unique coded alphanumeric characters. This unique alphanumeric identification will be maintained within the plant information network, and will list data pertaining to the waste container including waste numbers, and the major risk(s) associated with the waste.

Mixed waste generated through waste operations at IDF will be packaged based on the size of the waste, with the most common container being galvanized or aluminized 208-liter containers.

The container packaging and handling for the IDF are designed to maintain containment of the waste, limit storage intrusion, and limit human exposure to mixed waste. Unusual sized containers such as vitrified LAW packages will be handled by using cranes or other appropriate equipment.

Operations personnel will inspect each container to confirm appropriate documentation and compliance with the waste acceptance criteria before the container is placed in the IDF (refer to Chapter 3).

If containerized mixed waste must be opened (i.e., for confirmation sampling, repackaging, etc.), the container typically would be removed to an onsite treatment and/or storage unit or other approved location before being opened. The container would be sealed before being returned to the IDF.

4.2 Leachate Collection Tanks

The aboveground leachate collection tank supports the lined IDF landfill. The leachate collection tank will be operated in accordance with the generator provisions of WAC 173-303-200 and WAC 173-303-640 as referenced by WAC 173-303-200.

For informational purposes, the following is provided for an understanding of the operation of the Leachate Collection Tank. Procedures will be written to manage the leachate in accordance with WAC 173-303-200. The presence of leachate in the tank will be detected with instrumentation within the two stilling wells. The level instrument within the first stilling well monitors the depth of leachate in the tank. A second stilling well will have instrumentation for high-high and low-low alarm set-point trips. The leachate will be removed from the tank using a transfer pump.

4.3 Landfills

The following addresses the IDF lined landfill.

4.3.1 List of Wastes

IDF will receive mixed and/or dangerous waste.

Waste will be accepted in containers (e.g. drums, boxes, larger containers).

Waste streams acceptable at the IDF facility fall within the range of dangerous waste numbers identified in Chapter 1.0, Part A Form.

4.3.2 Liner System Exemption Requests

This permit documentation does not seek an exemption to liner system requirements.

4.3.3 Liner System, General Items

This section provides a general description of the liner system to be used for the IDF lined landfill.

The liner system was designed to prevent migration of leachate out of the lined landfill during the active life of the landfill. The Active Life will consist of the operational period and the closure/postclosure period. The liner system was designed to meet U.S. Environmental Protection Agency (EPA) requirements, as identified in RCRA Subtitle C requirements for hazardous waste disposal facilities (40 CFR 264), technical guidance documents (e.g., EPA 1985), and WAC-173-303-665. In addition, the liner system incorporates the following general functional requirements:

- Range of Operating Conditions--year-round operation, withstand construction, and long-term stresses.
- Degree of Reliability--function safely and effectively throughout operating and closure/postclosure period with minimum maintenance.
- Intended Life--operational phase plus closure/postclosure monitoring phase.

4.3.3.1 Liner System Description

The landfill liner system will comply with WAC 173-303-665 requirements for dangerous waste landfills. Figure 4.2 shows a typical design and includes the following components (from top to bottom).

- Operations layer: minimum 0.9-meter thick of native soil. This layer provides a working surface for equipment, protect the liner from mechanical damage, and prevent freezing of the underlying low-hydraulic conductivity soil layer. (Hydraulic conductivity is a measure of how rapidly a material can transmit water and is based on specific ASTM testing requirements.)
- Leachate collection and removal system (LCRS) contains a minimum 0.3-meter-thick drainage gravel layer with a hydraulic conductivity of at least 1×10^{-2} centimeter per second (sometimes including perforated drainage pipes). A nonwoven separation geotextile is located between the operations layer and the drainage gravel layer to minimize sediment (fine-soil) migration into the LCRS. A nonwoven cushion geotextile is located between the drainage gravel and the primary geomembrane to protect the primary geomembrane.

The LCRS liners collect and convey leachate to the LCRS sump for removal and include the following components.

- Primary geomembrane liner: this liner consists of high-density polyethylene (HDPE) because of its excellent resistance to expected chemicals (Chapter 1.0), nominal 60-mil thickness (54-mil minimum), which is textured (to improve stability against sliding). The geomembrane acts as a moisture barrier. Located immediately above the primary geomembrane the LCRS includes a perforated pipe that helps collect and guide water into the leachate collection sump. The perforated pipe is located along the centerline of the cell and provides high-flow path water to the primary collection sump.
- Primary geosynthetic clay liner (GCL): the GCL consisting of a high-swelling sodium synthetic mat containing bentonite with a hydraulic conductivity of 1×10^{-8} centimeter per second or less. This layer acts as an additional primary moisture barrier directly under the primary geomembrane.

The leak detection system (LDS) is similar to the LCRS except the composite drainage net (CDN) replaces the primary gravel layer, the geosynthetic clay liner (GCL) is placed directly under the secondary geomembrane liner only under the LDS sump and the perforated pipes are not be needed because very high flow capacities are not be required.

The purpose of this system is to collect any leachate that leaks through the primary liner system and convey the leachate to the LDS sump for removal. The LDS also serves as a secondary LCRS. The LDS liners will collect and convey leakage to the LDS sump and include the following components:

- Secondary geomembrane liner: same as primary geomembrane liner.
- Secondary geosynthetic clay liner: same as primary geosynthetic clay liner.
- Admix liner: a minimum 0.9-meter-thick layer of compacted soil/bentonite admixture with a hydraulic conductivity of 1 x 10^{-7} centimeter per second or less. The bentonite is high-swelling sodium bentonite. This layer acts as an additional moisture barrier directly under the secondary geosynthetic clay liner in the LDS sump area and the secondary geomembrane outside the LDS sump area.
- The secondary leak detection system (SLDS) consists of operations layer type fill for a foundation of the LDS admix layer, drainage gravel with a hydraulic conductivity of at least 1 x 10^{-2} centimeter per second adjacent to a perforated pipe, a composite drainage net (CDN) and tertiary geomembrane. A nonwoven separation geotextile is located between the operations layer type material and the drainage gravel to minimize sediment (fine-soil) migration into the SLDS piping. The purpose of this system is to provide access to the area immediately below the LDS sump area. The SLDS collects liquids resulting from construction water and potentially, liquid from other sources. The SLDS liners will convey collected liquids to the SLDS piping for monitoring and/or removal. (Note that the secondary leak detection system is not a design requirement of WAC 173-303-665, however DOE is adding the design feature pursuant to its authority under the Atomic Energy Act of 1954 (AEA) and not for the purposes of compliance with the dangerous waste regulations. Therefore, information regarding the design, construction, and operation of the secondary leak detection system is provided in this application as information only. Pursuant to AEA, DOE has sole and exclusive responsibility and authority to regulate the source, special nuclear and by-product material component of radioactive mixed waste at DOE-owned nuclear facilities. Source, special nuclear and by-product materials, as defined by AEA, are not subject to regulation under RCRA or the Hazardous Waste Management Act, by the State of Washington and are not be subject to State dangerous waste permit, orders, or any other enforceable instrument issued there under. DOE recognizes that radionuclide data may be useful in the development and confirmation of geohydrologic conceptual models. Radionuclide data contained herein is therefore provided as a matter of comity so the information may be used for such purposes).

4.3.3.1.1 Operations Layer

The purpose of the operations layer is to protect the underlying liner components from damage by equipment during lined landfill construction and operation. This layer also protects the admix layer from freezing and desiccation cracking.

Previous research and experience has shown that desiccation cracks can occur under geomembrane liners when either the liner is not in close contact with the compacted admix or when the liner is subjected to wide temperature fluctuations (Corser and Cranston 1991). The operations layer acts as a weight to keep the geomembrane in contact with the admix, thereby reducing the potential for water vapor to form in an underlying airspace. The operations layer also acts as an insulating layer, together with the dead air space trapped in the underlying drainage layers.

The operations layer material typically consists of onsite granular soil that is reasonably well graded. The material has a maximum particle size limit of 5.1 centimeters or less, to facilitate protection of the underlying layers.

4.3.3.1.2 Leachate Collection and Removal System

The LCRS is located below the operations layer and provides a flow path for the leachate flowing into the LCRS sump.

Between the operations layer and the underlying drainage gravel, a geotextile layer functions as a filter separation barrier. The geotextile prevents migration of fine soil and clogging of the drainage gravel. On the lined landfill floor the drain gravel is a minimum 0.3-meter-thick layer of washed, rounded to subrounded stone, with a hydraulic conductivity of at least 1 x 10^{-2} centimeter per second. In addition, a perforated high-density polyethylene drainage pipe placed within the drainage gravel accelerates leachate transport into the LCRS sump during high precipitation events. On the lined landfill floor, the drain gravel layer is underlain by a geotextile cushion resting on the primary high-density polyethylene geomembrane. The geotextile provides additional protection for the primary geomembrane on the floor of the landfill.

On the lined landfill sideslopes, the LCRS has a composite drainage net (CDN) layer composed of a geonet (which is a network of HDPE strands, interwoven and bonded to form a panel that provides a drainage pathway for fluids), with a layer of geotextile thermally bonded to each side. This CDN layer has a transmissivity of at least 3 x 10^{-5} meters squared per second. The CDN is used on the sideslopes to avoid problems associated with placement of clean granular material on slopes, thereby minimizing the potential for damaging the underlying liner system.

4.3.3.1.3 Primary Geomembrane Liner

The primary geomembrane liner acts both as an impermeable leachate barrier and as a flow surface, routing leachate to the primary sump. High-density polyethylene was used because of its high resistance to chemical deterioration. Generally, textured (roughened) geomembrane is used to maximize shear strength along adjacent interfaces and to reduce the potential for sliding of the liner system.

4.3.3.1.4 Primary Geosynthetic Clay Liner Layer

A primary geosynthetic clay liner (GCL) consists of a mat of bentonite placed between two geotextiles. The GCL is installed immediately beneath the primary high-density polyethylene liner on the floor of the lined landfill only. The purpose of this liner is to provide extra protection in the case of deterioration (such as stress cracking) of the primary geomembrane where operations will continue for several years.

The in-place hydraulic conductivity of the GCL is 1 x 10^{-8} centimeter per second or less, exceeding the WAC hydraulic conductivity requirement for the secondary soil liners. The upper surface of GCL provides a smooth uniform surface on which to place the overlying geomembrane liner.

4.3.3.1.5 Leak Detection System

The LDS provides the flow path for leachate flowing into the LDS sump. The following is a description of the system to be used in the IDF landfill.

The LDS has a CDN drainage layer on the floor, and a CDN drainage layer on the sideslopes. The CDN consist of a layer of geotextile thermally bonded to each side of the geonet. These materials and their configuration is similar to the LCRS described in Section 4.3.3.1.2, except for the absence of a drainage gravel layer and a perforated drainage pipe system on the floor of the lined landfill. The LDS will channel leachate that penetrates the primary liner system through the CDN into the leak detection sump.

The LDS serves as a secondary LCRS for the IDF. Leachate collected in the secondary sump will be measured to determine the leakage rate through the primary liner.

4.3.3.1.6 Secondary and Tertiary Geomembrane Liner

The secondary geomembrane liner, located underneath the LDS, is placed directly against the secondary compacted admix liner, except in the LDS sump area which includes a geosynthetic clay liner between the secondary geomembrane liner and the secondary compacted admix liner. For information only, the teriary geomembrane liner for the SLDS is placed directly against subgrade as per Section 4.3.3.1.8. The secondary and teriary geomembrane liners are similar to the primary geomembrane described in Section 4.3.3.1.3. The secondary geosynthetic clay liner material is similar to the primary geosynthetic clay liner described in Section 4.3.3.1.4.

4.3.3.1.7 Secondary Admix Liner

The secondary admix liner has a minimum 0.9-meter-thick compacted soil/bentonite admixture located immediately beneath the secondary high-density polyethylene liner, as required by WAC 173-303-665. The secondary admix liner typically consists of silty sand from local borrow sources mixed with a nominal 12 percent sodium bentonite, by dry weight. The in-place hydraulic conductivity of the admix liner is 1 x 10^{-7} centimeter per second or less, consistent with WAC requirements for secondary soil liners. The upper surface of the secondary admix liner is trimmed to the design grades and tolerances. The surface was rolled with a smooth steel-drum roller to remove all ridges and irregularities. The result is a smooth uniform surface on which to place the overlying geomembrane liner.

4.3.3.1.8 Subgrade/Liner System Foundation

The lined landfill in the IDF is founded in undisturbed native soils or material compacted to at least 95% of a standard proctor maximum density (determined by ASTM D698). The liner system foundation is discussed in further detail in Section 4.3.4.

4.3.3.1.9 Access Ramp

The lined landfill has an access ramp outside the lined portion of the landfill, minimizing damage to the liner system from vehicle traffic into the lined landfill. As the landfill expands the access ramp will be reconstructed to the south of each expansion in the landfill. The access ramp design could vary as the landfill expands.

4.3.3.1.10 Landfill Expansion

The initial phase of the IDF liner was complete at the north end of the landfill. As shown in Figure 4.1, construction of the initial IDF phase completed the liner system on the north sideslope and the excavated portions of the landfill floor, east sideslope, and west sideslope. The dashed line of Figure 4.1 across the south edge of the landfill floor denotes the southern extent of the landfill liner. The liner system will be installed to extend approximately 15 meters beyond the estimated toe of slope of the first phase waste placement. This extension will also allow waste haul vehicles to be staged or unloaded over a lined area. Termination detail for the south edge of the liner system is found in Appendix 4A, drawing H-2-830840. The south sideslope of the first phase of IDF is not lined to allow future expansion of the IDF. At the south end of the cells is a storm water berm/ditch with an infiltration area, which will capture clean runoff from the unlined south sideslope before it runs onto the lined landfill. The landfill floor slopes up 1% from north to south to allow adequate leachate collection capacity for a 25-year storm event. Each future liner construction project will connect to the south edge of the previously constructed liner and operations systems and extend the disposal area further to the south. With the expansion of the IDF in subsequent phases, access ramps for the previous phase will be destroyed and new ramps built on the south edge of the landfill.

4.3.3.2 Liner System Location Relative to High Water Table

The water table is located approximately 90 to 100 meters below the ground surface in the IDF. It is anticipated that the deepest point of the liner system is no greater than 20 meters below ground surface. Consequently, the liner systems is at least 69 meters above groundwater. The liner systems will not be affected by the water table because of this large elevational difference.

4.3.3.3 Loads on Liner System

The liner system experiences several types of stresses during construction, operation, and closure/postclosure periods. The following sections discuss the types of stress and analytical methods used to design the IDF liners.

4.3.3.3.1 Liner Stress

The geosynthetic liner components experience some stress particularly during installation and before placing waste in the lined landfill but also during the entire lifecycle.

The high-density polyethylene liner is temperature sensitive, expanding, and contracting as liner temperatures increase and decrease. Thermally induced stresses could develop in the liner if deployment and anchoring occur just before a significant decrease in the liner temperature. The operations layer is sufficiently thick to ensure liner stress remains below the yield strain and stress. Administrative procedures will prevent loading and backfilling of waste exceeding applicable thermal limits due to recent vitrification processes to avoid potential liner damage.

The drainage gravel has the potential to produce localized stress on the geomembrane liner during gravel placement with construction equipment. The geotextile cushion placed at the base of the drainage gravel protects the underlying geomembrane. A puncture analysis was performed to select a sufficiently thick cushion geotextile. This analysis incorporated expected construction vehicle ground pressures and design drainage gravel gradation listed in the construction specifications. If required, engineering controls such as independent foundations will be installed to minimize liner stress involved with large package disposal.

On the landfill sideslopes, tension induced by liner-component load transfer is not anticipated, because the liner interface effective shear strength angles are higher than the sideslope angles. The liner component interface strengths were determined by laboratory direct shear tests. Both static and dynamic stability analyses were performed, using standard methods, design accelerations, and factors of safety.

Stress on the geomembrane in the anchor trench also was evaluated during detailed design. Wind uplift and thermal expansion and contraction could cause stress in the geomembrane during construction. However, these stresses are not be a problem, because the stress is relatively low as compared to the tensile strength of the liner. In addition, these stresses are minimized by using sand bags to control liner position during liner panel placement and welding, as well as keeping the anchor trench open until the liner is stabilized with overlaying fill material. Placement of overlaying fill material is controlled to limit stress buildup in the liner. The stress is not present after construction, because of the weight and insulating properties of the operations layer.

4.3.3.3.2 Stress Resulting From Operating Equipment

Operations equipment provides a design load case on the IDF liner, which was analyzed as part of the IDF design (Appendix 4A). The analyses show that the 0.9-meter-thick operations layer dissipates stress produced by the operating equipment and is sufficient to protect the IDF liner system.

4.3.3.3.3 Stress from Maximum Quantity of Waste, Cover, and Proposed Closure/Postclosure Land Use

When the lined landfill is full and the cover system is in place, the liner system will experience a static load from the overlying waste, backfill, and cover materials. No significant increase in stresses on the liner system is anticipated from closure/postclosure land use. The maximum design load of material overlying the liner system includes an allowance for the cover system. Analyses include puncture protection of the geomembrane by the cushion geotextile, and decrease in transmissivity of CDN drainage layers. Materials were specified based on the ability of the materials to perform adequately under closure/postclosure loading conditions.

Dynamic stress on the liner system will result primarily from ground accelerations during seismic events. Both static and dynamic analyses were performed on the subgrade and liner components based on the finished configuration of the empty landfill. Under closure/postclosure conditions, the waste, backfill, and cover materials will tend to buttress the liner system, resulting in greater stability relative to the operational phase. All of the analyses verified adequate stability for the IDF.

4.3.3.3.4 Stresses Resulting From Settlement, Subsidence, or Uplift

The subgrade settlement produced by waste loading essentially will be elastic because of the coarse-grained, noncohesive, and drained nature of the soil. The subgrade rebounded during the excavation phase of construction and will settle as the landfill is filled. The compacted admix liner will consolidate under waste loads. The total settlement will be a combination of the subgrade elastic and the admix consolidation settlements.

These settlements were analyzed with standard methods during detailed design of the lined landfill. In general, differential settlements will be expected to occur primarily across the lined landfill sideslopes as the thickness of waste decreases from maximum to zero. The geosynthetic liner components were analyzed, the anticipated strains likely will not produce any appreciable stresses in the liner system.

The potential for subsidence-induced stress is believed to be negligible based on the following information:

- The soils underlying the IDF tend to be coarse-grained soils, sands, and gravels, in a relatively dense configuration that will not be subject to piping effects that could transport soil resulting in subsidence.
- The groundwater level is deep, at least 69.6 meters below the base of the lined landfill, and will not affect bearing soils.
- No natural voids, or man-made mining or tunneling has been noted. If the groundwater level was lowered substantially and consolidation occurred in the aquifer, local site-specific subsidence would be negligible because of the depth of the groundwater below the lined landfill.

The potential for stresses resulting from uplift on the liner system also is expected to be negligible. The seasonal groundwater level is very deep, and higher-elevation perched groundwater likely will not develop because of the absence of aquitards in the coarse-grained Hanford formation underlying the IDF. The coarse-grained nature of the Hanford formation also promotes rapid, primarily vertical, infiltration, which means it is unlikely that infiltration from outside the lined landfill boundary would be transported laterally underneath the landfill liner. Gas pressures similarly are unlikely to develop because of the absence of any organic material that could generate significant subsurface gas (from organic material decomposition) and the coarse-grained, highly permeable sands and gravels underlying the landfill.

4.3.3.3.5 Internal and External Pressure Gradients

Pressure gradients across the liner caused by liquids or gases will be expected to be negligible. Internal pressures due to liquids will be controlled by the leachate collection and removal system. Because leachate will be removed from the flat 50-foot by 50-foot LCRS sump in a timely manner, there will be minimal liquid head on the liner (less than 30.5 centimeters according to WAC regulations). Gas generated internally is expected to be minimal because waste is inorganic and non-reactive. However, any pre-closure internally generated gas will be vented through either the waste or the leachate collection system. The closure cover design will consider gas venting.

External pressures on the liner system is expected to be minimal. Gas pressures will be negligible because the subgrade soil contains no gas producing materials and is highly permeable, readily venting any potential gas to the atmosphere. External pressure from liquids is not anticipated because of the deep groundwater table and the highly permeable foundation soils.

4.3.3.4 Liner System Coverage

The liner system covers all soils underlying the lined landfill and extends over the crest of the sideslopes into the anchor trench (Figure 4.2, Detail 3).

4.3.3.5 Liner System Exposure Prevention

No geosynthetic or admix components of the liner system are exposed to the atmosphere. The minimum 0.9-meter-thick operations layer covers the entire lined landfill surface. This layer serves both as a physical protective barrier and as thermal insulation, protecting the admix layer from desiccation and frost damage.

Excessive erosion, such as gullying, will be repaired by replacing the eroded soil. Dust suppression agents will be used to prevent excessive wind erosion on the landfill sideslopes. The dust suppression agents will bind the surface of the operations layer and will minimize wind entrainment of soil.

4.3.4 Liner System, Foundation

The following sections discuss the foundations beneath the liner systems.

4.3.4.1 Foundation Description

At the IDF, the Hanford formation consists mainly of sand dominated facies with lesser amounts of silt dominated and gravel dominated facies. Where sands are present, these sands are underlain by the Hanford formation. Here, the Hanford formation has been described as poorly sorted pebble to boulder gravel and fine to course grained sand, with lesser amounts of interstitial and interbedded silt and clay.

The two geologic units pertinent to the IDF lined landfill are summarized as follows.

Recent eolian sand: The sand is light olive gray in color and has a density that is loose at the surface but becomes compact with depth. The sand has a fine to medium grain size and includes little to some nonplastic silt-sized fines. The deposit is homogeneous except for a distinguishable layer of volcanic ash in some locations.

Glaciofluvial flood deposit: This deposit has well graded mixtures of sands and gravels with trace to little nonplastic silt-sized particles. The gravel content can vary with depth, and the deposit can become predominantly gravel. This coarse-grained deposit is part of the Cold Creek Bar, which was formed during the Pleistocene Epoch by glacial outburst flooding.

4.3.4.2 Subsurface Exploration Data

Geological site investigations were used to support the detailed design of the landfill. The investigations consisted of a review of historical data, including well logs (Chapter 5.0), exploratory borings, and surface pit samples data. Because the foundation soils are relatively consistent over broad areas, the need for additional borings and geophysical investigations will be determined on a case-by-case basis. If boreholes are drilled, penetration test data will be collected to determine the strength of the foundation materials in situ.

4.3.4.3 Laboratory Testing Data

Laboratory testing will be performed on the surface soil samples and borings, both from the lined landfill site and from potential borrow source locations as follows. Testing will be performed to classify soils, provide input parameters to verify engineering analyses, and for preparing material and construction specifications. The following tests will be performed on the soil samples:

- Visual classification (ASTM D2487)--to classify soils
- Natural moisture content (ASTM D2216)--for input to engineering analyses and preparing construction specifications
- Particle size analysis (ASTM D422 or D1140/C136)--for classification and input to engineering analyses
- Moisture-density relationships (ASTM D698 or D1557)--for preparing compaction specifications

Laboratory testing will be performed according to the most recent versions of ASTM methods or other recognized standards. Additional tests will be performed as needed.

4.3.4.4 Engineering Analyses

The subgrade will be required to support the liner system and overlying materials (waste, fill, and cover) without excessive settlement, compression, or uplift that could damage the liner system. This section describes the design approach used to satisfy these criteria.

4.3.4.4.1 Settlement Potential

The subgrade settlement produced by waste loading essentially will be elastic because of the coarse-grained, noncohesive, and drained nature of the soil. The subgrade will rebound during the excavation phase of construction and will settle as the landfill is filled.

An elastic settlement analysis using standard methods was performed and results indicate the magnitude of the total and differential settlement is within performance limits.

4.3.4.4.2 Bearing Capacity

The bearing capacity of the subgrade soil will need to support structures such as leachate collection tanks. The construction specifications typically will require that the upper portion of the subgrade soil and all structural fill be moisture conditioned and compacted to at least 95 percent of the maximum standard Proctor dry density (ASTM D698). Maximum allowable bearing capacities for foundations have been established using standard geotechnical methods. Bearing capacities for the types of soils expected at the IDF typically are greater than the maximum expected loads from the support structures.

4.3.4.4.3 Stability of Lined Landfill Slopes

The lined landfill was constructed in eolian sand and the underlying coarse-grained Hanford formation. In granular, cohesionless, and drained soils such as these, the stability of the slope will be related primarily to the maximum slope angle. Both veneer and global stability analyses were performed to determine both static and dynamic sideslope stability. Results demonstrate adequate stability for the IDF throughout its design life.

4.3.4.4.4 Potential for Excess Hydrostatic or Gas Pressures

Because the seasonal high-water level is at least 69 meters below the base of the deepest lined landfill, no external hydrostatic pressure will be expected from this source. Because of the coarse-grained nature of the foundation soils, any infiltration of surface water around the perimeter of the lined landfill will be expected to travel primarily downward. Therefore, infiltration should not cause substantial pressure on the exterior of the liner system. Internal hydrostatic pressure from leachate will be negligible because the leachate will be removed from the lined landfill to limit head on the liner.

Gas pressure exerted externally on the liner system is expected to be negligible, because no gas-generating material (i.e., organic material) is expected in the foundation soils. If any gas were generated below the liner system, little pressure buildup would occur because of the unsaturated coarse-grained nature of the foundation soils, which would vent the gas to the atmosphere. Internal gas pressure buildup will not be anticipated, because wastes are generally inorganic and have low gas generating potential, and the leachate collection system will be vented to the atmosphere and dissipates any gas.

4.3.4.4.5 Seismic Conditions

Potential hazards from seismic events will include faulting, slope failure, and liquefaction. Disruption of the lined landfill by faulting is not considered a significant risk because (1) no major faults have been identified at the IDF (DOE/RW-0164) and (2) only one central fault at Gable Mountain on the Hanford Site shows evidence of movement within the last 13,000 years. The potential for slope failure is considered low, because granular materials typically have high strengths relative to the maximum sideslope angles expected for the lined landfill. Liquefaction will occur in loose, poorly graded granular materials that are subjected to shaking from seismic events. Saturated soils will be most susceptible because of high dynamic pore pressures that temporarily lower the effective stress. During this process, the soil particles will be rearranged into a denser configuration, with a resulting decrease in volume. The foundation materials at the IDF is not considered susceptible to liquefaction because the materials are well graded granular soils that are unsaturated and relatively dense.

The IDF support building (not sited within the TSD boundary) will be located in Zone 2B as identified in the Uniform Building Code (ICBO 1997).

4.3.4.4.6 Subsidence Potential

In general, subsidence of undisturbed foundation materials would be the result of dissolution, fluid extraction (water or petroleum), or mining. The potential for subsidence will be negligible at the IDF based on the following.

- The soils underlying the IDF are coarse-grained sands and gravels, in a relatively dense configuration, which are not subject to piping that can cause transport of soil and resulting subsidence.
- The groundwater level is deep, at least 69 meters below the base of the lined landfill, and does not affect bearing soils.
- The soil and rock types below the IDF are not soluble.
- No mining or tunneling has been noted. If the groundwater level was lowered substantially and consolidation occurred in the aquifer, local site-specific subsidence would be negligible because of the depth of the groundwater table below the lined landfill.

4.3.4.4.7 Sinkhole Potential

Borings in and around the IDF have not identified any soluble materials in the foundation soils or underlying sediments. Consequently, the potential for any sinkhole development is negligible.

4.3.5 Liner System, Liners

The following sections discuss the individual components of the IDF liner systems.

4.3.5.1 Synthetic Liners

As described in Section 4.3.3, the synthetic liners act as an impermeable barrier for leachate migration (Figure 4.2). The synthetic liners consist of high-density polyethylene material that make the liners resistant to chemical deterioration. Section 4.3.3 describes the synthetic liner system in detail.

4.3.5.2 Synthetic Liner Compatibility Data

During detailed design of the lined landfill, the composition of the expected leachate was estimated. Expected leachate composition was based on known waste composition, process information, leachate from other operating lined landfills, and similar sources of data. Leachate constituents were compared to manufacturers' chemical compatibility data for synthetic liner components. In addition, the results of previous chemical compatibility testing and studies were evaluated against leachate composition. Information gained from this evaluation was used to select a liner that will be compatible with the expected leachate.

Compatibility testing for leachate tank liner material is planned for construction. An immersion test program is included in the technical specifications for the tank liner (anticipated to be XR-5 material). The immersion-testing program will require the construction general contractor to submit tank liner samples to the design engineer for immersion testing as part of the submittal and certification process for the tank. Immersion testing will follow EPA 9090A (and ASTM) test protocols.

During landfill operation, the compatibility of waste receipts with the liner will be ensured. The compatibility of the waste constituents with the liner material will be established by laboratory testing if determined to be necessary, based on waste type and concentrations. Such tests will follow EPA Method 9090A or other appropriate methods. Test results will be evaluated using statistical methods and accepted criteria (based on past projects and agency acceptance) for liner/leachate compatibility.

4.3.5.3 Synthetic Liner Strength

As discussed in Section 4.3.3.3, the liner system will experience loads from several sources. During the detailed design process for the landfill, the strength of liner system materials was evaluated against these loads. The analysis indicated an adequate factor of safety for liner system materials.

Seams in geomembranes is a critical area; however, correct installation methods make the seams stronger than the surrounding material. Detailed installation and testing requirements will be included in the construction quality assurance plan (Section 4.3.7.3) to ensure that the liner is constructed properly. In addition, methods will be established to demonstrate adequate seam strength is achieved during installation.

Seaming requirements for the geotextiles and CDN: These materials were overlapped sufficiently to provide complete area coverage, and relatively light seams were used to hold the panels in position during construction, seam strength requirements for these materials will be negligible.

4.3.5.4 Synthetic Liner Bedding

The primary geomembrane liner is in contact with the GCL and geotextile cushion underlying the drainage gravel.

The secondary geomembrane liner is in direct contact with the compacted admix layer. This type of subgrade is typical for flexible geomembrane liners.

With respect to the drainage gravel and operations layers, the geomembranes are protected by overlying geotextile cushion or CDN layers. These geotextiles were designed to provide adequate protection during construction and operation to withstand the loads discussed in Section 4.3.3.3.

4.3.5.5 Soil Liners

The IDF landfill is lined with a minimum (0.9-meter thick) layer of compacted soil/bentonite mixture (admix) under the secondary geomembrane liner. This layer has an in-place hydraulic conductivity of less than 1×10^{-7} centimeter per second. The soil component of the admix is silty fine sand or similar material from areas near the IDF. Approximately 12 percent bentonite by dry weight was added to the fine soil to achieve sufficiently low hydraulic conductivity; however, the percent might vary.

4.3.5.5.1 Material Testing Data

Laboratory testing will be performed on soil liner materials to confirm input parameters for engineering analyses and for refining material and construction specifications.

Before constructing the lined landfill, a full-scale test fill of the admix material will be conducted. The primary purpose of the test fill will be to verify that the specified soil density, moisture content, and hydraulic conductivity values will be achieved consistently using proposed compaction equipment and procedures. In-place density will be measured using both the nuclear gauge (ASTM D2922) and sand cone (ASTM D1556) methods. In-place hydraulic conductivity will be determined from a two-stage infiltration from a borehole (ASTM D6391). Admix hydraulic conductivity will be estimated from thin-wall tube samples (ASTM D1587) obtained from the test fill and tested in the laboratory (ASTM D5084). Details of the test fill are presented in the Construction Quality Assurance Plan (Appendix 4B). During construction, field density (e.g., ASTM D2922, D2167, and/or D1556) and moisture content (ASTM D2216) will be measured periodically. Thin-wall tube samples (ASTM D1587) will be taken at regular intervals and will be tested for hydraulic conductivity (ASTM D5084). Additional details of field-testing during construction will be presented in the Construction Quality Assurance Plan.

Dispersion and piping in the admix are not considered likely because the hydraulic conductivity, and thus the flow velocity, will be very low, making it difficult to move the soil particles or otherwise disrupt the soil fabric. In addition, the admix will be well graded, so the component particles will tend to hold each other in place. Therefore, testing for these characteristics will not be necessary.

4.3.5.5.2 Soil Liner Compatibility Data

As discussed in Section 4.3.5.2, expected leachate composition was determined as part of detailed landfill design. The results of previous chemical compatibility testing and studies were evaluated against leachate composition to determine the effect of leachate on soil liner composition or hydraulic conductivity. The tests followed the procedures of ASTM D5084 (flexible wall parameter) and considered the effects of radiation on the soil liner materials.

4.3.5.5.3 Soil Liner Thickness

The IDF was designed to operate to minimize the leachate head over the liner systems.

Design of the primary liner system included an additional clay layer (the primary GCL layer, which was previously described in Section 4.3.3.1) underlying the primary HDPE geomembrane to further minimize liner leakage from the primary liner. Note that only a single geomembrane is required under WAC 173-303 for the primary liner.

Calculations evaluated the effectiveness of the primary soil liner as a barrier to leachate. Leakage analyses were performed for the primary liner system using EPA's Hydrologic Evaluation of Landfill Performance (HELP) Model (Schroeder et al. 1997). Estimated leakage rates were compared to the Action Leakage Rate (ALR, which is defined in WAC 173-303-665[8] as "the maximum design flow rate that the leak detection system ... can remove without the fluid head on the bottom liner exceeding 1 foot"), and were determined to be much lower than the ALR. This demonstrates the benefit of the GCL included in the primary bottom-lining system, which provides a composite lining system and minimizes actual leakage through the bottom primary lining system.

Overall, the IDF is designed to actively convey and collect leachate from the liner areas of the facility to minimize leachate buildup over the liners. Leachate is conveyed to the LCRS and LDS sumps for active removal from the facility. In addition, the LCRS sump area has been designed with a 6-inch-deep sump trough where the LCRS pumps are positioned to minimize the area of the sump that has a permanent liquid level (below the pump intake/shutoff elevation). Both the LCRS and LDS sump pumps will be operated throughout the Active Life of the facility and into the post-closure period until leachate generation has essentially ceased. By actively removing leachate from the IDF, head buildup is minimized, which in turn minimizes leakage through both the primary and secondary liner systems.

4.3.5.5.4 Soil Liner Strength

The expected loads on the liner system are discussed in Section 4.3.3.3. Significant stresses in the soil liner that were considered include (1) stresses from the weight of the liner system, (2) stresses on the interface with the overlying materials, and (3) stresses during construction.

Stresses will be present on the sideslopes from the weight of the operations layer and soil liner itself. Using material properties determined from laboratory testing, the stability of the soil liner was evaluated under both static and dynamic loading conditions. Standard methods of slope stability analysis were used. Interface strengths were found to provide adequate veneer stability for the liner system. Interface strength is the shear strength that occurs between layers of liner materials at their interface boundary, as established by ASTM test methods.

The primary concern during construction will be bearing failure caused by the weight of overlying soil components of the liner system (e.g., drainage gravel on the floor) and the construction equipment used to spread these materials. Strength parameters developed from laboratory testing and standard analytical methods were again used to determine that adequate stability and bearing capacity exist for the IDF liner system.

4.3.5.5.5 Engineering Report

An engineering report was prepared for the lined landfill as part of the definitive design document package. The report describes the design of the liner system and includes supporting calculations. The critical systems IDF Design Report is provided in Appendix 4A. The final IDF design report was prepared under the supervision of a professional engineer registered in Washington State.

4.3.6 Liner System, Leachate Collection and Removal System

The purpose of the leachate collection and removal system is to provide sufficient hydraulic conductivity and storage volume to collect, retain, and dispose of, in a timely manner, fluids falling on or moving through the waste. The primary leachate collection and removal system provides the preferential path along which the leachate flows into the primary sump. The secondary leachate collection and removal system (also called the leak detection system) is located between the primary and secondary geomembranes. The secondary leachate collection and removal system provides the preferential path along which any fluids leaking through the primary liner system flow to the secondary sump.

The collected leachate will be pumped to a leachate collection tank, screened and/or sampled, and transferred to a permitted treatment and disposal unit.

4.3.6.1 System Operation and Design

The lined landfill operates in a way that ensures the bottom liner is maintained as dry as possible, and the head on the top liner does not exceed 30.5 centimeters measured above the flat 50-foot-by-50-foot LCRS sump HDPE liner. In extreme conditions (i.e., in excess of a 25-year storm event), the head on the top liner could exceed 30.5 centimeters for short durations. The operating methodology, described in the following paragraphs, ensures that liquids on the bottom liner are removed continuously before liquids could accumulate and exceed 30.5 centimeters for the design storm event.

Both leachate collection systems operate either manually or automatically. When operated automatically, liquid level sensors will cycle the pumps on and off, in response to rising and falling leachate levels. The leakage rate through the top liner will be calculated to demonstrate that the leakage rate is less than the 'action leakage rate'. Data to support the leakage rate calculations will be obtained either from the flow totalizer in the secondary leachate collection pump discharge line or from the liquid level gauges. Collected leachate from the secondary leachate collection system is pumped to the leachate collection tank.

The design of the primary and secondary leachate collection systems is described in Section 4.3.3.1. System geometry was completed and material specifications were developed during the detailed design process. The leachate collection and removal system design will comply with WAC 173-303 requirements and applicable guidance.

Each sump has a thick layer of gravel designed to provide high hydraulic conductivity and storage capacity. Leachate is removed from the sumps by a pump installed in sideslope riser pipes. Pressure transducers monitor leachate level in the sumps and provide appropriate signals to the pump control system. All pumps and transducers are removable for maintenance, calibration, and related activities.

4.3.6.1.1 Primary System

The base of the leachate collection and removal system is defined by the primary geomembrane. On the floor of the lined landfill, the primary geomembrane is overlain by geotextile cushion, and the granular drainage layer. The granular drainage layer drains to the primary sump and a perforated pipe is located along the centerline of the cell to increase flow capacity to the primary sump. Geotextile layers at the top of the leachate collection and removal system prevent migration of fine soil particles into the gravel or geonet, thus prevent clogging. On the sideslopes, a CDN layer is over the geomembrane. The CDN includes bonded geotextiles on both sides of a geonet that increase the interface shear strength. Because of construction difficulties in placing a 30.5-cm thick gravel layer on 3:1 sideslopes, no drainage gravel was placed on the sideslopes.

The leachate collection and removal system is covered by the operations layer. The layer is a minimum 0.9-meter thick, and provides protection for the underlying liner and drainage materials. The operations layer covers both the landfill floor and the sideslopes.

The leachate collection and removal system was designed to accommodate the 25-year, 24-hour storm, as required by WAC regulations. However, the EPA recognizes the need to store temporarily leachate from such rare events (EPA 1985). Should a storm event that exceeds the 25-year, 24-hour storm event occur, the leachate collection and removal system sump was designed to store temporarily leachate at a depth greater than 30.5 centimeters, as opposed to the alternative of constructing an excessively large leachate collection tank.

The leachate collection and removal system sump is equipped with two sump pumps. One pump is a high capacity pump capable of rapid removal of large volumes of leachate, and is suitable for the transfer of batch quantities of leachate, and can handle the larger volumes of leachate anticipated from the 25-year, 24-hour storm event. The other pump is a low-capacity submersible pump located in the base of the sump. The sump pumps are located in a sump trough.

The sump trough was designed to contain the leachate below the intake of these pumps, within the smallest possible area, to minimize the residual leachate volume after each pumping cycle. The pumps are fabricated from stainless steel or other corrosion resistant material.

4.3.6.1.2 Leak Detection System

The base of the LDS is formed by the secondary geomembrane. The leak detection system is similar to the LCRS, except that the perforated collection pipe is not included. The perforated pipe is not be needed because high flow capacity is not required for the low leachate volumes.

The LDS drains to the LDS sump, which is located immediately below the LCRS sump. Because of the low volumes, the LDS is equipped with only one low-capacity submersible pump to meet WAC 173-303-665(8)(a).

4.3.6.1.3 Response Action Plan

In compliance with regulatory requirements, a response action plan (Appendix 4C) was prepared for the lined landfill. In accordance with EPA guidance, the action leakage rate was calculated as "the maximum design flow rate that the leak detection system can remove without the fluid head on the bottom liner exceeding 30.5 centimeters" (EPA 1992). If the action leakage rate were exceeded, DOE will do the following:

- Notify the appropriate regulatory authority in writing of the exceedence within 7 days of the determination.
- Submit a preliminary written assessment to the appropriate regulatory authority within 14 days of the determination, on the amount of liquids, likely sources of liquids, possible location, size, cause of any leaks, and short-term actions taken and planned.
- Determine to the extent practicable the location, size, and cause of any leak.
- Determine whether waste receipt should cease or be curtailed, whether any waste should be removed from the unit for inspection, repairs, or controls, and whether the unit should be closed.
- Determine any other short-term and/or long-term actions to be taken to mitigate or stop any leaks.
- Within 30 days after the notification that the action leakage rate has been exceeded, submit to the appropriate regulatory authority the results of the analyses specified in the following paragraphs, the results of actions taken, and actions planned. Monthly thereafter, as long as the flow rate in the leak detection system exceeds the action leakage rate, DOE will submit to the appropriate regulatory authority, a report summarizing the results of any remedial actions taken and actions planned.

The leachate will be analyzed for RCRA constituents as appropriate. A procedure will be in place to address details of analysis (i.e., analyses, constituents, test methods, etc.). If the analytical results on leakage fluids indicate that these constituents are present, and if the constituents can be traced to a particular type of waste placed in a known area of the lined landfill, it might be possible to estimate the location of the leak. In addition, waste packages might not undergo enough deterioration during the active life of the landfill to permit escape of the contents; the leachate might be clean or the composition too general to show a specific source location.

If the source location cannot be identified, large-scale removal of the waste and operations layer to find and repair the leaking area of the liner would be one option for remediation. However, this risks damaging the liner. In addition, waste would have to be handled, stored, and replaced in the landfill. Backfill would need to be removed from around any waste packages to accomplish this. If the waste packages were damaged during this process, the risk of accidental release might be high. For these reasons, large-scale removal of waste and liner system materials will not be a desirable option and will not be implemented except as a last resort.

The preferred alternative will depend on factors such as the amount of waste already in the landfill, the rate of waste receipt, the chemistry of the leachate (i.e., is it clean?), the availability of other disposal units, and similar considerations. Therefore, no single approach will be selected at this time. If necessary, an interim solution could be implemented while the evaluation and permanent remediation were performed. Examples of potential approaches include the following.

- The surface of the waste could be graded to direct run-off into a shallow pond. The surface would be covered with the low-hydraulic conductivity layer (geomembrane). Precipitation would be pumped or evaporated from the pond and would not infiltrate the waste already in the lined landfill. Waste would be placed only during periods of dry weather, and stored at other onsite TSD units at other times. This type of approach also could be used to reduce leakage immediately after the action leakage rate was exceeded, while other remediation options were evaluated.
- Partial construction of the final closure cover could begin earlier than planned. This would reduce infiltration into the lined landfill, and possibly reduce the leakage rate if the cover were constructed over the failed area.
- A layer of low-hydraulic conductivity soil could be placed over the existing waste, perhaps in conjunction with a geomembrane, to create a second 'primary' liner higher in the lined landfill. This new liner would intercept precipitation and allow its removal.
- A rigid-frame or air-supported structure could be constructed over the landfill to ensure that no infiltration occurs. Although costly, this approach could be less expensive than constructing a new landfill.

In general, the selected remediation efforts will be progressive. Those remediation methods that are judged the least difficult and the most cost effective will be used first. If these efforts are not effective, more difficult or expensive options would be used.

4.3.6.2 Equivalent Capacity

The CDN drainage layers used will be available commercially and will have equivalent flow capacity to a 30.5-centimeters layer of granular drainage material with a hydraulic conductivity of 1×10^{-2} centimeter per second.

4.3.6.3 Grading and Drainage

In accordance with EPA guidance, all areas of the lined landfill floor (except the sump bottoms) are graded at a slope of at least 2 percent towards the centerline of each cell. The centerline of each cell has a 1 percent slope lengthwise towards the sump, to facilitate drainage and avoid ponding on the liners. Grading tolerances have been established to ensure proper slope is maintained.

4.3.6.4 Maximum Leachate Head

The maximum head on the primary liner is less than 30.5 centimeters, except for rare storm events as discussed in Section 4.3.6.1 and the LCRS sump trough. The sump was sized and designed to provide adequate surge storage to prevent leachate build up on the primary liner.

4.3.6.5 System Compatibility

The primary and secondary leachate collection and removal systems is composed of inert geologic materials (sand and gravel), high-density polyethylene, and other geosynthetic materials such as polypropylene. As described in Section 4.3.5.2, the geosynthetics were evaluated for compatibility with the expected leachate. To ensure that the geosynthetics used in the lined landfill are similar chemically to those evaluated, manufacturers will be required to submit quality control certificates and other manufacturing information on all materials.

Before a new waste constituent, not previously analyzed (based on a dangerous waste number), is allowed in the lined landfill, the waste constituent will be evaluated for compatibility with the liner (e.g., identified in 9090A test results or other appropriate testing methods, etc.). Other materials could contact the leachate, for example:

- HDPE and Polyvinyl chloride (PVC) piping will be used.
- Polyvinyl chloride and other plastics in miscellaneous uses.
- Leachate tank will use a chemically resistant flexible geomembrane liner system.

Compatibility of these materials with the expected leachate was considered in the landfill liner system design. Compatibility of these materials will be of lesser concern, because items that consist of these materials will be located entirely within the containment area. Failure of these items would not result in a dangerous waste release, and the materials would be replaced or repaired.

4.3.6.6 System Strength

Stability of drainage layer, strength of piping, and prevention of clogging are discussed in the following sections.

4.3.6.6.1 Stability of Drainage Layers

As described in Sections 4.3.3.3 and 4.3.5.3, the stability of the liners and leachate collection and removal systems on the sideslopes was evaluated as part of detailed design (Appendix 4A). To provide sufficiently high shear strengths at the interfaces between geosynthetic components, textured geomembranes and thermally bonded CDNs are used.

Bearing capacity of the drainage and sump gravels is expected to be adequate, based on typical strength values for granular materials.

The transmissivity of the drainage layers under the combined load of the waste and cover was addressed in the design and will be adequate to support leachate removal.

4.3.6.6.2 Strength of Piping

The drainpipes in the primary drainage and sump gravel and sideslope riser pipes are high-density polyethylene pipe. During detailed design, the required wall thickness of the pipe was determined according to the manufacturer's recommendations and standard analytical methods used by the piping industry (Appendix 4A). In these analyses, the ultimate load (derived from the estimated weight of the waste and cover) was used, the allowable deflections were limited to 5 percent, and conservative values for soil modulus and lateral confinement were assumed.

4.3.6.7 Prevention of Clogging

The geotextiles that separate the drainage layers from adjacent soil layers was selected based on the ability of the geotextiles to retain the soil and to prevent the soil from entering the leachate collection and removal systems. In addition, the amount of fine material in the drainage and sump gravels was limited by specification to less than a few percent, and is not expected to cause clogging problems (Appendix 4A). Because the waste disposed in the lined landfill will be required to satisfy LDR (RCW 70.105.050(2), WAC 173-303-140, and 40 CFR 268), the amount of organic material is minimal, and consequently biologic clogging will not be a problem.

4.3.7 Liner System, Construction and Maintenance

Details relating to the liner system construction and maintenance are discussed in the following sections.

4.3.7.1 Material Specifications

Material specifications are provided in the following sections for each of the materials used in the liner system.

4.3.7.1.1 Synthetic Liners

As described in Section 4.3.3.1, both the primary and secondary geomembrane liners consist of high-density polyethylene. As described in Section 4.3.3.1.4, the primary barrier also contains a geosynthetic clay liner placed on the floor area only. Detailed specifications were prepared for the lined landfill as part of the design process.

4.3.7.1.2 Soil Liners

As described in Section 4.3.3.1, the soil liner consists of imported bentonite (expansive clay) blended with fine soil deposits on or next to the IDF. The fine soil was free of roots, woody vegetation, rocks greater than 2.54 centimeter in diameter, and other deleterious material. The bentonite content is dependent on the characteristics of the fine soil. Mixing was performed under carefully controlled conditions in a pugmill or other approved alternatives. The admix was placed and compacted to achieve an in-place hydraulic conductivity of 1×10^{-7} centimeter per second or less. The final surface of the soil liner was rolled smooth before placing the overlying geomembrane. Additional specifications were prepared for the lined landfill as part of the design process.

4.3.7.1.3 Leachate Collection and Removal System

Drainage and sump gravel consisted of hard, durable, rounded to subrounded material. The gravel was washed and the amount of fine material (i.e., passing the number 200 sieve) was limited to a few percent. The hydraulic conductivity of the gravel is 1×10^{-2} centimeter per second or greater. Additional specifications were prepared as part of the design process.

For geotextiles and geonets, the composition, thickness, transmissivity, unit weight, apparent opening size, strength, and other properties were determined during detailed design based on results of engineering analyses, experience, and industry standard approaches.

4.3.7.2 Construction Specifications

Construction requirements for major components of the lined landfill are summarized in the following sections.

4.3.7.2.1 Liner System Foundation

The excavated subgrade surfaces was moisture conditioned and compacted as required to achieve the specified compaction before placing the admix layer.

4.3.7.2.2 Soil Liners

The soil and bentonite was blended thoroughly and moisture conditioned so that the admix is uniform and homogeneous throughout. The admix layer was placed in loose lifts and compacted so that the compacted lift meets the requirements of the Construction Quality Assurance Plan. Each new lift of admix was kneaded into the previously placed lift. The methods for admix preparation, type of compaction equipment, number of passes, and other details of the placement process was determined by constructing a test fill section before placing admix in the lined landfill.

4.3.7.2.3 Synthetic Liners

To protect the overlying geomembranes, the admix surface is smooth and free of deleterious material. In all cases, the high-density polyethylene liner was deployed with the length of the roll parallel to the slope. Adjacent panels were overlapped and thermally seamed using fusion or extrusion methods. Seams were inspected continuously using air pressure tests. A vacuum box was used in areas where air pressure tests cannot be used (e.g., extrusion weld areas). Destructive seam tests (ASTM D4437) (peel and adhesion) were performed on samples taken at regular intervals. Placing the overlying geosynthetic layers when practicable will protect the geomembranes.

4.3.7.2.4 Leachate Collection and Removal Systems

Drainage and sump gravel was placed and spread carefully over the underlying geosynthetics using suitable equipment to prevent damage. Hauling and placing equipment will operate on a minimum thickness of soil above any geosynthetic layer to avoid damage. Geosynthetic layers in the leachate collection and removal system were deployed, overlapped, and joined (e.g., tying for geonets, sewing for geotextiles) according to standard industry practice and the manufacturers' recommendations. Drainage and riser pipes were installed in the landfill. Pipes were bedded carefully and the landfill was backfilled to provide adequate lateral support. Pumps and other mechanical components are installed according to manufacturers' recommendations.

4.3.7.3 Construction Quality Control Program

A construction quality assurance plan (Appendix 4B) will be used during lined landfill construction and establishes in detail the following in accordance with WAC 173-303-335:

Program must include observations, test, and measurements to ensure:

- Proper construction of all components of the liners, leachate collection and removal system.
- Conformity of all materials used in the design.

4.3.7.4 Maintenance Procedures for Leachate Collection and Removal Systems

The accessible components of the leachate collection and removal system will be maintained according to preventive maintenance methods. These methods will require periodic testing to prove that the equipment, controls, and instrumentation are functional and are calibrated properly. Testing intervals will be derived from applicable regulations and manufacturer's recommendations. All pumps and motors will be started or bumped monthly or at intervals suggested by the manufacturer, first, to demonstrate that the pumps and motors are functional and second, to move the bearing(s) so that the bearing surfaces do not seize or become distorted. Instruments will be calibrated annually or at intervals suggested by the manufacturer. When applicable, the preventive maintenance methods will include calibration instructions. The following instruments will require annual calibration:

- LCRS sump level indicator
- LDS sump level indicator

Other instrumentation inside the leachate handling and storage facilities will also require routine maintenance.

4.3.7.5 Liner Repairs during Operations

Because of the 0.9-meter-thick operations layer, damage to the liner system is not expected. If damage did occur, the operations layer could be removed laterally as far as required. Underlying geosynthetic and gravel layers will be removed until an undamaged layer is encountered. The damaged layers will be repaired and replaced from the lowest layer upwards using similar methods to those employed during construction. Most repairs to the geomembranes will be performed using a patch, which will be placed, welded, and tested by construction quality assurance personnel.

4.3.8 Run-On and Runoff Control Systems

Because of the sandy soils, small drainage area, and arid climate at the IDF, stormwater run-on and run-off will not be expected to require major engineered structures. Interceptor and drainage ditches will be adequate for run-on and run-off control. The 25-year, 24-hour precipitation event was the design storm used to size the lined landfill systems. Beyond this, surface water evaluation is highly site-specific, and appropriate analyses were performed as part of detailed design for the lined landfill.

4.3.8.1 Run-on Control System

Run-on will be controlled by drainage ditches or berms around the perimeter of the lined landfill. Any overland flow approaching the landfill will be intercepted by the ditches or berms and will be conveyed to existing drainage systems or suitable discharge points. All the drainage ditches or berms were designed to handle the peak 25-year flow from the potential drainage area. By using low channel slopes, design flow velocities in the ditches will be maintained below established limits for sand channels.

Between the landfill crest and the perimeter road, the area will be graded to provide drainage toward the perimeter road. The perimeter road will be sloped outward, at a grade of approximately 2 percent, to provide drainage away from the landfill. On the outside of the perimeter road, drainage ditches will be excavated to provide drainage away from the landfill.

4.3.8.1.1 Design and Performance

Design and performance details were determined for the landfill as part of the detailed design process.

4.3.8.1.2 Calculation of Peak Flow

Computation of design discharge for the drainage ditches or berms was performed using standard analytical methods, such as the Rational Method or the computer program HEC-1 (USACE 1981). The 25-year, 24-hour precipitation depth is 4.0 centimeters, based on precipitation data recorded from 1947 to 1969 (PNL-4622). The tributary area for each section of ditch or berm was based on local topography.

4.3.8.2 Runoff Control System

There will be no run-off from the lined landfill because the landfill will be constructed below grade. Any precipitation falling on the landfill will be removed by either evapotranspiration or the leachate collection and removal systems. Therefore, a run-off control system will not be needed.

4.3.8.3 Construction

The drainage ditches or berms around the lined landfill were constructed with conventional earthmoving equipment such as graders and small dozers.

4.3.8.4 Maintenance

The drainage ditches or berms require periodic maintenance to ensure proper performance. The most frequent maintenance activity, beyond periodic inspection, will be cleaning the ditches or berms to remove obstructions caused by windblown soil and vegetation (e.g., tumbleweeds). After rare storm events, regrading of the ditch bottom or repair of the berm might be required to repair erosion damage. This is expected to occur infrequently; however, inspections will be conducted after 25-year storm events or at least annually.

4.3.9 Control of Wind Dispersal

The IDF will use varied methods to prevent wind dispersal of mixed waste and backfill materials, depending on the waste form. Methods to prevent wind dispersal include containerizing, stabilizing, grouting, spray fixitants, and backfill. In other instances, the operating contractor implements a wind speed restriction during handling, and immediately backfills the waste to prevent wind dispersal.

4.3.10 Liquids in Landfills

Free liquids will not be accepted except as allowed by Chapter 3.0, Section 1.2. Waste received at the IDF must comply with waste acceptance requirements.

4.3.11 Containerized Waste

Containerized waste received in the IDF lined landfill will be limited to a maximum of 10 percent void space. Several inert materials (diatomaceous earth, sand, lava rock) will be used as acceptable void space fillers for waste that does not fill the container.

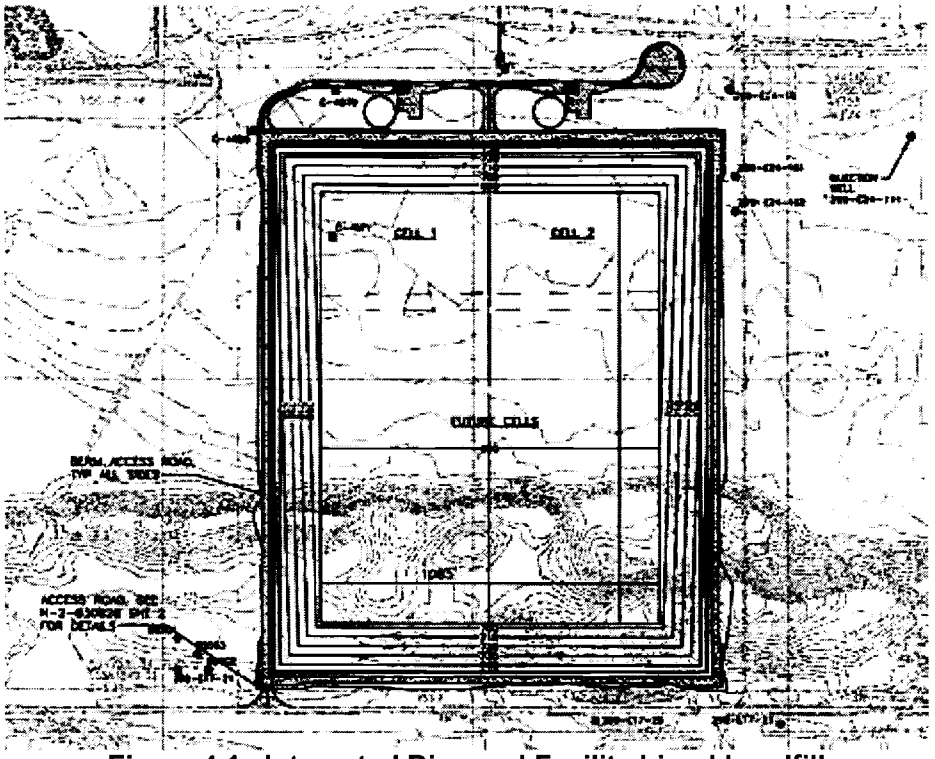

Figure 4.1. Integrated Disposal Facility Lined Landfill

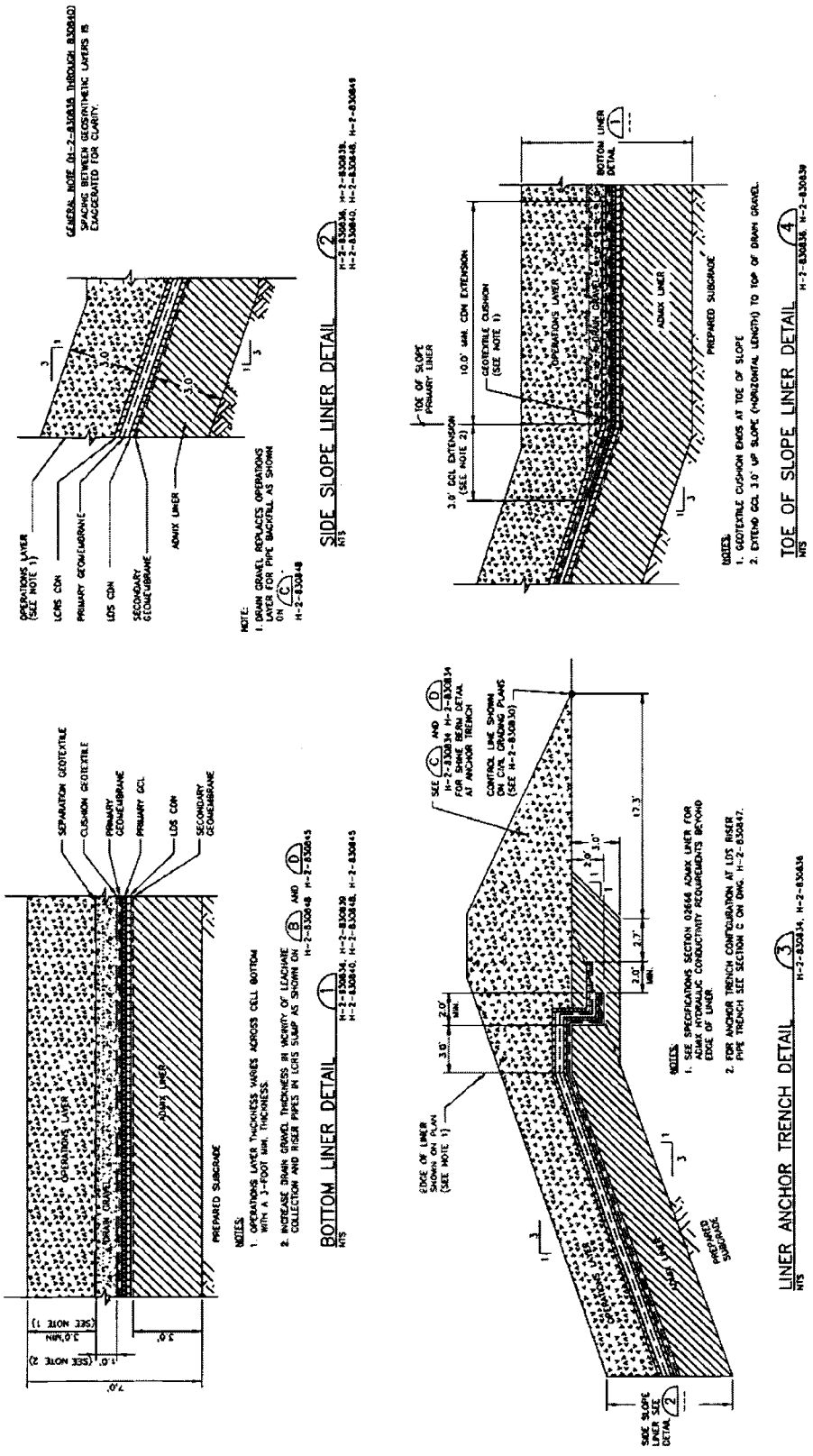

Figure 4.2. Example of a Typical Liner

APPENDIX 4A – SECTION 1
PHASE I CRITICAL SYSTEMS DESIGN REPORT

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APPENDIX 4A – SECTION 1
PHASE I CRITICAL SYSTEMS DESIGN REPORT

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Appendix D – Waste Placement

D.1 Waste Placement Plans

ACRONYMS AND ABBREVIATIONS

AASHTO	American Association of State Highway and Transportation Officials
Affiliate	CH2M HILL, Inc.
AFI	Air freeze index
ALR	Action leakage rate
AOS	Apparent opening size
ASCE	American Society of Civil Engineers
ASTM	American Society for Testing and Materials
AWWA	American Water Works Association
bgs	Below ground surface
CDN	Composite drainage net
CDR	Conceptual Design Report
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act
CFR	Code of Federal Regulations
CH2M HILL	CH2M HILL Hanford Group, Inc.
Design Report	IDF Phase I Critical Systems Design Report
DOE	U.S. Department of Energy
DBVS	Demonstration Bulk Vitrification System
Ecology	Washington State Department of Ecology
EPA	U.S. Environmental Protection Agency
FH	Fluor Hanford, Inc.
FLA	Full load amperage
FS	Factor of safety
FVNR	Full Voltage Non-Reversing
GCL	Geosynthetic clay liner
GFCI	Ground fault circuit interrupters
gpm	Gallons per minute
GRI	Geosynthetic Research Institute
HDPE	High-density polyethylene
HEC	Hydraulic Engineering Circular-1
HELP	Hydrologic Evaluation of Landfill Performance (Model)
HF	Hanford Facility
HMS	Hanford Meteorological Station
HVAC	Heating, ventilating, and air conditioning
I/O	Input/output
ICDF	INEEL CERCLA Disposal Facility (Idaho Falls, ID)
IDF	Integrated Disposal Facility (Hanford)
IEEE	Institute of Electrical and Electronic Engineers
IES	Integrated Engineering Software, Inc.
ILAW	Immobilized low-activity waste
INEEL	Idaho National Environmental Engineering Laboratory
LAN	Local area network
LCRS	Leachate collection and removal system
LDS	Leak detection system
LERF	Liquid Effluent Retention Facility (Hanford)
LLW	Low-level waste
MBPS	Megabits per second
MCC	Motor control center
MLLW	Mixed low-level wastes
NEC	National Electrical Code
NFPA	National Fire Protection Association
OIU	Operator interface unit

ORP	Office of River Protection
PC	Performance category
PICS	Process Instrumentation and Control Systems
PLCs	Programmable logic controllers
PNNL	Pacific Northwest National Laboratory
psi	Pounds per square inch
PVC	Polyvinyl chloride
QA	Quality Assurance
QC	Quality Control
RAP	Response Action Plan
RCRA	Resource Conservation and Recovery Act of 1976
RF	Radio frequency
RGS	Rigid galvanized steel
RPP	River Protection Project
SCADA	Supervisory control and data acquisition
SDR	Standard dimension ratio
SOW	Statement of work
SPT	Standard Penetration Testing
SSCs	Systems, structures, and components
STI	Soil Technology, Inc. (Bainbridge Island, Washington)
THW	Thermoplastic, vinyl insulated building wire; flame retardant, moisture and heat resistant, 75°C, dry and wet locations
TSD	Treatment Storage and Disposal facility
TRU	Transuranic waste (concentrations of transuranic radionuclides greater than or equal to 100nCi/g of the waste matrix)
UBC	Uniform Building Code
UPS	Uninterrupted power supply
USCS	Unified Soil Classification System
WAC	Washington Administrative Code
WSDOT	Washington State Department of Transportation
WTP	Waste Treatment and Immobilization Plant (Hanford)

1.0 INTRODUCTION

1.1 Purpose

The purpose of the Integrated Disposal Facility (IDF) is to develop the capability for near surface disposal of Immobilized Low-Activity Waste (ILAW) waste packages from the River Protection Project–Waste Treatment and Immobilization Plant (RPP-WTP). The IDF is essential in meeting the overall U.S. Department of Energy, Office of River Protection (ORP) mission to store, retrieve, treat, and dispose of the highly radioactive Hanford tank waste in an environmentally sound, safe, and cost-effective manner. The IDF will also provide capacity for disposal of mixed low-level waste (MLLW) and low-level (LLW) from the DBVS. The detailed design for the IDF Phase I Critical Systems landfill will finalize the design process for the:

- Landfill liner system
- Leachate removal system
- Leak detection system (LDS)

The IDF detailed design also involves completing all design work required for an operable landfill and supporting the Resource Conservation and Recovery Act of 1976 (RCRA) Part B permitting for the IDF.

This Phase I Critical Systems Design Report (the Design Report) provides documentation of engineering calculations, criteria, and information that have been developed as part of the IDF detailed design for Phase I. Specifically, the Design Report documents the following important design information:

- Identifies key design requirements for the project (Section 2).
- Summarizes studies on site conditions and investigations that have been used in the development of detailed design parameters for the critical systems (Sections 3 and 4).
- Presents detailed engineering analysis performed in the development of the Phase I Critical Systems design and updated during construction implementation (Section 5).
- Provides system component descriptions, references important construction quality assurance (QA) requirements, and describes important interfaces with non-critical systems (Section 6).
- Describes operating provisions that have influenced the development of the design including waste placement requirements, operational interfaces with other Hanford facilities, and leakage response action plan requirements (Section 7).

1.2 Scope

1.2.1 General

CH2M HILL, Inc. (Affiliate) is responsible for production of a cost-effective final design and to produce critical systems detailed design documents and construction specifications to facilitate RCRA permit approval of the IDF. The IDF technical requirements are found in the following documents:

- Immobilized Low-Activity Waste (ILAW) Project Definition Criteria, Revision 1 (RPP-7898)
- System Specifications for ILAW Disposal, Revision 3 (RPP-7307)
- Hanford Environmental Management Specification (DOE/RL-97-55)

Design products are to be prepared in compliance with the technical requirements, as well as with other specific procedures that are dictated by CH2M HILL Hanford Group, Inc. (CH2M HILL) requirements and outlined in the Statement of Work (SOW), *Integrated Disposal Facility Detailed Design Support* (Rev. 2, 2003), described in more detail under Section 2 of this Design Report. The overall design work includes reports, schedules, estimates, and other special services as specified in the SOW. As part of the design effort, the Affiliate will perform the following global tasks:

- Develop a conceptual layout and preliminary design drawings for the IDF. The IDF preliminary layout will depict a single expandable landfill system, with capability for segregation of RCRA regulated and non-regulated waste placement and segregated leachate management systems.
- Develop a detailed design that meets the requirements of the ILAW Project Definition Criteria and the ILAW System Specification.
- Develop the construction specifications for the detailed design.
- Ensure that there is full technical integration between all detailed design reports prepared for the detailed design of the IDF.
- Perform the design activities in accordance with all applicable regulatory requirements.

The design will implement the safety and health protection requirements imposed on the design by the SOW and the technical baseline criteria documents, and will comply with all applicable regulatory requirements for the project. It is important to note that although the design is for identified critical systems of the Phase I IDF, a preliminary safety evaluation was performed for the W-520 Project that identified no safety class items, including criticality safety (*Conceptual Design Report for ILAW Facility,* CH2M HILL, May 2001).

The timely completion of the critical system detail design of the IDF, in compliance with the RCRA permit approval process (Washington Administrative Code [WAC] 173-303-665 and 173-303-806[4][h]), is a critical component of the SOW. Drawings, construction specifications, and reports needed to obtain U.S. Department of Energy (DOE) certification and Washington State Department of Ecology (Ecology) approval of the IDF RCRA Part B permit is the overall goal of the project. The detailed design for the initial Phase I disposal landfill and the critical systems design include the liner system, the leachate collection system, and the LDS. The detailed design will produce an operable landfill design and support the IDF RCRA Part B permitting.

1.2.2 Design Report

The Design Report describes the key facility components and provides the design basis and detailed calculations that support the development of drawings and specifications. Key facility components that are described in the Design Report include:

- Facility layout (location, access roads and operational ramps, survey control system).
- Landfill geometry (disposal volume total and per disposal unit, disposal unit dimensions).
- Disposal unit grading design (foundation soils contour, lower admixture layer contour, operations layer cover contour).
- Grid point listing (grid point number, location, and elevation for all grid points required for construction of the IDF).
- Geosynthetic material design (primary geomembrane, secondary geomembrane, geotextile, and geocomposite drainage layer).
- Leachate collection and removal system (LCRS) and LDS design (sump design, removal system design–LCRS and LDS, leachate level monitoring system design, transfer pump as required to meet WAC-173-303-665(2)(h)(ii) to ensure that the leachate depth over the liner does not exceed 12 inches).
- Leachate temporary storage tank system design (tank volume, tank design, tank materials/ leachate compatibility, tank coating, tank secondary containment system), including electrical and power requirements necessary to support the leachate removal systems.
- Pump controls and instrumentation design (control, operations, monitoring, and control building design).
- Operational storm water management design.

- Backfill placement requirements and process (minimize void space, minimize subsidence of waste, placement and material requirements to ensure there are no adverse effects on the waste packages).
- Other facility designs identified as necessary to support the project completion.

The Design Report includes design calculations that are prepared in accordance with the requirements of procedure HNF-IP-0842 Vol. 4, Section 3.6 (July 30, 2002). Important calculations that are documented include:

Stability (liner side slope [each liner layer based on interface strength], requirements for verification for critical interface strengths, fill placement ramp, global stability of the overall design, and other relevant stability analysis).

Seismic analysis (side slope and global embankment stability under seismic loading, and seismic design of structures)

Bearing capacity (liner sub-grade soils and other relevant bearing capacity analysis)

Total settlement, differential settlement, and uplift analysis (foundations soils, compacted admixture layers, total settlement, top slope drainage evaluation, subsidence and sinkhole potential, uplift potential, and other relevant settlement analysis).

Admix liner analysis (liner admixture bearing capacity, admix liner specifications, desiccation cracking, and other relevant liner admixture analysis).

Geomembrane liner analysis (liner tension caused by thermal contraction/ expansion, anchor trench pullout analysis, puncture resistance, potential stress cracking, leachate compatibility, chemical and radiation resistance, mechanical degradation from operational traffic, and other relevant geomembrane analysis).

Drainage layers analysis (geotextile analysis and selection, geocomposite selection, drainage gravel selection analysis, and other relevant drainage analysis).

LCRS/LDS analysis (clogging prevention in LCRS, design of leachate collection sumps, design of high capacity and low capacity leachate removal pumping systems, design of leachate storage tank and secondary containment system, leachate depth monitoring system, design of leachate system control building, leachate compatibility of components in the LCRS, and other relevant leachate analysis).

Leachate system earth loading analysis (LCRS and LDS slope riser pipes, LCRS collection pipe, leachate transfer pipes, and other relevant system loading analysis).

Surface stormwater analysis (operations in-cell stormwater management, operations runon/runoff water management, site stormwater collection/evaporation management system, and other relevant storm water analysis).

Leachate production analysis (average annual leachate production, peak daily leachate production, leachate tank storage capacity, leachate transportation truck capacity, and trip frequency)

Action leakage rate (ALR) analysis (the maximum design flow rate that the secondary leachate collection, detection, and removal system can remove without the fluid head on the bottom liner exceeding one foot; calculation and justification of the maximum leachate infiltration rate through the primary liner system; a response action plan in case the maximum ALR is exceeded during operation of the IDF).

Updates to calculations that have occurred through the construction process, either during independent quality reviews of tank systems, in response to contractor's requests for information, or changes implemented during construction have been attached to the original calculations in the appendices.

Compliance matrices have been developed to demonstrate detailed design compliance with the applicable sections of the regulations (WAC 173-303) and with project-specific specifications, criteria, reports, codes, and standards.

Updates to the matrices that have resulted from the completion of construction activities and associated documentation are also provided. These matrices are presented in the Design Report in Appendix A.

1.3 Authorization

After careful consideration and evaluation, CH2M HILL elected to self-perform the IDF Phase I Critical Systems design. As such, the design is being performed as an inter-company work assignment by the Affiliate under the direction of CH2M HILL. CH2M HILL was authorized to self-perform the work by the U.S. Department of Energy, Office of River Protection (ORP), in a letter dated December 9, 2002.

CH2M HILL's Prime Contract Number with the ORP is DE-AC06-99RL14047. The inter-company work assignment is Contract 12317, Release 22, dated November 7, 2002.

1.4 General Facility Description

The IDF will consist of an expandable lined landfill located in the 200 East area on the Hanford Facility (HF). The landfill will be divided lengthwise into two distinct cells, one for disposal of low-level waste (LLW) and the other for disposal of mixed waste. The mission of the IDF will include the following functions:

- Provide an approved disposal facility for the permanent, environmentally safe disposition of ILAW packages that meets the environmental requirements and is approved by the DOE and Ecology.
- Receive ILAW from River Protection Project (RPP) tank operations and dispose this waste onsite. Receive waste from the DBVS and dispose this waste onsite.
- A more detailed discussion of waste types and the necessary storage volumes for these wastes is provided in Sections 5 and 6, respectively.
- The IDF will be constructed on 25 hectares of vacant land southwest of the Plutonium Uranium Extraction Facility (PUREX) Plant in the 200 East Area. The IDF will consist of a lined landfill that will be constructed in several phases. The landfill will be segregated into a RCRA permitted cell and a non-RCRA permitted cell. The scope of this permit is limited to the western cell of the landfill where the RCRA waste will be stored and disposed. The landfill is designed to accommodate four layers of vitrified LAW waste containers separated vertically by 0.9-meters of soil.
- This initial construction will start at the northern edge and the size is approximately 223 meters East/West by 233 meters North/South by 14 meters deep. At this initial size, IDF disposal capacity is 82,000 cubic meters of waste. Subsequent construction phase(s) will require a modification to the Part B Permit to be constructed after waste placement has progressed in the landfill to the point that additional disposal capacity is needed. This approach minimizes the open area susceptible to collection of rainwater and subsequent leachate
- The landfill is currently estimated at full build out to be up to 446 meters wide by 555 meters in length by up to 14 meters deep. The RCRA regulated portion of the landfill would be half of that at approximately 223 meters wide by 555 meters long by up to 14 meters deep providing a waste disposal capacity of up to 450,000 cubic meters.
- Both cells will have a RCRA C-compliant liner system that consists of an upper primary liner overlying a lower secondary liner. The upper liner will consist of a composite geomembrane liner and geosynthetic clay liner system on the bottom area, and a single geomembrane on the side slope. The secondary liner will consist of a composite geomembrane, overlying a 3-foot-thick soil admix liner. A LCRS and a LDS will overly the primary and secondary liner system, respectively. A Secondary Leak Detection System (SLDS) will be located below the clay liner, beneath the LDS sump.
- The IDF also will include a less than 90-day accumulation area of leachate for storage in two tanks, one per landfill half. The leachate storage tanks will be located at the north end, in close proximity to the lined landfill.

Each tank will be protected by secondary containment (double-lined tanks). Leak detection will be provided by monitoring of the secondary containment. The collected leachate will be stored and sampled before transfer to an onsite Treatment Storage and Disposal (TSD) unit or offsite TSD facility. The less than 90-day storage leachate collection tank will be operated in accordance with the generator provisions of WAC 173-303-200 and WAC 173-303-640, as referenced by WAC 173-303-200. The overall side development plan is shown in Figure 1-2.

- The landfill will be constructed in several phases. Starting at the northern edge, approximately one-third of the total length of the landfill will be constructed in Phase I. This will include the leachate collection system and 90-day accumulation tanks. The subsequent phases will be constructed after waste has been placed in the landfill and additional disposal capacity is needed. This approach will minimize the amount of open area susceptible to collection of rainwater and subsequent leachate.
- Before disposal, all waste will meet land disposal restriction requirements [Revised Code of Washington 70.105.050(2), WAC 173-303-140, and 40 Code of Federal Regulations (CFR) 268, incorporated by reference in WAC 173-303-140].
- Future landfill development and configuration within the IDF will be subject to change as disposal techniques improve or as waste management needs dictate. Additional IDF landfill development beyond the 62 acres will be subject to an approved permit modification, in accordance with the HF RCRA Permit (Ecology, 2001).
- Public access to the IDF will be restricted. Figure 1-3 depicts the normal transportation routes within the 200 East Area. Trucks typically will be used to transport waste to the IDF and will range in size from heavy-duty pickups to tractor-trailer rigs, depending on the size and weight of the load. In some cases, special equipment (such as transporters) will be used for unusual or unique loads. When special equipment is used, a prior evaluation will ensure that the equipment does not damage the roadways.
- Approximately 60 personnel will traverse this roadway, in personal vehicles in three shifts per 24 hours per week.

Figure 1-1. Integrated Disposal Facility Site Plan
Located in Chapter 1.0, Part A Form

Figure 1-2. Overall Site Development Plan/Transportation Routes
Located in Chapter 4.0, Figure 4.1

2.0 DESIGN REQUIREMENTS

Minimum design requirements for the IDF Phase I Critical Systems Design were provided by CH2M HILL in the SOW for Requisition # 92859, Integrated Disposal Facility Detailed Design Support, Revision 2, February 18, 2003. The IDF Phase I Critical Systems Design has been performed in compliance with all applicable design requirements, defined in Sections 2.1 through 2.7, and these requirements are:

- *Washington State Dangerous Waste Regulations* (WAC 173-303)
- *System Specification for Immobilized Low-Activity Waste Disposal System,* Revision 3 (RPP-7307)
- *ILAW Project Definition Criteria for Integrated Disposal Facility,* Revision 1 (RPP-7898)
- *Hanford Site Environmental Management Specification,* Revision 2 (DOE/RL-97-55)
- *Design Loads for Tank Farm Facilities* (TFC-ENG-STD-06, REV A)
- Technical baseline documents listed in Section 3.1 of the SOW
- Applicable national codes and standards

2.1 Washington State Dangerous Waste Regulations

The *Washington State Dangerous Waste Regulations* (WAC 173-303) implement Subtitle C of Public Law 94-580, the RCRA in the State of Washington. By conforming to the requirements of WAC 173-303, the design of the IDF Phase I Critical Systems also complies with the federal hazardous waste requirements contained in 40 CFR 264, *Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities.* Appendix A.1 provides a compliance matrix of where the applicable WAC 173-303 requirements are addressed in the IDF Phase I Critical Systems detailed design documents, or are addressed in documentation developed as a result of facility construction.

2.2 System Specification

The *System Specification for Immobilized Low-Activity Waste Disposal System*, Revision 3 (RPP-7307) contains the Level 1 system requirements for the Immobilized Low-Activity Waste Disposal System, of which the IDF is a part. Appendix A.2 provides a compliance matrix of where the applicable Level 1 system requirements are addressed in the IDF Phase I Critical Systems detailed design documents, or are addressed in documentation developed as a result of facility construction.

2.3 Project Definition Criteria

The *ILAW Project Definition Criteria for Integrated Disposal Facility*, Revision 1 (RPP-7898) contains the design criteria for the IDF, including requirements flow-down from RPP-7303, *System Specification for ILAW Disposal System*, and DOE/RL-97-55, *Hanford Site Environmental Management Specification.* Appendix A.3 provides a compliance matrix of where the applicable design criteria are addressed in the IDF Phase I Critical Systems detailed design documents, or are addressed in documentation developed as a result of facility construction.

2.4 Hanford Site Environmental Management Specification

The *Hanford Site Environmental Management Specification* (site specification), Revision 2 (DOE/RL-97-55) documents the top-level mission technical requirements for work involved in the Richland Operations Office, Hanford Site cleanup and infrastructure activities, under the responsibility of the DOE Office of Environmental Management. It also provides the basis for all contract technical requirements. Section 3.3.2, 200 Area Materials and Waste Management of the site specification contains the requirements for receiving and onsite disposal of ILAW from RPP tank operations. The documents, orders, and laws referenced in the site specification represent only the most salient sources of requirements. As such, the site specification is assumed to have no significant measurable requirements that would directly affect the IDF Phase I Critical Systems design.

2.5 Design Loads for Tank Farm Facilities

The *Design Loads for Tank Farm Facilities* (TFC-ENG-STD-06, REV A) defines the design requirements for systems, structures, and components (SSCs), and provides the minimum criteria for structural design and evaluation of SSCs. The standard establishes structural design loads and acceptance criteria for use in designing new SSCs. Figure 1 of this standard indicates that for new SSCs, structures and anchorage of systems and components are to be designed per DOE-STD-1020-02 and Section 3.0 of this standard. These were used for the design of the IDF Critical Systems facilities. The IDF Critical Systems facilities were defined by CH2M HILL as being Performance Category (PC)-1. The PC-1 requirements in this standard were used in the structural design of the facilities included in IDF Phase I Critical Systems.

2.6 Technical Baseline Documents

The technical baseline documents are listed in Section 3.1 of the SOW. These documents include the System Specification for Immobilized Low-Activity Waste Disposal System, ILAW Project Definition Criteria for Integrated Disposal Facility, Hanford Site Environmental Management Specification, and Design Loads for Tank Farm Facilities, discussed in the preceding sections.

2.7 National Codes and Standards

In addition to WAC 173-303, the system specification, project definition criteria, site specification, and tank farm design loads that are discussed above, the IDF Phase I Critical Systems design was guided by other applicable sections of accepted professional and industry standards. These included the following:

- Air Moving and Conditioning Association
- American Association of State Highway and Transportation Officials (AASHTO)
- American Concrete Institute
- American Galvanizers Association
- American Institute of Steel Construction
- American Iron and Steel Institute
- American National Standards Institute
- American Society for Testing and Materials (ASTM)
- American Society of Civil Engineers (ASCE)
- American Society of Heating, Refrigerating, and Air-Conditioning Engineers
- American Society of Mechanical Engineers
- American Water Works Association (AWWA)
- American Welding Society
- Building Officials and Code Administrators – Basic Building Code
- Code of Federal Regulations (CFR)
- Concrete Reinforcing Steel Institute (CRSI)
- Federal Standards
- Geosynthetic Research Institute (GRI)
- Hydraulic Institute Standards
- Institute of Electrical and Electronic Engineers (IEEE)
- International Conference of Building Officials – Uniform Building Code (UBC)
- Manufacturers Standardization Society
- Metal Building Manufacturers Association
- National Electrical Code (NEC)
- National Electrical Manufacturers Association

- National Fire Protection Association (NFPA)
- National Institute of Standards and Technology
- Occupational Safety and Health Administration
- Sheet Metal and Air Conditioning Contractors National Association
- Steel Door Institute
- Steel Structures Painting Council
- Specialty Steel Institute of North America
- The Aluminum Association, Inc.
- Underwriters Laboratories, Inc.
- Washington State Department of Transportation (WSDOT) Standard Specifications for Road, Bridge and Municipal Construction

3.0 SITE CONDITIONS

This section presents information on the Hanford Site and the area on the site where the IDF will be located. This information was obtained primarily from the *ILAW Preliminary Closure Plan for the Disposal Facility* (RPP-6911) and other Hanford Site data sources. It is intended to provide a general characterization of the IDF site conditions that are pertinent to the design of the IDF Phase I Critical Systems.

3.1 Geography

The following paragraphs briefly describe the geography of the IDF site and are prepared from information in the *ILAW Preliminary Closure Plan for the Disposal Facility* (RPP-6911).

3.1.1 Site Location

The location of the IDF is on the Hanford Central Plateau, in the 200 East Area within the Hanford Site boundary. The site identified for the IDF is 68 hectares (168 acres) of vacant and uncontaminated land, located southwest of the PUREX plant in the 200 East Area. It is bounded on the south by 1st Street and on the north by 4th Street.

3.1.2 Site Description

The IDF landfill will occupy approximately 25 hectares (62 acres) of the site identified for the facility. The remainder of the site will be used for soil stockpile, leachate storage tanks, operations support facilities, roads, parking areas, and open space. The IDF in Phase I will be approximately 11 hectares (28 acres). Phase I will be located at the north end of the IDF landfill and will include provisions for expansion to the south for future phases.

3.2 Meteorology and Climatology

The following paragraphs briefly describe the climate of the IDF site and are prepared from information in the *ILAW Preliminary Closure Plan for the Disposal Facility* (RPP-6911), which presented summary data from the Hanford Meteorological Station (HMS). Conditions at the HMS are considered similar to those at the IDF site. Detailed information is available in the *Hanford Site Climatological Data Summary 2001, with Historical Data* (Pacific Northwest National Laboratory, May 2002). The IDF Phase I Critical Systems is designed to operate in the climatic conditions reported in that document.

3.2.1 Precipitation

The site sits within the Pasco Basin, characterized as a semi-arid region because of its low annual precipitation levels. The basin receives 16 cm (6.3 inches) of annual average precipitation, with nearly half occurring in the winter months. Historical records indicate that the annual precipitation has varied from a low of 8 cm (3.1 inches) to a high of 30 cm (11.8 inches). Precipitation of 4 cm (1.56 inches) in 24 hours reportedly can be expected to occur once every 25 years.

However, based on the *Hanford Site Climatological Data Summary 2001*, a value of 1.28 inches was used for the 24-hour, 25-year precipitation in the IDF Phase I Critical Systems stormwater design analysis (see Appendix C.9). Total annual snowfall has varied from 0.8 cm to 110 cm (0.31 to 43.3 inches), with an average annual snowfall of 34 cm (24.4 inches).

3.2.2 Temperature

Temperature conditions for the site range from extremely cold during the winter months to extremely warm during the summer months. Local temperatures can reach -18 degrees C (0 degrees F) during some winter months. January is the coldest month, with an average temperature of -2 degrees C (29 degrees F). The lowest temperature ever recorded was -33 degrees C (-27 degrees F). During some summer months, daytime temperatures can exceed 40 degrees C (104 degrees F). July is the warmest month, with daily high and low temperatures averaging 33 and 25 degrees C (92 and 61 degrees F), respectively. The highest temperature ever recorded was 46 degrees C (115 degrees F).

3.2.3 Wind

Wind conditions can vary considerably throughout the year. The monthly average is about 10 kilometers/hour (6 miles/hour) during the winter and 15 kilometers/hour (9 miles/hour) during the summer. Wind speeds, especially during summer storm activity, can reach many times the average levels. The greatest peak gust was 130 kilometers/hour (81 miles/hour), recorded at 15 meters (50 feet) above the ground at the HMS.

3.2.4 Relative Humidity

The seasonal variation in the relative humidity is considerable, according to records of the HMS. The annual mean relative humidity recorded at HMS is approximately 54 percent, with the highest monthly average relative humidity (80 percent) occurring in December and the lowest monthly average relative humidity (32 percent) occurring in July. Daily relative humidity can change 20 to 30 percent between early morning and late afternoon, except in the winter months when changes are less pronounced.

3.3 Ecology

The following paragraphs briefly describe the ecology of the Hanford Site and are prepared from information in the *ILAW Preliminary Closure Plan for the Disposal Facility* (RPP-6911). The site consists of undeveloped land and is characterized as a shrub-steppe environment. This environment contains numerous plants and animal species, adapted to the regions semi-arid climate. Because of the aridity and low water-holding capacity of the soils, the productivity of both plants and animals is relatively low. The IDF site exhibits many of these same general characteristics, although to varying degrees.

3.3.1 Flora

The dominant plants on the Hanford Site are big sagebrush, rabbitbrush, cheatgrass, Russian thistle, and Sandberg's bluegrass, with cheatgrass providing half of the plant cover. Root penetration to depths of over 3 m has not been demonstrated in the 200 Areas. Rabbitbrush roots have been found only at a depth of 2.4 m (8 feet) near the 200 Areas.

3.3.2 Fauna

A variety of birds and mammals inhabit the Hanford Site. The most abundant nesting birds of the shrub-steppe at the site are the horned lark and western meadowlark. Significant populations of chukar and grey partridge inhabit the Hanford Site. The most abundant mammals at the site are mice, ground squirrels, gophers, voles, and cottontail rabbits. Larger animals include mule deer and elk. The coyote is the principal mammalian predator on the Hanford Site.

3.4 Geology

3.4.1 Regional Geology

The 200 East Area lies on the Cold Creek bar, a geomorphic remnant of the cataclysmic, glacial related floods of the Pleistocene Epoch. As the floodwaters raced across the lowlands of the Pasco Basin and Hanford Site, floodwaters lost energy and began to deposit sand and gravel. The 200 Area Plateau is one of the most prominent deposits. The 200 Area Plateau lies just southwest of one of the major flood channels across the Hanford Site that forms the topographic lowland south of Gable Mountain.

Borehole data provide the principal source of geologic, hydrologic, and groundwater information for the 200 East Area and the IDF site. Numerous boreholes (both vadose zone boreholes and groundwater monitoring wells) have been drilled in the 200 East Area for groundwater monitoring and waste management studies (Figure 3-1 shows the location of groundwater wells near the IDF site). However, data are limited within the IDF site, primarily because no previous construction or waste disposal activities have occurred in this part of the HF. Most boreholes in the 200 East area have been drilled using the cable tool method and either a hard tool or drive barrel to advance the hole. Some boreholes have been drilled by rotary and wire-line coring methods. More recently, boreholes in the area have been drilled, and in five cases cored, by percussion hammer methods. Geologic logs are based on examination of drill core, chips, and cuttings from these boreholes. Chip samples typically are taken at 1.5-meter (4.92 feet) intervals and routinely archived at the Hanford Geotechnical Sample Library.

3.4.2 Site Geology

The IDF site will be located south of the Gable Mountain segment of the Umtanum Ridge anticline and about 3 kilometers (1.86 miles) north of the axis of the Cold Creek syncline, that controls the structural grain of the basalt bedrock and the Ringold Formation. The basalt surface and Ringold Formation trend roughly southeast-northwest parallel to the major geologic structures of the site. As a result, the Ringold Formation and the underlying Columbia River Basalt Group gently dip to the south off the Umtanum Ridge anticline into the Cold Creek syncline.

Geologic mapping on the Hanford Site and examination of drill core and borehole cuttings in the area have not identified any faults in the vicinity of the IDF site (DOE/RW-0164). The closest known faults are along the Umtanum Ridge-Gable Mountain structure, north of the disposal site and the May Junction Fault east of the site (Figure 3-2).

3.4.2.1 Stratigraphy

The basalt and post-basalt stratigraphy for the IDF site is shown in Figure 3-3. Approximately 137 to 167 meters (449 to 548 feet) of suprabasalt sediments overlie the basalt bedrock at the site.

Basalt Bedrock. Previous studies (RHO-BWI-ST-14; Reidel and Fecht, 1994) have shown that the youngest lava flows of the Columbia River Basalt Group at the 200 East Area are those of the 10.5 million-year old Elephant Mountain Member. This member underlies the entire 200 East area and surrounding area, and forms the base of the suprabasalt aquifer. No erosional windows in the basalt are known or suspected to occur in the area of the IDF site.

Ringold Formation. Few boreholes penetrate the entire Ringold Formation at the IDF site, so available data are limited. The Ringold Formation reaches a maximum thickness of 95 meters (312 feet) on the west side of the site and thins eastward. The member of Wooded Island (Figure 3-3) is the only member of the Ringold Formation in the 200 East Area. The deepest Ringold Formation unit encountered is the lower gravel, unit A. Lying above unit A is the lower mud, and overlying the lower mud is an upper gravel, unit E. The sand and silt units of the members of Taylor Flat and Savage Island of the Ringold Formation are not present at the IDF site. Unit A and unit E are equivalent to the Pliocene-Miocene continental conglomerates (Reidel and Fecht, 1994). The lower mud is equivalent to the Pliocene-Miocene continental sand, silt, and clay beds (Reidel and Fecht, 1994).

Only three boreholes have penetrated unit A in the area of the IDF site. Unit A is 19 meters (62 feet) thick on the west side of the site and thins to the northeast. Unit A is partly to well-cemented conglomerate consisting of both felsic and basaltic clasts in a sandy matrix and is interpreted as fluvial gravel facies (Lindsey, 1996). There are minor beds of yellow to white interbedded sand and silt. Green-colored, reduced-iron stain is present on some grains and pebbles. Although the entire unit appears to be cemented, the zone produced abundant high-quality water in borehole 299-E17-21 (PNNL-11957, 1998).

Nineteen meters (62 feet) of the lower mud unit were encountered in one borehole at the IDF site (PNNL-11957, 1998). The uppermost one-meter or so consists of a yellow mud to sandy mud. The yellow mud grades downward into about 10 meters (33 feet) of blue mud. The blue mud, in turn, grades down into seven meters (23 feet) of brown mud with organic rich zones and occasional wood fragments. The lower mud unit is absent in the center of the site (northeast of borehole 299-E24-7 on Figure 3-4).

Unit E is described as a sandy gravel to gravelly sand. Unit E is interpreted to consist of as much as 15 meters (49 feet) of conglomerate, with scattered large pebbles and cobbles up to 25 centimeters (9.84 inches) in size in a sandy matrix. The gravel consists of both felsic and basaltic rocks that are well rounded, with a sand matrix supporting the cobbles and pebbles. Cementation of this unit ranges from slight to moderate. The upper contact of unit E is not identified easily at the IDF site. In the western part of the study area, unconsolidated gravels of the Hanford formation directly overly the Ringold Formation unit E gravels, making exact placement of the contact difficult. The dominance of basalt and the absence of cementation in the Hanford formation are the key criteria used to distinguishing these (PNNL-11957, 1998). In the central and northeast part of the area, unit E has been eroded completely. Unconsolidated gravels and sands typical of the Hanford formation replace unit E.

Unconformity at the Top of the Ringold Formation. The surface of the Ringold Formation is irregular in the area of the IDF site. A northwest-southeast trending erosional channel or trough is centered through the northeast portion of the site. The trough is deepest near borehole 299-E24-21 in the northern part of the site (PNNL-13652, 2001). This trough is interpreted as part of a larger trough under the 200 East Area, resulting from scouring by the Missoula floods.

Hanford formation. The Hanford formation is as much as 116 meters (381 feet) thick in and around the IDF site. The Hanford formation thickens in the erosional channel cut into the Ringold Formation and thins to the southwest along the margin of the channel.

At the IDF site, the Hanford formation consists mainly of sand dominated facies and less amounts of silt dominated and gravel dominated facies. The Hanford formation has been described as poorly sorted pebble to boulder gravel and fine- to coarse-grained sand, with lesser amounts of interstitial and interbedded silt and clay. In previous studies of the site (WHC-MR-0391, 1991), the Hanford formation was described as consisting of three units: an upper and lower gravel facies and a sand facies between the two gravelly units. The upper gravel dominated facies appears to be thin or absent in the immediate area of the IDF site (PNNL-12257, 1999; PNNL-13652, 2001; PNNL-14029, 2002).

The lowermost part of the Hanford formation encountered in boreholes at the IDF site consists of the gravel-dominated facies. Drill core and cuttings from boreholes 299-E17-21, 299-E17-22, 299-E17-23, 299-E17-25, and 299-E24-21 indicate that the unit is a clast-supported pebble- to cobble-gravel with minor amounts of sand in the matrix. The cobbles and pebbles almost are exclusively basalt, with no cementation. This unit pinches out west of the IDF site and thickens to the east and northeast (Figure 3-4). The water table beneath the IDF site is located in the lower gravel unit. The lower gravel unit is interpreted to be Missoula flood gravels, deposited in the erosional channel carved into the underlying Ringold Formation.

The upper portion of the Hanford formation consists of at least 73 meters (240 feet) of fine-to coarse-grained sand, with minor amounts of silt and clay and some gravelly sands.

Holocene Deposits. Holocene, eolian deposits cover the southern part of the IDF site. Caliche coatings on the bottom of pebbles and cobbles in drill cores through this unit are typical of Holocene caliche development in the Columbia Basin. The southern part of the IDF site is capped by a stabilized sand dune. The eolian unit is composed of fine- to coarse-grained sands with abundant silt, as layers and as material mixed with the sand.

Clastic Dikes. A clastic dike was encountered in borehole C3828, adjacent to well 299-E17-25 at the IDF site. Clastic dikes also have been observed in excavations surrounding the site (e.g., U.S. Ecology, the former Grout area, the 216-BC cribs, the Central Landfill, and the Environmental Restoration Disposal Facility [PNNL, BHI-01103]). In undisturbed areas such as the IDF site, clastic dikes typically are not observed because these are covered by wind blown sediments. The occurrence of a clastic dike in borehole C3828 suggests that these probably are present elsewhere in the subsurface at the disposal site.

3.4.3 Seismology

The IDF will be located in Zone 2B, as identified in the UBC (DOE/RL-91-28). The analyses in Sections 5.1 and 5.12 provide additional seismic detail for design of liner and structural systems.

No active faults, or evidence of a fault that has had a displacement during Holocene times, have been found on the Hanford Site (DOE/RL-91-28). The youngest faults recognized on the Hanford Site occur on Gable Mountain, over 4.5 kilometers (2.78 miles) north of the 200 East Area. These faults are Quaternary of age and are considered 'capable' by the Nuclear Regulatory Commission (DOE/RL-91-28).

3.5 Hydrology

The following paragraphs briefly describe the known hydrology conditions of the Hanford Site and most specifically the 200 Area Plateau where the IDF site is located. These are prepared from information in the *ILAW Preliminary Closure Plan for the Disposal Facility* (RPP-6911).

3.5.1 Surface Water

The IDF site is within the 200 East area, which is on a plateau above the Columbia River. The Columbia River runs generally to the east and swings around the site, lying about 8 miles northwest and northeast of the 200 East area. The project area is significantly higher than the Columbia River and is not in the river's floodplain.

The soils in the project area are sandy with high rates of infiltration. Most of the precipitation falling on the site infiltrates into the ground, and there are no significant long-term surface water features in the project area.

3.5.2 Groundwater

The geologic structure of the 200 East area is composed of multiple layers of sediments that range from sand, silt, volcanic ash, and clay to coarse gravels, cobbles, and conglomerates that overlay thick layers of basaltic lava. An unconfined aquifer exists in the lower part of the sedimentary sequence, overlaying the uppermost basalt layer. This relatively thin aquifer intercepts infiltration from the unsaturated zone above it. The aquifer under the IDF site is approximately 90 to 100 meters (300 to 330 feet) below the ground surface. Therefore, the groundwater table is well below the proposed bottom of the excavation for the IDF and is not expected to influence the facility. The recharge of water into the ground at the IDF site is expected to be small. This condition results primarily from the low levels of annual precipitation that occur in the region of the IDF as well as the rest of the Hanford Site. A more detailed description of groundwater beneath the IDF, developed from various site explorations performed in the site area, is presented below.

The unconfined aquifer under the IDF site occurs in the fluvial gravels of the Ringold Formation and flood deposits of the Hanford formation.

The thickness of the aquifer ranges from about 70 meters (230 feet) at the southwest corner of the site to about 30 meters (98 feet) under the northeast corner of the IDF site. The Elephant Mountain Member of the Columbia River Basalt Group forms the base of the unconfined aquifer (Figure 3-3).

The unsaturated zone beneath the land surface at the IDF site is approximately 100 meters (328 feet) thick and consists of the Hanford formation. The water level in boreholes in and around the site indicates that the water table is in the lower gravel sequence of the Hanford formation and at an elevation of approximately 123 meters (404 feet) above sea level. The water table is nearly flat beneath the IDF site. Table 3-1 gives water level information from wells near the site. The locations of the wells are shown on Figure 3-1. The latest water table map shows less than about 0.1 meter (3.94 inches) of hydraulic head across the IDF site (PNNL-13404, 2001).

The Ringold Formation lower mud unit occurs within the aquifer at the southwest corner of the IDF site (299-E17-21) but is absent in the central and northern parts of the site (299-E24-7 and 299-E24-21). The lower mud unit is known to be a confining or partly confining layer at places under the Hanford Site (PNNL-12261, 2000), and this might be the case under the southwest corner of the IDF site. Groundwater samples were collected and analyzed from above and below the lower mud unit during drilling of well 299-E17-21. Chemical parameters (pH, electrical conductivity, and Eh) were different in the two samples, suggesting that the lower mud is at least partly confining in the area. No contamination was found above or below the lower mud. An interpretation of the distribution and thickness of this stratum is shown in Figure 3-4. The surface of the lower mud unit is interpreted to dip gently to the southwest (PNNL-13652, 2001).

Hydrographs for selected wells near the IDF site are shown in Figures 3-5 and 3-6. Hydrographs for the older wells (299-E23-1, 299-E23-2, and 299-E24-7) show two maxima in the water level. These coincide with the operation of the PUREX Plant that operated between 1956 and 1972 and between 1983 and 1988. All the hydrographs show a decline in the water table during recent years. The rate of decline is between 0.18 and 0.22 meters (7.08 and 8.66 inches)/year and will take between 10 and 30 years to stabilize. The reason for the decline is the cessation of effluent discharge to the PUREX Plant and to the 216-B Pond System, centered northeast of 200 East area. Based on hindcast water table maps (PNNL, BNWL-B-360), the water table is expected to decline another 2 to 7 meters (7 to 23 feet) before reaching pre-Hanford Site elevations. The cessations of effluent discharge also are responsible for changing the direction of groundwater flow across much of the 200 East area.

Groundwater flow beneath the IDF site recently was modeled to be southeasterly (PNNL-13400, 2000). This direction differs from the easterly direction, predicted by the analysis of WHC-SD-WM-RPT-241 and other earlier reports. The southeasterly flow direction primarily is attributable to inclusion of the highly permeable Hanford formation sediments in the ancestral Columbia River/Missoula flood channel in the analysis. A southeasterly flow direction is reflected in the geographic distribution of the regional nitrate and tritium plumes in the south-central 200 East area (Figure 3-7) (PNNL-13788, 2002.). As stated in PNNL-13404 (2001), the water table gradient is too low to be used for determining flow direction or flow rate at the PUREX Plant cribs, immediately east of the IDF site.

Hydraulic conductivity directly beneath the IDF site was estimated from data collected during four slug tests at well 299-E17-21 and five slug tests of 299-E24-21. The interval tested at 299-E17-21 was the upper 7.8 meters (26 feet) of the unconfined aquifer from 101.3 to 109.1 meters (332 to 358 feet) depth. That portion of the aquifer is Hanford formation gravel, from 101.3 to 102.1 meters (332 to 335 feet) depth, and Ringold Formation unit E gravels, from 102.1 to 109.1 meters (335 to 358 feet) depth (PNNL-12257, 1999). The interval tested at well 299-E24-21 was entirely in the Hanford formation gravel sequence between 95.2 and 101.3 meters (312 and 332 feet) depth. The best-fit value to the data from 299-E17-21 indicated a hydraulic conductivity of about 68.6 meters (225 feet) per day (PNNL-12257, 1999), and that from 299-E24-21 suggested a hydraulic conductivity of 75 meters (246 feet) per day (PNNL-13652, 2001).

Table 3-1. Water Levels in Groundwater Wells in the Vicinity of the IDF Site

Well	Measure date	DTW ma	WT elev mb	Ref elev mc
299-E13-10	3/14/02	101.7	122.5	226.31
299-E17-12	3/14/02	100.0	121.1	221.09
299-E17-13	4/12/01	97.7	122.6	220.34
299-E17-17	4/12/99	97.8	122.8	220.54
299-E17-18	10/3/02	98.5	122.3	220.76
299-E17-20	4/9/97	97.1	123.2	220.33
299-E17-21	4/23/98	100.4	122.7	224.26
299-E17-22	5/20/02	98.1	122.5	220.59
299-E17-23	5/20/02	101.6	122.2	223.84
299-E17-25	5/21/02	98.3	126.7	225.03
299-E18-1	3/14/02	98.2	122.4	220.65
299-E18-3	6/27/96	97.8	123.4	221.20
299-E18-4	6/27/96	97.7	123.4	221.05
299-E19-1	3/22/88	100.4	124.9	225.26
299-E23-1	3/14/02	96.0	122.4	218.39
299-E23-2	12/20/94	97.2	123.5	220.77
299-E24-4	8/10/98	90.6	122.9	213.47
299-E24-7	6/11/97	96.2	123.2	219.34
299-E24-16	10/4/02	97.7	122.3	220.02
299-E24-17	4/7/97	97.36	122.9	220.16
299-E24-18	10/2/02	98.0	122.3	220.35
299-E24-21	3/22/01	95.4	122.6	217.85

a DTW = depth to water
b WT elev = elevation of water table (meters above mean sea level)
c Ref elev = reference elevation (meters above mean sea level, North American Vertical Datam 88 reference), generally top of well casing.

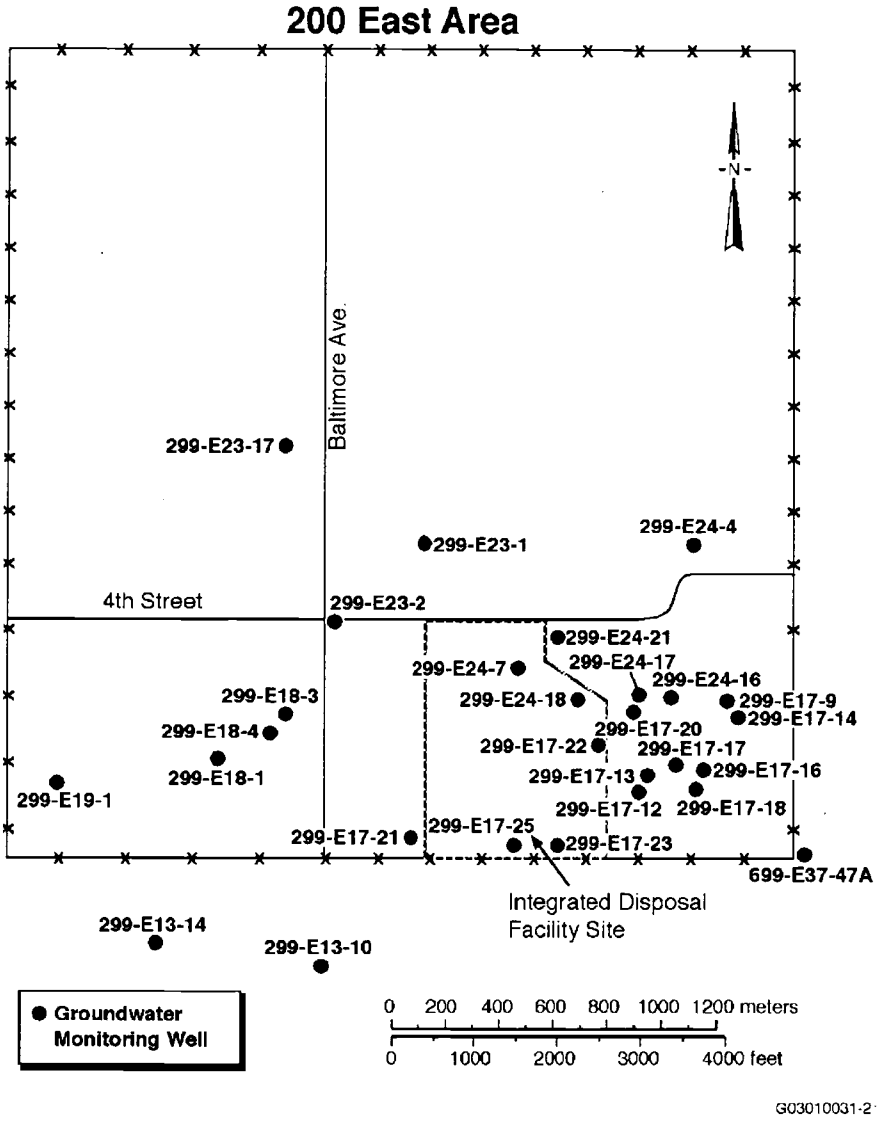


Figure 3-1. Location of the IDF and Nearby Boreholes

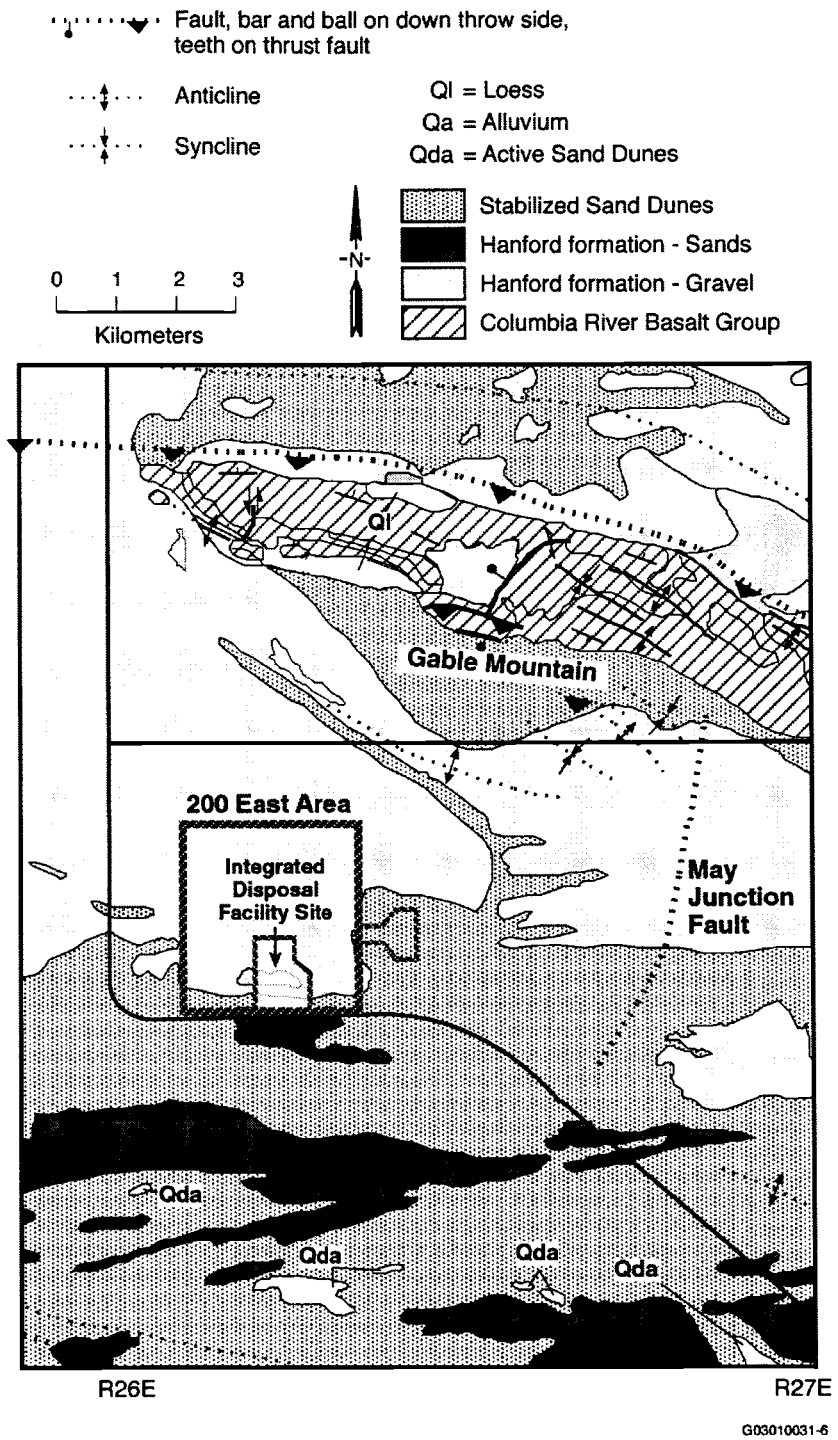


Figure 3-2. Geologic Map of the 200 East and 200 West Areas and Vicinity

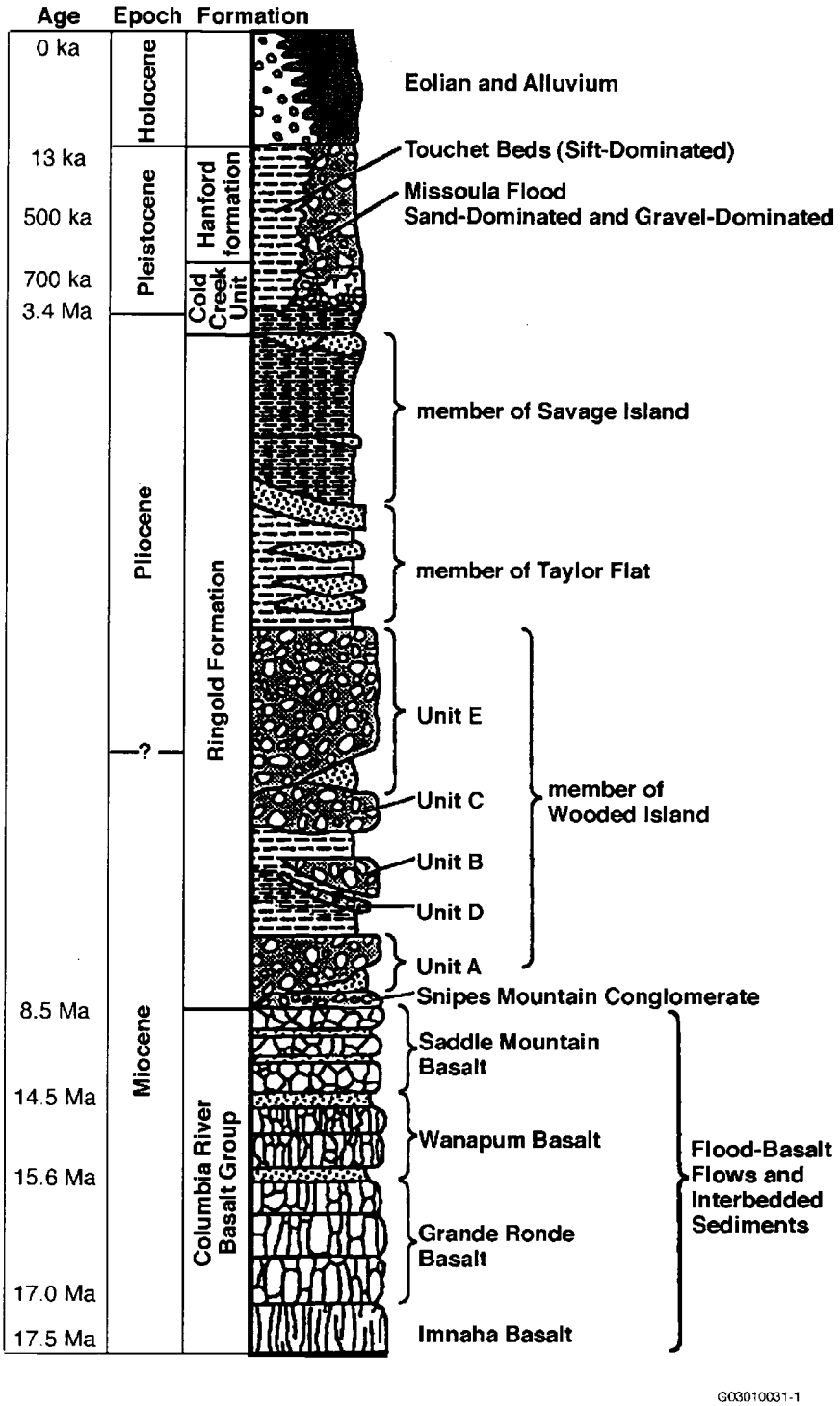


Figure 3-3. Stratigraphy of the Hanford Site

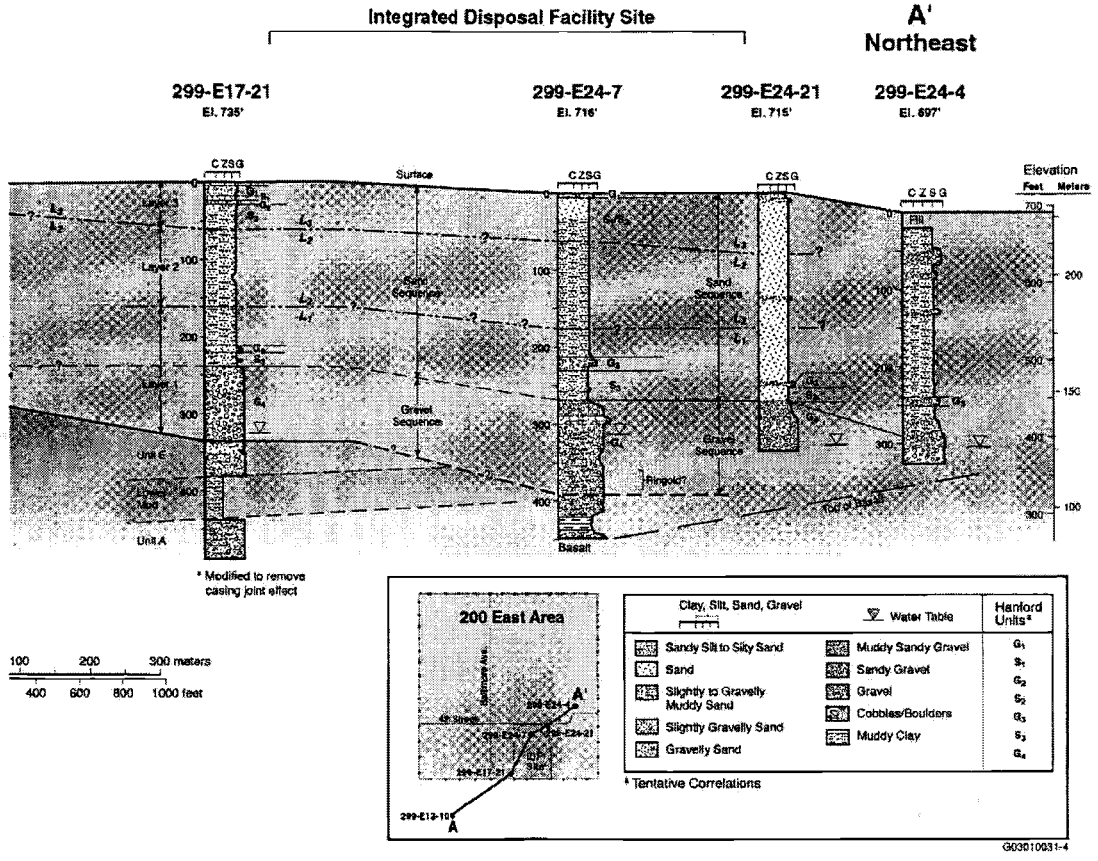
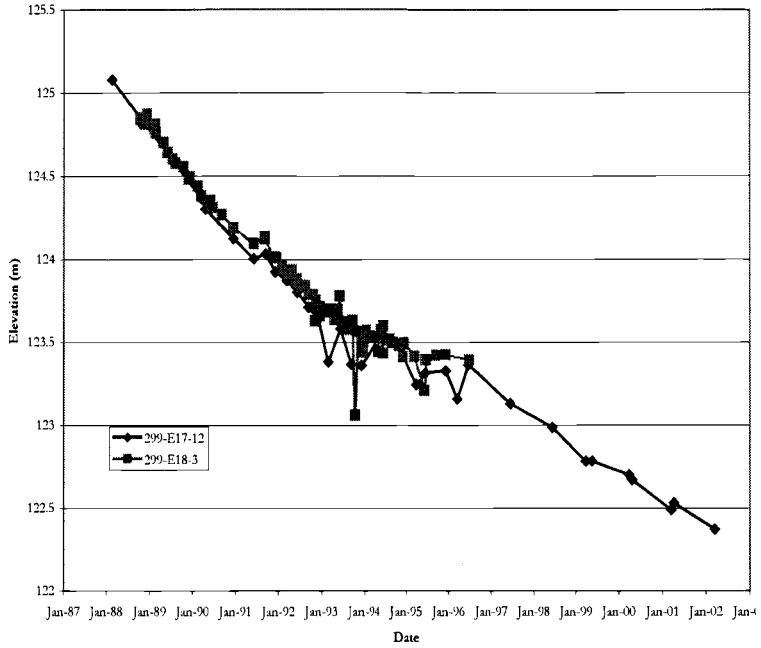


Figure 3-4. Cross-section through the IDF Site (refer to Figure 3-1 for boring exploration locations)



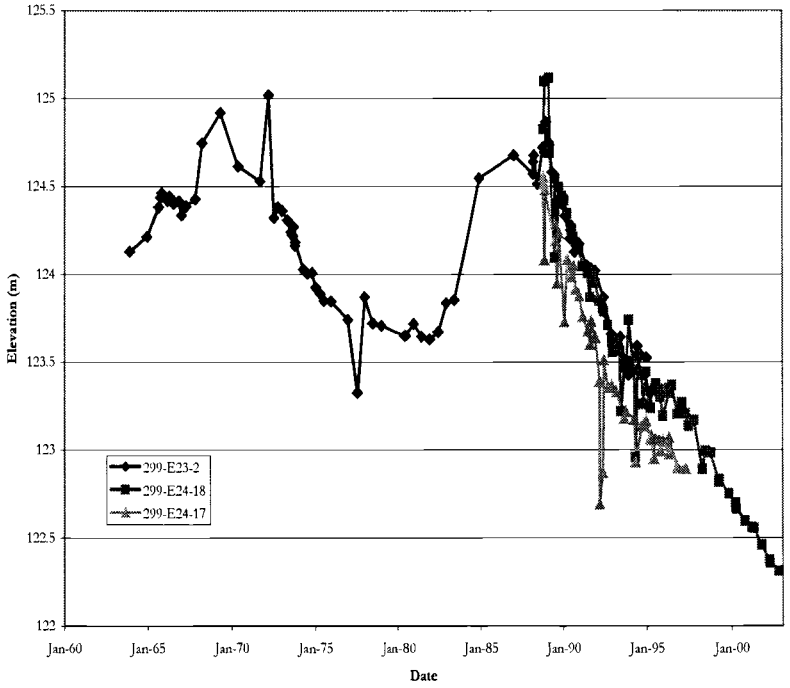


Figure 3-5. Hydrographs for Wells Near the IDF Site (1 and 2 of 3)

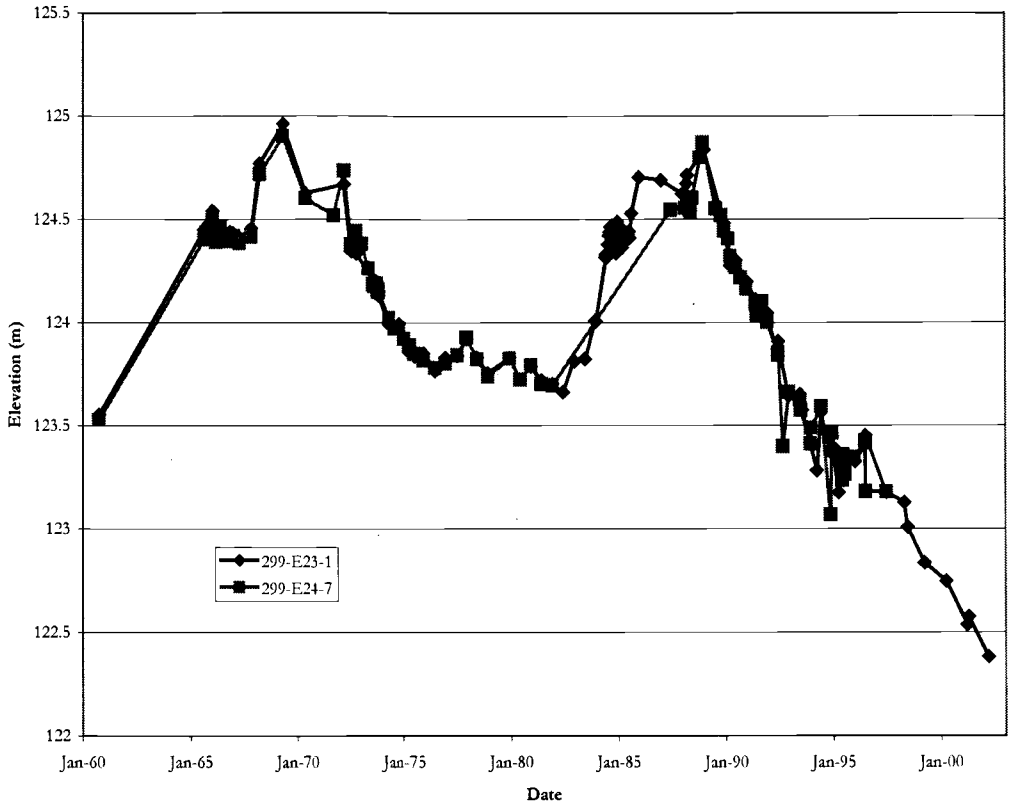


Figure 3-6. Hydrographs for Wells Near the IDF Site (3 of 3)

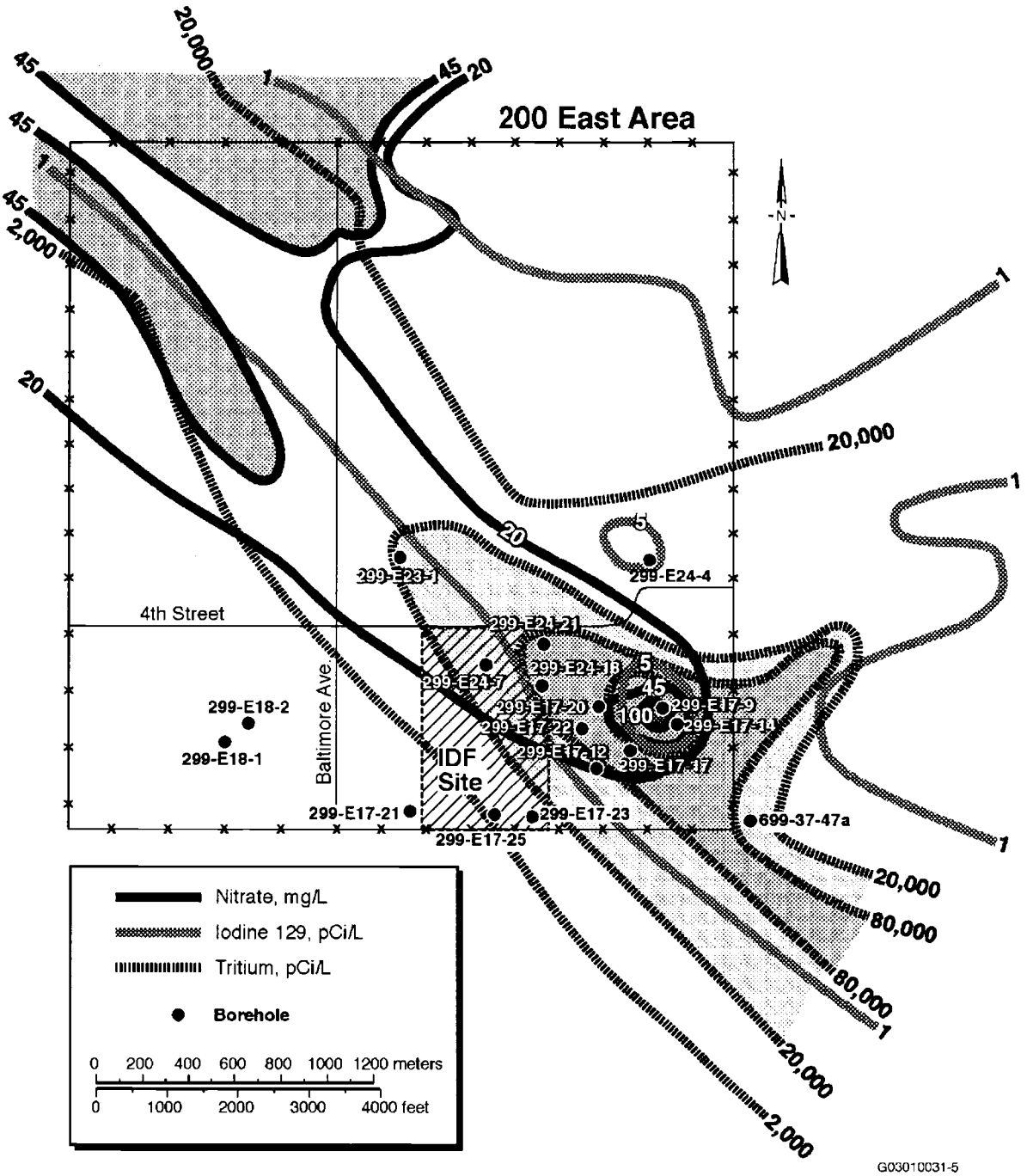


Figure 3-7. Contaminant Plume Map for the 200 East Area

4.0 SITE INVESTIGATION AND LABORATORY TEST PROGRAM

This section presents a summary of the existing, current, and planned explorations for the IDF, along with the laboratory test results for tests conducted during this design effort.

4.1 Field Explorations

This section discusses the existing and current soil explorations for the IDF. The generalized soil profile used in the analysis and design is presented; the engineering soil properties used for design are presented and discussed in Section 5 and related appendices.

4.1.1 Existing Explorations

Several field explorations have been conducted in the general area of the IDF, as discussed in Section 3.4.1 and presented in Figure 3-1. Figure 1-2 shows the current IDF footprint and the closest borings to the planned facility. As shown in Figure 1-2, with the exception of one boring, the existing explorations are all outside of the footprint of the IDF.

The existing field explorations at the IDF site have been conducted primarily for geologic and hydrogeologic characterization on a "big picture" scale. The existing explorations provide detailed information for the purposes for which they were conducted; however, from a geotechnical engineering perspective, the existing borings at or near the IDF site provides only general information, as discussed below.

Depth of Interest–In many cases, the explorations focused on providing detailed information for the entire soil column above the bedrock at the IDF site (300 or more feet below ground surface [bgs]). The primary depth of interest for detailed engineering and design purposes is the depth of the planned cell excavation (roughly 50 feet below the existing ground surface); for a few analyses, information about the material 25 to 50 feet below the base of the excavation is also important.

Type of Information–As intended, the existing explorations was generally focused on providing information for geologic characterization purposes. This focus differs from the key items generally required for geotechnical design, including Standard Penetration Testing (SPT) per ASTM requirements and classification by the Unified Soil Classification System (USCS) in both the field and the laboratory. For coarse-grained soils (sands and gravels), that make up the bulk of the native soil profile, *in situ* SPT in conjunction with grain-size data is the primary basis for determining geotechnical engineering parameters of the soil, such as shear strength. In all cases the SPT values were either not readily available or were conducted with non-standard equipment. Also, the existing grain size data and soil classifications, both for field and laboratory results were based on the Wentworth scale, which differs from the USCS scale at the gravel and fines divisions. These are the key division points for classifying coarse-grained soils. In particular, the break point for fines contents is important in determining the suitability of the excavated soils for use in the admix liner as well as for other on-site filling purposes. Many of the soils within the depth of interest for the IDF are near this classification break point.

Proximity to the IDF–As shown in Figure 1-2, in nearly all cases the explorations were located outside of the IDF footprint. The standard of practice for geotechnical engineering is to place explorations within or very close to the footprint of the proposed structure, if possible.

There have been several geotechnically focused explorations conducted for various projects at Hanford. The projects closest and/or most applicable to the IDF site are:

The Grout Vault project, located approximately one-half mile east of the IDF site (Dames and Moore, 1988).

The W-025 Project, a radioactive mixed-waste land disposal facility designed in accordance with RCRA Subtitle C design criteria, located several miles west of the IDF site (in Area 200W, Golder Associates, 1995, 1994a, 1994b, and 1988).

The RPP-WTP, location approximately 1 mile east of the IDF site (Shannon and Wilson, 2000 and 2001).

These projects all provide geotechnical engineering information; however, the closest site is one-half-mile from the IDF. The standard of care for geotechnical engineering is to either use existing geotechnically based information that is at the site and/or conduct site and project specific explorations. This is to verify that the soil conditions at the site are either still valid (no changes since the time of the existing explorations) or are consistent with existing data.

4.1.2 Current Explorations

Due to the limits of the geotechnical specific data, a subsurface exploration plan specific to the Phase I portion of the IDF was proposed. The suggested locations for the exploration are shown in Figure 1-2. This exploration is currently in planning.

During this design effort, a limited surface sampling plan was conducted at the locations shown in Figure 1-2. The surface samples were taken from the upper 2 to 3 feet of soil, primarily to provide samples for admix testing (to determine if the soils were suitable as a base soil), as well as to help fill in for the absence of a full exploration program at the time of this design effort. As shown in Figure 1-2, samples were taken from primarily from the dune sand borrow area within the IDF footprint (SD-1 through SD-4) and the active sand borrow area (SD-5) to the east of the IDF footprint. One surface sample (SD-6) was obtained from within the IDF Phase I limits.

4.1.3 Site Stratigraphy

In the absence of a comprehensive site and project specific geotechnical engineering data, the existing and current data discussed above was reviewed to determine appropriate soil profile and geotechnical parameters for use in engineering analysis and design. The stratigraphy and soil properties were generally selected conservatively to account for the uncertainty in the subsurface information. The general soil stratigraphy beneath the Phase I section of the IDF was assumed to be:

- 10 feet of Dune (Eolian) sand, overlying
- 50 feet of Upper Hanford sand, overlying

Lower Hanford sand to depth of interest.

It is expected that a greater depth of Dune sand exists in the southern portion of the IDF footprint (note topographic change in the southern one-third of the IDF footprint in Figure 1-2).

The engineering properties and parameters assumed for these soil units were based on the information provided in the geotechnical reports listed in the previous section. The individual values are discussed in Section 5 and related appendices.

4.1.4 Future Explorations

It is recommended that a comprehensive, geotechnically focused exploration program be completed, prior to construction, to verify that the assumptions made for soil stratigraphy and engineering properties are valid. A more comprehensive set of explorations is currently being planned. The planned locations for the additional explorations are shown in Figure 1-2, and include three explorations within the Phase I footprint and one exploration in the proposed sand borrow area.

4.2 Laboratory Testing

A limited laboratory testing program was conducted, using the soils collected during the surface sampling program discussed in Section 4.1.3. These samples were used to perform the index testing, admix testing, and geosynthetics interface shear testing.

4.2.1 Index Testing

Index testing was performed to evaluate the basic index and classification properties of the soil obtained from surface sampling program. This testing was conducted to provide data for comparison with both the soils used for the W025 admix liner and also for other soils that are considered for use as the base soil for the IDF project, as the final design and construction proceeds.

The laboratory testing was conducted by Soil Technology, Inc., (STI) of Bainbridge Island, Washington, under subcontract to the Affiliate. Test assignment and coordination was provided by the Affiliate. Index testing included the following ASTM tests:

- ASTM D422 – Test Method for Particle-Size Analysis of Soils (grain size and hydrometer analyses)
- ASTM D698 – Test Method for Laboratory Compaction Characteristics of Soil Using Standard Effort
- ASTM D1140 – Test Method for Amount of Material in Soils Finer then the No. 200 Sieve (P200 Wash)
- ASTM D1557 – Test Method for Laboratory Compaction Characteristics of Soil Using Modified Efforts
- ASTM D2216 – Test Method for Laboratory Determination of Water (Moisture) Content of Soil and Rock
- Compaction characteristics were also determined for a composite of the surface soils, as described in the next section.

4.2.2 Admix Testing Program

The admix testing program was developed to determine two key items:

- Percentage of sodium bentonite required to meet hydraulic conductivity requirements
- Appropriate moisture and density parameters to achieve the required hydraulic conductivity

Index testing of the admix soils was also conducted, as well as a consolidation test. The laboratory testing was conducted by STI. Tests were run in general accordance with the following:

- ASTM D422 – Test Method for Particle-Size Analysis of Soils (grain size and hydrometer analyses)
- ASTM D698B – Test Method for Laboratory Compaction Characteristics of Soil Using Standard Effort
- ASTM D1557 – Test Method for Laboratory Compaction Characteristics of Soil Using Modified Efforts
- ASTM D2216 – Test Method for Laboratory Determination of Water (Moisture) Content of Soil and Rock
- ASTM D2435 – Test Method for One-dimensional Consolidation Properties of Soils
- ASTM D4318 – Test Method for Liquid Limit, Plastic Limit, and Plasticity Index of Soils (Atterberg Limits)
- ASTM D5084 – Test Method for Measurement of Hydraulic Conductivity of Saturated Porous Materials Using a Flexible Wall Permeameter

The base soil for the admix testing was created by compositing SD-1 through SD-4 from the surface sampling program. This composite did not include SD-5, taken at the base of the existing sand borrow area (lower elevation than the other samples) that has slightly different properties than the remainder of the surface samples. SD-6 was not included at the time of the admix testing because it is not within the footprint of the planned borrow area. The base composite sample was labeled as COMP-1. This composite was then used to create the two other soils for admix testing:

- COMP-2: COMP-1 base soil mixed with 8 percent bentonite
- COMP-3: COMP-1 base soil mixed with 12 percent bentonite
- Moisture and density testing was conducted on all of the composite samples.

The initial hydraulic conductivity testing was conducted using eight and 12 percent bentonite (by weight), based on the results of the admix testing program conducted by Golder for the W025 Project (Golder, 1991b). The target laboratory hydraulic conductivity was less than 10-8 cm/sec when permeated with water. Testing was not conducted with leachate, as no actual leachate exists for the planned waste at this time. Golder Associates used a synthetic leachate to perform compatibility testing on the admix liner. Based on these results, they increased the bentonite percentage from 8 to 12 percent, hence the use of these values in these tests. Because the base soils are expected to be similar to that used by Golder for the W025 landfill, and until a more refined characterization of the IDF leachate is developed, the compatibility testing performed for the W025 project was considered applicable to the IDF project. Hydraulic conductivity testing was performed on all samples in flexible wall triaxial cells with backpressure saturation, in general accordance with ASTM D5084. An effective confining stress of 5 pounds per square in (psi) was applied to each test cell. Appendix B.1 includes the details for the test, including the inflow and outflow data used to confirm that each test had obtained a steady-state hydraulic conductivity value.

After the initial hydraulic conductivity testing was completed, additional samples were set up to determine the range of moisture and density parameters that are expected to produce the required hydraulic conductivity in the field.

As noted above, the samples used for the testing were gathered from the surface sampling program. Once a more comprehensive exploration program is conducted within the IDF footprint, the suitability of the soils within the excavation below a depth of 5 feet (upper 2-3 feet) can be examined for use as a base soil for the admix.

4.2.3 Geosynthetics Interface Shear Testing

A limited soil-to-geosynthetic interface shear testing program was conducted to determine the interface shear values between the operations soil and the composite drainage net (CDN), and the admix liner soils and the high-density polyethylene (HDPE). These interfaces are site specific because of the unique nature of the soils, hence their behavior in interface shear. The testing was conducted by Precision Geotechnical Laboratories in Anaheim, California. Soil samples collected during the surface sampling program were used for testing; GSE Lining Technologies, Inc. based in Houston, Texas provided the geosynthetics for testing.

The interface shear tests were conducted in general accordance with ASTM D5321–Standard Test Method for Determining the Coefficient of Soil and Geosynthetic or Geosynthetic and Geosynthetic Friction by the Direct Shear Method. The tests were conducted for both low (100 to 500 psf) and high (1000 to 8000 psf) normal stress levels, to account for the variation in normal stresses that will be applied across the lining system in the final landfill configuration. Both the peak and residual strength values were determined during testing. Additional details for the tests are presented with the test results in Appendix B.2.

Asperity testing was also conducted on the textured HDPE geomembrane, in general accordance with GRI-GM12 – *Asperity Height of Textured Geomembrane*. The purpose of the asperity testing was to establish a baseline roughness of the texturing of the HDPE geomembrane and for future assessments of the interface shear strength of other textured HDPE geomembrane products (e.g., from other manufacturers).

Site-specific interface shear testing was not conducted for geosynthetic-to-geosynthetic (such as CDN to geosynthetic clay liner [GCL]) interfaces in this phase of design, as these values are primarily a function of the manufactured product properties. A database of values for geosynthetic-to-geosynthetic interface testing was used to determine the appropriate interface shear values for design. During construction, the actual materials used on the site will be tested as part of the construction QC/QA, to ensure that the installed materials used onsite meet or exceed the interface shear strength values used in the design.

4.3 Laboratory Test Results

The results of the laboratory testing programs are summarized below and presented in Appendix B.1 and Appendix B.2.

4.3.1 Index Testing

The results of the index testing for the surface samples are presented in Table 4-1. The test results are included with the admix liner soils test results in Appendix B.1. Results of the index testing indicate that the grain size analyses for near-surface soil samples from locations SD-1 through SD-6 correlate well with data from the W025 base soil material. The W025 base soil was a dune sand (Eolian deposits) obtained from the upper 15 feet of site excavations. As discussed in Section 5.4, based on the results shown below and due to the limited nature of the near surface soil samples, the base soil is limited to the upper 5 feet of material excavated from the dune sand borrow area or the Phase I site excavation.

Table 4-1. Results of the Base Soil Index Testing

Test	Sample #	% Gravel	% Sand	% Fines	OMC, w_{opt} (%)	MDD, γ_{max} (pcf)
Grain Size Testing	SD#1			22.5	--	--
	SD#2		72.2	27.8	--	--
	SD#3			17.5	--	--
	SD#4		78.1	21.9	--	--
	SD#5	2.4	58.5	39.1	--	--
	SD#6		79.5	20.5	--	--
Standard Compaction	SD#6	--	--	--	14	106.6

OMC = optimum moisture content
MDD = maximum dry density

4.3.2 Admix Liner Soils Test Results

The results of the testing program conducted on the admix liner soils are summarized in Tables 4-2 and 4-3 and presented in detail in Appendix B.1. The associated placement and testing requirements during construction are also discussed in detail in Section 5.4.

Table 4-2. Results of the Admix Hydraulic Conductivity Testing

Sample ID	OMC (%)	MDD (pcf)	Remolded MC (%)	Remolded Wet Density (pcf)	Relative Compaction (%)	Saturated Hydraulic Conductivity (cm/sec)[a]	Gradient
COMP2-1	12.8[b]	117.2[b]	13.5	127	95	$2x10^{-8}$	11
COMP2-2	12.8[b]	117.2[b]	17.7	123	89	$4x10^{-8}$	10
COMP3-1	13.0[b]	115.5[b]	13.2	124	95	$<1x10^{-8}$	10
COMP3-2	13.0[b]	115.5[b]	17.4	122	90	$<1x10^{-8}$	10
COMP3-3	10.0[c]	126.3[c]	10.3	136	98	$<1x10^{-8}$	12
COMP3-4	10.0[c]	126.3[c]	14.2	139	96	$<1x10^{-8}$	10
COMP3-5	10.0[c]	126.3[c]	8	130	95	$<1x10^{-8}$	18
COMP3-6	13.0[b]	115.5[b]	10	115	91	$1x10^{-8}$	21
COMP3-7	10.0[c]	126.3[c]	10	123	89	$<1x10^{-8}$	20
COMP3-8	13.0[b]	115.5[b]	11	119	93	$<1x10^{-8}$	16

Abbreviations: OMC = optimum moisture content MDD = maximum dry density pcf = pounds per cubic foot
MC = moisture content
COMP 2 samples had 8 percent bentonite.
COMP-3 samples had 12 percent bentonite.
Average saturated hydraulic conductivity using tap water
Based on standard Proctor compaction curve (D698).
Based on modified Proctor compaction curve (D1557)

1

Table 4-3. Results of Admix Liner Soils Index Testing

Test	Sample #	% Gravel	% Sand	% Fines	LL (%)	PI (%)
Grain Size Testing	COMP-1	--	77.5	22.5	--	--
	COMP-2	--	70.6	29.4	--	--
	COMP-3	--	68.7	31.3	--	--
Atterberg Limits	COMP-2	--	--	--	40	17
	COMP-3	--	--	--	54	32

LL = Liquid Limit
PI = Plasticity Index

2 Consolidation testing conducted on the admix liner soils is presented with the rest of the results in
3 Appendix B.1. This results of this test were used for the settlement analysis discussed in Section 5.3.1.

4.3.3 Geosynthetics Interface Shear Tests

The results of the geosynthetic testing program are presented in Table 4-4; the results of the asperity are shown in Appendix B.2. The results are discussed in detail in Sections 5.1.1 and 5.1.3, and their related appendices (Appendix C.1.a and C.1.c, respectively).

Table 4-4. Summary of Geosynthetic Testing

	Test	Peak Friction Angle (°)	Peak Cohesion (psf)	Residual Friction Angle (°)	Residual Cohesion (psf)	Asperity	Comments
Low normal stress	Operations Soil-CDN Interface	29.6	205.9	24.6	205.4	--	– Test #1 – dry density = 92 pcf – w_c = 8.7%
Low normal stress	Admix Soil-HDPE Interface	33.3	94.4	33.5	56.8	--	– Test #3 – dry density = 110 pcf – w_c = 14%
High Normal	Operations Soil-CDN Interface	28.3	283.9	28	240.8	--	– Test #2 – dry density = 92 pcf – w_c = 8.7%
High Normal	Admix Soil-HDPE Interface	25.4	400.7	20.3	525.3	--	– Test #4 – dry density = 110 pcf – w_c = 14%
	Textured HDPE Asperity	--	--	--	--	23.5	Average value of two test results of 22 and 25.

As the final design progress and additional information is gathered for the admix soils and the operations soils, these results should be verified with additional testing. Testing during full scale construction is also planned to verify that the materials used in construction, both soils and geosynthetics, produce interface shear values at or greater than those used for design.

5.0 ENGINEERING ANALYSIS

This detailed Design Report finalizes the design for the landfill liner system, the leachate removal system, and the LDS. Engineering analysis components for each of these critical systems is presented in this section. A general description of system components is located in Section 5.6.1, that presents the primary and secondary liner systems that make up the major layers of the landfill (detailed system descriptions are presented in Section 6).

In preparation of the IDF design, a number of design requirements and criteria as presented in Section 2 have been considered. Compliance with these design requirements is provided in Appendix A. The specific criteria evaluated for the IDF design included:

- Slope stability
- Landfill bearing capacity
- Settlement and uplift analyses
- Admix liner
- Geosynthetic liner design
- Liner systems/leachate compatibility
- Drainage layer
- Leachate production
- Leachate collection system
- Surface stormwater

- Action leakage rate
- Building systems analyses
- Civil grading

5.1 Slope Stability

Slope stability for the IDF landfill was examined for liner veneer (side slope) stability, earthwork stability, waste/fill global stability. The analyses for each of these cases are summarized in the sections below; Appendices C.1.a through C.1.c presents the analyses and results in detail.

5.1.1 Liner Veneer (Side Slope) Stability

The veneer stability of the liner system on the side slopes was evaluated for the period prior to waste filling. The analysis examined the potential for sliding of the drainage and operations layers on the liner system before waste is placed.

The analyses were conducted using the weakest of the interface strengths of the various lining system components. The interface strengths were determined from regression analyses of data gathered from various sources, including site-specific test data completed to date. Based on the data (presented in Appendix C.1.a), the critical interface is the textured HDPE/CDN interface. Properties of the cover soil (operations layer) were determined from laboratory testing to date on the materials expected to be used for the operations layer.

Four loading conditions were examined:

- Dead load: self-weight of the lining system (including the first operations layer)
- Dead load + Equipment: self-weight of the lining system with an equipment load
- Dead load + Seepage: self-weight of the lining system with a seepage load (to account for fluid head in the leachate collection system); seepage loads were based on results from the leachate system hydraulic analyses
- Seismic Loading: self-weight of the lining system with seismic loading

The results of the analyses show that the lining system is stable for the conditions analyzed and no anchorage forces are required to meet the minimum factors of safety (1.5 for dead load only; 1.3 for equipment and seepage loading). A minimum interface friction of 25 degrees and cohesion of 0 psf is required to meet the minimum acceptable factors of safety. The slopes are also considered to be stable under seismic loading, based on comparing the calculated yield acceleration and with the design acceleration values provided in the design criteria by CH2M HILL (September, 2002), using the hazard classification assigned to the overall facility.

The critical interface friction values will be verified during construction to ensure that the system will be stable. The analyses and results are presented in full detail in Appendix C.1.a.

5.1.2 Earthwork Stability

The earthwork stability analysis covered the following three cases:

Excavation Case: This case covers the stability of the landfill slopes immediately after excavation and before placement of the lining system. Only static loading was considered since this is an interim configuration that will only exist for the construction period.

Ramp Case: This case covers the stability of the landfill slopes and access ramp at the south end of the cell, including equipment loading on the ramps. Both static and seismic loading were examined, as the access ramps are expected to be in use for a period of at least 10 years.

Dike Case: This case covers the stability of the perimeter dike (shine berm and access road) after construction of the dike and before final closure of the landfill.

Both static and seismic loading were examined, since the perimeter dike may be in place until the final cover system is completed (greater than 10 years).

Properties for the native soils are based on existing information, as a site-specific geotechnical engineering investigation program has not yet been completed for the IDF facility. When this investigation is completed, the results of this analysis (and any others that rely on the properties of the native soils) will be verified. Geometry used in the analyses is based on the civil plans (generally 3H:1V slopes with a few short 2H:1V slopes).

The results of the analyses show that the planned configurations of the landfill are stable under static loading (factor of safety [FS] greater than 1.3 and 1.5, depending on the case analyzed); the configurations are also considered seismically stable based on the criteria for the Hanford site. Full details on the analysis method, the input data, and the results are presented in Appendix C.1.b.

5.1.3 Waste/Fill Global Stability

This analysis examined the following conditions:

Phase I Full Build-Out: This case examined the stability of the waste mass in full build-out of the Phase I waste cell. The critical stability examined was the waste sliding on the lining system. Both static and seismic loading conditions were examined.

Final Configuration: This case examined the stability of the waste mass at the final configuration (entire IDF landfill completed) along the edge of the cover system. Only static loading conditions were examined, since this system is not being designed as part of the current effort.

Interim filling conditions and the internal stability of the waste mass were not examined. The internal waste mass stability will primarily be a function of the filling methodology. Possible filling plans for the waste are currently being developed.

For the analysis of the full build-out of Phase I, the critical interface strengths in the lining system were determined in the same way as for the veneer stability (regression analyses of existing and site specific testing data). A combination of peak and residual strengths were used, based on methodology currently being employed in the state of the practice. A final check was also made to confirm that the use of residual strengths in all locations resulted in a factor of safety greater than 1.0.

The results show that the system is stable for the configurations analyzed and for the interface friction values available at the time of the analyses (FS greater than 1.5 in static loading and yield acceleration greater than the 10,000-year event). The system also has a FS greater than 1.0 for the case of residual strengths in all locations. The critical interfaces are the HDPE-CDN on the side slopes (using residual strengths) and the HDPE-GCL on the base liner (using peak strengths) and the internal GCL strength (using residual strengths). These results should be verified when additional site-specific test data becomes available prior to and during construction.

Also, it should be noted that for the full Phase I build-out configuration, the most critical case appears to be a failure surface that is allowed to propagate through the waste mass. As noted previously, the waste mass was considered internally stable for this design effort. During final operations planning, the internal stability of the waste will be examined in conjunction with the proposed waste filling plan.

For the final configuration with the cover in place, the preliminary geometry and assumed cover system properties show that the configuration is stable under static loading (FS greater than 1.5) and the critical failure does not intersect the waste mass. Stability of the final configuration under both static and seismic loading should be examined in more detail as the final design develops for the final closure of the entire IDF facility.

A full discussion of the methodology, input data, and the results is presented in Appendix C.1.c.

5.2 Landfill Bearing Capacity

5.2.1 Subgrade Soil

Based on the available geotechnical data from other projects (as discussed in Section 4), the strength of the native subgrade soils beneath the landfill is expected to be greater than that for the operations layer or any of the liner system components. Greater strengths equate to higher bearing capacities, and hence, the bearing capacity of the subgrade soils within the landfill cell was not determined directly as they are not the controlling factor.

The bearing capacity of the subgrade soils beneath the supporting structures adjacent to the landfill cell was determined for the structural analyses, discussed under Section 5.12.1–Geotechnical Design Parameters, and the results of the analyses are presented in Appendix C.11.a.

5.2.2 Liner Soils

The soil layers in the lining system include the operations layer, drain gravel, and the admix liner soils. The admix liner soils will be placed beneath the geosynthetic lining system, and as such, loading on the admix liner soils is limited to the allowable loads for the GCL. The allowable loads for the GCL are much less than what the bearing capacity of the admix liner soils would be (the admix soils have much higher strengths, particularly for bearing pressures). The drain gravel will be placed just above the lining system; the shear strength and associated bearing capacity are also much greater than the GCL allowable values.

At the time of these calculations, structures that would cause bearing pressure were not yet determined. Hence, the bearing capacity for the operations soils was calculated for foundation widths from 1 to 10 feet and for 2 different shapes (square and strip). Properties for the operations soils were based on laboratory testing conducted to date; these properties will be verified during construction to ensure that the analyses results are valid.

For a factor of safety of 3, the allowable bearing capacities for the operations layer are presented in Table 5-1.

Table 5-1. Operations Soil Bearing Capacities

B, Foundation Width (feet)	q_{all}, square foundation (tsf)	q_{all}, strip foundation (tsf)
1	0.20	0.33
5	1.0	1.6
10	2.0	3.3

As the operations plans are further developed, these values can be updated for the planned structures (such as barrier walls). Details of the analyses are presented in Appendix C.2.

5.3 Settlement and Uplift Analyses

5.3.1 Settlement Analysis of Liner Foundation

The long term settlement of the soils supporting the geosynthetic liner system was estimated based on the maximum loading expected in the landfill at the final IDF completion. The two soil units examined were the admix liner soils and the native subgrade soils. For the admix soils, data from laboratory consolidation testing performed on samples available at the time of the analysis were used to determine the estimated settlements. Elastic methods were used to estimate the settlements of the subgrade soils.

As detailed in Appendix C.3, the estimated long term settlement over the lifetime of the landfill is 2.7 feet under the maximum loading.

5.3.2 Subsidence and Sinkhole Potential

Subsidence of undisturbed foundation materials is generally the result of dissolution, fluid extraction (water or petroleum), or mining. Subsidence is not expected to occur based on the following:

- The soils underlying the IDF are generally dense, coarse-grained, and well-graded sands and gravels that will not be subject to piping effects that could transport soil and result in subsidence. Also, sands and gravels are generally not susceptible to dissolution.
- The groundwater level is deep and will not affect bearing soils.
- The bedrock is basalt (volcanic), which is not generally susceptible to dissolution.
- No mining or tunneling has been reported in the areas beneath or surrounding the site for the IDF.
- Borings in and around the IDF have not identified any soluble materials in the foundation soils or underlying sediments. Consequently, the potential for any sinkhole development will be negligible.

5.3.3 Uplift Potential

The potential for uplift of the composite liner system is very low. The seasonal high-water level is over 200 feet below the base of the base of the landfill cell, so no external hydrostatic pressure is expected from this source. Perched groundwater is not expected to occur due to the absence of continuous aquitards (such as a clay layer) within the coarse-grained native soils at the IDF site. Any infiltration that does occur is expected to percolate rapidly to deeper soil layers.

Gas pressures are also expected to be negligible, as no gas-generating material (i.e., organic material) is expected in the foundation soils. Also, the subgrade soils are coarse grained and unsaturated, so any gas that might occur is expected to be rapidly dissipated.

5.4 Admix Liner

5.4.1 Mix Design

WAC 173-303-665(2)(h)(i)(B) requires that the lower component of a composite bottom liner be constructed of compacted soil material with an in-situ hydraulic conductivity no greater than 10-7 cm/sec. Because of the lack of naturally occurring soils on-site that could achieve this requirement, a test program was developed to determine the admixture requirements for a mixed soil design using on-site base soil from either the Phase I excavation or dune sand borrow area (see Drawing H-2-830826 for location) and sodium bentonite. Details of the base soil field exploration and admix testing program are provided in Section 4.

The results of the limited field exploration for base soil samples and subsequent admix testing program discussed in Section 4 show that a nominal bentonite content of 12 percent will meet the laboratory target hydraulic conductivity of less than 10-8 cm/sec when permeated with water. The laboratory target was established based on results of the soil liner/leachate compatibility study (Golder Associates, 1991b) for the W025 landfill. Details of Golder's study are discussed in Section 5.6. The W025 study concluded that the bentonite content of the admix should be increased from 8 percent (the minimum bentonite percent needed to achieve the required hydraulic conductivity) to 12 percent, to provide adequate resistance against high inorganic concentrations in the synthetic leachate for the W025 project. Index laboratory testing on the limited field exploration at the IDF site (surface sampling) established that the base soil for the IDF was similar to the W-025 project, as discussed in Section 4. Thus, until a more refined characterization of the IDF leachate is developed, the compatibility testing from the W025 testing is applicable to the IDF mix design.

Once initial hydraulic testing confirmed that an admix with 12 percent bentonite content could achieve the laboratory target value, additional samples were set up to evaluate a range of moisture and density parameters and their effect on hydraulic conductivity. The additional hydraulic conductivity tests were performed to define moisture content-density requirements for a range of compactive energy, as outlined by Daniel and Benson (1990). This data was being used to develop an "acceptable" zone of moisture and density for use by QC personnel during construction. The acceptable zone for the 12 percent admix is presented along with the admix design laboratory test results in Appendix B.1.

The acceptable zone was developed based on samples that achieved a hydraulic conductivity of less than 10-8 cm/sec. A lower bound of 95 percent relative compaction, based on Standard Proctor (ASTM D698) compactive effort, was established to ensure adequate shear strength levels. As indicated in the technical specifications (see Section 02666), the moisture-density range of the compacted admixes shall lie within a trapezoidal-shaped field with the following corners:

Moisture Content (%)	Dry Density (pcf)
8	126
14	126
12	110
19	110

Note that the minimum dry density of 110 listed above corresponds to approximately 95 percent of the maximum dry density for admix, as measured by ASTM D698.

5.4.2 Placement and Testing

The moisture-density requirements developed as part of the admix testing program will be included in the specifications for the admix liner (see discussion in Section 5.4.1, and technical specifications, Section 02666). The intent of the placement technical specifications is to help ensure that the admix liner will meet an in-place performance specification for hydraulic conductivity of less than 1x10-7 cm/sec. The contractor is responsible for developing and implementing compaction means and methods that will produce the required relative compaction.

The recommended nominal bentonite percentage (12 percent) and moisture-density parameters for the admix liner have been developed with a one order of magnitude factor of safety between laboratory and field values for hydraulic conductivity. The factor of safety is expected to account for two issues: (1) variations in the hydraulic conductivity between the laboratory soil amendment study and full-scale production, and (2) the laboratory samples were permeated with water rather than leachate, which could lead to a difference in the field hydraulic conductivity. However, factors such as base soil variability at the borrow source and field placement and construction are difficult to quantify until full-scale production begins for the admix liner. A test pad will be constructed as part the IDF construction to model the full-scale production. The purpose of the test pad is to determine acceptable processing, placement, and compaction methods that will produce a low-hydraulic conductivity admix liner with an *in situ* hydraulic conductivity of 10-7 cm/sec or less. The bentonite percentage and moisture content/density range may be modified if the preconstruction testing performed on the test pad indicates an *in situ* hydraulic conductivity greater than 10-7 cm/sec. Construction QA sampling and testing for the test pad is described in the Detailed Design Cell 1 Construction QA Plan (CH2M HILL, March 2004).

5.4.3 Freeze/Thaw

Compacted soil liners, such as the IDF admix liner, are known to be vulnerable to large increases in hydraulic conductivity due to freeze/thaw cycling; current data suggests that compacted soil bentonite admixtures may not be as vulnerable to damage as true clay liners (Kim and Daniel, 1992; Benson and Othman, 1993; Kraus et al., 1997). Existing laboratory data indicate that GCLs are less susceptible to damage from freeze/thaw conditions and therefore, do not undergo increases in hydraulic conductivity (Hewitt and Daniel, 1997; Kraus et al., 1997).

In order to provide adequate freeze/thaw protection for the admix liner and avoid potential damage to the GCL a protective soil cover can be used. The thickness of the protective soil cover should exceed the predicted freeze depth. For the IDF, protective soil cover is provided by the operations layer on the side slope (3 feet) and the drain gravel and operation layer (4 feet total) on the bottom liner.

The analysis was performed on the IDF lining system operations layer to determine the freeze depth or frost penetration for a probable freezing season during the 10-year expected period of waste filling.

Both a 10-year return period (90 percent probability on non-exceedance) and 20-year return period (95 percent probability on non-exceedance) air freeze index (AFI) were used to estimate maximum frost penetration depth in the operations layer. If the maximum frost penetration depth were less than the 3-foot minimum thickness operations layer over the lining system, the proposed operations layer thickness would be considered as adequate protection for exposure of the lining system to freeze-thaw cycles.

For the 10-year return AFI, the maximum freeze depth is estimated at 17 inches. For the 20-year return AFI, the maximum freeze depth is estimated at 21 inches. The maximum estimated freeze depths for both the 10-year and 20-year return period freezing seasons indicate that the proposed cover soil thicknesses provide more than adequate protection for the underlying admix liner and GCL from potential damage when subject to freeze-thaw cycles. Details of the freeze depth calculations are included in Appendix C.4.

5.5 Geosynthetic Liner Design

5.5.1 Geomembrane Liner Tension Caused By Thermal Contraction

The HDPE geomembrane for IDF lining system will be subject to temperature-induced tensile strain from expansion/contraction as the geomembrane is exposed to temperature fluctuation.

Strain on the liner was calculated using published values for the coefficient of linear thermal expansion for HDPE geomembrane (Koerner, 1998) and applying this to the maximum slope length. The maximum length is measured from the top of the slope, where liner is anchored, to the toe of the 3H:1V side slope. This is a conservative approach, as using the maximum slope length results in the maximum amount of expansion and strain on the liner. Additionally, a conservative temperature change of 40 degrees C (104 degrees F) was used in the analysis.

The maximum liner strain was estimated to be less than 0.5 percent, based on a maximum temperature change of 40 degrees C (104 degrees F). The estimated maximum of slack in the liner on the side slope is 8.6 inches. The corresponding amount of temperature induced stress is 566 psi. See Appendix C.5.a for supporting calculations.

As shown in the technical specifications, Section 02661 (Table 1), the elongation at yield for the geomembrane that will be used in the liner system is at least 12 percent, with a minimum tensile strength at yield of 2,000 psi. Therefore, the maximum anticipated strains are well below the yield tensile strain and stress for the HDPE geomembrane, and temperature-induced strain will have no adverse impact on lining system function.

It should be noted that temperature-induced strain is only applicable during the construction period when the HDPE geomembrane is exposed to temperature fluctuation. Once covered with 3 to 4 feet of cover soils (drain gravel and operations layer), the ambient temperature at the surface of the geomembrane will be more controlled and not subject to fluctuation.

During installation, care must be taken to allow for expansion/contraction of the HDPE geomembrane to minimize the development of wrinkles that could become future stress points under soil and waste loading. The technical specifications (see Section 02661) provide requirements for control of wrinkle development during liner deployment, including the limitation of working when the temperature is below 0 degrees C (32 degrees F) or above 40 degrees C (104 degrees F) without implementing installation procedures that address the environmental conditions.

5.5.2 Liner System Strain Due To Settlement

The barrier components (geomembrane and GCL) for the IDF lining system will be subject to settlement-induced tensile strains as the underlying soils, primarily the admix soil liner and the subgrade soil, settle over time. Strain within the lining system was calculated based on the results of the liner foundation settlement calculations (see Section 5.3 for settlement of foundation soil [subgrade] and admix liner).

The strain calculation assumed that all vertical settlement was translated into strain along the liner rather than just the vector component parallel to the liner. This is a conservative assumption that establishes an upper bound for liner strain.

The maximum liner strain was estimated to be less than 0.6 percent, based on a maximum estimate of 2.7 feet of settlement at the base of the lining system. See Appendix C.5.b for supporting calculations.

As shown the technical specification (Section 02661, Table 1), the elongation at yield for the geomembrane that will be used in the liner system is at least 12 percent. Based on studies of effect of differential settlement on GCLs (LaGatta et al., 1997), the limiting strain was defined as the strain in which an increase in hydraulic conductivity of the GCL was observed, which was taken as 5 percent.

Therefore, the maximum anticipated strains are well below the yield or limiting tensile strain for the barrier components of the lining system (geomembrane and GCL). Settlement-induced strain from foundation and admix soil settlement under maximum landfill content pressure will have no adverse impact on lining system function.

5.5.3 Anchor Trench Pullout Resistance

During construction, the geomembrane could experience pullout forces caused by thermal expansion/contraction or wind uplift. However, tension from thermal expansion and contraction is expected to be small (see Section 5.5.1), and the geosynthetics installer can use sand bags or other approved method to control wind uplift during installation.

After construction and placement of operation layer, the pullout forces on the geomembrane are expected to be negligible, as there is no tension force on the liner. As indicated in the veneer (side slope) stability analyses (see Section 5.1.1), the lining system interface strength exceeds the slope angle on the 3H:1V side slope. Thus, the pullout resistance requirements for the anchor trench are to support the self-weight of the geomembrane and other lining system components. Analyses for liner self-weight support requirements determined that the frictional resistance between geosynthetics exceeds the liner self-weight. Thus, no additional pullout resistance is needed at the anchor trench to support lining system self-weight.

Supporting calculations for the anchor trench design, as shown on Drawing H-2-830838, Detail 3, are included in Appendix C.5.c. Based on the calculations for the configuration shown in the drawing, a pullout resistance ranging from 1840 pound/foot (lb/ft) to 2440 lb/ft is estimated (depending on actual mobilized interface shear strength). The required minimum tensile yield strength for 60-mil HDPE geomembrane in the technical specifications (see Section 02661) is 1440 lb/ft (120 lb/in), which results in the estimated pullout resistance exceeding the geomembrane tensile yield strength. This situation is due primarily to the configuration of the shine berm, which helps to anchor the system. While it is generally not desired for the pullout resistance to exceed the yield strength, this is not expected to be a problem at the IDF, since, as discussed above, the potential causes for geomembrane tension have been addressed and there is not a scenario for mobilizing tensile or pullout forces on the lining system.

5.5.4 Puncture Resistance

The primary geomembrane in the IDF will be overlain by the LCRS. For the side slope lining system, the LCRS consists of a CDN (see Detail 2 on Drawing H-2-830838) that provides protection for the primary geomembrane from the overlying operations layer. A separate discussion of the CDN geotextile puncture resistance is provided in Section 5.7.2. For the bottom lining (floor) system, the LCRS consists of drain gravel overlying the geomembrane (see Detail 1 on Drawing H-2-830838). A geotextile cushion will be required between the drainage gravel and the geomembrane to prevent the gravel from puncturing the geomembrane. An analysis was performed to determine the weight of the geotextile fabric required to prevent geomembrane puncture either from operating equipment loads or from the combined static weight of the waste and final cover.

Koerner (1998) developed a method for estimating required geotextile thickness that considers the size and shape of the rock, as well as other factors that could decrease the long-term strength of the geomembrane. The equation used to determine puncture resistance is based on the mass per unit area of the geotextile and the protrusion height of the puncturing material.

Operating loads were estimated based on a melter transport trailer operating directly on the surface of the first operations layer. Static loads were estimated for the post-closure condition by using the weight of four layers of ILAW packages with cover soil and a 15-foot-thick closure cover, with a 2 percent grade to the center of the landfill. The static load was more than two times greater than the operating load, and therefore was used as the basis for the puncture analysis. Detail calculations for geomembrane puncture resistance and corresponding cushion geotextile requirements are included in Appendix C.5.d.

The proposed design specifies that the LCRS drainage gravel will have a gradation corresponding to WSDOT Standard Specification 9-03.12(4). This gradation has a maximum stone size of 1 inch. From the curves shown in the detailed calculations, the FS for a 12 oz/yd2 geotextile loaded by 1-inch angular rock is 4.5. For subrounded rock or gravel, this is more representative of the specified drain gravel, the FS increases to 8.9. The specified cushion geotextile (see technical specifications, Section 02371) has a nominal weight of 12 oz/sq yd, and therefore should be adequate to prevent geomembrane puncture. Koerner (1998) recommends a FS greater than 3.0 for the condition of packed stones on a geomembrane, such as would be the case for drain gravel over the geomembrane at the IDF.

5.5.5 Operational/Equipment Loading

The effects of loading on the GCL from construction and operational equipment and activities were examined. The maximum loads from the landfill waste itself were found to produce the highest loading on the geomembrane and the CDN; these materials were selected based on this maximum loading, as discussed in the previous sections.

The cases for construction equipment loading and operational loading on the GCL were examined, including the extreme loading case of the crane placing the heaviest waste loads at its maximum reach, a situation which produces very high pad loads. The expected loads were compared to the calculated allowable GCL bearing capacity to determine if the loads would have an effect on the GCL. The allowable GCL bearing capacity was determined from classical geotechnical theory and based on manufacturer's strength data.

The results of the analyses are presented in detail in Appendix C.5.e. For the construction loading, the analyses show that the specification requirements that limit construction loading are adequate to protect the GCL, based on the standard construction equipment anticipated to be used at the IDF and as examined in the calculations.

For the operational loading cases examined, the critical condition is the crane operating under an extreme condition. The minimum dunnage requirement for the crane pads is 60 square feet, or if square, a 7.7-foot by 7.7-foot dunnage pad. Lower loads will require less dunnage and can be calculated as detailed in Appendix C.5.e. As discussed in the appendix, dunnage requirements calculated in this way are appropriate as long as the lining system is functioning as intended (i.e., no moisture in the LDS). If moisture enters the LDS and the GCL becomes hydrated, the dunnage requirements will be increased by a factor of approximately 2.5.

It should also be noted that the primary purpose of the GCL in the IDF is not as a required lining system component (such as the geomembrane or the admix liner), but to "deflect" leachate from defects or pinholes in the primary geomembrane over the bottom area and longer-term storage areas (such as leachate sump trough), where the leachate head potential is greatest. The primary purpose of the primary GCL is to reduce the actual leakage rate into the LDS in the event of leak in the primary geomembrane. Given these considerations, the GCL should perform as intended under anticipated equipment and operational loading.

As the operations plans for the landfill are developed, loading values can be compared to the results shown in Appendix C.5.e to determine if the loads will affect the GCL.

5.6 Liner Systems/Leachate Compatibility

The purpose of this analysis is to demonstrate that the liner materials proposed for the IDF landfill are chemically compatible with the leachate. Certain materials deteriorate over time when exposed to chemicals that may be contained in hazardous leachate. It is important to anticipate the type and quality of the leachate that the landfill will generate and select compatible liner materials. Data collected from other similar low-level radioactive mixed waste and hazardous waste sites were used in conjunction with the anticipated IDF leachate concentrations to evaluate the allowable concentration of leachate constituents that could be in contact with the IDF landfill liner components.

5.6.1 Lining System Description

Detailed discussion of the lining system design elements is provided in Section 6. A summary is provided in this section to facilitate discussion with respect to the chemical and radiation resistance of the lining system components.

Drawing H-2-830838 (Detail 1) shows the bottom liner section consisting of the following components, from top to bottom:

- A 3-foot-thick operations layer
- A separation geotextile (polypropylene)
- A 1-foot-thick leachate gravel layer
- A minimum 12 oz/square yard cushion geotextile (polypropylene)
- A 60-mil (nominal thickness—see Section 6.3.2.1) textured primary HDPE geomembrane
- An internally-reinforced GCL
- A CDN drainage layer for primary leak detection/collection
- A 60-mil textured secondary HDPE geomembrane
- A 3-foot-thick low-hydraulic conductivity compacted admix (soil-bentonite) liner

For the bottom lining system, both the primary and secondary liners are a composite (geomembrane over admix liner or GCL) system. The addition of a GCL in the primary liner layer provides an extra measure of protection, exceeding the requirements of WAC 173-303-665(2)(h)(i), which stipulates a single geomembrane for the primary liner and composite for the secondary only. This will provide an extra measure of protection on the bottom flatter slopes of the IDF, where higher leachate head levels are more likely.

Drawing H-2-830838 (Detail 2) shows the side slope liner section consisting of the following components, from top to bottom:

- A 3-foot-thick operations layer
- A CDN drainage layer for primary leachate collection
- A 60-mil textured primary HDPE geomembrane
- A CDN drainage layer for primary leak detection/collection
- A 60-mil textured secondary HDPE geomembrane
- A 3-ft-thick low-hydraulic conductivity admix liner

The side slope lining system is a single geomembrane liner over a composite liner, meeting the requirements of WAC 173-303-665(2)(h)(i). The 3H:1V side slopes for the IDF will result in little or no leachate head build-up on the side slope lining system, thus eliminating the need for a lining system design that exceeds the WAC requirements.

In general, the liner system consists of two types of materials, geosynthetics and soil/bentonite mixtures (admix). The geomembranes, geotextiles, and CDN are manufactured from polymeric materials, such as HDPE, and polypropylene, made from synthetic polymers. The GCL consists of a bentonite layer sandwiched between two polypropylene geotextiles to assist in placement and construction. The admix liner is comprised mainly of silt to clay-sized particles, mixed with a silty sand base soil.

5.6.2 Leachate Characterization Assumptions

Several assumptions were made regarding the composition of the leachate concentrations and the applicability of previously conducted studies for this evaluation. Specifically, the studies considered directly applicable to this evaluation were:

- Geosynthetic and Soil Liner/Leachate Compatibility Studies for the W-025 Radioactive Mixed Waste Landfill in Hanford 200 West (Golder Associates, 1991a and 1991b; TRI, 1995; and WHC, 1995)
- Liner/Leachate Compatibility Study for the U.S. Department of Energy's Idaho National Engineering and Environmental Laboratory (INEEL) Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) Disposal Facility (ICDF) (DOE-ID, 2002).

Using these studies is considered appropriate for the following reasons:

- The leachate for the IDF is expected to have similar or lower concentrations of radionuclides than that used in the W025 facility study (since similar waste streams [other than ILAW] may be accepted).
- The leachate chemistry may be of similar composition to the W025 facility study (since similar waste streams [other than ILAW] may be accepted).
- Soils used in the W025 facility admix design are similar to those that will be used in the IDF admix design and will therefore be compatible.
- Similar technical specifications for the geosynthetics and admix liner used in the W025 facility design will be used in the IDF landfill design.
- A similar technical specification for a GCL used in the ICDF facility will be used in the IDF liner design.

5.6.2.1 Synthetic Leachate Concentrations for W-025 Landfill

The leachate generated for the W025 evaluation reflects both the waste materials and the stabilization agents used during waste preparation. Because the landfill will comply with waste acceptance criteria for WAC dangerous waste and RCRA facilities (as does the IDF), organic materials are not expected to be present in the waste after processing. The proposed geosynthetic materials are susceptible to damage from certain organic compounds but generally are not susceptible to damage from inorganic compounds, even with extreme pH values. As a result, the lack of organic materials results in a relatively benign leachate.

The source leachate generated for the W025 studies, was primarily based on the waste treatment and packaging approaches for W025. An aqueous solution of inorganic, with some organic compounds for conservative evaluation, was generated, resulting in a viscous, slurry-like mixture. This mixture was placed in a leaching column, and deionized water was introduced to simulate the effects of leachate generation. Although no organic components were anticipated in the waste, small quantities of benzene, methanol, and light machine oil were included to simulate the presence of organic compounds in the waste material.

The source leachate generated through the leachate column process was chemically analyzed with the following results:

- Concentrations of organics benzene and machine oil were below detection limits. Concentrations of methanol were detected, but at concentrations not considered aggressive for polyester or HDPE.
- Metals added to the waste were below the detection limits in the source leachate.
- Primary constituents of the source leachate were sodium cations and common inorganic anions, with a pH of 9.2.
- Based on these results, a synthetic leachate was generated for testing purposes. The source leachate formula resulted in a solution with total inorganics and dissolved salts of approximately 204,000 mg/L and pH of 9.2 using NaOH or HNO3, as required.

5.6.2.2 Simulated Irradiation Exposure for W-025 Landfill

Samples used to evaluate the effects of radiation were subjected to a 50,000-rad total dose of gamma radiation. This dose is expected to exceed the maximum level of radiation experienced by geosynthetic materials in the landfill under unfavorable conditions. Use of a total dose, rather than radiation type, is considered the primary factor causing damage to polymeric materials and is considered to adequately simulate actual IDF leachate conditions. Samples and leachate were irradiated together so that any synergistic effects would be seen. The following samples were included in the irradiation testing:

- Geomembrane
- Geotextile
- Geonet
- Admix (soil/bentonite mixture)

The synthetic leachate and radiation exposure developed from the W-025 studies were used as the basis of evaluation for the IDF lining system materials. Table 5-2 provides a comparison of the leachate concentrations for the W-025 project with other studies for which the U.S. Environmental Protection Agency (EPA) Test Method 9090 were performed on the lining system.

The ICDF project did not include EPA 9090 tests, however, a model for estimating leachate concentration based on the waste acceptance criteria for the project was developed. The maximum leachate concentrations and radiation exposure developed for the ICDF (DOE-ID, 2002) based on the anticipated waste design inventory were as follows:

- Organics–70 mg/l
- Inorganics–18,400 mg/l
- Radiation Exposure–12,000 rads

Table 5-2. EPA Test Method 9090 Compatibility Studies Comparison

Compatibility Study[a]	Type of Material Tested	General Composition of Leachate	9090[b] Test Concentrations or Radiation Exposure that Demonstrated Compatibility in Each Study
Hanford Liquid Effluent Retention Facility (LERF)	60-mil smooth HDPE from four manufacturers	Organics	16.25 mg/L
Hanford W-025 Landfill	**60-mil smooth HDPE**	**Inorganics** **Organic Leachate and Radiation Exposure** **pH**	**204,210 mg/L** **50,000 rads** **9.2**

Compatibility Study[a]	Type of Material Tested	General Composition of Leachate	9090[b] Test Concentrations or Radiation Exposure that Demonstrated Compatibility in Each Study
Hanford Grout Facility	60-mil smooth HDPE	Inorganics Organic Leachate and Radiation Exposure pH	368,336 mg/L 37,000,000 rads >14
Kettleman Hills Landfills	60-mil smooth HDPE	Organics Inorganics pH	93,040 mg/L 250,000 mg/L >12

a. Detailed compatibility test information is provided in Evaluation of Liner/Leachate Chemical Compatibility for the Environmental Restoration Disposal Facility report (USACE, 1995).
b. EPA Test Method 9090 "Compatibility Test for Wastes and Membrane Liners" (EPA, 1992c).

A review of the studies presented in Table 5-2 leads to the conclusion that the inorganic concentration developed for the W025 is somewhat conservative as it significantly higher than inorganic concentrations developed for the ICDF facilities. Other than the W-025 landfill, the ICDF is estimated to be most similar to the waste type to be received at the IDF of the studies included in Table 5-2. Nonetheless, the liner/leachate compatibility study for the IDF is based on the W025 synthetic leachate. Further analysis of the applicability of these leachate concentrations is recommended, if the conservative nature of this synthetic leachate requires costly revisions to the lining system to demonstrate compatibility.

5.6.3 Chemical and Radiation Resistance

Leachate will be generated from precipitation events and from water added to the waste for dust control and compaction purposes during operations. In reality, as the landfill nears the end of its operational life, concentrations of contaminants will decrease with time as the leachable waste mass is reduced. During the post-closure period, a robust landfill cover will significantly reduce infiltration, and the corresponding volume of leachate. Soluble contaminants leached from the waste will come in contact with the landfill bottom liner system during the operation period (approximately 10 years for each of the four planned phases) and minimum post closure period (30 years). The geosynthetics and admix lining system components may be in contact with soluble contaminants as long as contaminants are present in the landfill.

The expected chemical make up of the leachate for the IDF landfill was determined based on previously conducted compatibility studies (as discussed above) applicable to the same waste stream (the W025 studies), summarized as follows.

5.6.3.1 Geomembrane

HDPE geomembranes can deteriorate from contact with certain leachates, resulting in a decrease of elongation at failure, an increase in modulus of elasticity, a decrease in the stress at failure, and a loss of ductility.

Studies performed on polymer materials like HDPE show that their properties begin to change after absorbing ionizing radiation between 1,000,000 to 10,000,000 rads (Koerner et al., 1990). The HDPE geomembrane lining the bottom of the landfill will absorb ionizing radiation energy from the leachate generated in the landfill. Energy will be absorbed during the operational life of the landfill, as long as there are liquids with ionizing radionuclides in contact with the geomembranes.

Relevant compatibility studies on HDPE geomembranes have been performed for the W-025 Landfill (Golder, 1991a; TRI, 1995; WHC, 1995). The results of these studies indicate that a HDPE geomembrane will function well as a liner beneath the landfill waste.

EPA Method 9090 tests performed on HDPE geomembrane for the W-025 landfill, using the synthetic leachate solution (assumed representative of IDF leachate concentrations) resulted in no evidence of geomembrane deterioration. A comparison between the anticipated IDF landfill leachate (W-025 Landfill) and that used in compatibility tests for other facilities is summarized in Table 5-2.

Geomembrane samples tested for the W-025 facility did not produce measurable changes in the HDPE liner properties when irradiated for 120 days with a total dose of 50,000 rads. HDPE geomembranes are manufactured with additives, such as carbon black and antioxidants, to improve ductility and durability. The literature also indicates that these additives allow higher doses than standard HDPE material without additives (Kircher and Bowman, 1964). The literature indicates that thin films (i.e., 0.002 inches) of different types of HDPE material alone can become brittle when irradiated at doses between 4,400,000 and 78,000,000 rads. Studies performed using polymer materials, with carbon black and antioxidant additives, show that properties typically begin to change at a total radiation dose of between 1,000,000 and 10,000,000 rads (Koerner et al., 1990).

The manufacturers of the geosynthetic products proposed for the IDF landfill have published maximum allowable concentrations of various chemical compounds that can contact the HDPE geomembrane without adversely affecting its performance. The most recent recommended maximum concentrations of chemicals were obtained from the manufacturers of HDPE geomembrane (meeting the requirements for the IDF technical specifications). A list of the manufacturers' maximum allowable concentrations for specific leachate constituents for HDPE geomembrane and the GCL materials is shown on Table 5-3.

5.6.3.2 Geosynthetic Clay Liner (GCL)

The GCL underlying the geomembrane in the IDF landfill consists of processed sodium bentonite clay, sandwiched between two geotextile fabrics. Sodium bentonite is an ore comprised mainly of the montmorillonite clay mineral with broad, flat, negatively charged platelets that attract water, which hydrates the bentonite. The swelling provides the ability to seal around penetrations, giving the GCL its self-healing properties. A GCL product with Volclay-type sodium bentonite (manufactured by CETCO) is specified for installation at the landfill.

The compatibility of GCL materials is usually demonstrated by permeating the material with leachate and then determining its hydraulic conductivity. Typically, solutions with high concentrations of contaminants or pure products are allowed to permeate a sample under confining pressure and the saturated hydraulic conductivity of the material is determined using ASTM methods such as ASTM D5084. A significant increase in saturated hydraulic conductivity (approximately one order of magnitude) for a sample permeated with leachate, compared with a sample permeated with water, would be an indicator of incompatibility.

Based on review of the published studies (Ruhl and Daniel, 1997; Shackelford, et al., 2000; and EPA, 1995), GCLs perform well unless exposed to high concentrations of divalent cations, very acidic or basic solutions, or solutions with a low dielectric constant (such as gasoline). The leachate expected at the IDF will have a pH of 9.2, which is a mid-range pH. The studies further demonstrate that, when confined under a higher normal load (greater than 2000 psf) or if water is the first wetting liquid (Daniel et al., 1997), GCLs will perform well when exposed to high divalent cation concentrations. The GCL for the IDF lining system is expected to confine under normal loads in excess of 2000 psf as soon as the first lift or waste is placed.

No studies were identified that considered the long-term effects of radiation on the physical properties of GCL materials. Since long-term studies cannot be conducted, conservative radiation limitations have been employed. Low-hydraulic conductivity soils have been used at multiple DOE facilities containing radioactive waste. The only known potential adverse reaction that can occur with a GCL is high heat that could dry out the materials. The amount of radioactivity is expected to be low in the IDF landfill waste and will not generate a significant amount of heat that can desiccate the admix liner. Also, it is assumed that the ILAW packages will be cooled to ambient temperatures prior to placement with the cell.

It should be noted that the operations layer and drain gravel will provide a 3-foot buffer on the side slope and a 4-foot buffer between the liner system and waste for additional thermal protection, if needed.

Sodium bentonite is the primary clay mineral in a GCL that produces the low hydraulic conductivity and high swell potential. Exposure of sodium bentonite to liquids containing concentrated salts (such as brines), or divalent cation concentrations (such as Ca++ and Mg++), reduces the swelling potential and increases its hydraulic conductivity. Concentrated organic solutions (such as hydrocarbons) and strong acids and bases can break down the soil, which also increases hydraulic conductivity. The physical mechanism that causes these changes is a reduction of the thickness, and related absorption capacity, of the diffuse double layer of water molecules surrounding the clay minerals. This results in an effective decrease in the volume of the clay, since the water molecules are not attracted to the clay particles.

The GCL manufacturer allows the use of GCL with few restrictions on maximum chemical concentrations. Leachate concentrations for the IDF landfill (based on synthetic leachate from W025) have relatively high inorganics and dissolved salts. The W025 dissolved salt concentrations are above the manufacturers recommended concentration of 35,000 mg/L (see Table 5-3) (CETCO, 2001). As a point of reference, this concentration of dissolved salts is typical of seawater (USGS, 1989). However, the dissolved salt concentrations in the IDF leachate have been characterized as primarily sodium, and the synthetic leachate was comprised of entirely sodium salts, not the divalent cations such as Ca++ and Mg++, as assumed by the manufacturers. As such, the impact on GCL hydraulic conductivity should be less as compared to divalent cation solutions. Additionally, any effects of leachate degradation on the GCL would be minimized by hydration of the GCLs' sodium bentonite with relatively "fresh" water, allowing the GCL to swell initially and decrease hydraulic conductivity.

The rationale for use of the GCL in the IDF landfill primary liner is to "deflect" leachate from defects or pinholes in the geomembrane over the bottom area and longer-term storage areas (such as the leachate sump trough), where leachate head potential is greatest. The main purpose of the primary GCL is to reduce the actual leakage rate into the LDS in the event of leak in the primary geomembrane (see Section 5.10 and Appendix C.10). The GCL is expected to contact leachate only in the event of a leak in the primary geomembrane. These leachate collection and storage areas are subject to flushing throughout the active life of the landfill due to phased development and fill sequence, resulting in a more dilute leachate in leakage areas prior to attaining maximum leachate concentrations. Based on these considerations, the GCL and landfill liner system approach should perform as intended under the anticipated conditions.

5.6.3.3 Admix Liner

The admix layer consists of onsite silty sand mixed with processed bentonite amendment, similar to that used in the construction of GCLs. The swelling of sodium bentonite provides the ability to seal around soil particles, giving the admix a low hydraulic conductivity and self-healing properties. The compatibility of the admix layer with anticipated irradiation and leachate concentrations were evaluated previously as part of the W025 landfill design (Golder Associates, 1991b). The following summarizes the results of the compatibility testing for the admix layer that are directly applicable to the IDF landfill admix liner, since similar materials will be used in construction. More detailed discussion of the IDF admix liner design is provided in Section 5.4.

In the W025 study, samples of the admix were irradiated, similar to that conducted for the geomembrane layer, as discussed previously. Differences between irradiated and non-irradiated samples were not considered significant based on the results of testing.

The initial W025 admix design contained approximately 8 percent bentonite clay. Testing indicated an acceptable hydraulic conductivity of this admix after hydration in fresh water. However, when hydrated in leachate, some hydraulic conductivity test values were twice the allowable limit and, therefore, this admix formulation was not considered acceptable. This is the same leachate chemistry assumed for the IDF landfill.

It should be noted that there are two factors not considered in the W025 compatibility study (Golder Associates, 1991b) that would mitigate the impact of the synthetic leachate on the 8 percent admix samples, as listed below:

Effective stress for samples–hydraulic conductivity tests were performed with effective stresses of 5-10 psi across sample (equivalent to less than one full lift of ILAW packages). It is well documented that higher effective stresses will lower hydraulic conductivity and mitigate the effects of shrinking/cracking in clay under attack from chemicals. In reality, by the time any leachate contacts the lining system, there will be a substantial stress load on the liner that will mitigate the impacts of chemicals in leachate on the admix liner.

First wetting liquid–W025 tests were performed using both site water and synthetic leachate as the initial wetting fluid. It is well documented that if a clay soil is "attacked" by inorganics prior to saturation, the increase in hydraulic conductivity will be more dramatic than if water is first permeant. This was confirmed by W025 testing–there was an order of magnitude difference between samples with water as first wetting liquid as opposed to leachate. It is reasonable to expect something closer to water than concentrated leachate will be the first wetting liquid for the IDF admix liner.

Due to the results in the W025 testing showing greater than acceptable hydraulic conductivity in the admix when exposed to the W025 synthetic leachate, the bentonite percentage was increased from 8 to 12 percent. An admix containing 12 percent bentonite clay was permeated with synthetic leachate and tested with a resulting hydraulic conductivity that was 3 to 10 times lower than the maximum allowable limit (10-7 cm/sec). This admix formulation was considered acceptable with respect to W025 leachate compatibility and is applicable to the IDF. Thus, the technical specifications (see Section 02666) require a nominal 12 percent (range from 11 to 14 percent is acceptable) bentonite by weight for the admix liner. Consideration should be given to lowering the bentonite percentage upon further characterization of the IDF leachate and applicability of the mitigating factors discussed above.

5.6.3.4 Other Materials

Other materials for which compatibility needs to be addressed are the CDN and geotextiles (cushion, separation, and bonded to geonet of CDN). While these materials do not serve a barrier function, they provide either for removal of leachate or protection of the lining system and must continue to function when exposed to leachate.

During the W025 design, the effect of the synthetic leachate on the geonet core of the CDN and the geotextiles was evaluated (Golder Associates, 1991a). The study concluded that a geonet core comprised of HDPE provided adequate chemical and radiation resistance. For geotextiles, the study concluded that geotextiles made of polyester fabric were susceptible to degradation and recommended that geotextile material be limited to a more chemically resistant material such as polypropylene. The technical specifications for the IDF require that geotextiles be made from polypropylene (see Section 02371); thus, the geotextiles used for the IDF should have adequate chemical and radiation resistance.

Table 5-3. Maximum Allowable Concentrations in Leachate by Chemical Category for Geosynthetic Components

Chemical Category	Compatible Concentration for HDPE	Compatible Concentration for GCL	IDF Concentration Dose or Value
Organics	500,000[a] mg/L	500,000[b] mg/L	N/A
Acids and Bases	750,000[a] mg/L	500,000[b] mg/L	0[d] mg/L
Inorganic	500,000[a] mg/L	500,000[b] mg/L	204,000 mg/L[c]
Dissolved Salts	No Limit	35,000[a] mg/L	204,000 mg/L[c]
Strong Oxidizers	1,000 mg/L	No limit	0 [d] mg/L
Radionuclides	1,000,000[b] rads	No limit	50,000 rads[c]
PH	0.5 - 13.0[a]	0.5 - 13.0	9.2

a. Based on the typical manufacturers' maximum concentration of the list of constituents by the manufacturers.
b. Based on reported literature values.
c. Based on synthetic leachate formula for W-025
d. Strong acids, bases, or oxidizing compounds were not identified in the W-025 compatibility studies.

5.7 Drainage Layer

The drainage layer for the LCRS consists of three components: the separation geotextile, the CDN, and the drainage gravel. Analyses for the drainage layer required evaluation of these components.

5.7.1 Geotextile Analyses (Separation)

Analyses were preformed to verify that a separation geotextile between the operations layer and leachate collection drain gravel is required by evaluating natural graded filter criteria for these materials. Results indicated that natural filter criteria could not be achieved, thus a separation geotextile is required between the operations layer and drain gravel. Supporting natural filter calculations are included in Appendix C.6.a.

Analyses were conducted to determine the proper apparent opening size (AOS) and permittivity of the separation geotextile. Required AOS and permittivity were determined based on filter, fines retention, and clogging potential criteria. Results of these analyses were used to develop the technical specifications for the separation geotextile (see Section 02371). Supporting geotextile filter calculations are also included in Appendix C.6.A.

5.7.2 CDN Selection

The CDN selection was based on analysis of two design issues, CDN geotextile puncture resistance and CDN required transmissivity.

5.7.2.1 CDN Geotextile Puncture Resistance

The LCRS CDN layer at the IDF will be overlain by the operations layer on the 3H:1V side slope. The operations layer is allowed to contain a particle size up to 2 inches in dimension. An analysis was performed to determine if the geotextile bonded to geonet (to form the CDN) would be punctured by particles/rocks of this size.

The method developed by Koerner (1998) was used to calculate the puncture resistance. Koerner's method considers the size and shape of the rock, as well as other factors that could decrease the long-term strength of the geotextile. The two loading conditions examined were initial placement of the operations layer and the final depth of waste and closure cover. The geomembrane puncture resistance analysis (see Section 5.5.4) provides the details for the load analysis for these conditions. Detailed calculations for CDN geotextile puncture resistance and corresponding cushion geotextile requirements are included in Appendix C.6.b1.

Results of the analyses indicate that the required puncture resistance is 11.2 lbs. The minimum specified value for Type 1 geotextile (see technical specifications, Section 02371) is 65 lbs. Applying a partial safety factor of 2 gives a minimum resistance of 32.5 lbs. Therefore, the proposed geotextile bonded to the geonet of the CDN will resist puncture with a global safety factor of 2.9; it is adequate for resistance to puncture from the overlying operations layer under the pressure of maximum landfill contents pressure. Koerner (1998) recommends a minimum global safety factor of 2.0.

It should be noted that the results of this analysis are considered conservative because the analytical method assumes only a uniform particle size and does not take the surrounding soil matrix into consideration. This would effectively reduce the particle size by a considerable degree.

5.7.2.2 CDN Required Transmissivity

An additional selection criteria for the CDN is the required transmissivity (or flow rate) under design loading conditions. For the IDF two cases require analysis:

LDS CDN on bottom and side slope–For this case, the critical condition is to ensure that the transmissivity as required by WAC and EPA regulations (3 x 10-5 m2/sec) under the maximum load from the landfill contents can be achieved.

LCRS CDN on side slope only–There are actually two loading conditions for the LCRS CDN on the side slope. One is the open slope condition with operations layer only over the CDN, which is a low normal load (1,000 psf) condition. The second is in the filled condition, which is a high normal load (15,000 psf) condition. Based on the results of leachate production analyses using the Hydrologic Evaluation of Landfill Performance (HELP) model (see Section 5.8), the required transmissivity for the LCRS CDN is 6.5 x 10-5 m2/sec for the open slope condition and 1 x 10-5 m2/sec for the filled condition.

For each case, the approach was to compare the required transmissivity to typical manufacturer's data with test conditions (i.e., normal load and material boundary), similar to the design conditions. The allowable transmissivity (φ) was determined using guidance provided by GRI standard GC-8 (2001), *Determination of the Allowable Flow Rate of a Drainage Geocomposite*. The GRI-GC8 standard uses the following equation:

$$\varphi_{allow} = \varphi_{100\ hr\ test}/\text{Reduction Factors for intrusion, creep, chemical clogging and biological clogging}$$

The FS for design was then determined as follows:

$$FS = \varphi_{allow}/\varphi_{required}$$

Transmissivity data for the 100-hour test data was obtained from the manufacturer for both 200-mil and 250-mil thickness CDN for normal loads of both 1,000 psf and 15,000 psf. Test data was provided for a number of boundary conditions including flow tests between a geomembrane and a soil, as would be the case for the LCRS or LDS CDN. Test data used as the basis for the analyses are included with the calculations presented in Appendix C.6.b2.

Based on the analyses, a higher flow, thicker (250-mil minimum) CDN is required, due to the reduction of flow under the high normal loads in the final filling configuration. The technical specifications (see Section 02373) provide the required index values for the geonet core of the CDN as well as the CDN itself (with geotextile bonded to both sides of the geonet), based on the results of this analysis. The transmissivity requirements in the technical specifications are index values and not in-service condition values, as determined in this analysis. These index values are representative of testing that manufacturers typically perform in production and are correlated to design conditions using the approach outlined in GRI GC-8.

5.7.3 Drainage Gravel Selection

Section 02315 (Fill and Backfill) in the technical specifications requires that drain gravel meets the requirements of WSDOT 9-03.12(4) for gradation. The technical specifications also require a performance specification for a hydraulic conductivity greater or equal to 10-1 cm/sec.

Hydraulic conductivity of the specified drain gravel was estimated using two different empirical relationships. The most relevant of the two estimates a minimum hydraulic conductivity of 1 cm/sec, based on the specified gradation curve for WSDOT Gravel Backfill for Drains (9-03.12[4]). Supporting calculations are included in Appendix C.6.c.

The minimum estimated hydraulic conductivity for the drain gravel exceeds the required (by WAC and EPA regulations) hydraulic conductivity of 10-2 cm/sec by a factor or 100 to 1,000, and the performance specification hydraulic conductivity of 10-1 cm/sec by a factor of 10 to 100. This exceedance makes an allowance for two items: (1) it allows for the uncertainty in the empirical formulas used to predict hydraulic conductivity, and (2) it also allows for the potential long-term reduction in hydraulic conductivity in the drain gravel as fines from waste filling and the operations layer migrate into the gravel over time.

As part of Construction QA, testing it is recommended that samples of imported drain gravel be tested for conformance with the gradation and hydraulic conductivity requirements in the technical specifications.

5.8 Leachate Production

5.8.1 Leachate Production Analyses

Estimates of the amount of leachate produced during the development and operation of the IDF were needed to design the components of the leachate collection and conveyance system described in Section 5.9, and to provide information necessary when evaluating slope stability of the side slope and bottom liner systems. Leachate is produced when precipitation falls within the lined area and infiltrates vertically through the waste and/or bottom liner system. The amount of infiltration estimated to occur depends on the hydrologic processes and the relative fraction of precipitation that results as leachate and is collected by the leachate collection system.

The water balance components of the hydrologic process were estimated using EPA's *Hydrologic Evaluation of Landfill Performance (HELP) Model* (Schroeder et. al., 1997), a well known standard for water balance modeling. The HELP model has been widely used for evaluating hydrologic conditions and is the standard model used for providing information necessary for the design of landfill systems. Estimates of the water balance components of the hydrologic cycle provided by HELP include precipitation, evapotranspiration, surface water runoff, vertical percolation, soil moisture storage, and lateral drainage in soil layers.

The HELP model requires input of weather data, representing the conditions at the landfill location, soils data representing the various layers of cover soils, waste materials, and soils underlying the waste layers, and other design data used by the model for water balance calculations. A detailed description of the model and modeling inputs are included in Appendix C.7.

The development of the IDF from Phase I through Phase IV was considered to determine the maximum flow condition expected during development and operation of the landfill. That is, various combinations of open and interim closed phases were considered and the combination calculated to produce the maximum amount of leachate was chosen for analysis. The chosen combination was Phase I through III under interim closure condition and Phase IV in the open condition with little or no waste present. The flows from this condition were used to size the LCRS collection piping and pump systems.

Water balance components were taken directly from model output and a spreadsheet was used to calculate the volumes of leachate by multiplying the HELP output parameter by the area of the type of system modeled. For example, the lateral drainage estimated by the HELP model for the uncovered side slope condition in Phase IV development was multiplied by the total side slope area to determine the total volume of leachate from that area. A spreadsheet summarizing the estimated leachate flows is included in Appendix C.7.

The following modeling results were used for various aspects of design of the IDF systems:

LCRS collection system–Modeling results for the peak day event were used to size the leachate collection system piping that conveys flow to the LCRS systems. The peak day event, as predicted by HELP and referenced herein, was a 1.6-inch precipitation event. This event is approximately 25 percent higher than the 25 year, 24 hour peak day storm event of 1.28 inches (Appendix C.9), required by regulations to be used when complying with the maximum 12 inches of head over the liner (WAC 173-303-665, see Section 2). The spacing of the LCRS perforated collection piping and the properties of the drain gravel material that convey lateral drainage flows above the bottom liner geomembrane to the collection piping and LCRS sump area were checked to insure the maximum head buildup above the sump area of the liner system did not exceed the maximum allowed according to regulatory requirements, as outlined in Section 2.

LCRS pump and forcemain systems–Modeling results for the peak day event were used to size the LCRS high flow pump system that conveys flow to the leachate storage tanks and truck loadout facilities.

Average monthly flow rates plus one standard deviation (resulting in a conservatively-high expected flow rate) was used to design the LCRS low flow pump system for pumping from the IDF during average monthly conditions.

Leachate Collection Storage–Volumes for the peak day event and assumptions for the operational rate of removal of leachate from the tanks were used to size the storage tanks. Storage tank sizing is described in Section 5.9.2.2.

Liner system material properties and stability analyses–The lateral drainage layers of the side slope and bottom liner systems were checked to insure the transmissivity of the layers was sufficient to convey lateral flows and maintain less than the maximum head buildup over the liner system. The seepage height above the liner was used when checking the liner system for veneer stability.

5.9 Leachate Collection System

5.9.1 Earth Loading Analyses

5.9.1.1 Leachate System Loading Analyses for Piping within Phase I Liner Limits

Loading over the leachate system piping include all layers of soil materials, wastes, and anticipated traffic loading. The maximum loading occurs over the piping in the LCRS and LDS sump area, because of its low elevation and the height of material–both waste and soil layers–overlying the sumps. Loading calculations from the geosynthetic liner puncture resistance calculations described in Section 5.5.4 were modified to represent the maximum loading in the LCRS/LDS sump area. Other pipes in the Phase I area, including piping outside the sump and the side slope riser piping, will be subjected to less than the maximum loading. The maximum loading is listed in Appendix C.8.a, along with the calculations for pipe sizing required to withstand this anticipated pipe loading.

Pipe wall thickness was selected based on the maximum loading anticipated in the sump area such that the pipe will not fail due to excessive deflection, wall buckling, or wall crushing. All other piping in Phase I outside of the sump area was chosen with the same standard dimension ratio (SDR) to withstand the maximum load. Standard analysis methods, as recommended by the manufacturer of HDPE pipe made from PE3408 type resin, were used to evaluate pipe strength under loading. These standard methods are based on flexible pipe design practice as applied to HDPE piping. The manufacturer's recommended design analysis techniques are based on standard analysis techniques, including the Iowa formula (*Waste Containment Systems, Waste Stabilization, and Landfills Design and Evaluation*, Sharma and Lewis, 1994), with conservative factors of safety. The potential loss of strength due to the perforations in the perforated collection piping was assumed non-significant, based on actual test results of perforated pipe under similar load rates. The pipe material assumed is High Density Polyethylene PE3408 pipe with a cell classification of 345434C or better. The flexural modulus and material strength of the pipe was per manufacturer's published literature, based on this classification of pipe.

5.9.1.2 Leachate System Loading Analyses for Piping Outside of Phase I Liner Limits

Piping outside the Phase I liner area includes all underground piping between the crest pad building, combined sump, leachate transfer building, storage tank, and tanker truck load out facility (see Drawing H-2-830846). The civil road layout in these areas is generally configured to allow medium to light duty trucks, such as would be used for operations and maintenance activities. The leachate tanker truck accesses the concrete truck load pad only, and would not normally pass over any piping. However, the piping outside the Phase I Liner area was designed for H-20 semi-trailer type loading to be conservative. The same SDR pipe that used for the high loading within the Phase I liner limits as described in Section 5.9.1.1 was assumed for all piping exposed to earth and traffic loading outside of the Phase I liner limits. The expected pipe loading for H-20 loading plus earth load was compared to the loading used for designing the piping inside the Phase I liner limits and was found to be much lower. Since the pipe SDR is sufficiently strong for the maximum loading inside the Phase I limits, it will have more than sufficient strength for loading expected outside the Phase I limits. Calculations are included in Appendix C.8.a.

5.9.2 Leachate System Hydraulics Analyses

5.9.2.1 Leachate System Hydraulics Analyses

The leachate collection and conveyance system collects leachate that accumulates as a result of precipitation landing within the footprint of the cells, and it conveys the collected leachate from the cells to a storage tank or tanker truck. Perforated collection piping in the LCRS collects and conveys leachate from the bottom liner system and conveys it to a LCRS sump area in both cells. Lateral flow of leachate from the side slope and bottom liner areas also is conveyed directly to the sump area through a high permeability gravel layer and/or geosynthetic drainage net material. Submersible pumps in the LCRS sump and contained within perforated riser pipes convey leachate to the crest pad building and directly to the leachate storage tank or the tanker truck load facility. Hydraulics analysis was conducted to size the gravity flow piping of the LCRS collection piping and the pump and force main system from the sump area to the storage tank and tanker truck load facility. Sizing and design of leachate collection and conveyance systems were based on ultimate build out of the IDF through Phase IV. That is, the components installed as part of the Phase I design are sized for the ultimate configuration and flows estimated through Phase IV.

5.9.2.2 LCRS Gravity Flow Analyses

The LCRS perforated collection piping was sized using standard gravity flow analysis techniques. The pipe size (nominal 12-inch diameter) was chosen as double the minimum size required for cleanout of the pipe to insure any accumulation of fines would not significantly restrict the flow in the pipe, even though the drain gravel surrounding the pipe will have minimal fines present and geotextiles are present in the lining system to further restrict the migration of any fines. The maximum flow used for sizing was the maximum from the HELP predicted maximum day flow rate or the pump flow rate, based on the pump chosen to convey flow out of the cell.

Perforations in the pipe were sized to allow flow rates much higher than the required maximum flow rate out of the cell, with minimal head loss. This assumption was more conservative by virtue of the fact that the main LCRS collection pipe will only collect and convey a portion of the lateral drainage flow from the cell; the drain gravel and CDN will also convey a portion of the flow. Calculations are included in Appendix C.8.b.

5.9.2.3 Leachate System Pumps and Force Mains Analyses

The pump and forcemain systems for conveying leachate out of the cells and into the leachate storage tanks and to the tanker truck load out facility, and the design considerations for each are described below. Calculations are included in Appendix C.8.b.

LCRS pumps and forcemains–The LCRS pumps and forcemains convey leachate out of the cells to storage tanks or the tanker truck load areas. The criteria for pumping capacity is that the maximum head over the sump area of the cell will not be allowed to exceed 12 inches during the peak day event and during normal operations. To meet the requirement for not exceeding the 12-inch criteria for the peak day event, a LCRS high flow pump was sized to handle the expected peak day flow rate, as estimated and described in Section 5.8, Leachate Production. Hydraulic analyses were conducted to size the pump and forcemain piping according to standard practice to convey the maximum flow rate.

A LCRS low flow pump was sized to convey flow out of the cells under normal, monthly operations. The criteria established for the low flow pump was to convey the average monthly flow plus one standard deviation from the cells, assuming the pump could remove that amount of flow with less than continuous operation. The highest value of the average month plus one standard deviation was used for the maximum flow required of the pump. Under lower flow required conditions, the pump would operate near this rate, depending on the system curve head loss characteristics, but would run for a shorter length of time to remove the volume of leachate from the cell.

LDS pump and forcemain–The LDS pump and forcemain conveys flows from leakage through the LCRS sump area, if in the unlikely event any leakage occurs, to the storage tank or tanker truck load out facility. The LDS system is sized to convey the flow equal to the ALR (described in Section 5.11); however, this rate is so small that the pump capacity is much higher than necessary.

Leachate transfer pump to truckload and forcemain–Under normal operations, leachate conveyed out of the IDF will be routed to the leachate storage tank. Periodically the leachate will need to be conveyed to tanker trucks for transport to an offsite water treatment facility. A transfer pump is required to move water from the storage tank to the tanker truck loadout facility. The pump and forcemain were sized to convey approximately 250 gallons per minute (gpm), a rate commensurate with timely loading of the tanker trucks that have capacities equal to approximately 7,000 gallons. At 250 gpm, the tankers can be loaded quickly, depending on the operational requirements for moving leachate and making storage tank capacity available under high precipitation conditions and/or the condition when the storage tanks are at or near capacity. Storage and operations considerations are described in Section 5.9.2.4.

Combined sump pump and forcemain–The combined sump pump and forcemain must convey flow from the sump to the leachate storage tank. The flow criteria for this pump was set at approximately the same flow as the leachate transfer pump. This is based on the worst case scenario of the leachate transfer pump accidentally being left on when the tanker truck is filled, causing the full 250 gpm flow to overflow the truck, collect on the pad, and drain into the combined sump. Under less than maximum flow conditions, the pump would cycle when any leakage from other systems connected to the sump pump reached the level on control setting for the pump. In this case, the pump would cycle quickly to pump the small volume of the inner sump into the storage tank.

Crest pad building sump pump–A small sump pump is provided in the crest pad building to remove minor amounts of water in the sump from sampling activities or piping leaks. The nominal flow rate was chosen as a minimum of four gpm. The pump discharges into the main forcemain line to the storage tank or tanker truck load out facility.

The pump and forcemain piping systems were modeled using standard hydraulic analysis techniques. Actual pump curves for preliminary pump selections were input and the analyses conducted to determine the estimated run condition for the various operational conditions. For example, a pump was chosen for the LCRS high flow pump and forcemain system, and the analysis was run for the conditions of the pump conveying flow to the leachate storage tank and directly to the tanker truck load out facility. Different flow rates and system pressures resulted, based on the differences in the system curve for each flow path versus the pump curve characteristics. Pump cycle times were considered for the flow requirements and total removed volume. The manufacturer's recommendations for cycle times and other operating requirements, where applicable, were checked.

5.9.2.4 Leachate Collection Storage Analyses

The results of the leachate production analysis indicate a total of approximately 269,000 gallons of leachate must be removed from the IDF landfill within 24 hours after a peak storm event. A temporary storage tank for each cell was sized to store leachate generated by the associated cell. The leachate storage tank capacity is dependent on the flow rate of leachate into and out of the tank as well as a factor of safety.

The leachate production analysis indicates the worst case flow rate out of each cell into the associated tank would be 157 gpm (sum of the required flow rates of the high and low flow leachate pumps). The leachate transfer pump for each cell can fill a tanker truck at a maximum of 250 gpm; however, the limiting factor is how often a truck can be filled.

The calculation in Appendix C.8.c presents the method of determining the appropriate storage capacity of each leachate storage tank. The following leachate tanker truck loading activities were assumed:

- Tanker Capacity 7,000 gallons
- Number of tankers per cycle 1

- Hours per cycle (roundtrip) 2.4
- Hours per shift 8
- Shifts per day 1
- Leachate tank level prior to event 2 feet

The calculation indicates that each tank requires a maximum operational capacity of 375,000 gallons to maintain a safety factor of 1.5. The assumptions made in the calculation must be adhered to during operational activities to maintain the calculated safety factor.

5.10 Surface Stormwater

The surface stormwater analysis was done to determine the sizes of the surface stormwater facilities necessary for the IDF Phase I Critical Systems Design. The surface stormwater analysis is documented in detail in Appendix C.9.

The governing regulation is WAC 173-303-665(2) (c) and (d). This requires that the stormwater system be designed to prevent flow onto the active portion of the landfill during peak discharge from at least a 25-year storm. It also requires that the runoff management system be designed to collect and control at least the water volume resulting from a 24-hour, 25-year storm.

The primary purpose of the proposed stormwater facilities is to prevent stormwater runoff from areas adjacent to the two Phase I cells from entering the cells during Phase I operation. This will be done by collecting, conveying, and safely discharging stormwater from areas outside of the two Phase I cells that would otherwise run into these cells.

The Department of Ecology has issued State Waste Discharge Permit Number ST 4510 for industrial stormwater discharges to the ground through engineered land disposal structures on the Hanford site (ST 4510, Ecology, 1999; DOE/RL97-67 Revision 3, January 2000). Since the design for this project does include facilities for collecting stormwater runoff and discharging it to the ground, the permit was reviewed to determine whether it applied to these stormwater discharges. To be covered by this permit, the stormwater must be considered an industrial discharge that is collected in an engineered structure and is then discharged to the ground through an engineered structure. A stormwater discharge is an industrial discharge if the stormwater has the potential to come into contact with an industrial activity or is collected within an area of industrial activity. The purpose of the stormwater facilities that have been designed for this project is to prevent the stormwater from areas outside of the Phase I landfill from entering the landfill area. Therefore, the stormwater collected by these facilities would probably not be considered industrial stormwater. To be an engineered structure for the collection of stormwater, the structure has to be an impervious surface that is directly associated with industrial activities. The stormwater collection facilities designed for this project do not have impervious surfaces. Therefore, permit ST 4510 does not apply to the stormwater system designed for this project.

Stormwater facilities were designed only for the operation stage of Phase I and not for interim or final closure conditions. Therefore, no stormwater facilities have been designed for stormwater runoff from the Phase I cells after construction of their interim closure or final closure. Stormwater needs for the construction, operation, and closure of future phases were not considered.

No stormwater collection and conveyance facilities were analyzed and/or designed for any of the roads and support facilities that will be constructed as part of this project. The roads will be gravel surfaced, and stormwater that does run off the roads into adjacent areas will infiltrate. The stormwater from the roofs of the buildings will be caught in gutters and discharged to the ground surface via down spouts. The stormwater that falls on the leachate tanks will evaporate off the floating covers.

5.10.1 Existing Conditions

Under existing conditions, the area around the Phase I site slopes down gently from south to north at an average grade of approximately 0.5 percent.

The only area that may generate stormwater that can run into the Phase I excavation is the area that extends south from the excavation area to the crest of the sand dunes, located north of 1st Street (see drainage areas figure in Appendix C.9). This drainage area is moderately vegetated, primarily with large sage brush and grasses. The soils are generally sandy, with relatively high rates of infiltration. This area typically receives little precipitation. There is little to no runoff, and stormwater normally either infiltrates or is used by the vegetation. No existing drainage channels are apparent. The groundwater table is approximately 300 feet below the ground surface.

5.10.2 Proposed Stormwater Facilities

To prevent stormwater from the area south of the Phase I excavation from running overland into the excavation, a combination stormwater berm/ditch will be constructed south of the top of the south slope of the excavation. The south end of the excavation will be approximately 1,400 feet long, and the ground will be essentially flat. The berm/ditch will have a center high point and then slope down to the east and to the west (two discharge points). A berm will be constructed immediately south of the ditch. At the centerline of the excavation, the invert of the ditch will be at the existing ground surface, and the berm will form the south slope of the ditch. The ditch will be excavated, with a longitudinal slope of 0.5 percent to both the east and the west. This will be done in order to minimize the depth of the ditch at its east and west ends. Culverts will be installed at the east and west ditch ends to convey the flow under the access roads. The culverts will discharge into the east and west infiltration areas.

The base map does not show any areas where stormwater runoff from offsite areas may flow into the east or west boundaries of the Phase I excavation. However, if any offsite stormwater should flow toward these boundaries, the fill for the berm access road and the shine berm will prevent the stormwater from flowing into the excavation (see drainage areas figure in Appendix C.9). The intercepted stormwater will flow south along the toe of the fill and either infiltrate or flow overland to the north, away from the site at the north end of the berm access road.

The ground slopes away from the north end of the Phase I site, so there will be no offsite stormwater running toward the north Phase I boundary.

The Phase I liner will end north of the toe of the south slope of the Phase I excavation. In order to reduce potential leachate flows, a stormwater berm/ditch will be constructed just south of the south end of the liner. This berm/ditch will intercept and convey stormwater runoff from the unlined south slope and the unlined southern ends of the east and west slopes. The berm/ditch will be sloped to drain to the east. A stormwater pipe will convey the stormwater under the landing for the access ramp and will discharge to the excavation infiltration area. If this pipe ran straight from the ditch to the infiltration area, it would not have adequate cover. Therefore, a catch basin with a solid cover will be installed near the west end of the stormwater pipe. The invert of the pipe out of the catch basin will be lower than that of the pipe running into this catch basin. The stormwater pipe that will run from the catch basin to the excavation infiltration area will then have adequate cover. The excavation infiltration area will be excavated in the southeast corner of the excavation.

The south edge of the access ramp into the Phase I excavation and the south edge of the "flat" area at the bottom of the access ramp will serve as ditches. The access ramp will have a cross-slope of 2 percent down to the south. The "flat" area at the bottom of the access ramp will have a slope down to the south that varies between 1 and 3 percent. Adjacent to each of these will be the south slope of the excavation. Construction of a full V-shaped ditch along the south side of the access ramp and the "flat" area was considered. This idea was rejected because it would result in a larger excavation with the top of the Phase I south slope moved further south.

The stormwater facilities are shown on the Phase I Grading and Drainage Plan drawing (Drawing H-2-830830).

Stormwater runoff from the north, east, and west lined slopes of Phase I will run into the bottom lined area and will become leachate.

There are no provisions in the design of the Phase I critical systems to divert clean runoff from these side slopes and discharge it to the surface water system instead of the leachate system at this time. However, a rain curtain or other approach to reduce the amount of clean runoff from the lined area that enters the leachate system may be considered in the future.

5.10.3 Analysis

The surface stormwater analysis is documented in Appendix C.9 and is summarized below. Stormwater runoff flows were estimated for a 24-hour, 25-year design event, using the Soil Conservation Service curve number methodology as documented in *Urban Hydrology for Small Watersheds* (U.S. Department of Agriculture, June 1986) and the Hydraulic Engineering Cirular-1 (HEC-1) computer program (*Flood Hydrograph Package (HEC-1)*, U.S. Army Corps of Engineers, Hydrologic Engineering Center, revised June 1988). The precipitation data used was based on information from the *Hanford Site Climatological Data Summary 2001* (Pacific Northwest National Laboratory, May 2002). The ground at the project site is periodically frozen during the winter months, when the most precipitation falls. Therefore, it was assumed that the ground was frozen for the runoff flow calculations.

The peak flows (calculated using the HEC-1 model) were checked for reasonableness. The tabular and graphical methods in TR 55 were used to estimate peak 25-year flows for each of the drainage areas modeled in HEC-1. The results confirmed the reasonableness of the peak flows calculated by HEC-1.

The berm/ditches were designed to convey the peak 25-year flow with a minimum freeboard of one foot.

The infiltration areas were sized based on containing and infiltrating the runoff from the 24-hour, 25-year design event, without causing the water surface to extend above the upstream end of the culvert or stormwater pipe that will discharge to the infiltration area. No specific infiltration data have been collected at the IDF project site. However, infiltration rates have been determined for use at the Waste Treatment Plant (*Geotechnical Report Supplement No. 1*, Shannon and Wilson, April 2001). These infiltration rates were used in sizing each of the infiltration areas.

The culverts and stormwater pipes were designed to convey the peak 25-year flow with a maximum headwater to a diameter ratio of 1.25. Both inlet and outlet flow conditions were analyzed. The starting water surface for the outlet flow condition calculations were the maximum water surface elevation estimated for the associated infiltration area for the 24-hour, 25-year design event.

5.11 Action Leakage Rate (ALR)

5.11.1 LDS ALR

The ALR is defined in WAC 173-303-665(8) and the Final Rule (EPA 1992a, 40 CFR Part 264.222) as the "maximum design flow rate that the leak detection system...can remove without the fluid head on the bottom liner exceeding 1 foot". This calculation was performed to determine the ALR for the IDF lining system. The IDF consists of two cells, each with an area of approximately 8.5 acres.

In addition to determining the ALR, an estimate of actual leakage rate through the proposed primary bottom lining system is provided as a comparison to the calculated ALR. HELP modeling for the side slope indicates negligible head build-up on the side slopes (see Section 5.8), thus an estimation of the actual leakage rate was determined for the bottom primary lining system only.

EPA provides a formula (based on Darcy's Law for calculating this flow capacity), assuming that it originates from a single hole in the primary liner (EPA, 1992b). Calculations presented in Appendix C.10 provide details of the method of analysis and input data. The ALR calculations are dependent on the transmissivity value for the CDN. A value of 3 x 10-5 m2/sec was used in the ALR analysis (equivalent to the value required by WAC and EPA regulations for the LDS, Section 5.7.2). Calculations in Appendix C.6.b2 provide justification for the transmissivity used in the ALR analyses.

The results of the analyses indicate the ALR for each IDF cell is 206 gallons per acre per day (gpad) or approximately 1,800 gallons per day per cell. This ALR includes a factor of safety of 2 in accordance with EPA guidelines (EPA, 1992b).

It is also much lower than the capacity of the pump that removes liquid from the LDS. The estimated actual leakage rate for the composite primary lining system is 0.06 gpad (small defect) to 0.08 gpad (larger defect) for a composite liner with good intimate contact, and 0.3 gpad (small) to 0.4 gpad (large) for poor contact. Detailed calculations for both rates are presented in Appendix C.10.

The proposed primary composite lining system has a much lower estimated leakage rate than the ALR. This demonstrates the benefit of the GCL that is included in the primary bottom lining system, to provide a composite lining system and minimize actual leakage rate through the bottom primary lining system.

5.12 Building Systems Analyses

5.12.1 Geotechnical Design Parameters

The key geotechnical parameters and analyses for structural design of the supporting facilities for the Hanford IDF included the following:

- Bearing Capacity
- Settlement
- Modulus of Subgrade Reaction
- Earth Pressures
- UBC Seismic Soil Parameters

The methodologies, input data, and results for each of these categories of analysis are presented in detail in Appendix C.11.A.

5.12.2 Structural

5.12.2.1 Crest Pad Building Foundation Analysis, Pipe Bracing and Winch

The crest pad building foundation was analyzed as a concrete slab on an elastic foundation. The foundation was modeled with springs to model the vertical sub-grade reaction. The value of the vertical sub-grade reaction was provided by the geotechnical engineer. The applied loads and load combinations were input into Visual Analysis (version 4.0), a finite element program. The finite element analyses results include elastic settlement, moments, and shears values of the concrete slab. The results were then used to design slab depth and reinforcing.

Load reactions from the pre-engineered metal building were estimated using hand calculations and applied onto the concrete slab at the corners of the slab. It is a reasonable assumption that the frame loads from the pre-engineered metal building will only occur at the corner of the building, since the size of the building will not require any intermediate framing.

Loads and load combinations were used as required by TFC-ENG-STD-06, REV A. Performance category, PC-1 was used as specified and applied as applicable for both wind, seismic, and load combinations requirements.

In summary, the analyses results showed that an 8-inch thick slab sufficed with #5 reinforcing at 12-inch centers. The analyses results also showed that a 1 foot-10 inch edge thickening around the perimeter of the building would be sufficient. More detailed accounting of the analyses is presented in Appendix C.11.b1.

The pipe bracing and support for the small diameter PVC (polyvinyl chloride) piping included both gravity as well as lateral load resistance, due to a seismic event. The governing piping support is assumed a 6-foot-tall cantilever support, with the piping load and 50 pounds of lateral load applied to the top of the support. The 50 pound lateral load was used in lieu of the calculated seismic load because the calculated seismic load was only 19 pounds. Using a 50 pound lateral load gives the pipe support system greater rigidity. Detailed calculations of the pipe supports are included in Appendix C.11.b2.

The winch support was analyzed as a vertical cantilever that supports the winch and resists a total lateral load of 400 pounds.

A 400 pound lateral load was used since the entire gravity load of the pump and the hoses adds up to this weight. Therefore, using 400 pounds in the horizontal direction is conservative. Detailed calculations of the winch support are given in Appendix C.11.b3.

5.12.2.2 Leachate Transfer Building Foundation Analysis

As the leachate transfer building foundation is considered as a slab–on-grade, only hand calculations were performed. Foundation soil reactions were considered to be distributed linearly, then soil pressure distributions were applied to the concrete to calculated the moment and shear values for design of the concrete slab and reinforcing steel.

Load reactions from the pre-engineered metal building were estimated using hand calculations and applied onto the concrete slab along the perimeter of the slab.

Loads and load combinations were used as required by TFC-ENG-STD-06, REV A. Performance category, PC-1 was used as specified and applied as applicable for both wind, seismic, and load combinations requirements.

In summary, the analyses results showed that the 2-foot-6 inch-thick slab with #6 bars at 12-inch centers will suffice and appears to be overdesigned. The 2-foot-6-inch thickness is not based on concrete strength requirements but more for frost depth cover, simplifying the ground forming, and reinforcing bending requirements. Detailed calculations of the analyses are presented in Appendix C.11.c.

5.12.2.3 Leachate Tank Foundation Analysis

The leachate tank foundation is considered to be a concrete ringwall, per AWWA D103-97. The tank gravity loads, including both water load and tank dead loads, were considered in the design of the ringwall.

AWWA D103-97, Factory-Coated Bolted Steel Tanks for Water Storage is not listed in the TFC-ENG-STD-06, REV A. AWWA D100-96, Welded Steel Tanks for Water Storage, is listed; however, this standard does not apply, since the tank will be a bolted steel tank. Therefore, the tank will be designed per AWWA D103-97, Factory-Coated Bolted Steel Tanks for Water Storage.

The analysis of the concrete ringwall and reinforcing is based on the hoop tension on the ringwall from the surcharge of the liquid weight on the soil within the ringwall. In summary, a 4-foot-6-inch-deep by 1-foot-6-inch width ringwall with #7 at 12-inch-longitudinal reinforcing on each face of the ringwall will suffice. Detailed calculations of the analyses are presented in Appendix C.11.d.

5.12.2.4 Truck Loading Station Foundation Analysis and Leachate Loading

The Truck Loading Station foundation was analyzed as a concrete slab on an elastic foundation. The foundation was modeled with springs to model the vertical subgrade reaction. The value of the vertical subgrade reaction was provided by the geotechnical engineer. The applied loads and load combinations were input into Visual Analysis (version 4.0), a finite element program. The finite element analyses results include elastic settlement, moments, and shears values of the concrete slab. The results were then used to design slab depth and reinforcing.

Loads and load combinations were used as required by TFC-ENG-STD-06, REV A. As required, AASHTO HB-16 loading was used with an HS 20-44 load wheel pattern. For maximum axle load, 40,000 pounds was used instead of 32,000 pounds as required per HS 20-44. An impact factor was also applied as required by AASHTO HB-16.

The wheel pattern loading was arranged in three positions on the slab to yield the maximum moments and shears. Supporting calculations and further discussions are presented in Appendix C.11.e1.

The leachate loading support was analyzed as a post with a horizontal boom attached near the top of the post. The design load included the dead weight of the post, boom, and piping full of water. Wind loads were analyzed per ASCE 7-98.

In addition, the lateral load was compared with a 300-pound point load hanging vertically at the end of the boom. The lateral wind load governed for overall overturning at the base of the post; however, the 300-pound point load governed for the boom attachment to the post.

In summary, a 10-inch by 10-inch tube for the post, with a 6-inch by 6-inch tube as the horizontal boom welded to the post will suffice. The geotechnical engineer has verified that a 5-foot-6-inch-deep and 3-foot-diameter concrete encasement around the post will be sufficient for strength and stability. Supporting calculations and further discussions are presented Appendix C.11.e2.

5.12.3 Mechanical/Heating, ventilating, and air conditioning (HVAC)

5.12.3.1 Crest Pad and Leachate Transfer Building

Heating, ventilating, and air conditioning (HVAC) capacities were calculated for the crest pad and leachate transfer buildings. The temperature within the buildings must be controlled within a range to prevent freezing fluids in piping or overheating electronic devices. The HVAC components for the buildings were selected based on the criteria and calculations provided in Appendix C.11.f and C.11.g.

5.12.4 Electrical/I&C

This section introduces and summarizes the results of detailed electrical engineering calculations included in Appendix C.11.h.

- IDF leachate collection and handling crest pad facilities (two each)
- IDF leachate storage tank and leachate transfer facilities (two each)
- IDF truck loading facilities (two each)

5.12.4.1 Building Power Supply

Open Items

The Phase I Critical Systems 80% IDF design documents do not identify the following open items:

- Exact location of primary 13.8 kV, 3-phase tie-in
- Exact value of available primary short circuit current at primary tie-in location
- Exact length of primary extension
- Exact location, size, and impedance of utility step-down 13.8 kV – 480/277V three, phase, 4-wire pad mounted transformer(s)

These items are scheduled to be addressed during the next IDF Phase I Non-Critical design.

Assumptions

The following assumptions were made in order to complete the 80% engineering analysis.

- Assume electrical service gear inside each Cell 1 and Cell 2 crest pad building to be powered by separate pad mounted utility transformers.
- Assume pad mounted utility transformers to be rated 75 kVA and installed within 100 feet of respective Cell 1 and Cell 2 crest pad buildings.
- Assume each pad mounted utility transformer to be radial fed from a common 13.8 kV primary feeder.
- Assume each Cell 1 and Cell 2 leachate transfer building to be powered from electrical service gear, located inside respective crest pad buildings.
- Assume available short circuit at primary side of pad mounted utility transformer(s) to be 100 MVA with an (X/R) ratio equal to 8.
- Assume impedance of 75 kVA pad mounted utility transformer to be 3.2%Z, 2.42%IR, and 2.10%IX.
- Assume power factor and efficiency for all pump motors to be 85 percent and 82 percent, respectively.

- Assume 25 foot candles of lighting levels to be required for interior of each building.
- Assumptions will be reviewed and addressed during the next IDF Phase I Non-Critical design.

Method of Analysis

- Branch circuit, feeder and service calculations in accordance with NEC Code (2002)
- Short circuit analysis (per unit) in accordance with IEEE-Red Book, Standard 141 (1993)
- Grounding electrode analysis in accordance with IEEE-Green Book, Standard 142 (1991)
- Computer analysis by SKM PTW 32 (Power Tools for Windows, 2003)
- Building interior lighting zonal cavity method in accordance with Integrated Engineering Software, Inc. (IES) *Lighting Handbook* (2000)

Analysis Performed Includes

- Calculate and size service, feeder, and branch circuits, based upon demand and design loads.
- Calculate and size equipment, equipment bus amperage, protective devices, and motor overloads, based upon demand and design loads.
- Calculate and size power feeders and branch circuit wiring, based upon demand and design loads.
- Calculate short circuit ratings for equipment.
- Calculate feeder and branch circuit voltage drop, and power factor.
- Calculate building lighting system requirements.

Voltage Drop

Load flow steady state voltage drop calculations for all feeders were based upon an equipment 85 percent power factor. Wire size were calculated and selected so that circuits do not exceed total voltage drop from the source bus to the point of utilization, including feeders and branch circuits:

Service and sub feeders	2 percent	Heat trace from panels	1 percent
Lighting from panels	1 percent	Receptacles from panels	1 percent
Motors from motor control center (MCC)	1 percent	Instrumentation from panels	1 percent

Feeder and Equipment Sizing

Service, feeder, branch circuit conductor ampacity, and protection devices ratings are based upon applicable sections of the NEC (2002) including:

- Lighting Loads per Article 220: Lighting
- Receptacle Loads per Article 220.13: Non-dwelling Units
- Continuous Loads per Article 230: Service
- Motor Loads per Article 220:14 and 430: Motors
- Air Condition Load per Article 440.6: Refrigerant Motor Compressor
- Heat Loads per Article 200.15: Fixed Electric Space Heating
- Non-Coincident Loads per Article 220.21: Non-coincidental Loads
- Heat Trace per Article 427: Fixed Electric Heating Equipment for Pipelines and Vessels

Load Factors

The following table summarizes load factors applied for various equipment in accordance with appropriate sections of the NEC (2002), while determining demand and design load analysis:

1 **Table 5-4. Building Power Supply Load Factors**

Item	Panel and Service Load Analysis	Comment
Heater Loads*	100 percent full load amperage (FLA)	Branch circuit sized to 125 percent of FLA
Motor Loads	Sum of motor load (FLA) + 25 percent of largest motor (FLA)	Branch circuit sized to 125 percent of FLA
Receptacles	180 VA /outlet	Non-Continuous Load
Lighting	2 watts/sq.-ft or total connected (FLA), whichever is larger	Continuous Load
Cooling Loads*	100 percent FLA	Branch circuit sized to 125 percent of FLA
Demand Factors	**Demand Factor Percent**	
First 10 kVA	Non-Dwelling Receptacles	100 percent
Remainder over 10kVA	Non-Dwelling Receptacles	50 percent
Non-continuous Load		100 percent
Continuous Loads		125 percent
* Note: The largest of the non-coincidental heat and cooling loads are used for service sizing.		

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1 **Table 5-5. Input Data Typical for Cell 1 and Cell 2**

Description	Ratings	Comments
Pump 219(Y)-LH-P-202	1/2 HP @ 480V, 3-phase	Coincidental load
Pump 219(Y)-LH-P-203	7.5 HP @ 480V, 3-phase	Coincidental load
Pump 219(Y)-LH-P-204	1/2 HP @ 480V, 3-phase	Coincidental load
Pump 219(Y)-LH-P-205	1/3 HP @ 480V, 3-phase	Coincidental load
Pump 219(Y)-LH-P-207	3 HP @ 480V, 3-phase	Coincidental load
Pump 219(Y)1-LH-P-302	3 HP @ 480V, 3-phase	Coincidental load
Heater 219(Y)-LH-UH-001	3.3 kW @ 480V, 3-phase	Non-coincidental and continuous load*
Heater 219(Y)1-LH-UH-002	3.3 kW @ 480V, 3-phase	Non-coincidental and continuous load*
Air Condition 219(Y)-LH-AC-001	2.04 kVA @ 208V, 1-phase	Non-coincidental load
Air Condition 219(Y)1-LH-AC-002	.96 kVA @ 208V, 1-phase	Non-coincidental load
Control Panel 219(Y)-LH-CP-001	1.5 kVA @ 120V, L-N	Continuous load
Bldg. 219(Y) Lighting	71 kVA @ 120V, L-N	Continuous load
Bldg. 219(Y)1 Lighting	29 kVA @ 120V, L-N	Continuous load
Heat Trace 219(Y)201-LH-HT-001	77 kW @ 120V, L-N	Continuous load
Heat Trace 219(Y)201-LH-HT-002	77 kW @ 120V, L-N	Continuous load
Heat Trace 219(Y)1-LH-HT-003	77 kW @ 120V, L-N	Continuous load
Bldg. 219(Y) Receptacles	720 kVA @ 120V, L-N	180VA/ outlet
Bldg. 219(Y)1 Receptacles	360 kVA @ 120V, L-N	180VA/ outlet

Note: (Y) = A,E
Cell 1 (A), Cell 2 (E)
Heater Load is greater than AC load.

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1 **Table 5-6. Building Power Supply Results/Conclusions**

Description	Ratings
Bldg. 219(X) connected load @ 219(X)-LH-MCC-001	23 kVA connected – 26 kVA design for each crest pad building.
Bldg. 219(X) main service breaker size @ 219(X)-LH-MCC-001	100 amps
Bldg. 219(X) main service feeder to 219(x)-LH-MCC-001	3#1 TW, 1#1 TW (N)
Bldg. 219(X) service transformer	75 kVA, 480V, 3-phase, 4-wire
Bldg. 219(X)1 transfer bldg. feeder breaker size	50 amps
Bldg. 219(X)1 transfer bldg. feeder size	3#4 TW, 1#8 G
219(x)-LH-MCC-001 short circuit available	2,484 amps symmetrical
219(x)1-LH-SW-002 short circuit available	1,632 amps symmetrical
219(x)-LH-LP-001 short circuit available	1,177 amps symmetrical
219(x)1-LH-LP-002 short circuit available	1,068 amps symmetrical
219(X) –LH-LP-001 lighting panel rating	60 amps
219(X)1-LH-LP-002 lighting panel rating	60 amps
219(X)-LH-T-001 lighting panel transformer rating	15 kVA
219(X)1-LH-T-002 lighting panel transformer rating	15 kVA
219(X)-LH-P-203 LCRS high flow pump motor feeder size	3#12 TW, 1#12 G
219(X)-LH-P-202 LCRS low flow pump motor feeder size	3#12 TW, 1#12 G
219(X)-LH-P-204 LDS pump motor feeder size	3#12 TW, 1#12 G
219(X)-LH-P-205 sump pump motor feeder size	3#12 TW, 1#12 G
219(X)1-LH-P-302 transfer pump motor feeder size	3#12 TW, 1#12 G
219(X)-LH-P-207 combined sump pump motor feeder size	3#12 TW, 1#12 G
219(X)-LH-UH-001 unit heater feeder size	3#12 TW, 1#12 G
219(X)1-LH-UH-002 unit heater feeder size	3#12 TW, 1#12 G
219(X)-LH-AC-001 air condition feeder size	3#10 TW, 1#10 G
219(X)1-LH-AC-002 air condition feeder size	3#12 TW, 1#12 G
219(X)-LH-MD-001 motor damper feeder size	2#12 TW, 1#12 G
219(X)1-LH-MD-002 motor damper feeder size	2#12 TW, 1#12 G
219(Y)201-LH-HT-001 leachate storage tank heat trace feeder size	2#10 TW, 1#10 G
219(Y)201-LH-HT-002 leachate storage tank heat trace feeder size	2#10 TW, 1#10 G
219(Y)1-LH-HT-003 truck loading station heat trace feeder size	2#10 TW, 1#10 G
219(X)-LH-CP-001 main control panel feeder size	2#10 TW, 1#10 G
Note: (X) = A,E	

Recommendations

Building Power Supply

- Provide separate power distribution equipment (pad mount utility transformer, secondary service, and power distribution gear) for Cell 1 and Cell 2 in order to maximize redundancy.
- Install service rated motor control center inside each crest pad building for providing service entrance, branch, and sub-feeder distribution capability, and complete motor control for various process control systems.
- Power lighting, receptacle, and facility loads from 3-phase, 4-wire lighting panel installed in each building.
- Power instrumentation from surge protected distribution center mounted inside facility control panel.
- Ground Electrode System.
- Provide and install ground electrode system for service and each separately derived system that incorporates both ground ring, ground rod, and concrete encased building rebar.
- Provide ground bus inside Process Instrumentation and Control Systems (PICS) control panels and bond to common ground electrode system.
- Bond non-current carrying metallic structure to ground electrode system that has the potential of becoming energized by attached electrical devices such as metallic conduit systems, enclosures, storage tank structures, building metal framing and siding, and above grade metallic process equipment.

5.12.4.2 Crest Pad Building Lighting

Building lighting systems were based upon I.E.S Zonal Cavity method in order to maintain an average 25-foot-candle level for process interior of each building.

Note: Interior lighting levels are based upon *IES Lighting Handbook Indoor Industrial Areas Recommended Illuminance Levels* for interior activities inside work spaces where visual tasks of medium to large contrast are to be performed on occasional basis.

Note: Exterior entrance lighting levels are based upon *IES Lighting Handbook Outdoor Site/Area Recommended Illuminance Levels* for building exterior entrances frequently visited locations.

Open Items

None

Assumptions

The following assumptions were made when analyzing building lighting.

Reflectance for unfinished rooms:

Ceilings	50 percent reflectance
Walls	50 percent reflectance
Floors	20 percent reflectance

Maintenance factor (light loss factor), interior lighting:

Incandescent lighting	.80
Fluorescent lighting	.61
HPS lighting	.70

Maintenance factor (light loss factor), exterior lighting:

HPS lighting	.70

Inputs

Crest pad buildings are unfinished industrial buildings with interior dimensions of:

Room name: Cell 1 crest pad building	Ceiling height: 11 feet
Fixture type: fluorescent two-lamp	Mount height: 9 feet
Room size: width 16 feet and length 21 feet	Area: 336 square feet

Recommendations

- Provide fluorescent low temperature starting wrap-around industrial fixtures for interior lighting of buildings
- Use two lamps in six fixtures for 25-foot candles minimum
- Install low pressure sodium fixture at front entrance on north exterior wall

5.12.4.3 Leachate Transfer Building Lighting

Building lighting system was based upon I.E.S Zonal Cavity method in order to maintain an average 25-foot-candle level for process interior of each building.

Note: Interior lighting levels are based upon *IES Lighting Handbook Indoor Industrial Areas Recommended Illuminance Levels* for interior activities inside work spaces where visual tasks of medium to large contrast are to be performed on occasional basis.

Note: Exterior entrance lighting levels are based upon *IES Lighting Handbook Outdoor Site/Area Recommended Illuminance Levels* for building exterior entrances frequently visited locations.

Open Items

None

Assumptions

The following assumptions were made when analyzing building lighting.

Reflectance for unfinished rooms:

Ceilings	50 percent reflectance
Walls	50 percent reflectance
Floors	20 percent reflectance

Maintenance factor (light loss factor), interior lighting:

Incandescent lighting	.80
Fluorescent lighting	.61
HPS lighting	.70

Maintenance factor (light loss factor), exterior lighting

HPS lighting	.70

Inputs

Crest pad buildings are unfinished industrial buildings with interior dimensions of:

Room name: leachate transfer building	Ceiling height: eight feet
Fixture type: fluorescent two-lamp	Mount height: eight feet
Room size: width 10 feet and length 10 feet	Area: 100 square feet

Recommendations

- Provide fluorescent low-temperature starting wrap-around industrial fixtures for interior lighting of buildings
- Use two lamps in two fixtures for 25-foot candles minimum

- Install low pressure sodium fixture at front entrance on north exterior wall and low pressure sodium on south exterior wall

5.12.4.4 Uninterrupted Power Supply (UPS) Sizing

Uninterruptible power is provided and sized to provide 25 minutes minimum of continuous backup power to the PICS programmable logic controller (PLC), operator interface unit (OIU), and local area network communication equipment.

In the event of a power failure, UPS will maintain communication with remote monitoring sites (future) and insure safe shutdown of power sensitive PICS equipment.

Open Items

None

Assumptions

None

Table 5-7. Input Data Typical for Cell 1 and Cell 2 Control Panel Loads

Description	Ratings	Comments
PLC Power Supply	180 VA	Continuous load
OIU Power Supply	60 VA	Continuous load
Ethernet Switch Power Supply	44 VA	Continuous load
Total *1.25	**355 VA**	

Recommendations

Table 5-8. Fortress Runtimes for Typical Applications in Minutes

Load (VA)	50	100	200	300	400	500	600	750	900	1050	1250	1425	1800	2250
0520-1050U	200	125	63	42	31	24	19	14	11	9.5	-	-	-	-
0520-0750U	132	75	38	26	19	14	11	8.5	-	-	-	-	-	-

Provide 1050 VA 120 Volt- 120 Volt UPS to achieve the 25 minutes minimum of continuous backup, power in the event of a power failure. Additional capacity will compensate for battery cycling deprivation.

5.13 Civil Grading

5.13.1 Waste Volume, Cut/Fill and Stockpile Requirement Calculations

The IDF is designed to provide the waste volume requirements identified by CH2M HILL. Those requirements consist of an ultimate landfill capacity for 1,177,110 cubic yards of waste and a Phase I capacity of 213,515 cubic yards of waste.

The IDF is also designed to balance the cut and fill volumes of the project. The ultimate landfill layout on the project site provides this balance. The volume balance includes excavated material, which will be used for the construction of the closure cap. Since the closure cap will be selected and designed in the future, assumptions for the cap layout and construction were made.

With a phased construction approach planned for IDF and the fact that the material balance includes backfill to construct a closure cap for the ultimate landfill, a substantial volume of material will be stored in stockpiles at the completion of construction of Phase I landfill. The Phase I landfill design volumes for subgrade cut, admix liner, drain gravel, and operations layer material were calculated using a 3-D AutoCAD model of the landfill. These volumes were used to identify the stockpile requirements to store material once Phase I construction is complete.

Potential stockpile locations are identified on the project site plan. Calculations of these volumes are included in Appendix C.12.a. Calculations in Appendix C.12.a also present confirmation of the available waste volume and cut/fill balance.

5.13.2 Phase I Access Road and Ramp Cross Section Design

Two cross sections using granular material for base and top course were designed for the Phase I landfill access roads and the access ramp into the landfill. The design reflects the estimated wheel loads and vehicles to use the facility daily. Calculations presenting the development of these cross sections are included in Appendix C.12.b.

6.0 FACILITY DESIGN AND CONSTRUCTION

6.1 Facility Layout

6.1.1 Location

The IDF will be located approximately 1,400 feet east of Baltimore Avenue and directly north of 1st Street in the 200 East Area of the Hanford Site. Phase I of the IDF landfill will measure approximately 800 feet by 1,500 feet, with its north-south axis being the shorter dimension. Leachate handling facilities will be located immediately north of the Phase I cells. The excavated depth to subgrade (not including sump depressions) will range from approximately 44 to 51 feet. Excavation will be deepest at the landfill's north end, near the sumps and along the centerline of each cell. It will be shallowest at the southwest and southeast corners of Cells 1 and 2, respectively. Stockpile locations for excavated materials will be situated east and southeast of the Phase I landfill excavation. At the completion of Phase I construction, exposed surfaces of the stockpiles and disturbed areas will be covered with a layer of topsoil, then seeded and mulched. A borrow area of soil to supplement admix preparation is located south of the Phase I excavation location.

6.1.2 Access Roads and Ramps

For access to Phase I of the IDF, waste hauler and operations vehicles will follow an access road and travel north from 1st Street. All roads and ramps at the Phase I IDF site will be constructed with crushed surfacing material for the base and top courses. The access road from 1st Street will be aligned with the landfill's west berm access road. The road will also follow the alignment of the west access berm road for the future IDF cells.

The access road will lead north, approximately 1,000 feet from 1st Street to where it widens into an intersection. At this location, a turn to the east will lead down a 5 percent grade, 800-foot-long access ramp into the Phase I landfill. The access ramp slope was selected to allow use by both waste haul trucks and the melter transporter. The grade of the access road from 1st Street was also limited to a maximum of five percent for this same reason. The access ramp into the landfill and the access road from 1st Street to the intersection area will be both 30 feet wide.

At that base of the ramp into the landfill, there will be adequate room for waste haul vehicles to turn and move the waste into the cells. The liner system will be installed to extend approximately 50 feet south beyond the estimated toe of slope of Phase I waste placement. This extension will allow waste haul vehicles to be staged or unloaded over a lined area.

At the access road intersection, continuing north will lead up a short ramp and onto the berm access road. The berm access road will be 20 feet wide on the east and west sides of the landfill. The road will widen to 30 feet at the northwest and northeast corners of the landfill and along the landfill's north side.

The wider road in these areas will allow operations vehicles to traverse around road corners and the crest pad buildings.

The access road will continue from the northwest corner of the berm access road to the Cell 1 and Cell 2 leachate storage tank facilities. A cul-de-sac area will be provided just east of the Cell 2 leachate facilities to provide a turnaround area for operations vehicles and leachate tanker trucks.

A road will also be provided to allow operation vehicles to travel south between the leachate facilities and onto the berm access road at the centerline of IDF landfill.

Future projects are being planned to upgrade the 1st Street pavement and construct an operation building north of the IDF landfill. It is anticipated that these facilities will connect to access roads designed for the Phase I landfill.

Related to permanent access roads and their use, the construction contractor will be required by the project general requirements to submit a plan, which details their use during construction. This plan will address locations and limits of stripping/grubbing, construction haul roads, stockpile/borrow areas and other construction staging areas.

6.1.3 Survey Grids

Survey grids for this project use the Washington State Plane coordinate system (South Zone–feet, NAD83 Lambert Projection). Contours are based on 200 Area topographic mapping database, provided by Hanford HGIS Department and dated 1991. A 1-foot contour interval was used on the design drawings.

As part of the Phase I landfill design, construction control points were developed for landfill and sump subgrades as well as for the anchor trenches, stormwater facilities, and the finished grades for all roads and ramps. North and east coordinates and elevations for these points are included in a survey control table on Drawing H-2-830829, Sheet 2 of 2. The control points and lines between them will provide a location grid that will allow construction of the subgrade, liner system, operations layer, and the finished grades for the IDF.

6.2 Landfill Geometry

6.2.1 Waste Volumes and Types

6.2.1.1 Volume

Two key design criteria were provided by CH2M HILL concerning waste volumes:

- Phase I of the IDF should be designed to receive a waste volume of 213,515 cubic yards, which is equal to163,250 cubic meters. CH2M HILL identified the waste volume for placement in all phases of IDF (ultimate landfill size) as 1,177,110 cubic yards, or 900,000 cubic meters.
- Both the Phase I landfill and the ultimate landfill volumes should be sized for an air space, which includes 1.5 cubic yards of clean fill for every cubic yard of waste.
- Using these criteria, Phase I was designed to provide air space for placement of 533,620 cubic yards of waste and clean fill.

6.2.1.2 Waste Types

(Note: The disposal of MLLW other than ILAW, DBVS waste, and IDF generated waste is not permitted at this time by this permit.)

The IDF will receive waste types including ILAW, DBVS Waste, and LLW. These wastes include both contact and remote-handled wastes. As identified in the project kickoff meetings by CH2M HILL, the waste volumes (in cubic yards) are estimated to include the following:

Waste Type	Phase I	All Phases
ILAW	50,025	753,350
MLLW	57.550	146,485
LLW	105,940	277,275
Total	**213,515**	**1,177,110**

These volumes are based on waste forecast information provided by Fluor Hanford, Inc. (FH). The waste volume forecasts are updated by Hanford Site contractors on a regular basis. The volumes above represent an average between the FH 2002 Forecast and the FH 1999 (with EIS) Forecast. Short descriptions of the waste types are given below:

Immobilized Low-Activity Waste (ILAW)–The ILAW packages are stainless steel cylinders that have been filled with vitrified low-activity waste (physically similar to glass), sealed, and cooled. The source of these waste cylinders is the Waste Treatment and Immobilization Plant. The packages are 7.5 feet in height and 4 feet in diameter, and could weigh up to 22,050 pounds each.

Contact-Handled Mixed Low-Level Waste (CH MLLW)–This waste has a dose rate equal to or less than 200 mrem/h and contains radioactivity not classified as high-level waste, spent nuclear fuel or transuranic (TRU) waste (TRU is defined as concentrations of transuranic radionuclides greater than or equal to 100nCi/g of the waste matrix). The waste is also defined as dangerous (hazardous) waste in WAC 173-303.

Remote-Handled MLLW – This waste has a dose rate greater than 200 mrem/h and contains radioactivity not classified as high-level waste, spent nuclear fuel, or TRU waste. The waste is also defined as dangerous (hazardous) waste in WAC 173-303.

Low-Level Waste Category I (LLW I)–This waste contains radioactivity not classified as high-level waste, spent nuclear fuel, or TRU waste. The waste also meets the radionuclide limits for category I waste, defined in the *Hanford Site Solid Waste Acceptance Criteria* (RH, 1998). This waste may be comprised of either contact- or remote-handled waste considered low-activity waste with very low concentrations of long-lived radionuclides. This waste is not a dangerous (hazardous) waste as defined in WAC 173-303.

Low-Level Waste Category III (LLW III)–This waste contains radioactivity not classified as high-level waste, spent nuclear fuel, or TRU waste. The waste also exceeds the radionuclide limits for category I waste and meets the category III limits, defined in the *Hanford Site Solid Waste Acceptance Criteria* (FH, 1998). This waste may be comprised of either contact- or remote-handled waste considered moderate- to high-activity waste with low to moderate concentrations of long-lived radionuclides, in stabilized form that minimizes subsidence for a period of 1,000 years. This waste is not a dangerous (hazardous) waste as defined in WAC 173-303.

Remote-Handled LLW – This waste has a dose rate greater than 200 mrem/h and contains radioactivity not classified as high-level waste, spent nuclear fuel, or TRU waste. This waste is not a dangerous (hazardous) waste as defined in WAC 173-303.

6.2.2 Landfill Phases and Dimensions

The IDF will be a single, expandable RCRA Subtitle C disposal facility that provides ultimate capacity for 1,177,110 cubic yards (900,000 cubic meters) of waste. The facility is currently anticipated to be constructed in four phases. Phase I will have two cells. Only Phase I is being permitted at this time. Each cell has a floor width of approximately 543 feet and a lined floor length of 360 feet. The total floor width of the IDF will be 1,085 feet. Side slopes of the landfill will be 3:1 (horizontal: vertical). At the south end of the Phase I cells, there will be a stormwater berm/ditch system with an infiltration area. The south side of IDF will be unlined for Phase I.

IDF will be expanded by relocation of the landfill's unlined south slope from earlier phases and installation of liner system and operations layer. When expanded to its final configuration, the floor of IDF will be 1,385 feet long, measured along its north-south axis.

6.2.2.1 Depth and Length

The landfill depth for all phases of the IDF is set to accommodate four layers of ILAW waste packages, placed on end, and each layer will be covered with 3.3 feet of clean soil. In some cases, the waste packages received for placement in the mixed and low-level waste side of IDF will have heights that vary from the ILAW package dimensions. In these cases, waste heights will vary from the four layers of ILAW waste described. The total depth, measured from the top of the operations layer to the top of the cover layer over the fourth waste layer, will be 43.4 feet. This is sized for the 7.5-foot tall ILAW package dimension. However, other waste package types can be accommodated. The waste/clean fill depth (43.2 feet) will be uniform over the entire landfill floor, due to the operations layer and the top of the landfill both sloping up 1 percent from north to south. The operations layer will be flat in the east-west direction.

6.2.3 Materials Balance

The IDF was designed to achieve near soil balance. This will minimize excess soil stockpile at the end of the life of the IDF facility and minimize the cost of hauling offsite borrow material for construction. It is important to note that the soil balance was calculated for completing IDF through all its phases and the balance included soil required for construction of the final closure cap. The closure cap design was not part of the critical systems design, completed for this project.

Having a soil balance at the completion of all phases means that at the end of Phase I, a substantial amount (approximately 991,000 cubic yards) of material will be stockpiled onsite. The project design identified potential stockpile sites that were adequate in size for the material to be stockpiled. A portion of the stockpiled material will be used as clean fill during the waste placement in the Phase I cells. However, the stockpile will be replenished during the construction of cells for each subsequent IDF phase.

A description of the resulting soil cut and fill volumes can be found in Appendix C.12.a of this Design Report.

6.2.4 Erosion Control Measures

Permanent erosion control measures (for both wind and water caused erosion) will be provided for areas disturbed by Phase I construction.

Areas that are disturbed by the construction that are outside of the Phase I excavation will be stabilized with a 6-inch-thick layer of topsoil that will be seeded with grass. The south stormwater berm/ditch, the east and west infiltration areas, and the soil stockpiles will also be stabilized with topsoil and grass.

Geotextile and quarry spalls will be placed around each end of the culverts and the stormwater pipe to provide erosion protection.

Stormwater runoff will be conveyed along the south side of the access ramp and the south side of the flat area at the bottom of the access ramp, and will be discharged to the southwest corner of the excavation infiltration area. Road surfacing will reduce the erosion potential on the ramp and flat area. To prevent erosion of the south side slope adjacent to the ramp and flat area, a strip of erosion control matting will be installed on the south side slope, immediately adjacent to the ramp and flat area. Geotextile and quarry spalls will be placed in the southwest corner of the excavation pond in order to minimize the potential of erosion due to the stormwater that will be discharged from the south edge of the flat area to the top of the infiltration area.

Erosion control matting will also be placed on the shine berm to minimize the potential for wind erosion. The erosion control matting will be a plastic matting with an estimated service life at least equal to the 10-year period that the Phase I cells are expected to operate.

To reduce wind erosion, all of the side slopes of the Phase I excavation will be stabilized with a spray-on application of a soil stabilization material. Additional applications of the soil stabilization material may have to be done annually on the areas that remain exposed.

The contractor will also be required to prepare and implement a dust control plan for the construction.

6.3 Lining System Materials

6.3.1 Liner Selection Basis

WAC 173-303-665(2)(a)(i) requires submittal of an engineering report with the permit application under WAC 173-303-806(4) stating the basis for selecting the liner(s). The report must be certified by a licensed professional engineer. The intent of Section 6.3 of the Design Report is to satisfy this requirement of the WAC 173-303, Dangerous Waste Regulations.

Specific requirements to address as the basis for liner selection include:

- The liner must be constructed of materials that have appropriate chemical properties and sufficient strength and thickness to prevent failure due to pressure gradients (including static head and external hydrogeologic forces), physical contact with the waste or leachate to which they are exposed, climatic conditions, the stress of installation, and the stress of daily operation.
- The liner must be placed on a foundation or base that is capable of providing support to the liner and is able to resist pressure gradients above and below the liner to prevent failure of the liner due to settlement, compression, or uplift.
- The liner must be installed to cover all surrounding earth likely to be in contact with waste or leachate.
- The lining system must include a LCRS immediately above the liner that is designed, constructed, maintained, and operated to collect and remove leachate from the landfill. Design and operating conditions will ensure that the leachate depth over the liner does not exceed 1 foot. The LCRS shall be:
- Constructed of materials that are chemically resistant to the waste managed in the landfill and the leachate expected to be generated, and of sufficient strength and thickness to prevent failure under the pressures exerted by overlying wastes, waste cover materials, and any equipment used at the landfill.
- Designed and operated to function without clogging through the scheduled closure of the landfill.
- Engineering analyses were presented in Section 5 that address the above requirements for basis of lining selection. Of particular note is Section 5.6 that addresses lining system/leachate compatibility for all components of the lining system. Compatibility of the lining system components with the chemical and radiological constituents of the expected leachate is a critical aspect of the liner selection basis.
- Based on results of the engineering analyses presented in Section 5, the following liner sections are proposed for the IDF bottom (floor) and side slope lining systems. Section 6.3.2 provides a detailed discussion of the liner materials for the barrier components of the lining system, and Section 6.3.3 provides a detailed discussion of the liner materials for the drainage and protection components of the lining system.

Drawing H-2-830838 (Detail 1) shows the bottom liner section, consisting of the following components, from top to bottom:

- A 3-foot-thick operations layer
- A separation geotextile (polypropylene)
- A 1-foot-thick leachate collection drain gravel layer
- A minimum 12 oz/square yard cushion geotextile (polypropylene)
- A 60-mil textured primary HDPE geomembrane

- An internally-reinforced GCL
- A CDN drainage layer for the LDS
- A 60-mil textured secondary HDPE geomembrane
- A 3-foot-thick low-permeability compacted admix (soil-bentonite) liner

Drawing H-2-830838 (Detail 2) shows the side slope liner section, consisting of the following components, from top to bottom:

- A 3-foot-thick operations layer
- A CDN drainage layer for the LCRS
- A 60-mil textured primary HDPE geomembrane
- A CDN drainage layer for the LDS
- A 60-mil textured secondary HDPE geomembrane
- A 3-ft-thick low-permeability admix liner

6.3.2 Liner Materials – Barrier Components

6.3.2.1 Geomembranes

WAC 173-303-665(2)(h)(i) requires that the IDF lining system have both a primary and secondary geomembrane. The geomembrane for the IDF will serve as leachate barrier and as a flow surface routing leachate to the LCRS sump (for the primary geomembrane) or LDS sump (for the secondary geomembrane).

HDPE has been selected as the geomembrane liner material because it is generally acknowledged to have the highest chemical resistance of commercially-available liner materials, has been widely used at similar facilities, and has a high level of acceptance by regulatory agencies. Details of HDPE geomembrane compatibility with expected leachate is discussed in Section 5.6.

A nominal thickness of 60-mil has been selected for the HDPE geomembrane. A nominal thickness of 60-mil results in a minimal allowable thickness of 54-mil, as indicated in the technical specifications. Thus, 60-mil nominal thickness is the minimum required to achieve the 50-mil minimum thickness specified by Ecology guidance. Textured (roughened) geomembrane will be used to maximize shear strength along adjacent interfaces and to reduce the potential for sliding of the liner system. Analyses of the various stresses that the geomembrane is designed to withstand under construction and operational loads are presented in Section 5.5. Required material properties as a result of these analyses are included in the technical specifications.

Details of required HDPE geomembrane properties are provided in the technical specifications (see Section 02661).

6.3.2.2 GCL

A GCL will only be included in the primary bottom lining system. For the bottom lining system, both the primary and secondary liners will be a composite (geomembrane over admix liner or GCL) system. The addition of a GCL in the primary lining system will provide an extra measure of protection, exceeding the requirements of WAC 173-303-665(2)(h)(i) for a single geomembrane for the primary liner and composite for the secondary only. This will provide an extra measure of protection on the bottom flatter slopes of the IDF, where higher leachate head levels are more likely. A GCL will not be included on the side slope lining system. The 3H:1V side slopes for the IDF will result in little or no leachate head expected on the side slope lining system, thus eliminating the need for a lining system design that exceeds the WAC requirements.

Commercially-available reinforced GCL products consist of bentonite sandwiched between a woven and non-woven geotextile that are then needle-punched together. Other combinations of upper and lower, woven and non-woven geotextiles can also be manufactured and specified.

For the IDF lining system, a needle-punched, reinforced GCL with non-woven geotextiles on both sides was selected. This type GCL product was selected primarily because of the tensile strength requirements required for landfill global stability (Section 5.1.3). The tighter weave non-woven geotextile minimizes the amount of bentonite that migrates to the interface with the geomembrane, thus minimizing the potential to create a slip surface.

Details of required GCL properties are provided in the technical specifications (see Section 02667).

6.3.2.3 Admix Liner

Details of the admix design test program are provided in Sections 4.2.2 and 5.4.1. Placement and testing requirements are described in Section 5.4.2.

The admix liner will have a minimum 3-foot thick compacted soil/bentonite admixture and will be located immediately beneath the secondary HDPE geomembrane, as required by WAC 173-303-665(2)(h)(i)(B). The admix liner typically will consist of base soil mixed with a nominal 12 percent sodium bentonite, by dry weight. Mixing and processing of the base soil/bentonite admixture is required to be performed under carefully controlled conditions, using a pugmill operation.

The base soil for the admix liner will consist of natural soil, derived from the dune sand borrow area to the south of the Phase I cell (as shown on Drawing H-2-830828) or from within Phase I cell excavations. Based on the results of the limited field exploration for near surface base soil samples (discussed in Sections 4.1 and 4.2), base soil from either source will not be excavated below a depth of 5 feet bgs (after stripping) without further evaluation of the material suitability.

Base soils excavated from the dune sand borrow area or site excavation will meet the following requirements:

- The base soil will be free of roots, woody vegetation, frozen material, rubbish, and other deleterious material.
- Rocks greater than 1 inch in dimension will not comprise more than 2 percent by weight of the base soil.
- Base soil will have 20 percent minimum passing a No. 200 U.S. sieve.
- The in-place hydraulic conductivity of the admix liner will be 10-7 centimeters per second or less, consistent with WAC requirements for secondary soil liners. The upper surface of the admix liner will be trimmed to the design grades and tolerances. The surface will be rolled with a smooth steel-drum roller to remove all ridges and irregularities. The result will be a smooth, uniform surface on which to place the overlying geomembrane liner.
- Before production installation of the admix liner, a full-scale test pad of the admix liner will be conducted for both the bottom floor (horizontal) and side slope areas of the IDF. Details of the test pads are provided in the technical specifications (see Section 02666) and the IDF Construction QA Plan. The primary purpose of the test pad(s) will be to verify that the specified soil density, moisture content, and hydraulic conductivity values will be achieved consistently, using proposed compaction equipment and procedures. In-place density will be measured using both the nuclear gauge (ASTM D2922) and rubber balloon (ASTM D2167) or sand cone (ASTM D1556) methods. In-place hydraulic conductivity will be determined from a two-stage borehole permeameter (ASTM D6391). Admix liner hydraulic conductivity will be estimated from thin-wall tube samples (ASTM D1587) obtained from the test fill and tested in the laboratory (ASTM D5084). During construction, field density (e.g., ASTM D2922, D2167, and/or D1556) and moisture content (ASTM D2216) will be measured periodically. Thin-wall tube samples (ASTM D1587) will be taken at regular intervals and will be tested for hydraulic conductivity (ASTM D5084). Additional details of Construction QA testing and acceptance during admix liner test pad and production installation is provided in the IDF Construction QA Plan.

Details of required admix liner properties and placement requirements are provided in the technical specifications (see Section 02666).

6.3.3 Liner Materials–Drainage and Protection Components

6.3.3.1 Geotextiles

Two types and weights of geotextiles will be used in the IDF project. The separation (Type 1) geotextile has a nominal weight of 6 ounce/square yard and was selected based on the ability of the geotextile to retain the soil and to prevent the soil from entering the LCRS drain gravel. Required AOS and permittivity were determined based on filter, fines retention, and clogging potential criteria. The waste disposed in the IDF is expected to contain a minimal amount of organic material, and consequently, biologic clogging is not expected to be a problem.

The cushion (Type 2) geotextile has a nominal weight of 12 ounce/square yard and was selected based on providing the required cushion protection for geomembrane on the landfill bottom (floor). The drain gravel will have the potential to produce localized stress on the geomembrane liner during gravel placement with construction equipment and under the maximum static pressure from landfill contents at full waste height with final cover. A puncture analysis was performed to select a sufficiently thick geotextile to protect the liner. This analysis included the maximum load from landfill contents and final cover, expected construction vehicle ground pressures, and maximum drain gravel particle size listed in the technical specifications.

Both types of geotextiles are specified as non-woven needle-punched and made from polypropylene material. This material was selected because of its higher chemical resistance to the expected leachate (Golder Associates, 1991a).

Details of required geotextile properties are provided in the technical specifications (see Section 02371).

6.3.3.2 CDN

The CDN is a drainage geocomposite consisting of a HDPE geonet core with a layer of non-woven polypropylene geotextile thermally bonded to each side. The CDN selected for the IDF lining system has two drainage related functions. On the side slopes, it will function as the LCRS. A CDN is selected for the LCRS on the side slope to avoid construction stability problems associated with placement of clean granular material on slopes, thereby minimizing the potential for damaging the underlying liner system. Localized placement of drain gravel is required on side slopes (as shown on Drawing H-2-830848, Section C), to provide adequate backfill and bedding for leachate collection riser piping. On the side slope and bottom lining system, the CDN will function as the LDS.

Analyses were performed to evaluate the geotextile puncture requirements for the LCRS CDN on the side slope and the transmissivity requirements for both the LCRS and LDS CDN. These analyses and discussion are presented in Section 5.7.

The analyses for CDN geotextile puncture resistance determined that the specified geotextile is adequate for resistance to puncture from overlying operations layer, under the maximum static pressure from landfill contents.

The analyses for allowable transmissivity with applied reduction factors for intrusion, creep, and chemical and biological clogging determined that a higher flow, thicker (250 mil minimum) CDN is required, due to the reduction of flow under the high normal loads in the final filling configuration.

Details of required CDN properties are provided in the technical specifications (see Section 02373).

6.3.3.3 Drain Gravel

The LCRS for the bottom liner will be located below the operations layer and will provide a flow path for the leachate flowing into the LCRS sump and sump trough. Between the operations layer and the underlying drain gravel, a geotextile layer will function as a filter separation geotextile (as discussed in Section 6.3.3.1).

The separation geotextile will prevent migration of fine soil and clogging of the drain gravel. The gravel will be a minimum 1-foot thick layer of washed, rounded to subrounded stone, with a hydraulic conductivity of at least 10-2 cm/sec, as required by WAC 173-303-665(2)(h)(iii)(B). In addition, a slotted HDPE leachate collection piping will be placed within the drain gravel to accelerate leachate transport into the LCRS sump during high precipitation events. Slots on the leachate collection piping are sized to be compatible with the drain gravel gradation and particle sizes. Details of the leachate collection piping design are provided in Section 6.4.1.

Based on review of expected subsurface conditions for the IDF, it is not likely that material meeting drain gravel is available on or near the site. Thus, drain gravel will have to be an imported material. The technical specifications require that drain gravel meet the requirements of WSDOT Standard Specification 9-03.12(4) for gradation. The technical specifications also require a performance specification for a hydraulic conductivity greater or equal to 10-1 cm/sec.

As discussed in Section 5.7.3, the minimum estimated hydraulic conductivity for the drain gravel exceeds the required (by WAC regulations) hydraulic conductivity of 10-2 cm/sec by a factor or 100 to 1,000, and the performance specification hydraulic conductivity of 10-1 cm/sec by a factor of 10 to 100. This allows for uncertainty in the empirical formulas used to predict hydraulic conductivity, and the potential for long-term reduction in hydraulic conductivity in the drain gravel, if fines from waste filling and the operations layer migrate into this layer over time.

Details of required drain gravel material properties are provided in the technical specifications (see Section 02315).

6.3.3.4 Operations Layer

The purpose of the operations layer will be to protect the underlying lining system components from damage by equipment and waste canisters during IDF construction and operation. This layer also will protect the admix liner from freeze/thaw damage and desiccation cracking. This is especially the case on the side slopes, expected to be exposed (prior to waste placement) for longer duration than the bottom (floor) of the IDF cell.

The operations layer material typically will consist of onsite granular soil from the IDF Phase I excavation. The excavated material is expected to be a fine-grained sand to silty sand with traces of gravel. The technical specifications require the material to have a maximum particle size limit of two inches or less, and fines will be limited to maximum 25 percent fines (percent passing the U.S. No. 200 sieve). Based on review of expected subsurface conditions for the IDF excavation, the majority of soil excavated from the IDF Phase I excavation is expected to be suitable for use as operations layer without processing. As discussed in Section 4, additional geotechnical exploration within the IDF Phase I limits are recommended prior to construction to verify these findings.

Details of required operations layer material properties are provided in the technical specifications (see Section 02315).

6.4 Leachate Collection System

The leachate collection system for each cell in Phase I will consist of lateral flow media built into the cell's bottom and side slope liner system, a leachate collection pipe in the center of the cell, a sump at the north end of the cell where all leachate drains, pumps and leachate transfer piping to convey leachate out of the cell, and a network of piping and storage tanks for storing the leachate for later transfer to tanker trucks for offsite disposal. Below the bottom liner and under the LCRS sump area will be an LDS sump, pump, and associated piping. All components for Phase I of the leachate collection system are designed and configured for eventual full development of the IDF through Phase IV.

The type and configuration of the leachate collection system described below has been used successfully at other disposal facilities, and a very similar facility was recently (2002) implemented at the INEEL site near Idaho Falls, Idaho. This ICDF will accept waste with radioactive characteristics and is located in a region with dry weather conditions, similar to Hanford.

6.4.1 Leachate Collection Piping

6.4.1.1 Description

Lateral drainage media (drain gravel in the bottom liner section and CDN in the side slope section of each cell) will convey leachate by gravity to the leachate collection piping and to the LCRS sump area. The leachate collection piping system in both cells will have one 12-inch diameter HDPE slotted pipe running the length of the cell centerline from south to north. This main collector pipe will be sloped at 1 percent and will convey leachate from the south edge of the cell to the LCRS sump at the north end, where the bottom liner will intersect the side slope liner. The main collection pipe will change to solid pipe at the bottom of the side slope, continue up the side slope, and terminate at a cleanout, located just south of the crest pad building. Leachate in the sump will be collected through perforated pipes for the LCRS low flow and high flow pumps, which will be 12-inch and 18-inch HDPE slotted pipe, respectively. The riser pipes will protect the pumps and separate them from the surrounding drain gravel, allow removal and re-insertion of the pumps for maintenance, and provide a high inflow-rate screen for leachate to supply the pumps. A small-diameter pipe (housing a transducer to control the on/off levels for the pumps) will run from the LCRS sump up the side slope to the crest pad building.

The slotted portion of the riser pipes will extend from the toe of the side slope to the end of the LCRS sump area. The transducer pipe will also be slotted but for a shorter distance in the LCRS sump, whereupon it will be solid for the remaining distance to the crest pad building. A solid HDPE pipe (of the same diameter as the slotted portion of the pump riser pipes) will extend from the intersection of the side slope and bottom liner to the top of the shine berm where the pipes enter the crest pad building.

Pipe cleanouts will be provided at both ends of the main collection pipe in the center of each cell. The cleanout at the north end of Phase I, near the crest pad building, will be permanently available throughout the life of the IDF to allow access for cleaning and/or video inspection. The cleanout at the south end of the cell will also be available for cleaning and access, but only during the operation of Phase I. It will be removed and the Phase II collection pipe will be butt-fused to the pipe as the Phase II cell is brought online. Ultimately, a permanent cleanout will be installed at the south end of Phase IV, to allow cleaning and inspection of half of the collection pipe, with the other half being accessed by the permanent cleanout located at the crest pad building on the north side of Phase I.

Access to the riser pipes for cleanout or inspection, in the unlikely event this is needed, will be through the access points used for removal and re-insertion of the pumps within the crest pad building.

6.4.1.2 Design Considerations

The material chosen for piping within the Phase I lined area was HDPE , made of resin meeting the requirements of ASTM D3350 for PE 3408 material, with a cell classification of 345434C or higher. Design calculations were based on this material and pipe type, which is routinely used for leachate collection and disposal facilities and other applications. The pipe material is well suited for use in disposal facilities because of its high strength, high resistance to degradation from leachate constituents, and superior characteristics compared to all other readily available pipe materials. HDPE compatibility with leachate and the presence of radioactivity in the waste overlying the pipe were evaluated and discussed previously in Section 5.6.

The diameter of the riser pipes was chosen to provide ample clearance for the pumps to be inserted and removed on a routine basis, and specifically so that the pumps will have sufficient clearance when traveling through the angle points at the intersection of the bottom liner and side slope, and clearance at the radius transition from the side slope to the crest pad building. The pumps (described in Section 6.4.3) are specifically designed for this type of leachate collection system, where the riser pipes allow insertion of pumps down a side slope and into a sump area.

Lateral drainage media in the bottom liner and side slope liner, and the leachate collection piping system were chosen and configured to meet the regulatory requirement of no more than 12 inches of leachate head buildup over the sump area of the bottom liner as a result of a 25-year, 24-hour storm event.

The slots in the slotted pipe were designed to both be compatible with the granular material in the drain gravel and to allow a high rate of flow from the surrounding lateral drainage layers into the pipe. Slots were sized at 0.128 inches wide, with five rows of slots spaced equidistant around the perimeter of the pipe, and eleven slots per foot of pipe.

The thickness of the pipes expressed as the SDR (standard dimension ratio) was chosen to resist the highest estimated load for the IDF in its final configuration, including final cover and equipment loading (internal pressure was not a factor since the pipe will convey flow by gravity, and under the expected flow rates the pipes will only be partially full). A SDR of 17 was chosen for all piping to handle the maximum estimated load. In addition, a blanket of manufactured drain gravel will be placed around and to the sides of all collection piping and compacted to a firm, unyielding condition consistent with the soil modulus values used in the pipe loading calculations.

All piping will be butt-fused for maximum strength, and all fittings, whether available molded from the manufacturer or fabricated, will have the same or higher pressure rating than the pipe. During construction, piping will be butt-fused by certified technicians, using welding equipment approved by the manufacturer. All solid pipe will be pressure tested, even though the collection piping will see little or no internal pressure during gravity conveyance of leachate.

6.4.2 Leachate Transfer Piping

6.4.2.1 Description

At each cell, the leachate transfer piping will begin with the piping from the pumps in the LCRS and LDS sumps to the crest pad building. From the crest pad building, transfer piping will connect the leachate transfer building, storage tank, and tanker truck load facility. All underground transfer piping outside the Phase I liner limits will be double contained, that is the pressure pipe conveying leachate between various facilities will be contained in an outer pipe. The pressure pipe in the center of the double containment piping will be termed carrier pipe, while the outer pipe will be termed containment pipe. In the event of a leak in the carrier pipe, the containment pipe or leak detection pipes draining the containment pipes will convey the leakage to a combined sump facility for detection, sampling, and transfer. Any accumulation of leachate in the combined sump will be pumped through a transfer pipe to the storage tank. Piping within the crest pad building, transfer building, truck load facility, and combined sump, will not be double contained because the buildings or facilities will provide secondary containment and have sumps present to remove any leachate that accumulates as a result of leaking pipes or appurtenances. Leak detection pipes draining containment pipes and the leak detection pipe from the storage tank will be single pipes because they only will convey leakage and will not function as transfer piping (required to have double containment).

The transfer piping system also will include valves, fittings, flow meters, and other appurtenances necessary for operational functions for systems described in Sections 6.4.3, 6.4.4, and 6.4.5.

6.4.2.2 Design Considerations

All transfer piping outside of buildings will meet the same requirements as the HDPE pipe chosen for the leachate collection piping (described in Section 6.4.1). Single pipe and containment pipe exposed to earth and traffic loading will be SDR 17, while the carrier pipe, that will not be exposed to earth or traffic loading, will be SDR 21, with a pressure rating of 80 psi and a safety factor of 2 for the highest expected operating pressure in the system (SDR 17 piping has a pressure rating of 100 psi). All piping will be butt-fused except for the transfer piping from the LCRS and LDS sump pumps. This pipe will be HDPE, with quick release fittings to allow removal of the pumps from the sumps. Fittings will be pressure rated and re-useable. As the pumps are withdrawn from the sumps and moved up the riser pipes, each joint in the pipe will be unhinged to allow the pipe to be removed in 8-foot sections.

Piping inside buildings will be PVC, schedule 80, with solvent welded fittings. This pipe and classification is rated for higher pressure than required with a factor of safety of 8. PVC was chosen for application inside buildings because of its relative ease of fabrication with the solvent weld joint system.

Flange connections will be used between pumps and piping; valves and other appurtenances and piping; and joints between PVC and HDPE piping. Appurtenances will include air release valves to allow purging of any air trapped in the piping system, magnetic flow meters for measuring flow to the tanker truck load output and to and from the leachate storage tanks, and valves for flow control and diversion of flow between the various facilities. The flow control scheme and control logic for the transfer piping system are described in Section 6.4.5.

6.4.3 Leachate System Pumps

Three submersible leachate pumps will be required for each cell. For convenience and operational versatility, roller-mounted pumps were selected for all leachate removal facilities. The submersible pumps are standard stainless steel well pumps that have been installed within a screened stainless steel cylinder fitted with rollers. The configuration will allow the pumps to be installed from the crest pad building within riser piping that follows the slope of the landfill until the riser piping bends horizontally to terminate within the cell sump at the toe of slope. This type of pump can be lowered into the leachate sump through the riser pipe and removed as needed, using a winch mounted within the crest pad building. Each pump will have its foot valve removed to prevent freezing or retaining of the leachate in the pump discharge piping. Advantages of this type of pump include easy access for maintenance and inspection, no power equipment required to remove/install, and its small size will lend itself to being inserted within a curved riser pipe and evacuating nearly all of the leachate within the cell sump. Each pump will have the capability to pump either to the storage tank or truck loading station.

6.4.3.1 LCRS Pumps

Two of the three submersible pumps will be installed within the LCRS sump area of each cell above the primary liner. These pumps are required to maintain less than 12 inches of hydraulic head above the primary liner, per regulatory requirements. The pumps will be installed in a 6-inch depression within the LCRS, in order to minimize the area of permanent leachate storage at pump shutoff and allow full pump operation through the 12-inch maximum liner head zone over the primary liner. Only in the localized area of the LCRS sump depression will a maximum leachate head of 18 inches cover the primary liner. The leachate head over the primary liner will be maintained at or below 12 inches in the main sump area and throughout the landfill. One low-flow pump is required for typical pumping of leachate; a high-flow pump is necessary in the event that a large storm (24-hour, 25-year storm event) exceeds the capacity of the low-flow pump.

The selection of the low-flow pump was based on the average leachate flow from the landfill, determined in the leachate production analysis (Section 5.8.1). The analysis indicated that the maximum leachate flow, based on monthly data, is approximately 13 gpm. The hydraulics of the low-flow pump was modeled and a pump was selected, based on the hydraulic characteristics of the piping system and the required flow rate, determined in the leachate system hydraulics analysis (Section 5.9.2.1). An EPG Companies, Inc. (EPG) model WSD 3-3 (or equal) with a 0.5-horsepower motor was selected for the LCRS low-flow pump.

The selection of the high-flow pump was based on the 24-hour, 25-year storm event, determined in the leachate production analysis (Section 5.8.1). The analysis indicated that the high-flow pump capacity necessary to remove the leachate per regulatory guidelines is approximately 160 gpm. The hydraulics of the high-flow pump was modeled and a pump was selected, based on the hydraulic characteristics of the piping system and the required flow rate, determined in the leachate system hydraulics analysis (Section 5.9.2.1). An EPG model WSD 30-3 (or equal) with a 7.5-horsepower motor was selected for the LCRS high-flow pump.

6.4.3.2 LDS Pump

The third submersible pump will be installed within each cell in the LDS sump, under the primary liner and above the secondary liner.

This pump will detect and recover leachate that has leaked through the primary liner by pumping the leachate to the crest pad building. This pump was sized for low leachate generation flows.

The hydraulics of the LDS pump were modeled and a pump was selected that can produce 4 gpm, based on the hydraulic characteristics of the piping system and the required flow rate, identified in the leachate system hydraulics analysis (Section 5.9.2.1). An EPG model 1.5-3 (or equal) with a 0.5-horsepower motor was selected for the LDS pump.

6.4.3.3 Crest Pad Building Sump Pump

The sump pump within the crest pad building will be a submersible floor sump, activated by float switches within the floor sump. The function of the sump pump is to remove leachate that accumulates in the crest pad building as a result of unexpected spills or pipe leaks. The pump discharges water to the leachate storage tank via the crest pad building discharge piping.

The hydraulics of the sump pump was modeled and a pump was specified, based on the hydraulic characteristics of the piping system and the required flow rate identified in the leachate system hydraulics analysis (Section 5.9.2.1).

6.4.3.4 Leachate Transfer Pump

The leachate storage tank will be drained by using the leachate transfer pump, located in the leachate transfer building. The pump was sized to deliver a capacity of 250 gpm to the truck loading station, where it will discharge into a tanker truck. The typical volume allowed in a tanker truck is 7,000 gallons, corresponding to a loading time of approximately 30 minutes.

The hydraulics of the leachate transfer pump was modeled and a pump was selected, based on the hydraulic characteristics of the piping system and the required flow rate, identified in the leachate system hydraulics analysis (Section 5.9.2.1). A standard horizontal centrifugal pump, Paco model 30707 (or equal) with a 3-horsepower motor was selected for the leachate transfer pump.

6.4.3.5 Combined Sump Pump

The combined sump will be a 76-inch-diameter HDPE manhole with a 42 inch diameter HDPE manhole placed inside. The outer manhole will have a height of approximately 8 feet, and the inner manhole height will be approximately 6 feet. The secondary containment portion of all the buried HDPE pipelines, leachate tank, and leachate transfer building floor sump will drain to the annular space (leak detection chamber) between the two manholes. The leak detection chamber will include instrumentation to detect leachate and alarm accordingly. The sumps installed within the truck loading slab typically will collect precipitation that drains off the slab. The precipitation will be conveyed directly to the inner manhole of the combined sump, where the combined sump pump will be located. The combined sump pump then will pump the precipitation to the leachate storage tank.

The combined sump pump was conservatively sized for a capacity of 250 gpm. This large capacity was chosen based on an off-normal event that assumed the tanker truck was overtopped during leachate transfer activities, resulting in 250 gpm flowing into the inner sump. Another off-normal event considered was the remote possibility that the leachate tank primary liner failed catastrophically. This flow of leachate could eventually inundate the leak detection chamber and overflow into the inner manhole.

The hydraulics of the combined sump pump was modeled and a pump was selected based on the hydraulic characteristics of the piping system and the required flow rate, identified in the leachate system hydraulics analysis (Section 5.9.2.1). A Hydromatic model SB3S (or equal) with a 3-horsepower motor was selected for the combined sump pump.

6.4.4 Leachate Temporary Storage Tank

6.4.4.1 Tank Volume

A leachate temporary storage tank is required for each cell. The working capacity of each tank is 375,000 gallons that include a 1.5 safety factor. This volume is based on the results of the leachate production analysis (Section 5.8.1) and the leachate collection storage analyses (Section 5.9.2.4). The storage tank capacity is dependent on the net volume of leachate accumulation in the tank from flow into and out of the tank. The flow out of the tank via the leachate transfer pump is based on several assumptions, described in Section 5.9.2.4. Actual leachate transfer operations will affect the tank volume safety factor.

6.4.4.2 Tank Design

A bolted, corrugated steel tank, approximately 100 feet in diameter with a side wall height of 8 feet 2 inches, was selected for use as the leachate temporary storage tank. The tank will include a dual containment liner system that will act as the floor of the tank and will be bolted to the top of the tank side wall. The tank will be open-topped with a floating geomembrane cover to keep precipitation, debris, and wildlife from contacting the leachate.

The tank side wall will be bolted to a 1.5-foot thick, 4-foot-deep concrete ringwall to resist hydrostatic pressure of the leachate water. In addition, the top edge of the tank ringwall will include angle bracing, bolted around the tank perimeter to provide rigidity in the side wall to resist wind loads on the exterior of the tank. The maximum operating level of the tank is approximately 6 feet 2 inches; however, the tank is designed for a maximum water level of 8 feet 2 inches.

The inlet piping for the tank will be through the side wall of the tank. The inlets will all be located near the top of the tank, above the maximum leachate water operating level. This is to ensure that a siphon cannot develop in the inlet piping. Check valves will be installed throughout the system; however, if piping between the check valve and the tank leaked into the secondary containment system, there would not be an easy method of stopping the flow if the pipe was below the water surface of the tank.

The outlet pipe for the tank will be through the side wall, near the bottom of the tank. This method was chosen to provide a flooded suction for the leachate transfer pump that will provide added protection against pump damage.

6.4.4.3 Tank Liners

The tank liners will be constructed with an XR-5 geomembrane. XR-5 is a proprietary geomembrane manufactured by Seaman Corporation. XR-5 is the preferred liner of several tank manufacturers due to its higher strength properties and lower thermal expansion coefficient, as compared to HDPE geomembrane. As such, it is more readily constructible in the tank configuration, and it does not expand and contract as much as HDPE, so it's operating performance over the temperature range at Hanford should be improved. For the exposed condition at the IDF tanks, this is an important consideration. HDPE was considered for use as the tank liner system, but its high coefficient of expansion will not lend itself to the temperature extremes that the liner system will be subjected to and also it is not reinforced like the XR-5. The expansion and contraction of an HDPE liner exposed to the environment could put undue strain at the inlet and outlet connections as well as at the leak detection connection that could result in liner leakage.

Chemical compatibility of leachate with the liner system is also a consideration for liner material selection for the leachate storage tanks. As discussed in Section 5.6.3.1, compatibility testing on HDPE geomembrane was performed with synthetic leachate for the W-025 landfill with no evidence of geomembrane deterioration. With regard to leachate compatibility, XR-5 is comparable to HDPE in terms of compatibility with typical leachate constituents. The geomembrane manufacturer requires immersion testing for conclusive compatibility determination. Testing of this type has not been performed, but the manufacturer is confident that immersion testing results will be acceptable since XR-5 is generally comparable to HDPE.

To address the issue of chemical and radiation resistance for XR-5 with anticipated leachate constituents, an immersion test program is included in the technical specifications for the tank liner. Details are provided in Section 13205 of the technical specifications. This immersion testing program requires the construction general contractor to submit tank liner sample coupons to the design engineer for immersion testing, as part of the construction submittal process and certification of the tank liner.

In addition, it should be noted that leachate compatibility is not as critical an issue for the tank system as compared to the landfill liner system. The leachate tank liner system will be subject to continuous monitoring through the tanks' LDS, as is the landfill liner system. The difference is that the tank liners will be subject to routine maintenance and inspection that will be developed around liner warranty, performance observation, and manufacturer's requirements. Operation and maintenance procedures for the tank will be established that require that the tanks be drained, sediment removed, and the liner inspected for holes and seam integrity. Since liner performance guarantees are required in the technical specifications for the tank manufacturer for three years following installation, it is likely that the inspection program would be initially set up around this time frame and gradually be increased over the life cycle of the tank. Replacement of the leachate tank liner system is anticipated periodically throughout the life cycle of the landfill.

The tank lining system is a double-lined system. The primary and secondary tank liners will include a LDS beneath the primary tank liner. The LDS consists of a HDPE drainage net with a geotextile material, laminated to the drainage net that cushions the XR-5 liner. A geotextile material will also be used between the secondary liner and the inside face of the tank shell to create a cushion for the XR-5 against the tank shell and tank shell bolt heads. The bolt heads are also recessed for further liner protection.

6.4.4.4 Tank Leak Containment System

The HDPE drainage net between the primary and secondary liner will allow leachate that leaks through the primary liner to drain to the center of the tank. At the center of the tank under the secondary liner will be a depression in the underlying granular backfill that will form a shallow sump. The leak detection pipe will connect to the secondary liner at this sump location and convey leaking leachate to the leak detection chamber of the combined sump.

The tank inlet and outlet penetrations will be areas susceptible to leaks as a result of penetrations through the primary liner. Additional robust methods for sealing these locations were added over and above the typical manufacturer recommendations in an effort to make sure that these will not be points of leakage.

6.4.5 Pump Controls and System Instrumentation

The process and instrumentation diagrams for Cell 1 and Cell 2 are shown on Drawing H-2-830854, sheets 1 through 4. Detailed information regarding the instrumentation and control system, equipment listing, instrument listing, and loop descriptions can be found in the technical specifications, Section 13401 (Process Instrumentation and Control System).

6.4.5.1 Crest Pad Building

The leachate pumps within the landfill will be automatically controlled, based on leachate level set points within the cell sump. The level transducer that controls the LCRS pumps will be inserted into the sump via a slope riser pipe. The level transducer that controls the LDS pump is integral to the LDS pump. Leachate pumped by the leachate pumps will be monitored by a flow-indicating totalizer within the crest pad building. Controls will be in place to stop automatically the leachate pumps operation if alarm conditions are present for the leachate storage tank high-high level, leak alarm in the crest pad building sump, or a leak alarm in the combined sump.

The crest pad building sump pump will be automatically controlled by float switches within the building floor sump. In addition, a leak detection switch will be installed in the floor sump that will be capable of detecting small quantities of water in the sump before the float switches. This feature will add an extra level of conservatism to make sure unexpected spills are identified and controlled immediately.

Controls will be in place to stop automatically the crest pad building sump pump operation if alarm conditions are present for the leachate storage tank high-high level or for a leak alarm in the combined sump.

6.4.5.2 Leachate Transfer Building

The leachate transfer pump will be manually controlled except for automatic shut-off during specific alarm events. Controls will be in place to stop automatically the transfer pump operation if alarm conditions are present for the leachate storage tank low-low level or for leak alarm in the combined sump. Additional instrumentation (associated with the leachate transfer pump) will include a flow meter (measuring rate and total volume) and transmitter on the discharge of the leachate transfer pump. In addition, a local totalizer will be in the leachate transfer building to know exactly how much water is being transferred to the tanker truck. This totalizer will include a reset function to allow the total to be reset to zero, prior to every truck loading event.

6.4.5.3 Leachate Storage Tank

Instrumentation within the leachate storage tank will be contained within two vertical stilling wells that will penetrate through openings in the floating cover. The stilling wells will be small diameter pipe with perforations near the bottom that will allow the leachate within the stilling well to rise and fall with the level of the leachate in the tank. Analog instrumentation within one stilling well will provide a signal to the control system for alarm interlocks and constant monitoring of tank level. The second stilling well will contain discrete instrumentation for high-high and low-low alarm set point trips. The discrete instrumentation will provide conservatism in the off chance that the analog signal malfunctions, allowing the leachate level to reach extreme high or low levels.

6.4.5.4 Combined Sump

The combined sump pump will be automatically controlled by float switches within the inner manhole of the combined sump. Controls will also be in place to stop automatically the combined sump pump operation if alarm conditions are present for the leachate storage tank high-high level. A leak detection switch also will be installed within the leak detection chamber that will be capable of detecting a small quantity of water. The leak detection switch will provide a signal to the control system that automatically will shut down all the cell pumps except the combined sump pump. The pumps will be shut down because any one of the pipelines associated with the pumps could be leaking into the leak detection chamber. Operations will then need to determine which secondary containment pipeline supplied the water that drained into the leak detection chamber.

6.4.6 Process Instrument Control System (PICS)

6.4.6.1 Introduction

This section provides a summary of the PICS design and construction elements of the project, providing introduction and reference to the project layout and key design components for the following IDF facilities:

- IDF leachate collection and handling crest pad facilities (two each)
- IDF leachate storage tank and leachate transfer facilities (two each)
- IDF truck loading facilities (two each)

The PICS design identifies, specifies and integrates PICS components to automatically monitor and control IDF process control equipment and facilities including:

- LCRS
- LDS
- Crest pad and leachate transfer building environmental controls
- Leachate storage tank system

- Leachate transfer and truck loading system
- Combined sump system
- Secondary containment LDS

6.4.6.2 Key Design Components (Elements)

PICS design and construction elements of the project incorporate the following key PICS design components for each IDF facility:

- Instrumentation for continuous analog process monitoring
- Instrumentation for discrete process monitoring
- Instruments and programmed safety interlocks and alarming
- Programmable logic controller (PLC) system
- Operator Interface Unit (OIU)
- Communication Local Area Network (LAN)
- PICS application software
- Main and local control panels
- Uninterruptible power supply

6.4.6.3 Open Items

The IDF Phase I Critical Systems design documents do not identify the following items:

- Identification of communication LAN from IDF control panels to central supervisory control and data acquisition (SCADA)
- Extension of communication LAN from IDF control panels to central SCADA

These items are scheduled to be addressed during the IDF Phase I Non-Critical design of the project. As such, the following assumptions were made in order to complete IDF Phase I Critical Systems design:

- Assume 10/100 megabits per second (MBPS) Ethernet communication LAN from IDF control panels to central SCADA
- Assume fiber-optic multi-mode extension of communication LAN from IDF control panels to central SCADA

6.4.6.4 PICS Architectures

The PICS design identifies various architectures, designed to enable operators to locally and remotely interface and change program settings by the use of an Ethernet LAN. This document does not identify components and architectures to be provided and configured under the IDF Phase I Non-Critical design in order for personnel remote monitor and control processes over the LAN.

6.4.6.5 PICS Instrumentation Architecture

The PICS design identifies instrumentation architecture that consists of single variable level (submersible pressure), flow, and temperature elements and transmitters that provide continuous process data to PICS PLC and OIU architectures. Process signals from each instrument are monitored for the purpose of controlling, displaying, recording, and alarming all process data. PICS instrumentation will be wired directly into PLC input modules (i.e., Allen-Bradley 1746 I/O modules).

6.4.6.6 Instrumentation

The PICS design identifies all set-point adjustments as being programmed into the PLC via the OIU architecture. Field instruments incorporate the following signal types:

- Analog signals, current type: 4-20 mA dc signals conforming to ISA S50.1.1
- Transmitters type: 2-wire and 4-wire

- Transmitter load resistance capacity: Class L
- Fully isolated transmitters and receivers
- Discrete signals, voltage type: 24 VDC

6.4.6.7 Analog Instrumentation

The PICS design identifies flow analog instrumentation, consisting of electromagnetic flow elements and integral transmitters that will enable operators to monitor pump discharge flow for the following processes:

- Landfill LCRS pump discharge flow
- Landfill LDS pump discharge flow
- Leachate transfer truck loading station discharge flow

The PICS design identifies level analog instrumentation, consisting of submersible pressure transmitters that will enable operators to monitor liquid levels for the following:

- Landfill LCRS
- Landfill LDS
- Leachate storage tank system

The PICS design identifies temperature analog instrumentation, consisting of an element and transmitter that will enable operators to monitor temperature levels inside the following:

- Crest pad buildings
- Leachate transfer buildings

6.4.6.8 Discrete Instrumentation

The PICS design identifies level instrumentation, consisting of radio frequency (RF) admittance probes and transmitters that will enable operators to monitor discrete liquid levels inside the leachate storage tank system. The PICS design identifies level discrete instrumentation, consisting of magnetic float switches that will enable operators to monitor discrete liquid levels inside the following:

- Crest pad building sump
- Combine sump
- Combine sump interstitial

The PICS design identifies operator instrumentation, consisting of switches, indicating lights, and control relays that will enable operators to monitor the following discrete status:

- Crest pad building and control power status
- Landfill LCRS pumps ON/OFF, AUTO and FAIL status
- Landfill LDS pumps (on/off, auto, and fail) status
- Combined process sump pump (on/off, auto, and fail) status
- Leachate transfer pump (on/off, auto, and fail) status

6.4.6.9 PICS Programmable Logic Controller (PLC) Architecture

The PICS design identifies PLC architecture designed around Allen Bradley Ethernet small logic control technologies. PLC architecture consists of the following:

- PLC processor
- PLC input/output (I/O) modules
- PLC ancillary power supplies, chassis and cabling
- PLC application and development software and hardware

- The PLC processor is the microprocessor-based device that uses programmable ladder logic for implementing process monitoring and control, emulating the functions of conventional panel-mounted equipment such as relays, timers, counters, current switches, calculation modules, Proportional, Integral and Derivative controllers, stepping switches, and drum programmers.
- PLC(s) are programmed to interface with instrumentation and process motor control equipment. PICS PLC(s) are programmed to automatically operate (start/stop) all process control equipment as well as process flow totals, equipment runtime, operation alarms, equipment, and building status.
- Instrument architecture (analog and discrete control devices) interface with PLC via PLC I/O modules, installed in a common chassis with the PLC power supply.
- The type of I/O modules utilized include analog (4-20 mA) input, 24VDC discrete input, and 120VAC/ 24VDC discrete output.
- The PICS design identifies PLC application software that provides functions unique to the project and not provided by PLC system software alone, such as programmable controller ladder logic, math operations on input process variables (scaling, alarming, totalizing, comparisons).
- The PICS design identifies PLC standard system software packages that enable personnel to communicate and program PLC processor and configure I/O modules. PLC development and application software reside on the programming laptop from which the application is downloaded into the PLC processor.

The PICS design identifies communication protocols establishing data exchange between PLC, programming laptop, OIU architecture, and future remote SCADA as follows:

- Allen Bradley RS-232, RS-4585, and DF1
- Ethernet

6.4.6.10 PICS Operator Interface (OIU) Architecture

The PICS design identifies OIU architecture that allows operators to visually monitor process system data and interface with the facility's programmable logic controllers. OIU enables operators to view alarms and change process set points.

PICS OIU architecture is designed around Allen Bradley PanelView, communicating with PLC architecture over a communication local area network. OIU architecture includes:

- OIU assembly
- Local area network copper cabling
- OIU application and standard system software
- The PICS design identifies OIU application software that provides functions unique to the project and not provided by system software alone. These include, but are not limited to, programmable controller ladder logic, databases, reports, control strategies, graphical display screens, and operation scripts.
- The PICS design identifies OIU standard system software packages that enable personnel to communicate and program OIU. OIU application and standard system software reside on the programming laptop from which the application is downloaded into the OIU processor.

6.4.6.11 PICS Communication LAN Architecture

The PICS design identifies communication between PLC processors, OIU, programming laptop, and future IDF SCADA over a local area network consisting of a local 10/100 MBPS Ethernet switch, local PLC, OIU LAN drivers, and a cable system. The PLC processor and OIU are addressable over the LAN, allowing each device to share data and control points between each other and future devices.

6.4.6.12 Back Up Power

The PICS design identifies UPS mounted inside each main control panel. UPS(s) was sized to enable PLC and OIU networks to maintain monitoring of process control systems during a power failure as well as provide for an orderly shutdown. UPS does NOT power process control equipment such as solenoids, instruments, motorized valves, pumps, and motors.

6.4.6.13 Control Panels

The PICS design identifies the main control panel, mounted inside each crest pad building housing PLC processor and associated I/O modules, ancillary power supplies, termination devices, UPS, and control circuit protection devices. OIU and process flow and level indicators are mounted on front doors of control panels.

The PICS design identifies local control panels, integrating discrete level instrumentation, control relays, intrinsic safety relays, and providing interlock signals between PLC architecture and MCC pump controls.

6.5 Stormwater Management

The proposed stormwater system to be constructed just south of the south end of the Phase I excavation will intercept stormwater runoff from the area to the south for the 24-hour, 25-year storm event so that it will not flow into the Phase I excavation and will discharge the intercepted stormwater into the ground via infiltration. This system will consist of the south stormwater berm/ditch, two culverts, and the east and west infiltration areas. The berm will be two feet high above the existing ground surface. The minimum combined depth of the berm and ditch will be two feet. The ditch will be V-shaped with 3:1 side slopes. The culverts will be 18-inch-diameter, corrugated polyethylene pipe with smooth interior. Geotextile and quarry spalls will be placed around each end of the culverts to provide erosion protection. The east and west infiltration areas will have bottom lengths of 220 and 225 feet, respectively. Each of the infiltration areas will have a bottom elevation of 719 feet and a bottom width of 15 feet. In order to allow access for future maintenance into each of these infiltration areas, their north and south ends will be sloped at 15 percent and surfaced with quarry spalls placed on a geotextile.

The proposed stormwater system to be constructed at the south toe of slope within the Phase I excavation will intercept stormwater runoff from the unlined portions of the excavation for the 24-hour, 25-year storm event so that it will not flow into the active cells and will discharge the intercepted stormwater into the ground via infiltration. This system will consist of the excavation stormwater berm/ditch, a stormwater pipe, one catch basin, and the excavation infiltration area. There also will be a flow path along the south side of the access ramp that will continue along the south side of the flat area at the base of the access ramp and into the southwest corner of the excavation infiltration area. The south stormwater berm/ditch will slope to drain to the east. The combined depth of the berm and ditch will be two feet. The stormwater berm will be 2 feet high at its west end, and the corresponding depth of the ditch will be zero. The berm will gradually reduce in height as the depth of the ditch increases. The berm will end when the ditch depth reaches 2 feet. The ditch will be V-shaped with 3:1 side slope on the south and 2:1 side slope on the north. The stormwater pipe will be 18-inch-diameter corrugated polyethylene pipe with smooth interior. Geotextile and quarry spalls will be placed around each exposed end of the stormwater pipe to provide erosion protection. The catch basin will be used to lower the elevation of the stormwater pipe so that there will be adequate cover over the pipe for protection against wheel loads. The infiltration area will have a bottom elevation of 678 feet, a bottom width of 15 feet, and a bottom length of 50 feet. In order to allow access for future maintenance into this infiltration area, the west end will be sloped at 15 percent and surfaced with quarry spalls placed on a geotextile.

If the water builds up in the east or west infiltration area, it will eventually flow out of the north end of the infiltration area. The water would flow overland, north along the toe of the fill for the berm access road, and continue generally northward.

If the water builds up in the excavation infiltration area so that it extends into the ditch, then the operator will have to bring in a portable pump and pump the water into the east infiltration area.

Maintenance for each of the infiltration areas, the ditches, and the ends of each of the culverts and stormwater pipes will be primarily to remove accumulated sediment and debris.

6.6 Building Systems

6.6.1 Crest Pad Buildings

The crest pad building is designed as a pre-engineered, rigid frame metal building on a slab-on grade foundation. The building slab is separated into two portions. The lower portion of the slab is where the piping associated with the leachate pipe will be contained, and the higher slab is where the electrical and control equipment will be located. The slab where the leachate piping will be located is lowered to create a containment area for the leachate. Construction joints within this area have waterstops to ensure that leachate cannot egress through the construction joints. Additionally, a sump has been placed to drain the containment area, if required. The entire floor and sump area also is to be coated to provide even greater resistance to the leachate.

6.6.2 Leachate Transfer Buildings

The leachate transfer building is designed as a pre-engineered, self-framing metal building on a slab-on-grade foundation. The metal building is supported on an 8-inch curb that travels continuously around the exterior of the building. The curb is continuous, even through the door threshold, to provide a containment area for the leachate in case of spillage. In order to maintain conformance with building code requirements, a landing is used to eliminate the curb tripping hazard at the door threshold. Construction joints within this area have waterstops to ensure that leachate cannot egress through the construction joints. Additionally, a sump has been placed to drain the containment area, if required. The entire floor and sump area also is to be coated to provide even greater resistance to the leachate.

6.6.3 Truck Loading Station

The truck loading station is designed to receive trucks to load with leachate. The station is essentially a slab-on-grade. The station is designed to contain minor spillage of leachate by sloping the floor slab towards the center and using rounded curbs at the slab entrance and exits. Two sumps will be placed in the center of the station to drain the station as required. The entire floor and sump area also is to be coated to provide even greater resistance to the leachate.

6.7 Electrical Service and Lighting

6.7.1 Introduction

This section provides a summary of the electrical design and construction elements of the project, providing introduction and reference to the project layout and key design components for the following IDF facilities:

- IDF leachate collection and handling crest pad facilities (two each)
- IDF leachate storage tank and leachate transfer facilities (two each)
- IDF truck loading facilities (two each)

The electrical design identifies, specifies, and integrates power distribution systems that incorporate transformers, breaker panels, motor control, safety switches, conductors, and lighting for the safe, reliable, and maintainable operation of IDF process and facility equipment including:

- Process equipment (leachate collection and removal pump motors, leak detection pump motors, transfer pump motors, and instrumentation)
- Building facility equipment (lighting, power outlets, heating units, cooling fans, and building sump pumps)
- Personnel and equipment safety systems (standby egress lighting, process alarm lighting, surge protection, and process piping heat trace)
- Electrical design and installation shall be in accordance with NFPA 70 (NEC, 2002)

6.7.2 Key Design Components (Elements)

Key electrical design components (elements) for each IDF facility include:

- Electrical secondary service and monitoring
- Electrical secondary service and feeder protective device coordination
- Electrical secondary service ground electrode system
- Electrical service, equipment, and associated metal structures grounding
- Electrical low voltage motor control
- Facility maintenance outlets (standard, ground fault circuit interrupter [GFCI], weatherproof)
- Facility interior, exterior, and egress safety lighting
- Facility environmental control (heating and cooling)
- Facility hazardous classification
- Process equipment heat trace, ambient monitoring, and power indication
- Facility electrical system surge and phase protection
- Materials and methods of electrical construction (i.e., conduit, wire, control and safety device, and enclosure selection)

6.7.3 Open Items

IDF Phase I Critical System design documents do not identify the following primary and secondary electrical service items:

- Exact location of primary 13.8 kV, 3-phase tie-in
- Exact value of available primary short circuit current at primary tie-in location
- Exact length of primary extension
- Exact location, size, and impedance of utility step-down 13.8 kV – 480/277V, 3-phase, 4-wire pad mounted transformer(s)

6.7.4 Assumptions to Open Items

These items are scheduled to be addressed during the IDF Phase I Non-Critical design. As such, the following assumptions were made in order to complete the Phase I design:

- Assume electrical service gear inside each Cell 1 and Cell 2 crest pad building are powered by separate pad mounted utility transformers.
- Assume pad mounted utility transformers are rated 75 kVA and are installed within 100 feet of respective Cell 1 and Cell 2 crest pad buildings.
- Assume each pad mounted utility transformer is radial fed from a common 13.8 kV primary feeder.
- Assume each Cell 1 and Cell 2 leachate transfer building is powered from electrical service gear, located inside respective crest pad buildings.
- Assume utility short-circuit contribution to be 100 MVA at 13.8 kV, three-phase.

6.7.5 Crest Pad Building Electrical Secondary Service and Metering

Electrical design identified 480 volt, 3-phase, 4-wire secondary service cables eventually powering a service-rated MCC mounted inside each crest pad building.

Type	Designation	Configuration
Cell 1 Service rated MCC	219A-LH-MCC-001	480V, 3-ϕ, 3-wire, 4-wire
Cell 2 Service rated MCC	219E-LH-MCC-001	480V, 3-ϕ, 3-wire, 4-wire

The service-rated MCC will operate as a main service gear, power distribution center, and motor control assembly. A MCC distributes 480 volt, 3-phase power to the following 3-phase equipment:

- LCRS three-phase pump motors
- LDS three-phase pump motor
- Combine sump three-phase pump motor
- Crest pad building and leachate transfer building unit heaters
- Crest pad and leachate transfer lighting panel transformers

Secondary 3-phase power is monitored by phase loss and phase reversal protection relays mounted inside MCC(s). In the event of a phase loss or phase reversal condition, the protection relay will shunt the MCC main service breaker. With main service breaker shunted (open), a UPS mounted inside each PICS main control panel will continue the operation of voltage sensitive PICS equipment (i.e., PLC, OIU, local area network communication), allowing for future remote alarming (future SCADA) and the safe shutdown of sensitive equipment.

Incoming power is also monitored through the use of analog-style voltage and current meters. Operators will be able to observe operating status of incoming power by manually selecting analog-style voltage and current meters to Phase A, Phase B, and Phase C.

MCC associated gear (frame, bussing, and feeder protective devices) were sized to adequately and safely handle the calculated design and demand operating loads, and to safely withstand calculated short circuit interrupting currents.

6.7.6 Utilization Voltages

The electrical design identified utilization voltages for the following equipment and systems:

Equipment or System	Voltage, Phase
Lighting	120 V, 1-ϕ
Heat trace	120 V, 1-ϕ
Convenience outlets	120 V, 1-ϕ
Instrumentation control circuits	24 V DC
Motor control	120 V, 1-ϕ
Air conditioner	208 V, 1-ϕ
Motors, less than 1/3 hp	120 V, 1-ϕ
Motors, 1/3 hp and larger	480 V, 3-ϕ
Unit heaters	480 V, 3-ϕ
Instrument power	120V, 1-ϕ

6.7.7 Leachate Transfer Building Electrical Service

The electrical design identified three phase motor loads inside leachate transfer buildings as being powered from MCC, located inside each crest pad building. Power will be routed from MCC to service-rated disconnect, wire-way, enclosed breaker, and mini-power center (panel/transformer assembly), located inside each leachate transfer building.

Type	Designation	Configuration
Cell 1 service-rated disconnect	219A1-LH-SW-002	480V, 3-ϕ, 3-wire, 4-wire
Cell 2 service-rated disconnect	219E1-LH-SW-002	480V, 3-ϕ, 3-wire, 4-wire

6.7.8 Crest Pad and Leachate Transfer Building Lighting Panelboards

The electrical design identified lighting panel boards installed in each IDF facility to provide 120/208V 3-ϕ, 4-wire power to non-three-phase motor loads. Lighting panelboards will be fed from 480V-120/208V 3-ϕ, 4-wire step-down transformers. Lighting panelboards inside crest pad buildings will be mounted along with step-down transformers inside MCC. Lighting panelboards (mini-power centers, along with integral step-down transformers) inside leachate transfer buildings will be wall mounted.

Type	Designation	Configuration
Cell 1 crest pad building lighting panel	219A- LH-LP-001	120/208V, 3-ϕ, 4-wire
Cell 1 crest pad building lighting panel	219E- LH-LP-001	120/208V, 3-ϕ, 4-wire
Cell 1 leachate transfer building lighting panel	219A1- LH-LP-002	120/208V, 3-ϕ, 4-wire
Cell 2 leachate transfer building lighting panel	219E1- LH-LP-002	120/208V, 3-ϕ, 4-wire

Lighting distribution panelboards will provide 120 volt power to all single-phase equipment including:

- Building lighting
- Emergency lighting
- Receptacles
- Main control panel
- Instrumentation will be powered from surge-protected circuit breakers inside each crest pad building main control panel.
- Lighting distribution panelboards will provide 120/208 volt, single and three-phase power to equipment including the building air conditioner, and heat tracing for process piping.

6.7.9 Feeder and Branch Circuits

The electrical design identified feeder and branch circuit breakers and conductor's size, based upon connected and operating loads. Style of feeder and branch circuit breakers will be thermal-magnetic.

6.7.10 Raceways

480V power circuits–Standard rigid galvanized steel (RGS) in exposed locations, PVC conduit systems will be buried, RGS will be coated when conduits transition from below grade to above grade areas

120V power circuits–Standard RGS in exposed locations, PVC conduit systems buried, RGS coated when conduits transition from below grade to above grade areas.

6.7.11 Raceway Sizing, Selection, and Installation Guidelines

The electrical design identified conduit wire fill and size, based upon THW (thermoplastic, vinyl insulated building wire; flame retardant, moisture and heat resistant, 75°C, dry and wet locations) insulated conductors for wiring 600 volts and below. Minimum raceway sizes will be as follows in the designated locations:

Minimum Raceway Size:	Location:
3/4-inch	Exposed on walls and ceiling
3/4-inch	Concealed in frame construction and finished ceilings
1-inch	Underground for circuits below 600 volts, including instrumentation
3-inch	Fiber optic

The electrical design identified underground raceways assemblies as concrete duct bank constructed.

6.7.12 Wire and Cable

The electrical design identified stranded copper conductors that will be used for all wiring, except for lighting and receptacle circuits where solid copper will be used.

Minimum conductor size of No. 12 will be used for power and lighting branch circuits. Conductors installed in all branch circuits rated 100 amps or less was sized based upon NEC table for 60°C TW conductors.

- No. 12 AWG copper for lighting and receptacle branch circuits
- No. 10 AWG, minimum, wiring for all outdoor power circuits
- No. 14 AWG, minimum, for all instrumentation 24VDC discrete control and instrument power
- No. 16 AWG, minimum, shielded for all instrumentation 24VDC analog control

6.7.13 Convenience Receptacles

The electrical design identified weatherproof 20 amp duplex receptacles for indoor service, weatherproof GFCI 20 amp duplex receptacles for outdoor service.

6.7.14 Motor Control

The electrical design identified full voltage non-reversing (FVNR) combination motor starter assemblies, to be mounted inside MCC for each constant speed motor. FVNR combination motor starter assemblies will consist of thermal-magnetic, trip-molded case circuit breakers; full voltage combination starters; control power transformers; indicating lights; and control switches. All combination motor starters will be operated in AUTO mode by PICS.

6.7.15 Overload Protection

The electrical design identified each motor as being provided with thermal overload protection in all ungrounded phases. Each controller will be provided with overload heaters and controller-mounted relays with external manual reset.

6.7.16 Grounding

The electrical design identified the grounding electrode system for each IDF facility, integrating ground ring rods, and connection to building rebar. The electrical design identified electrical service neutral, and the neutrals of derived sources, electrical equipment, and PICS control panels that will be bonded to grounding electrode systems.

6.7.17 Equipment Grounding

The electrical design identified noncurrent-carrying parts of all electrical equipment, devices, panelboards, and metallic raceways that will be bonded to grounding system.

The electrical design identified noncurrent-carrying parts of all mechanical equipment, to which electrical components will be attached and may potentially become energized, that also will be bonded to the grounding system, including building metal structures and leachate storage tank.

All conduits that will be provided have an equipment grounding conductor.

6.7.18 Lighting

The electrical design identified lighting fixtures that will be installed at each IDF facility to maintain an average 25-foot candle inside each building, and 5-foot candles at entrance doorways.

Note: Interior lighting levels are based upon *IES Lighting Handbook Indoor Industrial Areas Recommended Illuminance Levels* for interior activities inside work spaces where visual tasks of medium to large contrast are to be performed on occasional basis.

Note: Exterior entrance lighting levels are based upon *IES Lighting Handbook Outdoor Site/Area Recommended Illuminance Levels* for building exterior entrances frequently visited locations.

6.7.19 Emergency Lighting System

The electrical design identified emergency illumination (battery-pack wall-mounted units or luminaries powered by integral battery-powered ballasts) that will be provided in all IDF facilities.

6.7.20 Circuiting and Switching

The electrical design identified interior process area lighting, switched to provide adequate lighting. Exterior building lighting will be controlled by photocells.

6.7.21 Heat Trace

The electrical design identified electrical heat trace for above grade process piping freeze protection. Heat trace cable will be the self-limiting type with the overall system controlled by an ambient control thermostat. Heat trace design incorporates circuit power indication.

6.7.22 Hazardous Classification

The electrical design identified the interior of the combined sump as Class 1, Division 2 group, C hazardous. The electrical design for the combined sump will incorporate materials and intrinsic safety devices compatible for the installation of electrical equipment in Class 1, Division 2, Group C hazardous locations.

6.8 Construction QA Requirements

The Construction QA Plan describes the QA activities for constructing the Phase I IDF. QA activities will be required during construction to ensure the following:

- Firm and stable foundation system for liners
- Stability of dikes or embankments
- Low permeability soil liners that inhibit contaminant migration
- Geosynthetic layers that function as either a hydraulic barrier or a drainage system, depending on intended function
- LCRS and LDS that remove leachate and control head on the lining systems
- The Construction QA Plan has been prepared to describe the activities that will be performed during construction of the lining system, leachate collection system, and operation layer of Cell 1 and Cell 2. The Construction QA Plan satisfies the regulatory requirements and guidance established in 40 CFR 264.19, the EPA technical guidance document, *Quality Assurance and Quality Control for Waste Containment Facilities* (EPA 1993), and WAC 173-303-335.

The specific physical components that the WAC requires the Construction QA Plan to address include:

- Foundations
- Dikes
- Low-permeability soil liners
- Geomembranes
- LCRS and LDS
- Final cover systems
- The WAC requires the Construction QA Plan to include the following:
- Identification of applicable units and how they will be constructed
- Identification of key personnel
- Description of inspection and sampling activities

The Construction QA Plan is intended to be implemented by an independent, qualified Construction QA certifying engineer, familiar with EPA's technical guidance document, *Quality Assurance and Quality Control for Waste Containment Facilities,* as well as the Construction QA Plan. The Construction QA certifying engineer will be supported by other Construction QA representatives, as necessary, to implement the requirements in the Construction QA Plan and document the work.

The Construction QA Plan establishes general administrative and documentation procedures that will be applicable for selected activities of construction. The Construction QA Plan addresses only those activities associated with the soils, geosynthetics, and related liner and leachate collection system piping components for the Phase I IDF landfill. Other aspects of construction, such as transmission piping, utilities, concrete, and storage tanks will require QA testing and oversight. These requirements are not mentioned in the Construction QA Plan, but they will be included in future construction inspection documents, accompanying the bid-ready drawings and specifications.

6.9 Interface with Non-Critical Systems

Critical systems for the Phase I IDF include three primary design components:

- Liner systems
- LCRS
- LDS

In addition, the Phase I IDF detailed design also involves completing all design work required for an operable landfill.

Non-critical systems for the Phase I IDF include the following components:

- Entrance facilities, including entrance area, scales, and staging areas
- Administration and control facilities
- Waste delivery access road improvements to the IDF site from the WTP
- Waste treatment and staging areas
- Gates and fences
- Utilities including fire protection, process water, electrical power, or instrumentation cables

The IDF Phase I Critical Systems design has been prepared to interface with these non-critical systems that are necessary for operational readiness for the IDF. The following discussion details interface elements of the current design with these non-critical systems.

There is the potential for the DOE to procure an independent contractor to provide operation and maintenance services for the IDF. These services could also include the detailed design and construction of a portion or all of the non-critical systems for the facility. If this should be the case, careful consideration will be given to these interface elements in the development of performance criteria that will be included as part of any contract package for these services.

6.9.1 Entrance Facilities

Entrance facilities will control the flow of waste into the IDF. These facilities will provide for waste delivery, inspection, check-in, and final authorization for disposal into the IDF. Typically, the location for the entrance facilities is adjacent to the in-bound access road, prior to reaching the disposal area. Other factors that can influence their location include access to existing utilities and other operational facilities such as waste treatment, soil stockpiles, or staging areas. Based on the current configuration planned for the IDF, there will be room for entrance facilities to the south of the Phase I disposal area, along the western access road. Typically, these facilities require connection to such utilities as fire protection, power, and process water. Utility interfaces are discussed later in this section.

Design criteria and detailed design elements for the IDF entrance facilities have not been developed. The overall mission for the facility has expanded from handling just the ILAW packages to other wastes including Waste from the DBVS and LLW materials. This may require the entrance facilities to have expanded capabilities for waste load staging, inspection, verification, and scaling, prior to release for disposal into the IDF. This could impact the location selected for the entrance facilities, since complete development of the IDF to its full capacity will leave little room to the south of the southern perimeter berm for the facility (refer to Drawing H-2-830827).

This could require the entrance facilities to be located along 1st Street, if a permanent initial location is desired. Otherwise, a more mobile entrance area could be developed and relocated along with phased development of the facility.

6.9.2 Administration and Control Facilities

Administration and control facilities will provide the control center for LCRS operations and monitoring, as well as monitoring for LDS and other emergency systems (fire, power interruption, and HVAC controls). The administration building will service facility operations, including waste tracking and record keeping systems as well as provide for staff needs including office facilities, lunch room, lockers, and storage. Other functions that may take place in this facility area include equipment maintenance, an equipment and staff decontamination area, and equipment storage.

The proposed location of the administration building is shown on Drawing H-2-830827, to the north of the leachate storage and handling area (north of the IDF Phase I development area). This location provides quick access to the leachate control buildings and storage tanks, as well as good interface with existing utilities that will come from existing facilities to the east and west of the IDF. Power and control/communications cables will connect the administration building to the leachate control buildings (crest pad buildings, leachate pump buildings, and leachate storage tanks), as well as to other leachate control structures including the combine manholes and truck loading stations for Cell 1 and Cell 2. Additional utilities will service the administration building including fire protection, process water, potable water, communications, and power. Calculations for power supply to future facilities are provided in the *Integrated Disposal Facility (IDF) Detailed Design: Site Utilities Design Report*, (RPP-18515, Revision 1).

Design criteria and detailed design need to be established for the administration building. The expanded mission of the IDF may influence existing criteria that have already been determined for this facility as provided in conceptual design documents for the original ILAW W-520 Project. Modular units may be considered for this facility.

6.9.3 Waste Delivery Access Road

The waste delivery from the WTP will access the IDF from 1st Street and enter the IDF along the western perimeter of the landfill. Waste delivery from other areas will access the facility from one of three gates (810, 812, or 815) to the 200 East Area, as discussed previously in Section 1 (refer to Figure 1-3). The Phase I access road is aligned horizontally with the proposed western berm of the complete IDF landfill. The vertical alignment of Phase I access road coordinates with the existing topography of the site between 1st Street and the Phase I landfill area, to minimize cut and fill requirements for this road construction. As such, the Phase I vertical alignment does not follow the vertical alignment of the future western perimeter berm of the landfill and will need to be modified in future expansion phases. All-weather pavement for the Phase I road as well as for 1st Street will need to be completed as part of non-critical design. It is anticipated that pavement will be asphalt concrete pavement.

Access for waste haul vehicles will require upgrades to 1st Street to be designed as part of non-critical systems. Design criteria for this upgrade will be based on the anticipated haul vehicles and wheel loads for the various wastes to brought to the facility. From the Phase I Critical Systems design, the melter transport vehicle represents the most restrictive design condition for the road in terms of axle load and radius/grade limitations. However, there are also substantial wheel loads and larger volumes for ILAW package transport vehicles and other MLW and LLW wastes.

It should be noted that there will be a significant grade differential between the southern end of the IDF perimeter berm and the existing 1st Street road grade. The western berm climbs at a uniform 1 percent grade to the south. As such, it will have an elevation of approximately 741 feet at the southern perimeter road. The existing grade of 1st Street at the western perimeter of the IDF is approximately 734 feet, and so 1st Street will need to be raised to make this transition and keep vertical road grades at a maximum of 5 percent to accommodate the melter transport vehicles.

6.9.4 Waste Treatment and Staging

Currently, no waste treatment facilities have been planned for the IDF. Consideration of waste treatment may be necessary as part of the IDF's expanded mission to take mixed wastes and low-level wastes from both onsite and offsite sources, depending on the waste acceptance criteria that are established for the facility. Waste staging areas are associated with waste receipt and inspection activities, as mentioned previously. Staging and storage areas may also be needed for waste treatment as well. Design of non-critical facilities may need to consider development of these waste treatment and staging areas.

During Phase I operation, there is adequate area south of the Phase I landfill area for treatment and staging. Some staging also can occur within the landfill itself that offers the advantage of occurring over lined areas with leachate collection systems in place. However, as wastes are placed and cell lifts become full, staging areas may be limited until new lifts are ready for waste placement. Regulatory requirements for waste staging and storage may also impact location and operational requirements for these areas.

6.9.5 Gates and Fences

The IDF is being developed within the 200 East area of the Hanford Site, that has controlled access with a perimeter fence and access control gates (refer to Figure 1-1). As such, it is currently not anticipated that additional fencing and gates will be required for access control to the facility. However, operationally it may be determined that a perimeter fence and additional gates may be warranted for isolation of the IDF from adjacent existing facilities and, if so, these need to be designed during implementation of non-critical design components. Site standards for fences and gates would be followed for this design.

6.9.6 Site Utilities

As mentioned previously, site utilities are included in non-critical systems design. Site utilities will interface with existing utilities that service facilities in the 200 East area. As such, substantial coordination will be required to locate these utilities, determine the best interface tie-in location, and bring these to the IDF site. Key utilities that are needed for the IDF include:

- Power to buildings and operating systems, as well as to area lighting
- Communication between administration building and operating systems, as well as from the IDF to other area networks
- Fire protection water
- Process (non-potable) water for operations and facility construction
- Potable water

Power requirements for leachate control and monitoring systems have been designed during this Phase I Critical Systems design. Access vaults to power and control systems are provided outside of both crest pad buildings (shown on Drawing H-2-830858). It is anticipated that the administration building will connect at these access vaults and will provide power for system operation and an Ethernet connection for controls. Transformer design for bringing power from the site to the administration building (and to leachate control facilities) will be performed during non-critical design, as will design of the Ethernet connection and administration control systems. Calculations for power supply to future facilities are provided in the Integrated Disposal Facility (IDF) Detailed Design: Site Utilities Design Report, (RPP-18515, Revision 1).

Utility corridors need to be developed to bring these utilities to facility areas. It is recommended that these corridors be developed outside of landfill embankment areas and access roads, to allow for uninterrupted waste placement and facility operation, for future landfill phase development, for protection of liner system anchor trenches, and for protection of utilities from heavy wheel loads. In addition, the future final cover of the IDF is located over the perimeter embankments and catches existing ground at the outside toe of the embankment.

7.0 OPERATING PROVISIONS

7.1 Waste Placement

7.1.1 Introduction

To establish a baseline for design, construction, and operation of the IDF, a plan for filling the landfill cells was developed. This plan was developed mainly to ensure that landfill configuration and size as proposed for the IDF Phase I Critical Systems were adequate for safe placement of the ILAW, waste from the DBVS, and LLW, both remote handle and contact handle, that will be placed in the Phase I development. The proposed configuration and size of the IDF Phase I landfill are identified in Section 6 of this report.

The drawings that show the waste placement plan are included in Appendix D.1. This plan was based on the concept of completely filling the first lift in both cells before beginning filling of the succeeding lift. The plan represents one approach to filling the cells within the proposed configuration. It is possible that other approaches, such as proceeding to a subsequent lift before completely filling the previous lift, also are workable, but development of the plan did not consider alternative methodologies to fill the cells. Development of the plan is also based on conformance with the operational procedures identified for the Base Alternative in Appendix K of the *Conceptual Design Report for Immobilized Low-Activity Waste Disposal Facility, Project W-520* (RPP-7908, Revision 0), (CDR).

This waste placement plan is intended to meet the applicable functional criteria identified in the *System Specification for Immobilized Low-Activity Waste Disposal System* (RPP-7303, Revision 3). "As low as reasonably achievable" principals (keeping radiation exposures to as low as reasonably achievable) are embodied in the waste placement plan that was developed. Because of the area available for waste disposal in each cell, the plan provides the capability to relocate filling operations to another area within each cell, if an event occurs that causes operations to halt temporarily, placing waste packages at the current working position. This will allow waste package placement to continue while the situation that caused the operations to cease is resolved.

7.1.2 Phase I Configuration

Under the proposed configuration for the IDF Phase I, there will be two cells, identical in size. One cell will be for disposal of ILAW and waste from the DBVS; the other cell will be for disposal of LLW. This waste placement plan proposes disposal of ILAW and DBVS waste Cell 1 and disposal of LLW in Cell 2. Provisions are included for disposal of both remote handle and contact handle waste in each cell.

The configuration of the IDF Phase I development as it will exist at the completion of construction, prior to beginning filling operations, is shown in Appendix D, Drawing D.1-1. The initial operations layer, placed as part of Phase I construction, will cover the entire bottom liner and LCRS. The top of the operations layer will be level in the east-west direction and slope down at 1 percent from the south to the north. The operations layer will extend up the west, north, and east side slopes. Access to the facility will be from 1st Street along the western site boundary. An access ramp from the southwest corner of Phase I will lead down the south excavation slope from the west side to the bottom of Phase I and connect to the top of the operations layer near the south east corner of Cell 2.

7.1.3 Waste Receipts

As stated in Section 6.2, the IDF will receive ILAW and Waste from the DBVS. The volumes stated in Section 6.2 are based on waste forecast information provided by FH. The waste volume forecasts are updated by Hanford Site contractors on a regular basis. Actual waste receipt rates at the IDF will likely vary from the estimated amounts. Depending on the receipt rate of ILAW and DBVS waste versus the receipt rate of LLW, each lift of Cell 1 and Cell 2 may fill at different rates. The waste placement plan can accommodate differing rates of waste receipt because filling in subsequent lifts in each cell could be begun at different times as soon as the prior lift was complete. The cell that has the higher waste receipt rate will fill faster than the other cell and will determine the time when subsequent phases of development will need to begin so that additional disposal capacity is available when it is needed.

7.1.4 General Waste Placement Procedures

The discussion of waste placement in this plan is based on placement of the uniform height ILAW packages using remote handle. Some adjustments may need to be made for the variable height LLW containers and for contact handle waste, but in general, the waste placement concept will be the same for all types of waste.

The configuration of IDF Phase I provides a height sufficient for four layers of ILAW packages, each covered with one meter of operations layer soil to provide shielding to operations personnel during waste package placement. LLW, which will be in variable height containers, can be accommodated within each of these four lifts. However, in some cases the LLW containers may exceed the lift height and, therefore, will not be completely covered by placement of the operations layer soil. In these cases, it may be necessary to mound cover soil around the individual projecting LLW containers to provide sufficient cover for shielding until they are completely covered by subsequent lifts.

Each lift will contain multiple ILAW package arrays that span the width of each cell. The packages will be placed in close-packet hexagonal arrays, with placement tolerance averaging 10 centimeters (4 inches) center to center. As the packages are placed in the cell, the array will proceed along the width of the cell. The earth cover will proceed shortly behind the advancing package array, the distance behind the front package limited by the repose slope of the fill soil. The array width (number of columns of packages) will be limited according to the amount of radiation generated by the total number of packages that can be exposed. The CDR indicates that even at some distance from the advancing array, the dose rate becomes a concern when the array approaches more than ten or twelve packages in width.

Off-loading of the ILAW packages and other waste containers will take place in the cell. A standard, manually operated, rubber-tired crane will off-load packages, move temporary shielding walls (concrete blocks), and place the interstitial fill between the packages using a hopper. In the CDR, the total weight of the shielding bell, package grapple, load cell, hooks, and other rigging is estimated at 20 metric tons (23 tons). The crane, as identified in the CDR, will be a Grove GMK 5100, a 108 metric ton (120 ton), rough terrain rubber-tired crane with a telescoping boom and a maximum reach of 15 meters (50 feet), with a load of 20 metric tons (23 tons). Pad loads could exceed 55 metric tons (60 tons) when placing an ILAW package at the maximum allowable reach. Dunnage required under each outrigger pad of the crane for lifts of this size has been determined to be 60 square feet, when operating directly on the base operations layer at its point of minimum thickness over the bottom liner system. Dunnage requirements for subsequent lifts would be less, but have not been determined. Refer to Section 5.5.5 and Appendix C.5.e of this Design Report for dunnage requirement calculations.

7.1.5 Moveable Shielding Wall

With off-loading operations in close proximity to the advancing package array, a moveable shielding wall will be set up between the crane and transporter operations and the placed packages (CDR, Drawing No. ES-W520-BASE). With the 15-meter (50-foot) maximum reach of the crane, the shield walls will have to be moved after every five rows of packages are placed. For a ten-package-wide-array, the wall will need to be relocated after fifty packages have been deposited, or about every eight days during Phase I.

To prevent the crane crew from receiving a high exposure rate, a new shield wall will be erected before the first shield wall is removed. A remote grappling system will be required to prevent rigging of the previously placed shield wall from causing high dose rates to operations personnel. Even then, the amount of time it will take to move the wall is estimated in the CDR to be 26 hours, four to five shifts, or a little less than two days when operating a full 24 hours per day.

An alternative to the movable shielding wall is to use contact handle waste to construct the shielding wall and to leave it in place after placement of each ILAW array rather than moving it. This can reduce operations labor and expenses. It can also result in the use of less cover soil because the space between the package arrays will be partly filled with contact handle waste, rather than with all soil.

This alternative needs to be considered further when developing the operations plan for operating the disposal facility.

7.1.6 Typical Array Size

The moveable shielding wall set up between the crane and transporter operations and the placed package configuration will limit the proximity of package placement to between 15 meters (50 feet) and 7.5 meters (25 feet) of the crane. The 7.5 meters (25 feet) usable range of the crane reach, working over the shielding wall, and the ten or twelve maximum package width (because of dose rate limitation) determines the nominal array size that can be placed by the crane from a single set point. The 1.22 meters (4 feet) diameter ILAW packages will be staggered in the array to minimize the space between the packages. A column that is five packages deep can fit within the 7.5 meters (25 feet) available range of the crane reach while working over the shield wall. A width of ten packages is within the reach of the crane and is less than the allowable limits for the dose rate. Allowing for a 10 centimeters (4 inches) average tolerance in package placement, the five-row by ten-package-wide array is roughly 6 meters (20 feet) deep by 13.3 meters (44 feet) wide. A typical array is shown in Appendix D, Drawing D.1-1.

7.1.7 Cover Soil

Prior to the shield wall being relocated, the crane will place interstitial soil material between the packages, using a hopper. The filling operation is expected to take about one shift, according to the CDR, using up the balance of the two days needed to move the shield wall. To make up the time spent moving the shield wall and placing the interstitial fill soil, the average rate of package placement will have to be increased to seven packages per day for five days, according to the CDR.

While the shield wall is being relocated, a soil cover will be placed over the packages from on top of the lift of previously placed packages. Dump trucks will drive over the previously covered portion of the array and back up to near the edge of the packages that are still exposed and dump a load of fill soil for spreading by a bulldozer. The soil will be spread over the top of the top and exposed side of the array. The side slope from soil, cascading off the top, will be formed in no less than 1.5 H: 1V for reasons of safety, and will use approximately a 5-meter (16-foot) wide space between lines of arrays. Approximately 300 cubic meters (400 cubic yards) of soil will be required to cover the top and side of the five-row-deep by ten-package-wide array. The cover soil will be held back from the advancing end of the array so that the toe of the cover soil does not extend beyond the outer package in the array. This will allow the next array to be placed in close proximity to the previous array. After the bulldozer spreads the soil to a somewhat uniform 1-meter-plus thickness over the packages, a sheepsfoot-style compactor will make several passes to consolidate the fill soil. The cover soil effort will take approximately 12 hours or two shifts, as estimated in the CDR, and will take place at the same time that the portable shield wall is being relocated.

7.1.8 Failed Melter Disposal Area

(Note: Disposal of failed melters is not permitted at this time by this permit.)

Failed melters can be disposed of as MLLW in Cell 1. A failed melter disposal area is provided on the bottom of Cell 1 at the southern toe of the waste lifts. Disposing of the failed melters in this area would eliminate placing them within the lifts along with the ILAW packages and other MLLW.

7.1.9 Access Ramps

Two 30-foot wide access ramps will be built into the south slope of the waste lifts to accommodate the movement of transport vehicles and equipment from one lift to the next. A third access ramp will be built through the north shine berm onto the top of the third lift to accommodate transport vehicles and equipment during the construction of Phase II, when the access ramp leading down the south excavation slope to the bottom of Phase I will be removed.

The access ramp into Cell 1 and the access ramp from the north side would have a maximum slope of 5 percent to accommodate failed melter transporters, if it becomes necessary to dispose of the melters in the waste lifts rather than in the designated area at the bottom of Cell 1. The access ramp into Cell 2 would have a maximum slope of 8 percent that would accommodate the ILAW, DBVS containers, and LLW waste transporters. The access ramps at the bottom of Phase I would have minimum outside turning radii of 75 feet, to accommodate the failed melter transporters. The dimensions of the access ramps provide flexibility to accommodate the various waste haul vehicles that could use the ramps.

7.1.10 Filling Lift 1

Filling of remote handle ILAW and DBVS waste in Cell 1 will begin in the northwest corner and proceed to the southeast. Filling of remote handle LLW in Cell 2 will begin in the northeast corner and proceed to the southwest. Filling of contact handle LLW will begin in the northwest corner of cell 2 and proceed southeast (see Appendix D, Drawing D.1-2). This filling approach places the remote handle wastes farthest apart from each other, with contact handle wastes between them, and eliminates the need for additional shielding provisions that would be necessary if the two remote handles wastes were located adjacent to each other. This filling approach will be continued in the three subsequent lifts.

Nearly all of Lift 1 can be filled with the crane and transporters, operating from the top of the first operations layer. A 5-meter (17-foot) wide separation will be maintained between Cell 1 and Cell 2 to separate the ILAW and DBVS waste from the LLW. This separation area will be filled with soil. Using a low permeability soil in this area will maximize separation of leachate between the two cells. Two access lanes (ramps) will be maintained into the cells for transporter access. The transporters can turn around within the cells until the packages are within 7.5 meters (25 feet) of the area needed for the unloading operations.

Before the space for filling Lift 1 from the top of the first operations layer is consumed, the two access ramps will be extended with soil and contact handle waste to the top of Lift 1. The crane and transporters will go to the top of Lift 1 and will finish placing the remainder of the Lift 1 waste packages from the top (see Appendix D, Drawing D.1-3). At this point, it will also be possible to begin using the failed melter disposal area (also shown on Drawing D.1-3).

7.1.11 Filling Lift 2

Lift 2 will be filled similarly to Lift 1 (see Appendix D, Drawing D.1-4). This filling approach will continue the pattern that was established in Lift 1. Nearly the entire lift can be filled with the crane and transporters operating on the top of Lift 1. The 5-meter (17-foot) wide soil-filled separation will be maintained between Cell 1 and Cell 2 to separate the ILAW and DBVS waste from the LLW. The two access ramps will be maintained into both cells for transporter access. The transporters can turn around within the cells until the packages are within 7.5 meters (25 feet) of the area needed for the unloading operations. Before the space for filling Lift 2 from the top of Lift 1 is consumed, the two access ramps will be extended with soil and contact handle waste to the top of Lift 2. The crane and transporters will go to the top of Lift 2 and will finish placing the remainder of the Lift 2 waste packages from the top (see Appendix D, Drawing D.1-5).

7.1.12 Filling Lift 3

Lift 3 will be filled similarly to Lift 2 (see Appendix D, Drawing D.1-6). Nearly the entire lift can be filled with the crane and transporters operating on the top of Lift 2. The 5-meter (17-foot) wide soil-filled separation will be maintained between Cell 1 and Cell 2 to separate the ILAW and DBVS waste from the LLW. Two access ramps will be extended into the cells for transporter access. The transporters can turn around within the cells until the packages are within 7.5 meters (25 feet) of the area needed for the unloading operations. Before the space for filling Lift 3 from the top of Lift 2 is consumed, the two access ramps will be extended with soil and contact handle waste to the top of Lift 3. The crane and transporters will go to the top of Lift 3 and will finish placing the remainder of the Lift 3 waste packages from the top (see Appendix D, Drawing D.1-7).

7.1.13 Filling Lift 4

Lift 4 will be filled similarly to the previous three lifts, but with a few differences (see Appendix D, Drawing D.1-8). Most of the lift can be filled with the crane and transporters operating on the top of Lift 3, using the access ramps from the south. However, only the easterly access ramp from the south is planned to be extended to the top of Lift 4 for transporter access. The westerly access ramp from the south will not be extended because, as shown on Appendix D, Drawing D.1-9, it would reach the top of Lift 4 too close to the west side slope to accommodate an adequate turning radius for the transport vehicles. The access ramp will be blocked by waste placement in Cell 1. However, with some minor adjustment in its location and/or increase in its slope, it will be possible to extend the access ramp into Cell 1, if desired. Also, at some point during the filling Lift 4, construction for Phase II to the south will begin, and the access road from the south will be removed from service.

Prior to the westerly access ramp becoming blocked with waste and the access road from the south removed for construction of Phase II, a third access ramp will be constructed from the north down onto the top of Lift 3 to provide additional access. This access ramp will maintain separation between Cell 1 and Cell 2, to separate the ILAW and DBVS waste from the LLW. The transporters can turn around within the cells until the packages are within 7.5 meters (25 feet) of the area needed for the unloading operations.

Before the space for filling Lift 4 from the top of Lift 3 is consumed, the easterly access ramp will be extended with soil and contact handle waste to the top of Lift 4, and the access ramp from the north will be graded out onto the top of Lift 4. The crane and transporters will go to the top of Lift 4 and will finish placing the remainder of the Lift 4 waste packages from the top (see Appendix D, Drawing D.1-9). Completion of Lift 4 will end the filling operations in Phase I. The configuration at the end of Lift 4, prior to placement of the final cover system, is shown on Appendix D, Drawing D.1-10.

7.1.14 Transitioning Between Lifts

As the available operating space in a lift gets smaller, operations efficiency will decrease to a point where it will become necessary to move part of the operations to the next lift before the active lift is completed. This will allow completion of each lift, using selected waste that will be easier to handle in the remaining space available on the lift. An example of this would be to use only contact handle waste to complete the filling of each lift while operating on the top of the lift that is being completed (see Appendix D, Drawings D.1-3, -5, -7, and -9) and sending all remote handle waste into the next lift.

7.1.15 Planning for Phase II and Operations During Phase II Construction

Phase II will need to be constructed and ready for operations sufficiently ahead of completion of filling operations in Lift 4 of Phase I to allow a smooth transition without operational constraints. Planning, design, and construction of Phase II may require several years. Phase II should be planned to be ready for operation at least six months, and preferably one year or more, before Lift 4 in Phase I is anticipated to be completed. This will provide a reasonable margin for changes in the incoming waste quantities and other variables while still having Phase II ready for operation, prior to reaching capacity in Phase I.

While Phase II is under construction, the access road on the west will be out of service for a period of time and the access ramp on the south into Phase I will be removed. During this time, it will be necessary for all waste transport vehicles to enter Phase I, using the access ramp on the north side. As currently designed, some access roads on the west and north sides of Phase I that normally would be used to reach the north access ramp might not accommodate all of the transport vehicles. In particular, the berm access road on the west side of Phase I and the access roads around the leachate storage tanks on the north do not have widths and turning radii as large as required by the waste transport vehicles. These roads would have to be widened and their turning radii increased to meet the requirements for transport vehicles, particularly the failed melter transporters.

7.2 Operational Interfaces

Operations and maintenance procedures will be prepared in the future as a separate project. These procedures will address operations, monitoring, and maintenance activities for the IDF.

This section of the Design Report presents important operational interfaces that have been identified by the design team. These interfaces should be considered during preparation of the operation and maintenance procedures. The interfaces are grouped by three categories–landfill excavation, liner system, and leachate handling system.

7.2.1 IDF Landfill Excavation and Related Subsystems

Operational interfaces for the landfill excavation and related subsystems include the following:

- Due to the containerized nature of the waste, the landfill is designed to be filled in a bottom-up fashion in four or more layers. The number of layers will depend on waste package size. Some waste packages may be larger in dimension than the ILAW packages. Operational procedures should be developed to accommodate various package sizes and their placement.
- Clean fill placement between waste packages must be done to minimize the potential for future consolidation and potential subsidence.
- Operations layer on side slopes of IDF will be monitored for material loss due to wind and water erosion. Lost material should be replaced. Annual application of spray-on type soil stabilization material to exposed areas of Phase I IDF should be considered.
- Shine berms should be monitored for erosion and height and should be repaired as necessary. Erosion control matting on the berm will be maintained and repaired or replaced if damage occurs.
- Stormwater control facilities should be maintained annually. Maintenance would include debris removal from the ditches and application of weed control. Periodically, if capacity of infiltration areas is diminished due to collection of fines, fines removal will be necessary. To maintain infiltration capacity, no other vehicle access should be allowed into these areas.
- Stormwater accumulation in the in-cell excavation infiltration area should be visually monitored. Pumping of the area may be necessary if accumulation becomes significant (near liner levels) in wet weather seasons. Periodically, if capacity of infiltration areas is diminished due to collection of fines, fines removal will be necessary. To maintain infiltration capacity, no other vehicle access should be allowed into these areas.
- Due to the heavy wheel loads on the access roads and ramps, gravel surfacing will be maintained with regular maintenance. Maintenance activities may include addition of more top course material, and grading and compaction of this material.
- Active faces of stockpiles will require periodic application of spray-on soil stabilization material.

7.2.2 IDF Liner System

Operational interfaces for the lining system include the following:

- Only equipment with ground pressures less than 4,400 lb/ft should be used for construction and maintenance on the side slopes, when operating directly on the operations layer. Bulldozers or other equipment may operate on the side slopes until a rain event in excess of 0.15 inches per hour occurs. In that event, equipment should be kept off the side slope (directly on the operations layer) and should not be permitted to operate on slopes until two hours after the end of the rainfall event. The precipitation event applies to both the lined slopes and the unlined slopes at the southern end of the Phase I cell.
- For equipment on ramps, equipment should be kept a minimum of 2 feet away from the edge of ramps, to avoid localized sloughing of the ramp edges.

- When operating equipment or placing waste on the operations layer above the lining system, care should be taken to avoid damaging the liner. Special care will be necessary for equipment operation on the side slopes.
- Any loads placed on the surface of the first operations layer must be examined to verify that they do not create loads on the lining system in excess of the allowable GCL bearing capacity. As an example, different types of waste other than canisters should be examined as the waste plan is more fully developed. Care should also be taken to avoid impact loading, such as dropping a canister.
- For static loading (such as for a barrier wall), refer to the discussion in Section 5.2 and Appendix C.2.
- For operational/equipment loading, refer to the discussion in Section 5.5.5 and Appendix C.5.e to determine applicable load limits and crane dunnage requirements.

The waste plan, as it is developed, should be followed for placement and density requirements. Any revisions to the proposed waste filling plan (discussed in Section 7.1) should be reviewed by the design engineer, to evaluate impacts on the waste/fill global stability analyses (Section 5.1.3 and Appendix C.1.c).

As part of the waste/fill global stability analyses, the waste mass was considered internally stable for this design effort. Internal waste mass stability is a function of the waste filling approach. There are numerous options available to stabilize the waste through operational methodologies, such as providing a greater soil buttress on the open 3:1 south slope. During subsequent design phases, the internal stability of the waste should be evaluated in conjunction with the waste filling plan.

7.2.3 IDF Leachate Handling System

Operational interfaces for the leachate handling system include the following:

- Coordinate with Liquid Effluent Retention Facility (LERF) for leachate hauling and removal of leachate from tanks to satisfy the 90-day accumulation period (Treatment capacities at LERF and leachate flows for critical periods should also be coordinated. See Section 5.9.2.4 for additional leachate hauling constraints.)
- Use leak detection history for leachate storage tanks, during the operation of IDF, to manage and plan for replacement of tank liner system and temporary storage required during its replacement.
- Periodic preventative inspection and maintenance for all rotating equipment should be scheduled.
- For leachate tanks floating covers, rain or snow will need to be pumped off with the manufacturer-included sump pump (mounted on side of tank). Water should not be allowed to accumulate except at the perimeter of the floating cover. Excessive water may prevent vent operation and cause mixing between precipitation water and leachate on top of the cover.
- An adequate store of critical spare electrical and mechanical parts should be maintained.
- All valves should be exercised at least annually.
- A small "contractor-type" trash pump with hose should be kept on hand that can be used to pump from the leak detection chamber within the combined sump to its inner sump.
- Periodically, test operation of the combined sump pump should be done.
- Annual testing of all leachate pumps for proper operation should be scheduled.
- Regular verification of level transducer calibration in cells should be done.
- Prior to winter months, proper operation of all heat tracing system should be checked.
- Periodic testing of all control relays, switches and contacts should be scheduled.
- Additional operational interface items will be developed, based on completion of design of the control system for the leachate handling system. This will be part of the IDF administration building design.

- Maintenance should be provided in accordance with manufacturer's recommendations.

7.3 Leakage Response Action Plan

WAC 173-303-665(9) regulations require the owner of the operator of a landfill unit to have an approved Response Action Plan (RAP) before receipt of waste. The RAP is a site-specific plan that establishes actions to be taken if leakage through the upper (primary) lining system of a landfill exceeds a certain rate. The intent of the RAP is to assure that any leachate that leaks through the primary lining system will not migrate out of the landfill into the environment.

A key element of the RAP is the ALR, a threshold value which triggers the responses described in the RAP, but below which no special actions are required. Because landfill liner systems have not yet been perfected, a small amount of leakage through the primary liner generally occurs, despite the use of best available materials, construction techniques, and QA procedures. (This leakage is collected by the LDS system and removed from the landfill.) Hence, the ALR is set at some level higher than normally expected leakage rates to serve as an indicator that the primary lining system is not functioning as expected. Exceeding the ALR may reflect serious failure of the primary lining system and indicates the need for investigation and possibly corrective action while the problem is still manageable.

This RAP has been prepared in accordance with requirements of WAC 173-303-665(9). The requirements for determining the ALR are contained in WAC 173-303-665(8) and EPA guidance document, *Action Leakage Rates for Leak Detection Systems* (EPA 1992a).

The following sections establish the ALR and discuss response actions to be taken if the ALR is exceeded.

7.3.1 Action Leakage Rate

Section 5.11 provides a detailed discussion of the analysis to determine the ALR into the LDS for the IDF. Based on this analyses, the ALR for each IDF cell is 206 gallons per acre per day, or approximately 1,800 gallons per day per cell (each cell area is approximately 8.5 acres). This value includes a factor of safety of 2 in accordance with EPA guidelines (EPA 1992b). It is also much lower than the LDS pump capacity. Details of the calculation are presented in Appendix C.10.

In accordance with WAC 173-303-665(8)(b), the flow rate used to determine if the ALR has been exceeded will be calculated as the average daily flow rate into the sump, expressed as gallons per acre per day (unless Ecology approves a different calculation). This calculation will be performed on a weekly basis during the active (operational) life of the landfill, and monthly after the landfill has been closed. Post-closure frequency may be reduced if only minimal amounts of leachate accumulate in the LDS sump. As outlined in WAC 173-303-665(4)(c)(ii), during post-closure monitoring, if the liquid level in the LDS sump stays below the pump operating level for two consecutive months, monitoring of the amount of liquid in the LDS sumps can be reduced to at least quarterly. If the liquid level in the LDS sump stays below the pump operating level for two consecutive quarters, monitoring of the amount of liquid in the LDS sumps can be reduced to at least semiannually. Pump operating level is defined as a liquid level approved by Ecology, based on pump activation level, sump dimensions, and level that minimizes head in the sump.

7.3.2 Response Actions

WAC 173-303-665(9) lists several required actions if the ALR is exceeded. In the event that the ALR is exceeded, DOE will:

- Notify Ecology in writing of the exceedance within 7 days of the determination.
- Submit a preliminary written assessment to Ecology within 14 days of the determination, as to the amount of liquids, likely sources of liquids, possible location, size, cause of any leaks, and short-term actions taken and planned.
- Determine, to the extent practicable, the location, size , and cause of any leak.

- Determine whether waste receipt should cease or be curtailed, whether any waste should be removed from the unit for inspection, repairs, or controls, and whether or not the unit should be closed.
- Determine any other short-term and longer-term actions to be taken to mitigate or stop any leaks.

Within 30 days after the notification that the ALR has been exceeded, submit to Ecology the results of the analyses specified in bullets 3, 4, and 5 of this section, the results of actions taken, and actions planned. Monthly thereafter, as long as the flow rate in the LDS exceeds the ALR, the owner or operator must submit to the regional administrator a report summarizing the results of any remedial actions taken and actions planned.

If the ALR is exceeded, the DOE will submit the required notifications to Ecology, as stated above. The EPA will also receive copies of this confirmation.

The leachate will be analyzed for chemical compounds and radionuclides. If the analytical results indicate that these constituents are present, and if the constituents can be traced to a particular type of waste stored in a known area of the landfill, then it may be possible to estimate the location of the leak. However, because the waste will meet land disposal restrictions, it will contain no free liquids and will be stabilized or solidified. In addition, the canister(s) or other type of waste package(s) may not undergo enough deterioration during the active life of the landfill to permit escape of its contents. For these reasons, it is possible that the leachate may be clean or the composition too general to indicate a specific source location.

If the source location cannot be identified, large-scale removal of the waste and operations layer to find and repair the leaking area of the liner would be one option for remediation. However, this procedure risks damaging the liner. In addition, waste would have to be handled, stored, and replaced in the landfill. Backfill would need to be removed from around the waste packages to accomplish this. If the waste packages are damaged during this process, the risk of accidental release may be high. For these reasons, large scale removal of waste and liner system materials is not considered a desirable option and will not be implemented except as a last resort.

The preferred options for remediation include covers and changes in landfill operating procedures. The preferred alternative will depend on factors such as the amount of waste already in the landfill, the rate of waste receipt, the chemistry of the leachate, the availability of other RCRA-compliant disposal facilities, and similar considerations. Hence, at this time no single approach can be selected. If the ALR is exceeded, potential options will be evaluated prior to selecting a remediation process. If necessary, an interim solution will be implemented while the evaluation and permanent remediation is performed. Examples of potential approaches include the following:

- The surface of the intermediate soil cover over the waste could be graded to direct runoff into a shallow pond. The surface would then be covered with a discardable, temporary geomembrane (e.g., 30-mil PVC or reinforced polypropylene). Precipitation water would be pumped or evaporated from the pond and would not infiltrate the waste already in the landfill. Waste packages would be placed only during periods of dry weather and stored temporarily at other times. This type of approach would also be used, if necessary, to reduce leakage during the time immediately after the ALR was exceeded, while other remediation options were being evaluated.
- If the landfill was nearly full, partial construction of the final closure cover might be an option. This would reduce infiltration into the landfill and possibly the leakage rate, if the cover were constructed over the failed area.
- A layer of low-permeability soil could be placed over the existing waste, perhaps in conjunction with a geomembrane, to create a second "primary" liner higher in the landfill. This new liner would intercept precipitation and allow its removal.

- A rigid-frame or air-supported structure could be constructed over the landfill to ensure that no infiltration occurred. Although costly, this approach might be less expensive than constructing a new landfill.

In general, the selected remediation efforts would be those that are easiest to implement, with more difficult or expensive options to be applied only if earlier approaches were not satisfactory.

8.0 REFERENCES

40 CFR 264. *Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities.*

American Water Works Association (AWWA). *Factory-Coated Bolted Steel Tanks For Water Storage.* D103-97. Denver, Colorado. 1998

Benson, C.H., and M.A. Othman. *Hydraulic Conductivity of Compacted Clay Frozen and Thawed In Situ. Journal of Geotechnical Engineering.* Vol. 119, No. 2. February 1993. p. 276.

CETCO. *Bent mat / Climax Technical Manual.* GCL manufacturer's publication. Colloid Environmental Technologies Company, Arlington Heights, Illinois. 2001.

CH2M HILL Hanford Group, Inc., *ILAW Preliminary Closure Plan for the Disposal Facility*, RPP-6911.

CH2M HILL Hanford Group, Inc., *Conceptual Design Report for ILAW Facility, Project W-520*, RPP-7908, Rev. 0, May 2001.

CH2M HILL Hanford Group, Inc., *System Specification for Immobilized Low-Activity Waste Disposal System.* Revision 3. RPP-7307.

CH2M HILL Hanford Group, Inc., *ILAW Project Definition Criteria for Integrated Disposal Facility.* Revision 1. RPP-7898.

CH2M HILL Hanford Group, Inc., *Design Loads for Tank Farm Facilities.* From TFC-ENG-STD-06, Rev A. Issued September 30, 2002.

CH2M HILL Hanford Group, Inc., *Integrated Disposal Facility Detailed Design Support.* Statement of Work, Rev 2. Req. #92859. February 18, 2003.

CH2M HILL Hanford Group, Inc., Integrated Disposal Facility Detailed Design Cell 1 Construction Quality Assurance Plan, RPP-18490, Attachment 1, Rev. 0, March 2004.

CH2M HILL Hanford Group, Inc., Integrated Disposal Facility (IDF) Detailed Design: Site Utilities Design Report, Revision 1, RPP-18515, September 2006.

Cutler-Hammer. *Consulting Application Catalog.* 12th edition. 2000.

Dames and Moore. *Geotechnical and Corrosion Investigation–Grout Vaults, Hanford, Washington.* Prepared for Kieser Engineers, Inc. October 10, 1988.

Daniel, D.E., and C. H. Benson. *Water Content-Density Criteria for Compacted Soil Liners. American Society of Civil Engineers Journal of Geotechnical Engineering.* Vol. 116, No. 12. 1990. pp. 1811-1830.

Daniel, D.E., and H.B. Scranton. *Report of 1995 Workshop on Geosynthetic Clay Liners.* EPA/600/R-96/149. June 1996.

Daniel, D.E, et al. *Laboratory Hydraulic Conductivity Testing on GCLs in Flexible Wall Permeameters. Testing and Acceptance Criteria for Geosynthetic Clay Liners.* ASTM 1308. 1997.

Design Loads for Tank Farm Facilities. TFC-ENG-STD-06, Revision A.

DOE-ID. *INEEL CERCLA Disposal Facility (ICDF) Liner Leachate Compatibility Study.* EDF-ER-278, Rev. 2. U.S. Department of Energy Idaho Operations Office, Idaho Falls, Idaho. May 2002.

DOE/RL-91-28. *Hanford Facility Dangerous Waste Permit Application, General Information Portion.* U.S. Department of Energy, Richland Operations Office, Richland, Washington. Updated periodically.

DOE/RL-97-55. *Hanford Site Environmental Management Specification.* Revision 2. U.S. Department of Energy, Washington, D.C.

DOE/RL-97-67. *Pollution Prevention and Best Management Practices Plan for State Waste Discharge Permits SR 4508, ST 4509, and ST 4510.* Revision 3. U.S. Department of Energy, Washington D.C. January 2000.

DOE/RW-0164. *Consultant Draft Site Characterization Plan Repository Location, Hanford Site, Washington.* U.S. Department of Energy, Washington, D.C.

EPA. *Action Leakage Rates for Leak Detection Systems.* EPA 530-R-92-004. Office of Solid Waste Management, Washington, D.C. January, 1992a.

EPA. *Liners and Leak Detection Systems for Hazardous Waste Land Disposal Units.* Published in the Federal Register, Vol. 57, No 19. January 29, 1992b.

EPA, *SW-846 Test Methods for Evaluating Solid Waste, Physical/Chemical Methods, Method 9090. Compatibility Test for Wastes and Membrane Liners.* U.S. Environmental Protection Agency Office of Solid Waste Management, Washington, D.C. 1992c.

Flour Hanford, Inc., *Hanford Site Solid Waste Acceptance Criteria (HSSWAC).* HNF-EP-0063, Rev. 5. Richland, Washington. 1998.

Geosynthetic Research Institute (GRI). *Determination of the Allowable Flow Rate of a Drainage Geocomposite.* Standard GC-8. Philadelphia, Pennsylvania. April 2001.

Giroud, J.P. *Equations for Calculating the Rate of Liquid Migration Through Composite Liners Due to Geomembrane Defects. Geosynthetics International.* Vol. 4, No. 3-4. 1997. pp.335-348.

Golder Associates. *Site Investigation Report, Non-Drag-Off Landfill Site, Low Level Burial Area No. 5, 200 West Area.* WHC-SD-W025-SE-001. Prepared for U.S. Department of Energy, Richland, Washington. 1989.

Golder Associates. *Geosynthetic Liner/Leachate Compatibility Testing in Support of Project W-025 Radioactive Mixed Waste Disposal Facility.* WHC-SD-W025-TRP-001. Prepared for Westinghouse Hanford Company, Richland, Washington. 1991a.

Golder Associates. *Soil Liner/Leachate Compatibility Testing in Support of Project W-025 Radioactive Mixed Waste Disposal Facility.* WHC-SD-W025-TRP-002. Prepared for Westinghouse Hanford Company, Richland, Washington. 1991b.

Golder Associates. *W025 Construction QA Report - Landfill #1.* WHC-SD-W025-RPT-001. Prepared for Westinghouse Hanford Company, Richland, Washington. 1994a.

Golder Associates. *Design Report, Project W-025, Radioactive Mixed Waste Land Disposal Facility.* Rev 1, Vol 1, SD-W025-FDR-001. 1994b.

Golder Associates. *W025 Construction QA Report - Landfill #2.* WHC-SD-W025-RPT-002. 1995.

Hewitt, R.D., and D.E. Daniel. *Hydraulic Conductivity of Geosynthetic Clay Liners After Freeze-Thaw. Journal of Geotechnical and Geoenvironmental Engineering.* Vol. 123, No. 4. April 1997. p. 305.

HNF-IP-0842 Vol. 4, Section 3.6. July 30, 2002.

IEEE. *Green Book.* Standard 142. 1991.

IEEE. *Red Book,* Standard 141, 1993.

Integrated Engineering Software, Inc., *Lighting Handbook*, 2000

Integrated Engineering Software, Inc., *Visual Analysis Version 4.0*. 2000.

Kircher, J.F., and R.E. Bowman. *Effects of Radiation on Material and Components.* Reinhold Publishing Corporation, New York, New York. 1964.

Kim, W.H., and D.E. Daniel. *Effects of Freezing on the Hydraulic Conductivity of a Compacted Clay. Journal of Geotechnical Engineering*. Vol. 118, No. 7. 1992. p. 1083.

Koerner, R.M., Y.H. Halse, and A.E. Lord, Jr. *Long-Term Durability and Aging of Geomembranes. Waste Containment.* ASTM Special Publication No. 26. American Society of Engineers, New York, New York. 1990.

Koerner, R.M. *Designing with Geosynthetics*. 4th edition. Prentice Hall, New Jersey. 1998.

Kraus, J.F., C.H. Benson, A.E. Erickson, and E.J. Chamberlain. *Freeze-Thaw and Hydraulic Conductivity of Bentonitic Barriers. Journal of Geotechnical and Geoenvironmental Engineering.* Vol. 123, No. 3. March 1997. p. 229.

LaGatta, M. D., B.T. Boardman, B.H. Cooley and D.E. Daniel. *Geosynthetic Clay Liners Subjected to Differential Settlement. Journal of Geotechnical and Geoenvironmental Engineering.* Vol. 123, No. 3. March 1997.p. 402.

Lindsey, K.A. *The Miocene to Pliocene Ringold Formation and Associated Deposits of the Ancestral Columbia River System, South-central Washington and North-central Oregon.* Open File Report 96-8. Washington State Department of Natural Resources, Division of Geology and Earth Resources, Olympia, Washington. 1996.

Mann, R.M., R.J. Pugh, P.D. Rittmann, N.W. Kline, J.A. Voogd, Y. Chen, C.R. Eiholzer, C.T. Kincaid, B.P. McGrail, A.H. Lu, G.F. Williamson, N.R. Brown, and P.E. LaMont. *Hanford Immobilized Low-Activity Tank Waste Performance Assessment.* DOE/RL-97-69. U.S. Department of Energy Richland Operations Office, Richland, Washington. 1998.

National Electric Code (NEC). Articles 220, 225, 430, 440, and 450. 2002.

Pacific Northwest National Laboratory (PNNL). *Clastic Injection Dikes of the Pasco Basin and Vicinity.* BHI-01103. Richland, Washington. Not dated.

Pacific Northwest National Laboratory (PNNL). *Selected Water Table Contour Maps and Well Hydrographs for the Hanford Reservation, 1944-1973.* BNWL-B-360. Richland, Washington. Not dated.

Pacific Northwest National Laboratory (PNNL). *Immobilized Low-Activity Waste Site Borehole 299-E17-21.* PNNL-11957. Richland, Washington. 1998.

Pacific Northwest National Laboratory (PNNL). *Geologic Data Package for 2001 Immobilized Low-Activity Waste Performance Assessment.* Rev 1. PNNL-12257. Richland, Washington. 1999.

Pacific Northwest National Laboratory (PNNL). *Revised Hydrogeology for the Suprebasalt Aquifer System, 200-East Area and Vicinity, Hanford Site.* PNNL-12261. Richland, Washington. 2000.

Pacific Northwest National Laboratory (PNNL). *Groundwater Flow and Transport Calculations Supporting the Immobilized Low-Activity Waste Disposal Facility Performance Assessment.* PNNL-13400. Richland, Washington. 2000.

Pacific Northwest National Laboratory (PNNL). *Hanford Site Groundwater Monitoring for Fiscal Year 2000.* PNNL-13404. Richland, Washington. 2001.

Pacific Northwest National Laboratory (PNNL). *Geologic and Wireline Borehole Summary from the Second ILAW Borehole (299-W24-21).* PNNL-13652. Richland, Washington. 2001.

Pacific Northwest National Laboratory (PNNL). *Hanford Site Groundwater Monitoring for Fiscal Year 2001.* PNNL-13788. M.J. Hartman, L.F. Morasch, and W.D. Webber. Richland, Washington. 2002.

Pacific Northwest National Laboratory (PNNL). *Geologic and Wireline Summaries from Fiscal Year 2002 ILAW Boreholes.* S.P. Reidel and A.M. Ho. PNNL-14029. Richland, Washington. 2002.

Pacific Northwest National Laboratory (PNNL). *Hanford Site Climatological Data Summary 2001, with Historical Data.* Prepared for the U.S. Department of Energy. May 2002.

Reidel, S.P., and K.R. Fecht. *Geologic Map of the Richland 1:100,000 Quadrangle, Washington.* Washington Division of Geology and Earth Resources Open File Report 94-8. Washington State Department of Natural Resources, Olympia, Washington. 1994.

RHO-BWI-ST-14. *Subsurface Geology of the Cold Creek Syncline.* Rockwell Hanford Operations, Richland, Washington.

Ruhl, Janice L., and David E. Daniel. *Geosynthetic Clay Liners Permeated with Chemical Solutions and Leachates. Journal of Geotechnical and Geoenvironmental Engineering.* Vol. 123, No. 3. 1997. p. 369.

Schroeder, Paul R., Tamsen S. Dozier, Paul A. Zappi, Bruce M. McEnroe, John W. Sjostrom, and R. Lee Peyton. *The Hydrologic Evaluation of Landfill Performance (HELP) Model, Engineering Documentation for Version 3.* Risk Reduction Engineering Laboratory, Office of Research and Development, U.S. Environmental Protection Agency, Cincinnati, Ohio. November 1997.

Shackelford, Charles D., Craig H. Benson, Takeshi Katsumi, Tuncer B. Edil, and L. Lin. *Evaluating the Hydraulic Conductivity of GCLs Permeated With Non-Standard Liquids. Geotextiles and Geomembranes.* Vol. 18. 2000. pp. 133-161.

Shannon & Wilson. *Final Report–Geotechnical Investigation* (River Protection Project–Waste Treatment Plant). Conducted for BNFL, Richland, Washington. May 2000.

Shannon and Wilson. *Geotechnical Report Supplement Structural Foundation Analysis.* No. 1. 22-1-01753-001 (River Protection Project–Waste Treatment Plant). Conducted for CH2M HILL Hanford Group, Inc., Richland, Washington. April 2001.

Sharma, Hari D., and Sangeeta P. Lewis. *Waste Containment Systems, Waste Stabilization, and Landfills Design and Evaluation.* John Wiley and Sons, Inc. 1994.

TRI/Environmental. *A Final Report for Laboratory Testing of Geomembrane for Waste Containment EPA Method 9090.* WHC-SD-WM-TRP-237. Prepared for Westinghouse Hanford Company, Richland, Washington. 1995.

U.S. Army Corps of Engineers. *Evaluation of Liner/Leachate Chemical Compatibility For the Environmental Restoration Disposal Facility.* Walla Walla, Washington. 1995.

U.S. Army Corps of Engineers. *Flood Hydrograph Package (HEC-1).* Hydrologic Engineering Center. Revised June 1988.

U.S. Department of Agriculture. *Urban Hydrology for Small Watersheds.* Technical Release 55. Soil Conservation Service, Engineering Division. June 1986.

U.S. Geological Survey. *Study and Interpretation of the Chemical Characteristics of Natural Water.* Water-Supply Paper 2254. Department of the Interior, Washington, D.C. 1989.

Washington Administrative Code (WAC). *Washington State Dangerous Waste Regulations.* Chapter 173-303.

Washington State Department of Ecology. State Waste Discharge Permit Number ST 4510. Issued to the U.S. Department of Energy. April 1, 1999.

Westinghouse Hanford Company (WHC). *Field Trip Guide to the Hanford Site.* WHC-MR-0391. Richland, Washington. 1991.

Westinghouse Hanford Company (WHC). *Contaminant Transport in the Unconfined Aquifer, Input to the Low-Level Tank Waste Interim Performance Assessment.* WHC-SD-WM-RPT-241. Richland, Washington. Not dated.

Westinghouse Hanford Company (WHC). High Density Polyethylene Liner Chemical Computability for Radioactive Mixed Waste Trenches, WHC-SD-WM-TI-714. Richland, Washington, 1995.

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APPENDIX 4A – SECTION 2
CRITICAL SYSTEMS TABLES & DATA SHEETS

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APPENDIX 4A – SECTION 2
CRITICAL SYSTEMS TABLES & DATA SHEETS

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02315 – Specification for Fill and Backfill

Part 1 – General

Work Included

This section describes placement and testing of fill and backfill in general areas of the site (including stockpiles).

References

The following is a list of standards, which may be referenced in this section:

	ASTM INTERNATIONAL (ASTM)
ASTM D75	Standard Practice for Sampling Aggregates
ASTM D422	Standard Test Method for Particle-Size Analysis of Soils
ASTM D698	Test Method for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft^3 (600 kN-m/m^3))
ASTM D1140	Standard Test Method for Amount of Material in Soils Finer than the No. 200 (75 micrometer) Sieve
ASTM D1556	Standard Test Method for Density and Unit Weight of Soil in Place by the Sand-Cone Method
ASTM D2216	Standard Test Method for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass.
ASTM D2922	Standard Test Methods for Density of Soil and Soil-Aggregate in Place by Nuclear Methods (Shallow Depth)
ASTM D3017	Standard Test Method for Water Content of Soil and Rock in Place by Nuclear Methods (Shallow Depth)

Definitions

Relative Compaction: Ratio, in percent, of as-compacted field dry density to laboratory maximum dry density as determined in accordance with ASTM D698.

Apply corrections for oversize material to maximum dry density.

Optimum Moisture Content: Determined in accordance with ASTM D698 specified to determine maximum dry density for relative compaction.

Prepared Ground Surface: Ground surface after completion of required demolition, clearing and grubbing, scalping of sod, stripping of topsoil, excavation to grade, and subgrade preparation.

Completed Course: A course or layer that is ready for next layer or next phase of Work.

Lift: Loose (uncompacted) layer of material.

Geosynthetics: Geotextiles, geocomposites, geosynthetic clay liner, or geomembranes.

Well-Graded: A mixture of particle sizes with no specific concentration or lack thereof of one or more sizes.

Does not define numerical value that must be placed on coefficient of uniformity, coefficient of curvature, or other specific grain size distribution parameters.

Used to define material type that, when compacted, produces a strong and relatively incompressible soil mass free from detrimental voids.

Influence Area: Area within planes sloped downward and outward at 60-degree angle from horizontal measured from:

- 1-foot outside outermost edge at base of foundations or slabs.
- 1-foot outside outermost edge at surface of roadways or shoulder.
- 0.5-foot outside exterior at spring line of pipes or culverts.

Imported Material: Materials obtained from sources offsite, suitable for specified use.

Standard Specifications: When referenced in this section, shall mean Standard Specifications for Road, Bridge, and Municipal Construction, as published by the Washington State Department of Transportation, 2002 edition, English units.

SLDS: Secondary Leak Detection System.

Permanent Stockpile: Stockpile of material that remains at the completion of construction.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Submit gradation test results for all imported materials from independent testing laboratory as specified in paragraph SOURCE QUALITY CONTROL in Part 2.

Stockpile Plan: Prior to any excavation activities associated with construction of the IDF, Construction General Contractor shall submit a Stockpile Plan for approval. The plan shall include the following information:

Scale drawing, using the project plans as a base, which shows the proposed location for stockpiles on the project site. Show all access roads around stockpiles. Address stockpile locations during construction of the IDF and permanent stockpiles, which will remain upon completion of construction activities for this project.

Stockpile layout drawings, which show the estimated location of toe of slope and top of slope for each stockpile. Drawings shall show plan and typical sections and shall be fully dimensioned.

Plan shall show how differing materials encountered during the excavation will be segregated for future use. This includes material for use as topsoil, admix base soil, and operations layer. Also, show a stockpile area for material to be used in the future as clean backfill during landfill operations by the Tank Farm Contractor.

Method by which stockpile compaction will be achieved.

Dust control for the stockpiles during active use and until grass is established.

Placing of topsoil, seeding, fertilizing, and mulching each stockpile after active use of stockpile is finished in accordance with Section 02920, RECLAMATION AND REVEGETATION.

Submittals–Approval Not Required

Information/Record (IR):

Qualifications of independent testing laboratory.

Qualifications of construction quality control personnel.

Construction quality control test reports

Sequencing and Scheduling

Complete applicable Work specified in Sections 02316, EXCAVATION, and 02319, SUBGRADE PREPARATION, prior to placing fill or backfill.

Permit Requirements

A backfill and fill permit is required for each backfill and fill work element. Construction General Contractor shall obtain from Tank Farm Contractor and post before starting backfill and fill work, as specified in Division 1, General Requirements.

Construction Quality Assurance:

The Construction General Contractor shall accommodate all CQA activities described herein and in the CPA Plan for this project. The CQA Plan is made part of these Specifications by reference.

The CQA Certifying Engineer shall determine in-place density and moisture content by any one or combination of the following methods: ASTM D2922, ASTM D3017, ASTM D1556, ASTM D2216, or other methods approved by the Construction Manager.

Testing requirements and locations will be determined by the CQA Certifying Engineer. Construction General Contractor shall cooperate with the CQA Certifying Engineer and testing work by leveling small test areas designated by the CQA Certifying Engineer. Backfill test areas at Construction General Contractor's sole expense. The CQA Certifying Engineer may have any material tested at any time, location, or elevation.

After Construction General Contractor makes repairs to any areas failing a test, the Construction General Contractor shall rerun appropriate tests, subject to the approval of the CQA Certifying Engineer, to demonstrate the area meets specifications, at the Construction General Contractor's sole expense.

The following minimum test schedule shall be assumed. Additional tests may be required as directed by CQA Certifying Engineer.

In-place density tests shall be made on the following minimum schedule:

Earth fill: One per 5,000 square feet (ft^2) per lift.

Structural Fill: One per 2,500 ft^2 per lift.

Subgrade Preparation: Four per acre.

Operations Layer (Outside Edge of Liner): One per 5,000 ft^2 per lift.

Operations Layer Material (SLDS): Two (2) per lift.

Standard Proctor (ASTM D698) laboratory density curves (five-point minimum) shall be performed for each material by the CQA Certifying Engineer. Samples of native materials used for embankment and backfill and samples of imported materials shall be taken at locations as specified by CQA Certifying Engineer.

Gradation tests (sieve analysis) shall be performed in accordance with ASTM D422 on operations layer material obtained from required excavations to demonstrate the materials meet the Specifications. Samples of operations layer material shall be taken from each 10,000 cubic yards of placed material in accordance with ASTM D75.

Part 2 – Products

Earthfill

Excavated material from required excavations and designated borrow sites, free from rocks larger than 4 inches in the greatest dimension, from roots and other organic matter, ashes, cinders, trash, debris, and other deleterious materials.

Structural Fill

Structural fill adjacent to concrete structures shall be as specified in Section 02320, TRENCH BACKFILL, for Pipe Bedding. Structural fill beneath concrete structures and beneath leachate storage tank shall conform to the requirements of Section 9-03 9(3) Crushed Surfacing-Top Course in the Standard Specifications.

Operations Layer

Meeting the requirements of earthfill above and having a maximum of 25 percent by weight passing the No. 200 U.S. sieve and a maximum particle size of 2 inches.

Drain Gravel

Material for drain gravel shall conform to the requirements of Section 9-03.12(4) of the Standard Specifications except material shall be subrounded to rounded gravel. Crushed rock and angular gravel shall not be allowed.

Crushed Surfacing

Material for crushed surfacing-base course and top course shall conform to the requirements in Section 9-03 9(3) of the Standard Specifications.

Quarry Spalls

Quarry spalls shall consist of broken stone free from segregation, seams, cracks, and other defects tending to destroy its resistance to weather and shall meet the following requirements for grading:

- Maximum Size: 8 inches
- 50 percent by weight shall be larger than 3 inches
- Minimum Size: 3/4 inch

Source Quality Control

Gradation tests performed in accordance with ASTM D422 by a qualified independent testing laboratory shall be made for imported materials on samples taken at place of production prior to shipment. Imported materials shall not be shipped without submittal approval. Samples of the finished product for gradation testing shall be taken from each 2,000 tons of prepared materials in accordance with ASTM D75. Test results shall be submitted to Construction Manager within 48 hours after sampling. Size distribution for imported quarry spalls material shall be determined in accordance with one of the methods described in ASTM D5519.

Base Soil

As specified in Section 02666, ADMIX LINER.

WATER FOR MOISTURE CONDITIONING:

See Section 02200, SITE PREPARATION, for raw water supply availability and requirements for proper compaction.

Part 3 – Execution

General

Keep placement surfaces free of water, debris, and foreign material during placement and compaction of fill and backfill materials.

Place and spread fill and backfill materials in horizontal lifts of uniform thickness as specified in paragraphs BACKFILL UNDER AND AROUND STRUCTURES and FILL, in a manner that avoids segregation, and compact each lift to specified densities prior to placing succeeding lifts. Slope lifts only where necessary to conform to final grades or as necessary to keep placement surfaces drained of water.

Do not place fill or backfill, if fill or backfill material is frozen, or if surface upon which fill or backfill is to be placed is frozen.

Tolerances:

Final Lines and Grades: Within a tolerance of 0.1-foot unless dimensions or grades are shown or specified otherwise.

Grade to establish and maintain slopes and drainage as shown. Reverse slopes are not permitted.

Settlement: Correct and repair any subsequent damage to structures, pavements, curbs, slabs, piping, and other facilities, caused by settlement of fill or backfill material.

Backfill Under and Around Structures

Under Facilities: Backfill with earthfill or structural fill, as shown on the Drawings, for each structure or facility. Place earthfill or structural fill in lifts of 6-inch maximum compacted thickness and compact each lift to minimum of 95 percent relative compaction as determined in accordance with ASTM D698.

Fill

Outside Influence Areas Beneath Structures, Slabs, Piping, and Other Facilities: Unless otherwise shown, place earthfill as follows:

- Allow for 6-inch thickness of topsoil where required.
- Maximum 8-inch thick lifts.
- Place and compact fill across full width of embankment.
- Compact to minimum 95 percent relative compaction.

Replacing Overexcavated Material

Replace excavation carried below grade lines shown as follows:

Beneath IDF Cell: Earthfill as specified herein.

Beneath Fill or Backfill: Same material as specified for overlying fill or backfill.

Beneath Structures and Roadways: Structural fill or earthfill as shown on the Drawings and specified herein.

Topsoil

Place topsoil on areas disturbed by construction and on permanent stockpile slopes in accordance with Section 02920, RECLAMATION AND REVEGETATION.

Stockpiling

Material shall be placed in permanent stockpiles as follows:

Place material in maximum 3-foot lifts and compact with a minimum four passes with earth-moving equipment. Uniformly route hauls truck traffic across the surface of each lift to aid in lift compaction.

Maximum slopes shall be 3H:1V. Minimum slopes shall be 3 percent to promote drainage.

Upper 2 feet of stockpile surface shall be placed in maximum 12-inch thick lifts and compacted to minimum 90 percent relative compaction as determined in accordance with ASTM D698.

Place 6-inch thick layer of topsoil on completed slopes in accordance with Section 02920, RECLAMATION AND REVEGETATION.

Permanent stockpiles shall be seeded, fertilized, and mulched when each stockpile is completed and as directed by the Engineer in accordance with Section 02920, RECLAMATION AND REVEGETATION.

Placing Crushed Surfacing

Place crushed surfacing base course and top course at locations shown on the Drawings. Placement shall conform to Section 4-04.3 of the Standard Specifications.

Thickness of the drain gravel on the cell floor shall be a minimum of 1.0 foot (increase drain gravel thickness in vicinity of LCRS collection and riser pipes as shown on drawings) and tolerances for the top of the drain gravel shall be grade to plus 0.2 foot.

Placing DRAIN GRAVEL and operations layer over geosynthetics:

Place material over geosynthetics as specified in Sections 02371, GEOTEXTILES; 02661, GEOMEMBRANES; and 02667, GEOSYNTHETIC CLAY LINER (GCL).

Compaction requirements for drain gravel on the cell floor, around pipes, and the sumps are specified in Section 02320, TRENCH BACKFILL.

Operations layer within lining system limits, except as specified for SLDS sump area, shall be placed in 12-inch thick lifts and track-walked into place with a minimum two passes with a Caterpillar D6M-LGP or equal. Operations layer material within SLDS sump area shall be placed in 12-inch thick lifts and compacted to 90 percent relative compaction. Operations layer placed outside edge of liner, such as for shine berm, shall be placed in maximum 8-inch thick lifts and compacted to 95 percent relative compaction.

Place material to the lines and grades shown and compact by tracking a minimum two passes with spreading equipment. Thickness of the operations layer shall be a minimum 3 feet and tolerances for top of operations layer shall be grade to plus 0.3 foot.

Quarry Spalls Placement

Quarry spalls shall be placed around the ends of stormwater pipes to provide erosion protection in accordance with the Plans and as directed by the Engineer. Quarry spalls shall be placed in such a manner that all relatively large stones are essentially in contact with each other and voids are filled with the finer materials to provide a well graded compact mass. Finished surface shall be free from irregularities. The stone shall be dumped on the ground in a manner that will ensure the stone attains its specified thickness in one operation. When dumping or placing, care shall be used to avoid damaging the underlying material. Stone shall not be dumped from height greater than 12 inches above surface. Material placement shall be started from the bottom of the installation, working toward edges. Geotextile damaged during the placement of quarry spalls shall be repaired at Construction General Contractor's sole expense.

Construction Quality Control

The Construction General Contractor shall perform in-place density and moisture content tests with own qualified personnel or with a qualified independent testing laboratory as specified in paragraph CONSTRUCTION QUALITY ASSURANCE, to be observed by the Construction Manager, on the following minimum schedule:

Material Placed by Stockpile (Upper 2 Feet): One per 10,000 ft^2 per lift.

Construction General Contractor shall submit qualifications of personnel or independent testing laboratory that will perform construction quality control.

END OF SECTION 0231

02319 – Specification for Subgrade Preparation

Part 1 – General

Work Included

This section describes requirements for preparation of subgrades in areas to receive fill.

References

The following is a list of standards, which may be referenced in this section:

ASTM INTERNATIONAL (ASTM)

ASTM D698	Test Method for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft^3 (600 kN-m/m^3))

Definitions

Optimum Moisture Content: As defined in Section 02315, FILL AND BACKFILL.

Prepared Ground Surface: Ground surface after completion of clearing and grubbing, scalping of sod, stripping of topsoil, excavation to grade, and scarification and compaction of subgrade.

Relative Compaction: As defined in Section 02315, FILL AND BACKFILL.

SLDS: Secondary Leak Detection System.

Subgrade: Layer of existing soil after completion of excavation to grade prior to placement of fill, roadway structure, or base for floor slab.

Proof Rolling: Testing of subgrade as specified herein to identify soft or loose zones requiring correction.

Sequencing and Scheduling

Complete applicable Work specified in Section 02316, EXCAVATION, prior to subgrade preparation.

Construction Quality Assurance

The CQA Certifying Engineer shall determine in-place density and moisture for subgrade preparation as specified in Section 02315, FILL AND BACKFILL, except for prepared subgrade for admix on sideslopes.

CQA requirements for geomembrane subgrade preparation are specified in Section 02661, GEOMEMBRANES.

Part 2 – Products (Not Used)

Part 3 – Execution

General

Keep subgrade free of water, debris, and foreign matter during compaction or proof rolling.

Bring subgrade to proper grade and cross-section as shown on the Drawings, and uniformly compact surface.

Maintain prepared ground surface in finished condition until next course is placed.

Prepared Subgrade for Roadway, Embankment, and Structures

After completion of excavation and prior to foundation, road fill, structural fill, or embankment construction, compact prepared subgrade to 95 percent relative compaction. Scarify and moisture condition subgrade soil as required to achieve specified compaction.

If soft or loose zones are found, correct as specified herein. Proof-roll subgrade with a fully loaded dump truck or equal to detect soft or loose subgrade or unsuitable material.

Landfill Bottom Floor Prepared Subgrade For Admix Liner

Prior to admix liner placement, subgrade shall be back bladed to remove loose soil. Low spots or erosion rills shall be backfilled with structural fill as specified herein. Compact prepared subgrade to 95 percent relative compaction. Scarify and moisture condition subgrade soil as required to achieve specified compaction. If soft or loose zones are found, correct as specified herein. Proof-roll subgrade with a vibratory drum roller or equal to detect soft or loose subgrade or unsuitable material.

Landfill Sideslope (3H: 1V) Prepared Subgrade For Admix Liner

Prior to admix placement, the subgrade shall be back bladed to remove all loose material produced by trimming operations. Low spots or erosion rills shall be backfilled with structural fill as specified herein. The trimmed surface shall be watered so that moisture penetrates a minimum of 3 inches into the subgrade. The trimmed and watered surface shall be track-walked by D6-LGP dozer or equivalent with a minimum 4 passes to produce a firm and stable subgrade. Visual monitoring (no in-place density testing is required) of the subgrade preparation on sideslopes will be performed by the CQA Certifying Engineer.

Prepared Subgrade for Geomembrane (Secondary And SLDS) And Secondary GCl

At completion of SLDS excavation and grading (SLDS geomembrane) or admix liner placement (secondary geomembrane and GCL), prepare the subgrade surface for geomembrane or GCL placement. The surface shall not have holes, depressions more than 1 inch in a 12-inch width, nor protrusions extending above the surface more than 1/2 inch. Roll surface with smooth-drum roller to form a firm stable base. Allow for leachate piping and sumps or features as shown on the Drawings.

Correction

Soft or Loose Subgrade:

Adjust moisture content and compact to meet density requirements, or

Over excavate and replace with suitable material from the excavation, as specified in Section 02315, FILL AND BACKFILL.

Unsuitable Material: Over excavate and replace with suitable material from the excavation, as specified in Section 02315, FILL AND BACKFILL. Dispose of unsuitable material excavation in accordance with Article DISPOSAL OF SPOIL in Section 02316, EXCAVATION.

END OF SECTION 02319

02371 – Table 1. Required Geotextile Properties

Value[a]

Property	Unit	Type 1	Type 2	Test Method
Mass/Unit Area	oz/yd^2	6.0[b]	12.0[b]	ASTM D5261 or D3776
Apparent Opening	U.S. Sieve	70 max opening	—	ASTM D4751
Size[b]		100 min opening		
Grab Strength	lb	140	300	ASTM D4632
Trapezoidal Tear				
Strength	lb	70		110
ASTM D4533				
Puncture Strength	lb	65	135	ASTM D4833
Permittivity	sec^{-1}	1.2	—	ASTM D4491
UV Resistance % strength	(500 hours)	retained 70	70	ASTM D4355

Notes:

[a]All values are minimum average values, except as noted.[b] Nominal values.

02373 – Table 1. Required Geonet Properties

Property	Qualifier	Unit	Value	Test
Polymer Composition	Minimum	% polyethylene	95	---
Resin Specific Gravity	Minimum	N/A	0.92	ASTM D1505
Carbon Black Content	Range	%	2 - 3	ASTM D1603 or D4218
Nominal Thickness	MARV	mils	250	ASTM D1777 or D5199
Nominal Transmissivity(1)	MARV	m^2/sec	3×10^{-3}	ASTM D4716

02373 – Table 2. Required CDN Properties

Property	Qualifier	Unit	Value	Test
Ply Adhesion	ARV	lb/in	1.0	ASTM D413 or GRI-GC7
Transmissivity(1)	MARV	m^2/sec	5×10^{-4}	ASTM D4716

Notes:
MARV = Minimum Average Roll Value.
ARV = Average Roll Value.

(1)The design transmissivity is the hydraulic transmissivity of the CDN measured using water at 70 degrees F ±3 degrees F with a hydraulic gradient of 0.1, under the compressive stress of 10,000 psf. Transmissivity value shall be measured between two steel plates 15 minutes after application of the confining stress in the machine direction.

02661 – Table 1. Required Geomembrane Properties 60-Mil Textured HDPE

Specified Property	Qualifier	Unit	Value	Test Method
Physical Properties				
Thickness	min. avg. value	mils	60	ASTM D5994
	minimum		54	ASTM D5994
Specific Gravity	minimum	`N/A	0.932	ASTM D1505
Melt Index	range	g/10 min	<1.1	ASTM D1238 condition 190/2.16
Asperity	min avg. value[1]	mils	10	GRI-GM12
Mechanical Properties				
Tensile Properties	(each direction)		(Type IV)	ASTM D638
Strength at yield	min. avg. value	lb/in	120	
Elongation at yield[2]	min. avg. value	%		12
Tear Resistance	min. avg. value	lb	42	ASTM D1004
Puncture Resistance	min. avg. value	lb	80	ASTM D4833
Carbon Black Content	Range	%	2-3	ASTM D1603 or D4218
Carbon Black Dispersion	Minimum[1] 8 of 10	category	1 or 2	ASTM D5596
Environmental Stress				
Crack	minimum[3]	hrs	200	ASTM D5397

[1] Of 10 readings, 8 out of 10 must be greater or equal to 7 mils, and lowest individual reading must be greater or equal to 5 mils. Provide data for both sides of textured geomembrane.

[2] Yield elongation is calculated using a gauge length of 1.3 inches.

[3] Minimum = mean minus 3 standard deviations from documented manufacturer's quality control (MQC) testing.

02661 – Table 2. Required Seam Properties HDPE Geomembranes

Property	Qualifier	Unit	Specified Value	Test Method
Shear Strength[1]	minimum	lb/in width	90% of tensile strength at yield as listed in tables in this section	ASTM D6392
Peel Adhesion	minimum	lb/in width	60% of tensile strength at yield as listed in tables in this section and FTB[2]	ASTM D6392

[1]Also called "Bonded Seam Strength".

[2]FTB = Film Tear Bond (failure occurs through intact geomembrane, not through seam).

Geomembrane Installer's Certification of Subsurface Acceptability

The geomembrane installer, _____
for the Integrated Disposal Facility (IDF), hereby certifies that the supporting prepared subgrade surfaces are acceptable for installation of the HDPE geomembrane lining system, the undersigned having personally inspected the condition of the constructed surfaces. This certification is for the areas shown on Attachment or defined as follows:

_____

_____

_____

_____

The condition of the supporting surfaces in the defined area meets or exceeds the minimum requirements for installation of the geomembrane.

Signed:_____	Signed:_____
Geomembrane Installer	Construction General Contractor
_____	_____
Date Signed	Date Signed

11305-01 – Sump Pump Data Sheet

Tag Numbers: 219A-LH-P-207, 219E-LH-P-207

Pump Locations and I.D.: Buildings 219A and 219E, Combined Sump Pump

Manufacturer and Model Number: (1) Hydromatic Pump Co., Model SB3S
(2) Barnes
(3) Or approved equal

Service Conditions

Liquid Pumped (Material and Percent): Leachate from low-level radioactive waste disposal facility and rain/snow melt

Pumping Temperature (Fahrenheit): Normal: 55 Max: 130 Min: 27

Specific Gravity at 60 Degrees F: 1.0 Viscosity Range: N/A pH: 5.0-9.3

Abrasive (Y/N): (fine/coarse soil particles) Possible Scale Buildup (Y/N): Y

Total suspended solids (mg/L): 200 (estimated)

Largest diameter solid pump can pass (inches): 0.5

Performance Requirements

Capacity (US gpm): Rated: 250
Total Dynamic Head (Ft): Rated: 19
Maximum Shutoff Pressure (Ft): 50
Min. Rated Pump Hydraulic Efficiency at Rated Capacity (%): 50
Max. Pump Speed at Rated Capacity (rpm): 1,750
Constant (Y/N): Y Adjustable (Y/N): N

Design and Materials

Pump Type: Heavy-Duty Nonclog (Y/N) Y

Volute Material: Cast Iron ASTM A48

Pump Casing Material: Cast Iron ASTM A48

Motor Housing Material: Cast Iron ASTM A48

Induction Drive Motor

Horsepower: 0.30 Voltage: 460 Phase: 3 Speed (rpm): 1,750

Service Factor: 1.15 Inverter Duty (Y/N): N

Motor nameplate horsepower shall not be exceeded at any head-capacity point on the pump curve.

Enclosure: Explosion-proof, submersible, Class 1, Div. 1 or Div. 2, Groups C and D

11305-02 – Sump Pump Data Sheet

Tag Numbers: 219A-LH-P-205, 219E-LH-P-205

Pump Locations and I.D.: Buildings 219A and 219E, Floor Sump

Manufacturer and Model Number: (1) Hydromatic Pump Co.

(2) Barnes

Service Conditions

Liquid Pumped: Leachate from low-level radioactive waste disposal facility

Pumping Temperature (Fahrenheit): Normal: 55 Max: 130 Min: 27

Specific Gravity at 60 Degrees F: 1.0 Viscosity Range: N/A pH: 5-9.3

Abrasive (Y/N): Y (fine/coarse soil particles) Possible Scale Buildup (Y/N): Y

Total suspended solids (mg/L): 200 (estimated)

Largest diameter solid pump can pass (inches): 0.5

Performance Requirements

Capacity (US gpm): Rated: 28

Total Dynamic Head (Ft): Rated: 14

Maximum Shutoff Pressure (Ft): 30

Min. Rated Pump Hydraulic Efficiency at Rated Capacity (%): 45

Max. Pump Speed at Rated Capacity (rpm): 1,750

Constant (Y/N): _____ (Y/N): _____

Design and Materials

Pump Type: Heavy-Duty Nonclog (Y/N): Y

Volute Material: Cast Iron ASTM A48

Pump Casing Material: Cast Iron ASTM A48

Motor Housing Material: Cast Iron ASTM A48

Induction Drive Motor

Horsepower: 0.30 Voltage: 460 Phase: 3 Speed (rpm): 1,750

Service Factor: 1.15 Inverter Duty (Y/N): N

Motor nameplate horsepower shall not be exceeded at any head-capacity point on the pump curve.

Enclosure: Submersible

11306-01 – Leachate Pump Data Sheet

Tag Numbers: 219A-LH-P-202, 219E-LH-P-202

Pump Locations and I.D.: Cell 1 LCRS Sump, Low Flow
Cell 2 LCRS Sump, Low Flow

Manufacturer and Model Number: (1) EPG Companies; Model WSD 3-3
(2) Or equal

Service Conditions

Liquid Pumped (Material and Percent): Leachate from low-level radioactive waste landfill

Pumping Temperature (Fahrenheit): Normal: 55 F Max: 130 F Min: 27 F

Specific Gravity at 60 Degrees F: 1.0 Viscosity Range: NA pH: 5.0-9.3

Abrasive (Y/N): Y (infrequent fine soil particles) Possible Scale Buildup (Y/N): Y

Total Suspended Solids (mg/l): 200 (estimated)

Performance Requirements at Primary Design Point

Capacity (US gpm): Rated: 13

Total Dynamic Head (Ft): Rated: 66

Min. Hydraulic Efficiency (%): 60

Maximum Shutoff Pressure (Ft): 90

Max. Pump Speed at Design Point (rpm): 3,450

Constant (Y/N): Y Adjustable (Y/N): N

Design and Materials

Design: Wheeled enclosure frame Back Pullout (Y/N): Y

Discharge Orientation: Center

Casing Materials: Type 304 SST

Case Wear Ring (Y/N): NA Material: NA

Impeller: Type: Closed Material: Type 304 SST

Impeller Wear Ring (Y/N): Y Material: E-Glide (engineered plastic) or equal

Shaft Material: Type 304 SST Shaft Sleeve Material: E-Glide or equal

Shaft Seal: Y Ring Material: E-Glide or equal Lubrication: Fluid

AFBMA B-10 Bearing Life (Hrs): NA Lubrication: NA

Drive Type: Direct Coupled

Induction Drive Motor

Horsepower: 0.5 Voltage: 460 Phase: 3

Speed (rpm): 3,450

Service Factor: 1.15 Inverter Duty (Y/N): N

Motor nameplate horsepower shall not be exceeded at any head-capacity point on the pump curve.

Enclosure: Submersible

11306-02 – Leachate Pump Data Sheet

Tag Numbers: 219A-LH-P-203, 219E-LH-P-203

Pump Location and I.D.: Cell 1 LCRS Sump, High Flow
Cell 2 LCRS Sump, High Flow

Manufacturer and Model Number: (1) EPG Companies; Model WSD 30-4
(2) Or equal

Service Conditions

Liquid Pumped (Material and Percent): Leachate from low-level radioactive waste landfill

Pumping Temperature (Fahrenheit): Normal: 55 F Max: 130 F Min: 27 F

Specific Gravity at 60 Degrees F: 1.0 Viscosity Range: NA pH: 5.0-9.3

Abrasive (Y/N): Y (infrequent fine soil particles) Possible Scale Buildup (Y/N): Y

Total Suspended Solids (mg/l): 200 (estimated)

Performance Requirements at Primary Design Point

Capacity (US gpm): Rated: 155

Total Dynamic Head (Ft): Rated: 118

Min. Hydraulic Efficiency (%): 60

Maximum Shutoff Pressure (Ft): 208

Max. Pump Speed at Design Point (rpm): 3,450

Constant (Y/N): Y Adjustable (Y/N): N

Design and Materials

Design: Wheeled enclosure frame (Y/N): Y

Discharge Orientation: Center

Casing Materials: Type 304 SST

Case Wear Ring (Y/N): NA Material: NA

Impeller: Type: Closed Material: Type 304 SST

Impeller Wear Ring (Y/N): Y Material: E-Glide (engineered plastic), or equal

Shaft Material: Type 304 SST Shaft Sleeve Material: E-Glide (engineered plastic), or equal

Shaft Seal: Y Ring Material: E-Glide or equal Lubrication: Fluid

AFBMA B-10 Bearing Life (Hrs): NA Lubrication: NA

Drive Type: Direct Coupled

Induction Drive Motor

Horsepower: 7.5 Voltage: 460 Phase: 3

Speed (rpm): 3,450

Service Factor: 1.15 Inverter Duty (Y/N): N

Motor nameplate horsepower shall not be exceeded at any head-capacity point on the pump curve.

Enclosure: Submersible

11306-03 – Leachate Pump Data Sheet

Tag Numbers: 219A-LH-P-204, 219E-LH-P-204

Pump Locations and I.D.: Cell 1 LDS Sump
Cell 2 LDS Sump

Manufacturer and Model Number: (1) EPG Companies; Model WSD 1.5-3
(2) Or equal

Service Conditions

Liquid Pumped (Material and Percent): Leachate from low-level radioactive waste landfill

Pumping Temperature (Fahrenheit): Normal: 55 F Max: 130 F Min: 27 F

Specific Gravity at 60 Degrees F: 1.0 Viscosity Range: NA pH: 5.0-9.3

Abrasive (Y/N) Y (infrequent fine soil particles) Possible Scale Buildup (Y/N): Y

Total Suspended Solids (mg/l): 200 (estimated)

Performance Requirements at Primary Design Point

Capacity (US gpm): Rated: 4

Total Dynamic Head (Ft): Rated: 65

Min. Hydraulic Efficiency (%): 60

Maximum Shutoff Pressure (Ft): 80

Max. Pump Speed at Design Point (rpm): 3,450

Constant (Y/N): Y Adjustable (Y/N): N

Design and Materials

Design: Wheeled enclosure frame Back Pullout (Y/N): Y

Discharge Orientation: Center

Casing Materials: Type 304 SST

Case Wear Ring (Y/N): NA Material: NA

Impeller: Type: Closed Material: Type 304 SST

Impeller Wear Ring (Y/N): Y Material: E-Glide (engineered plastic) or equal

Shaft Material: Type 304 SST Shaft Sleeve Material: E-Glide or equal

Shaft Seal: Y Ring Material: E-Glide or equal Lubrication: Fluid

AFBMA B-10 Bearing Life (Hrs): NA Lubrication: NA

Drive Type: Direct Coupled

Induction Drive Motor

Horsepower: 0.5 Voltage: 460 Phase: 3

Speed (rpm): 3,450

Service Factor: 1.15 Inverter Duty (Y/N): N

Motor nameplate horsepower shall not be exceeded at any head-capacity point on the pump curve.

Enclosure: Submersible

11312-01 – Horizontal End Suction Centrifugal Pump Data Sheet

Tag Numbers: _____

Pump Name: _____

Manufacturer and Model Number: (1) _____

(2) _____

Service Conditions:

Liquid Pumped (Material and Percent): _____

Pumping Temperature (Fahrenheit): Normal: 55 Max: 130 Min: 27

Specific Gravity at 60 Degrees F: 1.0 Viscosity Range: N/A pH: 5.0-9.3

Abrasive (Y/N): Y (fine soil particles) Possible Scale Buildup (Y/N): Y

Total suspended solids (mg/L): 200 (estimated)

Largest diameter solid pump can pass (inches): .25

Performance Requirements at Primary Design Point

Capacity (US gpm): Rated: 250

Total Dynamic Head (Ft): Rated: 25

Min. Hydraulic Efficiency (%): 75

Maximum Shutoff Pressure (Ft): 40

Max. Pump Speed at Design Point (rpm): 1,750

Constant (Y/N): Y Adjustable (Y/N): N

Design and Materials

ANSI (Y/N): Y Standard (Y/N): Y Design: Frame-mounted (Y/N): Y

Close-Coupled Casing (Y/N): N Back Pullout (Y/N): Y

Discharge Orientation: 12:00 Rotation (view from end coupling): CW

Shaft Seal: Packing (Y/N): N

Mechanical (Y/N): _____

Lubrication: Process Water

Drive Type: Direct-Coupled: _____ Belt _____ Adjustable Speed _____

Induction Drive Motor

Horsepower: _____ Voltage: _____ Phase: _____ Speed (rpm): 1,750

Service Factor: 1.15 Inverter Duty (Y/N): _____

Motor nameplate horsepower shall not be exceeded at any head-capacity point on the pump curve.

Enclosure: Totally enclosed fan cooled

Mounting Type: Horizontal _____ Nonreverse Ratchet (Y/N): _____

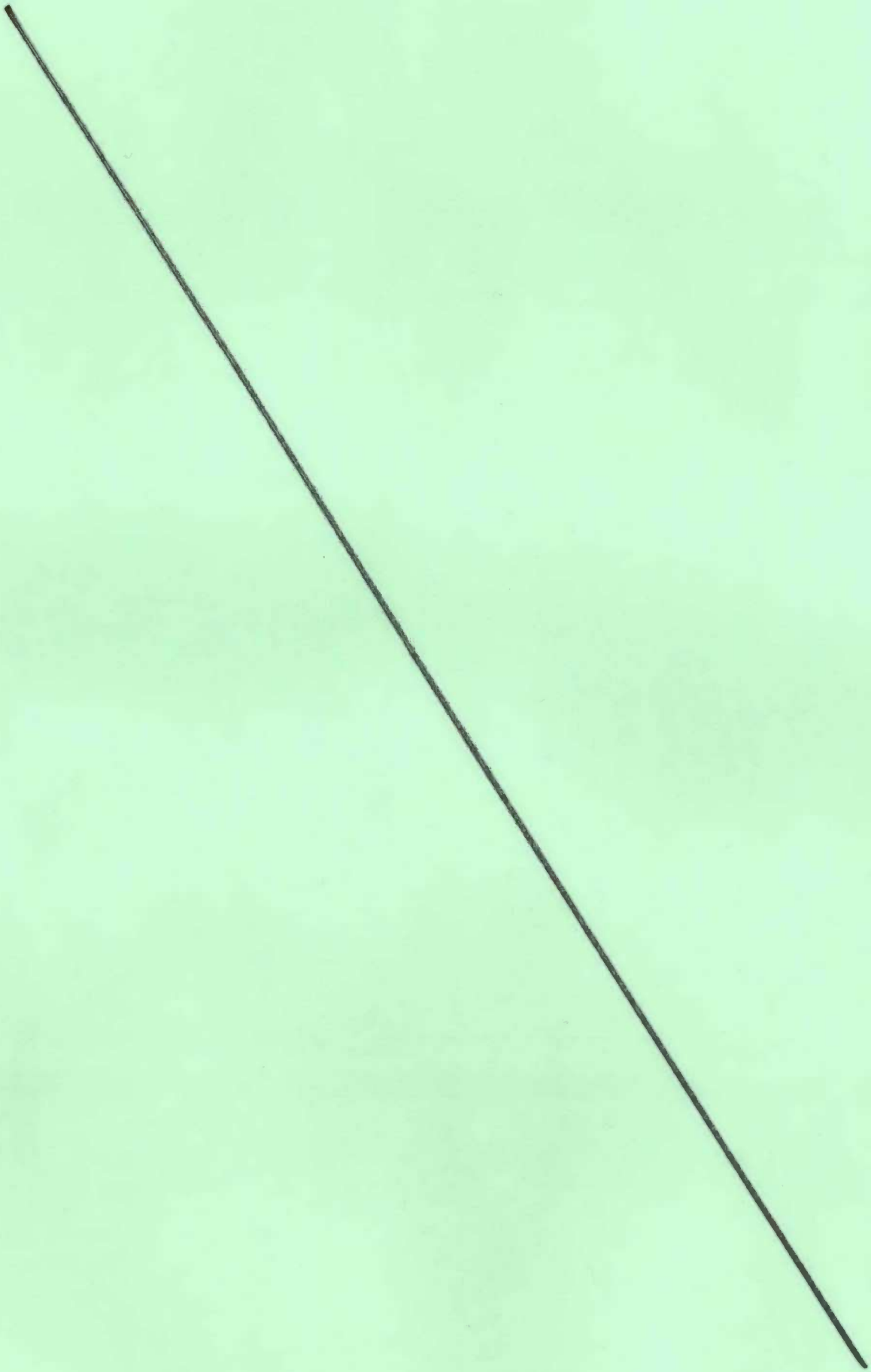
Testing

Pump Tests: Factory Functional (Y/N): _____ Field Performance (Y/N): N

Factory Hydrostatic Casing Pressure Test (Y/N): _____

Field Functional (Y/N): _____ Field Performance (Y/N): _____

Field Vibration (Y/N): N



APPENDIX 4A - SECTION 3
CRITICAL SYSTEMS DESIGN DRAWINGS

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APPENDIX 4A - SECTION 3
CRITICAL SYSTEMS DESIGN DRAWINGS

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Drawings redacted in electronic version. These documents may be viewed by appointment (509-372-7920) at the Washington State Department of Ecology Richland Office Library, 3100 Port of Benton Boulevard, Richland, Washington.

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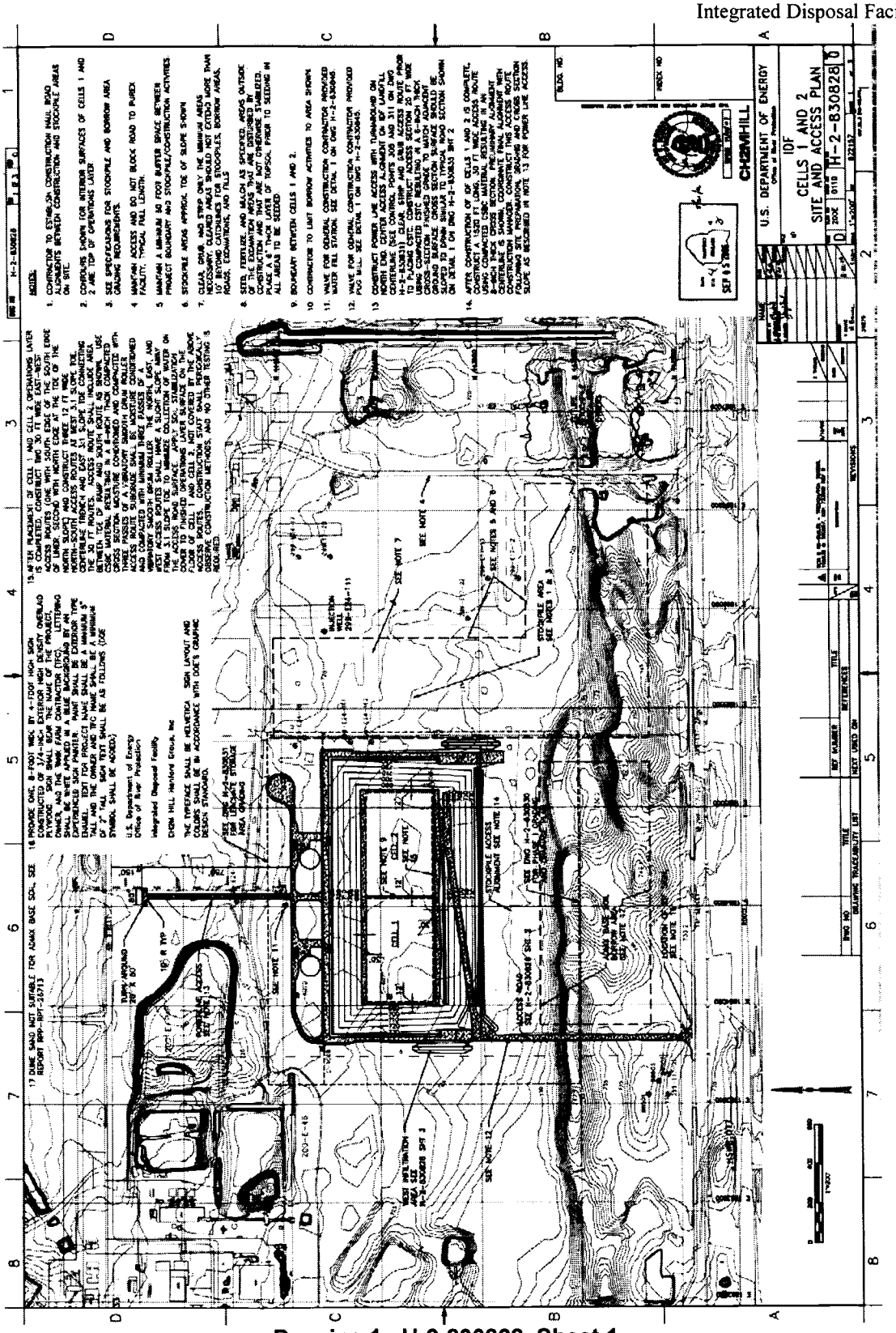
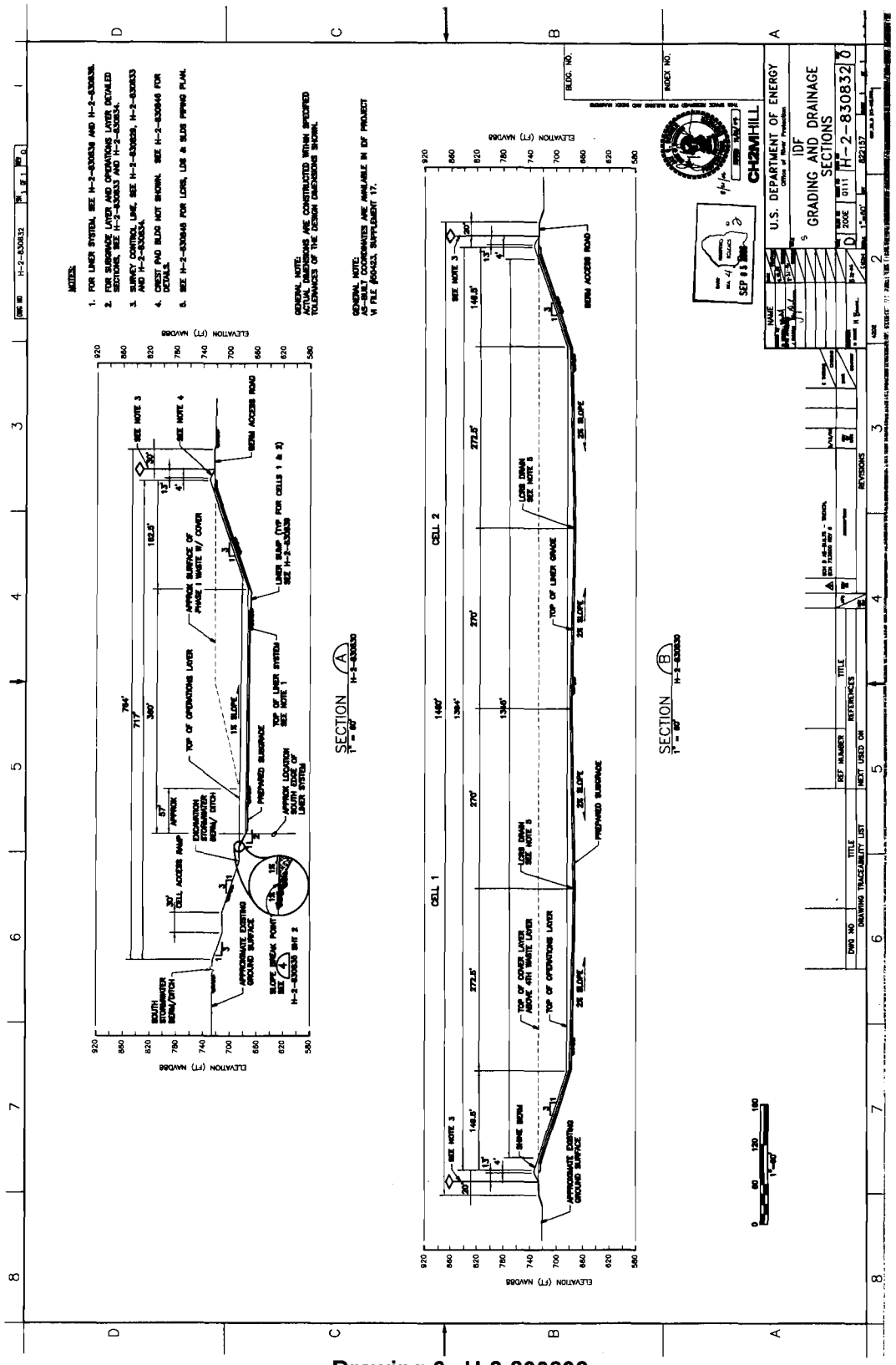
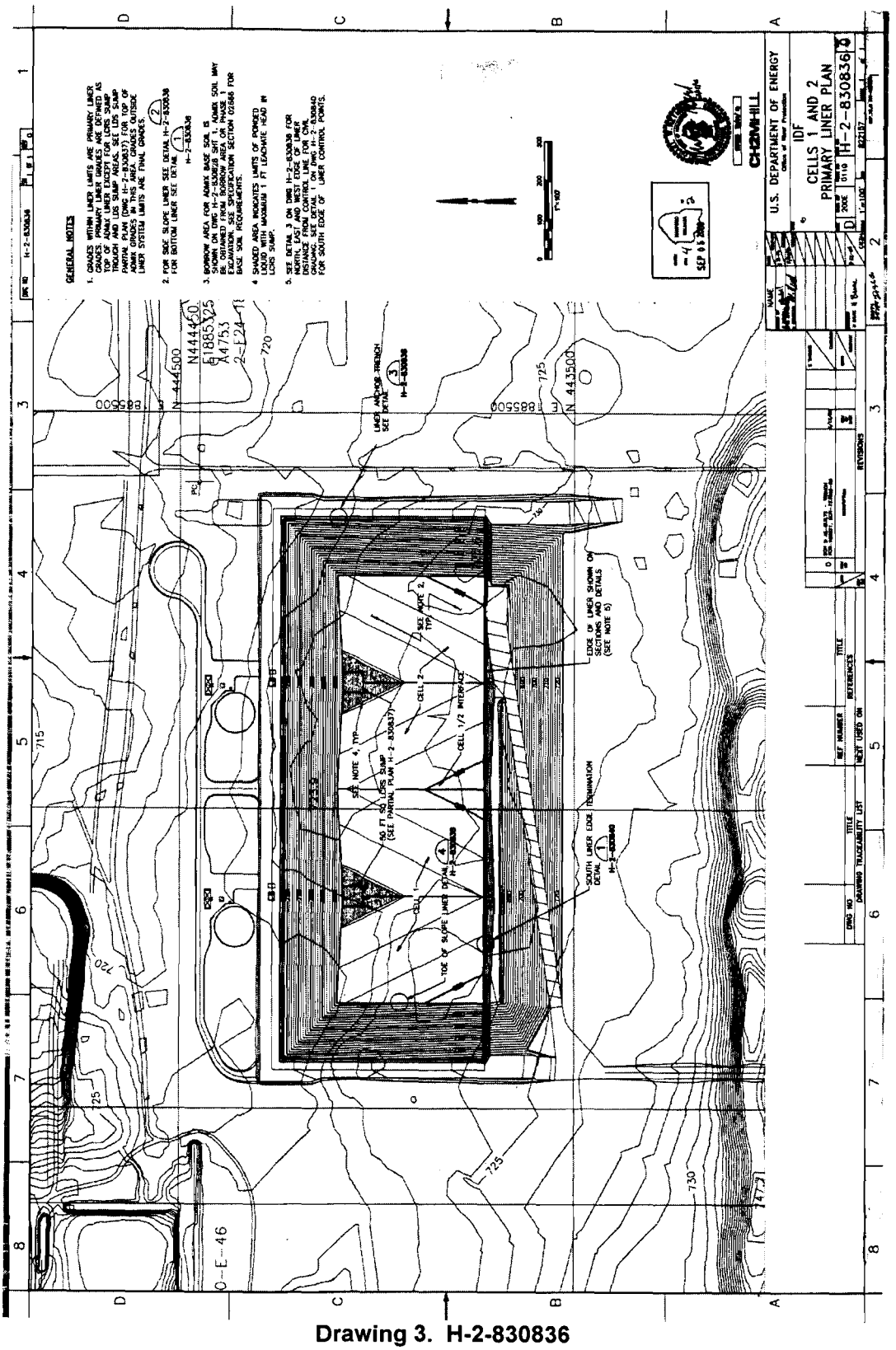


Drawing 1. H-2-830828, Sheet 1



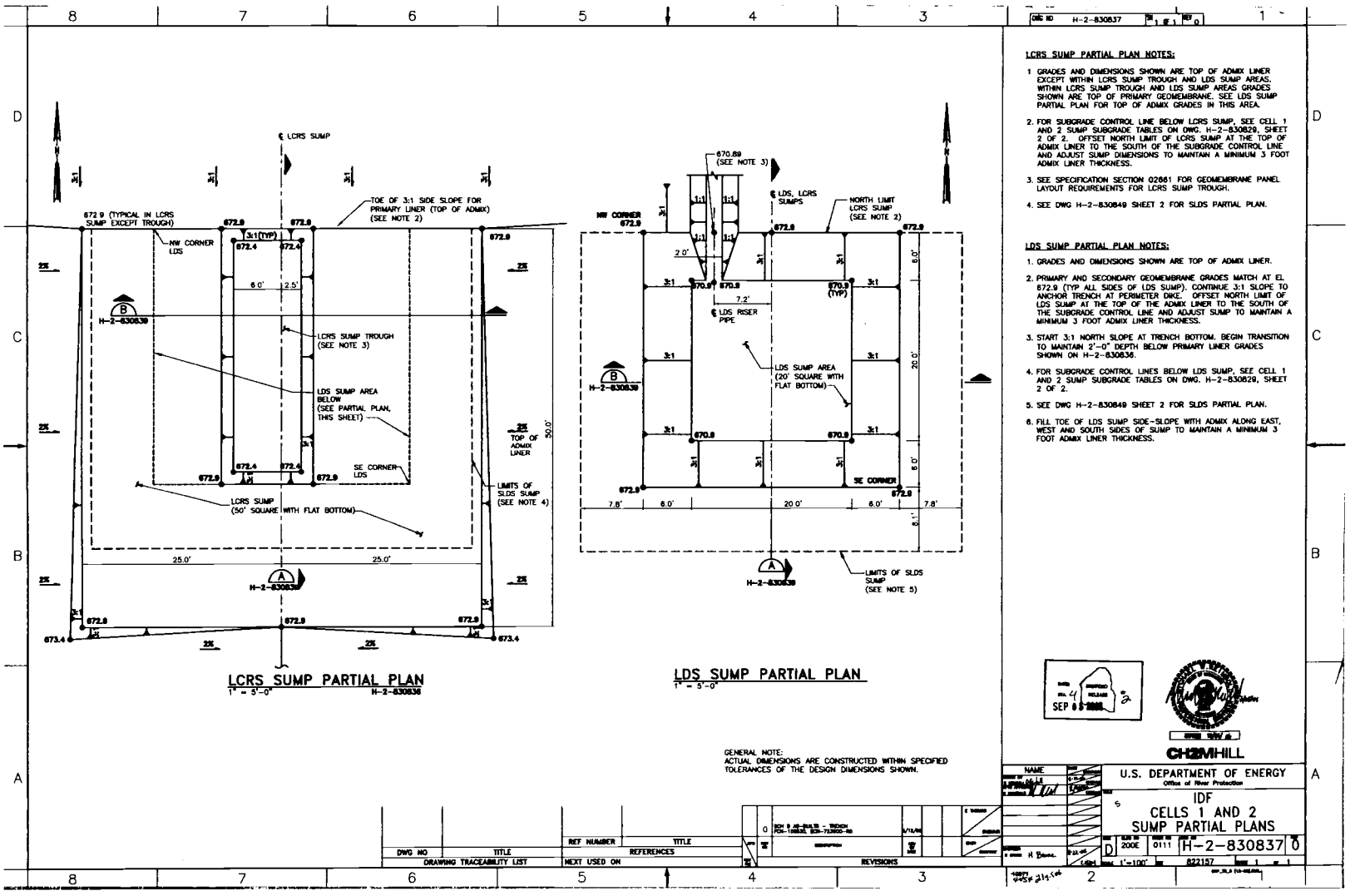
Drawing 2. H-2-830832



Drawing 3. H-2-830836

Appendix 4A.3.8

Drawing 4. H-2-830837



LCRS SUMP PARTIAL PLAN NOTES:

1. GRADES AND DIMENSIONS SHOWN ARE TOP OF ADMIX LINER EXCEPT WITHIN LCRS SUMP TROUGH AND LDS SUMP AREAS. WITHIN LCRS SUMP TROUGH AND LDS SUMP AREAS GRADES SHOWN ARE TOP OF PRIMARY GEOMEMBRANE. SEE LDS SUMP PARTIAL PLAN FOR TOP OF ADMIX GRADES IN THIS AREA.
2. FOR SUBGRADE CONTROL LINE BELOW LCRS SUMP, SEE CELL 1 AND 2 SUMP SUBGRADE TABLES ON DWG. H-2-830829, SHEET 2 OF 2. OFFSET NORTH LIMIT OF LCRS SUMP AT THE TOP OF ADMIX LINER TO THE SOUTH OF THE SUBGRADE CONTROL LINE AND ADJUST SUMP DIMENSIONS TO MAINTAIN A MINIMUM 3 FOOT ADMIX LINER THICKNESS.
3. SEE SPECIFICATION SECTION 02661 FOR GEOMEMBRANE PANEL LAYOUT REQUIREMENTS FOR LCRS SUMP TROUGH.
4. SEE DWG H-2-830849 SHEET 2 FOR SLDS PARTIAL PLAN.

LDS SUMP PARTIAL PLAN NOTES:

1. GRADES AND DIMENSIONS SHOWN ARE TOP OF ADMIX LINER.
2. PRIMARY AND SECONDARY GEOMEMBRANE GRADES MATCH AT EL. 672.9 (TYP ALL SIDES OF LDS SUMP). CONTINUE 3:1 SLOPE TO ANCHOR TRENCH AT PERIMETER DIKE. OFFSET NORTH LIMIT OF LDS SUMP AT THE TOP OF THE ADMIX LINER TO THE SOUTH OF THE SUBGRADE CONTROL LINE AND ADJUST SUMP TO MAINTAIN A MINIMUM 3 FOOT ADMIX LINER THICKNESS.
3. START 3:1 NORTH SLOPE AT TRENCH BOTTOM. BEGIN TRANSITION TO MAINTAIN 2'-0" DEPTH BELOW PRIMARY LINER GRADES SHOWN ON H-2-830836.
4. FOR SUBGRADE CONTROL LINES BELOW LDS SUMP, SEE CELL 1 AND 2 SUMP SUBGRADE TABLES ON DWG. H-2-830829, SHEET 2 OF 2.
5. SEE DWG H-2-830849 SHEET 2 FOR SLDS PARTIAL PLAN.
6. FILL TOE OF LDS SUMP SIDE-SLOPE WITH ADMIX ALONG EAST, WEST AND SOUTH SIDES OF SUMP TO MAINTAIN A MINIMUM 3 FOOT ADMIX LINER THICKNESS.

GENERAL NOTE:
ACTUAL DIMENSIONS ARE CONSTRUCTED WITHIN SPECIFIED TOLERANCES OF THE DESIGN DIMENSIONS SHOWN.

DRAWING TRACEABILITY LIST

DWG NO	TITLE	REF NUMBER	TITLE
		NEXT USED ON	REFERENCES

REVISIONS

NO	DESCRIPTION	BY	DATE
0	SCH 8 AS-BUILTS - TRENCH FOR 198832, SCH-713000-80		5/15/00

SEP 0 8 2000

CH2MHILL

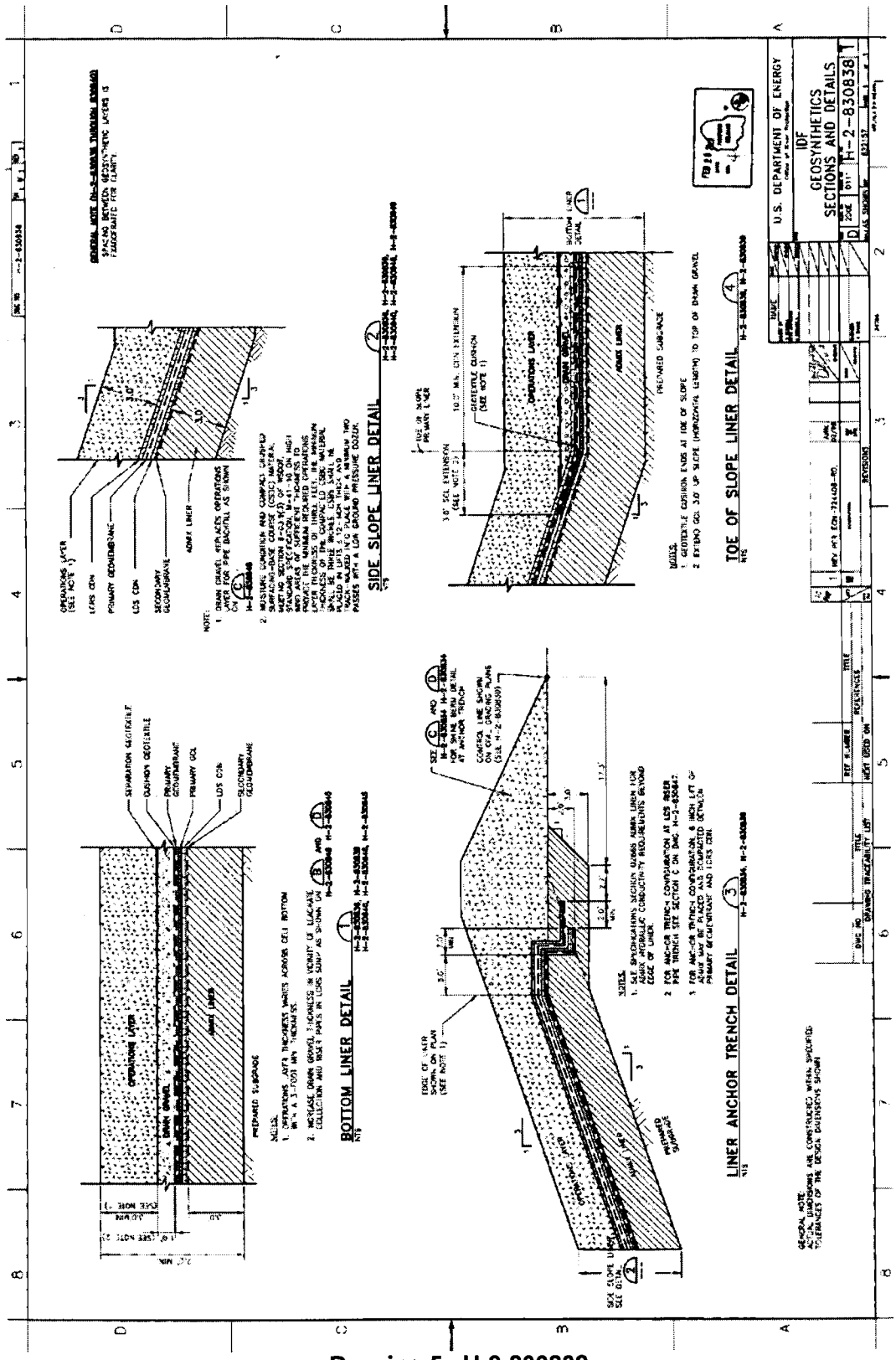
U.S. DEPARTMENT OF ENERGY
Office of River Protection

IDF
CELLS 1 AND 2
SUMP PARTIAL PLANS

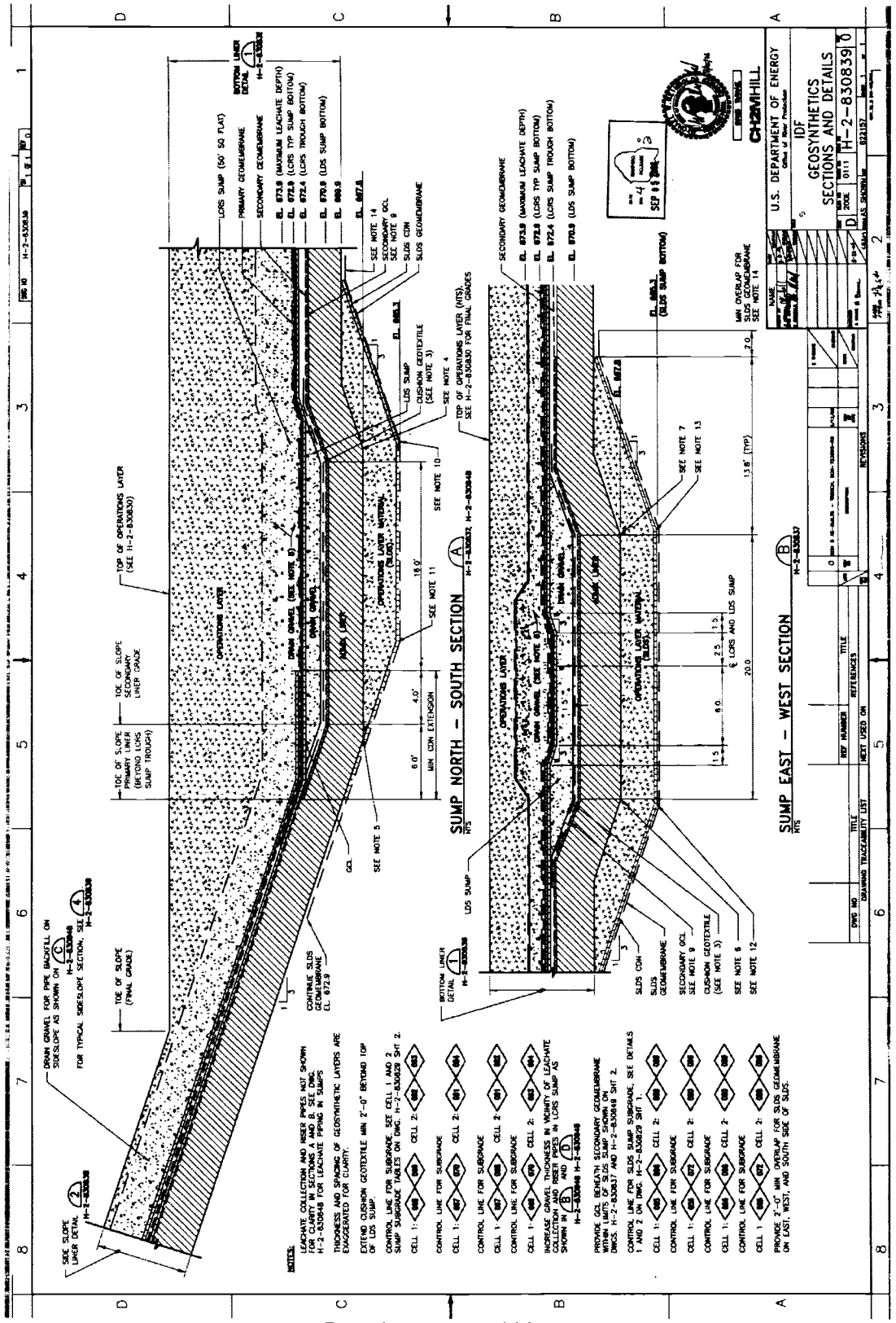
D | 200E | 0111 | H-2-830837 | 0

822157

Integrated Disposal Facility
WA7890008967

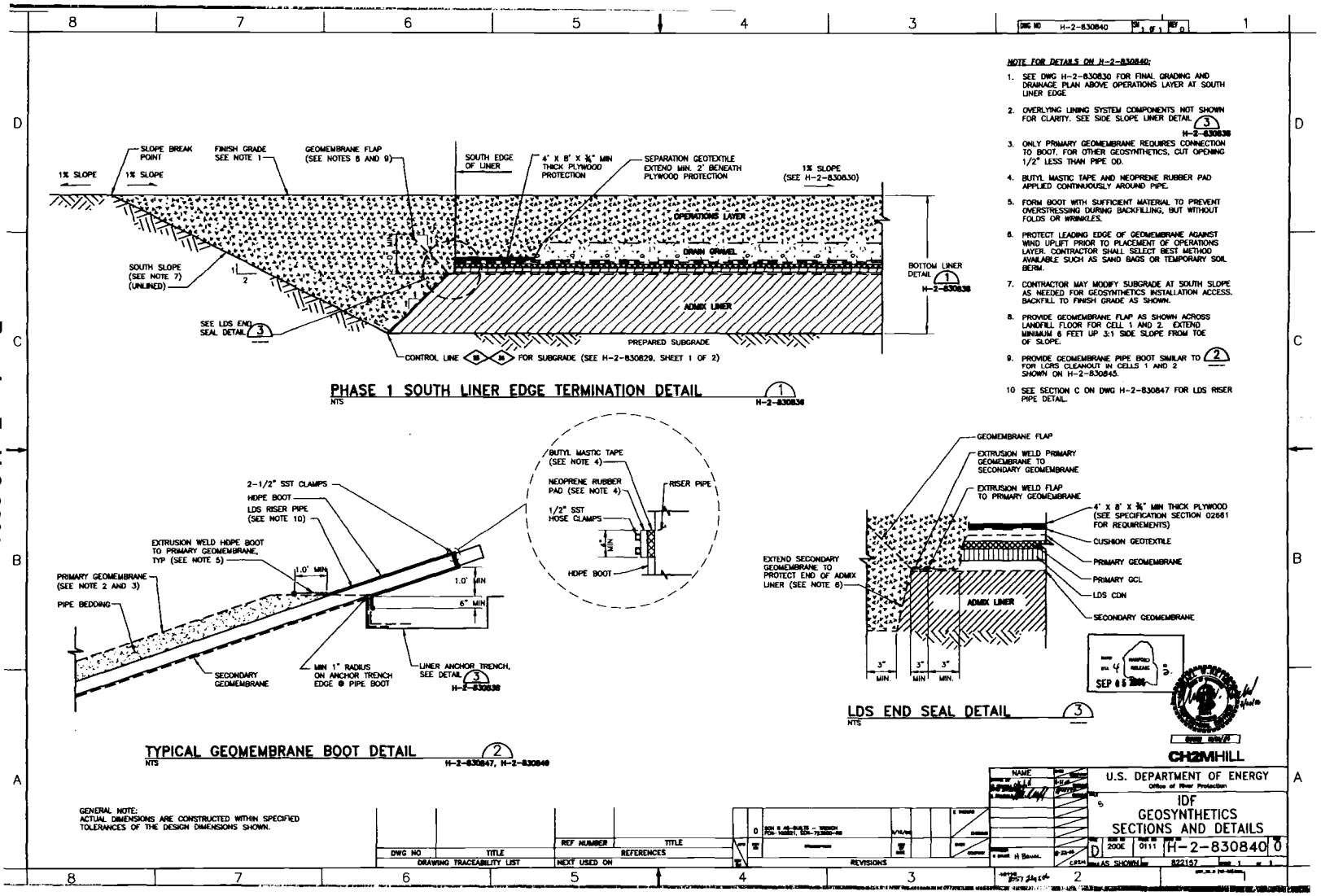


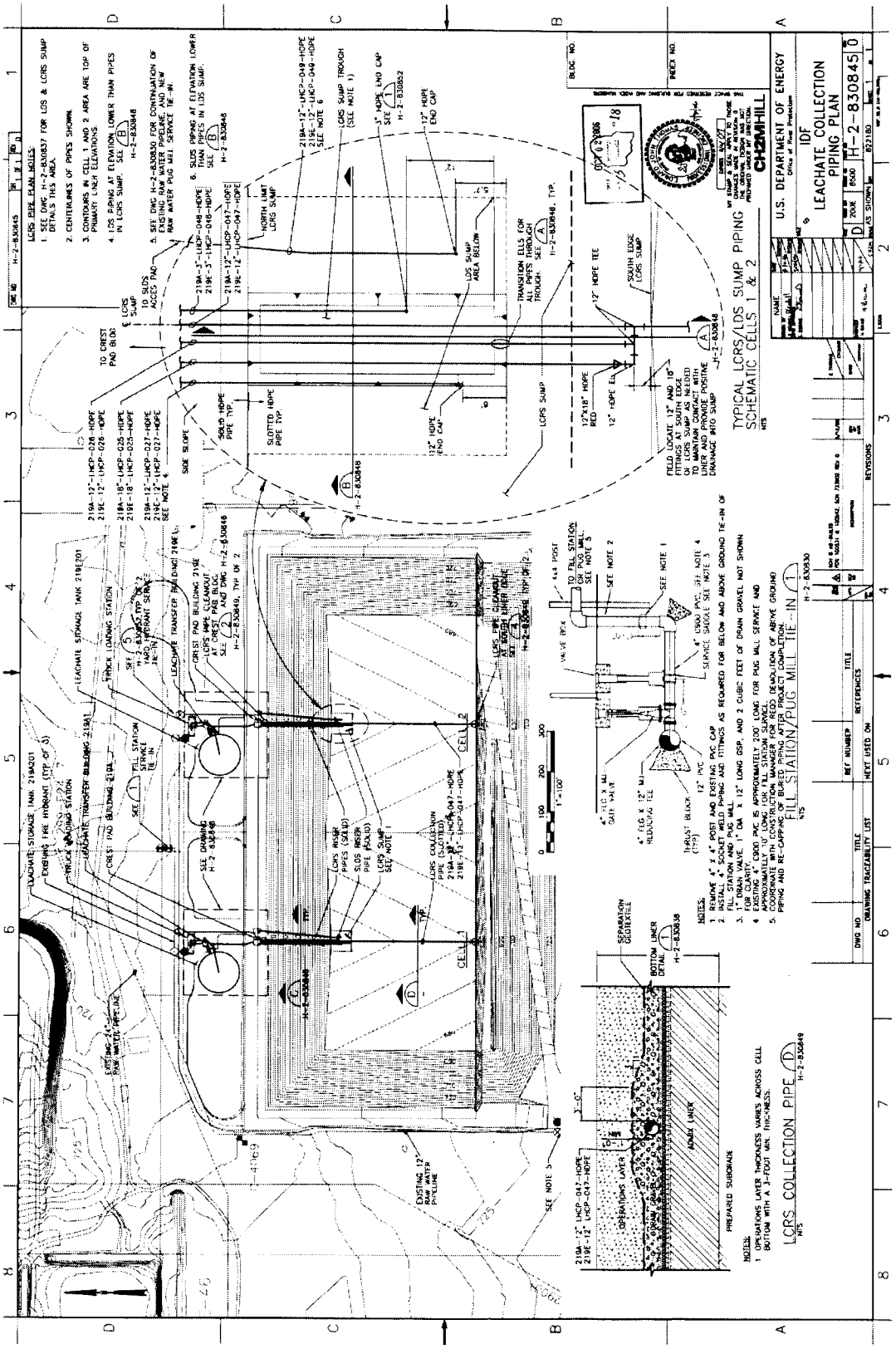
Drawing 5. H-2-830838



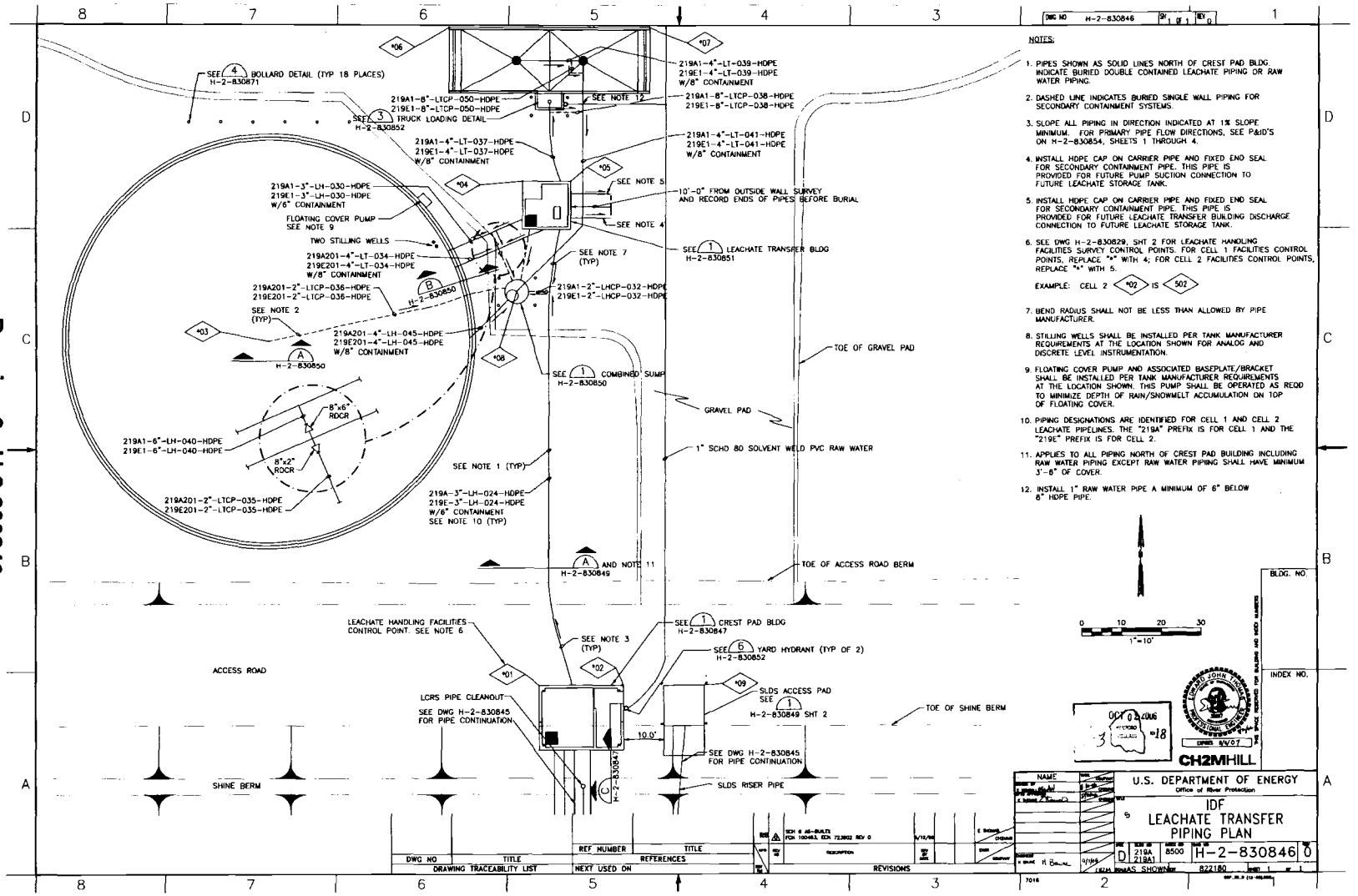
Drawing 6. H-2-830839

Drawing 7. H-2-830840

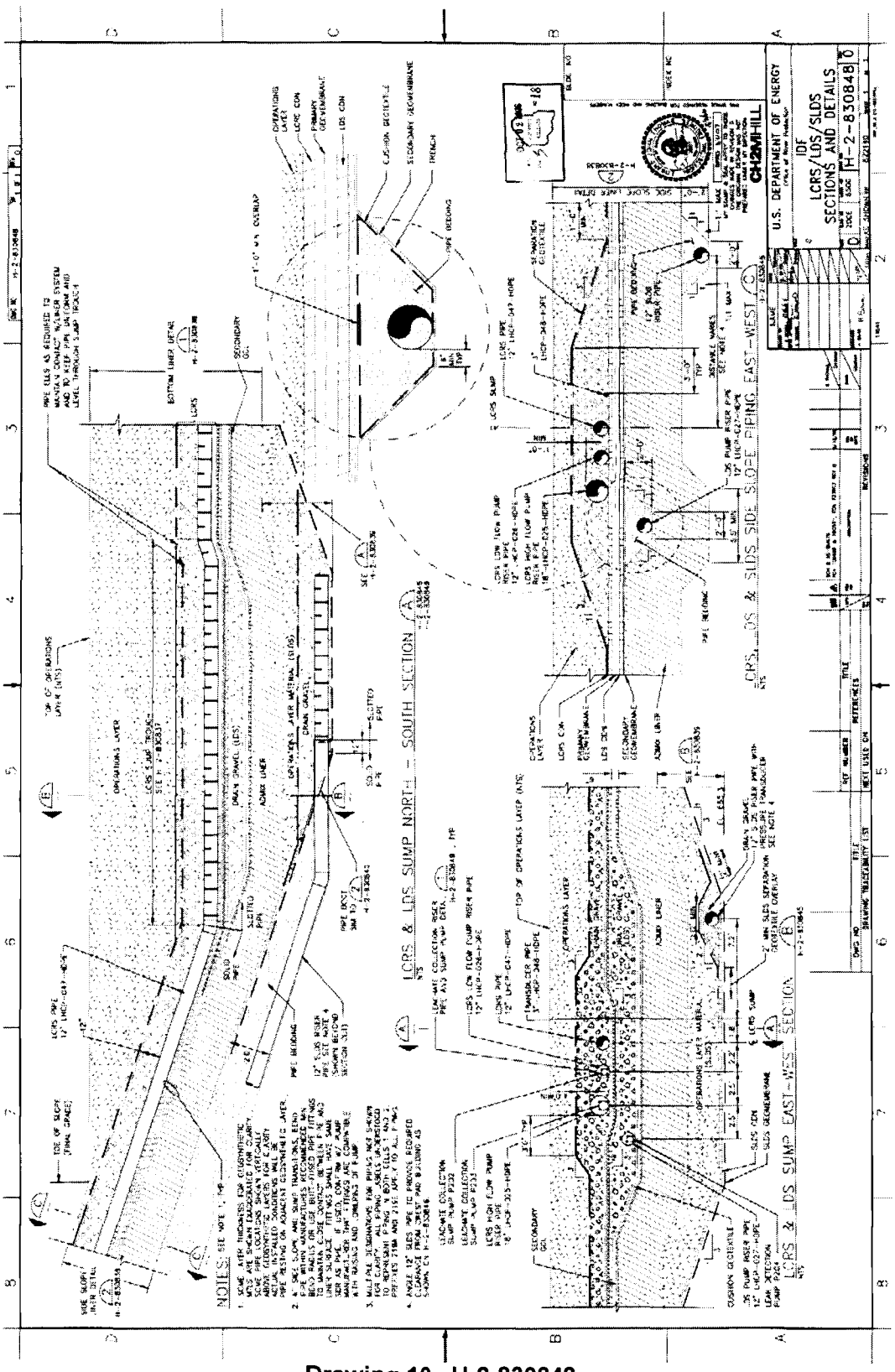




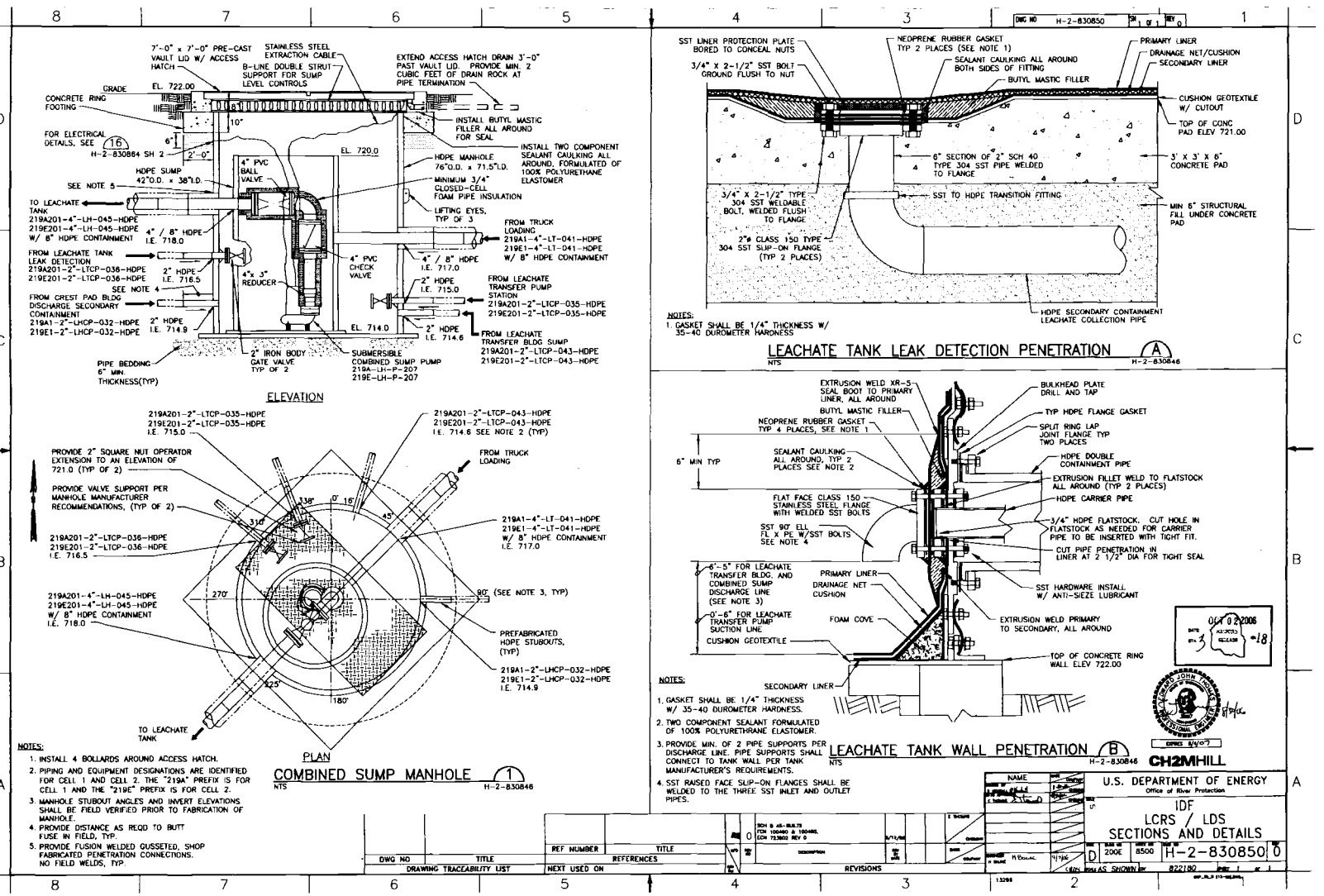
Drawing 8. H-2-830845

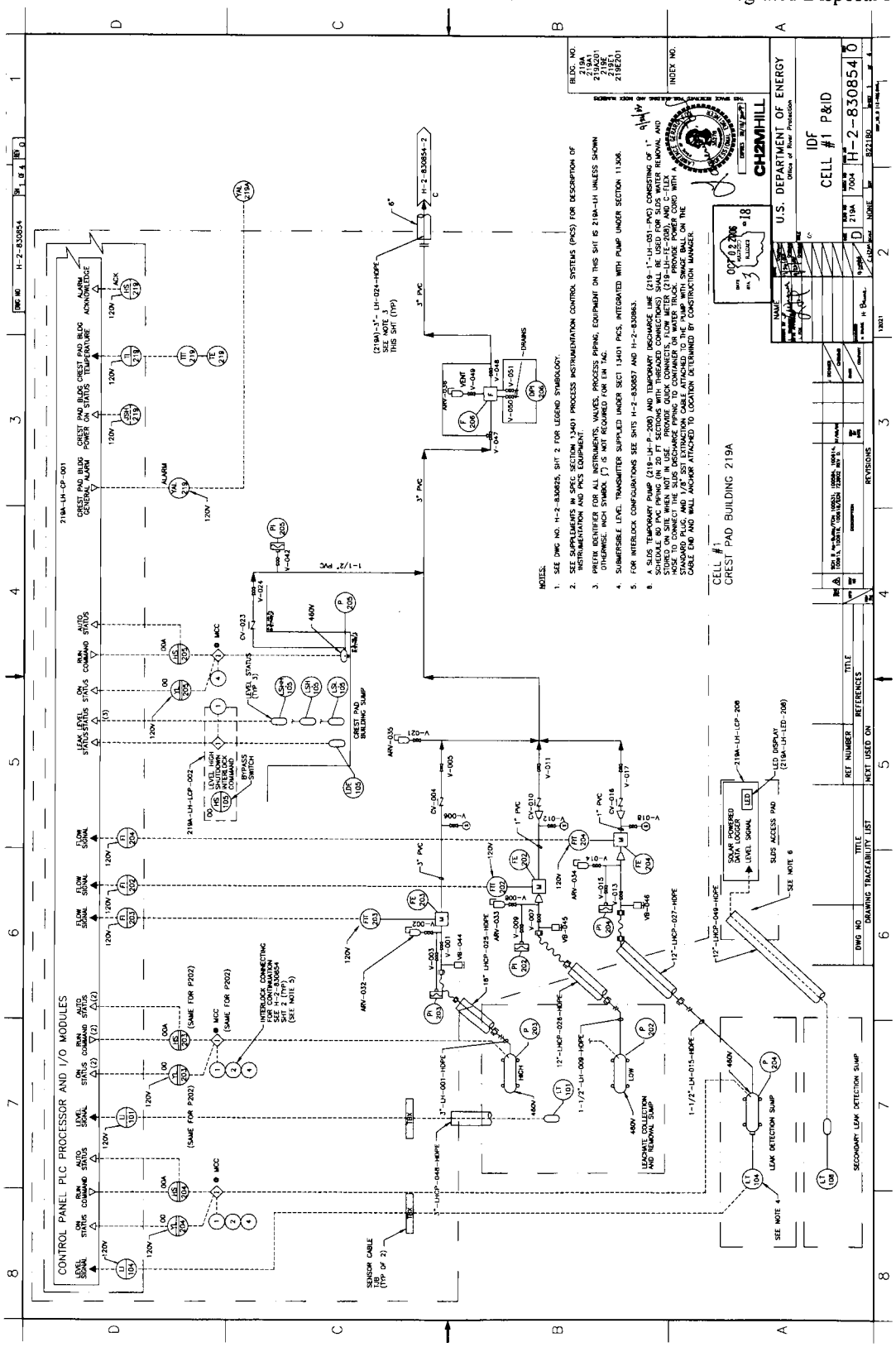


Drawing 9. H-2-830846

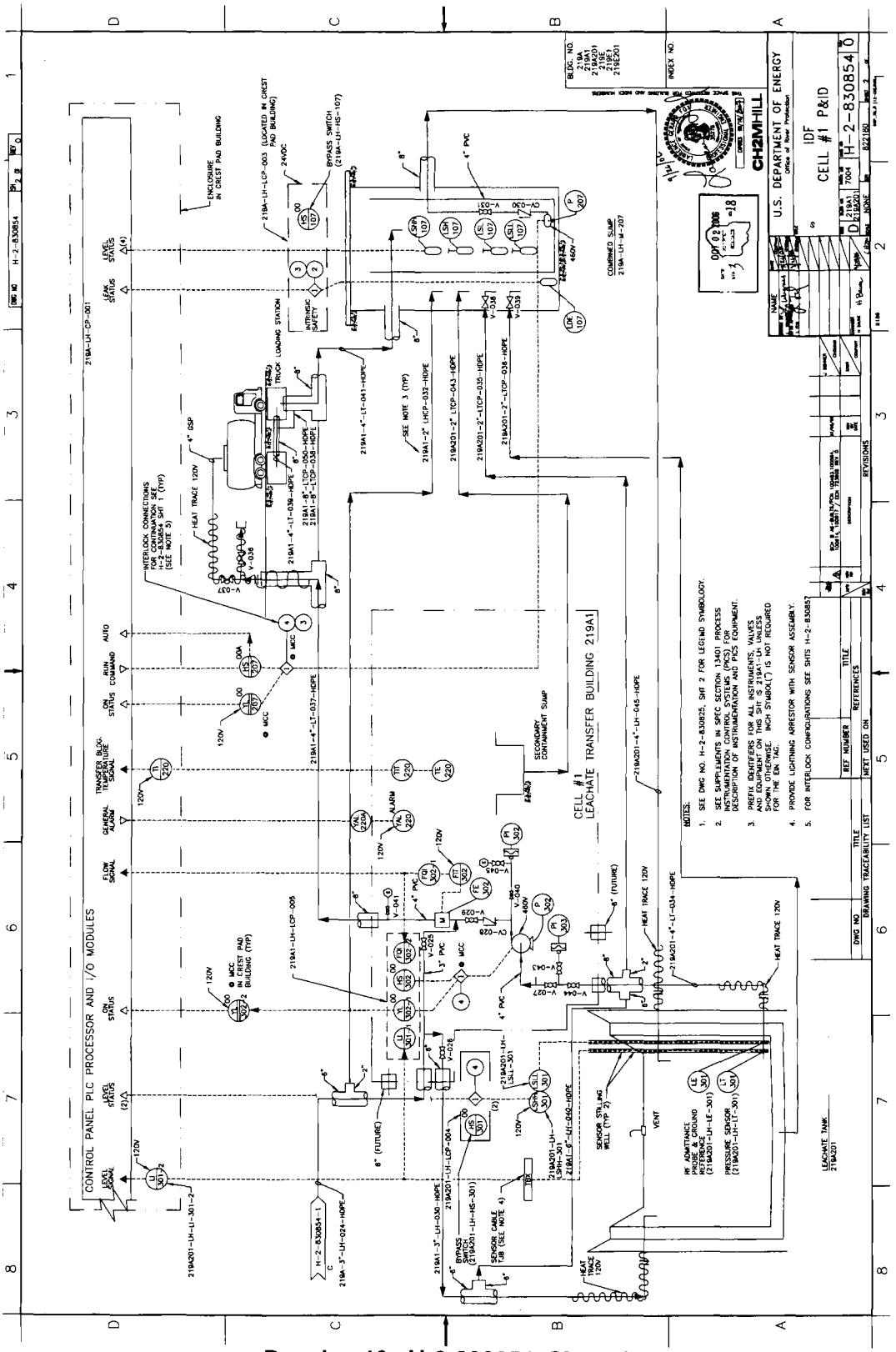


Drawing 10. H-2-830848

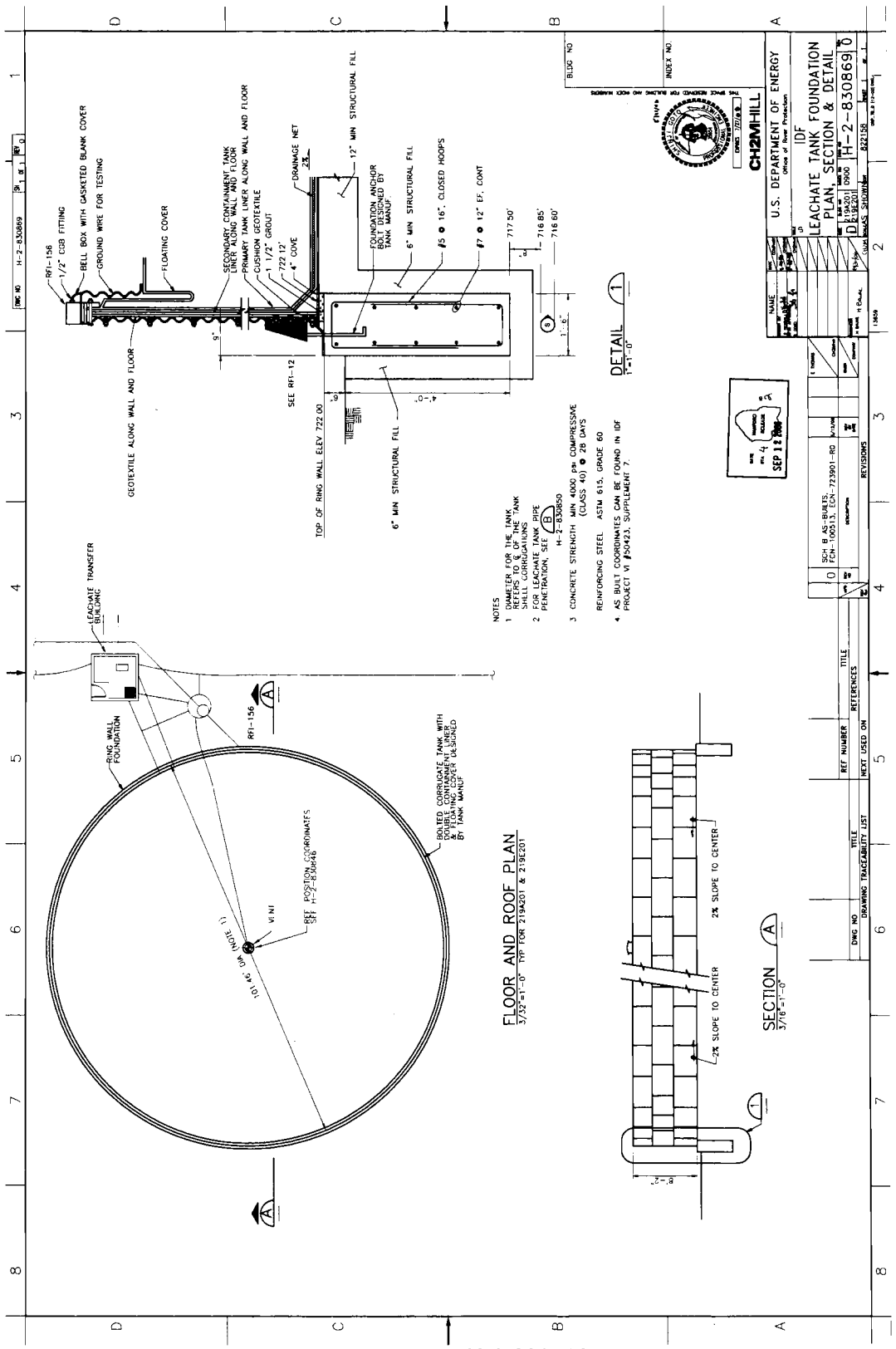

7'-0" x 7'-0" PRE-CAST VAULT LID W/ ACCESS HATCH
STAINLESS STEEL EXTRACTION CABLE
8-LINE DOUBLE STRUT SUPPORT FOR SUMP LEVEL CONTROLS
EXTEND ACCESS HATCH DRAIN 3'-0" PAST VAULT LID. PROVIDE MIN. 2 CUBIC FEET OF DRAIN ROCK AT PIPE TERMINATION
GRADE
EL. 722.00
CONCRETE RING FOOTING
INSTALL BUTYL MASTIC FILLER ALL AROUND FOR SEAL
FOR ELECTRICAL DETAILS, SEE 16 H-2-830864 SH 2
EL. 720.0
INSTALL TWO COMPONENT SEALANT CAULKING ALL AROUND, FORMULATED OF 100% POLYURETHANE ELASTOMER
HDPE MANHOLE 76"O.D. x 71.5"I.D.
HDPE SUMP 42"O.D. x 36"I.D.
4" PVC BALL VALVE
MINIMUM 3/4" CLOSED-CELL FOAM PIPE INSULATION
SEE NOTE 5
LIFTING EYES, TYP OF 3
TO LEACHATE TANK 219A201-4"-LH-045-HDPE 219E201-4"-LH-045-HDPE W/ 8" HDPE CONTAINMENT
4" / 8" HDPE I.E. 718.0
FROM TRUCK LOADING 219A1-4"-LT-041-HDPE 219E1-4"-LT-041-HDPE W/ 8" HDPE CONTAINMENT
4" / 8" HDPE I.E. 717.0
FROM LEACHATE TANK LEAK DETECTION 219A201-2"-LTCP-036-HDPE 219E201-2"-LTCP-036-HDPE
2" HDPE I.E. 716.5
4" PVC CHECK VALVE
8"x 3" REDUCER
SEE NOTE 4
FROM CREST PAD BLDG DISCHARGE SECONDARY CONTAINMENT 219A1-2"-LHCP-032-HDPE 219E1-2"-LHCP-032-HDPE
2" HDPE I.E. 714.9
2" HDPE I.E. 715.0
FROM LEACHATE TRANSFER PUMP STATION 219A201-2"-LTCP-035-HDPE 219E201-2"-LTCP-035-HDPE
EL. 714.0
2" HDPE I.E. 714.6
FROM LEACHATE TRANSFER BLDG SUMP 219A201-2"-LTCP-043-HDPE 219E201-2"-LTCP-043-HDPE
PIPE BEDDING 6" MIN. THICKNESS(TYP)
2" IRON BODY GATE VALVE TYP OF 2
SUBMERSIBLE COMBINED SUMP PUMP 219A-LH-P-207 219E-LH-P-207
ELEVATION
219A201-2"-LTCP-035-HDPE 219E201-2"-LTCP-035-HDPE I.E. 715.0
219A201-2"-LTCP-043-HDPE 219E201-2"-LTCP-043-HDPE I.E. 714.6 SEE NOTE 2 (TYP)
PROVIDE 2" SQUARE NUT OPERATOR EXTENSION TO AN ELEVATION OF 721.0 (TYP OF 2)
FROM TRUCK LOADING
PROVIDE VALVE SUPPORT PER MANHOLE MANUFACTURER RECOMMENDATIONS, (TYP OF 2)
219A1-4"-LT-041-HDPE 219E1-4"-LT-041-HDPE W/ 8" HDPE CONTAINMENT I.E. 717.0
219A201-2"-LTCP-036-HDPE 219E201-2"-LTCP-036-HDPE I.E. 716.5
90° (SEE NOTE 3, TYP)
219A1-4"-LH-045-HDPE 219E201-4"-LH-045-HDPE W/ 8" HDPE CONTAINMENT I.E. 718.0
PREFABRICATED HDPE STUBOUTS, (TYP)
219A1-2"-LHCP-032-HDPE 219E1-2"-LHCP-032-HDPE I.E. 714.9
TO LEACHATE TANK
PLAN
NOTES:
1. INSTALL 4 BOLLARDS AROUND ACCESS HATCH.
2. PIPING AND EQUIPMENT DESIGNATIONS ARE IDENTIFIED FOR CELL 1 AND CELL 2. THE "219A" PREFIX IS FOR CELL 1 AND THE "219E" PREFIX IS FOR CELL 2.
3. MANHOLE STUBOUT ANGLES AND INVERT ELEVATIONS SHALL BE FIELD VERIFIED PRIOR TO FABRICATION OF MANHOLE.
4. PROVIDE DISTANCE AS REQD TO BUTT FUSE IN FIELD, TYP.
5. PROVIDE FUSION WELDED GUSSETED, SHOP FABRICATED PENETRATION CONNECTIONS. NO FIELD WELDS, TYP.
COMBINED SUMP MANHOLE 1 H-2-830846
NTS
SST LINER PROTECTION PLATE BORED TO CONCEAL NUTS
NEOPRENE RUBBER GASKET TYP 2 PLACES (SEE NOTE 1)
PRIMARY LINER
3/4" X 2-1/2" SST BOLT GROUND FLUSH TO NUT
SEALANT CAULKING ALL AROUND BOTH SIDES OF FITTING
DRAINAGE NET/CUSHION
SECONDARY LINER
BUTYL MASTIC FILLER
CUSHION GEOTEXTILE W/ CUTOUT
TOP OF CONC PAD ELEV 721.00
6" SECTION OF 2" SCH 40 TYPE 304 SST PIPE WELDED TO FLANGE
3' X 3' X 6" CONCRETE PAD
3/4" X 2-1/2" TYPE 304 SST WELDABLE BOLT, WELDED FLUSH TO FLANGE
SST TO HDPE TRANSITION FITTING
MIN 8" STRUCTURAL FILL UNDER CONCRETE PAD
2"ø CLASS 150 TYPE 304 SST SLIP-ON FLANGE (TYP 2 PLACES)
HDPE SECONDARY CONTAINMENT LEACHATE COLLECTION PIPE
NOTES:
1. GASKET SHALL BE 1/4" THICKNESS W/ 35-40 DUROMETER HARDNESS
LEACHATE TANK LEAK DETECTION PENETRATION A H-2-830846
NTS
EXTRUSION WELD XR-5 SEAL BOOT TO PRIMARY LINER, ALL AROUND
BULKHEAD PLATE DRILL AND TAP
BUTYL MASTIC FILLER
TYP HDPE FLANGE GASKET
NEOPRENE RUBBER GASKET TYP 4 PLACES, SEE NOTE 1
SPLIT RING LAP JOINT FLANGE TYP TWO PLACES
6" MIN TYP
SEALANT CAULKING ALL AROUND, TYP 2 PLACES SEE NOTE 2
HDPE DOUBLE CONTAINMENT PIPE
EXTRUSION FILLET WELD TO FLATSTOCK ALL AROUND (TYP 2 PLACES)
FLAT FACE CLASS 150 STAINLESS STEEL FLANGE WITH WELDED SST BOLTS
HDPE CARRIER PIPE
3/4" HDPE FLATSTOCK. CUT HOLE IN FLATSTOCK AS NEEDED FOR CARRIER PIPE TO BE INSERTED WITH TIGHT FIT.
SST 90° ELL FL X PE W/SST BOLTS SEE NOTE 4
CUT PIPE PENETRATION IN LINER AT 2 1/2" DIA FOR TIGHT SEAL
6"-5" FOR LEACHATE TRANSFER BLDG. AND COMBINED SUMP DISCHARGE LINE (SEE NOTE 3)
PRIMARY LINER
DRAINAGE NET CUSHION
SST HARDWARE INSTALL W/ ANTI-SEIZE LUBRICANT
0'-6" FOR LEACHATE TRANSFER PUMP SUCTION LINE
FOAM COVE
EXTRUSION WELD PRIMARY TO SECONDARY, ALL AROUND
CUSHION GEOTEXTILE
TOP OF CONCRETE RING WALL ELEV 722.00
SECONDARY LINER
NOTES:
1. GASKET SHALL BE 1/4" THICKNESS W/ 35-40 DUROMETER HARDNESS.
2. TWO COMPONENT SEALANT FORMULATED OF 100% POLYURETHRANE ELASTOMER.
3. PROVIDE MIN. OF 2 PIPE SUPPORTS PER DISCHARGE LINE. PIPE SUPPORTS SHALL CONNECT TO TANK WALL PER TANK MANUFACTURER'S REQUIREMENTS.
4. SST RAISED FACE SLIP-ON FLANGES SHALL BE WELDED TO THE THREE SST INLET AND OUTLET PIPES.
LEACHATE TANK WALL PENETRATION B H-2-830846
NTS
CH2MHILL
U.S. DEPARTMENT OF ENERGY
Office of River Protection
IDF
LCRS / LDS
SECTIONS AND DETAILS
H-2-830850 0
DWG NO TITLE
DRAWING TRACEABILITY LIST
REF. NUMBER TITLE
REFERENCES
NEXT USED ON
REVISIONS



Drawing 12. H-2-830854, Sheet 1



Drawing 13. H-2-830854, Sheet 2



1 **Drawing 14. H-2-830869**

APPENDIX 4B
DETAILED DESIGN CELL 1 CONSTRUCTION QUALITY ASSURANCE PLAN

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APPENDIX 4B
DETAILED DESIGN CELL 1 CONSTRUCTION QUALITY ASSURANCE PLAN

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ACRONYMS

ASTM	American Society for Testing and Materials
CDN	Composite drainage net
CM	Construction Manager
CQA	Construction Quality Assurance
CQC	Construction Quality Control
DOE	U.S. Department of Energy
EPA	U.S. Environmental Protection Agency
GCL	Geosynthetic clay liner
GRI	Geosynthetic Research Institute
IDF	Integrated Disposal Facility
LCRS	Leachate collection and removal system
LDS	Leak detection system
NCR	Non-Conformance Report
ORP	Office of River Protection
OSHA	Occupational Safety and Health Administration
PM	Project Manager
QA	Quality assurance
QC	Quality control
RFI	Request for Information
SBL	Soil bentonite admix liner
SLDS	Secondary leak detection system
UCL	Upper control limit
WAC	Washington Administrative Code

SECTION 1 GENERAL

1.1 Introduction

This Construction Quality Assurance (CQA) Plan describes the quality assurance (QA) activities for constructing Phase I of the Integrated Disposal Facility (IDF) at the Hanford facility in Richland, Washington.

1.1.1 Applicable Units

QA activities will be required during construction of Cell 1 of Phase I to certify that the following construction activities are performed in accordance with the construction documents:

- Construction/preparation of foundation systems for liners.
- Construction of dikes or embankments.
- Construction of low-permeability soil liners.
- Construction of geomembranes.
- Construction of leachate collection and removal systems and leak detection systems.

This CQA Plan has been prepared to describe the activities that will be performed during construction of the lining system, leachate collection and leak detection systems, and operation layer of Cell 1. This CQA Plan is intended to satisfy the regulatory requirements and guidance established in 40 CFR 264.19, the U.S. Environmental Protection Agency's (EPA) technical guidance document, *Quality Assurance and Quality Control for Waste Containment Facilities* (EPA 1993), and Washington Administrative Code (WAC) 173-303-335.

This CQA Plan will be implemented by a CQA Officer (herein referred to as the CQA certifying engineer), a person familiar with EPA's technical guidance document, *Quality Assurance and Quality Control for Waste Containment Facilities* and this CQA Plan. The CQA certifying engineer will be supported by the number of CQA representatives necessary to implement the requirements in this CQA Plan and to document the work.

1.1.2 Scope

This CQA Plan establishes general administrative and documentation procedures that will be applicable for selected activities of construction. With respect to responsibilities, personnel qualifications, and specific inspection and testing activities, this CQA Plan addresses only those activities associated with the soils, geosynthetics, and related liner and leachate collection system piping components for the IDF.

The CQA requirements are divided into the following sections to provide quick access to CQA requirements for individual liner components:

- Soils CQA
- Geosynthetic Clay Liner CQA
- Geomembrane CQA
- Geotextile CQA
- Composite Drainage Net CQA
- Polyethylene Pipe and Fittings CQA
- CQA Documentation and Certification
- CQA Documentation and Certification

1.2 Project Organization

This section describes the anticipated project organization for the IDF construction activities. The following subsections address the organizations involved in the construction, their respective roles in construction activities, and the methods of interactions between organizations.

1.2.1 Responsibility and Authority

The organization chart for the IDF construction is shown in Figure 1-1. These personnel will be associated with two main entities that include the Tank Farm operating contractor and his agents and the construction general contractor and his personnel and/or subcontractors.

The project team consists of both full-time field personnel and part-time management personnel. The part-time management personnel will be onsite during the IDF construction periodically to monitor progress, attend meetings, resolve disputes, and ensure that the work is implemented in accordance with the construction drawings, technical specifications, CQA Plan, and the RCRA permit. The field personnel will consist of the key personnel onsite during construction. The solid lines on the organization chart represent project responsibilities such as scope, cost, and schedule. The dashed lines represent the functional responsibilities of staff for QA, design, and management. The responsibilities and reporting requirements for each project team member are described in the following sections.

1.2.1.1 Project Team

When the individuals identified below are designated to perform specific functions described in this CQA Plan, the reference to these individuals includes their designee or an alternate who can function on their behalf. The Department of Energy – Office of River Protection (DOE-ORP) Manager is the owner's representative and is responsible for project funding and overall project scope. The DOE-ORP manager and IDF project manager keep the regulatory agencies informed of IDF construction activities and progress.

1.2.1.2 IDF Project Manager (PM)

The IDF PM is an employee or agent of the Tank Farm operating contractor, has overall responsibility for the IDF construction, and interfaces with the DOE-ORP manager. The IDF PM directs the activities of the IDF project and field team staff, including the CM, design engineer, and the project engineer. Additionally, the IDF PM has overall responsibility for the achievement of quality. Functionally, the IDF PM reviews and approves quality assurance reports submitted by the IDF CQA certifying engineer.

1.2.1.3 IDF Project Engineer

The IDF project engineer is an employee or agent of the Tank Farm operating contractor and is responsible for providing technical support to the IDF project team. The IDF project engineer is supported by the design engineer for reviewing and/or preparing technical documents related to engineering design and analyses.

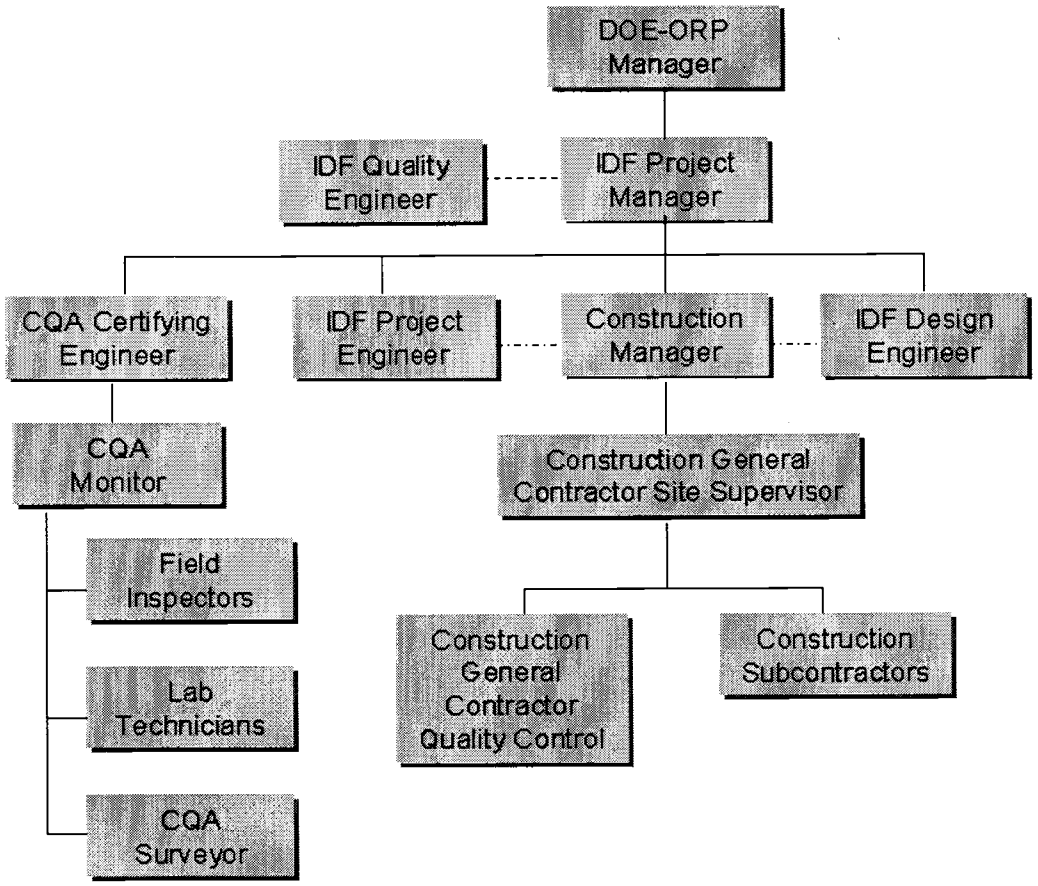

Figure 1-1. QA Organization Chart

1.2.1.4 IDF Quality Engineer

The IDF quality engineer is an employee or agent of the Tank Farm operating contractor and is independent from line management on the project. The IDF quality engineer provides overview and assessment of QA on the project. The IDF quality engineer provides feedback and assessment results to the IDF PM.

1.2.1.5 IDF Design Engineer

The IDF design engineer is an employee or agent of the Tank Farm operating contractor is responsible for reviewing and/or preparing technical documents related to the IDF design and construction. The design engineer prepares the construction drawings, technical specifications, and the CQA Plan. The IDF design engineer reports to the IDF PM and supports the IDF project engineer.

1.2.1.6 Field Team

1.2.1.7 IDF Construction Manager

The IDF CM is an employee or agent of the Tank Farm operating contractor and serves as the point of contact between the IDF construction general contractor and the IDF project team. All construction general contractor correspondence and direction flows through the CM. The CM oversees the daily construction field activities and is the onsite representative for the IDF PM.

1.2.1.8 CQA Certifying Engineer

The CQA certifying engineer is an employee or agent of the Tank Farm operating contractor who has the overall responsibility of implementing this CQA Plan and directly supervises the CQA monitor, field inspection team, and laboratory technicians. The CQA certifying engineer is responsible for preparation of an implementation plan that addresses how the CQA Plan is to be implemented, and how CQA work is to be performed, tracked, and coordinated, as well as how procedures outlined in this CQA Plan are to be followed. The implementation plan will be submitted to IDF project manager and CM for approval.

Functionally, the CQA certifying engineer submits certified CQA reports to the IDF CM for review and approval by the IDF PM. The CQA certifying engineer is a registered professional engineer in Washington and has the authority to provide a certification letter that the IDF is constructed in accordance with the approved CQA Plan, the approved plans and specifications, and any approved changes. The CQA certifying engineer also has the authority and responsibility to stop work and recommend remedial actions to the IDF PM.

1.2.1.9 Field Inspector

Field inspectors are employees or agents of the Tank Farm operating contractor and report to the CQA certifying engineer. The field inspector's function is to perform testing and observations, in accordance with this CQA Plan and under the direction of the CQA monitor and CQA certifying engineer.

1.2.1.10 Soils Laboratory Technicians

Laboratory technicians are employees or agents of the Tank Farm operating contractor and report to the CQA certifying engineer and provide the QA laboratory testing, required by this CQA Plan, as requested by the CQA monitor, and CQA certifying engineer.

1.2.1.11 CQA Surveyor

The CQA surveyor will be an employee or agent of the Tank Farm operating contractor and will be a registered land surveyor in the State of Washington.

1.2.1.12 CQA Monitor

The CQA monitor is an employee or agent of the Tank Farm operating contractor, reports directly to the CQA certifying engineer, and is a CQA representative, supported by the field inspection team and laboratory technician. The CQA monitor ensures that all CQA tests are performed in accordance with this CQA Plan and accepted procedures.

1.2.1.13 Construction General Contractor

The IDF construction general contractor is responsible for implementing the approved design by providing the necessary labor, equipment, materials, and all other resources necessary to construct the IDF.

1.2.1.14 Construction General Contractor Site Supervisor

The site supervisor is an employee or agent of the construction general contractor and is responsible for implementing the IDF construction activities. The site supervisor has overall responsibility for all construction activities related to the IDF, controls day-to-day construction tasks, and is the point of contact for construction general contractor field personnel. The site supervisor ensures the work is progressing in accordance with approved construction contract documents and the approved schedule.

1.2.1.15 Construction Subcontractors

Construction subcontractors include specialty companies, retained by the IDF construction general contractor, to perform specific work activities at the IDF such as earth moving, geosynthetic lining installation, piping, and building/tank installation. The construction subcontractors report directly to the construction general contractor site supervisor.

1.2.1.16 Construction General Contractor Quality Control

The construction general contractor provides a construction QC engineer who supports the site supervisor. The primary responsibility of the construction QC engineer is to ensure that the work is performed in accordance with the technical specifications and construction drawings. Specific duties of the construction QC engineer include activities such as preparing construction submittals, field documentation, and interfacing with the CQA certifying engineer.

1.2.2 Project Meetings

The various progress and status meetings that are anticipated to be held throughout the IDF construction are described below. The purpose of the meetings is to discuss work progress, planning, and other issues related to construction. A portion of these meetings can be dedicated to CQA issues, as necessary, to provide an opportunity for the CQA team to express concerns regarding quality, relay test results, and ensure good communication between all organizations involved in the construction of the IDF.

1.2.2.1 Pre-Construction Meeting

A pre-construction meeting will be scheduled prior to beginning construction activities for the IDF. At a minimum, the meeting will be attended by IDF staff including the PM, CM, project engineer, design engineer, as well as the construction general contractor site supervisor, and the CQA certifying engineer. A portion of the meeting will be dedicated to the discussion of QA issues. Suggested CQA topics will include, but not be limited to:

- Reviewing the responsibilities of each organization.
- Discussing the authority of agencies and project and field team members to order work stoppages.
- Reviewing lines of authority and communication for each organization.
- Providing each organization with all relevant CQA documents and supporting information.
- Familiarizing each organization with the CQA Plan and its role, relative to the design criteria, plans, and specifications.

- Discussing the established procedures or protocol for observations and tests, including sampling strategies.
- Discussing the established procedures or protocol for handling construction deficiencies, repairs, and re-testing, including "stop work" conditions.
- Reviewing methods for documenting and reporting inspection data.
- Reviewing methods for distributing and storing documents and reports.
- Reviewing work area security and safety protocol.
- Reviewing the proposed project schedule.
- Discussing procedures for the location and protection of construction materials and for the prevention of damage of the materials from inclement weather or other adverse events.
- Determining action items, assigning actionees, and recording minutes to be transmitted to meeting attendees.
- Discussing document control requirements and control of CQA records.
- Discussing control and protection of samples.

1.2.2.2 Daily Pre-Job Briefing

The construction general contractor will conduct daily pre-job briefings at the work area. The participants will include the construction field personnel, including lower tiered subcontractors and CQA representatives. The primary purpose of these meetings will be to address the day's planned activities. The CQA monitor will discuss CQA activities planned for that day and interface needs with the construction personnel. Suggested CQA topics are:

- Review the work location and activities for the day.
- Discuss the construction general contractor's personnel and equipment assignments for the day.
- Address scheduling of resources for upcoming work.
- Review any new test data.
- Discuss any potential construction problems, including unexpected subsurface conditions.
- Discuss CQA-planned activities and interface needs.

This meeting will be documented and the documentation will be retained on file by the CQA monitor.

1.2.2.3 Construction Progress Meetings

Weekly progress meetings will be held at the site to discuss construction progress. At a minimum, the weekly progress meetings will be attended by the IDF PM, CM, the site supervisor, and the CQA certifying engineer or CQA monitor. The purposes of the meeting are to:

- Review previous activities and accomplishments.
- Review claims, change orders, delays, and similar items.
- Review planned activities for the upcoming 2-week period.
- Finalize resolution of problems from previous meetings.
- Discuss potential problems with the work planned for the upcoming 2-week period.

Minutes will be recorded and transmitted to meeting attendees and other interested parties.

1.2.2.4 Non-Conformance Meetings

Meetings will be convened as necessary to address non-conformances discovered during inspection. Deficiencies observed during construction by the CQA representatives will be brought to the attention of the IDF CM and CQA certifying engineer and documented using the non-conformance reporting (NCR) procedures outlined in Section 8.1.4.

These deficiencies also will be tracked in the CQA representative's field logbook until resolution and included in the daily summary report. These documents will include the description of the deficiency and actions taken or to be taken to resolve.

1.2.3 Hold Points

Mandatory hold points will be established for certain key activities. At these points, the IDF construction general contractor will notify the CQA monitor or CQA certifying engineer that the layer or portion of a layer is ready for review. The hold points anticipated for the IDF would be at completion or partial completion of each of the following components:

- Prepared subgrade
- SLDS geomembrane and composite drainage net (CDN)
- SLDS riser pipe
- Soil bentonite admix soil liner
- Secondary GCL
- Secondary geomembrane
- LDS CDN
- LDS piping
- Primary geosynthetic clay liner (GCL)
- Primary geomembrane
- Cushion geotextile
- LCRS piping
- Drain gravel
- Separation geotextile
- Operations layer

On side slopes, a LCRS CDN would substitute for the cushion geotextile, drain gravel, and separation geotextile.

1.3 Personnel Qualifications and Training

This section describes the qualifications and training required for CQA personnel. All documentation relating to qualifications will be maintained with the project CQA records.

1.3.1 CQA Certifying Engineer

The CQA certifying engineer will have landfill construction certification experience. The CQA certifying engineer will, at a minimum, be a registered civil professional engineer in good standing in the State of Washington, possess a bachelor's degree in civil or construction engineering, geotechnical engineering, engineering geology, or a closely related discipline, and have sufficient practical, technical, and managerial experience to direct successfully the CQA activities discussed in this CQA Plan. The CQA certifying engineer's qualifications will be documented by training records and a professional resume showing significant field experience in landfill construction and low permeability soil-bentonite admix liner construction, having directed CQA activities at a minimum of three landfill construction projects or a minimum of 100 acres of combined landfill area certifying experience. The CQA certifying engineer will be familiar with the EPA technical guidance document, *Quality Assurance and Quality Control for Waste Containment Facilities* (EPA 1993). Qualification documentation will be reviewed by the IDF PM and IDF project engineer.

1.3.2 CQA Monitor

At a minimum, the CQA monitor will have a high school diploma and at least five years of construction-related experience, including at least three years of experience conducting CQA monitoring for earthwork construction (including a minimum of three landfill construction projects or a minimum of 50 acres of combined landfill area experience), or a bachelor of science degree from a four-year college or university and at least two years of experience conducting CQA monitoring for earthworks construction (including a minimum of three landfill construction projects). The CQA monitor must be capable of performing work with little or no daily supervision. The CQA monitor will be familiar with the EPA technical guidance document, *Quality Assurance and Quality Control for Waste Containment Facilities* (EPA 1993). Qualifications of the CQA monitor will be documented by training records and a professional resume, reviewed by the IDF PM and CQA certifying engineer.

1.3.3 Field Inspector

At a minimum, the field inspector will have a high school diploma and at least two years of construction-related experience, including at least one year of experience conducting CQA monitoring for earthwork construction, or will have a bachelor of science degree from a four-year college or university and at least six months of experience conducting field inspection for earthworks construction. The field inspector must be capable of routine engineering technician work, under general daily supervision. The field inspector will be familiar with the EPA technical guidance document, *Quality Assurance and Quality Control for Waste Containment Facilities* (EPA 1993). Qualifications of the field inspector will be documented by training records and a professional resume, reviewed by the IDF PM and CQA certifying engineer.

1.3.4 Soils Laboratory Technicians

Laboratory technicians will have at a minimum a high school diploma and at least five years of construction materials laboratory testing related experience, including at least three years of experience performing geotechnical laboratory tests for earthwork construction, including compacted low permeability soil-bentonite admix, or will have a bachelor of science degree from a four-year college or university and at least two years of experience performing geotechnical laboratory tests for earthwork construction, including low permeability soil-bentonite admix. The laboratory technician must be capable of routine laboratory tech work, under general daily supervision. Qualifications of laboratory technicians, including training records and professional resumes, will be reviewed by the IDF PM and CQA certifying engineer.

1.3.5 Geosynthetic Laboratory

The geosynthetic laboratory will be selected by the CQA certifying engineer and will provide the geosynthetic QA conformance testing required by this CQA Plan, as requested by the CQA monitor and/or CQA certifying engineer. The geosynthetics CQA laboratory will be unaffiliated with the materials supplier or manufacturer, or construction general contractor. The geosynthetics CQA laboratory will have at least five years of experience in testing geosynthetics and other relevant liner system components, and will be familiar with American Society for Testing and Materials (ASTM) and other applicable test standards.

1.4 Definitions Relating to Construction Quality Assurance

1.4.1 Construction Quality Assurance and Construction Quality Control

Construction Quality Assurance—A planned and systematic pattern of the means and actions designed to provide adequate confidence that items or services meet contractual and regulatory requirements, and will perform satisfactorily in service.

Construction Quality Control (CQC)—Those actions that provide a means to measure and control the characteristics of an item or service to meet contractual and regulatory requirements.

1.4.2 Use of the Terms in This Plan

The definitions used in the context of this CQA Plan are as follows:

- CQA refers to means and actions employed by the CQA representatives to assure conformity of liner system, LCRS, LDS, SLDS, and pipe preparation, production, and installation with this CQA Plan, the technical specifications, and the construction drawings. CQA will be provided by a third party, acting independently from the product manufacturer and construction general contractor.
- CQC refers to those actions taken by manufacturers, suppliers, or construction general contractor, including their designated representatives, to ensure that the materials and the workmanship meet the requirements of the technical specifications and the construction drawings.

1.5 References

1.5.1 Applicable Organizations

Organizations whose standards are referenced in the CQA Plan include:

- ASTM—American Society for Testing and Materials
- DOE—Department of Energy
- GRI—Geosynthetic Research Institute
- OSHA—Occupational Safety and Health Administration
- EPA—U.S. Environmental Protection Agency

1.5.2 Applicable Standards

Any reference to standards of any society, institute, association, or governmental agency will pertain to the edition in effect as of the date of this CQA Plan, unless stated otherwise.

Specific test standards for tests cited in the CQA Plan are provided in the technical specifications. These standards may be modified due to technological advances since compilation of the technical specifications. All such modifications are to be approved in accordance with change order procedures described in Section 8.1.5.

1.6 Construction Activities and Submittal Requirements

1.6.1 Construction Activities

This section describes the construction activities and submittal requirements that will be performed by the construction general contractor during the IDF construction. This CQA Plan only addresses selected activities of the Phase I construction.

In general, construction activities will consist of preparing the subgrade, installing the liner system, the leak detection systems (LDS and SLDS), the leachate collection and removal system (LCRS), and operations layer and necessary equipment to complete the landfill for waste acceptance. Construction will consist of these activities:

- Mobilizing construction equipment and personnel.
- Vendor data submittals.
- Installing sediment and erosion control.
- Preparing soil bentonite material.
- Excavation, embankment, fine grading of landfill subgrade, and sump construction.
- Constructing the secondary leak detection system (SLDS) sump.
- Constructing the soil bentonite admix liner (SBL).
- Dust control activities during construction.

- Placing the geosynthetics for the secondary liner.
- Constructing the leak detection system (LDS).
- Placing the geosynthetics for the primary liner.
- Constructing the leachate collection and removal system.
- Constructing the operations layer.
- Site restoration.
- Demobilization.

Prior to the start of construction activities, the CQA representatives will review and become familiar with all construction drawings, technical specifications, the CQA Plan, and RCRA permit. The CQA certifying engineer also will be familiar with the most recent construction schedule, so that adequate resources (i.e., laboratory, field testing equipment, staff, and CQA forms), including contingencies (e.g., backup equipment, alternate laboratory, and alternate CQA staff) for CQA activities, will be commensurate with the anticipated construction productivity and work schedule.

1.6.2 Submittal Requirements

The construction general contractor will provide the submittals required (listed in Table 1-1 in this section) to the IDF PM. Submittals will be provided far enough in advance of scheduled installation dates to allow time for reviews, possible revisions and resubmittals, placing orders, and securing delivery. The construction general contractor will identify, track, and disposition all required vendor data. The IDF PM will respond to each required submittal as stated in the technical specifications.

The submittals presented in Table 1-1 will be required as a minimum. A master submittal list will be provided as part of the contract documents.

1.6.3 Receipt Inspection Procedures

Inventory of manufactured materials used in lining system construction is detailed in Sections 3.1.4 (GCL), 4.1.4 (geomembrane), 5.1.4 (geotextiles), 6.1.4 (CDN) and 7.1.4 (polyethylene piping). The purpose of this section is to provide a general summary of the minimum requirements and procedures for receiving and controlling purchased materials, equipment, or services as required by the contract documents.

Procurement, receipt, and inspection of construction materials and equipment are the responsibilities of the construction general contractor, with verification by the CQA certifying engineer and IDF CM. Procedures specific to the IDF Phase I construction project will be prepared as part of the construction quality control (QC) plan, to be submitted by the construction general contractor.

Procedures to control receipt inspection will include the following, at a minimum:

- The contract documents will provide a master submittal list that identifies the materials, equipment, or services requiring receipt inspection. Upon delivery to the project site, the general construction contractor will attach secure and visible "Quality Hold for Inspection" tags to each item.
- All items, materials, and equipment that have been tagged will be stored in segregated areas, as identified in the contract documents. Items will be restricted from further use until all construction general contractor and CQA certifying engineer inspections are completed.
- Upon inspection if items, materials, or equipment held for inspection, the "Quality Hold for Inspection" tag will be removed and replaced with one of the following, as appropriate:
- Acceptance tag
- Non-conformance (red) tag
    a) Conditional use tag

- The construction general contractor may utilize only those items tagged as "Accepted" or "Conditional Use".
- Red-tagged materials will not be used in construction and will be moved to a segregated area or removed from the site.
- Conditional use tagged materials are restricted to use for specific conditions identified on the tag.
- Documentation of receipt inspection will be completed, maintained, and stored in a single location, in a secure and protected environment for the full performance period of the construction contract.

Table 1.1. Required Submittals

Submittal	Description	Requirement
Source Quality Control for Imported Materials (structural fill, drain gravel and crushed surfacing)	Gradation tests performed in accordance with ASTM D422 by a qualified independent test laboratory for imported materials on samples taken at place of production prior to shipment. Samples will be taken for gradation testing from every 2,000 tons of prepared materials, in accordance with ASTM D75.	Submitted by the construction general contractor and approved by the IDF PM prior to the shipment of material to the project site.
Geomembrane Installation Plan	Proposed layout drawings for each layer of geomembrane material. Geomembrane layout will show panel configuration, general dimensions, and seam locations.	Submitted by the construction general contractor and approved by the IDF PM prior to the installation of the respective geomembrane liner.
Subgrade surface acceptance	Certification in writing that the surface on which the geomembrane will be installed is acceptable to the installer. A certificate of acceptance will be provided by the construction general contractor to the CQA representative, who will then verify to the CQA certifying engineer that the deployment surface has been accepted immediately prior to commencement of geomembrane installation in the area under consideration.	Certificate signed by the installer and construction general contractor prior to installation of geomembrane over the subgrade.
GCL quality control certifications, test data and properties guarantee	Provide manufacturer's quality control (QC) test data for GCL material to be installed, including: Bentonite – suppliers' name and location, brand name, lot number, dated quality control information from supplier, manufacturer's test data verifying that bentonite meets manufacturer's specifications. GCL – written guarantee that GCL conforms to the technical specification requirements and test certificates for each production lot or 50,000 square feet of GCL material including roll numbers, test methods, and test results verifying compliance with the technical specification requirements for GCL.	Submitted by the construction general contractor prior to installation of the GCL material and approved by the IDF PM.

Submittal	Description	Requirement
Geomembrane quality control certifications, test data and properties guarantee	QC Testing shall be performed by manufacturer to demonstrate the geomembrane conforms to technical specification requirements. Prior to delivery of any geomembrane material, the manufacturer shall submit all required information listed in the technical specifications (Section 02661). QC Certification: Prior to shipment, the geomembrane manufacturer shall provide a quality control certificate for each roll of geomembrane. The quality control certificate shall be signed by a responsible party employed by the geomembrane manufacturer, such as the production manager. The quality control certificate shall include: Roll numbers and identification, resin lot, and batch numbers. Sampling procedures and results of quality control tests. As a minimum, results shall be given for thickness, asperity, tensile strength, and tear resistance in accordance with methods indicated in the technical specifications. Tests shall be conducted on each production lot of geomembrane or every 50,000 square feet, whichever results in the greater number of tests.	Submitted by the construction general contractor prior to installation of the geomembrane material and approved by the IDF PM.
Geotextile material certifications and test data	Provide manufacturer's QC test data for geotextile material to be installed, including: Geotextile – written guarantee that geotextile conforms to specification requirements, certification that manufacturer continuously inspected geotextile for presence of needles and found it to be needle-free, and test certificates for geotextile material including roll numbers, test methods, and test results verifying compliance with the technical specifications physical properties for geotextile. Frequency of manufacturer's QC testing shall be at the standard rate stated in the manufacturer's QC plan for each required property in the technical specifications.	Submitted by the construction general contractor prior to installation of the geotextile material and approved by the IDF PM.

Submittal	Description	Requirement
Composite drainage net material certifications and test data	Provide manufacturer's QC test data for composite drainage net material to be installed, including: Composite drainage net – manufacturer's specification measured using appropriate test methods, written guarantee that composite drainage net conforms to specification requirements, manufacturer's QC test data for the geotextile component as specified above for geotextile, and test certificates for composite drainage net material and geonet component including roll numbers, test methods, and test results verifying compliance with the technical specification requirements for composite drainage net and geonet. Frequency of manufacturer's QC testing shall be at the standard rate stated in the manufacturer's QC plan for each required property in the technical specifications.	Submitted by the construction general contractor prior to installation of the composite drainage net material and approved by the IDF PM.
Interface Shear Strength test data	Provide data prior to material shipment for the interface friction angle between the textured geomembrane and other materials (including CDN, GCL, and Admix Liner) directly in contact with the geomembrane as specified in Section 02661; and between the CDN and the operations layer as specified in Section 02373 of the technical specifications. Friction angle shall be determined by direct shear testing under fully saturated conditions (ASTM D5321 or D6243 for GCL interface) at low nominal normal loads of both 100, 250, and 500 pounds per square foot (psf), and high nominal normal loads of 2,000, 8,000, and 15,000 psf, except for the CDN/Operations Layer interface which shall be reported at low normal load only. Report results for both peak and large displacement (minimum 2 inches) strength. Perform two interface shear strength tests on each interface under each set of normal loads.	Submitted by the construction general contractor prior to geosynthetic material shipment, reviewed, and approved by the IDF PM for conformance with project strength requirements. Allow IDF PM a minimum 20 working days for this evaluation upon receipt of data.
Admix Liner Preparation and Placement Plan	Provide a detailed plan for preparation of the admix material, including a description of the equipment and procedures to be used, personnel qualifications, equipment calibration certificates and methods for monitoring bentonite additions and moisture conditioning. Also provide an admix liner placement plan to specify lift thickness control and to allow for required testing, described in the CQA Plan and technical specifications on the admix liner during placement operations.	Submitted by the construction general contractor prior to start of admix production for approval by IDF PM.

Submittal	Description	Requirement
Bentonite QC certificates and test data	Provide bentonite supplier's descriptive data, specification sheets, literature, and other data as necessary to fully demonstrate that the bentonite proposed for use in the admix complies with the requirements of the technical specifications. The manufacturer shall certify that the bentonite furnished complies with these Specifications. A certificate shall be submitted to the CQA Engineer for each railcar or every three truckloads of bentonite delivered.	Submitted by the construction general contractor prior to start of admix production for approval by IDF PM.
Polyethylene Pipe and Fittings	Provide manufacturer's QC test data for piping and fittings that will be installed on the landfill floors and slopes.	Submitted by the construction general contractor prior to installation of the pipe for approval by the IDF PM.

SECTION 2 SOILS CONSTRUCTION QUALITY ASSURANCE

This section discusses the CQA requirements for soil layers including fill placement, subgrade preparation, admix liner, drain gravel, and operations layer.

2.1 Fill Placement and Subgrade Preparation

This section of the CQA Plan addresses the soils components necessary to provide a prepared subgrade for the liner systems and specifies the soils CQA program to be implemented with regard to materials selection and evaluation, laboratory test requirements, field test requirements, and corrective action requirements.

2.1.1 Fill Placement and Compaction

The technical specifications will be followed for the stockpiling, placement, and compaction of earth fill and structural fill. The CQA monitor will monitor the fill placement and compaction to verify and document the following:

- The soil being placed meets the technical specifications requirements for earth fill and structural fill as determined by the test methods and frequencies specified within this CQA Plan and the source quality control submittals.
- The placement surface has been prepared as specified in the technical specifications.
- The compacted lift thickness is in accordance with the requirements of the technical specifications.
- The dry unit weight of the earth fill and structural fill meets specifications as determined by the test methods and frequencies described in Table 2-1 for earth fill and Table 2-2 for structural fill.
- Material placed in permanent stockpiles meets the appropriate specifications for earth fill or structural fill.

2.1.2 Construction Quality Assurance Evaluation

The frequency of soils testing for CQA purposes will conform to the minimum frequencies presented in Table 2-1 for earth fill and Table 2-2 for structural fill. Material properties will be determined from samples collected either immediately after placement or from stockpiles.

Nuclear density meter test methods will be used for the field-testing of the in situ dry unit weight of the in-place, compacted fill. Any settlement or other defects in the fill will be backfilled and compacted in accordance with the technical specifications.

Standard count calibrations will be conducted to monitor the aging of the nuclear density gauge sources in accordance with ASTM standards. Sand cone or drive sleeve tests will be conducted periodically to verify densities using the nuclear density gauge. Oven moisture content tests will be conducted and compared to field moisture content results to determine a field correction factor for moisture. Sand cone or drive sleeve tests and in situ moisture content tests will be performed at the frequencies specified in Tables 2-1 and 2-2.

If an in-place density test result fails to meet specifications, a confirmatory test will be performed immediately adjacent to the failed test. If the confirmatory test meets or exceeds specifications, a second confirmatory test will be performed at a second location immediately next to the failed test. If the second confirmatory test also meets or exceeds specifications, the area will be declared as meeting project specifications and the confirmatory tests will be reported. In the event that either confirmatory test fails to meet specifications, a CQA representative will determine the extent and nature of the defect by observations and/or additional testing, as necessary, to identify the limits of the area that does not meet project specifications.

If a defective area is discovered in the fill, a CQA representative will determine the extent and nature of the defect. If the defect is indicated by an unsatisfactory test result, the CQA representative will determine the extent of the defective area by additional tests, observations, a review of records, or other means that the CQA representative deems appropriate. If the defect is related to adverse site conditions, such as excessively wet soils or surface desiccation, the CQA representative will define the limits and nature of the defect by testing or observation. After the extent and nature of a defect is determined and remedied by the construction general contractor, the CQA representative will verify that the deficiency has been corrected by re-testing repaired areas before any additional work is performed by the construction general contractor in the area of the deficiency. All confirmatory tests, failing tests, and re-tests will be recorded in the CQA representative's field book or compaction testing form. The approximate location and elevation of each test will be recorded.

The CQA representative will document fill placement and compaction as determined by the test methods and frequency prescribed by this CQA Plan and will report any non-conformance in accordance with the non-conformance reporting procedures outlined in Section 8.1.4.

2.2 Prepared Subgrade

The CQA representative will verify and document that the prepared subgrade is constructed to the elevations and grades shown in the construction drawings, with subgrade meeting the requirements of the technical specifications as determined by the test methods and frequencies specified within this CQA Plan.

Upon completion of the excavation of the landfill, the CQA monitor will perform the following tasks:

- Inspect the subgrade on the side slopes and base of the landfill and note areas of weak or excessively weathered subgrade materials.
- Observe completion of excavation and subgrade compaction prior to foundation, fill, or liner placement.
- Observe the proof rolling of the base of the landfill and note areas that exhibit excessive rutting, heaving, or softening.
- Observe that the surface of the subgrade is free of debris, wet and soft areas, standing water, vegetation, mud, ice, or frozen material.
- Observe any excavation and backfilling operations associated with unsuitable material found in the prepared subgrade.

- Verify that a survey has been conducted to further verify that the subgrade grades and elevations conform to the construction drawings.
- Verify that the prepared subgrade material meets the requirements of the technical specifications as determined by the CQA testing methods and frequency in Table 2-3.
- Verify that sampling points in the prepared subgrade are plugged or backfilled so that the prepared subgrade meets the technical specifications.
- Document the location and volume of any unsuitable material removed from the prepared subgrade and report any non-conformance with the technical specifications in accordance with the non-conformance reporting procedures in Section 8.1.4

2.2.1 Layer Completion Certification

The construction general contractor will be required to notify the CQA representative when an area of prepared subgrade is complete prior to constructing the overlying layer. The construction general contractor can proceed with the overlying layer upon acceptance of the area of prepared subgrade by the CQA representative. The CQA certifying engineer will provide a certificate of layer completion to the construction general contractor and the IDF project engineer, certifying that the area is complete.

2.3 Soil Bentonite Admix Liner and Test Pads

The soil bentonite admix liner (SBL) is composed of a mixture of base soil and bentonite material. Two SBL test pads will include both a horizontal and a sloped test pad. The horizontal test pad will be constructed by using the same compaction methods as that used for the production SBL, to ensure the SBL is constructed to meet the minimum hydraulic conductivity requirements. The sloped test pad will be constructed on a sloping surface to verify that compaction methods (determined during the horizontal test pad) will be adequate for the side slopes of the landfill. If necessary, the technical specifications and/or CQA Plan may be modified, based on the results of the test pads.

2.3.1 Test Pads

Test pads will be constructed by the construction general contractor to determine acceptable placement and compaction methods to produce a low permeable SBL on a horizontal surface and on a 3H:1V side slope that satisfies the performance requirements of the technical specifications.

In addition, the mixing of the base soil and bentonite admixture using the pugmill will be tested to ensure adequate control of the ratio of admixture components as well as the homogeneity of the completed SBL mixture.

2.3.1.1 Construction Quality Assurance Evaluation

During test pad construction, the CQA representative will continuously observe and document the construction of the test pad. These guidelines will be followed to ensure that the test pad accurately represents the performance of the full-scale facility:

- Construction of the test pad will use the same soil material, design specifications, equipment, and procedures as proposed for the full-scale facility.
- The test pad length, width, and depth will be as required by the technical specifications and for the in-situ hydraulic conductivity test spacing.
- The number of lifts used to construct the test pad will be as required by the technical specifications.

The test pad will be constructed to allow determination of the relationship among density, moisture content, and method of compaction. Field variables can affect this relationship and must be carefully measured and controlled, both in the test pad and during construction of the full-scale liner. At a minimum, the following will be observed and documented:

- Track weight of base soil and bentonite during mixing operations.
- Mixing operation homogeneity.
- Test pad configuration and dimensions.
- Compaction equipment type, configuration, and weight.
- Number of passes and speed of the compaction equipment.
- Uncompacted and compacted lift thickness.
- Weather conditions, including ambient temperature, humidity, wind speed and direction, and precipitation.

The CQA representative will provide the necessary surveying and/or reference grid points, for adequately and expeditiously, determining the elevation and dimensions of the test pad, including each lift.

The CQA representative will be responsible for all testing, surveying, and documentation necessary to verify that the test pad performs in accordance with the technical specifications, and that the methods, equipment, and materials used can achieve the same results or better during full-scale construction.

Testing methods and frequencies will be as indicated in Table 2-4. Additional tests may be conducted at the direction of the CQA certifying engineer. All tests will be conducted in accordance with the methods and procedures specified in Table 2-4. Tests are separately identified in Table 2-4 which are intended to provide the following:

- Information Only, for use in evaluating overall methods, materials, or equipment.
- Pass/Fail, which have criteria established in the technical specifications which must be met.
- Calibration and Check, for use in calibrating instruments.

The CQA certifying engineer will compare the results of the test pad constructed on the level surface with the results of the test pad on the side slopes. The CQA certifying engineer will recommend changes to compaction methods, if necessary, to the IDF project engineer. The CQA certifying engineer will prepare an interim report, which summarizes the construction and testing of the test pads.

It is important to note that an acceptable zone has been established in the technical specifications for the allowable moisture content and density ranges that are applicable for the SBL admix to meet minimum permeability requirements. This zone may be adjusted as a result of the test pad data obtained during construction to reflect specific conditions observed based on the construction general contractor's proposed blending, placement, and compaction methods. With the range of placement moisture content and density allowed with this approach, minimum compaction effort (i.e., the number of passes a piece of compaction equipment needs to bring the admix into the allowable moisture/density zone) will vary based on material conditions and placement location. Minimum compaction effort recommendations will be developed by the CQA certifying engineer for application to both bottom slope and side-slope admix construction based on test pad results. However, these minimums should be considered as guidelines only and may need to be adjusted based on changes to admix properties (primarily moisture content), site conditions, and compaction location as needed to bring the admix into the required acceptable zone for compaction.

2.3.2 Soil Bentonite Admix Liner

The CQA team will verify and document that the SBL is placed to the elevations, grades, and thicknesses shown in the construction drawings, with bentonite-amended material meeting the requirements of the technical specifications as determined by the test methods and frequencies specified within this CQA Plan.

2.3.2.1 Construction Quality Assurance Evaluation

CQA testing will be performed during processing and placement of the SBL. The CQA team will conduct the processing and placement tests for the SBL as specified in Table 2-5. The maximum allowable percentage of failing tests is specified in Table 2-6.

Processing

The construction general contractor shall process and condition admix material using a central type pugmill plant as described in the technical specifications. Prior to amending the base soil with bentonite, a CQA representative will verify and document the following:

- Equipment and methods are the same or equivalent as determined from the test pad studies.
- All submittals have been reviewed and approved.
- The base soil source area (either onsite excavation or borrow area) has been approved by the design engineer, IDF PM, or IDF project engineer.
- The mixing equipment is suitable for amending base soils with bentonite.
- The base soil does not contain rocks with dimensions in excess of those required by the technical specifications.

During processing, the CQA representative will verify and document the following:

- The bentonite is in conformance with the technical specifications.
- Close observation of the base soil excavation and processing is performed by the field inspector.
- The processed SBL material meets the requirements of the technical specifications as determined by the CQA testing methods and frequency in Table 2-5.
- The moisture content and consistency of base soil allow bentonite to be mixed uniformly.
- Bentonite amendments are mixed uniformly with the base soil.
- The processed SBL material is stored, protected, and allowed to cure in accordance with the conditions and minimum requirements of the technical specifications.
- Calibration of the pugmill operation feed rate controls for bentonite, base soil, and water.
- The bentonite is mixed at the required application rate, established by the technical specifications as determined by the CQA testing methods and frequency in Table 2-5.

The CQA representative will document the properties of the processed soil bentonite material, as determined by the test methods and frequency prescribed by this CQA Plan, and will report any non-conformance with the technical specifications, following procedures outlined in Section 8.1.4.

The CQA representative will observe processing activities including base soil excavation, bentonite blending, and moisture conditioning.

The CQA representative will monitor the excavation of base soil from the approved borrow source or onsite excavations. Deleterious base soil or base soil not meeting the technical specifications will be identified and reported to the CQA certifying engineer and not allowed in the processing area.

CQA tests will be performed on the raw bentonite used in the SBL to verify conformance to the technical specifications. The CQA representative will collect samples of raw bentonite delivered to the site for testing. The CQA laboratory technician will conduct free swell, and grain size tests of the bentonite in accordance with Table 2-5. If the test results of a sample fail to meet specifications, a confirmatory test will be performed immediately subsequent to the failed test. If the confirmatory test meets or exceeds specifications, a second confirmatory test will be performed. If the second confirmatory test also meets or exceeds specifications, the bentonite will be declared as meeting project specifications and the confirmatory tests will be reported. In the event that either confirmatory test fails to meet specifications, the bentonite will be rejected and removed from the site.

The CQA representative will observe mixing and test the bentonite-amended soil, prior to placing it in the landfill.

Placement

Prior to the placement of the SBL, the CQA representative will verify and document the following:

- The test pads have been constructed with the approved liner material and production scale equipment to confirm placement and compaction procedures produce the required low-permeability admix for both on a horizontal surface and on a 3H:1V side slope.
- All or an approved portion of the prepared subgrade meets specifications as determined by the test requirements of this CQA Plan and the CQA certifying engineer has issued the completion certificate.
- The SBL material is free of roots, stumps, vegetation, or any other type of deleterious material that may impact the performance of the placed SBL.
- The SBL material does not contain stones with dimensions in excess of those required by the technical specifications.
- The SBL material meets or exceeds the requirements of the technical specifications as determined by the CQA testing methods and frequency in Table 2-5.
- The moisture content of the SBL material is uniform.

During placement and compaction of the SBL, the CQA Team will verify and document the following:

- Close observation of the placement and compaction of SBL material with earthmoving equipment is performed by the field inspectors. Inspectors to verify that means and methods are the same as those approved in the test pad process.
- The SBL material meets the requirements of the technical specifications as determined by the CQA testing methods and frequency in Table 2-5 and is within the maximum allowable failure rates in Table 2-6.
- The SBL is placed in accordance with the conditions and minimum requirements of the technical specifications.
- Each lift is compacted to the required thickness and minimum dry unit weight within the range of moisture contents established by the technical specifications as determined by the CQA testing methods and frequency in Table 2-5.
- Shelby tube samples are collected for laboratory permeability testing at the frequency specified in Table 2-5.
- Penetrations in the SBL at testing and sampling locations are repaired in accordance with the technical specifications.
- The SBL is maintained until it is covered by the geomembrane liner in accordance with the technical specifications.
- In areas of inaccessibility by the compactor, in areas of nonstandard SBL placement, and/or in areas of different compaction methods, more frequent testing will be performed due to thinner lift thicknesses to achieve equivalent compactive effort. Each lift, no matter how thin, will be tested for density and moisture in accordance with Table 2-5.

The CQA representative will document the properties of the SBL as determined by the test methods and frequency prescribed by this CQA Plan and will report any non-conformance in accordance with the non-conformance reporting, as outlined in Section 8.1.4.

The CQA representatives will collect samples immediately after a loose lift of SBL materials has been placed for property tests, prior to compaction. Once compacted, nuclear density gauge test methods will be used for testing the in situ compacted dry unit weight and moisture content of the SBL. Standard count calibration and moisture content tests will be used to calibrate the reading of the nuclear density gauge. Standard count calibration and in situ moisture content tests, using the oven dry method, will be performed at the frequencies specified in Table 2-5. The results of the oven dry moisture content tests will be compared with the field moisture content results to determine a field moisture correction factor.

The CQA representative will adjust the field moisture correction factor as test data is collected (i.e., moving average). The CQA representative will collect Shelby tube samples of the SBL for laboratory permeability tests as specified in Table 2-5.

If in-place density test results fail to meet specifications, a confirmatory test will be performed immediately adjacent to (within 3 ft of) the failed test. If the confirmatory test meets or exceeds specifications, a second confirmatory test will be performed at a second location immediately next to (within 3 ft of) the failed test. If the second confirmatory test also meets or exceeds specifications, the area will be declared as meeting project specifications and the confirmatory tests will be reported. In the event that either confirmatory test fails to meet specifications, additional testing will be performed to identify the limits of the area that does not meet project specifications. All confirmatory tests, failing tests, and re-tests will be recorded in the CQA representative's field book or compaction testing form. The approximate location and elevation of each test will be recorded.

Rapid laboratory permeability tests, such as the constant volume tests, will be used when possible to determine permeability. Once the sample has achieved the specified permeability, the test result will be reported immediately to the CQA certifying engineer. The number of failing tests will be less than the maximum percentage of failing tests specified in Table 2-6. The maximum percentage of failing tests are anticipated to cover laboratory or field recording mistakes, math errors, or other unknown circumstances that are not discovered until after the layer is covered with the succeeding layer(s). Otherwise, all failed tests will be corrected in the field as they are observed.

If a defective area is discovered in the SBL other than a failed in-place density test, the CQA representative will determine the extent and nature of the defect. If the defect is indicated by an unsatisfactory test result, the CQA representative will determine the extent of the defective area by additional tests, observations, a review of records, or other means that the CQA representative deems appropriate. If the defect is related to adverse site conditions, such as excessively wet soils or surface desiccation, the CQA representative will define the limits and nature of the defect by testing or observation. After the extent and nature of a defect is determined and remedied by the construction general contractor, the CQA representative will verify that the deficiency has been corrected by re-testing repaired areas before any additional work is performed by the construction general contractor in the area of the deficiency.

The testing frequency during the SBL construction may be increased or modified at the discretion of the CQA certifying engineer, when visual observations of construction performance indicate potential problems or when field experience with the proposed SBL material have been obtained.

During construction, the frequency of testing may be increased by the CQA representative during adverse weather conditions, if equipment breaks down, at the start and finish of grading, if the material fails to meet the requirements of the technical specifications, or if the extent of the work area is reduced.

The construction general contractor will repair all penetrations in the SBL resulting from sampling and other CQA activities, in accordance with the technical specifications. These perforations will be identified to the construction general contractor by the CQA representative. All repairs will be inspected by the CQA representative.

The construction general contractor will be required to use all means necessary to protect all prior work as well as all materials and completed work of other sections. In the event of damage, the construction general contractor will be required to make immediately all repairs and replacements necessary. The CQA representative will verify and document that all damages are repaired.

2.3.2.2 Layer Completion Certification

The construction general contractor will be required to notify the CQA representative when an area of SBL is complete, prior to constructing the overlying layer. The construction general contractor may begin placement of the overlying layer after acceptance of the SBL by the CQA certifying engineer.

The CQA certifying engineer will provide a certificate of layer completion to the construction general contractor and the IDF project engineer, certifying that the area is complete.

The CQA certifying engineer will ensure all CQA tests are complete and that all defective areas have been repaired and re-tested in accordance with this CQA Plan and the technical specifications. The certificate of layer completion will indicate that the SBL meets the low permeability requirement, based on laboratory tests and the thickness of the SBL meeting the minimum requirement specified in the technical specifications.

2.4 Drain Gravel

2.4.1 Conformance Evaluation

No CQA conformance material testing is planned for the drain gravel. Construction general contractor is required to submit gradation test results demonstrating conformance with required material properties as part of source quality control, in accordance with the technical specifications.

2.4.2 Placement and Compaction

The CQA representative will verify and document that the drain gravel is constructed to the elevations, grades, and thicknesses shown in the construction drawings, with material meeting the requirements of the technical specifications as determined by the test methods and frequencies specified within this CQA Plan.

Prior to the placement of the drain gravel, the CQA representative will verify and document that:

- The underlying geosynthetic layers are free of holes, tears, excessive wrinkles, or foreign objects.
- All work on underlying layers is complete and accepted by the CQA certifying engineer.

During placement and compaction of the drain gravel, the CQA representative will verify and document the following:

- Drain gravel material satisfies the requirements of the technical specifications as determined by the source quality control submittals.
- Drain gravel material is non-angular and free of material that could damage the underlying liner materials.
- Drain gravel material is spread during cooler portions of the day, unless otherwise approved by the CQA certifying engineer.
- Spreading and hauling equipment and operations are in compliance with material thickness and operations requirements, given in the technical specifications.
- If excessive wrinkles begin to develop in the underlying geosynthetics during gravel or sand placement or spreading, the wrinkles are worked out prior to continued placement operations.
- The drain gravel is placed in a manner that will not damage underlying geosynthetics, will minimize slippage of geosynthetic layers, and will not provide excess tensile stress on the geosynthetics, in accordance with the technical specifications.
- Close observation of the placement and compaction of drain gravel with earth moving equipment is performed.

2.4.3 Construction Quality Assurance Evaluation

No density tests will be conducted on the drain gravel. If the CQA representative suspects damage to pipes or underlying geosynthetic, the construction general contractor will be required to expose the potentially damaged materials and repair any observed damage.

2.4.4 Layer Completion Certification

The construction general contractor will be required to notify the CQA representative when an area of the LCRS or LDS drain gravel is complete, prior to constructing the overlying layer.

The construction general contractor may begin placing the overlying layer when the drain gravel is accepted by the CQA certifying engineer. The CQA certifying engineer will provide a certificate of layer completion to the construction general contractor and the IDF project engineer, certifying that the area is complete.

2.5 Operations Layer

The CQA representative will verify and document that the operations layer, including the operations layer material placed in the SLDS sump area, is constructed to the elevations, grades, and thicknesses shown in the construction drawings, with material meeting the requirements of the technical specifications as determined by the test methods and frequencies specified within this CQA Plan.

Prior to the placement of the operations layer, the CQA representative will verify and document the following:

- The underlying geosynthetic layer is free of holes, tears, excessive wrinkles, or foreign objects.
- All work on underlying layers is complete and accepted by the CQA certifying engineer.

During placement of the operations layer, the CQA representative will verify and document that:

- The soil is suitable and satisfies the requirements of the technical specifications as determined by the test methods and frequencies prescribed in Table 2-7.
- The operations soil is placed in accordance with the technical specifications and construction drawings.
- The lift thicknesses and total thickness of the operations layer agree with the requirements of the construction drawings.
- If excessive wrinkles begin to develop in the underlying geosynthetics during material placement or spreading, the wrinkles are worked out prior to continued placement operations.
- The operations layer is placed in a manner that will not damage underlying geosynthetics, will minimize slippage of geosynthetic layers, and will not provide excess tensile stress on the geosynthetics, in accordance with the technical specifications.
- Spreading and hauling equipment and operations are in compliance with material thickness and operations requirements given in the technical specifications.
- The operations layer is placed on the side slopes to the limits shown in the construction drawings.
- No operations layer material is placed or compacted during periods of unfavorable weather conditions, such as after heavy rains or snow, in accordance with requirements given in the technical specifications.

2.5.1 Conformance Evaluation

The test methods and frequencies for CQA conformance testing for the operations layer are specified in Table 2-7.

If damage to underlying geosynthetics is suspected, the CQA representative will require that the overlying operations layer material be removed to expose the geosynthetics.

The construction general contractor will be required to use all means necessary to protect all prior work, as well as all materials and completed work of other sections. In the event of damage, the construction general contractor will be required to make immediately all repairs and replacements necessary. The CQA representative will verify and document that all damages are repaired.

2.5.2 Layer Completion Certification

The construction general contractor will be required to notify the CQA representative when an area of the operations layer is complete. The CQA certifying engineer will provide a certificate of layer completion to the construction general contractor and the IDF project engineer, certifying that the area is complete.

2.6 Soil Surveying

A survey will be performed by or under the direction of a professional land surveyor registered in the State of Washington. The surveyor will independently survey the elevations and grades of the soil layers including, but not limited to:

- Top of prepared subgrade.
- Top of SBL.
- Top of LCRS drain gravel.
- Top of operations layer.

Surveys will be performed on the base, side slopes of the landfill, to confirm that the grades and elevations in the field agree with those shown in the construction drawings and with the minimum acceptable tolerances required in the technical specifications. The results of the survey, conducted by the surveyor, will be compiled in a report signed by the surveyor and the CQA certifying engineer.

The surveyor will be required to survey each soil layer of the liner system for the IDF landfill, in accordance with the requirements of this CQA Plan. A record drawing or tabular listing of surveyed points will be submitted to the CQA certifying engineer by the surveyor before the placement of the next liner system layer. The surveys will be conducted at a 50-ft grid across the entire area of the survey. The survey will include, but not be limited to, the following features of the landfill:

- Toe of slope.
- Crest of slope.
- Grade breaks.
- Anchor trench.
- SLDS, LDS and LCRS sumps.

Table 2.1. Minimum Frequency of Testing for CQA Evaluation of Earthfill

Test	Frequency	Standard Test Method
Material Properties		
Standard proctor or maximum index density for free-draining soil	1 per 20,000 yd3 (minimum 1 per source or soil type)	ASTM D698 or ASTM D4253
Placement		
In-place wet unit weight	1 per 5,000 ft2 per lift	ASTM D2922, D1556
In-place moisture content	1 per 5,000 ft2 per lift	ASTM D3017, D2216
Standard count calibration	1 per day of fill placement	ASTM D3017/D2922
Oven moisture contents (in situ moisture content)	1 per day of fill placement	ASTM D2216

Table 2.2. Minimum Frequency of Testing for CQA Evaluation of Structural Fill

Test	Frequency	Standard Test Method
Material Properties		
Standard proctor or maximum index density for free-draining soil	1 per 2,000 tons (minimum 1 per source or soil type)	ASTM 698 or ASTM 4253

Test	Frequency	Standard Test Method
Placement		
In place moisture content	1 per 2,500 ft2 per lift	ASTM D3017, D2216
In place dry unit weight	1 per 2,500 ft2 per lift	ASTM D2922, D1556
Standard count calibration	1 per day of fill placement	ASTM D3017/D2922
Oven moisture contents (in situ moisture content)	1 per day of fill placement	ASTM D2216

1 **Table 2.3. Minimum Frequency of Testing for CQA Evaluation of Prepared Subgrade**

Test	Frequency	Standard Test Method
Material Properties [a]		
Standard proctor or maximum index density for free-draining soil	1 per 250,000 ft2 (minimum 1 per source or soil type)	ASTM 698 or ASTM 4253
In Place [b]		
In-place wet unit weight	4 per acre (approx. 1 per 10,000 ft2)	ASTM D2922, D1556
In-place moisture content	4 per acre (approx. 1 per 10,000 ft2)	ASTM D3017, D2216
Standard count calibration	1 per day when in place tests are performed	ASTM D3017/D2922
Concrete block calibration	1 per day when in place tests are performed	ASTM D3017/D2922
Over moisture content (in situ moisture content)	1 per day when in place tests are performed	ASTM D2216

[a] Prior to subgrade excavation.
[b]. After reaching subgrade elevation.

2

1 **Table 2.4. Test Pad Testing Methods and Minimum Frequency**

Test	Frequency	Standard Test Method
Material Property		
Natural moisture content [a]	4 per each Base Soil type per test pad	ASTM D2216
Particle size distribution [b]	4 per each Base Soil type per test pad	ASTM D422
Standard proctor [a]	4 per each Base Soil type per test pad	ASTM D698
Bentonite dry fineness [b]	1 per test pad	Technical specification
Bentonite high swelling [b]	1 per test pad	Technical specification
Processing [b]		
Bentonite/Base Soil application rate	1 per working day of admix processing during test pad construction	Measure weight of base soil and bentonite entering pugmill during a given period of time
Maximum clod size	Periodic monitoring	Observation
Curing	1 per 12 hours	Observation
Pre Compaction [a]		
Lift thickness	1 per lift	Field measurement
Percent fines	1 per lift	ASTM D1140
Percent gravel	1 per lift	ASTM D422
Atterberg limits	1 per lift	ASTM D4318
Placement	Periodic monitoring	Observation
Post Compaction		
Lift thickness [b]	4 per lift	Field measurement
In-place moisture content [b]	4 per lift	ASTM D3017
In-place dry unit weight [b]	4 per lift	ASTM D2922
Shelby tube samples [b] (laboratory permeability)	1 per lift	ASTM D1587/ASTM D5084[c,d]
Number of passes [a]	Monitor each lift	Observation
Boutwell permeameter test [b]	5 for the horizontal test pad only	ASTM D6391, first stage only
Calibration and Check [e]		
Standard count calibration	1 per day of fill placement	ASTM D3017/D2922
Oven moisture contents (in situ moisture content)	1 per lift	ASTM D2216
In-place dry unit weight	1 per lift	ASTM D1556, D2167, or D2937

Notes:

[a] Tests for information only

[b] Pass/fail tests

[c] The average effective confining stress will be 5 psi.

[d] Rapid turnaround tests (Method F–Constant Volume) will be used when possible.

[e] Calibration check tests

1 **Table 2.5. Minimum Frequency of Testing for CQA Evaluation of SBL**

Test	Frequency	Standard Test Method
Bentonite		
Dry fineness	1 per lot	Technical specification a
High swelling	1 per lot	Technical specification
Processing		
Base soil excavation	Periodic monitoring	Observation
Base soil natural moisture content	1 per working day of hauling base material or per material color/consistency change	ASTM D2216
Base soil grain size	1 per working day of hauling base material or per material color/consistency change	ASTM D422
Bentonite/Base Soil application rate	1 per working day of admix processing	Measure weight of base soil and bentonite entering pugmill during a given period of time
Maximum clod size	Periodic monitoring	Observation
Curing c	1 per 12 hours	Observation
Pre-Compaction		
Lift thickness d	1 per 2,500 ft^2 per lift	Field measurement
Percent fines	1 per 1,000 yd^3 (minimum of 1 per day of placement)	ASTM D1140
Percent gravel	1 per 1,000 yd^3 (minimum of 1 per day of placement)	ASTM D422
Atterberg limits	1 per 1,000 yd^3 (minimum of 1 per day of placement)	ASTM D4318
Placement	Periodic monitoring	Observation
Post Compaction		
Lift thickness	5 per acre per lift	Full measurement
In place moisture content	5 per acre per lift	ASTM D3017
In place dry unit weight	5 per acre per lift	ASTM D2922
Shelby tube samples (laboratory permeability)	1 per acre per lift	ASTM D1587/ ASTM D5084 e, f
Number of passes g	Observe 1 per acre per lift	Observation
Construction oversight	Periodic monitoring	Observation
Calibration and Check		
Oven moisture content (per each nuclear gauge)	1 per 10 nuclear gauge moisture contents	ASTM D2216
Standard count calibration (per each nuclear gauge)	1 per day of placement	ASTM D2922/ASTM D3017

[a] The test method is described in the technical specification.
[b] Not used.
[c] Curing is stockpiling the SBL material for 12 hours to allow the bentonite to hydrate.
[d] A loose lift thickness is such that the compacted thickness is 6 inches or less.
[e] The average effective confining stress will be 5 psi.
[f] Rapid turnaround tests (Method F – Constant Volume) will be used when possible.
[g] A single pass is defined as forward and back.

Table 2.6. Maximum Allowable Percentage of Failed Tests for CQA Evaluation of SBL

Test	Maximum percentage
Percent gravel	5 percent not concentrated in one lift or one area
Clod size	10 percent not concentrated in one lift or one area
In place moisture content	3 percent not concentrated in one lift or one area, and no water content less than 2 percent or more than 3 percent of the specified value
In place dry unit weight	3 percent not concentrated in one lift or one area, and no dry unit weight less than 5 pounds per cubic foot (pcf) below the specified value
Shelby tube samples (laboratory permeability)	5 percent not concentrated in one lift or one area

Table 2.7. Minimum Frequency of Testing for CQA Evaluation of Operations Layer

Test	Frequency	Standard Test Method
Material Properties		
Standard proctor or maximum index density for free-draining soil	1 per 20,000 yd^3 (minimum 1 per source or soil type)	ASTM D698 or ASTM D4253
Sieve analysis	1 per 10,000 yd^3 placed (minimum 1 per source)	ASTM D422
In-Place (Outside edge of liner only)		
In-place wet unit weight	1 per 5,000 ft^2 per lift	ASTM D2922, D1556
In-place moisture content	1 per 5,000 ft^2 per lift	ASTM D3017, D2216
In-Place (SLDS sump)		
In-place wet unit weight	2 per lift	ASTM D2922, D1556
In-place moisture content	2 per lift	ASTM D3017, D2216

SECTION 3 GEOSYNTHETIC CLAY LINER CONSTRUCTION QUALITY ASSURANCE

3.1 Geosynthetic Clay Liner Manufacture and Delivery

3.1.1 Labeling

The CQA representative will verify and document that the GCL manufacturer has labeled each roll of GCL and includes the information required by the technical specifications. The CQA representative will examine GCL rolls upon delivery and deviation from the above requirements will be reported to the CQA certifying engineer prior to installation of the GCL.

3.1.2 Transportation and Handling

The CQA representative will observe and document that the type of GCL handling equipment used by the installer minimizes damage to the material. Upon delivery at the site, the CQA representative will conduct a visual inspection of all rolls for defects and for damage. This examination will be conducted without unrolling rolls unless visible defects or damages are found. The CQA representative will indicate to the CQA certifying engineer:

- Any rolls that need to be unrolled to allow for their inspection.
- Any rolls, or portions thereof, that need to be rejected and removed from the site because they have severe flaws.
- Any rolls that include minor repairable flaws.

3.1.3 Storage

The CQA representative will verify and document that storage of the GCL is in accordance with the technical specifications.

3.1.4 Inventory

All geosynthetic materials that arrive onsite will be inventoried. The inventory will include the specific roll numbers delivered with each shipment. The inventory will be compared to the QC testing information, supplied by the manufacturer to ensure that the material tested is the same material that was delivered to the site. Material for which QC testing data has been supplied will be sampled for conformance testing. Conformance samples may be obtained by the CQA representative at the manufacturing plant or taken upon delivery of the material to the site by a CQA representative. As shipments arrive at the site, a CQA representative will monitor the unloading operations and will inventory the material. Rolls selected for conformance testing will be set aside for sampling as soon as possible.

The CQA representative will record the following information, at a minimum, for each roll:

- **Manufacturer**–Indicate the manufacturer of the material that is being inventoried, which may not be the same as the installer.
- **Date of Inventory**–Date that the material was inventoried.
- **Date of Delivery**–Enter date when the truck arrived onsite, if known.
- **Truck Type**–Indicate type of truck used for shipping geosynthetics (covered or uncovered flatbed, box trailer).
- **Bill of Lading Number**–If the bill-of-lading is available, indicate number and date (also attach copy to inventory form).
- **CQA Representative**–Indicate name of CQA representative performing inventory.
- **Unloading Equipment**–Indicate the type and model number of the equipment unloading the geosynthetic material; also note any special attachments that are used to unload the material (stinger, straps, forks).

- **Weather Conditions**–Describe the weather conditions, including temperature, wind, cloud cover, and precipitation during unloading and conformance sampling operation.
- **Material Type**–Indicate type of geosynthetic material.
- **Roll Number**–Indicate each roll number that is written on the roll. (The roll numbers contain a variety of information regarding the material and the manufacturing process.)
- **Lot Number**–Lot number.
- **Roll (L × W)**–Indicate the roll width as written on the roll label; if two materials are bonded together (i.e., geonet/geotextile), obtain measurements for both materials.
- **Area (square feet)**–Indicate the total square footage of the roll.
- **Damage Remarks**–Document any visible damage to the roll; if possible, indicate if damage was present prior to unloading or if it occurred during unloading.

The CQA representative will immediately notify the IDF CM if a nonconforming or conditional use tag is attached to any of the inventoried items.

Items that are restricted from further use until the inspections have been completed will be clearly delineated by the CQA representative. Accepted materials will be kept separate or clearly delineated from inventoried and approved items, to the extent possible. The CQA representative will be responsible for coordinating with the construction general contractor during material delivery, so that the material is not moved more than necessary after it is unloaded and damage due to handling is minimized.

The CQA representative will perform the inventory immediately after the material arrives on the site to avoid delaying construction. The CQA representative will be responsible for verifying that only accepted material is installed at the IDF landfill and that all inventories and inspections are documented and maintained.

3.1.5 Quality Assurance Conformance Testing

Either at the manufacturer's plant or upon delivery of the rolls of GCL, the CQA representative will ensure that samples are removed at the specified frequency and forwarded to the Geosynthetics CQA Laboratory for testing, to verify and document conformance with the technical specifications.

Conformance samples will be taken across the entire width of the roll and will not include the first 3 feet along the length of the roll. Unless otherwise specified, samples will be 1.5 feet (minimum) long by the roll width. The CQA representative will mark the machine direction on the samples with an arrow.

Unless otherwise specified, samples will be taken at a rate of one per lot or one per 50,000 square feet, whichever is greater. These samples will be tested for:

- Index Flux (ASTM D5887).
- Bentonite Mass per Unit Area (ASTM D5993).
- Bentonite Swell Index Test (ASTM D5890).

The test will be conducted in accordance with the test procedure presented in the technical specifications.

The CQA representative will examine all results from laboratory conformance testing and compare the results to the specifications presented in the technical specifications. In addition, the CQA representative will report any non-conformance to the CQA certifying engineer as soon as practical after the test results become available.

The following procedure will apply whenever a sample fails a conformance test that is conducted by the Geosynthetics CQA Laboratory:

- The construction general contractor will be required to replace the roll (or rolls) of GCL not in conformance with the specifications with a roll, that meets the requirements of the technical specifications.

- The CQA representative will ensure that conformance samples are removed for testing by the Geosynthetics CQA Laboratory from the closest numerical roll on both sides of the roll from which the failing sample was obtained. These two samples must pass the above conformance tests. If either of these samples fails to meet the requirements, samples will be collected from the five numerically closest untested rolls on both sides of the failed samples and tested by the Geosynthetics CQA Laboratory. These ten samples must pass the above conformance tests. If any of these samples fail, a sample from every roll of GCL onsite and a sample from every roll that is subsequently delivered from the same manufacturer must be conformance tested by the Geosynthetics CQA Laboratory until the manufacturer has thoroughly demonstrated compliance with the above requirements to the sole satisfaction of the CQA certifying engineer. The costs of all such tests are to be borne by the construction general contractor.
- The CQA representative will document actions taken in conjunction with conformance test failures as outlined in Section 8.1.4 and report all actions to the CQA certifying officer.

3.2 Geosynthetic Clay Liner Installation

3.2.1 Surface Preparation

For fill surfaces that will underlay a GCL layer, the CQA representative will verify and document the following:

- The surface of the fill does not contain holes, ruts, protrusions, or other surface irregularities in excess of those dimensions specified by the technical specifications.
- The surface of the fill has been compacted to form a firm, stable base.
- The surface of the fill is free of any type of deleterious material that may cause damage to GCL, including debris, organic material, frozen soil, ice, and rocks.
- The surface of the fill is free of standing water or excessive moisture.
- The construction general contractor has certified in writing that the surface on which the GCL will be installed is acceptable.

The subgrade surface will be inspected immediately prior to commencement of GCL installation. If any change in the surface requires repair work, in accordance with the technical specifications, the construction general contractor will be responsible for repairing the fill surface.

A certificate of subgrade surface acceptance will be required from the construction general contractor. The CQA representative will verify that the subgrade is accepted by the GCL installer, immediately prior to commencement of GCL installation.

After the surface on which the GCL is to be installed has been accepted by the construction general contractor, it will be the CQA representative's responsibility to indicate to the CQA certifying engineer any change in the underlying layer that may, in accordance with the technical specifications, require repair work. If the CQA certifying engineer requires that repair work be done, it will be the responsibility of the construction general contractor to repair the underlying layer.

3.2.2 Anchor Trenches and Sumps

Prior to placement of geosynthetics in the anchor trenches or sumps, the CQA representative will verify and document the following:

- The sumps and anchor trenches are excavated to the grades and dimensions shown in the construction drawings. Any anomalies in the soil encountered during excavation will be brought to the attention of the IDF project engineer and removed as directed.
- The anchor trench excavation surface is prepared for installation of geosynthetics, with rounded corners, and free of loose soil or deleterious material.

After geosynthetics deployment into the anchor trench is complete, the CQA representative will verify and document that the backfill for the geosynthetic anchor trenches is placed and compacted in accordance with the technical specifications.

3.2.3 Geosynthetic Clay Liner Deployment

3.2.3.1 Field Panel Identification

A field panel is the unit area of GCL that is to be placed in the field (i.e., a field panel is a roll or a portion of roll cut in the field).

The CQA representative will track the placement location of each GCL panel by assigning an identification code (number or letter-number) or by an equivalent tracking method. The identification method will be agreed upon by the CQA certifying engineer and the construction general contractor. This field panel identification scheme will be as simple and logical as possible. (Note: manufacturing plant roll numbers are usually cumbersome and are not related to location in the field.) It will be the responsibility of the construction general contractor to ensure that each field panel placed is marked with the manufacturing plant roll number. The roll number will be marked in the center of the panel in a color to allow for easy inspection.

The CQA representative will establish a table or chart showing correspondence between manufacturing plant roll numbers and field panel identification codes. The field panel identification code will be used for all CQA records.

3.2.3.2 Field Panel Placement

Installation Schedule
The CQA representative will evaluate significant changes in the schedule, proposed by the construction general contractor, and will advise the CQA certifying engineer on the acceptability of that change. The CQA representative will verify and document that the condition of the underlying layer has not changed detrimentally during installation. Any damage to the surface of the underlying layer will be repaired by the construction general contractor, in accordance with the technical specifications.

Weather Conditions
The CQA representative will verify and document that GCL is not placed during inclement weather conditions, as specified in the technical specifications. Additionally, the CQA monitor will verify and document that the existing underlying layer has not been damaged by weather conditions.

Damage
The CQA representative will visually observe each panel, after placement, for damage. The CQA representative will inform the construction general contractor which panels, or portions of panels, are rejected, repaired, or accepted. Damaged panels or portions of damaged panels that have been rejected by the CQA representative will be marked, and their removal from the work area will be documented by the CQA representative.

Seam Overlap and Bentonite Seal
The construction general contractor will observe and document that the seam overlaps and bentonite material placed between panels, if required, along the seams meets specification guidelines. The CQA representative will verify overlap width and will observe bentonite seal placement.

3.2.3.3 Field Panel Protection

The CQA representative will observe and document that the GCL is completely covered with geomembrane or protective plastic cover at the end of each workday and protected from damage and hydration due to weather. The CQA representative will verify and document that equipment does not operate directly on the GCL and that a smooth rub sheet is used to maneuver textured geomembrane over the GCL to prevent damage to the GCL.

3.2.4 Defects and Repairs

Any defects and subsequent repairs will be documented, using NCR procedures outlined in Section 8.1.4.

3.2.4.1 Identification

All seams and non-seam areas of the GCL will be inspected by the CQA representative for evidence of defects, holes, contamination of geotextiles, displaced panels, premature hydration, and any sign of contamination by foreign matter. The CQA representative will observe and document repair procedures described below.

3.2.4.2 Repair Procedures

Prior to cover material placement, damage to the GCL will be identified and repaired by the installer.

Rip and Tear Repair (Flat Surfaces)
Rips or tears may be repaired by completely exposing the affected area, removing all foreign objects or soil, and by then placing a patch cut from unused GCL over the damage (damaged material may be left in place), with a minimum overlap of 12 inches on all edges.

Accessory bentonite will be placed between the patch edges and the repaired material at a rate of a quarter pound per lineal foot of edge, spread in a continuous 6-inch fillet.

Rip and Tear Repair (Slopes)
Damaged GCL material on slopes will be repaired by the same procedures as described above, however, the overlapped edges of the patch need to be wide enough to ensure the patch will keep its position during backfill or cover operations.

Displaced Panels
Displaced panels will be adjusted to the correct position and orientation. The adjusted panel will then be inspected for any geotextile damage or bentonite loss. Damage will be repaired by the above-described procedure.

Premature Hydration
If the GCL is subjected to premature hydration, the construction general contractor will notify the CQA certifying engineer for a site-specific determination as to whether the material is acceptable or if alternative measures must be taken to ensure the quality of the design dependent upon the degree of damage.

SECTION 4 GEOMEMBRANE CONSTRUCTION QUALITY ASSURANCE

4.1 Geomembrane Material

4.1.1 Labeling

The CQA representative will verify and document that the geomembrane manufacturer has labeled each roll of geomembrane and includes the information required by the technical specifications. The CQA representative will examine geomembrane rolls upon delivery and deviation from the requirements will be reported to the CQA certifying engineer, prior to installation of the geomembrane.

4.1.2 Transportation and Handling

Upon delivery at the site, the CQA representative will conduct a visual inspection of all rolls for defects and damage. This examination will be conducted without unrolling rolls unless visible defects or damage is found. The CQA representative will indicate the following to the CQA certifying engineer:

- Any rolls that need to be unrolled to allow for their inspection.
- Any rolls, or portions thereof, that need to be rejected and removed from the site because they have severe flaws.
- Any rolls that include minor repairable flaws.

4.1.3 Storage

The CQA representative will verify and document that storage of the geomembrane is in accordance with the technical specifications.

4.1.4 Inventory

All geosynthetic materials that arrive onsite will be inventoried in accordance with the technical specifications. The inventory will include the specific roll numbers delivered with each shipment. The inventory will be compared to the QC testing information supplied by the manufacturer to ensure that the material tested is the same material that was delivered to the site. Material for which QC testing data has been supplied will be sampled for conformance testing. Conformance samples may be obtained by the CQA representative at the manufacturing plant or taken upon delivery of the material to the site by a CQA representative. As shipments arrive at the site, a CQA representative will monitor the unloading operations and will inventory the material. Rolls selected for conformance testing will be set aside for sampling as soon as possible.

The CQA representative will record the following information, at a minimum, for each roll:

- **Manufacturer**–indicate the manufacturer of the material that is being inventoried, that may not be the same as the installer.
- **Date of Inventory**–Date that the material was inventoried.
- **Date of Delivery**–Enter date when the truck arrived onsite, if known.
- **Truck Type**–Indicate type of truck used for shipping geosynthetics (covered or uncovered flatbed, box trailer).
- **Bill-of-Lading Number**–If the bill-of-lading is available, indicate number and date (also attach copy to inventory form).
- **CQA Representative**–Indicate name of CQA representative performing inventory.
- **Unloading Equipment**–Indicate the type and model number of the equipment unloading the geosynthetic material; also note any special attachments that are used to unload the material (stinger, straps, forks).
- **Weather Conditions**–Describe the weather conditions, including temperature, wind, cloud cover, and precipitation during unloading and conformance sampling operation.
- **Material Type**–Indicate type of geosynthetic material (HDPE, geotextile, or geonet)

- **Roll Number**–Indicate each roll number that is indicated on the roll. (The roll numbers contain a variety of information regarding the material and the manufacture process.)
- **Lot Number**–Lot number as indicated.
- **Roll (L x W)**–Indicate the roll width as indicated on the roll label; if two materials are bonded together (i.e., geonet/geotextile), obtain measurements for both materials.
- **Area (square feet)**–Indicate the total square footage of the roll.
- **Damage Remarks**–Document any visible damage to the roll; if possible, indicate if damage was present prior to unloading or if it occurred during unloading.

Items that are restricted from further use until the inspections have been completed will be clearly delineated by the CQA representative. Accepted materials will be kept separate or clearly delineated from inventoried and approved items to the extent possible. The CQA representative will be responsible for coordinating with the construction general contractor during material delivery, so that the material is not moved more than necessary after it is unloaded and damage due to handling is minimized.

The CQA representative will perform the inventory immediately after the material arrives onsite to avoid delaying construction. The CQA representative will be responsible for verifying that only accepted material is installed at the IDF landfill, and that all inventories and inspections are documented and maintained.

4.1.5 Quality Assurance Conformance Testing

Either at the manufacturer's plant or upon delivery of the rolls of geomembrane, the CQA representative will ensure that samples are removed at the specified frequency and forwarded to the Geosynthetics CQA Laboratory for testing to verify and document conformance with the technical specifications.

Conformance samples will be taken by the CQA representative across the entire width of the roll and will not include the first 3 feet. Unless otherwise specified, samples will be 3 feet (minimum) long by the roll width. The CQA representative will mark the direction of the machine used to cut the samples with an arrow.

Unless otherwise specified, samples will be taken at a rate of one per lot or one per 50,000 square feet, whichever is greater. These samples will be tested for:

- Thickness (ASTM D5199 or D5994).
- Tensile characteristics (yield strength and elongation at yield, ASTM D638).
- Asperity (GRI GM-12).
- Puncture resistance (ASTM D4833).

Test will be conducted in accordance with the test procedure presented in the technical specifications. The CQA representative will examine all results from laboratory conformance testing and will report any non-conformance after the test results become available. The following procedure will apply whenever a sample fails a conformance test that is conducted by the CQA representative:

- The construction general contractor will be required to replace the roll (or rolls) of geomembrane in non-conformance with the technical specifications with a roll, that meets the technical specifications.
- The CQA certifying engineer will ensure that conformance samples are removed for testing by the Geosynthetics CQA Laboratory from the closest numerical roll on both sides of the failed roll. These two samples must pass the above conformance tests. If either of these samples fails, samples will be collected from the five numerically closest untested rolls on both sides of the failed sample and tested by the Geosynthetics CQA Laboratory. These ten samples must pass the above conformance tests.

If any of these samples fail, a sample from every roll of geomembrane onsite and every roll subsequently delivered from the same manufacturer must be conformance tested by the Geosynthetics CQA Laboratory, until the manufacturer has thoroughly demonstrated compliance with the above requirements to the sole satisfaction of the CQA certifying engineer. The costs of all such tests are to be borne by the construction general contractor.

4.1.6 Manufacturing Plant Site Visit

The manufacturer shall allow the CQA certifying engineer or his designated representative to visit the manufacturing plant, if the CQA certifying engineer so chooses. If possible, the visit shall be prior to or during the manufacturing of the geomembrane rolls for the specific project. The CQA Engineer or his designated representative shall review the manufacturing process, quality control, laboratory facilities, and testing procedures as described in the technical specifications (see Section 02661).

4.2 Installation

4.2.1 Surface Preparation

For SBL surfaces that will underlay a geomembrane layer, the CQA representative will verify and document the following:

- The surface of the subgrade or SBL does not contain holes, depressions, or protrusions in excess of those dimensions specified by the technical specifications.
- The surface of the subgrade or SBL has been rolled with a smooth-drum roller to form a firm stable base without ridges, wheel ruts, and surface irregularities.
- The surface of the subgrade or SBL is free of any type of deleterious material that may cause damage to geomembrane.
- The construction general contractor has certified in writing that the surface on which the geomembrane will be installed is acceptable.

The subgrade and SBL surface will be inspected immediately prior to commencement of geomembrane installation. If any change in the surface requires repair work, in accordance with the technical specifications, the construction general contractor will be responsible for repairing the surface. A certificate of subgrade surface acceptance will be required from the construction general contractor. The CQA representative will verify that the subgrade is accepted by the geomembrane installer immediately prior to commencement of geomembrane installation.

After the surface on which the geomembrane is to be installed has been accepted by the construction general contractor, it will be the CQA representative's responsibility to indicate to the CQA certifying engineer any change in the underlying layer that may, in accordance with the technical specifications, require repair work. If the CQA certifying engineer requires that repair work be done, it will be the responsibility of the construction general contractor to repair the underlying layer.

4.2.2 Anchor Trenches and Sumps

Prior to placement of geosynthetics in the anchor trenches or sumps, the CQA representative will verify and document the following:

- The excavation of the sumps and anchor trenches is performed in accordance with the technical specifications. Any anomalies in the soil encountered during excavation will be brought to the attention of the IDF project engineer and removed as directed.
- The anchor trench excavation surface is prepared for installation of geosynthetics with rounded corners and is free of loose soil or deleterious material.

After geosynthetics deployment into the anchor trench is complete, the CQA representative will verify and document that the backfill for the geosynthetic anchor trenches is placed and compacted in accordance with the technical specifications and construction drawings.

4.2.3 Geomembrane Deployment

4.2.3.1 Layout Drawing

The construction general contractor will be required to produce layout drawings that show the geomembrane panel configuration, dimensions, details, and seam locations. The layout drawings must be approved by the CQA certifying engineer, prior to the installation of the geomembrane.

4.2.3.2 Field Panel Identification

A field panel is the unit area of geomembrane that is to be seamed in the field (i.e., a field panel is a roll or a portion of roll cut in the field).

The CQA representative will verify that each field panel is given an identification code (number or letter-number) consistent with the layout plan. This identification code will be agreed upon by the CQA representative and the construction general contractor. This field panel identification code will be as simple and logical as possible. (Note: manufacturing plant roll numbers are usually cumbersome and are not related to location in the field.) It will be the responsibility of the construction general contractor to ensure that each field panel placed is marked with the manufacturing plant roll number. The roll number will be marked in the center of the panel in a color to allow for easy inspection.

The CQA representative will establish a table or chart showing correspondence between manufacturing plant roll numbers and field panel identification codes. The field panel identification code will be used for all CQA records.

4.2.3.3 Field Panel Placement

Location
The CQA representative will verify and document that field panels are installed at the locations and positions indicated in the construction general contractor's layout plan, as approved or modified by the CQA certifying engineer.

Installation Schedule
The CQA representative will evaluate significant changes in the schedule, proposed by the construction general contractor, and will advise the CQA certifying engineer on the acceptability of that change. The CQA representative will verify and document that the condition of the underlying layer has not changed detrimentally during installation. Any damage to the surface of the underlying layer will be repaired by the construction general contractor in accordance with the technical specifications.

The CQA representative will record the identification code, location, and date of installation of each field panel.

Weather Conditions
The CQA representative will verify and document that geomembrane is not placed during inclement weather conditions, as specified in the technical specifications. Additionally, the CQA representative will verify and document that the underlying layer has not been damaged by weather conditions.

Damage
The CQA representatives will visually observe each panel, after placement and prior to seaming, for damage (e.g., holes, blisters, and creases). The CQA representative will inform the construction general contractor which panels, or portions of panels, need to be rejected, repaired, or accepted. Damaged panels or portions of damaged panels that have been rejected by the CQA certifying engineer will be marked, and their removal from the work area will be documented by the CQA representative, using the NCR procedures outlined in Section 8.1.4.

4.2.4 Field Seaming

4.2.4.1 Seam Layout

The CQA certifying engineer will verify and document that the seam layout shown in the panel layout drawing is consistent with the technical specifications. A seam numbering system compatible with the panel numbering system will be agreed upon by the construction general contractor and CQA certifying engineer.

4.2.4.2 Seaming Equipment and Products

Processes approved by the technical specifications for field seaming are extrusion seaming; and fusion seaming. Proposed alternate processes will be required to be documented and submitted to the CQA certifying engineer for approval. The construction general contractor will be required to use a pyrometer to ensure that accurate temperatures of the extrudate and seamer nozzle are being achieved.

The extrusion seaming apparatus will be equipped with gauges, indicating the temperatures of the extrudate and nozzle. The construction general contractor will be required to provide to the CQA certifying engineer the manufacturer's certification that the extrudate is compatible with the geomembrane material and is comprised of the same resin as the geomembrane.

The CQA representative will log ambient temperatures, seaming apparatus temperatures, and extrudate temperatures or fusion seaming apparatus speeds. Ambient temperatures will be measured to verify compliance with the technical specifications.

4.2.4.3 Seam Preparation

The CQA certifying engineer will verify and document the following:

- Prior to seaming, the seam area is clean and free of moisture, dust, dirt, debris, and foreign material.
- Preparation of seams is in accordance with the technical specifications.

4.2.4.4 Weather Conditions for Seaming

The CQA representative will verify and document that weather conditions for seaming are within the limits specified in the technical specifications.

4.2.4.5 Trial Seams

The construction general contractor will be required to make trial seams on fragment pieces of geomembrane liner to verify that seaming conditions are adequate. The construction general contractor will be required to make and test trial seams at the frequency and in accordance with the methods specified in the technical specifications.

The CQA representative will observe all trial seam procedures. The trial seam samples will be assigned a number and marked accordingly by the CQA representative, along with the date, hour, ambient temperature, number of seaming unit, name of seamer, and pass or fail description. A sample of the trial seam will be retained by the CQA team until the construction of the liner is complete and the liner has been accepted by the CQA certifying engineer.

4.2.4.6 Nondestructive Seam Continuity Testing

Except as otherwise noted in the technical specifications, the construction general contractor will nondestructively test all field seams over their full length, in accordance with the technical specifications. The purpose of nondestructive tests is to check the continuity of seams. Continuity testing will be carried out as the seaming work progresses, not at the completion of all field seaming.

Nondestructive testing will not be permitted before sunrise or after sunset unless the construction general contractor demonstrates to the CQA certifying engineer that the construction general contractor has the capabilities to perform continuity testing under reduced light conditions. The CQA representative will perform the following tasks:

- Observe the continuity testing.
- Record location, date, test unit number, name of tester, and outcome of all testing
- Document and inform the construction general contractor of any required repairs.

The construction general contractor will be required to complete any required repairs, in accordance with the technical specifications. The CQA representative will perform the following tasks:

- Observe the repair and re-testing of the repair.
- Mark on the geomembrane that the repair has been made.
- Document the results.

The CQA representative will verify and document the procedures specified in the technical specifications where seams cannot be nondestructively tested. The location, date of visual observation, name of tester, and outcome of the test or observation will be recorded by the CQA representative and reported to the CQA certifying engineer.

4.2.4.7 Destructive Seam Testing

Concept

Destructive seam tests will be performed at selected locations. The purpose of these tests is to evaluate seam strength and integrity. Seam strength testing will be done as the seaming work progresses, not at the completion of all field seaming.

Location and Frequency

The CQA representative will select locations where seam samples will be cut out for laboratory testing at the frequency specified in the technical specifications (see Section 02661). In general, destructive tests will be located in non-critical areas, such as seam run-out areas or near three-panel intersections or other areas that will require a patch anyway. In addition, because extrusion welding may be limited on a daily basis, extrusion destructive samples may be welded after passing a trial seam on scrap material not used for construction. However, when significant lengths (greater than 100 feet) of seams or caps are extrusion welded, a destructive test of the weld will be taken.

Control charts will be used to track the performance of each welding machine and technician to allow for biased sampling, according to performance. An upper control limit (UCL) will be established to identify statistically the sources of test failures. Machines and technicians whose failure rates exceed the UCL will then be identified and destructively tested at twice the original frequency (one per 250 feet of seam length) to better monitor their performance. Once the failure rate drops back into compliance with the UCL, the original testing frequency will be reinstated. Machines and technicians whose failure notes are below the UCL will be identified to decrease the original frequency, as approved by the CQA certifying engineer.

The UCL is established based on the failure rate for all destructive tests plus three standard deviations with a ceiling of 3.5 percent. The ceiling is the maximum failure rate determined to be acceptable, as agreed upon jointly by the construction general contractor and CQA certifying engineer. The initial UCL will be calculated once a single machine or technician fails two destructive tests and will typically be updated daily with the most recent destructive testing results. Destructive tests tracking a failed destructive will not be included in the calculation of the failure rates.

Additional destructive test locations may be required during seaming operations. The necessity for such additional sampling and testing will be determined by CQA representatives and will be implemented when there is cause to suspect the presence of excess crystallinity, contamination, offset welds, or any other reason to suspect potentially defective seams. The location selection of the additional testing will be based on the CQA representative's judgment and observation of a suspected problem.

The construction general contractor will not be informed in advance of the locations where the seam samples will be taken.

Sampling Procedure

The construction general contractor will be required to cut samples, as directed by the CQA representative as the seaming progresses, in order to have laboratory test results before the geomembrane is covered by another material. The CQA representative will perform the following tasks:

- Observe sample cutting.
- Assign a number to each sample and mark it accordingly.
- Record the sample number and location on the panel layout drawing.
- Record the reason for taking the sample at this location (e.g., routine testing, suspicious feature of the geomembrane).

All holes in the geomembrane resulting from destructive seam sampling will be covered by the construction general contractor immediately after sampling and will be repaired in accordance with the repair procedures described in the technical specifications. The continuity of the new seams in the repaired area will be nondestructively tested, according to the technical specifications.

Size of Samples

At a given sampling location, two types of samples will be taken by the construction general contractor. First, two specimens for field-testing will be taken. Each of these specimens will be 1 inch wide by 6 to 12 inches long, with the seam centered parallel to the width. The distance between these two specimens will be approximately 42 inches. If both specimens pass the field test described in the technical specifications, a sample for laboratory testing will be taken.

The sample for laboratory testing will be required to be taken between the two specimens for field-testing. The destructive sample will be 12 inches wide by 42 inches long, with the seam centered lengthwise. The sample will be cut into three parts and distributed as follows:

- One portion to the construction general contractor, 12 inches long.
- One portion to the IDF CM for archive storage, 12 inches long.
- One portion to the CQA certifying engineer for CQA Laboratory testing, 18 inches long.

Final determination of the sample sizes will be made at the preconstruction meeting.

Field Testing

The two 1-inch-wide specimens, as specified above, will be required to be tested in the field by the CQA representative by tensiometer for peel and shear and need not to fail in the seam. If any field test sample fails to pass, the procedures outlined in the technical specifications will be followed.

The CQA representative will mark all samples and portions with their number, date, and time.

Geosynthetic Construction Quality Assurance Laboratory Testing

Laboratory destructive test samples will be packaged and shipped to the Geosynthetics CQA Laboratory by the CQA representative in a manner that will not damage the test sample. The CQA representative will store the archive samples until the completion of the project.

Testing will include "Shear Strength" and "Peel Strength" (ASTM D6392) with 1-inch-wide strip, tested at 2 inches per minute. The minimum acceptable values to be obtained in these tests are those indicated in the technical specifications. At least five specimens will be tested for each test method.

Specimens will be selected alternately by test from the samples (i.e., peel, shear, peel, and shear). At least four out of five of the specimens for each test must pass.

The laboratory will provide test results verbally to the CQA certifying engineer in a timely manner after they receive and test the samples. The CQA certifying engineer will review laboratory test results as soon as they become available and will inform the CQA certifying engineer of the test results.

Procedures for Destructive Test Failure

The procedures specified in the technical specifications will be required whenever a sample fails a destructive test, whether that test is conducted by the Geosynthetics CQA Laboratory or by field tensiometer. The CQA certifying engineer will verify and document that one of the options specified in the technical specifications is followed. The CQA representative will document all actions taken in conjunction with destructive test failures, including preparation of NCRs, as outlined in Section 8.1.4.

4.2.5 Defects and Repairs

4.2.5.1 Identification

All seams and non-seam areas of the geomembrane will be inspected by the CQA representative for evidence of defects, holes, blisters, undispersed raw materials, and any sign of contamination by foreign matter. Because light reflected by the geomembrane helps to detect defects, the surface of the geomembrane will be required to be clean at the time of examination. The geomembrane surface will be required to be swept or washed by the construction general contractor if the amount of dust or mud inhibits examination.

4.2.5.2 Evaluation

Each suspect location both in seam and non-seam areas will be required to be either non-destructively tested using the methods described in the technical specifications, or repaired as appropriate as determined by the CQA certifying engineer. Each location that fails the non-destructive testing will be marked by the CQA representative and will be required to be repaired by the construction general contractor. Materials will not be placed over geomembrane locations that have been repaired until the CQA representative has approved the repair.

4.2.5.3 Large Wrinkles

When seaming of the geomembrane is completed (or when seaming of a large area of the geomembrane is completed) and prior to placing overlying materials, the CQA representative will visually inspect the geomembrane for wrinkles. Based on the requirements of the technical specifications, the CQA representative will indicate to the construction general contractor, which wrinkles, if any, are to be cut, overlapped, and seamed to remove the wrinkle. The seam thus produced will be tested like any other seam.

4.2.5.4 Repair Procedures

Any portion of the geomembrane either exhibiting a flaw or failing a destructive or nondestructive test will be repaired by the construction general contractor in accordance with the applicable method specified in the technical specifications. An NCR will be prepared to document all flaws and failed tests, as outlined in Section 8.1.4. Each repair will be located and logged by the CQA representative.

4.2.5.5 Testing of Repairs

Each repair will be non-destructively tested, using the methods described in the technical specifications as appropriate. Repairs that pass the non-destructive test will be considered adequate. Large caps may be of sufficient extent to require destructive testing, at the discretion of the CQA certifying engineer. Failed tests will require the repair to be redone and re-tested until passing test results are obtained. The CQA representative will observe the non-destructive testing of repairs and will document the date of the repair and test outcome.

4.2.6 Appurtenances

The CQA representative will verify and document the following:

- Installation of the geomembrane around, and connection of geomembrane to, appurtenances have been made according to the technical specifications or manufacturer's recommendations.
- Extreme care is taken while seaming around appurtenances, since neither non-destructive nor destructive testing may be feasible in these areas.
- The geomembrane has not been visibly damaged while being connected to appurtenances.

The CQA representative will inform the CQA certifying engineer if the above conditions are not fulfilled.

4.3 Geomembrane Panel Layout Survey

A survey will be performed by or under the direction of a professional land surveyor registered in the State of Washington. The surveyor will independently survey the elevations and location of each panel intersection and destructive sample. The results of the survey conducted by the surveyor will be compiled in a report signed by the surveyor and the CQA certifying engineer.

The surveyor will be required to survey each geomembrane panel intersection and destructive sample location for the IDF landfill, in accordance with the requirements of this CQA Plan. A record drawing will be submitted to the CQA certifying engineer by the surveyor. The survey will include enough information to confirm that the geomembrane layout is in accordance with the panel layout and include, but not be limited to, the following information:

Geomembrane panel intersections:

- Destructive sample location and identification
- Edge of geomembrane liner
- Panel identification numbers

Each geomembrane layer will be surveyed including, but not be limited to:

- Secondary leak detection system geomembrane
- Secondary geomembrane
- Primary geomembrane

4.4 Layer Completion Certification

The construction general contractor will be required to notify the CQA representative when an area of geomembrane is complete, prior to constructing the overlying layer. The construction general contractor may place overlying layer after acceptance of geomembrane layer by the CQA Certifying Engineer. The CQA certifying engineer will provide a certificate of layer completion to the construction general contractor and the IDF project engineer, certifying that all CQA tests are complete and all defects have been repaired and tested.

SECTION 5 GEOTEXTILE CONSTRUCTION QUALITY ASSURANCE

5.1 Geotextile Material And Installation

5.1.1 Labeling

The CQA representative will verify and document that the geotextile manufacturer has labeled all rolls of geotextile with the information specified in the technical specifications. The CQA representative will examine rolls upon delivery, and any deviation from the requirements will be reported to the CQA certifying engineer. Geotextile rolls that are not labeled or that have illegible labels will be removed and disposed by the construction general contractor.

5.1.2 Transportation and Handling

The CQA representative will observe rolls of geotextile upon delivery at the site, and any deviation from the transportation and handling requirements specified in the technical specifications will be reported to the CQA certifying engineer. Any damaged rolls will be rejected by the CQA certifying engineer and required to be repaired or replaced by the construction general contractor.

5.1.3 Storage

The CQA representative will verify and document that storage of the geotextile is in accordance with the technical specifications.

5.1.4 Inventory

All geotextile materials that arrive onsite will be inventoried. The inventory will include the specific roll numbers delivered with each shipment. The inventory will be compared to the QC testing information, supplied by the manufacturer to ensure that the material tested is the same material that was delivered to the site. Material for which QC testing data has been supplied will be sampled for conformance testing. Conformance samples may be obtained by the CQA representative at the manufacturing plant or taken upon delivery of the material to the site by a CQA representative.

As shipments arrive at the site, a CQA representative will monitor the unloading operations and will inventory the material. Rolls selected for conformance testing will be set aside for sampling as soon as possible.

The CQA representative will record the following information, at a minimum, for each roll:

- **Manufacturer**–Indicate the manufacturer of the material that is being inventoried, that may not be the same as the installer.
- **Date of Inventory**–Date that the material was inventoried.
- **Date of Delivery**–Enter date when the truck arrived onsite, if known.
- **Truck Type**–Indicate type of truck used for shipping geosynthetics (covered or uncovered flatbed, box trailer).
- **Bill-of-Lading Number**–If the bill-of-lading is available, indicate number and date (also attach copy to inventory form).
- **CQA Representative**–Indicate name of CQA representative performing inventory.
- **Unloading Equipment**–Indicate the type and model number of the equipment unloading the geosynthetic material; also note any special attachments that are used to unload the material (stinger, straps, forks).
- **Weather Conditions**–Describe the weather conditions, including temperature, wind, cloud cover, and precipitation during unloading and conformance sampling operation.
- **Material Type**–Indicate type of geosynthetic material.
- **Roll Number**–Indicate each roll number that is indicated on the roll.
- **Lot Number**–Lot number.

- **Roll (L x W)**–Indicate the roll width as indicated on the roll label; if two materials are bonded together (i.e., geonet/geotextile), obtain measurements for both materials.
- **Area (square feet)**–Indicate the total square footage of the roll.
- **Damage Remarks**–Document any visible damage to the roll; if possible, indicate if damage was present prior to unloading or if it occurred during unloading.

Items that are restricted from further use until the inspections have been completed will be clearly delineated by the CQA representative. Accepted materials will be kept separate or clearly delineated from inventoried and approved items to the extent possible. The CQA representative will coordinate with the construction general contractor during material delivery so that the material is not moved more than necessary after it is unloaded and damage due to handling is minimized.

The CQA representative will perform the inventory immediately after the material arrives onsite to avoid delaying construction. The CQA representative will be responsible for verifying that only accepted material is installed at the IDF landfill and that all inventories and inspections are documented and maintained.

5.1.5 Conformance Testing

Either at the manufacturer's factory or upon delivery of the geotextile rolls, the CQA representative will ensure that samples are removed and forwarded to the Geosynthetics CQA Laboratory for testing to verify and document conformance with the requirements of the technical specifications. Conformance samples will be taken across the entire width of the roll and will not include the first 3 feet along the edge of the roll. Unless otherwise specified, samples will be 3 feet (minimum) long by the roll width. The CQA representative will mark the machine direction on the samples with an arrow.

Samples will be taken at a rate of one per material lot or one per 50,000 square foot, whichever is greater. These samples will be tested for the following:

- Permittivity (ASTM D4491, Type 1 only).
- Grab strength (ASTM D4632).
- Tear strength (ASTM D4533).
- Puncture strength (ASTM D4833).

The CQA representative will examine all results of laboratory conformance testing and report any non-conformance to the CQA certifying engineer as soon as results become available. The following procedure will apply whenever a sample fails a conformance test that is conducted by the Geosynthetics CQA Laboratory:

- The construction general contractor will replace the roll (or rolls) of geotextile not in conformance with the specifications with a roll, that meets the requirements of the technical specifications.
- The CQA representative will ensure that conformance samples are removed for testing by the Geosynthetics CQA Laboratory from the closest numerical roll on both sides of the roll from which the failing sample was obtained. These two samples must pass the above conformance tests. If either of these samples fails to meet the requirements, samples will be collected from the five numerically closest untested rolls on both sides of the failed sample and tested by the Geosynthetics CQA Laboratory. These ten samples must pass the above conformance tests. If any of these samples fail, a sample from every roll of geotextile onsite and a sample from every roll that is subsequently delivered from the same manufacturer must be conformance tested by the Geosynthetics CQA Laboratory, until the manufacturer has thoroughly demonstrated compliance with the above requirements to the sole satisfaction of the CQA certifying engineer. The costs of all such tests are to be borne by the construction general contractor.

The CQA representative will document actions taken in conjunction with conformance test failures and report all actions taken to the CQA certifying engineer. Failed tests will be documented using NCR procedures, outlined in Section 8.1.4.

5.1.6 Deployment

The construction general contractor will be required to handle all geotextile material in such a manner as to ensure that it is not damaged in any way.

It will be the CQA representative's responsibility to indicate to the CQA certifying engineer any change in the underlying layer that may, in accordance with the technical specifications, require repair work. If the CQA certifying engineer requires that repair work be done, it will be the responsibility of the construction general contractor to repair the underlying layer.

The CQA representative will verify and document compliance with the following:

- Just prior to geotextile placement, the layer that underlies the geotextile, if it is a geosynthetic, is clean and free of excessive amounts of dust, dirt, stones, rocks, or other obstructions that could potentially damage the liner system.
- In the presence of excessive wind, the geotextile is weighted with sandbags (or equivalent weight approved by the CQA representative).
- Geotextile is kept under tension to minimize the presence of wrinkles in the geotextile. If necessary, the geotextile is positioned by hand after being unrolled to minimize wrinkles.
- Geotextile is cut using a geotextile cutter approved by the geotextile manufacturer and the CQA representative. If in place, special care is taken to protect other materials (such as underlying geosynthetics) from damage that could be caused by the cutting of the geotextiles.
- The construction general contractor takes any necessary precautions to prevent damage to the underlying layers during placement of the geotextile.
- During placement of geotextile, care is taken not to entrap stones, excessive dust, or moisture that could damage the underlying layers, generate clogging of drains or filters, or hamper subsequent seaming.
- Geotextile is not left exposed for an excess of 14 days after placement, to prevent damage from exposure to ultraviolet radiation (sunlight). If the geotextile is exposed for more than 14 days, a temporary cover may be deployed for the duration of the delay or samples may be submitted to an independent testing laboratory to ensure that detrimental levels of UV degradation have not occurred. Test results shall be submitted to CQA certifying engineer for review and approval. Detrimental level of UV degradation is defined in the technical specifications (see Section 02371).

The CQA representative will document any non-compliance with the above requirements and report them to the CQA certifying engineer.

5.1.7 Seams and Overlaps

The CQA representative will verify and document that all geotextile seams are oriented and overlapped, in accordance with the technical specifications. The construction general contractor will be required to pay close attention at seams to ensure that no protective soil layer material could be inadvertently placed beneath the geotextile.

5.1.8 Repair

The CQA representative will verify and document that any holes or tears in the geotextile are repaired, in accordance with the requirements of the technical specifications. The CQA representative will document any noncompliance with the above requirements and report it to the CQA certifying engineer.

SECTION 6 COMPOSITE DRAINAGE NET CONSTRUCTION QUALITY ASSURANCE

6.1 Composite Drainage Net Material and Installation

6.1.1 Labeling

The CQA representative will verify and document that the composite drainage net manufacturer has labeled all rolls of composite drainage net as specified in the technical specifications. The CQA representative will examine rolls upon delivery, and any deviation from the above requirements will be reported to the CQA certifying engineer prior to installation of the composite drainage net.

6.1.2 Transportation and Handling

The CQA representative will observe rolls of composite drainage net upon delivery at the site, and any deviation from the requirements of the technical specifications will be reported to the CQA certifying engineer. Any damaged rolls will be rejected by the CQA representative and be required to be repaired or replaced by the construction general contractor.

6.1.3 Storage

The CQA representative will verify and document that the storage of the composite drainage net is in accordance with the technical specifications.

6.1.4 Inventory

All CDN that arrive onsite will be inventoried. The inventory will record the specific roll numbers delivered with each shipment. The inventory will be compared to the QC testing information supplied by the manufacturer, to ensure that the material tested is the same material that was delivered to the site. Material for which QC testing data has been supplied will be sampled for conformance testing. Conformance samples may be obtained by the CQA representative at the manufacturing plant or taken upon delivery of the material to the site by a CQA representative.

As shipments arrive at the site, a CQA representative will monitor the unloading operations and will inventory the material. Rolls selected for conformance testing will be set aside for sampling as soon as possible.

The CQA representative will record the following information, at a minimum, for each roll:

- **Manufacturer**–Indicate the manufacturer of the material that is being inventoried that may not be the same as the installer.
- **Date of Inventory**–Date that the material was inventoried.
- **Date of Delivery**–Enter date when the truck arrived onsite, if known.
- **Truck Type**–Indicate type of truck used for shipping geosynthetics (covered or uncovered flatbed, box trailer).
- **Bill-of-Lading Number**–If the bill-of-lading is available, indicate number and date (also attach copy to inventory form).
- **CQA Representative**–Indicate name of CQA representative performing inventory.
- **Unloading Equipment**–Indicate the type and model number of the equipment unloading the geosynthetic material; also note any special attachments that are used to unload the material (stinger, straps, forks).
- **Weather Conditions**–Describe the weather conditions, including temperature, wind, cloud cover, and precipitation during unloading and conformance sampling operation.
- **Material Type**–Indicate type of geosynthetic material (high-density polyethylene, geotextile, or geonet).
- **Roll Number**–Indicate each roll number that is written on the roll.
- **Lot Number**–Lot number as indicated.

- **Roll (L × W)**–Indicate the roll width as indicated on the roll label; if two materials are bonded together (i.e., geonet/geotextile), obtain measurements for both materials.
- **Area (square feet)**–Indicate the total square footage of the roll.
- **Damage Remarks**–Document any visible damage to the roll; if possible, indicate if damage was present prior to unloading or if it occurred during unloading.

Items that are restricted from further use until the inspections have been completed will be clearly delineated by the CQA representative. Accepted materials will be kept separate or clearly delineated from inventoried and approved items to the extent possible. The CQA representative will coordinate with the construction general contractor during material delivery so that the material is not moved more than necessary after it is unloaded and damage due to handling is minimized.

The CQA representative will perform the inventory immediately after the material arrives onsite to avoid delaying construction. The CQA representative will be responsible for verifying that only accepted material is installed at the IDF landfill and that all inventories and inspections are documented and maintained.

6.1.5 Conformance Testing

Either at the manufacturer's plant or upon delivery of the composite drainage net rolls, the CQA representative will ensure that samples are removed and forwarded to the Geosynthetics CQA Laboratory for testing, to verify and document conformance with the requirements of the technical specifications.

Conformance samples will be taken across the entire width of the roll and will not include the first 3 feet. Unless otherwise specified, samples will be 3 feet long (minimum) by the roll width. The CQA representative will mark the machine direction on the samples with an arrow.

Samples will be taken at a rate of one per lot or one per 50,000 square feet, except as noted otherwise below, whichever is greater. The geonets will be tested for the following:

- Polymer specific gravity (ASTM D1505).
- Thickness (ASTM D5199).
- Nominal transmissivity (ASTM D4716 – one per production lot).

The composite drainage nets will be tested for the following:

- Adhesion (GRI-GC7 or ASTM D413).
- Transmissivity (ASTM D4716 – one per production lot).

The CQA representative will examine all results from laboratory conformance testing and will report any non-conformance to the CQA certifying engineer as soon as the results become available.

The following procedure will apply whenever a sample fails a conformance test that is conducted by the Geosynthetics CQA Laboratory:

- The construction general contractor will be required to replace the roll (or rolls) of composite drainage net not in conformance with the specifications with a roll that meets the requirements of the technical specifications.
- The CQA representative will ensure that conformance samples are removed for testing by the Geosynthetics CQA Laboratory from the closest numerical roll on both sides of the failed roll. These two samples must pass the above conformance tests. If either of these samples fails, samples will be collected from the five numerically closest untested rolls on both sides of the failed sample and tested by the Geosynthetics CQA Laboratory. These ten samples must pass the above conformance tests.

If any of these samples fail, a sample from every roll of composite drainage net onsite and a sample from every roll that is subsequently delivered from the same manufacturer must be conformance tested by the Geosynthetics CQA Laboratory, until the manufacturer has thoroughly demonstrated compliance with the above requirements to the sole satisfaction of the CQA certifying engineer. The cost of such tests is to be borne by the construction general contractor.

The CQA representative will document actions taken in conjunction with conformance test failures and report all actions to the CQA certifying engineer. Failed tests will be documented using NCR procedures, outlined in Section 8.1.4.

6.1.6 Deployment

The construction general contractor will be required to handle all composite drainage net in such a manner as to ensure that it is not damaged.

The construction general contractor (responsible for composite drainage net installation) will be required to certify in writing that the surface on which the composite drainage net will be installed is complete and acceptable. A certificate of partial completion will be given by the construction general contractor to the CQA representative, who will then verify to the CQA certifying engineer that the deployment surface is complete, prior to commencement of composite drainage net installation.

After the surface on which the composite drainage net is to be installed has been accepted by the construction general contractor, the CQA representative will have responsibility to indicate to the CQA certifying engineer any change in the underlying layer that may, in accordance with the technical specifications, require repair work. If the CQA certifying engineer requires that repair work be done, it will be the responsibility of the construction general contractor to repair the underlying layer.

The CQA representative will verify and document compliance with the following:

- Just prior to composite drainage net placement, the layer that will underlie the composite drainage net is clean and free of excessive amounts of dust, dirt, stones, rocks, or other obstructions that could potentially damage the underlying layers or clog the drainage system.
- In the presence of excessive wind, the composite drainage net is weighted with sandbags (or equivalent weight approved by the CQA certifying engineer).
- Composite drainage net is kept under tension to minimize the presence of wrinkles in the composite drainage net. If necessary, the composite drainage net is positioned by hand after being unrolled, to minimize wrinkles.
- Composite drainage net is cut using a composite drainage net cutter, approved by the composite drainage net manufacturer and the CQA representative. If in place, special care is taken to protect other materials from damage that could be caused by the cutting of the composite drainage net.
- The construction general contractor takes all necessary precautions to prevent damage to the underlying layers during placement of the composite drainage net.
- Composite drainage net is not welded to geomembranes.
- During placement of clean composite drainage net, care is taken not to entrap stones, excessive dust, or moisture that could damage the underlying geomembrane, generate clogging of drains or filters, or hamper subsequent seaming.
- A visual examination of the composite drainage net is carried out over the entire surface, after installation, to ensure that no potentially harmful foreign objects, such as needles, are present.
- Composite drainage net is not left exposed for an excess of 14 days after placement, to prevent damage from exposure to ultraviolet radiation (sunlight).

The CQA representative will document any noncompliance with the above requirements and report it to the CQA certifying engineer.

6.1.7 Seams and Overlaps

The components of the composite drainage net (e.g., geotextile-geonet-geotextile) are not bonded together at the ends and edges of the rolls. The CQA representative will document that the composite drainage net is overlapped and secured in accordance with the technical specifications.

6.1.8 Repair

The CQA representative will verify that any holes or tears in the composite drainage net are repaired, in accordance with the technical specifications. The CQA representative will observe any repair, document any noncompliance with the above requirements, and report the noncompliance to the CQA certifying engineer. Repair areas will be documented using NCR procedures, outlined in Section 8.1.4.

SECTION 7 POLYETHYLENE PIPE AND FITTINGS CONSTRUCTION QUALITY ASSURANCE

7.1 Pipe and Fittings

The CQA representative will monitor the placement of the LCRS, LDS, and SLDS pipe, located on the IDF landfill floor and on the landfill slopes.

7.1.1 Labeling

The CQA representative will verify that the pipe is labeled with the information specified in the technical specifications. Any deviations from the labeling requirements will be reported to the CQA certifying engineer prior to pipe installation.

7.1.2 Transportation and Handling

The CQA representative will verify and document that the pipe and fittings are handled in accordance with the technical specifications. The CQA representative will visually inspect the pipe upon delivery at the site, and any deviations from the requirements of the technical specifications will be reported to the CQA certifying engineer.

7.1.3 Storage

The CQA representative will verify and document that storage of the pipe and fittings is in accordance with the technical specifications.

7.1.4 Inventory

The CQA representative will inventory the polyethylene piping and fitting, delivered to the site that will be installed at the bottom and on the slopes of the landfill. The CQA representative will perform the following tasks:

- Verify the material for conformance with the technical specifications and construction drawings.
- Verify slot dimensions for conformance with the technical specifications.
- Check the material for damage, mishandling, and adverse exposure.

Items that are restricted from further use until the inspections have been completed will be clearly delineated by the CQA representative. Accepted materials will be kept separate from inventoried and approved items, to the extent possible. The CQA representative will be responsible for coordinating with the construction general contractor during material delivery, to limit the material being moved more than necessary after it is unloaded and thereby minimizing damage due to handling.

The CQA representative will perform the inventory immediately after the material arrives onsite to avoid delaying construction. The CQA representative will be responsible for verifying that only accepted material is installed at the IDF landfill, and that all inventories and inspections are documented and maintained.

7.1.5 Conformance Testing

No conformance testing will be conducted on the materials delivered to the site.

7.1.6 Handling and Laying

The CQA representative will verify and document that the pipe is installed at the specified locations, grades, and angles, and that placement of backfill around and over the pipe is conducted in accordance with the requirements of the technical specifications and in a manner intended to prevent damage to the pipe.

The pipe and fittings will be carefully examined before installation by the CQA representative. The CQA representative will verify and document that cracks, damage, or defects are not present in the pipe and fittings in excess of that allowed by the technical specifications.

The CQA representative will also note the condition of the interior of pipes and fittings. Foreign material will be removed from the pipe interior before it is moved into final position. No pipe will be permitted to be placed until the CQA representative has observed the condition of the pipe. The CQA representative will document any deviation from the requirements and report it to the CQA certifying engineer.

7.1.7 Joints and Connections

Lengths of pipe will be required to be assembled into suitable installation lengths by the butt-fusion process. Butt-fusion refers to the butt joining of the pipe by softening the aligned faces of the pipe ends in a suitable apparatus and pressing them together under controlled pressure. The CQA representatives will spot-monitor butt-fusion welding operations to ensure that the construction general contractor follows the technical specifications for both slotted and solid pipes. The CQA representative will verify that internal weld beads have been removed from the horizontal and side slope sections of the LCRS, LDS, and SLDS riser pipes. The CQA representative will document any noncompliance with the requirements and report it to the CQA certifying engineer.

7.1.8 Surveying

A survey will be performed by or under the direction of a professional land surveyor registered in the State of Washington. The surveyor will independently survey the final elevation and alignment of the top of the pipe and fittings. Surveys will be performed on all pipe locations within the footprint of the landfill to confirm that the alignment and elevations in the field agree with those shown in the construction drawings. The results of the survey will be compiled in a report signed by the surveyor and the CQA certifying engineer.

The surveyor will be required to survey each pipe location within the IDF landfill, in accordance with the requirements of this CQA Plan. A record drawing will be submitted to the CQA certifying engineer by the surveyor before placement of the next liner system layer. The surveys will be conducted every 50 feet along the pipe alignment and appurtenances. The survey will include enough information to confirm that the following features of the landfill piping are constructed in accordance with the construction drawings:

- Beginning and end top of pipe elevations
- Connection location
- Grade breaks
- Riser pipes
- Sump extensions

The piping that will be surveyed will include, but not be limited to, the following:

- SLDS piping
- LDS piping
- LCRS piping

The CQA certifying engineer will approve the survey results for each layer before the subsequent component of the lining system is constructed.

SECTION 8 CONSTRUCTION QUALITY ASSURANCE DOCUMENTATION AND CERTIFICATION

8.1 Documentation and Certification

A major function of CQA is to properly and adequately document and certify the work. This section describes the minimum required documentation. The CQA certifying engineer may recommend to the IDF CM additional documentation for performing CQA tasks that are for certification. In addition, the CQA certifying engineer will prepare forms, field data sheets, sample labeling schemes, and chain-of-custody procedures and submit them to the IDF CM and IDF PM for approval, prior to construction.

8.1.1 Daily Reports

Daily reports will be completed by the CQA representatives when they are onsite. All CQA personnel will be assigned field books by the CQA certifying engineer that will be labeled with a unique number. The CQA representatives, including the CQA certifying engineer, will record all field observations and the results of field tests in their assigned field book. When not in use, all field books will be left in the field records file. After each book is filled (or at the end of the project), the field book will be returned to the CQA certifying engineer and routed to the project files.

Each page of the field book will be numbered, dated, and initialed by CQA personnel. At the start of a new work shift, CQA personnel will list the following information at the top of the page:

- Job name
- Job number
- Date
- Name
- Weather conditions
- Page number (if pages are not pre-numbered)

The remaining individual entries will be prefaced by an indication of the time at which they occurred. If the results of test data are being recorded on separate sheets, it will be noted in the field book. Entries in the field book will include, but not be limited to, the following information:

- Reports on any meetings held and their results.
- Equipment and personnel being used in each location, including construction general contractors.
- Descriptions of areas being observed and documented.
- Descriptions of materials delivered to the site, including any quality verification (vendor certification) documentation.
- Descriptions of materials incorporated into construction.
- Calibrations, or recalibrations, of test equipment, including actions taken as a result of recalibration.
- Decisions made regarding use of material and/or corrective actions to be taken in instances of substandard quality.
- Unique identifying sheet numbers of inspection data sheets and/or problem reporting and corrective measures reports used to substantiate the decisions described in the preceding item

At the end of each day, the field CQA monitor will summarize the day's activities on a daily field monitoring report form. The field report will include a brief summary of the day's activities and highlight any unresolved issues that must be addressed by the CQA certifying engineer or by CQA representatives the following day.

The daily field monitoring report will be filled out in triplicate. The CQA monitor will attach three copies of the field book notes for that day. The three copies will be distributed as follows:

- Original will be filed in field office.
- One copy will be transmitted to the CQA certifying engineer.
- One copy will be transmitted to the IDF CM.

The CQA certifying engineer will review and initial each summary field report before distributing to the project quality records and the IDF CM.

8.1.2 Inspection Data Sheets

All observed field and laboratory test data will be recorded on an inspection data sheet. At a minimum, each inspection data sheet will include the following information:

- Unique identifying sheet number for cross-referencing and document control.
- Description of the inspection activity.
- If appropriate, location of inspection activity or location from which the sample was obtained.
- Type of inspection activity and/or procedure used (reference to standard method when appropriate).
- Any recorded observation or test data, with all necessary calculations.
- Results of the inspection activity and comparison with specification requirements.
- Identification of any personnel involved in the inspection activity.
- Signature of the individual(s) performing the CQA representative activity and concurrence by the CQA certifying engineer.
- Identification of deficiencies and any required reinspections.

Forms used for the data sheets will be prepared and submitted to the IDF CM and IDF PM in accordance with this section. The data sheets will include, but are not limited to, the forms listed below:

- Sample log
- Compaction test result log
- Soil test result summary form
- Equipment calibration log

8.1.3 Record Drawing Maintenance

The construction general contractor will maintain a complete set of construction drawings labeled "Red-Line" as-built drawings. At the completion of the project, the as-built drawings pertaining to the work certified under this CQA Plan will be produced in electronic format and submitted to the CQA certifying engineer. The CQA certifying engineer will review the completed set of as-built drawings and certify the drawing set as the record drawings for the IDF.

8.1.4 Non-Conformance Reporting

Deficiencies/defects identified by in-process testing may be reworked in accordance with the technical specifications or CQA Plan to correct the deficiency without initiating the NCR process (i.e., failed compaction test or failed geomembrane destructive test), and in-process tests will be tracked by the CQA representative until it is corrected. A non-conformance is considered to be a deficiency in characteristics, documentation, or procedures that renders the quality of an item or activity unacceptable or indeterminate. All deficiencies, defects, damage, or test failures that are not corrected by in-process rework will be considered a non-conformance and will be documented on a Non-Conformance Report (NCR) form. The non-conformance will be referred to the IDF CM, for disposition and initiation of corrective action processes.

All NCR situations will be brought to the attention of the IDF CM for concurrence, prior to initiating the NCR.

Upon issuance of the non-conformance report, the IDF CM will notify the IDF design engineer, IDF quality engineer, and IDF PM that the report has issued. Other individuals, as directed by the IDF PM, will participate in NCR disposition, resolution, and corrective action processes as needed. All documentation relating to NCR situations will be retained in the project quality records.

8.1.5 Resolution of Contract Document Questions and Clarifications

Request for Information (RFI) forms will be provided to the CQA certifying engineer for the purpose of submitting written requests to the IDF CM, for assistance in understanding the design intent of the contract documents. The CM will determine whether the IDF design engineer's technical support staff will address the RFI.

RFIs initiated by the construction general contractor will be addressed by the IDF project engineer and CM, not by the CQA certifying engineer, and are not in the scope of the CQA Plan.

Any RFIs that result in contract document changes will be incorporated by the IDF CM and PM, following the procedures outlined in Section 8.1.6.

8.1.6 Construction Change Order and Contract Document Changes

Requests for changes to the technical specifications or construction drawings will be referred to the IDF CM and initiated as a change order. All change orders and resulting design changes will be approved by the appropriate project team member prior to implementation, as outlined in procedure HNF-IP-0842, Volume 4., Section 4.29 (Engineering Document Change Control Requirements). Requests for modifications to the CQA Plan will also be made by completing a change order to the IDF CM and procurement agent, with copies to the IDF quality engineer and IDF project engineer.

If, during the course of construction, questions arise regarding interpretation of the plans and/or specifications, the IDF CM will be contacted by the CQA certifying engineer. Any clarification of the construction drawings will be documented by a change order, if necessary, or by telephone conversation records or meeting minutes, and routed to the IDF design engineer, IDF CM, and IDF PM. The change order will also be routed to the project files.

8.1.7 Progress Reports

The CQA certifying engineer will prepare a summary progress report each week, or at time intervals established at the pre-construction meeting. At a minimum, this report will include the following information:

- A unique identifying sheet number for cross-referencing and document control.
- The date, project name, location, and other information.
- A summary of work activities accomplished during the progress reporting period.
- Identification of areas or items inspected and/or tested during the reporting period that are addressed by the report.
- A summary of the quality characteristics being evaluated, with appropriate cross-references to technical specifications and/or construction drawings.
- References to the technical specifications or construction drawings defining the acceptance criteria for each inspected characteristic.
- A summary of inspection and test results, failures, and re-tests.
- A summary of construction situations, deficiencies, and/or defects occurring during the progress reporting period.
- A summary of other problem resolutions and dispositions.
- The signature of the CQA certifying engineer.

The progress report will be submitted to the IDF PM no more than two days after the last reporting day in the progress report. Copies will also be submitted to the IDF PM, IDF quality engineer, and construction general contractor.

8.1.8 Final Documentation and Certification

All daily inspection summary reports, inspection sheets, problem identification and corrective measures reports, acceptance reports, change orders, NCRs, photographic records, progress reports, construction drawings, construction drawing revisions, and other pertinent documentation will be retained as permanent project quality records. At the completion of the project, a final CQA report that incorporates all such information, along with as-built drawings, will be prepared by the CQA certifying engineer and submitted to the IDF PM. The CQA certifying engineer will prepare an interim report for construction and testing of the test pads. A final CQA report and certification letter will be completed at the end of the construction that will fulfill the CQA certification requirements specified in WAC 173-303-335(4).

The CQA certifying engineer will coordinate the completion of the as-built record drawings that will be generated by a land surveyor licensed in the State of Washington. The as-built records will include scale drawings depicting depths, plan dimensions, elevations, fill thicknesses, and geosynthetic panel layouts. The report will include documentation of each construction component monitored by CQA personnel and will be signed, stamped, and certified by the CQA certifying engineer.

8.1.9 Storage of Records

During the construction of the IDF, the CQA certifying engineer will be responsible for all CQA documents. This includes the CQA certifying engineer's copy of the design criteria, plans, procedures, and specifications; the CQA Plan; and the originals of all the data sheets and reports. The field records will be kept in lockable, metal cabinets or on metal shelving within a facility, protected by a fire alarm and/or a communication system that provides fire department response and/or fire suppression systems; or, in an Underwriters Laboratory-listed, one-hour fire-rated cabinet. At the completion of the project, all completed documents will be routed to the project quality records.

8.1.10 Storage of Archive Construction Material Samples

The CQA certifying engineer will be responsible for storing construction material samples collected during the duration of the project.

The CQA certifying engineer will coordinate with the IDF PM and IDF CM on which samples will be archived at the completion of the project. All samples will be kept in small containers (i.e., 5-gallon plastic buckets). Each container will be labeled with the following information:

- Project name
- Date
- Sample I.D.
- Material type
- Point of contact

Control and protection of samples will be accomplished through the use of an index listing of samples. This index will identify each sample gathered and include the same information required for the sample containers. It will also identify where the sample is stored and person responsible for the sample storage, thus providing a documented record of each sample and methodology for verifying that all samples are available in storage and that no samples have been misplaced.

All samples will be stored neatly in a cool, dry location, approved by the CQA certifying engineer. The CQA certifying engineer will coordinate with the IDF PM and IDF CM to determine which sample will be archived at the project completion.

SECTION 9 REFERENCES

40 CFR 264.19. *Code of Federal Regulations,* Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities, Section 19, "Construction Quality Assurance Program". Office of the Federal Register. July 9, 1999.

ASTM. *1997 Annual Book of ASTM Standards,* Volume 4.08: Soil and Rock (I). American Society for Testing and Materials, Philadelphia, Pennsylvania. 1997.

CH2M HILL. *Drawings for the Integrated Disposal Facility (IDF) Detailed Design.* RPP-19941, Rev. 0. Prepared for CH2M HILL Hanford Group. February 2004.

CH2M HILL. *Specifications for the Integrated Disposal Facility (IDF) Detailed Design.* RPP-18489, Rev. 0. Prepared for CH2M HILL Hanford Group. February 2004.

Geosynthetic Research Institute (GRI). "Standard Test Method for Asperity Measurement of Textured Geomembranes Using a Depth Gage," Test Method GM12. Philadelphia, Pennsylvania. 2000.

U.S. Department of Energy. "Radioactive Waste Management," DOE O 435.1. August 28, 2001.

U.S. Environmental Protection Agency, Office of Research and Development. *Technical Guidance Document: Quality Assurance and Quality Control for Waste Containment Facilities*, EPA/600/R-93-182. Cincinnati, Ohio. 1993.

Washington Administrative Code (WAC) 173-303-335, *Construction Quality Assurance Program.*

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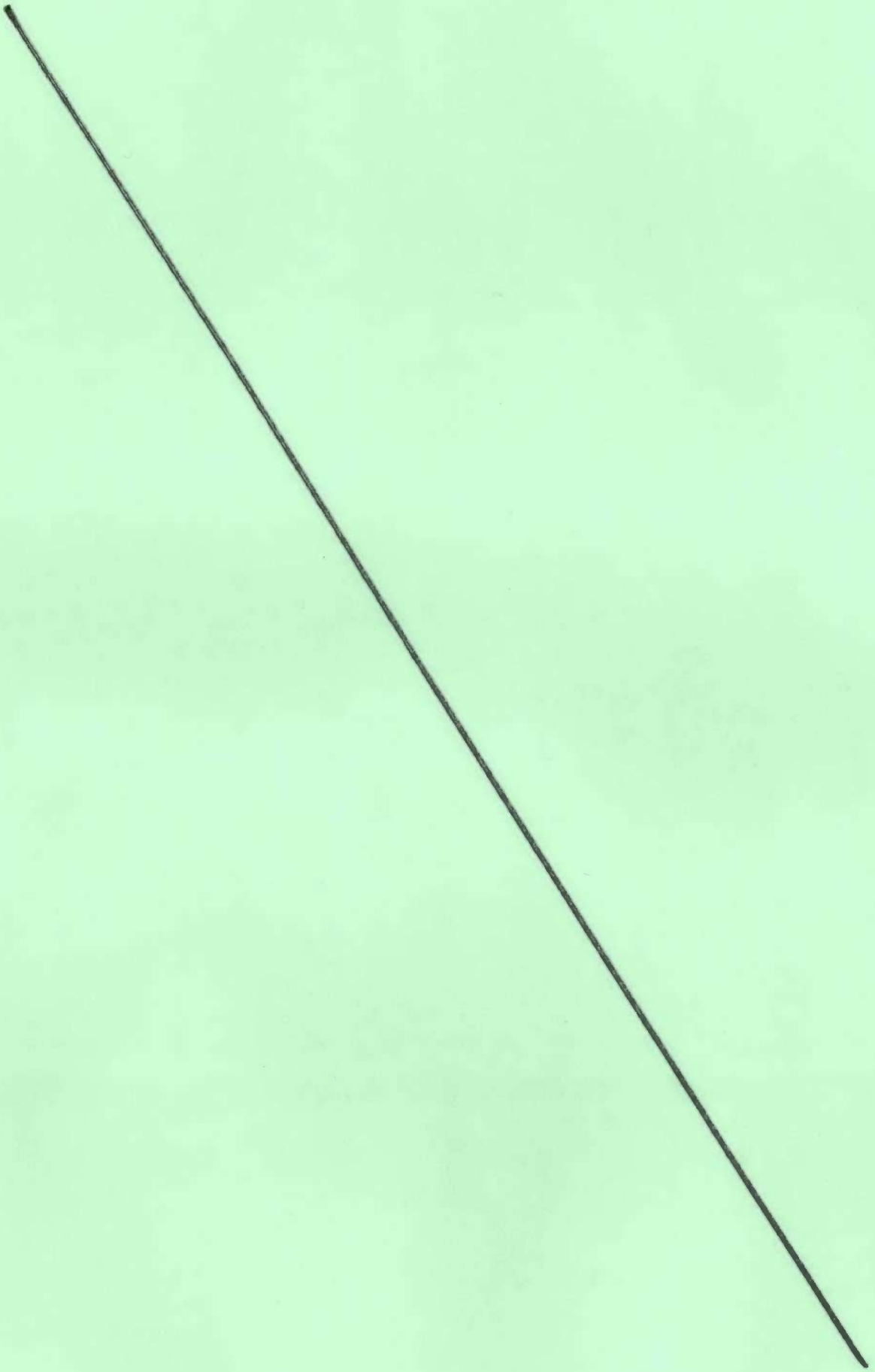

APPENDIX 4C
FACILITY RESPONSE ACTION PLAN

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APPENDIX 4C
FACILITY RESPONSE ACTION PLAN

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4C. FACILITY RESPONSE ACTION PLAN

4C.1. Leakage Response Action Plan

WAC 173-303-665(9) regulations require the owner of the operator of a landfill unit to have an approved Response Action Plan (RAP) before receipt of waste. The RAP is a site-specific plan that establishes actions to be taken if leakage through the upper (primary) lining system of a landfill exceeds a certain rate. The intent of the RAP is to assure that any leachate that leaks through the primary lining system will not migrate out of the landfill into the environment.

A key element of the RAP is the Action Leakage Rate (ALR), a threshold value which triggers the responses described in the RAP, but below which no special actions are required. Because landfill liner systems have not yet been perfected, a small amount of leakage through the primary liner generally occurs, despite the use of best available materials, construction techniques, and quality assurance procedures. (This leakage is collected by the LDS system and removed from the landfill.) Hence, the ALR is set at some level higher than normally expected leakage rates to serve as an indicator that the primary lining system is not functioning as expected. Exceeding the ALR may reflect serious failure of the primary lining system and indicates the need for investigation and possibly corrective action while the problem is still manageable.

This RAP has been prepared in accordance with requirements of WAC 173-303-665(9). The requirements for determining the ALR are contained in WAC 173-303-665(8) and the Environmental Protection Agency (EPA) guidance document, *Action Leakage Rates for Leak Detection Systems* (EPA 530-R-92-004).

The following sections establish the ALR and discuss response actions to be taken if the ALR is exceeded.

4C.1.1. Action Leakage Rate

Section 5.11 provides a detailed discussion of the analysis to determine the ALR into the LDS for the Integrated Disposal Facility (IDF). Based on this analyses, the ALR for the IDF permitted cell is 206 gallons per acre per day, or approximately 1,800 gallons per day per cell (each cell area is approximately 8.5 acres). This value includes a factor of safety of 2 in accordance with EPA guidelines (57 FR 19). It is also much lower than the LDS pump capacity. Details of the calculation are presented in Appendix C.10.

In accordance with WAC 173-303-665(8)(b), the flow rate used to determine if the ALR has been exceeded will be calculated as the average daily flow rate into the sump, expressed as gallons per acre per day (unless Ecology approves a different calculation). This calculation will be performed on a weekly basis during the active (operational) life of the landfill, and monthly after the landfill has been closed. Post-closure frequency may be reduced if only minimal amounts of leachate accumulate in the leak detection system sump. As outlined in WAC 173-303-665(4)(c)(ii), during post-closure monitoring, if the liquid level in the LDS sump stays below the pump operating level for two consecutive months, monitoring of the amount of liquid in the LDS sumps can be reduced to at least quarterly. If the liquid level in the LDS sump stays below the pump operating level for two consecutive quarters, monitoring of the amount of liquid in the LDS sumps can be reduced to at least semiannually. Pump operating level is defined as a liquid level approved by Ecology, based on pump activation level, sump dimensions, and level that minimizes head in the sump.

4C.1.2. Response Actions

WAC 173-303-665(9) lists several required actions if the ALR is exceeded. In the event that the ALR is exceeded, the United States Department of Energy (DOE) will:

- Notify Ecology in writing of the exceedance within 7 days of the determination.

- Submit a preliminary written assessment to Ecology within 14 days of the determination, as to the amount of liquids, likely sources of liquids, possible location, size, cause of any leaks, and short-term actions taken and planned.
- Determine, to the extent practicable, the location, size, and cause of any leak.
- Determine whether waste receipt should cease or be curtailed, whether any waste should be removed from the unit for inspection, repairs, or controls, and whether or not the unit should be closed.
- Determine any other short-term and longer-term actions to be taken to mitigate or stop any leaks.
- Within 30 days after the notification that the action leakage rate has been exceeded, submit to Ecology the results of the analyses specified in bullets 3, 4, and 5 of this section, the results of actions taken, and actions planned. Monthly thereafter, as long as the flow rate in the leak detection system exceeds the action leakage rate, the owner or operator must submit to the regional administrator a report summarizing the results of any remedial actions taken and actions planned.

If the ALR is exceeded, DOE will submit the required notifications to Ecology, as stated above. The EPA will also receive copies of this confirmation.

The leachate will be analyzed for Resource Conservation and Recovery Act (RCRA) constituents. If the analytical results indicate that these constituents are present, and if the constituents can be traced to a particular type of waste stored in a known area of the landfill, then it may be possible to estimate the location of the leak. However, because the waste will meet land disposal restrictions, it will contain no free liquids and will be stabilized or solidified, except as allowed by Appendix 3A, section 1.2. In addition, the canister(s) or other type of waste package(s) may not undergo enough deterioration during the active life of the landfill to permit escape of its contents. For these reasons, it is possible that the leachate may be clean or the composition too general to indicate a specific source location.

If the source location cannot be identified, large-scale removal of the waste and operations layer to find and repair the leaking area of the liner would be one option for remediation. However, this procedure risks damaging the liner. In addition, waste would have to be handled, stored, and replaced in the landfill. Backfill would need to be removed from around the waste packages to accomplish this. If the waste packages are damaged during this process, the risk of accidental release may be high. For these reasons, large scale removal of waste and liner system materials is not considered a desirable option and will not be implemented except as a last resort.

The preferred options for remediation include covers and changes in landfill operating procedures. The preferred alternative will depend on factors such as the amount of waste already in the landfill, the rate of waste receipt, the chemistry of the leachate, the availability of other RCRA-compliant disposal facilities, and similar considerations. Hence, at this time no single approach can be selected. If the ALR is exceeded, potential options will be evaluated prior to selecting a remediation process. If necessary, an interim solution will be implemented while the evaluation and permanent remediation is performed. Examples of potential approaches include the following:

- The surface of the intermediate soil cover over the waste could be graded to direct runoff into a shallow pond. The surface would then be covered with a discardable, temporary geomembrane (e.g., 30-mil PVC or reinforced polypropylene). Precipitation water would be pumped or evaporated from the pond and would not infiltrate the waste already in the landfill. Waste packages would be placed only during periods of dry weather and stored temporarily at other times. This type of approach would also be used, if necessary, to reduce leakage during the time immediately after the ALR was exceeded, while other remediation options were being evaluated.
- If the landfill was nearly full, partial construction of the final closure cover might be an option. This would reduce infiltration into the landfill and possibly the leakage rate, if the cover was constructed over the failed area.

- A layer of low-permeability soil could be placed over the existing waste, perhaps in conjunction with a geomembrane, to create a second "primary" liner higher in the landfill. This new liner would intercept precipitation and allow its removal.
- A rigid-frame or air-supported structure could be constructed over the landfill to ensure that no infiltration occurred. Although costly, this approach might be less expensive than constructing a new landfill.

In general, the selected remediation efforts would be those that are easiest to implement, with more difficult or expensive options to be applied only if earlier approaches were not satisfactory.

4C.2. References

EPA 530-R-92-004, *Action Leakage Rates for Leak Detection Systems*, U.S. Environmental Protection Agency, Office of Solid Waste Management, Washington, D.C., January 29, 1992.

57 FR 19, *Liners and Leak Detection Systems for Hazardous Waste Land Disposal Units*, U.S. Environmental Protection Agency, January 1992.

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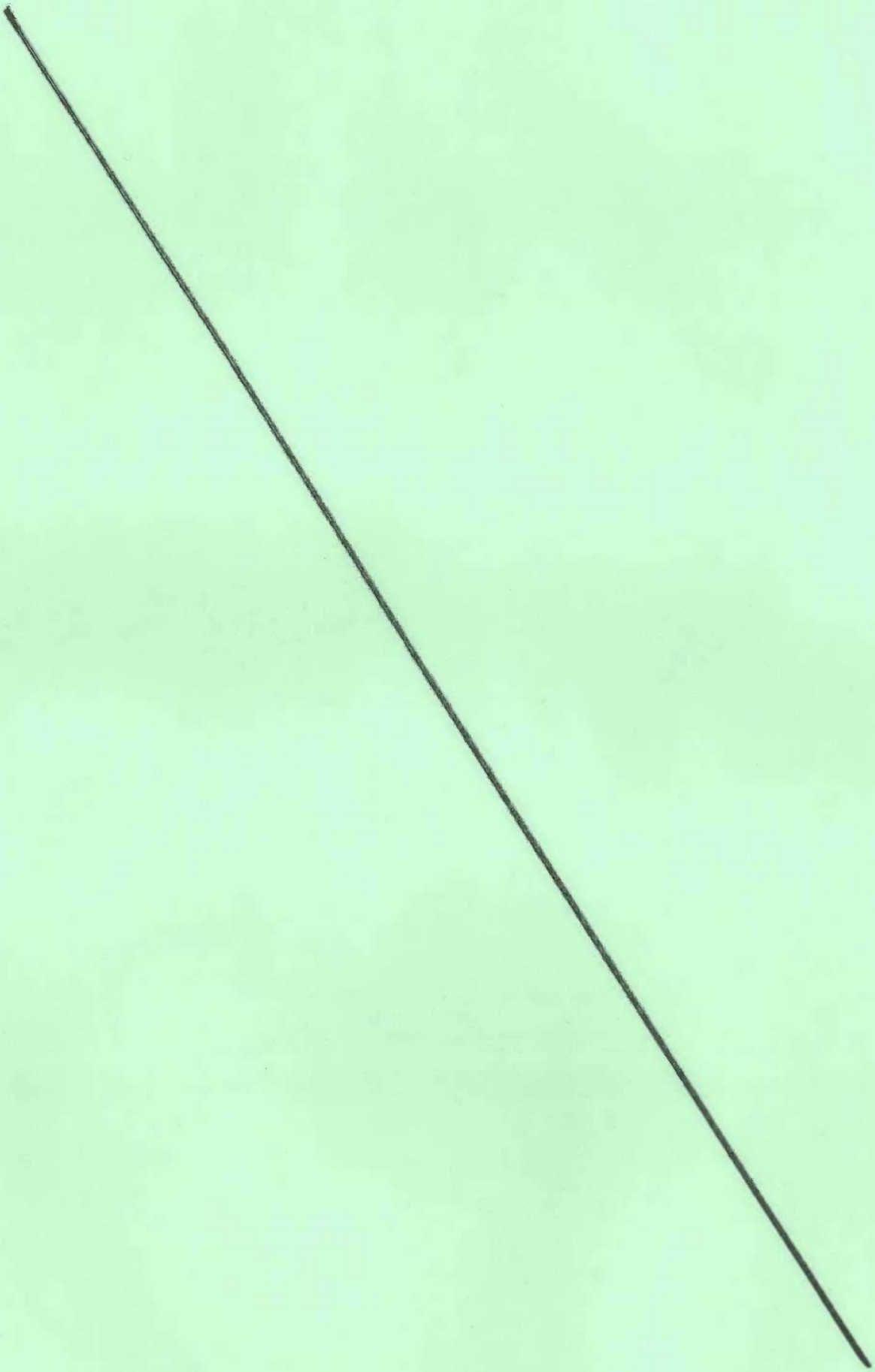

APPENDIX 4D
CONSTRUCTION SPECIFICATIONS (C-1)
RPP-18489, REV. 1

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APPENDIX 4D
CONSTRUCTION SPECIFICATIONS (C-1)
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1 DIVISION 1–GENERAL REQUIREMENTS

DIVISION 2–SITE CONSTRUCTION

SECTION 02200 – SITE PREPARATION SCHED. A & B

Part 1 – General

Work Included

This section describes requirements for clearing, grubbing and stripping of the IDF project area and associated facilities.

Definitions

Interfering or Objectionable Material: Trash, rubbish, and junk.

Clearing: Removal of existing vegetation and interfering or objectionable material lying on or protruding above ground surface.

Grubbing: Removal of vegetation and other organic matter including sagebrush, stumps, buried logs, and roots greater than 2 inches caliper to a depth of 6 inches below subgrade.

Stripping: Removal of topsoil and other organic matter. Strippings suitable for topsoil shall be limited to upper 6 inches.

Project Limits: Areas actually needed for site improvements, stockpiles, and borrow areas, as shown or specified, within which work is to be performed.

Raw Water Supply

Depending on the availability of connections to the raw water pipeline system, water for dust control may be obtained either from an existing water fill station on 4th Street near the west exit of the 200 East Area or from the raw water line that crosses the IDF site. A new raw water pipeline is being installed as part of Schedule A of the IDF project.

Whether water is obtained from the existing water fill station or from new facilities on the IDF site, use of connections to the raw water system shall be in accordance with guidance from the Hanford Site Water Purveyor. Use and discharge of raw water during construction for construction activities, including dust control and fire protection, shall be controlled through the Hanford Site Water Purveyor, and shall comply with WAC 173-200 and the State Waste Discharge Permit No. ST 4508. For any use of existing water fill station or a water connection on the IDF site, the total of all IDF project water usage from the raw water system shall not exceed 1,500 gpm. There is no guarantee that the 1,500 gpm will be available at all times or that will be available from a single location on the IDF water system. In addition, the maximum water use limitation may be periodically reduced by the Hanford Site Water Purveyor depending on the other Hanford Site demands on the raw water system and/or the availability of raw water pumping capacity within the Hanford raw water system. When using any connection to the raw water system, the Construction General Contractor shall take whatever means are necessary to operate fill stations and other connections in such a manner to prevent causing water hammer. If water usage from the system is found to cause water hammer, the Hanford Site Water Purveyor may require the use of orifice plates, changes to valve operation methods, and/or a reduction in water use at a water connection as necessary to eliminate the water hammer. For any onsite fill station, Construction General Contractor shall provide a water storage tank (minimum size 20,000 gallons) equipped with a float actuated fill valve to minimize demand surges on the Hanford Water Supply.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Site Preparation Plan: Prior to any mobilization of equipment to the site, the Construction General Contractor shall submit a Site Preparation Plan for approval. This plan shall include the following information as a minimum:

Detailed description of the proposed method for clearing, grubbing, and stripping the site. The plan shall identify those areas of the project site, which will be cleared, grubbed, and stripped. In addition, it will identify staging areas, stockpile areas, and the sequence in which the site preparation will be executed.

Dust Control Plan:

Prior to any earthwork activities on the project site such as site preparation and excavation, Construction General Contractor shall submit a Dust Control Plan for approval. The plan shall identify methods and equipment to minimize/control dust generation during all earthwork operations and include the following minimum requirements:

Continuous control of dust generation during excavation and backfill placement, etc.

Continuous control of wind-generated dust, including disturbed areas that are not being actively worked.

Keep haul roads watered to control dust.

A minimum two full water trucks (5,000 gallons each) for the duration of the project.

Apply water or other approved dust suppressants as minimum to keep visible dust to a minimum during execution of work.

Appropriate hand-held hose lines, sprinklers, and other equipment as needed to access and control non-vehicle access areas such as borrow and stockpile side slopes.

Maintain and protect native cover where possible, through minimization of site disturbance.

Limit access road development to minimum necessary to execute work.

Stabilization of inactive disturbed work areas by longer-term methods such as matting, tack and mulch or crusting agents.

Implementation of permanent stabilization on a regular basis when sufficient area exists for application or as needed to control dust.

General Construction Contractor shall use daily field reports to document dust control measures implemented and their effectiveness.

These dust control plan items are required to satisfy the requirements of Section 3.0 - Mitigation of Potential Dust Impacts from Construction Activities of the "Mitigation Action Plan for USDOE, Hanford Site, Immobilized Low-Activity Waste (ILAW) Disposal Site Construction (Project W-520)," prepared by PNNL for USDOE.

Submit details of raw water supply, storage, and water withdrawal limiting equipment as part of the Dust Control Plan.

Scheduling and Sequencing

The sequence of the activities listed below shall be followed by the Construction General Contractor for the site preparation work.

Initial site preparation activities shall commence only after Dust Control Plan and Site Preparation Plan have been approved.

Following the approved Site Preparation Plan, establish an adequate water supply source for dust control use.

After establishing an adequate water supply and sediment controls, proceed with site preparation activities as specified.

Part 2 – Products (Not Used)

Part 3 – Execution

General

Clear, grub, and strip only areas actually needed for stockpiles, borrow, or site improvements within limits shown and specified.

Extent of Site Preparation required for individual designated stockpile areas shall be as directed by the Construction Manager.

Do not injure or deface vegetation that does not require removal.

As an initial step in clearing the site, the General Construction Contractor shall remove all trash, rubbish, and junk from the site. This material shall be disposed in accordance with Division 1 requirements.

Clearing

Cut off shrubs, brush, weeds, and grasses to within 4 inches of ground surface.

Grubbing

Grub all areas where excavations, fill, roadways, structures, and ditches are to be placed.

Vegetation other than noxious weeds, removed by the clearing and grubbing, shall be placed in stockpile with the strippings to be used as topsoil. Place vegetation at the base of the strippings stockpile area and track with equipment to break apart and crush the material. Obtain Construction Manager approval of the vegetation placement.

Stripping

Strip all areas where excavations, borrow areas, stockpiles, fills, roadways, structures, and ditches are to be placed, to remove organic materials. Do not remove subsoil with topsoil.

Stockpile strippings from the upper 6 inches below ground surface after clearing and grubbing, meeting requirements for topsoil in Section 02920, RECLAMATION AND REVEGETATION, separately from other excavated material at either the designated stockpile area location shown on the Drawings, or other areas as approved by the Construction Manager.

IDF Raw Water Connections

General Construction Contractor shall take necessary steps to prevent freezing and/or damage to the IDF raw water system connections.

Disposal

Clearing and Grubbing Debris: Bury vegetation that is not suitable for topsoil at a designated area as directed by the Construction Manager. Disposal of the remaining interfering or objectionable material shall be in accordance with Division 1 requirements.

Strippings: Dispose of strippings that are unsuitable for topsoil as specified above for clearing and grubbing debris.

Burning Prohibited: No burning of any materials generated during the site preparation work will be allowed at the site.

END OF SECTION 02200

SECTION 02315 – FILL AND BACKFILL SCHED. A & B

Part 1 – General

Work Included

This section describes placement and testing of fill and backfill in general areas of the site (including stockpiles).

References

The following is a list of standards, which may be referenced in this section:

ASTM INTERNATIONAL (ASTM)

ASTM D75	Standard Practice for Sampling Aggregates
ASTM D422	Standard Test Method for Particle-Size Analysis of Soils
ASTM D698	Test Method for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft^3 (600 kN-m/m^3))
ASTM D1140	Standard Test Method for Amount of Material in Soils Finer than the No. 200 (75 micrometer) Sieve
ASTM D1556	Standard Test Method for Density and Unit Weight of Soil in Place by the Sand-Cone Method
ASTM D2216	Standard Test Method for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass.
ASTM D2922	Standard Test Methods for Density of Soil and Soil-Aggregate in Place by Nuclear Methods (Shallow Depth)
ASTM D3017	Standard Test Method for Water Content of Soil and Rock in Place by Nuclear Methods (Shallow Depth)

Definitions

Relative Compaction: Ratio, in percent, of as-compacted field dry density to laboratory maximum dry density as determined in accordance with ASTM D698.

Apply corrections for oversize material to maximum dry density.

Optimum Moisture Content: Determined in accordance with ASTM D698 specified to determine maximum dry density for relative compaction.

Prepared Ground Surface: Ground surface after completion of required demolition, clearing and grubbing, scalping of sod, stripping of topsoil, excavation to grade, and subgrade preparation.

Completed Course: A course or layer that is ready for next layer or next phase of Work.

Lift: Loose (uncompacted) layer of material.

Geosynthetics: Geotextiles, geocomposites, geosynthetic clay liner, or geomembranes.

Well-Graded:

A mixture of particle sizes with no specific concentration or lack thereof of one or more sizes.

Does not define numerical value that must be placed on coefficient of uniformity, coefficient of curvature, or other specific grain size distribution parameters.

Used to define material type that, when compacted, produces a strong and relatively incompressible soil mass free from detrimental voids.

Influence Area: Area within planes sloped downward and outward at 60-degree angle from horizontal measured from:

1-foot outside outermost edge at base of foundations or slabs.

1-foot outside outermost edge at surface of roadways or shoulder.

0.5-foot outside exterior at spring line of pipes or culverts.

Imported Material: Materials obtained from sources offsite, suitable for specified use.

Standard Specifications: When referenced in this section, shall mean Standard Specifications for Road, Bridge, and Municipal Construction, as published by the Washington State Department of Transportation, 2002 edition, English units.

SLDS: Secondary Leak Detection System.

Permanent Stockpile: Stockpile of material that remains at the completion of construction.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Submit gradation test results for all imported materials from independent testing laboratory as specified in paragraph SOURCE QUALITY CONTROL in Part 2.

Stockpile Plan: Prior to any excavation activities associated with construction of the IDF, Construction General Contractor shall submit a Stockpile Plan for approval. The plan shall include the following information:

Scale drawing, using the project plans as a base, which shows the proposed location for stockpiles on the project site. Show all access roads around stockpiles. Address stockpile locations during construction of the IDF and permanent stockpiles, which will remain upon completion of construction activities for this project.

Stockpile layout drawings, which show the estimated location of toe of slope and top of slope for each stockpile. Drawings shall show plan and typical sections and shall be fully dimensioned.

Plan shall show how differing materials encountered during the excavation will be segregated for future use. This includes material for use as topsoil, admix base soil, and operations layer. Also, show a stockpile area for material to be used in the future as clean backfill during landfill operations by the Tank Farm Contractor.

Method by which stockpile compaction will be achieved.

Dust control for the stockpiles during active use and until grass is established.

Placing of topsoil, seeding, fertilizing, and mulching each stockpile after active use of stockpile is finished in accordance with Section 02920, RECLAMATION AND REVEGETATION.

Submittals–Approval Not Required

Information/Record (IR):

Qualifications of independent testing laboratory.

Qualifications of construction quality control personnel.

Construction quality control test reports

Sequencing And Scheduling

Complete applicable Work specified in Sections 02316, EXCAVATION, and 02319, SUBGRADE PREPARATION, prior to placing fill or backfill.

Permit Requirements

A backfill and fill permit is required for each backfill and fill work element. Construction General Contractor shall obtain from Tank Farm Contractor and post before starting backfill and fill work, as specified in Division 1, General Requirements.

Construction Quality Assurance (CQA)

The Construction General Contractor shall accommodate all CQA activities described herein and in the CPA Plan for this project. The CQA Plan is made part of these Specifications by reference.

The CQA Certifying Engineer shall determine in-place density and moisture content by any one or combination of the following methods: ASTM D2922, ASTM D3017, ASTM D1556, ASTM D2216, or other methods approved by the Construction Manager.

Testing requirements and locations will be determined by the CQA Certifying Engineer. Construction General Contractor shall cooperate with the CQA Certifying Engineer and testing work by leveling small test areas designated by the CQA Certifying Engineer. Backfill test areas at Construction General Contractor's sole expense. The CQA Certifying Engineer may have any material tested at any time, location, or elevation.

After Construction General Contractor makes repairs to any areas failing a test, the Construction General Contractor shall rerun appropriate tests, subject to the approval of the CQA Certifying Engineer, to demonstrate the area meets specifications, at the Construction General Contractor's sole expense.

The following minimum test schedule shall be assumed. Additional tests may be required as directed by CQA Certifying Engineer.

In-place density tests shall be made on the following minimum schedule:

Earth fill: One per 5,000 square feet (ft^2) per lift.

Structural Fill: One per 2,500 ft^2 per lift.

Subgrade Preparation: Four per acre.

Operations Layer (Outside Edge of Liner): One per 5,000 ft^2 per lift.

Operations Layer Material (SLDS): Two (2) per lift.

Standard Proctor (ASTM D698) laboratory density curves (five-point minimum) shall be performed for each material by the CQA Certifying Engineer. Samples of native materials used for embankment and backfill and samples of imported materials shall be taken at locations as specified by CQA Certifying Engineer.

Gradation tests (sieve analysis) shall be performed in accordance with ASTM D422 on operations layer material obtained from required excavations to demonstrate the materials meet the Specifications. Samples of operations layer material shall be taken from each 10,000 cubic yards of placed material in accordance with ASTM D75.

Part 2 – Products

Earthfill

Excavated material from required excavations and designated borrow sites, free from rocks larger than 4 inches in the greatest dimension, from roots and other organic matter, ashes, cinders, trash, debris, and other deleterious materials.

Structural Fill

Structural fill adjacent to concrete structures shall be as specified in Section 02320, TRENCH BACKFILL, for Pipe Bedding.

Structural fill beneath concrete structures and beneath leachate storage tank shall conform to the requirements of Section 9-03 9(3) Crushed Surfacing-Top Course in the Standard Specifications.

Operations Layer

Meeting the requirements of earthfill above and having a maximum of 25 percent by weight passing the No. 200 U.S. sieve and a maximum particle size of 2 inches.

Drain Gravel

Material for drain gravel shall conform to the requirements of Section 9-03.12(4) of the Standard Specifications except material shall be subrounded to rounded gravel. Crushed rock and angular gravel shall not be allowed.

Crushed Surfacing

Material for crushed surfacing-base course and top course shall conform to the requirements in Section 9-03 9(3) of the Standard Specifications.

Quarry Spalls

Quarry spalls shall consist of broken stone free from segregation, seams, cracks, and other defects tending to destroy its resistance to weather and shall meet the following requirements for grading:

Maximum Size: 8 inches

50 percent by weight shall be larger than 3 inches

Minimum Size: 3/4 inch

Source Quality Control

Gradation tests performed in accordance with ASTM D422 by a qualified independent testing laboratory shall be made for imported materials on samples taken at place of production prior to shipment. Imported materials shall not be shipped without submittal approval. Samples of the finished product for gradation testing shall be taken from each 2,000 tons of prepared materials in accordance with ASTM D75. Test results shall be submitted to Construction Manager within 48 hours after sampling. Size distribution for imported quarry spalls material shall be determined in accordance with one of the methods described in ASTM D5519.

Base Soil: As specified in Section 02666, ADMIX LINER.

Water For Moisture Conditioning: See Section 02200, SITE PREPARATION, for raw water supply availability and requirements for proper compaction.

Part 3 – Execution

General

Keep placement surfaces free of water, debris, and foreign material during placement and compaction of fill and backfill materials.

Place and spread fill and backfill materials in horizontal lifts of uniform thickness as specified in paragraphs BACKFILL UNDER AND AROUND STRUCTURES and FILL, in a manner that avoids segregation, and compact each lift to specified densities prior to placing succeeding lifts. Slope lifts only where necessary to conform to final grades or as necessary to keep placement surfaces drained of water.

Do not place fill or backfill, if fill or backfill material is frozen, or if surface upon which fill or backfill is to be placed is frozen.

Tolerances:

Final Lines and Grades: Within a tolerance of 0.1-foot unless dimensions or grades are shown or specified otherwise.

Grade to establish and maintain slopes and drainage as shown. Reverse slopes are not permitted.

Settlement: Correct and repair any subsequent damage to structures, pavements, curbs, slabs, piping, and other facilities, caused by settlement of fill or backfill material.

Backfill Under and Around Structures

Under Facilities: Backfill with earthfill or structural fill, as shown on the Drawings, for each structure or facility. Place earthfill or structural fill in lifts of 6-inch maximum compacted thickness and compact each lift to minimum of 95 percent relative compaction as determined in accordance with ASTM D698.

Fill

Outside Influence Areas Beneath Structures, Slabs, Piping, and Other Facilities: Unless otherwise shown, place earthfill as follows:

Allow for 6-inch thickness of topsoil where required.

Maximum 8-inch thick lifts.

Place and compact fill across full width of embankment.

Compact to minimum 95 percent relative compaction.

Replacing Overexcavated Material

Replace excavation carried below grade lines shown as follows:

Beneath IDF Cell: Earthfill as specified herein.

Beneath Fill or Backfill: Same material as specified for overlying fill or backfill.

Beneath Structures and Roadways: Structural fill or earthfill as shown on the Drawings and specified herein.

Topsoil

Place topsoil on areas disturbed by construction and on permanent stockpile slopes in accordance with Section 02920, RECLAMATION AND REVEGETATION.

Stockpiling

Material shall be placed in permanent stockpiles as follows:

Place material in maximum 3-foot lifts and compact with a minimum four passes with earth-moving equipment. Uniformly route hauls truck traffic across the surface of each lift to aid in lift compaction.

Maximum slopes shall be 3H:1V. Minimum slopes shall be 3 percent to promote drainage.

Upper 2 feet of stockpile surface shall be placed in maximum 12-inch thick lifts and compacted to minimum 90 percent relative compaction as determined in accordance with ASTM D698.

Place 6-inch thick layer of topsoil on completed slopes in accordance with Section 02920, RECLAMATION AND REVEGETATION.

Permanent stockpiles shall be seeded, fertilized, and mulched when each stockpile is completed and as directed by the Engineer in accordance with Section 02920, RECLAMATION AND REVEGETATION.

Placing Crushed Surfacing

Place crushed surfacing base course and top course at locations shown on the Drawings. Placement shall conform to Section 4-04.3 of the Standard Specifications.

Thickness of the drain gravel on the cell floor shall be a minimum of 1.0 foot (increase drain gravel thickness in vicinity of LCRS collection and riser pipes as shown on drawings) and tolerances for the top of the drain gravel shall be grade to plus 0.2 foot.

Placing Drain Gravel And Operations Layer Over Geosynthetics

Place material over geosynthetics as specified in Sections 02371, GEOTEXTILES; 02661, GEOMEMBRANES; and 02667, GEOSYNTHETIC CLAY LINER (GCL).

Compaction requirements for drain gravel on the cell floor, around pipes, and the sumps are specified in Section 02320, TRENCH BACKFILL.

Operations layer within lining system limits, except as specified for SLDS sump area, shall be placed in 12-inch thick lifts and track-walked into place with a minimum two passes with a Caterpillar D6M-LGP or equal. Operations layer material within SLDS sump area shall be placed in 12-inch thick lifts and compacted to 90 percent relative compaction. Operations layer placed outside edge of liner, such as for shine berm, shall be placed in maximum 8-inch thick lifts and compacted to 95 percent relative compaction.

Place material to the lines and grades shown and compact by tracking a minimum two passes with spreading equipment. Thickness of the operations layer shall be a minimum 3 feet and tolerances for top of operations layer shall be grade to plus 0.3 foot.

Quarry Spalls Placement

Quarry spalls shall be placed around the ends of stormwater pipes to provide erosion protection in accordance with the Plans and as directed by the Engineer. Quarry spalls shall be placed in such a manner that all relatively large stones are essentially in contact with each other and voids are filled with the finer materials to provide a well graded compact mass. Finished surface shall be free from irregularities. The stone shall be dumped on the ground in a manner that will ensure the stone attains its specified thickness in one operation. When dumping or placing, care shall be used to avoid damaging the underlying material. Stone shall not be dumped from height greater than 12 inches above surface. Material placement shall be started from the bottom of the installation, working toward edges. Geotextile damaged during the placement of quarry spalls shall be repaired at Construction General Contractor's sole expense.

Construction Quality Control

The Construction General Contractor shall perform in-place density and moisture content tests with own qualified personnel or with a qualified independent testing laboratory as specified in paragraph CONSTRUCTION QUALITY ASSURANCE, to be observed by the Construction Manager, on the following minimum schedule:

Material Placed by Stockpile (Upper 2 Feet): One per 10,000 ft^2 per lift.

Construction General Contractor shall submit qualifications of personnel or independent testing laboratory that will perform construction quality control.

END OF SECTION 02315

SECTION 02316 – EXCAVATION SCHED. A & B

Part 1 – General

Work Included

This section describes all excavation necessary for completion of the Project, including excavation for structures, pipe trenches, and leachate sumps.

References

The following is a list of standards, which may be referenced in this section:

Code of Federal Regulations (CFR)

29 CFR 1926 OSHA Safety and Health Regulations for Construction; Subpart P – Excavations

Excavation Support And Safety

Install and maintain adequate excavation safety and shoring systems that meet the requirements of OSHA (29 CFR 1926 Subpart P) and Washington Industrial Safety and Health Act, Chapter 49.17 RCW and WAC 296-155, "Safety Standards for Construction Work" Part IV-Excavation, Trenching and Shoring, and all other applicable local, state, and federal regulations.

Design, provide, and maintain shoring, sheeting, and bracing as necessary to support the sides of excavations and to prevent detrimental settlement and lateral movement of existing facilities, adjacent property, and completed Work. Per OSHA requirements, 29 CFR 1926.652 requires that custom shoring installations shall be designed and stamped by a professional engineer licensed in the State of Washington.

Weather Limitations

Material excavated during inclement weather shall not be used as fill or backfill until after material drains and dries sufficiently for proper compaction.

Permit Requirements:

An excavation permit is required for each excavation work element. Construction General Contractor shall obtain from Tank Farm Contractor and post before starting excavation work, as specified in Division 1, General Requirements.

Part 2 – Products (Not Used)

Part 3 – Execution

General

Generally, excavate to lines, grades, and dimensions shown and as necessary to accomplish work. Excavate subgrade to within tolerance of minus 0.5 foot to plus 0.1 foot except where dimensions or grades are shown or specified as maximum or minimum.

If unexpected, debris foreign material of any kind (e.g., contaminated soil) or cultured properties (e.g., bones and artifacts) is exposed or encountered during excavation, the Construction General Contractor shall stop work in the affected area and notify the Construction Manager. Obtain approval from Construction Manager before resuming excavation.

Control dust from excavation activities as specified in approved Dust Control Plan. See Section 02200, SITE PREPARATION, for Dust Control Plan requirements.

If soft or loose subgrade zones are encountered at the bottom of the excavation after proof rolling, correct as specified in Section 02319, SUBGRADE PREPARATION.

Structure Excavation

Excavations for such structures as footings, foundations, slabs, and manholes shall be made to the depths shown on the Drawings and of sufficient width to allow adequate room for placement of structural fill setting and removing forms, installing accessories and inspection. Care shall be taken to prevent disturbing the bottom of the excavation. Excavation to final grade shall not be made until just before concrete forms are to be placed therein. Prepare bottom of structure excavation as specified in Section 02319, SUBGRADE PREPARATION, prior to placement of concrete foundations.

Trench And Sump Excavation

Trenches and sumps shall be of sufficient width to provide adequate room for workmen to perform any necessary service to the materials or items being installed therein and to permit proper compaction of the backfill.

Minimum Width of Trenches: As shown on Drawings.

Maximum Trench Width: Unlimited, unless otherwise shown or specified, or unless excess width will cause damage to existing facilities, adjacent property, or completed Work.

If wet or otherwise unsatisfactory soil is encountered in a trench excavation, at or below the trench bottom, it shall be brought to the attention of the Construction Manager and removed as directed. The bottom of the excavation shall then be brought to the required grade with stabilization as specified in Section 02320, TRENCH BACKFILL.

Temporary Stockpile Excavation

Always keep stockpile neat and orderly and work there in a systematic manner. Take necessary precautions to maintain existing erosion control measures and prevent offsite sediment releases.

When work is completed in the stockpile area, grade area to drain surface water runoff to appropriate collection and discharge points. Reclaim disturbed areas of stockpile as specified in Sections 02315, FILL AND BACKFILL, and 02920, RECLAMATION AND REVEGETATION.

Stockpiling Excavated Material

Stockpile excavated material that is suitable for use as embankment or backfill, as operations layer material, road gravel, or drain gravel, until material is needed. Place materials in stockpiles at the designated locations shown on the Drawings. Materials shall be placed in stockpiles as specified in Section 02315, FILL AND BACKFILL.

Confine stockpiles to within areas shown on Drawings. Do not obstruct roads or streets. Stockpiles should maintain clearance from existing permanent monitoring wells and disposed cribs as shown on the Drawings.

Do not stockpile excavated material adjacent to trenches and other excavations unless excavation sideslopes and excavation support systems are designed, constructed, and maintained for stockpile loads. The registered professional engineer responsible for the shoring design shall approve stockpile locations.

Do not stockpile excavated materials near or over existing facilities, adjacent property, or completed work, if weight of stockpiled material could induce excessive settlement. The registered professional engineer responsible for the shoring design shall approve stockpile locations.

Disposal of Spoil

Dispose of excavated materials, which are unsuitable or not needed for fill or backfill, in designated stockpile areas shown on the Drawings, or spoil disposal areas as directed by Construction Manager. Materials shall be placed as specified in Section 02315, FILL AND BACKFILL.

Trench Excavation For Geosynthetic Anchor Trenches

Geosynthetic anchor trench excavation shall be as specified in Section 02661, GEOMEMBRANES.

Construction Quality Control

Construction General Contractor provides adequate survey control to avoid unauthorized over excavation.

END OF SECTION 02316

SECTION 02317 – BORROW AREA EXCAVATION SCHED. B

Part 1 – General

Work Included

This section describes requirements for borrow excavation from the Admix Base Soil Borrow Area as a source of base soil for the admix. The Admix Base Soil Borrow Area is located within the IDF site boundary approximately 1,000 feet south of the Phase I area as shown on the Drawings.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Borrow Area Development Plan: Detailing development, operation, dust and erosion, mitigation measures, and reclamation of each borrow area.

Part 2 – Products (Not Used)

Part 3 – Execution

General

Clear, grub, and strip only areas actually needed for borrow within limits shown or specified.

Do not injure or deface vegetation that does not require removal.

Clearing, Grubbing, Stripping, And Disposal:

Clearing, grubbing, and stripping shall meet the requirements of Section 02200, SITE PREPARATION.

Disposal

Clearing and Grubbing Debris: As specified in Section 02200, SITE PREPARATION.

Strippings: As specified in Section 02200, SITE PREPARATION.

Borrow Area Operation

Borrow area shall be developed and operated in accordance with the mitigation measures identified in the approved Borrow Area Development Plan and these Specifications.

Always keep borrow pits neat and orderly, and work them in systematic manner. Continuously keep borrow pits graded to drain to a low point, and take necessary precautions to control erosion and prevent offsite sediment releases. Dewater as necessary to develop, operate, and reclaim each borrow area. Control dust as specified in approved Dust Control Plan. See Section 02200, SITE PREPARATION, for Dust Control Plan requirements.

Material meeting the requirements for base soil as specified in Section 02666, ADMIX LINER, shall be excavated from the Borrow Area. Base soil should not be obtained below a depth of 5 feet below existing ground surface (after stripping) without evaluation of the material suitability and authorization from the Construction Manager.

Do not excavate more borrow material than required for work. Leave surplus material in place.

Excavate material in an orderly manner to avoid inclusion of unacceptable material.

Reclamation

At the completion of borrow area excavation, grade borrow pits to drain to low point so that ponded surface water may be removed by pumping. Where practical, blend graded surfaces neatly with surrounding terrain at completion of borrow operations.

Final Slopes:

Maximum: 4H: 1V.

Minimum: 2 percent.

Do not use borrow pits for disposal, unless otherwise specified or shown.

Place 6-inch thick layer of topsoil and seed, fertilize, and mulch all disturbed areas as specified in Section 02920, RECLAMATION AND REVEGETATION.

END OF SECTION 02317

SECTION 02319 – SUBGRADE PREPARATION SCHED. B

Part 1 – General

Work Included

This section describes requirements for preparation of subgrades in areas to receive fill.

References

The following is a list of standards, which may be referenced in this section:

ASTM INTERNATIONAL (ASTM)

ASTM D698	Test Method for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft^3 (600 kN-m/m^3))

Definitions

Optimum Moisture Content: As defined in Section 02315, FILL AND BACKFILL.

Prepared Ground Surface: Ground surface after completion of clearing and grubbing, scalping of sod, stripping of topsoil, excavation to grade, and scarification and compaction of subgrade.

Relative Compaction: As defined in Section 02315, FILL AND BACKFILL.

SLDS: Secondary Leak Detection System.

Subgrade: Layer of existing soil after completion of excavation to grade prior to placement of fill, roadway structure, or base for floor slab.

Proof Rolling: Testing of subgrade as specified herein to identify soft or loose zones requiring correction.

Sequencing And Scheduling

Complete applicable Work specified in Section 02316, EXCAVATION, prior to subgrade preparation.

Construction Quality Assurance

The CQA Certifying Engineer shall determine in-place density and moisture for subgrade preparation as specified in Section 02315, FILL AND BACKFILL, except for prepared subgrade for admix on sideslopes.

CQA requirements for geomembrane subgrade preparation are specified in Section 02661, GEOMEMBRANES.

Part 2 – Products (Not Used)

Part 3 – Execution

General

Keep subgrade free of water, debris, and foreign matter during compaction or proof rolling.

Bring subgrade to proper grade and cross-section as shown on the Drawings, and uniformly compact surface.

Maintain prepared ground surface in finished condition until next course is placed.

Prepared Subgrade For Roadway, Embankment, And Structures

After completion of excavation and prior to foundation, road fill, structural fill, or embankment construction, compact prepared subgrade to 95 percent relative compaction. Scarify and moisture condition subgrade soil as required to achieve specified compaction.

If soft or loose zones are found, correct as specified herein. Proof-roll subgrade with a fully loaded dump truck or equal to detect soft or loose subgrade or unsuitable material.

Landfill Bottom Floor Prepared Subgrade For Admix Liner

Prior to admix liner placement, subgrade shall be back bladed to remove loose soil. Low spots or erosion rills shall be backfilled with structural fill as specified herein. Compact prepared subgrade to 95 percent relative compaction. Scarify and moisture condition subgrade soil as required to achieve specified compaction. If soft or loose zones are found, correct as specified herein. Proof-roll subgrade with a vibratory drum roller or equal to detect soft or loose subgrade or unsuitable material.

Landfill Sideslope (3h: 1v) Prepared Subgrade For Admix Liner

Prior to admix placement, the subgrade shall be back bladed to remove all loose material produced by trimming operations. Low spots or erosion rills shall be backfilled with structural fill as specified herein. The trimmed surface shall be watered so that moisture penetrates a minimum of 3 inches into the subgrade. The trimmed and watered surface shall be track-walked by D6-LGP dozer or equivalent with a minimum 4 passes to produce a firm and stable subgrade. Visual monitoring (no in-place density testing is required) of the subgrade preparation on sideslopes will be performed by the CQA Certifying Engineer.

Prepared Subgrade For Geomembrane (Secondary and SLDS) and Secondary GCL

At completion of SLDS excavation and grading (SLDS geomembrane) or admix liner placement (secondary geomembrane and GCL), prepare the subgrade surface for geomembrane or GCL placement. The surface shall not have holes, depressions more than 1 inch in a 12-inch width, nor protrusions extending above the surface more than 1/2 inch. Roll surface with smooth-drum roller to form a firm stable base. Allow for leachate piping and sumps or features as shown on the Drawings.

Correction

Soft or Loose Subgrade: Adjust moisture content and compact to meet density requirements, or

Over excavate and replace with suitable material from the excavation, as specified in Section 02315, FILL AND BACKFILL.

Unsuitable Material: Over excavate and replace with suitable material from the excavation, as specified in Section 02315, FILL AND BACKFILL. Dispose of unsuitable material excavation in accordance with Article DISPOSAL OF SPOIL in Section 02316, EXCAVATION.

END OF SECTION 02319

SECTION 02320 – TRENCH BACKFILL SCHED. A & B

Part 1 – General

Work Included

This section describes requirements for backfilling of trenches for pipe, conduit, and geosynthetics.

References

The following is a list of standards, which may be referenced in this section:

ASTM INTERNATIONAL (ASTM)

ASTM C136	Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
ASTM D698	Test Method for Laboratory Compaction Characteristics of Soil using Standard Effort (12,400 ft-lbf/ft^3 (600 kN-m/m^3))
ASTM D1140	Standard Test Method for Amount of Material in Soils Finer than the No. 200 (75 micrometer) Sieve
ASTM D4318	Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils

Definitions

Pipe Bedding: Granular material upon which pipes, conduits, cables, or duct banks are placed.

Imported Material: Material obtained by the Construction General Contractor from source(s) offsite.

Lift: Loose (uncompacted) layer of material.

Pipe Zone: Backfill zone that includes full trench width and extends from prepared trench bottom to an upper limit above top outside surface of pipe, conduit, cable or duct bank.

Prepared Trench Bottom: Graded trench bottom after stabilization and installation of bedding material.

Relative Compaction: The ratio, in percent, of the as-compacted field dry density to the laboratory maximum dry density as determined by ASTM D698. Corrections for oversize material may be applied to either the as-compacted field dry density or the maximum dry density.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Submit gradation test results for all imported materials as specified in paragraph SOURCE QUALITY CONTROL in Part 2.

Submittals–Approval Not Required

Vendor Information (VI): Locator ribbon product data.

Part 2 – Products

Locator Ribbon

Ribbon shall be 3 inches wide and shall be red for all electrical conduits, electrical cables, and telephone cables. Blue locator ribbon shall be used for all buried pipe in the raw water system. Yellow locator ribbon shall be used for the electrical duct bank from riser pole to the transformer. Purple locator ribbon shall be used for the buried leachate transfer lines.

Ribbon shall be tape manufactured by Reef Industries or Allen Markline or equal and shall have metal foil, which is completely encased in plastic and can be easily detected by metal detectors.

The ribbon shall be printed with the manufacturer's standard wording, "CAUTION ELECTRIC LINE BURIED BELOW", for all electrical conduits, phone lines, etc., "CAUTION BURIED PIPELINE BELOW", for all buried pipelines.

Trench Stabilization Material

Granular material from the excavation or stockpile meeting the requirements of structural fill as specified in Section 02315, FILL AND BACKFILL.

Pipe Bedding

Pipe bedding material for thermoplastic pipe shall be clean sand/gravel mixture free from organic matter and conforming to the following gradation:

Sieve Size	Percent Passing
3/4" square	100
3/8" square	70-100
U.S. No. 4	55-100
U.S. No. 10	35-100
U.S. No. 20	20-80
U.S. No. 40	10-55
U.S. No. 100	0-10
U.S. No. 200	0-5

All percentages are by weight.

Pipe Zone Material

Excavated granular material from required excavations, free from rocks, roots, and organic matter. The maximum particle size shall be 3/4 inch and the percent by weight passing the No. 200 U.S. sieve shall be a maximum 15 percent.

Pipe bedding may be used as substitute for pipe zone material.

Earth Backfill

Earthfill as specified in Section 02315, FILL AND BACKFILL.

Structural Fill

As specified in Section 02315, FILL AND BACKFILL.

Source Quality Control

As specified in Section 02315, FILL AND BACKFILL.

Part 3 – Execution

Trench Preparation

Water Control:

Promptly remove and dispose of water entering trench as necessary to grade trench bottom and to compact backfill and install manholes, pipe, conduit, direct-buried cable, or duct bank. Do not place concrete, lay pipe, conduit, direct-buried cable, or duct bank in water.

Remove water in a manner that minimizes soil erosion from trench sides and bottom.

Provide continuous water control until trench backfill is complete.

Remove foreign material and backfill contaminated with foreign material that falls into trench.

Trench Bottom

Firm Subgrade: Remove loose and disturbed material and trim off high areas and ridges left by excavating equipment. Tamp to provide a firm and unyielding subgrade. Allow space for bedding material if shown or specified.

Soft Subgrade: If subgrade is encountered that may require removal to prevent pipe settlement, notify Engineer. Engineer will determine depth of over excavation, if any, required.

Trench Stabilization Material Installation

Rebuild trench bottom with trench stabilization material.

Place material over full width of trench in 8-inch maximum, loose measurement lifts to required grade, providing allowance for bedding thickness.

Compact each lift to provide a firm, unyielding support for the bedding material prior to placing succeeding lifts.

Bedding

Place over the full width of the prepared trench bottom in two equal lifts when the required depth exceeds 8 inches.

Hand grade and compact each lift to provide a firm, unyielding surface.

Minimum Compacted Thickness: As shown on the Drawings. For leachate collection and riser pipes, there shall be no bedding between lining system and pipe.

Direct-Buried Cable: 3 inches.

Duct Banks:

2 inches.

Check grade and correct irregularities in bedding material.

Backfill Pipe Zone

Upper limit of pipe zone shall not be less than following:

Pipe: 12 inches, unless shown otherwise.

Conduit: 3 inches, unless shown otherwise.

Direct-Buried Cable: 3 inches, unless shown otherwise.

Duct Bank:

3 inches, unless shown otherwise.

Restrain pipe, conduit, cables, and duct banks as necessary to prevent their movement during backfill operations.

Place pipe zone material simultaneously in lifts on both sides of pipe and, if applicable, between pipes, conduit, cables, and duct banks installed in same trench.

Pipes 10 Inches and Smaller Diameter: First lift less than or equal to 1/2 pipe-diameter.

Pipes Over 10-Inch Diameter:

Maximum 8-inch, loose measurement lifts.

Thoroughly tamp each lift, including area under haunches, with handheld tamping bars supplemented by "walking in" and slicing material under haunches with a shovel to ensure that voids are completely filled before placing each succeeding lift.

After the full depth of the pipe zone material has been placed as specified, compact the material by a minimum of three passes with a vibratory plate compactor only over the area between the sides of the pipe and the trench walls.

Do not use power-driven impact compactors to compact pipe zone material.

Locator Ribbon Installation

Continuously install locator ribbon along centerline of all buried piping, at depth of 16 inches below ground surface unless shown otherwise on the Drawings. Coordinate with piping installation drawings.

Backfill Above Pipe Zone

General

Do not allow backfill to free fall into the trench or allow heavy, sharp pieces of material to be placed as backfill until after at least 2 feet of backfill has been provided over the top of pipe. Trench backfill using water-settling methods for compaction is not permitted.

Do not use power driven impact type compactors for compaction until at least 2 feet of backfill is placed over top of pipe. Hand-held jump jack type compactors are acceptable for compaction of backfill over top of pipe.

Backfill to grade with proper allowances for topsoil, road gravel subbase, and pavement thicknesses, wherever applicable.

Backfill around structures with same backfill as specified for adjacent trench unless otherwise shown or specified.

Backfill outside the Limits of Roadways, Utilities, and Other Facilities:

Place earthfill in lifts not exceeding 12-inch maximum, loose measurement thickness.

Mechanically compact each lift to a minimum of 90 percent relative compaction prior to placing succeeding lifts.

Backfill Under Facilities, Roadways, and Utilities: Backfill trench above the pipe zone with structural fill in lifts not exceeding 8 inches maximum, loose measurement thickness. Compact each lift to a minimum of 95 percent relative compaction prior to placing succeeding lifts.

Replacement of Topsoil

Where applicable, replace topsoil in top 6 inches of backfilled trench.

Maintain the finished grade of topsoil even with adjacent area and grade as necessary to restore drainage.

Drain Gravel Backfill For Leachate Collection (Slotted) Pipe, Riser Pipes, And Sumps

Use drain gravel as specified in Section 02315, FILL AND BACKFILL.

Drain gravel shall be placed in sumps by mechanical or hand methods that will not damage pipes or underlying geosynthetics. For areas within 3 feet of leachate collection pipe, riser pipe, and transducer pipe centerline, the Construction General Contractor shall place first lift of drain gravel on both sides of pipe in a 9-inch lift, and succeeding lifts shall be 6 inches maximum. The Construction General Contractor shall thoroughly tamp each lift, including area under haunches with handheld equipment and tools to ensure that voids are completely filled before placing each succeeding lift. After first lift and after the full depth of material has been placed as specified, the Construction General Contractor shall compact the material by a minimum of three passes with a hand-held vibratory plate compactor only over the area within 3 feet of pipe centerline.

For LDS Drain Gravel placement within the LDS Sump, outside of 3 feet from the leachate collection pipe centerline, material will be placed in 6-inch maximum lifts and compacted in lifts with a minimum of three passes of the hand-held plate compactor to ensure that firm and unyielding conditions are achieved throughout the entire gravel thickness in the sump. At all other locations, drain gravel shall be track-walked into place with a minimum two passes with a D6M-LGP bulldozer or equal. Equipment limitations and requirements for placing materials over geosynthetics are specified in Section 02315, FILL AND BACKFILL.

Backfill For Geosynthetic Anchor Trenches

Backfill with material as shown on the Drawings in loose lifts not exceeding 6 inches in thickness and compact using hand-operated equipment or mechanical equipment as approved by the Design Engineer to not less than 90 percent relative compaction.

Construction Quality Control

The Construction General Contractor shall perform in-place density and moisture content tests as specified in Section 02315, FILL AND BACKFILL, to be observed by the Construction Manager, on the following minimum schedule:

Backfill Above Pipe Zone: One per 500 linear feet per lift.

Maintenance of Trench Backfill

After each section of trench is backfilled, maintain the surface of the backfilled trench even with the adjacent ground surface until final surface restoration is completed.

Topsoil: Add topsoil where applicable and as necessary to maintain the surface of the backfilled trench level with the adjacent ground surface.

Other Areas: Add excavated material where applicable and keep the surface of the backfilled trench level with the adjacent ground surface.

Settlement of Backfill:

Settlement of trench backfill, or of fill or facilities constructed over trench backfill will be considered a result of defective compaction of trench backfill.

END OF SECTION 02320

SECTION 02371 – GEOTEXTILES SCHED. B

Part 1 – General

References

The publications listed below form a part of this Specification to the extent referenced. The publications are referred to in the text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D3776	Standard Test Method for Mass per Unit Area (Weight) of Fabric
ASTM D4355	Deterioration of Geotextiles from Exposure to Ultraviolet Light and Water (Xenon-Arc Type Apparatus)
ASTM D4491	Standard Test Methods for Water Permeability of Geotextiles by Permittivity
ASTM D4533	Standard Test Method for Trapezoid Tearing Strength of Geotextiles.
ASTM D4632	Standard Test Method for Grab Breaking Load and Elongation of Geotextiles
ASTM D4751	Standard Test Method for Determining Apparent Opening Size
ASTM D4833	Standard Test Method for Index Puncture Resistance of Geotextiles, Geomembranes, and Related Products
ASTM D5261	Standard Test Method for Measuring Mass per Unit Area of Geotextiles

Description

The Work includes manufacture, fabrication (if needed), supply, and installation of geotextiles associated with the lining of the disposal facility and other applications as shown on the Drawings. This section also applies to geotextiles used in geocomposite drainage layers [see Section 02373, COMPOSITE DRAINAGE NET (GEOCOMPOSITE)].

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Manufacturer's descriptive data, specification sheets, literature, and other data as necessary to fully demonstrate that those materials proposed for use comply with the requirements of these Specifications.

Manufacturer's Quality Control (QC) test data for geotextile physical properties, verifying compliance with these Specifications. Data shall include test results, methods, and roll numbers. Frequency of manufacturer's QC testing shall be at the standard rate stated in the manufacturer's QC plan.

Manufacturer's written certification that materials meet the requirements of these Specifications and that geotextile is continuously inspected for presence of needles and found to be needle-free.

Construction Quality Assurance (CQA)

Quality assurance procedures for geotextile are presented in the CQA Plan. CQA Plan requirements are discussed in Section 02661, GEOMEMBRANE. The Construction General Contractor shall accommodate all CQA activities described herein and in the CQA Plan for this project.

Prior to placing any materials over the installed geotextile, the Construction General Contractor shall allow time for acceptance of the Work as listed in the CQA Plan.

CQA Conformance Testing

Prior to deployment of the rolls of geotextile, the CQA Certifying Engineer will obtain samples at a frequency of one per production lot or 50,000 square feet of each material type, whichever results in the greater number of tests. The CQA Certifying Engineer will test the samples to determine conformance to both the design specifications and the list of certified properties.

As a minimum, the following tests will be performed on geotextiles (each type, except as noted):

Grab Strength: ASTM D4632.

Tear Strength: ASTM D4533.

Puncture Strength: ASTM D4833.

Permittivity: ASTM D4491 (Type 1 only).

The CQA Certifying Engineer shall be allowed to remove samples for testing and other activities. Sample dimensions, procedures, and frequency shall be the same as those specified in the CQA Plan. The Construction General Contractor shall assist the CQA Certifying Engineer as necessary in all sampling and testing activities.

Procedures for samples that fail conformance testing are outlined in the CQA Plan. The cost of additional conformance testing to demonstrate compliance of failed samples shall be borne by the Construction General Contractor.

Part 2 – Products

General

Types of Geotextiles:

Type 1 (separation) geotextile shall be 6 oz/yd^2 nominal weight and shall be used for separation of soil layers, in the geocomposite drainage layers, and at other locations as shown on the Drawings.

Type 2 (cushion) geotextile shall be 12 oz/yd2 nominal weight and shall be used for cushioning of geomembranes and at other locations as shown on the Drawings.

All geotextiles, regardless of type, shall be nonwoven, needle punched polypropylene.

Manufacturer: The geotextile manufacturer shall be a commercial entity normally engaged in manufacture of geotextiles for landfill applications.

Required Properties

Property Values:

Geotextile properties shall meet or exceed the values specified in Table 1, Required Geotextile Properties, contained in this section of the Specifications (Type 1 and Type 2 geotextiles).

The manufacturer shall provide test results for all properties listed in Table 1 (Type 1 and Type 2 geotextiles).

The manufacturer shall certify that the materials supplied meet the requirements of this Part (Type 1 and Type 2 geotextiles).

Integrity: Geotextiles shall retain their structure during handling, placement, and long-term service.

Transportation, Handling, and Storage

Geotextiles shall be supplied in rolls wrapped in covers and marked or tagged with the roll number. Each material roll shall include information to demonstrate material traceability through written documentation from the manufacturer and transport company. At a minimum, this information shall include the Manufacturer's Name, Product Identification, Lot Number, and Roll Dimension (Area and Width).

Transportation of the geotextiles to the site and all handling on site shall be the responsibility of the Construction General Contractor.

During shipment and storage, the geotextile shall be protected from mud, dirt, UV exposure, dust, puncture, cutting, or other damaging or deleterious conditions. Protective wrappings, which are damaged, shall be repaired or replaced, as necessary.

The Construction General Contractor shall be responsible for the storage of the geotextiles on site within the areas shown on the Drawings. The Construction General Contractor shall protect storage area(s) from theft, vandalism, passage of vehicles, etc.

Part 3 – Execution

General

Unacceptable Materials and Work: Materials and Work, which fail to meet the requirements of these Specifications, shall be removed and disposed of at the Construction General Contractor's expense. This includes geotextile rolls that are not labeled or where the label has deteriorated to the point of being illegible.

Handling And Placement

The Construction General Contractor shall handle all geotextiles in such a manner as to ensure that they are not damaged. Geotextile will be deployed on top of textured HDPE geomembrane in a manner that will not damage the geotextile. If necessary, use a smooth slip-sheet under the geotextile.

Place geotextiles in a manner that prevents folds and wrinkles. Folds or wrinkles shall be pulled smooth prior to seaming.

In the presence of wind, all exposed geotextiles shall be weighted with sandbags or equivalent. Sandbags shall be installed during placement and shall remain until replaced with cover material.

Geotextiles shall be cut using an approved geotextile cutter only. Special care shall be taken to protect underlying geosynthetic materials from damage during cutting.

During geotextile placement, care shall be taken not to entrap stones, excessive dust, or moisture that could damage the geomembrane, clog drains or filters, or hamper subsequent seaming.

Geotextiles shall be placed with the machine direction (long dimension) downslope or normal to the natural slope.

After installation and immediately prior to placing overlying materials, the geotextile shall be examined over its entire surface to ensure that no potentially harmful foreign objects, such as needles, are present. Any foreign objects encountered shall be removed, or the geotextile shall be replaced.

If light colored geotextile is used, precautions shall be taken against "snow blindness" of personnel.

After deployment, all geotextile shall be covered to prevent exposure to ultraviolet (UV) radiation (sunlight) within a maximum period of 14 days. If the geotextile is exposed for more than 14 days, a temporary cover may be deployed for the duration of the delay or samples may be submitted to an independent testing laboratory to ensure that detrimental levels of UV degradation have not occurred. Detrimental level of UV degradation is defined as greater than 10 percent loss of required geotextile properties listed in Table 1 for the following:

- Grab strength.
- Trapezoidal tear strength.
- Puncture strength.

Joints

Edge of roll seams are not required to be sewn and shall be overlapped a minimum of 6 inches. End of roll seams are not required to be sewn and shall be overlapped a minimum of 12 inches.

No end-of-roll seams shall be allowed on slopes 6H:1V and steeper. Overlaps shall be in the direction of flow with the upstream fabric on top of the downstream fabric.

On the landfill floor, no horizontal seam shall be closer than 3 feet to the toe of the slope or other areas of potential stress concentrations.

Areas to be seamed shall be clean and free of foreign material.

Repair

Any holes or tears in the geotextile shall be repaired as follows:

Remove any soil or other material, which may have penetrated the torn geotextile.

Replace torn areas and holes by placing a geotextile patch having dimensions of at least 12 inches greater than the tear or hole. The geotextile patch shall be sewn or heat bonded.

For repairs of the geotextile component of the CDN, a patch shall be heat bonded.

Materials In Contact With Geotextiles

The Construction General Contractor shall place all soil materials located on top of a geotextile in such a manner as to ensure that the following conditions are satisfied:

No damage to the geotextile.

Minimal slippage of the geotextile on underlying layers.

No excess tensile stresses in the geotextile.

END OF SECTION 02371

Table 1. Required Geotextile Properties

Value[a]

Property	Unit	Type 1	Type 2	Test Method
Mass/Unit Area	oz/yd^2	6.0[b]	12.0[b]	ASTM D5261 or D3776
Apparent Opening	U.S. Sieve	70 max opening	—	ASTM D4751
Size[b]		100 min opening		
Grab Strength	lb	140	300	ASTM D4632
Trapezoidal Tear				
Strength	lb	70		110
ASTM D4533				
Puncture Strength	lb	65	135	ASTM D4833
Permittivity	sec^{-1}	1.2	—	ASTM D4491
UV Resistance % strength	(500 hours)	retained 70	70	ASTM D4355

Notes:

[a]All values are minimum average values, except as noted.[b] Nominal values.

SECTION 02373 – COMPOSITE DRAINAGE NET (CDN) SCHED. B

Part 1 – General

References

The publications listed below form a part of this Specification to the extent referenced. The publications are referred to in the text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D413	Standard Test Method for Rubber Property Adhesion to Flexible Substrate
ASTM D1505	Standard Test Method for Density of Plastics by the Density-Gradient Technique
ASTM D1603	Standard Test Method for Carbon Black in Olefin Plastics
ASTM D1777	Standard Test Method for Measuring Thickness of Textile Materials
ASTM D4218	Test Method for Carbon Black Content in Polyethylene Compounds by the Muffle-Furnace Technique
ASTM D4716	Standard Test Method for Constant Head Hydraulic Transmissivity (In-Plane Flow) of Geotextiles and Geotextile Related Products
ASTM D5199	Standard Test Method for Measuring Nominal Thickness of Geotextiles and Geomembranes
ASTM D5321	Standard Test Method for Determining the Coefficient of Soil and Geosynthetic or Geosynthetic and Geosynthetic Friction by the Direct Shear Method

GEOSYNTHETIC RESEARCH INSTITUTE (GRI)

GRI-GC7	Determination of Adhesion and Bond Strength of Geocomposites description:

The work includes manufacture, fabrication (if needed), supply, and installation of geocomposite (hereinafter referred to as composite drainage net (CDN)) drainage layers associated with the lining of waste disposal facility. The CDN shall consist of a layer of geotextile thermally bonded to each side of a geonet. Requirements for geotextiles are contained in Section 02371, GEOTEXTILES, of these Specifications. Requirements for geonets and the finished CDN are contained in this section.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Manufacturer's descriptive data, specification sheets, literature, and other data as necessary to fully demonstrate that those materials proposed for use comply with the requirements of these Specifications.

Construction General Contractor shall submit required interface strength data, as specified in PART 2-PRODUCTS, prior to shipment of material to allow Engineer to evaluate if submitted material meets strength requirements for project design criteria. Allow Engineer 20 working days for this evaluation upon receipt of data.

Installation Plan: The Construction General Contractor shall submit a plan describing the proposed methods for CDN deployment, panel layout, seaming, repair, and protection. The plan shall include a quality assurance program for the Construction General Contractor's activities related to CDN installation.

Manufacturer's Quality Control (QC) test data for CDN composition and physical properties, verifying compliance with these Specifications. The data shall include roll numbers, test results, and test methods. Frequency of manufacturer's QC testing shall be at the standard rate stated in the manufacturer's QC plan.

Manufacturer's written certifications that CDN satisfy the requirements of these Specifications.

Construction Quality Assurance (CQA)

Quality assurance procedures for CDN installation are presented in the Construction Quality Assurance Plan (CQA Plan). CQA Plan requirements are discussed in Section 02661, GEOMEMBRANE. The Construction General Contractor shall accommodate all quality assurance activities described herein and in the CQA Plan for this project.

Prior to placing any materials over the installed CDN, the Construction General Contractor shall allow time for acceptance of the Work as listed in the CQA Plan.

CQA conformance testing: Upon delivery of the rolls of geonet and CDN, the CQA Certifying Engineer will obtain samples at a frequency of one per production lot or one per 50,000 square feet of each material type, whichever results in the greater number of tests; transmissivity shall be as noted below. The CQA Certifying Engineer will test the samples to determine conformance to both the design specifications and the list of certified properties.

CDN fabricated from non-conforming components shall be rejected at the Construction General Contractor's expense.

As a minimum, the following tests will be performed on geonets:

Polymer specific gravity (ASTM D1505).

Thickness (ASTM D5199).

Nominal transmissivity (ASTM D4716 – one per production lot).

As a minimum, the following tests will be performed on CDNs:

Adhesion (GRI-GC7 or ASTM D413).

Transmissivity (ASTM D4716 – one per production lot).

The CQA Certifying Engineer shall be allowed to remove samples of geonet and CDN for testing and other activities. Sample dimensions, procedures, and frequency shall be the same as those specified in the CQA Plan. The Construction General Contractor shall assist the CQA Certifying Engineer as necessary in all sampling and testing activities.

Procedures for samples that fail conformance testing are outlined in the CQA Plan. The cost of additional conformance testing to demonstrate compliance of failed samples shall be borne by the Construction General Contractor.

Part 2 – Products

General

Composition: The geonet shall be high-density polyethylene (HDPE), manufactured by extruding two crossing strands to form a bi-planar drainage net structure.

The CDN shall consist of Type 1 geotextile thermally bonded to each side of the HDPE geonet.

Manufacturer: The CDN manufacturer shall have a minimum of 5 years' experience as a commercial manufacturer of CDNs for landfill drainage applications.

Required Properties

Property Values

Geonet: Geonet properties shall meet or exceed the values specified in the table of required geonet properties contained in this section of the Specifications.

Geotextile: Geotextile properties shall meet or exceed the values specified in Section 02371, GEOTEXTILES, of these Specifications unless otherwise approved by the Engineer.

Finished CDN: CDN properties shall meet or exceed the values specified in the table of required CDN properties contained in this section of the Specifications.

Required Interface Shear Strength Data:

Provide data prior to material shipment for the interface friction angle between the CDN and textured geomembrane, and between the CDN and operations layer material, as specified in Article SUBMITTALS. Perform two interface shear strength tests on each interface. Friction angle shall be determined by direct shear testing under fully saturated conditions (ASTM D5321) at nominal normal loads of 100, 250, and 500 psf. Report results for both peak and large displacement (minimum 2 inches) strength.

The Engineer will review this data for conformance with project design strength requirements. Construction General Contractor shall not order material for shipment until approved by Engineer. Any product or material changes required as a result of inadequate strength data will be addressed by Change Order provided submitted material meet all other requirements of this section.

Manufacturer's Information: The manufacturer shall provide specification sheets, literature, and test results for all properties listed in these Specifications. The manufacturer shall certify that the materials supplied meet the requirements of this Part.

Integrity: Geonets and CDNs shall retain their structure during handling, placement, and long-term service.

Transportation, Handling, and Storage

Geonets and CDNs shall be supplied in rolls wrapped in covers and marked or tagged with the roll number. Each material roll shall include information to demonstrate material traceability through written documentation from the manufacturer and transport company. At a minimum, this information shall include the Manufacturer's Name, Product Identification, Lot Number, and Roll Dimension (Area and Width).

Transportation of the CDN to the site and all handling on site will be the responsibility of the Construction General Contractor.

During shipment and storage, the geonet and CDN shall be protected from mud, dirt, UV exposure, dust, puncture, cutting, or other damaging or deleterious conditions. Protective wrappings, which are damaged, shall be repaired or replaced, as necessary.

The Construction General Contractor shall be responsible for the storage of the CDN on site within the limits of construction. The Construction General Contractor shall protect storage area(s) from theft, vandalism, passage of vehicles, etc.

Part 3 – Execution

General

Unacceptable Materials and Work: Materials and Work, which fail to meet the requirements of these Specifications, shall be removed, disposed of, and replaced at the Construction General Contractor's expense.

Handling And Placement

The Construction General Contractor shall handle all CDNs in such a manner as to ensure that these materials are not damaged.

Clean geomembrane surface prior to placing CDN.

On slopes, CDN may be deployed over slip-sheets with the roll at the top of the slope. An alternative method is to secure the CDN and then roll it down slope in a manner to continually keep it in tension. If necessary, position the CDN after deployment to minimize wrinkles and remove the slip-sheet, if used.

Do not drag the CDN across textured geomembrane in any way that damages the geotextile component or reduces the friction at the geomembrane/CDN interface.

In the presence of wind, all exposed CDNs shall be weighted with sandbags or equivalent. Sandbags shall be installed during CDN placement and shall remain until replaced with cover material.

Unless otherwise specified, CDNs shall not be welded to geomembranes.

CDNs shall only be cut using approved cutting tools. Protect underlying geosynthetics when cutting.

The Construction General Contractor shall take any necessary precautions to prevent damage to underlying layers during placement of the CDN.

During placement of CDNs, care shall be taken not to entrap dirt or excessive dust that could cause clogging of the drainage system, and/or stones that could damage the adjacent geomembrane. Dirt or excessive dust entrapped in the CDN shall be cleaned prior to placement of the next material on top of it. Excessive dust is defined as any thickness greater than 20 mils (0.02 inch) within the geonet core of the CDN. In this regard, care shall be taken with the handling of sandbags, to prevent rupture or damage of the sandbag.

Tools shall not be left in the CDN.

After deployment, all CDN shall be covered within a maximum period of 14 days to prevent exposure of geotextile component to ultraviolet (UV) radiation (sunlight). See Section 02371, GEOTEXTILES, for requirements of geotextile exposed to UV radiation longer than 14 days.

Joining

Adjacent sections of CDN shall be overlapped according to the manufacturer's directions.

Overlaps shall be secured by spot welding or tying. Acceptable tying devices include strings, plastic fasteners, or polymer braid. Tying devices shall be white or yellow for easy observation. Metallic devices are not allowed.

Overlaps shall be secured every 5 feet along slopes and on the floor of the landfill. Overlaps shall be secured every 6 inches in anchor trenches. Along end-to-end seams, spot-weld, or tie each row at 6-inch intervals; stagger weld or ties between rows.

In joining CDNs, tearing the geotextile away from the geonet shall only be allowed at panels ends in order to seam same material components together and shall be minimized to the extent necessary to perform the required work.

No horizontal seams shall be allowed on side slopes except at roll ends.

If more than one layer of CDN is installed, joints shall be staggered.

Repair

Remove the damaged area of CDN.

Cut a piece of geonet to fit into the repair area. Geonet shall fit into repair area to form a flush surface with the CDN. Cut geonet so that ribs are in the same orientation as existing CDN.

Remove any dirt or other foreign material, which may have entered the CDN.

Place CDN patch over damaged area. Geonet component of patch shall be tied to in-place geonet component according to manufacturer's recommendations.

Place Type 1 geotextile over the exposed geonet component with an overlap of 4 inches of geotextile. Heat seam repair geotextile to existing geotextile.

Materials in Contact With CDNs

The Construction General Contractor shall place all soil materials located on top of a CDN layer in such a manner as to ensure that the following conditions are satisfied:

- No damage to the CDN.
- No slippage of the CDN on underlying layers.
- No excess tensile stresses in the CDN.

END OF SECTION 02373

Table 1. Required Geonet Properties

Property	Qualifier	Unit	Value	Test
Polymer Composition	Minimum	% polyethylene	95	---
Resin Specific Gravity	Minimum	N/A	0.92	ASTM D1505
Carbon Black Content	Range	%	2 - 3	ASTM D1603 or D4218
Nominal Thickness	MARV	mils	250	ASTM D1777 or D5199
Nominal Transmissivity(1)	MARV	m^2/sec	3×10^{-3}	ASTM D4716

Table 2. Required CDN Properties

Property	Qualifier	Unit	Value	Test
Ply Adhesion	ARV	lb/in	1.0	ASTM D413 or GRI-GC7
Transmissivity(1)	MARV	m^2/sec	5×10^{-4}	ASTM D4716

Notes:

MARV = Minimum Average Roll Value.

ARV = Average Roll Value.

(1)The design transmissivity is the hydraulic transmissivity of the CDN measured using water at 70 degrees F ±3 degrees F with a hydraulic gradient of 0.1, under the compressive stress of 10,000 psf. Transmissivity value shall be measured between two steel plates 15 minutes after application of the confining stress in the machine direction.

SECTION 02500 – RAW WATER CONVEYANCE PIPING – GENERAL SCHED. A & B

Part 1 – General

Summary

This section is for furnishing and installing raw water piping and associated components.

References

The following is a list of standards, which may be referenced in this section:

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 301	Standard Specification for Structural Concrete

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C110/A21.10	Ductile-Iron and Gray-Iron Fittings, 3 in. Through 48 in. (75 mm Through 1200 mm), for Water and Other Liquids
AWWA C115/A21.15	Flanged Ductile-Iron Pipe with Ductile-Iron or Gray-Iron Threaded Flanges
AWWA C207	Steel Pipe Flanges for Waterworks Service - Sizes 4 in. Through 144 in. (100 mm Through 3,600 mm)
AWWA C210	Liquid-Epoxy Coating Systems for the Interior and Exterior of Steel Water Pipelines
AWWA C213	Fusion-Bonded Epoxy Coating for the Interior and Exterior of Steel Water Pipelines
AWWA C217	Cold-Applied Petroleum Tape and Petroleum Wax Tape Coatings for the Exterior of Special Sections, Connections, and Fittings for Buried Steel Water Pipelines
AWWA C219	Bolted, Sleeve-Type Couplings for Plain-End Pipe
AWWA C221	Fabricated Steel Mechanical Slip-Type Expansion Joints
AWWA C606	Grooved and Shouldered Joints

ASTM INTERNATIONAL (ASTM)

ASTM A497	Standard Specification for Steel Welded Wire Fabric, Deformed, for Concrete Reinforcement
ASTM A615/A615M	Standard Specification for Deformed and Plain Billet-Steel Bars for Concrete Reinforcement
ASTM C94/C94M	Standard Specification for Ready-Mixed Concrete
ASTM C150	Standard Specification for Portland Cement

NSF INTERNATIONAL (NSF)

NSF 61B	Drinking Water System Components - Health Effects

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Detailed pipe fabrication drawings showing pipe details, special fittings and bends, dimensions, coatings, and other pertinent information.

Documentation for pipe pressure class.

Submittals–Approval Not Required

Information/Record (IR)

Submit results of pressure leakage testing for raw water conveyance piping.

Submit results for materials and test certificate after completion of the system in accordance with NFPA 24 Chapter 9.

Delivery, Storage, And Handling

In accordance with manufacturer's recommendations.

Marking at Plant:

Mark each pipe and fitting at plant. Include date of manufacture, manufacturer's identification, specification standard, diameter of pipe, dimension ratio, pipe class, and other information required for type of pipe.

Pipe, specials, and fittings received at Project site in damaged condition will not be accepted.

Gasket Storage:

Store rubber gaskets in cool, well-ventilated place, and do not expose to direct rays of sun. Do not allow contact with oils, fuels, petroleum, or solvents.

Store and support pipe securely to prevent accidental rolling and to avoid contact with mud, water, or other deleterious materials.

Handling:

Pipe shall be handled with proper equipment in a manner to prevent distortion or damage. Use of hooks, chains, wire ropes, or clamps that could damage pipe, damage coating or lining, or kink and bend pipe ends is not permitted.

Use heavy canvas, or nylon slings of suitable strength for lifting and supporting materials.

Lifting pipe during unloading or lifting into trench shall be done using two slings placed at quarter point of pipe section. Pipe may be lifted using one sling near center of pipe, provided pipe is guided to prevent uncontrolled swinging and no damage will result to pipe or harm to workmen. Slings shall bear uniformly against pipe.

Pipe and fittings shall not be stored on rocks or gravel, or other hard material that might damage pipe. This includes storage area and along pipe trench.

Part 2 – Products

Pipe

As specified in Section 02502, RAW WATER DUCTILE IRON PIPE AND FITTINGS, Section 02509, RAW WATER POLYVINYL CHLORIDE (PVC) PRESSURE PIPE AND FITTINGS, and the Carbon Steel Pipe and Galvanized Steel Pipe Data Sheets attached to this section.

Joints

As specified in Section 02502, RAW WATER DUCTILE IRON PIPE AND FITTINGS, Section 02509, RAW WATER POLYVINYL CHLORIDE (PVC) PRESSURE PIPE AND FITTINGS, and the Carbon Steel Pipe and Galvanized Steel Pipe Data Sheets attached to this section.

Flexible Lock Couplings

General:

Couplings shall be rated for 173 PSI.

Buried, bolted, sleeve-type couplings shall be lined and coated with fusion-bonded epoxy in accordance with AWWA C213.

Locking pins shall be integral to coupling, number and diameter of pins per manufacturer's standard.

For Pipe with Plain-Ends:

Bolted, sleeve-type coupling, in accordance with AWWA C219.

Manufacturer of couplings shall observe same quality control requirements as specified in AWWA C221 for fabrication of pipe expansion joints.

Manufacturers and Products: Dresser; Style 167.

Service Saddles

Double strap type with minimum strap width of 2 inches.

Straps shall be Type 304 stainless steel. Saddles shall be ductile iron, epoxy-coated, 10 mils minimum thickness.

Minimum Pressure Rating: 200 psi.

Flanges, Flange Gaskets, and Bolting Materials

As specified in individual raw water specifications for each piping material.

Concrete For Thrust Blocks

Thrust Block Concrete: As specified in Section 03301, CONCRETE.

Reinforcing Steel: ASTM A615/A615M, Grade 60 deformed bars.

Welded Wire Fabric: ASTM A497.

Formwork: Plywood; earth cuts may be used as approved by Construction Manager.

Locator Ribbon

As specified in Section 02320, TRENCH BACKFILL.

Pipe Bedding And Pipe Zone Material

Granular material as specified in Section 02320, TRENCH BACKFILL.

Trench Stabilization Material:

As specified in Section 02320, TRENCH BACKFILL.

Part 3 – Execution

General

Installation shall be in accordance with NFPA 24 Standard for the Installation of Private Service Mains and Their Appurtenances.

Notify Construction Manager at least 2 weeks prior to field fabrication of pipe or fittings.

Furnish feeler gauges of proper size, type, and shape for use during installation for each type of pipe furnished.

Distributing Materials: Place materials along trench only as will be used each day, unless otherwise approved by Construction Manager. Placement of materials shall not be hazardous to traffic or to general public, obstruct access to adjacent property, or obstruct others working in area.

Examination

Verify size, material, joint types, elevation, and horizontal location of existing pipeline to be connected to new pipeline or new equipment.

Inspect size and location of structure penetrations to verify adequacy of wall pipes, sleeves, and other openings.

Damaged Coatings and Linings: Repair using coating and lining materials in accordance with manufacturer's instructions.

Preparation Of Trench

Prepare trench as specified in Section 02316, EXCAVATION.

Unless otherwise permitted by Construction Manager, maximum length of open trench shall not exceed 500 feet.

Installation

General

Join pipe and fittings in accordance with manufacturer's instructions, unless otherwise shown or specified.

Install individual pipe lengths in according to approved lay diagram. Misplaced pipe shall be removed and replaced.

Inspect pipe and fittings before installation, clean ends thoroughly, remove foreign matter and dirt from inside.

Flanged Joints: Install perpendicular to pipe centerline.

Bolt Holes:

Straddle vertical centerline, aligned with connecting equipment flanges or as shown on Drawings.

Use torque-limiting wrenches to provide uniform bearing and proper bolt tightness.

Flange Type: Use flat-faced flange when joining with flat-faced ductile or cast iron flange.

Couplings:

Install in accordance with manufacturer's written instructions.

Before coupling, clean pipe holdback area of oil, scale, rust, and dirt.

Remove pipe coating, if necessary, to obtain smooth surface.

Clean gaskets before installation.

If necessary, lubricate with gasket lubricant for installation on pipe ends.

Tighten coupling bolts progressively; drawing up bolts on opposite sides gradually until bolts have uniform tightness.

Buried Pressure Pipe

Placement:

Keep trench dry until pipe laying and joining is completed.

Exercise care when lowering pipe into trench to prevent twisting or damage to pipe.

Excavate trench bottom and sides of ample dimensions to permit proper joining, visual inspection, and testing of entire joint.

Prevent foreign material from entering pipe during placement.

Close and block open end of last laid pipe section when placement operations are not in progress and at close of day's work.

In general, lay pipe upgrade with bell ends pointing in direction of laying.

Deflect pipe at joints for pipelines laid on a curve using unsymmetrical closure of spigot into bell. Pipe joints shall be pushed together in straight alignment and then deflected. If joint deflection of standard pipe lengths will not accommodate horizontal or vertical curves in alignment, provide:

Shorter pipe lengths.
Special mitered joints.
Standard or special fabricated bends.
Check gasket position with feeler gauge to assure proper seating.
After joint has been made, check pipe alignment and grade.
Place sufficient pipe zone material to secure pipe from movement before next joint is installed.
Prevent uplift and floating of pipe prior to backfilling.

Tolerances:

Deflection From Horizontal Line: Maximum 2 inches.

Deflection From Vertical Line: Maximum 1 inch.

Joint Deflection:

Maximum of 75 percent of manufacturer's recommendation.
Horizontal position of pipe centerline on alignment around curves maximum variation of 1 foot from position shown.

Cover Over Top of Pipe: Minimum 3.5 feet, unless otherwise shown.

Disposal of Excess Excavated Material: As specified in Section 02316, EXCAVATION.

Thrust Restraint

Location: At pipeline tees, plugs, caps, bends, and locations where unbalanced forces exist.

Thrust Blocking

Quantity of Concrete:

Sufficient to cover bearing area of pipe and provide required soil-bearing area as shown on Drawings. For vertical bends, concrete shall be sufficient to provide required volume as shown on Drawings.
Place blocking so pipe and fitting joints are accessible for repairs.
Place concrete in accordance with Section 03301, CONCRETE.

Corrosion Protection

Buried Pipe:

As specified in the individual specifications following this section.
Notify Construction Manager at least 3 days prior to start of surface preparation, coating application, and corrosion protection work.

Placement Of Pipe Locator Ribbon

Place pipe locator ribbon in accordance with Section 02320, TRENCH BACKFILL.

Pipe Bedding And Zone Material

Place pipe bedding and pipe zone material in accordance with Section 02320, TRENCH BACKFILL.

Construction Quality Control

Pressure Leakage Testing for Raw Water Ductile Iron and PVC: As specified in the individual piping Specification(s) following this section.

Pressure Leakage Testing for Raw Water Carbon Steel Pipe and Fittings and Valves Upstream of 12-Inch Gate Valve:

Shop Test: After fabrication of carbon steel spools (as specified in carbon steel pipe supplement at the end of this section) and prior to lining or coating, blind flanges shall be installed on each flange. One blind flange shall be tapped as necessary for a test port. A hydrostatic leak test shall be performed at 173 psig for 30 minutes; no visible leaks shall be allowed.

Field Test: An in-service leak test shall be performed with excavation open. Retighten bolts on fittings/valves as necessary and within manufacturer's recommendations to stop any visible leaks.

Pressure Leakage Testing for Raw Water Galvanized Steel Pipe and Malleable Iron Fittings: Perform in conjunction with leakage test for raw water ductile iron and PVC.

Supplements

Supplement 1—Carbon Steel Pipe and Fittings.

Supplement 2—Galvanized Steel Pipe and Malleable Iron Fittings.

END OF SECTION 02500

Table 1. Carbon Steel Pipe and Fittings

Item	Size	Description
Pipe	24"	Black carbon steel, ASTM A106, Grade B seamless or ASTM A53, Grade B seamless or ERW. Threaded, butt-welded, and flanged joints: Schedule 20 (24" OD, 0.375" wall thickness) For pressure testing purposes of 24" OD flange x plain end spools, initially construct one flange x flange spool. After completion of pressure testing per Article CONSTRUCTION QUALITY CONTROL of this Specification section, the flange x flange spool shall be cut in half. Each half shall then be coated, lined, and installed per the Drawings. Other flange x flange spools shall be pressure tested prior to coating and lining per Article CONSTRUCTION QUALITY CONTROL.
Joints		Flanged at valves as shown
Flanges		Forged carbon steel, ASTM A105/A105M, ANSI B16.5 Class 150 slip-on or welding neck; weld neck bore to match pipe internal diameter. Welding shall conform to AWS D1.1, AWWA C206, approved welding procedures, and referenced welding codes. In case of conflict, AWS D1.1 shall govern.
Bolting	All	Carbon steel ASTM A193/A193M, Grade B7 studs and ASTM A194/A194M, Grade 2H hex head nuts. When mating flange on equipment is cast iron and gasket is flat ring, provide ASTM A307, Grade B hex head bolts and ASTM A563, Grade A heavy hex nuts.
Gaskets	All	General Service and Oil/Gas: 1/16" thick compressed nonasbestos composition flat ring type. Garlock, Style 3000; Manville, Style 978.
Coating		General: Holdback of, and coating shall be as follows: For flex couplings, 8 inches. Epoxy Coating: Coating system for pipe and flanges (except machined surfaces) shall conform to AWWA C210. Paint system for coating shall be Tnemec Series 141 (Color WH03) or approved equal. Completed coating shall have a total dry film thickness of 16 mils minimum. Application of epoxy coatings shall be in accordance with federal, state, and local regulations. Coatings shall be shop-applied, except for field repairs and holdback areas. Materials shall be suitable for temperatures that may be encountered in Project location, and for time of year when materials are being applied and pipeline is being installed. Coating shall be tested in accordance with AWWA C210. Defects shall be repaired in accordance with AWWA C210. Furnish and apply in field, coat tar epoxy paint, Koppers 300M, or equal, on holdback areas after installation of flex couplings and at damaged coating locations.
Lining		Epoxy Lining: Conform to AWWA C210. Lining shall be an epoxy system suitable for potable water service (listed by National Sanitation Foundation Standard 61). Paint system for lining shall be Tnemec Series 141 (Color WH03) or approved equal. Completed lining shall have a total dry film thickness of 10 mils minimum.

Table 2. Galvanized Steel Pipe and Malleable Iron Fittings

Item	Size	Description
Pipe		Galvanized carbon steel, ASTM A106, Grade B seamless or ASTM A53, Grade B seamless or ERW.
	2" and smaller	Schedule 80.
	2-1/2" through 6"	Schedule 40
Joints	All	Threaded or flanged at valves and equipment.
Fittings		Threaded: 150- or 300-pound galvanized malleable iron, ASTM A197 or ASTM A47, dimensions in accordance with ANSI B16.3.
Flanges		Galvanized forged carbon steel, ASTM A105/A105M, ANSI B16.5 Class 150 or Class 300, threaded, 1/16-inch raised face.
Unions		Threaded malleable iron, ASTM A197 or A47, 300-pound WOG, brass to iron seat, meeting the requirements of ANSI B16.3.
Bolting		Flanges: Carbon steel ASTM A307, Grade A hex head bolts, and ASTM A563, Grade A hex head nuts.
Gaskets	All flanges	Flanged, Water and Sewage Service: 1/8 inch thick, red rubber (SBR), hardness 80 (Shore A), rated to 200 degrees F, conforming to ANSI B16.21, AWWA C207, and ASTM D1330, Grades 1 and 2.
Thread	2" & smaller	Teflon tape or joint compound that is insoluble in
Lubricant		Water

SECTION 02502 – RAW WATER DUCTILE IRON PIPE AND FITTINGS SCHED. A

Part 1 – General

Summary

This section is for furnishing and installing 6-inch ductile iron raw water piping and associated components.

References

The following is a list of standards that may be referenced in this section:

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO T99	Standard Specification for the Moisture-Density Relations of Soils Using a 2.5 kg (5.5LB) Hammer and a 305mm (12 in.) Drop

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C104	Cement-Mortar Lining for Ductile Iron Pipe and Fittings for Water
AWWA C105	Polyethylene Encasement for Ductile Iron Pipe Systems
AWWA C110	Ductile Iron and Grey Iron-Fittings, 3-inch through 48-inch
AWWA C111	Rubber-Gasket Joints for Ductile Iron Pressure Pipe and Fittings.
AWWA C115	Flanged Ductile Iron Pipe with Ductile Iron and Grey Iron Fittings
AWWA C150	Thickness Design of Ductile-Iron Pipe
AWWA C151	Ductile-Iron Pipe. Centrifugally Cast, for Water
AWWA C153	Ductile Iron Compact Fittings, 3-inch through 24-inch and 54-inch through 64-inch for Water Service
AWWA C207	Steel Pipe Flanges for Waterworks Service, Sizes 4-inch Through 144-inch (100mm through 3600mm)
AWWA C600	Installation of Ductile-Iron Water Mains and Their Appurtenances
AWWA C606	Grooved End, Shouldered Joints

ASTM INTERNATIONAL (ASTM)

ASTM A307	Standard Specification for Carbon Steel Bolts and Studs, 60,000 psi Tensile Strength
ASTM A563	Standard Specification for Carbon and Alloy Steel Nuts
ASTM B16.21	Standard Specification for Nonmetallic Flat Gaskets for Pipe Flanges
ASTM D882	Standard Test Method for Tensile Properties of Thin Plastic Sheeting
ASTM D1330	Standard Specification for Rubber Sheet Gaskets
ASTM D1922	Standard Test Method for Propagation Tear Resistance of Plastic Film and Thin Sheeting by Pendulum Method
ASTM D2000	Standard Classification System for Rubber Products in Automotive Applications
ASTM D4976	Standard Specification for Polyethylene Plastics Molding and Extrusion Materials

DUCTILE IRON PIPE RESEARCH INSTITUTE (DIPRA)

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Flushing Procedure: The flushing procedure shall outline the method and materials for handling flush water, i.e., pipe, valves, fittings for filling pipeline and similar for discharging flush water, method for restraining aboveground pipe, and preventing erosion damage.

Field Hydrostatic Testing Plan: Submit at least 15 days prior to testing and at minimum, include the following:

- Testing dates.
- Piping systems and section(s) to be tested.
- Method of isolation.
- Method of conveying water from source to system being tested.
- Calculation of maximum allowable leakage for piping section(s) to be tested.

Certifications of Calibration: Approved testing laboratory certificate if pressure gauge for hydrostatic test has been previously used. If pressure gauge is new, no certificate is required.

As-Built Survey Data: Before final acceptance of raw water piping system, provide as-built locations of systems and components, showing pressure pipelines, including grade breaks or alignment, horizontal and vertical locations of all utility crossings, finished grade profile on all alignments, and valve locations. The submitted data shall be certified by a Washington licensed land surveyor.

Shop Drawings: Marking plan and details of standard pipe section showing dimensions, pipe joints, fitting and special fitting pressure rating and thickness, size, coating and lining data.

Submittals–Approval Not Required

Information/Record (IR): Hydrostatic test documentation form and results.

Part 2 – Products

Materials

Pipe:

General:

- Centrifugally cast, grade 60-42-10 iron.
- Meet requirements of AWWA C150, C153, and C111.
- Lined and coated as specified.
- Pipe wall thickness Class 50.
- Pipe wall thickness of threaded pipe for a flanged pipe end shall be minimum special thickness Class 53 from 12-inch to 54-inch diameter pipe in accordance with AWWA C115.
- Grooved end pipe, for all pipe diameters, shall be minimum Special Class 53.
- Pipe shall be new and recently manufactured. Refurbished pipe shall not be provided.

Joints:

Push-On Joint: Rated at minimum working pressure equal to pipe material design.

Restrained Joint: Manufactured proprietary joint that mechanically restrains pipe to adjoining pipe.

Manufacturers and Products:

- American Cast Iron Pipe; Flex-Ring and Lok-Ring.
- Pacific States Pipe; Thrust-Lock.

U.S. Pipe; TR Flex.

Mechanical Wedge Action Type Joint:

Use only in areas where adjoining to fixed points where laying length is determined in field.

Prior to purchase and installation, type and application of this joint shall be approved by Construction Manager.

Use of setscrews for restraint or field-lock gaskets shall not be allowed.

Flanged Joint: Threaded 250 psi working pressure ductile iron flanges conforming to AWWA C115.

Fittings:

Ductile Iron, Push-On, Flanged, or Restrained Joint: In accordance with AWWA C110, at 250 psi minimum working pressure for 3- to 24-inch fittings and 150-psi minimum working pressure for 30- to 48-inch fittings.

Mechanical Joint Fittings: In accordance with AWWA C111.

Fittings shall be new and recently manufactured. Refurbished fittings will not be accepted.

Welded Outlet: Only weld to pipe in manufacturer's shop.

Lining: Pipe and fittings for clean water applications shall be cement lined and asphaltic seal coated in accordance with AWWA C104.

Coating: Asphaltic type, 1 mil thick, in accordance with AWWA C151, C115, C110, and C153.

Bolting: Bolts for flanged connections shall be carbon steel, ASTM A307, Grade A hex bolts and ASTM A563, Grade A hex head nuts.

Gaskets: Gaskets for flat faced 150 and 250 psi working pressure flanges shall be 1/8 inch thick, red rubber (SBR), hardness 80 (Shore A), rated to 200 degrees F, conforming to ANSI B16.21, AWWA C207, and ASTM D1330, Grades 1 and 2.

Part 3 – Execution

Examination

Inspect pipe and fittings to ensure no cracked, broken, or otherwise defective materials are being used.

Preparation

Trench Grade:

Grade bottom of trench by hand to specified line and grade, with proper allowance for pipe thickness and pipe base, when specified. Trench bottom shall form a continuous and uniform bearing and support for pipe between bell holes.

Before laying each section of pipe, check grade and correct irregularities found. Grade may be disturbed for removal of lifting tackle.

Bell (Joint) Holes: At each joint, dig bell holes of ample dimensions in bottom of trench, and at sides where necessary, to permit joint to be made properly and to permit easy visual inspection of entire joint.

Installation

General:

Provide and use proper implements, tools, and facilities for safe and proper prosecution of work.

Lower pipe, fittings, and appurtenances into trench, piece by piece, by means of a crane, slings, or other suitable tools and equipment, in such a manner as to prevent damage to pipe materials, protective coatings, and linings.

Do not drop or dump pipe materials into trench.

Cleaning Pipe and Fittings:

Remove lumps, blisters, and excess coal tar coating from bell and spigot ends of each pipe. Wire brush outside of spigot and inside of bell and wipe clean, dry, and free from oil and grease before pipe is laid.

Wipe ends of mechanical joint pipe and fittings and of rubber gasket joint pipe and fittings clean of dirt, grease, and foreign matter.

Laying Pipe:

Direction of Laying: Lay pipe with bell end facing in direction of laying. For lines on an appreciable slope, face bells upgrade at discretion of Construction Manager.

Mechanical Joint, Push-On Joint, and Restrained Joint Pipe:

After first length of pipe is installed in trench, secure pipe in place with approved backfill material tamped under and along sides to prevent movement. Keep ends clear of backfill. After each section is jointed, place backfill as specified to prevent movement.

Take precautions necessary to prevent floating of pipe prior to completion of backfill operation.

When using movable trench shield, take necessary precautions to prevent pipe joints from pulling apart when moving shield ahead.

Do not allow foreign material to enter pipe while it is being placed in trench.

Close and block open end of last laid section of pipe to prevent entry of foreign material or creep of gasketed joints when laying operations are not in progress, at close of day's work, or whenever workers are absent from job.

Joining Push-On Joint Pipe and Mechanical Joint Fittings:

Join pipe with push-on joints and mechanical joint fittings in strict accordance with manufacturer's recommendations.

Provide special tools and devices, such as, special jacks, chokers, and similar items required for installation.

Lubricate pipe gaskets using lubricant furnished by pipe manufacturer. No substitutes will be permitted.

Clean ends of fittings of dirt, mud, and foreign matter by washing with water and scrubbing with a wire brush, after which, slip gland and gasket on plain end of pipe. If necessary, lubricate end of pipe to facilitate sliding gasket in place, then guide fitting onto spigot of pipe previously laid.

Cutting Pipe:

General: Cut pipe for inserting valves, fittings, or closure pieces in a neat and workmanlike manner without damaging pipe or lining and to leave a smooth end, at right angles to axis of pipe.

Pipe: Cut pipe with milling type cutter or saw. Do not flame cut.

Dressing Cut Ends: Dress cut end of mechanical joint pipe to remove sharp edges or projections, which may damage rubber gasket. Dress cut ends of push-on joint pipe by beveling, as recommended by manufacturer.

Field Welding:

Use of field-welded outlets will not be allowed. Welding for outlets shall be performed only in pipe manufacturer's shop.

Field installed outlets may be installed with saddle approved by Construction Manager. Opening in pipe shall be machined cut and not with cutting torch.

Field welding of bars for restrained joint systems will not be allowed. All welding shall be performed in pipe manufacturer's shop.

Line and Grade:

Minimum Pipe Cover: 3.5 feet, unless otherwise indicated.

Install pipe to uniform grades and minimize high points.

Maintain pipe grade between invert elevations.

Deviations exceeding 6 inches from specified line or 1 inch from specified grade will not be allowed without express approval of Construction Manager.

Pipeline sections that are not installed to elevations shown or installed as approved by Construction Manager shall be reinstalled to proper elevation.

Construction Quality Control

Cleaning:

Following assembly and testing, and prior to final acceptance, flush pipelines with water at 2.5 fps minimum flushing velocity until foreign matter is removed.

If impractical to flush large diameter pipe at 2.5 fps, clean pipe by use of pipe pig as approved by Construction Manager. Multiple passes of pipe pig may be required to adequately clean line.

Remove accumulated debris through blowoffs 2 inches and larger or by removing spools and valves from piping.

Pipeline Hydrostatic Test:

General:

Notify Construction Manager in writing 5 days in advance of testing. Construction Manager will coordinate with other parties required to witness testing. Perform testing in presence of Construction Manager.

Test newly installed pipelines. Using water as test medium, pipes shall successfully pass a leakage test prior to acceptance.

Furnish testing equipment and perform tests in manner satisfactory to Construction Manager. Testing equipment shall provide observable and accurate measurements of leakage under specified conditions.

Isolate new pipelines that are connected to existing pipelines.

Conduct tests on entire pipeline after trench has been backfilled. Testing may be done prior to placement of asphaltic concrete or roadway structural section.

Construction General Contractor may, if field conditions permit and as determined by Construction Manager, partially backfill trench, and leave joints open for inspection and conduct an initial service leak test. Hydrostatic test shall not, however, be conducted until backfilling has been completed.

Procedure:

Maximum filling velocity shall not exceed 0.25 foot per second, calculated based on the full area of pipe.

Expel air from pipe system during filling. Expel air through air release valve or through corporation stop installed at high points and other strategic points.

Test pressure shall be 173 psi as measured at low point of pipeline.

Apply and maintain specified test pressure with hydraulic force pump. Valve off piping system when test pressure is reached.

Maintain hydrostatic test pressure continuously for 2 hours minimum, adding additional make-up water only as necessary to restore test pressure.

Determine actual leakage by measuring quantity of water necessary to maintain specified test pressure for duration of test.

If measured leakage exceeds allowable leakage or if leaks are visible, repair defective pipe section and repeat hydrostatic test.

Identify method of disposing of water after system testing.

Allowable Leakage: Maximum allowable leakage shall not exceed amount stated in AWWA C600.

END OF SECTION 02502

SECTION 02509 – RAW WATER POLYVINYL CHLORIDE (PVC) PRESSURE PIPE AND FITTINGS SCHED. A & B

Part 1 – General

Summary

This section is for furnishing and installing 1-inch and 4-inch Schedule 80 solvent weld, 4-inch and 12-inch bell and spigot, raw water piping and associated components.

References

The following is a list of standards, which may be referenced in this section:

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C110	Ductile-Iron and Gray-Iron Fittings, 3 inch Through 48 inch (75 mm Through 1200 mm), for Water and Other Liquids
AWWA C605	Underground Installation of Polyvinyl Chloride (PVC) Pressure Pipe and Fittings for Water
AWWA C900	Polyvinyl Chloride (PVC) Pressure Pipe and Fabricated Fittings, 4 inch Through 12 inch (100 mm Through 300 mm), for Water Distribution
AWWA C905	Polyvinyl Chloride (PVC) Pressure Pipe and Fabricated Fittings, 14 inches Through 48 inches (350 mm Through 1,200 mm) for Water Transmission and Distribution
AWWA C907	Polyvinyl Chloride (PVC) Pressure Fittings for Water, 4 inch Through 8 inch (100 mm Through 200 mm)
AWWA Manual M23	PVC Pipe - Design and Installation

ASTM INTERNATIONAL (ASTM)

ASTM D1784	Standard Specification for Rigid Poly (Vinyl Chloride) (PVC) Compounds and Chlorinated Poly (Vinyl Chloride) (CPVC) Compounds
ASTM D1785	Standard Specification for Poly (Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120
ASTM D2241	Standard Specification for Rigid Poly (Vinyl Chloride) (PVC) Pressure-Rated Pipe (SDR Series)
ASTM D2321	Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications
ASTM D2467	Standard Specification for Socket-Type Poly (Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80
ASTM D2564	Standard Specification for Solvent Cements for Poly (Vinyl Chloride) (PVC) Plastic Piping Systems
ASTM D2672	Standard Specification for Joints for IPS PVC Pipe Using Solvent Cement
ASTM D2855	Standard Practice for Making Solvent-Cemented Joints with Poly (Vinyl Chloride) (PVC) Pipe and Fittings
ASTM D3139	Standard Specification for Joints for Plastic Pressure Pipes Using Flexible Elastomeric Seals

NSF INTERNATIONAL (NSF)

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Flushing Procedure: The flushing procedure shall outline the method and materials for handling flush water, i.e., pipe, valves, fittings for filling pipeline and similar for discharging flush water, method for restraining aboveground pipe, and preventing erosion damage.

Qualifications:

Solvent Welders: List of solvent welders and current test records for solvent welder(s) for field solvent welding.

Hydrostatic Testing Plan: Submit at least 15 days prior to testing and at minimum, include the following:

- Testing dates.
- Piping systems and section(s) to be tested.
- Method of isolation.
- Method of conveying water from source to system being tested.
- Calculation of maximum allowable leakage for piping section(s) to be tested.

Certification of Calibration: Approved testing laboratory certificate if pressure gauge for hydrostatic test has been previously used. If pressure gauge is new, no certificate is required.

As-Built Survey Data: Before final acceptance of raw water piping system, provide as-built locations of systems and components, showing pressure pipelines, including grade breaks or alignment, horizontal and vertical locations of all utility crossings, finished grade profile on all alignments, and valve locations. The submitted data shall be certified by a Washington licensed land surveyor.

Shop Drawings: Drawings showing pipe diameter, pipe class, and fitting details.

Submittals–Approval Not Required

Information/Record (IR): Hydrostatic test documentation form and results.

Part 2 – Products

Materials

Bell and Spigot Pipe:

PVC, conforming to requirements of AWWA C900.

DR shall be 14, 200-psi pressure class.

Joints:

- Rubber gasketed.
- Conform to AWWA C900.

Fittings: Ductile iron, conforming to AWWA C153 or C110.

Solvent Weld PVC Pipe and Fittings: As specified on Piping Data Sheet located at the end of this section as a supplement.

Part 3 – Execution

Installation of Schedule 80 Solvent Weld Pipe

In accordance with Section 02500, RAW WATER CONVEYANCE PIPING - GENERAL.

Installation of C900 Bell And Spigot Pipe

In accordance with AWWA C605.

Joints:

Rubber Gasketed: In accordance with manufacturer's written instructions.

Restrained Joint Systems: In accordance with manufacturer's written instructions.

Pipe Bending for Horizontal or Vertical Curves: Radius of curves shall not exceed 75 percent of manufacturer's recommended values.

Use blocks or braces at pipe joints to ensure axial deflection in gasketed or mechanical joints does not exceed allowable deflection.

Maximum Joint Deflection: 75 percent of manufacturer's recommended values.

Construction Quality Control

Cleaning

Following assembly and testing, and prior to final acceptance, flush pipelines with water at 2.5 fps minimum flushing velocity until foreign matter is removed.

If impractical to flush large diameter pipe at 2.5 fps, clean pipe by use of pipe pig as approved by Construction Manager. Multiple passes of pipe pig may be required to adequately clean line.

Remove accumulated debris through blowoffs 2 inches and larger or by removing spools and valves from piping.

Pipeline Hydrostatic Test:

General:

Notify Construction Manager in writing at least 5 days in advance of testing. Construction Manager will coordinate with other parties required to witness testing. Perform testing in presence of Construction Manager.

Using water as test medium, all newly installed pipelines shall successfully pass hydrostatic leakage test prior to acceptance.

Conduct field hydrostatic test on buried piping after trench has been completely backfilled. Testing may, as approved by Construction Manager, be done prior to placement of asphaltic concrete or roadway structural section.

Construction General Contractor may, if field conditions permit and as approved by Construction Manager, partially backfill trench, and leave joints open for inspection and conduct initial service leak test. Final field hydrostatic test shall not, however, be conducted until backfilling has been completed as specified above.

Install temporary thrust blocking or other restraint as necessary to prevent movement of pipe and protect adjacent piping or equipment. Make necessary taps in piping prior to testing.

Wait a minimum of 5 days after concrete thrust blocking is installed to perform pressure tests. If high-early strength cement is used for thrust blocking, wait may be reduced to 2 days.

Prior to test, remove or suitably isolate appurtenant instruments or devices that could be damaged by pressure testing.

Procedure:

Furnish testing equipment, as approved by Construction Manager, which provides observable and accurate measurements of leakage under specified conditions.

Maximum Filling Velocity: 0.25 foot per second calculated based on full area of pipe.

Expel air from piping system during filling.

Test Pressure: 173 psi as measured at low point of pipeline.

Apply and maintain specified test pressure with hydraulic force pump. Valve off piping system when test pressure is reached.

Maintain hydrostatic test pressure continuously for 2 hours minimum, adding make-up water only as necessary to restore test pressure.

Determine actual leakage by measuring quantity of water necessary to maintain specified test pressure for duration of test.

Identify method of disposing water after system testing.

Maximum Allowable Leakage:

$$L = \frac{ND(P)^{1/2}}{7400}$$

where:

L = Allowable leakage, in gallons per hour.

N = Number of joints in tested line.

D = Nominal diameter of pipe, in inches.

P = Average test pressure during leakage test, in pounds per square inch.

Supplements

Supplement 1—Solvent Weld Polyvinyl Chloride (PVC) Pipe and Fittings.

END OF SECTION 02509

Table 1. Solvent Weld Polyvinyl Chloride (PVC) Pipe and Fittings

Item	Size	Description
Pipe	All	Schedule 80 PVC: Type I, Grade I or Class 12454-B conforming to ASTM D1784 and ASTM D1785. Pipe shall be manufactured with 2 percent titanium dioxide for ultraviolet protection. Threaded Nipples: Schedule 80 PVC.
Fittings	All	Schedule to Match Pipe Above: ASTM D2466 and ASTM D2467 for socket-weld type and Schedule 80 ASTM D2464 for threaded type. Fittings shall be manufactured with 2 percent titanium dioxide for ultraviolet protection.
Joints	All	Solvent socket-weld except where connection to threaded valves and equipment may require future disassembly.
Flanges	All	One piece, molded hub type PVC flat face flange in accordance with Fittings above, 125-pound ANSI B16.1 drilling.
Bolting	All	ASTM A193/A193M Type 316 stainless steel Grade B8M hex head bolts and ASTM A194/ A194M Grade 8M hex head nuts.
Gaskets	All	Flat Face Mating Flange: Full faced 1/8-inch thick ethylene propylene (EPR) rubber. Raised Face Mating Flange: Flat ring 1/8-inch ethylene propylene (EPR) rubber, with filler gasket between OD of raised face and flange OD to protect the flange from bolting moment.
Solvent Cement	All	As recommended by the pipe and fitting manufacturer conforming to ASTM D2564.
Thread Lubricant	All	Teflon tape

SECTION 02631 – CATCH BASINS SCHED. B

Part 1 – General

References

The following is a list of standards that may be referenced in this section:

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO M105	Standard Specification for Gray Iron Castings
AASHTO M198	Standard Specification for Joints for Circular Concrete Sewer and Culvert Pipe Using Flexible Watertight Gaskets

ASTM INTERNATIONAL (ASTM)

ASTM A536	Standard Specification for Ductile Iron Castings
ASTM C387	Standard Specification for Packaged, Dry, Combined Materials for Mortar and Concrete
ASTM C478	Standard Specification for Precast Reinforced Concrete Manhole Sections

Part 2 – Products

Precast Units

Precast units shall conform to ASTM C478 except dimensions shall be as shown on the Drawings.

The joints shall be the tongue-and-groove type or the shiplap type, sufficiently deep to prevent lateral displacement. Precast concrete units shall be furnished with knockouts or cutouts.

Rubber gaskets for use in joints of precast catch basin sections shall conform to the applicable requirements of AASHTO M198. Rubber gasket material shall be stored in a clean, cool place, protected from sunlight and contaminants. They shall be protected from direct sunlight at all times except during actual installation.

Concrete risers for extensions shall be a maximum of 6 inches high and of same quality as sections. Risers shall be reviewed by Engineer before installation.

Mortar

Standard premixed mortar conforming to ASTM C387, Type S, or proportion 1 part Portland cement to 2 parts clean, well-graded sand which will pass a 1/8-inch screen. Admixtures may be used not exceeding the following percentages of weight of cement: Hydrated lime, 10 percent; diatomaceous earth or other inert materials, 5 percent. Consistency of mortar shall be such that it will readily adhere to concrete.

Rings And Covers

Castings for catch basin rings shall be gray-iron conforming to the requirements of AASHTO M105, Grade 30B. Covers shall be ductile iron conforming to ASTM A536, Grade 80-55-06.

Catch basin rings and covers shall meet the strength requirements of Federal Specification RR-F-621E. All mating surfaces shall be machine finished to ensure a nonrocking fit.

The horizontal surface and inside vertical recess face of the ring, and the horizontal seating surface and vertical outside edge of the cover, shall be machine finished to the following tolerances:

Ring: +3/32 inch to -3/32 inch.

Cover: +3/32 inch to -3/32 inch.

All catch basin rings and covers shall be identified by the name or symbol of the manufacturer and country of casting origin. This identification shall be in a plainly visible location when the ring and cover are installed. Ductile iron shall be identified by the following, "DUC" or "DI". The manufacturer's identification and material identification shall be adjacent to each other and shall be minimum 1/2-inch to maximum 1-inch high letters, recessed to be flush with the adjacent surfaces.

Part 3 – Execution

Excavation And Backfill

The excavation for all catch basins shall be sufficient to leave 1 foot in the clear between their outer surfaces and the earth bank. Backfilling of catch basins shall be done in accordance with the provisions of Section 02320 TRENCH BACKFILL. Backfilling around the work will not be allowed until the concrete and mortar have thoroughly set. Any excavation safety systems shall meet the requirements of Section 02316, EXCAVATION.

Placing Precast Units

If material in bottom of trench is unsuitable for supporting unit, excavate and backfill to required grade with 3-inch minus, clean, pit-run material. Set units to grade at locations shown.

Ladder rungs shall be grouted in the precast concrete walls. Rungs shall be uniformly spaced at 12 inches and be vertically aligned.

The ends of all pipes shall be trimmed flush with the inside walls.

Rubber gaskets may be used in tongue-and-groove joints of precast units. All other joints and all openings cut through the walls shall be grouted and watertight.

If gaskets are used, handling of the precast units after the gasket has been affixed shall be done carefully to avoid disturbing or damaging the gasket or contaminating it with foreign material. Care shall be exercised to attain proper alignment before the joints are entirely forced home. During insertion of the tongue or spigot, the units shall be partially supported to minimize unequal lateral pressure on the gasket and to maintain concentricity until the gasket is properly positioned.

Catch basins shall be watertight.

Extensions

Install extensions to height determined by Engineer. Lay risers in mortar with sides plumb and tops to grade. Joints shall be sealed with mortar, with interior and exterior troweled smooth. Prevent mortar from drying out and cure by applying a curing compound. Extensions shall be watertight.

Installation Of Rings And Covers

Set rings and covers at elevations indicated or as determined in field and in conformance with Drawings.

Rings may be cast in, or shall be set in mortar.

The cover of a catch basin shall not be grouted to final grade until the final elevation of the adjacent ground surface has been established. Covers shall be seated properly to prevent rocking.

Cleaning

Upon completion, clean each structure of all silt, debris, and foreign matter.

END OF SECTION 02631

SECTION 02632 – STORMWATER PIPING SCHED. B

Part 1 – General

References

The following is a list of standards, which may be referenced in this section and any supplemental Data Sheets:

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO M294	Standard Specifications for Corrugated Polyethylene Pipe, 300- to 1200-mm Diameter

ASTM INTERNATIONAL (ASTM)

ASTM F477	Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe

Submittals–Approval Not Required

Information/Record (IR): Catalog and manufacturer's data sheets for stormwater pipe and fittings.

Part 2 – Products

Pipe And Fittings

Pipe and fittings for culverts and stormwater pipes shall be corrugated polyethylene (CPE) in accordance with the following:

Item	Description
Pipe	AASHTO M294
Pipe Stiffness (Minimum)	In accordance with specified AASHTO Specification Profile Type S
Joints	Bell and spigot, gasketed type and water-tight. Gaskets ASTM F477
Fittings	Manufacturer's standard; same stiffness as adjacent pipe.
Source Quality Control	In accordance with specified AASHTO Specification.
Factory Testing	Pipe lengths used for deflection testing shall be destroyed after testing.

Part 3 – Execution

Installation Of Pipe, Fittings, And Appurtenances

General:

- Pipe laying shall proceed upgrade with spigot ends pointing in direction of flow.
- Excavate bell holes at each joint to permit correct assembly and inspection of entire joint.
- Pipe invert may deviate from line or grade up to 1/2 inch for line and 1/4 inch for grade, provided that finished pipeline will present a uniform bore, and such variation does not result in a level or reverse sloping invert, or less than minimum slope shown.
- Pipe bedding shall form continuous and uniform bearing and support for pipe barrel between joints. Pipe shall not rest directly on bell or pipe joint.
- Prevent entry of foreign material into gasketed joints.

Plug or close off pipes that are stubbed off for manhole, concrete structure, or for connection by others, with temporary watertight plugs.

Trench excavation and placement of pipe bedding and pipe zone materials shall be in accordance with Section 02320, TRENCH BACKFILL.

Any excavation safety systems shall be in accordance with Section 02316, EXCAVATION.

Pipe Cleaning:

Prior to final acceptance and final inspection of the stormwater pipes by Construction Manager, flush and clean all stormwater pipes and catch basins. Remove all accumulated construction debris, rocks, gravel, sand, silt, and other foreign material. If necessary, use mechanical rodding or bucketing equipment.

Upon Construction Manager's final inspection of the stormwater pipes, if any foreign matter is still present in the system, reflush and clean the sections and portions of the lines as required.

END OF SECTION 02632

SECTION 02661 – GEOMEMBRANES SCHED. B

Part 1 – General

References

The publications listed below form a part of this Specification to the extent referenced. The publications are referred to in the text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D638	Standard Test Method for Tensile Properties of Plastics
ASTM D1004	Standard Test Method for Initial Tear Resistance of Plastic Film and Sheeting
ASTM D1238	Standard Test Method for Flow Rates of Thermoplastics by Extrusion Plastometer
ASTM D1505	Standard Test Method for Density of Plastics by the Density-Gradient Technique
ASTM D1603	Standard Test Method for Carbon Black in Olefin Plastics
ASTM D4218	Determination of Carbon Black Content in Polyethylene Compounds by the Muffle-Furnace Technique
ASTM D4833	Standard Test Method for Index Puncture Resistance of Geotextiles, Geomembranes, and Related Products
ASTM D5199	Standard Test Method for Measuring the Nominal Thickness of Geosynthetics
ASTM D5321	Standard Test Method for Determining the Coefficient of Soil and Geosynthetic or Geosynthetic and Geosynthetic Friction by the Direct Shear Method
ASTM D5397	Evaluation of Stress Crack Resistance of Polyolefin Geomembrane Using Notched Constant Tension Load Test (Appendix A, Single Point)
ASTM D5596	Microscopic Evaluation of the Dispersion of Carbon Black in Polyolefin Geosynthetics
ASTM D5994	Standard Test Method for Measuring Core Thickness of Textured Geomembranes
ASTM D6243	Standard Test Method for Determining the Internal and Interface Shear Resistance of Geosynthetic Clay Liner by the Direct Shear Method
ASTM D6392	Standard Test Method for Determining the Integrity of Nonreinforced Geomembrane Seams Produced Using Thermo-Fusion Methods
	GEOSYNTHETIC RESEARCH INSTITUTE (GRI)
GRI-GM12	Asperity Measurement of Textured Geomembranes Using a Depth Gage
GRI-GM13	Test Properties, Testing Frequency and Recommended Warranty for High Density Polyethylene (HDPE) Smooth and Textured Geomembrane

Description

The work includes IDF manufacture, fabrication (if needed), supply, and installation of geomembrane for the lining system, and for other geomembrane applications, as shown on the Drawings. Geomembrane is also referred to as flexible membrane liner (FML).

Definitions

CQA Certifying Engineer: Engineer providing independent oversight and responsible for implementing the CQA Plan. Independent is defined as an organization that operates separately from the Construction General Contractor, DOE-ORP, and the Tank Farm Contractor.

Engineer: Design Engineer for the IDF, providing technical design support during construction.

Construction Manager: Construction coordinator overseeing the IDF construction activities in the field and the Tank Farm Contractor's onsite technical representative.

Construction General Contractor: Responsible for overall construction activities and operations, including Construction Subcontractors.

Installer: Construction Subcontractor responsible for installation of geosynthetics (geomembrane, GCL, CDN, and geotextiles).

LCRS: Leachate collection and removal system.

LDS: Leak detection system.

SLDS: Secondary Leak Detection System.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Manufacturer's descriptive data, specification sheets, literature, and other data as necessary to fully demonstrate that those materials proposed for use comply with the requirements of these Specifications.

Construction General Contractor shall submit required interface strength data as specified in PART 2-PRODUCTS prior to shipment of material to allow Engineer to evaluate if submitted material meets strength requirements for project design criteria. Allow Engineer 20 working days for this evaluation upon receipt of data.

Installation Plan:

The Construction General Contractor shall submit an installation plan describing the proposed methods for geomembrane deployment, panel layout, seaming, repair, and protection. The installation plan shall provide for no field seam locations within the LCRS sump trough under the leachate collection and riser pipes. Construction General Contractor shall orient panel layout such that one full panel width spans the LCRS sump trough. The plan shall also include a quality control program for the Construction General Contractor's activities related to geomembrane installation.

Manufacturer's Quality Control (QC) test data for geomembrane composition and properties as specified in paragraph MANUFACTURING QUALITY CONTROL - POLYETHYLENE GEOMEMBRANES.

Manufacturer's QC certification as specified herein.

The Construction General Contractor shall submit Geomembrane Installer's organizational and seaming personnel qualifications, and other as required to provide the information described in these Specifications.

The Construction General Contractor shall submit Geomembrane Installer's Certificate of Subgrade Acceptability to the CQA Certifying Engineer as specified herein.

Calibration certification for construction quality control test equipment.

Submittals–Approval Not Required

Information/Record (IR):

Documentation of test results from construction quality control testing.

Construction Quality Assurance (CQA)

CQA Plan: A CQA Plan has been prepared as part of the landfill design. The CQA Plan discusses the testing procedures that will be followed by the CQA Certifying Engineer during installation of the geosynthetics (geotextile, CDN, geomembrane, GCL) and the documentation of the process. The CQA Plan is made part of these Specifications by reference. The Construction General Contractor shall conform to the requirements of the CQA Plan for all aspects of the geosynthetics, including submittals, supply, storage, installation, testing, documentation, covering, and protection.

Quality assurance procedures are presented in the CQA Plan. The Construction General Contractor shall accommodate all quality assurance activities described in this section and in the CQA Plan for this project.

Prior to placing any materials over the installed geomembrane, the Construction General Contractor shall allow time for acceptance of the Work as listed in the CQA Plan.

CQA Conformance Testing

Upon delivery of the rolls of geomembrane, the CQA Certifying Engineer will obtain samples at a frequency of one per production lot or one per 50,000 square feet of geomembrane, whichever results in greater number of tests. The CQA Certifying Engineer will test the samples to determine conformance to both the design specifications and the list of guaranteed properties.

As a minimum, tests to determine the following characteristics will be performed on geomembranes:

- Thickness (ASTM D5994).
- Tensile characteristics (yield strength, elongation at yield; ASTM D638).
- Asperity (GM-12).
- Puncture resistance (ASTM D4833).

Where optional procedures are noted in the test method, the requirements of these Specifications will prevail.

Sampling Procedures:

Samples will be taken across the entire width of the roll and will not include the first 3 feet. Unless otherwise specified, samples will be 3 feet long by the roll width. The CQA Certifying Engineer will mark the machine direction on the samples with an arrow.

The CQA Certifying Engineer shall be allowed to remove samples for testing and other activities. Sample dimensions, procedures, and frequency shall be the same as those specified in the CQA Plan. The Construction General Contractor shall assist the CQA Certifying Engineer as necessary in all sampling and testing activities.

Procedures for samples that fail conformance testing are outlined in the CQA Plan. The cost of additional conformance testing to demonstrate compliance of failed samples shall be borne by the Construction General Contractor.

Warranty

The geomembrane manufacturer(s) shall provide warranties on all geomembrane materials installed at the project site. The warranties shall be provided to the Construction General Contractor as purchaser with the Tank Farm Contractor named as beneficiary and shall be signed by an authorized representative of the geomembrane manufacturer. The terms of the warranties shall, at a minimum, include the provisions contained in the most recent version of GRI Test Method GM-13.

Part 2 – Products

General

Type: High-density polyethylene (HDPE). Unreinforced, 60-mil nominal thickness, textured both sides.

Manufacturer: The HDPE geomembrane manufacturer(s) shall have a minimum of 5 years of experience as a commercial manufacturer of HDPE geomembranes for landfill applications. Examples of potential manufacturers include: GSE Lining Technology, Inc., Houston, TX; and Poly-Flex, Inc., Grand Prairie, TX. Use of these examples is not intended to restrict potential manufacturers.

Required Properties - Polyethylene Geomembranes

Use of Recycled Polymer: The raw material shall be new polyethylene resin containing no more than two percent clean recycled polymer by weight. Two percent-recycled polymer shall not include any finished sheet material that has actually seen some type of service performance. Regrind, reworked, or trim materials in the form of chips or edge strips that have not actually seen some type of use may be added, if the material is from the same manufacturer and is the same formulation as the geomembrane being produced.

Resin Properties: The resin shall meet the following Specifications:

HDPE:

Resin Specific Gravity (ASTM D1505): >0.932.

Melt Index (ASTM D1238 Condition 190/2.16): <1.1 g/10 min.

Finished Sheet Properties: The physical, mechanical, and environmental properties of the finished sheet shall meet or exceed the values specified in Table 1 contained in this part of the Specifications. Where applicable, values in Table 1 are Minimum Average Values.

Required Interface Shear Strength Data:

Provide data prior to material shipment for the interface friction angle between the textured geomembrane and other materials (including CDN, GCL, and Admix Liner) directly in contact with the geomembrane as specified in Article SUBMITTALS. Perform two interface shear strength tests on each interface under each set of normal loads.

Friction angle shall be determined by direct shear testing under fully saturated conditions (ASTM D5321 or D6243 for GCL interface) at both low normal loads of 100, 250, and 500 pounds per square foot (psf), and high normal loads of 2,000, 8,000, and 15,000 psf. Report results for both peak and large displacement (minimum 2 inches) strength.

The Engineer will review this data for conformance with project strength requirements. Construction General Contractor shall not order material for shipment until approved by Engineer. Any product or material changes required as a result of inadequate strength data will be addressed by Change Order provided submitted material meet all other requirements of this section.

Manufacturing Quality Control - Polyethylene Geomembranes

Quality Control Testing: Quality control testing shall be carried out by the manufacturer to demonstrate that the geomembrane meets the Specifications in this section. Additional testing may be carried out for purposes of determining conformance by the CQA Certifying Engineer. If the results of the manufacturer's and the CQA Certifying Engineer's testing differ significantly, the testing shall be repeated by the CQA Certifying Engineer, and the manufacturer shall be allowed to monitor this testing. The results of this latter series of tests will prevail, provided that the applicable test methods have been followed.

Required Information: Prior to the delivery of any geomembrane material, the manufacturer shall submit the following information:

The origin (resin supplier's name, resin production plant), identification (brand name, number), and production date of the resin.

A list of quantities and descriptions of materials other than the base polymer, which comprise the geomembrane.

Copies of the quality control certificates issued by the resin supplier.

Reports on the tests conducted by the manufacturer to confirm that the quality of the resin used to manufacture the geomembrane satisfy these Specifications.

A statement that recycled polymer (if any) is clean and does not exceed 2 percent by weight.

A properties sheet including, at a minimum, all specified properties, measured using test methods indicated in these Specifications, or equivalent.

Reports on the tests, including sampling procedures, conducted by the manufacturer to confirm that the geomembrane meets the Specifications.

A certification that property values given in the properties sheet are guaranteed by the geomembrane manufacturer.

QC Certification: Prior to shipment, the geomembrane manufacturer shall provide a quality control certificate for each roll of geomembrane. The quality control certificate shall be signed by a responsible party employed by the geomembrane manufacturer, such as the production manager. The quality control certificate shall include:

Roll numbers and identification, resin lot, and batch numbers.

Sampling procedures and results of quality control tests. As a minimum, results shall be given for thickness, asperity, tensile strength, and tear resistance in accordance with methods indicated in these Specifications. Tests shall be conducted on each production lot of geomembrane or every 50,000 square feet, whichever results in the greater number of tests.

Manufacturing Plant Visit: The manufacturer shall allow the CQA Certifying Engineer or his designated representative to visit the manufacturing plant, if the CQA Certifying Engineer so chooses. If possible, the visit shall be prior to or during the manufacturing of the geomembrane rolls for the specific project. The CQA Certifying Engineer or his designated representative shall review the manufacturing process, quality control, laboratory facilities, and testing procedures. During the visit, visiting personnel will also:

Confirm that the measurements of properties by the manufacturer are properly documented and test methods used are acceptable.

Spot inspect the rolls and confirm that they are free of holes, blisters, or any sign of contamination by foreign matter.

Review packaging and transportation procedures to confirm that these procedures are not damaging the geomembrane.

Confirm that roll packages have a label indicating the name of the manufacturer, type of geomembrane, thickness, and roll number.

If applicable, confirm that extrusion rods and/or beads are derived from the same base resin type as the geomembrane.

The geomembrane manufacturer shall accommodate these activities.

Extrudate

Extrudate for Fusion Welding of HDPE Geomembrane: Formulated from same HDPE resin as geomembrane and shall meet applicable physical property requirements.

Field-Fabricated Boots

Pipes and other structures penetrating the lining system shall be sealed to the geomembrane with fabricated boots made of the same material and workmanship as the lining system geomembrane. The flange portion of each boot shall match the angle of the slope or bottom, be sealed to the geomembrane, and fit smoothly without folds or stretching of the material.

Sealant Caulking

Where shown on the Drawings, the caulking used shall be a one-component sealant formulated of butyl rubber and other selected ingredients, equivalent to Biddle Co., St. Louis, MO, Butylgrip Sealant, or as recommended by the manufacturer of the geomembrane materials.

Stainless Steel Clamps

As indicated on the Drawings, clamps shall be used to secure the HDPE geomembrane to pipes, poles, or risers that are intended to protrude through the cover. One-half-inch wide clamps shall meet or exceed specifications for "Make-a-Clamp" as manufactured by Breeze Clamp Products Division, Federal Laboratory, Inc., Saltsburg, PA.

Butyl Mastic Tape

Shall be as manufactured by Tremco, Cleveland, OH; or of a type recommended by HDPE geomembrane manufacturer.

Neoprene Rubber Pad

As indicated on the Drawings, neoprene rubber shall be used as compression strip beneath the stainless steel clamps (ASTM D2240). One-half-inch thick neoprene rubber shall be 35-to 40 durometer hardness, as supplied by Aero Rubber Co., Inc., Bridgeview, IL, or approved equal. Cut to a continuous 2-inch wide piece of neoprene to form the gasket. Neoprene rubber contact cement recommended by the supplier shall be used to bond butt ends of joined strips and to bond neoprene rubber in position on surface. Butt joints in neoprene strips shall be offset from adjacent joints by at least 6 inches.

Tensiometer For Field Testing

Motor driven with jaws capable of traveling at measured rate of 2 inches per minute. Equipped with gauge, which measures force in unit pounds exerted between jaws.

Plywood Sheeting

Use APA rated sheeting EXT for protection of the HDPE geomembrane at termination edges on south side of Phase I.

Part 3 – Execution

General:

Personnel Qualifications - Polyethylene Geomembranes:

Installer Organization: At a minimum, the Construction General Contractor organization shall have successfully completed at least five projects consisting of installation of at least 5,000,000 square feet (total) of HDPE liner. Projects shall include RCRA landfills.

Seaming Personnel: All personnel performing seaming operations shall be qualified by experience or by successfully passing seaming tests similar to those described in this section. The superintendent and lead welder foreman shall have experience seaming a minimum of 1,000,000 square feet of polyethylene geomembrane using the same type of seaming apparatus proposed for use on this project. These individuals shall provide direct supervision over less experienced seamers.

No field seaming shall take place without one of these individuals being present. Key personnel are defined as the superintendent, foreman, and lead welder. Key personnel shall be full time employees of the Geosynthetics Installer.

Applicability: The primary and secondary geomembranes shall be installed at the locations, lines, and grades shown on the Drawings. All geomembranes shall be installed in accordance with these Specifications and the CQA Plan.

Installation Plan: Prior to beginning geomembrane installation, the Construction General Contractor shall submit a plan describing the proposed size, number, position, and sequence of geomembrane panel placement, and location of field seams.

Subgrade Surface Preparation - Polyethylene Geomembranes

The Construction General Contractor shall be responsible for preparing the subgrade surface of the soil bentonite liner for the geomembrane. Prepare the underlying soil surface as specified in Section 02319, SUBGRADE PREPARATION, and as approved by the Geomembrane Installer.

The Geomembrane Installer shall certify in writing that the surface on which the geomembrane will be installed is acceptable. The certificate of acceptance shall be given by the Installer to the Construction General Contractor prior to commencement of geomembrane installation in the area under consideration. The CQA Certifying Engineer shall be given a copy of this certificate by the Construction General Contractor. The form for Geomembrane Installer certification is provided as Supplement to this Specification. Submittal of this form only applies to soil surfaces underlying the geomembrane. In this case Geomembrane Installer Certification of Subsurface Acceptability is only required for the surface on which the secondary and SLDS geomembrane shall be installed.

After the subgrade surface has been accepted by the Installer, it shall be the Installer's responsibility to indicate to the Construction General Contractor any change in the subgrade surface condition that may require repair work.

Special care shall be taken to avoid desiccation cracking or freezing of the admix liner. Specifications for allowable desiccation cracking of soil liner and repair measures are contained in Section 02666, ADMIX LINER. The surface of the admix liner shall be maintained in the required condition throughout the course of geomembrane installation.

Anchor Trench Excavation and Backfilling

The anchor trench shall be excavated to the lines and widths shown on the design Drawings, prior to geomembrane placement. The corners of the trench shall be rounded so as to avoid sharp bends in the geomembrane. No loose soil shall be allowed to underlie the geomembrane in the anchor trench. Backfill with material as shown on the Drawings and compact as specified in Section 02320, TRENCH BACKFILL.

Geomembrane Placement - Polyethylene Geomembranes

Field Panel Identification: A field panel is the unit area of geomembrane, which is to be seamed in the field. Two cases are defined:

If the geomembrane is fabricated into panels in a factory, a field panel is a factory panel or a portion of factory panel cut in the field.

If the geomembrane is not fabricated into factory panels, a field panel is a roll or a portion of roll cut in the field.

It will be the responsibility of the CQA Certifying Engineer to assign each field panel an "identification code" (number or letter-number) consistent with the layout plan. This identification code shall be agreed upon by the Construction Manager, Installer, and CQA Certifying Engineer.

This field panel identification code shall be as simple and logical as possible. (Note that roll numbers assigned in the manufacturing plant are usually cumbersome and are not related to location in the field.)

The CQA Certifying Engineer will establish a table or chart showing correspondence between roll numbers, factory panels, and field panel identification codes. The field panel identification code shall be used for all quality assurance records, including Installer's quality control (QC) testing.

Field Panel Placement:

Location: Field panels shall be installed at the locations indicated in the Installer's layout plan, as approved or modified.

Installation Schedule: In general seaming of geomembrane will be performed the same day as deployment. However, at the discretion of Geosynthetic Installer, seaming may be carried over to the following workday.

Placement Conditions:

Geomembrane placement shall not proceed at an ambient temperature below 32 degrees F or above 104 degrees F as measured 6 inches above the geomembrane surface unless installation procedures approved by the CQA Certifying Engineer are in place to address environmental conditions. Geomembrane placement shall not be done during any precipitation, in the presence of excessive moisture (e.g., fog, dew), in an area of ponded water, or in the presence of excessive winds. Placement methods shall prevent damage to underlying soil liner or geosynthetic materials.

Factors such as expansion, contraction, overlap at seams, anchorage requirements, seaming progress, and drainage shall be considered. Textured-surface sheets shall be aligned in a manner, which maximizes their frictional capabilities along the slope. Maneuver sheets of geomembrane into place in a manner, which prevents wrinkles, folds, or similar distress, which can damage the geomembrane or prevent its satisfactory alignment or seaming. A smooth-surface HDPE geomembrane rub sheet shall be used when placing textured HDPE geomembrane over underlying GCL. The rub sheet shall be maintained in good condition without tears, rough edges, holes, or scuff marks that can catch, displace, or otherwise disturb the underlying GCL, or the overlying geomembrane.

Damage: Damaged panels or portions of damaged panels, which have been rejected, shall be removed from the work area. Any repairs shall be made according to procedures described in this Part of the Specifications.

Exposed Geomembrane Protection: After panel deployment, all geomembrane, except those shown as permanently exposed on the Drawings, shall be covered to within a maximum period of 20 working days to minimize exposure to temperature cycles that cause expansion/contraction of the geomembrane and desiccation of the underlying admix liner. Exposure in excess of 15 days, Construction General Contractor will peel back leading panel of geomembrane on a routine basis (established by the CQA inspector) for CQA to inspect for desiccation of admix liner. Any observed desiccation observed outside of specification tolerance will be repaired and the geomembrane covered immediately. Additional exposure areas may be required by CQA to verify complete repair areas required. Geomembrane panels shall be covered by other geosynthetic components of the lining system or overlying soil cover materials as shown on the Drawings.

Field Seaming - Polyethylene Geomembranes

Seaming Equipment and Products: Approved processes for field seaming are extrusion welding and fusion welding, except that use of extrusion welding shall be limited to areas (such as sumps or repairs) where fusion welding cannot be employed. Proposed alternate processes shall be documented and submitted by the Installer to the Construction Manager and CQA Certifying Engineer for approval. Only equipment, which has been specifically approved by make and model, shall be used.

Extrusion Process:

The extrusion-welding machine shall be equipped with gages capable of measuring the temperature at the nozzle or the preheat temperature.

The Installer shall provide documentation regarding the extrudate and shall certify that the extrudate is compatible with these Specifications and is comprised of the same resin type as the geomembrane sheeting.

The Installer shall comply with the following:

The Installer shall maintain on-site a sufficient number of spare operable seaming machines (at least one at all times) to ensure continuous operation.

The equipment used for seaming shall not be likely to damage the geomembrane.

The extruder shall be purged prior to beginning a seam until all heat-degraded extrudate has been removed from the barrel.

The electric generator shall be placed on a smooth base such that no damage occurs to the geomembrane.

Grinding shall be completed no more than 1 hour prior to seaming.

A smooth insulating plate or fabric shall be placed beneath the hot welding machine after usage.

The geomembrane shall be protected from damage in heavily trafficked areas.

Fusion Process: The fusion-welding machines shall be automated vehicular-mounted devices. The fusion-welding machines shall be equipped with gages giving the pertinent temperatures.

The Installer shall comply with the following:

The Installer shall maintain on site a sufficient number of spare operable seaming machines (at least one at all times) to ensure continuous operations.

The equipment used for seaming shall not be likely to damage the geomembrane.

The electric generator shall be placed on a smooth base such that no damage occurs to the geomembrane.

A smooth insulating plate or fabric shall be placed beneath the hot welding machine after usage.

The geomembrane shall be protected from damage in heavily trafficked areas.

If a build-up of moisture is observed prior to seaming a movable protective layer shall be used directly below each overlap of geomembrane to be seamed to prevent buildup of moisture between the sheets.

Seam Layout: In general, seams shall be oriented parallel to the line of maximum slope, i.e., oriented up and down, not across, the slope to the maximum extent practical. In corners and odd-shaped geometric locations, the number of seams shall be minimized. No seams shall be permitted within the LCRS sump trough for leachate collection and riser pipes. One full panel width shall span the LCRS sump trough.

On the landfill floor:

No horizontal seam shall be less than 5 feet from the toe of the slope, or other area of potential stress concentrations.

Over the LCRS, LDS, and SLDS sump areas in each cell, no horizontal seam shall be placed less than 150 feet from the toe of the north slope for a distance of at least 100 feet in each direction from the LCRS and LDS sump centerline.

Seams shall be aligned to produce the fewest possible number of wrinkles and "fishmouths".

A seam numbering system consistent with the panel numbering system shall be utilized.

Weather Conditions for Seaming: The allowable weather conditions for seaming are as follows:

Unless authorized in writing by the Construction Manager, no seaming shall be attempted at ambient temperatures below 32 degrees F or above 104 degrees F as measured 6 inches above the geomembrane surface.

The geomembrane shall be dry, protected from wind, and free of dust.

If the Installer wishes to use methods, which may allow seaming at ambient temperatures below 32 degrees F, the Installer shall certify in writing that the quality of the seams welded at these temperatures is the same as the quality of seams welded at temperatures above 32 degrees F. In addition, if the Installer wishes to seam at ambient temperatures below 32 degrees F, the following conditions shall be satisfied in addition to the general seaming procedures:

For extrusion welding, preheating shall be performed. Preheating may be waived by the Construction Manager if it is demonstrated to the satisfaction of the CQA Certifying Engineer that welds of equivalent quality may be obtained without preheating at the expected temperature of installation.

Sheet grinding, if required, may be performed before preheating.

Observe all areas of the geomembrane that have been preheated to determine that they have not been subjected to excessive melting.

Confirm that geomembrane surface temperatures have not decreased below the minimum specified for welding, due to wind or other adverse conditions. Wind protection for the seam area may be required.

Trial seams, as described in paragraph Trial Seams of this section, shall be made in the immediate area where seaming will occur, under the same ambient temperature and preheating conditions as the actual seams. New trial seams shall be made if the ambient temperature decreases by more than 5 degrees F from the previous trial seam conditions. Such new trial seams shall be conducted as soon as seams in progress during the temperature drop have been completed.

Additional destructive seam tests, as described in paragraph Destructive Seam Strength Testing of this section, shall be performed at intervals of 250 to 500 feet of seam length at the CQA Certifying Engineer's discretion.

The Installer shall provide sample coupons cut from each end of the seam.

Seam Preparation:

Cleaning: Prior to seaming, the seam area shall be clean and free of moisture, dust, dirt, debris of any kind, and foreign material. Special attention shall be paid to cleaning the existing geomembrane at tie-in locations.

Overlap: Cross slope, seams on both the trench floor and sideslopes shall be overlapped so that liquids are not trapped, i.e., seams shall be shingled downslope. If seam overlap grinding is required, the process shall be completed according to the geomembrane manufacturer's instructions within 1 hour of the seaming operation, and in a way, that does not damage the geomembrane. Panels of geomembrane shall have a finished overlap of a minimum of 3 inches for extrusion welding and 5 inches for fusion welding.

Use of Solvents: No solvent or adhesive shall be used.

Temporary Bonding: The procedure used to temporarily bond adjacent panels together shall not damage the geomembrane; in particular, the temperature of hot air at the nozzle of any spot welding apparatus shall be controlled such that the geomembrane is not damaged.

General Seaming Procedure: The general seaming procedure used by the Installer shall be as follows:

Seaming shall extend to the outside edge of panels to be placed in the anchor trench.

If required, a firm substrate shall be provided by using a flat board, a conveyor belt, or similar hard surface directly under the seam overlap to achieve proper support.

If seaming operations are carried out at night, adequate illumination shall be provided.

"Fishmouths" or wrinkles at the seam overlaps shall be cut along the ridge of the wrinkle in order to achieve a flat overlap. The cut "fishmouths" or wrinkles shall be seamed, and any portion where the overlap is inadequate shall then be patched with an oval or round patch of the same geomembrane extending a minimum of 6 inches beyond the cut in all directions.

Construction Quality Control Testing - Polyethylene Geomembranes

General: Testing requirements specified herein are intended for the Construction General Contractor (and Geomembrane Installer) during geomembrane installation. Testing requirements for the CQA Certifying Engineer are provided in the CQA Plan.

Trial Seams:

Trial seams shall be made on fragment pieces of geomembrane liner to verify that seaming conditions are adequate. A trial seam shall be made prior to each seaming period (maximum of 6 hours) for each seaming machine used that day. Also, each seamer shall make at least one trial seam each day. Trial seams shall be made under the same conditions as actual seams.

The trial seam sample shall be at least 2 feet long by 1 foot wide (after seaming) with the seam centered lengthwise. Seam overlap shall be as indicated in this Part.

Test three specimens for peel and two specimens for shear. Each specimen shall be at least 1 inch wide and shall be cut from the trial seam sample by the Installer. The specimens shall be tested respectively in shear and peel using a field tensiometer, and they shall not fail in the seam. All trial seam specimens must meet the minimum requirements of Table 2, Required Seam Properties, for trial seam acceptance. If a specimen fails, the entire operation shall be repeated. If the additional specimen fails, the seaming apparatus and seamer shall not be accepted and shall not be used for seaming until the deficiencies are corrected and two consecutive successful full trial seams are achieved.

All test equipment shall be in calibration and conform to manufacturer's specifications. The Installer shall provide the Construction Manager and CQA Certifying Engineer with current calibration certificates.

Nondestructive Seam Continuity Testing:

General:

The Installer shall nondestructively test all field seams over their full length using a vacuum test unit, air pressure test (for double fusion seams only), or other approved method (i.e., spark test). Vacuum testing and air pressure testing are described below. The purpose of the nondestructive test is to check the continuity of seams. It does not provide any information on seam strength. Continuity testing shall be done as the seaming work progresses. Nondestructive testing will not be permitted before sunrise or after sunset unless the Construction General Contractor demonstrates to the CQA Certifying Engineer their capability to perform testing under reduced light conditions. Any seams, which fail nondestructive testing, shall be repaired in accordance with these Specifications. Seams, which cannot be nondestructively tested because of seam geometry, shall be double welded or capped.

All test equipment shall be in calibration and conform to manufacturer's specifications. The Installer shall submit current calibration certificates.

Vacuum Testing: The equipment shall be comprised of the following:

A vacuum box assembly consisting of a rigid housing, a transparent viewing window, a soft neoprene gasket attached to the bottom, port hole or valve assembly, and a vacuum gage.

A steel vacuum tank and pump assembly equipped with a pressure controller and connections.

A rubber pressure/vacuum hose with fittings and connections.

A bucket and wide paint brush.

A soapy solution.

The following procedures shall be used:

Energize the vacuum pump and reduce the tank pressure to a minimum of 5 inches of mercury.

Wet a strip of geomembrane approximately 12 inches wide by 48 inches long with the soapy solution.

Place the vacuum box over the test area.

Close the bleed valve and open the vacuum valve.

Ensure that a leak tight seal is created.

For a period of not less than 10 seconds, examine the geomembrane through the viewing window for the presence of soap bubbles.

If no bubbles appear coming from the seam after 10 seconds, close the vacuum valve and open the bleed valve, move the box over the next adjoining area with a minimum 3-inch overlap, and repeat the process.

All areas where soap bubbles appear shall be marked and repaired in accordance with this Part.

Air Pressure Testing: The following procedures are applicable only to those processes, which produce a double seam with an enclosed air channel. All double seams with an enclosed air channel shall be air pressure tested. The equipment shall be comprised of the following:

An air pump (manual or motor driven) capable of generating and sustaining a pressure of 60 to 65 psi.

A rubber hose with fittings and connections.

A sharp hollow needle, or other approved pressure feed device.

A calibrated pressure gage capable of reading pressures up to 65 psi.

The following procedures shall be used:

Seal both ends of the seam to be tested.

Insert needle with pressure gage, or other approved pressure feed device, into the air channel created by the fusion weld.

Energize the air pump and pressurize the channel to a minimum 25 psi for a 1/2-inch wide channel, or 55 psi for a 1-inch wide channel. Close the valve and sustain the pressure for a minimum of 5 minutes.

If loss of pressure exceeds 2 psi, or does not stabilize, locate faulty area and repair in accordance with this section. If significant changes in geomembrane temperature occur during the test (e.g., due to cloud cover), the test shall be repeated after the geomembrane temperature has stabilized.

Cut end of seam opposite to the pressure gage and observe that the pressure drops. If the pressure does not drop, locate the obstruction(s) in the seam, repair, and retest seam.

Remove needle or other approved pressure feed device and repair seam.

Destructive Seam Strength Testing:

General:

Destructive seam tests shall be performed at selected locations. The purpose of these tests is to evaluate seam strength. Seam strength testing shall be done as the seaming work progresses. The samples shall meet the requirements of Table 2, Required Seam Properties.

All test equipment shall be in calibration and conform to manufacturer's specifications. The Installer shall submit current calibration certificates.

Each sample shall be tested for bonded seam shear and peel strength by an independent testing laboratory.

Test at least five specimens for each seam test method (shear and peel). Four out of five specimens must meet the minimum requirements of Table 2, Required Seam Properties, for field seam acceptance.

Location and Frequency:

Destructive seam samples shall be obtained from actual fabricated field seams as work progresses, not at the completion of field seaming. The CQA Certifying Engineer will select locations where seam samples will be removed.

Sampling frequency shall be a minimum of one sample per 500 feet of seam length per welding machine (this minimum frequency shall be determined as an average taken from all the panels, including welds for caps), or a minimum of two samples per factory panel, whichever gives the largest number of samples. If agreed by all parties (Construction General Contractor, Construction Manager, and the CQA Certifying Engineer) the frequency of destructive seam testing may be reduced to one sample per 1,000 feet of seam if test results and other nondestructive seam tests appear adequate for assuring seam quality. If, based on the specified test frequency, a destructive test location should fall within the LCRS or LDS sump area (as shown on the Drawings); the distance between destructive tests shall be reduced to relocate the destructive test location outside the sump area.

Sampling Procedures:

Samples shall be cut by the Installer as the seaming progresses in order to provide laboratory test results before completion of installation. The CQA Certifying Engineer shall assign a number to each sample, mark it accordingly, and record the sample location on the layout drawing.

All holes in the geomembrane resulting from destructive seam sampling shall be immediately repaired in accordance with repair procedures. The continuity of the new seams in the repaired area shall be tested as described in this Part.

Sample Size: The samples shall be minimum 12 inches wide by minimum 42 inches long with the seam centered lengthwise. One 1-inch wide strip shall be cut from each end of the samples, and these shall be tested in the field as described below. The remaining sample shall be distributed as follows:

One portion (minimum 12 inches by 12 inches) to the Installer for laboratory testing at his discretion.

One portion (minimum 12 inches by 12 inches) to the Construction Manager for archive storage.

One portion (minimum 12 inches by 18 inches) to the CQA Certifying Engineer for laboratory testing.

Field Testing:

The two 1-inch wide strips described above shall be tested in the field by tensiometer for peel and shear and shall not fail in the seam. If any test sample fails to pass, then the procedures outlined below (Procedures for Areas Failing Destructive Tests) shall be followed.

The CQA Certifying Engineer will mark all samples and portions with its number. The CQA Certifying Engineer will also record the date and time, ambient temperature, number of seaming unit, name of seamer, welding apparatus temperatures and pressures, and pass or fail descriptions, and attach a copy to each sample portion.

Procedures for Areas Failing Destructive Tests: The following procedures shall apply whenever a sample fails a destructive test, whether that test is conducted by the independent testing laboratory, the Installer's laboratory, or by field tensiometer. The Installer has two options:

The Installer shall cap the seam between any two passing test locations, or

The Installer shall trace the seam to two intermediate locations 10 feet minimum from the point of the failed test in each direction and take a small sample for an additional field test at each location. If these additional samples pass the test, then full samples shall be taken for laboratory testing.

If these laboratory samples pass the tests, then the seam shall be capped between these locations. If either sample fails, then the sampling and testing process shall be repeated to establish the zone over which the seam shall be capped.

All acceptable capped seams shall be bounded by two locations from which samples passing CQA laboratory destructive tests have been taken. In cases where the length of the capped seam exceeds 150 feet, a sample of the capping seam shall be taken and shall pass destructive testing as described in this Part.

Repairs - Polyethylene Geomembranes

General:

Any portion of the geomembrane exhibiting a flaw or failing a destructive or nondestructive test shall be repaired. All repairs shall be conducted in accordance with this Part. All repairs shall be subjected to the nondestructive seam testing procedures described in this Part.

Each patch or other type of repair will be numbered and recorded.

Repair Procedures:

Patching, used to repair large holes, tears, undispersed raw materials, and contamination by foreign matter.

Grinding and rewelding, used to repair small sections (typically with a maximum length of no more than several inches) of extruded seams.

Spot welding or seaming used to repair pinholes.

Capping, used to repair large lengths of failed seams or areas where large wrinkles or fish mouths have been cut to flatten the geomembrane sheet.

Topping, used to repair areas of inadequate seams, which have an exposed edge. Topping shall be limited to an aggregate length of no more than 3 m (10 feet) on any given seam.

Removing bad seam and replacing with a strip of new material welded into place, used with large lengths of fusion seams.

For all repair methods, the following provisions shall be satisfied as applicable:

Surfaces of the geomembrane, which are to be repaired, shall be abraded no more than one hour prior to the repair.

All surfaces shall be clean and dry at the time of the repair.

Patches or caps shall extend at least 6 inches beyond the edge of the defect, and all corners of patches shall be rounded with a radius of at least 3 inches.

The geomembrane below large caps shall be appropriately cut to avoid water or gas collection between the two sheets.

Verification of Repairs: Each repair shall be numbered and recorded. Each repair shall be nondestructively tested using the methods described in this Part. Large caps may be of sufficient extent to require destructive test sampling. Repairs that fail nondestructive or destructive tests shall be redone and retested until a passing test is obtained.

Protection Of Termination Edges

Along the south termination of the geomembrane, and along any termination edges of the membrane that may be exposed or buried for extended periods of time prior to their joining to adjacent subsequent sections, the Construction General Contractor shall protect leading edges with protective (sacrificial) layers of cushion geotextile and plywood sheet as shown on the Drawings.

Materials In Contact With Geomembrane

The requirements of this Part are intended only to assure that the installation of other materials does not damage the geomembrane. Additional requirements as established in other sections of these Specifications are necessary to assure that systems built with these other materials are constructed in such a way as to provide proper performance. Material requirements for operations layer and drain gravel are specified in Section 02315, FILL AND BACKFILL.

Requirements of this Part apply to geomembranes that are directly in contact with overlying soil or are covered with a layer of geotextile or geocomposite.

Do not place granular materials in manner that will cause wrinkles to fold over or become confined to form a vertical ridge. Maximum wrinkle height shall be 4 inches and spacing between wrinkles shall be greater than 10 feet prior to placement of granular materials over the geomembrane.

Minimum Thickness: Equipment used for spreading granular material shall not be driven directly on the geomembrane. A minimum thickness of 1 foot of granular material shall be maintained between spreading equipment and the geomembrane. A minimum thickness of 3 feet of granular material shall be maintained between rubber-tired hauling vehicles and the geomembrane. Construction haul vehicles shall have a maximum ground contact pressure of 25 psi.

Spreading Equipment: Equipment used for spreading granular material shall be a light low ground pressure dozer (such as a wide-pad Caterpillar D6M LGP or lighter), low ground pressure excavator (Bucyrus-Erie 325H with 0.91-m [36-in] wide treads or lighter), or approved equal, with a maximum ground contact pressure of 5 psi.

Spreading Operations:

Spreading equipment operating on soil materials shall not spin their tracks, make sharp turns, or make sharp, rapid starts or stops. Soil materials shall be pushed carefully from previously placed material and not dumped directly onto geosynthetics except for the drain gravel in the LDS sump and operations layer material in SLDS sump. This material shall be carefully dumped onto the cushion geotextile or SLDS CDN from a maximum height of 24 inches.

The spreading operation on the sideslope (3H:1V) shall begin at the lower elevations and shall proceed either upslope or laterally at about the same elevation such that a full layer of granular material is always covering the geomembrane downslope from the area being covered. In no case shall the lift thickness be less than the stated minimum. Material shall be placed in such a manner that no air is trapped underneath the geomembrane. Provide and maintain a means of continuously observing the depth of granular materials such as by freestanding markers until placement is complete, at intervals of 50 feet maximum each way. Sharpened stakes or methods that could damage the geomembrane will not be allowed.

Lining System Acceptance - Polyethylene Geomembranes

The Installer shall retain all ownership of and responsibility for the geosynthetics in the lining system until acceptance by the Construction Manager.

The geosynthetic lining system will be accepted by the Construction Manager when all of the following requirements have been satisfied:

The installation is finished.

Verification of the adequacy of all field seams and repairs, including associated testing, is complete.

A written construction report, including "as built" drawings and all other installation documents, has been prepared by the CQA Certifying Engineer, sealed by a registered professional engineer, and received by the Construction Manager.

Supplements

The supplements listed below, following "END OF SECTION", are a part of this Specification.

Table 1. Required Geomembrane Properties, 60-mil Textured HDPE.

Table 2. Required Seam Properties.

Geomembrane Installer's Certification of Subsurface Acceptability.

END OF SECTION 02661

Table 1. Required Geomembrane Properties 60-Mil Textured HDPE

Specified Property	Qualifier	Unit	Value	Test Method
Physical Properties				
Thickness	min. avg. value	mils	60	ASTM D5994
	minimum		54	ASTM D5994
Specific Gravity	minimum	`N/A	0.932	ASTM D1505
Melt Index	range	g/10 min	<1.1	ASTM D1238 condition 190/2.16
Asperity	min avg. value[1]	mils	10	GRI-GM12
Mechanical Properties				
Tensile Properties	(each direction)		(Type IV)	ASTM D638
Strength at yield	min. avg. value	lb/in	120	
Elongation at yield[2]	min. avg. value	%		12
Tear Resistance	min. avg. value	lb	42	ASTM D1004
Puncture Resistance	min. avg. value	lb	80	ASTM D4833
Carbon Black Content	Range	%	2-3	ASTM D1603 or D4218
Carbon Black Dispersion	Minimum[1] 8 of 10	category	1 or 2	ASTM D5596
Environmental Stress				
Crack	minimum[3]	hrs	200	ASTM D5397

[1] Of 10 readings, 8 out of 10 must be greater or equal to 7 mils, and lowest individual reading must be greater or equal to 5 mils. Provide data for both sides of textured geomembrane.

[2] Yield elongation is calculated using a gauge length of 1.3 inches.

[3] Minimum = mean minus 3 standard deviations from documented manufacturer's quality control (MQC) testing.

Table 2. Required Seam Properties Hdpe Geomembranes

Property	Qualifier	Unit	Specified Value	Test Method
Shear Strength[1]	minimum	lb/in width	90% of tensile strength at yield as listed in tables in this section	ASTM D6392
Peel Adhesion	minimum	lb/in width	60% of tensile strength at yield as listed in tables in this section and FTB[2]	ASTM D6392

[1]Also called "Bonded Seam Strength".

[2]FTB = Film Tear Bond (failure occurs through intact geomembrane, not through seam).

Geomembrane Installer's Certification of Subsurface Acceptability

The geomembrane installer, _____
for the Integrated Disposal Facility (IDF), hereby certifies that the supporting prepared subgrade surfaces are acceptable for installation of the HDPE geomembrane lining system, the undersigned having personally inspected the condition of the constructed surfaces. This certification is for the areas shown on Attachment or defined as follows:

_____

_____

_____

_____

The condition of the supporting surfaces in the defined area meets or exceeds the minimum requirements for installation of the geomembrane.

Signed:_____ Signed:_____
Geomembrane Installer Construction General Contractor

Date Signed_____ Date Signed_____

SECTION 02666-ADMIX LINER SCHED. B

Part 1 – General

References

The publications listed below form a part of this Specification to the extent referenced. The publications are referred to in the text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D698	Test Method for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbs/ft^3) (600 Kn-m/m^3)
ASTM D422	Method for Particle-Size Analysis of Soils
ASTM D2216	Method for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
ASTM D6391	Standard Test Method for Field Measurement of Hydraulic Conductivity Limits of Porous Materials Using Two Stages of Infiltration from a Borehole

Description

This section describes the low permeability admix that will be used in the liner of the disposal facility. In addition requirements for base soil in the lining system are specified.

The admix liner is an admixture that consists of natural base soil which is mixed with bentonite and moisture conditioned.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

The Construction General Contractor shall submit a detailed plan for preparation of the admix material, including a description of the equipment and procedures to be used, personnel qualifications, and methods for monitoring bentonite additions and moisture conditioning. This plan shall be approved by the Engineer prior to the start of admix production.

The Construction General Contractor shall submit an admix liner placement plan to specify lift thickness control and to allow for required testing, specified herein and described in the CQA Plan, on the soil liner during placement operations. This plan shall be approved by the Engineer prior to the start of admix placement.

Supplier's descriptive data, specification sheets, literature, and other data as necessary to fully demonstrate that the bentonite proposed for use in the admix complies with the requirements of these Specifications. The manufacturer shall certify that the bentonite furnished complies with these Specifications. A certificate shall be submitted to the CQA Certifying Engineer for each railcar or every three truckloads of bentonite delivered.

Certificates for equipment calibration.

Construction Quality Assurance

Construction Quality Assurance testing will be provided by the CQA Certifying Engineer and shall be performed in accordance with the CQA Plan. The Construction General Contractor shall make allowances for sampling and testing by the CQA Certifying Engineer in both his production operations and schedule.

Prior to placing any materials over the completed admix liner, the Construction General Contractor shall allow time for acceptance of the Work as listed in the CQA Plan. The Construction General Contractor shall accommodate all CQA testing and sampling activities, as specified in the CQA Plan (i.e., in addition to the QC testing), and shall repair sample locations as specified herein.

Definitions

Grain Size: Determined by ASTM D422.

Imported Material: Meets requirements of this Specification and is obtained offsite and transported to site.

Natural Moisture Content: Determined by ASTM D2216.

Optimum Moisture Content: Determined in accordance with ASTM D698 specified to determine maximum dry density for relative compaction.

Admix: Natural material (base soil), as specified in this section, that when mixed with bentonite.

Admix Liner: Compacted liner component consisting of admix materials, designed, formulated, and constructed to provide low-permeability barrier against infiltration of liquids or contaminants.

Tolerances

Thickness of the admix liner shall be a minimum of 3 feet and tolerances for the top of admix grade shown on Phase I Primary Liner Plan shall be minus 0.1 foot to plus 0.3 foot. The top surface of the admix liner along the north-south centerline of each cell shall be graded to provide a minimum slope of 1.0% for the LCRS collection pipe installation. The minimum required thickness of the soil liner layer shall be maintained. The as-built elevations of the underlying prepared subgrade shall be used as the basis for determining the final elevation of the soil liner layer.

Part 2 – Products

Bentonite

The bentonite shall be Bara-Kade 90, manufactured by Bentonite Performance Materials, Inc.; or approved equal. Do not provide calcium bentonites or chemically treated sodium bentonites.

The bentonite to be used in the admix shall consist of a commercially prepared material and shall have the following index properties:

High Swelling: Ability of 2 grams of base bentonite, when mechanically reduced to minus 100 mesh, to swell in water to an apparent volume of 20 cubic centimeters or more when added gradually to 100 cubic centimeters of distilled water contained in graduated cylinder.

Dry Fineness:

65 percent minimum passing No. 200 U.S. Sieve.

The Construction General Contractor shall provide suitable containers on site to store bentonite in a dry condition prior to use.

Base Soil

The base soil for the admix liner shall consist of natural soil derived from the admix base soil borrow area shown on the Drawings, as specified in Section 02317, BORROW AREA EXCAVATION, or from site excavations. Base soil from either source shall not be excavated below a depth of 5 feet below ground surface (after stripping) without evaluation of the material suitability and approval from the Engineer. This material may be temporarily stockpiled at the stockpile area as shown on the Drawings, or within processing area within Phase I as approved by the Construction Manager.

Base soil excavated from dune sand borrows area or site excavation shall meet the following requirements: The base soil shall be free of roots, woody vegetation, frozen material, rubbish, and other deleterious material. Rocks greater than 1 inch in dimension shall not comprise more than 2 percent by weight of the base soil. Base soil shall have 20 percent minimum passing No. 200 U.S. Sieve. Base soil shall be screened or otherwise processed if necessary to meet this requirement.

For evaluating compliance with these requirements, test results shall be considered acceptable when the average value of the data satisfies the associated criterion. Testing and sampling frequency for base soil compliance is provided in the CQA Plan.

Admix Liner Material (Admix)

Composition: The admix shall consist of the base soil mixed with a nominal bentonite content of 12 percent by dry weight of base soil by dry weight. The acceptable range for bentonite content shall be a minimum of 11 percent and maximum of 14 percent of base soil by dry weight. The bentonite percentage and moisture content range may change as a result of preconstruction testing performed on the test pad as described in the CQA Plan, and may be modified by the Construction General Contractor with the approval of the Construction Manager, Engineer, and the CQA Certifying Engineer at any time during the admix processing to reflect changes in the base soil or other components. The moisture content and bentonite dispersion in the admix shall be uniform and homogenous. The finished admix shall be a uniform homogenous material.

CQA Testing: The admix shall be prepared by the Construction General Contractor and tested by the CQA Certifying Engineer in accordance with the CQA Plan. The Construction General Contractor shall make the admix stockpiles available to the CQA Certifying Engineer at all times for sampling, testing, or visual observation.

Raw Water Supply: See Section 02200, SITE PREPARATION, and the Drawings for information on raw water supply availability and requirements for admix processing and admix liner placement and compaction.

Part 3 – Execution

Admix Processing

Bentonite Use Monitoring: Record weight of bentonite used and volume of admix produced each day.

Processing Using Pugmill:

The Construction General Contractor shall process and condition admix material using a central type pugmill plant prior to compaction. The pugmill shall have automated controls to continuously control the established proportions of bentonite and water as ratios of the base soil. It shall have provisions to easily change the proportions. It shall be capable of maintaining a constant time of mixing and varying the rate of discharge so that the degree of mixing can be controlled if necessary to achieve complete mixing.

The Construction General Contractor shall provide all necessary equipment and labor to operate the pugmill, load material into pugmill, offload admix, and stockpile admix.

Equipment: Admix shall be prepared using a pugmill with the following characteristics and ancillary equipment:

Continuous mixing pugmill. Blades shall be adjustable for angular position on shafts and shall be reversible to retard flow of mix.

Belt scales on base soil, bentonite, and finished product belts.

Feed rate meters and totalizers for bentonite, base soil, and water.

Production rate meters and totalizers for finished product.

Variable speed hydraulic supply water pumps capable of producing 500 tons per hour of admixture.

Calibration:

After setting up the pugmill, it shall be calibrated to determine the accuracy of the feed rate for each material being mixed. When the feed controls are set at any desired rate, the measured accuracy shall be within 1 percent by weight of the indicated feed rate for the item being mixed.

The accuracy will be determined by operating each feed control separately and collecting and weighing the material over a given period of time as determined by the CQA Certifying Engineer. The material shall be collected in a dump truck (or appropriate container) supplied by the Construction General Contractor, and the truck will be weighed by the Construction General Contractor and verified by the CQA Certifying Engineer before and after loading. During admix production; the CQA Certifying Engineer may request a recalibration of the feed rate for each material as described above.

All measuring equipment shall be calibrated and calibration certificates provided to the CQA Certifying Engineer prior to starting admix production.

Preparation Requirements:

The base soil shall be processed through a pugmill to add bentonite at the specified rate, to add additional water if required, and to provide mechanical mixing action required to homogeneously blend the bentonite and water into the mix. Additional mixing by rotovator or other approved means shall be performed at the Construction General Contractor's sole expense as required to further break down the soil clumps or if additional mixing is needed to achieve a homogenous blend of soil, water, and bentonite. The admix shall be broken down in size sufficiently to result in at least 80 percent of the soil clumps broken down to 1/2 inch in maximum size. Clods are defined as dry hard particles in the admix that cannot be remolded by hand pressure.

If mixing is found to be insufficient to produce a thoroughly blended, uniform mixture of base soil and bentonite, or the base soil and bentonite are not being mixed in the specified proportions, the Construction General Contractor shall stop production of admix material. The Construction General Contractor shall not restart production and installation of admix liner until procedures and equipment have been modified so that the specified material is produced. Admix liner that is installed without complete mixing or the correct percentage of bentonite shall be removed and modified by the Construction General Contractor to meet the Specifications. After being modified to meet the Specifications, the material may be reinstalled in the lining. Additional work and delays caused by inadequate or incorrect mixing shall be performed at the Construction General Contractor's sole expense. The Construction General Contractor shall not change the bentonite application rate unless directed to do so by the Engineer in writing.

Admix shall be processed and allowed to cure at least 12 hours prior to placement. The Construction General Contractor shall be responsible for maintaining the moisture content of the admix within the specified limits. Admix that does not meet Specifications shall not be reused as feed stock unless approved by the Engineer and CQA Certifying Engineer.

Test Pad(s)

Test pads for the admix liner shall be constructed as specified in Article ADMIX LINER PLACEMENT AND COMPACTION, by the Construction General Contractor to determine acceptable placement and compaction methods to produce a low-permeability admix liner that satisfies the requirements of this section. Both a horizontal and sideslope test pad shall be constructed. The location of the test pads will be designated by the Construction Manager and the CQA Certifying Engineer.

Test Pad Material: The Construction General Contractor will prepare a sufficient quantity of soil for the test pad in accordance with the requirements of Article ADMIX LINER MATERIAL. All specified procedures for mixing, conditioning, and stockpiling of the soil material will be followed.

Horizontal Test Pad Construction:

The test pad will be constructed on a horizontal surface within the limits of the IDF in an area representative of conditions beneath the waste disposal cells. The pad will be located in a well-drained area to prevent surface water intrusion or saturation of the test pad soils.

The test pad location will be cleared and grubbed, and the subgrade will be compacted in the same manner anticipated for construction beneath the waste disposal cells. Prior to placement of the test pad materials, the Contractor's Site Superintendent and the CQA Certifying Engineer will evaluate the condition of the subgrade; areas containing potentially unsuitable materials will be replaced, or another location will be selected for the test pad.

So that the test pad will accurately represent the performance of the full-scale facility, the following guidelines will be followed:

- Construction of the test pad will use the same soil materials, design specifications, equipment, and procedures as proposed for the full-scale facility.
- The test pad will be constructed at least four times wider than the construction compactor drum width to be used for the full-scale facility and allow for installation of field permeability testing per method ASTM D6391 or 50 feet minimum (whichever is greater). This is required to ensure a sufficient representative area for testing, avoiding the edges of the test pad. The test pad may be subdivided into "lanes" to facilitate evaluation of different compaction methods; however, the width of any individual lane shall be no less than twice the width of the construction compactor drum equipment.
- The test pad will be long enough to allow construction equipment to achieve normal operating speed before reaching the area that will be used for testing or 80 feet minimum (whichever is greater).
- The test pad will be constructed with at least six lifts to evaluate the methodology used to tie lifts together. Lift thickness will be as described in Article ADMIX PLACEMENT AND COMPACTION, and the total thickness of the test pad will be at least 3 feet.
- The test pad constructed will include the removal and replacement of a portion of the soil to evaluate the method proposed for repairing defective portions of the full-scale liner as specified in Article REPAIR OF ADMIX LINER.

Sideslope Test Pad Construction:

The sideslope test pad will be constructed on a 3H:1V sideslope (within the lined area of Phase I) to evaluate compaction methods and performance on the sideslope. Field permeability testing is not required for sideslope test pad. Sideslope test pad will be used to demonstrate that compaction and placement methods to achieve acceptable moisture and density requirements can be achieved.

So that the test pad will accurately represent the performance of the full-scale facility, the following guidelines will be followed:

- Construction of the test pad will use the same soil materials, design specifications, equipment, and procedures as proposed for the full-scale facility.
- The test pad will be constructed at least four times wider than the widest piece of construction equipment to be used for the full-scale facility or 40 feet minimum (whichever is greater). This is required to ensure a sufficient representative area for testing, avoiding the edges of the test pad. The test pad may be subdivided into "lanes" to facilitate evaluation of different compaction methods; however, the width of any individual lane shall be no less than twice the width of the widest piece of construction equipment.
- The test pad will be long enough to allow construction equipment to achieve normal operating speed before reaching the area that will be used for testing or 80 feet minimum (whichever is greater).
- The test pad will be constructed with at least six lifts to evaluate the methodology used to tie lifts together. Lift thickness will be as described in Article SOIL BENTONITE PLACEMENT AND COMPACTION, and the total thickness of the test pad will be at least 3 feet.

Demonstrate the Following During Test Pad(s) Construction:

- Base soil/bentonite mixing process prior to compaction.
- Compaction equipment type, configuration, and weight.

The method used to break down clods before compaction and maximum resulting clod size.

The speed of compaction equipment traveling over the test pad.

Moisture content of soil bentonite at time of compaction.

Lift thicknesses (compacted), compaction procedures, and number of passes for proposed compaction equipment.

Dry unit weight achieved and measured by field density testing.

Hydraulic conductivity of compacted test fill on undisturbed samples (Shelby Tubes) as described in the Construction Quality Assurance (CQA) Plan.

Field permeability of compacted test fill using ASTM D6391 (horizontal test pad only) as described in the CQA Plan.

Excavate at least four holes, each 3 feet square, through each completed pad for observation, sampling, and testing of compacted material. These holes shall be used for the purpose of demonstrating repair methods as specified herein.

No admix liner shall be placed until the associated test pad has been constructed and the results from all test methods indicate that the admix liner will satisfy the permeability requirements specified in this section. Testing for each test pad shall be as described in the CQA Plan. At the completion of the test pad(s), the CQA Certifying Engineer, as described in the CQA Plan, will prepare an interim report with recommendations for compaction and placement methods to be applied to the full-scale admix liner construction.

After all testing has been completed and approved, the material in the test pad can be used by the Construction General Contractor for liner construction provided that the material satisfies the requirements of these Specifications.

Subgrade Preparation

As specified in Section 02319, SUBGRADE PREPARATION, Article PREPARED SUBGRADE FOR ADMIX LINER.

Admix Liner Placement and Compaction

Lift Thickness: Admix liner material, as specified in Article ADMIX LINER MATERIAL, shall be placed in loose lifts, and compacted such that the compacted lift thickness is 6 inches or less (within a tolerance of 0.1 foot). However, the first lift of admix liner placed over subgrade soils may be placed and compacted to a maximum thickness of 8 inches (within a tolerance of 0.1 foot).

Placement methods shall prevent excessive mixing of admix liner with subgrade soil.

Compaction: The intent of this Specification is that admix liner shall be produced to meet an in-place performance specification of less than 1×10^{-7} cm/sec hydraulic conductivity within the limits of edge of liner shown on the Drawings. See paragraph Outside Edge of Liner in this Article for compaction and hydraulic conductivity requirements beyond edge of liner. The Construction General Contractor is responsible to develop and use compaction methods that produce the required relative compaction.

The moisture-density ranges of the compacted admix shall lie within a trapezoidal-shaped field with the following corners:

Moisture Content (%)	Dry Density (pcf)
8	126
12	110
14	126
19	110

Hydraulic Conductivity: The in-place compacted admix liner shall achieve a saturated hydraulic conductivity as listed below:

All field (in-situ) test results shall be 1 x 10^{-7} cm/sec or less. Field (in-situ) hydraulic conductivity tests will be performed only on admix liner of the horizontal test pad.

All hydraulic conductivity during admix production placement will be verified on undisturbed Shelby tube samples (see ASTM D1587) obtained from in-place admix liner per the CQA plan and then tested in laboratory using methods as described in ASTM D5084.

Hydraulic conductivity will be verified on undisturbed samples from completed areas of the liner as described in the CQA Plan. The arithmetic average of laboratory test results on undisturbed Shelby tube samples shall be less than or equal to 5 x 10^{-8} cm/sec, with no individual test result exceeding 1 x 10^{-7}cm/sec. The acceptable values for laboratory test results will be verified or adjusted prior to admix liner production placement based on the correlation of field and laboratory hydraulic conductivity test results from the horizontal test pad. The final laboratory hydraulic conductivity requirement will be approved by the CQA Certifying Engineer and documented in the horizontal test pad report.

Outside Edge of Liner: Compact admix liner as specified herein. Performance specification for admix hydraulic conductivity do not apply to admix placed beyond edge of liner.

Uniformity: The compacted soil distribution and gradation throughout the liner shall be free from lenses, pockets, streaks, layers, or material differing substantially in texture, moisture content, dry density, or gradation from surrounding material. The admix liner material shall be free of organic debris, frozen material, rubbish, construction debris, and other deleterious material. Any soil containing unacceptable material shall be removed and discarded in the permanent stockpile, placed in accordance with Section 02315, FILL AND BACKFILL.

Moisture Conditioning:

The moisture content of the admix liner shall be uniform throughout each lift prior to and during compaction of the material. If the moisture content of a lift of compacted admix liner falls below the acceptable limit during placement operations, the Construction General Contractor shall moisture condition the dry soil and re-compact the lift prior to placement of additional lifts. If the moisture content of a lift of compacted soil exceeds the acceptable limit due to precipitation or over watering, the Construction General Contractor, before placement of additional lifts, shall either allow the wet soil to dry back or remove the wet soil. If the admix liner material cannot be conditioned to meet the placement specifications, the material shall be removed and replaced with new admix liner.

When the final lift of admix liner placement will be interrupted for more than a few hours or when precipitation is imminent, as determined by the Contractor's Site Superintendent, the lift surface shall be sealed with a smooth drum roller to prevent excessive moisture infiltration. This surface shall be scarified with a rotovator, or other equivalent equipment, immediately prior to resuming soil placement. The Construction General Contractor shall verify that existing moisture content is within the range specified in Article ADMIX LINER PLACEMENT AND COMPACTION, prior to resumption of soil placement activities.

Placement Equipment: The Construction General Contractor shall place layers of the admix liner to form a continuous monolithic material. All admix liner shall be placed and compacted with a self-propelled pegfoot or padfoot roller compactor having a minimum operating weight of 68,000 pounds. Smaller compaction equipment may be used in limited areas as necessary provided that the required moisture/density, lift bonding, and hydraulic conductivity can be achieved. Hydraulic conductivity performance specification for the admix liner will be verified in areas where the lighter equipment is used. Hauling and spreading equipment will not be considered as compaction equipment. The compactor feet shall be sufficiently long to knead (bond) new lifts into previously placed lifts.

The feet shall be kept free of large amounts of dried soil that might restrict foot penetration or become incorporated into the soil lift. The top of each lift may be scarified with a rotovator, or other equivalent equipment or procedures, prior to placing the subsequent lift. The final lift of admix liner may be compacted with a smooth drum roller provided that all other requirements are met.

Provide a smooth soil surface on the final lift prior to placement of the HDPE geomembrane as specified in Article SURFACE FINISHING.

Tie-in Areas: Where new admix liner is tied in to existing admix of a previous day's placement, any areas of the existing admix which are soft, cracked, or otherwise unsuitable shall be removed until acceptable material is exposed. Where new admix will be placed, the surface of the existing admix liner shall be scarified and moisture conditioned as described in this section. New admix liner shall be placed in accordance with the requirements of this section and shall be thoroughly kneaded into the existing admix liner to form a monolithic mass free of seams or other discontinuities.

Placement Method: Admix liner may be placed on the sideslopes in either horizontal lifts (along the contour) or in lifts parallel to the slope (up and down the slope). If admix liner is placed parallel to the slope, compaction equipment shall not spin their wheels or in any other way disturb the previously placed lifts. If this occurs, the Construction General Contractor shall place all of the admix liner in horizontal lifts.

Restrictions: Production, mixing, and stockpiling of admix or native clay soil shall be restricted to the area shown on the Drawings or within the Phase I footprint as approved by the Construction Manager.

Surface Finishing

The surface of the admix liner shall be trimmed to the design grades and tolerances as shown on the Drawings. The surface of the admix liner shall be rolled with a smooth-drum roller to remove all ridges and surface irregularities as specified in Section 02319, SUBGRADE PREPARATION. All wheel ruts in excess of depths specified in Section 02319, SUBGRADE PREPARATION, on the surface of the admix liner shall be repaired by the Construction General Contractor prior to placement of the geomembrane. Acceptable methods for repair of the admix liner are specified in Article REPAIR OF ADMIX LINER.

Maintenance

The Construction General Contractor shall maintain the admix liner surface in a condition suitable for geomembrane installation until the surface is covered. The admix liner shall be protected from desiccation or excessive moisture. This may be accomplished by periodic watering, exclusion of traffic, placement of a temporary removable plastic cover, or other methods. Desiccation cracks larger than 1 inch deep or 0.25 inch wide shall be excavated to the full depth of the crack and repaired as specified in Article REPAIR OF ADMIX LINER. In the event that the geomembrane cannot be installed within 12 hours after placement of the final admix liner lift, the final lift of admix liner shall be constructed 4 to 6 inches thicker than required and cut to finish grade immediately before geomembrane deployment.

The Construction General Contractor shall take measures to prevent the admix liner from freezing. Lifts of admix liner shall not be placed on frozen surfaces. Geomembrane shall not be placed on a surface, which is frozen or has been frozen and thawed until directed by the Construction Manager and the CQA Certifying Engineer.

Repair of Admix Liner

The Construction General Contractor shall repair the surface of any areas identified to be out of tolerance. The size of the repair area shall be as required to remove and/or repair defective areas of the admix liner. Repair as follows:

Remove soil that does not meet specifications.

Scarify surface and spray with water.

Place additional approved admix material.

Compact soil with self-propelled pegfoot or padfoot type compactor as described above.

Trim and roll the surface as described above to design grades and tolerances.

Alternative methods for repair of the admix liner will be allowed if submitted by the Construction General Contractor and approved by the Engineer and Construction Manager.

Construction General Contractor will repair small holes (up to a maximum 6-inch diameter) resulting from sampling and other CQA activities. Such holes shall be repaired by backfilling with admix liner or powdered bentonite material in lifts of no more than 2-inch thickness and hand tamping with a steel rod or other suitable device to firmly compact each lift.

Construction Quality Assurance and Acceptance

Testing and criteria for admix liner acceptance is provided in the CQA Plan, which is made part of these Specifications by reference.

END OF SECTION 02666

SECTION 02667 - GEOSYNTHETIC CLAY LINER (GCL) SCHED. B

Part 1 – General

References

The publications listed below form a part of this Specification to the extent referenced. The publications are referred to in the text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D4632	Standard Test Method for Grab Breaking Load and Elongation of Geotextiles
ASTM D4643	Standard Test Method for Determination of Water (Moisture) Content of Soil by the Microwave Oven Method
ASTM D4833	Standard Test Method for Index Puncture Resistance of Geotextiles, Geomembranes and Related Products
ASTM D5084	Standard Test Method for Measurement of Hydraulic Conductivity of Saturated Porous Materials Using a Flexible Wall Permeameter
ASTM D5887	Standard Test Method for Measurement of Index Flux Through Saturated Geosynthetic Clay Liner Specimens Using a Flexible Wall Permeameter
ASTM D5890	Standard Test Method for Swell Index of Clay Mineral Component of Geosynthetic Clay Liner
ASTM D5891	Standard Test Method for Fluid Loss of Clay Component of Geosynthetic Clay Liners
ASTM D5993	Standard Test Method for Measuring Mass per Unit of Geosynthetic Clay Liners

Description

The Work includes supply and installation of geosynthetic clay liners (GCLs) for the landfill lining system as shown on the Drawings.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Submit manufacturer's descriptive data, specification sheets, literature, and other data as necessary to fully demonstrate that the materials proposed for use comply with the requirements of these Specifications.

Submit manufacturer's quality control test results, written properties certification guarantee, and QC certificates, as specified in PART 2-PRODUCTS.

Construction Quality Assurance (CQA)

Quality assurance procedures are presented in the CQA Plan. CQA Plan requirements are discussed in Section 02661, GEOMEMBRANES. The Construction General Contractor shall accommodate all quality assurance activities described herein and in the Construction Quality Assurance Plan (CQA Plan) for this project.

Prior to placing any materials over the installed GCL, the Construction General Contractor shall allow time for acceptance of the Work as listed in the CQA Plan.

CQA Conformance Testing

Prior to deployment of the GCL, CQA personnel will remove samples and forward them to an approved geosynthetics laboratory for testing to document conformance to both the design specifications and the list of guaranteed properties.

Unless otherwise specified, samples shall be taken at a rate of one per lot or one per 50,000 square feet, whichever results in the greater number of tests. Samples shall be taken from any portion of the roll that has not been damaged. Unless otherwise specified, samples shall be 3 feet long by the roll width. The machine direction shall be marked on the samples with an arrow.

As a minimum, the following tests shall be performed on GCL:

1. Bentonite Swell Index: ASTM D5890.
2. GCL Index Flux: ASTM D5887.
3. Bentonite Mass per Unit Area: ASTM D5993.

The CQA Certifying Engineer shall be allowed to remove samples of GCL for testing and other activities. Sample dimensions, procedures, and frequency shall be the same as those specified in the CQA Plan. The Construction General Contractor shall assist the CQA Certifying Engineer as necessary in all sampling and testing activities.

Procedures for samples that fail conformance testing are outlined in the CQA Plan. The cost of additional conformance testing to demonstrate compliance of failed samples shall be borne by the Construction General Contractor.

Part 2 – Products

GCL

GCL shall be BENTOMAT(R) DN as manufactured by Colloid Environmental Technologies Co. (CETCO), Arlington Heights, IL; or Bentofix NWL as manufactured by GSE Lining Systems, Houston, TX; or approved equal, and shall meet the following requirements:

Property	Requirement	Test Method
Bentonite Mass/Area, lb/sq ft at 0%		
moisture content, MinARV	0.75	ASTM D5993
Bentonite Fluid Loss, mL, MaxARV	18	ASTM D5891
Bentonite Swell Index, mL/2g, MinARV	24	ASTM D5890
Grab Strength, lb, Tested Dry, MinARV	150	ASTM D4632
Peel Strength, lb, Tested Dry, MinARV	15	ASTM D4632
Index Flux, $m^3/m^2/sec$, MaxARV	$1x10^{-8}$	ASTM D5887
• 2 psi Water Head Pressure		
• 5 psi Effective Confining Pressure		
Permeability with Water, cm/sec, MaxARV	$5x10^{-9}$	ASTM D5084
• 2 psi Water Head Pressure		
• 5 psi Effective Confining Pressure		
Finished GCL Roll Width, Feet, MinARV	14	Linear Measurement
Finished GCL Roll Length, Feet, MinARV	150	Linear Measurement

The bentonite in the GCL shall be a sodium montmorillonite clay.

The GCL shall be manufactured so that the bentonite shall be continuously contained throughout the GCL and to support the geotextiles so that no displacement of the bentonite occurs when the material is unrolled, moved, cut, torn, or punctured. GCL products that utilize an alternate edge system with grooves cut in seam overlap areas are permitted subject to approval of the Engineer.

Any adhesive used shall be inert, nontoxic, and water-soluble. GCL materials made without the use of adhesives shall be stabilized to contain the granular bentonite by a process such as needle punching or stitching through the top and bottom layers of geotextile and the bentonite.

Encapsulating geotextile materials shall be polypropylene, consisting of two nonwoven geotextile components, which are needle-punched together. The nonwoven components of the GCL shall have a nominal mass per unit area of 6 ounces per square yard needle-punched geotextile.

Quality control testing shall be carried out by the manufacturer to demonstrate that the GCL meets the specifications in this section. Tests shall be conducted on each production lot of GCL or every 50,000 square feet, whichever results in the greater number of tests.

The manufacturer shall provide a written guarantee that the GCL has the properties listed on the specification sheet.

The GCL manufacturer shall provide a quality control certification that the GCL has the properties listed on the specification sheet for each roll of GCL shipped to the project site. The quality control certificate shall be signed by a responsible party employed by the manufacturer, such as the production manager. The quality control certificate shall include:

1. Roll numbers and production lot identification.
2. Results of manufacturer quality control tests.
3. Results of bentonite supplier quality control tests for bentonite used in GCL production.

Accessory Bentonite

Accessory bentonite for seaming shall be as recommended by the GCL manufacturer.

Transportation, Handling, and Storage

Transportation of the GCL shall be the responsibility of the manufacturer, and the Construction General Contractor. All handling on site shall be the responsibility of the Construction General Contractor.

Upon delivery at the site, the Construction General Contractor shall observe the surfaces of all rolls for defects and for damage. This inspection shall be conducted without unrolling rolls unless defects or damages are found or suspected. The Construction General Contractor will determine:

Rolls, or portions thereof, which should be rejected and removed from the site because they have severe flaws.

Rolls that are not properly labeled. No unlabelled rolls shall be used for any application. Unlabelled rolls shall be removed from the site and replaced at the Construction General Contractor's expense.

The Construction General Contractor shall be responsible for the storage of the GCL onsite. The Construction General Contractor shall provide storage space in a location as shown on the Drawings or as approved by the Construction Manager such that on-site transportation and handling are optimized to the extent possible. Storage space shall be protected from theft, vandalism, passage of vehicles, etc. Stored GCLs shall be protected from moisture and other damaging conditions in accordance with the manufacturer's recommendations.

Part 3 – Execution

General

Install GCLs at the locations, lines, and grades shown on the Drawings. All GCLs shall be installed in accordance with these Specifications.

Materials and work, which fail to meet the requirements of these Specifications, shall be removed and disposed of at the Construction General Contractor's expense. This includes GCL rolls that are not labeled or where the label has deteriorated to the point of being illegible.

Handling And Placement

The Construction General Contractor shall handle and deploy all GCLs in such a manner as to ensure that they are not damaged.

Surface Preparation – Primary GCL

For the IDF project, primary GCL will be deployed over the CDN surface. Primary GCL shall be placed over a firm, unyielding surface. Wrinkle height in the underlying LDSCDN shall be minimized as to allow primary GCL deployment on a flat unyielding surface. Maximum wrinkle height for geosynthetics is specified in Section 02661, GEOMEMBRANES.

Surface Preparation – Secondary GCL

As specified in Section 02319, SUBGRADE PREPARATION.

Deployment

GCL shall be deployed so that seams run up and down (not across) the slope.

Prior to placement of cover material over the GCL and HDPE geomembrane, the moisture content of the bentonite component of the GCL shall not exceed 100 percent.

Only areas of GCL suspected of exposure to excessive moisture, in the judgment of the CQA Certifying Engineer, shall be sampled for moisture content. GCL panels with bentonite component moisture content greater than 100 percent shall be removed and replaced at Construction General Contractor's expense, regardless of the source of moisture, including adsorption from subgrade soil and/or condensation under the HDPE geomembrane or temporary plastic cover.

Any wrinkles in excess of the maximum wrinkle height specified in Section 02661, GEOMEMBRANES, shall be reduced to below specified height by adjusting and smoothing the GCL after placement.

GCL shall not be deployed during precipitation or in the presence of moisture, ponded water, snow, or in other situations that could cause premature hydration of the bentonite. Any GCL that hydrates prematurely shall be removed and replaced at the Construction General Contractor's expense.

The panels shall be placed to provide an overlap of 6 inches on longitudinal (edge of roll) seams, regardless of slope steepness. The panels shall be placed to provide an overlap of 24 inches on transverse (end of roll) seams for slopes flatter than 6H:1V. No transverse seams shall be allowed on slopes 6H:1V and steeper.

No more GCL shall be deployed than can be covered with geomembrane or other protective layer the same day.

Provide protection from wind uplift as necessary using sandbags or other method that will not damage the GCL.

Overlapping GCL Panels

Overlap marks 6 inches from the panel edge shall be marked longitudinally on the GCL to assist in obtaining the proper overlap.

Prior to lapping, remove all dirt, gravel, or other debris from the overlap area. Apply 1/4 pound of accessory per linear foot of seam. Lap areas that have been contaminated by soil and/or sand shall receive additional accessory bentonite in the amount of 1/4 pound per linear foot evenly spread across the longitudinal seam area. GCL products with alternate edge treatment system with grooves cut in the seam overlap area that eliminate the requirement for accessory bentonite, are permitted for edge of roll seams with prior approval by the Engineer. Accessory bentonite shall be required for end of roll seams.

End of roll overlap on slopes less than 6H:1V shall be shingled so that the direction of flow is from the top panel onto the bottom panel.

On slopes 6H:1V and steeper, the panels shall be placed with the long dimension (length) continuous from the crest to the toe and the upper end anchored in a trench with soil backfill as shown on the Drawings.

Repairs

Replace or repair damaged or hydrated areas of GCL.

Place a patch of GCL that extends at least 12 inches beyond the edges of the damaged area in all directions.

Overlap areas shall conform to requirements for seams described above.

Placement Of Overlying Materials On Geosynthetic Clay Lining

The GCL shall be completely covered with HDPE geomembrane or temporary plastic cover and protected at the end of each shift or workday. The Construction General Contractor shall be fully responsible to protect the GCL from damage, shrinkage, or prehydration and shall replace all affected materials at the Construction General Contractor's sole expense.

To prevent premature hydration or shrinkage in hot weather, only the amount of GCL that can be anchored, inspected, repaired, and covered with HDPE geomembrane or temporary plastic cover in the same day shall be installed.

Equipment used to install the overlying materials shall not operate directly on the GCL.

Construction General Contractor shall use a "rub sheet" of smooth HDPE geomembrane between the GCL and textured HDPE geomembrane to prevent damage to the GCL while maneuvering the textured HDPE geomembrane into position for seaming. Construction General Contractor shall develop method(s) of removing rub sheet that, after maneuvering textured HDPE geomembrane into place, prevents damage to toe underlying GCL.

Overlying materials shall be placed over the GCL and HDPE geomembrane as specified in Section 02661, GEOMEMBRANES.

END OF SECTION 02667

SECTION 02920 – RECLAMATION AND REVEGETATION SCHED. A & B

Part 1 – General

Summary

This section includes, but is not limited to, stabilization measures to prevent wind and water caused erosion of areas disturbed by the construction.

References

ASTM INTERNATIONAL (ASTM)

ASTM D586 Standard Test Method for Ash in Pulp, Paper, and Paper Products

Submittals–Approval Required

See Section 01300, SUBMITTALS, for procedures.

Manufacturer's technical data and installation recommendations for erosion control matting, including type and spacing of anchorage devices.

Manufacturer's written certification that wood fiber mulch product contains less than 250 parts per million boron, is nontoxic to plant and animal life, and satisfies the specified organic matter content as determined by ASTM D586.

Submittals–Approval Not Required

Information/Record (IR):

Tackifier and Mulch: Submit manufacturer's information and/or product data sheets for the tackifier selected for use on this project. Include confirmation of the application rate, which will be used. Submit Material Safety Data Sheets for the tackifier and any dye used in the tackifier and mulch application.

Installation warranty.

Part 2 – Products

Materials:

Topsoil: Strippings that are free from toxic minerals, noxious weeds, and other objectionable material shall be used for topsoil in accordance with Section 02200, SITE PREPARATION. Vegetation shall be removed during clearing and grubbing in accordance with Section 02200, SITE PREPARATION. The removed vegetation, other than noxious weeds, shall be reduced to pieces that are no larger than 1 inch in any dimension and shall be incorporated uniformly into the strippings. Large clods, hard lumps, rocks 2 inches in diameter and larger, and litter shall be removed from the topsoil.

Topsoil shall be stockpiled in accordance with Section 02200, SITE PREPARATION.

Grass Seed: Grass seed shall be crested wheatgrass var. Nordan. The grass seed shall conform to the standards for "Certified" grade seed or better as outlined by the State of Washington Department of Agriculture "Rules for Seed Certification," latest edition.

Seed shall be furnished in standard containers on which shall be shown the following information:

Common name of seed.

Lot number.

Net weight.

Percentage of purity.

Percentage of germination.

Percentage of weed seed content and inert material clearly marked in accordance with applicable state and federal laws.

The maximum allowable noxious weed percentage (by weight) is 0.5 percent. The maximum allowable inert percentage is 7 percent.

Upon request, the Construction General Contractor shall furnish to the Construction Manager duplicate copies of a statement signed by the vendor certifying that each lot of seed has been tested by a recognized seed testing laboratory within 6 months before the date of delivery on the project. Seed, which has become wet, moldy, or otherwise damaged in transit or storage will not be accepted.

Fertilizer: Fertilizer shall be either fertilizer Type A or fertilizer Type B.

Fertilizer Type A shall be an organic product developed from byproducts of the manufacture of various antibiotics, enzymes, and proteins. Fertilizer Type A shall provide a slow release of organically bound nutrients including nitrogen, potassium, and phosphorous. It shall have a minimum analysis (nutrient ratio) of 7-2-3 (nitrogen-phosphorous-potassium) with pH in the range of 5.3 to 6.0. Fertilizer Type A shall be sterilized and weed free. Fertilizer Type A shall be supplied in dried, granulated form with the dried weight, contents, and chemical analysis clearly marked on each bag. One suggested product name for fertilizer Type A is Biosol Mix, Rocky Mountain Bio-Products, Inc., Edwards, CO.

Fertilizer Type B shall be an organic product manufactured from seedmeal (60 percent), protein-derived, and fortified with calcium-rich (300 pounds per ton) composted chicken manure. Fertilizer Type B shall provide a slow release of organically bound nutrients including nitrogen, potassium, and phosphorous. It shall have a minimum analysis (nutrient ratio) of 6-4-1 (nitrogen-phosphorous-potassium) with pH in the range of 5.3 to 6.0. Fertilizer Type B shall be biodegradable, nonpolluting, nonvolatile, nontoxic, sterilized and weed free, and contain no heavy metals or salts. Fertilizer Type B shall be supplied in dried, pelletized form with the dried weight, contents, and chemical analysis clearly marked on each container. One suggested product name for fertilizer Type B is Fertile-Fibers Nutrimulch™, Quattro Environmental, Coronado, CA.

Wood Fiber Mulch: Wood fiber mulch shall be produced from natural or recycled (pulp) fiber, such as wood chips or similar wood materials, or from newsprint, corrugated cardboard, or a combination of these processed materials. The fibers shall not contain any rock, metal, or plastic. It shall be suitable for hydromulching and shall be treated with a nontoxic green dye to facilitate inspection of the placement of the material. It shall be manufactured in such a manner that after addition and agitation in slurry tanks with water, the fibers in the material will become uniformly suspended to form a homogenous slurry. When hydraulically sprayed on the ground, the material shall allow the absorption and percolation of moisture. The product shall contain less than 250 parts per million boron and shall be nontoxic to plant and animal life. The organic matter content shall be at least 93 percent on any oven-dry basis as determined by ASTM D586. The moisture content shall be no more than 15 percent as determined by oven-dried weight. Each package of the wood fiber mulch shall be marked by the manufacturer to show the dried weight. Wood fiber mulch shall be added to the tackifier at the rate of 1,500 pounds per acre minimum.

Straw Mulch: Straw mulch shall be air dried straw free of noxious weeds and other materials detrimental to plant life. Straw shall be seasoned before baling or loading. Straw mulch so provided shall be suitable for spreading with mulch blower equipment.

Tackifier: The tackifier shall be an organic guar tackifier derived form natural organic plant sources or a 100 percent polyacrylamide. The tackifier used shall contain no growth or germination inhibiting materials. The guar based tackifier shall be applied at a rate of 60 pounds per acre minimum. If polyacrylamide is used as the tackifier instead of guar, it shall be applied at 5 pounds per acre minimum.

Erosion Control Matting:

Erosion control matting shall be used to prevent erosion of soil due primarily to wind. Erosion control matting shall be a long-life dense matting composed of nylon fiber, polyolefin fiber, or polyester fibers.

The matting shall be of a consistent thickness with the fiber evenly distributed over the entire area of the matting. The fibers shall be encased between two layers of heavy polypropylene or polyolefin netting. The fibers and the netting shall be stitched top to bottom to form a three-dimensional matrix using polyester or polyolefin thread. All components of the erosion control matting shall be stabilized against ultraviolet degradation and inert to chemicals normally encountered in a natural soil environment. The erosion control matting shall have a minimum thickness of 0.5 inch and a minimum weight of 10 ounces per square yard. Three suggested product names for erosion control matting are Landlok ECRM 450, Synthetic Industries, Chattanooga, TN; P300, North American Green, Evansville, IN; and Recylex TRM, American Excelsior Company, Arlington, TX.

Anchorage devices for erosion control matting shall be as recommended by the manufacturer of the erosion control matting and as approved by the Construction Manager.

Soil Stabilization Cover: Soil stabilization cover shall be applied on the finished grade inside side slopes of the Phase I excavation to reduce wind and water caused erosion. Soil stabilization cover shall be a waterborne copolymer emulsion consisting of nonflammable concentrated PVA liquid copolymer with acrylic base having 60 percent solids. On drying the soil stabilization cover shall from a colorless transparent net-like film. Such film shall have permeability to allow exchange of air and moisture and have an effective life of at least 1 year. The copolymer shall not re-emulsify when cured. The liquid copolymer emulsion shall be nontoxic to plants and animals. One suggested product name for soil stabilization cover is Marloc, Reclamare Company, Des Moines, WA.

Part 3 – Execution

Placing Topsoil

A 6-inch thick layer of topsoil, or as otherwise ordered by the Construction Manager, shall be evenly spread over all areas where material has either been excavated from or has been placed in that are to be seeded, including all borrow areas and permanent stockpiles.

Topsoil shall not be placed when the ground or topsoil is frozen, excessively wet, or in the opinion of the Construction Manager in a condition detrimental to the work.

Upon physical completion of the work, remaining topsoil shall be stockpiled at the location shown on the Drawings. The permanent topsoil stockpile shall than be seeded, fertilized, and mulched.

Seeding, Fertilizing, And Mulching

General: Areas to be seeded, fertilized, and mulched are indicated on the Drawings. No seeding, fertilizing, and mulching shall be done within the Phase I excavation, unless directed otherwise by the Construction Manager. Areas to be seeded, fertilized, and mulched include at a minimum the east and west infiltration areas, the berm and ditch located south of the Phase I excavation, soil stockpiles that will remain after the completion of the construction, and borrow areas. Other areas outside of the Phase I excavation that are disturbed by the construction and are not otherwise stabilized shall be seeded, fertilized, and mulched as directed by the Construction Manager.

Season of Work: Seeding shall be done between September 1 and March 1. Specific ideal seeding times within this window shall be as required for proper seedbed preparation.

Weed Control: Areas to be seeded shall be maintained reasonably free of weeds. Weeds shall be kept from going to seed.

Seedbed Preparation:

Soil shall be tilled to a minimum depth of 6 inches. The seedbed shall be firm below seeding depth and well pulverized and loose on top. It shall be free of clods and weeds. Tillage shall leave cross-slope furrows. Seedbed preparation shall not be performed when soil conditions are not suitable for tilling: too dry, too wet, frozen, etc.

Areas to be seeded that have not either had material excavated from them or placed in them shall not receive any seedbed preparation unless directed otherwise by the Construction Manager. The seedbed preparation would destroy any existing soil crust. Existing soil crust provides erosion protection.

Application of Seed and Fertilizer: Seeding and fertilizing shall be done closely following seedbed preparation and shall not be done during windy weather or when the ground is frozen or excessively wet. The Construction General Contractor shall notify the Construction Manager not less than 24 hours in advance of any seeding operation and shall not begin the work until areas prepared or designated for seeding have been approved. Following the Construction Manager's approval, seeding of the approved areas shall begin immediately.

Seed and fertilizer shall be applied by one of the following methods:

Hydroseeding: Use a hydroseeder that utilizes water as the carrying agent, and maintains continuous agitation through paddle blades. It shall have an operating capacity sufficient to agitate, suspend, and mix into a homogeneous slurry the specified amount of seed and water or other material. Distribution and discharge lines shall be large enough to prevent stoppage and shall be equipped with a set of hydraulic discharge spray nozzles that will provide a uniform distribution of the slurry. Seed and fertilizer may be applied in one application provided that the fertilizer is placed in the hydroseeder tank no more than 1 hour prior to application.

Hand Broadcasting:

Apply fertilizer first. The seed shall be incorporated into the top 1/4 inch of soil by hand raking or other method that is approved by the Construction Manager.

Wood fiber mulch shall be added as a tracer to visibly aid uniform application. The application rate of wood fiber mulch used as a tracer shall not exceed 250 pounds per acre.

Seed shall be applied at a rate of 10 to 12 pounds pure live seed per acre. Fertilizer shall be applied at a rate of 1,000 pounds per acre.

Mulching:

Straw mulch shall be evenly applied at a rate of 1.0 ton per acre within 48 hours after seeding. Mulching shall not be performed when wind interferes with mulch placement. Distribution of straw mulch material shall be by means of a mulch spreader that utilizes forced air to blow mulch material on the seeded areas. In spreading straw mulch, the spreader shall not cut or break the straw into short stalks. Straw mulch may be spread by hand over areas that were seeded by hand. Straw mulch shall be crimped into the soil to a depth of 2 inches and with no more than one pass of the equipment.

In areas where it is not possible to crimp the straw mulch into the soil, tackifier shall be applied over the straw mulch. The method of application for tackifier shall be in accordance with the manufacturer's instructions. The tackifier application rate shall be as specified herein. Tackifier shall be sprayed over mulch, seed, and fertilizer. The Construction Manager shall indicate which areas (if any) shall have the straw mulch held down with tackifier.

Mulching can be accomplished by either hydromulching with wood fiber mulch, or by mulch spreader (or by hand) with straw mulch. Using hydromulching, wood fiber mulch will be added to the tackifier at the rate of 1,500 pounds per acre, in addition to the mulch used as tracer for hydroseeding.

Protection: Traffic over seeded areas shall be prohibited.

Installation Warranty:

The warranty period for seeding, fertilizing, and mulching will begin upon the date of acceptance of the completed installation. The installation shall be considered complete by the Construction Manager upon satisfactory completion of the initial inspection, which is described below. Acceptance will be certified in writing by the Construction Manager.

Seeded areas shall be guaranteed by the Construction General Contractor for a period of 1 year. Mulch coverage will be used to evaluate the materials and workmanship of the application of seed, fertilizer, and mulch. Seed, fertilizer, and mulch will be reapplied one time only as directed by the Construction Manager at the Construction General Contractor's expense in areas where the coverage does not meet the following criteria. Three inspections of mulch coverage will occur:

Initial inspection will occur between 1 to 3 business days following completion of the installation for the purposed of Construction Manager acceptance. Mulch coverage must equal 100 percent of the area over which it was spread before Construction Manager acceptance will occur.

Mulch coverage will be inspected 60 days after Construction Manager acceptance at which time mulch coverage must equal 100 percent of the area over which it was spread.

A final inspection will occur 30 days prior to the end of the warranty period. At this time, mulch coverage must equal at least 80 percent of the area over which it was spread.

Placing Erosion Control Matting

Erosion control matting shall be installed in the locations shown on the Plans and as directed by the Construction Manager. The erosion control matting shall be securely anchored to resist the wind. The erosion control matting shall be installed following the manufacturer's recommendations and the following minimum requirements. Where more than one strip of erosion control matting is required, it shall overlap the adjacent matting a minimum of 6 inches. The ends of the erosion control matting shall overlap a minimum of 6 inches with the uphill section on top.

Placing Soil Stabilization Cover

Soil stabilization cover shall be applied with hydroseeding equipment in two passes of opposite directions. Copolymer shall be applied at a minimum rate of 200 gallons per acre; dilution rate for copolymer and water shall be per the manufacturer's recommendation. Wood fiber mulch shall be added as a tracer to visibly aid uniform application. The application rate of wood fiber mulch used as a tracer shall not exceed 250 pounds per acre.

Soil stabilization cover shall be applied when adequate weather conditions for proper curing, as determined by the manufacturer, are anticipated. The Construction General Contractor shall apply soil stabilization cover only to finish graded areas unless directed otherwise by Construction Manager.

END OF SECTION 02920

DIVISION 3–CONCRETE

SECTION 03301 - CONCRETE SCHED. A & B

Part 1 – General

Work Included

This section is a ready-mix concrete and reinforcing bar specification for concrete.

References

The following is a list of standards, which may be referenced in this section:

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 117	Standard Specifications for Tolerances for Concrete Construction and Materials
ACI 301	Specifications for Structural Concrete for Buildings
ACI 305R	Hot Weather Concreting
ACI 306.1	Standard Specification for Cold Weather Concreting
ACI 318/318R	Building Code Requirements for Reinforced Concrete
ACI 347	Formwork for Concrete

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A185	Standard Specification for Steel Welded Wire Fabric, Plain, for Concrete Reinforcement.
ASTM A615	Standard Specification for Deformed and Plain Billet-Steel Bars for Concrete Reinforcement.
ASTM C31	Standard Practice for Making and Curing Concrete Test Specimens in the Field.
ASTM C33	Standard Specification for Concrete Aggregates.
ASTM C39	Standard Test Methods for Compressive Strength of Cylindrical Concrete Specimens.
ASTM C94	Standard Specification for Ready-Mix Concrete.
ASTM C150	Standard Specification for Portland Cement.
ASTM C260	Standard Specification for Air-Entraining Admixtures for Concrete.
ASTM C309	Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete.
ASTM C494	Standard Specification for Chemical Admixtures for Concrete.
ASTM C618	Standard Specification for Fly Ash and Raw or Calcined Natural Pozzolan for Use as a Mineral Admixture in Portland Cement Concrete.
ASTM D994	Standard Specification for Preformed Expansion Joint Filler for Concrete (Bituminous Type).

CONCRETE REINFORCING STEEL INSTITUTE (CRSI)

CRSI	Manual of Standard Practice.
	Recommended Practice for Placing Reinforcing Bars.

1997 UNIFORM BUILDING CODE (UBC)

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Shop Drawings:

Reinforcing steel in accordance with CRSI Manual of Standard Practice and ACI SP-66, Detailing Manual.

Curing compound data.

Complete data on the concrete mix, including aggregate gradations and admixtures, in accordance with ASTM C94.

Statement identifying aggregates reactivity.

Concrete mix design signed by a qualified mix designer that is a licensed professional engineer in the State of Washington.

Submittals–Approval Not Required

Information/Record (IR):

Qualifications of independent qualified testing laboratory for concrete construction quality control.

Manufacturer's application instructions for curing compound.

Statement for batch plant currently certified by the National Ready Mixed Concrete Association. If batch plant cannot be certified, then Construction General Contractor shall demonstrate to the Construction Manager's satisfaction by providing additional testing as determined by the Construction Manager such that the concrete may be determined to be acceptable. These test results shall be submitted to Construction Manager.

Certification for proper functioning of concrete transport trucks including blades, drum rotators, counters, and other components. Certification shall be acceptable only for concrete batch plants certified by the National Ready Mixed Concrete Association. Batch plants that are not certified shall require concrete transport trucks to undergo independent assessment to the satisfaction of the Construction Manager.

Ready-mix delivery tickets for each truck in accordance with ASTM C94.

Concrete crack repair epoxy injection statement of qualifications for manufacturer's site representative, injector applicator, and injector pump operating technician. Submit only if crack repair is required.

Pour slip shall be required prior to any concrete placement to serve as a checklist between the Construction Manager and the Construction General Contractor. This checklist will be used to document Construction General Contractor's readiness for concrete placement. Pour slip form shall be provided to the Construction General Contractor from the Construction Manager.

Results of construction quality control testing.

Environmental Requirements

Do not place concrete when the ambient temperature is below 40 degrees F or approaching 40 degrees F and air temperature less than 40 degrees F for the first 7 days, without special protection to keep concrete above 40 degrees F.

Do not use curing compound where solvents in the curing compounds are prohibited by state or federal air quality laws.

Form sealer shall be a ready-to-use water based material formulated to reduce or eliminate surface imperfections, containing no mineral oil or organic solvents. Environmentally safe, meeting local, state, and federal regulations.

Part 2 – Products

Form Materials

All formwork shall conform to the guidelines in ACI 347.

Forms for Exposed Finish Concrete:

Provide continuous, straight, smooth, exposed surfaces. Furnish in largest practicable sizes to minimize number of joints. Provide form material with sufficient thickness to withstand pressure of newly-placed concrete without visible bow or deflection:

Plywood shall comply with American Plywood Association, grade "EXT-DFPA PLYFORM" or better.

Forms for Unexposed Finish Concrete: Form concrete surfaces, which will be, unexposed in finished structure with plywood, lumber, or metal.

Form Coatings: Provide commercial formulation form-coating compounds that will not bond with, stain nor adversely affect concrete surfaces, and will not impair subsequent treatments of concrete surfaces.

Form Ties:

Steel form ties with conical or spherical spreader insert that will leave no metal closer than 1 inch to concrete surface after tie ends are removed. Wire ties shall not be used.

Water stop ties shall be perpendicular to water stop and symmetrical about the center of tie and designed to prevent rotation or disturbance of center portion of tie during removal of ends and to prevent water leaking along the tie. Maximum spacing shall be 12 inches, and all corners shall be tied.

Concrete Materials

Portland Cement: Cement shall conform to ASTM C150, Type I-II. The cement shall contain no more than 0.60 percent by weight of alkalies calculated as (Na2O + 0.658 K2O).

Pozzolans: Pozzolans (fly ash) shall conform to ASTM C618 Class C or Class F, except that the loss on ignition (LOI) shall be less than 2 percent.

Aggregate: Fine and coarse aggregate shall conform to ASTM C33. Maximum coarse aggregate size shall conform to ACI 318, paragraph 3.3.2. Unless otherwise specified, maximum aggregate size shall be 1-1/2 inches. Aggregate shall be nonpotentially reactive in accordance with ASTM C33, Appendix XI, paragraph X1.1.

Mixing Water: Potable having no pronounced taste or odor, and containing no deleterious materials.

Air-Entraining Agents (AEA): ASTM C260.

Water-Reducing Admixtures: If water-reducing admixtures are used, they shall conform to ASTM C494, Type A, and contain no more than 1 percent chloride ions.

Calcium Chloride: Calcium chloride is not permitted.

Reinforcing Steel

Deformed Bars: ASTM A615, Grade 60. Welding of reinforcing shall not be permitted.

Supports for Reinforcement: Provide supports for reinforcement including bolsters, chairs, spacers, and other devices for spacing, supporting and fastening reinforcing in place. Use wire bar type supports complying with CRSI recommendations, or approved substitute. Use supports with sand plates or horizontal runners where base material will not support chair legs. Pumice blocks, adobe, bricks, rocks, etc. are not acceptable for rebar or wire mesh supports.

Ancillary Materials

Concrete Crack Repair Epoxy Injection Manufacturers:

Contech Group, Seattle, WA, or Portland, OR.

Sika Group, Lindhurst, NJ.

Euclid Chemical Co., Cleveland, OH.

Expansion Joint Filler: ASTM D994, 1/2 inch thick, or as shown.

Nonshrink Grout:

Color: To match concrete.

Manufacturers and Products

Master Builder Co., Cleveland, OH; Master Flow 928.
Euclid Chemical Co., Cleveland, OH; Hi-flow Grout.

Curing Compound:

Material: Water-based curing compound in accordance with ASTM C309, Type I or Type 1D, with additional requirement that the moisture loss not exceed 0.040 gram per centimeter squared per 72 hours.

Manufacturers and Products:

Master Builders Co.; Masterkure 200W.
Euclid Chemical Co.; Super Diamond Clear Vox.

Water Stop: Extruded elastomeric plastic compound with basic resin to be polyvinyl chloride.

Manufacturers and Products:

Vinylex Corp., Knoxville, TN; Catalog No. 03250/VIN, RB6-38H.
A. C. Horn, Inc., Beltsville, MD; Catalog No. CSP-162, Type 9 (6-inch by 3/8-inch).

Hydrophilic Water Stop: Material shall be a non-bentonite hydrophilic rubber compound. Material shall be a combination of chloroprene rubber and chloroprene rubber modified to impart hydrophilic properties.

Manufacturers and Products:

Greenstreak Plastic Products, St. Louis, MO; Hydrotite CJ-1020-K with Leakmaster LV-1 adhesive and sealant.
Adeka Ultra Seal, JLM Associates, Spearfish, SD; MC-2010M with 3M-2141 adhesive and P-201 sealant.

Red Coloring for Electrical Duct Encasement: Commercial grade red iron oxide, 3 pounds per sack of cement.

Concrete Slab Coating

As specified in Section 09900, PAINT COATING SYSTEMS.

Proportioning and Design of Mixes

Mix Design: Prepare design mixes for each type and strength of concrete by either laboratory trial batch or field experience methods as specified in ACI 318 and 1997 UBC. The more stringent requirements of ACI 318 and 1997 UBC shall apply. Existing mix design test records shall be acceptable only if the facility is certified by National Ready Mixed Concrete Association. Uncertified concrete batch plant shall require testing from a qualified independent testing agency to develop the test and number of tests for an acceptable standard deviation to be developed.

Design mixes to provide normal weight concrete with the following specified 28-day compressive strengths, minimum, as indicated on drawings and schedules:

Class 30: 3,000 psi (non-structural concrete elements such as sidewalks, guard posts, fences, post and pole foundations, conduit encasement, and thrust blocks).

Class 40: 4,000 psi (structural concrete).

Class 50: 5,000 psi (truck loading pad).

See CONSTRUCTION QUALITY CONTROL of this specification for acceptance criteria.

Concrete mixes incorporating pozzolan (fly ash) shall contain a minimum of 15 percent fly ash by weight and the maximum 25 percent fly ash of the total cementitious materials.

Class 40 and Class 50 air content shall be 4 to 6 percent when tested in accordance with ASTM C231.

Follow manufacturer's recommendations for addition of water reducers.

Class 40 and Class 50: Concrete shall be air-entrained and shall incorporate the usage of high-range water reducer. Concrete mix design shall contain fly ash. Add air entraining agent (AEA) at the manufacturer's prescribed rate to result in concrete at point of placement having air content complying with ACI 301.

Unless specifically stated otherwise, water-cement ratio (or water-cement plus fly ash ratio) shall control amount of total water added to concrete as follows:

Water-Cement Ratio	**Maximum W/C Ratio**	**Maximum W/C Ratio**
Coarse Aggregate Size	**w/Superplasticizer**	**w/o Superplasticizer**
1-1/2"	0.40	0.45
1"	0.40	0.45
3/4"	0.40	0.45

Slump Range at Site(Class 40 and Class 50):

4-1/2 inches minimum, 8 inches maximum for concrete with a high range water reducing admixture.

3 inches minimum and 5 inches maximum for concrete without high range water reducing admixture.

Combined Aggregate Gradation:

Select one of the gradations shown in the following table.

Combined Gradation Limits:	Limits shown are for coarse aggregates and fine aggregates mixed together (combined).			
		Combined Gradation		
		Percentage Passing		
	Sieve Sizes	1-1/2" Max.	1" Max.	3/4" Max.
	2"	- 100	-	-
	1-1/2"	95 - 100	- 100	-
	1"	65 - 85	90 - 100	- 100
	3/4"	55 - 75	70 - 90	92 – 100
	1/2"	-	-	68 – 86
	3/8"	40 - 55	45 - 65	57 – 74
	No. 4	30 - 45	31 - 47	38 – 57
	No. 8	23 - 38	23 - 40	28 – 46
	No. 16	16 - 30	17 - 35	20 – 36
	No. 30	10 - 20	10 - 23	14 – 25
	No. 50	4 - 10	2 - 10	5 – 14
	No. 100	0 - 3	0 - 3	0 – 5
	No. 200	0 - 2	0 - 2	0 – 2

Minimum Cement Content (or Combined Cement Plus Fly Ash Content When Fly Ash is Used):

517 pounds per cubic yard for concrete with 1-1/2-inch maximum size aggregate.

540 pounds per cubic yard for 1-inch maximum size aggregate.

564 pounds per cubic yard for 3/4-inch maximum size aggregate.

Increase cement content or combined cement plus fly ash content, as required to meet strength requirements and water-cement ratio.

Mixing And Delivery (Class 40 and Class 50)

The manufacture and delivery of all concrete shall conform to ASTM C94 except as modified herein. Hand-mixed concrete is prohibited.

When concrete arrives at the jobsite with slump below that suitable for placing, as indicated by the Specification, water may be added only if the maximum permissible water-cement ratio and the maximum permissible slump is not exceeded. Any water thus added to bring the slump within required limits shall be injected in such a manner that uniformity requirements are met. Water shall be incorporated by additional mixing equal to at least half of the total mixing required or 30 drum revolutions at rated mixing speed, whichever is more. Additional AEA may be introduced during this mixing period if necessary to meet Specifications. Neither water nor AEA shall be added to the batch at any later time.

Concrete uniformity shall meet the requirements of ASTM C94 except as modified herein. After final mixing is complete, visible lumps, nonconformance to uniformity requirements, or failure to meet specified slump, entrained air, and temperature requirements shall be considered cause for rejecting the remainder of the load. In addition, failure of the ready-mix truck drum to meet uniformity requirements will be deemed cause for rejection of the mixing equipment until adequate repairs have been made.

Discharge of the concrete shall be completed within 1-1/2 hours, or before the drum has revolved 300 revolutions, whichever comes first, after the introduction of mixing water to the cement and aggregates unless special approved time delay admixtures are used. Coordinate time delay admixture information with manufacturer and Construction Manager prior to placing concrete. In hot weather or under conditions contributing to quick stiffening of the concrete a time limit less than 1-1/2 hours may be designated by the Construction General Contractor.

Additional high-range water-reducing admixtures (superplasticizer) may be added to the mixer at the jobsite using manufacturer-approved dispensing when unexpected delays cause too great of slump loss.

Concrete that is rejected for failure to meet any of the above requirements will be evaluated by the Engineer and may be removed and replaced at the expense of the Construction General Contractor.

Hot or Cold Weather Concreting: Methods and means of batching, mixing, and delivery of concrete in hot or cold weather shall comply with ACI 305R or ACI 306.1.

Part 3 – Execution

Formwork

Unless otherwise shown on the drawings, all forms shall be straight and plumb, rigid and mortar tight. All forms shall be braced, tied, and supported sufficiently to maintain their required position during and after the placing of concrete. Joints shall be sufficiently tight to prevent mortar leakage. Where shown on the Drawings, embedded items shall be placed in forms to shape edges or surfaces to that of the concrete members. All formwork shall conform to the guidelines in ACI 347.

All exposed corners of concrete shall be chamfered 3/4 inch.

Form Removal:

Formwork Not Supporting Weight of Concrete: This formwork may be removed after cumulatively curing at not less than 50 degrees F for 32 hours after placing concrete, provided concrete is sufficiently hard not to be damaged by form removal or subsequent operations. Curing must then continue through the minimum curing period.

Formwork Supporting Weight of Concrete: This formwork may not be removed until concrete has attained its 28-day design compressive strength, except as permitted under "Early Loading of New Concrete" as specified below.

Early Loading of New Concrete: Early loading of concrete structures shall comply with requirements of ACI 318, Section 6.2. When construction loading is proposed before concrete has achieved its 28-day design strength, structural calculations and concrete strength test data shall be submitted and approved by the Construction Manager prior to loading.

Form Sealer:

Material: Surface sealer will not bond with, stain, or adversely affect concrete surfaces, and will not impair subsequent treatments of concrete surfaces when applied to forms.

Manufacturers and Products:

Master Builders, Inc.; Rheofinish.

Burke Chemicals; Burke Release No. 1.

Placing Reinforcing Steel:

Unless otherwise specified, place reinforcing steel in accordance with CRSI Recommended Practice for Placing Reinforcing Bars and ACI 301.

Splicing:

Follow ACI 318/318R.

Use lap splices, unless otherwise shown or permitted in writing by Engineer.

Stagger splices in adjacent bars where indicated.

Placing Concrete

Place concrete in accordance with ACI 301.

Prior to placing concrete, remove water from excavation and debris and foreign material from forms. Check reinforcing steel for proper placement and correct discrepancies.

Before depositing new concrete on old concrete, clean surface using sandblast or bushhammer or other mechanical means to obtain a 1/4-inch rough profile, and pour a cement-sand grout to minimum depth of 1/2-inch over the surface. Proportion 1 part cement to 2.5 parts sand by weight.

Place concrete as soon as possible after leaving mixer, without segregation or loss of ingredients, without splashing forms or steel above, and in layers not over 2 feet deep. Place within 1-1/2 hours after adding cement to mix.

Eight feet maximum vertical drop to final placement, when not guided with chutes or other devices to prevent segregation due to impact with reinforcing.

Concrete shall be placed near its final location to avoid segregation.

Cold Weather Placing: Protect concrete work from damage or reduced strength, which could be caused by frost, freezing, or low temperatures, in compliance with ACI 306.1 and as specified herein. Minimum concrete temperature as placed and maintained shall be 55 degrees F, or as required by ACI-306.1, Table 3.2.1.

Hot Weather Placing:

When hot weather conditions that would seriously impair quality and strength of concrete exist, place concrete in compliance with ACI 305 and as specified herein:

Cool mixing drum and/or ingredients before mixing to maintain concrete temperature below 90 degrees F at time of placement.

Compaction

Vibrate concrete as follows:

Apply approved vibrator at points spaced not farther apart than vibrator's effective radius.

Apply close enough to forms to vibrate surface effectively but not damage form surfaces.

Vibrate until concrete becomes uniformly plastic.

Vibrator must penetrate fresh placed concrete and into previous layer of fresh concrete below.

Vibrator shall not be used to move concrete.

Construction Joints

Locate as shown or as approved.

Provide waterstops in construction joints as indicated and with other Construction General Contractor-required construction joints that is approved by the Engineer.

Maximum Spacing Between Construction Joints: 40 feet.

Installation of Embedded Items

Set and build into work anchorage devices and other embedded items required for other work that is attached to, or supported by cast-in-place concrete. Secure all such items firmly in position.

Finishing

Floor Slabs and Tops of Walls:

Finish slabs to grades shown on Drawings.

Screed surfaces to true level planes.

After initial water has been absorbed, float with wood float and trowel with steel trowel to smooth finish free from trowel marks.

Do not absorb wet spots with neat cement.

Unexposed Slab Surfaces: Screed to true surface, bull float with wood float, and wood trowel to seal surface. Finish surfaces to grades shown on Drawings.

Tolerances: Floors shall not vary from level or true plane more than 1/4-inch (plus or minus) in 10 feet when measured with a straightedge. Floors shall conform to grades shown on Drawings. Conform to ACI 117.

Exterior Slabs and Sidewalks:

Bull float with wood float, wood trowel, and lightly trowel with steel trowel.

Finish with broom to obtain nonskid surface.

Finish exposed edges with steel edging tool.

Mark walks transversely at 5-foot intervals with jointing tool.

Finishing And Patching Formed Surfaces

Smooth Form Finish (SmFm):

Provide as-cast smooth form finish for formed concrete surfaces that are exposed to view, or that are covered with a coating material applied directly to concrete, or a covering material bonded to concrete such as waterproofing, damp proofing, painting, or other similar system.

Produce smooth form finish (SmFm) by selecting form material to impart a smooth, hard, uniform texture and arranging them orderly and symmetrically with a minimum of seams. Repair and patch defective areas with fins or other projections completely removed and smoothed.

Cut out honeycombed and defective areas.

Cut edges perpendicular to surface at least 1 inch deep. Do not feather edges. Soak area with water for 24 hours.

Patch with nonshrink grout.

Finish surfaces to match adjacent concrete.

Keep patches damp for minimum 7 days or spray with curing compound to minimize shrinking.

Fill form tie holes with Nonshrink Grout.

Concrete Protection And Curing

General:

Protect freshly placed concrete from injurious action by sun, rain, wind, flowing water, mechanical injury, and premature drying for not less than seven (7) consecutive days after placement.

Protect concrete against damage from frost or freezing for a minimum of 3 days. Provisions of ACI 306.1 shall apply for cold weather unless otherwise specified.

Remove and replace concrete damaged by freezing.

Curing Methods: Perform curing of concrete by one or more of the following methods:

Moist Curing: Cover concrete surfaces with moisture retaining cover for curing period. Exposed horizontal concrete surfaces may be covered with sand or other approved material and kept wet for the required period. Wood forms shall be kept sufficiently wet at all times to prevent the forms from separating at the joints and the concrete from drying.

Membrane Curing: Concrete surfaces to receive membrane curing shall be treated with a curing compound as specified or otherwise approved. The curing compound shall be applied in strict accordance with the directions of the manufacturer of the compound.

Temperature, Wind, and Humidity:

Cold Weather:

When the mean daily outdoor temperature is less than 40 degrees F, the temperature of the concrete surface shall be maintained between 55 and 90 degrees F for the required curing period. When necessary, arrangements for heating, covering, insulating, or housing the concrete work shall be made in advance of placement and shall be adequate to maintain the required temperature without injury due to concentration of heat. Combustion heaters shall not be used during the first 24 hours unless precautions are taken to prevent exposure of the concrete to exhaust gases that contain carbon dioxide. If early loading is anticipated during cold weather, provide temperature protection to ensure necessary strength development.

The concrete surface temperature requirements (based on section thickness) in ACI 306.1 may be used in lieu of the 55 degrees F minimum specified before.

If concrete surface temperatures as measured by the inspecting agency are below the minimum curing temperature but meet the freeze protection requirements, the concrete curing period shall be extended to ensure adequate strength is developed. The extension time shall be at least equivalent to the time period in which temperatures were too low.

Hot Weather: The concrete surfaces shall be kept below 100 degrees F for the curing period. When necessary, provision for windbreaks, shading, fog spraying, sprinkling, ponding, or wet covering with a light colored material shall be made in advance of placement, and such protective measures shall be taken as quickly as concrete hardening and finishing operations will allow.

Rate of Temperature Change:

Changes in temperature of the air immediately adjacent to the concrete during and immediately following the curing period shall be kept as uniform as possible and shall not exceed 5 degrees F in any 1-hour or 50 degree F in any 24-hour period.

Use curing compound only where approved by Construction Manager. Cure formed surfaces with curing compound applied in accordance with manufacturer's directions as soon as forms are removed and finishing is completed.

Water Stops: Plastic and Hydrophilic

Install in accordance with manufacturer's instructions.

Repairing Concrete Cracks

Cracks requiring repair shall be determined by the Engineer and shall be repaired using epoxy injection. Method of epoxy injection shall be approved by the Construction Manager prior to application.

Construction Quality Control

Construction General Contractor Supplied Testing: The Construction General Contractor shall provide the necessary testing and monitoring services for the following:

Construction General Contractor shall procure the services of a qualified independent testing laboratory to control or monitor the production, transportation, placement, protection, curing, or temperature of the concrete as specified herein. Construction General Contractor shall submit qualifications of independent testing laboratory.

Evaluation of Concrete Field Strength:

Provide adequate facilities for safe storage and proper curing of concrete test cylinders onsite for first 24 hours, and for additional time as may be required before transporting to test lab.

Provide concrete for testing of slump, air content, and for making cylinders from the point of discharge into forms. When concrete is pumped, Samples used shall be taken from discharge end of pump hose.

Evaluation will be in accordance with ACI 301, Chapter 17, and Specifications.

Specimens shall be made, cured, and tested in accordance with ASTM C31 and ASTM C39.

Pumped Concrete: Take concrete samples for slump (ASTM C143) and test cylinders (ASTM C31 and C39).

Reject concrete represented by cylinders failing to meet strength and air content specified. For Class 30 concrete, reject concrete represented by cylinders failing to meet strength specified.

END OF SECTION 03301

DIVISION 4–MASONRY (NOT USED)

DIVISION 5–METALS

SECTION 05500 – METAL FABRICATIONS AND CASTINGS SCHED. A & B

Part 1 – General

References

The following is a list of standards, which may be referenced in this section:

THE ALUMINUM ASSOCIATION, INC. (AA)

The Aluminum Design Manual

AMERICAN GALVANIZERS ASSOCIATION (AGA)

Inspection of Products Hot-Dip Galvanized After Fabrication

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC S329	Allowable Stress Design Specification for Structural Joints using ASTM A325 or A490 Bolts

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Stainless Steel Types

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI A10.11	Safety Requirements for Personnel and Debris Nets
ANSI A14.3	Ladders - Fixed - Safety Requirements
ANSI B1.1	Unified-inch Screw Threads (UN and UNR Thread Form)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1	Structural Welding Code - Steel
AWS D1.2	Structural Welding Code - Aluminum
AWS D1.6	Structural Welding Code - Stainless Steel

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	Specification for Carbon Structural Steel
ASTM A48	Specification for Gray Iron Castings
ASTM A53/A53M	Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A108	Specification for Steel Bars, Carbon, Cold-Finished, Standard Quality
ASTM A123/A123M	Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A143	Practice for Safeguarding Against Embrittlement of Hot-Dip Galvanized Structural Steel Products and Procedure for Detecting Embrittlement
ASTM A153/A153M	Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A193/A193M	Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service
ASTM A194/A194M	Specification for Carbon and Alloy Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both
ASTM A240/A240M	Specification for Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels

ASTM A276	Specification for Stainless Steel Bars and Shapes
ASTM A278	Specification for Gray Iron Castings for Pressure-Containing Parts for Temperatures Up to 650 Degree
ASTM A283/A283M	Specification for Low and Intermediate Tensile Strength Carbon Steel Plates
ASTM A307	Specification for Carbon Steel Bolts and Studs, 60,000 PSI Tensile
ASTM A325	Specification for Structural Bolts, Steel, Heat Treated 120/105 ksi Minimum Tensile Strength
ASTM A380	Practice for Cleaning, Descaling, and Passivation of Stainless Steel Parts, Equipment, and Systems
ASTM A384	Practice for Safeguarding Against Warpage and Distortion During Hot-Dip Galvanizing of Steel Assemblies
ASTM A385	Practice for Providing High-Quality Zinc Coatings (Hot-Dip)
ASTM A489	Specification for Carbon Steel Lifting Eyes
ASTM A500	Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
ASTM A501	Specification for Hot-Formed Welded and Seamless Carbon Steel Structural Tubing
ASTM A563	Specification for Carbon and Alloy Steel Nuts
ASTM A653	Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A780	Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM A786/A786M	Specification for Hot-Rolled Carbon, Low-Alloy, High-Strength Low-Alloy, and Alloy Steel Floor Plates
ASTM A793	Specification for Rolled Floor Plate, Stainless Steel
ASTM A967	Specification for Chemical Passivation Treatments for Stainless Steel Parts
ASTM A992/A992M	Specification for Steel for Structural Shapes for Use in Building Framing
ASTM B209	Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B308/B308M	Specification for Aluminum-Alloy 6061-T6 Standard Structural Profiles
ASTM B429	Specification for Aluminum-Alloy Extruded Structural Pipe and Tube
ASTM B632/B632M	Specification for Aluminum-Alloy Rolled Tread Plate
ASTM D1056	Specification for Flexible Cellular Materials - Sponge or Expanded Rubber
ASTM F436	Specification for Hardened Steel Washers
ASTM F468	Specification for Nonferrous Bolts, Hex Cap Screws, and Studs for General Use
ASTM F593	Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs
ASTM F594	Specification for Stainless Steel Nuts
ASTM F844	Specification for Washers, Steel, Plain (Flat), Unhardened for General Use

ASTM F1554 Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength

INTERNATIONAL CONFERENCE OF BUILDING OFFICIALS (ICBO)

Evaluation Reports for Concrete and Masonry Anchors

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910.27 Fixed Ladders

29 CFR 1926.105 Safety Nets

29 CFR 1926.502 Fall Protection Systems Criteria and Practices

SPECIALTY STEEL INDUSTRY OF NORTH AMERICA (SSINA)

Specifications for Stainless Steel

Design Guidelines for the Selection and Use of Stainless Steel

Stainless Steel Fabrication

Stainless Steel Fasteners

Definitions

Submerged: Location at or below top of wall of open water-holding structure, such as a basin or channel, or wall, ceiling or floor surface inside a covered water-holding structure, or exterior belowgrade wall or roof surface of water-holding structure, open or covered.

Exterior Area: Location not protected from the weather by a building or other enclosed structure.

Interior Wet Area: Location inside building or structure where floor is sloped to a sump, floor drains or gutters and is subject to liquid spills or washdown, or where wall, floor, or roof slab is common to a water-holding or earth-retaining structure.

Interior Dry Area: Location inside building or structure where floor is not subject to liquid spills or washdown, nor where wall or roof slab is common to a water-holding or earth-retaining structure.

Corrosive Area: Containment area or area exposed to delivery, storage, transfer, or use of chemicals.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Shop Drawings:

Metal fabrications, including welding, shop and field weld WPSs, PQRs, and fastener information.

Specific instructions for concrete anchor installation, including drilled hole size, preparation, placement, procedures, and instructions for safe handling of anchoring systems.

Submit source quality control data specified herein.

Samples: Color samples of abrasive stair nosings.

Submittals–Approval Not Required

Information/Record (IR):

Concrete and Masonry Drilled Anchors:

Manufacturer's product description and installation procedures.

ICBO evaluation report.

Adhesive Anchor Installer Certification.

Welding:

WPSs and WPQs.
CWI credentials and CWI reports.
Welding documentation.

U-Channel Concrete Inserts:

Manufacturer's product description.
Allowable load tables.

Ladders:

Certification of load and fatigue tests.
Passivation method for stainless steel members.
Documentation of construction quality control inspections specified herein.

Qualifications

Adhesive Anchor Installers: Trained and certified by manufacturer.

Galvanized Coating Applicator: Company specializing in hot-dip galvanizing after fabrication and following procedures of Quality Assurance Manual of the American Galvanizers Association.

Welding: WPSs and WPQs in accordance with AWS D1.1.

Delivery, Storage, and Handling

Insofar as practical, factory assemble items specified herein. Assemblies that due to necessity have to be shipped unassembled shall be packaged and tagged in manner that will protect materials from damage and will facilitate identification and field assembly.

Package stainless steel items in a manner to provide protection from carbon impregnation.

Protect painted coatings and hot-dip galvanized finishes from damage due to metal banding and rough handling. Use padded slings and straps.

Store fabricated items in dry area, not in direct contact with ground.

Part 2 – Products

General

Unless otherwise indicated, meet the following requirements:

Item	ASTM Reference
Steel Shapes and Plates	A36/A36M or A992
Steel Pipe	A501 or A53/A53M, Type E or S, Grade B
Structural Steel Tubing	A500, Grade B
Stainless Steel:	
Bars and Angles	A276, AISI Type 316
Shapes	A276, AISI Type 304
Steel Plate, Sheet, and Strip	A240/A240M, AISI Type 316
Bolts, Threaded Rods, Anchor Bolts, and Anchor Studs	F593, AISI Type 316, Condition CW
Nuts	F594, AISI Type 316, Condition CW

Item	ASTM Reference
Steel Bolts and Nuts:	
Carbon Steel	A307 bolts, with A563 nuts
High-Strength	A325, Type 1 bolts, with A563 nuts
Anchor Bolts and Rods	F1554, Grade 55, with weldability supplement S1
Eyebolts	A489
Threaded Rods	A36/A36M
Flat Washers (Unhardened)	F844
Flat and Beveled Washers (Hardened)	F436
Thrust Ties for Steel Pipe:	
Threaded Rods	A193/A193M, Grade B7
Nuts	A194/A194M, Grade 2H
Plate	A283/A283M, Grade D
Welded Anchor Studs	A108, Grades C-1010 through C-1020
Aluminum Plates and Structural Shapes	B209 and B308/B308M, Alloy 6061-T6
Aluminum Bolts and Nuts	F468, Alloy 2024-T4
Cast Iron	A48, Class 35

Bolts, Washers, and Nuts: Use stainless steel, hot-dip galvanized steel, zinc-plated steel, and aluminum material types as indicated in FASTENER SCHEDULE at end of this section.

Anchor Bolts and Anchor Bolt Sleeves

Cast-In-Place Anchor Bolts: Headed type, unless otherwise shown on Drawings.

Material type and protective coating as shown in FASTENER SCHEDULE at end of this section.

Anchor Bolt Sleeves:

Plastic:

Single unit construction with corrugated sleeve.

Top of sleeve shall be self-threading to provide adjustment of threaded anchor bolt projection.

Material: High density polyethylene.

Manufacturer: Sinco Products, Inc., Middletown, CT. (800-243-6753).

Fabricated Steel: ASTM A36/A36M.

Concrete Drilled Anchors

General

AISI Type 316 stainless, hot-dip galvanized, or zinc-plated steel, as shown in FASTENER SCHEDULE at end of this section.

Current evaluation and acceptance reports by ICBO.

Wedge Anchors:

Manufacturers and Products:

ITW Ramset/Red Head, Wood Dale, IL; Trubolt Wedge Anchor.
Hilti, Inc., Tulsa, OK; Kwik-Bolt II Stud Anchor.
Powers Rawl, New Rochelle, NY; Power-Stud Anchor.
Simpson Strong-Tie Co., Inc., Pleasanton, CA; Wedge-All Anchor.
Wej-It Corp., Tulsa, OK; ANKRtite Wedge Anchor.
U.S. Anchor, Pompano Beach, FL; Kingpin Wedge Anchor.

Expansion Anchors: Self-drilling anchors, snap-off or flush type, zinc-plated.

Nondrilling Anchors: Flush type for use with zinc-plated or stainless steel bolt, or stud type with projecting threaded stud.

Manufacturers and Products:

ITW Ramset/Red Head, Wood Dale, IL; Multi-Set II Drop-In and Self Drill Anchor.
Hilti, Inc., Tulsa, OK; Hilti HDI Drop-In Anchor.
Powers Rawl, New Rochelle, NY; Steel Drop-In Anchor.
Simpson Strong-Tie Co., Inc., Pleasanton, CA; Drop-In Anchor.

Sleeve Anchors:

Manufacturers and Products:

ITW Ramset/Red Head, Wood Dale, IL; Dynabolt Hex Nut Sleeve Anchor.
Powers Rawl, New Rochelle, NY; Hex Head Power-Bolt Anchor.
Simpson Strong-Tie Co., Inc., Pleasanton, CA; Sleeve-All Hex Head Anchor.
Wej-It Corp., Tulsa, OK; Wej-It Sleeve Anchor.

Adhesive Anchors:

Threaded Rod:

ASTM F593 stainless steel threaded rod, diameter as shown on Drawings.
Length as required, to provide minimum depth of embedment.
Clean and free of grease, oil, or other deleterious material.

Adhesive:

Two-component, designed to be used in adverse freeze/thaw environments, with gray color after mixing.

Cure Temperature, Pot Life, and Workability:

Compatible for intended use and environmental conditions.
Nonsag, with selected viscosity base on installation temperature and overhead application where applicable.

Packaging and Storage:

Disposable, self-contained cartridge system capable of dispensing both components in the proper mixing ratio and fitting into a manually or pneumatically operated caulking gun.
Store adhesive cartridges on pallets or shelving in covered storage area, in accordance with manufacturer's written instructions.

Cartridge Markings:

Include manufacturer's name, product name, material type, batch or serial number, and adhesive expiration date.

Dispose of cartridges if shelf life has expired.

Manufacturers and Products:

ITW Ramset/Red Head, Wood Dale, IL; Epcon Ceramic 6 Epoxy or A7 Adhesive Anchor System.

Hilti, Inc., Tulsa, OK; HIT Doweling Anchor System, HIT HY 150 (HIT HY 20 for hollow masonry).

Powers Rawl, New Rochelle, NY; Power Fast Epoxy Injection Gel Cartridge System.

Simpson Strong-Tie Co., Inc., Pleasanton, CA; Epoxy-Tie Adhesive ET or Acrylic-Tie Adhesive. (Use only Acrylic-Tie Adhesive for temperatures below 40 degrees F.)

Covert Operations, Inc., Long Beach, CA; CIA-Gel 7000 Epoxy Anchors.

U.S. Anchor, Pompano Beach, FL; Ultrabond 1.

Unitex, Kansas City, MO; Pro-Poxy 300 and Pro-Poxy 300 Fast Epoxy Adhesive Anchors.

Diversified Fastening Systems of America, Charles City, Iowa; DFS Wedge Anchors.

Adhesive Threaded Inserts: Stainless steel, internally threaded insert.

Manufacturer and Product: Hilti, Inc., Tulsa, OK; HIS-R Insert with HIT HY 150 adhesive.

Welded Anchor Studs

Headed anchor studs (HAS) or threaded anchor studs (TAS), as indicated on Drawings.

Carbon Steel: ASTM A108, Standard Quality Grades 1010 through 1020, inclusive either semikilled or killed aluminum or silicon dioxidation, unless indicated otherwise.

Stainless Steel: ASTM F593, AISI Type 316, Condition CW, where indicated.

Manufacturers:

Nelson Stud Welding, FabriSteel Co., Elyria, OH.

Stud Welding Associates, Inc., Elyria, OH.

Embedded Steel Support Frames For Floor Plate And Grating

Steel angle support frames to be embedded in concrete shall be stainless steel, ASTM A276, AISI Type 316, unless indicated otherwise.

Welded anchors for stainless steel support frames shall also be stainless steel.

Abrasive Nosing For Stairs

Unless otherwise shown on Drawings, furnish flush type abrasive nosing on stairs.

Nosing Components:

Homogeneous epoxy abrasive, with minimum 50 percent aluminum oxide content, formed and cured upon an extruded aluminum base.

Epoxy abrasive shall extend over and form curved front edge of nosing.

Base of Nosing: Extruded aluminum alloy, 6063-T5, heat-treated.

Anchoring System: Double-set anchors consisting of two rows of integrally extruded anchors.

Size: 3 inches wide by 1/4 to 3/8 inch thick by length as shown.

Color: Selected by Tank Farm Contractor from manufacturer's standard color range.

Manufacturers and Products:

Wooster Products, Inc., Wooster, OH; Spectra Type WP3C.

American Safety Tread Co., Inc., Helena, AL; Type FA-311D.

Fabrication

General:

Fabricate as shown and in accordance with AISC Specification for Structural Steel Buildings and AISC Code of Standard Practice for Steel Buildings and Bridges.

Finish exposed surfaces smooth, sharp, and to well-defined lines.

Grind cut edges smooth and straight. Round sharp edges to small uniform radius. Grind burrs, jagged edges, and surface defects smooth.

Fit and assemble in largest practical sections for delivery to site.

Materials:

Use steel shapes, unless otherwise noted.

Steel to be hot-dip galvanized: Limit silicon content to less than 0.04 percent or to between 0.15 and 0.25 percent.

Welding:

Weld connections and grind exposed welds smooth. When required to be watertight, make welds continuous.

Welded fabrications shall be free from twisting or distortion caused by improper welding techniques.

Steel: Meet fabrication requirements of AWS D1.1, Section 5.

Stainless Steel: Meet requirements of AWS D1.6.

Welded Anchor Studs:

Prepare surface to be welded and weld with stud welding gun in accordance with AWS D1.1, Section 7, and manufacturer's instructions.

Complete welding before applying finish.

Painting:

Coat all fabricated carbon steel as specified in Section 09900, PAINT COATING SYSTEMS, unless otherwise indicated.

Do not apply protective coating to galvanized steel anchor bolts or galvanized steel welded anchor studs and stainless steel anchor bolts, unless indicated otherwise.

Galvanizing:

Fabricate steel to be galvanized in accordance with ASTM A143, ASTM A384, and ASTM A385. Avoid fabrication techniques that could cause distortion or embrittlement of the steel.

Provide venting and drain holes for tubular members and fabricated assemblies in accordance with ASTM A385.

Remove welding slag, splatter, burrs, grease, oil, paint, lacquer, and other deleterious material prior to delivery for galvanizing.

Remove by blast cleaning or other methods surface contaminants and coatings not removable by normal chemical cleaning process in the galvanizing operation.

Hot-dip galvanize steel members, fabrications, and assemblies after fabrication in accordance with ASTM A123/A123M.

Hot-dip galvanize bolts, nuts, washers, and hardware components in accordance with ASTM A153/A153M. Oversize holes to allow for zinc alloy growth. Shop assemble bolts and nuts.

Galvanized steel sheets in accordance with ASTM A653.

Galvanize components of bolted assemblies separately before assembly. Galvanizing of tapped holes is not required.

Accessories: Furnish as required for a complete installation. Fasten by welding or with stainless steel bolts or screws.

Source Quality Control

Visually inspect all fabrication welds and correct any deficiencies.

Steel: AWS D1.1, Section 6 and Table 6.1, Visual Inspection Acceptance Criteria.

Stainless Steel: AWS D1.6.

Welded Anchor Studs: AWS D1.1 or AWS D1.6 as applicable.

Part 3 – Execution

Installation Of Metal Fabrications

General

Install metal fabrications plumb or level, accurately fitted, free from distortion or defects.

Install rigid, substantial, and neat in appearance.

Install manufactured products in accordance with manufacturer's recommendations.

Obtain Construction Manager approval prior to field cutting steel members or making adjustments not scheduled.

Cast-In-Place Anchor Bolts

Accurately locate and hold anchor bolts in place with templates at the time concrete is placed.

Use anchor bolt sleeves for location adjustment and provide two nuts and one washer per bolt of same material as bolt.

Minimum Bolt Size: 1/2-inch diameter by 12 inches long, unless otherwise shown.

Concrete and Masonry Drilled Anchors

Begin installation only after concrete or masonry to receive anchors has attained design strength.

Install in accordance with manufacturer's instructions.

Provide minimum embedment, edge distance, and spacing as follows, unless indicated otherwise by anchor manufacturer's instructions or shown otherwise on Drawings:

Anchor Type	Min. Embedment (bolt diameters)	Min. Edge Distance (bolt diameters)	Min. Spacing (bolt diameters)
Wedge	9	6	12
Expansion and Sleeve	4	6	12
Adhesive	9	9	13.5

Use only drill type and bit type and diameter recommended by anchor manufacturer. Clean hole of debris and dust with brush and compressed air.

When embedded steel or rebar is encountered in the drill path, slant drill to clear obstruction. If drill must be slanted more than 10 degrees to clear obstruction, notify Construction Manager for direction on how to proceed.

Adhesive Anchors:

Do not install adhesive anchors when temperature of concrete is below 40 degrees F (25 degrees F for Simpson Strong-Tie Acrylic-Tie Adhesive) or above 100 degrees F.

Remove any standing water from hole with oil-free compressed air. Inside surface of hole shall be dry where required by manufacturer's instructions.

Do not disturb anchor during recommended curing time.

Do not exceed maximum torque as specified in manufacturer's instructions.

Electrolytic Protection

Aluminum and Galvanized Steel:

Coat surfaces of galvanized steel fabricated and aluminum items to be in direct contact with concrete, grout, masonry, or dissimilar metals, as specified in Section 09900, PAINT COATING SYSTEMS, unless indicated otherwise.

Do not apply protective coating to galvanized steel anchor bolts or galvanized steel welded anchor studs, unless indicated otherwise.

Allow coating to dry before installation of the material.

Protect coated surfaces during installation.

Should coating become marred, prepare and touch up in accordance with paint manufacturer's written instructions.

Stainless Steel:

During handling and installation, take necessary precautions to prevent carbon impregnation of stainless steel members.

After installation, visually inspect stainless steel surfaces for evidence of iron rust, oil, paint, and other forms of contamination.

Remove contamination in accordance with requirements of ASTM A380 and A967.

Brushes used to remove foreign substances shall utilize only stainless steel or nonmetallic bristles.

After treatment, visually inspect surfaces for compliance.

Repair Of Galvanized Steel

Conform to ASTM A780.

For minor repairs at abraded areas, use sprayed zinc conforming to ASTM A780.

For flame cut or welded areas, use zinc-based solder, or zinc sticks, conforming to ASTM A780.

Use magnetic gauge to determine that thickness is equal to or greater than the base galvanized coating.

Fastener Schedule

Provide fasteners as follows:

Service Use and Location	Product	Remarks
Anchor Bolts Cast Into Concrete for Structural Steel Column Base Plates		
Interior Dry Areas	Hot-dip galvanized steel headed anchor bolts, unless indicated otherwise	
Exterior and Interior Wet Areas	Stainless steel headed anchor bolts	
Anchor Bolts Cast Into Concrete for Equipment Bases		
Interior Dry Areas	Stainless steel headed anchor bolts, unless otherwise specified with equipment	
Anchor Bolts Cast Into Concrete for Metal Fabrications and Structural Components		
Interior Dry Areas	Stainless steel headed anchor bolts	
Drilled Anchors for Metal Components to Cast-in-Place Concrete (e.g., Ladders, Handrail Posts, Electrical Panels, and Equipment)		
Interior Dry Areas	Zinc-plated or stainless steel wedge or expansion anchors	
Submerged, Exterior, Interior Wet, and Corrosive Areas	Adhesive stainless steel anchors	
Connections for Structural Steel Framing		
Exterior and Interior Wet and Dry Areas	High-strength steel bolted connections	Use hot-dipped galvanized high-strength bolted connections for galvanized steel framing members
Connections for Steel Fabrications		
Exterior and Interior Wet and Dry Areas	Stainless steel bolted connections	
All Others		
Exterior and Interior Wet and Dry Areas	Stainless steel fasteners	

Antiseizing Lubricant: Use on all stainless steel threads.

Do not use adhesive anchors to support fire-resistive construction or where ambient temperature will exceed 120 degrees F.

Construction Quality Control

Construction General Contractor shall inspect concrete drill anchor installation to verify compliance with anchor size, embedment, edge length, and spacing as specified herein and shown on the Drawings. Provide inspection documentation to Construction Manager.

END OF SECTION 05500

DIVISION 6–WOODS AND PLASTIC (NOT USED)

DIVISION 7–THERMAL AND MOISTURE PROTECTION

SECTION 07210 - BUILDING INSULATION SCHED. B

Part 1 – General

References

The following is a list of standards, which may be referenced in this section:

ASTM INTERNATIONAL (ASTM)

ASTM C578	Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation.
ASTM C665	Standard Specification for Mineral-Fiber Blanket Thermal Insulation for Light Frame Construction and Manufactured Housing.
ASTM D4397	Standard Specification for Polyethylene Sheeting for Construction, Industrial, and Agricultural Applications.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Product Data: Submit product data indicating compliance with the requirements of this section.

Material Storage

Store off ground and keep dry at all times. Protect against weather condensation and damage.

Part 2 – Products

Materials

Mineral/Glass Fiber Blanket/Batt Insulation: ASTM C665, Type III, Class B, fiberglass batts with vinyl vapor retarder; R-30 for roof and R-19 for walls.

Rigid Insulation: ASTM C578, Type IV, extruded polystyrene; R-value as shown.

Vapor Retarder: ASTM D4397 plastic sheeting, 6 mils minimum.

Part 3 – Execution

Installation

Batt Insulation:

Install in accordance with the manufacturer's instructions.

Fasten flanges to the sides of framing members with the vapor retarder facing the warm side. Fit tightly to ensure a continuous seal.

Where electrical outlets, ducts, pipes, vents, or other utility items occur, place insulation on the cold weather side of the obstruction.

Provide fasteners, adhesive, tape, and sealant as recommended by insulation manufacturer.

Vapor Retarder:

Apply to inside face of exterior wall and ceiling framing in sheets as large as possible, lapping all joints 6 inches and sealing with sealant and tape recommended by manufacturer.

Fit tightly and seal around all penetrations.

Replace torn and punctured sheets.

Repair minor tears or holes with tape.

Repair by replacement major tears or holes that require more than a 6-inch length of tape to repair.

Rigid Insulation:

Install with fasteners or adhesive recommended by manufacturer.

Butt joints tightly together.

Where thicker than 2 inches, install in two layers, staggering all joints.

Cleanup and Protection

Remove from site all containers, wrappings, and scrap insulation material. Leave floors broom clean.

Protect installed insulation from tears or other damage until covered with finish material. Replace damaged material.

END OF SECTION 07210

DIVISION 8–DOORS AND WINDOWS (NOT USED)

DIVISION 9–FINISHES

SECTION 09900 - PAINT COATING SYSTEMS SCHED. A & B

Part 1 – General

References

The following documents and others referenced therein form part of Contract to the extent designated. Referenced documents are those current, unless otherwise indicated.

ASTM INTERNATIONAL (ASTM)

ASTM D412	Test Methods for Vulcanized Rubber and Thermoplastic Rubbers and Thermoplastic Elastomers - Tension
ASTM D570	Test Method for Water Absorption of Plastics
ASTM D638	Test Method for Tensile Properties of Plastics
ASTM D714	Test Method for Evaluating Degree of Blistering of Paints
ASTM D772	Test Method for Evaluating Degree of Flaking (Scaling) of Exterior Paints
ASTM D1653	Test Methods for Water Vapor Transmission of Organic Coating Films
ASTM D3912	Test Method for Chemical Resistance of Coatings Used in Light-Water Nuclear Power Plants
ASTM D4060	Test Method for Abrasion Resistance of Organic Coatings by the Taber Abraser
ASTM D4082	Test Method for Effects of Gamma Radiation on Coatings for Use in Light-Water Nuclear Power Plants
ASTM D4259	Standard Practice for Abrading Concrete
ASTM D4263	Test Method for Indicating Moisture Content in Concrete by the Plastic Sheet Method
ASTM D4541	Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers
ASTM D5139	Sample Preparation for Qualification Testing of Coatings to be Used in Nuclear Power Plants
ASTM D5144	Guide for Use of Protective Coating Standards in Nuclear Power Plants
ASTM E84	Test Method for Surface Burning Characteristics of Building Materials

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 255	Method of Test of Surface Burning Characteristics of Building Materials

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC-SP 3	Power Tool Cleaning
SSPC-SP 6	Commercial Blast Cleaning

Submittals –Approval Required

See Section 01300, SUBMITTALS, for submittal procedures

List of Materials: Before delivery, submit colors and location to be used and manufacturer catalog data sheets and charts showing adequate information to substantiate compliance to the requirements of this section.

Submittal shall also enumerate percentage of volatile and nonvolatile materials, percentage of component parts of each type of material, and the conversion factors to determine dry film thickness from applied wet film thickness. Also, submit Material Safety Data Sheets (MSDS) for materials proposed to be used.

Installer's Certificate (Decontaminable Coatings Only): Before application, submit documentation that the application crew has been certified by the coating system manufacturer as qualified to apply the selected coating system. As an alternative to crew certification, a submittal documenting onsite training by a technical representative from the coating manufacturer would be acceptable.

Cleaning and Disposal Plan: Before application, submit a plan for proper collection, storage, and disposal of all materials spotted or soaked with paint, oil, solvents, and other flammable waste materials. The plan shall also address handling and disposal of empty cans. The plan shall address both daily cleanup requirements and cleanup at the completion of the coatings application activities.

Submit documentation and test results from construction quality control testing specified herein.

Delivery, Storage, And Handling

Obtain inspection and acceptance by Tank Farm Contractor before opening containers or removing labels.

Project Conditions

Environment for Coating: Coat exterior surfaces only when ambient and surface temperatures are within the range recommended by the coating manufacturer for the respective coating, which is within 40 to 120 degrees F, and ambient temperature is a minimum of 5 degrees F above the dewpoint.

Part 2 – Products

Materials

Shop Primer for Carbon Steel Assemblies: Ameron-Amerlock 400.

Decontaminable Coatings for all Metal and Concrete: (Service Level II as defined in ASTM D5144)

Decontaminability: Evaluation of coating systems decontaminability as noted in ASTM D5144 Section 5.4, is difficult to quantify. To determine the coating systems decontamination properties, the coating system shall be tested for chemical resistance properties as addressed in this specification.

Radiation Tolerance: Coatings applied to the specified thickness shall demonstrate tolerance to a total accumulated dose in air of 6×10^7 Rads of gamma radiation in accordance with ASTM D4082. Test samples shall be prepared in accordance with ASTM D5139.

Physical Properties: Base and finish coatings shall have the physical property strengths shown in the tables below as determined by the respective test method.

For Rigid Decontaminable Coatings (High Solids Epoxies):

Test	Method	Results
Adhesion to Substrate	ASTM D4541	Steel-Minimum 900 psi Concrete-8% of concrete compressive strength
Elongation at break at 75°F		Minimum 5%
Water Absorption or	ASTM D570 (24 hr)	Maximum 0.5%
Moisture Vapor Transmission	ASTM D1653	Maximum 8 gm/m^2
Wear Resistance (Finish or lost Top Coat Only)	ASTM D4060, 1,000 cycles, 1,000 g weight, CS-17 wheel	Less than 175 mg

Chemical Resistance: The coating system shall be resistant to the standard decontamination solutions listed in ASTM D3912, Figure 1. Chemical resistance testing shall be in accordance with ASTM D3912, or an equivalent standard, for occasional splash and spillage service, except test samples shall be prepared in accordance with ASTM D5139. Submit manufacturer's chemical resistance test plan, including procedure for exposing coating samples for evaluating occasional splash and spillage conditions, for evaluation and approval. Criteria for acceptance shall be based on the following:

Flaking: As evaluated in accordance with ASTM D772, flaking and peeling shall not be permitted

Blistering:

As evaluated in accordance with ASTM D714, blisters shall be limited to size 4, 6, or 8, and a frequency no more than a "few."

Delamination will not be permitted.

Slight discoloration will be permitted.

Coating shall be volatile organic content (VOC) compliant with a maximum VOC of 2.9 lbs/gal.

Fire Characteristics:

Coatings used shall not develop significant qualities of toxic or other harmful products of combustion when exposed to fire. Coatings shall have a UL (ASTM E84/NFPA 255) flame spread rating of 25 or less and smoke developed rating of 50 or less.

Coatings shall be repairable for cracks appearing through the applied-coated surface to the substrate and for chips and flaking due to mechanical damage.

Coating shall have a design life of 12 years. In addition to radiation tolerance requirements, coating shall be resistant to humidity ranging from 0 to 100 percent and an ambient air temperature range of 120 degrees F to minus 32 degrees F, with a maximum 24-hour differential of 52 degrees F.

Primers, thinners, and coating accessory materials shall be produced or approved for use by the same manufacturer as the finish coating system.

Decontaminable Coating Systems (Epoxies), or Approved Substitute:

Manufacturer Product	Ameron
Steel and Concrete Primer	Amerlock 400
Concrete Surfacer	Nu-Klad 114A
Base Coating	Amerlock 400

Intermediate and Finish Coating PSX 700 Siloxane

Joint Sealant: "Nu-Klad 750A/760A" by Ameron, or approved substitutes, shall be used.

Scrim Cloth: Glass fiber reinforcing fabric as recommended by coating system manufacturer.

Part 3 – Execution

Examination

Examine surfaces scheduled to receive coatings for conditions that will adversely affect execution, performance, or quality of work, and that cannot be put into acceptable condition through preparatory work.

Report conditions that could adversely affect proper application of coatings, in writing, to Construction Manager. Do not begin surface preparation or coating application until defects have been corrected and conditions have been made suitable.

Preparation

Before application, remove as much dust and debris as possible from space or area to receive coating to allow for proper installation.

Pre-Priming:

Ferrous Metal and Carbon Steel:

Remove oil, grease, welding fluxes, and other surface contaminants prior to blast cleaning.

Prepare shop assemblies in accordance with SSPC-SP 6. Remove abrasive residue and dust, and prime within 4 hours after preparation. Apply minimum of 3.0 mil of shop primer in accordance with manufacturer's recommendations.

Select type and size of abrasive to produce a surface profile that meets coating manufacturer's recommendations for particular primer to be used.

Prepare field erections in accordance with SSPC-SP 3 or SSPC-SP 6. Remove abrasive residue and dust.

Surface Preparation:

Shop and field surface cleaning and surface preparation requirements for all substrates shall be in accordance with the manufacturer's written instructions and these Specifications. Where the specified degree of surface preparation differs from the manufacturer's recommendations, the more stringent shall apply.

Concrete shall be at least 30 days old before coating is applied.

Prior to application of coating system or surfacer to new concrete surfaces, perform a plastic sheet test in accordance with ASTM D4263. The test shall be initiated in the afternoon and completed the following morning. The absence of condensation on the test sheet shall indicate the concrete is ready to have the coating system applied. Document test results.

Clean new concrete surfaces to be coated by Abrasive Blast Cleaning Procedures in accordance with ASTM D4259.

Prepare or repair construction joints, shrinkage cracks, and other non-expanding cracks, gaps, or crevices in the surface to be coated, in accordance with coating manufacturer's recommendations. Scratches, cracks, holes, pinholes, and abrasions shall be cut back to proper key and filled with surfacer.

Post-Priming: Feather abrasions, chips, skips, and holidays occurring in prime coat by sanding, and recoat with material and color to minimum dry film thickness specified.

Previously coated surfaces shall be recoated only after existing film is completely dry. Some coating systems require the application of succeeding coats within a set time frame for it to properly adhere to the previous coat. Should the time frame recommended by the coating manufacturer be exceeded, prepare the base coat as recommended by the coating manufacturer.

Protection:

Provide and install drop cloths, shields, and other protective devices required to protect surfaces adjacent to areas being coated. Keep spatter, smears, droppings, and over-run of coating materials to a minimum and remove as coating work progresses.

Protect coating from rain until dry to touch.

Upon completion of each coating application, protect coated surfaces from physical damage or chemical contamination.

Application

Apply coating materials in accordance with manufacturer's recommendations.

Apply with equipment recommended by coating manufacturer.

Number of Coats, Film Thickness:

Apply the minimum number of coats specified without regard to coating thickness. Additional coats may be required to obtain minimum required paint thickness, depending on method of application, differences in manufacturers' products, and atmospheric conditions.

Maximum film build per coat shall not exceed coating manufacturer's recommendations.

Give particular attention to edges, angles, flanges, and other similar areas, where insufficient film thickness is likely to be present, and ensure proper millage in these areas.

Sealant Application:

Rigid Coating Systems:

After pre-primer is installed (see Article COAT SCHEDULE), apply sealant to expansion joints at the coated surface boundary. Mask limits of joint to provide a neat appearance. Roughen contact surfaces with sandpaper. Prime and install sealant in accordance with manufacturer's instructions.

Identify each coat of opaque material by its relation to color of finish coat. Prime coat shall be darkest tint of specified color with each succeeding coat lighter, up to finish coat, which shall be color, tint, and sheen specified in Article COAT SCHEDULE or as shown on the Drawings. Tints of identical coats of identical color and material shall not vary.

Recoat and repair as necessary for compliance with the Specifications.

Cleaning

Collect and dispose of materials spotted or soaked with paint, oil, or solvents, and other flammable waste materials daily in accordance with the coating manufacturer's recommendations. Minimize volume of potentially contaminated solids and liquids that must be disposed.

Salvageable brushes, rollers, spatulas, and spray equipment shall be thoroughly cleaned after use and shall contain no oils, thinners, or other residue after cleaning.

Dispose of empty cans at end of each shift in accordance with the cleaning and disposal plan.

At completion of coating work, remove and dispose of materials, containers, rags, cloths, brushes, equipment, and miscellaneous other debris in accordance with the cleaning and disposal plan. Clean up spills and report, if required, in accordance with the cleaning and disposal plan.

Construction Quality Control

The Construction General Contractor shall perform the field tests specified herein with properly calibrated instruments. All testing shall be performed and recorded by personnel trained in the use of the test instruments.

Thickness Testing:

Measure coating thickness on steel with a properly calibrated, magnetic type dry film thickness gauge (as manufactured by Nordson; or approved equal).

Measure the wet film thickness (WFT) of each coat of material with a notched WFT gage (Nordson 790-015) at a minimum of five evenly spaced points for each 100 square feet of surface area or portion thereof to verify the application will provide the specified minimum dry film thickness.

Adhesion Testing:

Adhesion testing is required where the specified decontaminable coatings system is applied to concrete.

Perform adhesion testing at each field or shop location where surfaces are prepared and coatings are applied.

After surface preparation and coating application procedures have been observed and approved by the paint manufacturer's representative, select one representative location for an adhesion pull test. If the adhesion pull test does not meet the specified requirement, perform additional pull tests to determine the area of inadequate adhesion. Remove and replace coatings with inadequate adhesion.

If changes are observed in the shop or field application procedures that may affect coating adhesion, Construction Manager may require additional adhesion tests.

Construction General Contractor shall provide all test equipment required for adhesion testing.

Repair all coatings damaged by adhesion testing in accordance with the coating manufacturer's directions.

Inspection: Perform tests to ascertain that coating materials have been applied as specified in this section. Document test results. Document surface preparation, application of all coats of material, and performance of wet and dry film thickness testing in accordance with this section.

Perform water tightness test of concrete sumps in accordance with Component Construction Acceptance Test as provided in the Construction Inspection Plan, RPP-18490, Rev. 0.

Application Schedule

Concrete: Coat exposed concrete surfaces of building slabs, truck loading slab, sump, and other concrete surfaces as shown on the Drawings.

Carbon Steel:

Coat all exposed carbon steel with the coating system specified in Article COATING SCHEDULE.

Coat concrete embedded anchor bolts and concrete embedded structural steel with carbon steel coating system, except delete intermediate and finish coats. Repair coating on anchor bolt threads after nuts are installed in accordance with manufacturer's recommendations.

See Section 13122, METAL BUILDING SYSTEMS, for painting requirements associated with prefabricated steel buildings and Section 13205, LINED BOLTED STEEL LIQUID STORAGE TANKS, for painting requirements associated with the bolted steel tank.

Aluminum and Galvanized Steel in Contact with Concrete: Coat aluminum and galvanized steel in contact with concrete with carbon steel coating system, primer only. Provide a minimum dry film thickness of 4 mils.

Coat Schedule

Coat	Description	Color (*see Note 1)	Minimum Dry Film µm (mils)
Concrete – Rigid Coating System (Epoxies):			
Pre-Prime	Amerlock 400-thinned approx. 20-25% with #65 thinner.	N/A	Enough to seal surface
Joint Sealer	NuKlad 760A	N/A	N/A
Surfacer	NuKlad 114A	N/A	As required to fill voids
Prime and Base	Amerlock 400	*	Two coats: Prime 75 (3.0) Base 100 (4.0)
	PSX 700 Siloxane	*	
Intermediate and Finish. Between coats wipe entire surface with clean thinner #65, #12, or equal.			Two coats: Intermediate 100 (4.0) Finish 100 (4.0) (See Note 2)
Vitrogrit crushed glass abrasive	#29 Round Mesh (See Note 7)		
Carbon Steel:			
Shop and Field Primer	Amerlock 400, see Note 3	*	75 (3.0)
Base	Amerlock 400, see Note 3	*	100 (4.0)
Intermediate	See Note 4 PSX 700 Siloxane	*	100 (4.0)
Finish	PSX 700 Siloxane		100 (4.0)
PVC (exterior UV exposed):			
Finish	Carbocrylic 3359 DTM (See notes 5 and 6)		100 (4.0)

Notes:

1. Contrast each coat from primer-darker to finish-lighter. Finish coat to be off-white to white, except bollards, truck loading connection piping support, and manhole lids, which shall be yellow.
2. Verify recoating times between intermediate and finish coats with the coating manufacturer if relative humidity is less than 40 percent.
3. Prime and base coats may be applied in single 180 µm coat if base coat is self-priming.
4. For anchor bolts, delete intermediate and finish coats.
5. Prepare PVC by hand tool cleaning (scuff sand) using minimum 80 grit sandpaper in all directions.

6. For PVC associated with stilling wells at leachate tanks, coat only PVC exposed to UV above tank floating cover.
7. For the Truck Loading concrete pad, apply #29 round mesh VitroGrit crushed glass abrasive in between the intermediate and final coat in accordance with manufacturer's recommendation. After the intermediate coat is applied spread the sand in excess over the wet surface and let dry. After the surface has dried, sweep or vacuum the excess sand and apply the final coat.

END OF SECTION 09900

DIVISION 10–SPECIALTIES (NOT USED)

DIVISION 11–EQUIPMENT

SECTION 11305 - COMBINED AND BUILDING SUMP PUMPS SCHED. B

Part 1 – General

References

The following is a list of standards that may be referenced in this section:

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI).

ASTM INTERNATIONAL (ASTM)

ASTM A48 Standard Specification for Gray Iron Castings

ASTM A576 Standard Specification for Steel Bars, Carbon, Hot-Wrought, Special Quality

HYDRAULIC INSTITUTE STANDARDS (HIS)

NATIONAL ELECTRIC CODE (NEC)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 National Electric Code

UNDERWRITERS LABORATORIES INC. (UL)

Definitions

Terminology pertaining to pumping unit performance and construction shall conform to ratings and nomenclature of Hydraulic Institute Standards.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Make, model, weight, and horsepower of each equipment assembly.

Complete catalog information, descriptive literature, specifications, and identification of materials of construction.

Performance data curves showing head, capacity, horsepower demand, and pump efficiency over entire operating range of pump, from shutoff to maximum capacity. Indicate separately head, capacity, horsepower demand, overall efficiency, and minimum submergence required at guarantee point.

Power and control wiring diagrams, including terminals and numbers.

Complete motor nameplate data, as defined by NEMA, from motor manufacturer.

Functional testing plan demonstrating compliance with requirements specified herein.

Submittals–Approval Not Required

Information/Record (IR): Results of construction quality control testing.

Vendor Information (VI):

Manufacturer's printed installation instructions, operations, and maintenance data, including preventative maintenance tasks and frequencies for performance of those tasks.

Suggested spare parts list to maintain equipment in service for period of 5 years. Include list of special tools required for checking, testing, parts replacement, and maintenance with current price information.

Factory finish system.

Extra Materials

Furnish for each pump: One set mechanical seals.

Part 2 – Products

Supplements

Specific requirements are attached to this section as supplements.

Components

Pump equipment shall consist of pump(s) complete with motor(s), anchoring brackets, power cable(s), and pump lifting cable(s).

Lifting Arrangement: 2 feet minimum, stainless steel chain, and one "grip-eye". Attach chain permanently to pump and access platform with stainless steel wire rope. "Grip-eye" will be capable of being threaded over and engaging links of stainless steel chain so pump and motor may be lifted with "grip-eye" and independent hoist.

Components – Typical:

Oil chamber between seals shall be equipped with drain and inspection plug. Plug shall have positive anti-leak seal and shall be easily accessible from outside.

Motor nameplate horsepower shall not be exceeded at any head-capacity point on pump curve.

Pump motor and sensor cables shall be suitable for submersible pump application and cable sizing shall conform to NEC specifications for pump motors. Cable shall be of sufficient length to reach junction boxes without strain or splicing.

Cable Entry System:

Junction chamber and motor shall be separated by stator lead sealing gland or terminal board that shall prevent foreign material entering through pump top.

Utilize cable with factory-installed sealing gland with nonshrink epoxy seal system.

O-ring compression seal between sealing gland and cable entry point shall also be acceptable.

Accessories

Equipment Identification Plate: 16-gauge stainless steel with 1/4-inch die-stamped equipment tag number securely mounted in readily visible location.

Lifting Lugs: Equipment weighing over 100 pounds.

Factory Finishing

Manufacturer's standard enamel finish.

Part 3 – Execution

Installation

Install in accordance with manufacturer's printed instructions.

Connect suction and discharge piping without imposing strain to pump flanges.

Construction Quality Control

Construction General Contractor shall perform functional testing in accordance with approved test plan. Functional testing shall be performed in presence of Construction Manager or representative designated by the Construction Manager. Notify Construction Manager in writing at least 5 days in advance of testing.

Functional Test: Conduct on each pump.

Flow Output: Measured by plant instrumentation and/or storage volumes.

Amp Draw:

Verify motor current agrees with motor nameplate.

Test for continuous 30-minute period.

Supplements

The supplements listed below, following "END OF SECTION", are part of this Specification.

Sump Pump Data Sheet, 11305-01.

Sump Pump Data Sheet, 11305-02.

END OF SECTION 11305

Sump Pump Data Sheet, 11305-01

Tag Numbers: 219A-LH-P-207, 219E-LH-P-207

Pump Locations and I.D.: Buildings 219A and 219E, Combined Sump Pump

Manufacturer and Model Number: (1) Hydromatic Pump Co., Model SB3S
(2) Barnes
(3) Or approved equal

Service Conditions

Liquid Pumped (Material and Percent): Leachate from low-level radioactive waste disposal facility and rain/snow melt

Pumping Temperature (Fahrenheit): Normal: 55 Max: 130 Min: 27

Specific Gravity at 60 Degrees F: 1.0 Viscosity Range: N/A pH: 5.0-9.3

Abrasive (Y/N) (fine/coarse soil particles) Possible Scale Buildup (Y/N): Y

Total suspended solids (mg/L) 200 (estimated)

Largest diameter solid pump can pass (inches) 0.5

Performance Requirements

Capacity (US gpm): Rated: 250

Total Dynamic Head (Ft): Rated: 19

Maximum Shutoff Pressure (Ft): 50

Min. Rated Pump Hydraulic Efficiency at Rated Capacity (%): 50

Max. Pump Speed at Rated Capacity (rpm): 1,750

Constant (Y/N): Y Adjustable (Y/N): N

Design And Materials

Pump Type: Heavy-Duty Nonclog (Y/N) Y

Volute Material: Cast Iron ASTM A48

Pump Casing Material: Cast Iron ASTM A48

Motor Housing Material: Cast Iron ASTM A48

Induction Drive Motor

Horsepower: 0.30 Voltage: 460 Phase: 3 Speed (rpm): 1,750

Service Factor: 1.15 Inverter Duty (Y/N): N

Motor nameplate horsepower shall not be exceeded at any head-capacity point on the pump curve.

Enclosure: Explosion-proof, submersible, Class 1, Div. 1 or Div. 2, Groups C and D

Sump Pump Data Sheet, 11305-02

Tag Numbers: 219A-LH-P-205, 219E-LH-P-205

Pump Locations and I.D.: Buildings 219A and 219E, Floor Sump

Manufacturer and Model Number: (1) Hydromatic Pump Co.

(2) Barnes

Service Conditions

Liquid Pumped: Leachate from low-level radioactive waste disposal facility

Pumping Temperature (Fahrenheit): Normal: 55 Max 130 Min 27

Specific Gravity at 60 Degrees F: 1.0 Viscosity Range: N/A pH: 5-9.3

Abrasive (Y/N) Y (fine/coarse soil particles) Possible Scale Buildup (Y/N): Y

Total suspended solids (mg/L) 200 (estimated)

Largest diameter solid pump can pass (inches) 0.5

Performance Requirements

Capacity (US gpm): Rated: 28

Total Dynamic Head (Ft): Rated: 14

Maximum Shutoff Pressure (Ft): 30

Min. Rated Pump Hydraulic Efficiency at Rated Capacity (%): 45

Max. Pump Speed at Rated Capacity (rpm): 1,750

Constant (Y/N): _____ (Y/N): _____

Design And Materials

Pump Type: Heavy-Duty Nonclog (Y/N) Y

Volute Material: Cast Iron ASTM A48

Pump Casing Material: Cast Iron ASTM A48

Motor Housing Material: Cast Iron ASTM A48

Induction Drive Motor

Horsepower: 0.30 Voltage: 460 Phase: 3 Speed (rpm): 1,750

Service Factor: 1.15 Inverter Duty (Y/N): N

Motor nameplate horsepower shall not be exceeded at any head-capacity point on the pump curve.

Enclosure: Submersible

SECTION 11306 - LEACHATE PUMPS SCHED. B

Part 1 – General

General

Provide multi-stage, centrifugal, submersible pumps specifically designed for landfills and sideslope installations. Pumps shall be designed for pumping contaminated water and leachate. Provide all necessary pump appurtenances including lifting cable for lowering and removing the pump, power cable, a minimum 4-wheel system at each end of the pump specifically designed for transporting the pump in HDPE butt-fused carrier pipe, outlet pipe attachments and flex hose as necessary, and all other fittings or accessories required for a complete and fully functional installation.

The pump and all associated appurtenances shall be designed by the pump manufacturer to operate as a fully functional and reliable pump system. Provide a pump system capable of operating unattended with a high degree of reliability with multiple cycles per day.

Provide vent valve system, if necessary, to purge air from pumps to prevent pump air lock. Vacuum air release valves are provided in system piping at top of riser.

Provide quick-couple fitting at end of pump where outlet pipe attaches.

Remove pump discharge check valve or drill hole in check valve to prevent water from accumulating above pump outlet. Pump shall be fully capable of operating with check valve removed.

Provide stainless steel tag numbers and mounting fasteners and engrave with the equipment tag number and model number for each pump.

Note that pump control will be accomplished through software programming and the PLC mounted in the system control panels (by others) located in each Crest Pad Building.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Make, model, weight, and horsepower of each equipment assembly.

Complete catalog information, descriptive literature, specifications, and identification of materials of construction.

Performance data curves showing head, capacity, horsepower demand, and pump efficiency over the entire operating range of the pump, from shutoff to maximum capacity. Indicate separately the head, capacity, horsepower demand, overall efficiency, and minimum submergence required at the guarantee point.

Functional testing plan demonstrating compliance with requirements specified herein.

Detailed mechanical and electrical drawings showing the equipment dimensions, size, and locations of connections and weights of associated equipment.

Detailed catalog information, descriptive literature, and specifications of all components associated with pump removal system.

Power and control wiring diagrams, including terminals and numbers.

Complete motor nameplate data, as defined by NEMA, motor manufacturer.

Results of source quality control testing.

Submittals–Approval Not Required

Information/Record (IR):

Manufacturer's certification of factory testing to establish conformance with specified requirements. Certification must include certificates of calibration traceable to a nationally recognized standards organization such as National Institute of Standards and Technology (NIST).

Special shipping, storage and protection, and handling instructions.

Manufacturer's Certificate of Proper Installation.

Results of field quality control testing.

Vendor Information (VI):

Suggested spare parts list to maintain the equipment in service for a period of 1 year. Include a list of special tools required for checking, testing, parts replacement, and maintenance with current price information.

List special tools, materials, and supplies furnished with equipment for use prior to and during startup and for future maintenance.

Manufacturer's printed installation instructions, including recommended preventative maintenance tasks and frequencies for performance of those tasks.

Operation and maintenance manual.

Part 2 – Products

Pumps

All major components shall be Type 304 stainless steel including the housing, fasteners, shaft, diffuser chamber, and impeller(s). Components shall be highly corrosion resistant and suitable for contaminated water and leachate service. Gaskets, O-rings, and seals shall have compatibility properties equivalent to Viton material as a minimum.

Pump bearings shall have better heat and wear resistance than Teflon bearings.

Provide power and stainless steel cable as recommended by manufacturer and to the length and configuration as shown on the Drawings.

Motors

Provide sealed pump motors suitable for continuous submerged service. Provide continuous motor leads without splices along the full length of the discharge pipe. Leads shall be fully insulated with chemical and waterproof insulation properties. Provide motor designed for continuous duty and multiple cycle times of 60 starts per hour. Motors shall have thermal overload protection.

Source Quality Control

Construction General Contractor shall perform source quality testing at the factory as specified herein. Notify Construction Manager in writing at least 10 days in advance of testing. Construction Manager will coordinate with personnel required to witness testing.

Factory Tests and Adjustments: Test all equipment actually furnished.

Factory Test Report: Include test data sheets, curve test results, performance test logs.

Performance Test

Conduct on each pump.

Perform under simulated operating conditions, at a minimum of six operating points on the pump curve.

Test for a continuous 30-minute period without malfunction. Check for excessive or abnormal vibrations and correct deficiencies.

Test Log: Record the following:

Total head.

Flow measured by factory instrumentation and/or storage volumes.

Average distance from suction well water surface to pump discharge centerline for duration of test.

Pump discharge pressure converted to feet of liquid pumped and corrected to pump discharge centerline.

Driving motor voltage and amperage measured for each phase.

Adjust or modify units and retest if necessary.

Pump Removal System:

Provide one common manual pump removal winch and winch support for raising and lowering the LCRS low flow and secondary leak detection pump and associated discharge piping. Provide one manual pump removal winch and winch support for raising and lowering the high flow pump and associated discharge piping. Winches shall be brake (spur gear) winch with automatic brake: Thern Model No. 4032PB (low flow/secondary leak detection) and M452B (high flow), or equal. Provide winch with sufficient capacity to handle required length of lifting cable (stainless steel wire rope). Lifting cable shall include and the winch accommodate a swage ball fitting cable end for quick connect/disconnect.

Winch shall be mounted on structural steel support. The winch support shall only be installed for pump installation/removal. Three sets of drop-in anchors shall be installed in floor that match the winch support base bolt pattern. Each set of drop-in anchors shall be located to align the winch support with each HDPE riser pipe such that the lifting cable does not rub the HDPE riser pipe during pump installation/removal. Coordinate location of the drop-in anchors with Construction Manager.

Provide two manual hose reels for separately handling the leachate pump power and level transducer cables. Hose reels shall be heavy-duty hand crank reels with adjustable spool rotation drag and spool lock pins. Hose reels shall be Reelcraft Model No. C33118LI, or equal. Each reel shall be located as required for proper alignment with appropriate riser pipe. Drop-in anchors shall be installed in floor that match hose reel "feet" bolt pattern for both reels. Bolts shall be left in place to prevent accumulation of dirt, etc., in anchor threads. Bolts shall be tapered flathead to eliminate tripping hazard.

Level Sensors:

Provide level sensors integral to Cell 1 and Cell 2 LDS leachate pump (two total) as shown on the Drawings. Level elements shall be designed and constructed for landfill leachate service, i.e., fully submersible and chemically resistant.

The level sensor shall include a transmitter with built-in temperature compensation and an accuracy of plus or minus 1.0 percent. Sensor output shall be a conditioned compensated 4 to 20 mA signal.

The sensor control cable shall be shielded to prevent signal disruption and include a vent tube for atmospheric pressure compensation. Control cables shall include polyurethane jacket and Kevlar tension members.

Level sensors shall be mounted on the pump housing and be field serviceable without having to disassemble the pump.

Part 3 – Execution

Installation

Install in accordance with manufacturers' printed instructions and manufacturers' representatives' guidance and recommendations.

Construction Quality Control

Construction General Contractor shall perform functional testing in accordance with approved testing plan. Functional testing shall be performed in the presence of the Construction Manager. Notify Construction Manager in writing at least 5 days in advance of testing.

Function Tests:

Prior to the pump and level transducer insertion tests identified in the Component CAT procedures, verify the LCRS, LDS, and SLDS riser pipe transition from the side slope to the horizontal portion of the riser pipe is adequate for leachate pump and level transducer insertion. Perform the pump/level transducer insertion tests prior to backfilling and after the riser pipe installation is finished from within the sump to a location approximately 50 feet up the side slope. Perform the pump/level transducer insertion tests using the associated leachate pump or level transducer for each riser. In addition to the level transducer insertion for the SLDS riser pipe, test the transition using the LDS low-flow leachate pump as well. If actual level transducers or leachate pumps are not available at the time of the testing, "dummy" level transducers and pumps can be used per approval from CHG Construction Manager.

After complete installation of the side slope riser pipe from the sump to the crest pad building, verify exact length of pump discharge and level transducer piping required by using a long tape measure to measure actual dimension. Test the insertion and extraction of each pump from the side slope riser pipe and into the crest pad buildings. Perform testing while the perforated carrier pipe sections in the sumps are exposed to allow observation of the pump removals and insertions from the carrier pipe.

Test the pumps under simulated conditions using a temporary tank located at the bottom of the landfill. Place pump in the tank and connect temporary flexible hose between the pump and discharge pipe routed up the side slope surface and between the discharge pipe and riser connection in Crest Pad Building. Keep the tank full to supply adequate water to the pumps during the pump test. Record amp draw readings.

Supplements

The supplements listed below, following "END OF SECTION," are a part of this Specification.

Data Sheets:

Supplement 1—Leachate Pump Data Sheet, 11306-01.

Supplement 2—Leachate Pump Data Sheet, 11306-02.

Supplement 3—Leachate Pump Data Sheet, 11306-03.

Supplement 4—Leachate Pump Data Sheet, 11306-04.

END OF SECTION 11306

Leachate Pump Data Sheet, 11306-01

Tag Numbers: 219A-LH-P-202, 219E-LH-P-202

Pump Locations and I.D.: Cell 1 LCRS Sump, Low Flow

Cell 2 LCRS Sump, Low Flow

Manufacturer and Model Number: (1) EPG Companies; Model WSD 3-3

(2) Or equal

Service Conditions

Liquid Pumped (Material and Percent): Leachate from low-level radioactive waste landfill

Pumping Temperature (Fahrenheit): Normal: 55 F Max: 130 F Min: 27 F

Specific Gravity at 60 Degrees F: 1.0 Viscosity Range: NA pH: 5.0-9.3

Abrasive (Y/N) Y (infrequent fine soil particles) Possible Scale Buildup (Y/N): Y

Total Suspended Solids (mg/l): 200 (estimated)

Performance Requirements At Primary Design Point

Capacity (US gpm): Rated: 13

Total Dynamic Head (Ft): Rated: 66

Min. Hydraulic Efficiency (%): 60

Maximum Shutoff Pressure (Ft): 90

Max. Pump Speed at Design Point (rpm): 3,450

Constant (Y/N): Y Adjustable (Y/N): N

Design And Materials

Design: Wheeled enclosure frame Back Pullout (Y/N) Y

Discharge Orientation: Center

Casing Materials: Type 304 SST

Case Wear Ring (Y/N) NA Material: NA

Impeller: Type: Closed Material: Type 304 SST

Impeller Wear Ring (Y/N): Y Material: E-Glide (engineered plastic) or equal

Shaft Material: Type 304 SST Shaft Sleeve Material: E-Glide or equal

Shaft Seal: Y Ring Material: E-Glide or equal Lubrication: Fluid

AFBMA B-10 Bearing Life (Hrs): NA Lubrication: NA

Drive Type: Direct Coupled

Induction Drive Motor

Horsepower: 0.5 Voltage: 460 Phase: 3

Speed (rpm): 3,450

Service Factor: 1.15 Inverter Duty (Y/N) N

Motor nameplate horsepower shall not be exceeded at any head-capacity point on the pump curve.

Enclosure: Submersible

Leachate Pump Data Sheet, 11306-02

Tag Numbers: 219A-LH-P-203, 219E-LH-P-203

Pump Location and I.D.: Cell 1 LCRS Sump, High Flow

Cell 2 LCRS Sump, High Flow

Manufacturer and Model Number: (1) EPG Companies; Model WSD 30-4

(2) Or equal

Service Conditions

Liquid Pumped (Material and Percent): Leachate from low-level radioactive waste landfill

Pumping Temperature (Fahrenheit): Normal: 55 F Max: 130 F Min: 27 F

Specific Gravity at 60 Degrees F: 1.0 Viscosity Range: NA pH: 5.0-9.3

Abrasive (Y/N) Y (infrequent fine soil particles) Possible Scale Buildup (Y/N): Y

Total Suspended Solids (mg/l): 200 (estimated)

Performance Requirements At Primary Design Point

Capacity (US gpm): Rated: 155

Total Dynamic Head (Ft): Rated: 118

Min. Hydraulic Efficiency (%): 60

Maximum Shutoff Pressure (Ft): 208

Max. Pump Speed at Design Point (rpm): 3,450

Constant (Y/N): Y Adjustable (Y/N): N

Design and Materials

Design: Wheeled enclosure frame (Y/N) Y

Discharge Orientation: Center

Casing Materials: Type 304 SST

Case Wear Ring (Y/N) NA Material: NA

Impeller: Type: Closed Material: Type 304 SST

Impeller Wear Ring (Y/N): Y Material: E-Glide (engineered plastic), or equal

Shaft Material: Type 304 SST Shaft Sleeve Material: E-Glide (engineered plastic), or equal

Shaft Seal: Y Ring Material: E-Glide or equal Lubrication: Fluid

AFBMA B-10 Bearing Life (Hrs): NA Lubrication: NA

Drive Type: Direct Coupled

Induction Drive Motor

Horsepower: 7.5 Voltage: 460 Phase: 3

Speed (rpm): 3,450

Service Factor: 1.15 Inverter Duty (Y/N) N

Motor nameplate horsepower shall not be exceeded at any head-capacity point on the pump curve.

Enclosure: Submersible

Leachate Pump Data Sheet, 11306-03

Tag Numbers: 219A-LH-P-204, 219E-LH-P-204

Pump Locations and I.D.: Cell 1 LDS Sump

Cell 2 LDS Sump

Manufacturer and Model Number: (1) EPG Companies; Model WSD 1.5-3

(2) Or equal

Service Conditions

Liquid Pumped (Material and Percent): Leachate from low-level radioactive waste landfill

Pumping Temperature (Fahrenheit): Normal: 55 F Max: 130 F Min: 27 F

Specific Gravity at 60 Degrees F: 1.0 Viscosity Range: NA pH: 5.0-9.3

Abrasive (Y/N) Y (infrequent fine soil particles) Possible Scale Buildup (Y/N): Y

Total Suspended Solids (mg/l): 200 (estimated)

Performance Requirements At Primary Design Point

Capacity (US gpm): Rated: 4

Total Dynamic Head (Ft): Rated: 65

Min. Hydraulic Efficiency (%): 60

Maximum Shutoff Pressure (Ft): 80

Max. Pump Speed at Design Point (rpm): 3,450

Constant (Y/N): Y Adjustable (Y/N): N

Design And Materials

Design: Wheeled enclosure frame Back Pullout (Y/N) Y

Discharge Orientation: Center

Casing Materials: Type 304 SST

Case Wear Ring (Y/N) NA Material: NA

Impeller: Type: Closed Material: Type 304 SST

Impeller Wear Ring (Y/N): Y Material: E-Glide (engineered plastic) or equal

Shaft Material: Type 304 SST Shaft Sleeve Material: E-Glide or equal

Shaft Seal: Y Ring Material: E-Glide or equal Lubrication: Fluid

AFBMA B-10 Bearing Life (Hrs): NA Lubrication: NA

Drive Type: Direct Coupled

Induction Drive Motor

Horsepower: 0.5 Voltage: 460 Phase: 3

Speed (rpm): 3,450

Service Factor: 1.15 Inverter Duty (Y/N) N

Motor nameplate horsepower shall not be exceeded at any head-capacity point on the pump curve.

Enclosure: Submersible

Leachate Pump Data Sheet, 11306-04

Tag Numbers: 219-LH-P-208

Pump Locations and I.D.: Cell 1 SLDS Sump

Cell 2 SLDS Sump

Manufacturer and Model Number: (1) EPG Companies; Model WSD 1.5-4

(2) Or equal

Service Conditions

Liquid Pumped (Material and Percent): Leachate from low-level radioactive waste landfill

Pumping Temperature (Fahrenheit): Normal: 55 F Max: 130 F Min: 27 F

Specific Gravity at 60 Degrees F: 1.0 Viscosity Range: NA pH: 5.0-9.3

Abrasive (Y/N) Y (infrequent fine soil particles) Possible Scale Buildup (Y/N): Y

Total Suspended Solids (mg/l): 200 (estimated)

Performance Requirements At Primary Design Point

Capacity (US gpm): Rated: 4

Total Dynamic Head (Ft): Rated: 65

Min. Hydraulic Efficiency (%): 60

Maximum Shutoff Pressure (Ft): 80

Max. Pump Speed at Design Point (rpm): 3,450

Constant (Y/N): Y Adjustable (Y/N): N

Design and Materials

Design: Wheeled enclosure frame Back Pullout (Y/N) Y

Discharge Orientation: Center

Casing Materials: Type 304 SST

Case Wear Ring (Y/N) NA Material: NA

Impeller: Type: Closed Material: Type 304 SST

Impeller Wear Ring (Y/N): Y Material: E-Glide (engineered plastic) or equal

Shaft Material: Type 304 SST Shaft Sleeve Material: E-Glide or equal

Shaft Seal: Y Ring Material: E-Glide or equal Lubrication: Fluid

AFBMA B-10 Bearing Life (Hrs): NA Lubrication: NA

Drive Type: Direct Coupled

Induction Drive Motor

Horsepower: 0.5 Voltage: 120 Phase: 1

Speed (rpm): 3,450

Service Factor: 1.15 Inverter Duty (Y/N) N

Motor nameplate horsepower shall not be exceeded at any head-capacity point on the pump curve.

Enclosure: Submersible

Note: An adequate length of power cord shall be supplied. Standard plug shall be provided on the power cord.

SECTION 11312 - HORIZONTAL END SUCTION CENTRIFUGAL PUMPS SCHED. B

Part 1 – General

References

The following is a list of standards, which may be referenced in this section:

AMERICAN BEARING MANUFACTURERS' ASSOCIATION (ABMA)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

- Type 416 Stainless Steel
- Type 1035 Steel
- Type 1045 Carbon Steel
- Type 4140 Alloy Steel

ASTM INTERNATIONAL (ASTM)

ASTM A48	Standard Specification for Gray Iron Castings
ASTM A53/A53M	Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A276	Standard Specification for Stainless Steel Bars and Shapes
ASTM A576	Standard Specification for Steel Bars, Carbon, Hot-Wrought, Special Quality
ASTM B62	Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM B148	Standard Specification for Aluminum-Bronze Sand Castings
ASTM B584	Standard Specification for Copper Alloy Sand Castings for General Applications

HYDRAULIC INSTITUTE STANDARDS

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 112	Standard Test Procedure for Polyphase Induction Motors and Generators

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

NEMA MG 1	Motors and Generators

Definitions

Terminology pertaining to pumping unit performance and construction shall conform to the ratings and nomenclature of the Hydraulic Institute Standards.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Make, model, weight, and horsepower of each equipment assembly.

Complete catalog information, descriptive literature, specifications, and identification of materials of construction.

Performance data curves showing head, capacity, horsepower demand, and pump efficiency over the entire operating range of the pump, from shutoff to maximum capacity. Indicate separately the head, capacity, horsepower demand, overall efficiency, and minimum submergence required at the guarantee point.

Detailed mechanical and electrical drawings showing the equipment dimensions, size, and locations of connections and weights of associated equipment.

Functional testing plan demonstrating compliance with requirements specified herein.

Power and control wiring diagrams, including terminals and numbers.

Complete motor nameplate data, as defined by NEMA, motor manufacturer.

Results of source quality control testing.

Submittals–Approval Not Required

Information/Record (IR):

Special shipping, storage and protection, and handling instructions.

Manufacturer's Certificate of Proper Installation.

Results of field quality control testing.

Vendor Information (VI):

Suggested spare parts list to maintain the equipment in service for a period of 5 years. Include a list of special tools required for checking, testing, parts replacement, and maintenance with current price information.

List special tools, materials, and supplies furnished with equipment for use prior to and during startup and for future maintenance.

Manufacturer's printed installation instructions.

Operation and maintenance data, including recommended preventative maintenance tasks and frequencies for performance of those tasks.

Factory finish system data sheets.

Extra Materials

Furnish for each pump:

Complete set packing.

Complete set bearings.

Complete set gaskets and O-ring seals.

Complete set of shaft sleeves.

Complete set keys, dowels, pins, etc.

Complete mechanical seal.

Impeller.

Impeller shaft.

Impeller wear ring.

Head shaft.

One complete set of any special tools required to dismantle pump.

Part 2 – Products

General

Coordinate pump requirements with drive manufacturer and be responsible for pump and drive requirements.

Where adjustable speed drives are required, furnish a coordinated operating system complete with pump, drive, and speed controller.

Supplements

Some specific requirements are attached to this section as supplements.

Accessories

Equipment Identification Plate: 16-gauge stainless steel with 1/4-inch die-stamped equipment tag number securely mounted in a readily visible location.

Lifting Lugs: Equipment weighing over 100 pounds.

Anchor Bolts: Galvanized, sized by equipment manufacturer, 1/2-inch minimum diameter, and as specified in Section 05500, METAL FABRICATIONS AND CASTINGS.

Factory Finishing

Manufacturer's standard enamel finish.

Source Quality Control

Construction General Contractor shall perform source quality control testing at the factory as specified herein. Notify Construction Manager at least 10 days in advance of testing.

Performance Test: Perform manufacturer's standard motor test on equipment.

Part 3 – Execution

Installation

Install in accordance with manufacturer's printed instructions.

Level base by means of steel wedges (steel plates and steel shims). Wedge taper not greater than 1/4 inch per foot. Use double wedges to provide a level bearing surface for pump and driver base. Accomplish wedging so that there is no change of level or springing of the baseplate when the anchor bolts are tightened.

Adjust pump assemblies such that the driving units are properly aligned, plumb, and level with the driven units and all interconnecting shafts and couplings. Do not compensate for misalignment by use of flexible couplings.

After pump and driver have been set in position, aligned, and shimmed to proper elevation, grout the space between the bottom of the baseplate and the concrete foundation with a poured, nonshrinking grout. Remove wedges after grout is set and pack void with grout.

Connect suction and discharge piping without imposing strain to pump flanges.

Anchor Bolts: Accurately place using equipment templates and as specified in Section 05500, METAL FABRICATIONS AND CASTINGS.

Construction Quality Control

Construction General Contractor shall perform field quality control testing in accordance with approved testing plan. Functional testing shall be performed in the presence of the Construction Manager. Notify Construction Manager in writing at least 5 days in advance of testing.

Functional Tests:

Conduct on each pump.

Test for a continuous 1/2-hour period without malfunction.

Test Log: Record the following:

Total head.

Capacity.

Flow measured by factory instrumentation and/or storage volumes.

Average distance from suction well water surface to pump discharge centerline for duration of test.

Pump discharge pressure converted to feet of liquid pumped and corrected to pump discharge centerline.

Driving motor voltage and amperage measured for each phase.

Alignment: Test complete assemblies for correct rotation, proper alignment and connection, and quiet operation.

Operating Temperatures: Monitor bearing areas on pump and motor for abnormally high temperatures.

Manufacturer's Services

Manufacturer's Representative: Present at site or classroom designated by Tank Farm Contractor, for minimum person-days listed below, travel time excluded:

Person-day for installation assistance and inspection.

Person-day for functional and performance testing and completion of Manufacturer's Certificate of Proper Installation.

Supplements

The supplements listed below, following "END OF SECTION," are a part of this Specification.

Pump Data Sheet, 11312-01

END OF SECTION 11312

Horizontal End Suction Centrifugal Pump Data Sheet, 11312-01

Tag Numbers: _____

Pump Name: _____

Manufacturer and Model Number: (1) _____

(2) _____

Service Conditions

Liquid Pumped (Material and Percent): _____

Pumping Temperature (Fahrenheit): Normal: 55 Max 130 Min 27

Specific Gravity at 60 Degrees F: 1.0 Viscosity Range: N/A pH: 5.0-9.3

Abrasive (Y/N) Y (fine soil particles) Possible Scale Buildup (Y/N): Y

Total suspended solids (mg/L) 200 (estimated)

Largest diameter solid pump can pass (inches) .25

Performance Requirements At Primary Design Point

Capacity (US gpm): Rated: 250

Total Dynamic Head (Ft): Rated: 25

Min. Hydraulic Efficiency (%): 75

Maximum Shutoff Pressure (Ft): 40

Max. Pump Speed at Design Point (rpm): 1,750

Constant (Y/N): Y Adjustable (Y/N): N

Design And Materials

ANSI (Y/N) Y Standard (Y/N) Y Design: Frame-mounted (Y/N) Y

Close-Coupled Casing (Y/N) N Back Pullout (Y/N) Y

Discharge Orientation: 12:00 Rotation (view from end coupling): CW

Shaft Seal: Packing (Y/N) N

Mechanical (Y/N) _____

Lubrication: Process Water

Drive Type: Direct-Coupled: _____ Belt _____ Adjustable Speed _____

Induction Drive Motor

Horsepower: _____ Voltage: _____ Phase: _____ Speed (rpm): 1,750

Service Factor: 1.15 Inverter Duty (Y/N) _____

Motor nameplate horsepower shall not be exceeded at any head-capacity point on the pump curve.

Enclosure: Totally enclosed fan cooled

Mounting Type: Horizontal _____ Nonreverse Ratchet (Y/N) _____

Testing

Pump Tests: Factory Functional (Y/N) _____ Field Performance (Y/N) N

Factory Hydrostatic Casing Pressure Test (Y/N) _____

Field Functional (Y/N) _____ Field Performance (Y/N) _____

Field Vibration (Y/N) N

DIVISION 12–FURNISHINGS (NOT USED)

DIVISION 13–SPECIAL CONSTRUCTION

SECTION 13122 - METAL BUILDING SYSTEMS SCHED. B

Part 1 – General

Work Included

The Construction Subcontractor shall furnish and install four prefabricated pre-engineered metal building, complete, as shown on the Drawings and as specified herein.

References

The following Codes and Standards, including others referenced therein, form a part of this Section to the extent specified herein:

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC	Specification for Structural Steel for Buildings – Allowable Stress Design (ASD)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI	Specification for the Design of Cold-Formed Steel Structural Members

ASTM INTERNATIONAL (ASTM)

ASTM A36	Standard Specification for Carbon Structural Steel
ASTM A53	Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A325	Standard Specification for Structural Bolts, Steel, Heat-Treated, 120/105 ksi Minimum Tensile Strength
ASTM A500	Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
ASTM A501	Standard Specification for Hot-Formed Welded and Seamless Carbon Steel Structural Tubing
ASTM A529	Standard Specification for High-Strength Carbon-Manganese Steel of Structural Quality
ASTM A570	Standard Specification for Steel, Sheet and Strip, Carbon, Hot-Rolled, Structural Quality
ASTM A572	Standard Specification for High-Strength, Low-Alloy Columbium-Vanadium Structural Steel
ASTM A607	Standard Specification for Steel, Sheet and Strip, High-Strength, Low-Alloy, Columbium or Vanadium, or Both, Hot-Rolled, and Cold-Rolled
ASTM F959	Standard Specification for Compressible-Washer-Type Direct Tension Indicator for Use with Structural Fasteners

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1	Structural Welding Code – Steel

METAL BUILDING MANUFACTURERS ASSOCIATION (MBMA)

Recommended Design Practices Manual, for applicable loads and load combinations

Metal Building Systems Manual, for collateral loads

INTERNATIONAL CONFERENCE OF BUILDING OFFICIALS (ICBO)

UBC, Uniform Building Code

STEEL DOOR INSTITUTE (SDI)

SDI 100 Recommended Specifications for Standard Steel Doors and Frames

SDI 117 Manufacturing Tolerances Standard Steel Doors and Frames

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Shop Drawings:

Manufacturer's Standard Details and Structural Calculations: Clearly mark those portions that apply to specific Project and those parts that do not apply.

Manufacturer's Literature and Technical Data: Drawings and Specifications for proposed metal building system.

Painting System: Specifications including paint manufacturer's name, product trade-name, and preparation for shop and field coats. Provide minimum 20 color samples for Tank Farm Contractor to choose.

Drawings Stamped by Engineer Registered in the State of Washington and Prepared Specifically for this Project:

Materials and Details: Show materials, details of components (including doors and other accessories), finishes, fastenings, methods of joining, sealants, anchor bolt, shear angle, and baseplate details including all sizes and dimensions, size and location of structural members and bracing, wall structural members, bracing, openings, and structural wind columns as required.

Calculations Stamped by Engineer Registered in the State of Washington: Complete structural stress and deflection analysis of structural components and connections; consider prying action of bolts should proposed design use bolted moment-resistant connections in main frames. Provide anchor bolt calculations and separately tabulate anchor bolt reaction for all case loads and load combinations.

Samples: Minimum 2-inch by 3-inch metal for components requiring color selection.

Submit documentation of construction quality control testing as specified herein.

Submittals–Approval Not Required

Information/Record (IR):

Statements of Qualification:

Documentation of past 5 years' experience record to include project name, location, type and date completed, building manufacturer and owner's contact person.

Certification of approval by manufacturer.

Vendor Information (VI):

Manufacturer's written instructions for shipping, handling, storage, protection and erection or installation of building and components.

Manufacturer's certification or proof of current membership in Metal Building Manufacturer's Association (MBMA).

Manufacturer's Certificate of Proper Installation.

Operations and maintenance manual.

Qualifications

Provide prefabricated metal buildings as produced by a manufacturer who is regularly engaged in fabrication of pre-engineered metal structures of type and quality indicated. All components shall be provided from one manufacturer.

Warranties

Warranty shall begin at the time of Substantial Completion.

The roofing and siding shall be warranted for a minimum of 5 years against wind damage, leakage, paint fade, chipping, peeling, attachment and rusting. Warranty shall include labor and materials for replacement of defective panels. Warranty shall not be pro-rated over 5-year period.

Finish on metal roof and wall panels, flashing, and trim will not chalk, crack, check, blister, peel, flake, chip, or loose adhesion shall be warrantied for 5 years.

Part 2 – Products

Manufacturer

Dimensions of the 12-foot by 12-foot buildings are based on Panl-Line Building System as manufactured by Butler Manufacturing Co. Dimension and sizes of the 21-foot by 16-foot buildings are based on the Parkline Building Systems, Inc. (Type AL), and VP Buildings, Inc. If an "or equal" building manufacturer is submitted and approved by the Engineer, all dimensions and clearances shall be taken as minimums for evaluation of submittal. Construction Subcontractor shall be responsible for all adjustments required to plans as a consequence of changing building manufacturer. All shop drawings and special process procedures as welding, painting and structural bolting, shall be submitted for approval and shall be stamped by a registered professional engineer licensed to practice in the State of Washington.

Type: The metal building shall be a prefabricated, weather-tight, free-standing building having a structural steel frame. The 12-foot by 12-foot buildings shall be self-framing buildings incorporating diagonal bracing. The 21-foot by 16-foot buildings shall be rigid frame in all walls with no diagonal bracing in the wall. The roof slope and the eave height shall be at as specified on Construction Drawings.

Design Loads

The building shall be designed for the following applied loads in addition to dead load:

Roof Live Loads: Roof covering shall be designed for 20 psf uniformly distributed load.

Roof Snow Load: Ground snow load is 15 psf, $C_e = 10$, $I = 0.8$ designed per ASCE 7.

Wind Loads: The wind load on the structure shall be designed per ASCE 7 and DOE STD-1020-02 using a 3-second gust wind velocity of 85-mph wind speed. Design and calculate according to the ASCE 7 exposure Class "C" with an Importance Factor = 1.0.

Seismic Loads: Seismic loads shall be determined and applied in accordance with the UBC Zone 2B, Importance Factor = 1.0, Soil S_c. Out-of-plane system stability, nonstructural components, and equipment shall be evaluated using UBC 1632.

Collateral Loads: All additional dead loads, other than the weight of the metal building system, such as fire sprinklers, mechanical HVAC systems, electrical systems, and ceilings. Collateral loads shall be a minimum of 10 pounds per square foot as defined in the Metal Building Systems Manual published by the MBMA.

Maximum Deflection: Deflection shall be limited to L/240 for DL and DL+LL for all building components.

Combination of Loads: Combined loads shall be as prescribed in the UBC.

Building Code Requirement: Design building, roof system, roof overhang including support framing, roof and wall panels, and fasteners for horizontal and uplift wind loads and earthquake forces.

Materials

Hot-Rolled Structural Shapes: Conform to ASTM A36 or A529.

Tubing or Pipe: Conform to ASTM A500, Grade B; ASTM A501, or ASTM A53.

Members Fabricated from Plate or Bar Stock: 42,000 psi minimum yield strength; Conform to ASTM A529, A570, or A572.

Members Fabricated by Cold Forming: Conform to ASTM A607, Grade 50.

Galvanized Steel Sheet: Conform to ASTM A446 with G90 coating. "Class" to suit building manufacturer's standards.

Structural Framing Components

Rigid Frames:

Rigid frames shall be hot-rolled structural steel, factory welded, and shop painted. Furnish complete with attachment plates, bearing plates, and splice members. Factory drilled for bolted field assembly.

Length of span and spacing of frames shall be as shown on Drawings except slight roof slope variations are acceptable to meet manufacturer's standard.

Wind Bracing: No "x" type rod bracing shall be used in bays where bracing would cross windows or door openings, or where the interior of the exterior walls are to be finished. Use portal frames where bracing is required at window or door openings.

Secondary Framing: Purlins, eave girts, girts, flange and sag bracings shall be "Z" or "C" roll formed sections no pre-punched for fasteners, and shall be shop prime painted. Roof purlins shall be spaced a maximum of 5-foot 0-inch O.C. Base channel, sill angle, purlin spacers; minimum 14-gauge cold-formed steel; and shall be shop prime painted.

Anchor Bolts: The anchor bolts for the rigid frames shall be carbon steel and designed by the pre-engineered building manufacturer. Location and placement shall be coordinated with the foundation rebar shown on the Drawings. Any changes in rebar placement shall be brought to the attention of the Construction Subcontractor and engineering calculations shall be provided taking into account the changed rebar location.

Bolts: Bolts shall be ASTM A325 in quantities necessary for design loads and connection details. Provide zinc- or cadmium-plated units when in direct contact with panels. Direct tension indicators shall conform to ASTM F959.

Fabrication:

Shop fabricate to the indicated size and section, complete with base plates, bearing plates, and other plates as required for erection, welded in place, and with all required holes for anchoring or connections shop drilled or punched to template dimensions.

Shop connections shall be power riveted, bolted, or welded.

Field connections shall be bolted. Install high strength threaded fasteners in accordance with "Specifications for Structural Joints Using ASTM A325 or A490 Bolts."

Weld Construction:

Comply with AWS D1.1 for procedures, appearance and quality of welds, and methods used in connecting welding work. Welding shall not be performed at the project site.

Construction General Contractor shall provide Certified Weld Inspector (CWI) to perform visual examination of all off-site welds in accordance with AWS D1.1, Section 6. Document weld acceptance on Construction Subcontractor Weld History.

Shop Painting:

Surfaces to be primed shall be cleaned of loose mill scale, rust, dirt, oil, grease, and other matter precluding paint bond. Follow procedures of SSPC-SP3 for power tool cleaning, SSPC-SP7 for brush-off blast cleaning, and SSCP-SP1 for solvent cleaning.

Prime structural steel primary and secondary framing members with manufacturer's standard rust-inhibitive primer having over 50 percent rust-inhibitive pigment, such as organic zinc. No lead or chromate will be allowed.

Prime galvanized members, after phosphoric acid pretreatment, with zinc dust-zinc oxide primer.

Roofing And Siding

General: Provide roofing and siding sheets formed to general profile or configuration as specified. Provide flashings, closers, fillers, ridge covers, and other sheet metal accessories, factory formed of same material and finish as roofing and siding. Factory-applied baked enamel, in color selected by the Engineer.

Roof Panels:

The Interlocking-Standing Seam Roof Covering shall carry an Underwriters' Laboratories, Inc., Uplift Classification of not less than Class 90 and shall consist of material not less than 24-gauge aluminized coated steel with Kynar finish on exterior face. The panels shall be installed with the ribs upstanding and parallel to the roof slope.

All longitudinal interlocking ribs as well as any transverse end laps shall be properly sealed, according to the manufacturer's instructions, with non-drying sealant.

The roof panels shall be secured to each structural support by a steel clip concealed between the adjacent male and female ribs and fastened under that panel's weather surface. Clip shall be long enough to allow Styrofoam thermal spacer on top of purlin.

Penetrations through the roof panel by fasteners shall be limited to only those required at the rake eaves, at end laps and at the ridge. All exposed fasteners shall be fitted with weather-seal washers of hydrocarbon-based elastomer (synthetic rubber) with a compatible metal backing.

Thermal (break) spacers shall be provided continuously at each structural support to minimize thermal conductivity. The thermal spacer shall be a continuous Styrofoam strip, 3 inches by 1 inch thick.

Wall Panels Exterior: The interlocking-ribbed wall covering shall consist of panels of not less than 24 U.S. gauge, fluoropolymer enamel finished, aluminized coated steel with male and female ribs. The wall panels shall be applied to the structural framing with the interlocking ribs toward the interior of the structure. The interlocking ribs shall be secured at the base, at each intermediate girt, and at the support at which it terminates, by means of concealed fasteners, thus eliminating any through-wall fastening. Trim finish to match wall panel.

All interior fasteners, i.e., screws, bolts and nuts, etc., shall be of carbon steel having a protective coating of either zinc or cadmium.

Interior Liner Panels: Interior wall liner panels shall be provided throughout the building on all perimeter walls. The panels shall be 24 gauge, white with concealed fasteners. All panel joints shall be provided with sealer along the edges of each panel. The liner panels shall function as a vapor barrier. Length of panels shall be full height with no horizontal joints. Finish shall be as described below.

Sealing Tape: Sealing tape shall be 100 percent solids, pressure sensitive grey polyisobutylene compound tape with release paper backing. Not less than 1/2 inch wide and 1/4 inch thick, nonsag, nontoxic, nonstaining and permanently elastic.

Joint Sealant: Joint sealant shall be one-part elastomeric; polyurethane, polysulfide, or silicon rubber as recommended by building manufacturer.

Ice Stops: Provide ice stops to prevent snow and ice damage to gutters. Ice stops shall be "ICEJAX" as manufactured by Snowjax Inc., Mechanicsburg, Pennsylvania, or approved equal. "ICEJAX" shall be adhered with Loctite "Depend", or approved equal, to metal roof panels.

Rain Gutter and Downspouts: The rain gutter shall be continuous along the eaves of the building. The gutter shall be a surface mounted type with downspout size and number as called for by the building manufacturer or as shown on the drawings. Gutter shall be minimum 5 x 5 inches in cross section. Gutter and downspouts shall be standard design as manufactured by Metal Building Manufacturer, or approved equal. Gutter shall be installed with 1/4 inch per 10-foot 0-inch slope to downspout. Factory finish to match wall panels.

Insulation And Vapor Retarder

As specified in Section 07210, BUILDING INSULATION.

Doors

Steel Doors: 1-3/4-inch doors, conforming to ANSI/SDI 100, with manufacturer's standard. Provide exterior doors with top and bottom edges finished flush. Provide doors of materials and ANSI/SDI 100 grades and models specified below, or as indicated on drawings and schedules.

Exterior Doors: Unless otherwise indicated, Grade III, extra heavy duty, Model 2 (seamless) design), minimum 16 gauge galvanized steel sheet faces.

Door Frames

Provide metal frames for doors and other openings according to ANSI/SDI 100 and of types and styles as shown on drawings and schedules. Conceal fastenings unless otherwise indicated. Frames shall be No. 14 USS gage or heavier cold-rolled steel sheet. Form exterior frames of hot dip galvanized steel. Fabricate frames with mitered and welded corners.

Available manufacturers of steel doors include the following:

AMWELD Building Products Div.
Ceco Corp.
Curries
Fenestra
Republic Builders Products Corp.
Steelcraft Mfg. Co.

Thermal-Rated (Insulating) Assemblies: At all exterior locations, provide doors, which have been fabricated as thermal insulating door and frame assemblies and tested in accordance with ASTM C 236 or ASTM C 976. Unless otherwise indicated, provide assemblies with maximum apparent U factor for thermal-rated assemblies is 0.24 Btu/hr (ft^2) degrees F.

Heating, Ventilating, And Air Conditioning System

As specified in Section 15500, HEATING, VENTILATING, AND AIR CONDITIONING SYSTEM.

Fixed Louvers

Material: Factory finish to match wall panels.

Free Airflow: Minimum 50 percent.

Weather Projection: Drainage-type louver.

Insect Screen: Manufacturer's standard 14-Ga to 18-Ga galvanized steel wire mesh screen.

Pipe Penetrations

For pipe penetrations through the roof use a "DEKTITE" pipe flashing unit as manufactured by ITW Buildex, or approved equal. Provide a stainless steel hose clamp for positive sealing of flashing to pipe.

Wall Penetrations

Provide opening as required by HVAC air conditioning manufacturer.

Electrical And Lighting

As specified in Section 16005, ELECTRICAL.

Part 3 – Execution

Erection

Framing: Erect structural framing true to line, level and plumb, rigid and secure. Level base plates to a true even plane with full bearing to supporting structures, set with double-nutted anchor bolts. Use a non-shrinking grout as specified in Section 03301, CONCRETE, to obtain uniform bearing and to maintain a level base line elevation. Moist cure grout for not less than 7 days after placement.

Bracing:

Install diagonal rod or angle bracing in roof as required.

Diagonal/rod bracing shall not interfere with ceiling purlins.

Install portal frame bracing in sidewalls as specified.

Framed Openings: Provide shapes of proper design and size to reinforce opening and to carry loads and vibrations imposed, including equipment furnished under mechanical or electrical work. Securely attach to building structural frame.

Roofing and Siding

General:

Install panels and associated items for neat and weather tight enclosure. Avoid "panel creep" or application not true to line. Protect factory finish from damage.

Provide weather seal under ridge cap. Flash and seal roof panels at eave, swaged joints and rake with manufacturer's standard rubber, neoprene, or other closures to exclude weather.

Roof Sheets:

Provide sealant tape at lapped joints of ribbed or fluted roof sheets, and between roof sheeting and accessories.

Apply sealant tape continuous to clean, dry surface of weather side of fastenings on end laps and on sidelaps of corrugated or nesting type, ribbed or fluted panels and elsewhere to make weatherproof to driving rains.

Wall Sheets:

Apply elastomeric sealant continuous between metal base channel (sill angle) and concrete foundation and elsewhere as necessary for waterproofing. Handle and apply sealant and backup in accordance with sealant manufacturer's recommendations.

Align bottoms of wall panels. Fasten flashings, trim around openings, etc., with self-tapping screws.

Provide small quantities of paint material and touch-up coatings damaged during construction in accordance with the manufacturer's direction.

Sheet Metal Accessories: Install louvers and other sheet metal accessories in accordance with manufacturer's recommendations for positive anchorage to building and weather tight mounting.

Interior Wall Liner Panels: Install all wall liner panels as shown on the Drawings.

Certification: The Construction Subcontractor shall submit a certified statement that all standing seam metal roofing, flashings, rain gutter and downspout, wall panels, structural framing, and anchor bolts have been installed in strict accordance with the manufacturer's printed instructions and this specification.

Door Installation: Fit hollow metal doors accurately in frames, within clearance specified in SDI-100.

Hardware Schedule

Group No. 2:

Butts: 1-1/2 pair McKinney T4A3386 4.5 x 4.5 x BHMA 630.

Lockset: 1 Best 84-7-C-15D-S3 x BHMA 626.

Closer: 1 LCN P4041 x BHMA 673.

Rain Drip: 1 Pemko 346C.

Kick Plate: 1 SST–10 inches high by 0.05 inch thick.

Weather-stripping: 1 set Pemko 319CN x S88 x BHMA 628.

Door Bottom: 1 Pemko 430CRL x BHMA 628.

Threshold: 1 Pemko 254X4AFG x BHMA 628.

Construction Quality Control

High Strength Bolted Connections: Construction General Contractor shall provide special inspections to verify field connections with high strength bolts are installed in accordance with plans and specifications and AISC requirements.

END OF SECTION 13122

SECTION 13205 – LINED BOLTED STEEL LIQUID STORAGE TANKS SCHED. B

Part 1 – General

References

The publications listed below form a part of this Specification to the extent referenced. The publications are referred to in the text by basic designations only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA D103	Factory-Coated Bolted Steel Tanks for Water Storage
AWWA D130	Flexible-Membrane-Lining and Floating-Cover Materials for Potable Water Storage

ASTM INTERNATIONAL (ASTM)

ASTM A446	Steel Sheet, Zinc-Coated by the Hot-Dip Process, Structural quality
ASTM A525	General Requirements for Steel Sheet, Zinc-Coated by the Hot-Dip Process
ASTM D413	Standard Test Methods for Rubber Property-Adhesion to Flexible Substrate
ASTM D751	Standard Test Method for Coated Fabrics

FEDERAL STANDARDS (FS)

FS 5100	Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power-Driven Metal Woodworking Machinery

INTERNATIONAL CONFERENCE OF BUILDING OFFICIALS

UBC	Uniform Building Code

Description

This Specification sets the minimum standards for design and construction of two lined, bolted liquid storage tanks. The tanks shall be constructed from corrugated galvanized steel panels bolted together such that no field welding or onsite coating is required. The system shall provide an interior geosynthetic fabric to protect the factory fabricated membrane liner. A tank primary and secondary liner system shall be utilized.

Qualifications

Tank Manufacturer: At least five tanks presently in service, of similar size and character required for this Project, and minimum of 5 years' satisfactory operation.

Tank Installer: Certified by tank manufacturer that installer is qualified to do the work.

Registered Professional Engineer: Licensed in the state of Project with training and expertise in tank system design and installation. Able to recognize signs of potential tank system failure during the intended operating life of the tank. Able to assess and interpret information on the waste to be stored in the tank and the waste compatibility with the materials used for the tank and piping system.

Installation Inspector: Knowledge of the physical sciences and the principals of engineering acquired by a professional education and related practical experience. Trained and experienced in the proper installation of tank systems or components. Certified by tank manufacturer that the inspector is qualified and experienced in type of Work to be performed.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Statements of Qualifications:

Tank manufacturer.

Tank installer.

Registered professional engineer.

Installation inspector.

Tank Secondary and Primary Liners and Floating Cover:

Material Samples: Within 15 days from Notice to Proceed: Samples of the materials proposed for use. Submit fifty (50) sample coupons, each 8 inches by 10 inches in size, for use by the Engineer to conduct leachate compatibility testing.

Manufacturer's Data: Manufacturer's descriptive data, specifications sheets, literature, and other data as necessary to fully demonstrate that those materials proposed for use comply with the requirements of these Specifications.

Installation Plan: Submit an installation plan for the liners and cover describing the proposed methods for liner and cover deployment, panel layout, seaming, repair, and protection. The plan shall also include a quality control program for the Construction General Contractor's activities related to liner and cover materials installation.

Factory Fabrication Inspection Data (Source Quality Control): Submit documentation of factory inspection as specified herein.

Drawings:

Tank and Equipment: Detailed drawings for tanks, anchor bolts and anchor bolt chains, and equipment, such as wall construction, pipe connections, floating cover, floating cover water removal system, secondary containment system, and stilling wells for installation of level controls shall be stamped by the Registered Professional Engineer. Level controls are provided by others (see Section 13401, PICS). Drawings shall include a complete list of equipment and materials, including manufacturer's descriptive and technical literature, and installation instructions.

Calculations: Stamped by the Registered Professional Engineer. Complete structural stress analysis of structural components and connections and anchorage system to the concrete ringwall. Include anchor bolt reaction for all load cases and load combinations.

Design Assessment Report:

A written report providing the results of the tank system design assessment prepared and certified by the Registered Professional Engineer attesting that the tanks furnished under this section of the Specifications has sufficient structural integrity and is acceptable for the storing and treating of dangerous waste.

The assessment report shall contain the following:

Site map of the facility showing the proposed location of the tank system within the overall facility.

A sketch of the tank system including connected piping and fittings. Individual tanks shall be clearly labeled.

Structural design standards and criteria used with reference to applicable industry standards and recommended practice codes. Include all calculations for tanks and anchoring. Tank shell shall be designed based on full tank. Design parameters used in calculations shall be clearly indicated and labeled on clarifying sketches. Seismic considerations that are appropriate to the seismic risk zone shall be accounted for in the calculations.

Assessment of the compatibility of the leachate to be stored in the tank with the tank system materials. Show that the characteristics of the leachate to be stored are compatible with the material properties of the tank system, including material properties of the interior lining. Include the results of the chemical compatibility testing provided by the Engineer in this assessment.

Description and assessment of the secondary containment system, results of primary liner and secondary liner leak detection surveys, and collection of releases into the secondary containment system; compatibility of the materials in the secondary containment system with the leachate to be stored in the tank; strength of secondary containment system to withstand stresses from static head during a release, climatic conditions, nearby vehicle traffic, and daily operations; description of the leak detection system that will detect the failure of the primary containment structure or the presence of any release of leachate or accumulated liquid in the secondary containment system within 24 hours; a description of the corrosion protection for the exterior surface of the tank.

Assessment of ancillary equipment as shown on the Drawings (piping, fittings, flanges, valves, and pumps) associated with the tank including support and protection against damage and excessive stress due to excessive settlement, vibration, expansion, or contraction. Verify that peak flows and internal stresses are within the design limits specified by the manufacturer of the ancillary equipment.

The recommended inspection schedule once the tank is placed in service based on the performance of similarly designed tank systems operating under similar conditions.

A statement by the Registered Professional Engineer certifying that the tank system has been adequately designed and that the tank system has sufficient structural strength to ensure that it will not collapse, rupture, or fail under the design conditions. The certification shall include the following statement:

"I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations."

The Registered Professional Engineer's signature and stamp must be placed below the certification statement.

Submittals–Approval Not Required

Vendor Information (VI):

Installation: Tank Installation Instructions.

O&M Manual:

Tank Materials: Submit operating and maintenance instructions prior to completion of the Project. The manual shall include the manufacturer's cut sheets, parts lists, and a brief description of all equipment and their operating features. Maintenance instructions shall include all routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guide, including recommended preventative maintenance tasks and frequencies for performance of those tasks.

Information/Record (IR): Submit documentation of construction quality control as specified herein.

Installation Inspection Report: A written report prepared by the Installation Inspector or the Registered Professional Engineer documenting the results of the tank system installation inspection. The installation inspection report shall contain the following:

The as-built site plan showing the location of the installed tank system.

An as-built drawing of the installed tank system including connected piping. Individual tanks shall be clearly labeled with ID numbers.

Inspection notes, photographs, and any other material used to document inspection activities.

An assessment of the tank system for structural damage or inadequate construction/installation including weld breaks, punctures, damage to protective coatings, cracks, and corrosion, and documentation of any defects discovered in materials, equipment, or installation procedures and measurements taken to correct these defects.

Documentation of tightness testing results demonstrating the tank system is tight prior to placing it in service.

A statement certifying the proper installation of the tank system liner, signed by the liner installer's representative.

A signed and dated statement by the Installation Inspector or Registered Professional Engineer certifying the proper installation of the tank system. The certification shall include the following statement:

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Leachate Compatibility Testing

The Engineer will conduct leachate compatibility testing on the tank liner and cover material samples submitted by the Construction General Contractor in accordance with EPA SW 846, Method 9090, or ASTM D5322 and D5747 procedures. At the completion of the testing, the Engineer will evaluate the testing data for conformance with the project requirements and approve or reject the material. The Engineer will provide the results of the evaluation and approval or rejection to the Construction General Contractor within 180 days after receipt of material samples. Construction General Contractor shall not order materials or proceed with fabrication until after receiving results and approval from the Engineer. Any product or material changes required as a result of inadequate leachate compatibility results will be addressed by Change Order, provided that the submitted material meets all other requirements of this section.

Delivery and Storage

All materials and equipment delivered and placed in storage shall be stored with protection from the weather, excessive humidity, and excessive temperature variation; and dirt, dust, or other contaminants. The tank components shall be shipped in crate(s) or pallet(s) designed to prevent physical damage to the tank coating, linings, and structural components.

Warranty

The tank shall have a 3-year warranty from the date of Substantial Completion covering workmanship, materials, all steel components, and the liners and floating cover system. The warranty shall provide for correction, or, at the option of the Tank Farm Contractor, removal and replacement of Work specified in this Specification section found defective during the period of the warranty.

The Construction General Contractor shall provide the manufacturer's written warranty for the liners and cover. The warranty shall be provided to the Construction General Contractor as purchaser with the Tank Farm Contractor named as beneficiary and shall be signed by an authorized representative of the liner and cover manufacturer. The warranty shall guaranty the liner and cover material for the above-stated period against:

Manufacturing Defects

Deterioration due to ozone, ultraviolet, and other exposure to the elements, including the stored leachate.

Defects in material and factory seams.

Defects resulting from installation.

Part 2 – Products

Manufacturer

Dimensions are based on bolted steel tank as manufactured by Environetics, Inc. All dimensions and clearances shall be taken as minimum if an "or equal" tank manufacturer is submitted and approved by the Engineer. Construction General Contractor shall be responsible for all adjustments required to Drawings as a consequence of changing tank manufacturer.

Standard Products

Materials and equipment shall be the standard products of a manufacturer regularly engaged in the manufacture of such products and shall essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening. Equipment shall be supported by a service organization that is, in the opinion of the Engineer, reasonably convenient to the site. The items specified under this section shall be furnished by constructors having experience and regular practice in the design, fabrication, and construction of steel tanks.

Tank Size Requirements

Number of Tanks Required: Two.

Required Diameter: 101.46 feet.

Required Height: 8.17 feet.

Cover Required: Yes.

Leachate Volume: 375,000 gallons.

Top Capacity Level (TCL): 6.20 feet above tank floor.

Design

Design shall be in accordance with the requirements of AWWA D103.

Design Loads:

Specific Gravity: The tank shall be designed for liquids with a specific gravity of 1.00.

Earthquake: The tank shall be designed for Seismic Zone 2B per UBC and AWWA D103, site amplification soil profile C, I=1.0. For seismic, use leachate top capacity level (TCL).

Wind Force: The tank shall be designed to the greater requirements of a 100-mph wind with pressure loads as calculated with AWWA D103 or an 85-mph with pressure loads determined using ASCE 7, 1998. Wind force calculations shall include wind analysis with an empty tank case as well as a full tank case.

Hydrostatic Pressure: Design tank for static pressure to top of tank shell height.

Leachate Characteristics for Leachate Compatibility Testing:

Based on previous testing, the synthetic leachate for leachate compatibility testing of the tank liner material will have the following characteristics:

Chemical Compound	Concentration (g/l)
$NaNO_3$	185.0
Na_2SO_4	11.86
NaF	1.46
Na_2CO_3	3.45
$NaHCO_3$	2.44
pH	9.2 ± 0.1, using NaOH or HNO_3 as required

Tank Components

The tank and liner system shall consist of the following components: Corrugated steel wall panels with anchor embedded in concrete ringwall, geotextile base and wall buffer, secondary containment liner, drainage net, primary liner and floating cover, pipe connections, tank ladder, and piping for level and leak detection measurement.

Corrugated Steel Wall Panels:

The tank walls shall be constructed from individual rings of corrugated, hot-dip galvanized steel, mill-rolled to finished diameter. Sheet materials shall be ASTM A446, Grade D; or equal. Sheet materials shall be mill galvanized to ASTM A525, Class G90 standards, or equal. Shell plate thickness shall be based on AWWA D103 structural requirements. Wall plate thickness shall be a minimum of 12 gauge. Provide wind stiffeners as required by design.

Sheet materials shall receive an electrostatically applied, thermally cured, polyester powder coat finish. The coating shall be applied in two coats with a minimum dry film thickness of 5.0 mils. The finished coating shall be white in color.

Anchor Bolts: Shall conform to the requirements of AWWA D103.

Bolted Joints: Structural bolts conform to the requirements of AWWA D103 and shall be zinc electroplated. Bolted joints shall utilize a minimum two vertical rows as required to withstand structural loads.

Geotextile Base and Wall Buffer: An 8-ounce geotextile polypropylene nonwoven needle-punched fabric shall be placed on the tank floor and wall as a buffer for the liner. The fabric shall be designed to protect the liner from irregular surfaces on the tank wall. The material properties shall conform with the following:

Fabric Property	Unit	Test Method	Value
Grab Tensile Strength	lb	ASTM D4632	203
Grab Elongation	%	ASTM D4632	50
Puncture Strength	lb	ASTM D4833	130
Mullen Burst	psi	ASTM D3786	400
Trapezoid Tear Strength	lb	ASTM D4533	80
Permittivity[a]	sec^{-1}	ASTM D4491	1.5
Water Flow Rate[a]	gpm/sq ft	ASTM D4491	110
AOS	sieve	ASTM D4751	100
UV Resistance	% strength @ 500 hrs	ASTM D4355	70

(a) Minimum average roll values (MARV) for these secondary physical properties shall not exceed specified values.

Tank Secondary and Primary Liners and Floating Cover:

The liners and floating cover shall be fabricated from polyester reinforced polymeric alloy. Sheet materials shall have UV resistance and weathering qualities and conform to the following properties:

Physical Property	Physical Values	Test Method
Thickness	30.0 mils min.	ASTM D751
Weight	30.0 ± 2 oz/yd^2	ASTM D751
Tear Strength	35/35 lb_f min.	ASTM D4533, Trapezoid Tear
Breaking Yield Strength	550/550 lb_f min.	ASTM D751, Grab Tensile
Low Temperature	Pass @ -30°F	ASTM D2136, 4 hr – 1/8" mandrel
Dimensional Stability	1.5% max.	ASTM D1204, 212°F – 1 h reach direction
Adhesion - Heat Sealed	35 lb_f/2 in min.	ASTM D751, Dielectric Weld
Seam	Dead Load - Seam Shear 2 in seam, 4 hr,	ASTM D751
Strength	1 in strip 210 lb_f @ 70°F 105 lb_f @ 160°F	
Bursting Strength	650 lb_f min.	ASTM D751
Ball Tip	800 lb_f typical	
Hydrostatic Resistance	800 psi min.	ASTM D751 Method A
Blocking Resistance	# 2 Rating max.	ASTM D751 (180°F/82°C)

Physical Property	Physical Values	Test Method
Adhesion - Ply	15 lb_f/in min.	ASTM D413
or Film Tearing		
Bond		
Bonded Seam Strength	550 lb_f min.	ASTM D751 as modified by
NSF 54		
Abrasion Resistance	2,000 cycles (min.)	ASTM D3389
before fabric exposure	(H-18 Wheel, 1,000 g load)	
50 mg/100 cycles max		
weight loss		
Weathering Resistance	8,000 hrs (min.) –	ASTM G23 (Carbon-Arc)
No appreciable changes or stiffening or cracking of coating		
Water Absorption	0.025 kg/m2 max.	ASTM D471 Section 12, @ 70°F/21°C
	7 days, 0.14 kg/m2 max.	@ 212°F/100°C
Wicking Shelter-Rite® Procedure	1/8 in max.	
Puncture Resistance	250 lbf min.	ASTM D4833
Coefficient of Thermal	8 x 10^{-6} in/in/°F max.	ASTM D696

EXPANSION/CONTRACTION

Secondary and Primary Tank Liners: Shall be fabricated in a controlled factory environment into complete liners or large prefabricated panels. Size shall be limited to 3,000 pounds for ease of installation.

Tank Liner Source Quality Control: The tank liner(s) and cover shall be fabricated from standard width sheeting into a full-size fitted liner by means of minimum 1-inch wide dielectric and 2-inch wide thermal welds. The liner(s) and cover shall be thoroughly inspected by the fabricator for flaws in materials or fabrication prior to shipment. Inspection shall be performed by 100 percent visual inspection and proprietary inflation-light test methods. Construction General Contractor shall provide documentation of factory inspections to the Construction Manager.

Floating Cover: The cover shall be designed to comply with applicable AWWA D103 design standards. The cover shall incorporate closed cell polyethylene foam floatation elements in cover membrane pockets to provide the required buoyancy, drainage, and wind stability. The cover shall incorporate a center vent to evacuate air during fill cycle. The manufacturers shall provide a pump system with pump to remove excess surface water. Floating cover shall drain towards the perimeter of the tank where excess water may be collected and pumped out. Cover vent shall include a base plate to prevent tipping due to wind or snow load.

Drainage Net: The floor area of the tank shall be covered with fitted panels of high density polyethylene (HDPE) drainage net with a geotextile laminated to both sides of the drainage net to prevent clogging and to provide a cushion for the HDPE drainage net against the tank liners. The drainage net shall be installed between the primary and secondary liners to convey liquids between the liners to a leak detection sump. Properties for the drainage net and geotextile are as follows:

Physical Properties	Test Method	Physical Value
Combined:		
Transmissivity, m/sec	ASTM D4716	4×10^{-5}
Drainage Net Component:		
Transmissivity, m/sec	ASTM D4716	1×10^{-3}
Thickness, mill	ASTM D1777	200
Density g/cm^3	ASTM D105	0.94
Tensile Strength, lb/in	ASTM D5034/5035	45
Carbon Black Content, %	ASTM D1603	2.0
Geotextile Component:		8 oz/yd^2
Thickness, mill	ASTM D5199	90
Grab Tensile, lb	ASTM D4632	210
Puncture Strength, lb	ASTM D4833	135 ± 5 lbs
AOS, US Sieve	ASTM D4751	80
Flow Rate, gpm/ft	ASTM D4491	110 ± 10 gpm/ft
UV Resistance, % retained	ASTM D4355	70

Pipe Connections: Pipe fittings and connections shall be in accordance with manufacturer's requirements for double containment connections. Location of pipe connections shall be as shown on the Drawings.

Tank Ladder: Provide a hot-dipped galvanized steel ladder for access to the floating cover pump. The ladder shall be attached at the top of the tank wall and at its base to the concrete ringwall. The ladder shall be of sufficient height to allow viewing of the floating cover system and its pump. The ladder shall be located adjacent to the tank level element installation.

Tank Level and Leak Detection Measurement:

Provide as part of tank construction two 2-inch diameter (Schedule 80 PVC) internal (stilling wells) that extend the whole interior operating height of tank, for the purpose of facilitating the installation of a submersible pressure transmitter (in one pipe), and a multipoint level sensor (in the other pipe). Level measurement instrumentation provided under Section 13401, PROCESS INSTRUMENTATION AND CONTROL SYSTEMS (PICS), and installed by Construction General Contractor under this section.

Construction General Contractor shall furnish and install all necessary equipment and personnel to properly support installation of measurement devices (i.e., PVC flanges, straps, and gaskets).

Foundation: Tank shell to bear on a Type 1 concrete ringwall per AWWA D103 as shown on the Drawings. A 1-1/2-inch minimum space between the tank bottom and the top of the ringwall shall be filled with a nonshrink grout as specified in Section 03301, CONCRETE. Cane fiber joint filler shall not be used. Ringwall design is shown on Drawings.

Part 3 – Execution

General

Tank construction shall be in accordance with AWWA D103.

Tank Installation

Field erection of lined bolted steel tanks, including, but not limited to, shell plates, pipe connections, awning, primary and secondary containment, and floating cover, shall be in strict accordance with the manufacturer's recommendations including their guidance on environmental factors that could affect the tank installation.

Construction Quality Control

The Construction General Contractor shall establish and maintain a quality control system to assure compliance with contract requirements and shall maintain records of its quality control for all operations including, but not limited to the following:

- Inspection of materials delivered to project site against approved material data.
- Storage and handling of materials.
- Finished appearance.
- Completion of required testing.

Copies in duplicate of these records and tests, as well as records of corrective action taken when results are unsatisfactory, shall be furnished to the Construction Manager within 24 hours following the inspection or test.

Tank System Installation Inspection:

The Construction General Contractor shall provide the services of an Installation Inspector or Registered Professional Engineer to provide full-time supervision of the installation of the storage tanks. No work shall be performed without the presence in the field of the Installation Inspector or Registered Professional Engineer. The Installation Inspector or Registered Professional Engineer shall observe and verify that correct materials and procedures are used for the following activities:

- Visual inspection and testing.
- Subgrade and foundation preparation.
- Placement and compaction of backfill.
- Placement of reinforcing steel and anchor bolts.
- Concrete placement.
- Placement of shop-fabricated tank parts.
- Erection of field-erected tank parts.
- Installation of tank liner systems. Tank liner inspection requirements are specified herein.
- Installation of piping, pumping, and other ancillary equipment.
- Tightness testing.

Tank Liner Inspection:

Visual Inspection: 100 percent visual inspection along all seams of the liners.

Air Jet Inspection: 100 percent air jet inspection of all seams.

Any required repairs shall be corrected in accordance with the manufacturer's recommendations. Results of all testing shall be provided to the Construction Manager.

Electronic Leak Location Survey: Prior to installing the floating cover, complete an electronic leak location survey of the secondary and primary liners.

Floating Cover Drainage Test: Add water on floating cover to test that water drains towards the perimeter. Manufacturer to revise ballast if water does not drain to allow removal of excess water. Demonstrate proper operation of floating cover pump system to remove excess water.

Tank Tightness Testing: Upon completion of tank installation, the tank shall be visually inspected for any signs of physical damage. Any questionable areas shall be repaired in accordance with the manufacturer's instructions. The tank shall be filled with water and let stand for a period of not less than 2 days. The Construction General Contractor shall maintain a level not less than 7.2 feet for the duration of 2 days. Following the 2 days, the Construction General Contractor shall cyclically change the tank water level at a constant rate from 0.5 foot to 7.2 feet for four cycles over the next 28 days. During the 30 days, there shall be no signs of leakage from a defect in the primary liner to the secondary containment system of the tank. Any leaks discovered by this test shall be corrected by the Construction General Contractor in accordance with the manufacturer's recommendations. The tank system shall be successfully tested before it is accepted. Results of all testing shall be provided to the Construction Manager.

END OF SECTION 13205

SECTION 13401 – PROCESS INSTRUMENTATION AND CONTROL SYSTEMS (PICS) SCHED. B

Part 1 – General

UL And NRTL Compliance

Materials manufactured within the scope of UL or another nationally recognized testing laboratory (NRTL) shall conform to UL or NRTL standards and have an applied UL or NRTL listing mark. References to UL throughout this section imply conformity with UL or NRTL standards and guidelines.

PICS control panels shall be manufactured, assembled, tested, approved, and clearly labeled in accordance with UL 508A when required, prior to delivery to construction site.

Approval by Authority Having Jurisdiction (AHJ): As specified in Section 16005, ELECTRICAL.

Work Includes

Engineering, furnishing, installing, calibrating, adjusting, testing, documenting, starting up, and Tank Farm Contractor training for a complete Process Instrumentation and Control System for plant.

Detailed Design: PICS as shown and specified includes functional and performance requirements and component specifications. Complete detailed PICS design.

Major Cell No. 1 and Cell No. 2 components and controls to integrate into PICS and to program include:

- Crest Pad Building Control Panel, PLC, and Operator Interface Assemblies.
- Crest Pad Building Sump, Combined Sump, Leachate Tank, and Leachate Transfer Building Transfer Pump Local Control Panel Assemblies.
- Leachate Collection and Removal, and Leak Detection System Pump Control.
- Leachate Collection and Removal, and Leak Detection System Pump Discharge Flow and Flow Totalization.
- Leachate Collection and Removal System Continuous Level Measurement.
- Leak Detection System Continuous Level Measurement.
- Leachate Storage Tank Continuous Level Measurement.
- Leachate Storage Tank, Crest Pad Building Sump, Carrier Pipe, and Combined Sump Leak Detection Chamber Discrete Level Measurement.
- Crest Pad Building Sump Discrete Level Measurement, and Pump Control.
- Combined Sump Discrete Level Measurement, and Pump Control.
- Interlock Control between Crest Pad Building Sump and Leachate Collection, and Removal and Leak Detection System Pump Controls.
- Crest Pad and Leachate Transfer Building Continuous Temperature Measurement.
- Leachate Transfer Pump Control and Flow Measurement, and Flow Totalization.
- Crest Pad Building Discrete Power Measurement.

Definitions

Abbreviations

CAT: Construction Acceptance Test.

CP: Control Panel.

FDT: Factory Demonstration Test.

LCP: Local Control Panel.

MCC: Motor Control Center.

OIU: Operator Interface Unit.

PCT: PICS Continuity Test.

PFT: PICS Functionality Test.

PLC: Programmable Logic Controller.

SLC: Small Programmable Logic Controller.

Rising/Falling: Terms used to define actions of discrete devices about their set points.

Rising: Contacts change state when an increasing process variable rises through set point.

Falling: Contacts change state when a decreasing process variable falls through set point.

Signal Types:

Analog Signals, Current Type:

4 to 20 mA dc signals conforming to ISA S50.1.

Unless otherwise indicated for specific PICS Subsystem components, use the following ISA 50.1 options:

Transmitter Type: Number 2, two-wire.

Transmitter Load Resistance Capacity:

Class L.

Fully isolated transmitters and receivers.

Analog Signals, Voltage Type:

1 to 5 volts dc within control panels only.

Discrete signals, two-state logic signals using dc or 120V ac sources as indicated.

Special Signals: Other types of signals used to transmit analog and digital information between field elements, transmitters, receivers, controllers, and digital devices.

Instrument Tag Numbers: In accordance with DOE-RL Standards.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Shop Drawings:

General:

Shop Drawings, full-scaled details, wiring diagrams, catalog cuts, and descriptive literature.

Identify proposed items and options. Identify installed spares and other provisions for future work (e.g., reserved panel space; unused components, wiring, and terminals).

Bill of Materials: List of required equipment.

Group equipment items as follows:

I&C Components: By component identification code.

Other Equipment: By equipment type.

Data Included:

Equipment tag number.

Description.

Manufacturer, complete model number, and all options not defined by model number.

Quantity supplied.

Component identification code where applicable.

Catalog Cuts: I&C Components, Electrical Devices, and Mechanical Devices:

Catalog information, mark to identify proposed items and options.

Descriptive literature.

External power and signal connections.

Scaled drawings showing exterior dimensions and locations of electrical and mechanical interfaces.

Component Data Sheets: Data sheets for I&C components.

Format and Level of Detail:

In accordance with ISA-S20.

Include component type identification code and tag number on data sheet.

Specific features and configuration data for each component:

Location or service.

Manufacturer and complete model number.

Size and scale range.

Set points.

Materials of construction.

Options included.

Name, address, and telephone number of manufacturer's local office, representative, distributor, or service facility.

Panel Construction Drawings:

Scale Drawings: Show dimensions and location of panel mounted devices, doors, louvers, and subpanels, internal and external.

Panel Legend: List front of panel devices by tag numbers, nameplate inscriptions, service legends, and annunciator inscriptions.

Bill of Materials: List devices mounted within panel that are not listed in panel legend. Include tag number, description, manufacturer, and model number.

Construction Details: NEMA rating, materials, material thickness, structural stiffeners and brackets, lifting lugs, mounting brackets and tabs, door hinges and latches, and welding and other connection callouts and details.

Construction Notes: Finishes, wire color schemes, wire ratings, wire and terminal block, numbering and labeling scheme.

Panel Control Diagrams: For discrete control and power circuits.

Diagram Type: Ladder diagrams. Include devices, related to discrete functions that are mounted in or on the panel and that require electrical connections. Show unique rung numbers on left side of each rung.

Item Identification: Identify each item with attributes listed.

Wires: Wire number and color. Cable number if part of multiconductor cable.

Terminals: Location (enclosure number, terminal junction box number, or MCC number), terminal strip number, and terminal block number.

Discrete Components:

Tag number, terminal numbers, and location ("FIELD", enclosure number, or MCC number).

Switching action (open or close on rising or falling process variable), set point value and units, and process variable description (e.g., Sump Level High).

Relay Coils:

Tag number and its function.

On right side of run where coil is located, list contact location by ladder number and sheet number. Underline normally closed contacts.

Relay Contacts:

Coil tag number, function, and coil location (ladder rung number and sheet number).

Show each circuit individually. No "typical" diagrams or "typical" wire lists will be permitted.

Ground wires, surge protectors, and connections.

Panel Wiring Diagrams: Show point-to-point and terminal-to-terminal wiring within panel.

Loop Diagrams:

Individual wiring diagram for each analog or pulse frequency loop.

Conform to the ISA S5.4 Standards.

Drawing Size:

Individual 11-inch by 17-inch sheet for each loop.

Divide each loop diagram into areas for panel face, back-of-panel, and field.

Show:

Terminal numbers, location of dc power supply, and location of common dropping resistors.

Switching contacts in analog loops and output contacts of analog devices. Reference specific control diagrams where functions of these contacts are shown.

Tabular summary on each diagram.

Transmitting Instruments: Output capability.

Receiving Instruments: Input impedance.

Loop Wiring Impedance:

Estimate based on wire sizes and lengths shown.

Total loop impedance.

Reserve output capacity.

Conduit and cable schedule names.

Interconnecting Wiring Diagrams:

Diagrams, device designations, and symbols in accordance with NEMA ICS 1.

Diagrams shall bear electrical Construction Subcontractor's signature attesting diagrams have been coordinated with Division 16, ELECTRICAL.

Show:

Electrical connections between equipment, consoles, panels, terminal junction boxes, and field mounted components.

Component and panel terminal board identification numbers, and external wire and cable numbers.

Circuit names matching Conduit and Cable Schedule.

Intermediate terminations between field elements and panels for, e.g., to terminal junction boxes and pull boxes.

Pull boxes.

Factory Demonstration Test (FDT): Provide FDT documentation for control panels.

Installation Details:

Include modifications or further details required to adequately define installation of I&C components.

List of spares, expendables, test equipment and tools.

Submittals–Approval Not Required

Information/Record (IR): For PICS equipment, provide Manufacturer's Certificate of Proper Installation and readiness for operation.

Tank Farm Contractor Training Plan: In accordance with Article TRAINING.

Construction Quality Control Test Data: Provide documentation of PICS Continuity Test (PCT) and PICS Functionality Test (PFT).

Operation and Maintenance (O&M) Manuals:

Content and Format:

Complete sets O&M manuals, including recommended preventative maintenance tasks and frequencies for performance of those tasks.

Sufficient detail to allow operation, removal, installation, adjustment, calibration, maintenance and purchasing replacements for each PICS component.

Final versions of Legend and Abbreviation Lists.

Include:

Process and Instrumentation Diagrams: One reproducible copy of revised P&ID to reflect as-built PICS design.

Refer to paragraph Shop Drawings for the following items:

Bill of Materials.

Catalog Cuts.

Component Data Sheets.

Panel Control Diagrams.

Panel Wiring Diagrams, one reproducible copy.

Loop Diagrams, one reproducible copy.

Interconnecting Wiring Diagrams, one reproducible copy.

Device O&M manuals for components, electrical devices, and mechanical devices include:

Operations procedures.

Installation requirements and procedures.

Maintenance requirements and procedures including recommended preventative maintenance tasks and frequencies for performance of those tasks.

Troubleshooting procedures.

Calibration procedures.

Internal schematic and wiring diagrams.

Component Calibration Sheets from field quality control calibrations.

List of spares, expendables, test equipment and tools provided.

List of additional spares, expendables, test equipment and tools recommended.

Factory Demonstration Test (FDT), PICS Continuity Test (PCT), and PICS Functionality Test (PFT) Submittals:

Preliminary Test Procedures: Outlines of proposed tests, forms, and checklists.

Final Test Procedures: Proposed test procedures, forms, and checklists.

Test Documentation: Copy of signed off test procedures when tests are completed.

Application Software Submittal and Design Workshops:

Location: There shall be a minimum of six (6) workshops held at the Tank Farm Contractor's facility (or by video and audio conferencing) during the course of the project.

Objective: To provide a vehicle by which the Tank Farm Contractor is able review and comment on PLC, OIU, communication hardware, standard software, and application software submittals and application software development.

Documentation: Application software supplier shall summarize resolutions reached in each workshop, including cost and schedule impacts and distribute copies to Tank Farm Contractor.

Order and minimum topics to be covered in each workshop:

Applications Software Design Workshop (kick off) that establishes project processes, including:
Workshop objectives.
Submittal process.
Review Work Sequence and schedule.

Loop Specifications, P&ID Review Workshop:

Application Software Supplier use P&IDs and Specifications to present how the proposed control system design and Applications Software will meet the functional requirements specified herein.

At the completion of workshop Applications Software Supplier modifies as necessary Loop Specifications.

Submit finalized Loop Specification along with an outline of any application software cost and schedule impacts.

PLC Software Standards Submittal Workshop: PLC Software Standards shall be developed in a Software Standards Workshop. Ladder diagram standards for commonly used functions, including the following:

Objective: To develop, implement, and review implementation of PLC Software Standards in ladder logic programming.

Ladder diagram standards for commonly used functions, including the following:
High and low process variable alarm checking.
Instrument failure alarm detection.
Equipment start/stop control.
Equipment failure detection.
Equipment run time.
Leak detection and equipment interlocks.
Signal filtering.
Flow totalization.
Alarm routines.

Interface with OIU.

Memory mapping, data transfer (read/write, remote set point adjustment, pump control and alarm management).

Submit for review ladder logic programming for each PLC including: descriptive ladder logic, cross references, memory map and point databases.

OIU Standard Workshop:

Objective: To develop, implement, and review implementation of OIU standards with Tank Farm Contractor.

Design Products and Topics to be Finalized:

OIU and PLC integration.

OIU tag naming conventions.

Process, set point, and runtime graphics.

Display paging and navigation.

Dynamic Objects:

Pumps, valves, gates, compressors, etc.

Equipment control through pop-up windows.

General data entry through the OIU.

Dynamic Objects:

Pumps, valves, gates, process indicators, indicators with alarms, data entry, controller face plate, and tanks.

Security.

Alarm Management.

Minimum OIU Design Products and Topics to be Finalized for Each OIU:

Eight (8) Process Graphics.

Eight (8) Pop-Up Equipment Operation Control Graphics.

One (1) Alarm Summary Process Control Graphic.

One (1) Alarm History Process Control Graphic.

One (1) Equipment Runtime Process Control Graphic.

One (1) Analog Process Summary Control Graphic.

Submit for review OIU programming and development for each OIU computer including: memory mapping, database structures, graphic displays, and alarms.

Delivery, Storage, and Handling

Provide site and warehouse storage facilities for PICS equipment.

Prior to shipment, include corrosive-inhibitive vapor capsules in shipping containers, and related equipment as recommended by the capsule manufacturer.

Prior to installation, store items in dry indoor locations. Provide heating in storage areas for items subject to corrosion under damp conditions.

Cover panels and other elements that are exposed to dusty construction environments.

Electrical equipment (valves, instruments, sensors, enclosures) shall be wired complete and in accordance with the manufacturer's wiring diagrams and instructions.

Completed wiring diagrams shall be incorporated in the O&M submittal.

Environmental Requirements

Standard Environmental Requirements: Unless otherwise noted, provide equipment for continuous operation in these environments:

Freestanding Panel and Consoles:

Inside: NEMA 12.

Smaller Panels and Assemblies (that are not Freestanding):

Inside: NEMA 4X.

All Other Locations: NEMA 4X.

Field Elements: Outside.

Special Environmental Requirements:

Design panels for continuous operation in environments listed:
Building Sump Local Control Panel to be installed inside Cell No. 1 and Cell No. 2 Crest Pad Buildings.
Transfer Pump Local Control Panel to be installed inside Cell No. 1 and Cell No. 2 Leachate Transfer Buildings.
Leachate Storage Tank Local Control Panel to be installed outdoors adjacent to Cell No. 1 and Cell No. 2 Leachate Storage Tanks.
Combined Sump Local Control Panel to be installed inside Cell No. 1 and Cell No. 2 Crest Pad Buildings.
Control Panel to be installed inside Cell No. 1 and Cell No. 2 Crest Pad Buildings.

Environmental Design Requirements: Environmental conditions are defined below:

Inside:

Temperature: 10 to 30 degrees C.

Relative Humidity: 15 to 90 percent noncondensing.

NEC Classification: Nonhazardous.

Outside:

Temperature: Minus 40 to 40 degrees C.

Relative Humidity: 15 to 90 percent noncondensing.

NEC Classification: Nonhazardous (except for interior of Combined Sump Assemblies).

Snow Accumulation: 5 inches.

Sequencing and Scheduling:

Activity Completion: The following is a list of key activities and their completion criteria:

Shop Drawings: Reviewed and approved.

Factory Demonstration Testing of Control Panels: Reviewed and accepted.

Hardware Delivery: Hardware delivered to site and inventoried by Tank Farm Contractor.

PCT: Completed and required test documentation accepted.

PFT: Completed and required test documentation accepted.

PICS Substantial Completion: When Construction Manager issues Certificate of Substantial Completion.

Prerequisites:

All PICS Submittals have been completed.

PICS has successfully completed FDT and PFT.

Tank Farm Contractor training plan is on schedule.

All spares, expendables, and test equipment have been delivered to Tank Farm Contractor.

PICS Acceptance: When Construction Manager issues a written notice of Final Payment and Acceptance.

Prerequisites:

Certificate of Substantial Completion issued for PICS.

Punch-list items completed.

Final revisions to O&M manuals accepted.

Maintenance service agreements for PICS accepted by which shall satisfy the following requirements:

Duration of 2 years unless negotiated with Tank Farm Contractor.

Start on date of PICS acceptance, as identified in Section 13401, PICS, Article SEQUENCING AND SCHEDULING, Paragraph PICS Acceptance.

Performed by factory trained service engineers with experience on PICS systems to be maintained.

All materials and labor for preventive maintenance and visit site bimonthly.

All materials and labor for demand maintenance with coverage 8:00 a.m. to 5:00 p.m., Monday through Friday.

Response Time: Service engineer shall be onsite within 24 hours of request by Tank Farm Contractor.

Spare Parts:

If not stocked onsite, delivered to Site within 24 hours from time of request.

Repair or replace all components or software found to be faulty.

Replace and restock within 1 month, onsite spare parts and expendables used for maintenance. Provide list of items used and replaced.

Submit records of inspection, maintenance, calibration, repair, and replacement within 2 weeks after each visit to site.

Telephone Support: Coverage 8:00 a.m. to 5:00 p.m., Monday through Friday.

Software Subscription: 2-year support per Section 13401, PICS, Supplements.

Prerequisite Activities and Lead Times: Do not start the following key Project activities until the prerequisite activities and lead times listed below have been completed and satisfied:

Activity	Prerequisites and Lead Times
Submittal reviews by Engineer and Tank Farm Contractor	Tank Farm Contractor acceptance of Submittal breakdown and schedule.
Hardware purchasing, fabrication, and assembly.	Associated shop drawing Submittals completed
Shipment	Completion of PICS Shop Drawing and Quality Control Submittals, preliminary O&M manuals, and Factory Demonstration Testing.
PCT	PCT procedures completed; notice 3 weeks prior to start.
Tank Farm	Tank Farm Contractor training plan completed.
Contractor Training	
PFT	Startup, Tank Farm Contractor training, and PFT procedures completed; notice 4 weeks prior to start.

Part 2 – Products

General

The general functions of the PICS are as depicted on the Drawings. The PICS Contractor shall provide a full-featured system that is complete, calibrated, and fully operational.

Like Equipment Items: Use products of one manufacturer and of the same series or family of models to achieve standardization for appearance, operation, maintenance, spare parts, and manufacturer's services.

Implement all same or similar functions in same or similar manner. For example, control logic, sequence controls, and display layouts.

Loop Specifications

Location: Article SUPPLEMENTS.

Organization: By unit process and loop number.

Functional Requirements for Control Loops: Shown on Drawings, in Panel Control Diagrams, and Process and Instrumentation Diagrams (P&ID). P&ID format and symbols are in accordance with ISA S5.1, except as specified or shown on Drawings.

Supplemented by Loop Specifications.

Subheadings for Each Loop:

Functions: Clarifies functional performance of loop, including abstract of interlocks.

Components: Lists major components for each loop. Information listed include: Tag numbers.

Component Identification Codes: Alphanumeric codes of required components. Refer to Component Specification referenced in Article SUPPLEMENTS.

Component Names and Options: Required to tailor general Component Specifications to specific application. For example, special materials, mounting, size, unit range, scale, set points, and controller options.

I&C Components

Components for Each Loop: Major components for each loop are listed in Instrument List referenced in Article SUPPLEMENTS. Furnish all equipment that is necessary to achieve required loop performance.

Component Specifications: Generalized specifications for each type of component are located in Article SUPPLEMENTS.

Nameplates And Tags

Panel Nameplates: Enclosure identification located on the enclosure face.

Location and Inscription: As shown.

Materials: Laminated plastic attached to panel with stainless steel screws.

Letters: 1/2-inch white on black background, unless otherwise noted.

Component Nameplates—Panel Face: Component identification located on panel face under or near component.

Location and Inscription: As shown.

Materials: Laminated plastic attached to panel with stainless steel screws.

Letters: 3/16-inch white on black background, unless otherwise noted.

Component Nameplates—Back of Panel: Component identification located near component inside of enclosure.

Inscription: Component tag number.

Materials: Adhesive backed, laminated plastic.

Letters: 3/16-inch white on black background, unless otherwise noted.

Legend Plates for Panel Mounted Pushbuttons, Lights, and Switches:

Inscription: Refer to:

Table under paragraph Standard Pushbutton Colors and Inscriptions.
Table under paragraph Standard Light Colors and Inscriptions.
P&IDs in Drawings.

Materials: Engraved plastic, keyed legend plates. Secured to panel by mounting nut for pushbutton, light, or switch.

Letters: Black on gray or white background.

Service Legends: Component identification nameplate located on face of component.

Inscription: As shown.

Materials: Adhesive backed, laminated plastic.

Letters: 3/16-inch white on black background, unless otherwise noted.

Nametags: Component identification for field devices.

Inscription: Component tag number.

Materials: 16-gauge, Type 304 stainless steel.

Letters: 3/16-inch imposed.

Mounting: Affix to component with 16- or 18-gauge stainless steel wire or stainless steel screws.

Electrical Requirements

In accordance with Division 16, ELECTRICAL.

I&C and electrical components, terminals, wires, and enclosures: UL recognized or UL listed.

Wires Within Enclosures:

ac Circuits:

Type: 600-volt, Type SIS stranded copper.

Size: For current to be carried, but not less than No. 14 AWG.

Analog Signal Circuits:

Type: 600-volt stranded copper, twisted shielded pairs.

Size: No. 16 AWG, minimum.

Other dc Circuits:

Type: 600-volt, Type SIS stranded copper.

Size: For current carried, but not less than No. 18 AWG.

Special Signal Circuits: Use manufacturer's standard cables.

Wire Identification: Numbered and tagged at each termination.

Wire Tags:

Snap-on or slip-on PVC wire markers with legible machine printed markings and numbers. Adhesive or taped-on tags are not acceptable.

Wires entering or leaving enclosures, terminate and identify as follows:

Analog and discrete signal, terminate at numbered terminal blocks.

Special signals, terminated using manufacturer's standard connectors.

Identify wiring in accordance with Division 16, ELECTRICAL.

Terminal Blocks for Enclosures:

Wire spare PLC I/O points to terminal blocks.

One wire per terminal for field wires entering enclosures.

Maximum of two wires per terminal for 18-WG wire for internal enclosure wiring.

Spare Terminals: 20 percent of all connected terminals, but not less than 5 per terminal block.

General:

Connection Type: Screw compression clamp.

Compression Clamp:

Complies with DIN-VDE 0611.

Hardened steel clamp with transversal groves that penetrate wire strands providing a vibration-proof connection.

Guides strands of wire into terminal.

Screws: Hardened steel, captive and self-locking.

Current Bar: Copper or treated brass.

Insulation:

Thermoplastic rated for minus 55 to plus 110 degree C.
Two funneled shaped inputs to facilitate wire entry.

Mounting:

Standard DIN rail.
Terminal block can be extracted from an assembly without displacing adjacent blocks.

End Stops: Minimum of one at each end of rail.

Wire Preparation: Stripping only permitted.

Jumpers: Allow jumper installation without loss of space on terminal or rail.

Marking System:

Terminal number shown on both sides of terminal block.
Allow use of preprinted and field marked tags.
Terminal strip numbers shown on end stops.
Mark terminal block and terminal strip numbers as shown on Panel Control Diagrams and Loop Diagrams.

Terminal Block, General-Purpose:

Rated Voltage: 600V ac.

Rated Current: 30 amp.

Wire Size: No. 22 to No. 10 AWG.

Rated Wire Size: No. 10 AWG.

Color: Grey body.

Spacing: 0.25 inch, maximum.

Test Sockets: One screw test socket 0.079-inch diameter.

Manufacturer and Product: Entrelec; Type M4/6.T.

Terminal Block, Ground:

Wire Size: No. 22 to No. 12 AWG.

Rated Wire Size: No. 12 AWG.

Color: Green and yellow body.

Spacing: 0.25 inch, maximum.

Grounding: Ground terminal blocks electrically grounded to the mounting rail.

Manufacturer and Product: Entrelec; Type M4/6.P.

Terminal Block, Blade Disconnect Switch:

Rated Voltage: 600V ac.

Rated Current: 10-amp.

Wire Size: No. 22 to No. 12 AWG.

Rated Wire Size: No. 12 AWG.

Color: Grey body, orange switch.

Spacing: 0.25 inch, maximum.

Manufacturer and Product: Entrelec; Type M4/6.SN.T.

Terminal Block, Fused, 24V dc:

Rated Voltage: 600V dc.

Rated Current: 16-amp.

Wire Size: No. 22 to No. 10 AWG.

Rated Wire Size: No. 10 AWG.

Color: Grey body.

Fuse: 0.25 inch by 1.25 inches.

Indication: LED diode 24V dc.

Spacing: 0.512 inch, maximum.

Manufacturer and Product: Entrelec; Type M10/13T.SFL.

Terminal Block, Fused, 120V ac:

Rated Voltage: 600V ac.

Rated Current: 16-amp.

Wire Size: No. 22 to No. 10 AWG.

Rated Wire Size: No. 10 AWG.

Color: Grey body.

Fuse: 0.25 inch by 1.25 inches.

Indication: Neon Lamp 110V ac.

Leakage Current: 1.8 mA, maximum.

Spacing: 0.512 inch, maximum.

Manufacturer and Product: Entrelec; Type M10/13T.SFL.

Terminal Block, Fused, 120V ac, High Current:

Rated Voltage: 600V ac.

Rated Current: 35 amps.

Wire Size: No. 18 to No. 8 AWG.

Rated Wire Size: No. 8 AWG.

Color: Grey.

Fuse: 13/32 inch by 1.5 inches.

Spacing: 0.95 inch, maximum.

Manufacturer and Product: Entrelec; Type MB10/24.SF.

Grounding of Enclosures: Furnish copper isolated ground bus. Take car to ensure that this bus is connected to the safety ground bus at only one point.

Single Point Ground for Each Analog Loop:

Group and connect shields in following locations:

Control Panel.

Ground terminal block rails to ground bus.

Analog Signal Isolators: Furnish signal isolation for analog signals that are sent from one enclosure to another and where required to provide proper function. Do not wire in series instruments on different panels, cabinets, or enclosures.

Power Distribution Within Panels:

Feeder Circuits:

One or more 120V ac, 60-Hz feeder circuits as shown on Drawings.

Make provisions for feeder circuit conduit entry.

Furnish terminal blocks for termination of wires.

Power Panel:

Furnish main circuit breaker and a circuit breaker on each individual branch circuit distributed from power panel.

Locate to provide clear view of and access to breakers when door is open.

Breaker Sizes: Coordinate such that fault in branch circuit will blow only branch breaker but not trip the main breaker.

Branch Circuit Breaker: Select size of circuit breaker to suit load at 250V ac.

Breaker Manufacturers and Products: Allen-Bradley 1492-GH.

Circuit Wiring: P&IDs and Control Diagrams on Drawings show function only. Use following rules for actual circuit wiring:

Devices on Single Circuit: 20, maximum.

Multiple Units Performing Parallel Operations: To prevent failure of any single branch circuit from shutting down entire operation, do not group all units on same branch circuit.

Branch Circuit Loading: 12 amperes continuous, maximum.

Panel Lighting and Service Outlets:

Put on separate 15-amp, 120V ac branch circuit.

Provide 120-volt ac plugmold for panel components with line cords.

Signal Distribution:

Within Panels: 4 to 20 mA dc signals may be distributed as 1 to 5V dc.

Outside Panels:

Isolated 4 to 20 mA dc only.

All signal wiring in twisted shielded pairs.

Between Panels: 4 to 20 mA dc signals isolated by current signal isolators.

Signal Switching:

Use dry circuit type relays or switches.

No interruption of 4 to 20 mA loops during switching.

Switching Transients in Associated Signal Circuit:

4 to 20 mA dc Signals: 0.2 mA, maximum.

1 to 5V dc Signals: 0.05V, maximum.

Current Signal Isolators: Solid state three- and four-way isolation of the input signal, two output signals, and external power supply.

Features:

Zero and span trim adjustments using 15-turn potentiometers.

Calibration independent of load.

Signal Interface:

Input: 4 to 20 mA dc maximum impedance: 75 ohms.

Output: Two 4 to 20 mA dc. Capable of drives output load impedance up to 1,050 ohms independent of supply voltage to isolator.

Enclosure: NEMA 1, unless otherwise noted.

Mounting: DIN rail, unless otherwise noted.

Power: 115V ac, unless otherwise noted.

Manufacturer: Moore ECT Isolators; or approved equal.

Intrinsic Safety: Programmable three-channel switching amplifier with intrinsically safe input circuits, used to isolate and transfer discrete signals from Class I, Class II, or Class III hazardous location to a nonhazardous location.

Inputs: Three-channel dry contact inputs to switching amplifier.

Outputs: Three-channel SPDT dry relay contact outputs, each selectable to be (N.O.) or (N.C.) Output function dependent upon input condition.

Indications: Two-color switching status LED for each channel. "Yellow" LED when output relay is energized. "Green" LED with power ON status, "Red" LED for Fault Condition.

Supply Voltage: 10-30 VDC.

Power Consumption: >2 watts.

Output contact Ratings: 500 VA/60W.

Approvals and Certifications: FM approved, and CSA Certified.

Manufacturer and Product: TURK MD13-231Ex0-R/24VDC or equal.

Relays:

General:

Relay Mounting: Plug-in type socket.

Relay Enclosure: Furnish dust cover.

Socket Type: Screw terminal interface with wiring.

Socket Mounting:

Rail.

Provide hold-down clips.

Control Circuit Switching Relay, Nonlatching:

Type: Compact general-purpose plug-in.

Contact Arrangement: 3 Form C contacts.

Contact Rating: 10A at 28V dc or 240V ac.

Contact Material: Silver cadmium oxide alloy.

Coil Voltage: As noted or shown.

Coil Power: 1.2 watts (dc), 1.75VA (ac).

Expected Mechanical Life: 10,000,000 operations.

Expected Electrical Life at Rated Load: 100,000 operations.

Indication Type: Neon or LED indicator lamp.

Push to test button.

Manufacturer and Product: Allen-Bradley; 700-HA Series.

For all 11-pin relays use Allen-Bradley 700-HN203. For 8-pin relays, use Allen-Bradley 700-HN203.

Control Circuit Switching Relay, Latching:

Type: Dual coil mechanical latching relay.

Contact Arrangement: 2 Form C contacts.

Contact Rating: 10A at 28V dc or 120V ac.

Contact Material: Silver cadmium oxide alloy.

Coil Voltage: As noted or shown.

Coil Power: 2.7 watts (dc), 5.3VA (ac).

Expected Mechanical Life: 500,000 operations.

Expected Electrical Life at Rated Load: 50,000 operations.

Manufacturer and Product: Potter and Brumfield; Series KB/KBP.

Control Circuit Switching Relay, Time Delay:

Type: Adjustable time delay relay.

Contact Arrangement: 3 Form C contacts.

Contact Rating: 10A at 240V ac.

Contact Material: Silver cadmium oxide alloy.

Coil Voltage: As noted or shown.

Operating Temperature: Minus 10 to 55 degrees C.

Repeatability: Plus or minus 0.5 percent.

Timing Module: Solid state multifunction plug-in module. Plugs into socket to add timing feature to general purpose relay.

Manufacturer and Products: Allen-Bradley 700-HT1 for ac, 700-HT2 for dc.

Power Supplies:

Furnish to power instruments requiring external dc power, including two-wire transmitters and dc relays.

Convert 120V ac, 60-Hz power to dc power of appropriate voltage(s) with plus or minus 0.05 percent voltage regulation and ripple control to assure that instruments being supplied can operate within their required tolerances.

Provide output over voltage and over current protective devices to:

Protect instruments from damage due to power supply failure.

Protect power supply from damage due to external failure.

Enclosures:

NEMA 1 in accordance with NEMA 250.

Mount such that dissipated heat does not adversely affect other components.

Fuses: For each dc supply line to each individual two-wire transmitter.

Type: Indicating.

Mount so fuses can be easily seen and replaced.

Resistors: All resistors used to derive a 1-5V dc signal from a 4-20 mA dc signal shall be 250 ohm, ±1 percent, 3 watts, axial lead, non-inductive wire wound, welded construction, silicone coated, 1,000V ac dielectric. Vishay-Dale RS-2B-NS or equal. 250 ohms is a standard value in this line, and use of a resistance other that 250 ohms is not acceptable.

Internal Panel Lights for Freestanding Panels:

Type: Switched 100-watt fluorescent back-of-panel lights.

Quantity: One light for every 4 feet of panel width.

Mounting:

Inside and in the top of back-of-panel area.

Protective metal shield for lights.

Service Outlets for Freestanding Panels:

Type: Three-wire, 120-volt, 15-ampere, GFI duplex receptacles.

Quantity:

For Panels 4 Feet Wide and Smaller: One.

For Panels Wider Than 4 Feet: One for every 4 feet of panel width, two minimum per panel.

Mounting: Evenly spaced along back-of-panel area.

Standard Pushbutton Colors and Inscriptions: Use following color code and inscriptions for pushbuttons, unless otherwise noted in Instrument List, Article SUPPLEMENTS.

Tag Function	Inscription(s)	Color
OO	ON	Red
	OFF	Green
OC	OPEN	Red
	CLOSE	Green
OCA	OPEN	Red
	CLOSE	Green
	AUTO	White
OOA	ON	Red
	OFF	Green
	AUTO	White
MA	MANUAL	Yellow
	AUTO	White
SS	START	Red
	STOP	Green
RESET	RESET	Red
EMERGENCY STOP	EMERGENCY STOP	Red
Unused or Noninscribed Buttons		Black.

Standard Light Colors and Inscriptions: The following table gives the inscriptions for service legends, and the lens colors for indicating lights.

Tag Function	Inscription	Color
ON	ON	Red
OFF	OFF	Green
OPEN	OPEN	Red
CLOSED	CLOSED	Green
LOW	LOW	Green
FAIL	FAIL	Amber
HIGH	HIGH	Red
AUTO	AUTO	White
MANUAL	MANUAL	Yellow
LOCAL	LOCAL	White
REMOTE	REMOTE	Yellow

Lettering Color:

Black on white and amber lenses.

White on red and green lenses.

Fabrication

General: Panels with external dimensions and instruments arrangement as shown on Drawings.

Panel Construction and Interior Wiring:

In accordance with the National Electrical Code, state and local codes, NEMA, ANSI, UL, and ICECA.

Fabricate panels, install instruments, wire, and plumb, at the PICS factory.

Electrical Work: In accordance with Division 16, ELECTRICAL.

Shop Assembly: No panel assembly other than correction of minor defects or minor transit damage shall be done on panels at site.

UL Label for Enclosures: UL label stating "Listed Enclosed Industrial Control Panel."

Wiring Within PICS Panels:

Routed through slotted PVC wiring duct with mating cover.

Hinge Wiring:

Secure at each end so that bending or twisting will be around longitudinal axis of wire. Protect bend area with sleeve.

Arrange wiring neatly, cut to proper length, and remove surplus wire.

Abrasion protection for wire bundles, which pass through holes or across edges of sheet metal.

Connections to Screw Type Terminals:

Locking-fork-tongue or ring-tongue lugs.

Use manufacturer's recommended tool with required sized anvil to make crimp lug terminations and to avoid crossovers at a 90 degree angle.

Wires terminated in a crimp lug, maximum of one.

Lugs installed on a screw terminal, maximum of two.

Connections to Compression Clamp Type Terminals:

Strip, prepare, and install wires in accordance with terminal manufacturer's recommendations.

Wires installed in a compression screw and clamp, maximum of one for field wires entering enclosure, otherwise maximum of two, or quantity as approved by manufacturer.

Splicing and tapping of wires, allowed only at device terminals or terminal blocks.

Terminate 24V dc and analog terminal blocks separate from 120V ac circuit terminal blocks.

Separate analog and dc circuits by at least 6 inches from ac power and control wiring, except at unavoidable crossover points and at device terminations.

Arrange wiring to allow access for testing, removal, and maintenance of circuits and components.

Plastic Wire Ducts Fill: Do not exceed manufacturer's recommendation.

Temperature Control:

Freestanding Panels:

Nonventilated Panels: Size to adequately dissipate heat from equipment mounted inside panel or on panel.

Ventilated Panels:

Provide all ventilated panels with louvers and fans with filters or other cooling means as required to maintain internal temperature between 40 degrees F to 90 degrees F.

For panels with backs against wall, furnish louvers on top and bottom of panel sides.

For panels without backs against wall, furnish louvers on top and bottom of panel back.

Louver Construction: Stamped sheet metal.

Ventilation Fans:

Furnish where required to provide adequate cooling.

Create positive internal pressure within panel.

Fan Motor Power: 120 volts, 60-Hz ac, thermostatically controlled.

Air Filters: Washable aluminum, Hoffman Series A-FLT.

Refrigerated System: Furnish where heat dissipation cannot be adequately accomplished with natural convection or forced ventilation. Smaller Panels (that are not freestanding): Size to adequately dissipate heat from equipment mounted inside panel or in panel face.

Freestanding Panel Construction:

Materials: Sheet steel, unless otherwise shown on Drawings with minimum thickness of 12-gauge, unless otherwise noted.

Panel Fronts:

Fabricated from a single piece of sheet steel, unless otherwise shown on Drawings.

No seams or bolt heads visible when viewed from front.

Panel Cutouts: Smoothly finished with rounded edges.

Stiffeners: Steel angle or plate stiffeners or both on back of panel face to prevent panel deflection under instrument loading or operation.

Internal Framework:

Structural steel for instrument support and panel bracing.

Permit panel lifting without racking or distortion.

Lifting rings to allow simple, safe rigging and lifting of panel during installation.

Adjacent Panels: Securely bolted together so front faces are parallel.

Doors: Full height, fully gasketed access doors where shown on Drawings.

Latches: Three-point, Southco Type 44.

Handles: "D" ring, foldable type.

Hinges: Full length, continuous, piano type, steel hinges with stainless steel pins.

Rear Access Doors: Extend no further than 24 inches beyond panel when opened to 90-degree position.

Front and Side Access Doors: As shown on Drawings.

Nonfreestanding Panel Construction:

Based on environmental design requirements required and referenced in Article ENVIRONMENTAL REQUIREMENTS, provide the following:

For panels listed as inside:

Enclosure Type: NEMA 12 in accordance with NEMA 250.

Materials: Steel.

For all other panels:

Enclosure Type: NEMA 4X in accordance with NEMA 250.

Materials: Type 316 stainless steel.

Metal Thickness: 14-gauge, minimum.

Doors:

Rubber-gasketed with continuous hinge.
Stainless steel lockable quick-release clamps.

Manufacturers:

Hoffman Engineering Co.
H. F. Cox.

Factory Finishing:

Enclosures:

Stainless Steel and Aluminum: Not painted.

Nonmetallic Panels: Not painted.

Steel Panels:

Sand panel and remove mill scale, rust, grease, and oil.
Fill imperfections and sand smooth.
Prepare metal and paint panel interior and exterior with one coat of epoxy coating metal primer, two finish coats of two-component type epoxy enamel.
Sand surfaces lightly between coats.

Dry Film Thickness: 3 mils, minimum.

Color: Light gray.

Manufacturer's standard finish color, except where specific color is indicated. If manufacturer has no standard color, finish equipment with light gray color.

Corrosion Protection

Corrosion-Inhibiting Vapor Capsule Manufacturers:

Northern Instruments; Model Zerust VC.
Hoffmann Engineering Co; Model A-HCI.

Source Quality Control

Factory Demonstration Testing (FDT):

Scope: Test PICS control panels to demonstrate panel assemblies are operational, prior to shipment:

Location: PICS factory.

Loop-Specific Functions: Demonstrate proper functions for each control loop, as shown on P&IDs and as required.

Make following documentation available to Construction Manager both before and after FDT:

- Master copy of FDT procedures.
- List of equipment to be tested including make, model, and serial number.
- Equipment and loop verification sheets signed by PICS Construction Subcontractor showing that each equipment and loop has been tested and has functioned properly.

Part 3 – Execution

Examination

For equipment not provided by PICS, but that directly interfaces with the PICS, verify the following conditions:

- Proper installation.
- Calibration and adjustment of positioners and I/P transducers.
- Correct control action.
- Switch settings and dead bands.
- Opening and closing speeds and travel stops.
- Input and output signals.
- Report discrepancies to the Construction Manager.

Installation

Material and Equipment Installation: Retain a copy of manufacturers' instructions at site, available for review at all times.

Electrical Wiring: As specified in Division 16, ELECTRICAL

Removal or Relocation of Materials and Equipment:

- Remove from site materials that were part of the existing facility but are no longer used, unless otherwise directed by Construction Subcontractor to deliver to Construction General Contractor.
- Repair affected surfaces to conform to type, quality, and finish of surrounding surface.

Construction Quality Control

Testing:

- Onsite testing shall be required for each major process instrumentation and control system in accordance with this section and submitted/accepted test procedures. Provide personnel and equipment in support of PICS Continuity (PCT) and PICS Functionality (PFT) testing.
- Tests shall be performed to demonstrate that each function is implemented and operational. These tests are electrical component tests to be performed in advance of facility-wide construction acceptance testing (CAT). CAT shall be performed in accordance with Division 1 requirements. Copies of all tests shall be submitted as specified herein.

Startup and Testing Team:

- Thoroughly inspect installation, termination, and adjustment for components and systems.
- Complete onsite tests.
- Complete onsite training.
- Provide startup assistance.

PICS Continuity Test (PCT) Inspections and Calibrations: Prior to startup, inspect and test to ensure that entire PICS is ready for operation.

Loop/Component Inspections and Calibrations:

Check PICS for proper installation, calibration, and adjustment on a loop-by-loop and component-by-component basis.

Prepare component calibration sheet for each active component (except simple hand switches, lights, gauges, and similar items).

Project name.

Loop number.

Component tag number.

Component code number.

Manufacturer for elements.

Model number/serial number.

Summary of functional requirements, for example:

Indicators and recorders, scale and chart ranges.

Transmitters/converters, input and output ranges.

Computing elements' function.

Switching elements, unit range, differential (fixed/adjustable), reset (auto/manual).

Calibrations, for example:

Analog Devices: Actual inputs and outputs at 0, 10, 50, and 100 percent of span, rising and falling.

Discrete Devices:

Actual trip points and reset points.

Space for comments.

These inspections and calibrations will be witnessed by the Construction Manager or designated representative(s).

PICS Functionality Test (PFT):

General:

Test all PICS elements to demonstrate that PICS satisfies all requirements.

Test Format: Cause and effect.

Person conducting test initiates an input (cause).

Specific test requirement is satisfied if correct result (effect) occurs.

Procedures, Forms, and Checklists:

Conduct tests in accordance with, and documented on, Tank Farm Contractor accepted procedures, forms, and checklists.

Describe each test item to be performed.

Have space after each test item description for sign off by appropriate party after satisfactory completion.

Required Test Documentation: Test procedures, forms, and checklists. All signed by Construction Manager and Construction General Contractor.

Conducting Tests:

Provide special testing materials, equipment, and software.

Wherever possible, perform tests using actual process variables, equipment, and data.

If it is not practical to test with real process variables, equipment, and data, provide suitable means of simulation.

Define simulation techniques in test procedures.

Coordinate PICS testing with Construction Manager and affected Construction Subcontractors.

Test Requirements:

Once facility has been started up and is operating, perform a witnessed PFT on complete PICS to demonstrate that it is operating as required. Demonstrate each required function on a paragraph-by-paragraph and loop-by-loop basis.

Perform local and manual tests for each loop before proceeding to remote and automatic modes.

Where possible, verify test results using visual confirmation of process equipment and actual process variable. Unless otherwise directed, exercise and observe devices supplied by others, as needed to verify correct signals to and from such devices and to confirm overall system functionality. Test verification by means of disconnecting wires or measuring signal levels is acceptable only where direct operation of plant equipment is not possible.

Make updated versions of documentation required for PFT available to Construction Manager at site, both before and during tests.

Make one copy of O&M manuals available to Construction Manager at the site both before and during testing.

Training

General:

Provide an integrated training program to meet specific needs of Tank Farm Contractor's personnel in accordance with submitted and accepted training plan.

Include training sessions, classroom and field, for managers, engineers, operators, and maintenance personnel.

Provide instruction on two working shifts as needed to accommodate the Tank Farm Contractor's personnel schedule.

Tank Farm Contractor reserves the right to make and reuse video tapes of training sessions.

Provide reference handouts that cover the course content for all personnel attending any course or training session.

Operations and Maintenance Training:

Include a review of O&M manuals and survey of spares, expendables, and test equipment.

Use equipment similar to that provided or currently owned by Tank Farm Contractor.

Provide training suitable for instrument technicians with at least a 2-year associate engineering or technical degree, or equivalent education and experience in electronics or instrumentation.

Operations Training:

Training Session Duration: One 8-hour instructor days.

Number of Training Sessions: Two.

Location: Site.

Content: Conduct training on loop-by-loop basis.

Loop Functions: Understanding of loop functions, including interlocks for each loop.

Loop Operation: For example, adjusting process variable set points, AUTO/MANUAL control transfer, AUTO and MANUAL control, annunciator acknowledgement and resetting.

Interfaces with other control systems.

Maintenance Training:

Training Session Duration: One 8-hour instructor days.

Number of Training Sessions: One.

Location: Project site.

Content: Provide training for each type of component and function provided.

Loop Functions:

Understanding details of each loop and how they function.

Component calibration.

Adjustments:

For example, controller tuning constants, current switch trip points, and similar items.

Troubleshooting and diagnosis for components.

Replacing lamps, fuses.

Component removal and replacement.

Periodic maintenance.

Cleaning/Adjusting

Repair affected surfaces to conform to type, quality, and finish of surrounding surface.

Cleaning:

Prior to closing system using tubing, clear tubing of interior moisture and debris.

Upon completion of Work, remove materials, scraps, and debris from interior and exterior of equipment.

Protection

Protect enclosures and other equipment containing electrical, instrumentation and control devices, including spare parts, from corrosion through the use of corrosion-inhibiting vapor capsules.

Periodically replace capsules in accordance with capsule manufacturer's recommendations. Replace capsules just prior to Final Payment and Acceptance.

Supplements

Supplements listed below, following "END OF SECTION," are part of this Specification.

Supplement 1—Instrument Listing for Cell No. 1 and Cell No. 2.

Supplement 2—Component Specifications.

Supplement 3—PLC Input and Output List.

Supplement 4—Loop Specifications.

Supplement 5—PLC and OIU Application Software Setpoints for Cells No. 1 and No. 2.

END OF SECTION 13401

Instrument Listing for Cell No. 1 & Cell No. 2

Instrument Listing Sorted by Equipment Number

Item	Rev	Tag 1 Area	Tag 2 Process	Tag 3 ISA	Tag 4 Eqt.#	Description	Description	Drawing	Component Number	Process Ranges	Eng. Units	Detail	Comments
1	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Enclosure	H-2-830855	Reference 13401 PICS	NA		Reference Controls on Dwg. H-2-830857 sheet 1	48"W x 20"D x 72"High NEMA 12
2	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Operator Interface Unit	H-2-830855	Y50	NA			Ethernet Communication
20	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Main and Sub Breakers	H-2-830855	Reference 13401 PICS	NA			
21	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	High Density Breakers	H-2-830855	Reference 13401 PICS	NA			
22	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Surge Protection	H-2-830855	Reference 13401 PICS	NA			
23	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	24 V dc Power Supplies	H-2-830855	Reference 13401 PICS	NA			Size power supplies for all control loops and various local control panel power
24	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	PLC, I/O, Power Supply and Chassis	H-2-830855	Y50	NA			
24	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	PLC Programming and Communication Software	H-2-830855	Y50	NA			
24	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Programming Laptop	H-2-830855	Y50	NA			
25	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Ethernet Switch and Mounting Bracket	H-2-830855	Y50	NA			
26	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Wiring Duct	H-2-830855	Reference 13401 PICS	NA			
27	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Wiring Terminal Strips Analog	H-2-830855	Reference 13401 PICS	NA			
28	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Interposing Relays	H-2-830855	Reference 13401 PICS	NA			
29	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Wiring Terminal Strips Discrete and Power	H-2-830855	Reference 13401 PICS	NA			
3	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Locking Mechanism	H-2-830855	Reference 13401 PICS	NA			

Instrument Listing for Cell No. 1 & Cell No. 2

Instrument Listing Sorted by Equipment Number

Item	Rev	Tag 1 Area	Tag 2 Process	Tag 3 ISA	Tag 4 Eqt.#	Description	Description	Drawing	Component Number	Process Ranges	Eng. Units	Detail	Comments
30	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Cooling Fan	H-2-830855	Reference 13401 PICS	NA			
30	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Temperature Thermostat	H-2-830855	Reference 13401 PICS	NA		Reference Controls on Dwg. H-2-830857 sheet 2	
31	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Programming Receptacle	H-2-830855	Reference 13401 PICS	NA			
32	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Uninterruptible Power Supply	H-2-830855	Y40	NA		Reference Controls on Dwg. H-2-830857 sheet 2	1050 VA, 120V in - 120V out The UPS system has been disconnected and removed and may be reinstalled in the future.
33	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Incandescent Lighting	H-2-830855	Reference 13401 PICS	NA			
34	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Signal Isolators	H-2-830855	Reference 13401 PICS	NA		Reference Controls on Dwg. H-2-830857 sheet 2	
96	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Alarm Dialer	H-2-830855	Y51	NA	NA		
97	1	219(Y)	LH	CP	001	Crest Pad Building Control Panel	Cellular Modem (CDMA)	H-2-830855	Y52	NA	NA		
40	1	219(Y)	LH	LCP	002	Crest Pad Building Sump Local Control Panel	Enclosure	H-2-830855	Reference 13401 PICS	NA		Reference Controls on Dwg. H-2-830857 sheet 2	16"W x 8"D x 16"H NEMA 4X with internal relays, terminals
41	1	219(Y)	LH	LCP	003	Combined Sump Instrinsic Safety Local Control Panel	Enclosure	H-2-830855	Reference 13401 PICS	NA		Reference Controls on Dwg. H-2-830857 sheet 2	16"W x 8"D x 16"H NEMA 4X with internal relays, terminals
68	1	219(Y)201	LH	LCP	004	Leachate Storage Tank Local Control Panel	Enclosure	H-2-830855	Reference 13401 PICS	NA		Reference Controls on Dwg. H-2-830857 sheet 2	16"W x 8"D x 16"H NEMA 4X with internal relays, terminals
42	1	219(Y)1	LH	LCP	005	Leachate Transfer Pump Local Control Panel	Enclosure	H-2-830855	Reference 13401 PICS	NA		Reference Controls on Dwg. H-2-830857 sheet 2	16"W x 8"D x 16"H NEMA 4X with internal relays, terminals
91	1	219(Y)	LH	LCP	208	Secondary Leachate Collection System Local	Enclosure	H-2-830854 sheet 1 of 4	Reference 13401 PICS	NA	NA	H-2-830849	16" W x 18" H x 9" D NEMA 4X - fiberglass

Instrument Listing for Cell No. 1 & Cell No. 2

Instrument Listing Sorted by Equipment Number

Item	Rev	Tag 1 Area	Tag 2 Process	Tag 3 ISA	Tag 4 Eqt.#	Description	Description	Drawing	Component Number	Process Ranges	Eng. Units	Detail	Comments
						Control Panel						sheet 2 of 2	reinforced polyester, with Data logger, input/output channel, signal converter, communications, rechargeable battery, and charge controller
95	1	219(Y)	LH	LED	208	Secondary Leachate Collection System Local Control Panel	Loop Powered Process Indicator	H-2-830854 sheet 1 of 4		NA	NA	H-2-830849 sheet 2 of 2	Red Lion, Model CUB4LP
4	1	219(Y)	LH	LI	101	Leachate Collection and Removal System	Panel Mount Level Indicator	H-2-830854 sheet 1 of 4	S27	0 to 26.75	Inches		
43	1	219(Y)	LH	LT	101	Leachate Collection and Removal System	Submersible Pressure Transducer	H-2-830854 sheet 1 of 4	L42	0 to 26.75	Inches	Detail No. 6 on Dwg. H-2-830854	Sensor supplied with termination enclosure (TBX)
5	1	219(Y)	LH	LI	104	Leak Detection System	Panel Mount Level Indicator	H-2-830854 sheet 1 of 4	S27	0 to 26.75	Inches		
44	1	219(Y)	LH	LT	104	Leak Detection System	Submersible Pressure Transducer	H-2-830854 sheet 1 of 4	L42	0 to 26.75	Inches	-	Sensor supplied with termination enclosure (TBX) under this section and installed by pump vendor
45	1	219(Y)	LH	HS	105	Crest Pad Building Sump Control Panel	Local Control Panel Mount Handswitch	H-2-830854 sheet 1 of 4	M12	NA	On/ Off		Bypass Operation
46	1	219(Y)	LH	LDE	105	Crest Pad Building Sump	Leak Detection Sensor	H-2-830854 sheet 1 of 4	L109	Actuate elevation 720.5	Feet	Detail No. 4 on Dwg. H-2-830854	1/4" stem actuation from bottom
47	1	219(Y)	LH	LSH	105	Crest Pad Building Sump High	Level Float	H-2-830854 sheet 1 of 4	L8	Actuate elevation 722.0	Feet	Detail No. 1 on Dwg. H-2-830854	
48	1	219(Y)	LH	LSHH	105	Crest Pad Building Sump High High	Level Float	H-2-830854 sheet 1 of 4	L8	Actuate elevation 723.0	Feet	Detail No. 1 on Dwg. H-2-830854	
49	1	219(Y)	LH	LSL	105	Crest Pad Building Sump Low	Level Float	H-2-830854 sheet 1 of 4	L8	Actuate elevation 721.0	Feet	Detail No. 1 on Dwg. H-2-830854	

Instrument Listing for Cell No. 1 & Cell No. 2

Instrument Listing Sorted by Equipment Number

Item	Rev	Tag 1 Area	Tag 2 Process	Tag 3 ISA	Tag 4 Eqt.#	Description	Description	Drawing	Component Number	Process Ranges	Eng. Units	Detail	Comments
50	1	219(Y)	LH	HS	107	Combined Sump Instrinsic Safety Local Control Panel	Local Control Panel Mount Handswitch	H-2-830854 sheet 2 of 4	M12	NA	On/ Off		Bypass Operation
51	1	219(Y)	LH	LDE	107	Combined Sump Pump	Leak Detection Sensor	H-2-830854 sheet 2 of 4	L109	Actuate elevation 714.2	Feet	Detail No. 4 on Dwg. H-2-830854	1/4" stem actuation from bottom
52	1	219(Y)	LH	LSH	107	Combined Sump High	Level Float	H-2-830854 sheet 2 of 4	L8	Actuate elevation 718.0	Feet	Detail No. 1 on Dwg. H-2-830854	
53	1	219(Y)	LH	LSHH	107	Combined Sump High High	Level Float	H-2-830854 sheet 2 of 4	L8	Actuate elevation 719.0	Feet	Detail No. 1 on Dwg. H-2-830854	
54	1	219(Y)	LH	LSL	107	Combined Sump Low	Level Float	H-2-830854 sheet 2 of 4	L8	Actuate elevation 717.0	Feet	Detail No. 1 on Dwg. H-2-830854	
55	1	219(Y)	LH	LSLL	107	Combined Sump Low Low	Level Float	H-2-830854 sheet 2 of 4	L8	Actuate elevation 715.0	Feet	Detail No. 1 on Dwg. H-2-830854	
92	1	219(Y)	LH	LT	108	Secondary Leachate Collection System	Submersible Pressure Transducer	H-2-830854 sheet 1 of 4	L42	0 to 26.75	Inches	Detail No. 6 on Dwg. H-2-830856	Signal interface shall be 12V dc supply instead of 24V dc
94	1	219(Y)	LH	FE	208	Secondary Leachate Collection System	Flow Meter	H-2-830854 sheet 1 of 4		1 to 20	GPM		Battery powered flow meter (Omega FTB 790 Series or approved equal).
7	1	219(Y)	LH	FI	202	Leachate Collection and Removal System Low Flow Pump	Panel Mount Flow Indicator	H-2-830854 sheet 1 of 4	S27	0 to 25	GPM		0-30 PSI Range
56	1	219(Y)	LH	FIT	202	Leachate Collection and Removal System Low Flow Pump	In-Line Flow Magmeter	H-2-830854 sheet 1 of 4	F4	0 to 25	GPM	Section A on Dwg. H-2-830847	0-30 PSI Range with integral transmitter
57	1	219(Y)	LH	HS	202	Leachate Collection and Removal System Low Flow Pump	Motor Control Handswitch	H-2-830854 sheet 1 of 4	M12	NA			Provided under Section 16440
84	1	219(Y)	LH	PI	202	Leachate Collection and Removal System Low Flow Pump	Pressure Gauge	H-2-830854 sheet 1 of 4	P4/P6	0 to 30	PSI		0-30 PSI Range with Diaphragm Seal

Instrument Listing for Cell No. 1 & Cell No. 2

Instrument Listing Sorted by Equipment Number

Item	Rev	Tag 1 Area	Tag 2 Process	Tag 3 ISA	Tag 4 Eqt.#	Description	Description	Drawing	Component Number	Process Ranges	Eng. Units	Detail	Comments
58	1	219(Y)	LH	YL	202	Leachate Collection and Removal System Low Flow Pump	Motor Control On Indicator	H-2-830854 sheet 1 of 4	M12	NA			Provided under Section 16440
8	1	219(Y)	LH	FI	203	Leachate Collection and Removal System High Flow Pump	Panel Mount Flow Indicator	H-2-830854 sheet 1 of 4	S27	0 to 250	GPM		0-60 PSI Range
59	1	219(Y)	LH	FIT	203	Leachate Collection and Removal System High Flow Pump	In-Line Flow Magmeter	H-2-830854 sheet 1 of 4	F4	0 to 250	GPM	Section A on Dwg. H-2-830847	0-60 PSI Range with integral transmitter
60	1	219(Y)	LH	HS	203	Leachate Collection and Removal System High Flow Pump	Motor Control Handswitch	H-2-830854 sheet 1 of 4	M12	NA			Provided under Section 16440
85	1	219(Y)	LH	PI	203	Leachate Collection and Removal System High Flow Pump	Pressure Gauge	H-2-830854 sheet 1 of 4	P4/P6	0 to 60	PSI		0-60 PSI Range with Diaphragm Seal
61	1	219(Y)	LH	YL	203	Leachate Collection and Removal System High Flow Pump	Motor Control On Indicator	H-2-830854 sheet 1 of 4	M12	NA			Provided under Section 16440
9	1	219(Y)	LH	FI	204	Leak Detection System Pump	Panel Mount Flow Indicator	H-2-830854 sheet 1 of 4	S27	0 to 15	GPM		0-15 PSI Range
62	1	219(Y)	LH	FIT	204	Leak Detection System Pump	In-Line Flow Magmeter	H-2-830854 sheet 1 of 4	F4	0 to 15	GPM	Section A on Dwg. H-2-830847	0-15 PSI Range with integral transmitter
63	1	219(Y)	LH	HS	204	Leak Detection System Pump	Motor Control Handswitch	H-2-830854 sheet 1 of 4	M12	NA			Provided under Section 16440
86	1	219(Y)	LH	PI	204	Leak Detection System Pump	Pressure Gauge	H-2-830854 sheet 1 of 4	P4/P6	0 to 15	PSI		0-15 PSI Range with Diaphragm Seal
64	1	219(Y)	LH	YL	204	Leak Detection System Pump	Motor Control On Indicator	H-2-830854 sheet 1 of 4	M12	NA			Provided under Section 16440
65	1	219(Y)	LH	HS	205	Crest Pad Building Sump Pump	Motor Control Handswitch	H-2-830854 sheet 1 of 4	M12	NA			Provided under Section 16440
89	1	219(Y)	LH	PI	205	Crest Pad Building Sump Pump	Pressure Gauge	H-2-830854 sheet 1 of 4	P4/P6	0 to 15	PSI		0-15 PSI Range with Diaphragm Seal
66	1	219(Y)	LH	YL	205	Crest Pad Building Sump Pump	Motor Control On Indicator	H-2-830854 sheet 1 of 4	M12	NA			Provided under Section 16440

Instrument Listing for Cell No. 1 & Cell No. 2

Instrument Listing Sorted by Equipment Number

Item	Rev	Tag 1 Area	Tag 2 Process	Tag 3 ISA	Tag 4 Eqt.#	Description	Description	Drawing	Component Number	Process Ranges	Eng. Units	Detail	Comments
93	1	219(Y)	LH	DPI	206	Crest Pad Building	Wall-Mounted Differential Pressure Indicator	H-2-830854 sheet 1 of 4	Reference 15100 (LEACHATE FILTERS)	0 TO 15	PSID		With re-settable drag pointer
67	1	219(Y)	LH	HS	207	Combined Sump Pump	Motor Control Handswitch	H-2-830854 sheet 2 of 4	M12	NA			Provided under Section 16440
69	1	219(Y)	LH	YL	207	Combined Sump Pump	Motor Control On Indicator	H-2-830854 sheet 2 of 4	M12	NA			Provided under Section 16440
12	1	219(Y)	LH	HS	219	Crest Pad Building	Control Panel Mount Switch	H-2-830854 sheet 1 of 4	M12	NA			Alarm Acknowledge Switch
70	1	219(Y)	LH	JSH	219	Crest Pad Building	Power Relay	H-2-830854 sheet 1 of 4	Reference 13401 PICS	NA			Power Relay mounted inside Control Panel
10	1	219(Y)	LH	TI	219	Crest Pad Building	Panel Mount Temp Indicator	H-2-830854 sheet 2 of 4	S27	-40 to 104	°F		
71	1	219(Y)	LH	TIT	219	Crest Pad Building	Temperature Transmitter	H-2-830854 sheet 1 of 4	T3	-40 to 104	°F		With integral transmitter
72	1	219(Y)	LH	YAL	219	Crest Pad Building	Alarm Light	H-2-830854 sheet 1 of 4	M31	NA			
11	1	219(Y)1	LH	TI	220	Leachate Transfer Building	Panel Mount Temp Indicator	H-2-830854 sheet 2 of 4	S27	-40 to 104	°F		
73	1	219(Y)1	LH	TIT	220	Leachate Transfer Building	Temperature Transmitter	H-2-830854 sheet 2 of 4	T3	-40 to 104	°F		With integral transmitter
74	1	219(Y)1	LH	YAL	220	Leachate Transfer Building	Alarm Light	H-2-830854 sheet 2 of 4	M31	NA			
88	1	219(Y)201	LH	HS	301	Leachate Storage Tank Local Control Panel	Local Control Panel Mount Handswitch	H-2-830854 sheet 2 of 4	M12	NA	On/ Off		Bypass Operation
75	1	219(Y)201	LH	LSHH	301	Leachate Storage Tank	Level Switch	H-2-830854 sheet 2 of 4	L1A	Actuate elevation 728.33	Feet	Detail No. 2 on Dwg. H-2-830854	
76	1	219(Y)201	LH	LSLL	301	Leachate Storage Tank	Level Switch	H-2-830854 sheet 2 of 4	L1A	Actuate elevation 722.7	Feet	Detail No. 2 on Dwg. H-2-830854	
77	1	219(Y)201	LH	LT	301	Leachate Storage Tank	Submersible Pressure Transducer	H-2-830854 sheet 2 of 4	L42	0 to 108	Inches	Detail No. 3 on Dwg. H-2-830854	Sensor supplied with termination enclosure (TBX) and Lightning

Instrument Listing for Cell No. 1 & Cell No. 2

Instrument Listing Sorted by Equipment Number

Item	Rev	Tag 1 Area	Tag 2 Process	Tag 3 ISA	Tag 4 Eqt.#	Description	Description	Drawing	Component Number	Process Ranges	Eng. Units	Detail	Comments Arrestor
78	1	219(Y)201	LH	LI	301-1	Leachate Storage Tank	Local Control Panel Mount Level Indicator	H-2-830854 sheet 2 of 4	S27	0 to 108	Inches		Provide Signal Isolator, and Lightning Arrestor
6	1	219(Y)201	LH	LI	301-2	Leachate Storage Tank	Panel Mount Level Indicator	H-2-830854 sheet 2 of 4	S27	0 to 108	Inches		Provide Signal Isolator
79	1	219(Y)1	LH	HS	302	Leachate Transfer Pump	Panel Mount Motor Control Handswitch	H-2-830854 sheet 2 of 4	M12	NA	On/ Off		Motor On/Off control switch
87	1	219(Y)1	LH	PI	302	Leachate Transfer Pump	Pressure Gauge (Load)	H-2-830854 sheet 2 of 4	P4/P6	0 to 15	PSI		0-15 PSI Range with Diaphragm Seal
90	1	219(Y)1	LH	PI	303	Leachate Transfer Pump	Pressure Gauge (Suction)	H-2-830854 sheet 2 of 4	P4/P6	0 to 100	Inches		0-100 Inches Range with Diaphragm Seal
80	1	219(Y)1	LH	FIT/FQI	302-1	Leachate Transfer Pump	In-Line Flow Magmeter	H-2-830854 sheet 2 of 4	F4	0 to 300	GPM	Section A on Dwg. H-2-830851	0-15 PSI Range. Flow and Total Integral with same indicator.
81	1	219(Y)1	LH	YL	302-1	Leachate Transfer Pump	Local Control Panel Mount Motor Control On Indicator	H-2-830854 sheet 2 of 4	M12	NA			
82	1	219(Y)1	LH	FQI	302-2	Leachate Transfer Pump	Local Control Panel Mount Flow Totalizer Indicator	H-2-830854 sheet 2 of 4	S27	0 to 10,000	Gal		Flow and Total Integral to same meter. Provided Signal Isolator
83	1	219(Y)1	LH	YL	302-2	Leachate Transfer Pump	Motor Control On Indicator	H-2-830854 sheet 2 of 4	M12	NA			Provided under Section 16440
(Y) = A for Cell No. 1						For Cell No. 1 reference corresponding H-2-830854 sheet 1 of 4 and sheet 2 of 4.							
(Y) = E for Cell No. 2						For Cell No. 2 reference corresponding H-2-830854 sheet 3 of 4 and sheet 4 of 4.							

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Component Specifications

F4 Flow Element and Transmitter, Electromagnetic:

General:

Function: Measure, indicate, and transmit the flow of a conductive process liquid in a full pipe.

Type:

Electromagnetic flowmeter, with operation based on Faraday's Law, utilizing the pulsed dc type coil excitation principle with high impedance electrodes.

Full bore meter with magnetic field traversing entire flow-tube cross section.

Unacceptable are insert magmeters or multiple single point probes inserted into a spool piece.

Parts: Flow element, transmitter, interconnecting cables, and mounting hardware. Other parts as noted.

Service:

Stream Fluid:

As noted.

Suitable for liquids with a minimum conductivity of 5 microS/cm and for demineralized water with a minimum conductivity of 20 microS/cm.

Flow Stream Descriptions: If and as described below.

Operating Temperature:

Element:

Ambient: Minus 5 to 140 degrees F, typical, unless otherwise noted.

Process: Minus 5 to 140 degrees F, typical, unless otherwise noted.

Transmitter:

Ambient: Minus 5 to 140 degrees F, typical, unless otherwise noted.

Storage: 15 to 120 degrees F, typical, unless otherwise noted.

Performance:

Flow Range: As noted.

Accuracy: Plus or minus 0.5 percent of rate for all flows resulting from pipe velocities of 2 to 33 feet per second.

Turndown Ratio: Minimum of 10 to 1 when flow velocity at minimum flow is at least 1 foot per second.

Features:

Zero stability feature to eliminate the need to stop flow to check zero alignment.

No obstructions to flow.

Very low pressure loss.

Measures bi-directional flow.

Process Connection:

Meter Size (diameter inches): As noted.

Connection Type: 150-pound ANSI raised-face flanges or wafer style depending on meter size, unless otherwise noted.

Flange Material: Carbon steel, unless otherwise noted.

Power (Transmitter): 120V ac, 60-Hz, unless otherwise noted.

Element:

Meter Tube Material: Type 304 or 316 stainless steel, unless otherwise noted.

Liner Material:

Teflon, unless otherwise noted.
Low activated waste leachate solution.

Liner Protectors: Covers (or grounding rings) on each end to protect liner during shipment.

Electrode Type: Flush or bullet nose as recommended by the manufacturer for the noted stream fluid.

Electrode Material: Type 316 stainless steel or Hastelloy C, unless otherwise noted.

Grounding Ring:

Required, unless otherwise noted.

Material: Type 316 stainless steel, unless otherwise noted.

Enclosure: NEMA 4X, minimum, unless otherwise noted.

Transmitter:

Mounting: Integral, unless otherwise noted.

Display:

Required, unless otherwise noted.
Digital LCD display, indicating flow rate and total.

Bi-directional Flow Display:

Required, unless otherwise noted.
Forward flow rate.
Forward, net totalization.

Parameter Adjustments: By keypad or non-intrusive means.

Enclosure: NEMA 4X, minimum, unless otherwise noted.

Empty Pipe Detection: Drives display and outputs to zero when empty pipe

Signal Interface (at Transmitter):

Analog Output:

Isolated 4 to 20 mA dc for load impedance from 0 to at least 500 ohms minimum for 24V dc supply.
Supports Superimposed Digital HART protocol.

Cables:

Types: As recommended by manufacturer.

Lengths: As required to accommodate device locations.

Built-in Diagnostic System:

Features:

Field programmable electronics.

Self-diagnostics with troubleshooting codes.

Ability to program electronics with full scale flow, engineering units, meter size, zero flow cutoff, desired signal damping, totalizer unit digit value, etc.

Initial flow tube calibration and subsequent calibration checks.

Factory Calibration:

Calibrated in an ISO 9001 and NIST certified factory.

Factory flow calibration system must be certified by volume or weight certified calibration devices.

Factory flow calibration system shall be able to maintain calibration flow rate for at least 5 minutes for repeatability point checks.

Factory Ready for Future In situ Verifications: If noted.

Original meter parameter values available from vendor by request.

Accessories:

In situ Verification System: If noted.

Quantity:

One complete system provided for the project.

Verifies quantitatively that the meter and signal converter's present condition is the same as originally manufactured.

Physical access to the flow-tube not required.

Meet standards established by the National Testing Laboratory.

Tests and stores over 50-meter parameters related to primary coils, electrodes, interconnecting cable and signal converter.

Verification standard shall be plus or minus 1 percent of wet calibration for meters produced using the calibration verification service, or plus or minus 2 percent for standard meters.

Windows-based software

Primary Simulation System: If noted.

Quantity:

One complete system provided for the project.

Verifies proper operation of the signal converter by simulating the flow meter's output signal.

Generates pulsed dc excitation signal with a reference voltage of 70 mV.

Generated signal ranges from 0 to 99 percent (0 to 32.8 feet per second) with a resolution of 0.1 percent.

Switch selectable for forward, reverse and zero flow rate.

Verifies various input and output signals.

Manufacturers:

Krohne Electromagnetic Integral Systems: Aqua Flux Flowmeter (size: 3/8 to 120 inches).

Endress & Hauser, Inc. Flow Measuring System: Promag 50/53W (size: 1 to 78 inches).

Invensys Foxboro (includes IMT 25 Series Intelligent Magnetic Flow Transmitter): 9100A Series Flanged Body Flow Tubes (size: 1 to 78 inches).

L1A Multipoint Level Element and Switches, Admittance:

General:

Function: Operate switches at two separate, distinct, preset product levels in a vessel.

Type: Admittance using low power radio frequency circuit.

Parts: Element and electronics unit. For remote mount, interconnecting cable.

Service: Fluid as noted.

Performance:

Set Points: As noted.

Temperature:

Operating range minus 40 to 140 degrees F.

Unaffected by coating buildup on element.

Features:

Electronics Unit:

Filtering: Built-in RFI protection.

Fail-Safe Contacts: Field convertible switch action.

Enclosure Type: Explosion-proof and weatherproof (NEMA 4).

Electronics Mounting: Integrally with element, unless otherwise noted.

When remote, provide cable with length as required to accommodate device locations.

Response Time: 20 milli-seconds standard, or as noted.

Element:

Type: Probe rod.

Insertion Length: As required to achieve noted set points.

Material: 316 stainless steel, unless otherwise noted.

Rating: Element and cable intrinsically safe.

Grounding Element: Required for nonmetallic tank applications.

Process Connection: 3/4-inch NPT unless otherwise noted.

Signal Interface: Contacts, 3 DPDT rated 5A continuous at 120V ac, minimum.

Power: 120V ac 50/60-Hz, or as noted.

Manufacturers and Products:

Drexelbrook; Model 506-3100.

Princo.

Endress & Hauser, Inc.

L8 Level Switch, Float:

General:

Function: Actuate contact at preset liquid level.

Type: Direct-acting float with an enclosed mercury switch and integral cable.

Service: Liquid; low activated waste leachate solution, unless otherwise noted.

Performance:

Set Point: As noted.

Differential: 1-inch maximum.

Temperature: 0 to 180 degrees F.

Features:

Entire Assembly: Watertight and impact-resistant.

Float Material and Size: Polyethylene/foam filled; 4.5-inch diameter tear drop.

Cable:

Combination support and signal.
Length as noted or as necessary per mounting requirements.
Type SO nitrile PVC jacket, AWG No. 18/2 or No. 18/4.

Mounting:

Pipe:

Cable-to-pipe clamp, corrosion-proof cable for 1-inch pipe.
Pipe-to-wall bracket for 1-inch pipe.

Suspended Type: As noted.

Signal Interface:

Switch Type: Mercury tilt.

Switch Contacts:

Isolated, rated 4.5A continuous at 120V ac.
As required (for example 1NO, 1NO+1NC) to meet functional requirements, or as shown.

Manufacturers and Products:

Consolidated Electric Co.; Model LS.
Anchor Scientific; Roto-Float, Type P/Type S.

L42 Level Element/Transmitter, Submersible, Wastewater:

General:

Function: Measure and transmit a signal proportional to level.

Type: Totally submersible pressure sensor (loop powered).

Parts: Sensor, interconnecting cable, sensor termination enclosure.

Service:

Fluid: Wastewater, unless otherwise noted.

Performance:

Process Range:

As noted.
Provide fixed factory range such that noted process range is between 40 and 80 percent of fixed factory range.

Accuracy: 0.25 percent of full scale.

Temperature, Operating: Minus 4 to plus 140 degrees F.

Overpressure:

Range dependent.
4X for ranges of 5 psig and above (to a maximum of 2,000 psi).
Greater than 4X for ranges below 5 psig.

Long Term Stability: Plus or minus 0.1 percent full scale/year, typical.

Features:

Sensor:

Silicon sensing element.
Titanium body.

Diaphragm: Titanium isolation, unless otherwise noted.

Pressure Connection:

Depth cone with radial inlet holes.
NEMA 6 rating (submersible to 2,300 feet).

Temperature Compensation: Plus 30 to 80 degrees F.

Dimensions:

Not to exceed 8.5L by 1.1 diameter, inches, nominal.
Loop powered, 9-30V dc.
Open face with perforated Protective Plate: Not to exceed 1.35 diameter, inches.

Interconnecting Cable:

Length:

As required.
Polyurethane sheathed.
Kevlar strain relief cord.
Integral vent tube.

Sensor Termination Enclosure:

Enclosure:

NEMA 4X, PVC/polycarbonate.
Desiccant module.
Micro filter.

Wall and 2-Inch Pipe Mounting Kit: Required, unless otherwise noted.

Lightning Arrestor(s): Required, unless otherwise noted.

Signal Interface: 4 to 20 Ma dc output, for load impedance of 0 to 750 ohms, minimum for 24V dc supply without load adjustment.

Area Classification: Intrinsically safe; certified for use in Class 1, Division 1, Groups A, B, C, and D atmospheres.

Manufacturers:

Druck; Type PTX 1830 with STE110.

Pressure Systems, Inc. KPSI; Series 720 with Series 815 Aneroid Bellows and Series 840 Junction Box.

Or approved equivalent.

L109 Level Detection Switch, Rises on Stem:

General:

Function: Actuate contact at preset liquid level.

Type: Direct acting; rises on stem.

Service: Liquid, water, wastewater, unless otherwise noted.

Performance:

Set point as noted.

Switch Actuation Point: Approximately 3/4-inch distance from end of stem to weighted support collar.

Operating Temperature Range: Minus 40 to plus 110 degrees F.

Features:

Assembly Material: Brass stem, Buna N Float, and Type 316 stainless steel wetted parts.

Float Size: 2-inch diameter.

Mounting: Suspension cable with compact-sized float, slosh shield, and weighted collar suspended in standpipes or sumps for leak detection.

Signal Interface:

Switch Type: Magnetic reed switch.

Switch Contacts:

SPST Isolated, rated at 20 VA.
NC (by inverting float on unit stem).

Cable and Lead Wires: No. 22 AWG, 25 feet of length of PVC jacketed cable.

Manufacturer and Product: GEMS; Specialty Switches Liquid Level Switch, Model LS-750.

M12 Hand Switch and Light, Oiltight, Round:

General:

Function: Select, initiate, and display discrete control functions.

Type: Heavy-duty, oiltight, industrial.

General Features:

Mounting: 30.5 mm single round hole. Panel thickness 1/16 inch to 1/4 inch.

Legend Plate: Standard size square style aluminum field and black markings, unless otherwise noted. Markings as shown.

Configuration: Light, pushbutton, or switch as noted or shown.

Light Features:

Lights: 6V ac lamps and integral transformer for operation from 120V ac, unless otherwise noted.

Lens Color: Color as specified under PANEL, STANDARD LIGHT COLOR AND INSCRIPTIONS, or as noted.

Pushbutton and Switch Features:

Guard: Full guard with flush button, unless otherwise noted.

Operator: Black pushbutton, black non-illuminated knob on switch, unless otherwise noted.

Boot: None, unless otherwise noted.

Signal Interface:

Contact Block:

Type: Silver-coated butting, unless otherwise noted.

Rating: 10 amps continuous at 120V ac or as noted.

Sequence: Break-before-make, unless otherwise shown.

Arrangement: Normally open or normally closed as shown, or perform functions noted.

Terminals: Screw with strap clamp, unless otherwise noted.

NEMA Rating: NEMA 4, watertight and dusttight and NEMA 13, oiltight.

Manufacturers/Models:

Allen-Bradley; Bulletin 800T.
Eaton Corp.; Cutler-Hammer, Type 10250T.
Square D Co.; Class 9001, Type K.

M31 Warning Light, Indoor/Outdoor:

General:

Function: Visual alarm.

Type: Rotating reflector or flashing bulb.

Parts: Light and spare bulbs.

Performance:

Temperature, Operating: Minus 35 to 190 degrees F.

Flash Rate: Nominally 90 per minute.

Features:

Dome Color: Amber, unless otherwise noted.

Lamp Life: 200 hours.

Lamp: Incandescent/25 watts.

Enclosure:

Type: Water-resistant closed cell neoprene gasket.

Mounting: Wall bracket, unless otherwise noted.

UL Listing: Indoor/outdoor use.

Power: 120V ac, 50/60-Hz.

Spare Bulbs: Two for each light.

Manufacturers:

Federal Signal; Model 225.

Benjamin Electric Manufacturing; Series KL-4000.

P4 Pressure Gauge:

General:

Function: Pressure indication.

Type: Bourdon tube or bellows.

Performance:

Scale Range: As noted.

Accuracy: Plus or minus 0.50 percent of full scale for Bourdon tube. Plus or minus 2-1-2% of span (ASME B 40.1 Grade A).

Features:

Liquid Filled:

Required unless otherwise noted.

Glycerin fill, unless otherwise noted.

Dial: 4-1/2-inch diameter, unless otherwise noted.

Case Material: Black phenolic plastic, unless otherwise noted.

Element Material: Phosphor-bronze, unless otherwise noted.

Pointer: Micrometer-adjustable.

Movement: Stainless steel, Teflon coated bearings, rotary geared.

Window: Glass, unless otherwise noted.

Socket Materials:

Brass, unless otherwise noted.

Threaded reinforced polypropylene front ring for easy zero adjustment.

Case Type: Solid front with solid wall between window and element. Rear of case, gasketed pressure relief.

Process Connection:

Mounting: Lower stem, unless otherwise noted.

Size: 1/2 inch, unless otherwise noted.

Connection Type: Threaded (NPT).

Manufacturers and Products:

Ashcroft; Duragauge Model 1279/1379.

Ashcroft LP Bellows Gauge Model 1188.

Weksler; Royal Process Gauge Model AAXX.

Ametek U.S. Gauge; Solfrunt Model 19XX.

P6 Pressure Seal, Diaphragm:

General:

Function: Isolate sensing element from process fluid.

Type: Fluid filled, corrosion resistant.

Service:

Pressure: Same as associated sensor.

Temperature: As noted.

Features:

Material Lower Housing: Type 316 stainless steel, unless otherwise noted.

Diaphragm Material: Type 316 stainless steel, unless otherwise noted. Bleed screw in upper housing.

Fill Fluid: As noted. Factory filled and assembled when possible.

Process Connections:

Instrument: 1/2-inch female NPT, unless otherwise noted.

Process: 1/2-inch female NPT, unless otherwise noted.

Connection Material: Compatible with pressure indicator and process lines.

Manufacturers:

Ametek, Mansfield and Green Division; Type SG.
Ashcroft; Type 101.
Ashcroft; Type 741.

S27 Indicator, Digital Panel:

General:

Function: Display analog signal, or totalize analog signal, and display engineering units.

Type: 7-segment digital, horizontal edgewise.

Performance:

Range: As noted, engineering units as noted.

Accuracy:

Plus or minus 0.1 percent full scale.
Temperature, Operating: 32 to 120 degrees F.

Features:

Digits: 4-1/2; 0.56-inch high minimum; 7-segment LED, gas plasma, or vacuum fluorescent.

Decimal Point: Field selectable.

Input Impedance: 100 ohms maximum.

Service Legend: Permanent, display of engineering units.

Response Time: 1 second maximum to 0.1 percent accuracy.

Signal Interface: 4 to 20 mA dc.

Enclosure:

Type: NEMA 4X.

Mounting: Panel; approximately 2-inch high, 4-inch wide, 5-inch deep.

Power: 120V ac, 50/60-Hz unless otherwise noted.

Manufacturers:

Red Lum Controls.
Action Instruments.
Analogic.
Moore Industries.

T3 Temperature Element and Transmitter, Resistance:

General:

Function: Measure the temperature of ambient, and transmit analog signal proportional to temperature.

Type: RTD.

Parts: Element and transmitter.

Service:

Process: As noted.

Process Temperature Range: As noted.

Element:

Type:

Single-element, unless otherwise noted
Three-wire, RTD.
Platinum, 100 ohm nominal at 0 degrees C.

Performance:

Accuracy: Greater of plus or minus 4 degrees F or plus or minus 0.75 percent of reading.

Features:

Dimensions:

1/4-inch diameter.
Length to accommodate thermowell insertion and extension lengths.
Spring-loaded element when well is used.

Sheath: Type 316 Stainless Steel, unless otherwise noted.

Process Operating Temperature Range: Minus 320 to 900 degrees F, unless otherwise noted.

Terminal Connection Head: General purpose, NEMA 4 weatherproof, unless otherwise noted.

Maximum Temperature: 220 degrees F, unless otherwise noted.

Thermowell Connection: Union Coupler, unless otherwise noted.

Sensitive Length: 1.6 inch minimum, measured from closed end.

Transmitter:

Ambient Operation Conditions.

Temperature: minus 40 to 140 degrees F, with display.

Relative Humidity: 0 to 100 percent, noncondensing.

Type: Two-wire, powered by a remote power supply.

Performance:

Accuracy: Greater of plus or minus 0.7 degree F or plus or minus 0.06 percent of span.

Response Time: 1.2 second 90 percent response time for 80 percent input step, with minimum damping.

Electrical Safety: Standard unless otherwise noted.

Features:

Indicator:

Three line LCD, unless otherwise noted.
Automatic reference junction compensation.

Failsafe Mode:

User configurable ON, unless otherwise noted.
Downscale, unless otherwise noted.

Electric Damping: 1.2 seconds.

Signal Interface: 4 to 20 mA dc

Power: 24V dc external power supply.

Digital Communication: HART.

One HART communicator to be supplied for all HART capable transmitters, if not already supplied under another Specification section.

Enclosure:

Materials: Epoxy coated, low-copper aluminum, unless otherwise noted.

Type: NEMA 4X.

Mounting:

Wall, as noted.
For wall, provide stainless steel mounting set, unless otherwise noted.

Manufacturers and Products:

Foxboro; RTT20 Series Transmitter with PR Series RTD and Thermowell.
Rosemount; 78 Series Platinum RTD and Model 644H Transmitter.

Y40 Uninterruptible Power Supply System:

General:

Function: Provides isolated, regulated uninterrupted ac output power during a complete or partial interruption of incoming line power.

Major Parts: Inverter, a battery charger, sealed battery.

Performance:

Capacity: As noted.

Input Power: 120V ac single-phase/60 Hz, unless otherwise noted.

Connections: As noted.

Output Power: 120V ac single-phase/60 Hz, unless otherwise noted.

Connections: As noted.

On-line Efficiency: 85 percent minimum, unless otherwise noted.

Backup Runtime:

Full Load: 9 minutes minimum, unless otherwise noted.

Half Load:

24 minutes minimum, unless otherwise noted.

Continuous no-break power with no measurable transfer time.

Sine-Wave Output Power Regulation:

Plus or minus 5 percent or less total harmonic distortion.

Meet or exceed CSA C22.2 No. 107.1 for harmonic distortion.

Voltage Regulation: Plus or minus 3 percent nominal.

Operating Temperature: 0 to 40 degrees C (32 to 104 degrees F).

Lightning and Surge Protection:

Pass lightning standard ANSI/IEEE C62.41 Categories A and B test.

2000 to 1 attenuation of input spike.

Isolation:

True separately derived power source as per NEC Article 250-5d with output neutral bonded to ground.

Complete from line.

Less than 2 pF effective input to output capacitance.

Features:

Enclosure:

Floor mounted cabinet, unless otherwise noted.

RS232 external interface with full-duplex output capable of:

Remote monitoring of meter functions and alarm conditions.

Remote diagnostic testing.

Remotely set point display and adjustment.

Manufacturers:

Best Power, FERRUPS Uninterruptible Power System.

Controlled Power.

American Power Conversion; Back-UPS Pro.

Y50 Programmable Logic Controller and Operator Interface Unit System:

General:

Function: Microprocessor based system configured, assembled, and programmed in order to implement the safe automatic control and measurement of process control equipment.

System incorporates programmable logic controllers, processors, power supplies, operator interface units, communication hardware, programming and development software, and cables, and programming laptop.

Programmable Logic Controller (PLC):

Function: Used for process monitoring and control by emulating functions of conventional panel mounted equipment such as relays, timers, counters, current switches, calculation modules, PID controllers, stepping switches, and drum programmers.

PLC Parts: Central processing unit (CPU), power supply, local input/output modules, local (chassis/rack) controllers, I/O terminals board and termination cable assemblies, and factory assembled programming laptop, ETHERNET and OIU communication interconnecting cables.

PLC Central Processing Unit (CPU) Specifications:

Type: Microprocessor, 16-bit minimum.

Memory: 32K words.

I/O Capacity: 4096 inputs, 4096 outputs.

Standard RAM with lithium battery for 2 years backup.

Scan Time: 0.9 ms/1K ladder logic.

Communications:

Two communication ports, RS-232/RS485 and 10BASE-T Ethernet channel.

10 Mbps communications – TCP/IP protocol.

RS-232 and DH-485 Communication protocols.

Instruction Set: Timers and Counters.

Math: Signed integer and floating-point math including add, subtract, multiply, divide, square root, exponent, and compare.

Register Operations: Shift registers, bit shift, bit set, bit clear, data move and data format conversion.

Process Loop Control: User configurable direct or reverse acting PID loop control computation with the capability of both AUTO and MANUAL modes of operation, remote access to controller tuning constants.

Real Time Clock: Date and time set and compare.

Miscellaneous: Jump or skip to a label, one shot, quantity drums, pre-configured analog alarm functions, subroutines, quantity.

Environment:

Operating Temperature: 0 to 55 degrees C (32 to 131 degrees F).

Storage Temperature: -25 to 70 degree C (-13 to 158 degrees F).

Relative Humidity: (noncondensing) 5 to 95 percent at 0 to 55 degrees C (32 to 131 degrees F).

Heat Dissipation: 15 Watts.

Agency Approvals and Standards:

UL listed.

CSA certified.

or another state approved agency.

Random Access Memory (RAM):

Type: CMOS type.

Word Size: 16 bits, minimum.

Battery Backup: 24 months, minimum.

Memory Size: Sufficient to implement all applications software plus 50 percent spare.

Read only memory (ROM) for controller's operating system and diagnostics.

Memory Protection: Keylock switch.

Manufacture and Product: Allen-Bradley 1747-L552.

PLC Power Supply: One unit for each input/output base assembly.

Voltage: 120/220 volts (user selectable), 60 Hz input; 24 VDC output.

Mounting: Integral with PLC chassis.

Manufacture and Product: Allen-Bradley 1747-P4.

PLC Input/Output: Complete input/output system specifications:

Discrete Input Modules:

Voltage: 24 VDC.

Operating Power: 2 watts.

Points per Module: 16 maximum.

LED status indicator for each point.

Isolation: Between input point and PLC, 1,500 volts rms.

Discrete Output Modules:

Voltage: 24VDC.

Operating Power: 2 watts.

Load Rating: 2 amps continuous.

Isolation: Between PLC and output point, 1,500 volts rms.

Points per Module: 16 maximum.

LED status indicator for each point.

Isolated Discrete Output Modules:

Type: Isolated Form C relay.

Voltage: 120 volts, 60-Hz.

Isolated Outputs per Module: 8 Maximum.

Load Rating: 2 amps continuous.

Operating Power: 2.5 watts.

LED status indicator and fuse for each point.

Analog Input and Output Modules:

Voltage: 24 volts dc.

Power: 3 watts.

Differential Analog Points Per Module: 8 maximum.

Isolated Analog Output Points Per Module: 8 maximum.

Isolation: Between PLC and I/O point and between I/O points, 1,500 volts rms.

Analog Input Resolution: 12 bits minimum.

Analog Output Resolution: 12 bits minimum.

Manufacturer and Series: Allen-Bradley 1746 Series.

Operator Interface Unit:

Function: Panel mounted terminal unit with color video display screen and keypad, which enable an operator to monitor and interface with the process control system programmable logic controller. OIU linked with PLC over ETHERNET network.

Type: Microprocessor based device and programmable using Microsoft Windows based development software. *(Note: PICS PLC and OIU design is based upon the Allen-Bradley SLC-5/05E programmable logic controller and the Allen-Bradley Panelview 600 operator interface unit).*

Parts: Central processing unit (CPU), power supply, video display touch screen, keypad, Ethernet and printer ports.

Specifications:

Electrical: DC Power Supply Limits: 85 to 264 VAC AC Power, Power Consumption 60 VA maximum.

Mechanical: Enclosure NEMA Type 12/13, 4X (Indoor use only), LED Indicators – "Green" COMM, "Red" FAULT.

Display: Active Matrix Thin Film resistor (TFT) with cold cathode fluorescent (CCF) backlight.

Size: 4.54 x 3.4 in.

Pixels: 320 x 234.

Touch Cells: 128 (16 columns x 8 rows).

Touch Cell Size: (20 x 29 pixels).

Terminal Memory: total application flash memory 240K bytes (application screens)

Environment:

Operating Temperature: -0 to 55 degrees C (32 to 131 degrees F).

Storage Temperature: -25 to 70 degrees C (-13 to 158 degrees F).

Relative Humidity: (noncondensing) 5 to 95 percent at 0 to 55 degrees C (32 to 131 degrees F).

Heat Dissipation: 32 Watts.

Agency Approvals and Standards: UL, CSA certified, or another state approved agency.

Manufacturer and Model: Panelview600 or equal.

Software Packages:

PLC Programming: Microsoft Windows based RSLogix500 programming and communication software (RSLinx) with master disk, most recent revisions, and 2-year support.

OIU Programming: Microsoft Windows based Panelbuilder32 development software with master disk, most recent revisions, and 2-year support.

Ethernet Switch:

Function: Mixed Media 10/100 Base T 8 port modular fiber switch with 4 port fiber module and 4 port RJ45 dual speed module.

Specifications: Address Table 24K nodes with address aging.

Cooling Method: Internal 9-CFM fan.

Filtering and Forwarding Rate: 16-port aggregate, 2380K packets per second.

Latency, 100 Mbps: 5 µs + packet time; 10 Mbps: 15 µs + packet time, Packet Buffers 8 MB dynamic.

Processing Type: Store and forward with IEEE 802.3x full-duplex flow control

Standards: IEEE 802.3: 10BASE-T, 10BASE-FL; IEEE 802.3u: 100BASE-TX, 100BASE-FX.

Connectors:

LE1401A, (1) power.

LE1419C: (4) pairs of SC.

LE1425C: (4) RJ-45.

Indicators Chassis: Power; Per port: LK: ON when link is operational; Act: ON with port activity; FDX/HDX: ON for full-duplex mode, OFF for half-duplex mode; 100/10: ON for 100 Mbps, OFF for 10 Mbps

Power Input: 110–240 VAC, 47–63 Hz, internal, autosensing; 20 W.

Size: 1.75 inches high (1U) by 17 inches wide by 9 inches deep (4.4 x 43.2 x 22.9 cm); weight: 2.5 pounds (1.1 kg).

Agency Approvals and Standards: UL, CSA or another state approved agency.

Manufacturer and Product: Black Box LE1401A; or equal.

Programming Notebook (Laptop) Computer:

Function: Notebook computer used to implement, test, and store all PLC and OIU application software programming. Install and configure all PLC and OIU vendor software packages and licenses onto laptop. Complete and save application software to notebook computer and to backup R/W CD(s).

Specifications:

Processor 2650: Intel Pentium 4-M processor at 2.0 GHz, 512 KB cache.

Memory: 128 MB DDR SDRAM standard, upgradable to 512 MB maximum, SDRAM configurations include one of 128, 192, 256, 384 or 512 MB.

I/O Ports:

25-hole pin parallel connector.

15-pin monitor connector.

6-pin PS/2-style keyboard, mouse, and keypad.

2-USB (Universal Serial Bus) compliant 4-pin connectors.

RJ-11 connector for modem.

RJ-45 connector for connection to Ethernet multimedia switch.

Chassis:

14.1-inch XGA Display: Height: 36 mm (1.42-inch); width: 328 mm (12.9-inch); depth: 275 mm (10.8-inch); weight: 7.25 lbs. with CD, floppy and battery.

Display: Displays 15-inch SXGA+ TFT active-matrix display with 1400 x 1050 resolution; height: 38 mm (1.5-inch); width: 332 mm (13.1-inch); depth: 275 mm (10.8-inch).

Power: Lithium Ion battery, AC Adapter: Input voltage: 90 to 135 VAC and 164 to 264 VAC.

Slots: Connectors: (1) Type I or Type II card, 3.3 and 5 V cards supported, Warm-swap Capable.

Graphics: 16MB DDR 4X AGP NVIDIA® GeForce2TM.

Storage: 20 GB4 Ultra ATA hard drive.

Optical Devices:

Fixed Bay integrated in left side with CD-RW: 24x/10x/24x max.

Removable Media: Fixed Floppy drive standard.

Communication Devices: Network Interface Cards, Integrated 10/100 network interface card.

Modems: Standard: Internal 56K5 capable v.92 Fax modem.

Software & Accessories: Microsoft® Windows® 2000 or XP Professional Small Business most recent version. Insure compatibility between platform and vendor software packages prior to installation.

Utilities: Norton AntiVirusTM 2003, introductory version.

Manufacturer and Model: Dell Inspiron 2650 or equal.

Y51 AutoDialer:

General:

Function: The Auto-dialer shall be a solid state component capable of bidirectional communication with the PLC network via cellular phone.

Environmental:

Operating Temperature Range: Minus 6°C to 54°C.

Relative Humidity: 0 to 95%, noncondensing.

Case: Suitable for mounting inside panel enclosure.

Features:

System Security: Access protected by security codes.

Alarm Monitoring: Continuously monitors all IDF alarms.

Remote Data-Table Access: Monitor or alter any data table location on demand via front panel or telephone.

Protocol: Allen-Bradley DFl.

Latency:

01-10 seconds.

Rotary pulse or tone dialing, keyboard selectable.

Dial up to 16 different numbers, each up to 60 digits long.

FCC Registered Part 68, "Ringer Equivalence": 0.3A.

Alarm Acknowledgment shall be Touch-Tone key or by calling back.

Autodialer shall be compatible with most cellular telephone systems.

Provide Serial Cable for use with AB SLC.

Autocall Test.

Nonvolatile Program Memory Retention.

Local Data Logging.

Primary Power:

Voltage: 120 VAC, 50/60 Hz

Standby Supply Current: 300mA.

Operational Supply Current: 460mA.

Battery backup: The product is to contain its own gel cell rechargeable battery, which is automatically kept charged when AC power is present. The system shall operate on battery power for a minimum of 13 continuous hours in the event of AC power failure.

Surge protection: All power, phone line, dry contact, and analog signal inputs shall be protected at the circuit board to IEEE Standard 587, category B (6,000 volts open circuit/3,000 amps closed circuit).

Diagnostics:

The unit shall include user commands to execute diagnostics of the PLC network to determine the health of the network. The unit shall inform the user of the length of scan time for the set of all configured remote channels.

AC Power failure.

Remote (PLC) Channel Monitoring: failure of the active serial communications channels.

Input Monitoring: with configurable aux contacts.

Warranty: The autodialer shall be covered by not less than a two (2) year warranty covering parts and labor performed at the Factory.

Agency Approvals: FCC Part 68.

Manufacturer and Product: RACO Verbatim Gateway Alarm Dialer or equivalent.

Y52 Cellular Modem (CDMA):

General:

Function: The Fixed Wireless Terminal shall provide wireless connectivity for standard telephone equipment.

Environmental:

Operating temperature range: -10°C to +50°C.

Storage temperature range: -40°C to + 60°C.

Relative Humidity: 5% to 95%.

Case: Suitable for mounting inside panel enclosure.

Features:

Compatible with popular supplementary services including caller ID, call waiting, 3-way calling, and call forwarding.

Single jack or dual jack for voice and fax operation.

PSTN Emulation.

Support up to five phones (5 REN).

Emergency battery backup.

Automatic end-of-dialing (no SEND key).

LED Indicators: shall include power/battery status, signal status, message status, and ON/OFF hook status.

Primary Air Interface Standard:

TIA/EIA/IS-2000A.

Primary Transmit Power:

200 mW (23dBm).

Frequency Ranges: Transmit/Receive:

CDMA800 824-849 MHz 869-894 MHz.

PCS (CDMA) 1900 1850-1910 MHz 1930-1990 MHz.

Primary Power:

Voltage: 110 - 230 VAC, 50/60 Hz.

Battery backup: One (1) 6-volt, 4 AH lead acid rechargeable battery.

Connectors:

Two RJ-11 interface jacks for telephone, Group 3 analog fax, or analog data.

TNC antenna connector (50 ohms).

Antennae:

Gain: 8.0 dBi minimum

Frequency Bands: 824 - 1000 and 1700 - 2000 MHz.

VSWR: Transmit sub-bands < 1.5:1, Receive sub-bands < 2.0:1.

Front to Back Ratio: (F/B Ratio) > 16 dB (nom).

Feed power handling: 10 W.

Nominal input impedance: 50 Ohms.

Connector: TNC male on short coax (30 - 50 cm).

Polarization: Linear (vertical when clamped to vertical pole).

Mounting: Pole mount, maximum diameter: 50 mm (2").

Model Number: HGD-0 "High Gain Log Periodic Dipole Array".

Warranty: The Cellular modem shall be covered not less than a fifteen (15) month warranty covering parts and labor performed at the Factory.

Agency Approvals:

FCC Part 15/Part 22/Part 24 Class B Compliance.

Manufacturer and Product: Phonecell® SX4E CDMA or approved equivalent.

Y178A Programmable Data Logger, Integral Solar Powered:

General:

Function: Programmable data logger and system capable of interfacing with process measurements loop powered analog device(s) for the purpose of measuring, collecting, storing and serially transferring process level data with externally connected programming and storage device (i.e. laptop computer).

Type: Microprocessor based.

Parts: Data logger, input/output channel, signal converter, communications, rechargeable battery, solar panel, charge controller, enclosure, and programming and data retrieval software. Vendor shall provide complete system meeting the intent of these specifications and contract drawings.

Service: Interface with Level/Element Transmitter, Submersible, Wastewater (Reference Section 13401 PICS Component L42 for detail information, Druck PTX 1830 and STE110 Level Element/Transmitter Submersible 12 V dc compatible or equal).

Performance:

Environmental:

Operating Temperature: Minus 67 degrees F to plus 185 degrees F (minus 25 to plus 85 degrees C), with gel cell battery, minus 40 degrees F to plus 140 degrees F (minus 40 to plus 60 degrees C).

Relative Humidity: 5 to 95 percent, non-condensing.

Central Processing Unit (CPU):

Processor: Hitachi 6303.

Memory:

Table based memory structure 62,000 data points.
Real time clock.
Diagnostic LEDs.
12-bit A/D converter.
16kB active program.
128kB operating and flash final storage.

Comm Port: 9-pin D type connector for RS232 interface with PC; up to 9600 baud selectable baud rates.

Manufacturer and Model: Campbell Scientific; CR510 Series with extended temperature test.

Input/Output Channel:

Analog Inputs:

Two (2) differential or four (4) single ended configured.

Range: 0 to 2.5 V dc.

Accuracy: +/-0.25 percent.

Resolution: 0.33 uV.

Sample Rates: 2.72 ms adjustable.

Manufacturer and Model: Campbell Scientific; Model CR510.

Terminal Input Channel Converter:

Convert 4-20mA to datalogger input range of 0 to 2.5V dc.

Shunt Resistor: 100 ohms.

Tolerance: +/- 0.01 percent.

Power: 0.25 watt.

Manufacturer and Model: Campbell Scientific; Model CURS100.

DC Power Supply:

Solar Panel, Regulator and Mounts:

Solar panel converting sunlight to DC power.

Voltage Peak: 16.8 volts.

Current at Peak, amps: 1.19 amps.

Peak Power: 20 watts.

Manufacturer and Model: Campbell Scientific; Model MSX20R.

Rechargeable Power Supply with Batteries:

Rechargeable power supply capable of recharging batteries from solar power.

Sealed rechargeable batteries providing 7.0 AHRS nominal output at 68 degrees F (20 degrees C).

Float charged by solar panel.

Manufacturer and Model: Campbell Scientific; Model PS100.

Enclosure: NEMA 4X - fiberglass reinforced polyester (FRP) 16 inches wide by 18 inches high by 9 inches deep (minimum) sized to accommodate datalogger, input/output channels, DC power supply, and sensor terminations.

Enclosure Mounting: Wall or pipe mounting as required.

Power Requirements: 5.4V dc to 16V dc.

Software Language: Pakbus operating system, and Microsoft® Windows® 2000 or XP based Datalogger Support and editing software and licenses complete and latest versions.

Support: Two (2) eight-hour days in support of installation, startup, calibration, and testing of complete system.

Training: One (1) eight-hour day in support of Tank Farm Contractor training on operations, setup, data retrieval, and troubleshooting of complete system.

Warranty: Three (3) years complete system beginning after system installation is complete and tested.

Manufacturer: Campbell Scientific.

Item	Rev	Tag 1 Area	Tag 2 Process	Tag 2 ISA	Tag4 Loop	Description	P&ID	Engineering Units	Voltage/ Current	Address	Typical Wiring Diagram	Notes
1	1	219(Y)	LH	FIT	203	Leachate Collection and Removal System High Flow Pump Flow	H-2-830854 Sheet 1 of 4	GPM	4-20mA	I:01/00	No. 1	
2	1	219(Y)	LH	FIT	202	Leachate Collection and Removal System Low Flow Pump Flow	H-2-830854 Sheet 1 of 4	GPM	4-20mA	I:01/01	No. 1	
3	1	219(Y)	LH	FIT	204	Leak Detection System Pump Flow	H-2-830854 Sheet 1 of 4	GPM	4-20mA	I:01/02	No. 1	
4	1	219(Y)	LH	LT	101	Leachate Collection and Removal System Level	H-2-830854 Sheet 1 of 4	Inches	4-20mA	I:01/03	No. 2	
5	1	219(Y)	LH	LT	104	Leak Detection System Level	H-2-830854 Sheet 1 of 4	Inches	4-20mA	I:01/04	No. 2	
6	1	219(Y)	LH	TIT	219	Crest Pad Building Temperature	H-2-830854 Sheet 1 of 4	Fahrenheit	4-20mA	I:01/05	No. 3	
7	1	219(Y)201	LH	LT	301	Leachate Storage Tank Level	H-2-830854 Sheet 1 of 4	Inches	4-20mA	I:01/06	No. 2/ No. 6	Add Signal Isolator
8	1	219(Y)1	LH	FIT	302	Leachate Transfer Pump Discharge Flow	H-2-830854 Sheet 1 of 4	GPM	4-20mA	I:01/07	No. 1/ No. 6	Add Signal Isolator
9	1	219(Y)1	LH	TIT	220	Leachate Transfer Building Temperature	H-2-830854 Sheet 1 of 4	Fahrenheit	4-20mA	I:02/00	No. 3	
10	1	219(Y)	LH			Spare Input			4-20mA	I:02/01		Wire in spare inputs
11	1	219(Y)	LH			Spare Input			4-20mA	I:02/02		Wire in spare inputs
12	1	219(Y)	LH			Spare Input			4-20mA	I:02/03		Wire in spare inputs

Item	Rev	Tag 1 Area	Tag 2 Process	Tag 2 ISA	Tag4 Loop	Description	P&ID	Engineering Units	Voltage/ Current	Address	Typical Wiring Diagram	Notes
13	1	219(Y)	LH			Spare Input			4-20mA	I:02/04		Wire in spare inputs
14	1	219(Y)	LH			Spare Input			4-20mA	I:02/05		Wire in spare inputs
15	1	219(Y)	LH			Spare Input			4-20mA	I:02/06		Wire in spare inputs
16	1	219(Y)	LH			Spare Input			4-20mA	I:02/07		Wire in spare inputs
17	1	219(Y)	LH	HS	203	Leachate Collection and Removal System High Flow Pump Auto Status	H-2-830854 Sheet 1 of 4	On/Off/Auto	24V dc	I:03/00	No. 4	
18	1	219(Y)	LH	YL	203	Leachate Collection and Removal System High Flow Pump On Status	H-2-830854 Sheet 1 of 4	On/Off	24V dc	I:03/01	No. 4	
19	1	219(Y)	LH	HS	202	Leachate Collection and Removal System Low Flow Pump Auto Status	H-2-830854 Sheet 1 of 4	On/Off/Auto	24V dc	I:03/02	No. 4	
20	1	219(Y)	LH	YL	202	Leachate Collection and Removal System Low Flow Pump On Status	H-2-830854 Sheet 1 of 4	On/Off	24V dc	I:03/03	No. 4	
21	1	219(Y)	LH	HS	204	Leak Detection System Pump Auto Status	H-2-830854 Sheet 1 of 4	On/Off/Auto	24V dc	I:03/04	No. 4	
22	1	219(Y)	LH	YL	204	Leak Detection System Pump On Status	H-2-830854 Sheet 1 of 4	On/Off	24V dc	I:03/05	No. 4	

Item	Rev	Tag 1 Area	Tag 2 Process	Tag 2 ISA	Tag4 Loop	Description	P&ID	Engineering Units	Voltage/ Current	Address	Typical Wiring Diagram	Notes
23	1	219(Y)	LH	LSH	105	Crest Pad Building Sump Level High	H-2-830854 Sheet 1 of 4	High/Normal	24V dc	I:03/06	No. 4	
24	1	219(Y)	LH	LSHH	105	Crest Pad Building Sump Level High High	H-2-830854 Sheet 1 of 4	High-High /Normal	24V dc	I:03/07	No. 4	
25	1	219(Y)	LH	LSL	105	Crest Pad Building Sump Level Low	H-2-830854 Sheet 1 of 4	Low/Normal	24V dc	I:03/08	No. 4	
26	1	219(Y)	LH	LDE	105	Crest Pad Building Sump Leak Detector	H-2-830854 Sheet 1 of 4	Leak/Normal	24V dc	I:03/09	No. 4	
27	1	219(Y)	LH	JSH	219	Crest Pad Building Power Status	H-2-830854 Sheet 1 of 4	Normal/Fail	24V dc	I:03/10	No. 4	
28	1	219(Y)	LH	HS	205	Crest Pad Building Sump Pump Auto Status	H-2-830854 Sheet 1 of 4	On/Off/Auto	24V dc	I:03/11	No. 4	
29	1	219(Y)	LH	YL	205	Crest Pad Building Sump Pump On Status	H-2-830854 Sheet 1 of 4	On/Off	24V dc	I:03/12	No. 4	
30	1	219(Y)	LH	HS	219	Crest Pad Building General Alarm Acknowledge Status	H-2-830854 Sheet 1 of 4	Acknowledge/ Off	24V dc	I:03/13	No. 4	
31	1	219(Y)	LH			Spare	H-2-830854 Sheet 1 of 4		24V dc	I:03/14	No. 4	Wire in spare inputs
32	1	219(Y)	LH			Spare	H-2-830854 Sheet 1 of 4		24V dc	I:03/15	No. 4	Wire in spare inputs
33	1	219(Y)	LH			Spare	H-2-830854 Sheet 1 of 4		24V dc	I:04/00	No. 4	Wire in spare inputs
34	1	219(Y)201	LH	LSHH	301	Leachate Storage Tank Level Switch	H-2-830854 Sheet 2 of 4	High High/Normal	24V dc	I:04/01	No. 4	
35	1	219(Y)201	LH	LSLL	301	Leachate Storage Tank Level Switch	H-2-830854 Sheet 2 of 4	Low Low /Normal	24V dc	I:04/02	No. 4	

Item	Rev	Tag 1 Area	Tag 2 Process	Tag 2 ISA	Tag4 Loop	Description	P&ID	Engineering Units	Voltage/ Current	Address	Typical Wiring Diagram	Notes
36	1	219(Y)	LH			Spare			24V dc	I:04/12	No. 4	Wire in spare inputs
37	1	219(Y)1	LH	YL	302-2	Leachate Transfer Pump On Status	H-2-830854 Sheet 2 of 4	On/Off	24V dc	I:04/04	No. 4	
38	1	219(Y)	LH	LSHH	107	Combined Sump Level Switch	H-2-830854 Sheet 2 of 4	High High/Normal	24V dc	I:04/05	No. 4	
39	1	219(Y)	LH	LSH	107	Combined Sump Level Switch	H-2-830854 Sheet 2 of 4	High /Normal	24V dc	I:04/06	No. 4	
40	1	219(Y)	LH	LSL	107	Combined Sump Level Switch	H-2-830854 Sheet 2 of 4	Low/Normal	24V dc	I:04/07	No. 4	
41	1	219(Y)	LH	LSLL	107	Combined Sump Level Switch	H-2-830854 Sheet 2 of 4	Low Low/Normal	24V dc	I:04/08	No. 4	
42	1	219(Y)	LH	LDE	107	Combined Sump Leak Detector	H-2-830854 Sheet 2 of 4	Leak/Normal	24V dc	I:04/09	No. 4	
43	1	219(Y)	LH	HS	207	Combined Sump Pump Auto Status	H-2-830854 Sheet 2 of 4	On/Off/Auto	24V dc	I:04/10	No. 4	
44	1	219(Y)	LH	YL	207	Combined Sump Pump On Status	H-2-830854 Sheet 2 of 4	On/Off	24V dc	I:04/11	No. 4	
45	1	219(Y)	LH			Spare			24V dc	I:04/12	No. 4	Wire in spare inputs
46	1	219(Y)	LH			Spare			24V dc	I:04/13	No. 4	Wire in spare inputs
47	1	219(Y)	LH			Spare			24V dc	I:04/14	No. 4	Wire in spare inputs
48	1	219(Y)	LH			Spare			24V dc	I:04/15	No. 4	Wire in spare inputs

Item	Rev	Tag 1 Area	Tag 2 Process	Tag 2 ISA	Tag4 Loop	Description	P&ID	Engineering Units	Voltage/ Current	Address	Typical Wiring Diagram	Notes
49	1	219(Y)	LH	YS	203	Leachate Collection and Removal System High Flow Pump Start Command	H-2-830854 Sheet 1 of 4	Start/Stop	120 Vac	O:05/00	No. 4	Note: Interlock Control
50	1	219(Y)	LH	YS	202	Leachate Collection and Removal System Low Flow Pump Start Command	H-2-830854 Sheet 1 of 4	Start/Stop	120 Vac	O:05/01	No. 4	Note: Interlock Control
51	1	219(Y)	LH	YS	204	Leak Detection System Pump Start Command	H-2-830854 Sheet 1 of 4	Start/Stop	120 Vac	O:05/02	No. 4	Note: Interlock Control
52	1	219(Y)	LH	YS	219	Crest Pad Building General Alarm	H-2-830854 Sheet 1 of 4	Normal/Fail	120 Vac	O:05/03	No. 5	
53	1	219(Y)	LH	YS	205	Crest Pad Building Sump Pump Start Command	H-2-830854 Sheet 1 of 4	Start/Stop	120 Vac	O:05/04	No. 4	Note: Interlock Control
54	1	219(Y)	LH	YS	207	Combined Sump Pump Start Command	H-2-830854 Sheet 2 of 4	Start/Stop	120 Vac	O:05/05	No. 4	Note: Interlock Control
55	1	219(Y)1	LH	YS	220	Leachate Transfer Building General Alarm	H-2-830854 Sheet 2 of 4	Normal/Fail	120 Vac	O:05/06	No. 4	Note: Interlock Control
56	1	219(Y)	LH			Spare Output			24V dc	O:05/07	No. 5	
57	1	219(Y)	LH			Spare Output			24V dc	O:06/00	No. 4	Wire in spare outputs
58	1	219(Y)	LH			Spare Output			24V dc	O:06/01	No. 4	Wire in spare outputs
59	1	219(Y)	LH			Spare Output			24V dc	O:06/02	No. 4	Wire in spare outputs

Item	Rev	Tag 1 Area	Tag 2 Process	Tag 2 ISA	Tag4 Loop	Description	P&ID	Engineering Units	Voltage/ Current	Address	Typical Wiring Diagram	Notes
60	1	219(Y)	LH			Spare Output			24V dc	O:06/03	No. 4	Wire in spare outputs
61	1	219(Y)	LH			Spare Output			24V dc	O:06/04	No. 4	Wire in spare outputs
62	1	219(Y)	LH			Spare Output			24V dc	O:06/05	No. 4	Wire in spare outputs
63	1	219(Y)	LH			Spare Output			24V dc	O:06/06	No. 4	Wire in spare outputs
64	1	219(Y)	LH			Spare Output			24V dc	O:06/07	No. 4	Wire in spare outputs
65	1	219(Y)	LH	HS	001	Handswitch in Interlock Relay Cabinet						
66	1	219(Y)	LH	YL	001	Indicator Light in Interlock Relay Cabinet						

(Y) = A for Cell No. 1. For Cell No. 1 reference corresponding H-2-830854 sheet 1 of 4 and sheet 2 of 4.

(Y) = E for Cell No. 2. For Cell No. 2 reference corresponding H-2-830854 sheet 3 of 4 and sheet 4 of 4.

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Loop Descriptions

Supplement 4 PLC and OIU Application Software

Document Objectives

This Supplement No. 4 to Specification Section 13401, PROCESS INSTRUMENTATION AND CONTROL SYSTEMS (PICS) provides the basis for estimating the minimum level of effort required to implement PLC and OIU application software for Cell No. 1 and Cell No. 2 Integrated Disposal Facilities (IDF).

Unless otherwise directed by the Tank Farm Contractor during Application Software Submittal and Design Workshops, this document provides guidelines for the programming and testing of all PLC and OIU application software. *Note: Issues related to Application Software Submittal and Design Workshops are discussed in Section 13401, PROCESS INSTRUMENTATION AND CONTROL SYSTEMS (PICS).*

This Supplement No. 4 shall be used in conjunction with Contract Documents (drawings and specifications, and supplements) and relevant PICS supplier technical information and manuals while developing application software.

PICS Supplements shall be provided to Construction General Contractor in electronic format upon request.

Reference Management Directive TFC–MD–034, Rev. A, attached to this supplement for directives on HMI (OIU) configurations.

PLC And OIU Program Layout

1. Programmable Logic Controller (PLC) Application Software: Provide PLC application software as necessary for a fully functional and operable PLC system in accordance with the project design. PLC programming includes, but shall not be limited to the following:
   a. PLC application software development and installation
   b. PLC equipment configuration
   c. PLC interface with Operator Interface Unit (OIU)
   d. System testing
   e. PLC application software debugging and trouble-shooting
   f. System hardware and software documentation
   g. PLC system startup
   h. PLC system training.
2. ETHERNET Address: For more information on SLC500 Ethernet Addressing see Allen-Bradley Quick Start Ethernet SLC500 Processor > Publication 1747-10.4.
   a. Host Name: (To be assigned by Tank Farm Contractor)
   b. IP Address: (To be assigned by Tank Farm Contractor)
   c. Subnet Masking: 255.255.255.0
   d. Gateway address: None
3. PLC Memory Allocation: Provide PLC programming and documentation with logical, structured groupings of PLC memory registers. Allocate memory within each group to allow for 50 percent spare capacity. Examples of functional groupings shall be:
   a. Inputs > I:01/00
   b. Outputs > O:01/00
   c. Control Operation Bits > B3:0 through B3: 200

d. Timers > T4:0 through T4:100
e. Counter > C5:0 through C5:100
f. Sequencers > R6:0 through R6:100
g. Ethernet Diagnostic Files > N7:00 through N7:50
h. Scaled Analog Inputs > F8:00 through F8:49
i. Process Calculations, Runtime, and Flow Totalization > F8:50 through F8:100
j. Process Set-points, Comparisons, and Scaling Ranges for Analog Inputs > F9:0 through F9:100
k. Reserve for future SCADA Discrete Write Commands to Processor > N10:0 through N10:100
l. Reserve for future SCADA Analog Write Commands from Processor > F11:0 through F11:100
m. Reserve for future SCADA Discrete Read Status from Processor > N12:0 through N12:100
n. Reserve for future SCADA Analog Read Status from Processor > F13:0 through F13:100
o. Reserve for future PID Control Files > MG14:0 or N14:0
p. Reserve for future Messaging > N15
q. Other Diagnostic Files > N16

4. PLC Programming Sub-routines: Divide PLC program into logical sub-routines. Each sub-routine shall contain the programming for a process or set of equipment. Examples of logical subroutines include:

a. Ethernet Communication
b. Process flow, level, and temperature signal scaling, totalizing and alarming
c. Landfill leachate collection and removal system pump control
d. Landfill leak detection system pump control
e. Landfill crest pad building sump pump monitoring
f. Landfill crest pad building miscellaneous control
g. Leachate storage tank level measurement and leak detection
h. Leachate collection and transmission line leak detection
i. Leachate transfer pump control and flow totalization
j. Combined sump pump control and leak detection
k. Leachate transfer building miscellaneous control

5. PLC Program Documentation: Each PLC program shall be fully annotated with descriptive notations that define the functions associated with each program element. Provide the following minimum level of program documentation.

a. Each program element shall have a descriptive name associated.
b. Each program sub-routine and article associated within a major program shall be proceeded by a description of the sub-routine and article function.
c. Each major program shall be proceeded by a description of the program and a listing of the associated program sub-routines and articles.

6. The program documentation shall be developed using the PLC suppliers standard program documentation software. The PLC program documentation shall also include a complete cross reference listing of all program elements and their location within each program.

7. Program OIU to enable secured access to process set-points and control modes of operation. OIU shall be configured with process status display screens, set-point entry, equipment control mode, equipment runtime, flow totaling, alarm status, and alarm enable/disable screens.

Process set-point and equipment control mode shall be implemented using RETURN ENTRY at the control panel mounted OIU.

8. Program OIU with three (3) security levels: engineer, operator, and system administrator. Engineer shall have access to all displays. Operator shall have access to all process status and alarm display screens but not control mode, alarm enable/disable, and set-point screens. Administrator shall have access to process status, and alarm enable/disable screens.

9. Configure and link OIU Main Menu, Sub-Menu and Data screens in a logical fashion allowing Operator(s) to logically move through each process control loop, with the ability to access data entry set-point and control modes.

10. Program and configure alarm screens for each alarm as described in Section 3.2 PLC Loop Descriptions of this Supplement. OIU alarm screens shall enable operator(s) to visually identify and acknowledge all alarm conditions for each process control loop. Alarms shall be acknowledged at local alarm display ONLY.

11. Program and configure Autodialer for each alarm as described in Section 3.2 PLC Loop Descriptions of this Supplement. Autodialer alarm options shall enable operator(s) to identify and acknowledge all alarm conditions for each process control loop. Alarms shall be acknowledged at Autodialer ONLY.

PLC & OIU Programming

PLC Standard Functions

1. General — PICS shall implement functions for the purpose of program standardization and minimum PLC application software capability. Functions may not be applicable to every program control loop or network.

2. PLC Standard Function No. 1 — Equipment Run Timers: For each equipment ON status input, track the runtime of the associated equipment item. Accumulate equipment runtime in tenths of hours using retentive timers with no timer reset due to equipment stoppage or power loss. Provide a command interface to allow reset or adjustment of equipment run time.

3. PLC Standard Function No. 2A — Equipment Failure Detection: Provide for each equipment item controlled by the PLC for which the PLC receives ON and AUTO or REMOTE status signals. Monitor for a fail condition when the unit is in the AUTO or REMOTE mode. A failure condition shall be activated when the unit is commanded by the PLC to run, and the PLC does not receive the unit's ON status signal after a 10-second delay period. When a failure condition is detected, inhibit the PLC run command to the unit. Reset the failure condition when the unit is taken out of AUTO or REMOTE mode.

4. PLC Standard Function No. 2B — Equipment Flow Failure Detection: Provide for each pump controlled by the PLC for which the PLC receives a PLC receives ON and AUTO or REMOTE, status and a PUMP DISCHARGE flow signal. Monitor for a fail condition when the unit is in the AUTO or REMOTE mode. A failure condition shall be activated when the unit is commanded by the PLC to run, and the PLC does not receive or maintain a positive flow value for a 10-second delay period. When a failure condition is detected, inhibit the PLC run command to the unit. Reset the failure condition when the unit is taken out of AUTO or REMOTE mode.

5. PLC Standard Function No. 3 — Analog Signal Scaling: Provide program logic to scale process analog signals to engineering units that correspond to the calibrated range of the field sensor. Store the scaled value in a PLC memory register for use in process control programming and for transmission to the OIU. Program PLC to monitor each analog input and ALARM in the event a process signal is out of the 4-20mA range. A process signal out-of-range ALARM condition shall prevent associated process control from taking place.

Program PLC and OIU such that in the event a process signal is out-of-range, data point on OIU shall change to color YELLOW enabling personnel to know the data is incorrect.

6. PLC Standard Function No. 4 — Analog alarms: Provide the capability to detect and annunciate alarms, such as high-high, high, low, and low-low for analog process signals. The PLC shall continuously compare the scaled process signal to a set-point value stored in PLC register memory. If the scaled process signal exceeds the alarm set-point, the PLC shall initiate an alarm signal to the OIU system and to external annunciation equipment as noted in the Process Control Loop Descriptions. The analog alarm function shall also have the following capabilities:

   a. Provide time delays on all alarm set-points. When the PLC detects the scaled process signal exceeding set-point, initiate the alarm delay timer. At the end of the time delay, activate the alarm.

   b. Unless specified in the Loop Functional Descriptions, all analog alarms shall be self-resetting. The PLC shall reset the alarm condition when the scaled process signal value is within the set-point value plus a 2 percent dead band.

   c. Where noted in the Loop Functional Descriptions, provide alarm ENABLE/DISABLE program logic that respond to commands from the OIU. Suppress the activation of alarms when commanded to the DISABLE mode.

   d. Provide program logic to receive, and store in PLC register memory, alarm set-point commands from the OIU.

7. PLC Standard Function No. 5 — Discrete alarms: Provide adjustable alarm delay timers on all discrete alarm input signals monitored by the PLC. Upon detection of the alarm condition, the PLC shall initiate the time delay. At the end of the time period, the PLC shall communicate the alarm condition to the OIU system and to external annunciation equipment as noted in the Process Control Loop Descriptions. The discrete alarm function shall also have the following capabilities:

   a. Unless specified in the Loop Functional Descriptions, all discrete alarms shall be self-resetting. The PLC shall reset the alarm condition when the alarm input signal resets.

   b. Where noted in the Loop Functional Descriptions, provide alarm ENABLE/DISABLE program logic that respond to commands from the OIU. Suppress the activation of alarms when commanded to the DISABLE mode.

8. PLC Standard Function No. 7 — Process Flow Totalizers: For all process flow signals provide program logic in the PLC to accumulate flow totals and flow total RESET. The totaled value stored in the PLC shall be transmitted to the OIU for display and historical logging. PLC shall increment flow total when pump ON status and minimum flow status is confirmed.

9. PLC Standard Function No. 8 — Pump Restart Time Delay: Provide adjustable delay timers in PLC program so as to prevent rapid restart of all pumps. An adjustable timer for each pump's AUTO start control logic, shall begin to countdown once a pump has stopped operating. PLC shall not execute AUTO start control logic for each pump until respective adjustable delay timer's accumulated value is equal to preset value of 30 seconds, except for Pumps P-205 and P-207, which will have a 10-second delay period.

10. PLC Standard Function No. 9 — Power Failure: Program PLC to prevent PLC from executing process equipment automatic control during a power failure condition.

PLC Loop Descriptions

Note: Loop Descriptions are typical for both Cell No. 1 and Cell No. 2 process equipment.

1. Landfill Leachate Collection and Removal System (Loops 101, 202 and 203). Reference P&ID Drawing H-2-830854 sheet 1 of 4.

a. General — Landfill leachate collection and removal system consists of one collection sump, high and low flow collection pumps, one common level sensor, two flow transmitters, pump discharge piping, and ancillary valves. Pumps shall extract leachate from landfill collection and removal system sump and pump leachate to an above grade leachate storage tank via an underground double walled containment pipe. PLC shall monitor discharge flow from each pump via flow transmitters installed in each pump's discharge line. PLC shall automatically compute total extracted leachate from sump. All process variables and equipment status shall be displayed locally via panel mounted OIU and digital indicators.

b. Program — Program PLC to implement continuous monitoring of liquid levels inside landfill leachate collection and removal sump and provide automatic operation of high and low flow pumps. Program PLC to automatically operate pumps ON/OFF based upon liquid levels inside the landfill leachate collection and removal system sump. PLC shall automatically operate pumps ON/OFF in order to insure that liquid levels over landfill liner - as measured from the bottom of the sump - do not exceed 12-inches. The PLC shall be programmed to automatically turn off the low flow pump when the high flow pump is called to run.

c. AUTO Operation — Automatic operation requires operator(s) place each pump's respective ON-OFF-AUTO control switch (located at MCC) to AUTO mode. Pump operation and alarm set-points shall be entered at the control panel mounted OIU. The pumps shall operate between separate start and stop level set-points.

d. Process Control Set-points — Operator(s) shall enter the following control set-points at the control panel mounted OIU:

(1) High flow pump start level in inches

(2) Low flow pump start level in inches

(3) High flow pump stop level in inches

(4) Low flow pump stop level in inches

(5) Landfill leachate collection and removal system high-high level alarm in inches

(6) Landfill leachate collection and removal system high level alarm in inches

e. REMOTE Operation — None.

f. ON (manual) Operation — ON operation requires the following steps be taken by operator(s):

(1) Operator(s) shall place high and low flow pump's respective ON-OFF-AUTO control switch (located at MCC) to ON mode.

g. Interlocks — Interlocks shall prevent operation of leachate collection and removal pumps in the event of a leachate storage tank high-high level alarm condition, or a leak alarm condition in either the landfill crest pad building sump, leachate storage tank or combined sump.

h. Alarms — Program PLC to monitor process and alarm for the following minimum conditions:

(1) Landfill leachate collection and removal system high-high level

(2) Landfill leachate collection and removal system high level

(3) Landfill leachate collection and removal system high flow pump fail

(4) Landfill leachate collection and removal system low flow pump fail

(5) Landfill leachate collection and removal system high flow pump flow fail

(6) Landfill leachate collection and removal system low flow pump flow fail

(7) Landfill leachate collection and removal system level signal fail

(8) Landfill leachate collection and removal system high flow pump flow signal fail

(9) Landfill leachate collection and removal system low flow pump flow signal fail

i. Runtime and Flow Totaling — Program PLC to implement runtime and flow totals for each leachate collection and removal system pump.

PLC increments equipment runtime and process flows when respective pump ON status and minimum flow status is confirmed. PLC control logic shall be programmed to allow runtime and flow value reset and adjustment.

2. Landfill Leak Detection System (Loops 104 and 204). Reference P&ID Drawing H-2-830854 sheet 1 of 4.

a. General — Landfill leak detection system consists of one leak detection sump, collection pump, one level sensor, flow transmitter, pump discharge piping , and ancillary valves. Leak detection system pump shall extract leachate from landfill leak detection sump and pump this leachate to an above grade leachate storage tank via an underground double walled containment pipe. PLC shall monitor discharge flow from leak detection pump via flow transmitter installed in pump's discharge line. PLC shall automatically compute total extracted leachate from sump. All process variables and equipment status shall be displayed locally via panel mounted OIU and digital indicators.

b. Program — Program PLC to implement continuous monitoring of liquid levels inside landfill leak detection system chamber and provide automatic operation of leak detection pump. Program PLC to automatically operate leak detection pump ON/OFF based upon liquid levels inside leak detection sump.

c. AUTO Operation — Automatic operation requires operator(s) place leak detection pump's ON-OFF-AUTO control switch (located at MCC) to AUTO mode and enter sump pump operation control set-points at the control panel mounted OIU. The landfill leak detection system pump shall operate between separate start and stop level set-points.

d. Process Control Set-points – Operator(s) shall enter the following control set-points at the control panel mounted OIU:

(1) Leak detection system pump start level in inches

(2) Leak detection system pump stop level in inches

(3) Landfill leak detection system high-high level alarm in inches

(4) Landfill leak detection system high level alarm in inches

e. REMOTE Operation — None.

f. ON (manual) Operation — ON (manual) Operation requires the following steps be taken by operator(s):

(1) Operator(s) shall place landfill leak detection pump's ON-OFF-AUTO control switch (located at MCC) to ON mode.

g. Interlocks — Interlocks shall prevent operation of leachate leak detection pump in the event of a leachate storage tank high-high level alarm condition, or a leak alarm condition in either the landfill crest pad building sump, leachate storage tank, or combined sump.

h. Alarms — Program PLC to monitor process and alarm for the following minimum conditions:

(1) Landfill leak detection system high-high level

(2) Landfill leak detection system high level

(3) Landfill leak detection system pump fail

(4) Landfill leak detection system pump flow fail

(5) Landfill leak detection system level signal fail

(6) Landfill leak detection system flow signal fail

i. Runtime and Flow Totaling — Program PLC to implement runtime and flow totals for landfill leak detection pump. PLC increments equipment runtime and process flows when respective pump ON status and minimum flow is confirmed. PLC control logic shall be programmed to allow runtime and flow value reset and adjustment.

3. Landfill Crest Pad Building Sump System (Loops 105, 205). Reference P&ID Drawing H-2-830854 sheet 1 of 4.

a. General — Landfill crest pad building sump system consists of a building sump, sump pump, level floats, leak detection sensor, and pump discharge piping. Sump pump extracts and pumps liquids from building sump to an above ground leachate storage tank. Landfill crest pad building sump pump operates ON/OFF based upon actuation of level floats inside the landfill crest pad building sump. PLC shall monitor leak detection sensor. All process variables and equipment status shall be displayed locally via panel mounted OIU and digital indicators.

b. Program — Program PLC to monitor sump float switches and provide automatic operation of landfill crest pad building sump pump between high and low level status. In the event of high-high level condition, PLC shall alarm. In the event of a leak alarm condition PLC shall prevent operation of landfill leachate collection and leak detection pumps.

c. AUTO Operation — Automatic operation requires operator(s) place sump pump's ON-OFF-AUTO control switch (located at MCC) to AUTO mode. The sump pump shall operate between start and stop levels. Operator(s) shall place sump leak detection ENABLE/DISABLE control switch (located at local sump control panel) to ENABLE leak detection control logic.

d. Process Control Set-points — None

e. REMOTE Operation — None

f. ON (manual) Operation — ON (manual) Operation requires the following steps be taken by operator(s):

(1) Operator(s) shall place sump pump's ON-OFF-AUTO control switch (located at MCC) to ON mode.

g. Interlocks — Interlocks shall prevent operation of landfill crest pad building sump pump in the event of a leachate storage tank high-high level alarm condition, or a leak alarm condition in either the leachate storage tank, or combined sump.

h. Alarms — Program PLC to monitor process and alarm for the following minimum conditions:

(1) Landfill crest pad building sump level high

(2) Landfill crest pad building sump level high-high

(3) Landfill crest pad building sump leak detected

(4) Landfill crest pad building sump pump fail

i. Runtime — PLC shall provide runtime for sump pump. PLC increments equipment runtime when respective pump ON status is confirmed. PLC control logic shall be programmed to allow runtime value reset and adjustment.

4. Leachate Storage Tank System (Loops 300, 301). Reference P&ID Drawing H-2-830854 sheet 2 of 4.

a. General — Leachate storage tank system consists of an above grade tank with an outer steel wall, an interstitial leak detection chamber, floating cover, level sensors, leak detection sensors, and a transfer pump suction connection. Liquid is pumped to leachate storage tank from landfill leachate collection and leak detection pumps, landfill crest pad building sump pump, and combined sump. A level sensor located inside a stilling well - mounted inside the storage tank - provides continuous liquid levels measurements to PLC. This level sensor is backed-up by additional discrete level switches. All process variables and equipment status shall be displayed locally via panel mounted OIU and digital indicators.

b. Program — Program PLC to implement continuous monitoring of liquid levels inside leachate storage tank. Level measurements shall be displayed at control panel inside landfill crest pad building and local control panel inside leachate transfer building.

Program PLC to monitor level high-high and low-low switches which provide backup safety measurement and control for the continuous level sensor.

c. AUTO Operation — None.

d. Process Control Set-points:

(1) Leachate storage tank high-high level in inches

(2) Leachate storage tank high level in inches

(3) Leachate storage tank low level in inches

(4) Leachate storage tank low-low level in inches

e. REMOTE Operation — None.

f. ON (manual) Operation — None.

g. Interlocks — Interlocks shall prevent operation of landfill leachate collection and removal and leak detection pumps, crest pad sump pump, and combined sump pump in the event of a leachate storage tank high-high level condition alarm condition. Interlocks shall prevent operation of transfer pump in the event of a storage tank low-low level condition.

h. Alarms — Program PLC to monitor process and alarm for the following minimum conditions:

(1) Leachate storage tank high-high level

(2) Leachate storage tank high level

i. Runtime – None.

5. Combined Sump Pump System (Loops 107, 207). Reference P&ID Drawing H-2-830854 sheet 2 of 4.

a. General — Combined sump pump system consists of a sump assembly with dual chambers (pump chamber and leak detection chamber), sump pump, level floats, and a leak detection sensor. Liquids from truck loading station, gravity drain to the sump pump chamber. Liquids from containment piping, gravity drain to leak detection chamber. All process variables and equipment status shall be displayed locally via panel mounted OIU and digital indicators.

b. Program — Program PLC to implement automatic operation of combined sump pump. Combined sump pump shall operate ON/OFF based upon actuation of discrete floats mounted inside sump: sump high-high alarm level, sump pump high (start) level, sump pump low (stop) level and sump pump low-low (alarm) level. Program PLC to alarm when high-high level float switch is activated or low-low level float is actuated and sump pump ON status exists. Program PLC to monitor operation of leak detection sensor. Sump floats and leak detection sensor shall be hardwired to intrinsic safety relay modules.

c. AUTO Operation — Automatic operation requires operator(s) place combined sump pump ON-OFF-AUTO control switch (located inside transfer building) to AUTO mode. Sump pump shall operate between sump pump start and stop levels.

d. Process Control Set-points — None.

e. REMOTE Operation — None.

f. ON (manual) Operation — ON (manual) Operation requires the following steps be taken by operator(s):

(1) Operator(s) shall place sump pump ON-OFF-AUTO control switch (located inside transfer building) to ON mode.

(2) Pump selected to ON mode shall continue to operate until sump pump low-low level is reached.

g. Interlocks — Interlocks shall prevent operation of combined sump pump in the event of a leachate storage tank high-high level alarm condition.

Interlocks shall prevent operation of landfill leachate collection and leak detection pumps, and landfill crest pad building sump pump in the event of a combined sump leak alarm condition.

h. Alarms — Program PLC to monitor process and alarm for the following minimum conditions:

(1) Combined sump high-high level

(2) Combined sump low-low level

(3) Combined sump pump fail

(4) Combined sump leak detected

i. Runtime — Program PLC to provide runtime. PLC shall increment equipment runtime when pump ON status is confirmed. Implement PLC control logic necessary to allow local runtime value reset and adjustment.

6. Leachate Transfer Pump System (Loop 302). Reference P&ID Drawing H-2-830854 sheet 2 of 4.

a. General — Leachate transfer pump system consists of a transfer pump with its suction line connected to leachate storage tank, discharge piping, flow meter, and a truck loading quick disconnect. System allows operator(s) to pump liquids from leachate storage tank to storage trucks. All process variables and equipment status shall be displayed locally via panel mounted OIU and digital indicators.

b. Program — Program PLC to monitor totalized flow from leachate storage tank to truck loading station for the purpose of determining amount of liquids being trucked off-site. Transfer pump operation is manual, with interlocks preventing transfer pump operation during a storage tank low-low level condition.

c. AUTO Operation — None

d. Process Control Set-points — None

e. REMOTE Operation — None

j. ON (manual) Operation — ON (manual) Operation requires the following steps be taken by operator(s):

(1) Operator(s) shall place transfer pump ON-OFF control switch (located inside transfer building) to ON mode. Operator(s) shall monitor totalized flow as a means of determining duration of transfer operation.

f. Interlocks — Interlocks shall prevent transfer pump operation during a storage tank low-low level condition.

g. Alarms— Program PLC to monitor process and alarm for the following minimum conditions:

(1) Truck Loading Station flow signal fail.

(2) Transfer Pump low flow fail.

h. Flow Totaling — PLC shall provide flow totals for truck loading station. PLC control logic shall allow for flow reset and value adjustment.

7. Control Panel Power Monitoring System (Loop 219). Reference P&ID Drawing H-2-830854 sheet 1 of 4.

a. General — Control panel power monitoring system consists of a control relay mounted inside landfill crest pad building control panel. Power to landfill crest pad building control panel is monitored by this control relay. In the event of a power failure, control relay drops out removing input status to PLC. The PLC (which is powered by a UPS) senses the loss of signal, and prevents automatic operation of process equipment. All process variables and equipment status shall be displayed locally via panel mounted OIU and digital indicators.

b. Program — Program PLC to monitor power to control panel. In the event of a power failure, PLC program shall remove run command from all process equipment so as to prevent nuisance alarming and equipment failure. Program PLC to stagger start equipment after control power returns to normal.

c. AUTO Operation — None.

d. Process Control Set-points — None.

e. REMOTE Operation — None.

f. ON (manual) Operation — None.

g. Interlocks — Interlocks prevent operation of equipment in the event of loss of power to control panel.

h. Alarms — Program PLC to monitor and alarm control panel power for following minimum conditions:

(1) Control Panel Power fail.

i. Runtime — None.

8. Crest Pad and Leachate Transfer Building Temperature Monitoring System (Loops 219, 220). P&ID drawings H-2-830854 sheets 1 & 2 of 4.

a. General — Crest pad and leachate transfer building temperature monitoring system consists of an ambient temperature transmitter which monitors the effective operation of building heating and cooling equipment. All process variables and equipment status shall be displayed locally via panel mounted OIU and digital indicators.

b. Program — Program PLC to monitor crest pad and leachate transfer building temperature levels and alarm in the event of temperature alarm condition signifying heating and cooling equipment failure.

c. Automatic Operation — None.

d. Process Control Set-points:

(1) Landfill crest pad building temperature level high alarm in degrees F

(2) Landfill crest pad building temperature level low alarm in degrees F

(3) Leachate transfer building temperature level high alarm in degrees F

(4) Leachate transfer building temperature level low alarm in degrees F

e. REMOTE Operation — None.

f. ON (manual) Operation — None.

g. Interlocks — None.

h. Alarms — Program PLC to monitor process and alarm for the following minimum conditions:

(1) Landfill crest pad building temperature level high alarm

(2) Landfill crest pad building temperature level low alarm

(3) Landfill crest pad building temperature signal fail alarm

(4) Leachate transfer building temperature level high alarm

(5) Leachate transfer building temperature level low alarm

(6) Leachate transfer building temperature signal fail alarm

i. Runtime — None.

9. Building General Alarm System (Loops 219, 220). Reference P&ID Drawing H-2-830854 sheet 1, 2 of 4.

a. General — Building general alarm system consists of alarm lights installed on the exterior wall of the landfill crest pad building and leachate transfer building which are operated by the PLC. Alarm light provides a visual method of notifying operator(s) of a general alarm condition.

b. Program — Program PLC to monitor all process loops and initiate general alarm in the event of an alarm condition. General alarm shall continue until process alarm condition is acknowledged by operator(s). Acknowledge shall be initiated at pushbutton or OIU on landfill crest pad building control panel. Program PLC to receive acknowledgement during an alarm condition, to turn off general alarm light, and be able to initiate a new general alarm in the event of a new and subsequent alarm condition.

c. AUTO Operation — None.

d. Process Control set-points – None.

e. REMOTE Operation — None.

f. ON (manual) Operation — None.

g. Interlocks — None.

h. Alarms — (See previous loops).

i. Runtime — None.

OIU Displays and Layout

1. General — OIU shall be configured with process displays, data entry, runtime, and alarm screens. Process set-points and control modes shall be one-shot commands from the OIU.
2. Process display screens shall have a P&ID style of presentation. Displays shall include text for equipment and process identification. Data entry points and screens shall have a TABLE style of presentation.
3. Program OIU to enable secured access to process set-points and control modes of operation. OIU shall be configured with process status display screens, set-point entry, equipment control mode, equipment runtime, flow totaling, alarm status, and alarm enable/disable screens. Process set-point and equipment control mode shall be implemented using RETURN ENTRY at the control panel mounted OIU.
4. Program OIU with three (3) security levels: engineer, operator, and system administrator. Engineer shall have access to all displays. Operator shall have access to all process status and alarm display screens but not control mode, alarm enable/disable, and set-point screens. Administrator shall have access to process status, and alarm enable/disable screens.
5. Configure and link OIU Main Menu, Sub-Menu and Data screens in a logical fashion allowing operator(s) to logically move through each process control loop, with the ability to access data entry set-point and control modes as described in this document.
6. Program and configure alarm screens for each alarm as described in this document. OIU alarm screens shall enable operator(s) to visually identify and acknowledge all alarm conditions for each process control loop. Alarms shall be acknowledged at local alarm display ONLY. Acknowledge shall not be broadcast across network from OIUs.
7. Alarm Identifications:
   a. Process in Alarm and Unacknowledged: Flashing
   b. Process in Alarm and Acknowledged: Steady
8. Configure and link displays in a logical fashion allowing operator(s) to logically access all processes and select the following:
   a. IDF Process System Overviews (Displaying process levels, flows, and temperature for each system)
   b. Disposal Facility Equipment Runtime and Runtime Reset Display

c. Disposal Facility Process Control Set-point Display(s)
d. Disposal Facility Process Alarm Set-point Display(s)
e. Disposal Facility Process Flow Totalization and Flow Totalization Reset Display
f. Integrated Disposal Facility Alarm Display

Reference Management Directive TFC–034, Rev. A

USQ #04-0016-AA

CH2M HILL Hanford Group, Inc.	**Manual**	**Management Directive**
	Revision	**TFC-MD-034, REV A**
GRAPHIC PROCESS DISPLAY CRITERIA FOR HUMAN-MACHINE INTERFACES	**Page**	**260 of 3**
	Issue Date	**January 9, 2004**
	Effective Date	**January 12, 2004**
	Expiration Date	**March 1, 2004**
FUNCTIONAL AREA MANAGER:		**D. C. Lowe**
DOCUMENT OWNER:		**R. E. Larson**

1.0 PURPOSE AND SCOPE

The purpose of this management directive (MD) is to establish the requirements for standardized criteria for new graphic process displays (human-machine interfaces) that are being designed or acquired for process monitoring and control systems of CH2M HILL Hanford Group, Inc. (CH2M HILL) facilities and managed projects.

The requirements of this MD are applicable to new monitoring and control system designs (Human-Machine Interfaces) for CH2M facilities and to projects that have not completed an Acceptance Test Procedure (ATP). Application of the MD requirements to existing human-machine interfaces systems and human-machine interfaces systems with a completed ATP will be determined, planned, and performed in accordance with the resolution of PER-2003-4039.

2.0 IMPLEMENTATION

This management directive is effective on the date shown in the header and will remain in effect until a new Graphic Process Display Criteria for Human-Machine Interface Standard is issued.

3.0 DIRECTIVE

The following requirements that are based on the national standard ISA-S5.5-1985, "Graphic Symbols for Process Displays," are the Tank Farm Contractors (TFC) design requirement for new process monitoring and control systems (Human-Machine Interfaces).

3.1 General

The following general symbol usage standards apply:

1. The graphic process display will follow requirements specified in ISA-S5.5 standard.
2. The use of outline and solid (filled) symbol forms to indicate status is as follows:
   - An outline symbol form indicates an off, stopped, or nonactive state.
   - A solid (filled) symbol form indicates an on, running, or active state.
   - Status designation by use of solid or outline forms are particularly applicable to the rotating equipment and valves and actuators. In depicting valve position, use solid to show open (material flowing or active) and outline to show closed (material stopped or nonactive).
   - A symbol may be partially filled or shaded to represent the characteristic of the contents of a vessel, e.g. level, temperature, etc.

3.2 Color

The following standards are to be applied to the application of color to process displays:

1. The following color application guidelines are to be followed:
   - The number of colors in one display should be limited to the minimum necessary (typically 4 or less) to satisfy the process interface objectives of the display.
   - Compatible color combinations should be used.
   - Use color as a redundant indictor along with text, symbol, shape, size, reverse video, blinking, and intensity coding to preserve communication with individuals having limited color perception.
   - Colors are not to be used to indicate quantitative value.
2. Colors are to be assigned consistent with Tables 1-4.

Table 1. Unique Component Colors (e.g., Leak Detectors, Tanks)

(Outline colors and text shall be black (unless otherwise noted), display background to be tan or light blue)	
State	**Fill Color**
Out of Service	Black
In Service – Unselected	White
In Service – Selected (No Alarm)	Green
Alarming (Unacknowledged)	Flash Red
Alarm (Acknowledged)	Solid Red
Caution Alarm (Unacknowledged)	Flash Yellow (Amber)
Caution Alarm (Acknowledged)	Solid Yellow (Amber)
Not Used (e.g., bypassed)	Border is white, fill color as noted above

Table 2. Pipe Colors

(Outline colors and text shall be black (unless otherwise noted), display background to be tan or light blue)	
State	**Fill Color**
Out of Service	Black
In Service – Unselected	White
In Service – Selected (No Alarm)	Grey
In Service – Selected (with Air or Fluid Flow)	Green
Encasement Leak Detector Alarming (Unacknowledged)	Red with Flashing Text
Encasement Leak Detector Alarm (Acknowledged)	Solid Red
Caution Alarm (Unacknowledged)	Flash Yellow (Amber)
Caution Alarm (Acknowledged)	Solid Yellow (Amber)
Not Used (e.g., bypassed)	Border is white, fill color as noted above

Table 3. Valves Port Colors

(Outline colors and text shall be black (unless otherwise noted), display background to be tan or light blue)	
State	**Fill Color**
Out of Service	Black
In Service – Unselected	Grey*
In Service – Selected/Operable: Open Port (No Alarm)	Green
In Service – Selected/Operable: Shut Port (No Alarm)	White*
In Service – Operating: Transition	Yellow (All Ports)
Valve Position Alarm (Unacknowledged)	Flash Red
Valve Position Alarm (Acknowledged)	Solid Red
Caution Alarm	Flash Yellow (Amber)
Not Used (e.g., bypassed)	Border is white, fill color as noted above

* The grey color used for the unselected valve is for contrast with an unselected pipe (white), which assists the operator in distinguishing the pipe from the valve (operator influence from display screen tests). The white color used to identify a shut port on a valve is in contrast with the valve open ports (green) and with the selected pipe (grey) which assists the operator in distinguishing the pipe from the valve.

Table 4. Pump Colors

(Outline colors and text shall be black (unless otherwise noted), display background to be tan or light blue)		
State	**Fill Color**	**Text**
Out of Service	Black	--
In Service – Unselected	White	--
In Service – Selected: Pump Off and Power Off (No Alarm)	Grey	Stopped
In Service – Selected: Pump Off and Power On (No Alarm)	Yellow	Stopped
In Service – Selected: Pump On (Rotating)	Green	Running
Pump Alarm (Unacknowledged)	Flash Red	Stopped
Pump Alarm (Acknowledged)	Solid Red	Stopped
Caution Alarm (Unacknowledged)	Flash Yellow (Amber)	Caution

4.0 RECORDS

No records are generated in the performance of this management directive.

5.0 REFERENCES

Instrument Society of America (ISA) S5.5-1985, "Graphic Symbols for Process Displays."

PLC AND OIU APPLICATION SOFTWARE SETPOINTS FOR CELLS NO. 1 AND NO. 2										
Rev	**Tag 1**	**Tag 2**	**Tag 3**	**Tag 4**	**System Description**	**Setpoint Description**	**Drawing**	**Process Ranges**	**Eng. Units**	**Comments**
	Area	**Process**	**ISA**	**Eqt #**						
1	219(Y)	LH	LAHH	101	Leachate Collection and Removal System	High High Alarm Level Setpoint	H-2-830854 sheet 1 of 4	12.0 inches above LCRS 50' x 50' sump floor.	Inches	Actuate elevation 673.90
1	219(Y)	LH	LAH	101	Leachate Collection and Removal System	High Alarm Level Setpoint	H-2-830854 sheet 1 of 4	11.0 inches above LCRS 50' x 50' sump floor	Inches	Actuate elevation 673.82, activates slightly above elevation that LCRS High Flow pump On activates
1	219(Y)	LH	LAHH	104	Leak Detection System	High High Alarm Level Setpoint	H-2-830854 sheet 1 of 4	12.0 inches above LDS sump floor	Inches	Actuate elevation 671.90
1	219(Y)	LH	LAH	104	Leak Detection System	High Alarm Level Setpoint	H-2-830854 sheet 1 of 4	11.0 inches above LDS sump floor	Inches	Actuate elevation 671.82
1	219(Y)	LH	FALL	202	Leachate Collection and Removal System Low Flow Pump	Low Flow Pump Flow Fail Setpoint	H-2-830854 sheet 1 of 4	5	gpm	Half of Design Flowrate of 10 gpm
1	219(Y)	LH	FALL	202	Leachate Collection and Removal System Low Flow Pump	Low Flow Pump Flow Fail Timer Setpoint	H-2-830854 sheet 1 of 4	180	Seconds	Conservative approximation for time needed to fill empty discharge piping plus 20%. Can refine during testing.
1	219(Y)	LH	LAM	202	Leachate Collection and Removal System	Low Flow Pump Start Level Setpoint	H-2-830854 sheet 1 of 4	6.00 inches above LCRS 50' x 50' sump floor	Inches	Actuate elevation 673.40
1	219A	LH	LAL	202	Leachate Collection and Removal System	Low Flow Pump Stop Level Setpoint	H-2-830854 sheet 1 of 4	0.65 inch above LCRS 50' x 50' sump floor	Inches	Actuate elevation 672.95
1	219E	LH	LAL	202	Leachate Collection and Removal System	Low Flow Pump Stop Level Setpoint	H-2-830854 sheet 3 of 4	1.75 inches above LCRS 50' x 50' sump floor	Inches	Actuate elevation 673.04

PLC AND OIU APPLICATION SOFTWARE SETPOINTS FOR CELLS NO. 1 AND NO. 2										
Rev	**Tag 1**	**Tag 2**	**Tag 3**	**Tag 4**	**System Description**	**Setpoint Description**	**Drawing**	**Process Ranges**	**Eng. Units**	**Comments**
	Area	**Process**	**ISA**	**Eqt #**						
1	219(Y)	LH	LAM	203	Leachate Collection and Removal System	High Flow Pump Start Level Setpoint	H-2-830854 sheet 1 of 4	10.8 inches above LCRS 50' x 50' sump floor	Inches	Actuate elevation 673.80
1	219A	LH	LAL	203	Leachate Collection and Removal System	High Flow Pump Stop Level Setpoint	H-2-830854 sheet 1 of 4	4.25 inches above LCRS 50' x 50' sump floor	Inches	Actuate elevation 673.22
1	219E	LH	LAL	203	Leachate Collection and Removal System	High Flow Pump Stop Level Setpoint	H-2-830854 sheet 3 of 4	9.25 inches above LCRS 50' x 50' sump floor	Inches	Actuate elevation 673.67
1	219(Y)	LH	FALL	203	Leachate Collection and Removal System High Flow Pump	High Flow Pump Flow Fail Setpoint	H-2-830854 sheet 1 of 4	74	gpm	Half of Design Flowrate of 147 gpm
1	219(Y)	LH	FALL	203	Leachate Collection and Removal System High Flow Pump	High Flow Pump Flow Fail Timer Setpoint	H-2-830854 sheet 1 of 4	40	Seconds	Conservative approximation for time needed to fill empty discharge piping plus 20%. Can refine during testing.
1	219(Y)	LH	FALL	204	Leak Detection System	Leak Detection Pump Flow Fail Setpoint	H-2-830854 sheet 1 of 4	1	gpm	Half of Design Flowrate of 2 gpm
1	219(Y)	LH	FALL	204	Leak Detection System	Leak Detection Pump Flow Fail Timer Setpoint	H-2-830854 sheet 1 of 4	880	Seconds	Conservative approximation for time needed to fill empty discharge piping plus 20%. Can refine during testing.
1	219(Y)	LH	LAM	204	Leak Detection System	Leak Detection Pump Start Level Setpoint	H-2-830854 sheet 1 of 4	10.8 inches above LDS sump floor	Inches	Actuate elevation 671.80

PLC AND OIU APPLICATION SOFTWARE SETPOINTS FOR CELLS NO. 1 AND NO. 2										
Rev	**Tag 1**	**Tag 2**	**Tag 3**	**Tag 4**	**System Description**	**Setpoint Description**	**Drawing**	**Process Ranges**	**Eng. Units**	**Comments**
	Area	**Process**	**ISA**	**Eqt #**						
1	219(Y)	LH	LAL	204	Leak Detection System	Leak Detection Pump Stop Level Setpoint	H-2-830854 sheet 1 of 4	5.25 inches above LDS sump floor	Inches	Actuate elevation 671.34
1	219(Y)	LH	TAH	219	Crest Pad Building	Temperature High Alarm Setpoint	H-2-830854 sheet 1 of 4	90	Fahrenheit	AC will allow 85 degrees F max temp
1	219(Y)	LH	TAL	219	Crest Pad Building	Temperature Low Alarm Setpoint	H-2-830854 sheet 1 of 4	45	Fahrenheit	Heater will allow 50 degrees F low temp
1	219(Y) 1	LH	TAH	220	Transfer Building	Temperature High Alarm Setpoint	H-2-830854 sheet 1 of 4	90	Fahrenheit	AC will allow 85 degrees F max temp
1	219(Y) 1	LH	TAL	220	Transfer Building	Temperature Low Alarm Setpoint	H-2-830854 sheet 1 of 4	45	Fahrenheit	Heater will allow 50 degrees F low temp
1	219(Y) 201	LH	LAHH	301	Leachate Storage Tank	High High Alarm Level Setpoint	H-2-830854 sheet 1 of 4	74.4 inches above tank floor at wall	Inches	Actuate elevation 728.33, Activates at same elev as Hardware HH switch (at max operating capacity 6.2'), allowing 2 ft of freeboard
1	219(Y) 201	LH	LAH	301	Leachate Storage Tank	High Alarm Level Setpoint	H-2-830854 sheet 1 of 4	Activates at 24 inches above tank floor at wall	Inches	Actuate elevation 724.13, reserving 4.2 ft operating capacity for design storm event
1	219(Y) 201	LH	LAL	301	Leachate Storage Tank	Low Alarm Level Setpoint	H-2-830854 sheet 1 of 4	Activates just before falling water level reaches Hardware LL (7.8 inches above tank floor at wall) setpoint	Inches	Actuate elevation 722.78 (approximately 1 - 9,000 gallon truck load above the Low Low Setpoint)

PLC AND OIU APPLICATION SOFTWARE SETPOINTS FOR CELLS NO. 1 AND NO. 2										
Rev	**Tag 1**	**Tag 2**	**Tag 3**	**Tag 4**	**System Description**	**Setpoint Description**	**Drawing**	**Process Ranges**	**Eng. Units**	**Comments**
	Area	**Process**	**ISA**	**Eqt #**						
1	219(Y) 201	LH	LALL	301	Leachate Storage Tank	Low Low Alarm Level Setpoint	H-2-830854 sheet 1 of 4	6 inches above tank floor at wall	Inches	Activates at same elev as Hardware LL switch, activates just before falling water level reaches 6 inches above tank floor (722.63), assuring transfer pump does not run dry
1	219(Y) 1	LH	FALL	302-1	Leachate Transfer Pump	Transfer Pump Low Flow Fail Setpoint	H-2-830854 sheet 1 of 4	125	gpm	Half of Design Flowrate at 250 gpm
1	219(Y) 1	LH	FALL	302-1	Leachate Transfer Pump	Transfer Pump Low Flow Fail Timer Setpoint	H-2-830854 sheet 1 of 4	10	Seconds	Conservative approximation for time needed to fill empty discharge piping plus 20%. Can refine during testing.
(Y) = A for Cell No. 1 (Y) = E for Cell No. 2					For Cell No. 1 reference corresponding H-2-830854 sheet 1 of 4 and sheet 2 of 4. For Cell No. 2 reference corresponding H-2-830854 sheet 3 of 4 and sheet 4 of 4.					

DIVISION 14 (NOT USED)

1 **DIVISION 15 (NOT USED)**

SECTION 15021 – HIGH DENSITY POLYETHYLENE (HDPE) PIPE SCHED. B

Part 1 – General

Summary

This section is for furnishing and installing leachate piping and associated components.

References

The publications listed below form a part of this Specification to the extent referenced. The publications are referred to in the text by basic designation only. Recognizing some requirements of the references cited below may not be applicable, the Engineer shall judge the applicability of compliance with the references not specifically addressed herein. In the event of a conflict between the text of this Specification and the references cited herein, the text of this Specification shall take precedence or as directed by the Engineer.

ASTM INTERNATIONAL (ASTM)

ASTM D792	Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement.
ASTM D1248	Specification for Polyethylene Plastics Molding and Extrusion Materials.
ASTM D1505	Standard Test Method for Density of Plastics by the Density-Gradient Technique.
ASTM D2513	Specification for Thermoplastic Gas Pressure Pipe, Tubing, and Fittings.
ASTM D3350	Specification for Polyethylene Plastics Pipe and Fitting Materials.
ASTM F714	Standard Specification for Polyethylene Plastic Pipe (SDR-PR) Based on Outside Diameter.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 192.285	Plastic pipe; qualifying persons to make joints.

Description

Pipe: This section includes all high density polyethylene (HDPE) pipe used in the cells including but not limited to:

- Leachate collection piping on floor and cleanout access pipes on the slopes of the trench.
- Leachate discharge piping, leak detection piping, and associated riser pipes.
- Double containment piping outside the cell (e.g., leachate force main and drain lines) and elsewhere as shown on the Drawings.

Submittals–Approval Required

- See Section 01300, SUBMITTALS, for submittal procedures.
- Manufacturer's design guidance that includes spacer maximum spacing, attachment method, orientation and material type.
- Manufacturer's certificates of compliance for all pipe and fittings. Certificates shall acknowledge that pipe and fittings meet the requirements of the Specifications.
- Descriptive literature about the fusion equipment to be used and certification from the pipe supplier or manufacturer that the joining technician(s) is certified and experienced in heat fusion joining of HDPE pipe. Certification shall contain the following minimum information:
  - Name of technician.
  - Date of certification.

Statement by the pipe supplier that the technician is certified in the means and methods of joining the supplier's pipe and fittings using butt fusion techniques.

Make(s) and model(s) of fusion equipment the technician is certified to join pipe with.

Submittals–Approval Not Required

Information/Record (IR):

Catalog and manufacturer's data sheets for HDPE pipe and fittings.

Catalog and manufacturer's data sheets, electrofusion couplers, mechanical cutters, and appurtenances.

Part 2 – Products

All HDPE pipe and fittings shall conform with additional applicable requirements defined in the Piping Schedule in Section 15060, PIPING-GENERAL.

HDPE Pipe

Resin: HDPE pipe shall be manufactured from first quality extra-high molecular weight, high density polyethylene resin containing no more than 2 percent clean recycled polymer by weight. Resin shall meet or exceed the requirements of ASTM D3350 for PE3408 material with a cell classification of 345434C or higher. Alternate cell classifications are acceptable if one or more of the 6 numbers in the cell classification is greater than the minimum. Pipe shall be rated PE3408. Pipe and fittings shall be in compliance with schedule attached as supplement (see Attachment 1, HIGH DENSITY POLYETHYLENE (HDPE) PIPE) or as shown on the Drawings.

Quality: The pipe shall have uniform wall thickness and shall be uniform in color, opacity, density, and other physical properties. Pipe shall be homogeneous throughout and free of visible cracks, holes, blisters, bubbles, undispersed raw materials, or any contamination by foreign matter. Any pipe with nicks, scrapes, or gouges deeper than 10 percent of the nominal wall thickness shall be rejected.

Form: Pipe may be supplied in a continuous extruded seamless piece or in sections.

Manufacturer's Certificates of Compliance:

The manufacturer shall submit a Certificate of Compliance of the HDPE pipe supplied for the IDF project, which will include that the pipe is grade PE3408 and the identity of the cell classification per ASTM D3350.

HDPE pipe SDR shall be as indicated on the Piping Schedule in Section 15060, PIPING-GENERAL.

Fittings:

Fittings shall conform to the requirements of Article HDPE PIPE of this section, shall be compatible with components of the double containment system, and HDPE manholes where required.

Polyethylene fittings shall be from the same manufacturer as the pipe (or approved equal), molded or fabricated from polyethylene pipe and shall have the same or numerically smaller SDR than pipe connecting to the fitting. Fittings shall follow requirements in Attachment 1.

All reducing tees shall be factory molded if available as a standard item by any manufacturer having pipe meeting this section. If not available as a standard item, branch saddle reducing tees shall be used. Reducers shall be shop manufactured. Fabricated branch connections will not be allowed if branch saddle connections are listed in the manufacturer's catalog.

All molded polyethylene fittings shall have the same or higher pressure rating as the pipe when installed in accordance with the latest technical specifications. All fabricated polyethylene fittings shall have the same or higher pressure rating as the adjoining pipe when installed in accordance with the manufacturer's recommendations.

Double Containment Pipe

Pipe Materials: Both carrier pipe and containment pipe shall meet the requirements of Article HDPE PIPE of this section.

Configuration: Double containment pipe shall consist of a carrier pipe installed within a containment pipe. All pipe and fittings shall provide an annular space between the carrier and containment pipes to accommodate possible flow of fluid from the carrier pipe.

Support Spacers: Support spacers shall be manufactured from nonmetallic, corrosion-resistant material with the same or better chemical compatibility properties as the HDPE pipe. Spacer intervals and attachment method to the carrier pipe shall be in accordance with the manufacturer's recommendations. Spacing shall be reduced if required to maintain the annulus between the carrier and containment pipes and shall be positioned to allow for unrestricted passage of possible flow of fluid from the carrier pipe. Spacers shall be chamfered at both ends to allow for removal of carrier pipe. Materials and systems used to secure the spacers to the pipe shall have the same or better chemical compatibility properties as the HDPE pipe.

Fittings: Fittings shall conform to the requirements of Article HDPE PIPE of this section and shall be compatible with components of the single wall HDPE pipe where required (except for trench riser pipes).

Slotted Pipe

Leachate Collection Piping: Leachate collection and leak detection piping on the floor of the cells and elsewhere as shown on the Drawings shall be slotted. Cleanout access pipes and leachate transmission piping shall not be slotted.

In addition to meeting all other requirements of this section, slotted pipe shall have slots 0.128 inch wide and 1.25 inches long, in five places equidistant around the pipe for 12-inch and 18-inch diameter pipes. Slots shall provide a minimum of 9 square inches of open area per linear foot of pipe for 12-inch and 18-inch diameter pipes. For 3-inch diameter slotted pipe, the pipe supplier shall propose a configuration of 0.128-inch wide slots that provides a maximum of 7 square inches of open area per linear foot of pipe. Slotted pipes shall be free of cutting debris from the slot cutting process.

Perforated pipe with circular drill holes is not allowed.

Part 3 – Execution

General

All HDPE pipe and fittings shall be installed in conformance with applicable code requirements referenced in Section 15060, PIPING-GENERAL.

Dimensions

Piping dimensions shown on the Drawings are approximate. It is the Construction General Contractor's responsibility to furnish and install piping of the proper dimensions, which will properly fit with the connecting elements, pipes, fittings, pumps, etc.

Installation

Pipe shall be handled and stored in such a manner as to ensure a sound, undamaged condition.

Pipe shall be cut in a neat, workmanlike manner using a mechanical cutter that will not damage the pipe.

Joining of HDPE pipe to HDPE pipe shall be accomplished by thermal butt fusion joint; no solvent welding or adhesive welding shall be allowed. Electrofusion couplings shall only be allowed when access to piping is restricted and only as approved by the Engineer. Slotted leachate collection piping shall be joined with thermal butt fusion joints. Pipe shall be joined per ASTM D2657 and manufacturer's recommendations.

Single butt fusion welds shall be used to create pipe sections as long as practicable. Fabricated pipe sections and fittings may be joined by the double butt fusion process.

During installation, the pipe shall not be pulled across sharp projections that could cause gouges, kinks, or other types of damage. To minimize "snaking" due to thermal expansion, protect pipe from direct sunlight, or limit unrestrained length of pipe during installation.

Allowance for Thermal Expansion/Contraction:

HDPE has a coefficient of thermal expansion of 1.2 x 10-4 ft/ft/deg F. Buried HDPE pipe shall be installed with excess length between anchor points such that contraction caused by temperature drop to 40 degrees F will produce the length of pipe between two points shown on the Drawings. Amount of excess pipe depends on temperatures of pipe at the time of installation, according to Table 1 for buried piping:

TABLE 1

Installation Temperature (degrees F)	Excess Pipe Length (in./100 ft)
50	1.4
60	2.9
70	4.3
80	5.8
90	7.2
100	8.6
120	11.5

Installation temperature is of the pipe material and not ambient air temperature. Measure installation temperature with a strip thermometer laid directly on the pipe. Verify temperate and excess pipe length required immediately before burial.

Placement of Buried Pipes:

Excavate trench bottom and sides of ample dimensions to permit visual inspection and testing of entire flange, valve, or connection.

The pipe shall not be dropped into the trench. Exercise care when lowering pipe into trench to prevent twisting or damage to pipe. The full length of the pipe shall be firmly bedded on the trench bottom.

The pipe shall be bedded in such a way as to maintain grade with a tolerance of -0.0 percent, +0.5 percent with a uniform, constant grade and no localized low spots.

Pipe Base and Pipe Zone:

As specified in Section 02320, TRENCH BACKFILL.

Keep trench dry until pipe laying and joining are completed.

Prevent foreign material from entering pipe during placement.

Close and block open end of last laid pipe section when placement operations are not in progress and at close of day's work.

Install closure sections and adapters for gravity piping at locations where pipe laying changes direction.

After joint has been made, check pipe alignment and grade.

Place sufficient pipe zone material to secure pipe from movement before next joint is installed.

Prevent uplift and floating of pipe prior to backfilling.

Place pipe along pipe runs starting at one end and moving towards the other to avoid joints that will not be feasible with butt fusion.

Tolerances: Horizontal position of pipe centerline on alignment around curves maximum variation of 1.0 foot from position shown.

Pipe Cover:

Minimum 2 feet 6 inches from finished elevation of overlying material, unless otherwise shown.

Temporarily close pipe ends as required to avoid introducing dirt or other foreign material into the pipe.

Trenching and backfilling operations shall be conducted in accordance with the requirements of Section 02320, TRENCH BACKFILL, for utility trenching. If trenching is used, underlying materials shall not be disturbed or damaged in anyway. Backfilling operations shall ensure that no voids are present under or at the sides of the pipe. Backfill shall initially be placed to the top of the pipe, then hand compacted. The remainder of the trench shall then be backfilled and compacted by hand or with a power tamper only.

On the floor of the cell, pipe may be placed directly on geosynthetic layers prior to placing drainage gravel. Placement of gravel around pipes shall be by hand unless otherwise approved by the Engineer. Placement operations shall ensure that no voids are present under or at the sides of the pipe. Placement operations shall not disturb the position of the pipe.

Where flanged joints are used, the bolts shall be evenly torqued using a crossing pattern to gradually tighten the lug nuts. Torque values shall be as recommended by the flange manufacturer. Flanged joints shall be retorqued after one hour or more has passed. Apply anti-seize compound on all threaded surfaces before tightening.

Flaws (minor imperfections, damaged areas, etc.) in HDPE pipe with a depth of 10 percent or less of the nominal wall thickness will not require repair or replacement. In double containment systems, carrier pipe with flaws deeper than 10 percent of the wall thickness shall be replaced. Single pipe or containment pipe with flaws between 10 and 25 percent of the wall thickness shall be repaired in accordance with the pipe manufacturer's recommendations. The Construction General Contractor shall certify in writing that the repaired area will have material properties that meet or exceed those of intact pipe. Any pipe with flaws deeper than 25 percent of the nominal wall thickness shall be rejected.

All valves and equipment shall be supported independently from pipe. Anchor valves such that turning moment resulting from their operation will not be transmitted to pipe.

Special Precautions at Flanges: Polyethylene pipe connected to heavy fittings, manholes, and rigid structures shall be supported in such a manner that no subsequent relative movement between polyethylene pipe at flanged joint and rigid structures is possible.

Butt-fusion shall be performed in accordance with pipe manufacturer's recommendations as to equipment and technique.

Weld Beads: Remove internal weld beads from the side slope risers and horizontal sections of slotted pipe where the LCRS and LDS pumps will be placed and the horizontal and vertical sections of the LCRS level transducer pipe. Remove all plastic debris from inside pipe.

Slotted Pipe: Slotted pipe shall be cut and joined so that full contact is made around the entire circumference of the weld. Partial weld contact because of joints through a slot row is not acceptable.

Locator Ribbon

Locator ribbon shall be installed as specified in Section 02320, TRENCH BACKFILL.

Identification Ribbon

Underground pipelines, except for pipelines inside the Phase I liner limits, shall be identified by use of a plastic ribbon or stencil no less than 3 inches in width with a message printed on the ribbon, which identifies the actual pipeline contents. Marking tapes or stencils shall be placed on existing lines where they are exposed by trenching operations. The ribbon shall be wrapped around the pipeline at no less than 1 wrap per 3 feet of run. The plastic ribbon/stencil shall be color coded in accordance with the Piping Schedule.

Cleaning

Clean all piping as required in Section 15060, PIPING-GENERAL, to remove all foreign materials including dirt, grease, and other matter.

Construction Quality Control (Acceptance Testing)

Per Section 15992, PIPING LEAKAGE TESTING, and the Piping Schedule in Section 15060, PIPING-GENERAL.

END OF SECTION 15021

Attachment 1, High Density Polyethylene (HDPE) Pipe

Item	Size	Description
General	All	Pipe lengths, fittings, and flanged connections to be joined by thermal butt-fusion shall be of the same type, grade, and class of polyethylene compound and supplied from the same raw material supplier.
Pipe		Pipe SDR shall be AS INDICATED ON THE Piping Schedule in Section 15060, PIPING-GENERAL. Protection shall be provided against ultraviolet light degradation using carbon black, not less than 2 percent well dispersed in the resin. Pipe wall thickness shall reflect the required SDR* and diameter, as shown in Table 8, ASTM F714. *SDR: standard dimension ratio = OD/thickness
Fittings	6-inch and smaller 8-inch and larger	Molded fittings, butt fusion joined, conforming to ASTM D3261. Molded if manufactured as a standard item or same as pipe, butt fusion joined, conforming to ASTM D3350.
Electrofusion		Rigid, straight coupler constructed from injection-
Couplers		molded polyethylene with embedded heating coils as manufactured by Central Plastics; or equivalent.
Flanges		ASTM A351 Type 316/CF8M stainless steel, 150-pound, ANSI B16.5 standard, convoluted back-up ring with one-piece polyethylene molded flange adaptor ends, same rating pressure as pipe.
Bolting		Stainless steel, ASTM A193/A193M Grade B8M studs and ASTM A194/A194M Grade 8M hex head nuts. Manufacturer's recommended anti-seize compound on all threads. Washers shall be same material as bolts.
Gaskets		Flat ring, 1/8-inch Viton.

SECTION 15022 – HIGH DENSITY POLYETHYLENE MANHOLES SCHED. B

Part 1 – General

References

The following is a list of standards, which may be referenced in this section:

ASTM INTERNATIONAL (ASTM)

ASTM D1248 Specification for Polyethylene Plastics Molding and Extrusion Materials.

ASTM D3350 Specification for Polyethylene Plastics Pipe and Fitting Materials.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Shop Drawings:

Product data sheets for make and model.

Complete catalog information, descriptive literature, specifications, construction drawings, and identification of materials of construction.

Provide calculations indicating diameter and wall thickness of each manhole is acceptable per design criteria specified.

Submittals–Approval Not Required

Information/Record (IR): Submit results of manhole leakage testing.

Part 2 – Products

HDPE Manholes

The manhole shall be manufactured by the fabrication of high density polyethylene (HDPE pipe). The riser shall be made of HDPE plastic compound meeting the requirements of Type III, Class C, Category 5, Grade P34 as defined in ASTM D1248. The cell classification shall be 345434C per ASTM D3350. Alternate cell classifications are acceptable if one or more of the six numbers in the cell classification are greater than the minimum. Pipe shall be rated PE3408.

Flatstock shall meet or exceed ASTM D1248 requirements for Type III, Class C, Category 3, Grade G5.

HDPE pipe and flatstock used to fabricate the HDPE manhole shall meet all product requirements of Section 15021, HIGH DENSITY POLYETHYLENE (HDPE) PIPE. HDPE pipe for 76-inch diameter manhole shall be SIDR 39.0. HDPE pipe for 42-inch diameter manhole shall be SDR 21.

Welding rods, connecting couplings, pipe collars and other materials, as required to complete the installation, shall be of the same plastic as the flatstock.

Piping and appurtenances shall meet the requirements of Section 15021, HIGH DENSITY POLYETHYLENE (HDPE) PIPE, and Section 15060, PIPING–GENERAL.

Access hatch shall be precast vault lid with spring-assisted galvanized plate cover with locking latch; cover shall be weather tight with channel drain. Utility Vault Model 77-2-332P; or approved equal.

Wall thickness and diameter of each manhole shall be verified assuming the following design criteria:

Dimensions specified and depth of manholes identified on Drawings.

Lateral load transmitted to 76-inch diameter manhole from concrete ring footing shown on Drawing H-2-830850.

Weight of precast utility vault equal to 4,020 pounds.

Load on utility vault equal to 100 pounds per square foot (vault will not have traffic loads).

No groundwater forces acting on 76-inch diameter manhole.

42-inch manhole full of water.

Part 3 – Execution

Fabrication

Inlet and outlet piping shall be installed as shown on Drawings by fabricator prior to delivery to site.

The manhole shall be fabricated with the minimum number of welds practical.

All welds shall be heat fused in accordance with manufacturer's recommendations on equipment specifically designed for welding thermoplastic sheets or extrusion welded by precertified welders.

Installation

Manholes shall be handled and stored according to manufacturer's recommendations and in such a manner as to ensure a sound undamaged condition.

Excavation and backfilling operations shall be conducted in accordance with Section 02316, EXCAVATION, and Section 02315, FILL AND BACKFILL.

Joining of HDPE field piping to HDPE manhole pipe stubouts shall be done by thermal butt or socket fusion, no solvent or adhesive welding shall be allowed. HDPE welding shall be by qualified and approved welders as identified in Section 15021, HIGH DENSITY POLYETHYLENE (HDPE) PIPE.

Install piping, appurtenances, and pipe hangers and supports in accordance with Section 15021, HIGH DENSITY POLYETHYLENE (HDPE) PIPE, and Section 15060, PIPING-GENERAL.

Install access hatch in accordance with Drawings.

Construction Quality Control – Leakage Testing

After all HDPE manhole pipe stubout connections have been completed for inner and outer HDPE manholes, and prior to backfilling, perform separate leakage tests for inner and outer manholes. Each manhole shall be completely filled with water and no leaking shall be allowed from any HDPE joints for a period of 8 hours. General Construction Contractor shall provide necessary shoring during leakage testing of outer manhole per manufacturer's recommendations to compensate for lack of backfill.

END OF SECTION 15022

SECTION 15060 – PIPING-GENERAL SCHED. B

Part 1 – General

Summary

This section is for furnishing and installing leachate piping and associated components.

References

The following is a list of standards, which may be referenced in this section and any supplemental Data Sheets:

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B1.20.1	Pipe Threads, General Purpose (Inch)
ANSI B16.1	Cast Iron Pipe Flanges and Flanged Fittings
ANSI B16.3	Malleable Iron Threaded Fittings
ANSI B16.5	Pipe Flanges and Flanged Fittings
ANSI B16.11	Forged Fittings, Socket-Welding and Threaded
ANSI B16.21	Nonmetallic Flat Gaskets for Pipe Flanges
ANSI B16.42	Ductile Iron Pipe Flanges and Flanged Fittings, Classes 150 and 300

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B36.10M	Welded and Seamless Wrought Steel Pipe

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C153/ A21.53	Ductile-Iron Compact Fittings 3 Inches Through 24 Inches and 54 Inches Through 64 Inches, for Water Service

ASTM INTERNATIONAL (ASTM)

ASTM A153	Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A307	Standard Specification for Carbon Steel Bolts and Studs, 60,000 psi Tensile Strength
ASTM A536	Standard Specification for Ductile Iron Castings
ASTM A563	Standard Specification for Carbon and Alloy Steel Nuts
ASTM D1248	Standard Specification for Polyethylene Plastics Molding and Extrusion Materials
ASTM D1784	Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds
ASTM D1785	Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120
ASTM D2467	Standard Specification for Socket-Type Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80
ASTM D2564	Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems
ASTM D3261	Standard Specification for Butt Heat Fusion Polyethylene (PE) Plastic Fittings for Polyethylene (PE) Plastic Pipe and Tubing

ASTM D3350 Standard Specification for Polyethylene Plastics Pipe and Fittings Materials

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Product data sheets, complete catalog information, descriptive literature, specifications, and identification of materials of construction.

Laboratory Testing Equipment: Certified calibrations, manufacturer's product data, and test procedures.

Qualifications:

Solvent Welders: List of solvent welders and current test records for solvent welder(s) for field solvent welding.

Submittals–Approval Not Required

Information/Record (IR): Manufacturer's Certification of Compliance.

Part 2 – Products

Piping

High Density Polyethylene Piping:

As specified in Section 15021, HIGH DENSITY POLYETHYLENE (HDPE) PIPE.

Others as specified on Piping Data Sheet(s) and Piping Schedule located at the end of this section as Supplement.

Diameters Shown:

Standardized Products: Nominal size.

Fabricated Steel Piping (Except Cement-Lined): Outside diameter, ASME B36.10M.

Joints

Flanged Joints:

Flat-faced carbon steel or alloy flanges when mating with flat-faced cast or ductile iron flanges.

Higher pressure rated flanges as required to mate with equipment when equipment flange is of higher pressure rating than required for piping.

Threaded Joints:

NPT taper pipe threads in accordance with ANSI B1.20.1.

Mechanical connections of high density polyethylene pipe to auxiliary equipment such as valves, pumps, tanks, and other piping systems shall be through flanged connections consisting of products as specified in Section 15021, HIGH DENSITY POLYETHYLENE (HDPE) PIPE, and bolts and nuts of sufficient length to show a minimum of three complete threads when the joint is made and tightened to manufacturer's standard. Retorque nuts after 4 hours.

Gasket Lubricant

Lubricant shall be supplied by pipe manufacturer and no substitute or "or-equal" will be allowed.

Double Wall Containment Piping System

As specified in Section 15021, HIGH DENSITY POLYETHYLENE (HDPE) PIPE.

Flexible Hose

Flexible hose required for connection of leachate pump discharge piping to crest pad piping manifold and at truck loading station shall be Royalflex, vinyl nitrile by Boston Industrial Products, or equal, rated at a minimum of 125 psi.

A male and female quick-connect coupling shall be factory or shop installed on each length of flexible hose with stainless steel banding. 1-1/2-inch quick connect couplings for flexible hose and leachate pump discharge piping within the slope riser pipe shall be fiberglass-filled polypropylene, 3-inch quick connect couplings shall be stainless steel, each size shall include stainless steel rings, arms, and pins. Stainless steel quick connects shall have minimum pressure rating of 150 psig. Anti-galling compound shall be used for threaded stainless steel connections.

Vent And Drain Valves

Pipeline 2-Inch Diameter and Smaller: 1/2-inch vent, 1-inch drain, unless shown otherwise.

Pipelines 2-1/2-Inch Diameter and Larger: 3/4-inch vent, 1-inch drain, unless shown otherwise.

Fabrication

Flanged pipe shall be fabricated in the shop, not in the field, and delivered to the site with flanges in place and properly faced. Threaded flanges shall be individually fitted and machine tightened on matching threaded pipe by the manufacturer.

Finishes

Factory prepare, prime, and finish coat in accordance with Pipe Data Sheet(s) and Piping Schedule.

Galvanizing:

Hot-dip applied, meeting requirements of ASTM A153.

Electroplated zinc or cadmium plating is unacceptable.

Stainless steel components may be substituted where galvanizing is specified.

Locator Ribbon

As specified in Section 02320, TRENCH BACKFILL.

Insulation

Piping:

Combined Sump (Discharge Piping Only) and Piping Requiring Heat Trace:

Material: Flexible elastomeric pipe insulation, closed cell structure, 3/4 inch thick.

Temperature Rating: Minus 40 degrees F to 180 degrees F.

Nominal Density:

6 pcf.

Conductivity in accordance with ASHRAE 90.1 and minimum of 0.27 BTU-in/hr-ft2 degrees F at 75 degrees F per ASTM C177 or ASTM C518.

Minimum water vapor transmission of 0.10 perm-inch per ASTM E96.

Seal joints with manufacturer's adhesive.

Flame Spread Rating: Less than 25 per ASTM E84.

Manufacturers and Products:

Rubatex: R-180-FS.

Armstrong: Armaflex AP.

Piping and Insulation Cover: Aluminum jacket 0.016-inch thick.

Part 3 – Execution

Examination

Verify size, material, joint types, elevation, horizontal location, and pipe service of existing pipelines to be connected to new pipelines or new equipment.

Inspect size and location of structure penetrations to verify adequacy of wall pipes, sleeves, and other openings.

Preparation

Inspect pipe and fittings before installation, clean ends thoroughly, and remove foreign matter and dirt from inside.

Damaged Coatings and Linings: Repair using original coating and lining materials in accordance with manufacturer's instructions.

Installation-General

Join pipe and fittings in accordance with manufacturer's instructions, unless otherwise shown or specified.

Remove foreign objects prior to assembly and installation.

Flanged Joints: Install perpendicular to pipe centerline.

Bolt Holes:

Straddle vertical centerlines, aligned with connecting equipment flanges or as shown.

Use torque-limiting wrenches to ensure uniform bearing and proper bolt tightness.

Plastic Flanges: Install annular ring filler gasket at joints of raised-face flange.

Raised-Face Flanges:

Use flat-face flange when joining with flat-faced ductile or cast iron flange.

Verify compatibility of mating flange to adapter flange gasket prior to selecting grooved adapter flanging.

Threaded flanged joints must be shop fabricated and delivered to jobsite with flanges in-place and properly faced.

Threaded and Coupled Joints:

Conform with ANSI B1.20.1.

Produce sufficient thread length to ensure full engagement when screwed home in fittings.

Countersink pipe ends, ream and clean chips and burrs after threading.

Make connections with not more than three threads exposed.

Lubricate male threads only with thread lubricant or tape as specified on Piping Data Sheets.

High Density Polyethylene Piping: As specified in Section 15021, HIGH DENSITY POLYETHYLENE (HDPE) PIPE.

Installation-Exposed Piping

Piping Runs:

Parallel to building or column lines and perpendicular to floor, unless shown otherwise.

Piping upstream and downstream of flow measuring devices shall provide straight lengths as required for accurate flow measurement.

Group piping wherever practical at common elevations; install to conserve building space and not interfere with use of space and other work.

Unions or Flanges: Provide at each piping connection to equipment or instrumentation on equipment side of each block valve to facilitate installation and removal.

Install piping so that no load or movement in excess of that stipulated by equipment manufacturer will be imposed upon equipment connection; install to allow for contraction and expansion without stressing pipe, joints, or connected equipment.

Piping Clearance (unless otherwise shown):

Over Walkway and Stairs: Minimum of 7 feet 6 inches, measured from walking surface or stair tread to lowest extremity of piping system including flanges, valve bodies or mechanisms, insulation, or hanger/support systems.

Between Equipment or Equipment Piping and Adjacent Piping: Minimum 3 feet 0 inches, measured from equipment extremity and extremity of piping system including flanges, valve bodies or mechanisms, insulation, or hanger/support systems.

From Adjacent Work:

Minimum 1 inch from nearest extremity of completed piping system including flanges, valve bodies or mechanisms, insulation, or hanger/support systems.

Do not route piping in front of or to interfere with access ways, ladders, stairs, platforms, walkways, openings, doors, or windows.

Head room in front of openings, doors, and windows shall not be less than the top of the opening.

Do not install piping containing liquids or liquid vapors in transformer vaults.

Do not route piping over, around, in front of, in back of, or below electrical equipment including controls, panels, switches, terminals, boxes, or other similar electrical work.

Installation-Double Wall Containment Piping System

Install as specified in Section 15021, HIGH DENSITY POLYETHYLENE (HDPE) PIPE.

Installation-Buried Pipe

Placement: In accordance with Section 15021, HIGH DENSITY POLYETHYLENE (HDPE) PIPE.

Slab, Floor, Wall, and Roof Penetrations

Application and Installation: As shown on Drawings.

Wall Pipe Installation: Support wall pipes securely by framework to prevent contact with reinforcing steel and tie wires.

Branch Connections

Do not install branch connections smaller than 1/2-inch nominal pipe size, including instrument connections, unless shown otherwise.

When line of lower pressure connects to a line of higher pressure, requirements of Piping Data Sheet for higher pressure rating prevails up to and including the first block valve in the line carrying the lower pressure, unless otherwise shown.

Threaded Pipe Tap Connections:

Welded Steel or Alloy Piping: Connect only with welded threadolet or half-coupling as specified on Piping Data Sheet.

Limitations: Threaded taps in pipe barrel are unacceptable.

Cleaning

Following assembly and testing, and prior to final acceptance, flush pipelines (except as stated below) with water at 2.5 fps minimum flushing velocity until foreign matter is removed.

The up-slope riser pipe and slotted pipe shall be cleaned internally after deburring by pulling cotton pillows attached to ropes through pipe repeatedly until no debris comes out of pipe with pillow. Water flushing shall not be permitted. Pipe ends shall be covered after fabrication and at the end of shifts to avoid foreign materials from entering pipe.

If impractical to flush large diameter pipe at 2.5 fps, clean in-place from inside by brushing and sweeping, then flush or blow line at lower velocity.

Insert cone strainers in flushing connections to attached equipment and leave in-place until cleaning is complete.

Remove accumulated debris through drains 2 inches and larger or by removing spools and valves from piping.

Field Finishing

Notify Construction Manager at least 3 days prior to start of any surface preparation or coating application work.

Locator Ribbon

Locator ribbon shall be installed as specified in Section 02320, TRENCH BACKFILL.

Pipe Identification

Exposed Piping

In general, all exposed piping shall be color coded and identified in accordance with ANSI A-13-1. It is the intent of this standard that the identification method of aboveground piping is by English text that allows the contents to be readily identified. Flow direction should be also shown by arrows.

All piping and equipment shall be identified in accordance with established site standards.

In addition to the requirements specified herein, all pipelines and standard equipment shall be color coded and identified with beaded chain or steel cable stainless steel tags displaying the pipe or equipment number as shown on the Drawings. The tags shall be fabricated from 300 series austenitic stainless steel metal strips 3/4 inch wide, 24-gauge minimum thickness, with 3/16-inch high letters stamped on the metal surface. Any pipes entering or leaving a building shall be tagged adjacent to floor or wall penetration. The tags shall be attached to the pipe or austenitic equipment with austenitic stainless steel bead chain or austenitic stainless steel cable. When tagging valves, the bead chain shall be attached to the valve stem or yoke.

Construction Quality Control – Leakage Testing

As specified in Section 15992, PIPING LEAKAGE TESTING.

Supplements

Supplement 1—Polyvinyl Chloride (PVC) Pipe and Fittings.

Supplement 2—Galvanized Steel Pipe and Malleable Iron Fittings.

Supplement 3—Piping Schedule.

END OF SECTION 1506

Polyvinyl Chloride (PVC) Pipe & Fittings

Item	Size	Description
Pipe	All	Schedule 80 PVC: Type I, Grade I or Class 12454-B conforming to ASTM D1784 and ASTM D1785. Pipe shall be manufactured with 2 percent titanium dioxide for ultraviolet protection. Threaded Nipples: Schedule 80 PVC.
Fittings	All	Schedule to Match Pipe Above: ASTM D2466 and ASTM D2467 for socket-weld type and Schedule 80 ASTM D2464 for threaded type. Fittings shall be manufactured with 2 percent titanium dioxide for ultraviolet protection.
Joints	All	Solvent socket-weld except where connection to threaded valves and equipment may require future disassembly.
Flanges	All	One piece, molded hub type PVC flat face flange in accordance with Fittings above, 125-pound ANSI B16.1 drilling
Bolting	All	ASTM A193/A193M Type 316 stainless steel Grade B8M hex head bolts and ASTM A194/ A194M Grade 8M hex head nuts.
Gaskets	All	Flat Face Mating Flange: Full faced 1/8-inch thick ethylene propylene (EPR) rubber. Raised Face Mating Flange: Flat ring 1/8-inch ethylene propylene (EPR) rubber, with filler gasket between OD of raised face and flange OD to protect the flange from bolting moment.
Solvent Cement	All	As recommended by the pipe and fitting manufacturer conforming to ASTM D2564.
Thread Lubricant	All	Teflon Tape.

Galvanized Steel Pipe & Malleable Iron Fittings

Item	Size	Description
Pipe		Galvanized carbon steel, ASTM A106, Grade B seamless or ASTM A53, Grade B seamless or ERW.
	2" and smaller	Schedule 80.
	2-1/2" through 6"	Schedule 40.
Joints	All	Threaded or flanged at valves and equipment.
Fittings		Threaded: 150- or 300-pound galvanized malleable iron, ASTM A197 or ASTM A47, dimensions in accordance with ANSI B16.3.
Flanges		Galvanized forged carbon steel, ASTM A105/A105M, ANSI B16.5 Class 150 or Class 300, threaded, 1/16-inch raised face.
Unions		Threaded malleable iron, ASTM A197 or A47, 300-pound WOG, brass to iron seat, meeting the requirements of ANSI B16.3.
Bolting		Flanges: Carbon steel ASTM A307, Grade A hex head bolts and ASTM A563, Grade A hex head nuts.
Gaskets	All flanges	Flanged, Water and Sewage Service: 1/8 inch thick, red rubber (SBR), hardness 80 (Shore A), rated to 200 degrees F, conforming to ANSI B16.21, AWWA C207, and ASTM D1330, Grades 1 and 2.
Thread Lubricant	2" & smaller	Teflon tape or joint compound that is insoluble in water.

Piping Schedule Legend

Service Code	
LH	Leachate Handling
LHCP	Leachate Handling Containment Pipe
LT	Leachate Transfer
LTCP	Leachate Transfer Containment Pipe
Service	
CPB	Crest Pad Building
CS	Combined Sump
IC	In-Cell
LTB	Leachate Transfer Building
OC	Outside-Cell
TL	Truck Loading
Exposure	
BUR	Buried
EXP	Exposed
SUB	Submerged
Material	
GSP	Galvanized Steel Pipe
HDPE	High Density Polyethylene
PVC	Polyvinyl Chloride
Pressure Test	
H	Hydrostatic
P	Pneumatic
NA	Not Applicable
Joint Type	
BF	Butt Fused
FL	Flanged
SW	Solvent Weld
TH	Threaded
QC	Quick Connect

1 Piping Schedule

Service Code	Service	Size(s) (In.)	Exposure	Piping Material	Joint Type	Specification Section	Test Type	Test Pressure (psi)[1]	Remarks
LH	IC	All	BUR	HDPE	BF, QC	15021	H	65* LCRS–Low Flow 100* LCRS–High Flow 65* LDS	SDR 11
LHCP	IC	3"	BUR	HDPE	BF	15021	P	8	SDR 11, slotted portion of pipe not to be tested
LHCP	IC	12" & 18"	BUR	HDPE	BF	15021	P	8	SDR 17, slotted portion of pipe not to be tested
LH	CPB CS	All	EXP	PVC	SW, FL, TH	15060	H	75	SCHD 80
LH LT	OC	3"	BUR	HDPE	BF	15021	H	65, except 75 for LH-30 and LH-24	SDR 11
LH LT	OC	4"	BUR	HDPE	BF	15021	H	65	SDR 17
LHCP LTCP	OC	2"	BUR	HDPE	BF	15021	P	8	SDR 11 for all except SDR 9 for 219A(E)201-2"-LTCP-043-HDPE
LHCP LTCP	OC	6" & 8"	BUR	HDPE	BF	15021	P	8	SDR 17
LT	LTB	All	EXP	PVC	SW, FL, TH	15060	H	75	SCHD 80
LT	TL	All	EXP	GSP	FL, TH	15060	H	65, except 75 for LT-37	SCHD 40

*Test pressures measured at Crest Pad Building. Isolate any appurtenances not rated for test pressure such as vacuum breakers and pressure gages.

Notes: 1. Test pressures shown are for a minimum allowable pressure. Maximum test pressures cannot exceed pressure rating of the pipe during the expansion phase of the pressure test.

2

SECTION 15100 – VALVES, OPERATORS, AND FILTER SCHED. A & B

Part 1 – General

Summary

This section is for furnishing and installing valves, filters, and associated components for leachate and raw water systems. Valves and accessories for raw water are identified as such. Other requirements apply to leachate and raw water systems.

References

The following is a list of standards, which may be referenced in this section:

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B16.1	Cast Iron Pipe Flanges and Flanged Fittings

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C509	Resilient-Seated Gate Valves for Water and Sewerage Systems
AWWA C550	Protective Epoxy Interior Coatings for Valves and Hydrants

ASTM INTERNATIONAL (ASTM)

ASTM A276	Standard Specification for Stainless and Heat-Resisting Steel Bars and Shapes
ASTM A351	Standard Specification for Castings, Austenitic, Austenitic-Ferric (Duplex), for Pressure-Containing Parts
ASTM B61	Standard Specification for Steam or Valve Bronze Castings
ASTM B62	Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM B98	Standard Specification for Copper-Silicon Alloy Rod, Bar, and Shapes
ASTM B127	Standard Specification for Nickel-Copper Alloy (UNS N04400) Plate, Sheet, and Strip
ASTM B139	Standard Specification for Phosphor Bronze Rod, Bar, and Shapes
ASTM B164	Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire
ASTM B194	Standard Specification for Copper-Beryllium Alloy Plate, Sheet, Strip, and Rolled Bar
ASTM B584	Standard Specification for Copper Alloy Sand Castings for General Applications
ASTM D429	Test Methods for Rubber Property—Adhesion to Rigid Substrates
ASTM D1784	Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Shop Drawings:

Product data sheets for make and model.

Complete catalog information, descriptive literature, specifications, and identification of materials of construction.

Manufacturer's Certificate of Compliance for butterfly valves; full compliance with AWWA C504.

Submittals–Approval Not Required:

Information/Record (IR): Documentation of construction quality control testing as specified herein.

Part 2 – Products

General

Valve to include operator, actuator, hand wheel, chain wheel, extension stem, floor stand, worm and gear operator, operating nut, chain, wrench, and accessories for a complete operation.

Valve to be suitable for intended service. Renewable parts not to be of a lower quality than specified.

Valve same size as adjoining pipe.

Valve ends to suit adjacent piping.

Size operator to operate valve for the full range of pressures and velocities.

Valve to open by turning counterclockwise.

Factory mount operator, actuator, and accessories.

Provide nametag for each valve. Nametag shall include valve tag number and be constructed of 16-Gauge Type 304 stainless steel, letters shall be 3/16-inch imposed, affix to valve with 16- or 18-gauge stainless steel wire.

Materials

Brass and bronze valve components and accessories that have surfaces in contact with liquids other than leachate to be alloys containing less than 16 percent zinc and 2 percent aluminum. Valves in service on leachate lines shall have no bronze, brass, or copper wetted parts.

Approved alloys are of the following ASTM designations:

B61, B62, B98 (Alloy UNS No. C65100, C65500, or C66100), B139 (Alloy UNS No. C51000), B584 (Alloy UNS No. C90300 or C94700), B164, B194, and B127.

Stainless steel Alloy 18-8 may be substituted for bronze.

Factory Finishing

Exposed Valves:

Manufacturer's standard corrosion-resistant coating suitable for intended service.

Safety isolation valves and lockout valves with handles, hand wheels, or chain wheels "safety yellow."

Epoxy Lining and Coating:

Use where specified for individual valves described herein.

In accordance with AWWA C550 unless otherwise specified.

Either two-part liquid material or heat-activated (fusion) material except only heat-activated material if specified as "fusion" or "fusion bonded" epoxy.

Minimum 7-mil dry film thickness except where limited by valve operating tolerances.

Valves

Ball Valves

Type V330 PVC Ball Valve 2 Inches and Smaller: Rated 150 psi at 73 degrees F, with ASTM D1784, Type I, Grade 1 polyvinyl chloride body, ball, and stem, end entry, double union design, solvent-weld socket ends, elastomer seat, Viton or Teflon O-ring stem seals, to block flow in both directions.

Manufacturers and Products:

Nibco; True-Bloc.

ASAHI America; Duo-Bloc.

Or approved equal.

Type V331 PVC Ball Valve 3 and 4 Inches: Rated 150 psi at 73 degrees F, with ASTM D1784 Type I, Grade 1 polyvinyl chloride full port body, Teflon seat, Viton O-ring stem, face and carrier seals, end entry design with dual union, solvent-weld socket ends, or single union ball valve with flanged ends drilled to ANSI B16.1.

Manufacturers and Products:

Nibco.

ASAHI America.

Or approved equal.

PVC 3-Way Ball Valve 2 Inches and Larger: Valves shall be three-way type with port option necessary to either direct flow through leachate filter or bypass flow around leachate filter. Valve configuration shall allow filter replacement simultaneously with flow through filter bypass. Rated 150 psi at 73 degrees F, with ASTM D1784, Type I, Grade 1 polyvinyl chloride body, ball, and stem, double union design, solvent weld socket ends, or flanged ends drilled to ASME B 16.5, Class 150, elastomer seat, Viton or Teflon O-ring stem seals, full ported ball.

Manufacturers and Products:

Spears; True Union 2000.

Or approved equal.

Check and Flap Valve:

Type V609 PVC Self-Closing Check Valve 3 Inches and Smaller: ASTM D1784, Type I, Grade 1, PVC body, rated at 150 psi, Viton seats and seals, stainless steel spring.

Manufacturer and Product:

PLAST-O-MATIC; Series CKS.

PVC Swing Check Valve 4 Inch and Larger:

PVC body, flanged, rated at 150 psi at 70 degrees F minimum, EPDM elastomer seal.

Manufacturers and Product:

Hayward Model 5167.040

Self-Contained Automatic Valves:

Air Release Valve 1/2 Inch to 16 Inches:

1/2-inch through 3-inch NPT inlets and outlets, 4 inch and larger ANSI B16.1 flanged inlet with plain outlet and protective hoods.

Rated 150 psi working pressure, PVC body, EPDM seals.

Manufacturer and Product: IPEX; Series VAFV.

Vacuum Breaker Valve: 1/2-inch NPT inlet and outlet, PVC body, EPDM diaphragm, working pressure 100 psi.

Manufacturer and Product: PLAST-O-MATIC; Series VBM.

Gate Valves:

Type V100 Gate Valve 3 Inches and Smaller (Raw Water): All-bronze, screwed bonnet, single solid wedge gate, nonrising stem, rated 125-pound SWP, 200-pound WOG.

Manufacturers and Products:

Stockham; B103, threaded end.

Crane; 438, threaded end.

Type V130 Resilient Seated Gate Valve, 2 Inches to 12 Inches (Combined Sump and Truck Loading Station):

Iron body, resilient seat, bronze mounted, ANSI Class 125 flanged ends, nonrising stem in accordance with AWWA C509, design working water pressure 200 psig for 2 inches through 12 inches, full port, fusion-epoxy coated inside and outside per AWWA C550.

Provide 2-inch operating nut and operator extensions for the gate valves on the secondary containment piping that drains into the combined sump. Provide handwheel operators for the gate valves at the truck loading station.

Manufacturers and Products:

M&H Valve; AWWA C509.

U.S. Pipe; Metroseal.

Type V130 Resilient Seated Gate Valve, 4 Inches to 12 Inches (Raw Water):

Iron body, resilient seat, bronze mounted, ANSI Class 125 flanged ends, nonrising stem in accordance with AWWA C509, design working water pressure 200 psig for 2 inches through 12 inches, full port, fusion-epoxy coated inside and outside per AWWA C550.

Provide post indicating assembly with detachable crank handle for 12-inch gate valve at raw water tie-in.

Manufacturers and Products:

U.S. Pipe; Metroseal.

Clow Corp; Model F-61XX.

Butterfly Valves:

General:

Valves specified as AWWA C504 to be in full compliance with AWWA C504 and following requirements:

Suitable for throttling operations and infrequent operation after periods of inactivity.

Elastomer seats, which are bonded or vulcanized to the body, shall have adhesive integrity of bond between seat and body assured by testing, with minimum 75-pound pull in accordance with ASTM D429, Method B.

Bubble-tight with rated pressure applied from either side.

No travel stops for disc on interior of body.

Self-adjusting V-type or O-ring shaft seals.

Isolate metal-to-metal thrust bearing surfaces from flowstream.

Valve actuators to meet the requirements of AWWA C504.

Type V530 Butterfly Valve 4 Inches to 24 Inches for Fire Protection Service (Raw Water):

UL Listed and FM Approved, flanged style, AWWA C504 Class 150B valve with cast iron body, aluminum-bronze disc, stainless steel stem, EPDM seat, geared operator with highly visible position indicator and detachable crank handle.

For buried service, provide post indicating assembly with detachable crank handle.

Manufacturers and Product: Pratt; PIVA.

Miscellaneous Valves:

Type V930 Fire Hydrant (Raw Water): Hydrants shall be dry-barrel type conforming to AWWA C502 with valve opening at least 5 inches in diameter and designed so that the flange at the main valve seat can be removed with the main valve seat apparatus remaining intact, closed, and reasonably tight against leakage and with a breakable valve rod coupling and breakable flange connections located no more than 8 inches above the ground grade. Hydrants shall have a 6-inch bell connection, two 2-1/2-inch hose connections, and one 4-1/2-inch pumper connection. Outlets shall have American National Standard fire-hose coupling threads. Working parts shall be bronze. Design, material, and workmanship shall be similar and equal to the latest stock pattern ordinarily produced by the manufacturer. Hydrants shall be painted with one coat of red iron oxide, zinc oxide primer conforming to SSPC Paint 25, and two finish coats of silicone alkyd paint conforming to SSPC Paint 21; color shall be safety yellow. Caps and chains shall be furnished.

Manufacturers and Products: Clow Medallion.

Type V931 Yard Hydrant (Raw Water): Non-freeze yard hydrant with 3/4-inch hose connection. Constructed of manufacturer's standard materials, rated for minimum pressure of 125 psi.

Manufacturers and Products: Wade 8610; or equal.

Accessories

Operating Wrenches (Raw Water): Two each T-handled galvanized operating wrenches for 2-inch square nut operator, 6 feet long.

Manufacturers and Products:

Mueller; No. A-24610.

Clow No.; F-2520.

Two each T-handled galvanized forked operating keys for cross handled valves, 7 feet long.

Cast Iron Valve Box (Raw Water): Designed for traffic loads, sliding type, with minimum of 6-inch ID shaft.

Box: Cast iron with minimum depth of 9 inches.

Lid: Cast iron, minimum depth 3 inches, marked WATER.

Extensions: Cast iron.

Operators

Manual Operator:

General:

Operator force not to exceed 40 pounds under any operating condition, including initial breakaway. Gear reduction operator when force exceeds 40 pounds.

Operator self-locking type or equipped with self-locking device.

Position indicator on quarter-turn valves.

Worm and gear operators one-piece design worm-gears of gear bronze material. Worm hardened alloy steel with thread ground and polished. Traveling nut type operators threader steel reach rods with internally threaded bronze or ductile iron nut.

Exposed Operator:

Galvanized and painted hand wheels.

Lever operators allowed on quarter-turn valves 8 inches and smaller.

Valve handles to take a padlock, and wheels a chain and padlock.

Buried Operator:

Buried service operators on valves larger than 2-1/2 inches shall have a 2-inch AWWA operating nut. Buried operators on valves 2 inches and smaller shall have cross handle for operation by forked key. Enclose moving parts of valve and operator in housing to prevent contact with the soil.

Design buried service operators for quarter-turn valves to withstand 450 foot-pounds of input torque at the FULLY OPEN or FULLY CLOSED positions, grease packed and gasketed to withstand a submersion in water to 10 psi.

Buried valves shall have extension stems, bonnets, and valve boxes.

Leachate Filters

Stainless steel filter housings rated for 150 psi minimum at 120 degrees F and 160 gpm shall be installed as shown on the piping details. Each filter housing shall be installed with (12) 5 micron filter cartridges, each cartridge shall be 40 inches long. The filter housing shall have a mill finish and include installation of a vent valve with drain tubing, drain valves with drain tubing, and filter housing stand. The filter assembly shall have no more than a 10 psi pressure drop across the housing and clean filter cartridges when operating at 160 gpm of water flow. The filter inlet and outlets shall be ASME B16.5, Class 150 flanges. Furnish and install filter cartridges within filter housing, in addition provide a minimum of 72 additional filter cartridges for future filter changeouts.

The filter assembly shall include a wall mounted differential pressure indicator. The differential pressure indicator shall tie-in to the filter housing drain ports. The differential pressure indicator shall include a resettable drag pointer for indication of maximum differential pressure achieved, 4.5-inch dial, and a range of 0-15 psid.

Copper tubing shall be used for differential pressure indicator. Tubing shall be 1/4-inch seamless copper tubing conforming to ASTM B75. Wall thickness, diameter tolerances, and compression type brass fittings shall be in accordance with ASTM B251.

Manufacturers and Products:

Filter Assembly: GE Osmonics; NDV Series w/Hytrex Filter Cartridges (5 micron, 40-inch length).

Differential Pressure Indicator: Wika; 700.05.

Part 3 – Execution

Installation

Flange Ends:

Flanged valve boltholes shall straddle vertical centerline of pipe.

Clean flanged faces, insert gasket and bolts, and tighten nuts progressively and uniformly; do not over tighten.

Screwed Ends:

Clean threads by wire brushing or swabbing.

Apply joint compound.

Valve Orientation:

Install operating stem vertical when valve is installed in horizontal runs of pipe having centerline elevations 4 feet 6 inches or less above finished floor, unless otherwise shown.

Install operating stem horizontal in horizontal runs of pipe having centerline elevations between 4 feet 6 inches and 6 feet 9 inches above finish floor, unless otherwise shown.

Install a line size ball valve and union upstream of each solenoid valve, in-line flow switch, or other in-line electrical device, excluding magnetic flowmeters, for isolation during maintenance.

Locate valve to provide accessibility for control and maintenance. Install access doors in finished walls and plaster ceilings for valve access.

Construction Quality Control

Valves shall be tested prior to leak testing pipelines.

Construction General Contractor shall perform test that valves open and close smoothly.

Construction General Contractor shall count and record number of turns to open and close valve; account for any discrepancies with manufacturer's data.

END OF SECTION 15100

SECTION 15140 – PIPING SUPPORT SYSTEMS SCHED. B

Part 1 – General

Summary

This section is for furnishing and installing leachate piping support systems.

References

The following is a list of standards, which may be referenced in this section:

ASTM INTERNATIONAL (ASTM)

ASTM A525	Standard Specification for General Requirements for Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process

BUILDING OFFICIALS AND CODE ADMINISTRATORS (BOCA)

Basic Building Code

INTERNATIONAL CONFERENCE OF BUILDING OFFICIALS (ICBO)

Uniform Building Code

MANUFACTURERS' STANDARDIZATION SOCIETY (MSS)

SP 58	Pipe Hangers and Supports-Materials, Design and Manufacture
SP 69	Pipe Hangers and Supports-Selection and Application
SP 89	Pipe Hangers and Supports-Fabrication and Installation

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Shop Drawings:

Drawings of each piping support system to scale shown, locating each support, brace, hanger, guide, component, and anchor. Identify support, hanger, guide, and anchor type by catalog number and shop drawing detail number.

Revisions to support systems resulting from changes in related piping system layout or addition of flexible joints.

Definitions

Ferrous Metal: Iron, steel, stainless steel, and alloys with iron as principal component.

Wetted or Submerged: Submerged, less than 1 foot above liquid surface, below top of channel wall, under cover or slab of channel or tank, or in other damp locations.

Design Requirements

General:

Piping Smaller than 30 Inches: Supports are shown only where specific types and locations are required; additional pipe supports may be required.

Meet requirements of MSS SP 58, MSS SP 69, and MSS SP 89.

Pipe Support Systems:

Support Load: Dead loads imposed by weight of pipes filled with water, except air and gas pipes, plus insulation.

Seismic Load: Seismic performance category forces with seismic loads in accordance with local codes.

Safety Factor: Minimum of 5.

Maximum Support Spacing and Minimum Rod Size:

Steel or Ductile Iron Piping:

Maximum Support/ Pipe Size	Minimum Rod Size Hanger Spacing	Single Rod Hangers
1-inch and smaller	6 feet	1/4-inch
1-1/2-inch thru 2-1/2-inch	8 feet	1/4-inch
3-inch and 4-inch	10 feet	3/8-inch

Plastic and Fiberglass Piping:

Maximum Support Spacing: As recommended by manufacturer for flow temperature in pipe. Pipe insulation shall be included in the selection of maximum pipe support spacing.

Minimum Hanger Rod Sizing: Same as listed for steel pipe.

Framing Support System:

Beams: Size such that beam stress does not exceed 25,000 psi and maximum deflection does not exceed 1/240 of span.

Column Members: Size in accordance with manufacturer's recommended method.

Support Loads: Calculate using weight of pipes filled with water.

Maximum Spans:

Steel and Ductile Iron Pipe, 3-Inch Diameter and Larger: 10-foot centers, unless otherwise shown.

Other Pipelines and Special Situations: May require supplementary hangers and supports.

Electrical Conduit Support: Include in design of framing support system.

Anchoring Devices: Design, size, and space support anchoring devices, including anchor bolts, inserts, and other devices used to anchor support, to withstand shear and pullout loads imposed by loading and spacing on each particular support.

Vertical Sway Bracing: 10-foot maximum centers, or as shown.

Part 2 – Products

General

When specified items are not available, fabricate pipe supports of correct material and to general configuration indicated by catalogs.

Special support and hanger details are shown for cases where standard catalog supports are inapplicable.

Materials:

Wetted and Submerged: Stainless steel.

Atmospheric Exposed: Galvanized or painted steel.

Hangers

Clevis Type:

MSS SP 58, Type 1 or 6.

Grinnell; Figure 104 or 260.

B-Line; Figure B3198 or B3100.

Hinged Split-Ring Pipe Clamp:

MSS SP 58, Type 6 or 12.

Grinnell; Figure 104.

B-Line; Figure B3198H.

Hanger Rods, Clevises, Nuts, Sockets, and Turnbuckles: In accordance with MSS SP 58.

Attachments:

I-Beam Clamp: Concentric loading type, MSS SP 58, Type 21, 28, 29, or 30, which engage both sides of flange.

Concrete Insert: MSS SP 58, Type 18, continuous channel insert with load rating not less than that of hanger rod it supports.

Saddle Supports

Pedestal Type: Schedule 40 pipe stanchion, saddle, and anchoring flange.

Nonadjustable Saddle:

MSS SP 58, Type 37 with U-bolt.

Grinnell; Figure 259.

B-Line; Figure B3090.

Adjustable Saddle:

MSS SP 58, Type 38 without clamp.

Grinnell; Figure 264.

B-Line; Figure B3093.

Wall Brackets

Welded Steel Bracket:

MSS SP 58, Type 33 (heavy-duty).

Grinnell; Figure 199.

B-Line; Figure B3607.

One-Hole Clamp: Grinnell; Figure 126.

Channel Type:

Unistrut.

Kin-Line.

Pipe Clamps

Riser Clamp:

MSS SP 58, Type 8.

Grinnell; Figure 261.

B-Line; Figure B3373 or approved equal.

Channel Type Support Systems

Material:

Galvanized: Pre-galvanized in accordance with ASTM A525, Class G90, or hot-dip galvanized after fabrication.

Stainless Steel: Type 304 stainless steel.

Channel Size: 12-gauge, 1-5/8-inch wide series minimum.

Members and Connections: Design for all loads with safety factor of 5.

Manufacturers and Products:

Kin-Line; Series CI3812.
Unistrut; Series P3200.
B-Line; B2000 Series, or approved equal.

Anchoring Systems

Material:

Wetted and Submerged: Stainless steel.

Atmospheric Exposed: Galvanized.

Size: Sized by equipment manufacturer, 1/2-inch minimum diameter.

Shop/Factory Finishing

Prepare, prime, and finish coat in accordance with:

Surface preparation with abrasive blast or centrifugal wheel blast (SP10).

Paint with:

One coat, 2.5 minimum dry film thickness (MDFT) of Epolon rust inhibitor primer.
One coat, 2.5 MDFT Epolon Multi-Mill Epoxy.
One coat, 1.5 MDFT Acrolon II, No. 2200 Series.

Part 3 – Execution

Installation

General:

Install support systems in accordance with MSS SP 69, Pipe Hangers and Supports-Selection and Application and MSS SP 89, Pipe Hangers and Supports-Fabrication and Installation, unless shown otherwise.

Support piping connections to equipment by pipe support and not by the equipment.

Support large or heavy valves, fittings, and appurtenances independently of connected piping.

Support no pipe from the pipe above it.

Support pipe at changes in direction or in elevation, adjacent to flexible joints and couplings, and where shown.

Do not install pipe supports and hangers in equipment access areas or bridge crane runs.

Brace hanging pipes against horizontal movement by both longitudinal and lateral sway bracing.

Install lateral supports for seismic loads at all changes in direction.

Repair mounting surfaces to original condition after attachments are made.

Standard Pipe Supports:

Horizontal Suspended Piping:

Single Pipes: Adjustable swivel-ring, splint-ring, or clevis hangers.

Grouped Pipes:

Trapeze hanger systems.

Furnish galvanized steel protection shield and oversized hangers for all insulated pipe.

Furnish precut sections of rigid insulation with vapor barrier at hangers for all insulated pipe.

Horizontal Piping Supported From Walls:

Single Pipes: Wall brackets or wall clips attached to wall with anchors. Clips attached to wall mounted framing also acceptable.

Stacked Piping:

Wall-mounted framing system and clips acceptable for piping smaller than 3-inch minimal diameter.

Piping clamps, which resist axial movement of pipe through support not acceptable.

Wall-mounted piping clips not acceptable for insulated piping.

Horizontal Piping Supported From Floors:

Stanchion Type:

Pedestal type; adjustable with stanchion, saddle, and anchoring flange.

Use yoked saddles for piping whose centerline elevation is 18 inches or greater above the floor and for all exterior installations.

Floor Mounted Channel Supports:

Use for piping smaller than 3-inch nominal diameter running along floors and in trenches at piping elevations lower than can be accommodated using pedestal pipe supports.

Attach channel framing to floors with anchor bolts.

Attach pipe to channel with clips or pipe clamps.

Vertical Pipe: Support with wall brackets and base elbow or riser clamps on floor penetrations.

Standard Attachments:

To Steel Beams: I-beam clamp or welded attachments.

To Concrete Walls: Concrete inserts or brackets or clip angles with anchor bolts.

Field Finishing

Paint atmospheric exposed surfaces of black and hot-dip galvanized steel components as specified in Article SHOP/FACTORY FINISHING.

END OF SECTION 15140

SECTION 15500 – HEATING, VENTILATING, AND AIR CONDITIONING SYSTEMS SCHED. B

Part 1 – General

References

The following is a list of standards, which may be referenced in this section:

AIR MOVING AND CONDITIONING ASSOCIATION (AMCA)

AMCA 99	Air Movement and Control Association Standards Handbook
AMCA 210	Laboratory Methods of Testing Fans for Aerodynamic Performance Rating
AMCA 300	Reverberant Room Method for Sound Testing of Fans
AMCA 2401	Impeller Diameters and Outlet Areas for Centrifugal Fans and Metric Equivalents

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 52	Method of Testing Air-Cleaning Devices Used in General Ventilation for Removing Particulate Matter
ASHRAE 90A	Energy Conservation in New Building Design

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 112	Standard Test Procedure for Polyphase Induction Motors and Generators

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1-	Motors and Generators

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

Guidelines for Seismic Restraints of Mechanical Systems

HVAC Testing, Adjusting, and Balancing Manual

UNDERWRITERS LABORATORIES INC. (UL)

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Shop Drawings:

Complete specifications, descriptive drawings, catalog cuts, and descriptive literature that include make, model, dimensions, weight of equipment, and electrical schematics for the following products:

Air conditioning units.

Unit heaters.

Motorized dampers.

Complete performance data that indicate full compliance with the Specifications.

Recommended procedures for protection and handling of equipment and materials prior to installation.

Manufacturer's certification of factory testing to establish conformance with specified requirements for the unit heater and air conditioning unit.

For motors specified to be energy efficient type, certified copy of test report for identical motor tested, in accordance with NEMA MG 1-12.53a and IEEE Standard 112, Test Method B, showing full load efficiency.

Detailed information on structural, mechanical, electrical, or other modifications necessary to adapt the arrangement or details shown to the equipment furnished.

Submittals–Approval Not Required

Information/Record (IR):

List of recommended spare parts for equipment and materials specified.

Manufacturer's warranty.

Operations and maintenance manuals including recommended preventative maintenance tasks and frequencies for performance of those tasks.

Extra Materials

Furnish, tag, and box for shipment and storage the following spare parts:

Filters: Four complete sets per unit.

Special Guarantee

Manufacturer shall provide standard warranty.

Part 2 – Products

General:

Heating Equipment: Minimum operating efficiencies as specified in Chapter 6 of ASHRAE Standard 90A, and the State of Washington Energy Code.

Wall-Mounted Air-Cooled Packaged Air Conditioning Units (For Equipment Identification Numbers See Supplement)

General:

Packaged through-the-wall air conditioning unit.

Cooling section.

Heating section.

Controls.

Fans.

Filters.

All contained in a standard weatherproof enclosure.

UL listed.

Enclosure:

Zinc-coated steel finished with manufacturer's standard baked enamel paint.

Adjustable discharge grille.

Return grille.

Permanent filter.

Internal sound attenuation.

Controls with adjustable thermostat.

Fan speed switch with HIGH/LOW manual selections.

Heating Section:

Low-density electric heating elements.
Built-in overheat protection.

Cooling Section:

Hermetic compressor.
Air-cooled condenser coil.
Evaporator coil.
Drain pan with drain line connections.
Direct-drive evaporator.
Condenser fans.
Fan motors with integral overload protection.
Operating and safety controls.
Operating charges of refrigerant and oil.

Capacity (219A-LH-AC-001 and 219E-LH-AC-001): 600 cfm at fan medium speed, at 0.3 inch of water column static pressure, minimum outside air 100 cfm, cooling capacity 25,110 Btuh total, 17,530 Btuh sensible, 85/72 degrees F DB/WB entering air temperature, and 100 degrees F ambient, 208-volt, single-phase power supply, MCA17, breaker size 20 amps.

Capacity (219A1-LH-AC-002 and 219E1-LH-AC-002): 360 cfm at 0.3 inch of water column static pressure, minimum outside air 100 cfm, cooling capacity 11,840 Btuh total, 8,130 Btuh sensible, 85/72 degrees F DB/FB entering air temperature, and 100 degrees F ambient, 208-volt, single-phase power supply, MCA 8 amps, breaker 15 amps.

Manufacturers and Products:

Bard:

Model (219A-LH-AC-001 and 219E-LH-AC-001): WA241-A-00-EXXXXA.

Model (219A1-LH-AC-002 and 219E1-LH-AC-002): WA121-A-00-EXXXXJ.

Or approved equal.

Electric Unit Heater (For Equipment Identification Numbers See Supplement)

General:

Heater shall be installed and wired in accordance with the manufacturer's recommendations.
Unit heater shall be UL listed.

Casing:

Fabricated of die-formed, heavy-gauge steel and finished in high gloss, baked enamel.
Supply air shall be drawn through a stamped louver periphery evenly across the heating element.
Discharge air shall be through an outward drawn Venturi.
Cabinet shall have adjustable discharge louvers.
Cabinet shall be furnished with an access door.
Wiring diagram shall be permanently attached to the inside at the access door.

Elements:

Elements shall be high mass, all steel tubular finned type, copper brazed.

Elements shall be centrally located and installed in fixed element banks.

Motor:

Motor shall be totally enclosed, all angle industrial rated.

Bearings shall be sealed and permanently lubricated.

Fan:

Fan blades shall be of the axial flow type.

Fan speed shall not exceed 1,600 rpm.

Wiring:

Unit heater shall be factory prewired.

Unit heater shall have balanced phases.

Unit heater shall be equipped with automatic reset thermal overload.

Controls: Wall-mounted thermostat.

Manufacturers and Products:

Capacity: 3.3 kW, 460 volts, three-phase, horizontal discharge.

Trane; UHEC-033DACA.

Relief Louver (For Equipment Identification Numbers See Supplement)

Extruded aluminum frame.

Double drainable blades.

Bird screen.

Size: Louver size as shown on Drawings.

Manufacturer And Product: Ruskin; Model ELF 375DD.

Damper Actuator (For Equipment Identification Numbers See Supplement)

Line voltage actuator.

Spring return.

UL listed.

NEMA 2 housing.

60-inch-pound torque.

Sequence of Operation: Actuator shall be interlocked with an economizer. Actuator shall be energized when economizer is on, and shall be closed when economizer is off.

Manufacturer and Product: Belimo; NF120US, 120 volts, single-phase, less than 60 seconds return time.

Sequence Of Control

Air Conditioning Units:

Air conditioning unit shall be started by the signal from wall-mounted thermostat if the room temperature rises above 85 degrees F.

Thermostat set point is 80 degrees F.

If the room temperature drops below 70 degrees F, the modulating type return, outside and exhaust air dampers will be adjusted to maintain room temperature.

When outside air temperature is between 50 degrees F and 65 degrees F, air conditioning unit will provide a free cooling.

Unit Heaters: Unit heater will be operating in the ON-OFF sequence to maintain 50 degrees F temperature.

Part 3 – Execution

Installation

Install equipment and systems in accordance with manufacturers' instructions.

Packaged Wall-Mounted Air Conditioning Units:

Mount unit in accordance with manufacturer's instructions.
Provide access for maintenance.
Seal watertight to wall.

Electric Unit Heater: Install in accordance with recommendations of NFPA 90A.

Supplements

The supplement listed below, following "END OF SECTION," is a part of this Specification.

Equipment Identification Numbers.

END OF SECTION 15500

Equipment Identification Numbers

Location	Equipment Name	Equipment Number
Cell No. 1 Crest Pad Building	Air Conditioning Unit	219A-LH-AC-001
	Unit Heater	219A-LH-UH-001
	Relief Louver with Motorized Damper	219A-LH-MD-001
Cell No. 1 Leachate Transfer	Air Conditioning Unit	219A1-LH-AC-002
	Building Unit Heater	219A1-LH-UH-002
	Relief Louver with Motorized Damper	219A1-LH-MD-002
Cell No. 2 Crest Pad Building	Air Conditioning Unit	219E-LH-AC-001
	Unit Heater	219E-LH-UH-001
	Relief Louver with Motorized Damper	219E-LH-MD-001
Cell No. 2 Leachate Transfer	Air Conditioning Unit	219E1-LH-AC-002
	Building Unit Heater	219E1-LH-UH-002
	Relief Louver with Motorized Damper	219E1-LH-MD-002

SECTION 15992 – PIPING LEAKAGE TESTING SCHED. B

Part 1 – General

Summary

This section is for leak testing (construction quality control) leachate piping and associated components.

Submittals–Approval Required:

See Section 01300, SUBMITTALS, for submittal procedures.

Testing Plan:

Submit prior to testing and include at least the information that follows.
Testing dates.
Piping systems and section(s) to be tested.
Test type.
Method of isolation.
Sample of test report form.

Certifications of Calibration: Testing equipment.

Submittals–Approval Not Required

Information/Record (IR): Certified Test Report.

Provide Manufacturer's calibration recommendations and current calibrations of pressure gauge(s).

Part 2 – Products (Not Used)

Part 3 – Execution

Preparation

Notify Construction Manager in writing 5 days in advance of testing. Perform testing in presence of Construction Manager.

Pressure Piping:

Install temporary thrust blocking or other restraint as necessary to protect adjacent piping or equipment and make taps in piping prior to testing.
Prior to test, remove or suitably isolate appurtenant instruments or devices that could be damaged by pressure testing.
Items that do not require testing include: Piping between wetwells and wetwell isolation valves, tank overflows to atmospheric vented drains, tank atmospheric vents, and slotted piping.
Test section may be filled with water and allowed to stand under low pressure prior to testing.

Other Piping:

Perform testing of other pipe service types using the same methods outlined for pressure piping.

Hydrostatic Test

General: Hydrostatic testing shall be performed on all single-wall pipe, inner carrier pipes, and all PVC piping.

Fluid: Clean water of such quality to prevent corrosion of materials in piping system.

Test Pressure:

Per Section 15060, Piping–General

Exposed Piping:

Perform testing on installed piping prior to application of insulation.

Prior to pressure testing, check all manually operated valves for smooth operation and count and record number of turns to open and close each valve.

Maximum Filling Velocity: 0.25 feet per second, applied over full area of pipe.

Vent piping during filling. Open vents at high points of piping system or loosen flanges, using at least four bolts, or use equipment vents to purge air pockets.

HDPE Piping

Test Procedure: The test procedure consists of an initial expansion phase and then the test phase. Prior to the test procedure the test medium and pipe test section shall be allowed time to equalize in temperature. Testing shall not be allowed if temperatures of the test medium or pipe test section exceed 100 degrees F.

Maintain the test pressure for a period of 3 hours during the initial expansion phase by adding water as needed.

At the beginning of the test phase after the initial expansion phase, reduce pressure by 10 psi. Maintain this test pressure for a period of 1 hour.

Under no circumstances shall the testing be allowed to exceed 8 hours.

Acceptance Criteria:

The test phase is passed and the pressure test is acceptable if the pressure remains steady (within 5 percent of the test phase beginning pressure) for 1 hour and there are no indications, visible or otherwise, of leakage.

If acceptance criteria is not met, any leakage points shall be fixed and any other changes made to the piping system as necessary. Retest and repeat until acceptance criteria is met.

Empty pipe of water prior to final cleaning.

Buried Piping: Test piping using the same procedure as outlined for exposed piping as described above.

PVC and Non-HDPE Piping:

Perform testing on installed piping prior to application of insulation.

Prior to pressure testing, check all manually operated valves for smooth operation and count and record number of turns to open and close each valve.

Maximum Filling Velocity:

0.25 foot per second, applied over full area of pipe.

Vent piping during filling. Open vents at high points of piping system or loosen flanges, using at least four bolts, or use equipment vents to purge air pockets.

Maintain hydrostatic test pressure continuously for 30 minutes, minimum, and for such additional time as necessary to conduct examinations for leakage. No fluid shall be added to the system, and system shall not drop below 95 percent of the test pressure during the test period.

Examine exposed joints and connections for leakage.

No loss of fluid allowed. Find any leakage points, fix, and retest as specified.

Empty pipe of water prior to final cleaning or disinfection.

Pneumatic Test

General: Pneumatic testing shall be performed for outer pipe of double-wall HDPE piping and atmospheric drains.

Double-Wall Pipe: Inner carrier pipe shall be full of water when outer containment pipe is tested to prevent damage to carrier pipe and riser pipes.

Equipment:

Gauges shall be calibrated within manufacturer's recommended frequency and calibration shall be current.

Install gauges, air piping manifolds, and valves at ground surface.

Provide pressure release device, such as rupture disc or pressure relief valve, to relieve pressure at 10 psi or less.

Restrain plugs used to close lines to prevent blowoff.

Procedure:

Maintain test pressure for 10 to 60 minutes but not for more than 60 minutes.

Slowly introduce air into pipe section until internal air pressure reaches required test pressure. (A maximum 5% gauge loss is acceptable.)

Allow 2 minutes minimum for air temperature to stabilize.

Examine exposed joints and connections for leakage using liquid bubble tests or other method to determine source of leakage approved by the construction manager.

No loss in pressure allowed. Find any leakage points, fix, and retest as specified.

Defective Piping Sections: Replace or test and seal individual joints, and retest as specified.

END OF SECTION 15992

1 DIVISION 16 – ELECTRICAL

SECTION 16005 – ELECTRICAL

Part 1 – General

UL And NRTL Compliance

Materials manufactured within the scope of UL or another nationally recognized testing laboratory (NRTL) shall conform to UL or NRTL standards and have an applied UL or NRTL listing mark. References to UL throughout this section imply conformity with UL or NRTL standards and guidelines.

Electrical system process control panels shall be manufactured, assembled, tested, approved, and clearly labeled in accordance with UL 508A, prior to delivery to construction site.

Approval By Authority Having Jurisdiction (AHJ)

Provide all work in accordance with NFPA 70, National Electrical Code, ANSI C2 National Electrical Safety Code (NESC), and where required by Hanford CH2M HILL authority having jurisdiction (AHJ), as defined under Division 1, material and equipment shall be labeled or listed by a nationally recognized testing laboratory or other organization acceptable to AHJ, in order to provide a basis for approval under NEC.

All material and equipment shall be tested after installation by a qualified testing firm (as specified in Section 16080, ELECTRICAL TESTING), or other organization acceptable to AHJ, in order to provide a basis for approval under NEC. Construction General Contractor is responsible for providing qualified testing firm and shall coordinate with the Construction Manager for NEC inspection services.

All material and equipment shall be provided with a visibly attached label by a nationally recognized testing laboratory or other organization acceptable to AHJ, prior to delivery to construction site.

All motor control panels shall be provided in accordance with UL 508 and with a visibly attached label by a nationally recognized testing laboratory or other organization acceptable to AHJ, prior to delivery to construction site.

All equipment shall be installed per manufacturer's instructions, with NFPA 70, NESC, and with other applicable requirements.

All electrical work including conduit, wiring, and terminal and splice connections shall be accomplished by a state-registered and approved electrical journeyman or a state-registered and approved electrical apprentice under supervision of state-registered and approved electrical journeyman.

Electrical Description Of Work

Schedule A Work:

Provide and install primary aerial cables, power poles, fuse-cutouts, supports, primary riser conductors, and ancillary equipment as needed to extend existing primary power circuit C8–L6 from 4th street to new IDF pad mount service transformer as shown on Drawings and in accordance with Section 16312, OVERHEAD ELECTRICAL DISTRIBUTION.

Provide and install IDF pad mount service transformer as shown on Drawings and in accordance with Section 16270, Oil-Filled Pad Mounted Transformer.

Provide site preparation, pad mount slab and concrete work, and grounding as necessary to facilitate utility vault and pad mount service transformer installation.

For primary (15kV) conductors, which route from riser pole fuse cut-outs to pad mount service transformer, reference CONDUCTOR OVER 600 VOLTS this Section.

Fluor Hanford Electrical Utilities will make final aerial conductor connections to existing primary circuit C8–L6, and primary connections to pad mount service transformer.

Coordinate installation of all primary service equipment, testing, and secondary metering with Construction Manager, who will contact Fluor Hanford Electrical Utilities (FH EU).

Schedule B Work:

Provide and install power conduits and cables to electrical service gear for each crest pad and leachate transfer building. Power conduits and cables shall route from crest pad building electrical service gear to power handhole(s) and service transformer as shown on Drawings.

FH EU will make final secondary conductor connections within low voltage compartment of pad mount service transformer after cables are routed under this schedule of work.

Coordinate with FH EU installation of secondary cables and energizing of secondary service equipment.

Provide and install communication conduits and cables for each crest pad and leachate transfer building communication service. Communication conduits and cables shall route below grade from crest pad building to communication handhole(s) as shown on Drawings.

Provide and install electrical service gear for each crest pad and leachate transfer building including: motor control center (MCC), MCC integrally mounted lighting panel and transformer, and grounding electrode system.

Motor control center shall provide 480V, three-phase, three-wire power to pump motors, power outlets, building heaters, air monitoring transformer and panel assemblies, and motor starters.

Lighting panel and transformer shall provide 208/120V, three-phase, four-wire power for instruments, lighting, receptacles, small motor loads, and miscellaneous panels.

Lighting panel and transformer(s) shall provide 120/240V, single-phase, three-wire power for air monitoring equipment and ancillary lighting and receptacles.

Provide and install grounding electrode system at each crest pad and leachate transfer building. Bond service gear, lighting transformer, power and communication panels, and metal structures (i.e., leachate storage tank, buildings) to grounding electrode system.

Provide and install power conduits and cables to the following three-phase equipment:

- Building unit heaters.
- Leachate collection and removal and leak detection system pumps.
- Combined sump pumps.
- Leachate transfer pumps.
- Portable generator power outlets.

Provide and install power conduits and cables to the following single-phase equipment:

- Building interior and exterior lighting.
- Building receptacles.
- Building control panels.
- Building air conditioning units.
- Heat tracing.
- Air monitoring equipment.

Provide and install control and signal conduits and cables to the following instrumentation:

- Process local control panels.
- Building temperature transmitters.
- Building sump level floats and panels.
- Building ventilation thermostats.

Leachate collection and removal and leak detection system pump flow meters and submersible pressure transmitters.

Storage tank level transmitters and switches.

Leachate collection carrier pipe leak detection level switches, mounted in combined sumps' interstitial spaces.

Combined sump level floats.

Leachate transfer flow meters and transmitters.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Product Data:

Primary and secondary service entrance and metering equipment.

Boxes and device plates.

Junction and pullboxes.

Precast utility vaults, manholes and handholes.

Wiring devices.

Panelboards and mini-power centers.

Circuit breakers and switches.

Motor-rated switches.

Control devices, terminal blocks, and relays.

Contactors.

Transformers.

Support and framing channels.

Nameplates and nameplate schedule.

TVSS equipment.

Volt and current meters.

Conduit, fittings, and accessories.

Wireways.

Conductors, cable, and accessories.

Motors:

Nameplate data, detailed information on any special features.

Grounding materials.

Motor Controls: Arrangement drawings, ratings, schematic and wiring diagrams, bill-of-materials, nameplate schedule, manufacturer information on components.

Local Control Panels:

Arrangement drawings, schematic and wiring diagrams, bill of materials, nameplate schedule, manufacturer information on components.

Luminaires.

Factory test reports.

Submittals–Approval Not Required

Information/Record (IR):

Field test reports.

Signed permits indicating Work is acceptable to regulatory authorities having jurisdiction.

Operation and Maintenance Data:

Provide for all equipment, as well as each device having features that can require adjustment, configuration, or maintenance, in accordance with Division 1, including recommended preventative maintenance tasks and frequencies for performance of those tasks.

Minimum information shall include manufacturer's preprinted instruction manual, one copy of the approved submittal information for the item, tabulation of any settings, and copies of any test reports.

Environmental Conditions

Provide equipment and conduit systems approved for installing in the following environmental conditions:

Climatic and Geographic Site Conditions:

Site Elevation: 1,000 feet.

Relative Humidity: 90 percent maximum at 30 degrees F dry bulb, 15 percent minimum at 60 degrees F dry bulb.

Uniform Building Code: Seismic Zone 2B.

Temperature: 105 degrees F max. 0 degrees F min.

Enclosures and Environmental Conditions:

Provide and install NEMA 250 Type 4X, Type 304 stainless steel (corrosion resistant, wash down protection) enclosures in process mechanical and wash down indoor locations unless otherwise noted within this section.

Provide and install NEMA 250 Type 3 (dust, rain and ice protection) enclosures in outdoor locations unless otherwise noted in this section.

Provide NEMA 250 Type 12 (dust protection) enclosures for indoor dry protected locations unless otherwise noted in this section.

Labeling: Install permanent labels on all electrical panels, cabinets, disconnects, motor starters, major equipment or components, receptacles, and switches.

Part 2 – Products

General

Products shall comply with all applicable provisions of NFPA 70.

Like Items of Equipment: End products of one manufacturer in order to achieve standardization for operation, maintenance, spare parts, and manufacturer's service.

Equipment and Devices Installed Outdoors or in Unheated Enclosures: Capable of continuous operation within ambient temperature ranges identified under ENVIRONMENTAL CONDITIONS in this section.

Hazardous Areas: Products shall be acceptable to the regulatory authority having jurisdiction for the interior of the combined sumps. Class 1, Division 2, Groups C and D.

Equipment Finish: Manufacturer's standard finish color, except where specific color is indicated.

Service Entrance

Coordinate all service entrance work with Construction Manager, who will in turn coordinate with Fluor Hanford Electrical Utilities, contact person: Cris Carlson, P.E., 509-521-2823.

Utility Metering

Watt-hour Meter: Socket type, for three-phase, 4-wire wye service, self-contained, with relay option board having output for watt KYZ pulses, 480 volt, class 200, form 16S, Elster alpha plus, type A1D+.

Meter Socket: Provide with manual circuit closing blocks, 7 terminal, 200 amp, 600 volt, Milbank type U3517-XL.

Meter Base Hub: Provide for standard RL opening, 2-inch, Milbank type A7517.

Lighting And Power Distribution Panelboard

NEMA PB 1, NFPA 70, and UL 67.

Panelboards and Circuit Breakers: Suitable for use with 75 degrees C copper wire at full NFPA 70, 75 degrees C ampacity.

Short-Circuit Current Equipment Rating: Fully rated; series connected unacceptable.

Rating: Applicable to a system with available short-circuit current of 10,000 amperes rms symmetrical at 208/120 volts and 120/240 volts.

Ground Fault Circuit Interrupter (GFCI): UL Class A GFCI, 5-mA trip, 10,000-amp interrupting capacity circuit breakers.

Ground Fault Equipment Protection (GFEP): 30-mA trip, 10,000-amp interrupting capacity circuit breaker, UL listed for equipment ground fault protection.

Interior Panelboard: NEMA 250, Type 12 unless otherwise noted.

Material: Code-gauge, hot-dip galvanized sheet steel, with reinforced steel frame.

Wiring Gutter: Minimum 4 inches square; both sides, top and bottom.

Front: Fastened with adjustable clamps.

Interior:

Factory assembled; complete with circuit breakers.

Capable of circuit breaker replacement without disturbing adjacent circuit breakers or without removing main bus.

Spaces: Cover openings with easily removable metal cover.

Circuit Directory: Metal frame with transparent plastic face and enclosed card on interior of door.

Bus Bar:

Material:

Copper and/or tin-plated copper full sized throughout length.

Provide for mounting of future circuit breakers along full length of bus regardless of number of units and spaces shown. Machine, drill, and tap as required for current and future positions.

Neutral bus with at least two (neutral and ground) terminal screws for each circuit.

Note: Do not install multiwire branch circuits that share common neutral. Install neutral for each 120-volt branch circuit.

Lugs and Connection Points:

Suitable for copper conductors.

Solderless main lugs for main, neutral, and ground bus bars.

Bolt together and rigidly support bus bars and connection straps on molded insulators.

Circuit Breakers:

NEMA AB 1 and UL 489.

Thermal-magnetic, quick-make, quick-break, molded case, of indicating type showing ON/OFF and TRIPPED positions of operating handle.

Noninterchangeable, in accordance with NFPA 70.

Locking: Provisions for handle padlocking, unless otherwise shown.

Type:

Bolt-on circuit breakers in all panelboards.

Multipole circuit breakers designed to automatically open all poles when an overload occurs on one pole.

Do not substitute single-pole circuit breakers with handle ties for multipole breakers.

Do not use tandem or dual circuit breakers in normal single-pole spaces.

Ground Fault Circuit Interrupter (GFCI):

Equip with conventional thermal-magnetic trip and ground fault sensor rated to trip in 0.025 second for a 5-mA ground fault (UL 943, Class A sensitivity).

Sensor with same rating as circuit breaker and a push-to-test button.

Manufacturers:

Square D

Cutler-Hammer

General Electric

Allen-Bradley

Mini-Power Center (MPC)

General: Transformer, primary and secondary main circuit breakers, and secondary panelboard section enclosed in NEMA 250, Type 3 enclosure.

Transformer:

Type: Dry, self-cooled, encapsulated.

Insulation: Manufacturer's standard, with UL 1561 temperature rise.

Full Capacity: 2-1/2 percent voltage taps, two above and two below normal voltage.

Primary Voltage: 480, three-phase; 480 single-phase as shown.

Secondary Voltage: 208/120 volts, three-phase, four-wire; 120/240 volts, single-phase, three-wire as shown.

Size: 7.5 kVA and 15 kVA as shown.

Panelboard: UL 489, fully rated.

Type: Thermal-magnetic, quick-make, quick-break, indicating, with noninterchangeable molded case circuit breakers.

Number and Breaker Ampere Ratings: Refer to Panelboard Schedule.

Manufacturers:

Square D Co.

Cutler-Hammer.

General Electric Co.

Lighting And Power Distribution Stepdown Transformer (0-600 Volts)

Type: Self-cooled, two-winding.

UL 1561 and NEMA ST 20.

Insulation Class/Temperature Rise: 115 Degrees F.

Core and Coil:

30 kVA or Less: Encapsulated.

Voltage Taps: Full capacity, 2-1/2 percent, two above and two below normal voltage rating.

Sound Level: Not to exceed NEMA ST 20 levels.

Vibration isolators to minimize and isolate sound transmission.

Manufacturers:

Square D
Cutler-Hammer/Westinghouse
General Electric.

Local Control Panels

Enclosure:

Reference ENVIRONMENTAL CONDITIONS in this section.

Minimum Metal Thickness: 14 gauge.

Doors: Rubber gasketed with continuous hinge.

Incandescent Light: Hand switch controlled, 100-watt.

Receptacle: Breaker protected 120-volt, 15-amp duplex.

Finish: Internal and external surfaces (NEMA 250, Type 12 only):

Sand panel; remove mill scale, rust, grease, and oil.
Fill imperfections and sand smooth.
Paint with one coat of epoxy coating metal primer, two finish coats of two-component type epoxy enamel.
Sand surfaces lightly between coats.

Final Dry Film Thickness: Minimum 3 mils.

Size panels to adequately dissipate heat generated by equipment mounted in or on panel.

Manufacturers:

Hoffman.
H. F. Cox.

Wiring:

Power and Control Wiring: 600-volt class, insulated, stranded copper.

Size: Minimum No. 14 AWG enclosed in either sheet metal raceway or plastic wiring duct.

Signal Circuit Wiring: Twisted shielded pairs minimum No. 16 AWG, separated at least 6 inches from power wiring.

Identification: Permanent heat impregnated polyvinyl chloride (PVC) alpha-numeric labels.

Safety Switches

UL 98 listed for use and location of installation.

Type: Visible blade, fusible.

Class: Heavy-duty.

Enclosures: Reference ENVIRONMENTAL CONDITIONS in this section.

Circuit Breaker, Individual, 0 To 600 Volts:

UL 489 listed for use at location of installation.

Minimum Interrupt Rating: As shown.

Thermal-magnetic, quick-make, quick-break, indicating type, showing ON/OFF and TRIPPED indicating positions of operating handle.

Suitable for use with 75 degrees C wire at full NFPA 70, 75 degrees C ampacity.

Locking: Provisions for padlocking handle.

Enclosure: Reference ENVIRONMENTAL CONDITIONS in this section.

Interlock: Enclosure and switch shall interlock to prevent opening cover with breaker in the ON position.

Manufacturers:

Square D Co.

Cutler-Hammer.

General Electric Co.

Allen-Bradley.

Fused Switch, Individual, 0 To 600 Volts:

UL 98 listed for use and location of installation.

NEMA KS 1 and UL 98 Listed for application to system with available short-circuit current as shown.

Quick-make, quick-break, motor rated, load-break, heavy-duty (HD) type with external markings clearly indicating ON/OFF positions.

Suitable for use with 75 degrees C wire at full NFPA 70, 75 degrees C ampacity.

Fuse mountings shall reject Class H fuses and accept only current-limiting fuses specified.

Enclosure: Reference ENVIRONMENTAL CONDITIONS in this section.

Interlock: Enclosure and switch to prevent opening cover with switch in ON position.

Manufacturers:

Square D Co.

Cutler-Hammer.

General Electric Co.

Fuse, 0 To 600 Volts

Current-limiting, with 42,000 max ampere rms interrupting rating.

Provide to fit mountings specified with switches and features to reject Class H fuses.

Motor and Transformer Circuits, 0 to 600 Volts:

Amperage: 0 to 600.

UL 198E, Class RK-1, dual element, with time delay.

Manufacturers and Products:

Bussmann: Type LPS-RK.
Littelfuse, Inc.: Type LLS-RK.

Motor and Transformer Circuits, 0 to 250 Volts:

Amperage: 0 to 600.

UL 198E, Class RK-1, dual element, with time delay.

Manufacturers and Products:

Bussmann; Type LPN-RK.
Littelfuse, Inc.; Type LLN-RK.

Feeder and Service Circuits, 0 to 600 Volts:

Amperage: 0 to 600.

UL 198E, Class RK-1, dual element, with time delay.

Manufacturers and Products:

Bussmann; Type LPS-RK.

Littelfuse, Inc.; Type LLS-RK.

Magnetic Control Relays

NEMA ICS 2, Class A600 (600 volts, 10 amperes continuous, 7,200VA make, 720VA break), machine tool type with field convertible contacts.

Smaller Magnetic Control Relays: Reference Section 13401, PROCESS INSTRUMENTATION AND CONTROL SYSTEMS (PICS).

Time Delay Relay

Industrial Relay Rated: 150 volts, 5 amps continuous, (3600 VA make, 360 VA break).

Solid-state electronic, field convertible ON/OFF delay.

Two Form-C contacts (minimum).

Repeat accuracy plus or minus 2 percent.

Timer Adjustment: Multiple adjustable ranges, including 1 to 60 seconds, unless otherwise shown.

Manufacturers:

Omron.
Cutler-Hammer.
General Electric Co.
Allen-Bradley.

Elapsed Time Meters

Type: Synchronous motor driven, 0 to 99,999.9 hours range, nonreset, suitable for semiflush, panel mounting.

Manufacturers:

General Electric Co.

Veeder-Root.

Phase Monitor Relay

Voltage and phase monitor relay shall drop out on loss of phase, or phase reversal.

Contacts: Single-pole, double-throw, 10 amperes, 120/240V ac. Where additional contacts are shown or required, provide magnetic control relays.

Adjustable trip and time delay settings.

Transient Protection: 1,000V ac.

Mounting: Multipin plug-in socket base.

Accessories: Provide properly sized and rated line isolating switches and fuses for each phase monitored.

Manufacturer:

Square D Co.

Cutler-Hammer.

General Electric Co.

Allen-Bradley.

Transient Voltage Surge Suppressor (TVSS) Equipment

General: Units shall be suitable for the service voltage and configuration (phases and wires) shown.

Protection Modes:

Normal, differential, and common.

Bipolar or bi-directional.

Ratings:

Short-circuit current rating shall equal or exceed that of protected distribution equipment. Surge Voltage Rating (SVR) shall not exceed those specified under UL 1449 for the associated nominal system voltage. Maximum Allowable Continuous Operating Voltage (MCOV) shall be at least 115 percent of the nominal system voltage.

Unit shall be UL-listed.

Provide status indicators for unit ON-LINE and unit operation NORMAL.

Provide common alarm contact output.

Provide fusible disconnect switch (integral with TVSS unit, where available) where not shown connected via branch circuit device of protected distribution equipment.

Minimum Enclosure Rating: NEMA 250, Type 12.

Type 2 TVSS:

Requirements: Designed for critical loads at service equipment (Category C3/B3) or distribution panelboard (Category C2/B3) locations. Unit shall utilize voltage-matched Silicon Avalanche Suppressor Diode (SASD) technology. Unit shall utilize modular, plug-in suppressor design.

Manufacturer and Product: Transtector; Model Apex III (nonservice entrance distribution panelboard) or Apex IV (service equipment).

Volt And Current Meters

Voltmeter, Panel Type:

NEMA 250 Type 12.
Nominal 3-1/2 inch model.
90 degree scale; accuracy of plus or minus 2 percent.

Manufacturer: Same as Manufacturer of Motor Control Center.

Voltmeter Switch:

NEMA 250 Type 12.
Rotary cam type with pistol grip handle engraved escutcheon.
Four-position, phase-to-phase, and OFF.]

Manufacturer: Same as Manufacturer of Motor Control Center.

Ammeter, Panel Type:

NEMA 250 Type 12.
Nominal 3-1/2-inch model.
90 degree scale; accuracy of plus or minus 2 percent.

Manufacturer: Same as Manufacturer of Motor Control Center.

Ammeter Switch:

NEMA 250 Type 12.
Rotary cam type with pistol grip handle engraved escutcheon.
Four-position, three-phase currents, and OFF.

Manufacturer: Same as Manufacturer of Motor Control Center.

Conduit And Fittings

Rigid Galvanized Steel Conduit (RGS)

ANSI C80.1.

Fittings: Threaded type.

Galvanize by hot-dipping, electroplating, sherardizing, or metalizing process, including fittings.

Polyvinyl Chloride Conduit (PVC):

Rigid, Schedule 40, NEMA TC 2.
UL 651 listed for concrete encased, direct burial, concealed and direct sunlight exposed use.
UL 651 listed and marked for use with conductors having 90 degrees C insulation.

Fittings: NEMA TC 3, for intended use.

Flexible Metal Liquid-Tight Conduit:

UL 1 listed for liquid-tight service.
Galvanized steel, flexible conduit covered with extruded PVC jacket.

Termination: Nylon bushings or bushings with steel or malleable iron body and insulated throat and sealing O-ring.

Interior Conduit Sealing:

Spare conduits should be capped in panels and handholes.
Conduits inside handholes will be sealed using spray insulating foam or other material, which is non-deleterious to conductors and cables.

Conduits inside panels will be sealed using insulating foam or other material, which is non-deleterious to conductors and cables.

Conduit Sealing Fitting:

Restrict the passage of gasses, vapors, or flames from one portion of the electrical installation to another at atmospheric pressure and normal ambient temperatures.

In conduit systems when leaving Class 1, Division 2 hazardous locations.

Manufacturers and Products:

Appleton; Type EYF, EYM, or ESU.

Crouse-Hinds; Type EYS or EZS.

Fitting Sealing Compound: Form a seal around each electrical conductor and between them and inside of the sealing fitting to restrict the passage of gases, vapors, or flames through the sealing fitting.

Manufacturers and Products:

Appleton; Kwiko.

Crouse-Hinds; Chico.

Identification Devices: Conduit tags.

Material: Permanent, nylon.

Shape: Round.

Conduit Designation: Pressure stamped, embossed or engraved.

Support and Framing Channels

Carbon Steel Framing Channel:

Material: Rolled, mild strip steel, 12-gauge, ASTM A570, Grade 33.

Finish: Hot-dip galvanized after fabrication.

Paint-Coated Framing Channel: Carbon steel framing channel with electro-deposited rust inhibiting acrylic or epoxy paint.

Manufacturers:

B-Line Systems, Inc.

Unistrut Corp.

Aickinstrut.

Precast Utility Vaults, Manholes and Handholes

Concrete Strength: Minimum 3,000 psi compressive, in 28 days.

Loading: AASHTO H-20, in accordance with ASTM C857.

Drainage: Slope floors toward drain points, leaving no pockets or other nondraining areas.

Raceway Entrances:

Provide on all four sides along with pulling eyes.

For raceways to be installed under this Contract, provide knockout panels or precast individual raceway openings.

At entrances where raceways are to be installed by others, provide minimum 12-inch high by 24-inch wide knockout panels for future raceway installation.

Handhole Frames and Covers:

Material: Steel, hot-dipped galvanized.

Cover Type: Solid, torsion spring of checkered diamond design.

Cover Loading: AASHTO H-20.

Cover Designation: Burn by welder, on upper side in integral letters, minimum 2 inches in height, appropriate titles:

Above 600 Volts: ELECTRIC HV.

600 Volts and Below: ELECTRIC LV.

Instrumentation, Communication: Signal.

Hardware: Steel, hot-dip galvanized.

Furnish knockout for ground rod in each handhole.

Manufacturers:

Utility Vault Co.
Penn-Cast Products, Inc.
Concrete Conduit Co.
Associated Concrete Products, Inc.
Pipe, Inc.

Conductors 600 Volts And Below

Material: Annealed copper.

Insulation:

No. 8 AWG and Smaller: Type THW, THWN or XHHW conductors may be utilized at Construction General Contractor's option, subject to code requirements.

No. 6 AWG and Larger: Type XHHW.

Direct Buried: Type XLPE-USE.

Flexible Cord and Cable: Type SO, 600 volts.

Signal: Type 3, No. 16 AWG twisted, shielded pair instrumentation cable, 45-mil PVC outer jacket, 600-volt rating.

Type:

Control Conductor No. 14 AWG and Smaller: Stranded.

Power Conductors No. 10 AWG and Smaller: Solid or stranded.

Power Conductors No. 8 AWG and Larger: Stranded.

Type 3: No. 16 AWG stranded (copper seven-stranded)

Conductors Above 600 Volts (Schedule A Work Only)

Ethylene-Propylene Rubber (EPR) Insulated Cable:

Extrusion: Single-pass, triple-tandem, of conductor screen, insulation, and insulation screen.

Type: 15kV, tape shielded UL 1072, Type MV-90.

Conductors: Copper concentric lay Class B round stranded in accordance with ASTM B3, ASTM B8, and ASTM B263.

Conductor Screen: Extruded, semi-conducting ethylene-propylene rubber in accordance with NEMA WC 71 and AEIC CS 6.

Insulation: 133 percent insulation level, ethylene-propylene rubber (EPR), containing no polyethylene in accordance with NEMA WC 71, and AEIC CS 6.

Insulation Thickness: 220-mil, 15 kV, nominal.

Insulation Screen: Thermosetting, semi-conducting ethylene-propylene rubber (EPR), extruded directly over insulation in accordance with NEMA WC 74, and AEIC CS 6.

Metallic Shield: Uncoated, 5-mil, copper shielding tape, helically applied with [17-1/2] percent minimum overlap.

Jacket: Extruded polyvinyl chloride (PVC) compound applied over the metallic shield in accordance with NEMA WC 71.

Operating Temperature: 90 degrees C continuous normal operations, 130 degrees C emergency operating conditions, and 250 degrees C short-circuit conditions.

Manufacturers:

Okonite Co.
Pirelli Wire and Cable.
BICC.
Southwire Co.

Accessories For Conductors Above 600 Volts (Schedule A Work Only)

Termination Kits:

Capable of terminating 15 kV, single-conductor, polymeric-insulated tape shielded cables plus a shield ground clamp.

Capable of producing a termination with a current rating equal to, or greater than, the cable ampacity, meeting Class 1 requirements of IEEE 48.

Capable of accommodating any form of cable shielding or construction without the need for special adapters or accessories.

Manufacturers:

Raychem.
3M Co.

Elbow Connector Systems: Molded, peroxide-cured, EPDM-insulated, Class 15 kV, 95kV BIL, 200A, 10,000A rms load-break elbows as shown having all copper current-carrying parts in accordance with ANSI 386.

Protective Caps: Class 15 kV, 95 kV BIL, 200 amperes, with molded EPDM insulated body.

Insulated Standoff Bushings: Class 15kV, 95kV BIL, 200 amperes, complete with EPDM rubber body, stainless steel eyebolt with brass pressure foot, and stainless steel base bracket.

Bushing Inserts: 15kV, 95kV BIL, 200A, load-break with EPDM rubber body and all-copper, current-carrying parts.

Manufacturers:

Cooper Industries.
Elastimold.

Cable Lugs:

In accordance with NEMA CC1.
Rated 15kV of same material as conductor metal.

Manufacturers and Products, Uninsulated Compression Connectors and Terminators:

Burndy, Hydent.
Thomas & Betts; Color-Keyed.
ILSCO.

Terminal Blocks and Enclosures

Provide enclosures for all indoor and outdoor terminal block applications in accordance with ENVIRONMENTAL CONDITIONS in this section..

Type: Compression screw clamp, with current bar providing direct contact with wire and yoke, with individual rail mounted terminals.

Yokes and Clamping Screws: Zinc-plated, hardened steel.

Rating: 600V ac.

Pushbuttons and Selector Switches

NEMA ICS 2, Type 600.

Type: Heavy-duty, oiltight.

Lockout: Pushbuttons and selector switches shall lock in OFF position wherever lockout provisions are indicated.

Nameplates:

Individual, large, laminated plastic.
Function indicated.
Pushbutton station nameplates shall indicate the drive controlled.

Manufacturers and Models:

Square D; Type T.
Cutler-Hammer; Type 10250T.
General Electric.

Luminaires

Specific requirements relating to fixture type, lamp type, poles, and mounting hardware are located in the Luminaire Schedule attached to this section.

Receptacles

NEMA WD 1 and FS W-C-596.

Specification Grade:

Type: Three-wire grounding, with screw type terminals suitable for No. 10 AWG wire. Contact to be made on two sides of each inserted blade without detent.

Number of Poles: Two.

Rating: 125 volts, NEMA WD 1, Configuration 5-20R, 20 amps.

Base: Phenolic composition.

Color: Gray.

Special Outlets

Weatherproof outdoor heavy duty circuit breaking receptacle assembly and housing.

One matching plug with cord-grip features for each special purpose outlet.

Rating: 100-amp rating, 600 volts, three-phase, three-wire with ground (four-pole) as required for anticipated purpose.

Manufacturer and Model:

Crouse-Hinds, Arktite Style 2 AREA10425.

Or equal.

Switches

NEMA WD 1 and FS W-S-896E.

Totally enclosed, ac type, quiet tumbler switches, with screw terminals.

Capable of control of 100 percent tungsten filament and fluorescent lamp loads.

Rating: 20 amps, 120/277 volts (single and double-pole as required).

Color: Gray.

Boxes

Small Standard Boxes: NEMA 250, Type 1, minimum 2 inches deep, unless shallower required by structural conditions.

Large Galvanized Steel Boxes: NEMA 250, Type 12 unless otherwise noted.

14-gauge, with full access screw covers mounted with corrosion-resistant machine screws.

Large Cast Metal Boxes: NEMA 250, Type 4, (Type 7 for combined sumps), cast malleable iron, with hot-dip galvanized finish.

Neoprene gasketed, watertight, with cast metal covers, stainless steel screws, and drilled and tapped conduit entrances.

Handholes: Reinforced cast concrete boxes sized to provide adequate working space as required by standard procedures and NFPA 70.

Nonmetallic:

Box: PVC.

Cover: PVC, weatherproof, with stainless steel screws.

Manufacturer and Product: Carlon; Type FS or FD, with Type E98 or E96 covers.

Large Nonmetallic Box: NEMA 250, Type 4X.

Box: High-impact, fiberglass-reinforced polyester or engineered thermoplastic, with stability to high heat.

Cover: Hinged with clamps.

Hardware and Machine Screws: ASTM A167, Type 316 stainless steel.

Conduit hubs and mounting lugs.

Manufacturers and Products:

Crouse-Hinds; Type NJB.
Carlon; Series N, C, or H.
Robroy Industries.

Metal Wireways

Meet requirements of UL 870.

Type: Steel-enclosed, with removable, hinged cover.

Rating: Reference ENVIRONMENTAL CONDITIONS in this section.

Finish: Gray, baked enamel.

Manufacturers:

Circle AW.
Hoffman.
Square D.

Cover Plates

Metal:

Material: Specification grade, one-piece, stainless steel.

Thickness: Minimum 0.40-inch nominal.

Finish: No. 302/304 satin.

Mounting Screws: Oval head, stainless steel, to match plate.

Cast Metal:

Material: Malleable ferrous, with gaskets.

Mounting Screws: Oval head, stainless steel.

Weatherproof Device Plates:

Material: Cast metal, gasketed, weatherproof, with individual cap over each opening held with stainless steel springs.

Finish: Stainless steel or fiberglass reinforced plastic.

Mounting Screws: Stainless steel.

Grounding

General

Grounding shall be in compliance with NFPA 70 and ANSI C2.

Ground electrical service neutral at service entrance equipment to supplementary grounding electrodes.

Ground each separately derived system neutral to nearest effectively grounded building structural steel member or separate grounding electrode.

Bond together system neutrals, service equipment enclosures, exposed noncurrent-carrying metal parts of electrical equipment, metal raceways, ground conductor in raceways and cables, receptacle ground connections, metal piping systems, and metal structures which may become energized by attached electrical devices (i.e., leachate storage tank, metal frame of buildings).

Shielded Instrumentation Cables:

Ground shield of instrumentation cables at PLC end only, using drain wire connected to terminal block that is connected to an isolated instrument ground. Isolated instrument ground terminals block is located inside PLC control panel enclosure.

Insulate ungrounded end of all shielded instrumentation cables' shield with shrink tubing for a distance of 1/2 inch either side of the end of the outer jacket.

Wire Connections:

Ground Conductors: Install in conduit containing power conductors and control circuits.

Nonmetallic Raceways and Flexible Tubing:

Install equipment grounding conductor and bond at both ends.

Connect ground conductors to raceway grounding bushings.

Bond all equipment grounding conductors to equipment ground bus and equipment enclosures as required by the NEC.

Bolt connections to equipment ground bus.

Bond grounding conductors to metallic enclosures at each end, and to intermediate metallic enclosures.

Junction Boxes: Furnish materials and connect to equipment grounding system with grounding clips mounted directly on box, or with 3/8-inch machine screws.

Motor Grounding: Extend equipment ground bus via grounding conductor installed in motor feeder raceway; connect to motor frame.

Nonmetallic Raceways and Flexible Tubing: Install an equipment grounding conductor and bond at both ends.

Motors Less Than 10 hp: Furnish compression, spade-type terminal connected to conduit box mounting screw.

Circuits 20 Amps or Above: Tap motor frame or equipment housing; install solderless terminal with minimum 5/16-inch diameter bolt.

Grounding Conductors:

Equipment: Solid or stranded copper with green, Type USE/RHH/RHW-XLPE or THHN/THWN, insulation.

Direct Buried: Bare stranded copper.

Isolated Instrument Ground: Stranded copper with green insulation with yellow stripe or yellow phasing tape at all ends.

Ground Rod:

Material: Copper.

Diameter: Minimum 3/4 inch.

Length: 10 feet.

Connectors:

Exothermic Weld Type:

Outdoor Weld: Suitable for exposure to elements or direct burial.

Indoor Weld: Utilize low-smoke, low-emission process.

Manufacturers: Erico Products, Inc., Cadweld and Cadweld Exolon.

Compression Type:

Compress-deforming type; wrought copper extrusion material.
Single indentation for conductors 6 AWG and smaller.
Double indentation with extended barrel for conductors 4 AWG and larger.
Barrels prefilled with oxide-inhibiting and antiseizing compound and sealed.

Manufacturers:

Burndy Corp.
Thomas and Betts Co.
ILSCO Corp.

Mechanical Type: Split-bolt, saddle, or cone screw type; copper alloy material.

Manufacturers:

Burndy Corp.
Thomas and Betts Co.
ILSCO Corp.

Grounding Wells:

Ground rod box complete with cast iron riser ring and traffic cover marked GROUND ROD.

Manufacturers and Products:

Christy Co.; No. G5.
Lightning and Grounding Systems, Inc.; I-R Series.

Part 3 – Execution

General

All work shall be performed in a neat and workman-like manner and shall comply with all applicable provisions of NECA 5055 standards and practices.
Install materials and equipment in hazardous areas in a manner acceptable to regulatory authority having jurisdiction for the hazardous area indicated.
Ground equipment, enclosures, and complete conduit system securely in accordance with applicable sections of NFPA 70.

Panelboards and Mini-Power Centers

Install securely, plumb, in-line and square with walls.
Install top of cabinet 6 feet above floor, unless otherwise shown.
Provide typewritten circuit directory for each panelboard.

Cabinet Location/Type:

Industrial Use in Areas Not Otherwise Classified: Reference ENVIRONMENTAL CONDITIONS in this section.

Transient Voltage Surge Suppression (TVSS) Equipment

Install in accordance with manufacturer's instructions, including lead length, overcurrent protection, and grounding.

Motor Starter

Field adjust trip settings of motor starter magnetic, trip-only circuit breakers in accordance with manufacturer's instructions.

Conduit And Fittings

General:

Conduit system shall be carefully planned with proper attention to details before starting the work. Do not install crushed or deformed raceways. Replace any raceway that has been damaged after installation.

Raceways that are installed so as to form a moisture trap are not allowed.

Prevent plaster, dirt, or trash from lodging in raceways, boxes, fittings, and equipment during the course of construction. Clear clogged raceways of obstructions.

All conduit runs shall be made parallel to or perpendicular to the lines of the building.

Secure conduits entering cabinets, pull boxes or outlet boxes with galvanized locknuts and bushings, on both sides of box wall.

Identify conduits at each terminus using conduit and cable schedule designations.

Applications:

Exposed Exterior: Type RGS.

Concrete Embedded: Type PVC.

Direct Buried: Type PVC inside concrete duct bank.

Vertical Runs Through Slab on Grade: Convert PVC conduit to RGS wrapped with watertight adhesive plastic tape.

PVC Bends: Bends in PVC runs shall be incorporated using RGS. RGS wrapped with watertight adhesive plastic tape.

Final Connection to Motors:

Conduit Size 4 Inches or Less: 18-inch minimum, 60-inch maximum length of flexible liquid-tight metal conduit.

Penetrations:

Conduits penetrating fire-rated walls shall be sealed with a compound approved by UL and appropriate to the fire rating of the wall.

Flash and counterflash conduits penetrating roofing membrane.

Seal penetrations with oakum or expandable plastic compound.

Provide sleeves and chases where conduits pass through floors or walls. Finish to match adjacent surfaces.

Provide escutcheon plates where exposed conduits pass through walls, floors or ceilings.

Conduits from the combined sump area shall be sealed with a compound approved by UL, and appropriate for conduits in hazardous areas entering nonhazardous areas.

Slab-On-Grade or Direct Buried:

Install horizontal runs below floor slab. Horizontal runs within slab shall not be permitted.

Field wrap RGS conduit and joints installed below slab or direct buried with 0.010-inch thick pipe wrapping plastic tape applied with a 50 percent overlay, or factory apply a plastic resin, epoxy, or coal-tar coating system.

Exposed Raceways:

Install parallel or perpendicular to walls, structural members, or intersections of vertical planes and ceilings.

Underground Duct Banks:

All underground duct banks shall be installed in locations shown on drawings, enclosed in a red concrete casing as specified in Section 03301, CONCRETE. The concrete casing shall also enclose all standard conduit bends or elbows. All underground ducts shall have steel reinforcement in sizes as shown on the drawings.

Excavate the trenches as specified in Section 02316, EXCAVATION, to provide elevation on top of concrete envelope as shown on drawings. After trenches are excavated and graded, the duct shall be laid in rows on plastic spacers or approved equals.

Spacers shall be placed so that each section of duct is supported at intervals as specified in NFPA 70 (NEC). Concrete shall then be placed per Section 03301, CONCRETE, until the ducts are covered to the required depth and leveled, leaving NOT less than 4 inches of concrete over top tier of ducts. Backfill shall be in accordance with Section 02320, TRENCH BACKFILL.

Changes in Direction of Runs:

Make with symmetrical bends or cast metal fittings.

Bends and offsets shall be made with a hickey or conduit bending machine.

Supports:

Provide pipe straps, wall brackets, conduit clamps, conduit hangers, threaded C-clamps with retainers, or ceiling trapeze.

Install suitable braces for conduit, junction boxes, light fixtures and other electrical equipment as needed for seismic support.

Securely and rigidly fasten in place.

Maximum Interval: 10 feet.

Precast Utility Vaults, Manholes and Handholes

Excavate, shore, brace, backfill, and final grade in accordance with Section 02316, EXCAVATION and Section 02320, TRENCH BACKFILL.

Do not install until final raceway grading has been determined.

Install such that raceways enter at nearly right angles and as near as possible to one end of wall, unless otherwise shown.

Conductors

Conduit system shall be complete prior to drawing conductors.

Lubricate prior to drawing into conduit. Lubrication type shall be as approved by conductor manufacturer.

Connections: Pressure type solderless, complete with insulator and security ring.

Control Circuits:

Where multiple units perform parallel operations, do not group all devices on same branch circuit.

Do not exceed the ampacity of the branch circuit, or 12 amperes continuous.

Terminate feeder and interconnecting conductors between panel mounted equipment and external equipment at numbered terminal blocks.

Identification:

Where two or more conduits run to a single outlet box, color code each circuit as a guide in making connections.

Carry colors continuously throughout the system.

Do not install multiwire branch circuits that share a common neutral.

Identify conductors, cables at each terminus, and handhole using conduit and cable schedule designations.

Colors:

Confirm and utilize the existing Hanford field center color coding system as shown below:

Conductor Origin: 480Y/277-volt, three-phase system. Transformers, panels, switchboard, etc.

Phase A	Red
Phase B	Yellow
Phase C	Blue
Neutral	White or Gray
Equipment Ground	Green (or bare)

Conductor Origin: 208Y/120-volt, three-phase system. Transformers, panels, switchboard, etc.

Phase A	Black
Phase B	Purple
Phase C	Brown
Neutral	White or Gray
Equipment Ground	Green (or bare)

Conductor Origin: 120/240-volt, single-phase system. Transformers, panels, switchboard, etc.

Hot Number 1	Black
Hot Number 2	Brown
Neutral	White or Gray
Equipment Ground	Green (or bare)

Conductor Origin: DC system. Instruments, control panels, etc.

DC+	Red
DC-	Black

Conductors Above 600 Volts (Schedule A Work Only)

Do not splice conductors.

Single Conductor Cable Terminations:

Coordinate all terminations with FH EU.

Make terminations with termination kits, in accordance with kit manufacturer's instructions. Install terminations as continuous operation in accessible locations under clean, dry conditions.

Provide heat shrinkable stress control and outer nontracking insulation tubings, high relative permittivity stress relief mastic for insulation shield cutback treatment, and a heat-activated sealant for environmental sealing plus a ground braid and clamp.

Install terminals or connectors acceptable for type of conductor material used.

Provide shield termination and grounding for all terminations.

Provide necessary mounting hardware, covers, and connectors.

Where elbow connectors are specified, install in accordance with manufacturer's instructions.

Connections and Terminations:

Install uninsulated crimp connectors and terminators for instrumentation, control, and power circuit conductors No. 4 AWG through No. 2/0 AWG.

Give 2 working day's notice to FH EU prior to making terminations.

Terminal Blocks

Install for termination of all control circuits leaving or entering equipment, panels, or boxes.

Luminaires

Install luminaires and poles in accordance with manufacturer's recommendations.

Install plumb and true.

Provide swivel type hangers and canopies to match pendant mounted fixtures.

Furnish all lamps and clean the reflectors, the diffusers, and the lamps before closing up the fixtures.

Boxes

Support to the structure, independent of conduit attachment.

Boxes installed belowgrade shall be installed flush with finished grade.

Boxes and covers in paved areas, roadways, or walkways shall be suitable for weights to which they may be subjected.

Box Extensions: Not permitted.

Classified Hazardous Areas: Boxes shall be applicable for location.

Cover Plates

Shall fit tightly to box.

Shall not extend beyond sides of box on surface mounted boxes, unless covers have no sharp corners or edges.

Trench Backfill

In accordance with Section 02320, TRENCH BACKFILL.

Protection Following Installation

Protect materials and equipment from corrosion, physical damage, and the effects of moisture on insulation.

Cap conduit runs during construction with manufactured seals.

Close openings in boxes or equipment during construction.

Energize space heaters furnished with equipment.

Construction Quality Control

In accordance with Section 16080, ELECTRICAL TESTING, and as specified herein.

Circuit Balance: Confirm the balance of electrical load between phases on three-phase panelboards and motor control centers after installation. Notify Construction Manager of current unbalances 10 percent and greater.

Voltage Testing: When installation is complete and facility is in operation, check voltage at point of termination of electric supply system to project.

Check voltage amplitude and balance between phases for loaded and unloaded conditions.

Record supply voltage for 24 continuous hours. If unbalance exceeds 1 percent, or if voltage varies throughout the day and from loaded to unloaded conditions more than plus or minus 4 percent of nominal, make written request to Tank Farm Contractor to correct condition.

Equipment Line Current:

Check line current in each phase for each piece of equipment.

If electric utility makes adjustments to supply voltage magnitude or balance, make line current check after adjustments are made.

Inspection of Low Voltage Cables, 600 Volts Maximum (Note: FH EU shall inspect and test all cables rated above 600 volts):

Inspect each individual exposed power cable for physical damage, proper connections in accordance with Section 16080, ELECTRICAL TESTING.

Electrical Tests for Conductors (600 Volts and Below):

Prior to final connection and energizing of power and control circuits, conduct an insulation resistance test to determine insulation integrity in accordance with Section 16080, ELECTRICAL TESTING.

Ground Electrode Test:

Inspect grounding connections prior to any backfill of cables in accordance with Section 16080, ELECTRICAL TESTING.

Maximum ground electrode resistance shall be 3 ohms. Add maximum 2 additional ground rods spaced 6 feet apart if 3 ohms is not achieved.

Supplements

The supplement listed below, following "END OF SECTION," is a part of this Specification.

Supplement 1—Luminaire Schedule.

END OF SECTION 16005

Luminaire Schedule						
Type	**Voltage**	**Description**	**Manufacturer**	**Catalogue No.**	**Lamp**	**Mounting Type**
1	120	Heavy duty industrial 4-foot fluorescent fixture with 2 lamps with low temp electronic starting ballast(s).	Holophane	7200-4-12-LT Fluorescent Prismatite or equal.	2-40W R.S. T12, 0°F starting (48")	Pendant with chains and surface mount
2	120	Standby light NiCaD battery operated.	Holophane	C1-6N-25-W-WCHY-2 Cortez A1	2-12 watts 6-volt Halogen	Wall mount
3	120	WallPack wall mount HPS with integral photocell.	Holophane	WallPack WL2K-070HP-12-BK—F1-LAMP-PC	1-70W HPS	Wall mount
4	120	Pole mount outdoor flood wet location duty. Single and double 2U configuration as indicated.*	Holophane	Predator Floodlight PF-250HP-12-K-W-1-B-CR[2]	250-watt HPS	Pole mount round tapered steel poles, single and 2U configurations SPRT20J/1/SG SPRT20J/2/SG
5	120	Crestwood outdoor post light with HPS fixture and cover*	Holophane	Crestwood CW-24-15AHP-12-GR-CA	1-150W HPS	Round tapered steel galvanized pole 10 foot SPRT10J/SG
Note: *Install added vertical light cutoff "shades."						

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SECTION 16055 – PIPE HEAT TRACING SCHED. B

Part 1 – General

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Product Data: Manufacturer's descriptive literature.

Plastic Pipe Installations: Output adjustment factors for heating tape for the services indicated.

Pipe heat loss calculations for each pipe size to be heat traced.

Submittals–Approval Not Required:

Information/Record (IR):

Field Testing: Submit inspection/test report on insulation resistance per CONSTRUCTION QUALITY CONTROL section of this Specification.

Part 2 – Products

System Design Requirements

Design Heating Load:

Heating load to be calculated based upon a 100-degree F delta, 20 mph wind if pipes are located outdoors, insulation as specified in Section 15060, PIPING-GENERAL, and shall include a 10 percent safety factor.

Heat loss calculations shall be based on IEEE 515, Equation 1, Page 19.

Electrical Heating Tape

Cable: Auto-trace, self-limiting, parallel circuit construction consisting of continuous inner core of variable resistance conductive heating material between two parallel copper bus wires. Provide tinned copper braid for PVC, FRP, and stainless steel pipe applications.

UL Listing: Listed as self-limiting pipe tracing material for pipe freeze protection application in ordinary conditions.

Maximum Maintenance Temperature: 150 degrees F (65 degrees C).

Maximum Intermittent Temperature: 185 degrees F (85 degrees C).

Minimum Maintenance Temperature: Minus 40 degrees F (-40 degrees C).

Service Voltage: As indicated by branch circuits provided for heat tracing on the Drawings. All heat trace circuits shall be powered by ground fault interrupter type (GFEP) circuit breakers (30 mA).

Manufacturers and Products:

Raychem; Chemelex BTV or BRV-C.

Thermon; FLX-BC or FLX-OJ.

Nelson; CL1-J1 or L1-J1.

Connection System

Rating: NEMA 250, Type 4 and Factory Mutual approved.

Operating Monitor Light: Furnish with each circuit power connection kit to indicate when heat tracing is energized.

Manufacturers and Products:

Power Connection Kit:

Raychem; Chemelex AM-BC.
Thermon; PCA-COM.
Nelson; PLT-BC.

Splice Kit:

Raychem; Chemelex AM-BS.
Thermon; PCS-COM.
Nelson; PLT-BS.

Tee Kit:

Raychem; Chemelex AM-B4.
Thermon; Tee Snap.
Nelson; PLT-BY.

End Seal Kit:

Raychem; Chemelex AM-E.
Thermon; ET-6C or ET-8C.
Nelson; LT-ME.

Pilot Light:

Raychem; Chemelex AM-L.
Thermon; VIL-4C.
Nelson; LT-L.

Pipe Adapter Kit:

Raychem; Chemelex AM-P or AM-T.
Thermon; included with power connection kit.
Nelson; LT-P.

Securing Tape

Plastic Piping Systems:

Type: Aluminum foil coated adhesive tape.

Manufacturers and Products:

Raychem; Chemelex AT-180.
Thermon; AL-20P.
Nelson; AT-50.

Metallic Piping Systems:

Type: Glass or polyester cloth pressure sensitive tape.

Manufacturers and Products:

Raychem; Chemelex GS54 or GT66.
Thermon; PF-1.
Nelson; GT-6 or GT-60.

Ambient Thermostat

Type: Adjustable setting (15 to 140 degrees F), set to 40 degrees F, so as to switch cable off when ambient temperature exceeds 40 degrees F.

Sensor: Fluid-filled probe.

Enclosure: Epoxy-coated NEMA 250, Type 4X aluminum enclosure with exposed hardware of stainless steel.

Switch: SP-DT, UL or FM listed, rated 22 amps, 125 to 250V ac.

Manufacturers and Products:

Raychem; Chemelex Model AMC-1A.

Thermon; B4X-15140.

Nelson; TX-4X140.

Part 3 – Execution

Installation

General:

Install in accordance with the manufacturer's instructions and recommended practices.

Provide insulation as specified in Section 15060, PIPING-GENERAL, over all pipe heat tracing.

Ground metallic structures or materials used for support of heating cable or on which it is installed in accordance with applicable codes.

Wiring between power connection points of heat tracing cable branch lines shall be provided by heat tracing system supplier.

Provide end of circuit pilot lights on heat tracing circuits for buried piping.

Electrical Heating Tape:

Determine required length of electrical heating tape by considering length of circuit, number and type of fittings and fixtures, design heating load, and heating tape output.

Where design heating load exceeds heating tape capacity, install by spiraling.

Derate heating tape capacity when installed on plastic piping.

Install on services as follows (reference Drawings H-2-830854, Sheets 2 and 4):

Service	Piping Material	Placement	Location
Cell No. 1	3" LH-030-HDPE	Heat trace and insulate exposed portion of piping from tank inlet connection to about 2 feet below grade.	Cell No. 1 and Cell No. 2
Cell No. 2	3" LH-030-HDPE		Leachate Storage Tank(s) Process Inlet Connections.
Cell No. 1	4" LT-034-PVC	Heat trace and insulate exposed portion of piping from tank inlet connection to about 2 feet below grade.	Cell No. 1 and Cell No. 2
Cell No. 2	4" LT-034-PVC	Connections	Outlet
Cell No. 1	4" LH-045-HDPE	Heat trace and insulate exposed portion of piping from tank inlet	Cell No. 1 and Cell No. 2
Cell No. 2	4" LH-045-HDPE	Connection to about 2 feet below grade.	Leachate Storage Tank(s) Process Inlet Connections.
Cell No. 1	4"LT-037-HDPE	Heat trace and insulate exposed portion of piping from above grade	Cell No. 1 and Cell No. 2 Truck
Cell No. 2	4"LT-037-HDPE	Connection fittings to about 2 feet below grade, including 8" containment piping.	Loading Station(s) Process to Truck Connections.

Wrap heat trace cable once every 33 inches under pipe insulation. Install additional heating tape at bolted flanges, valves, pipe supports, and other fittings and fixtures as recommended by supplier, but not less than the following:

Item	Heating Tape Length (min. feet)
Bolted flanges (per pair)	Two times pipe diameter
Valves	Four times valve length
Pipe hanger or support penetrating insulation	Three times pipe diameter

Heat Tracing Circuits: Limit individual lengths of heat tracing circuits such that maximum single circuit capacity is 20 amps when starting the circuit at 40 degrees F. Provide multiple 20-amp circuits as required at individual heat tracing locations.

Thermostats: Install in accordance with manufacturer's instructions and as approved by Engineer.

For each group of heat traced circuit, install one ambient thermostat.

Construction Quality Control

In accordance with Section 16080, ELECTRICAL TESTING, and as specified herein.

Test each circuit with 500-volt insulation tester between circuit and ground with neutrals isolated from ground.

Insulation Resistance: Minimum 1,000 megohms per 1,000 feet.

END OF SECTION 16055

SECTION 16080 – ELECTRICAL TESTING SCHED. A &B

Part 1 – General

Fluor Hanford Electrical Utilities (FH EU) shall test pad mounted transformer and conductors rated above 600 volts. Construction General Contractor shall coordinate FH EU testing with Construction Manager.

Onsite testing shall be required for each major electrical system as specified herein in the presence of representatives for the authority having jurisdiction (AHJ). Tests shall be performed to demonstrate that each function is implemented and operational. These tests are electrical component tests to be performed in advance of facility-wide construction acceptance testing (CAT). CAT shall be performed in accordance with Division requirements.

Provide personnel and equipment in support of Section 13401, PROCESS INSTRUMENTATION AND CONTROL SYSTEMS (PICS), Operation Readiness (ORT) and Performance Acceptance (PAT) testing.

Approval By Authority Having Jurisdiction (AHJ)

As specified in Section 16005, ELECTRICAL.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Qualifications: Submit Testing Firm Qualifications as specified herein.

Submittals–Approval Not Required

Information/Record (IR):

Submit 30 days prior to performing inspections or tests:

- Schedule for performing inspection and tests.
- List of references to be used for each test.
- Sample copy of equipment and materials inspection form(s).
- Sample copy of individual device test form.
- Sample copy of individual system test form.
- Submit within 30 days after completion of test:
- Test or inspection reports and certificates for each electrical item tested.

Operation and Maintenance Data:

After test or inspection reports and certificates have been reviewed by Engineer and returned, insert a copy of each in operation and maintenance manual.

Testing Firm Qualifications

Employer of engineers and technicians regularly engaged in testing and inspecting of electrical equipment, installations, and systems.

Supervising engineer accredited as Certified Electrical Test Technologist by National Institute for Certification of Engineering Technologists (NICET), or International Electrical Testing Association and having a minimum of 5 years' testing experience on similar projects.

Technicians certified by NICET or NETA.

Registered professional engineer to provide comprehensive project report outlining services performed, results of such services, recommendations, actions taken, and opinions.

In compliance with OSHA Title 29, Part 1907 criteria for accreditation of testing laboratories.

Sequencing And Scheduling

Perform inspection and electrical tests after equipment has been installed. Construction General Contractor shall coordinate NEC required inspections and Fluor Hanford Electrical Utilities (FH EU) required testing and inspections with Construction Manager. Tank Farm Contractor will provide services of an NEC inspector.

Perform tests with apparatus de-energized whenever feasible.

Inspection and electrical tests on energized equipment are to be:

- Scheduled with Construction Manager prior to de-energization.
- Minimized to avoid extended period of interruption to the operating plant equipment.

Notify Construction Manager at least 24 hours prior to performing tests on energized electrical equipment.

Part 2 – Products

Test Equipment/Instrumentation Requirements:

Test equipment shall have an operating accuracy equal to, or greater than, requirements established by NETA ATS.

Test instrument calibration shall be in accordance with NETA ATS.

Part 3 – Execution

General:

Tests and inspection shall establish that:

- Electrical equipment is operational within industry and manufacturer's tolerances.
- Installation operates properly.
- Equipment is suitable for energization.
- Installation conforms to requirements of Contract Documents and NFPA 70, NFPA 70E, and ANSI C2.
- Perform inspection and testing in accordance with NETA ATS, industry standards, and manufacturer's recommendations.
- Adjust mechanisms and moving parts for free mechanical movement.
- Adjust adjustable relays and sensors to correspond to operating conditions, or as recommended by manufacturer.
- Verify nameplate data for conformance to Contract Documents.
- Realign equipment not properly aligned and correct unlevelness.
- Properly anchor electrical equipment found to be inadequately anchored.
- Tighten accessible bolted connections, including wiring connections, with calibrated torque wrench to manufacturer's recommendations, or as otherwise specified.
- Clean contaminated surfaces with cleaning solvents as recommended by manufacturer.
- Provide proper lubrication of applicable moving parts.

Inform Construction Manager of working clearances not in accordance with NFPA 70.

Investigate and repair or replace:

- Electrical items that fail tests.
- Active components not operating in accordance with manufacturer's instructions.
- Damaged electrical equipment.

Electrical Enclosures:

- Remove foreign material and moisture from enclosure interior.
- Vacuum and wipe clean enclosure interior.
- Remove corrosion found on metal surfaces.
- Repair or replace, as determined by Construction Manager, door and panel sections having dented surfaces.
- Repair or replace, as determined by Construction Manager, poor fitting doors and panel sections.
- Repair or replace improperly operating latching, locking, or interlocking devices.
- Replace missing or damaged hardware.

Finish:

- Provide matching paint and touch up scratches and mars.
- If required due to extensive damage, as determined by Construction Manager, refinish the entire assembly.
- Replace fuses and circuit breakers that do not conform to size and type required by the Contract Documents.

Dry Type Transformers

Visual and Mechanical Inspection:

- Physical and insulator damage.
- Proper winding connections.
- Bolt torque level in accordance with NETA ATS, Table 10.1, unless otherwise specified by manufacturer.
- Defective wiring.
- Proper operation of fans, indicators, and auxiliary devices.
- Removal of shipping brackets, fixtures, or bracing.
- Free and properly installed resilient mounts.
- Cleanliness and improper blockage of ventilation passages.
- Verify that tap-changer is set at correct ratio for rated output voltage under normal operating conditions.
- Verify proper secondary voltage phase-to-phase and phase-to-ground after energization and prior to loading.

Electrical Tests:

Insulation Resistance Tests:

Applied megohmmeter dc voltage in accordance with NETA ATS, Table 7.2.3 for each:

- Winding-to-winding.
- Winding-to-ground.

10-minute test duration with resistances tabulated at 30 seconds, 1 minute, and 10 minutes.

Results temperature corrected in accordance with NETA ATS, Table 7.2.4.

Temperature corrected insulation resistance values equal to, or greater than, ohmic values established by manufacturer.

Insulation resistance test results to compare within 1 percent of adjacent windings.

Perform tests and adjustments for fans, controls, and alarm functions as suggested by manufacturer.

Low Voltage Cables, 600 Volts Maximum:

Visual and Mechanical Inspection:

Inspect Each Individual Exposed Power Cable No. 8 and Larger For:

- Physical damage.
- Proper connections in accordance with single-line diagram.
- Cable bends not in conformance with manufacturer's minimum allowable bending radius where applicable.
- Color coding conformance with specifications.
- Proper circuit identification.

Mechanical Connections For:

- Proper lug type for conductor material.
- Proper lug installation.
- Bolt torque level in accordance with NETA ATS, Table 10.1, unless otherwise specified by manufacturer.

Shielded Instrumentation Cables For:

- Proper shield grounding.
- Proper terminations.
- Proper circuit identification.
- Continuity test by ohmmeter method to ensure proper cable connections.

Control Cables For:

- Proper termination.
- Proper circuit identification.
- Continuity test by ohmmeter method to ensure proper cable connections.

Cables Terminated Through Window Type CTs: Verify that neutrals and grounds are terminated for correct operation of protective devices.

Electrical Tests for Conductors No. 8 and Larger:

Insulation Resistance Tests:

- Utilize 1,000-volt dc megohmmeter for 600-volt insulated conductors.
- Test each conductor with respect to ground and to adjacent conductors per IEEE 118 procedures for 1 minute.
- Evaluate ohmic values by comparison with conductors of same length and type.
- Investigate values less than 50 megohms.
- Continuity test by ohmmeter method to ensure proper cable connections.

Safety Switches, 600 Volts Maximum

Visual and Mechanical Inspection:

- Proper blade pressure and alignment.
- Proper operation of switch operating handle.
- Adequate mechanical support for each fuse.
- Proper contact-to-contact tightness between fuse clip and fuse.

Cable connection bolt torque level in accordance with NETA ATS, Table 10.1.

Proper phase barrier material and installation.

Verify that fuse sizes and types correspond to one-line diagram.

Perform mechanical operational test and verify electrical and mechanical interlocking system operation and sequencing.

Electrical Tests:

Insulation Resistance Tests:

Applied megohmmeter dc voltage in accordance with NETA ATS, Table 10.2.

Phase-to-phase and phase-to-ground for 1 minute on each pole.

Insulation resistance values equal to, or greater than, ohmic values established by manufacturer.

Contact Resistance Tests:

Contact resistance in microhms across each switch blade and fuse holder.

Investigate deviation of 50 percent or more from adjacent poles or similar switches.

Molded And Insulated Case Circuit Breakers

General: Inspection and testing limited to circuit breakers rated 70 amperes and larger and to motor circuit protector breakers rated 30 amperes and larger.

Visual and Mechanical Inspection:

Proper mounting.

Proper conductor size.

Feeder designation according to nameplate and one-line diagram.

Cracked casings.

Connection bolt torque level in accordance with NETA ATS, Table 10.1.

Operate breaker to verify smooth operation.

Compare frame size and trip setting with circuit breaker schedules or one-line diagram.

Verify that terminals are suitable for 75 degrees C rated insulated conductors.

Electrical Tests:

Insulation Resistance Tests:

Utilize 1,000-volt dc megohmmeter for 480- and 600-volt circuit breakers and 500-volt dc megohmmeter for 240-volt circuit breakers.

Pole-to-pole and pole-to-ground with breaker contacts opened for 1 minute

Pole-to-pole and pole-to-ground with breaker contacts closed for 1 minute.

Test values to comply with NETA ATS, Table 10.2.

Contact Resistance Tests:

Contact resistance in microhms across each pole.

Investigate deviation of 50 percent or more from adjacent poles and similar breakers.

Current Injection Test to Verify:

Long-time minimum pickup and delay.

Short-time pickup and delay.

Instantaneous pickup by run-up or pulse method.

Trip characteristics of adjustable trip breakers shall be within manufacturer's published time-current characteristic tolerance band, including adjustment factors.

Trip times shall be within limits established by NEMA AB 4, Table 5-3.

Instantaneous pickup value shall be within values established by NEMA AB 4, Table 5-4.

Instrument Transformers

Visual and Mechanical Inspection:

Visually Check Current, Potential, and Control Transformers For:

Cracked insulation.

Broken leads or defective wiring.

Proper connections.

Adequate clearances between primary and secondary circuit wiring.

Verify Mechanically That:

Grounding and shorting connections have good contact.

Withdrawal mechanism and grounding operation, when applicable, operate properly.

Verify proper primary and secondary fuse sizes for potential transformers.

Electrical Tests:

Current Transformer Tests:

Insulation resistance test of transformer and wiring-to-ground at 1,000 volts dc for 30 seconds.

Polarity test.

Potential Transformer Tests:

Insulation resistance test at test voltages in accordance with NETA ATS, Table 7.1.1 for 1 minute on:

Winding-to-winding.

Winding-to-ground.

Polarity test to verify polarity marks or H1-X1 relationship as applicable.

Insulation resistance measurement on instrument transformer shall not be less than that shown in NETA ATS, Table 7.1.1.

Utility Metering:

Testing to be conducted by FH EU.

Grounding Systems:

Visual and Mechanical Inspection:

Equipment and circuit grounds in motor control center and panelboard assemblies for proper connection and tightness.

Ground bus connections in motor control center and panelboard assemblies for proper termination and tightness.

Effective transformer core and equipment grounding.

Accessible connections to grounding electrodes for proper fit and tightness.

Accessible exothermic-weld grounding connections to verify that molds were fully filled and proper bonding was obtained.

Electrical Tests:

Fall-Of-Potential Test:

In accordance with IEEE 81, Section 8.2.1.5 for measurement of main ground system's resistance.

Main ground electrode system resistance to ground to be no greater than 3 ohms.

Two-Point Direct Method Test:

In accordance with IEEE 81, Section 8.2.1.1 for measurement of ground resistance between main ground system, equipment frames, and system neutral and derived neutral points.

Equipment ground resistance shall not exceed main ground system resistance by 0.25 ohm.

AC Induction Motors

General: Inspection and testing limited to motors rated 1/3 hp and larger.

Visual and Mechanical Inspection:

- Proper electrical and grounding connections.
- Shaft alignment.
- Blockage of ventilating air passageways.

Operate Motor and Check For:

- Excessive mechanical and electrical noise.
- Overheating.
- Correct rotation.
- Check vibration detectors, resistance temperature detectors, or motor inherent protectors for functionability and proper operation.
- Excessive vibration.
- Check operation of space heaters.

Electrical Tests:

Insulation Resistance Tests:

In accordance with IEEE 43 at test voltages established by NETA ATS, Table 10.2 for:

- Motors 200 hp and less for 1-minute duration with resistances tabulated at 30 and 60 seconds.
- Insulation resistance values equal to, or greater than, ohmic values established by manufacturers.
- Insulation resistance test on insulated bearings in accordance with manufacturer's instructions.
- Measure running current and voltage, and evaluate relative to load conditions and nameplate full-load amperes.

Low Voltage Motor Control

Visual and Mechanical Inspection:

- Proper barrier and shutter installation and operation.
- Proper operation of indicating and monitoring devices.
- Proper overload protection for each motor.
- Improper blockage of air cooling passages.
- Proper operation of drawout elements.
- Integrity and contamination of bus insulation system.

Check Breaker and Kirk Key Interlocking System By:

- Closure attempt of breaker when associated Kirk key is in place.
- Open attempt of breaker when associated Kirk key is in place.
- Closure attempt of breaker when associated Kirk key is not in place.
- Open attempt of breaker when associated Kirk key is not in place.

Check Door and Device Interlocking System By:

- Closure attempt of device when door is in OFF or OPEN position.
- Opening attempt of door when device is in ON or CLOSED position.

Check Nameplates for Proper Identification Of:

- Equipment title and tag number with latest one-line diagram.
- Control switches.
- Pilot lights.
- Control relays.
- Circuit breakers.
- Verify that fuse and circuit breaker sizes and types conform to Contract Documents.
- Verify that current and potential transformer ratios conform to Contract Documents.

Check Bus Connections for High Resistance by Low Resistance Ohmmeter: Ohmic value to be zero.

Check Operation and Sequencing of Electrical and Mechanical Interlock Systems By:

- Closure attempt for locked open devices.
- Opening attempt for locked closed devices.
- Key exchange to operate devices in OFF-NORMAL positions.
- Verify performance of each control device and feature furnished as part of the motor control center.

Control Wiring:

- Compare wiring to local and remote control, and protective devices with elementary diagrams.
- Check for proper conductor lacing and bundling.
- Check for proper conductor identification.
- Check for proper conductor lugs and connections.
- Exercise active components.

Inspect Contactors For:

- Correct mechanical operations.
- Correct contact gap, wipe, alignment, and pressure.
- Correct torque of all connections.
- Compare overload settings and voltage ratings with full-load current for proper size.
- Compare fuse motor protector and circuit breaker with motor characteristics for proper size.

Electrical Tests:

Insulation Resistance Tests:

- Applied megohmmeter dc voltage in accordance with NETA ATS, Table 10.2.
- Bus section phase-to-phase and phase-to-ground for 1 minute on each phase.
- Contactor phase-to-ground and across open contacts for 1 minute on each phase.

Starter section phase-to-phase and phase-to-ground on each phase with starter contacts closed and protective devices open.

Test values to comply with NETA ATS, Table 10.2.

Overpotential Tests:

Maximum applied ac or dc voltage in accordance with NETA ATS, Table 7.1.2.

Phase-to-phase and phase-to-ground for 1 minute for each phase of each bus section.

Test results evaluated on pass/fail basis.

Current Injection Through Overload Unit at 300 Percent of Motor Full-Load Current and Monitor Trip Time:

Trip time in accordance with manufacturer's published data.

Investigate values in excess of 120 seconds.

Control Wiring Tests:

Apply secondary voltage to control power and potential circuits.

Check voltage levels at each point on terminal boards and each device terminal.

Insulation resistance test at 1,000 volts dc on control wiring except that connected to solid state components.

Insulation resistance to be 1 megohm minimum.

Operational test by initiating control devices to affect proper operation.

Low Voltage Surge Arrestors

Visual and Mechanical Inspection:

Adequate clearances between arrestors and enclosures.

Ground connections to ground bus and electrode.

Electrical Tests:

Varistor Type Arrestors:

Clamping voltage test.

Rated RMS voltage test.

Rated dc voltage test.

Varistor arrestor test values in accordance with ANSI C62.33, Sections 4.4 and 4.7.

END OF SECTION 16080

SECTION 16270 – OIL-FILLED PAD MOUNTED TRANSFORMERS SCHED. A

Part 1 – General

References

The following is a list of standards, which may be referenced in this section:

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C57.12.00	Standard General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers
ANSI C57.12.22	Pad-Mounted, Compartmental-Type, Self-Cooled, Three-Phase Distribution Transformers with High-Voltage Bushings, 2,500 kVA and Smaller
ANSI C57.12.26	Pad-Mounted, Compartmental-Type, Self-Cooled, Three-Phase Distribution Transformers for Use with Separable Insulated High Voltage Connectors
ANSI C57.12.28	Switchgear and Transformers - Pad-Mounted Equipment, Enclosure Integrity
ANSI C57.12.90	Standard Test Code for Liquid Immersed Distribution, Power, and Regulating Transformers
ANSI 386	Standard for Separable Insulated Connector Systems for Power Distribution Systems Above 600V

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC. (IEEE)

IEEE C62.11	Metal-Oxide Surge Arrestors for Alternating-Current Power Circuits (>1 KV)

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Product Data:

Descriptive information.
Dimensional drawings.
Transformer nameplate data.
Schematic and connection diagrams.
Factory test reports certified.

Submittals–Approval Not Required

Information/Record (IR):

Operation and maintenance data.
Material Safety Data Sheet (MSDS) for Envirotemp FR3™ Fluid.
Submit documentation and test results from construction quality control testing.

Extra Materials

Furnish, tag, and box for shipment and storage and deliver prior to 30 percent Project completion the following spare parts, special tools, and materials:

One quart of paint to match color and quality of equipment final shop finish.
Two spare fuse links for each fuse size.

Part 2 – Products

Source Quality Control (Factory Tests)

Design, test, and assemble in accordance with applicable standards of ANSI C57.12.00, C57.12.22, C57.12.26, and C57.12.90.

Production tests in accordance with ANSI C57.12.90 and C57.12.00, Section 8 and Table 16.

Dielectric test in accordance with ANSI C57.12.26.

Manufacturers

Cooper Power System.

Cutler-Hammer.

Square D Co.

General Electric.

General

Integral Unit: Compartmental type unit consisting of transformer, oil-filled tank, and high and low voltage terminating compartments, assembled on a common structural base.

Anchor Bolts: Type 316 stainless steel, sized by equipment manufacturer, and as specified in Section 05500, METAL FABRICATIONS AND CASTINGS.

Transformer

kVA Rating: 112.5.

Primary Voltage: 13.8 kV line-to-line, three-phase, 60 Hz.

Secondary Voltage: 480/277 volts, three-phase, four-wire, 60 Hz.

BIL Rating:

95 BIL for 15 kV insulation class transformers.

30 BIL for secondary.

Temperature Rise: 65 degrees C above 30 degrees average ambient with maximum ambient not to exceed 40 degrees C.

Impedance: 3.2 percent.

Coolant: Normally formulated, hydro-refined oil free of PCB chemical, FMRC approved, UL classified less-flammable Envirotemp FR3, in accordance with FMRC 3990. Fluid shall have a minimum open cup fire point of 350 degrees C and a minimum 5-day BOD (SM5210B) of 200 ppm. Manufacturer is to provide information on the transformers nameplate that it is "NON-PCB" along with "manufacturer's name and type of insulating fluid."

Primary Taps:

Full capacity, two 2-1/2 percent below and two 2-1/2 percent above, rated voltage.

Externally operated no-load tap changer.

Provisions for locking handle in any position.

Coil Conductors: Copper windings.

Delta-wye transformers wound on triplex cores.

Sound Level: In accordance with manufacturer's standards.

Enclosure

In accordance with ANSI C57.12.28 requirements.

Welded carbon steel transformer tank, with cooling panels when required, and lifting eyes.

12-gauge sheet steel terminal compartment enclosure having no exposed screws, bolts, or other fasteners that are externally removable.

Corrosion Protection and Color: Base(s) and cabinet(s) of the transformer shall be corrosion resistant and shall be fabricated of steel. Provide insulating and corrosion resistance undercoating on base of transformer. Paint bases, cabinets, and tanks Munsell 7GY3.29/1.5 green. The Munsell color notation is specified in ASTM D1535.

Terminal Compartments

General: ANSI C57.12.28, enclosed high and low voltage compartments side by side, separated by steel barrier, bolted to transformer tank.

Doors:

Individual, full-height, air-filled.

Low voltage door with three-point latching mechanism, vault type handle, and single padlocking provision.

High voltage door fastenings inaccessible until low voltage door has been opened.

Door Bolts:

Hex-head type.

Lift-off, stainless steel hinges and door stops.

Removable front sill to facilitate rolling or skidding over conduit stub ups.

Recessed lock pocket, with steel door release bolt adjacent to secondary compartment door handle.

High Voltage Compartment:

Deadfront in accordance with ANSI C57.12.26 type construction.

Protective fuses.

High voltage bushings.

Transformer grounding pad.

Surge arrestors with barriers.

Four-position, oil-immersed type switch to permit closed transit in loop feed and sectionalizing position sectionalizing load-break switch.

Parking stands.

Low Voltage Compartment:

Livefront in accordance with ANSI C57.12.26 type construction.

Low voltage bushings.

Grounding pad.

Stainless steel equipment nameplate.

Liquid level gauge.

1-inch upper filter press and filling plug.

Drain valve with sampling device.

Dial type thermometer.

Pressure relief valve.

Pressure relief device, self-resealing with indicator.
Pressure-vacuum gauge.
Mounting provision for current and potential transformers.
Nameplate per Nameplate C, Table 9, IEEE C57.12.00.

Bushings

High Voltage:

Deadfront Termination:

Universal bushing well rated at 15 kV in accordance with ANSI 386.
Bushings externally clamped and front removable.
Rated for 200 amperes continuous, 95 kV BIL.
Standoff brackets located adjacent to bushings.

Low Voltage:

Molded epoxy bushing clamped to tank with 4 hole spade type terminals.
Rated 150 percent of continuous full-load current, 30 BIL, 600 volts.
Internally connected neutral extending to neutral bushing.

High Voltage Switching

Internal, oil-immersed, gang-operated load-break, manually operated switches.
Hot stick operated handle located in high voltage compartment.
Capable of operating at full-load current.

Feed Switch: Four-position, oil-immersed type switch to permit closed transition loop feed and sectionalizing. Switch shall be rated at 15 kV, 95 kV BIL, with a continuous current rating and load-break rating of 200 amperes, and a make-and-latch rating of 10,000 rms amperes symmetrical. The switch handle shall be located in the high-voltage compartment and lockable in the ON/OFF position. Switch shall be similar to the Cooper Power Systems, Electrical Apparatus 800-64, Sectionalizing Switches, "T" blade switch.

High Voltage Protection

Combination Oil-Immersed Bayonet Expulsion and Current Limiting Fuses:

Accessibility:

Bayonet expulsion fuse accessible through primary compartment.
Current-limiting fuse accessible through tank handhole.

Expulsion Fuse for Low Current Faults: Interrupting capacity of 1,800 amperes rms asymmetrical.

Current Limiting for High Current Faults:

Interrupting capacity of 50,000 amperes rms symmetrical.
Bayonet fuse externally replaceable with hot stick.
Bayonet fuse links shall be dual sensing for both high currents and high oil temperature in order to provide thermal protection to the transformer.
Coordinate transformer protection with expulsion fuse clearing low-current faults and current-limiting fuse clearing high-current faults beyond the interrupting rating of the expulsion fuse.
In order to eliminate or minimize oil spills, the bayonet fuse assembly shall include an oil retention valve inside the housing, which closes when the fuse holder is removed and an external drip shield.

Warning shall be conspicuously displayed within the high-voltage compartment cautioning against removing or inserting fuses unless the load-break switch is in the OPEN position and the tank pressure has been released.

Bayonet Fuse Assembly: 150 kV BIL.

Oil-Immersed Current-Limiting Fuses: NEMA C37.47; 50,000 rms amperes symmetrical interrupting rating at the system voltage specified.

Surge Arrestors

Metal-Oxide, Varistor Type:

Insulated body, elbow type, 18 kV in accordance with IEEE C62.11.

Installed in high voltage compartment.

Connected to transformer high voltage bushing wells.

Tank Grounding Pads

Low Voltage Compartments:

Connected together with bare No. 2/0 stranded copper conductors.

Wye low voltage neutral internally connected with link and brought out to insulated low voltage bushing externally grounded to tank.

Low voltage neutral connected to externally mounted insulating bushing in low voltage compartment and grounded to tank with removable strap.

Tap Changer Warning Sign

Red laminated plastic, engraved to white core.

Engrave to read DO NOT OPERATE WHEN TRANSFORMER ENERGIZED.

Mount above tap changer handle.

Part 3 – Execution

General

Prepare subgrade for utility vault for pad mounted transformer as specified in Section 02319, SUBGRADE PREPARATION, paragraph "Prepared Subgrade for Roadway, Embankment and Structures."

Secure to mounting pads with anchor bolts.

Install plumb and longitudinally in alignment with pad or adjacent building wall.

Ground neutrals and enclosures in accordance with applicable codes and as shown on the Drawings.

Construction Quality Control

In accordance with Section 16080, ELECTRICAL TESTING.

Adjustments

Adjust voltage taps to obtain rated output voltage under normal operating load conditions.

END OF SECTION 16270

SECTION 16270-1 – SINGLE PHASE OIL-FILLED PAD MOUNTED TRANSFORMERS SCHED. B

Part 1 – General

References

The following is a list of standards which may be referenced in this section:

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C57.12.00	Standard General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers
ANSI C57.12.21	Pad-Mounted, Compartmental-Type, Self-Cooled, Single-Phase Distribution Transformers with High-Voltage Bushings, 167 kVA and Smaller
ANSI C57.12.25	Pad-Mounted, Compartmental-Type, Self-Cooled, Single-Phase Distribution Transformers for Use with Separable Insulated High Voltage Connectors
ANSI C57.12.28	Switchgear and Transformers - Pad-Mounted Equipment, Enclosure Integrity
ANSI C57.12.90	Standard Test Code for Liquid Immersed Distribution, Power, and Regulating Transformers
ANSI 386	Standard for Separable Insulated Connector Systems for Power Distribution Systems Above 600V

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC. (IEEE)

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Product Data:

Descriptive information.
Dimensional drawings.
Transformer nameplate data.
Schematic and connection diagrams.
Factory test reports certified.

Submittals–Approval Not Required

Information/Record (IR):

Operation and maintenance data, including recommended preventative maintenance tasks and frequencies for performance of those tasks.
Material Safety Data Sheet (MSDS) for Envirotemp FR3™ Fluid.
Submit documentation and test results from construction quality control testing.

Extra Materials

Furnish, tag, and box for shipment and storage and deliver prior to 30 percent Project completion.

Part 2 – Products

Source Quality Control (Factory Tests)

Design, test, and assemble in accordance with applicable standards of ANSI C57.12.00, C57.12.21, C57.12.25, and C57.12.90.

Production tests in accordance with ANSI C57.12.90, Section 8 and Table 16.

Dielectric test in accordance with ANSI C57.12.25.

Manufacturers

Cooper Power System; Maxishrub type.

ABB.

General

Integral Unit: Compartmental type unit consisting of transformer, oil-filled tank, and high and low voltage terminating compartment, assembled on a common base.

Anchor Bolts: Type 316 stainless steel, sized by equipment manufacturer, and as specified in Section 05500, METAL FABRICATIONS AND CASTINGS.

Transformer

kVA Rating: 75.

Primary Voltage: 13.8 kV line-to-line, single-phase, 60 Hz.

Secondary Voltage: 120/240 volts, single-phase, three-wire, 60 Hz.

BIL Rating:

95 BIL for 15 kV insulation class transformers.

30 BIL for secondary.

Temperature Rise: 65 degrees C above 30 degrees average ambient with maximum ambient not to exceed 40 degrees C.

Impedance: 3.2 percent, minimum.

Coolant: Normally formulated, hydro-refined oil free of PCB chemical, FMRC approved, UL classified less-flammable Envirotemp FR3, in accordance with FMRC 3990. Fluid shall have a minimum open cup fire point of 350 degrees C and a minimum 5-day BOD (SM5210B) of 200 ppm. Manufacturer is to provide information on the transformers nameplate that it is "NON-PCB" along with "manufacturer's name and type of insulating fluid."

Primary Taps:

Full capacity, two 2-1/2 percent below and two 2-1/2 percent above, rated voltage.

Externally operated no-load tap changer.

Provisions for locking handle in any position.

Coil Conductors: Copper windings, insulated with B-stage, epoxy coated, diamond pattern insulating paper, thermally cured under pressure.

Sound Level: In accordance with manufacturer's standards.

Enclosure

In accordance with ANSI C57.12.28 requirements.

Welded carbon steel sealed transformer tank, with domed tank cover , and recessed stainless steel lifting eyes.

12-gauge sheet steel terminal compartment enclosure having no exposed screws, bolts, or other fasteners that are externally removable.

Corrosion Protection and Color: Base(s) and cabinet(s) of the transformer shall be corrosion resistant and shall be fabricated of steel.

Provide insulating and corrosion resistance undercoating on base of transformer. Paint bases, cabinets, and tanks Munsell 7GY3.29/1.5 green. The Munsell color notation is specified in ASTM D1535.

Terminal Compartment

General: ANSI C57.12.28, enclosed high and low voltage assemblies side by side, bolted to transformer tank.

Door: Hinged door with stainless steel hinge pins and barrels.

Door Bolts: Recessed, captive, penta head type.

Removable front sill to facilitate rolling or skidding over conduit stub ups.

Floating lock pocket.

High Voltage Assembly:

- Deadfront in accordance with ANSI C57.12.25 type construction.
- Protective fuses.
- High voltage bushings.
- Transformer grounding pad.
- Surge arrestors with barriers.
- Parking stands.

Low Voltage Assembly:

- Livefront in accordance with ANSI C57.12.25 type construction.
- Low voltage bushings.
- Grounding pad.
- Stainless steel equipment nameplate.
- Liquid level gauge.
- Oil filling plug.
- Drain valve with sampling device.
- Dial type thermometer.
- Pressure relief device, self-resealing with indicator.
- Pressure-vacuum gauge.
- Nameplate per Nameplate A, IEEE C57.12.00.

Bushings

High Voltage:

Deadfront Termination:

- Universal bushing well rated at 15 kV in accordance with ANSI 386.
- Bushings externally clamped and front removable.
- Rated for 200 amperes continuous, 95 kV BIL.
- Standoff brackets located adjacent to bushings.

Low Voltage:

- Molded epoxy bushing clamped to tank with 4 hole spade type terminals.
- Rated 150 percent of continuous full-load current, 30 BIL, 600 volts.

High Voltage Protection

Combination Oil-Immersed Bayonet Expulsion and Current Limiting Fuses:

Accessibility:

Bayonet expulsion fuse accessible through primary compartment.

Current-limiting fuse accessible through tank handhole.

Expulsion Fuse for Low Current Faults: Interrupting capacity of 1,800 amperes rms asymmetrical.

Current Limiting for High Current Faults:

Interrupting capacity of 50,000 amperes rms symmetrical.

Bayonet fuse externally replaceable with hot stick.

Bayonet fuse links shall be dual sensing for both high currents and high oil temperature in order to provide thermal protection to the transformer.

Coordinate transformer protection with expulsion fuse clearing low-current faults and current-limiting fuse clearing high-current faults beyond the interrupting rating of the expulsion fuse.

In order to eliminate or minimize oil spills, the bayonet fuse assembly shall include an oil retention valve inside the housing, which closes when the fuse holder is removed and an external drip shield.

Warning shall be conspicuously displayed within the high-voltage compartment cautioning against removing or inserting fuses unless the load-break switch is in the OPEN position and the tank ressure has been released.

Bayonet Fuse Assembly: 150 kV BIL.

Oil-Immersed Current-Limiting Fuses: NEMA C37.47; 50,000 rms amperes symmetrical interrupting rating at the system voltage specified.

Tank Grounding Pads

Low Voltage Compartment:

Low voltage neutral connected to externally mounted insulating bushing in low voltage compartment and grounded to tank with removable strap.

Tap Changer Warning Sign

Red laminated plastic, engraved to white core.

Engrave to read DO NOT OPERATE WHEN TRANSFORMER ENERGIZED.

Mount above tap changer handle.

Adjustments

Adjust voltage taps to obtain rated output voltage under normal operating load conditions.

Part 3 – Execution

General

Prepare subgrade for utility pad mounted transformer as specified in Section 02319, SUBGRADE PREPARATION, paragraph "Prepared Subgrade for Roadway, Embankment and Structures."

Secure to mounting pads with anchor bolts.

Install plumb and longitudinally in alignment with pad or adjacent building wall.

Ground neutrals and enclosures in accordance with applicable codes and as shown on the Drawings.

Construction Quality Control

In accordance with Section 16080, ELECTRICAL TESTING.

END OF SECTION 16270-1

SECTION 16312 – OVERHEAD ELECTRICAL DISTRIBUTION SCHED. A

Part 1 – General

References

The following is a list of standards which may be referenced in this section:

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C2	National Electrical Safety Code
ANSI C12.7	Watthour Meter Sockets
ANSI C12.10	Watthour Meters
ANSI C12.11	Instrument Transformers for Revenue Metering, 10 kV BIL Through 350 kV (0.6 kV NSV Through 69 kV NSV)
ANSI C12.13	Electronic Time-of-Use Registers for Electricity Meters
ANSI C12.15	Electricity Metering Solid-State Demand Registers for Electromechanical Watthour Meters
ANSI C12.16	Solid-State Electricity Meters
ANSI C29.2	Insulators - Wet-Process Porcelain and Toughened Glass - Suspension Type
ANSI C29.3	Wet Process Porcelain Insulators - Spool Type
ANSI C29.4	Wet-Process Porcelain Insulators - Strain Type
ANSI C29.5	Wet-Process Porcelain Insulators - Low- and Medium-Voltage Types
ANSI C29.6	Wet-Process Porcelain Insulators, High-Voltage Pin Type
ANSI C29.7	Wet Process - Porcelain Insulators - High-Voltage Line-Post Type
ANSI C37.42	High Voltage Expulsion Type Distribution Class Fuses, Cutouts, Fuse Disconnecting Switches and Fuse Links
ANSI C57.12.20	Transformers – Overhead Type Distribution Transformers, 500 kVA and Smaller: High-Voltage, 34500 Volts and Below; Low-Voltage, 7970/13800Y Volts and Below
ANSI C57.12.28	Switchgear and Transformers - Pad-Mounted Equipment - Enclosure Integrity
ANSI O5.1	Wood Poles

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.11	Forged Fittings, Socket-Welding and Threaded
ASME D3487	Mineral Insulating Oil Used in Electrical Apparatus

AMERICAN WOOD-PRESERVERS' ASSOCIATION (AWPA)

AWPA C1	All Timber Products - Preservative Treatment by Pressure Processes
AWPA C4	Poles - Preservative Treatment by Pressure Processes
AWPA C25	Crossarms, Pressure Treatment

ASTM INTERNATIONAL (ASTM)	
ASTM A53	Pipe, Steel, Black and Hot-Dipped Zinc-Coated, Welded and Seamless
ASTM A153/A153M	Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A167	Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
ASTM A475	Zinc-Coated Steel Wire Strand
ASTM B1	Hard-Drawn Copper Wire
ASTM B2	Medium-Hard-Drawn Copper Wire
ASTM B3	Soft or Annealed Copper Wire
ASTM B8	Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft
ASTM B228	Concentric-Lay-Stranded Copper-Clad Steel Conductors
ASTM B231/231M	Concentric-Lay-Stranded Aluminum 1350 Conductors
ASTM B232/B232M	Concentric-Lay-Stranded Aluminum Conductors, Coated Steel-Reinforced (ACSR)
ASTM B397	Concentric-Lay-Stranded Aluminum-Alloy 5005-H19 Conductors
ASTM B399/B399M	Concentric-Lay-Stranded Aluminum-Alloy 6201-T81 Conductors
ASTM D117	Electrical Insulating Oils of Petroleum Origin
INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC. (IEEE)	
IEEE 48	Standard Test Procedures and Requirements for Alternating-Current Cable Terminations 2.5 kV through 765kV
IEEE C37.30	High-Voltage Switches
IEEE C37.60	Overhead, Pad Mounted, Dry Vault, and Submersible Automatic Circuit Reclosers and Fault Interrupters for AC Systems
IEEE C37.63	Overhead, Pad Mounted, Dry-Vault, and Submersible Automatic Line Sectionalizers for AC Systems
IEEE C57.12.00	General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers
IEEE C57.12.90	Test Code for Liquid-Immersed Distribution, Power, and Regulating Transformers and Guide for Short-Circuit Testing of Distribution and Power Transformers
IEEE C57.13	Instrument Transformers
IEEE C62.11	Metal-Oxide Surge Arrestors for Alternating Current Power Circuits
NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)	
NEMA WC 70	Standard for Non-shielded Power Cables Rated 2000V or Less for the Distribution of Electrical Energy
INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)	
NETA ATS	Electrical Power Distribution Equipment and Systems
NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)	
NFPA 70	National Electrical Code

RURAL UTILITIES SERVICE (RUS)

RUS 202-1	List of Materials Acceptable for Use on Systems of RUS Electrification Borrowers
RUS 1728F-700	Wood Poles, Stubs, and Anchor Logs
RUS 1728H-701	Wood Crossarms (Solid and Laminated Transmission Timbers and Pole Keys)

UNDERWRITERS LABORATORIES INC. (UL)

UL 6	Rigid Metal Conduit
UL 510	Polyvinyl Chloride Polyethylene and Rubber Insulating Tape

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Product Data:

Conductors.
Insulators.
Wood poles and crossarms.
Utility vault (reference Section 16005, ELECTRICAL).
Cutouts.
Surge arrestors.
Guy strand and guards.
Anchors and anchor rods.
Ground rods.
Conduit.

Submittals–Approval Not Required:

Information/Record (IR):

Test Reports:

Acceptance checks and tests.
Ground resistance test reports.

Certificates:

Wood poles.
Wood crossarms.

Delivery, Storage, And Handling

Pole Line Material Storage:

Poles that will be stored longer than 2 weeks shall be stored on supports at least 1 foot aboveground in accordance with ANSI O5.1. Strength and spacing of supports, and manner of stacking shall produce no noticeable distortion in poles.

Construction hooks, tongs, or other sharp tools shall not be used on the treated portion of poles. Do not use pointed tools capable of producing indentations of more than 1 inch in depth. Nails and holes are not permitted in the top of poles.

Cable Inspection: Upon delivery to construction site, cable and reels shall be inspected for shipping damage such as:

Marks caused by improper lifting equipment or techniques.

Breaks or cuts in outer covering.

Damaged jacket or insulation.

Reel damage from mishandling.

Cable Testing: Upon delivery to construction site and prior to installation, Construction General Contractor shall perform DC over-potential tests on new cable.

Cable Reel Storage:

Reels shall be stored with flanges resting on hard surface or pallets to prevent sinking into the ground.

Reel flanges shall not touch cable on other reels.

Reels shall not be stored on their sides; they shall be stored with reel axis horizontal.

Cable ends shall be taped or capped to prevent entrance of moisture.

Material stored at construction site shall be located to prevent damage from weather and adjacent construction operations.

Cable Reel Handling:

Slings and forklifts shall not contact cable or protective covering.

A spreader bar shall be used when lifting reel with bar and sling.

Reels shall not be dropped.

Part 2 – Products

Material And Equipment

Consider materials specified herein or shown on Drawings which are identical to materials listed in RUS 202-1 as conforming to requirements.

Wood Pole

Machine trimmed by turning western red cedar, cut from live timber in accordance with ANSI O5.1.

Poles shall be butt-treated by manufacturer in accordance with AWPA C7, using AWPA P8 and P9 preservatives.

Each pole shall be given single top cut at 30-degree angle with normal to axis of pole and at right angles to sweep. Gains shall be cut so roof will be at right angles to line and sweep of pole will be in line. Roofs and gains shall be brush-treated by manufacturer with specified preservative. Each gain shall fit crossarm tightly. Boltholes shall not be more than 1/16 inch oversize.

Quality of each pole shall be ensured with "WQC" (wood quality control) brand on each piece, or by an approved inspection agency report.

Wood Crossarm

Conform to RUS 1728H-701. Straight-gained Douglas fir, free from twists to within 0.1 inch per foot length, with bends and twists in one direction.

Pressure treat crossarms with pentachlorophenol, chromated copper arsenate (CCA), or ammoniacal copper arsenite (ACA).

Treatment shall conform to AWPA C25.

Crossarm Braces: Provide wood crossarm braces as indicated.

Hardware

Hot-dip galvanized, conforming to ASTM A153/A153M.

Insulator

Provide wet-process porcelain insulators, which are radio interference free.

Line Post Type Insulators: ANSI C29.7, Class 4.

Suspension Insulators: ANSI C29.2, Quantity one per cable connection assembly, Class 4.

Spool Insulators: ANSI C29.3, Class 53-2.

Guy Strain Insulators: Porcelain, ANSI C29.4, Class 4, except provide fiberglass type when used with underground terminal or when other interference problems exist.

Pin Insulators: ANSI C29.5, Class 55-5.

Overhead Conductor

Conductor of bare copper, ASTM B1, ASTM B2, and ASTM B3, hard-drawn, medium-hard-drawn, and soft-drawn, ASTM B8, stranded, aluminum conductor steel reinforced (ACSR), ASTM B232/B232M, of size and type indicated.

Guys

Guy Strands: ASTM A475, extra-high strength, Class A or B, galvanized strand steel cable. Guy strand shall be 3/8 inch in diameter with ultimate breaking strength as shown on the Drawings. Provide guy terminations designed for use with the particular strand and developing at least the ultimate breaking strength of the strand.

Round Guy Guard: Vinyl or PVC material, yellow colored, 8-feet long, and shatter resistant at sub-zero temperatures.

Guy Attachment: Thimble eye.

Anchor And Anchor Rod

Anchor shall be concrete cone anchor presenting holding area indicated on Drawings as a minimum. Anchor rod shall be twin thimble-eye, 3/4-inch diameter by 9-feet long. Anchor and anchor rod shall be hot-dip galvanized.

Grounding

Rod:

Copper clad steel at least 3/4 inch in diameter and 10 feet long.

Hard, clean, smooth, continuous, surface throughout length of rod.

Die-stamp each near top with name or trademark of manufacturer and length of rod in feet.

Wire:

Soft drawn copper wire ground conductor, minimum No. 4 AWG.

Ground wire protector may be either PVC or half round wood molding. Wood molding shall be fir, pressure treated in accordance with AWPA C25, or shall be cypress or cedar.

Surge Arrestor

IEEE C62.11, metal oxide, polymer housing, surge arrestor arranged for equipment mounting. RMS voltage rating shall be 18 kV. Arrestor shall be Heavy-Duty Distribution class.

Fused Cutout

Nonloadbreak open type construction rated 100 amperes, 15 kV, 110 kV BIL, with a minimum 10,000 amperes symmetrical interrupting rating conforming to ANSI C37.42.

Fuses shall be of "6T" Link type, size as specified by Fluor Hanford Electric Utility (FHEU). Fuse cutouts shall be equipped with mounting brackets suitable for the indicated installations.

Conduit Riser And Conductor

Rigid galvanized steel conduit conforming to UL 6. Provide conductors (600 volts and above) as specified in Section 16120, CONDUCTORS.

Porcelain Insulator Type Terminator:

Comply with requirements of IEEE 48, Class 1, except that requirements of design tightness test need not be met.

Shall not exude any insulating filler compound under either test or service.

Consist of porcelain insulator, copper cable connector-hoodnut assembly and copper aerial lug as required, metal body and supporting bracket, sealed cable entrance, internal stress relief device for shielded cable, and insulating filler compound or material.

Electrical Tapes

Tapes shall be UL listed for electrical insulation and other purposes in wire and cable splices. Termination, repair, and miscellaneous purpose electrical tapes shall comply with UL 510.

Caulking Compound

Compound for Sealing Conduit Risers:

Puttylike consistency, workable with hands at temperatures as low as 35 degrees F.

Shall not slump at 300 degrees F and shall not harden materially when exposed to air.

Shall readily caulk or adhere to clean surfaces of material with which it is designed to be used.

Shall have no injurious effects upon workmen or upon materials.

Part 3 – Execution

Installation

General: Provide overhead pole line installation conforming to requirements of ANSI C2 for Grade C construction of overhead lines in medium loading districts and NFPA 70 for overhead services. Consider street, alleys, roads and drives "public." Pole configuration shall be as indicated on Drawings.

Pole Setting: Provide pole holes at least as large at top as at bottom and large enough to provide 4 inch clearance between pole and side of hole.

Pole Setting Depths:

Length of Pole (feet)	Setting in Soil (feet)	Setting in Solid Rock (feet)
20	5.0	3.0
25	5.5	3.5
30	5.5	3.5
35	6.0	4.0
40	6.0	4.0
45	6.5	4.5
50	7.0	4.5
55	7.5	5.0
60	8.0	5.0

Setting in Soil, Sand, and Gravel: Applying where the following occurs:

Where pole holes are in soil, sand, or gravel or any combination of these.

Where soil layer over solid rock is more than 2 feet deep.

Where hole in solid rock is not substantially vertical.

Where diameter of hole at surface of rock exceeds twice the diameter of pole at same level. At corners, dead ends, and other points of extra strain, poles 40 feet or more long shall be set 6 inches deeper.

Backfill: Thoroughly tamp pole backfill for full depth of hole and mound excess fill around pole.

Setting Poles: Set poles so that alternate crossarm gains face in opposite directions, except at terminals and dead ends where gains of last two poles shall be on side facing terminal or dead end. On unusually long spans, set poles so that crossarm comes on side of pole away from long span. Where pole top pins are used, they shall be on opposite side of pole from gain, with flat side against pole.

Alignment of Poles: Set poles in alignment and plumb except at corners, terminals, angles, junctions, or other points of strain, where they shall be set and raked against strain. Set not less than 2 inches for each 10 feet of pole length above grade, nor more than 4 inches for each 10 feet of pole length after conductors are installed at required tension. When average ground run is level, consecutive poles shall not vary more than 5 feet in height. When ground is uneven, poles differing in length shall be kept to a minimum by locating poles to avoid highest and lowest ground points. If it becomes necessary to shorten pole, a piece shall be sawed off top and shall be treated and capped. Holes shall be dug large enough to permit proper use of tampers to full depth of hole.

Pole Cap: Provide plastic pole caps with 1/4-inch sealing rings and 4 nailing tabs. Fill sealing area with either a bituminous, elastigum roof cement, or an acceptable preservative paste to level of sealing ring to eliminate possibility of condensation. Place on pole top and nail each tab down with 1-1/4-inch nail. Pole caps are not necessary for ACA/CCA treated poles, unless they are shortened.

Cutting of Wood Poles: Where new gains or holes are required, paint gains with preservative compound as recommended by the pole manufacturer. Plug unused or abandoned holes using treated wood dowel pins.

Do not cut the tops of wood poles, except under very exceptional conditions, and only upon approval of Construction Manager. If cutting is deemed necessary, pole top shall be capped. Do not cut butt of wood poles.

Anchor and Guy: Place anchor in line with strain. Length of guy lead (distance from base of pole to top of anchor rod) shall be as indicated.

Setting Anchor: Set anchor in-place with anchor rod aligned with, and pointing directly at, guy attachment on pole with anchor rod projecting 6 to 9 inches out of ground to prevent burial of rod eye.

Setting Guy Strand:

Complete anchor and guy installation, dead end to dead end, and tighten guy before wire stringing and sagging is begun on that line section.

Provide strain insulator at a point on guy strand 8 feet, minimum, from ground and 6 feet, minimum, from surface of pole.

Hardware: Provide hardware with washer against wood and with nut and lock nut applied wrench tight. Provide locknut on threaded hardware connection. Locknut shall be M-F style and not palnut style.

Grounding: Conform to ANSI C2. Ground fused switches and lightning arrestors. Bond together pole line hardware separated by less than 2 inches.

Ground Rod Connection:

On pole lines by exothermic weld or by using compression connector for ground wire or wire to rod connection.

Exothermic welds strictly in accordance with manufacturer's written recommendations.

Welds which have puffed up or which show convex surfaces indicating improper cleaning, are not acceptable.

No mechanical connectors are required at exothermic weldments.

Compression connector shall be type that uses hydraulic compression tool to provide correct pressure.

Provide tools and dies recommended by compression connector manufacturer.

Embossing die code or similar method shall provide visible indication that connector has been fully compressed on ground wire.

Grounding and Grounded Connections:

Where no primary or common neutral exists, surge arrestors and frames of equipment operating at over 750 volts shall be bonded together and connected to a dedicated grounding electrode.

Where no primary or common neutral exists, transformer secondary neutral bushing, secondary neutral conductor, and frames of equipment operating at under 750 volts shall be bonded together and connected to a dedicated grounding electrode.

When a primary or common neutral exists, connect all grounding and grounded conductors to common grounding electrode.

Protective Molding: Protect grounding conductors that are run on surface of wood poles by wood molding or plastic molding of equal mechanical strength extending from ground line throughout communication and transformer spaces.

Conductors: Prevent nicking, kinking, gouging, flattening, or otherwise deforming or weakening conductor or impairing its conductivity. Remove damaged sections of conductor and splice conductor.

Splices: Conductor splices, as installed, shall exceed ultimate rated strength of conductor and shall be of type recommended by conductor manufacturer. No splice shall be permitted within 10 feet of a support.

Ties: Provide ties on pin insulators tight against conductor and insulator and ends turned down flat against conductor so that no wire ends project.

Reinstalling: Existing conductors to be reinstalled or resagged shall be strung to "final" sag table values indicated for particular conductor type and size involved.

New Installation: String new conductors to "initial" sag table values indicated for conductor type and size of conductor and ruling span indicated.

Aluminum Protection: Protect ACSR conductors by armor rod at pin insulators and by flat aluminum wire at attachments made of galvanized or coated iron or steel.

Fittings: Dead end fittings, clamp or compression type, shall conform to written recommendations of conductor manufacturer and shall develop full ultimate strength of conductor.

Aluminum Connections: To copper or other material using only splices, connectors, lugs, or fittings designed for that specific purpose.

Riser: Secure conduit on pole by two hole galvanized steel pipe straps spaced no more than 10 feet apart and within 3 feet of any outlet or termination. Ground metallic conduit.

Construction Quality Control

Wood Crossarm Inspection: Furnish inspection report from independent inspection agency, approved by the Tank Farm Contractor, stating that offered products comply with applicable AWPA and RUS standards. The RUS approved Quality Mark "WQC" on each crossarm will be accepted, in lieu of inspection reports, as evidence of compliance with applicable AWPA treatment standards.

Acceptance Checks and Tests:

Notify Fluor Hanford Electrical Utility (FHEU) 5 working days prior to start of checking and testing. FHEU will test 15 kV cables and transformers.

Perform in accordance with manufacturer's recommendations and include the following visual and mechanical inspections, and electrical tests, performed in accordance with NETA ATS.

Grounding System:

Visual and Mechanical Inspection: Inspect ground system for compliance with Drawings and Specifications.

Electrical Tests: Perform ground-impedance measurements utilizing fall-of-potential method. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of single ground rod, perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground testing megger in accordance with manufacturer's instructions to test each ground or group of grounds. Instrument shall be equipped with a meter reading directly in ohms or fractions thereof to indicate ground value of ground rod or grounding systems under test.

Report: Before energizing electrical equipment, submit the measured ground resistance of grounding system.

Include the test method and test setup (i.e., pin location) used to determine ground resistance and soil conditions at time measurements were made.

Devices Subject to Manual Operation: Operate at least three times, demonstrating satisfactory operation each time.

Follow-Up Verification: Upon completion of acceptance checks and tests, show, by demonstration in service, that circuits and devices are in operating condition and properly performing intended function.

END OF SECTION 16312

SECTION 16440 – LOW VOLTAGE MOTOR CONTROL SCHED. B

Part 1 – General

UL Compliance

Products manufactured within scope of Underwriters Laboratories shall conform to UL Standards and have an applied UL Listing Mark.

Approval By Authority Having Jurisdiction (AHJ)

As specified in Section 16005, ELECTRICAL.

Electrical Description of Work

Provide the following 480-volt, three-phase, four-wire with ground, 600-amp service rated motor control centers in accordance with this Specification and Contract Drawings:

219A-LH-MCC-001 to be installed inside the Cell No. 1 Crest Pad Building 219A.

219E-LH-MCC-001 to be installed inside the Cell No. 2 Crest Pad Building 219E.

MCC(s) shall be provided with the following minimum features in accordance with this Specification and Contract Drawings:

Main Incoming Service Breaker, with service neutral terminal connection in main breaker section only.

Portable Generator Power Outlet Feeder Breaker and Kirk Key (ed) arrangement with main incoming service breaker.

Phase Loss and Reversal Protection Relay and Alarm Lighting.

Transient Voltage Surge Suppressor (TVSS).

Volt and Current Meters and Selector Switches.

Lighting Panel.

Lighting Panel Transformer.

NEMA 1 Size Motor Starters (incorporating: breaker, control power transformer, fuses, contactor, overloads, indicators, control switches, elapsed time meter, terminal strips, wiring, and locking mechanism).

NEMA Feeder Breakers.

Three Sections.

Time Delay and Magnetic Control Relays.

Interlocking control relays, terminals, and lining assembled together in unit compartment.

Submittals–Approval Required

See Section 01300, SUBMITTALS, for submittal procedures.

Product Data:

Itemized bill-of-material.

Descriptive information.

Dimensional drawings.

Conduit entrance locations.

Bus data.

Protective Devices:

Copies of time-current characteristics.
Operational description.
Anchoring instructions and details.

Typed Tabulation:

Motor name; tag (equipment) numbers as shown on Drawings.
Motor horsepower.
Nameplate full load current.
Measured load current and voltage.
Heater catalog number.
Protective device trip settings.
Attach above typed, tabulated data to a copy of starter manufacturer's overload selection tables for the starters provided.

Control Diagrams:

NEMA ICS 2, Section 322.08 Type I.
Wiring Type B.
In addition to standard NEMA control diagrams, provide the following:
Remote control devices.
Remote indication and/or pilot lights.
Interconnections and interlocking circuits between starter and remote equipment.
Remote sensors.
Tag numbers associated with all control devices and equipment.
One-line diagrams.
Schematic (elementary) diagrams.
Outline diagrams.

Submittals–Approval Not Required

Information/Record (IR):

Manufacturer's installation instructions.
Operation and maintenance data, including recommended preventative maintenance tasks and frequencies for performance of those tasks.

Submit documentation and test results for construction quality control testing.

Packing And Shipping:

Shipping Splits: Established by Construction Subcontractor to facilitate ingress of equipment to final installation location within the building.

Part 2 – Products (Reference Section 16005, Electrical)

Manufacturers

Square D.
Cutler-Hammer.
General Electric.
Allen-Bradley.

Motor Control

General:

Like Items of Equipment: End product and responsibility of one manufacturer.

Make adjustments as necessary to wiring, conduit, disconnect devices, motor starters, branch circuit protection, and other affected material or equipment to accommodate motors actually provided under this Contract.

Controllers: NEMA ICS 2, Class B.

Electronic Overload Protection:

Programmable solid-state electronic overload relay with integral CT(s) for monitoring three-phase current and voltage, thereby providing motor overload, phase reversal and phase loss protection. In the event of an alarm condition, electronic overload relay will de-energize fail safe alarm contact to motor control circuit.

Ratings:

Voltage Range (L1, L2-L3): 480 volts 50/60 Hz.

Current Range: Specific range compatible with motor FLA operation.

Power Consumption: 10 watts.

Trip Circuit: Form C SPDT (N.O and N.C) contacts, 5 amps 120 volts.

Measurements: Voltage, current, and timing.

Relay Trip: Standard Class 20.

Manual Reset.

Mount within starter unit.

Manufacturer:

Square D.
Allen-Bradley.
General Electric.
Cutler-Hammer.

Control Transformer:

Two winding, 120-volt secondary, primary voltage to suit.
Two current-limiting fuses for primary circuit.
One fuse in secondary circuit.
Mount within starter unit.
Suitable for use with 75 degrees C copper wire at full NFPA 70, 75 degrees C ampacity.
Lifting lugs on all equipment and devices weighing over 100 pounds.

Operating Conditions:

Ambient Temperature: Maximum 40 degrees C.

Equipment to be fully rated without any derating for operating conditions listed in Section 16005, ELECTRICAL.

Enclosures: In accordance with NEMA 250 and ANSI C57.12.28.

Equipment Finish:

Electrocoating process applied over a rust-inhibiting phosphated base coating.

Exterior Color: Manufacturer's standard.

Manually Operated Starter, Fractional Horsepower:

Rating: 16 amperes continuous at 277 volts maximum, or horsepower rated for the voltage and horsepower of the load served.

Single-phase, nonreversing, full voltage with overload protection.

Toggle operated.

Enclosure: Reference Section 16005, ELECTRICAL, Article ENVIRONMENTAL CONDITIONS.

Pilot LED Light: Red.

Handle guard/lock-off attachment.

Combination Full-Voltage, Magnetic Starter:

Rating: Hp rated at 600 volts, UL labeled for 42,000 amperes fault current withstand capacity with overload protection.

Three-phase, nonreversing, full voltage.

Control: HAND/OFF/AUTO selector switch.

Disconnect Type: Motor circuit protector.

Enclosure: Reference Section 16005, ELECTRICAL, Article ENVIRONMENTAL CONDITIONS.

Pilot LED Lights: Red—ON and Green—OFF.

Padlockable operating handle.

Kirk Key Interlocks

Provide Kirk Key interlocks for one main and one portable generator plug breaker arrangement in each MCC.

Provide engraved plate on MCC which describes Kirk Key breaker arrangement and operation as described herein.

Operation:

- One unique key available for MCC main and portable generator breaker locks (i.e., Kirk keys for additional MCC(s) shall not be identical).
- One of the two breakers CLOSED at any one time.
- Breaker must be opened before key can be removed and inserted.
- Key must be inserted and operated before breaker can be CLOSED.

Motor Control Centers

General:

Motor Control Center to be manufactured and provided as a complete UL-approved assembly that includes the following major components specified under this section and Section 16005, ELECTRICAL:

- Motor starters with electronic overload protection relays.
- Feeder and main breakers.
- Power monitoring.

Lighting and power distribution panelboard.

Lighting and power distribution stepdown transformer.

Transient Voltage Surge Suppressor (TVSS).

Interlocking control relays.

In accordance with NEMA ICS 2, UL 845, and UL 508/508A.

Voltage Rating: 600 volts.

Short Circuit Rating: 42,000 minimum amperes rms symmetrical for entire motor control center as a complete assembly.

All controllers, main and branch circuit breakers, wire connections, and other devices to be front mounted and accessible unless otherwise noted.

NEMA ICS 2, Section 322.08.

Class: I.

Type: B.

Wire remote control and signal circuits to separate terminal board in each motor starter compartment.

Enclosure:

Type: NEMA 250 Type 12 unless otherwise rated.

Vertical Section Dimensions: 90 inches high, 20 inches wide, 20 inches deep.

Construction:

Sheet steel reinforced with channel or angle irons.

Butt sections flush, end-to-end against similar section without bolts, nuts, or cover plates causing interference.

Removable top cover plates and bottom cover plates.

Section Mounting: Removable formed-steel channel sills and lifting angles to meet specified seismic requirements.

Horizontal Wiring Compartments: Accessible from front, full width, top and bottom.

Vertical Wiring Compartment: Full height, isolated from unit starters with separate door.

Unit Compartment: Individual compartments separated by steel barriers for each starter, feeder, or other unit capable of being wired from front without unit removal.

Compartment Doors: Separate hinged doors for each starter, feeder, or other unit.

Door Interlocking: Interlock starter and feeder doors mechanically so doors cannot be opened with unit energized. Provide defeater mechanism to allow intentional access at any time.

External disconnect handles, padlockable in OFF position.

Cable Entrance: Incoming service enters from bottom; control and feeder circuits enter from top and bottom.

Bus:

Horizontal Power Bus:

Three-phase tin-plated, fully insulated, copper, entire width of control center, rated 600 amperes.

Construct to allow future extension of additional sections.

Pressure type solderless lugs for each incoming line cable.

Isolated from top horizontal wireway.

Provide Belleville washers on bus connection bolts.

Vertical Power Bus:

Three-phase tin-plated, fully insulated, copper, full height of section, rated 300 amperes.

Sandwich type bus insulation providing deadfront construction with starter units removed except for bus stab openings.

Insulated and isolated barrier complete with shutters.

Provide Belleville washers on bus connection bolts.

Neutral Bus: 50 percent neutral, copper-tin-plated main breaker section only.

Ground Bus:

Copper, tin-plated, 33 percent minimum of phase bus ampacity, entire width of control center.

Provide Belleville washers on bus connection bolts.

Bus Bracing: 42,000 minimum amperes rms symmetrical.

Motor Controller Unit:

Provide indicated individual components and control devices including pushbuttons, selector switches, indicating lights, control relays, time delay relays, and elapsed time meters as specified in this section.

Construction:

Drawout combination type with stab connections for starters NEMA ICS, Size 4 and smaller.

Readily interchangeable with starters of similar size.

Pull-apart unit control wiring terminal boards on all units.

Starters: NEMA ICS 2, Section 322.08 standard rating, except none smaller than NEMA ICS, Size 1.

Rating: Hp rated at 600 volts, UL labeled for 42,000 amperes fault current withstand capacity with overload protection.

Three-phase, nonreversing.

Disconnect Type: Thermal magnetic as shown. Motor circuit protector may be substituted, properly sized and adjusted.

Combination Full Voltage, Magnetic Starter:

Control: ON/OFF/AUTO selector switch. As shown.

Pilot LED Lights: Red—ON; Green—OFF.

Padlockable operating handle when de-energized.

Unit door interlocked to prevent opening when disconnect is in closed position.

Mechanical interlocked to prevent placing disconnect in ON position when unit door is open.

Minimum Dimensions: 12 inches high by full section width, less vertical wireway.

Disconnecting Device: In each starter, control circuit disconnect to de-energize circuits in unit which are not de-energized by starter power disconnect device.

Padlockable in OPEN position.

Circuit Breaker:

Meeting the requirements of NEMA AB1 and UL 489.
Molded case with manufacturer's recommended trip setting for maximum motor protection.
Thermal-magnetic trip or magnetic trip only as shown.
Tripping indicated by operating-handle position.
Interrupting capacity required for connection to system with short circuit capacity indicated.

Motor Overload, Phase Reversal and Loss Protection:

Programmable solid-state electronic overload relay with internal CTs for monitoring three-phase current and voltage, thereby providing motor overload, phase reversal, and phase loss protection.

Make overload adjustments based upon motor FLA.

Make voltage adjustments based upon incoming voltage nominal readings.

Control Unit:

Disconnecting Device: Capable of de-energizing external source control circuits in unit.

Control Devices: As indicated and as specified in Section 16005, ELECTRICAL.

Control Wiring:

Minimum wire size No. 14 AWG copper.
Permanent sleeve type markers with wire numbers applied to each end of wires.
Terminate current transformer leads on shorting type terminal blocks.

Feeder Unit and Main Protective Device:

Construction: As specified in paragraph Motor Controller Unit.

Incoming Service Feeder: Cable entering section at bottom.

Molded Case Circuit Breaker:

In accordance with NEMA AB1 and UL 489.
Main, feeder, and motor protective device.
UL labeled as suitable for service entrance.
Thermal-magnetic trip and interrupting capacity required for connection to system with short circuit capacity indicated.
Indicate tripping by operating-handle position.
Suitable for use with 75 degrees C copper wire at full NEC 75 degrees C ampacity.

Reset Timer:

Timing Method: Solid state with LCD display.

Mounting: Semi-flush, panel.

Contacts: 5-amp, 120-volt.

Manufacturers and Products:

Square D.
Cutler-Hammer.
General Electric.

Magnetic Contactor:

UL listed.

Electrically operated, electrically held.

Main Contacts:

NEMA B600 contacts.

Electrically held.

Silver alloy with wiping action and arc quenchers.

NEMA Size 0 or 1 as required for the motor controller.

Three-pole.

Control: Two-wire.

One normally open and one normally closed auxiliary contact rated 10 amperes at 480 volts.

Manufacturers and Products:

Allen-Bradley.

Square D Co.; Type F.

Cutler-Hammer.

Pushbutton, Indicating Light and Selector Switches:

Contact Rating: NEMA ICS 2, Type A600.

Selector Switch Operating Lever: Standard.

Indicating Lights: Push-to-test, LED, full voltage.

Pushbutton Color:

ON or START: Black.

OFF or STOP: Red.

Pushbuttons and selector switches lockable in OFF position where indicated.

Legend Plate:

Material: Aluminum.

Engraving: 11 characters/spaces on one line, 14 characters/spaces on each of two lines, as required, indicating specific function.

Letter Height: 7/64 inch.

Manufacturers:

Square D Co.

Cutler-Hammer.

General Electric.

Allen-Bradley.

Nameplates:

Provide nameplates per Hanford standards.

Laminated plastic; white, engraved to black core.

Provide for each motor control center and each unit.

Engrave with inscription shown on single-line diagram.

Provide blank nameplates on spaces for future units.

Attach with stainless steel panhead screws on face of control center.

Factory Testing: NEMA ICS 1, Section 109, or UL 486A if not specified by the manufacturer.

Part 3 – Execution

Installation

Install equipment in accordance with NEMA ICS 2.3, Submittal Drawings, and Manufacturer's Instructions and Recommendations.

Secure equipment to mounting pads with anchor bolts of sufficient size and number adequate for specified seismic conditions. Reference Section 13122, METAL BUILDING SYSTEMS, Part 2, Article DESIGN LOADS, for information on seismic loading. Install suitable braces from MCC to building structural members for seismic support.

Install equipment plumb and in longitudinal alignment with pad or wall.

Coordinate terminal connections with installation of secondary feeders.

Grout mounting channels into floor or mounting pads.

Retighten current-carrying bolted connections and enclosure support framing and panels to manufacturer's recommendations.

Circuit Breakers (Magnetic-Trip-Only)

Field adjust trip settings of motor starter magnetic-trip-only circuit breakers.

Adjust to approximately 11 times motor rated current in accordance with NEC 430-52.

Determine motor rated current from motor nameplate following installation.

Overload Relay

Select and install overload relay settings after the actual nameplate full-load current rating of motor has been determined.

Motor Data

Provide typed, self-adhesive label attached outside each motor starter enclosure door displaying the following information with plastic black and white lettering, minimum 1/2-inch size:

- Motor served by tag number and equipment name.
- Nameplate horsepower.
- Motor code letter.
- Full load amperes.
- Service factor.
- Installed overload relay heater catalog number.

Construction Quality Control

In accordance with Section 16080, ELECTRICAL TESTING.

Manufacturer's Services

Furnish manufacturer's representative in accordance with Section 01640, MANUFACTURERS' SERVICES, for the following services at jobsite for minimum person-days listed below, travel time excluded:

- Person-day for installation assistance, and inspection of installation.
- Person-day for functional and performance testing.

END OF SECTION 16440

CHAPTER 5.0
GROUNDWATER MONITORING

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CHAPTER 5.0
GROUNDWATER MONITORING

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5.0 GROUNDWATER MONITORING

The Integrated Disposal Facility (IDF) will be an Resource Conservation and Recovery Act (RCRA)-compliant landfill (i.e., a double lined trench with leachate collection system). This chapter describes the groundwater monitoring plan for the IDF and addresses the requirements of RCRA, as described in 40 CFR 264, Subpart F, by reference of WAC 173-303-645(3). Figure 5.1 shows the location of the IDF and surrounding groundwater wells in the 200 East Area. This chapter is designed to meet final status detection-level groundwater monitoring requirements for the IDF. This groundwater monitoring plan is based on the application of a modified data quality objectives (EPA QA/G-4) process to a conceptual model, and the most recent evaluations of groundwater hydrology and chemistry at the site.

This plan describes the characteristics of the waste to be disposed in the IDF and the site geology and hydrology used to design and operate the monitoring well network and to interpret the groundwater data. The historic groundwater chemistry from wells near the IDF site is provided. Much of the information pertaining to waste characterization is taken from HNF-4921 and that pertaining to hydrogeology from PNNL-11957, PNNL-12257, PNNL-13652, and PNNL-14029.

The plan includes a description of network well locations, well construction, sample constituents, and sampling frequency for detection-level groundwater monitoring. Procedures for determination of compliance point groundwater quality also are included. Finally, this plan provides the basis for rapid development of a compliance monitoring plan if a validated exceedance of an indicator parameter is found. This plan controls initial baseline monitoring and subsequent detection level monitoring only for the IDF.

Source, special nuclear, and byproduct materials as defined by the *Atomic Energy Act of 1954*, as amended, are regulated at the United States Department of Energy (DOE) facilities exclusively by DOE acting pursuant to its AEA authority. These materials are not subject to regulation by the State of Washington. All information contained herein and related to, or describing AEA-regulated materials and processes in any manner may not be used to create conditions or other restrictions set forth in any permit, license, order, or any other enforceable instrument. DOE asserts that pursuant to the AEA, it has sole and exclusive responsibility and authority to regulate source, special nuclear and by-product materials at DOE-owned nuclear facilities. Information contained herein on radionuclides is provided for process description purposes only.

5.1 Exemption from Groundwater Protection Requirement

An exemption is not requested.

5.2 Interim Status Period Groundwater Monitoring Data

The IDF will be a new facility constructed in the 200 East Area. Interim status groundwater monitoring is not applicable.

5.3 Aquifer Identification

The following sections discuss geology and hydrology.

5.3.1 Geology of the IDF Site

The 200 East Area lies on the Cold Creek bar, a geomorphic remnant of the cataclysmic, glacial related floods of the Pleistocene Epoch. As the floodwaters raced across the lowlands of the Pasco Basin and Hanford Site, floodwaters lost energy and began to deposit sand and gravel. The 200 Area Plateau is one of the most prominent deposits. The 200 Area Plateau lies just southwest of one of the major flood channels across the Hanford Site that forms the topographic lowland south of Gable Mountain.

Borehole data provide the principal source of geologic, hydrologic, and groundwater information for the 200 East Area and the IDF site.

Numerous boreholes (both vadose zone boreholes and groundwater monitoring wells) have been drilled in the 200 East Area for groundwater monitoring and waste management studies (Figure 5.1 shows the location of groundwater wells near the IDF site.) However, data are limited within the IDF site primarily because no previous construction or waste disposal activities have occurred in this part of the Hanford Facility. Most boreholes in the 200 East Area have been drilled using the cable tool method and either a hard tool or drive barrel to advance the hole. Some boreholes have been drilled by rotary and wire line coring methods. More recently, boreholes in the area have been drilled, and in five cases cored, by percussion hammer methods. Geologic logs are based on examination of drill core, chips, and cuttings from these boreholes. Chip samples typically are taken at 1.5-meter intervals and routinely archived at the Hanford Geotechnical Sample Library.

5.3.1.1 Structural Framework

The IDF site will be located south of the Gable Mountain segment of the Umtanum Ridge anticline and about 3 kilometers north of the axis of the Cold Creek syncline, which controls the structural grain of the basalt bedrock and the Ringold Formation. The basalt surface and Ringold Formation trend roughly southeast-northwest parallel to the major geologic structures of the site. As a result, the Ringold Formation and the underlying Columbia River Basalt Group gently dip to the south off the Umtanum Ridge anticline into the Cold Creek syncline.

Geologic mapping on the Hanford Site and examination of drill core and borehole cuttings in the area have not identified any faults in the vicinity of the IDF site (DOE/RW-0164). The closest known faults are along the Umtanum Ridge-Gable Mountain structure north of the disposal site and the May Junction fault east of the site (Figure 5.2).

5.3.1.2 Stratigraphy

The basalt and post-basalt stratigraphy for the IDF site is shown in Figure 5.3. Approximately 137 to 167 meters of suprabasalt sediments overlie the basalt bedrock at the site.

Basalt Bedrock. Previous studies (RHO-BWI-ST-14; Reidel and Fecht 1994) have shown that the youngest lava flows of the Columbia River Basalt Group at the 200 East Area are those of the 10.5 million year old Elephant Mountain Member. This member underlies the entire 200 East Area and surrounding area and forms the base of the suprabasalts aquifer. No erosional windows in the basalt are known or suspected to occur in the area of the IDF site.

Ringold Formation. Few boreholes penetrate the entire Ringold Formation at the IDF site so available data are limited. The Ringold Formation reaches a maximum thickness of 95 meters on the west side of the site and thins eastward. The member of Wooded Island (Figure 5.3) is the only member of the Ringold Formation in the 200 East Area. The deepest Ringold Formation unit encountered is the lower gravel, unit A. Lying above unit A is the lower mud unit and overlying the lower mud unit is upper gravel, unit E. The sand and silt units of the members of Taylor Flat and Savage Island of the Ringold Formation are not present at the IDF site. Unit A and unit E are equivalent to the Pliocene-Miocene continental conglomerates (Reidel and Fecht 1994). The lower mud unit is equivalent to the Pliocene-Miocene continental sand, silt, and clay beds (Reidel and Fecht 1994).

Only three boreholes have penetrated unit A in the area of the IDF site. Unit A is 19 meters thick on the west side of the site and thins to the northeast. Unit A is partly to well cemented conglomerate consisting of both felsic and basaltic clasts in a sandy matrix and is interpreted as a fluvial gravel facies (Lindsey 1996). There are minor beds of yellow to white interbedded sand and silt. Green colored, reduced-iron stain is present on some grains and pebbles. Although the entire unit appears to be cemented, the zone produced abundant high quality water in borehole 299-E17-21 (PNNL-11957).

Nineteen meters of the lower mud unit were encountered in one borehole at the IDF site (PNNL-11957). The upper most 1-meter or so consists of a yellow mud to sandy mud. The yellow mud grades downward into about 10 meters of blue mud.

The blue mud, in turn, grades down into 7 meters of brown mud with organic rich zones and occasional wood fragments. The lower mud unit is absent in the center of the site (northeast of borehole 299-E24-7 on Figure 5.4).

Unit E is described as a sandy gravel to gravelly sand. Unit E is interpreted to consist of as much as 15 meters of conglomerate with scattered large pebbles and cobbles up to 25 centimeters in size in a sandy matrix. The gravel consists of both felsic and basaltic rocks that are well rounded with a sand matrix supporting the cobbles and pebbles. Cementation of this unit ranges from slight to moderate. The upper contact of unit E is not identified easily at the IDF site. In the western part of the study area, unconsolidated gravels of the Hanford formation directly overly the Ringold Formation unit E gravels, making exact placement of the contact difficult. The dominance of basalt and the absence of cementation in the Hanford formation are the key criteria used to distinguishing these (PNNL-11957). In the central and northeast part of the area, unit E has been eroded completely. Unconsolidated gravels and sands typical of the Hanford formation replace unit E.

Unconformity at the Top of the Ringold Formation. The surface of the Ringold Formation is irregular in the area of the IDF site. A northwest-southeast trending erosional channel or trough (the Columbia River/Missoula flood channel) is centered through the northeast portion of the site. The trough is deepest near borehole 299-E24-21 in the northern part of the site (PNNL-13652). This trough is interpreted as part of a larger trough under the 200 East Area resulting from scouring by the Missoula floods. Borehole 299-E17-21, located at the southwest corner of the IDF site, is at the west side of the channel where approximately 46 meters of Ringold Formation have been removed and replaced by Hanford formation gravels. Boreholes 299-E17-25 and 299-E17-23, located along the southeastern edge of the Site, are near the deepest portion of the channel where it is interpreted that almost all of the Ringold Formation has been eroded. At this location, the water table in the channel is interpreted to be 52 meters above the basalt, which forms the floor of the channel. The surface of basalt rises to the north where the water table is approximately 27 meters above the basalt at the northeast corner of the site near borehole 299-E24-21.

Hanford formation. The Hanford formation is as much as 116 meters thick in and around the IDF site. The Hanford formation thickens in the erosional channel cut into the Ringold Formation and thins to the southwest along the margin of the channel.

At the IDF site, the Hanford formation consists mainly of sand dominated facies with lesser amounts of silt dominated and gravel dominated facies. The Hanford formation has been described as poorly sorted pebble to boulder gravel and fine- to coarse-grained sand, with lesser amounts of interstitial and interbedded silt and clay. In previous studies of the site (WHC-MR-0391), the Hanford formation was described as consisting of three units: an upper and lower gravel facies and a sand facies between the two gravelly units. The upper gravel dominated facies appears to be thin or absent in the immediate area of the IDF site (PNNL-12257, PNNL-13652, and PNNL-14029).

The lowermost part of the Hanford formation encountered in boreholes at the IDF site consists of the gravel-dominated facies. Drill core and cuttings from boreholes 299-E17-21, 299-E17-22, 299-E17-23, 299-E17-25, and 299-E24-21 indicate that the unit is a clast-supported pebble- to cobble gravel with minor amounts of sand in the matrix. The cobbles and pebbles almost are exclusively basalt with no cementation. This unit pinches out west of the IDF site and thickens to the east and northeast (Figure 5.4). The water table beneath the IDF site is located in the lower gravel unit. The lower gravel unit is interpreted to be Missoula flood gravels deposited in the erosional channel carved into the underlying Ringold Formation.

The upper portion of the Hanford formation consists of at least 73 meters of fine- to coarse-grained sand with minor amounts of silt and clay and some gravelly sands.

Holocene Deposits. Holocene, eolian deposits cover the southern part of the IDF site. Caliche coatings on the bottom of pebbles and cobbles in drill cores through this unit are typical of Holocene caliche development in the Columbia Basin.

The southern part of the IDF site is capped by a stabilized sand dune. The eolian unit is composed of fine- to coarse-grained sands with abundant silt, as layers and as material mixed with the sand.

Clastic Dikes. A clastic dike was encountered in borehole C3828, adjacent to well 299-E17-25, at the IDF site. Clastic dikes also have been observed in excavations surrounding the site [e.g., US Ecology, the former Grout area, the 216-BC cribs, the Central Landfill, and the Environmental Restoration Disposal Facility (BHI-01103)]. In undisturbed areas, such as the IDF site, clastic dikes typically are not observed because these are covered by windblown sediments. The occurrence of a clastic dike in borehole C3828 suggests that these probably are present elsewhere in the subsurface at the disposal site. The IDF excavation will be geologically mapped to document the occurrence of any clastic dikes that may exist at the site.

5.3.2 Groundwater Hydrology

The unconfined aquifer under the IDF site occurs in the fluvial gravels of the Ringold Formation and flood deposits of the Hanford formation. The thickness of the aquifer ranges from about 70 meters at the southwest corner of the site to about 30 meters under the northeast corner of the IDF site. The Elephant Mountain Member of the Columbia River Basalt Group forms the base of the unconfined aquifer (Figure 5.4).

The unsaturated zone beneath the land surface at the IDF site is approximately 100 meters thick and consists of the Hanford formation. The water level in boreholes in and around the site indicates that the water table is in the lower gravel sequence of the Hanford formation and at an elevation of approximately 123 meters above sea level. The water table is nearly flat beneath the IDF site. Table 5.1 gives water level information from wells near the site. The locations of the wells are shown on Figure 5.1. The latest water table map shows less than about 0.1 meter of hydraulic head differential across the IDF site (Figure 5.5).

The Ringold Formation lower mud unit occurs within the aquifer at the southwest corner of the IDF site (299-E17-21) but is absent in the central and northern parts of the site (299-E24-7 and 299-E24-21). The lower mud unit is known to be a confining or partly confining layer at places under the Hanford Site (PNNL-12261) and this might be the case under the southwest corner of the IDF site. Groundwater samples were collected and analyzed from above and below the lower mud unit during drilling of well 299-E17-21. Chemical parameters (pH, electrical conductivity, and Eh) were different in the two samples suggesting that the lower mud is at least partly confining in the area. No contamination was found above or below the lower mud. An interpretation of the distribution and thickness of this stratum is shown in Figure 5.4. The surface of the lower mud unit is interpreted to dip gently to the southwest (PNNL-13652).

Hydrographs for selected wells near the IDF site are shown in Figure 5.6. Although the water table is extremely flat in the area of the IDF, hydrographs suggest that groundwater flow has had an easterly component throughout the 1990s and has not significantly changed due to cessation of discharges to the 216-B Pond system. Hydrographs for the older wells (299-E23-1, 299-E23-2, and 299-E24-7) show two maxima in the water level. These coincide with the operation of the PUREX Plant, which operated between 1956 and 1972 and between 1983 and 1988. All the hydrographs show a decline in the water table during recent years. The rate of decline is between 0.18 and 0.22 meter per year and will take between 10 and 30 years to stabilize. The reason for the decline is the cessation of effluent discharge to the 216-B Pond System, which is centered northeast of 200 East Area. Based on hindcast water table maps (BNWL-B-360), the water table is expected to decline another 2 to 7 meters before reaching pre-Hanford Site elevations. The cessations of effluent discharge also are responsible for changes in the direction of groundwater flow across much of the 200 East Area.

Groundwater flow beneath the IDF site recently was modeled to be southeasterly (PNNL-13400). This direction differs from the easterly direction predicted by the analysis of WHC-SD-WM-RPT-241 and other earlier reports.

The southeasterly flow direction primarily is attributable to inclusion of the highly permeable Hanford formation sediments in the ancestral Columbia River/Missoula flood channel in the analysis. A southeasterly flow direction is reflected in the geographic distribution of the regional nitrate plume and in the distribution of other constituents under the south-central 200 East Area (PNNL-14187, 1 of 2, 2 of 2). As stated in PNNL-13404 (1 of 2, 2 of 2), the water table gradient is too low to be used for determining flow direction or flow rate at the PUREX Plant cribs immediately east of the IDF site.

Hydraulic conductivity directly beneath the IDF site was estimated from data collected during four slug tests at well 299-E17-21 and five slug tests of 299-E24-21. The interval tested at 299-E17-21 was the upper 7.8 m of the unconfined aquifer from 101.3 to 109.1 m depth. That portion of the aquifer is Hanford formation gravel from 101.3 to 102.1 m depth and Ringold Formation unit E gravels from 102.1 to 109.1 m depth (PNNL-11957). The interval tested at well 299-E24-21 was entirely in the Hanford formation gravel sequence between 95.2 and 101.3 m depth. The best fit value to the data from 299-E17-21 indicated a hydraulic conductivity of about 68.6 meters per day (PNNL-11957) and from 299-E24-21 suggested a hydraulic conductivity of 75 meters per day (PNNL-13652).

5.4 Contaminant Plume Description

Although no groundwater monitoring has been done for the IDF, groundwater monitoring has been done in support of RCRA permitting activities and in support of other activities in the area. The results of that monitoring show that a regional nitrate plume exists beneath the IDF site (PNNL-14187, 1 of 2, 2 of 2). In the south-central 200 East Area, the plume extends in a northwest - southeast direction along the axis of the Columbia River/Missoula flood channel eroded into the Ringold Formation sediments. The channel is filled with more transmissive Hanford formation sediments.

5.4.1 Groundwater Contamination

Nitrate, associated with past-practice activities in 200 East Area, is a general groundwater chemistry parameter and is not a contaminant of concern for the IDF. However, the distribution of existing nitrate in the groundwater gives an indication of the general groundwater flow direction and the influence that adjacent sites might have on the IDF.

High nitrate concentrations found near liquid waste disposal facilities located outside the IDF site that received effluent from the PUREX Plant are decreasing steadily with time. The highest nitrate concentration found in 2002 was 170,000 µg/L in well 299-E17-9 at the 216-A-36B crib and the crib is thought to be the source of the nitrate. The drinking water standard for nitrate is 45,000 µg/L (nitrate ion).

Nitrate in well 299-E24-18, just inside the east boundary of the IDF site, decreased from a high of 86,300 µg/L in 1990 to a low of 17,000 µg/L in 1993, reflecting the cessation of PUREX Plant operations in 1988. Since 1993, nitrate has increased to 48,300 µg/L in 2003 (Figure 5.7). The reason for the increase is not understood. One possibility is related to changing groundwater flow direction. During PUREX Plant operations, flow direction was probably to the northwest because of effluent discharges to the B Pond System and PUREX Plant cribs, and nitrate contamination might have spread to the northwest during that period. Subsequently, liquid discharges to the B Pond System and PUREX Plant cribs have ceased and the flow direction in the area of the IDF site apparently has returned to the southeast direction. With that change, higher levels of nitrate contaminated groundwater might be returning to the area from the northwest.

Except for an anomalous value of 82,600 µg/L in 1988, nitrate concentration in well 299-E24-7 was steady and ranged between 12,800 and 35,400 µg/L between 1985 and 1996 when the well was last sampled (Figure 5.7). The last two measured values from 1995 and 1996 were 26,000 µg/L. Farther southwest, nitrate detected in 1998 in well 299-E17-21 in Ringold unit E was 23,600 µg/L.

5.4.2 Vadose Zone Contamination

Very little characterization and monitoring of the soil have been done at the IDF site because no major construction or waste disposal activities have occurred in this part of the Hanford Site.

Implementation of the *Integrated Disposal Facility Preoperational Monitoring Plan* (RPP-6877) has begun and characterization activities will occur during the next few years. The *Integrated Disposal Facility Preoperational Monitoring Plan* (RPP-6877) has a strong emphasis on vadose zone characterization and deferred groundwater monitoring to this groundwater monitoring plan. Vadose zone information resulting from preoperational monitoring will be included, if applicable, in updates to this groundwater monitoring plan.

The *Integrated Disposal Facility Preoperational Monitoring Plan* (RPP-6877) identified three areas near the IDF site that might have had an influence on the vadose zone beneath the site. These are the 218-E-1 Burial Ground and an unplanned release associated with the burial ground; the coal ash pile in the northwest part of the site; and a transfer line along the northern part of the west boundary of the IDF site (RPP-6877). Work was outlined in the *Integrated Disposal Facility Preoperational Monitoring Plan* to determine whether these three areas had introduced contamination to the site. Appropriate results from preoperational monitoring will be incorporated into this groundwater monitoring plan as results become available and as revisions are needed.

In addition to these facilities, the 216-A-38-1, 216-A-45, and 216-A-10 cribs and the 299-E24-111 injection well are located east of the IDF site. The 216-A-38-1 crib never was used (DOE/RL-92-04). The 299-E24-111 injection well never received any waste (DOE/RL-92-04). The 216-A-45 and the 216-A 10 cribs both received large quantities of liquid waste (DOE/RL-92-04). Because these latter two facilities are more than 200 meters from the IDF site, it is unlikely these facilities have affected the soil beneath the IDF site. Data from the vadose zone in IDF wells drilled along the east side of the site support this.

5.5 Detection Monitoring Program

Because the IDF has not been constructed, no contaminants have been released to the ground or to the groundwater.

5.5.1 Indicator Parameters, Waste Constituents, Reaction Products to be Monitored

5.5.1.1 Regulated Constituents

The regulated constituents for this groundwater monitoring plan are the constituents identified on the IDF Part A Form.

5.5.1.2 Monitoring Parameters

The parameters to be routinely monitored are listed in Table 5.2. These parameters include the indicator parameters and supplemental parameters.

The indicator parameters will be used to monitor for hazardous constituents reaching the groundwater as a result of IDF operations. Only the indicator parameters are subject to the statistical methods described in Section 5.5.4.7. Total organic carbon and total organic halides are indicator parameters selected to monitor impacts of RCRA regulated organic constituents on the groundwater quality. Specific conductance is selected as an indicator parameter to monitor impacts of metals and anions on groundwater quality. pH is a general indicator of groundwater quality. Specific conductance and pH are measured in the field at the time of sampling. Chromium is included as an indicator parameter because hexavalent chromium is one of the more mobile of the regulated metals to be disposed of at the IDF and should be one of the first constituents to enter groundwater if the regulated facility impacts groundwater.

Analyses of alkalinity, anions, and metals are to provide supplemental data on general groundwater chemistry beneath the IDF. This information aids data interpretation and quality control. Supplemental parameters will not be used in statistical evaluations. Turbidity is analyzed at the well just before sampling and provides an indication of the groundwater condition at the time of sampling.

For the first year of monitoring, all parameters listed in Table 5.2 will be monitored twice each quarter to determine background concentrations.

After the first year, indicator and supplemental parameters will be monitored semi-annually. In addition, field measurements of temperature and turbidity will be made at each sampling event.

During the first sampling event at each well for the first year of monitoring, samples will be collected for analysis of the indicator parameters, the supplemental parameters, and the Appendix IX constituents (40 CFR 264) included in IDF Part A Form. After the first sampling event, samples will be collected for analysis of indicator parameters and supplemental parameters only.

After the first year of sampling, if an indicator parameter suggests there is an impact to groundwater, additional samples will be collected to verify the initial results. If a statistically significant increase in any indicator parameter is confirmed, analyses will be made for the regulated parameters in IDF Part A Form.

Monitoring for baseline conditions was completed for the indicator parameters in April 2006 and for the complete Appendix IX list in January 2007. Semi-annual monitoring has continued since that time with the collection of four independent samples each semiannual period. During the Pre-Active life, sampling will continue at the IDF with the collection of one sample each year to maintain the baseline. When the IDF becomes operational, sampling will revert to four independent samples collected each semiannual period.

5.5.1.3 Dangerous Waste Characterization

This section describes the waste to be disposed in the IDF and gives background information on how the constituents of concern (regulated constituents) and indicator parameters were selected.

5.5.1.3.1 Volume of the Waste Package

The IDF will be a single, expandable disposal facility constructed to RCRA Subtitle C standards, half of which is for disposal of mixed waste the other half will be for disposal of low-level waste. Initial capacity for mixed waste disposal is 82,000 cubic meters of waste with an ultimate capacity of up to 450,000 cubic meters of waste. Disposal capacity beyond the initial 82,000 cubic meters will require a modification to the Part B Permit. The mixed waste types to be disposed in the IDF include vitrified Low Activity Waste (LAW) from the RPP-WTP and DBVS. Additionally, mixed waste generated by IDF operations will be disposed of in IDF.

The vitrified LAW will be mostly silicate glass monoliths. The RPP-WTP packages nominally measure approximately 1.22 m diameter by 2.3 m high and the DBVS package nominally measure approximately 2.4 m wide by 3.1 m high by 7.3 m long. Vitrified LAW will be remote handled.

If other forms of immobilized LAW are considered in the future, this monitoring plan will be amended.

Mixed waste generated through waste operations at IDF will be packaged based on the size of the waste, with the most common container being galvanized or aluminized 208 liter containers.

5.5.1.3.2 Composition of the Waste Packages

HNF-4921 provides detailed estimates for the inventory of hazardous chemicals in the vitrified LAW feed and in the vitrified LAW package. The composition of the vitrified LAW package was estimated in HNF-4921 based on:

1) The Tank Waste Retrieval System Characterization Program tank-by-tank Best Basis Inventories.
2) The latest U.S. Department of Energy, Office of River Protection (DOE/ORP) guidance.
3) The requirements for waste retrieval and vitrification.
4) Available information from waste treatment plant contractors, and (5) proposed operating scenarios for retrieval of waste from Double Shell Tanks (DSTs) and Single Shell Tanks (SSTs).

5.5.1.4 Behavior of Constituents

Almost all of the regulated constituents for the IDF show some degree of retardation in the vadose zone and in the saturated zone.

Table 5.3 indicates the range of expected behaviors in the subsurface at the IDF for selected regulated constituents. The constituents in Table 5.3 were selected by comparing the expected constituents in the vitrified LAW package (from HNF-4921) and the historical inventories of the Hanford Site low-level burial grounds (from WHC-MR-0008 and WHC-SD-EN-AP-015) to 40 CFR 264, Appendix IX (see IDF Part A Form). The mobilities and solubilities in Table 5.3 give an estimated range for the properties of the constituents of concern.

5.5.1.5 Detectability

The detection limits in groundwater for each RCRA regulated constituent and the indicator parameters are given in Table 5.4.

5.5.2 Groundwater Monitoring Program

The following sections provide a description of wells, equipment decontamination, representative samples, and monitoring wells that are not upgradient.

5.5.2.1 Description of Wells

The groundwater monitoring well network for the IDF ultimately will have eight wells: three hydraulically upgradient of the facility and five hydraulically downgradient. The downgradient wells will be placed to sample groundwater passing the point of compliance. The point of compliance at the IDF site is a plane connecting the groundwater monitoring wells along the southern and eastern sides of the site in accordance with WAC 173-303-645(6), which states "The point of compliance is a vertical surface located at the hydraulically downgradient limit of the waste management area that extends down into the uppermost aquifer underlying the regulated unit". The monitoring network will consist of existing and new, downgradient wells to complete the monitoring network. All wells will be WAC 173-160 compliant.

Three upgradient wells will be used for the IDF monitoring network. Two of these wells (299-E18-1 and 299-E24-21) are existing wells. Upgradient well 299-E24-21 was installed in March 2001 for characterization of the IDF site. The well, located at the northeast corner of the site (Figure 5.8), was constructed to RCRA standards as per WAC 173-160. Well 299-E18-1 was installed in 1988 as part of the 2101-M RCRA monitoring network. The well currently has 2 to 3 meters of water above the bottom of the screened interval.

The third upgradient well will be a new well located at the northwest corner of the IDF (Figure 5.8). The well will be constructed to RCRA standard as per WAC 173-160 and screened at the water table.

Three of the downgradient wells are existing wells (299-E17-22, 299-E17-23, and 299-E17-25) that were installed as WAC 173-160 compliant wells in 2002. Their location is shown in Figure 5.8. The remaining two downgradient wells will be installed in a sequence coordinated with the IDF operations.

Three phases of trench construction are assumed for the purposes of this monitoring plan. Excavation for the first phase is scheduled for September 2004 and a new phase is planned for every ten subsequent years. Changes in the planned operations of the IDF will be reflected in changes to this groundwater monitoring plan as needed.

The first new downgradient well will be installed along the eastern side of the facility (Figure 5.8) at least one year before the IDF receives waste. The second new downgradient well will be installed along the southern boundary of the Site at least one year before the third phase of waste disposal becomes operational. Both wells will be installed such that at least one year of background data can be obtained prior to the associated operational phase becoming active. Figure 5.8 shows the sequence for both groundwater well construction and waste disposal. The locations of all existing and new wells in the IDF monitoring network are noted on the figure.

The placement of the wells for the IDF monitoring network was based on professional judgment. The efficiency of the resulting groundwater monitoring network was evaluated using a simple two dimensional, horizontal transport model called the monitoring efficiency model (MEMO) (Wilson et al. 1992). The model estimates the efficiency of a monitoring network at the point of compliance. The model simulates a contaminant plume originating from a series of grid points within the disposal facility using the Domenico-Robbins method (Domenico and Robbins, 1985). The model calculates both advective flow and dispersive flow in two dimensions and determines whether the resulting plume will be detected by a monitoring well before the plume travels some selected distance beyond the disposal facility boundary. The selected distance is termed the buffer zone. (A longitudinal dispersivity of 95 meters and horizontal dispersivity of 9.5 meters were used to evaluate the monitoring network in Figure 5.8.) Outputs from the model are the monitoring efficiency and a map of the disposal facility showing areas where leaks would not be detected under the given site-specific parameters provided as input to the model. Monitoring efficiency is defined as the ratio of the area within a disposal facility from which a release likely would be detected to the total area of the disposal facility, expressed as a percentage.

The monitoring efficiency calculated by the MEMO model for the proposed monitoring network is 100% for phase I, 98% for phase II, and 99% for phase III (Figure 5.8).

All wells for the IDF site will be constructed to meet WAC 173-160 requirements. The wells will be protected at the surface with a concrete pad, protective posts, a protective outer casing, and locking cap. The casing and screen will be stainless steel, an appropriate filter pack for the screen slot size will be used, and an annular seal of bentonite and cement will be emplaced. All wells will be screened at the water table with 10.6 meter long screens, which will accommodate the greatest possible future decrease in water level. The wells will be developed and dedicated sampling pumps will be installed.

New wells will be surveyed with a down hole gyroscope at the time of construction to determine any deviation from vertical so that corrections can be made to subsequent water level measurements. Gyroscope surveys will also be conducted on existing wells in the network prior to IDF operations.

5.5.2.2 Equipment Decontamination

Drilling equipment will be decontaminated using high temperature and pressure [82°C (180°F) and greater than 70.3 kg/cm^2 (1,000 psi)] washing with an approved cleaning solution. The equipment will be rinsed with clean water. The procedure is specified in controlled manuals.

Equipment for collecting soil samples during drilling for later chemical analysis and for measuring the water table will be decontaminated according to established methods. The methods call for washing equipment with phosphate free detergent, rinsing three times with reverse osmosis/de-ionized water, rinsing once with 1M or 10% nitric acid (glass or stainless steel equipment only), rinsing three more times with reverse osmosis/de-ionized water, and a final rinse with chromatograph grade hexane. Equipment will be dried for 50 minutes at 100°C (212°F). After drying, equipment will be wrapped in unused aluminum foil and sealed with tape.

No decontamination of groundwater sampling equipment will be necessary because each well will have a dedicated pump.

5.5.2.3 Representative Samples

No groundwater chemistry data specific to the IDF site are available. Sample representativeness will be addressed after collection of the first year of background data.

5.5.2.4 Locations of Background Groundwater Monitoring Wells that are not Upgradient

All background groundwater monitoring wells at the IDF are located upgradient.

5.5.3 Background Values

Groundwater background (baseline) has not been established for the IDF site. Background data will be determined before construction of the site using the wells described previously (Section 5.5.2.1) for the use of upgradient vs. downgradient comparisons (Section 5.5.4.7).

5.5.3.1 Plan for Establishing Groundwater Quality Data

Well location, sampling frequency, sampling quantity, and background values are discussed in the following sections.

5.5.3.1.1 Well Locations

Groundwater monitoring wells in the IDF monitoring network were described in Section 5.5.2.1 and their locations are shown on Figure 5.8.

5.5.3.1.2 Sampling Frequency

Eight background samples will be collected during the first year of monitoring from phase I wells. Two samples will be collected quarterly for one year. For the new well needed for phase III operations, two samples will be collected quarterly for one year before phase III is operational. For all wells, two independent samples will be collected each quarter, one per month for 2 consecutive months followed by a month of non-sampling. This sequence will be repeated each quarter during the first year of monitoring. Section 5.5.3.1.3 provides frequency logic.

5.5.3.1.3 Sampling Quantites

The performance of the statistical method proposed for the IDF is evaluated by the following two goals:

- To have adequate statistical power to detect real contamination when contamination occurs.
- To keep the network wide Type I error (across all constituents and wells being tested) at an acceptably low level (approximately 5%). [Note that the Type I error in the detection monitoring stage equates to the false positive rate, that is, the probability that the test will indicate contamination has occurred although no contamination has truly occurred.]

The statistical power and the network-side false positive rate of a test depend on several factors, including the background sample size, the type of proposed test, and the number of comparisons. All other factors being equal, the larger the sample size is (i.e., the number of background samples), the greater the statistical power is. Therefore, as recommended in EPA/530-R-93-003, at least eight independent samples will be collected from each well for background purposes. This is a sufficient number of samples to establish a reliable background (EPA/530-R-93-003) and meets the regulations in WAC 173-303-645(9)(d).

5.5.3.1.4 Background Values

The default method of analysis of variance (ANOVA) will be used to detect any impact on groundwater quality at the IDF where the mean of the measurements from compliance (downgradient) wells is compared to the mean of the distribution of background data from the upgradient wells. The details of the method are described in Section 5.5.4.7.1.

5.5.4 Sampling, Analysis and Statistical Procedures

Sample collection, sample preservation and transfer/shipment, analytical procedures, chain of custody and additional requirements for compliance point monitoring are discussed in the following sections.

5.5.4.1 Sample Collection

Groundwater sampling procedures, sample collection documentation, sample preservation and transfer/shipment, and chain-of-custody requirements are described in subcontractor operating procedures/manuals and in a quality assurance project plan for the Hanford Groundwater Performance Assessment Project.

Quality requirements for sampling activities, including requirements for procedures, containers, transport, storage, chain of custody, and records requirements, are specified in a statement of work (SOW) to subcontractors. To ensure that samples of known quality are obtained, the subcontractor will be required to use contractor controlled procedures based on standard methods for groundwater sampling whenever possible. The procedures will be reviewed for technical quality and consistency. In addition, periodic assessments of sample collection activities will be performed to ensure further that procedures are followed to maintain sample quality and integrity. The following is a brief description of the sampling requirements.

Samples generally will be collected after three casing volumes of groundwater are withdrawn or after the field parameters pH, temperature, and specific conductance have stabilized. Field parameters are measured in a flow through chamber. Turbidity should be equal to or below 5 NTU (nephelometric turbidity units) before sample collection if possible. Sample preservatives will be added to the collection bottles in the laboratory before their use in the field. Samples to be analyzed for metals will be filtered in the field to ensure results represent dissolved metals and do not include particulates (40 CFR 136.3). Duplicates, trip blanks, and field equipment blanks will be collected as part of the general quality control program.

Water level measurements will be made each time a well is sampled. Procedures developed in accordance with the techniques described in American Society for Testing and Materials (ASTM 1988), Garber and Koopman (1968), and U. S. Geological Survey (1977) will be followed to measure water levels. Water levels will be measured primarily with laminated steel electrical sounding tapes, although graduated steel tapes are used occasionally.

5.5.4.2 Sample Preservation and Shipment

Sample preservation will be done in accordance with existing procedures. A chemical preservative label will be affixed to the sample container listing the specific preservative. The brand name, lot number, concentration, and date opened of the preservatives will be recorded. A calibrated dispenser or pipette will be used to dispense preservatives. Appropriate measures will be taken to eliminate any potential for cross contamination.

Sample packaging and transfer/shipping will be done in accordance with subcontract procedures. Samples will be labeled and sealed with evidence tape, wrapped with bubble wrap, and placed in a Department of Transportation approved container with coolant (if required). Hazardous samples will have packaging parameters determined by associated hazards. A chain of custody will accompany all samples.

5.5.4.3 Analytical Procedures

The methods for analysis of chemical constituents in groundwater will conform to *Test Methods for Evaluating Solid Wastes: Physical/Chemical Methods, 3rd Ed.* (SW-846); *Methods for Chemical Analysis of Water and Wastes* (EPA-600/4-79-020) or other EPA methods; and the *Annual Book of ASTM Standards* (American Society for Testing and Materials, 1986). The methods used to obtain routine data results are presented in Table 5.4.

5.5.4.3.1 Data Storage and Retrieval

All contract analytical laboratory results will be submitted by the laboratory to be loaded into the Hanford Environmental Information System (HEIS) database. Most data are received from the laboratory in electronic form, and will be loaded electronically. Parameters measured in the field will be entered into HEIS either manually or through electronic transfer. Hard copy data reports are received for records storage. Data from the HEIS database will be retrieved for data validation, data reduction, and trend analysis. Copies of supporting analytical data will be sent yearly to Pacific Northwest National Laboratory (PNNL) for storage.

5.5.4.3.2 Data Verification and Validation

Verification of analytical data provided by the subcontracted laboratory will be performed in accordance with established procedure. This procedure includes checks for: (1) completeness of hardcopy deliverable, (2) condition of samples on receipt by the laboratory, (3) problems that arose during the analysis of the samples, and (4) correct reporting of results. The procedure also describes the actions to be taken if data are incomplete or deficient.

Verification and validation of groundwater chemistry data will be performed according to established procedures. Data will be reviewed quarterly to assure the data are complete and representative. The review will include evaluation of quality control data (e.g., field blanks, duplicates, and laboratory blanks) and a technical review by a project scientist familiar with the hydrogeology of the site. The technical review might include comparison of recent data to historical trends and comparison of related constituents. Suspect data will be investigated through the data review process in accordance with established procedures and will be flagged in the database.

5.5.4.3.3 Reporting

Groundwater chemistry and water level data will be reviewed after each sampling event and will be available in the HEIS database. The results of the statistical evaluation and associated information will be submitted to Ecology quarterly in Hanford Site groundwater monitoring reports.

If statistically, significant evidence of contamination is determined (after waste has been introduced to the facility and after the confirmation re-sampling evaluation process) for one or more of the indicator parameters at any monitoring well at the compliance point, and if the owner or operator decides not to make a false positive claim, the following will be performed.

- Notify Ecology in writing within 7 days of the finding indicating which chemical parameters or dangerous waste constituents have shown statistically significant evidence of contamination.
- Determine whether dangerous constituents are present and, if so, in what concentration.
- The owner or operator might re-sample within 1 month and repeat the analysis for those compounds detected in the above (i.e., second bullet). The resample data will be compared with the trigger value.
- Submit an application for a permit modification, if necessary, to establish a compliance monitoring program to Ecology in 90 days or within the time agreed to in writing by Ecology.

The dangerous constituents detected, either in the initial analysis or in the second confirmation analysis, will form the basis for compliance monitoring.

In case of a false positive claim [as allowed by WAC 173-303-645(9)(g)(vi)], the following will apply.

- Notify Ecology in writing within 7 days of the finding (i.e., exceedance) and indicate that a false positive claim will be made.
- Submit a report to Ecology within 90 days or within the time agreed to in writing by Ecology. This report should demonstrate that a source other than the regulated unit caused the contamination or that the contamination resulted from an error in sampling, analysis, evaluation, or natural variation in groundwater chemistry.
- Submit an application for a permit modification, if necessary, to make any appropriate changes to the detection monitoring program within 90 days or within the time agreed to in writing by Ecology.
- Continue to monitor in accordance with the detection monitoring program.
- Submit an application for a permit modification, if the detection monitoring program is determined to no longer satisfy the requirements [of WAC 173-303-645(9)], to make any appropriate changes to the program within 90 days or within the time agreed to in writing by Ecology.

5.5.4.4 Chain of Custody

The procedures used for chain-of-custody control of samples are documented in existing manuals. The procedure requires that each transfer of custody shall be documented by the signatures of the custodian relinquishing the samples and the custodian receiving the samples, as well as the time and date of transfer. The laboratory custodian will sign and date the chain-of-custody form upon receipt of the samples at the laboratory.

5.5.4.5 Additional Requirements for Compliance Point Monitoring

This section describes sampling frequency and determination of groundwater quality for the samples from the groundwater monitoring network. Compliance data will be compared to baseline data collected from the upgradient wells and a determination of impacts to groundwater will be made using the proposed ANOVA method (explained in Section 5.5.4.7.1).

5.5.4.5.1 Sampling Frequency

Under final status regulations, the default sampling procedure states that a sequence of at least four samples from each well (background and compliance wells) must be collected at least semiannually during detection monitoring at an interval that ensures, to the greatest extent technically feasible, that an independent sample is obtained [40 CFR 264.97(g)(1) and (2), WAC 173-303-645(8)(g)(i) and (ii), and (9)(d)].

The default sampling procedures are adopted for the IDF Active life as follows: four independent samples from each groundwater monitoring well will be sampled for the indicator parameters (Table 5.2) semiannually during the active life of the regulated unit (including the closure period), one per month for four consecutive months followed by two months of non-sampling. The mean of the measurements from the downgradient wells will be compared semiannually to the mean of the distribution of the background data using ANOVA.

Semi-annual monitoring has been accomplished at the IDF since January 2007 with the collection of four independent samples each semiannual period. During the Pre-Active life, sampling will continue at the IDF with the collection of one sample each year to maintain the baseline. During Active life, sampling will revert to four independent samples collected each semiannual period described above.

5.5.4.5.2 Compliance Point Groundwater Quality Values

The groundwater quality data collected from the groundwater monitoring wells will be compared to the mean of the background data from upgradient wells for each constituent by ANOVA. If the mean is calculated from transformed baseline data (logarithmic transformation or nonparametric approach), then the monitoring data will be transformed accordingly; otherwise, the original monitoring data will be used in the comparisons.

During detection monitoring, data verification will be applied in case of an initial exceedance. For ANOVA test, if the test of hypothesis of equal means for all wells fails, *post hoc* comparisons are needed to determine which compliance well(s) is (are) contaminated. This will be done by comparing concentration differences (called contrasts in the ANOVA and multiple comparison framework) between each compliance well with the background wells (EPA/530-SW-89-026). If the contaminated compliance well(s) is (are) determined by *post hoc* comparisons, verification sampling will be implemented for the constituent(s) in question. Verification sampling is needed to determine if the exceedance is an artifact caused by an error in sampling, analysis, or statistical evaluation or an actual variation in groundwater chemistry. A collection of at least four measurements from the re-sampled compliance well(s) is required to perform ANOVA test on comparison with the mean of the background data (EPA/530-R-93-003). Adequate time should elapse to ensure statistical independence between the original measurements and the re-sample measurements, which is assured by the sampling frequency proposed in Section 5.5.4.5.1.

The existing nitrate plume beneath the IDF site is described in Section 5.4.1. Nitrate is not included in IDF Part A Form and, therefore, is not a constituent of concern for the IDF.

Existing groundwater conditions will be monitored by the indicator parameters and supplemental constituents as described in Section 5.5.1. Specific conductance will respond to nitrate so that any changes in the nitrate concentration will be reflected by changes in the indicator parameter specific conductance.

Anion analysis is one of the supplemental constituents to be monitored at the IDF site. Anion analysis will determine the nitrate concentration. Therefore, through comparison of regression lines of specific conductance and nitrate (Zar, 1999) and/or contaminant source analysis (Gibbons, 1994), it can be determined whether any change in specific conductance is due to a change in nitrate. If a change in specific conductance is due to a change in nitrate, then that specific conductance change is not attributed to the IDF. If, however, a statistically significant change in specific conductance is not attributable to nitrate, verification sampling will occur as described above.

5.5.4.6 Annual Determination

Groundwater flow rate and flow direction at the IDF site will be determined annually for the uppermost aquifer. Flow rate will be determined by calculation using the groundwater gradient, and the Darcy flow equation, $v_h = K_h i_h / n_e$, where v_h is the horizontal groundwater velocity, K_h is the horizontal hydraulic conductivity, i_h is the horizontal hydraulic gradient, and n_e is the effective porosity. Effective porosities used at Hanford Site RCRA regulated units are on the order of 0.1 to 0.3 (PNNL-14187, 1 of 2, 2 of 2); effective porosity might be determined specifically for the IDF from hydrologic tests.

Hydraulic gradients will be determined from measurements of water levels.

5.5.4.7 Statistical Determination

This section describes the method of statistical evaluation and the statistical procedures to indicate whether dangerous waste or dangerous waste constituents from the IDF might have entered the groundwater in the uppermost aquifer. These evaluations will be made as soon as practicable after validation of the full data set from each sampling event.

The monitoring program periodically will re-evaluate the statistical tests being used. The methods described will be reviewed during and after background, data are collected to ensure the methods are the most appropriate, considering site conditions.

The goal of a RCRA final status detection-monitoring program [WAC 173-303-645(9)] is to monitor for indicator parameters that provide a reliable indication of the presence of dangerous constituents in groundwater in the uppermost aquifer beneath the site. This is accomplished by testing for statistically significant changes in concentrations of indicators in downgradient wells relative to baseline values. The default statistical method ANOVA is proposed for the detection monitoring program of the IDF. The proposed statistical method is consistent with EPA/530-SW-89-026, EPA/530-R-93-003, and WAC 173-303-645.

The number of tested constituents will be limited to the indicators to maintain a sufficiently low false-positive rate (EPA/530-R-93-003, page 62; Gibbons 1994, page 16). Verification sampling is an integral part of the statistical design to lower the overall false-positive rate and determine whether the difference between background and compliance-point data is an artifact caused by an error in sampling, analysis, or statistical evaluation (Section 5.5.4.5.2).

5.5.4.7.1 Statistical Procedure

In accordance with WAC 173-303-645(8)(h), acceptable statistical methodology includes analysis of variance (ANOVA), tolerance intervals, prediction intervals, control charts, test of proportions, or other statistical methods approved by Ecology. The type of monitoring, the nature of the data, the proportions of non-detects, and spatial and temporal variations are some of the important factors to be considered in the selection of appropriate statistical methods. The EPA default method ANOVA will be implemented for the IDF site to compare the differences of means of the measurements from upgradient and downgradient wells.

The detailed discussions of the ANOVA test can be found in EPA/530-SW-89-026 and statistical textbooks (Gilbert, 1987; Casella and Berger, 1990; Davis, 2002), and can be executed using commercial statistical software such as SAS or SYSTAT. Under WAC 173-303-645(8)(i)(ii), the proposed statistical method must comply with the performance standard, that is, for a multiple comparisons procedure the Type I error level must be no less than 0.05, and maintained at the level of no less than 0.01 for individual well comparisons. By definition, Type I error is the false rejection rate of the null hypothesis (H_0) of the statistical test. In detection or compliance monitoring, the statistical test is defined as H_0: no release, i.e., the means of the distributions from upgradient and downgradient wells are the same, and the alternative (H_a) evidence of release, e.g., "clean until proven contaminated" (EPA/530-R-93-003). Therefore, the proposed statistical method must comply with the requirement of maintaining Type I error, which equates false positive rate in the stage of detection monitoring at approximate 5% level. As described in EPA/530-SW-89-026, ANOVA procedures have the advantages of combining multiple downgradient into a single statistical test, thus enabling the network-wide false positive rate for any single constituent (not multiple constituents) to be kept at 5 percent, and also maintain reasonable power for detecting contamination.

The details of the ANOVA procedures are described as follows (EPA/530-SW-89-026):

- First, check the proportion of non-detects of the measurements from the upgradient and downgradient wells. When the proportion of non-detects is less than 15%, the non-detects will be reported as one-half the minimum detection limit or practical quantitation limit, and proceed with parametric ANOVA analysis. When the proportion of non-detects is greater than 15%, non-parametric ANOVA analysis will be used for comparing the means of downgradient and upgradient wells.
- Evaluate the distributions of the measurements from the upgradient and downgradient wells. The assumptions with parametric ANOVA test are the residuals are normally distributed with equal variance. The normality of the distribution the residuals can be checked using coefficient of variation, plotting the data on probability plot, and/or Shapiro-Wilk's test (EPA/530-SW-89-026; Gibbons, 1994). The assumption of normality usually can be met by log-transforming the data or by other Box-Cox transformations. When the assumptions of normality and lognormality cannot be justified, the non-parametric ANOVA method will be used for the IDF. Bartlett's test can be used in checking equality, or homogeneity, of variances.
- The parametric ANOVA procedures include:
  - Assume a monitoring network with k wells, and total number of observations N. First, compute well total, well mean, and well residuals (observations subtracted by well mean) for each well, and grand total and mean of all observations (all wells). The well residuals are used to check the assumption of normality.
  - Compute the sum of squares of difference between well means and the grand mean, SS_{wells} that is a measure of the variability between wells with (k-1) degrees of freedom.
  - Compute the total sum of squares of differences between all observations and the grand mean, SS_{total}, which is a measure of the variability in all observations with (N-1) degrees of freedom.
  - Compute the sum of squares of differences of observations within wells from the well means, SS_{error}, which is a measure of the variability within wells with (N-k) degrees of freedom calculated by the following subtraction:

    $$SS_{error} = SS_{total} - SS_{wells}$$

  - Test the hypothesis of equal means for all k wells by computing F value with the means squares of differences:

    $$F = MS_{wells} / MS_{error}$$

where the means of squares are the sums of squares divided by the associated degrees of freedom, that is, $MS_{wells} = SS_{wells} / (k-1)$, and $MS_{error} = SS_{error} / (N-k)$. Compare the F value to the tabulated F statistics with (k-1) and (N-k) degress of freedom at the 5% significance level (EPA/530-SW 89-026, Appendix B, Table 2). If the calculated F value exceeds the tabulated F statistics, the null hypothesis of equal well means is rejected. Proceed with test of contrasts in the next step. Otherwise, the hypothesis of equal means is accepted that there is no significant difference between the concentrations at k wells (upgradient and downgradient wells), that is, no evidence of contamination.

- If the hypothesis of equal well means is rejected, contrasts (concentration differences between a compliance well and background wells) will be tested for each compliance well to determine which compliance well(s) is (are) contaminated. Bonferroni t-statistics will be computed to determine if the significant F value is due to difference between background and compliance wells. Assume that of the k wells, k_b are background (upgradient) wells, and k_c are compliance (downgradient) wells (i.e., $k_b + k_c = k$). Each of the k_c compliance wells is compared to the mean of the background wells as the following steps:
  - Compute the mean m_b from the k_b background wells with a total of n_b samples.
  - Compute the difference D_i between the mean from the i^{th} compliance well and the mean from the background wells.
  - Compute the standard error of the difference from the i^{th} compliance well with n_i observations as:

    $SE_i = [MS_{error} (1/n_b + 1/n_i)]^{1/2}$

    where MS_{error} is computed previously as the measure of variability within wells.
  - Obtain the t-statistics from Bonferroni's t-table (EPA/530-SW-89-026, Appendix B, Table 3) with a significance level of (α=0.05/k_c) but no less than 0.01 (for individual comparison) and (N-k) degrees of freedom. The critical value for the i^{th} compliance well is defined as $C_i = SE_i \times t$.
  - If the difference D_i exceeds the critical value C_i, conclude that the mean of the i^{th} compliance well is significantly higher than the mean of the background wells. Otherwise, conclude that the well is not contaminated.
- The one-way non-parametric ANOVA tests the null hypothesis that the data from each well come from the same continuous distribution and hence have the same median. The procedures, called the Kruskal-Wallis test, include the following steps:
  - Assume the monitoring network as defined previously with a total of N observations from k wells (kb background wells and kc compliance wells). Rank all N observations from least (1) to greatest (N). Let the background wells be group 1, and denote the compliance wells as group 2 to (kc+1). (one group per compliance well).
  - Compute the sum (Ri) and the average (mi) of the ranks of the ni observations in the ith group.
  - Compute the Kruskal-Wallis statistics (H) as

$$H = \left[\frac{12}{N(N+1)} \sum_{i=1}^{kc+1} \frac{R_i^2}{n_i} \right] - 3(N+1)$$

- Compare the calculated H value to the tabulated chi-squared value with kc degrees of freedom (EPA/530-SW-89-026, Appendix B, Table 1). The null hypothesis of equal medians is rejected when the calculated H value exceeds the tabulated critical value.

- When the null hypothesis of equal medians is rejected, compute the critical difference C_i for each compliance well to the background data (group 1 with n_b observations):

$$C_i = Z_{(0.05/kc)}\left[\frac{N(N+1)}{12}\times\left(\frac{1}{n_b}+\frac{1}{n_i}\right)\right]^{1/2}$$

Where $Z_{(0.05/kc)}$ is the upper $(0.05/k_c)$ percentile from the standard normal distribution (EPA/530-SW-89-026, Appendix B, Table 4). If there are more than five compliance wells ($k_c > 5$), use $Z_{0.01}$, the upper one-percentile from the standard normal distribution ($Z_{0.01}$=2.32) for individual comparison (WAC 173-303-645(8)(i)(ii)).

- Compute the difference ($D_i = m_i - m_1$) of average rank m_i (i=2 to k_c+1) for each compliance well to the background (m_1). Compare the difference D_i to the critical value C_i for each compliance well. If D_i exceeds C_i, conclude that the median of the i^{th} compliance well is significantly higher than the background median.

- As monitoring continues, the background data will be updated periodically (e.g., every year or two) to incorporate the new data from upgradient wells. This updating process will continue for the life of the monitoring program. Prior to updating older background data with more recent results, a two-sample t-test will be run to compare the older concentration levels with the concentrations of the proposed update samples. If the t-test does not show a significant difference at the 5 percent significant level, proceed to re-estimate the baseline parameters by including the more recent data. If the t-test does show a significant difference, the newer data will not be included as background unless some specific factors can be, identified explaining why background levels at the IDF site have naturally changed (EPA/530-R93-003).

Formal testing for outliers will be done when an observation of the background data seems inconsistently high (by orders of magnitude) compared to the rest of the data set in order to avoid the artificial increase of the mean of the background data and a corresponding increase of the false negative rate. Statistical methods such as the Grubbs' method (Grubbs, 1969), the box-and-whisker plot (Ostle and Malone, 1988), EPA guidance (EPA/530-SW-89-026, p. 11-14) and/or American Society for Testing and Materials guidance (ASTM 1996) will be used for testing outliers. The outliers must be checked to determine if the measurements are in error and need to be corrected or excluded from calculating the background mean. If no specific error is found, the measurements must be retained in the data.

A statistically significant exceedance over background (baseline) levels only indicates that the new measurement in a particular monitoring well for a particular constituent is inconsistent with chance expectations based on the available sample of background (baseline) measurements. Any statistical result must be supported by other information to determine if a waste disposal facility has impacted groundwater (ASTM 1996).

5.5.4.7.2 Results

Sampling and analysis results are reviewed at least semiannually (i.e., after each sampling event) and are available in HEIS. The DOE will submit results of statistical evaluations to Ecology.

5.5.5 Compliance Monitoring Program

A compliance monitoring program that satisfies requirements set forth in WAC 173-303-645(10) will be established for the IDF if detection-level monitoring reveals statistically significant evidence of dangerous waste contamination from sources within the regulated unit. If compliance monitoring is required, DOE will submit a revised monitoring plan to Ecology specifying dangerous constituents to be monitored, sampling and analysis protocols, statistical evaluation methods, etc. In the compliance monitoring program, the dangerous constituents or parameters will be compared to concentration limits specified in the facility permit as discussed in WAC 173-303-645(5) during the compliance period.

The RCRA regulations [WAC 173-303-645(9)(g)] state that if a statistical exceedance occurs in a downgradient well, the entire network immediately must be resampled and analyzed for the constituents in Appendix IX of 40 CFR 264. This sampling would be conducted in parallel with a required permit modification. Appendix IX is an extensive list including a wide variety of volatile and semivolatile organic compounds and trace metals. It is prudent to narrow the analyte list to the specific exceedance event; e.g., if the exceeding contaminant is total organic halides, the project would analyze for the halogenated hydrocarbons most likely to be present in the area. Results of the resampling will form the basis for returning to detection monitoring or designing a compliance monitoring program.

5.5.6 Corrective Action Program

If, at a point of compliance (a well), dangerous constituents of concern are measured in the groundwater at concentrations that exceed the applicable groundwater concentration limit, Ecology must be notified in 7 days, and an application to modify the permit to include a corrective action plan must be sent to Ecology within 90 days or within the time agreed to by Ecology. A description of the groundwater monitoring plan, including all additional corrective actions that are appropriate for a corrective action program will be prepared and submitted to Ecology when the need for corrective action first is identified.

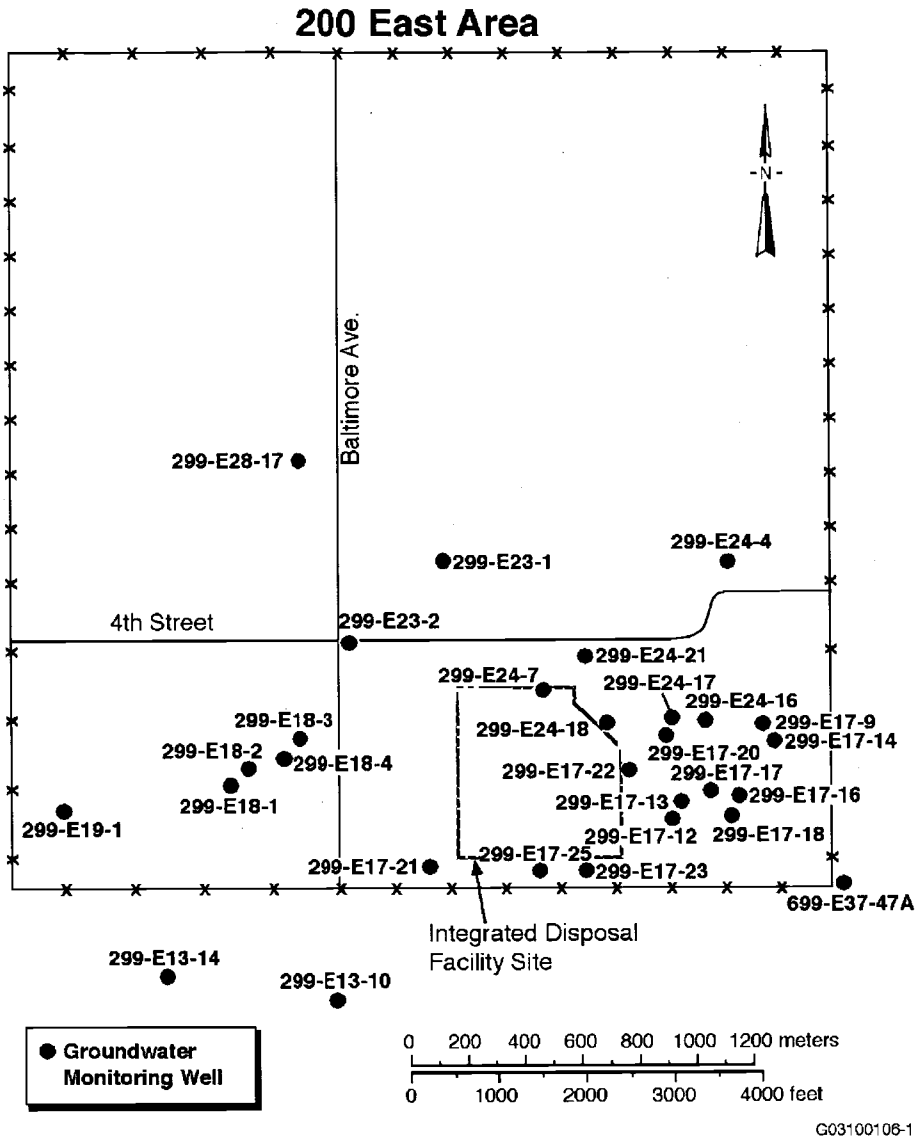


Figure 5.1. Location of the IDF and Nearby Boreholes

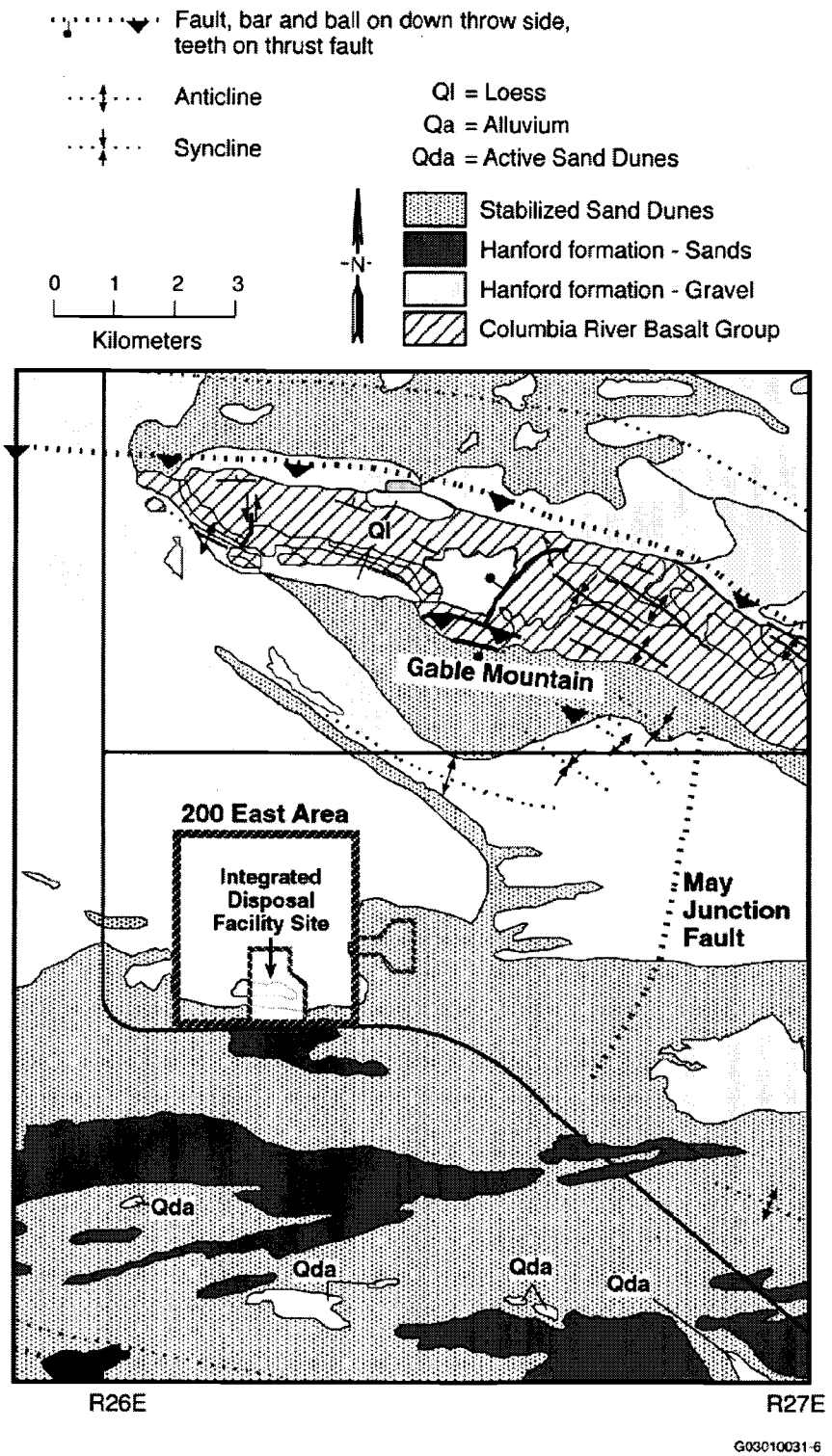


Figure 5.2. Geologic Map of the 200 East and 200 West Areas and Vicinity

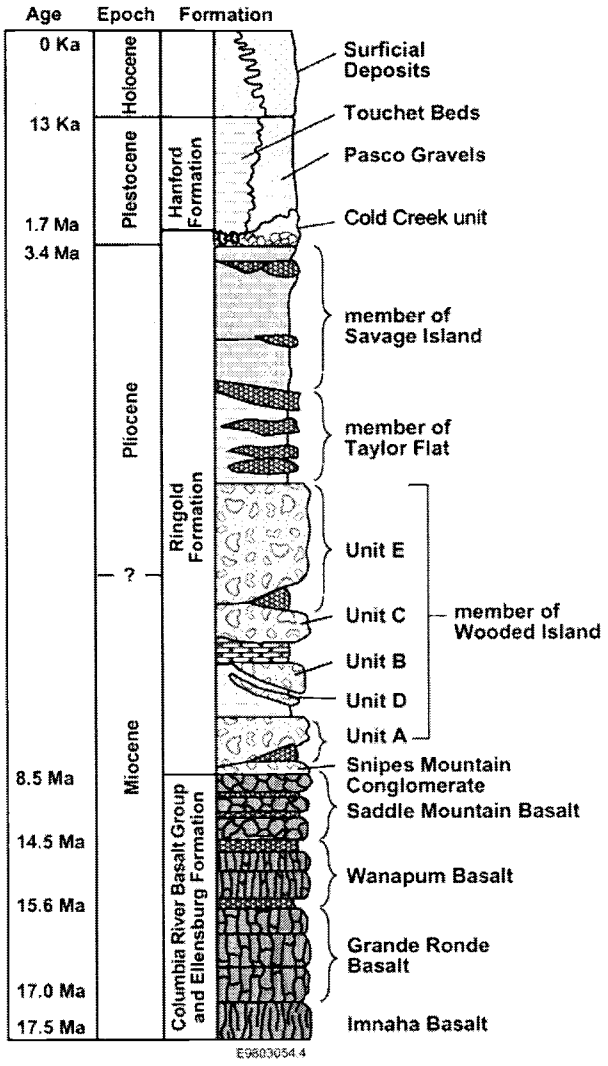


Figure 5.3. Stratigraphy of the Hanford Site

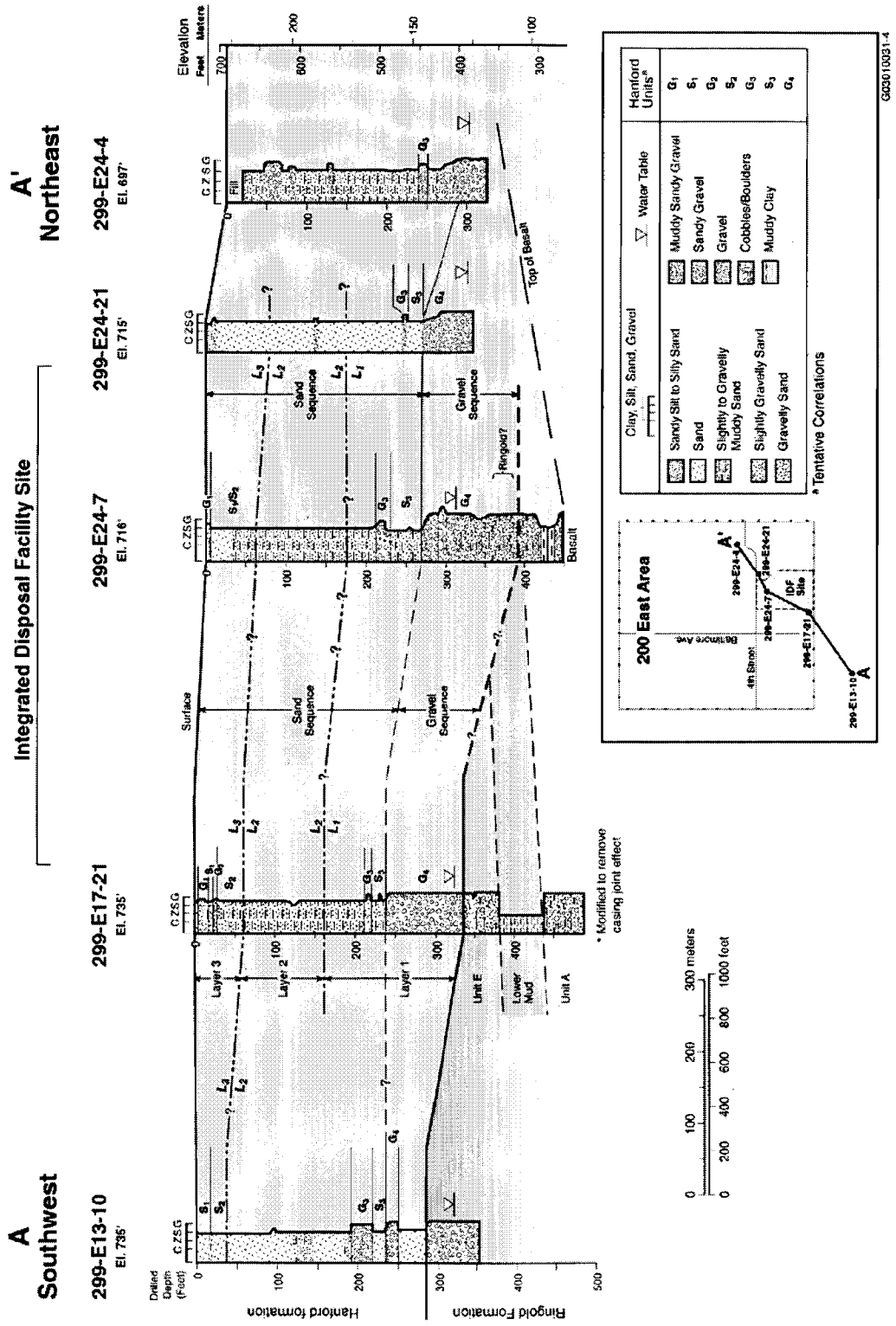


Figure 5.4. Cross-Section through the IDF Site

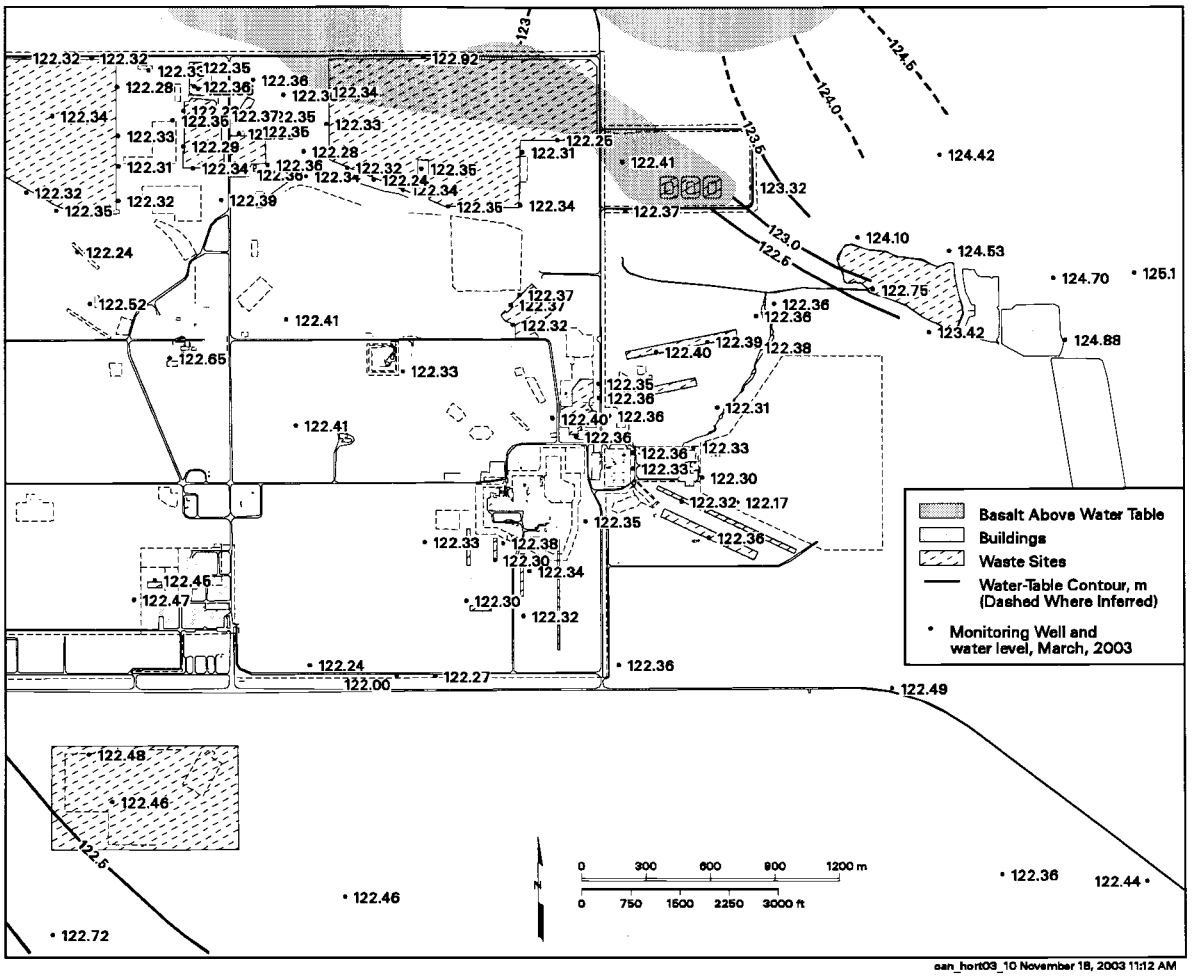
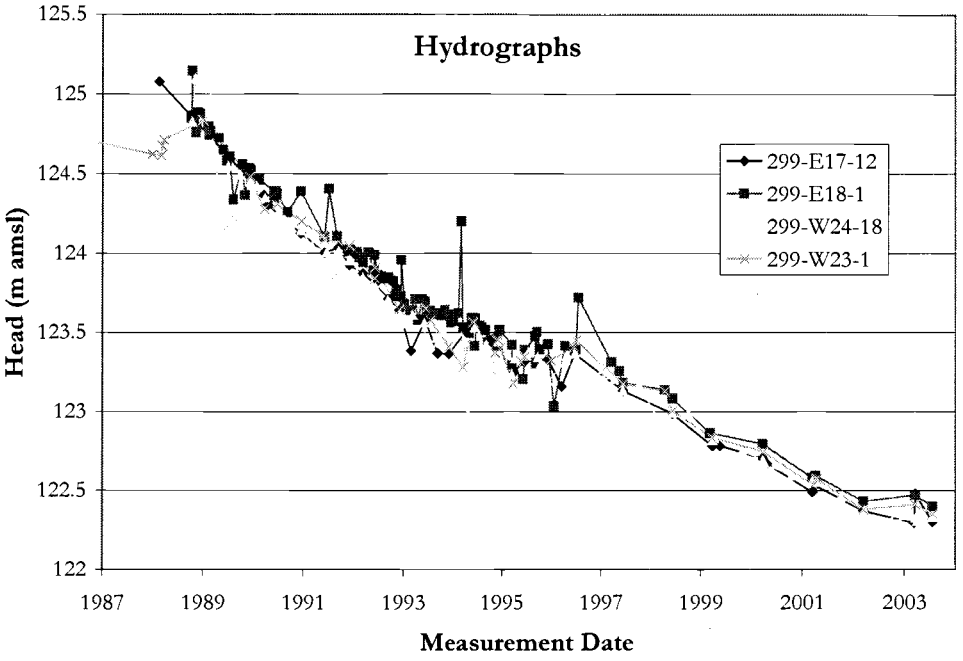


Figure 5.5. Water Table Map for the Hanford Site 200 East Area



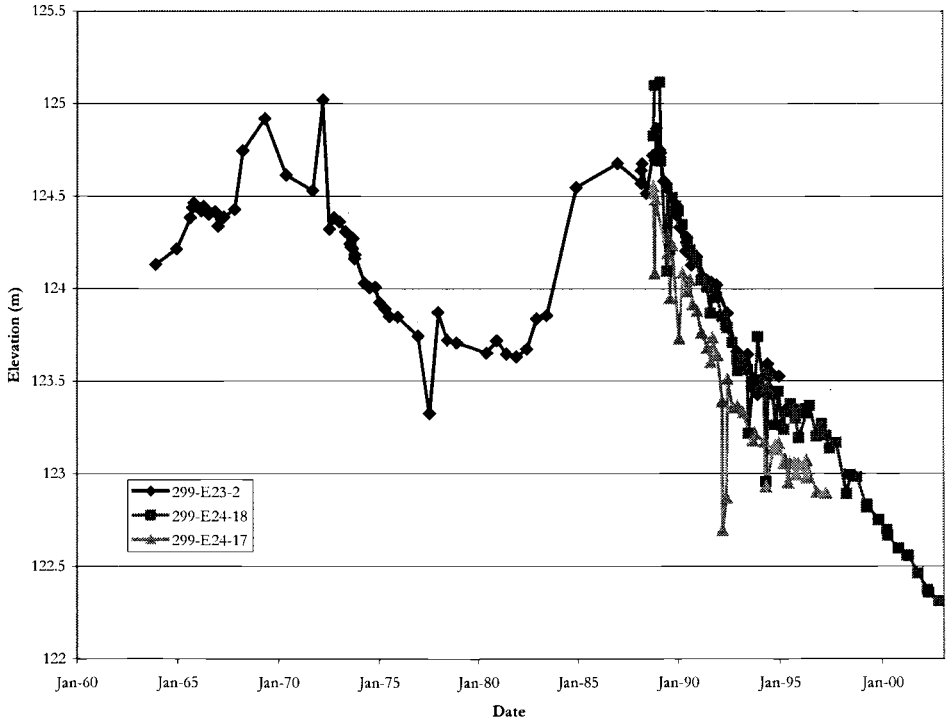


Figure 5.6. Hydrographs for Wells Near the IDF Site

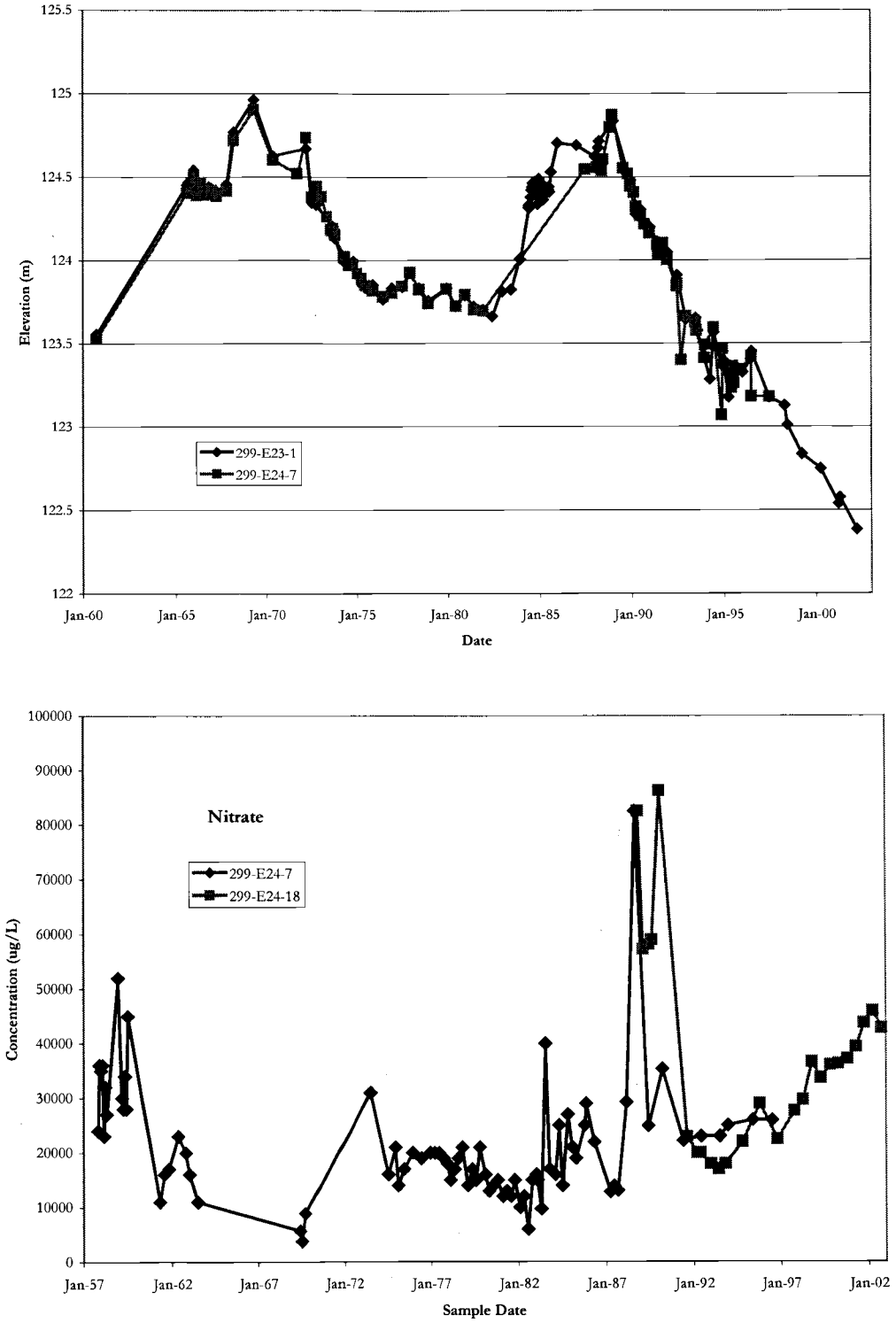


Figure 5.7. Concentration versus Time for Nitrate in Wells 299-E24-7 and 299-E24-18

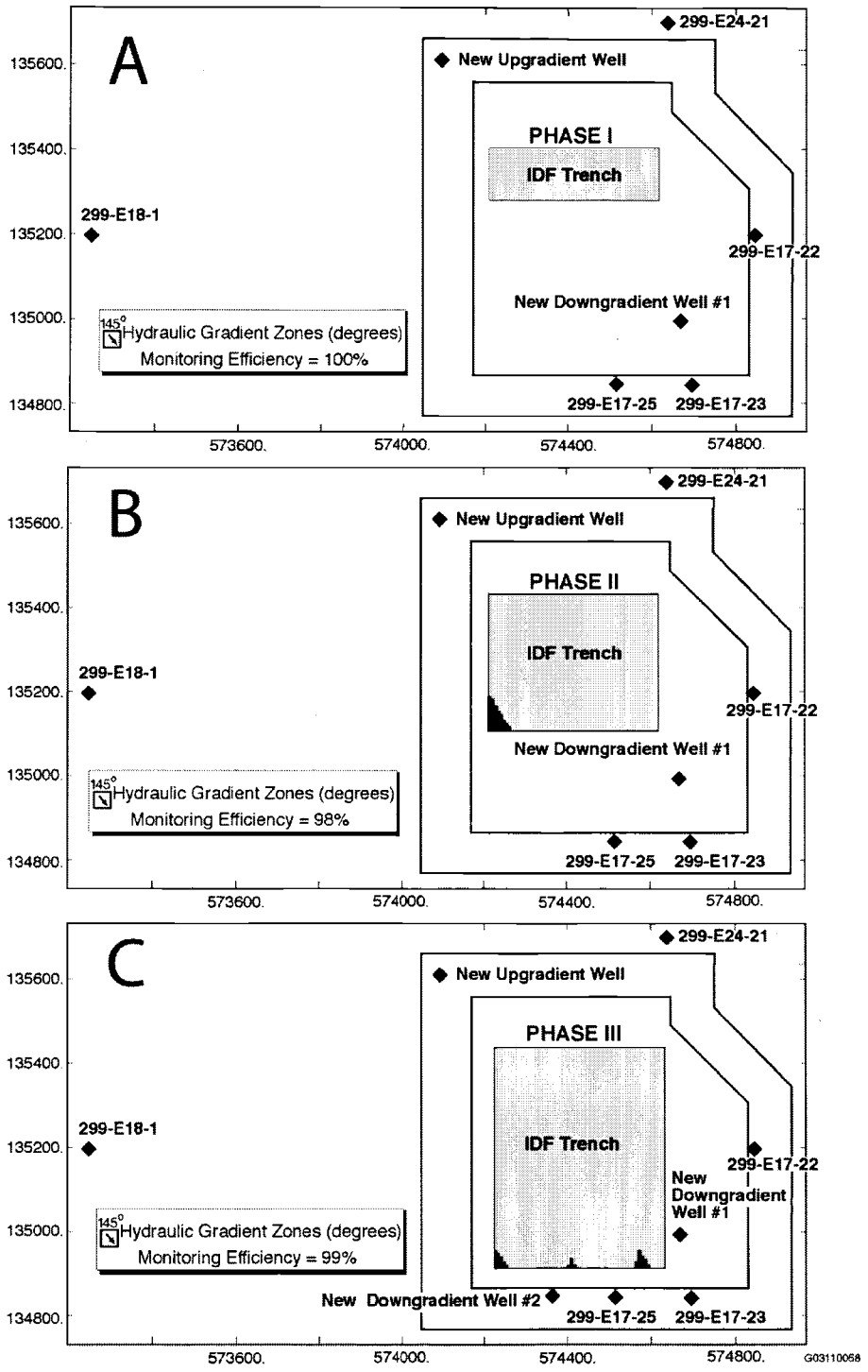


Figure 5.8. Sequence for Installation of Downgradient Monitoring Wells at the IDF

Areas in black are areas from which leaks will not be detected with the array of monitoring wells shown.

A. Area used for disposal and associated monitoring wells for construction phase I;
B. Area used for disposal and associated monitoring wells for construction phase II;
C. Area used for disposal and associated monitoring wells for construction phase III.

Table 5.1. Water Levels in Groundwater Wells in the Vicinity of the IDF Site

Well	Measure date	DTW m[a]	WT elev m[b]	Ref elev m[c]
299-E13-10	03/14/02	101.7	122.5	226.31
299-E17-12	03/14/02	100.0	121.1	221.09
299-E17-13	04/12/01	97.7	122.6	220.34
299-E17-17	04/12/99	97.8	122.8	220.54
299-E17-18	10/03/02	98.5	122.3	220.76
299-E17-20	04/09/97	97.1	123.2	220.33
299-E17-21	04/23/98	100.4	122.7	224.26
299-E17-22	05/20/02	98.1	122.5	220.59
299-E17-23	05/20/02	101.6	122.2	223.84
299-E17-25	05/21/02	98.3	126.7	225.03
299-E18-1	03/14/02	98.2	122.4	220.65
299-E18-3	06/27/96	97.8	123.4	221.20
299-E18-4	06/27/96	97.7	123.4	221.05
299-E19-1	03/22/88	100.4	124.9	225.26
299-E23-1	03/14/02	96.0	122.4	218.39
299-E23-2	12/20/94	97.2	123.5	220.77
299-E24-4	08/10/98	90.6	122.9	213.47
299-E24-7	06/11/97	96.2	123.2	219.34
299-E24-16	10/04/02	97.7	122.3	220.02
299-E24-17	04/07/97	97.36	122.9	220.16
299-E24-18	10/02/02	98.0	122.3	220.35
299-E24-21	03/22/01	95.4	122.6	217.85

[a] DTW = depth to water
[b] WT elev = elevation of water table (meters above mean sea level)
[c] Ref elev = reference elevation (meters above mean sea level, North American Vertical Datum 88 reference), generally top of well casing.

Table 5.2. Monitored Constituents for the IDF

Indicator parameters	Supplemental constituents
Chromium (filtered)	Alkalinity
Specific conductance (field)	Anions
Total organic carbon	ICP metals
Total organic halides	Turbidity (field)
pH (field)	

Table 5.3. Expected Behavior of Selected Regulated Constituents/Materials for the IDF

Constituent/material	Expected charged state	Expected mobility[1] (K_d)	Comments
Organics			
Acetonitrile	N/A	High (0.16)	Miscible with water (Howard Volume IV, 1993)
Carbon tetrachloride	N/A	High (0.60); 0.29 (DOE/RL-93-99)	Moderately soluble in water (805 mg/L) (Howard, Volume II,1990)
Creosote[2]	N/A	High (0.03 to 0.06)[3]	Relatively low solubility in water. Naphthalene solubility in water (31.7 mg/L [Howard, Volume 1, 1989]). Anthracene solubility in water (0.03 to 0.5 mg/L[Mackay et al, Volume II, 1992])
Dioxane	N/A	High (0.01)	Miscible with water (Howard, Volume II, 1990)
Ethylene glycol	N/A	Unknown[4]	Miscible with water (Howard, Volume II, 1990)
Naphthalene		Moderate (4 to 10); 1.4 (DOE/RL-93-99)	Sparingly soluble in water (31.7 mg/L [Howard, Volume I, 1989]).
Polychlorinated biphenyls	N/A	Low (20 to 100); 440 to 2,300 (DOE/RL-93-99)	Low solubility in water. 0.01 to 1 mg/L as Aloclors (Mackay et al. 1992); 0.27 to 1.45 mg/L (WHC-SD-EN-TI-201)
Tetrachloroethylene	N/A	High (2.1); 0.22 (DOE/RL-93-99)	Moderately soluble in water (1,503 mg/L) (Howard, Volume II, 1990)
Toluene	N/A	High (0.37 to 1.8); 0.18 (DOE/RL-93-99)	Moderately soluble in water (535 mg/L) (Howard, Volume II, 1990)
Trichloroethylene	N/A	High (1.0); 0.1 to 1.0 (WHC-SC-EN-TI-201); 0.11 (DOE/RL-93-99)	Moderately soluble in water (1,100 mg/L) (Howard, Volume II, 1990)
Vinyl chloride	N/A	High (0.004); 0.056 (DOE/RL-93-99)	Moderately soluble in water (2,763 mg/L) (Howard, Volume I, 1989)
Inorganics			
Antimony	Cation (Sb^{+2})	Moderate (0 to 40, best estimate: 20 [DOE/RL-93-99])	Moderately soluble (best estimate): 1,000 mg/L (DOE/RL-93-99)
Arsenic	Anion (AsO_4^{-5})	High , 0 (DOE/RL-93-99)	Moderately soluble (best estimate): 1,000 mg/L (DOE/RL-93-99)
Barium	Cation (Ba^{+2})	Moderate, 20 to 200, best estimate: 50 (DOE/RL-93-99)	Low solubility (best estimate): 1 mg/L (DOE/RL-93-99)
Beryllium	Cation (Be^{+2})	Moderate, 15 to 200, best estimate: 20 (DOE/RL-93-99)	Solubility unknown. Best estimate: 1 mg/L
Cadmium	Cation (Cd^{+2})	Moderate, 15 to 30, best estimate: 23 (DOE/RL-93-99)	Sparingly soluble. Best estimate: 25 mg/L (DOE/RL-93-99)
Chromium	Anion (CrO_4^{-2})	High (0.0 to 1.02 [PNNL-13895); 0.001 (WHC-SC-EN-TI-201)	Low solubility: 0.5 to 10 mg/L (WHC-SC-EN-TI-201)
Lead	Cation (Pb^{+2})	Low (1,330 to 469,000 [PNNL-13895])	Low solubility: 287 µg/L in Hanford Site groundwater (PNL-9791)
Mercury	Cation (Hg^{+2})	Moderate, best estimate: 30 (DOE/RL-93-99)	Solubility unknown. Best estimate: 1 mg/L (DOE/RL-93-99)
Nickel	Cation (Ni^{+2}) $Ni(OH)_2$ $NiCO_3$	Low (48 to 337 [PNNL-13895)	Low solubility: 1.9 mg/L in Hanford Site groundwater (PNL-9791)

Table 5.3. Expected Behavior of Selected Regulated Constituents/Materials for the IDF

Constituent/material	Expected charged state	Expected mobility[1] (K_d)	Comments
Selenium	Anion (SeO_4^{-6})	High (3 to 10 [PNNL-13895]) (3 to 8 PNNL-11966)	Moderately soluble. Best estimate: 1,000 mg/L (DOE/RL-93-99)
Silver	Cation (Ag^+)	Moderate, 20 to 30, best estimate: 25 (DOE/RL-93-99)	Sparingly soluble (best estimate): 25 mg/L (DOE/RL-93-99).

1 N/A = Not applicable

2 [1] Unless cited in the column, K_d (partition coefficient) values were calculated from Koc (normalized sorption coefficient) values
3 obtained from either the Handbook of Environmental Fate and Exposure Data for Organic Chemicals series (Volumes I-IV) (P.H.
4 Howard, ed) or the Illustrated Handbook of Physical-Chemical Properties and Environmental Fate for Organic Chemicals series
5 [Mackay et al. 1992a, 1992b]. For all organics (except carbon tetrachloride), the calculation assumes an organic carbon content
6 for Hanford Site soil of 1.0%. The value of organic carbon assumed is conservative recognizing that the organic carbon content
7 of most Hanford Site soil falls considerably below this value. However, applying this level of conservatism also recognizes that
8 mineral-driven sorption likely plays a role in organic constituent mobility for Hanford Site soils with organic carbon content at or
9 below 0.1% (PNNL-13560). A calculation of a K_d value using acetonitrile as an example is as follows. The literature estimated
10 value of K_{oc} for acetonitrile is 16 (Howard 1993).

11 $K_d = f_{oc} \times K_{oc}$ where f_{oc}= the mass fraction of organic carbon in the soil.

12 K_d (acetonitrile) = $0.01 \times 16 = 0.16$.

13 [2] Creosote is a coal tar distillate containing high quantities of naphthalene and anthracene (Lewis, R.J., Sr. 1993).

14 [3] Because creosote is predominately a mixture of naphthalene and anthracene (footnote 2), assumed K_{oc} values for naphthalene
15 (Howard 1989) and anthracene (Mackay et al., Volume II) in calculating a K_d range for creosote.

16 [4] This constituent has a low octanol/water partition coefficient indicating that its adsorption to soil would be low (Howard,
17 Volume II, 1990)
18

Table 5.4. Analytical Methods and Method Detection Limits for Regulated Constituents and Indicator Parameters

Class of Compounds	Analytical Methods [1]	Method Detection Limit [3] (ug/L)
Metals	Trace Metals:	
	SW 846, Method 6010 or	0.18 - 44.8[2]
	SW 846, Method 6020 or	0.042–8.5
	EPA/600/R-94/111, Method 200.8	.05–50
	Arsenic:	
	SW 846, Method 6010 or	50
	SW 846, Method 6020 or	2
	EPA/600/R-94/111, Method 200.8	0.40
	Cadmium:	
	SW 846, Method 6010 or	4
	SW 846, Method 6020 or	0.86–2.3
	EPA/600/R-94/111, Method 200.8	0.10
	Chromium:	
	SW 846, Method 6010 or	4
	SW 846, Method 6020 or	1.9–3.1
	EPA/600/R-94/111, Method 200.8	0.5
	Lead:	
	SW 846, Method 6010 or	27
	SW 846, Method 6020 or	0.49
	EPA/600/R-94/111, Method 200.8	0.10
	Mercury:	
	SW 846, Method 6020 or	.093
	SW 846 Method 7470 or	0.1
	EPA/600/R-94/111, Method 200.8	0.05
	Selenium:	
	SW 846, Method 6010 or	30
	SW 846, Method 6020 or	1
	EPA/600/R-94/111, Method 200.8	0.30
	Thallium:	
	SW 846, Method 6010 or	32
	SW 846, Method 6020 or	0.6
	EPA/600/R-94/111, Method 200.8	0.10
Semi-Volatile Organics	SW 846, Method 8041 or	Not available
	SW 846, Method 8040	2.0 – 3.7[2]
	SW 846, Method 8270	0.24 – 50[2]
Pesticides/Polychlorinated Biphenyls	SW 846, Method 8081 (Pesticides)	0.0034 –1.9[2]
	SW 846, Method 8082 (PCBs)	0.14–0.49[2]
Herbicides	SW 846, Method 8151	.085–84[2]
Volatile Organic Compounds	SW 846, Method 8260 (VOAs)	.04–100[2]
Dioxins	SW 846, Method 8290	.00000067–.000005[2]
General Chemistry	Cyanide:	
	SW 846, Method 9012 or	2.0–2.4
	Standard Methods 4500-CN or	4
	600/4-79-020, Method 335.2	4
	Sulfide:	
	SW 846, Method 9030	180–730[2]

Table 5.4. Analytical Methods and Method Detection Limits for Regulated Constituents and Indicator Parameters

Class of Compounds	Analytical Methods [1]	Method Detection Limit [3] (ug/L)
Alkalinity	EPA-600/4-79-020, Method 310.1 & 310.2, Standard Methods 2320	850 – 2500[4]
Anions	EPA-600/R-93-100, Method 300.0	5.1–4430[2]
pH	Company specific	Not applicable
Specific conductance	EPA-600/R-93-100, Method 120.1	Not applicable

[1] Changes to the Analytical Methods require prior approval per WAC 173-303-830, Appendix I, C.2.

[2] Detection limit varies according to specific compound. The range of method detection limits for all compounds detected by the specific analytical method is given.

[3] Method detection limits are based on historical values reported by the analytical laboratories, where available. MDLs may vary by laboratory and are updated periodically.

[4] This MDL is based on Method 310.1, which was used previously. No technical difference is found between Method 310.1 and SM 2320, except the SM covers more information on the principles of the method.

5.6 REFERENCES

Public Laws

42 USC 2011 et seq. (1954). *Atomic Energy Act of 1954*. As amended, Ch. 1073, 68 Stat.919; available online at http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr0980/v1/sr0980v1.pdf

42 USC 6901 et seq. *Resource Conservation and Recovery Act of 1976*, as amended, Public Law 94-580, 90 Stat. 2795; available online at http://www.epa.gov/lawsregs/laws/rcra.html

Code of Federal Regulations

40 CFR 136.3 (5.5.4.1) Code of Federal Regulations, Title 40, Part 136. Whole Effluent Toxicity: Guidelines Establishing Test Procedures for Chemical Analysis of Pollutants.

40 CFR 264, Code of Federal Regulations, Title 40, Part 264, Subpart F. *Standards for Owners of Hazardous Waste Treatment, Storage, and Disposal Facilities*.

Washington Administrative Code

WAC 173-303-645, Washington Administrative Code. *Releases from Regulated Units*. Olympia, Washington.

Others

ASTM. (1986). *Annual Book of ASTM Standards*. American Society for Testing and Materials, West Conshohocken, Pennsylvania; available for purchase at http://www.astm.org/bookstore/bos

BHI-01103. (1999). *Clastic Injection Dikes of the Pasco Basin and Vicinity*, Rev. 0, Fecht, K.R., Bechtel Hanford, Inc., Richland, Washington.

BNWL. (1974). BNWL-B-360. *Selected Water Table Contour Maps and Well Hydrographs for the Hanford Reservation, 1944 - 1973*. Kipp, K.L. and Mudd, R.D., Pacific Northwest Laboratories, Richland, Washington.

DOE. (1988). DOE/RW-0164. *Repository Location, Hanford Site, Washington*, Vols. 1,2,3,4,5,6,7,8,9, U.S. Department of Energy, Office of Civilian Radioactive Waste Management, Washington, D.C.

DOE-RL. (1993). DOE/RL-92-04, Revision 0 (Sections 1 of 2, 2 of 2). *PUREX Source Aggregate Area Management Study Report*. U.S. Department of Energy, Richland Operations, Richland, Washington.

DOE-RL. (1994). DOE/RL-93-99. *Remedial Investigation and Feasibility Study Report for the Environmental Restoration Disposal Facility*, U.S. Department of Energy, Richland, Washington.

Domenico, P. A. and G. A. Robbins. (1985). *A New Method of Contaminant Plume Analysis.* Groundwater, Vol. 23, No. 4.

EPA. (1979). *Methods for Chemical Analysis of Water and Wastes.* EPA-600/4-79-020, as revised, U.S. Environmental Protection Agency, Washington, D.C.

EPA. (1986). SW-846. *Test Methods for Evaluating Solid Wastes: Physical/Chemical Methods,* 3rd Edition, as amended. U.S. Environmental Protection Agency, Office of Solid Waste and Emergency Response, Washington, D.C.

EPA. (2006). EPA QA/G-4. *Guidance on Systematic Planning Using the Data Quality Objectives Process*, U.S. Environmental Protection Agency, Office of Environmental Information, Washington, D.C.

Garber, M. S. and Koopman, F. C. (1968). *Methods of Measuring Water Levels in Deep Well: U.S. Geological Survey.* TRWI, Book 8, Chapter A-1. U. S. Government Printing Office, Washington, D. C.

Gibbons, R. D. (1994). *Statistical Methods for Groundwater Monitoring*. John Wiley & Sons, New York.

Gilbert, R. O. (1987). *Statistical Methods for Environmental Pollution Monitoring*, Van Nostrand Reinhold, New York.

HNF. (1999). HNF-4921, Rev. 0. *Immobilized Low Activity Tank Waste Inventory Data Package.* Wootan, D. W., Fluor Daniel Northwest, Inc., Richland, Washington.

Howard, P.H., Ed. (1989). *Handbook of Environmental Fate and Exposure Data for Organic Chemicals: Volume I, Large Production and Priority Pollutants*. Lewis Publishers, Chelsea, MI.

Howard, P.H., Ed. (1990) *Handbook of Environmental Fate and Exposure Data for Organic Chemicals: Volume II, Solvents 2*. Lewis Publishers, Chelsea, MI.

Howard, P.H., Ed. (1993). *Handbook of Environmental Fate and Exposure Data for Organic Chemicals:* Volume IV, Solvents and Chemical Intermediates. P.H. Howard, ed. Lewis Publishers, Ann Arbor, MI.

Lindsey, K.A. (1996). *The Miocene to Pliocene Ringold Formation and Associated Deposits of the Ancestral Columbia River System, South-central Washington and North-central Oregon.* Open File Report 96-8, Washington State Department of Natural Resources, Division of Geology and Earth Resources, Olympia, Washington.

Mackay, D., Shiu, W.Y., and Ma, K.C. (1992). *Illustrated Handbook of Physical-Chemical Properties and Environmental Fate for Organic Chemicals,* Volume I: Monoaromatic Hydrocarbons, Chlorobenzenes, and PCB's. Lewis Publishers, Ann Arbor, MI.

Mackay, D., Shiu, W.Y., and Ma, K.C. (1992). *Illustrated Handbook of Physical-Chemical Properties and Environmental Fate for Organic Chemicals,* Volume II: Polynuclear Aromatic Hydrocarbons, Polychlorinated Dioxins, and Dibenzofurans. Lewis Publishers, Ann Arbor, MI.

PNL. (1992). PNL-9791. *Estimation of the Release and Migration of Nickel Through Soils and Groundwater at the Hanford Site 218-E-12B Burial Ground.* Rhoads, K., Bjornstad, B.N., Lewis, R.E., Teel, S.S., Cantrell, R.J., Serne, L.H., Smoot, J.L, Kincaid, C.T., and Wurstner, S.K., Pacific Northwest Laboratory, Richland, Washington.

PNNL. (1998). PNNL-11957. *Immobilized Low-Activity Waste Site Borehole 299-E17-21.* Reidel, S.P., Reynolds, K.D., and Horton, D.G. Pacific Northwest National Laboratory, Richland, Washington.

PNNL. (1999). PNNL-12257, Rev. 1. *Geologic Data Package for 2001 Immobilized Low-Activity Waste Performance Assessment,* Reidel, S.P. and Horton, D.G., Pacific Northwest National Laboratory, Richland, Washington.

PNNL. (2000). PNNL-12261. *Hydrogeology for the Suprabasalt Aquifer System, 200-East Area and Vicinity, Hanford Site, Washington.* Williams, B.A., Bjornstad, B.N., Schalla, R., and Webber, W.D. Pacific Northwest National Laboratory, Richland, Washington.

PNNL. (2000). PNNL-13400. *Groundwater Flow and Transport Calculations Supporting the Immobilized Low-Activity Waste disposal Facility Performance Assessment.* Bergeron, M.P. and Wurstner, S.K., Pacific Northwest National Laboratory, Richland, Washington.

PNNL. (2000). PNNL-13404, Sections 1 of 2, 2 of 2. Hanford Site Groundwater Monitoring for Fiscal Year 2009. Hartman, M.J., Morasch, L.F., Webber, W.D., Pacific Northwest National Laboratory, Richland, Washington.

PNNL. (2001). PNNL-13560. *Assessment of Carbon Tetrachloride Groundwater Transport in Support of the Hanford Carbon Tetrachloride Innovative Technology Demonstration Program.* Truex, M.J., Murray, C.J., Cole, C.R., Cameron, R.J., Johnson, M.D., Skeen, R.S., and Johnson, C.D. Pacific Northwest National Laboratory, Richland, Washington.

PNNL. (2001). PNNL-13652. *Geologic and Wireline Borehole Summary from the Second ILAW Borehole (299-W24-21).* Reidel, S.P., Horton, D.G., and Valenta, M.M. Pacific Northwest National Laboratory, Richland, Washington.

PNNL. (2002). PNNL-13895. *Hanford Contaminant Distribution Coefficient Database and User's Guide.* Cantrell, K.J., Serne, R.J. and Last, G.V., Pacific Northwest National Laboratory, Richland, Washington.

PNNL. (2002). PNNL-14029. *Geologic and Wireline Summaries from Fiscal Year 2002 ILAW Boreholes.* Reidel, S.P., and Ho, A.M., Pacific Northwest National Laboratory, Richland, Washington.

PNNL. (2003). PNNL-14187, Sections 1 of 2, 2 of 2. *Hanford Site Groundwater Monitoring for Fiscal Year 2002.* Hartman, M.J., Morasch, L.F., and Webber, W.D., Pacific Northwest National Laboratory, Richland, Washington.

Reidel, S. P. and Fecht, K. R. (1994). *Geologic Map of the Richland 1:100,000 Quadrangle, Washington.* Washington Division of Geology and Earth Resources Open File Report 94-8. Washington State Department of Natural Resources, Olympia, Washington.

RHO. (1981). RHO-BWI-ST-14. Chapter 3, *Wanapum and Saddle Mountains Basalts of the Cold Creek Syncline Area.* Reidel, S.P., and Fecht, K.R. In, *Subsurface Geology of the Cold Creek Syncline.* Myers, C.W. and S.M. Price, editors, Rockwell Hanford Operations, Richland, Washington.

RPP-6877. (2004). *Integrated Disposal Facility Preoperational Monitoring Plan,* Rev. 1, Horton, D.F., Reidel, S.P., and Chien, Yi-Ju, Pacific Northwest National Laboratory for Bechtel Hanford, Inc., Richland, Washington.

U. S. Geological Survey. (1977). *National Handbook of Recommended Methods for Water Data Acquisition.* Office of Water Data Coordination, Reston, Virginia. Available at http://pubs.usgs.gov/chapter11

WHC. (1989). WHC-MR-0008. *Low-Level Burial Grounds Database.* Poremba, B.E., Westinghouse Hanford Company, Richland, Washington.

WHC. (1992). WHC-MR-0391. (1992). *Field Trip Guide to the Hanford Site.* Reidel, S.P., Lindsey, K.A., and Fecht, K.R., Westinghouse Hanford Company, Richland, Washington.

WHC. (1993). WHC-SD-EN-TI-201. *Screening Performance Assessment/Risk Assessment for the Proposed Environment Restoration Disposal Facility (ERDF),* Westinghouse Hanford Company, Richland, Washington.

WHC. (1999). WHC-SD-EN-AP-015. *Revised Groundwater Monitoring Plan for the 200 Areas Low-Level Burial Grounds*, Last, G.V. and Bjornstad, B.N., Westinghouse Hanford Company, Richland, Washington.

Grubbs, F.E. (February 1969). *Procedures for Detecting Outlying Observations in Samples*, Technometrics, Vol. 11, No. 1, pp. 1-21.

Ostle B., Malone L.C. (1988). *Statistics in Research: Basic Concepts and Techniques for Research Workers*. 4th ed, Iowa State Press, ISBN 081381569X

Wilson, C. R., Einberger, C. M., Jackson, R. L., and Mercer, R. B. (1992). *Design of Ground-Water Monitoring Networks Using the Monitoring Efficiency Model (MEMO)*. Ground Water, Vol. 30, pp. 965 - 970.

Zar, Jerrold, H. (1999). *Biostatistical Analysis*, Prentice-Hall, Inc., New Jersey, pp. 360 – 388, ISBN: 0131008463.

CHAPTER 6.0
PROCEDURES TO PREVENT HAZARDS

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CHAPTER 6.0
PROCEDURES TO PREVENT HAZARDS

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6.0 PROCEDURES TO PREVENT HAZARDS

This chapter discusses security, inspection schedules, preparedness and prevention requirements, preventive procedures, structures, equipment, and prevention of reaction of ignitable, reactive, and incompatible waste at the Integrated Disposal Facility (IDF). The requirements in this chapter that address activities involving the receipt and disposal of dangerous waste as defined in WAC 173-303-040 shall be applied during the Active Life of the IDF. Active Life of a facility means the period from the initial receipt of dangerous waste at the facility until the department receives certification of final closure (WAC 173-303-040). The requirements of this chapter that do not apply to receipt and disposal of dangerous waste as defined in WAC 173-303-040, shall be implemented by the Permittees during the Pre-Active Life of IDF. Pre-Active Life is not defined in the regulations, but refers to the facility maintenance period between final construction and the start of Active Life.

The IDF is designed and will be operated to minimize exposure of the general public and operating personnel to disposed waste. Shielding, contamination control, control of toxic or dangerous material, and safety and security procedures will be used to keep exposure as low as reasonably achievable (ALARA).

6.1 Security

The following sections describe the security measures, equipment, and warning signs to be used to control entry to the IDF. A discussion of Hanford Facility security is provided in Attachment 33, Section 6.1.

6.1.1 Security Procedures and Equipment

The following sections describe the 24-hour surveillance system, barrier, and warning signs to be used to provide security and control access to the IDF.

6.1.1.1 24-Hour Surveillance System

The Hanford Facility is a controlled-access area [refer to Attachment 33, Section 6.1].

6.1.1.2 Barrier and Means to Control Entry

Because the IDF is located within the portion of the Hanford Facility controlled by the 24-hour surveillance system, WAC 173-303-310(2)(c) does not apply.

6.1.1.3 Warning Signs

Signs will be visible from all angles of approach, and legible from a distance of at least 7.6 meters. Each active area used for disposal will be posted with a sign, in English, reading, *DANGER-UNAUTHORIZED PERSONNEL KEEP OUT* or an equivalent legend.

6.1.2 Waiver

A waiver of the security procedures and equipment requirements for the IDF was not requested. Therefore, the requirements of WAC 173-303-310(1)(a) and (b) are not applicable.

6.2 Inspection Plan

This section describes the method and schedule for inspections of the IDF. These inspections help to ensure that situations do not exist that might cause or lead to the release of waste to the environment, degradation of safety equipment and/or systems, or that might pose a threat to human health. Abnormal conditions identified by inspections must be corrected.

6.2.1 General Inspection Requirements

The content and frequency of inspections are described in this section. Inspection discrepancies are documented on inspection checklists and log sheets. The schedule and inspection records will be kept in the inspection logbooks and retained by the IDF operations personnel. Inspection records will be retained in accordance with Permit Condition II.I.1 and contain the following information:

- Date and time of inspection.
- Printed name and the hand written signature of the inspector.
- Notation of the observations made.
- An account of spills or discharges in accordance with WAC 173-303-145.
- Date and nature of any repairs or remedial actions taken.

The inspection checklists consist of a listing of items that are assessed during each inspection. A yes/no response will be made for each listed item. A 'yes' response means that the item is in compliance with the conditions stated on the checklist. Any problems identified during the inspection, as indicated by a 'no' response on the checklist, will be reported immediately to the IDF operations supervisor.

6.2.1.1 Types of Problems

Types of problems looked for during an inspection in Pre-Active Life are in Table 6.2. Types of problems looked for during an inspection in Active Life are in Table 6.3. Once the IDF begins to receive dangerous waste, the requirements in Table 6.2 are no longer applicable. Each day mixed waste containers and/or bulk waste are handled within the IDF; an operator will perform a daily inspection of areas subject to spills (e.g., loading and unloading areas and waste handling areas).

6.2.1.2 Frequency of Inspections

Table 6.2 provides inspection frequencies during the Pre-Active Life. Tables 6.1, 6.2, and 6.3 provide inspection frequencies during the Active Life. For clarification, areas with operations that may result in spills are described below

Each step in the waste placement operation occurs in the landfill over the double High Density Polyethylene (HDPE) liner system that provides containment of any spill from the waste handling operation.

Waste Handling Operations involve the following:

- Unloading of the waste shipment in the landfill Placement of the cover soil over the waste container

During Active Life leachate,[1] movement occurs within the double-contained leachate handling system. There is a potential for a leachate spill on the concrete containment slab of the Crest Pad Building, Leachate Transfer Building, and/or the Leachate Loading Truck Pad.

Leachate Handling Operations involve the following:

- Pumping leachate from the collection sumps to the Crest Pad Building.
- Activities within the Crest Pad Building.
- Transfer of leachate to and from the double-lined Leachate Tanks.
- All activities that occur in the Leachate Transfer Building.
- Pumping of leachate to a tanker truck on the Truck Loading Pad.

Liquid handling operations involve the following:

- The Secondary Leak Detection System (SLDS) is similar to the Leak Detection System (LDS), except that it is equipped with liquid level indication instrumentation only. A low-capacity submersible pump can be inserted into the SLDS sump if required. Pumping of liquid from the collection sump to the small, portable container on the SLDS Pad may be required. Collected liquid in the SLDS that may be construction water and/or liquid from other sources.

[1] WAC 173-303-040 defines "Leachate" as any liquid, including any components suspended in the liquid that has percolated through or drained from dangerous waste.

(Note that the secondary leak detection system is not a design requirement of WAC 173-303-665, however United States Department of Energy (DOE) has added the design feature pursuant to its authority under the Atomic Energy Act of 1954 (AEA) and not for the purposes of compliance with the dangerous waste regulations. Therefore, information regarding the design, construction, and operation of the SLDS is provided for information only. Pursuant to AEA, DOE has sole and exclusive responsibility and authority to regulate the source, special nuclear and by-product material component of radioactive mixed waste at DOE-owned nuclear facilities. Source, special nuclear and by-product materials, as defined by AEA, are not subject to regulation under Resource Conservation and Recovery Act (RCRA) or the Hazardous Waste Management Act, by the State of Washington and are not be subject to State dangerous waste permit, orders, or any other enforceable instrument issued there under. DOE recognizes that radionuclide data may be useful in the development and confirmation of geohydrologic conceptual models. Radionuclide data contained herein is therefore provided as a matter of comity so the information may be used for such purposes).

6.2.2 Schedule for Remedial Action for Problems Revealed

The operating organization will remedy any problems revealed by the inspection on a schedule that prevents hazards to human health and the environment. Where a hazard is imminent or already has occurred immediate action will be taken. Immediate actions will be implemented based on ALARA considerations, availability of supplies, equipment, and personnel.

6.2.3 Specific Process or Waste Type Inspection Requirements

The following sections detail the inspections to be performed at the IDF.

6.2.3.1 Container Inspection

On receipt, operations personnel will confirm appropriate documentation by inspecting each mixed wasted container for disposal and compliance with the container receipt inspection criteria (Chapter 3.0) before the mixed waste is placed in the IDF.

If present, off-specification waste and vitrified waste requiring cooling in storage will be subject to the specific items and/or problems noted during weekly container inspection (Table 6.1) include the following:

- Condition of trench floor and sides.
- Container structural integrity.
- Containers closed.
- At a minimum, 76.2 centimeters aisle spacing.
- Corrosion of containers.
- Evidence of spills or leaks.
- Container labels and markings in place, legible, and unobscured.
- Areas in and around stored waste are free of combustibles (e.g., tumbleweeds).
- Waste separations such as tape, rope, chain or other cordon mechanism are intact.

If present, transport vehicles containing off specification waste or vitrified waste requiring cooling will be subject to the specific items and/or problems noted during weekly inspection include the following:

- Transport vehicle structural integrity.
- At a minimum, 76.2 centimeters aisle spacing between transporters.
- Evidence of spills or leaks.
- Areas in and around transport vehicles are free of combustibles (e.g. tumbleweeds).
- Separations such as tape, rope, chain or other cordon mechanism are intact.

Transport vehicles will not be subject to an individual container inspection within the transporter. Records of inspection will be maintained as detailed in Section 6.2.1.

6.2.3.2 Landfill Inspection

The IDF will be inspected according to the frequencies in Table 6.2 during Pre-Active Life and in accordance with Table 6.3 during Active Life.

6.2.3.2.1 Run-on and Runoff Control System

A run-on control system is installed around the perimeter of each lined trench (Chapter 4.0). The system consists of a berm along the outer margin of each trench that prevents run-on from entering the trench. All run-on control system berms are inspected quarterly (Table 6.2) and after storms for signs of deterioration, malfunction, or improper operation. During Active Life, any precipitation that falls between the run-on control berm and the edge of the trench excavation eventually might flow into the primary leachate control and removal system sump and will be treated as leachate.

6.2.3.2.2 Leak Detection System

During pre-active life, the leak detection system will be monitored quarterly and after storms (Table 6.2) for the amount of liquid removed. To calculate the action leakage rate, measurements are needed to be collected over five consecutive days each quarter. The action leakage rate will be determined for the quarter using these measurements collected during one five day work week each quarter.

During Active life (Table 6.3), leak detection for lined trench at the IDF is accomplished by the following:

- Monitoring liquid level above the secondary liner.
- Monitoring liquid levels above primary liner.
- Inspecting for the presence of liquids after significant precipitation events.
- Verifying certain gauges and instruments are in current calibration; calibration is performed annually or more frequently at intervals suggested by the manufacturer (Chapter 4.0, Section 4.3.7.4).
- Recording secondary sump levels on a daily action leakage rate calculation sheet (Figure 6.1).

If the action leakage rate (Chapter 4.0, Appendix 4C) has not been exceeded, the liner system will be functioning properly.

6.2.3.2.3 Wind Dispersal Control System

During pre-active life, berms will be inspected quarterly and after storms to ensure the berms are functioning properly (Table 6.2).

During active life (Table 6.3), waste is inspected on receipt for evidence of damage, corrosion, or deterioration that might lead to dispersal of the contents.

Unpackaged or bulk waste with any potential for wind dispersal is covered or sprayed with fixative after being placed in a trench.

In addition, unpackaged or bulk waste handling operations are suspended in winds exceeding 24 kilometers per hour unless specifically approved by operations supervisors. The supervisor only would grant approval to operate in winds over 24 kilometers per hour after determining that the risk to human health or the environment would be diminished by completing the work activity, or that the nature and form of the waste handling activity was such that the wind speed would have no significant impact.

6.2.3.2.4 Leachate Collection and Removal System

During pre-active life, the Leachate Collection and Removal System is inspected quarterly and after storms (Table 6.2) for the presence of liquids, and that the system is functioning properly.

During active-life (Table 6.3), Liquids in the Leachate Collection and Removal System and Leak Detection System are monitored daily to ensure the action leakage rate (Chapter 4.0, Appendix 4A) is not exceeded and will be inspected per Table 6.2. In addition, a flow meter is used to check if the amount of actual leachate pumped corresponds to the amount accumulated in the leachate collection tank. This check will verify the proper function of the leachate collection and removal sump pumps with each use.

6.3 Preparedness and Prevention Requirements

Section 6.3.1 describes the preparedness and prevention measures to be implemented during Pre-Active Life. Sections 6.3.2 through 6.3.7 describe the preparedness and prevention measures taken at the IDF during Active Life.

6.3.1 Pre-Active Life Preparedness and Prevention

During Pre-Active Life, the Permittees will comply with Permit Attachment 4, Hanford Emergency Management Plan (DOE/RL-94-02) as applicable for a facility that does not contain dangerous waste. An emergency coordinator will be assigned to IDF who will manage and control all aspects of the initial facility response when an emergency occurs.

6.3.2 Equipment Requirements

The following sections describe the internal and external communications systems and the emergency equipment required.

6.3.3 Internal Communication

There is one building, MO-518, equipped to support communications. Immediate emergency instruction to personnel working at the IDF will be provided by cellular telephones.

6.3.4 External Communications

Personnel at the IDF will have voice communication or equivalent (e.g., hand signals) during work assignments to maintain external communications with shift supervisors. Supervision will contact the Hanford Facility emergency telephone number (911) (373-3800 for cellular telephones) if assistance is needed in the field.

6.3.5 Emergency Equipment

Emergency equipment will be available for use at the IDF. A list of equipment is included in the contingency plan (Addendum J.1, Pre-Active Life, and Addendum J.2, Active Life).

The Hanford Facility relies primarily on the Hanford Fire Department to control fires. Emergency equipment will not be located at IDF trenches. Portable fire extinguishers will be carried on IDF operations vehicles. Attachment 4, *Hanford Emergency Management Plan*, (DOE/RL-94-02) identifies the trained firefighting and emergency medical personnel and equipment.

6.3.6 Water for Fire Control

- Hanford Fire Department trucks as described in Permit Attachment 4, *Hanford Emergency Management Plan*, (DOE/RL-94-02); and fire hydrants described in Addendum J.1 and Addendum J.2 supply water for fire control at the IDF.

6.3.7 Aisle Spacing Requirements for Off-Specification Waste

Aisle spacing during off-specification and cooling vitrified waste storage operations is sufficient to allow the movement of personnel and fire protection equipment in and around the containers. This aisle spacing meets the requirements of WAC 173-303-340(3). Inspection aisle space must be at least 76.2 centimeters. During off-specification storage operations, rows of containers are placed no more than two containers wide in accordance with WAC 173-303-630(5)(c). Aisle spacing requirements will be applied to transport vehicles but not to the waste within the transport vehicles.

6.4 Preventive Procedures, Structures, and Equipment

The following sections will apply during the Active Life for the IDF and describe preventive procedures, structures, and equipment.

6.4.1 Unloading Operations

Methods used to prevent release of waste during unloading operations will be employed as follows.

- Waste will be inspected according to the receipt inspection criteria (Chapter 3.0).
  - If waste fails the inspection, it will be designated as an off-specification waste and could be placed in the storage area or returned to the generator.
- Containers and bulk waste will be handled by appropriate equipment (i.e., crane) during unloading.
- Path from loading area to trench area will be clear of obstructions.

Spills will be managed as identified in the contingency plan (Addendum J.2).

Containers and bulk waste will be staged at the waste unloading area no longer than necessary for placement into the landfill. Administrative procedures may prevent immediate unloading and backfilling of waste containers. Containers might be left in the transporters as needed to resolve the administrative procedure requirements or to support the operational schedule before containers are placed into the landfill. The transfer vehicle containing vitrified waste requiring cooling may be temporarily placed in the storage area prior to unloading for disposal.

6.4.2 Runoff

The waste in the IDF will be placed below the land surface; thus, the IDF is designed to prevent run-off of precipitation that might have come in contact with the waste. The land surface is relatively level, so trenches have only internal drainage. The minimal amounts of precipitation that accumulate are contained within the trench.

The IDF trench is designed to channel run-on liquid away from the trench. Precipitation that percolates to the bottom of the trench is captured in the leachate collection system and is managed as rainwater during Pre Active Life. During Active Life, these liquids will be managed as multi-source leachate waste.

6.4.3 Water Supplies

The design and operation of the IDF during Active Life is intended to minimize the generation of potentially contaminated leachate and to prevent leachate migration into groundwater resources in the local area. All activities performed during Active Life (Chapter 4.0) or Pre-active Life is designed to protect local water supplies.

Activities that prevent contamination of water supplies or groundwater will include the following:

- Placement of waste in lined trenches
  - Run-on and run-off will be controlled
  - Leak detection systems will be used
  - Leachate will be collected and managed as waste
  - Inspections will be performed
- Placement of backfill will occur after a layer of waste has been placed in the trench.

6.4.4 Equipment and Power Failure

Electrical power is required for the landfill. Electricity supplies power to the sump pumps used to remove accumulated leachate from the primary and secondary liners. Electricity outages will be restored as soon as possible. Backup equipment will be acquired if necessary to provide electrical service. Failed equipment will be repaired or replaced as soon as possible.

6.4.5 Personal Protection Equipment

Personnel will be trained in the use of applicable personal protection equipment. The protective clothing required for Active Life will vary depending on the form and content of the waste.

6.5 Prevention of Reaction of Ignitable, Reactive, and Incompatible Waste

The waste acceptance criteria will prohibit the disposal of ignitable, reactive, and incompatible waste at the IDF. Waste acceptance criteria (Chapter 3.0) will ensure that the required treatment has been performed before the waste is disposed in the IDF.

Waste stream compatibility (i.e., compatibility between individual waste streams and compatibility between waste streams and landfill design and construction parameters) will be assessed on a case-by-case basis. Criteria for assessing and determining compatibility will be identified in either the facility Waste Acceptance Criteria, Waste Analysis Plan, or other protocol or procedure as appropriate (Chapter 3.0) for further discussion of waste stream compatibility.

AVERAGE DAILY ACTION LEAKAGE RATE CALCULATION

Operating Day __/__/__ ____Gallons

Operating Day __/__/__ ____Gallons

Operating Day __/__/__ ____Gallons

Operating Day __/__/__ ____ Gallons

Operating Day __/__/__ ____ Gallons

Operating Day __/__/__ ____ Gallons

Operating Day __/__/__ ____ Gallons

Secondary Sump Total Volume **TOTAL** ____ Gallons

(DIVIDE TOTAL VOLUME BY 7)

AVERAGE DAILY ACTION LEAKAGE RATE: ____ Gallons

NOTIFY IDF Operations Supervisor if Average Daily Action Leakage Rate is GREATER than XXX Gallons

---

Repairs or remedial action taken:

Operator's Printed Name: _____ Date _____

Operator's Signature: _____ Time _____ hrs

Operations Supervisor Printed Name: _____ Date _____

Operations Supervisor Signature: _____ Time _____ hrs

Figure 6.1. Typical Average Daily Action Leakage Rate Calculation

Table 6.1. Container Storage Inspections

Requirement Description	Inspection Frequency	Types Of Problems
-630(6) Containers	Weekly	Leaking Containers Deteriorating containers
-630(6) Containment System	Weekly	Deteriorating containment system
-395(1)(d) Ignitable or reactive waste	Not Applicable	Not Applicable

Table 6.2. Landfill Inspections during Pre-Active Life***

Requirement Description	Inspection Frequency	Types Of Problems
-665(4)(b)(i) Run-on and run-off control	Quarterly and after storms*	Deterioration, malfunction, or improper operation
-665(4)(b)(ii) Wind dispersal control systems	Quarterly and after storms*	Proper functioning
-665(4)(b)(iii) Leachate collection and removal systems	Quarterly and after storms*	Presence of liquids; proper functioning
-665(4)(c)(i) Leak detection system sump	Quarterly and after storms*	Amount of liquids removed
Secondary leak detection system sump**	Quarterly **	Presence of unexpected liquid volume**
Security "Danger unauthorized personnel keep out" signs	Quarterly	Signs are posted and legible
Areas subject to spills	Daily when any activities may take place that have a potential for a spill or release to occur	Evidence of spills

*A storm is any atmospheric disturbance with either wind gust of 56.3 kilometers per hour (35 miles per hour) or greater, or precipitation of 0.5 inch or greater within a 24-hour period.

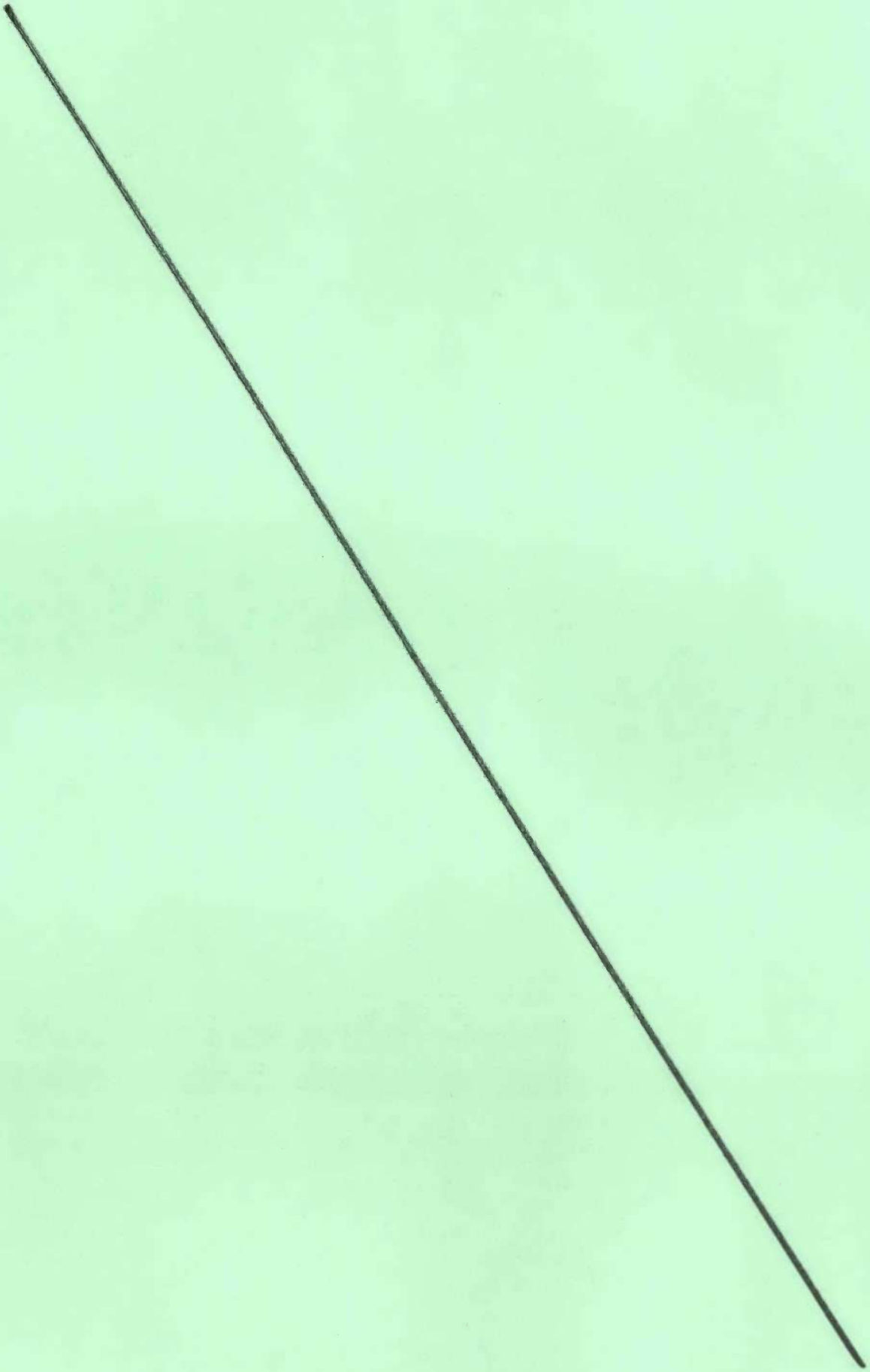
Note: Secondary leak detection system is not a design requirement of WAC 173-303-665, however DOE is adding the design feature pursuant to its authority under the Atomic Energy Act of 1954 (AEA) and not for the purposes of compliance with the dangerous waste regulations. Therefore, formation regarding the design, construction, and operation of the secondary leak detection system is provided for information only. Pursuant to AEA, DOE has sole and exclusive responsibility and authority to regulate the source, special nuclear and by-product material component of radioactive mixed waste at DOE-owned nuclear facilities. Source, special nuclear and by-product materials, as defined by AEA, are not subject to regulation under RCRA or the Hazardous Waste Management Act, by the State of Washington and are not be subject to State dangerous waste permit, orders, or any other enforceable instrument issued there under. DOE recognizes that radionuclide data may be useful in the development and confirmation of geohydrologic conceptual models. Radionuclide data contained herein is therefore provided as a matter of comity so the information may be used for such purposes.* Once the IDF begins to receive dangerous waste, the requirements in Table 6.2 are no longer applicable, and inspection requirements will be as provided in Table 6.3

Table 6.3. Landfill Inspections during Active Life

Requirement Description	Inspection Frequency	Types Of Problems
-665(4)(b)(i) Run-on and run-off control	Weekly and after storms*	Deterioration, malfunction, or improper operation
-665(4)(b)(ii) Wind dispersal control systems	Weekly and after storms*	Proper functioning
-665(4)(b)(iii) Leachate collection and removal systems	Weekly and after storms*	Presence of leachate; proper functioning
-665(4)(c)(i) Leak detection system sump	Weekly and after storms*	Amount of liquids removed
Secondary leak detection system sump**	Monthly**	Presence of unexpected liquid volume**
Security devices: "Danger unauthorized personnel keep out" signs	Weekly	Signs are posted and legible
Areas subject to spills	Daily when waste management activities having a potential for a spill to occur	Evidence of spills

*A storm is any atmospheric disturbance with either wind gust of 56.3 kilometers per hour (35 miles per hour) or greater, or precipitation of 0.5 inch or greater within a 24-hour period.

**Note: Secondary leak detection system is not a design requirement of WAC 173-303-665, however DOE is adding the design feature pursuant to its authority under the Atomic Energy Act of 1954 (AEA) and not for the purposes of compliance with the dangerous waste regulations. Therefore, formation regarding the design, construction, and operation of the secondary leak detection system is provided for information only. Pursuant to AEA, DOE has sole and exclusive responsibility and authority to regulate the source, special nuclear and by-product material component of radioactive mixed waste at DOE-owned nuclear facilities. Source, special nuclear and by-product materials, as defined by AEA, are not subject to regulation under RCRA or the Hazardous Waste Management Act, by the State of Washington and are not be subject to State dangerous waste permit, orders, or any other enforceable instrument issued there under. DOE recognizes that radionuclide data may be useful in the development and confirmation of geohydrologic conceptual models. Radionuclide data contained herein is therefore provided as a matter of comity so the information may be used for such purposes.



CHAPTER 8.0
PERSONNEL TRAINING

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CHAPTER 8.0
PERSONNEL TRAINING

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8.0 PERSONNEL TRAINING [H]

This chapter discusses personnel training requirements based on WAC 173-303 and the Hanford Facility Resource Conservation and Recovery Act (RCRA) Permit (Permit). Permit Condition II.C (Personnel Training), contains training requirements applicable to Hanford Facility personnel and non-Facility personnel. Compliance with these requirements at the Integrated Disposal Facility (IDF) is demonstrated by information contained in Permit Attachment 5, and this chapter. This chapter supplements Permit Attachment 5.

8.1 Outline of Introductory and Continuing Training Programs [H-2]

The introductory and continuing training programs are designed to prepare personnel to manage and maintain the treatment, storage, and disposal (TSD) unit in a safe, effective, and environmentally sound manner. In addition to preparing personnel to manage and maintain TSD units under normal conditions, the training programs ensure that personnel are prepared to respond in a prompt and effective manner should abnormal or emergency conditions occur. Emergency response training is consistent with the description of actions contained in Addendum J.1, Contingency Plan, Pre-Active Life.

Introductory training includes general Hanford Facility training and TSD unit-specific training. General Hanford Facility training is described in Permit Attachment 5, and is provided in accordance with Permit Condition II.C.2. TSD unit-specific training is provided to Hanford Facility personnel allowing personnel to work unescorted. Hanford Facility personnel cannot perform a task for which they are not properly trained, except to gain required experience while under the direct supervision of a supervisor or coworker who is properly trained. Hanford Facility personnel assigned the job title of Emergency Coordinator and alternates to this position performing tasks described in WAC 173-303-360 (e.g., Building Emergency Directors) are thoroughly familiar with applicable contingency plan documentation, operations, activities, location, and properties of all waste handled, location of all records, and the unit/building layout.

Continuing training meets the requirements for WAC 173-303-330(1)(b) and includes general Hanford Facility training and TSD unit-specific training. General Hanford Facility training is the same as described for introductory training. TSD unit-specific training provides an annual review of emergency response training and an annual review of training necessary to ensure TSD unit operations are in compliance with WAC 173-303.

8.2 Description of Training Plan

In accordance with Permit Condition II.C.3, the unit-specific portion of the *Hanford Facility Dangerous Waste Permit Application* must contain a description of the training plan. The plan is written to comply with WAC 173-303-330. Written training plan documentation is maintained outside of the *Hanford Facility Dangerous Waste Permit Application* and the Permit. Therefore, changes made to the written training plan documentation are not subject to the Permit modification process. The training plan will be maintained as part of the operating records of the facility and will be available to regulators upon request.

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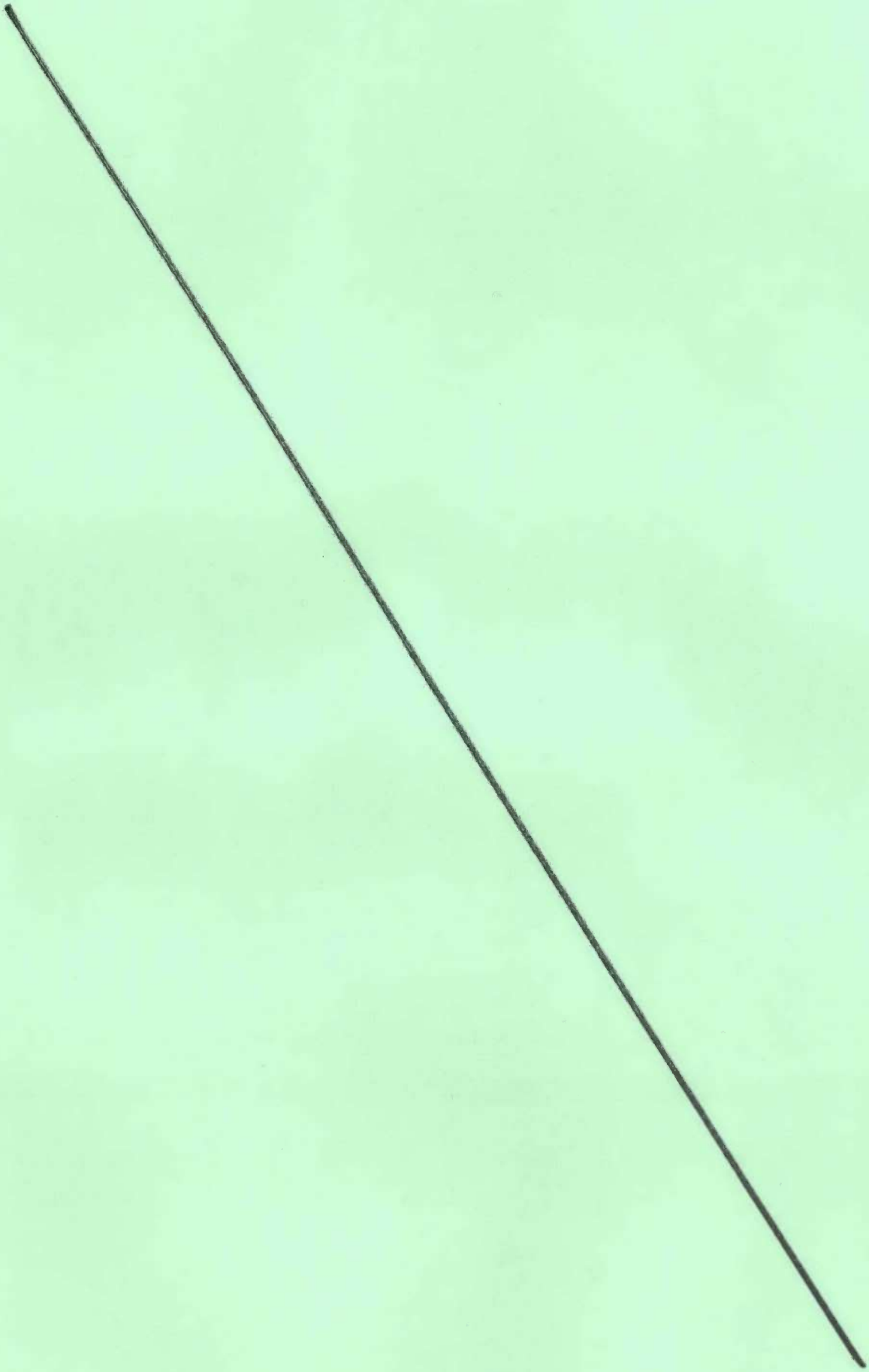

CHAPTER 11.0
CLOSURE

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CHAPTER 11.0
CLOSURE

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FIGURE

Figure 11.1. Typical Hanford Site Landfill Cover Design 11.10

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11.0 CLOSURE

This chapter discusses preclosure, closure, and post closure activities for the Integrated Disposal Facility (IDF). This closure plan complies with WAC 173-303-610 and represents the baseline for closure.

The IDF has been constructed on 25 hectares of vacant land southwest of the Plutonium Uranium Extraction Facility (PUREX) Plant in the 200 East Area [see the topographic map on file at the Department of Ecology library (3100 Port of Benton Boulevard, Richland, WA 99354)]. The landfill is segregated into a Resource Conservation and Recovery Act (RCRA) permitted side and a non-RCRA permitted side. The scope of this permit is limited to the western side of the landfill where the RCRA waste will be placed. The waste containers and bulk waste that meet the IDF waste acceptance criteria will be inventoried, and disposed in this lined landfill. Leachate collected from the lined landfill will be transferred to leachate collection tanks located in proximity to the landfill for subsequent treatment.

A more detailed discussion of IDF waste types and the identification of the IDF processes and equipment are provided in Chapters 3.0 and 4.0, and attendant appendices. The IDF only will accept and dispose waste containers and bulk waste that meet the IDF waste acceptance criteria, RCRA and Land Disposal Restriction (LDR).

The closure process will be the same for partial closure or closure of the entire IDF. The remainder of this chapter describes the performance standards that will be met, and the closure/post closure activities that will be conducted.

11.1 Closure Plan

Waste containers and bulk waste that meet the IDF waste acceptance criteria will be disposed in the lined landfill that complies with WAC 173-303-665 standards (Chapter 4.0). The IDF will be closed according to current applicable WAC 173-303 regulations, United States Department of Energy (DOE) requirements, best management practices, and will be integrated with the overall cleanup activities performed under the Tri-Party Agreement (HFFACO).

The disposal landfill cover will be designed and located to comply with WAC 173-303-665(6) and WAC 173-303-610. The specification and/or variation for other cover designs will be provided at the time of closure once a hazard(s) has been defined.

11.2 Closure Performance Standards

Closure requirements found in WAC 173-303-665(6), incorporated by reference, and detailed here in Chapter 11.0 of the IDF portion of the permit, will make up the closure performance standards for the IDF.

11.3 Preclosure Activities

Preclosure activities could include, at a minimum, placing interim or final covers over the filled portions of the landfill as the landfill is expanded to accept more waste. Placement of covers over the filled portions might be deferred until closure of all the IDF. Once a decision is made to construct the final cover over the landfill, a closure cover design will be used that satisfies the dangerous waste disposal requirements defined in WAC 173-303.

The selection of a final cover design has not been identified. Figure 11-1 shows an example of a typical Hanford Site landfill cover design. Design(s) will include features to satisfy the minimum requirements found in WAC 173-303-665(6).

11.4 Maximum Extent of Operation

The maximum process design capacity of the IDF conservatively is calculated to be 100 hectare-meters, which is 1,000,000 cubic meters (Chapter 1.0, Part A, Form, Section III). The IDF landfill will be segregated into a RCRA permitted side of 50 hectare-meters and a non-RCRA permitted side of 50 hectare-meters.

11.5 Decontaminating Equipment and Structures

All ancillary equipment and its secondary containment, and instrumentation (e.g., level-indicating devices, leak detection devices, pumps, piping) meet the definition of "debris" as defined in WAC 173-303-040. Items in direct contact with mixed waste are assumed to meet the definition of "hazardous debris" as defined in WAC 173-303-040.

Currently, three options are available for treating hazardous debris. The first option is to treat the debris using one of the three debris treatment technologies-extraction, destruction, or immobilization-as described in 40 CFR 268.45. If the hazardous debris is treated using approved extraction or destruction technologies, the debris is no longer required to be managed as a dangerous waste as long as the debris does not exhibit a characteristic of a dangerous waste. If hazardous debris contaminated with a listed waste is treated using an immobilization technology, it remains a listed waste, even after the LDR treatment standards are met unless Ecology makes a case-by-case determination that the debris "no longer contains" a mixed waste. In effect, by making this "contained-in" determination on a case-by-case basis, Ecology will be setting clean closure standards in accordance with the closure performance standards of WAC 173-303-610(2)(a)(ii).

The second option is to treat the hazardous debris to meet the constituent-specific LDR treatment standard for the waste or waste-specific constituents contaminating the debris; however, such debris, even after treatment, may be considered a dangerous waste under the dangerous waste regulations and may require management at a facility permitted to manage dangerous waste.

The third option involves obtaining a "contained-in determination" for the hazardous debris, thereby rendering the waste "non-hazardous" for those waste-specific-listed constituents that fall below Model Toxic Control Act (MTCA) method B risk-based health limits. Moreover, it must be proven that the debris does not designate as a characteristic waste under WAC-173-303.

11.5.1 Contaminated Soil

Contaminated soil could be generated as a result of spill cleanup. Since the majority of IDF operations will be performed within secondary containment (see Chapters 4.0 and 6.0) the potential for spilling dangerous waste into the surrounding soil is low. Contaminated soil generated as a result of a dangerous waste spill will be managed pursuant to WAC-173-303-200.

Once the soil is designated, appropriate treatment and disposal or storage options will be determined and implemented.

A contained-in determination could also be sought for contaminated soil generated as a result of a spill. For contaminated media the contained-in policy requires that a statistically based sampling plan be used for obtaining the data to support a contained-in demonstration. The contained-in policy does not require that the waste be analytically nondetectable for it to be considered nondangerous. However, the analytical results must prove that the listed constituents in the soil are below health-based limits as provided in WAC 173-303-610(2)(b)(i) and that the soil does not exhibit any dangerous waste characteristics (i.e., soil does not designate for D codes). If approved by Ecology, this could allow waste that falls below specific health-based levels to be disposed of without requiring treatment

11.6 Closure of Landfill Units

Closure of the IDF will be consistent with the closure requirements specified in WAC-173-303-665(6) and WAC 173-303-610. The cover design(s) will satisfy the requirements of WAC 173-303-665(6).

11.6.1 Cover Design

The cover could consist of several layers constructed on top of a native soil base. A generalized cross-section of an example cover is shown on Figure 11-1. It is assumed that before construction of the final cover, the waste form would be stabilized appropriately.

11.6.1.1 Grade Layer

The surface of the landfill would be graded and/or shaped, if necessary, to match the slope of the desired low-permeability layer. Additional soil would be placed over the landfill to achieve the required cover grade. This grade layer could taper from zero thickness near the edge of the cover boundary to perhaps several meters at the center of the cover; the thickness would depend on the lateral dimensions of the particular cover and the grade of the cover.

11.6.1.2 Low-Permeability Layer

The selection of an appropriate material for this layer would be based on the hazard that is to be isolated. The low-permeability layer will be the primary barrier in preventing soil and/or water from migrating into the waste zone and meet WAC 173-303-655(6)(v) "Have a permeability less than or equal to the permeability of any bottom liner system or natural sub soils present".

11.6.1.3 Drainage Layer

The drainage layer would conduct any water that percolates through the overlying layers laterally to the drainage ditch. Thus, the drainage layer would prevent hydraulic pressure from building up directly on the low-permeability liner, and thereby eliminate one set of forces that would drive moisture through the primary moisture control barrier.

11.6.1.4 Plant, Animal, and Human Intrusion Layer (optional)

The performance objectives for the permanent isolation surface barrier are summarized as follows:

- Function in a semiarid to sub-humid environment.
- Limit the recharge of water through the waste to near zero amounts [0.05 centimeter per year (1.6×10^{-9} centimeters per second)].
- Be maintenance free.
- Minimize the likelihood of plant, animal, and human intrusion.
- Limit the exhalation of noxious gases.
- Minimize erosion-related problems.
- Meet or exceed WAC 173-303-665(6) cover performance requirements.
- Isolate waste for 1,000 years.

To satisfy the intrusion performance objective, an optional layer would be included in the design of barriers that require the additional human and/or biointrusion protection to reduce either the environmental or human health risk.

11.6.1.5 Graded Filter Layer

A graded filter consisting of crushed rock overlaid by sand would be placed on the plant, animal, and human intrusion layer if incorporated into the design, or directly over the drainage layer. The graded filter would serve to separate the surface soil layer from the drainage layer. A geotextile would be placed on the top of the graded filter to decrease the potential for fine material to enter the filter and drainage zone. The geotextile would be permeable, allowing drainage, and would not support a standing head of water.

11.6.1.6 Surface Soil Layer

The two most important factors in engineering the surface soil thickness would be the assignment of the water retention characteristics for soil and climate information. Surface soil would be placed over the geotextile to intercept, store, recycle water, and prevent damage to the underlying structure from natural and synthetic processes.

11.6.1.7 Vegetative Cover

The vegetative cover would perform three functions. First, the plants would return water stored in the surface soil back to the atmosphere, significantly decreasing net infiltration and reducing the amount of moisture available to penetrate the cover. Second, the vegetation would stabilize the surface soil component of the cover against wind and water erosion. Finally, the vegetative cover would restore the appearance of the land to a more natural condition and appearance.

A mixture of seeds would be used to establish vegetation. The seed types would be selected based on resistance to drought, rooting density, and ability to extract water.

11.6.2 Wind Erosion

The principal hazard associated with wind erosion is the thinning of the cover surface soil layer. This in turn potentially could lead to breaching of the moisture barriers, gradually allowing larger quantities of water to reach the waste. The engineering approaches to mitigating wind erosion of the cover would be (1) designing the surface soil layer with an appropriate total thickness to compensate for future soil loss that might result from wind erosion, (2) establishing a vegetative cover on the surface to reduce wind erosion, and (3) including an appropriate coarse material (admix) in the upper layer of the surface soil to form an armor layer.

11.6.3 Water Erosion

The potential hazard associated with water erosion is the same as that for wind erosion, namely the loss of soil from the top or surface layer.

Several of the following engineering approaches could be adopted to minimize the potential for water erosion:

- Limiting the surface slopes.
- Providing run-on control with the sideslope drainage ditches.
- Compacting the surface soil in a way that promotes significant infiltration rather than excessive run-off.
- Properly designing the sideslopes to prevent gullying.
- Establishing a vegetative cover to slow surface run-off.
- Incorporating coarse material (pea gravel admix) in the upper portion of the surface soil layer to help form an erosion-resistant armor.
- Limiting flow path lengths through the use of vegetation and admix.

The cover design would be evaluated for potential erosion damage from overall soil erodibility, sheet flow, and gullying.

11.6.4 Deep-Rooted Plants

The following design features could minimize the potential for problems with deep-rooted plants.

- The surface soil (top two layers) would retain most of the precipitation, because the underlying drainage layer would have significantly higher permeability and much less water retention capacity. Therefore, it is expected that vegetation preferentially would occupy the surface soil layer and not have an affinity for growing into the drier underlying layers.
- The thickness of the surface soils would be sized to promote the development of semiarid deep-rooted perennial grasses and to discourage the development of deep-rooting intrusive species.

11.7 Schedule for Closure

As stated previously, closure of the IDF will be a complex process. At the time of closure, this closure plan will be updated to reflect the current closure plan schedule per WAC 173-303-830, Appendix I. In addition, when a closure date is established, a revised closure plan and closure schedule will be submitted to Ecology that contains detailed information regarding specific activities and implementation timeframes.

11.8 Extension for Closure

An extension for closure request is anticipated to complete the closure/post closure process of the IDF.

11.9 Postclosure Plan

Because of the long active life of the IDF, a comprehensive post closure plan will be developed when closure becomes imminent or when 200 Areas cleanup activities prescribed by the Tri-Party Agreement require integration.

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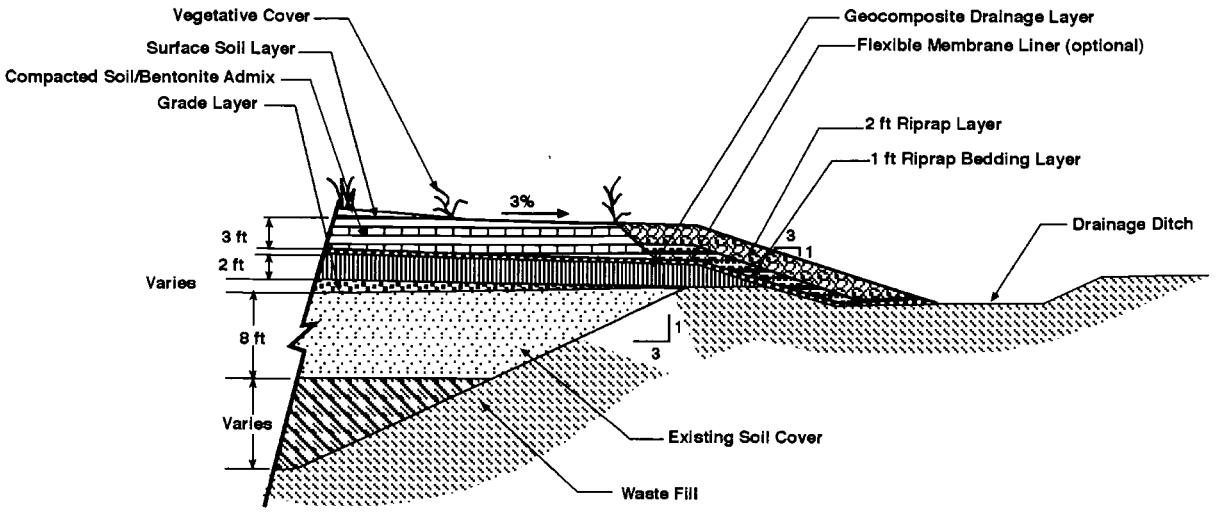

Notes:

1. Drawing not to scale.
2. Cover shown for unlined trench.
   Similar configuration for lined trench.

To convert feet (ft) to meters, multiply by 0.3048.

H00040105.2
M0105-2.1
5/31/01

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3 **Figure 11.1. Typical Hanford Site Landfill Cover Design**

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CHAPTER 13.0
OTHER FEDERAL AND STATE LAWS

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OTHER FEDERAL AND STATE LAWS

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13.0 OTHER FEDERAL AND STATE LAWS [J]

Generally, the laws applicable to the IDF include, but might not be limited to, the following:

- *Atomic Energy Act of 1954*
- *Federal Facility Compliance Act of 1992*
- *Clean Air Act of 1977*
- *Safe Drinking Water Act of 1974*
- *Emergency Planning and Community Right-to-Know Act of 1986*
- *Toxic Substances Control Act of 1976*
- *National Historic Preservation Act of 1966*
- *Endangered Species Act of 1973*
- *Fish and Wildlife Coordination Act of 1934*
- *Federal Insecticide, Fungicide, and Rodenticide Act of 1975*
- *Hazardous Materials Transportation Act of 1975*
- *National Environmental Policy Act of 1969*
- *Washington Clean Air Act of 1967*
- *Washington Water Pollution Control Act of 1945*
- *Washington Pesticide Control Act of 1971*
- *State Environmental Policy Act of 1971.*
- *Letter ,C.J .Paperiello NRC to J.E Kinzer DOE/RL "Classification of Hanford Low-Activity Tank Waste Fraction" dated June 9th 1997*

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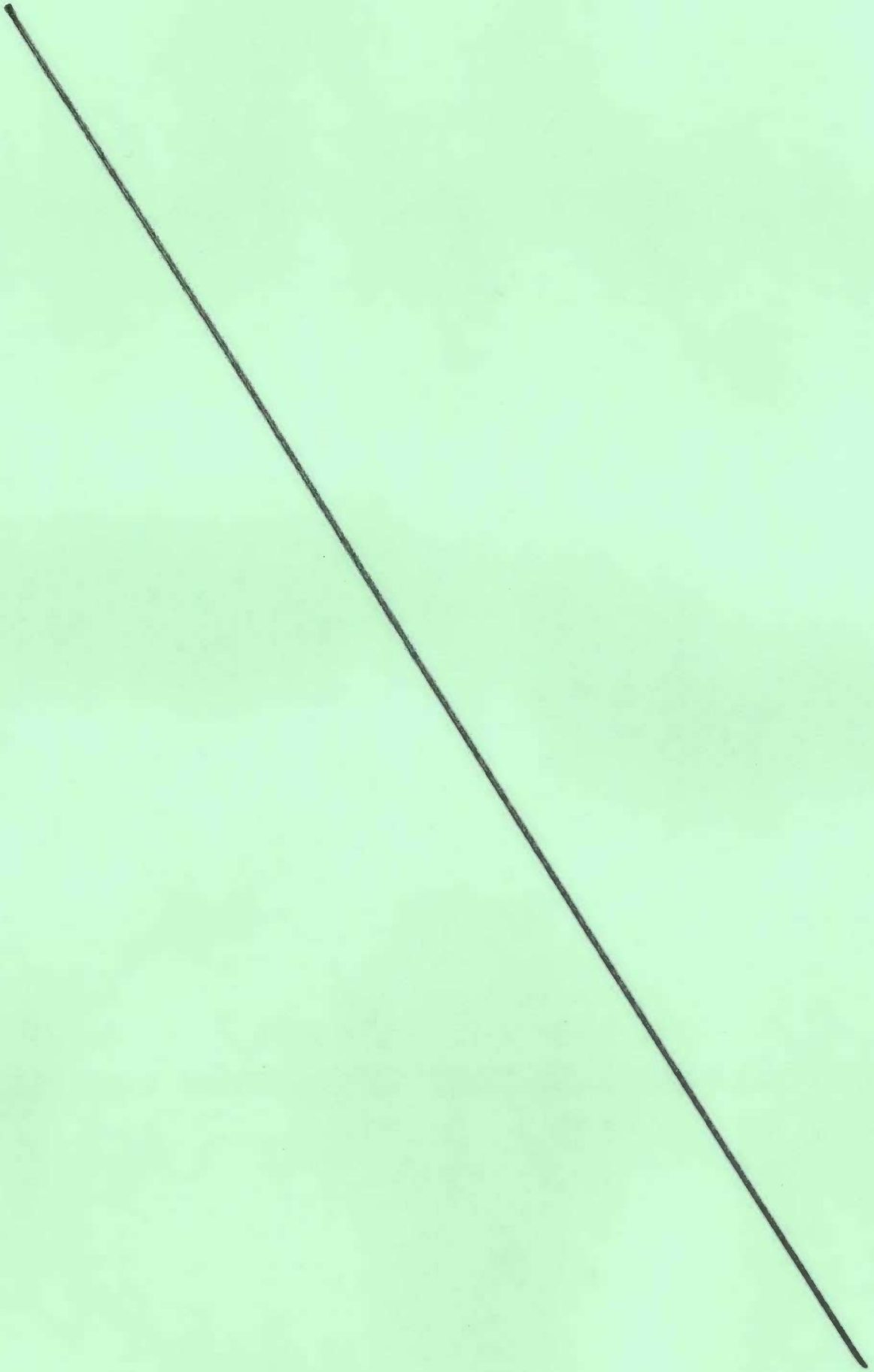

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Title: Addendum J.1: Contingency Plan—Pre-Active Life

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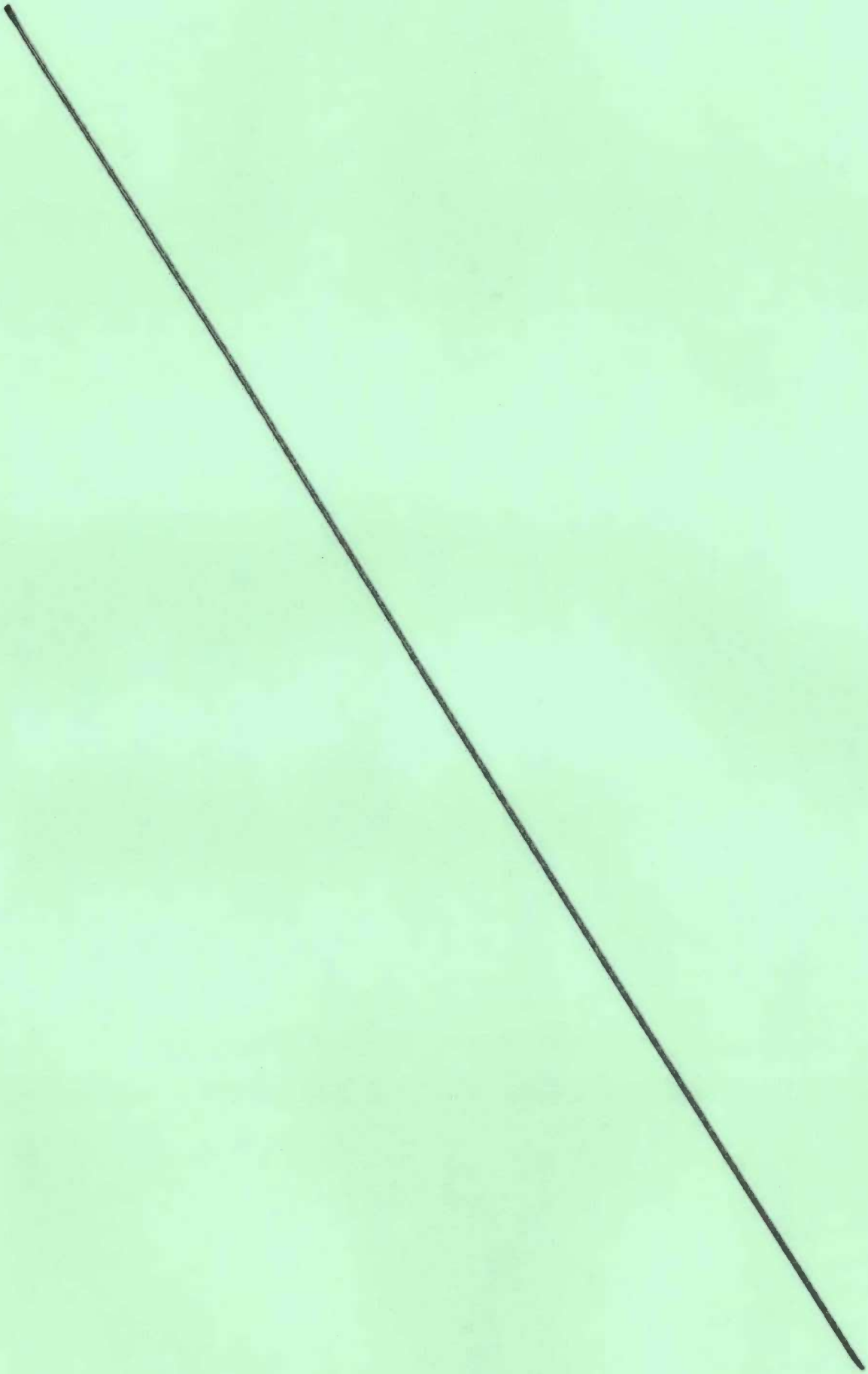

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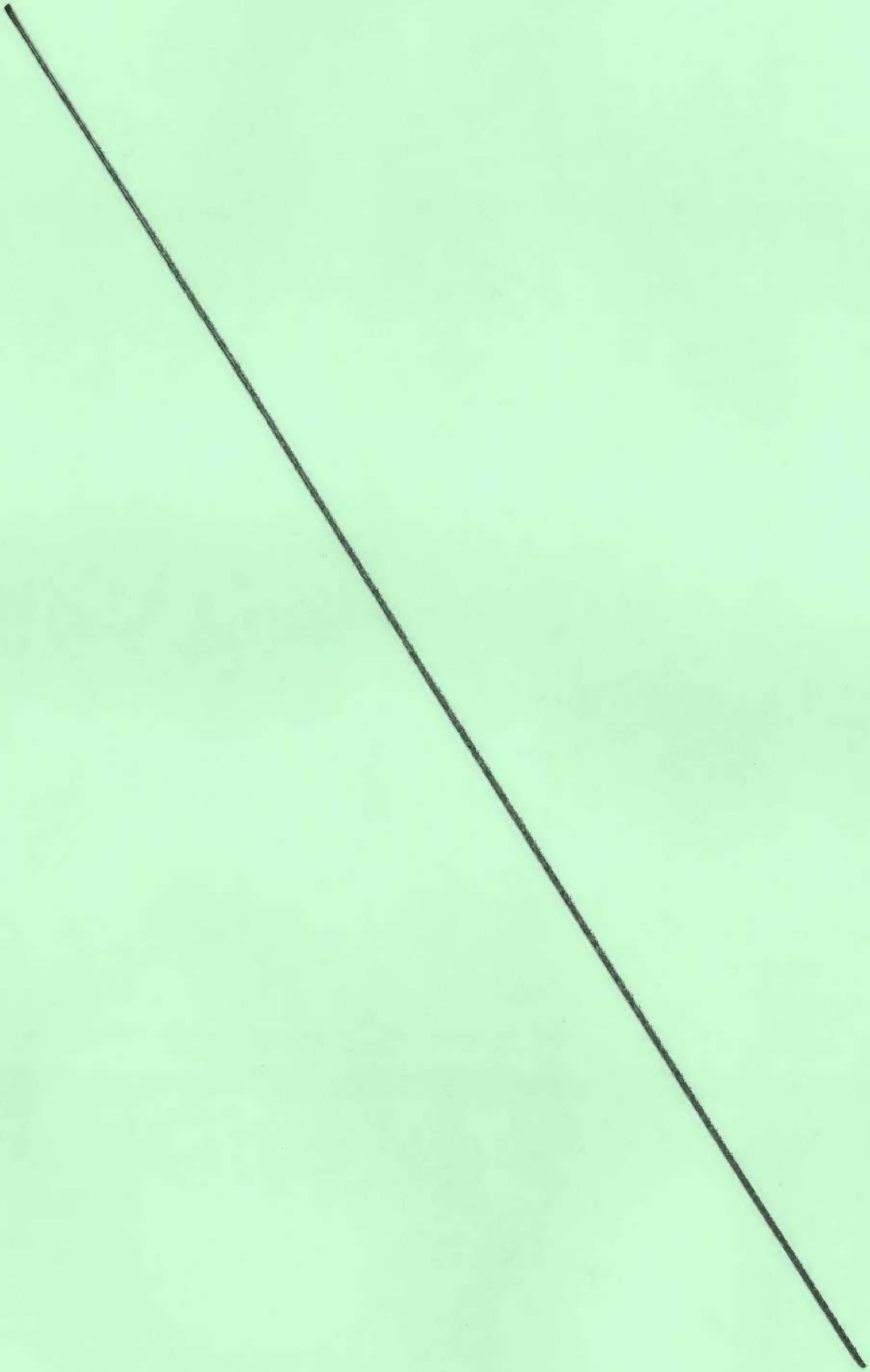

Liquid Effluent Retention Facility (LERF) & 200 Area Effluent Treatment Facility (ETF)

Change Control Log

Change Control Logs ensure that changes to this unit are performed in a methodical, controlled, coordinated and transparent manner. Each unit addendum will have a "**Last Modification Date**" which represents the last date the portion of the unit has been modified. The "**Modification Number**" represents Ecology's method for tracking the different versions of the permit. This log will serve as an up to date record of modifications and version history of the unit.

Last modification to Liquid Effluent Retention Facility & 200 Area Effluent Treatment Facility **June 30, 2015**

Addenda	Last Modification Date	Modification Number
Conditions	06/30/2015	
A. Part A Form	03/31/2015	
B. Waste Analysis Plan	06/30/2015	
C. Process Information	12/31/2014	
D. Groundwater Monitoring Plan	04/29/2014	
E. Security	06/30/2011	
F. Preparedness and Prevention	04/08/2014	
G. Personnel Training	06/30/2015	
H. Closure Plan	06/30/2011	
I. Inspection	04/08/2014	
J. Contingency Plan	06/30/2015	

PART III, OPERATING UNIT GROUP 3 PERMIT CONDITIONS
LIQUID EFFLUENT RETENTION FACILITY & 200 AREA EFFLUENT TREATMENT FACILITY

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PART III, OPERATING UNIT GROUP 3 PERMIT CONDITIONS
LIQUID EFFLUENT RETENTION FACILITY & 200 AREA EFFLUENT TREATMENT FACILITY

UNIT DESCRIPTION

The Liquid Effluent Retention Facility (LERF) and 200 Area Effluent Treatment Facility(200 Area ETF) consists of an aqueous waste treatment system that provides treatment, storage integral to the treatment process, and storage of secondary wastes from the treatment process for a variety of aqueous mixed waste. The 200 Area ETF is located in the 200 East Area. Aqueous wastes managed by the 200 Area ETF include process condensate from the LERF and 200 Area ETF and other aqueous waste generated from onsite remediation and waste management activities.

The LERF consists of three lined surface impoundments, or basins. Aqueous waste from LERF is pumped to the 200 Area ETF for treatment in a series of process units, or systems, that remove or destroy essentially all of the dangerous waste constituents. The treated effluent is discharged to a State-Approved Land Disposal Site (SALDS) north of the 200 West Area, under the authority of a Washington State Waste Discharge Permit (Ecology 2000) and 200 Area ETF Delisting (40 Code of Federal Regulations (CFR) 261, Appendix IX, Table 2). Construction of the LERF began in 1990. Waste management operations began at LERF in April 1994. Construction of the 200 Area ETF began in 1992. Waste management operations began at 200 Area ETF in November of 1995.

This Chapter provides unit-specific Permit conditions applicable to the dangerous waste management units for LERF and 200 Area ETF.

LIST OF ADDENDA SPECIFIC TO OPERATING UNIT GROUP 3

Addendum A	Part A Form, dated March 31, 2015
Addendum B	Waste Analysis Plan, dated June 30, 2015
Addendum C	Process Information, dated December 31, 2014
Addendum D	Groundwater Monitoring, approved April 29, 2014
Addendum E	Security Requirements, dated, June 30, 2011
Addendum F	Preparedness and Prevention, dated April 8, 2014
Addendum G	Personnel Training, dated June 30, 2015
Addendum H	Closure Plan, dated June 30, 2011
Addendum I	Inspection Requirements, dated April 8, 2014
Addendum J	Contingency Plan, dated June 30, 2015

DEFINITIONS

State and federal delisting actions: The state delisting action pursuant to Washington Administrative Code (WAC) 173-303-910(3), August 8, 2005, and the federal delisting action appearing in 40 CFR 261, Appendix IX, Table 2 applicable to the United States, Department of Energy, Richland, Washington.

ACRONYMS

LERF and 200 Area ETF	200-Area Liquids Processing Facility

III.3.A COMPLIANCE WITH UNIT-SPECIFIC PERMIT CONDITIONS

III.3.A.1 The Permittees will comply with all Permit Conditions in this Chapter and its Addendums and Chapters with respect to dangerous waste management and dangerous waste management units in LERF and 200 Area ETF, in addition to requirements in Permit Part I and Part II.

III.3.B GENERAL WASTE MANAGEMENT

III.3.B.1 The Permittees are authorized to accept dangerous and/or mixed waste for treatment in dangerous waste management units that satisfies the waste acceptance criteria in Permit Addendum B according to the waste acceptance procedures in Permit Addendum B. [WAC 173-303-300]

III.3.B.2 The Permittees are authorized to manage dangerous and/or mixed wastes physically present in the dangerous waste management units in LERF and 200 Area ETF as of the effective date of this Permit according to the requirements of Permit Condition III.15.B.1.

III.3.B.3 The Permittees are authorized to treat and/or store dangerous/mixed waste in the dangerous waste management units in LERF and 200 Area ETF according to the following requirements:

III.3.B.3.a The Permittees are authorized to treat, and store as necessary in support of treatment, dangerous waste in the 200 Area ETF tank systems identified in Permit Addendum C, Section C.2, and Section C.4 according to the Permit Conditions of this Chapter.

III.3.B.3.b The Permittees are authorized to store and treat those dangerous and/or mixed waste identified in Permit Addendum C, Section C.3, in containers according to the requirements of this Chapter. All container management activities pursuant to this Permit Condition will take place within the container storage areas or within the 200 Area ETF process area identified in Permit Addendum C, Figures C.3 and C.4.

III.3.B.3.c Treatment in containers authorized by Permit Condition III.3.B.3.b is limited to decanting of free liquids, and addition of sorbents to free liquids. The Permittees will ensure that sorbents are compatible with wastes and the containers. Sorbents will be compliant with the requirements of WAC 173-303-140(4)(b)(iv), incorporated by reference.

III.3.B.3.d The Permittees are authorized to treat aqueous waste in LERF Basins (Basins 42, 43 and 44) subject to the following requirements:

III.3.B.3.d.1 Following treatment in a LERF basin, aqueous wastes must be treated in 200 Area ETF according to Permit Conditions III.3.B.3.a through c.; [40 CFR 268.4(2)(iii), incorporated by reference by WAC 173-303-140]

III.3.B.3.d.2 The Permittees must ensure that for each basin, either supernatant is removed on a flow-through basis, to meet the requirement of 40 CFR 268.4(a)(2)(ii) incorporated by reference by WAC 173-303-140, or incoming waste is shown to not contain solids by either: (1) sampling results showing the waste does not contain detectable solids, or (2) filtering through a 10 micron filter;[WAC 173-303-815(2)(b)(ii)]

III.3.B.4 The Permittees will maintain the physical structure of the LERF and 200 Area ETF as documented in the applicable sections of Permit Addendum C, Section C.2. [WAC 173-303-630(7), WAC 173-303-640(3), WAC 173-303-640(4)]

III.3.B.5 The Permittees are authorized to use treated effluent for recycle/makeup water purposes at the 200 Area ETF as outlined in Permit Addendum C, Section C.2.5.5, and the letters

dated August 19, 2005, EPA Region 10 to Keith A. Klein; and August 8, 2005, Department of Ecology to Keith A. Klein. [WAC 173-303-815 (2)(b)(ii)]

III.3.B.6 The Permittees will maintain and operate systems for the 200 Area ETF documented in Permit Addendum C, Section C.2.5 as necessary for proper operation of the 200 Area ETF, compliance with the conditions of this Permit, and protection of human health and the environment. For purposes of this Permit Condition, the Monitor and Control System documented in Permit Addendum C, Section C.2.5.1, is considered to include all indicators, sensors, transducers, actuators and other control devices connected to but remote from the centralized monitor and control system (MCS) computer.

III.3.B.7 The Permittees must complete the following requirements prior to acceptance for treatment in 200 Area ETF aqueous waste streams with listed waste numbers subject to the requirements of the State and Federal delisting: [WAC 173-303-815(2)(b)(ii)]

III.3.B.7.a The Permittees will prepare a written waste processing strategy according to the requirements of the State and Federal Delisting Actions Conditions (1)(a)(ii) and (1)(b), incorporated by reference, and Permit Addendum B, Section B.2.2.2.

III.3.B.7.b The waste processing strategy required by Permit Condition III.3.B.7.a, must document the proposed processing configuration for the 200 Area ETF, operating conditions for each processing unit, and the expected treated effluent characteristics based on the process model and treatability envelope data required by State and Federal Delisting Conditions (1)(a)(ii) and (1)(b).

III.3.B.7.c The written waste processing strategy required by Permit Condition III.3.B.7.a must demonstrate that the projected treated effluent characteristics satisfy the delisting exclusion limits in State and Federal Delisting Condition (5) of the state and federal delisting actions, and the discharge limits of the State Discharge Permit ST-4500.

III.3.B.7.d The Permittees will place a copy of the written waste processing strategy required by Permit Condition III.3.B.7.a in the Hanford Facility Operating Record, LERF and 200 Area ETF file as part of the documentation of waste streams accepted for management at the 200 Area ETF.

III.3.B.8 Treatment of aqueous waste streams in the 200 Area ETF with listed waste numbers that are subject to the requirements of the state and federal delisting actions must comply with the requirements of State and Federal Delisting Condition (1)(c), incorporated by reference. [WAC 173-303-815 (2)(b)(ii)]

III.3.B.9 The Permittees will manage treated effluent in the final verification tanks according to the requirements of the State and Federal Delisting Conditions (3) and (5), incorporated by reference. [WAC 173-303-815 (2)(b)(ii)]

III.3.B.10 The Permittees will manage treated effluent from the 200 Area ETF according to the requirements of the State Waste Discharge Permit ST 4500 and State and Federal Delisting Condition (7). [WAC 173-303-815(2)(b)(ii)]

III.3.B.11 The Permittees will ensure compliance with treatment standards (40 CFR 268, incorporated by reference by WAC 173-303-140) applicable to treated effluent prior to discharge to the State Authorized Land Disposal Site (SALDS), the delisting criteria at 40 CFR 261, Appendix IX, Table 2, and the corresponding state-approved delisting (dated August 8, 2005, all incorporated by reference). Sampling and analysis necessary for these demonstrations must meet the corresponding requirements in Permit Addendum B. [WAC 173-303-140, WAC 173-303-815 (2)(b)(ii)]

III.3.C	**WASTE ANALYSIS**	
III.3.C.1	The Permittees will comply with requirements in Permit Addendum B for sampling and analysis of all dangerous and/or mixed waste required by conditions in this Chapter. [WAC 173-303-300]
III.3.C.2	The Permittees will have an accurate and complete waste profile as described in Permit Addendum B, Section B.2.1.2, for every waste stream accepted for management in LERF and 200 Area ETF dangerous waste management units. [WAC 173-303-380 (1)(a), (b)]
III.3.C.3	The Permittees will place a copy of each waste profile required by Permit Condition III.15.C.2 in the Hanford Facility Operating Record, LERF and 200 Area ETF file required by Permit Condition II.I.1.j. [WAC 173-303-380 (1)(a), (b)]
III.3.C.4	The Permittees will make a copy of the waste profile required by Permit Condition III.15.C.2 available upon request. [WAC 173-303-380 (1)(a), (b)]
III.3.C.5	Records and results of waste analysis described in this Permit will be maintained in the Hanford Facility Operating Record, LERF and 200 Area ETF file required by Permit Condition II.I.1.b. [WAC 173-303-380 (1)(a), (b)]
III.3.D	**RECORDKEEPING AND REPORTING**	
III.3.D.1	The Permittees will place the following into the Hanford Facility Operating Record, LERF and 200 Area ETF file required by Permit Condition II.I.1:
III.3.D.1.a	Records required by WAC 173-303-380 (1)(k), and -(o) incorporated by reference.
III.3.D.1.b	Records and results of waste analysis, waste determinations (as required by Subpart CC) and trial tests required by WAC 173-303-300, General waste analysis, and by 40 CFR §264.1034,§264.1063, §264.1083, §265.1034, §265.1063, §265.1084, §268.4(a), and §268.7; [WAC 173-303-310(2)]
III.3.D.1.c	An inspection log, summarizing inspections conducted pursuant to Permit Condition III.3.H.1; [WAC 173-303-380(1)(e)]
III.3.D.1.d	Records required by the State and Federal Delisting Condition (6), incorporated by reference; [WAC 173-303-815 (2)(b)(ii)]
III.3.E	**SECURITY**	
III.3.E.1	The Permittees comply with the Security requirements specific to the LERF and 200 Area ETF in Addendum E and Permit Attachment 3 as required by Permit Condition II.M. [WAC 173-303-310(2)]
III.3.F	**PREPAREDNESS AND PREVENTION**	
III.3.F.1	The Permittees will comply with the Preparedness and Prevention requirements specific to LERF and 200 Area ETF in Addendum F. [WAC 173-303-340]
III.3.G	**CONTINGENCY PLAN**	
III.3.G.1	The Permittees will comply with Addendum J, Contingency Plan, in addition to the requirements of Permit Condition II.A when applicable. [WAC 173-303-350]
III.3.H	**INSPECTIONS**	
III.3.H.1	The Permittees will comply with Addendum I in addition to the requirements of Permit Condition II.X. [WAC 173-303-320]

III.3.I TRAINING PLAN

III.3.I.1 The Permittees will include the training requirements described in Addendum G of this Chapter specific to the dangerous waste management units and waste management activities at LERF and 200 Area ETF into the written training plan required by Permit Condition II.C.

III.3.J GENERAL REQUIREMENTS

III.3.J.1 The Permittees will comply with the requirements of WAC 173-303-395(1), incorporated by reference, for prevention of reaction of ignitable, reactive, or incompatible wastes.

III.3.K CLOSURE

III.3.K.1 The Permittees will close dangerous waste management units in the LERF and 200 Area ETF in accordance with Addendum H, Closure Plan, and Permit Condition II.J. [WAC 173-303-610(3)(a)]

III.3.L POST CLOSURE – RESERVED

III.3.M CRITICAL SYSTEMS – RESERVED

III.3.N RESERVED

III.3.O CONTAINERS

III.3.O.1 Container Storage and Treatment Unit Standards

III.3.O.1.a As part of or in addition to the requirements of Permit Condition III.3.B.2, the Permittees will ensure the integrity of container storage secondary containment and the chemically resistant coating described in Addendum C, Section C.3.4.1 as necessary to ensure any spills or releases to secondary containment do not migrate to the underlying concrete or soils.

III.3.O.1.a.1 Include documentation of any damage and subsequent repairs in the Hanford Facility Operating Record, LERF and 200 Area ETF file required by Permit Condition II.1.I.

III.3.O.2 Container Management Standards

III.3.O.2.a The Permittees will maintain and manage wastes in accordance with the requirements of Addendum C, Section 4.3.2, and Section 4.3.2. [WAC 173-303-630(2)]

III.3.O.2.b The Permittees will label containers in accordance with the requirements of Addendum C, Section C.3.2, and Section C.3.3. [WAC 173-303-630(3)]

III.3.O.2.c The Permittees will comply with the requirements for managing wastes in containers in WAC 173-303-630(5), incorporated by reference.

III.3.O.2.d The Permittees will ensure wastes are compatible with containers and with other wastes stored or treated in containers within the 200 Area ETF according to the requirements of Addendum C, Section C.3.4.3. [WAC 173-303-630(4), WAC 173-303-630(9)]

III.3.O.2.e The Permittees may treat wastes in containers via decanting of free liquids and addition of sorbents. The Permittees may not use addition of sorbents for purposes of changing the treatability group of a waste with respect to the land disposal restriction standards of 40 CFR 268, incorporated by reference by WAC 173-303-140.

III.3.O.2.f The Permittees will remove any accumulated liquids from container storage areas in 200 Area ETF according to the requirements of Addendum C, Section C.3.4.2, to ensure containers are not in contact with free liquids and to prevent overflow of the container storage area secondary containment.

III.3.O.2.g The Permittees will comply with the requirements for air emissions from containers in Addendum C, Section C.6.3.2. [WAC 173-303-692]

III.3.P TANK SYSTEMS

III.3.P.1 Tank System Requirements

III.3.P.1.a The Permittees will develop a schedule for conducting integrity assessments (IA). The schedule will meet the requirements of Addendum C, Section C.4.2, and consideration of the factors in WAC 173-303-640(2)(e) or WAC 173-303-640(3)(b) as applicable:

III.3.P.1.b The Permittees will maintain a copy of the schedule required by Permit Condition III.3.P.1.a, in the Hanford Facility Operating Record, LERF and 200 Area ETF file, and conduct periodic integrity assessments according to the schedule. The Permittees will document results of integrity assessments conducted according to the IA in the Hanford Facility Operating Record, LERF and 200 Area ETF file.

III.3.P.1.c For existing tank systems, if a tank system is found to be leaking, or is unfit for use, the Permittees must follow the requirements of WAC 173-303-640(7), incorporated by reference. [WAC 173-303-640(3)(b)]

III.3.P.2 Tank System Operating Requirements

III.3.P.2.a The Permittees will comply with the requirements of WAC 173-303-640(5)(a), incorporated by reference.

III.3.P.2.b The Permittees will comply with the requirements of Addendum C, Section C.4.5.2. [WAC 173-303-640(5)(b)]

III.3.P.2.c The Permittees will comply with the requirements of Addendum C, Section C.4.6. [WAC 173-303-640(5)(d)]

III.3.P.2.d The Permittees will comply with the requirements of WAC 173-303-640(7), incorporated by reference, in response to spills or leaks from tanks systems at 200 Area ETF. [WAC 173-303-640(5)(c)]

III.3.P.2.e The Permittees will ensure that the Waste Processing Strategy required by Permit Condition III.3.B.7.a, provides for the immediate treatment or blending of waste accepted for management at the 200 Area ETF such that the resulting waste or mixture is no longer reactive or ignitable when further managed in 200 Area ETF tank systems. [WAC 173-303-640(9)]

III.3.P.2.f The Permittees will comply with the requirements of WAC 173-303-640(10), incorporated by reference.

III.3.Q SURFACE IMPOUNDMENTS

III.3.Q.1 The Permittees will maintain the three LERF basins according to the requirements of WAC 173-303-650 (2)(f), incorporated by reference.

III.3.Q.2 The Permittees will operate the LERF basins according to the requirements of Addendum C, Section C.5.3, and Addendum I, Section I.2.2.3.1 to prevent over-topping. [WAC 173-303-650 (2)(c)]

III.3.Q.3 The Permittees will develop and maintain, and operate the LERF basins to ensure that any flow of waste into the impoundment can be immediately shut off in the event of overtopping or liner failure. [WAC 173-303-650 (2)(d)]

III.3.Q.4 The Permittees will comply with the requirements of WAC 173-303-650 (2)(g), incorporated by reference.

III.3.Q.5 The Permittees will comply with the requirements of WAC 173-303-650 (4)(b), incorporated by reference.

III.3.Q.6 The Permittees will comply with the requirements of WAC 173-303-650 (4)(c), incorporated by reference. The certification required by this Permit Condition must be provided to Ecology no later than seven calendar days after the date of the certification. A copy of the certification will be placed in the Hanford Facility Operating Record, LERF and 200 Area ETF file required by Permit Condition II.I.1. [WAC 173-303-650 (4)(c)]

III.3.Q.7 The Permittees will comply with the requirements of WAC 173-303-650(5)(b), incorporated by reference, in response to events in WAC 173-303-650(5)(a), incorporated by reference.

III.3.Q.8 The Permittees will comply with the requirements of WAC 173-303-650(5)(d) for any LERF basin that has been removed from service in accordance with Permit Condition III.3.Q.7 that the Permittees will restore to service. [WAC 173-303-650(5)(d)]

III.3.Q.9 The Permittees will close any LERF basin removed from service in accordance with the requirements of Permit Condition III.3.Q.7 or a basin that cannot be repaired or that the Permittees will not to return to service. [WAC 173-303-650(5)(e)]

III.3.Q.10 The Permittees will comply with the requirements of Addendum C, Section C.5.10 with respect to management of ignitable or reactive wastes in the LERF basins. [WAC 173-303-650(7)]

III.3.Q.11 The Permittees can place incompatible wastes and materials in the same LERF basin only if in compliance with the requirements of WAC 173-303-395(1)(b), (c). [WAC 173-303-650(8)]

III.3.Q.12 The Permittees will use the action leakage rate in Addendum C, Section C.5.8, for operation of LERF basins, and comply with the requirements of WAC 173-303-650(10)(b). [WAC 173-303-650(10)]

III.3.Q.13 The Permittees will comply with the requirements of WAC 173-303-650(11), incorporated by reference.

III.3.Q.14 The Permittees will comply with the requirements of 40 CFR 264, Subpart CC, incorporated by reference by WAC 173-303-692.

III.3.R GROUNDWATER

III.3.R.1 The Permittees will comply with the requirements of Addendum D, Groundwater Monitoring Plan. [WAC 173-303-645]

III.3.R.2 All wells constructed pursuant to this Permit will be constructed in compliance with Chapter 173-160 WAC incorporated by reference through WAC 173-303-645 (8)(c).

III.3.R.3 Update the Groundwater Monitoring Network

III.3.R.3.a The Permittees will install an additional downgradient monitoring well E-26-15 as identified in Addendum D, Groundwater Monitoring Plan by December, 2016.

III.3.R.3.b Within 60-days of the well installation, the Permittees will submit a Class 2 Permit modification [WAC 173-303-830 Appendix I, C.1.a] to update Addendum D and include the additional monitoring well into the groundwater monitoring network.

III.3.R.3.c Concurrently with the permit modification request, the Permittees will submit a revised "Liquid Effluent Retention Facility Characterization Report" for the additional monitoring well that includes:

1) Well construction in accordance with WAC 173-303-645(8)(c)
2) Well screen placement in the upper aquifer in accordance with WAC 173-303-645(8)(a)
3) Hydrogeologic conditions, stratigraphy and hydraulic conductivity, derived from geologist observations of borehole archive samples, down hole gamma logging, and aquifer slug tests in accordance with WAC 173-303-645(8)(a)(i)(A)
4) Drilling and sampling details in accordance with WAC 173-303-645(8)(d)
5) Borehole corrections (e.g., precision surveys, gyroscopic corrections, and barometric response corrections) to ensure adequate hydraulic understanding considering the very small gradient in accordance with WAC 173-303-645(8)(f)
6) Geochemical comparison of the water quality with other existing wells to ensure anticipated representative conditions in accordance with WAC 173-303-645(8)(a)(ii)
7) Document surface location as required by WAC 173-303-645(6)

III.3.R.3.c.1 Groundwater sample results from the new well (E-26-15) and the existing wells for all constituents in the Addendum D, Groundwater Monitoring Plan for the Liquid Effluent Retention Facility,

III.3.R.3.c.2 Results of evaluating final well development data and drilling logs,

III.3.R.3.c.2.a A well use designation (e.g., upgradient or downgradient).

WASHINGTON STATE DEPARTMENT OF ECOLOGY

Addendum A Part A Form

Date Received	Reviewed by: ER Skinnland	Date: 03/31/2015
Month Day Year	Approved by: PK Skinnland for Suzanne Dahl	Date: 03/31/2015

I. This form is submitted to: (place an "X" in the appropriate box)

☒	Request modification to a final status permit (commonly called a "Part B" permit)
☐	Request a change under interim status
☐	Apply for a final status permit. This includes the application for the initial final status permit for a site or for a permit renewal (i.e., a new permit to replace an expiring permit).
☐	Establish interim status because of the wastes newly regulated on: (Date)
	List waste codes:

II. EPA/State ID Number

WA7890008967

III. Name of Facility

US Department of Energy – Hanford Facility

IV. Facility Location (Physical address not P.O. Box or Route Number)

A. Street

825 Jadwin

City or Town	State	ZIP Code
Richland	WA	99352

County Code (if	County Name
005	Benton

B. Land Type	C. Geographic Location Latitude (degrees, mins, secs)	Longitude (degrees, mins, secs)	D. Facility Existence Date Month Day Year
F	Refer to TOPO Map (Section XV.)		03 02 1943

V. Facility Mailing Address

Street or P.O. Box

P.O. Box 450

City or Town	State	ZIP Code
Richland	WA	99352

VI. Facility contact (Person to be contacted regarding waste activities at facility)		
Name (last)	**(first)**	
Smith	Kevin	
Job Title	**Phone Number (area code and number)**	
Manager	(509) 372-2315	
Contact Address		
Street or P.O. Box		
P.O. Box 450		
City or Town	**State**	**ZIP Code**
Richland	WA	99352

VII. Facility Operator Information	
A. Name	**Phone Number**
Department of Energy Owner/Operator Washington River Protection Solutions, LLC Co-Operator for LERF & 200 Area ETF*	(509) 372-2315 (509) 372-9138*
Street or P.O. Box	
P.O. Box 450 P.O. Box 850 *	

City or Town	State	ZIP Code
Richland	WA	99352

B. Operator Type	F

C. Does the name in VII.A reflect a proposed change in operator?	☐ Yes ☒ No Co-Operator* change
If yes, provide the scheduled date for the change:	Month: 03 Day: 30 Year: 2015

D. Is the name listed in VII.A. also the owner? If yes, skip to Section VIII.C.	☐ Yes ☒ No

VIII. Facility Owner Information	
A. Name	**Phone Number (area code and number)**
Kevin W. Smith, Operator/Facility-Property Owner	(509) 372-2315
Street or P.O. Box	
P.O. Box 450	

City or Town	State	ZIP Code
Richland	WA	99352

B. Owner Type	F

C. Does the name in VIII.A reflect a proposed change in owner?	☐ Yes ☒ No
If yes, provide the scheduled date for the change:	Month Day Year

IX. NAICS Codes (5/6 digit codes)													
A. First							**B. Second**						
5	6	2	2	1		Waste Treatment & Disposal	9	2	4	1	1	0	Administration of Air & Water Resource & Solid Waste Management Programs
C. Third							**D. Fourth**						
5	4	1	7	1		Research & Development in the Physical, Engineering, & Life Sciences							

X. Other Environmental Permits (see instructions)

A. Permit Type	B. Permit Number	C. Description
E	TSCA0-3-102	TSCA approval, 40 CFR 761
E	WCM-127	40 CFR 761.61(c), TSCA risk-based approval 2003-10-22
E	AOP00-0-05-006	Title V Air Operating Permit
E	DE07NW-P-003	WAC 173-400 & -460 Criteria & Toxics approval
E	FF-01-146	WAC 246-247 Radioactive Air Emissions approval, LERF Basin 44, EU 146, AIR 12-305
E	FF-01-147	WAC 246-247 Radioactive Air Emissions approval, LERF Basin 43, EU 147, AIR 12-305
E	FF-01-148	WAC 246-247 Radioactive Air Emissions approval, LERF Basin 42, EU 148, AIR 12-305
E	FF-01-301	WAC 246-247 Radioactive Air Emissions approval, ETF, EU 301, AIR 12-305
U	ST 4500	WAC 173-216, State Waste Discharge Permit for the 200 Area Effluent Treatment Facility State-Approved Land Disposal Site
U	ST 4511	WAC 173-216, State Waste Discharge Permit Program, Sitewide Permit for miscellaneous streams

XI. Nature of Business (provide a brief description that includes both dangerous waste and non-dangerous waste areas and activities)

Construction of the Liquid Effluent Retention Facility (LERF) began in 1990. Waste management operations began at LERF in April 1994. Construction of the 200 Area Effluent Treatment Facility (ETF) began in 1992. Waste management operations began at ETF in November of 1995.

The LERF and ETF comprise an aqueous waste treatment system located in the 200 East Area that provides storage and treatment for a variety of aqueous mixed waste. This aqueous waste includes process condensate from the 242-A Evaporator and other aqueous waste generated from onsite remediation and waste management activities.

The LERF consists of three lined surface impoundments, or basins. Aqueous waste from LERF is pumped to the ETF for treatment in a series of process units, or systems, that remove or destroy dangerous waste constituents. The treated effluent is discharged to a State-Approved Land Disposal Site (SALDS) north of the 200 West Area, under the authority of a Washington State Waste Discharge Permit (ST4500) and the Final Delisting (40 CFR 261, Appendix IX, Table 2)

Sludge that accumulates in the bottoms of ETF process tanks is removed periodically and placed into containers. The waste is solidified by decanting the supernate from the container and the remainder of the liquid is allowed to evaporate, or absorbents are added, as necessary, to address the residual liquid. The process design capacity for treatment of waste in containers (T04) is 18,927 liters per day.

EXAMPLE FOR COMPLETING ITEMS XII and XIII (shown in lines numbered X-1, X-2, and X-3 below): A facility has two storage tanks that hold 1200 gallons and 400 gallons respectively. There is also treatment in tanks at 20 gallons/hr. Finally, a one-quarter acre area that is two meters deep will undergo *in situ vitrification.*

Section XII. Process Codes and Design Capacities								Section XIII. Other Process Codes								
Line Number		A. Process Codes (enter code)			B. Process Design Capacity		C. Process Total Number of Units	Line Number		A. Process Codes (enter code)			B. Process Design Capacity		C. Process Total Number of Units	D. Process Description
					1. Amount	2. Unit of Measure (enter code)							1. Amount	2. Unit of Measure (enter code)		
X	1	S	0	2	1,600	G	002	X	1	T	0	4	700	C	001	In situ vitrification
X	2	T	0	3	20	E	001									
X	3	T	0	4	700	C	001									
	1	S	0	4	88,500,000	L	003		1	T	0	4	18,927	V	001	container treatment
	2	T	0	2	88,500,000	V	003		2							
	3	S	0	2	9,849,350	L	019		3							
	4	T	0	1	817,646	V	019		4							
	5	S	0	1	147,630	L	003		5							
	6	T	0	4	18,927	V	001		6							
	7								7							
	8								8							
	9								9							
1	0							1	0							
1	1							1	1							
1	2							1	2							
1	3							1	3							
1	4							1	4							
1	5							1	5							
1	6							1	6							
1	7							1	7							
1	8							1	8							
1	9							1	9							
2	0							2	0							
2	1							2	1							
2	2							2	2							
2	3							2	3							
2	4							2	4							
2	5							2	5							

XIV. Description of Dangerous Wastes

Example for completing this section: A facility will receive three non-listed wastes, then store and treat them on-site. Two wastes are corrosive only, with the facility receiving and storing the wastes in containers. There will be about 200 pounds per year of each of these two wastes, which will be neutralized in a tank. The other waste is corrosive and ignitable and will be neutralized then blended into hazardous waste fuel. There will be about 100 pounds per year of that waste, which will be received in bulk and put into tanks.

Line Number	A. Dangerous Waste No. (enter code)	B. Estimated Annual Quantity of Waste	C. Unit of Measure (enter code)	D. Processes (1) Process Codes (enter)	D. Processes (2) Process Description [If a code is not entered in D (1)]
X 1	D 0 0 2	400	P	S 0 1 T 0 1	
X 2	D 0 0 1	100	P	S 0 2 T 0 1	
X 3	D 0 0 2				Included with above
1	D 0 0 1	106,940,410	K	S 0 4 T 0 2	
2	D 0 0 2		K	S 0 4 T 0 2	
3	D 0 0 3		K	S 0 4 T 0 2	
4	D 0 0 4		K	S 0 4 T 0 2	
5	D 0 0 5		K	S 0 4 T 0 2	
6	D 0 0 6		K	S 0 4 T 0 2	
7	D 0 0 7		K	S 0 4 T 0 2	
8	D 0 0 8		K	S 0 4 T 0 2	
9	D 0 0 9		K	S 0 4 T 0 2	
10	D 0 1 0		K	S 0 4 T 0 2	
11	D 0 1 1		K	S 0 4 T 0 2	
12	D 0 1 8		K	S 0 4 T 0 2	
13	D 0 1 9		K	S 0 4 T 0 2	
14	D 0 2 2		K	S 0 4 T 0 2	
15	D 0 2 8		K	S 0 4 T 0 2	
16	D 0 2 9		K	S 0 4 T 0 2	
17	D 0 3 0		K	S 0 4 T 0 2	
18	D 0 3 3		K	S 0 4 T 0 2	
19	D 0 3 4		K	S 0 4 T 0 2	
20	D 0 3 5		K	S 0 4 T 0 2	
21	D 0 3 6		K	S 0 4 T 0 2	
22	D 0 3 8		K	S 0 4 T 0 2	
23	D 0 3 9		K	S 0 4 T 0 2	
24	D 0 4 0		K	S 0 4 T 0 2	
25	D 0 4 1		K	S 0 4 T 0 2	

EPA/State ID Number	W	A	7	8	9	0	0	0	8	9	6	7

Continuation of Section XIV. Description of Dangerous Waste

Line Number	A. Dangerous Waste No. (enter code)	B. Estimated Annual Quantity of Waste	C. Unit of Measure (enter code)	D. Process	
				(1) Process Codes (enter)	(2) Process Description [If a code is not entered in D (1)]
26	D 0 4 3		K	S 0 4 T 0 2	
27	F 0 0 1		K	S 0 4 T 0 2	
28	F 0 0 2		K	S 0 4 T 0 2	
29	F 0 0 3		K	S 0 4 T 0 2	
30	F 0 0 4		K	S 0 4 T 0 2	
31	F 0 0 5		K	S 0 4 T 0 2	
32	F 0 3 9		K	S 0 4 T 0 2	
33	W T 0 1		K	S 0 4 T 0 2	
34	W T 0 2		K	S 0 4 T 0 2	
35	U 2 1 0		K	S 0 4 T 0 2	
36	D 0 0 1	106,940,410	K	T 0 1	
37	D 0 0 2		K	T 0 1	
38	D 0 0 3		K	T 0 1	
39	D 0 0 4		K	T 0 1	
40	D 0 0 5		K	T 0 1	
41	D 0 0 6		K	T 0 1	
42	D 0 0 7		K	T 0 1	
43	D 0 0 8		K	T 0 1	
44	D 0 0 9		K	T 0 1	
45	D 0 1 0		K	T 0 1	
46	D 0 1 1		K	T 0 1	
47	D 0 1 8		K	T 0 1	
48	D 0 1 9		K	T 0 1	
49	D 0 2 2		K	T 0 1	
50	D 0 2 8		K	T 0 1	
51	D 0 2 9		K	T 0 1	
52	D 0 3 0		K	T 0 1	
53	D 0 3 3		K	T 0 1	
54	D 0 3 4		K	T 0 1	
55	D 0 3 5		K	T 0 1	

56	D	0	3	6		K	T	0	1							
57	D	0	3	8		K	T	0	1							
58	D	0	3	9		K	T	0	1							
59	D	0	4	0		K	T	0	1							
60	D	0	4	1		K	T	0	1							
61	D	0	4	3		K	T	0	1							
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63	F	0	0	2		K	T	0	1							
64	F	0	0	3		K	T	0	1							
65	F	0	0	4		K	T	0	1							
66	F	0	0	5		K	T	0	1							
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68	W	T	0	1		K	T	0	1							
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72	D	0	0	2		K	S	0	2							
73	D	0	0	3		K	S	0	2							
74	D	0	0	4		K	S	0	2							
75	D	0	0	5		K	S	0	2							
76	D	0	0	6		K	S	0	2							
77	D	0	0	7		K	S	0	2							
78	D	0	0	8		K	S	0	2							
79	D	0	0	9		K	S	0	2							
80	D	0	1	0		K	S	0	2							
81	D	0	1	1		K	S	0	2							
82	D	0	1	8		K	S	0	2							
83	D	0	1	9		K	S	0	2							
84	D	0	2	2		K	S	0	2							
85	D	0	2	8		K	S	0	2							

86	D	0	2	9		K	S	0	2						
87	D	0	3	0		K	S	0	2						
88	D	0	3	3		K	S	0	2						
89	D	0	3	4		K	S	0	2						
90	D	0	3	5		K	S	0	2						
91	D	0	3	6		K	S	0	2						
92	D	0	3	8		K	S	0	2						
93	D	0	3	9		K	S	0	2						
94	D	0	4	0		K	S	0	2						
95	D	0	4	1		K	S	0	2						
96	D	0	4	3		K	S	0	2						
97	F	0	0	1		K	S	0	2						
98	F	0	0	2		K	S	0	2						
99	F	0	0	3		K	S	0	2						
100	F	0	0	4		K	S	0	2						
101	F	0	0	5		K	S	0	2						
102	F	0	3	9		K	S	0	2						
103	W	T	0	1		K	S	0	2						
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105	U	2	1	0		K	S	0	2						
106	D	0	0	1	153,932	K	S	0	1						Includes Debris
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111	D	0	0	6		K	S	0	1						Includes Debris
112	D	0	0	7		K	S	0	1						Includes Debris
113	D	0	0	8		K	S	0	1						Includes Debris
114	D	0	0	9		K	S	0	1						Includes Debris
115	D	0	1	0		K	S	0	1						Includes Debris
116	D	0	1	1		K	S	0	1						Includes Debris

117	D	0	1	8		K	S	0	1						Includes Debris
118	D	0	1	9		K	S	0	1						Includes Debris
119	D	0	2	2		K	S	0	1						Includes Debris
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122	D	0	3	0		K	S	0	1						Includes Debris
123	D	0	3	3		K	S	0	1						Includes Debris
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137	F	0	3	9		K	S	0	1						Includes Debris
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148	D	0	0	8		K	T	0	4						Includes Debris

149	D	0	0	9		K	T	0	4						Includes Debris
150	D	0	1	0		K	T	0	4						Includes Debris
151	D	0	1	1		K	T	0	4						Includes Debris
152	D	0	1	8		K	T	0	4						Includes Debris
153	D	0	1	9		K	T	0	4						Includes Debris
154	D	0	2	2		K	T	0	4						Includes Debris
155	D	0	2	8		K	T	0	4						Includes Debris
156	D	0	2	9		K	T	0	4						Includes Debris
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166	D	0	4	3		K	T	0	4						Includes Debris
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171	F	0	0	5		K	T	0	4						Includes Debris
172	F	0	3	9		K	T	0	4						Includes Debris
173	W	T	0	1		K	T	0	4						Includes Debris
174	W	T	0	2		K	T	0	4						Includes Debris
175	U	2	1	0		K	T	0	4						Includes Debris
176															
177															
178															
179															
180															

181															
182															
183															
184															

XV. Map

Attach to this application a topographic map of the area extending to at least one (1) mile beyond property boundaries. The map must show the outline of the facility; the location of each of its existing and proposed intake and discharge structures; each of its dangerous waste treatment, storage, recycling, or disposal units; and each well where fluids are injected underground. Include all springs, rivers, and other surface water bodies in this map area, plus drinking water wells listed in public records or otherwise known to the applicant within ¼ mile of the facility property boundary. The instructions provide additional information on meeting these requirements.

Topographic map is located in the Ecology Library

XVI. Facility Drawing

All existing facilities must include a scale drawing of the facility (refer to instructions for more detail).

XVII. Photographs

All existing facilities must include photographs (aerial or ground-level) that clearly delineate all existing structures; existing storage, treatment, recycling, and disposal areas; and sites of future storage, treatment, recycling, or disposal areas (refer to instructions for more detail).

XVIII. Certifications

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

Operator Name and Official Title (type or print) Kevin W. Smith, Manager U.S. Department of Energy Office of River Protection	**Signature**	**Date Signed** 1/15/15
Co-Operator* Name and Official Title (type or print) L. David Olson President and Project Manager Washington River Protection Solutions, LLC	**Signature** L D Olson	**Date Signed** 12/19/14
Co-Operator – Address and Telephone Number* P.O. Box 850 Richland, WA 99352 (509)372-9138		
Facility-Property Owner Name and Official Title (type or print) Kevin W. Smith, Manager U.S. Department of Energy Office of River Protection	**Signature**	**Date Signed** 1/15/15

Comments

Changes from Revision 3A to 4

Section V, "Facility Mailing Address"

- Updated P.O. Box from 550 to 450.

Section VI, "Facility Contact"

- Updated from "Matthew McCormick" to "Kevin Smith".
- Updated phone number of facility contact.
- Updated P.O. Box address of facility contact.

Section VII, "Facility Operator Information, A. Name"

- Updated Owner/Operator phone number.
- Updated Co-Operator name and phone number.
- Updated P.O. Box addresses for Owner and Co-Operator.

Section VII, "Facility Operator Information, C"

- Updated schedule for transition date to 03/30/2015.

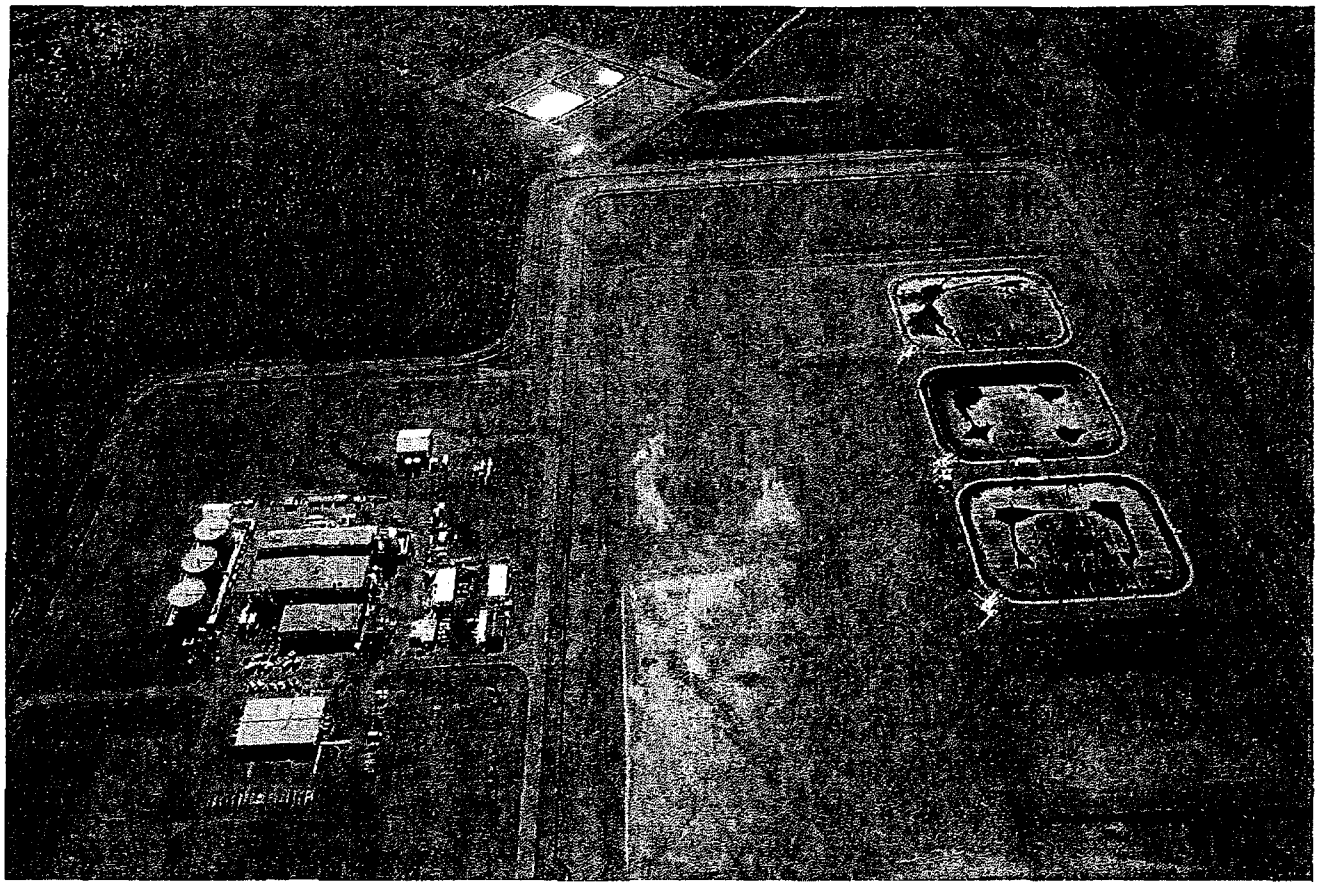
Section VIII, "Facility Owner Information, A. Name"

- Updated from "Matthew S. McCormick" to "Kevin W. Smith".
- Updated phone number of facility owner.
- Updated PO Box address of facility owner.

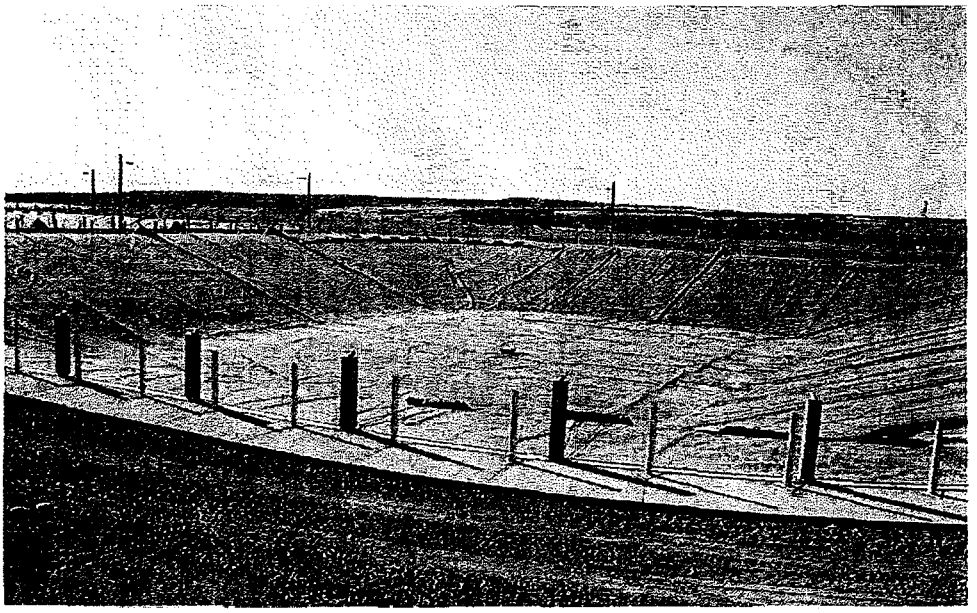
Section XVIII, "Certifications"

- Updated Operator from "Matthew S. McCormick" of "Richland Operations Office" to "Kevin W. Smith" of "Office of River Protection".
- Updated Co-Operator from "John C. Fulton" of "CH2M Hill Plateau Remediation Company" to "L. David Olson" of Washington River Protection Solutions, LLC".
- Updated Co-Operator title from "Chief Executive Officer" to "Project Manager".
- Updated Co-Operator P.O. Box address and phone number.
- Updated Facility-Property Owner from "Matthew S. McCormick" of "Richland Operations Office" to "Kevin W. Smith" of "Office of River Protection".

Liquid Effluid Effluent Retention Facility & 200 Area Effluent Treatment Facility



Liquid Effluent Retention Facility



Typical Basin Photo Taken 1992

200 Area Effluent Treatment Facility

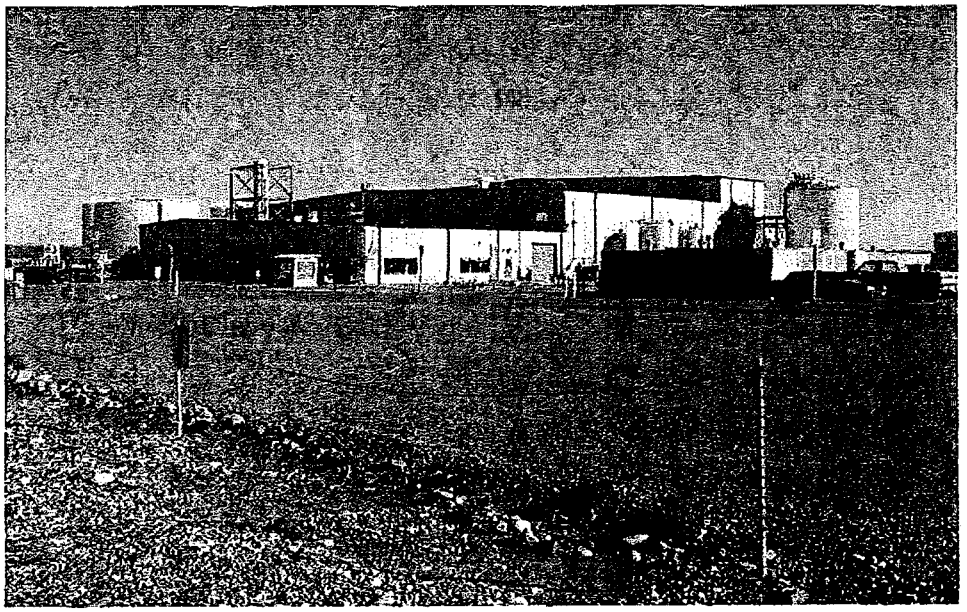
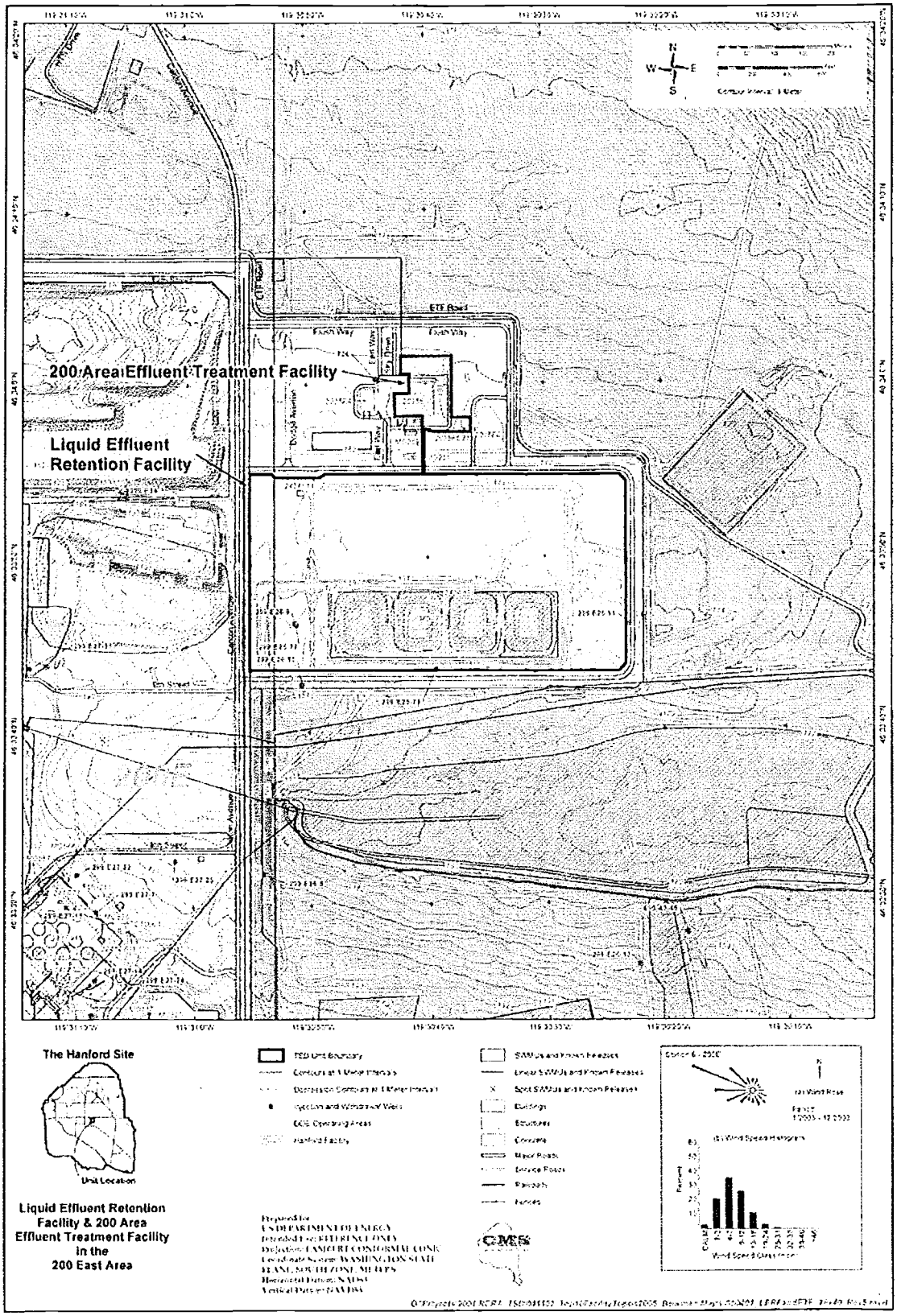
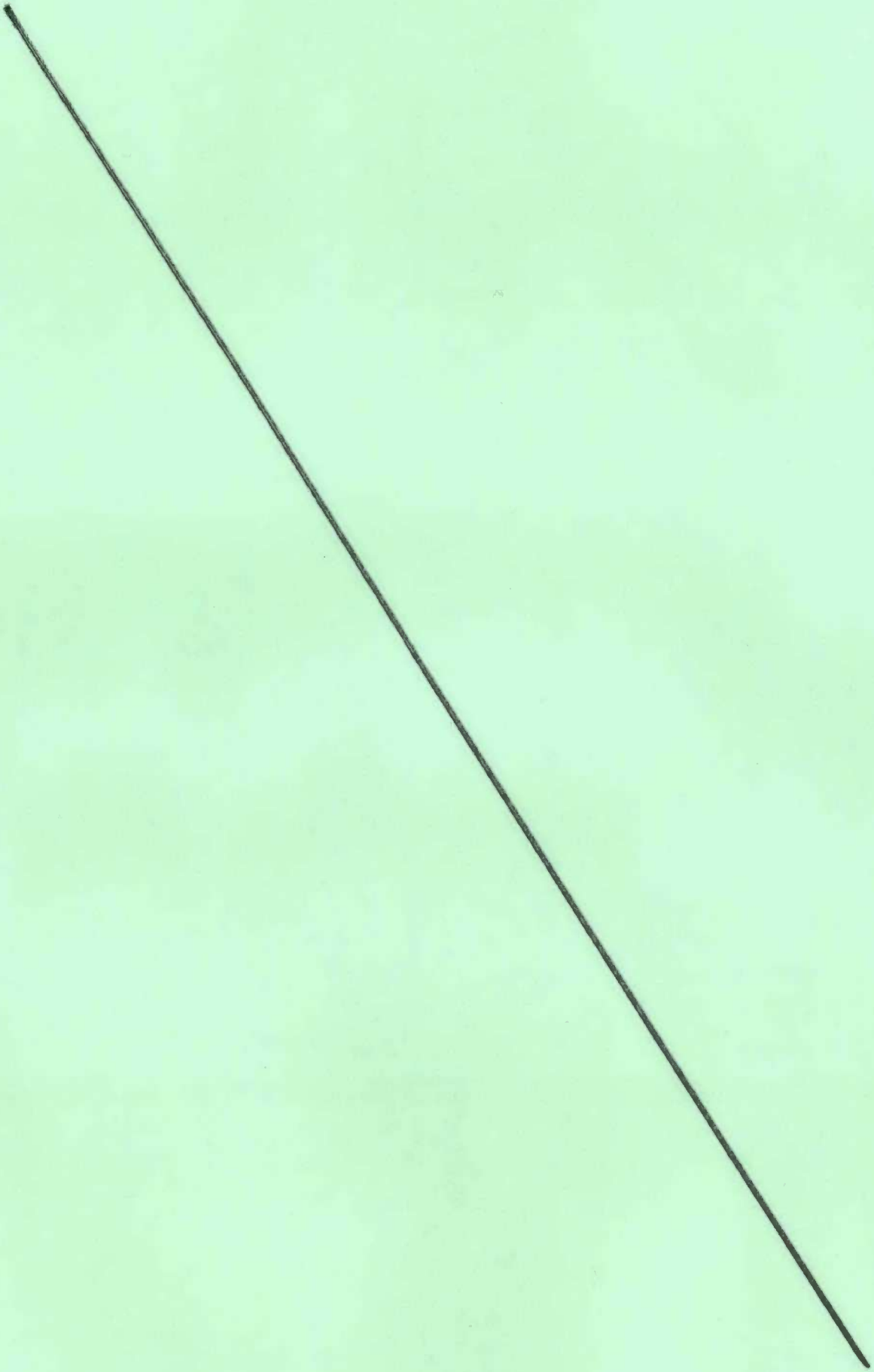


Photo Taken 2005


200 Area Effluent Treatment Facility
Liquid Effluent Retention Facility
ETF Road
The Hanford Site
Unit Location
Liquid Effluent Retention Facility & 200 Area Effluent Treatment Facility in the 200 East Area
N
W
E
S



ADDENDUM B
WASTE ANALYSIS PLAN

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ADDENDUM B
WASTE ANALYSIS PLAN

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B. WASTE ANALYSIS PLAN

Metric Conversion Chart

Into metric units			Out of metric units		
If you know	Multiply by	To get	If you know	Multiply by	To get
Length			Length		
inches	25.40	millimeters	millimeters	0.0393	inches
inches	2.54	centimeters	centimeters	0.393	inches
feet	0.3048	meters	meters	3.2808	feet
yards	0.914	meters	meters	1.09	yards
miles	1.609	kilometers	kilometers	0.62	miles
Area			Area		
square inches	6.4516	square centimeters	square centimeters	0.155	square inches
square feet	0.092	square meters	square meters	10.7639	square feet
square yards	0.836	square meters	square meters	1.20	square yards
square miles	2.59	square kilometers	square kilometers	0.39	square miles
acres	0.404	hectares	hectares	2.471	acres
Mass (weight)			Mass (weight)		
ounces	28.35	grams	grams	0.0352	ounces
pounds	0.453	kilograms	kilograms	2.2046	pounds
short ton	0.907	metric ton	metric ton	1.10	short ton
Volume			Volume		
fluid ounces	29.57	milliliters	milliliters	0.03	fluid ounces
quarts	0.95	liters	liters	1.057	quarts
gallons	3.79	liters	liters	0.26	gallons
cubic feet	0.03	cubic meters	cubic meters	35.3147	cubic feet
cubic yards	0.76456	cubic meters	cubic meters	1.308	cubic yards
Temperature			Temperature		
Fahrenheit	subtract 32 then multiply by 5/9ths	Celsius	Celsius	multiply by 9/5ths, then add 32	Fahrenheit
Force			Force		
pounds per square inch	6.895	kilopascals	kilopascals	1.4504×10^{-4}	pounds per square inch

Source: *Engineering Unit Conversions*, M. R. Lindeburg, P.E., Second Ed., 1990, Professional Publications, Inc., Belmont, California.

B.1 Introduction

In accordance with the regulations set forth in the Washington State Department of Ecology (Ecology) *Dangerous Waste Regulations*, Washington Administrative Code (WAC) 173-303-300, this waste analysis plan (WAP) has been prepared for operation of the Liquid Effluent Retention Facility (LERF) and the 200 Area Effluent Treatment Facility (200 Area ETF) located in the 200 East Area on the Hanford Site, Richland, Washington.

The purpose of this WAP is to ensure that adequate knowledge as defined in WAC 173-303-040, is obtained for dangerous and/or mixed waste accepted by and managed in LERF and 200 Area ETF. This WAP documents the sampling and analytical methods, and describes the procedures used to obtain this knowledge. This WAP also documents the requirements for generators sending aqueous waste to the LERF or 200 Area ETF for treatment. Throughout this WAP, the term generator includes any Hanford Site source, including treatment, storage, and disposal (TSD) units, whose process produces an aqueous waste.

LERF consists of three surface impoundments, which provide treatment and storage. The 200 Area ETF includes a tank system, which provides treatment and storage, and a container management area, which provides container storage and treatment. Additionally, this WAP discusses the sampling and analytical methods for the treated effluent (treated aqueous waste) that is discharged from 200 Area ETF as a non-dangerous, delisted waste to the State Approved Land Disposal Site (SALDS). Specifically, the WAP contains sampling and analysis requirements including quality assurance/quality control requirements, for the following:

- **Influent Waste Acceptance Process** - determines the acceptability of a particular aqueous waste at the LERF or 200 Area ETF pursuant to applicable Permit conditions, regulatory requirements, and operating capabilities prior to acceptance of the waste at the LERF or 200 Area ETF for treatment or storage. This includes documenting that wastes accepted for treatment at ETF are within the treatability envelope required by the Final Delisting 200 Area ETF, Permit Condition 1.a.i. Refer to Section B.2.
- **Special Management Requirements** - identifies the special management requirements for aqueous wastes managed in the LERF or 200 Area ETF. Refer to Section B.3.
- **Influent Aqueous Waste Sampling and Analysis** - describes influent sampling and analyses used to characterize an influent aqueous waste to ensure proper management of the waste and for compliance with the special management requirements. Also includes rationale for analyses. Refer to Section B.4.
- **Treated Effluent Sampling and Analysis** - describes sampling and analyses of treated effluent (i.e., treated aqueous waste) for compliance with Washington State Waste Discharge Permit, No. ST 4500 (Ecology 2000); and Final Delisting 200 Area ETF 40 CFR 261, Appendix IX, Table 2 and the corresponding State Final Delisting issued pursuant to WAC 173-303-910(3) limits. Also includes rationale for analyses. Refer to Section B.5.
- **200 Area ETF Generated Waste Sampling and Analysis** - describes the sampling and analyses used to characterize the secondary waste streams generated from the treatment process and to characterize waste generated from maintenance and operations activities. Also includes rationale for analyses. Characterization and designation of wastes generated from maintenance and operations activities are conducted pursuant to WAC 173-303-170 and are not subject to the permit requirements of WAC 173-303-800. These descriptions are included in this WAP for purposes of completeness, but are not enforceable conditions of this WAP or the permit. Refer to Section B.6.
- **Quality Assurance and Quality Control** - ensures the accuracy and precision of sampling and analysis activities. Refer to Section B.7.

This WAP meets the specific requirements of the following:

- Land Disposal Restrictions Treatment Exemption for the LERF under 40 CFR 268.4, U.S. Environmental Protection Agency (EPA), December 6, 1994 (EPA 1994).
- Final Delisting 200 Area ETF [40 CFR 261, Appendix IX, Table 2.
- Corresponding State Final Delisting issued pursuant to WAC 173-303-910(3).
- Washington State Waste Discharge Permit (No. ST 4500), as amended.
- Hanford Facility Dangerous Waste Permit (Permit) WA7890008967, as amended.

The Permit conditions of the Washington State Waste Discharge Permit (No. ST 4500) are included in this WAP for completeness, as well as generator requirements for designation of wastes generated by LERF and 200 Area ETF from operation and maintenance activities. The Washington State Waste Discharge Permit (No. ST 4500) Conditions are not within the scope of Resource Conservation and Recovery Act (RCRA) or WAC 173-303 or subject to the permit requirements of WAC 173-303-800. Therefore, revisions of this WAP that are not governed by the requirements of WAC 173-303 will not be considered as a modification subject to review or approval by Ecology. Any other revisions to this WAP will be incorporated through the Permit modification process as necessary to demonstrate compliance with requirements of this Permit, including Permit Conditions I.E.7 and I.E.8.

B.1.1 Liquid Effluent Retention Facility and Effluent Treatment Facility Description

The LERF and 200 Area ETF comprise an aqueous waste treatment system located in the 200 East Area. Both LERF and 200 Area ETF may receive aqueous waste through several inlets. 200 Area ETF can receive aqueous waste through three inlets. First, 200 Area ETF can receive aqueous waste directly from the LERF. Second, aqueous waste can be transferred from the Load-in Station to 200 Area ETF. Third, aqueous waste can be transferred from containers (e.g., carboys, drums) to the 200 Area ETF through either the Secondary Waste Receiving Tanks or the Concentrate Tanks. The Load-in Station is located just east of 200 Area ETF and currently consists of three storage tanks and a pipeline that connects to either LERF or 200 Area ETF through fiberglass pipelines with secondary containment.

The LERF can receive aqueous waste through four inlets. First, aqueous waste can be transferred to LERF through a dedicated pipeline from the 200 West Area. Second, aqueous waste can be transferred through a pipeline that connects LERF with the 242-A Evaporator. Third, aqueous waste also can be transferred to LERF from a pipeline that connects LERF to the Load-in Station at 200 Area ETF. Finally, aqueous waste can be transferred into LERF through a series of sample ports located at each basin.

The LERF consists of three lined surface impoundments with a nominal capacity of 29.5 million liters each. Aqueous waste from LERF is pumped to 200 Area ETF through a double walled fiberglass pipeline. The pipeline is equipped with leak detection located in the annulus between the inner and outer pipes. Each basin is equipped with six available sample risers constructed of 6-inch-perforated pipe. A seventh sample riser in each basin is dedicated to influent waste receipt piping, and an eighth riser in each basin contains liquid level instrumentation. Each riser extends along the sides of each basin from the top to the bottom of the basin. Detailed information on the construction and operation of the LERF is provided in Addendum C, Process Information.

200 Area ETF is designed to treat the contaminants anticipated in process condensate from the 242-A Evaporator and other aqueous wastes from the Hanford Site. Section B.1.2 provides more information on the sources of these wastes.

The capabilities of 200 Area ETF were confirmed through pilot plant testing. A pilot plant was used to test surrogate solutions that contained constituents of concern anticipated in aqueous wastes on the Hanford Site. The pilot plant testing served as the basis for a demonstration of the treatment capabilities of 200 Area ETF in the *200 Area Effluent Treatment Facility Delisting Petition* (DOE/RL-92-72).

200 Area ETF consists of a primary and a secondary treatment train (Figure B.1). The primary treatment train removes or destroys dangerous and mixed waste components from the aqueous waste.

In the secondary treatment train, the waste components are concentrated and dried into a powder. This waste is containerized, and transferred to a waste treatment, storage, and/or disposal (TSD) unit.

Each treatment train consists of a series of operations. The primary treatment train includes the following:

- Surge tank
- Filtration
- Ultraviolet light oxidation (UV/OX)
- pH adjustment
- Hydrogen peroxide decomposition
- Degasification
- Reverse osmosis (RO)
- Ion exchange
- Final pH adjustment and verification

The secondary treatment train uses the following:

- Secondary waste receiving
- Evaporation (with mechanical vapor recompression)
- Concentrate staging
- Thin film drying
- Container handling
- Supporting systems

A dry powder waste is generated from the secondary treatment train, from the treatment of an aqueous waste. The secondary waste treatment system typically receives and processes by-products generated from the primary treatment train. However, in an alternate operating scenario, some aqueous wastes may be fed to the secondary treatment train before the primary treatment train.

The treated effluent is contained in verification tanks where the effluent is sampled to confirm that the effluent meets the delisting criteria. Under 40 CFR 261, Appendix IX, Table 2, the treated effluent from 200 Area ETF is considered a delisted waste; that is, the treated effluent is no longer a listed dangerous waste subject to the hazardous waste management requirements of RCRA provided that the delisting criteria are satisfied and the treated effluent does not exhibit a dangerous characteristic. The treated effluent is discharged under the Washington State Waste Discharge Permit (No. ST 4500) as a nondangerous, delisted waste to the SALDS, located in the 600 Area, north of the 200 West Area. A portion of the treated wastewater from the Verification Tanks is recycled as service water throughout the facility; for example, it is used to dilute bulk acid and caustic to meet processing needs, thereby reducing the demand for process water.

B.1.2 Sources of Aqueous Waste

200 Area ETF was intended and designed to treat a variety of mixed wastes. However, process condensate from the 242-A Evaporator was the only mixed waste initially identified for storage and treatment in the LERF and 200 Area ETF. As cleanup activities at Hanford progress, many of the aqueous wastes generated from site remediation and waste management activities are sent to the LERF and 200 Area ETF for treatment and storage. A brief discussion of waste streams that may be managed by LERF and 200 Area ETF in the future may be found in the 200 Area ETF Delisting Petition (DOE/RL-92-97).

Prior to management of any new waste streams, it may be necessary to modify this WAP through the permit modification process to ensure that adequate knowledge of such new waste streams is available prior to management of them in LERF and 200 Area ETF.

The 242-A process condensate is a dangerous waste because it is derived from a listed, dangerous waste stored in the Double-Shell Tank (DST) System. The DST waste is transferred to the 242-A Evaporator where the waste is concentrated through an evaporation process. The concentrated slurry waste is returned to the DST System, and the evaporated portion of the waste is recondensed, collected, and transferred as process condensate to the LERF.

Other aqueous wastes that are treated and stored at the LERF and 200 Area ETF include, but are not limited to the following Hanford wastes:

- Contaminated groundwater from pump-and-treat remediation activities such as groundwater from the 200-UP-1 Operable Unit.
- Purgewater from groundwater monitoring activities.
- Water from deactivation activities, such as water from the spent fuel storage basins at deactivated reactors (e.g., N Reactor).
- Laboratory aqueous waste from unused samples and sample analyses.
- Leachate from landfills, such as the Environmental Restoration Disposal Facility.
- Any dilute waste, which may be accepted for treatment and within the scope of wastewaters that maybe delisted under terms of the revised delisting (40 CFR 261, Appendix IX, Table 2).

Most of these aqueous wastes are accumulated in batches in a LERF basin for interim storage and treatment through pH and flow equalization before final treatment in 200 Area ETF. However, some aqueous wastes, such as 200-UP-1 Groundwater, maybe treated on a flow through basis in LERF en route to 200 Area ETF for final treatment. The constituents in these aqueous wastes are common to the Hanford Site and were considered in pilot plant testing or in vendor tests, either as a constituent or as a family of constituents. According to the 200 Area ETF Delisting, Permit Condition 1.a.i, all wastes accepted for treatment at 200 Area ETF must be within a specified treatability envelope that ensures that wastes will be within the treatment capability of 200 Area ETF.

B.2 Influent Waste Acceptance Process

Throughout the acceptance process, there are specific criteria required for an influent waste (i.e., aqueous waste) to be accepted at the LERF and/or 200 Area ETF. These criteria are identified in the following sections and summarized in Table B.2. The process of accepting a waste into the LERF and 200 Area ETF systems involves a series of steps, as follows.

- **Waste information:** The generator of an aqueous waste works with LERF and 200 Area ETF personnel to provide characterization data of the waste stream (Section B.2.1).
- **Waste management decision process:** LERF and 200 Area ETF management decision is based on a case-by-case evaluation of whether an aqueous waste stream is acceptable for treatment or storage at LERF and the 200 Area ETF. The evaluation has two categories:
  - **Regulatory acceptability:** a review to determine if there are any, regulatory concerns that would prohibit the storage or treatment of an aqueous waste in the LERF or 200 Area ETF; e.g., treatment would meet permit conditions that would comply with applicable regulations.
  - **Operational acceptability:** an evaluation to determine if there are any operational concerns that would prohibit the storage or treatment of an aqueous waste in the LERF or 200 Area ETF and storage of treatment residuals; e.g., determine treatability and compatibility or safety considerations (Section B.2.2.2).

B.2.1 Waste Information

When an aqueous waste stream is identified for treatment or storage in the LERF or 200 Area ETF, the generator is required to characterize the waste stream according to the requirements in Section B.2.1.1 and document the results of characterization on an aqueous waste profile sheet. This requirement is the first waste acceptance criterion.

The LERF and 200 Area ETF personnel work with the generators to ensure that the necessary information is collected for the characterization of a waste stream (i.e., the appropriate analyses or adequate knowledge), and that the information provided on the waste profile sheet is complete. The completed waste profile sheet is maintained in the Hanford Facility Operating Record, LERF and 200 Area ETF File according to Permit Condition II.I.1.j.

B.2.1.1 Waste Characterization

Because the constituents in the individual aqueous waste streams vary, each waste stream is characterized and evaluated for acceptability on a case-by-case basis. The generator is required to designate an aqueous waste, which generally will be based on analytical data. However, a generator may use knowledge to substantiate the waste designation, or for general characterization information. Examples of acceptable knowledge include the following:

- Documented data or information on processes similar to that which generated the aqueous waste stream
- Information/documentation that the waste stream is from specific, well documented processes, e.g., F-listed wastes
- Information/documentation that sampling/analyzing a waste stream would pose health and safety risks to personnel
- Information/documentation that the waste stream does not lend itself to collecting a laboratory sample for example, wastewater collected (e.g., sump, tank) where the source water characterization is documented. Typically, these circumstances occur at decommissioned buildings or locations, not at operating units.

When a generator performs characterization of a dangerous and/or mixed waste stream based on knowledge, LERF and 200 Area ETF personnel review the knowledge as part of the waste acceptance process to ensure the knowledge satisfies the definition of *knowledge* in WAC 173-303-040. Specifically, LERF and 200 Area ETF personnel review the generator's processes to verify the integrity of the knowledge, and determine whether the knowledge is current and consistent with requirements of this is WAP. LERF and 200 Area ETF management or their designee determines the final decision on the adequacy of the knowledge. The persons reviewing generator process knowledge and those making decisions on the adequacy of knowledge are trained according to the requirements of Addendum G, Personnel Training.

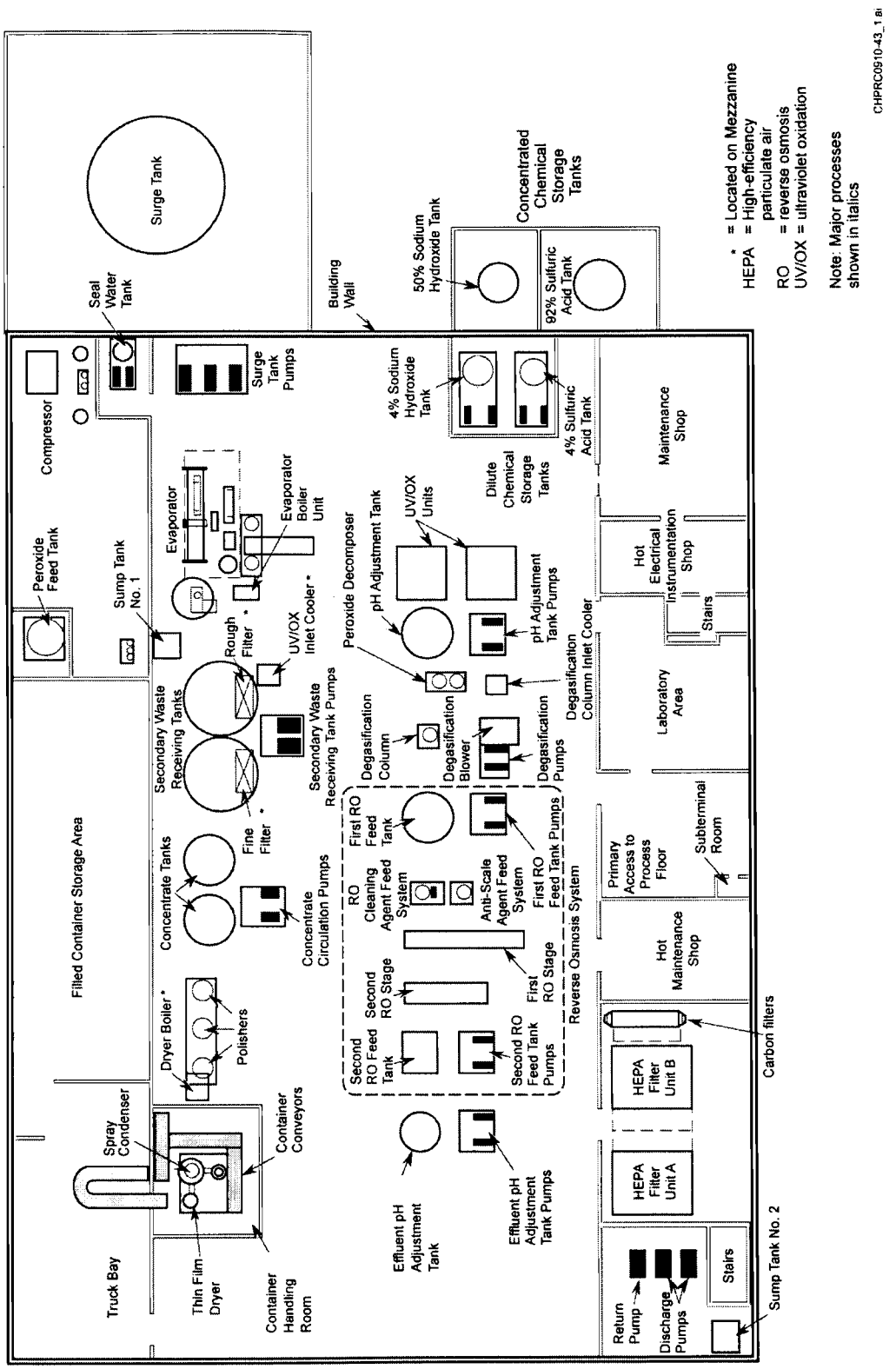


Figure B.1. 200 Area Effluent Treatment Facility Floor Plan

The generator is also responsible for identifying Land Disposal Restrictions (LDRs) treatment standards applicable to the influent aqueous waste as part of the characterization, as required under 40 CFR 268.40 incorporated by reference by WAC 173-303-140. Because the 200 Area ETF main treatment train is a Clean Water Act, equivalent treatment unit [40 CFR 268.37(a)] incorporated by reference by WAC 173-303-140, generators are not required to identify underlying hazardous constituents for characteristic wastes pursuant to 40 CFR 268.9, incorporated by reference by WAC 173-303-140, for wastewaters (i.e., <1 percent total suspended solids and <1 percent total organic carbon). The 200 Area ETF secondary waste (e.g., powder) reflects a change in LDR treatability group (i.e., wastewater to non-wastewater) so there is a new LDR point of generation, at which point any characteristic and associated underlying hazardous constituents must be identified. Therefore, generators of a non-wastewater may be required to identify underlying hazardous constituents for characteristic wastes pursuant to 40 CFR 268.9, incorporated by reference by WAC 173-303-140.

When analyzing an aqueous waste stream for LERF and 200 Area ETF waste acceptance characterization, a generator is required to use the target list of parameters identified in Table B.3, of this WAP. This requirement is in addition to any analysis required for purposes of designation under WAC 173-303-070. These data are used by LERF and 200 Area ETF to verify the treatability of an aqueous waste stream, and to develop a treatment plan for the waste after acceptance. Refer to Table B.6, for the corresponding analytical methods. The generator may use knowledge in lieu of some analyses, as determined by LERF and 200 Area ETF management or their designee, if the knowledge satisfies the definition of *knowledge* in WAC 173-303-040. For example if a generator provides information that the process generating an aqueous waste does not include or involve organic chemicals, analyses for organic compounds likely would not be required. Additional analyses could be required if historical information and/or knowledge indicate that an aqueous waste contains constituents not included in the target list of parameters.

The characterization and historical information are documented in the waste profile sheet, which is discussed in the following section and is part of the Hanford Facility Operating Record, LERF and 200 Area ETF File according to Permit Condition II.I.

B.2.1.2 Aqueous Waste Profile Sheet

The waste profile sheet documents the characterization of each new aqueous waste stream. The profile includes a detailed description of the source, volume, waste designation and applicable LDR treatment standards, and physical nature (wastewater or non-wastewater) of the aqueous waste. For an aqueous waste to be accepted for treatment or storage in the LERF or 200 Area ETF, each new waste stream generator is required to complete and provide this form to LERF and 200 Area ETF management. Each generator also is required to provide the analytical data and/or knowledge used to designate the aqueous waste stream according to WAC 173-303-070 and to determine the chemical and physical nature of the waste.

The LERF and ETF management determine whether the information on the waste profile sheet is sufficient according to the criteria above. The LERF and 200 Area ETF management use this information to evaluate the acceptability of the aqueous waste stream for storage and treatment in the LERF and 200 Area ETF, and to determine if the secondary waste generated from treatment is acceptable for storage at the 200 Area ETF and has a defined path forward to final disposal.

B.2.2 Waste Management Decision Process

All aqueous waste under consideration for acceptance must be characterized using analytical data and/or knowledge. This information is used to determine the acceptability of an aqueous waste stream. The LERF and 200 Area ETF Facility Manager or their designee is responsible for making the decision to accept or reject an aqueous waste stream. The management decision to accept any aqueous waste stream is based on an evaluation of regulatory acceptability and operational acceptability. Each evaluation uses acceptance criteria, which were developed to ensure that an aqueous waste is managed in a safe, environmentally sound, and in compliance with this Permit. The following sections provide detail on the acceptance evaluation and the acceptance criteria.

An aqueous waste stream could be rejected for one of the following reasons:

- The paperwork and/or laboratory analyses from the generator are insufficient
- Discrepancies with the regulatory and operational acceptance criteria cannot be reconciled, including:
  - An aqueous waste is not allowed under the current Washington State Waste Discharge Permit (No. ST 4500) or 200 Area ETF Delisting, and LERF and 200 Area ETF management elect not to pursue an amendment, or the Permit and Delisting cannot be amended (Section B.2.2.1)
  - An aqueous waste is incompatible with LERF liner materials or with other aqueous waste in LERF and no other management method is available (Section B.2.2.2.2).
- Adequate storage or treatment capacity is not available.

B.2.2.1 Regulatory Acceptability

Each aqueous waste stream is evaluated on a case-by-case basis to determine if there are any regulatory concerns that would preclude the storage or treatment of a waste in the LERF or 200 Area ETF based on the criteria in Sections B2.2.1.1 and B.2.2.1.2. Before an aqueous waste can be stored or treated in either the LERF or 200 Area ETF, the waste designation must be determined. Information on the waste designation of an aqueous waste is documented in the waste profile sheet. This information is used to confirm that treating or storing the aqueous waste in the LERF or 200 Area ETF is allowed under and in compliance with WAC 173-303, Permit (WA7890008967), 200 Area ETF Delisting in 40 CFR 261, Appendix IX, Table 2, the corresponding State-Issued Delisting, and the Washington State Waste Discharge Permit (No. ST 4500) for 200 Area ETF.

B.2.2.1.1 Dangerous Waste Regulations, State and Federal Delisting Actions, and Permits

Before an aqueous waste stream is sent to the LERF or 200 Area ETF, the generator will characterize and designate the stream with the appropriate dangerous/hazardous waste numbers according to WAC 173-303-070. Addendum A, the 200 Area ETF Delisting and the corresponding State-Issued Delisting identify the specific waste numbers for dangerous/mixed waste that can be managed in the LERF and 200 Area ETF. Dangerous waste designated with waste numbers not specified in these documents cannot be treated or stored in the LERF or 200 Area ETF, unless the documents are appropriately modified.

Additionally, aqueous wastes designated with listed waste numbers identified in the 200 Area ETF Delisting and the corresponding State-Issued Delisting will be managed in accordance with the conditions of the delisting, or an amended delisting.

B.2.2.1.2 State Waste Permit Regulations/Permit

Compliance with the Washington State Waste Discharge Permit (No. ST 4500), constitutes another waste acceptance criterion. In accordance with the permit conditions of the Washington State Waste Discharge Permit (No. ST 4500), the constituents of concern in each new aqueous waste stream must be identified. The waste designation and characterization data provided by the generator are used to identify these constituents. The Washington State Waste Discharge Permit (No. ST 4500), defines a constituent of concern in an aqueous waste stream, under the conditions of the Discharge Permit, as any contaminant with a maximum concentration greater than one of the following:

- Any limit in the Washington State Waste Discharge Permit (No. ST 4500)
- Groundwater Quality Criteria (WAC 173-200)
- Final Delisting level (40 CFR 261, Appendix IX, Table 2)
- The corresponding State-Issued Delisting

- Background groundwater concentration as measured at the SALDS disposal site. The practical quantification limit (PQL) is used for the groundwater background concentration for constituents not analyzed or not detected in the SALDs background data.

The Permit conditions of the Washington State Waste Discharge Permit (No. ST 4500), also require a demonstration that 200 Area ETF can treat the constituents of concern to below discharge limits.

B.2.2.2 Operational Acceptability

Because the operating configuration or operating parameters at the LERF and 200 Area ETF can be adjusted or modified, most aqueous waste streams generated on the Hanford Site can be effectively treated to below Delisting and Discharge Permit limits. Because of this flexibility, it would be impractical to define numerical acceptance or decision limits. Such limits would constrain the acceptance of appropriate aqueous waste streams for treatment at the LERF and 200 Area ETF. The versatility of the LERF and 200 Area ETF is better explained in the following examples:

- The typical operating configuration of 200 Area ETF is to process an aqueous waste through the UV/OX unit first, followed by the RO unit. However, high concentrations of nitrates may interfere with the performance of the UV/OX. In this case, 200 Area ETF could be configured to process the waste in the RO unit prior to the UV/OX unit.
- For a small volume aqueous waste with high concentrations of some anions and metals, the approach may be to first process the waste stream in the secondary treatment train. This approach would prevent premature fouling or scaling of the RO unit. The liquid portion (i.e., untreated overheads from 200 Area ETF evaporator and thin film dryer) would be sent to the primary treatment train.
- An aqueous waste with high concentrations of chlorides and fluorides may cause corrosion problems when concentrated in the secondary treatment train. One approach is to adjust the corrosion control measures in the secondary treatment train. An alternative may be to blend this aqueous waste in a LERF basin with another aqueous waste, which has sufficient dissolved solids, such that the concentration of the chlorides in the secondary treatment train would not pose a corrosion concern.
- Some metal salts (e.g., barium sulfate) tend to scale the RO membranes. In this situation, descalants used in the treatment process may be increased.
- Any effluent that does not meet these limits in one pass through 200 Area ETF treatment process is recycled to 200 Area ETF for re-processing.

There are some aqueous wastes, whose chemical and physical properties preclude that waste from being treated or stored at the LERF or 200 Area ETF. Accordingly, an aqueous waste is evaluated to determine if it is treatable, if it would impair the efficiency or integrity of the LERF or 200 Area ETF, and if it is compatible with materials in these units. This evaluation also determines if the aqueous waste is compatible with other aqueous wastes managed in the LERF.

The waste acceptance criteria in this category focus on determining treatability of an aqueous waste stream, and on determining any operational concerns that would prohibit the storage or treatment of an aqueous waste stream in the LERF or 200 Area ETF. The chemical and physical properties of an aqueous waste stream are determined as part of the waste characterization, and are documented on the waste profile sheet and compared to the design of the units to determine whether an aqueous waste stream is appropriate for storage and treatment in the LERF and 200 Area ETF. All decisions and supporting rationale and data will be documented in the Hanford Facility Operating Record, LERF and 200 Area ETF File according to Permit Condition II.I.

B.2.2.3 Special Requirements Pertaining to Land Disposal Restrictions

Containers of 200 Area ETF secondary waste are transferred to a storage or final disposal unit, as appropriate (e.g., the Central Waste Complex or to the Environmental Restoration Disposal Facility).

200 Area ETF personnel provide the analytical characterization data and necessary process knowledge for the waste to be managed by the receiving staff, and the appropriate LDR documentation.

The following information on the secondary waste is included on the LDR documentation provided to the receiving unit:

- Dangerous waste numbers (as applicable)
- Determination on whether the waste is restricted from land disposal according to the requirements of 40 CFR 268 incorporated by reference by WAC 173-303-140 (i.e., the LDR status of the waste)

The waste tracking information associated with the transfer of waste

- Waste analysis results.

Generally, the operating parameters or operating configuration at the LERF or 200 Area ETF can be adjusted or modified to accommodate these properties. However, in those cases where a treatment process or operating configuration cannot be modified, the aqueous waste stream will be excluded from treatment or storage at the LERF or 200 Area ETF. Additionally, an aqueous waste stream is evaluated for the potential to deposit solids in a LERF basin (i.e., whether an aqueous waste contains sludge or could precipitate solids). This evaluation will also consider whether the blending or mixing of two or more aqueous waste streams will result in the formation of a precipitate. However, because the waste streams managed in the LERF and 200 Area ETF are generally dilute, the potential for mixing waste streams and forming a precipitate is low; no specific compatibility tests are performed. Filtration at the waste source could be required before acceptance into LERF. Waste streams with the potential to form precipitates in LERF or that cannot be blended with other waste streams to avoid precipitate formation are not accepted for treatment at LERF and 200 Area ETF. The Load-in Facility has the ability to perform filtration on incoming waste streams going to both the LERF and 200 Area ETF Load in. See additional discussions of precipitate formation and compliance with LDR requirements in Section B.3. Similar filtration requirements could apply to aqueous waste fed directly to 200 Area ETF without interim treatment in LERF.

To determine if an aqueous waste meets the criterion of treatability, specific information is required. Treatability of a waste stream is evaluated from characterization data provided by the generator as verified through the waste acceptance process, the 200 Area waste acceptance criteria, and the treatability envelope for the 200 Area ETF as documented in Tables C.1 and C.2 of the November 29, 2001 delisting petition. Generators will also provide characterization data to identify those physical and chemical properties that would interfere with, or foul 200 Area ETF treatment process in consultation with LERF and 200 Area ETF representatives. In some instances, knowledge that meets the definition of *knowledge* in WAC 173-303-040 is used for purposes of identifying a chemical or physical property that would be of concern. For example, the generator could provide knowledge that the stream has two phases (an oily phase and an aqueous phase). In this case, if the generator could not physically separate the two phases, the aqueous waste stream would be rejected because the oily phase could compromise some of the treatment equipment. Typically, analyses for the following parameters are required to evaluate treatability and operational concerns:

- total dissolved solids
- total organic carbon
- total suspended solids
- specific conductivity
- pH
- alkalinity
- ammonia
- barium
- calcium
- chloride
- fluoride
- iron
- magnesium
- nitrate
- nitrite
- phosphate
- potassium
- silicon
- sodium
- sulfate

These constituents are identified in Table B.2, which is the list of target analytes used for waste characterization and waste acceptance evaluation.

B.2.2.3.1 Compatibility

Corrosion Control. Because of the materials of construction used in 200 Area ETF, corrosion is generally not a concern with new aqueous waste streams. Additionally, these waste streams are managed in a manner that minimizes corrosion. To ensure that a waste will not compromise the integrity of 200 Area ETF tanks and process equipment, each waste stream is assessed for its corrosion potential as part of the compatibility evaluation. This assessment usually focuses on chloride and fluoride concentrations; however, the chemistry of each new waste also is evaluated for other parameters that could cause corrosion.

Compatibility with Liquid Effluent Retention Facility Liner and Piping. As part of the acceptance process, the criteria of compatibility with the LERF liner materials are evaluated for each aqueous waste stream. This evaluation is performed using knowledge (as defined by WAC 173-303-040) of constituent concentrations in the aqueous waste stream or using constituent concentrations obtained by analyzing the waste stream for the constituents identified in Table B.1 using the analytical methods for these constituents in Section B.9. Then, the constituent concentrations in the waste stream are compared to the decision criteria in Table B.1. If all constituent concentrations are below the decision criteria, then the waste stream is considered compatible with the LERF liner and may be accepted for treatment. Otherwise, the waste stream is considered incompatible with the LERF liner, and it cannot be accepted for treatment in the LERF basins. However, a waste stream may still be acceptable for treatment in ETF if it is fed directly to ETF, bypassing the LERF Basins. Results of this evaluation are documented in the Hanford Facility Operating Record, LERF and 200 Area ETF File according to Permit Condition II.I. The rational for establishing the liner compatibility constituents and decision criteria in Table B.1 is as follows: The high-density polyethylene liners in the LERF basins potentially are vulnerable to the presence of certain constituents that might be present in some aqueous waste. Using EPA SW-846, Method 9090, the liner materials were tested to evaluate compatibility between aqueous waste stored in the LERF and synthetic liner components. Based on the data from the compatibility test and vendor data on the liner materials, several constituents and parameters were identified as potentially harmful (at high concentrations) to the integrity of the liners. From these data and the application of safety factors, concentration limits in Table B.1 were established.

The strategy for protecting the integrity of a LERF liner is to establish upfront that an aqueous waste is compatible before the waste is accepted into LERF. Characterization data on each new aqueous waste stream are compared to the limits outlined in Table B.1 to ensure compatibility with the LERF liner material before acceptance into the LERF.

Before a waste stream is processed at the 242-A Evaporator, the generator reviews DST analytical data and a process condensate profile is developed to ensure the process condensate is compatible with the LERF liner. For flow through aqueous wastes like the 200-UP-1 Groundwater, characterization data will be obtained and reviewed every two years to ensure that liner compatibility is maintained.

In some instances, knowledge may be adequate to determine that an aqueous waste is compatible with the LERF liner. When knowledge is used, it must satisfy the definition of *knowledge* in WAC 173-303-040. In those instances where knowledge is adequate, the waste characterization would likely not require analysis for these parameters and constituents. Storm water is an example where knowledge is adequate to determine that this aqueous waste is compatible with the LERF liner.

Compatibility with Other Waste. Some aqueous wastes, especially small volume streams, are accumulated in the LERF with other aqueous waste. Before acceptance into the LERF, the aqueous waste stream is evaluated for its compatibility with the resident aqueous waste(s). The evaluation focuses on the potential for an aqueous waste to react with another waste (40 CFR 264, Appendix V, *Examples of Potentially Incompatible Wastes*) including formation of any precipitate in the LERF basins.

However, the potential for problems associated with commingling aqueous wastes is very low due to the dilute nature of the wastes; this evaluation confirms the compatibility of two or more aqueous wastes from different sources. Compatibility is determined by evaluating parameters such as pH, ammonia, and chloride. No specific analytical test for compatibility is performed.

If it is determined that an aqueous waste stream is incompatible with other aqueous waste streams, alternate management scenarios are available. For example, another LERF basin that contains a compatible aqueous waste(s) might be used, or the aqueous waste stream might be fed directly into 200 Area ETF for treatment. In any case, potentially incompatible waste streams are not mixed, and all aqueous waste is managed in a way that precludes a reaction, degradation of the liner, or interference with 200 Area ETF treatment process.

B.2.3 Periodic Review Process

In accordance with WAC 173-303-300(4)(a), an influent aqueous waste will be periodically reviewed as necessary to ensure that the characterization is accurate and current. At a minimum, an aqueous waste stream will be reviewed in the following situations.

- The LERF and 200 Area ETF management have been notified, or have reason to believe that the process generating the waste has changed.
- The LERF and 200 Area ETF management note an increase or decrease in the concentration of a constituent in an aqueous waste stream, beyond the range of concentrations that was described or predicted in the waste characterization.
- Waste streams will be reviewed every two years.

In these situations, LERF and 200 Area ETF management will review the available information. If existing analytical information is not sufficient, the generator may be asked to review and update the current waste characterization, to supply a new WPS, or re-sample and re-analyze the aqueous waste, as necessary. Other situations that might require a re-evaluation of a waste stream are discussed in the following sections.

B.2.4 Record/Information and Decision

The information and data collected throughout the acceptance process, and the evaluation and decision on whether to accept an influent aqueous waste stream for treatment or storage in the LERF or 200 Area ETF are documented as part of Hanford Facility Operating Record, LERF and 200 Area ETF File pursuant to Permit Condition II.I. Specifically, the Hanford Facility Operating Record, LERF and 200 Area ETF File contains the following components on a new influent aqueous waste stream:

- The signed WPS for each aqueous waste stream and analytical data.
- Knowledge used to characterize a dangerous/mixed waste (under WAC 173-303), and information supporting the adequacy of the knowledge.
- The evaluation on whether an aqueous waste stream meets the waste acceptance criteria, including:
  - The evaluation for regulatory acceptability including appropriate regulatory approvals.
  - The evaluation for LERF liner compatibility and for compatibility with other aqueous waste.

Table B.1. General Limits for Liner Compatibility

Chemical Family	Constituent(s) or Parameter(s)[1]		Limit (mg (sum of constitue concentr
Alcohol/glycol	1-butanol		500,000
Alkanone[3]	acetone,		200,000
Alkenone[4]	none targeted		N/A
Aromatic/cyclic hydrocarbon	acetophenone, benzene, carbozole, chrysene, cresol, di-n-octyl phthalate, diphenylamine, isophorone, pyridine, tetrahydrofuran		2000
Halogenated hydrocarbon	arochlors, carbon tetrachloride, chloroform, hexachlorobenzene, lindane (gamma-BHC), hexachlorocyclopentadiene, methylene chloride, p-chloroaniline, tetrachloroethylene, 2,4,6-trichlorophenol		2000
Aliphatic hydrocarbon	none targeted		N/A
Ether	dichloroisopropyl ether		2000
Other hydrocarbons	acetontrile, carbon disulfide, n-nitrosodimethylamine, tributyl phosphate		2000
Oxidizers	none targeted		NA
Acids, Bases, Salts	ammonia, cyanide, anions, cations		100,000
pH	pH		0.5 < pH <

[1]Analytical methods for the parameters and constituents are provided in Section B.9
[2]Analytical data are evaluated using the following 'sum of the fraction' technique. The individual constituent concentration is evaluated against the compatibility limit for its chemical family. The sum of the evaluations must be less than 1. pH is not part of this evaluation.

$$\sum_{n=1}^{i} \left(\frac{\text{Conc}_n}{\text{LIMIT}_n}\right) \leq 1$$

[3]Ketone containing saturated alkyl group(s)
[4]Ketone containing unsaturated alkyl group(s)
Where 'i' is the number of organic constituents detected
mg/L = milligrams per liter
NA = not applicable

Table B.2. Waste Acceptance Criteria

General Criteria Category	Criteria Description
1. Characterization	A. Each generator must provide an aqueous waste profile.
	B. Each generator must designate the aqueous waste stream.
	C. Each generator must provide analytical data and/or knowledge.
2. Regulatory acceptability	A. The LERF and 200 Area ETF can store and treat influent aqueous wastes with waste numbers identified in Addendum A for the LERF and 200 Area ETF, and the 200 Area ETF Delisting, 40 CFR 261, Appendix IX, Table 2.
	B. The aqueous waste must comply with conditions of the Discharge Permit.
3. Operational acceptability	A. Determine whether an aqueous waste stream is treatable, considering: 1. Whether the removal and destruction efficiencies on the constituents of concern will be adequate to meet the Discharge Permit and Delisting levels 2. Other treatability concerns; analyses for this evaluation may include: total dissolved solids — iron total organic carbon — magnesium total suspended solids — nitrate specific conductivity — nitrite alkalinity — phosphate ammonia — potassium barium — silicon calcium — sodium chloride — sulfate fluoride — pH
	B. Determine whether an aqueous waste stream is compatible, considering: 1. Whether an aqueous waste stream presents corrosion concerns with respect to ETF; analysis may include chloride and fluoride 2. Whether an aqueous waste stream is compatible with LERF liner materials, compare characterization data to the liner compatibility limits (Table B.1). 3. Whether an aqueous waste stream is compatible with other aqueous waste(s), 40 CFR 264, Appendix V, comparison will be used.

B.3 Special Management Requirements

Special management requirements for aqueous wastes that are managed in the LERF or 200 Area ETF are discussed in the following section.

B.3.1 Land Disposal Restriction Compliance at Liquid Effluent Retention Facility

Because LERF provides treatment through flow and pH equalization, a surface impoundment treatment exemption from the land disposal restrictions was granted in accordance with 40 CFR 268.4, and WAC 173-303-040. This treatment exemption is subject to several conditions, including a requirement that the WAP address the sampling and analysis of the treatment 'residue' [40 CFR 268.4(a)(2)(i) and WAC 173-303-300(5)(h)(i) and (ii)] to ensure the 'residue' meets applicable treatment standards. Though the term 'residue' is not specifically defined, this condition further requires that sampling must be designed to represent the "sludge and the supernatant" indicating that a residue may have a sludge (solid) and supernatant (liquid) component.

Solid residue is not anticipated to accumulate in a LERF basin for the following reasons:

- Aqueous waste streams containing sludge would not be accepted into LERF under the acceptance criteria of treatability (Section B.2.2.2.1).
- No solid residue was reported from process condensate discharged to LERF in 1995.
- The LERF basins are covered and all incoming air first passes through a breather filter.
- No precipitating or flocculating chemicals are used in flow and pH equalization.
- Multiple waste streams managed in a single LERF basin are evaluated for the formation of precipitates. Wastes that would form precipitates are not accepted for treatment at LERF.

Therefore, the residue component subject to this condition is the supernatant (liquid component). Additionally, an aqueous waste stream is evaluated for the potential to deposit solids in a LERF basin (i.e., an aqueous waste that contains suspended solids). If necessary, filtration at the waste source could be required before acceptance into LERF. Therefore, the residue component in LERF subject to this condition is the supernatant (liquid component). The contingency for removal of solids will be addressed during closure in Addendum H, Closure Plan.

The conditions of the treatment exemption also require that treatment residues (i.e., aqueous wastes), which do not meet the LDR treatment standards "must be removed at least annually" [40 CFR 268.4(a)(2)(ii) incorporated by reference by WAC 173-303-140]. To address the conditions of this exemption, an influent aqueous waste is sampled and analyzed and the LDR status of the aqueous waste is established as part of the acceptance process. The LERF basins are then managed such that any aqueous waste(s), which exceeds an LDR standard is removed annually from a LERF basin, except for a heel of approximately 1 meter. A heel is required to stabilize the LERF liner. The volume of the heel is approximately 1.9 million liters.

B.4 Influent Aqueous Waste Sampling and Analysis

The following sections provide a summary of the sampling procedures, frequencies, and analytical parameters for characterization of influent aqueous waste (Section B.2) and in support of the special management requirements for aqueous waste in the LERF (Section B.3).

B.4.1 Sampling Procedures

With a few exceptions, generators are responsible for the characterization, including sampling and analysis, of an influent aqueous waste. Process condensate is either sampled at the 242-A Evaporator or accumulated in a LERF basin following a 242-A Evaporator campaign and sampled. Other exceptions will be handled on a case-by-case basis and the Hanford Facility Operating Record, LERF and 200 Area ETF File will be maintained at the unit for inspection by Ecology. The following section discusses the sampling locations, methodologies, and frequencies for these aqueous wastes. For samples collected at the LERF and 200 Area ETF, unit-specific sampling protocol is followed. The sample containers, preservation materials, and holding times for each analysis are listed in Section B.9.

B.4.1.1 Batch Samples

In those cases where an aqueous waste is sampled in a LERF basin, samples are collected from four of the six available sample risers located in each basin, i.e., four separate samples. When LERF levels are low, fewer than four samples can be taken if the sampling approach is still representative. Though there are eight sample risers at each basin, one is dedicated to liquid level instrumentation and another is dedicated as an influent port. Operating experience indicates that four samples adequately capture the spatial variability of an aqueous waste stream in the LERF basin. Specifically, sections of stainless steel (or other compatible material) tubing are inserted into the sample riser to an appropriate depth. Using a portable pump, the sample line is flushed with the aqueous waste and the sample collected. The grab sample containers typically are filled for volatile organic compounds (VOC) analysis first, followed by the remainder of the containers for the other parameters.

Several sample ports are also located at 200 Area ETF, including a valve on the recirculation line at 200 Area ETF surge tank, and a sample valve on a tank discharge pump line at 200 Area ETF Load-in Station. All samples are obtained at the LERF or 200 Area ETF are collected in a manner consistent with SW-846 procedures (EPA as amended).

B.4.2 Analytical Rationale

As stated previously, each generator is responsible for designating and characterizing an aqueous waste stream. Accordingly, each generator samples and analyzes an influent waste stream using the target list of parameters (Table B.3) for the waste acceptance process. At the discretion of the LERF and ETF management, a generator may provide knowledge in lieu of some analyses as discussed in Section B.2.1.1. The LERF and ETF personnel will work with the generator to determine which parameters are appropriate for the characterization.

The analytical methods for these parameters are provided in Section B.9. All methods are EPA methods satisfying the requirements of WAC 173-303-110(3). Additional analyses may be required if historical information and knowledge indicate that an influent aqueous waste contains constituents not included in the target list of parameters. For example, if knowledge indicates that an aqueous waste contains a parameter that is regulated by the Groundwater Quality Criteria (WAC 173-200), that parameter(s) would be added to the suite of analyses required for that aqueous waste stream.

The analytical data for the parameters presented in Table B.3, including VOC, Semi-volatile Organic Compound (SVOC), metals, anions, and general chemistry parameters are used to define the physical and chemical properties of the aqueous waste for the following:

- Set operating conditions in the LERF and ETF (e.g., to determine operating configuration , refer to Section B.2.2.2)
- Identify concentrations of some constituents which may also interfere with, or foul ETF treatment process (e.g., fouling of the RO membranes, refer to Section B.2.2.2)
- Evaluate LERF liner and piping material compatibility
- Determine treatability to evaluate if applicable constituents in the treated effluent will meet Discharge Permit and Delisting limits
- Estimate concentrations of some constituents in the waste generated in the secondary treatment train (i.e., dry powder waste).

Table B.3. Target Parameters for Influent Aqueous Waste Analyses

VOLATILE ORGANIC COMPOUNDS	SEMIVOLATILE ORGANIC COMPOUNDS
Acetone	Acetophenone
Acetonitrile	Cresol (o, p, m)
Benzene	Dichloroisopropyl ether (bis(2-chloropropyl)ether)
1-Butanol	Di-n-octyl phthalate
Carbon disulfide	Diphenylamine
Carbon tetrachloride	Hexachlorobenzene
Chloroform	Hexachlorocyclopentadiene
Methylenechloride	Isophorone
Tetrachloroethylene	Lindane (gamma-BHC)
Tetrahydrofuran	N-nitrosodimethylamine
	Pyridine
	Tributyl phosphate
	2,4,6-Trichlorophenol

TOTAL METALS		ANIONS
Arsenic	Magnesium	Chloride
Barium	Mercury	Fluoride
Beryllium	Nickel	Nitrate
Cadmium	Potassium	Nitrite
Calcium	Selenium	Phosphate
Chromium	Silicon	Sulfate
Copper	Silver	**GENERAL CHEMISTRY PARAMETERS**
Iron	Sodium	Ammonia
Lead	Vanadium	Cyanide
	Zinc	pH
		Total suspended solids
		Total dissolved solids
		Total organic carbon
		Specific conductivity

B.5 Treated Effluent Sampling and Analysis

The treated aqueous waste, or effluent, from 200 Area ETF is collected in three 2,940,000-liter verification tanks before discharge to the SALDS. To determine whether the Discharge Permit early warning values, enforcement limits, and the Delisting criteria are met, the effluent routinely is sampled at the verification tanks. The sampling and analyses performed are described in the following sections.

B.5.1 Rationale for Effluent Analysis Parameter Selection

The parameters measured in the treated effluent are required by the following regulatory documents:

- Delisting criteria from the 200 Area ETF Delisting (40 CFR 261, Appendix IX, Table 2)
- Corresponding State Final Delisting issued pursuant to WAC 173-303-910(3)
- Effluent limits from the Washington State Waste Discharge Permit (No. ST 4500)
- Early warning values from the Washington State Waste Discharge Permit (No. ST 4500)

The 200 Area ETF Delisting provides two testing regimes for the treated effluent. Initial verification testing is performed when a new influent waste stream is processed through the 200 Area ETF.

For each 200 Area ETF influent waste stream, the first generated verification tank must be sampled and analyzed for all delisting constituents and conductivity. Subsequent verification sampling and analysis of all delisting parameters is performed on every 15th tank of that 200 Area ETF influent waste stream. If the concentration of any analyte is found to exceed a Washington State Waste Discharge Permit (No. ST 4500), enforcement limit or a Delisting criterion, the contents of the verification tank are reprocessed and/or re-analyzed. The next verification tank generated is also sampled for all delisting constituents. If the concentration of any analyte exceeds an early warning value, an early warning value report is prepared and submitted to Ecology.

B.5.2 Effluent Sampling Strategy: Methods, Location, Analyses, and Frequency

Effluent sampling methods and locations, the analyses performed, and frequency of sampling are discussed in the following sections.

B.5.2.1 Effluent Sampling Method and Location

Samples of treated effluent are collected and analyzed to verify the treatment process using 200 Area ETF specific sampling protocol. These verification samples are collected at a sampling port on the verification tank recirculation line. Section B.9 presents the sample containers, preservatives, and holding times for each parameter monitored in the effluent.

B.5.2.2 Analyses of Effluent

The parameters required by the current Washington State Waste Discharge Permit (No. ST 4500), and Final Delisting 200 Area ETF, conditions are presented in Table B.4. The analytical methods and PQLs associated with each parameter are provided in Section B.9. The methods and PQLs are equivalent to those used in the analysis of influent aqueous waste.

B.5.2.3 Frequency of Sampling

Treated effluent is tested for all parameters listed in Table B.4 on a frequency satisfying the permit conditions of the Washington State Waste Discharge Permit (No. ST 4500), and the 200 Area ETF Delisting. This effluent must meet the Washington State Waste Discharge Permit (No. ST 4500), and 200 Area ETF Delisting limits associated with these parameters. Grab samples are collected from each verification tank.

During operation of 200 Area ETF, if one or more of the constituents exceeds a Delisting criterion, the Delisting conditions require:

- The characterization data and processing strategy of the influent waste stream be reviewed and changed accordingly to ensure the contents of subsequent tanks do not exceed the Delisting criteria
- The contents of the verification tank are recycled for additional treatment. The contents that are recycled are resampled after treatment to ensure no constituents exceed a Delisting criteria
- The contents of the following verification tank are sampled for compliance with the Delisting criteria.
- Treated effluent that does not meet Washington State Waste Discharge Permit (No. ST 4500) is not discharged to the SALDS until the tank has been retreated and/or reanalyzed.

B.6 Effluent Treatment Facility Generated Waste Sampling and Analysis

The wastes discussed in this section include the wastes generated at 200 Area ETF and are managed in the container storage areas of 200 Area ETF. This section describes the characterization of the following secondary waste streams generated within 200 Area ETF:

- Secondary waste generated from the treatment process, including the following waste forms:
  - dry powder waste

- concentrate tanks slurry
- sludge removed from process tanks

- Waste generated by operations and maintenance activities
- Miscellaneous waste generated within 200 Area ETF.

For each waste stream described, a characterization methodology and rationale are provided, and sampling requirements are addressed.

B.6.1 Secondary Waste Generated from Treatment Processes

The following terms used in this Section, including powder, dry powder, waste powder, and dry waste powder, are equivalent to the term 'dry powder waste'.

A dry powder waste is generated from the secondary treatment train, from the treatment of an aqueous waste. Waste is received in the secondary treatment train in waste receiving tanks where it is fed into an evaporator. Concentrate waste from the evaporator is then fed to a concentrate tank. From these tanks, the waste is fed to a thin film dryer and dried into a powder, and collected into containers. The containers are filled via a remotely controlled system. The condensed overheads from the evaporator and thin film dryer are returned to the surge tank to be fed to the primary treatment train.

Occasionally, salts from the treatment process (e.g., calcium sulfate and magnesium hydroxide) accumulate in process tanks as sludge. Because processing these salts could cause fouling in the thin film dryer, and to allow uninterrupted operation of the treatment process, the sludge is removed and placed in containers. The sludge is dewatered and the supernate is pumped back to 200 Area ETF for treatment.

The secondary treatment system typically receives and processes the following by-products generated from the primary treatment train:

- Concentrate from the first RO stage
- Backwash from the rough and fine filters
- Regeneration waste from the ion exchange system
- Spillage or overflow collected in the process sumps.

In an alternate operating scenario, some aqueous wastes may be fed to the secondary treatment train before the primary treatment train.

B.6.1.1 Special Requirements Pertaining to Land Disposal Restrictions

Containers of 200 Area ETF secondary waste are transferred to a storage or final disposal unit, as appropriate (e.g., the Central Waste Complex or to the Environmental Restoration Disposal Facility). 200 Area ETF personnel provide the analytical characterization data and necessary knowledge for the waste to be managed by the receiving staff, and for the appropriate LDR documentation.

The following information on the secondary waste is included on the LDR documentation provided to the receiving unit:

- Dangerous waste numbers (as applicable)
- Determination on whether the waste is restricted from land disposal according to the requirements of 40 CFR 268 incorporated by reference by WAC 173-303-140 (i.e., the LDR status of the waste)

The waste tracking information associated with the transfer of waste

- Waste analysis results.

B.6.1.2 Sampling Methods

The dry powder waste and containerized sludge are sampled from containers using the principles presented in SW-846 (EPA as amended) and ASTM Methods (American Society for Testing Materials), as referenced in WAC 173-303-110(2). The sample container requirements, sample preservation requirements, and maximum holding times for each of the parameters analyzed in either matrix are presented in Section B.9.

Concentrate tank waste samples are collected from recirculation lines, which provide mixing in the tank during pH adjustment and prevent caking. The protocol for concentrate tank sampling prescribes opening a sample port in the recirculation line to collect samples directly into sample containers. The sample port line is flushed before collecting a grab sample. The VOC sampling typically is performed first for grab samples. Each VOC sample container will be filled such that cavitation at the sample valve is minimized and the container has no headspace. The remainder of the containers for the other parameters will be filled next.

Table B.4. Rationale for Parameters to be Monitored in Treated Effluent

Parameter	(Cas No.)	200 Area ETF Delisting[1]	Discharge Permit[2]	
			Enforcement Limit	Early Warning Value
VOLATILE ORGANIC COMPOUNDS				
Acetone	(67-64-1)	X		
Acetonitrile	(75-05-8)	X		
Benzene	(71-43-2)	X		X
1-Butanol	(71-36-3)	X		
Carbon disulfide	(75-15-0)	X		
Carbon tetrachloride	(56-23-5)	X	X	
Chloroform	(67-66-3)			X
Methylene Chloride	(75-09-2)		M	
Tetrachloroethylene	(127-18-4)		X	
Tetrahydrofuran	(109-99-9)	X		X
SEMIVOLATILE ORGANIC COMPOUNDS				
Acetophenone	(98-86-2)		X	
Carbazole	(86-74-8)	X		
p-Chloroaniline	(106-47-8)	X		
Chrysene	(218-01-9)	X		
Cresol (total)	(1319-77-3)	X		
Dichloroisopropyl ether (bis(2-chloroisopropyl)ether)	(108-60-1)	X		
Di-n-octyl phthalate	(117-84-0)	X		
Diphenylamine	(122-39-4)	X		
Hexachlorobenzene	(118-74-1)	X		
Hexachlorocyclopentadiene	(77-47-4)	X		
Isophorone	(78-59-1)	X		
Lindane (gamma-BHC)	(58-89-9)	X		
N-nitrosodimethylamine	(62-75-9)	X	X	
Pyridine	(110-86-1)	X		

Table B.4. Rationale for Parameters to be Monitored in Treated Effluent				
			Discharge Permit[2]	
Parameter	(Cas No.)	200 Area ETF Delisting[1]	Enforcement Limit	Early Warning Value
Tributyl phosphate	(126-73-8)	X		
2,4,6-Trichlorophenol	(88-06-2)	X		
PCBs				
Aroclor 1016	(12674-11-2)	X		
Aroclor 1221	(11104-28-2)	X		
Aroclor 1232	(11141-16-5)	X		
Aroclor 1242	(53469-21-9)	X		
Aroclor 1248	(12672-29-6)	X		
Aroclor 1254	(11097-69-1)	X		
Aroclor 1260	(11096-82-5)	X		
TOTAL METALS[3]				
Arsenic	(7440-38-2)	X	X	
Barium	(7440-39-3)	X		
Beryllium	(7740-41-7)	X	X	
Cadmium	(7440-43-9)	X		X
Chromium	(7440-47-3)	X	X	
Copper	(7440-50-8)			X
Lead	(7439-92-1)	X		X
Mercury	(7439-97-6)	X		X
Nickel	(7440-02-0)	X		
Selenium	(7782-49-2)	X		
Silver	(7440-22-4)	X		
Vanadium	(7440-62-2)	X		
Zinc	(7440-66-6)	X		
ANIONS				
Chloride	(16887-00-6)		X	
Fluoride	(16984-48-8)	X		
Nitrate (as N)	(14797-55-8)		X	
Nitrite (as N)	(1479765-0)		X	
Sulfate	(14808-79-8)		X	
OTHER ANALYSES				
Ammonia	(7664-41-7)	X	X	
Cyanide	(57-12-5)	X		
Total dissolved solids				X
Total organic carbon			X	
Total suspended solids			X	
Specific conductivity			M	

1 [1]Parameters required by the current conditions of the 200 Area ETF Delisting, 40 CFR 261, Appendix IX, Table 2,70 FR 44496
2 (EPA 2005)

3 [2]Parameters required by the current conditions of the State Waste Discharge Permit, No. ST 4500

[3]Metals reported as total concentrations
X = Rationale for measuring this parameter in treated effluent
M = Monitor only; no limit defined
PCBs = polychlorinated biphenyls

B.6.1.3 Sampling Frequency

When designation or identification of applicable LDR treatment standards of the 200 Area ETF secondary waste cannot be based on influent characterization data or knowledge as described in Section B.6.1.1, 200 Area ETF secondary waste is sampled on a batch basis. A batch is defined as any volume of aqueous waste that is being treated under consistent and constant process conditions.

When personnel exposures are of concern, one representative sample will be collected from the concentrate tank, if waste from the concentrate tank. The sample will be analyzed for the appropriate parameters identified in Table B.5 based on the needs identified from evaluating influent waste analysis data. If sampling of the concentrate tank is not technically practicable for purposes of designating the powder, direct sampling of the dry powder will be used to make determinations on the dry powder. The dry powder or concentrate tanks will be resampled in the following situations:

- Change in influent characterization.
- Change in process chemistry, as indicated by in-line monitoring of conductivity and pH.
- The LERF and 200 Area ETF management have been notified, or have reason to believe that the process generating the waste has changed (for example, a source change such as a change in the well-head for groundwater that significantly changes the aqueous waste characterization).
- The LERF and 200 Area ETF management note an increase or decrease in the concentration of a constituent in an aqueous waste stream, beyond the range of concentrations that was described or predicted in the waste characterization.

B.6.2 Operations and Maintenance Waste Generated at the 200 Area Effluent Treatment Facility

Operation and maintenance of process and ancillary equipment generates additional routine waste. These waste materials are segregated to ensure proper handling and disposition, and to minimize the commingling of potentially dangerous waste with nondangerous waste. The following waste streams are anticipated to be generated during routine operation and maintenance of 200 Area ETF. This waste might or might not be dangerous waste, depending on the nature of the material and its exposure to a dangerous waste.

- Spent lubricating oils and paint waste from pumps, the dryer rotor, compressors, blowers, and general maintenance activities.
- Spent filter media and process filters.
- Spent ion exchange resin.
- High Efficiency Particulate Air (HEPA) filters.
- UV light tubes.
- RO membranes.
- Equipment that cannot be returned to service.
- Other miscellaneous waste that might contact a dangerous waste (e.g., plastic sheeting, glass, rags, paper, waste solvent, or aerosol cans).

These waste streams are stored at 200 Area ETF before being transferred for final treatment, storage, or disposal as appropriate.

This waste is characterized and designated using knowledge (from previously determined influent aqueous waste composition information); analytical data; and material safety data sheets (MSDS) of the chemical products present in the waste or used (the data sheets are maintained at 200 Area ETF). Sampling of these waste streams is not anticipated; however, if an unidentified or unlabeled waste is discovered, that waste is sampled. This 'unknown' waste is sampled and analyzed for the parameters in Table B.5 as appropriate, and will be designated according to Washington state regulatory requirements. The specific analytical methods for these analyses are provided in Section B.9.

B.6.3 Other Waste Generated at the 200 Area Effluent Treatment Facility

There are two other potential sources of waste at 200 Area ETF: spills and/or overflows, and discarded chemical products. Spills may be subject to the requirements of Permit Condition II.E. Spilled material that potentially might be dangerous waste generally is either containerized or routed to 200 Area ETF sumps where the material is transferred either to the surge tank for treatment or to the secondary treatment train. In most cases, knowledge and the use of MSDSs are sufficient to designate the waste material. If the source of the spilled material is unknown and the material cannot be routed to 200 Area ETF sumps, a sample of the waste is collected and analyzed according to Table B.5, as necessary, for appropriate characterization of the waste. Unknown wastes will be designated according to Washington State regulatory requirements at WAC 173-303-070. The specific analytical methods for these analyses are provided in Section B.9.

A discarded chemical product waste stream could be generated if process chemicals, cleaning agents, or maintenance products become contaminated or are otherwise rendered unusable. In all cases, these materials are appropriately containerized and designated. Sampling is performed, as appropriate, for waste designation.

Table B.5. 200 Area Effluent Treatment Facility Generated Waste - Sampling and Analysis

Parameter[1]	Rationale
• Total solids or percent water[2]	• Calculate dry weight concentrations
• Volatile organic compounds[3]	• LDR - verify treatment standards
• Semivolatile organic compounds[3]	• LDR - verify treatment standards
• Metals (arsenic, barium, cadmium, chromium, lead, mercury, selenium, silver)	• Waste designation • LDR - verify treatment standards
• Cation and anions of concern	• Address receiving TSD unit waste acceptance requirements
• pH	• Waste designation

1 For influent and concentrate tank samples, the total sample (solid plus liquid) is analyzed and the analytical result is expressed on a dry weight basis. The result for toxicity characteristic metal and organic is divided by a factor of 20 and compared to the toxicity characteristic (TC) constituent limits [WAC 173-303-090(8)]. If the TC limit is met or exceeded, the waste is designated accordingly. All measured parameters are compared against the corresponding treatment standards.

2 Total solids or percent water are not determined for unknown waste and dry powder waste samples and are analyzed in maintenance waste and sludge samples, as appropriate (i.e., percent water might not be required for such routine maintenance waste as aerosol cans, fluorescent tubes, waste oils, batteries, etc., or sludge that has dried).

3 VOC and/or SVOC analysis of secondary waste is required unless influent characterization data and knowledge indicate that the constituent will not be in the final secondary waste at or above the LDR.

LDR = land disposal restrictions

TSD = treatment, storage, and/or disposal

B.7 Quality Assurance/Quality Control

The following quality assurance/quality control (QA/QC) plan for LERF and 200 Area ETF is provided as required by WAC 173-303-810(6) and follows the guidelines of EPA QA/G-5.

B.7.1 Project Management

The following sections address project administrative functions and approaches.

B.7.1.1.1 Project Organization

Overall management of the LERF/200 Area ETF is performed by the Facility Manager, who is responsible for safe operation of the facility, including implementation of this QA/QC plan and compliance with applicable permits and regulations. The Facility Manager also provides retention of project records in accordance with this plan. Assisting the Facility Manager is an Environmental Field Representative that monitors compliance, reviews new requirements and regulations, and interfaces with EPA and Ecology. Also assisting the Facility Manager is a QA representative who is responsible for implementing the QA program at the facility.

Reporting to the Facility Manager are several support groups. The Operations group consists of trained personnel who operate the plant, including operators performing sampling activities such as collection, packaging, and transportation of samples to the laboratory. The Maintenance group is responsible for performing calibrations and preventative maintenance on facility equipment, including pH, conductivity, and flow meters required by environmental permits. The Engineering group monitors the process with online instruments and sampling for process control. The Engineering group also performs waste acceptance, and environmental compliance activities, including scheduling sampling, generating data forms, and reviewing data.

B.7.1.2 Special Training

Individuals involved in sampling, analysis, and data review will be trained and qualified to implement safely the activities addressed in this WAP and QA/QC plan. Training will conform to the training requirements specified in WAC 173-303-330 and the LERF/200 Area ETF Dangerous Waste Training Plan (Addendum F). Training records will be maintained in accordance with Section B.7.1.3 of this WAP.

B.7.1.3 Documentation and Records

Sample records are documented as part of the Hanford Facility Operating Record, LERF and 200 Area ETF File pursuant to Permit Condition II.I. These documents and records include the following:

- Training
- Chains of Custody for all regulatory sampling performed by LERF and 200 Area ETF
- Data Summary Reports
- QA/QC reports
- Assessment reports
- Instrument inspection, maintenance, and calibration logs

B.7.2 Data Quality Parameters and Criteria

Data quality parameters are listed by EPA QA/G-5S, *Guidance for Choosing a Sampling Design for Environmental Data Collection* as:

- Purpose of Data Collection (e.g. determining if a parameter exceeds a threshold level).
- Spatial and Temporal Boundaries of Study.
- Preliminary Estimation of Sample Support (volume that each sample represents).

- Statistical Parameter of Interest (e.g. mean, percentile, percentage).
- Limits on Decision Error/Precision (e.g. false acceptance error, false rejection error).

The parameters for the first four bullets (limits, sample points, frequency of samples, etc.) are already established in the permits, delisting petition, and this WAP. The focus of this QA/QC plan is on limits on decision error/precision.

The data quality parameters were chosen to ensure Limits on Decision Error/Precision are appropriate for purposes of using the data to demonstrate compliance with permits, delisting exclusion limits, and this WAP. The principal quality parameters are precision, accuracy, representativeness, comparability, and completeness. Secondary data parameters of importance include sensitivity and detection levels. The data quality parameters and the data acceptance criteria are discussed below.

B.7.2.1 Precision

Precision is a measure of agreement among replicate measurements of the same property, under prescribed similar conditions. Precision is expressed in terms of the relative percent difference (RPD) for duplicate measurements. QA/QC sample types that test precision include field and laboratory duplicates and spike duplicates. The RPDs for laboratory duplicates and/or matrix spike duplicates will be routinely calculated.

$$\text{RPD} = (100)\textit{absolute value of}\left(\frac{\text{sample result} - \text{duplicate sample result}}{\text{average of sample result} + \text{duplicate sample result}}\right)$$

Matrix spike duplicates are replicates of matrix spike samples that are analyzed with every analytical batch that contains an ETF treated effluent sample. The precision of the analytical methods are estimated from the results of the matrix spike (MS) and the matrix spike duplicate (MSD) for selected analytes. Matrix spike analyses cannot be performed for certain analytical methods, including conductivity, pH, and total dissolved solids. Duplicate analyses are used to determine the RPD for these methods. The precision acceptance criteria are specified in Table B.6.

B.7.2.2 Accuracy

Accuracy assesses the closeness of the measured value to an accepted reference value. Accuracy of analytical results is typically assessed using matrix spikes. A matrix spike is the addition of a known amount of the analyte to the sample matrix being analyzed. Accuracy is expressed as a percent recovery of the spiked samples.

$$\text{Percent Recovery} = 100\left(\frac{\text{matrix spike sample result} - \text{sample result}}{\text{spiked amount}}\right)$$

Matrix spike analyses cannot be performed on certain analytical methods, including conductivity, pH, and total dissolved solids. The percent recovery for the laboratory control standard samples demonstrates that these methods are working properly and gives an estimate of the method's accuracy. The percent recovery will be routinely calculated.

Accuracy criteria are established to provide confidence that the result is below the action level. Therefore the closer the result is to the action level the higher the degree of accuracy needed. The upper and lower accuracy acceptance criteria are specified in Table B.6. The criteria are reasonable values based on previous analysis of constituents in the delisting exclusion, or similar constituents.

B.7.2.3 Representativeness

Representativeness expresses the degree to which data accurately and precisely represent selected characteristics of a parameter at a sampling point or process condition.

Because of the matrix being analyzed, dilute aqueous solution, it is not expected that representativeness will be of concern, except when there are potential for changes to process conditions such as the facility influent concentrations or waste processing strategy. Sampling due to these changes in process conditions is addressed in Section B.6.1.3 of this WAP.

The representativeness of a sample may be compromised by the presence of contaminants introduced in the field or the laboratory. To determine if contamination may be present, a blank sample of reagent water is analyzed. A method blank is performed by the laboratory on every batch of 20 samples being analyzed at the same time. The presence of a constituent in the sample and the blank sample indicates contamination has occurred.

B.7.2.4 Completeness

Completeness is a measure of the amount of valid data obtained from a measurement system, expressed as a percentage of the number of valid measurements that were planned to be collected. Lack of completeness is sometimes caused by loss of a sample, loss of data, or inability to collect the planned number of samples. Incompleteness also occurs when data are discarded because they are of unknown or unacceptable quality. Since most regulatory sampling events performed by LERF/200 Area ETF involve a single sample, all analysis must be complete and valid.

B.7.2.5 Comparability

Comparability is the confidence with which one data set can be compared to another. Comparability is achieved by using sampling and analytical techniques, which provide for measurements that are consistent and representative of the media and conditions measured. In laboratory analysis, the term comparability focuses on method type, holding times, stability issues, and aspects of overall analytical quantitation.

B.7.2.6 Sensitivity and Detection Levels

Sensitivity is the measure of the concentration at which an analytical method can positively identify and report analytical results. Sensitivity represents the maximum value for a detection level that will reasonably assure the results are below the established limits. The analytical method selected by LERF/200 Area ETF should have a detection level for each constituent that is below the sensitivity. The preferred detection level is the practical quantitation limit (PQL), which is lowest concentration that can be reliably measured during routine laboratory conditions. If the method PQL cannot meet the sensitivity for some constituents, the minimum concentration or attribute that can be measured by a method (method detection limit) or by an instrument (instrument detection limit) may be used. The sensitivity levels, specified in Table B.6, are derived from the delisting limits, water discharge limits, and uncertainty values, which are based on the required precision and accuracy for each constituent.

B.7.3 Data Generation and Acquisition

The following section addresses QA requirements for data generation and acquisition.

B.7.3.1 Sampling Method

LERF/200 Area ETF samples required by the permits and delisting are collected as grab samples. Sampling for the purpose of waste designation of secondary waste is performed using grab, composite, thief, scoop, or composite liquid waste sampler (COLIWASA). The selection of the sample collection device depends on the type of sample, the sample container, the sampling location, and the nature and distribution of the waste components. In general, the methodologies used for specific materials correspond to those referenced to WAC 173-303-110(2). The selection and use of the sampling device is supervised or performed by a person thoroughly familiar with the sampling requirements.

The following protocol applies to all sampling methods:

- All containers will be filled within as short a time period as reasonably achievable.

- Volatile Organic Analysis (VOA) sample containers will be filled first, and prior to any subdividing of a composited sample.
- VOA samples consisting of a set of two or more sample containers will be filled sequentially. The sample containers are considered equivalent and given identical sampling times.
- All VOA sample containers must have no headspace and be free of trapped air bubbles.
- Grab sample protocol includes:
  - Sample lines should be as short as reasonably achievable and free of traps and pockets in which solids might settle.
  - The sample line should be flushed before sampling with a minimum volume equivalent to three times the sample line volume.
  - Contamination to the sample from contact with the internal and external surfaces of the tap should be minimized.

Thief and COLIWASA samplers are used to sample liquid waste containers such as drums. Scoop samplers are used to sample powder waste generated in the thin-film dryer. Sample requirements for these samples include:

- Thief or COLIWASA sampler, the sampler should be lowered into the liquid slowly so the level of the liquid inside and outside the sampler tube remain about the same.
- When lifting the thief or COLIWASA sampler from the solution, the outside should be wiped down, or the excess water allowed to drip off, before filling the sample container.

B.7.3.2 Sample Handling, Custody, and Shipping

The proper handling of sample bottles after sampling is important to ensure the samples are free of contamination and to demonstrate the samples have not been tampered with.

B.7.3.2.1 Chain-of-Custody

Evidence of collection, shipment, receipt at the laboratory, and laboratory custody until disposal will be documented using a chain-of-custody form. The chain-of-custody form will, as a minimum identify sample identification number, sampling date and time, sampling location, sample bottle type and number, analyses to be performed, and preservation method.

The operations person who signs as the collector on the chain of custody is the first custodian of the samples. A custodian must maintain continuous custody of sample containers at all times from the time the sample is taken until delivery to the laboratory or until delivery to a common carrier for shipment to an off-site location. Custody is maintained by any of the following:

- The custodian has the samples in view, or has placed the samples in locked storage, or keeps the samples within a secured area (e.g., controlled by authorized personnel only), or has applied a tamper-indicating device, such as evidence tape, to the sample containers or shipping containers.
- The custodian has taken physical possession of the samples or the shipping containers sealed with an intact tamper-indicating device, such as evidence tape.

B.7.3.2.2 Sample Preservation, Containers, and Holding Time

Table B.6 lists the sample container, preservation method, and holding time requirements for different types of analyses. These parameters are based on the requirements of 40 CFR 136, Table II.

B.7.3.3 Instrument Calibration and Preventive Maintenance

LERF/200 Area ETF uses instruments to monitor operations and meet regulatory requirements. This includes continuous pH and conductivity monitors required by facility permits and delisting. All instruments are calibrated according to frequencies and tolerances established by the LERF/200 Area ETF engineering group.

Calibrations and other maintenance actions are scheduled and tracked by LERF/200 Area ETF maintenance group using a preventive maintenance database. Measuring and test equipment used for instrument calibration is controlled, calibrated at specified intervals, and maintained to establish accuracy limits.

B.7.4 Assessment and Oversight

Quality programs can only be effective if meaningful assessments are performed to monitor and respond to issues associated with program performance. Routine assessment of data is performed as part of the validation process discussed in Section B.7.5.1.

B.7.4.1 Assessments and Response

Management assessments are conducted by first line management and subject matter experts, focusing on procedural adequacy, compliance, and overall effectiveness of the program. Management assessments of the sample program typically include the LERF and 200 Area ETF QA representative. Each management assessment has a performance objective or lines of inquiry. Examples may include personnel training, proper performance of sample custody, or completeness of sampling records.

B.7.4.2 Reports to Management

Results of performance assessments, including any issues identified, are provided to the LERF and 200 Area ETF Facility Manager in a written report. The Facility Manager is responsible to correct all findings from the report.

B.7.5 Verification and Validation of Analytical Data

The data verification and validation processes will ensure that the data resulting from the selected analytical method are consistent with requirements specified in this QA/QC plan.

B.7.5.1 Data Verification

The primary data reporting will be by electronic data systems. Data verification will be performed on laboratory data packages that support environmental compliance to ensure that their content is complete and in order. A review of the data package will be performed to ensure that:

- The data package contains the required technical information.
- Deficiencies are identified and documented.
- Identified deficiencies are corrected by the laboratory and the appropriate revisions are made.
- Deficient pages are replaced with the laboratory corrections.
- A copy of the completed verification report is placed in the data file.

B.7.5.2 Data Validation

Data validation ensures that the data resulting from analytical measurements meet the quality requirements specified in the QA/QC plan. Data validation will be performed on data packages that support environmental compliance.

The following are included in data validation:

- Chain-of-Custody (COC) – Verify the COC shows unbroken custody from sampling through receipt at the laboratory.
- Request analysis – Review the sample results to verify the requested analysis was performed. If an alternate method was used, verify permit-required detection limits were met.
- Holding times – Review the sample results to verify the analyses were performed within required holing times and where applicable, extraction times.

- Blank – Review the results of trip, field, and equipment blank samples to verify the sample results are not compromised by contamination.
- Laboratory QC – Verify the laboratory QC was completed and there are no outstanding problems

B.8 References

DOE/RL-92-72, *200 Area Effluent Treatment Facility Delisting Petition,* Revision 1, 1993, U.S. Department of Energy, Richland Operations Office, Richland, Washington.

Ecology, 1996b, *The Washington State Department of Ecology (Ecology) Regulatory Interpretation of the Liquid Effluent Retention Facility (LERF) Land Disposal Restriction Exemption*, letter from Washington State Department of Ecology to T. Teynor, U.S. Department of Energy and A. DiLiberto, Westinghouse Hanford Company, September 9, 1996.

Ecology, 2000, *Washington State Waste Discharge Permit* (No. ST 4500), as amended, for 200 Area Effluent Treatment Facility, Hanford Facility, Washington State Department of Ecology, Olympia, Washington, August 1, 2000.

EPA, 1983, *Methods for Chemical Analysis of Water Wastes*, EPA-600/4-79/020, (as amended), National Exposure Research Laboratory, Cincinnati, Ohio

EPA, *Test Methods for Evaluating Solid Waste Physical/Chemical Methods, SW-846* (Third Edition, as amended), U.S. Environmental Protection Agency, Office of Solid Waste and Emergency Response, Washington, D.C.

EPA, 1994, *Liquid Effluent Retention Facility (LERF) Land Disposal Restrictions Treatment Exemption-Regulatory Interpretation EPA/Ecology ID No: WA7890008967*, letter from U.S. Environmental Protection Agency, Region 10 to J. Hennig, U.S. Department of Energy, December 6, 1994.

EPA, 2005, 200 Area ETF Delisting [Exclusion], issued to U.S. Department of Energy, 40 CFR 261, Appendix IX, Table 2 (70 FR 44496, August 3, 2005), U.S. Environmental Protection Agency, Washington, D.C.

B.9 Analytical Methods, Sample Containers, Preservative Methods, and Holding Times

Table B.6. Sample and Analysis Criteria for Influent Aqueous Waste and Treated Effluent

Parameter	Analytical Method[1]	Method PQL Sensitivity[2]	Accuracy/ Precision for Method[3] (percent)	Sample container[4]/ Preservative[4]/ Holding time[5]
VOLATILE ORGANIC COMPOUNDS				
Acetone	SW-846 8260	40	60-120 / 20	Sample container 3 x 40-mL amber glass with septum Preservative HCl to pH<2; 4°C Holding time 14 days
Acetonitrile		820	60-120 / 20	
Benzene		5	60-120 / 20	
1-Butanol		1600	60-120 / 20	
Carbon Disulfide		1500	60-120 / 20	
Carbon tetrachloride		5	60-120 / 20	
Chloroform		5	50-130 / 20	
Methylene chloride		5	50-150 / 20	
Tetrachloroethylene		5	65-140 / 20	
Tetrahydrofuran		100	60-120 / 20	
SEMIVOLATILE ORGANIC COMPOUNDS				
Acetophenone	SW-846 8270	10	70-110 / 25	Sample container 4 x 1-liter amber glass Preservative 4°C Holding time 7 days for extraction; 40 days for analysis after extraction
Carbazole		110	50-120 / 25	
p-Chloroaniline		76	50-120 / 25	
Chrysene		350	50-120 / 25	
Cresol (o, p, m)		760	50-120 / 25	
Di-n-octyl phthalate		300	50-120 / 25	
Diphenylamine		350	50-120 / 25	
Hexachlorobenzene		2	50-120 / 25	
Hexachlorocyclopentadiene		110	50-120 / 25	
Isophorone		2600	50-120 / 25	
Lindane (gamma-BHC)		1.9	50-120 / 25	
N-nitrosodimethylamine		12	50-120 / 25	
Pyridine		15	50-120 / 25	

Table B.6. Sample and Analysis Criteria for Influent Aqueous Waste and Treated Effluent				
Parameter	**Analytical Method[1]**	**Method PQL Sensitivity[2]**	**Accuracy/ Precision for Method[3] (percent)**	**Sample container[4]/ Preservative[4]/ Holding time[5]**
Tributyl phosphate		76	50-120 / 25	
2.4.6-Trichlorophenol		230	50-120 / 25	
POLYCHLORINATED BIPHENYLs (PCBs)				
Aroclor-1016	SW-846 8082	0.4	50-110 / 25	Sample container 4 x 1-liter amber glass Preservative 4°C Holding time 1 year for extraction; 1 year for analysis after extraction
Aroclor-1221		0.4	50-110 / 25	
Aroclor-1232		0.4	50-110 / 25	
Aroclor-1242		0.4	50-110 / 25	
Aroclor-1248		0.4	50-110 / 25	
Aroclor-1254		0.4	50-110 / 25	
Aroclor-1260		0.4	50-110 / 25	
TOTAL METALS				
Arsenic	EPA-600 200.8	11	70-130 / 20	Sample container 1 x 0.5-liter plastic/glass Preservative 1:1 HNO_3 to pH<2 Holding time 180 days; mercury 28 days
Cadmium		5	70-130 / 20	
Chromium		20	70-130 / 20	
Copper		70	70-130 / 20	
Lead		10	70-130 / 20	
Mercury		2	70-130 / 20	
Selenium		20	70-130 / 20	
Barium	SW-846 6010/ EPA-600 200.7	1200	75 - 125 / 20	
Beryllium		34	75 - 125 / 20	
Calcium		200	75 - 125 / 20	
Iron		100	75 - 125 / 20	
Magnesium		400	75 - 125 / 20	
Nickel		340	75 - 125 / 20	
Potassium		10,000	75 - 125 / 20	
Silicon		580	75 - 125 / 20	
Silver		83	75 - 125 / 20	
Sodium		2500	75 - 125 / 20	
Vanadium		120	75 - 125 / 20	

Table B.6. Sample and Analysis Criteria for Influent Aqueous Waste and Treated Effluent

Parameter	Analytical Method[1]	Method PQL Sensitivity[2]	Accuracy/ Precision for Method[3] (percent)	Sample container[4]/ Preservative[4]/ Holding time[5]
Zinc		5100	75 - 125 / 20	
GENERAL CHEMISTRY				
Chloride	EPA-600 300.0	1000	70-130 / 20	Sample container 1 x 60-mL plastic/glass Preservative 4°C Holding time 28 days; nitrate and nitrite 48 hours
Fluoride		880	70-130 / 20	
Formate		1250	70-130	
Nitrate (as N)		100	70-130 / 20	
Nitrite (as N)		100	70-130 / 20	
Phosphate		1500	70-130 / 20	
Sulfate		10,000	70-130 / 20	
Ammonia (as N)	EPA-600, 300.7	40	70-130 / 20	Sample container 1 x 50-mL glass or plastic Preservative H_2SO_4 to pH<2; 4°C Holding time 28 days
Cyanide	EPA-600 335.2/335.3	350	70-130 / 20	Sample container 1 x 250-mL glass or plastic Preservative NaOH to pH>12; 4°C Holding time 14 days
Alkalinity	EPA-600 310.1/310.2	ND	ND	Sample container 1 x 50-mL glass or plastic Preservative 4°C Holding time 14 days
Total dissolved solids	EPA-600 160.1	ND	ND	Sample container 1 x 500-mL glass or plastic Preservative 4°C Holding time 7 days

Table B.6. Sample and Analysis Criteria for Influent Aqueous Waste and Treated Effluent

Parameter	Analytical Method[1]	Method PQL Sensitivity[2]	Accuracy/ Precision for Method[3] (percent)	Sample container[4]/ Preservative[4]/ Holding time[5]
Total suspended solids	EPA-600 160.2	ND	ND	Sample container 1 x 1-L glass or plastic Preservative 4°C Holding time 7 days
Specific conductivity	EPA-600 120.1 (in lab)	ND	ND	Sample container 1 x 50-mL glass or plastic Preservative 4°C Holding time 28 days
pH[7]	EPA-600 150.1	ND	ND	Sample container 1 x 60-mL glass or plastic Preservative None Holding time Analyze immediately
Total organic carbon	SW-846 9060	ND	ND	Sample container 1 x 250-mL amber glass Preservative H_2SO_4 to pH<2; 4°C Holding time 28 days

[1]SW-846 or EPA-600 methods are presented unless otherwise noted. Other methods might be substituted if the applicable PQL can be met.

[2]ST-4500 required method PQL or Delisting Exclusion condition 2 report sensitivity/detection level, whichever is lower. Units are parts per billion unless otherwise noted.

[3]Accuracy/precision used to confirm or re-establish MDL

[4]Sample bottle, volumes, and preservatives could be adjusted, as applicable, for safety reasons

[5]Holding time = time between sampling and analysis

[7]pH monitored in influent aqueous waste only

L = liter
mL = milliliter
NA = not applicable
ND = not determined
MDL = method detection level
PQL = practical quantitation limit
RL = reporting limit

Table B.7. Sample Containers, Preservative Methods, and Holding Times for 200 Area ETF Generated Waste				
Parameter	**Analytical Method**	**Method PQL**	**Accuracy/ Precision for Method (percent)**	**Sample container[1]/ Preservative[1]/ Holding time[2]**
Liquid Matrix				
For methods other than total solids, analyze using the methods and QA/QC in Table B.6. For each method, analyze the target compound list				
Total solids	EPA-600 160.3	ND	ND	Sample container 1 x 500-mL glass or plastic Preservative – 4°C Holding time –7 days
Solid Matrix				
Volatile organic compounds (combined method target compound lists)	SW-846 8260	Refer to Table B.6	Refer to Table B.6	Sample container 1 x 40-mL amber glass with septum Preservative –4°C Holding time –14 days
Semivolatile organic compounds (method target compound list)	SW-846 8270	Refer to Table B.6	Refer to Table B.6	Sample container 1 x 125-mL amber glass Preservative –4°C Holding time –14 days for extraction; 40 days for analysis after extraction
PCBs (method target compound list)	SW-846 8082	Refer to Table B.6	Refer to Table B.6	Sample container Amber glass – 50 g of sample Preservative –4°C Holding time –1 year for extraction; 1 year for analysis after extraction
RCRA Metals (method target compound list)	EPA-600 200.8	Refer to Table B.6	Refer to Table B.6	Sample container glass or plastic – 10 g of sample Preservative –none, mercury 4°C Holding time –180 days; mercury 28 days
Total Metals (method target compound list)	SW-846 6010	Refer to Table B.6	Refer to Table B.6	
Anions (method target compound list)	EPA-600 300.0	Refer to Table B.6	Refer to Table B.6	Sample container glass or plastic –25 g of sample Preservative –none Holding time –6 months for extraction; 28 days for analysis after extraction, nitrate and nitrite 48 hours for analysis after extraction

Table B.7. Sample Containers, Preservative Methods, and Holding Times for 200 Area ETF Generated Waste

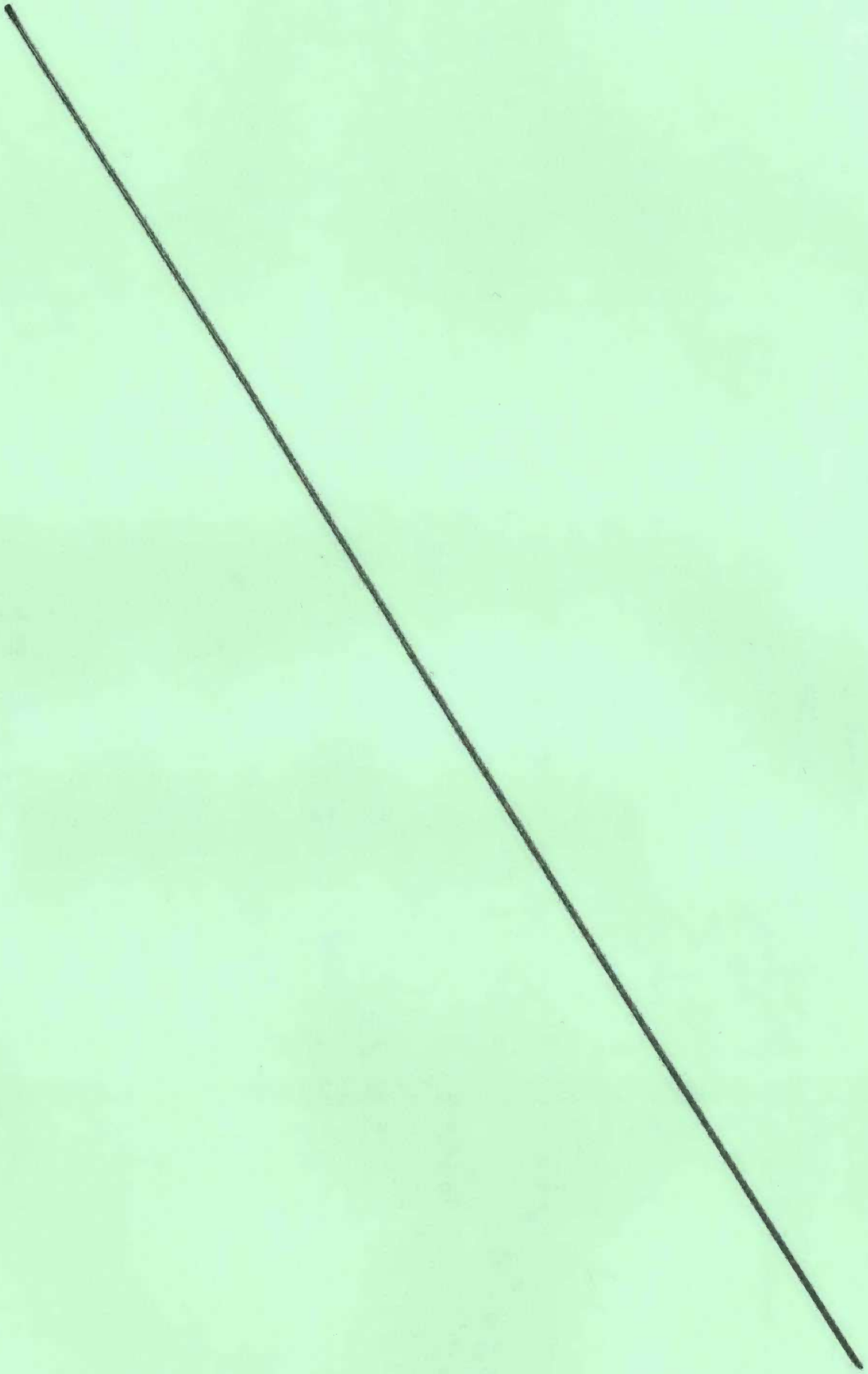
Parameter	Analytical Method	Method PQL	Accuracy/ Precision for Method (percent)	Sample container[1]/ Preservative[1]/ Holding time[2]
Ammonia	EPA-600 300.7	Refer to Table B.6	Refer to Table B.6	Sample container glass or plastic – 25 g of sample Preservative –none Holding time –6 months for extraction; 28 days for analysis after extraction
pH	SW-846 9045	ND	ND	Sample container glass or plastic – 50 g of sample Preservative –none Holding time –none
Toxicity Characteristic Leaching Procedure[3]	SW-846 1311	NA	NA	Sample container Refer to specific method being performed after TCLP – 125 g of sample Preservative –None (after TCLP, preserve extract per method being performed) Holding time –Metals: 180 days for TCLP extraction, mercury 28 days for TCLP extraction SVOA: 14 days for TCLP extraction (after TCLP, refer to specific methods for time for analysis after extraction)

[1] Sample bottle, volumes, and preservatives could be adjusted, as applicable, for safety reasons

[2] Holding time equals time between sampling and analysis

[3] Extraction procedure, as applicable; extract analyzed by referenced methods [WAC 173-303-110(3)(c)]

g = grams
NA = not applicable
PQL = practical quantitation limit
mL = milliliter
ND = not determined
TCLP = toxicity characteristic leaching procedure



ADDENDUM C
PROCESS INFORMATION

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ADDENDUM C
PROCESS INFORMATION

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C. PROCESS INFORMATION

This addendum provides a detailed discussion of the Liquid Effluent Retention Facility (LERF) and 200 Area Effluent Treatment Facility (200 Area ETF) processes and equipment. The LERF and 200 Area ETF comprise an aqueous waste treatment system located in the 200 East Area that provides storage and treatment for a variety of aqueous mixed waste. This aqueous waste includes process condensate from the 242-A Evaporator and other aqueous waste generated from onsite remediation and waste management activities.

The LERF consists of three lined surface impoundments, or basins. Aqueous waste from LERF is pumped to the 200 Area ETF for treatment in a series of process units, or systems, that remove or destroy essentially all of the dangerous waste constituents. The treated effluent is discharged to a State-Approved Land Disposal Site (SALDS) north of the 200 West Area, under the authority of a Washington State Waste Discharge Permit (Ecology 2000) and the Final Delisting (40 CFR 261, Appendix IX, Table 2).

Both LERF and 200 Area ETF waste processing operations are controlled in a central Control room located in the 200 Area ETF building. The ETF control room is staffed continuously during ETF processing operations. Processing operations are defined as when liquid transfers of any sort are occurring to/from/within the LERF and ETF or when wastes are being treated at ETF[1]. Examples of processing operations include, but are not limited to, when liquid waste are transferred to/from the LERF basins [see section C.1], during active liquid waste treatment/processing at the ETF (e.g., liquid waste treatment in tanks and liquid waste movement between primary and secondary treatment train processes and/or other ETF tanks [see Section C.2]), and liquid waste receipts at the load-in station [see Section C.2.1]). Section C.2.5.1 describes the centralized computer system (i.e., monitor and control system or MCS) that is located at the ETF control room and other locations at the ETF facility. This MCS monitors the performance of the ETF operations and records alarms from various equipment as described in this Addendum C and Addendum I, Inspection Requirements. At times when processing operations are not occurring, the ETF Control room is not manned continuously, and alarms are monitored daily as specified in Addendum I.

C.1 Liquid Effluent Retention Facility Process Description

Each of the three LERF basins has an operating capacity of 29.5-million liters. The LERF receives aqueous waste through several inlets including the following:

- A pipeline that connects LERF with the 242-A Evaporator.
- A pipeline from the 200 West Area.
- A pipeline that connects LERF to the Load-In Station at the 200 Area ETF.
- A series of sample ports located at each basin.

Figure C.1 presents a general layout of LERF and associated pipelines. Aqueous waste from LERF is pumped to the 200 Area ETF through one of two double-walled fiberglass transfer pipelines. Effluent from the 200 Area ETF also can be transferred back to the LERF through one of these transfer pipelines. These pipelines are equipped with leak detection located in the annulus between the inner and outer pipes. In the event that these leak detectors are not in service, the pipelines are visually inspected during transfers for leakage by opening the secondary containment drain lines located at the 200 Area ETF end of the transfer pipelines.

Each basin is equipped with six available sample risers constructed of 6-inch perforated pipe. A seventh sample riser in each basin is dedicated to influent aqueous waste receipt piping (except for aqueous waste received from the 242-A Evaporator), and an eighth riser in each basin contains liquid level instrumentation.

[1] *Liquid transfers does not include standard facility operations of liquid recirculation (e.g. for pump seals), sanitary water and cooling water, and outdoor rainwater management activities.*

Each riser extends along the sides of each basin from the top to the bottom of the basin and allows samples to be collected from any depth. Personnel access to these sample ports is from the perimeter area of the basins. A catch basin is provided at the northwest corner of each LERF basin for aboveground piping and manifolds for transfer pumps. Aqueous waste from the 242-A Evaporator is transferred through piping which ties into piping at the catch basins. Under routine operations, a submersible pump is used to transfer aqueous waste from a LERF basin to the 200 Area ETF for processing or for basin-to-basin transfers. This pump is connected to a fixed manifold on one of four available risers.
Each basin consists of a multilayer liner system supported by a concrete anchor wall around the basin perimeter and a soil-bentonite clay underlayment. The multilayer liner system consists of a primary liner in contact with the aqueous waste, a layer of bentonite carpet, a geonet, a geotextile, a gravel layer, and a secondary liner that rests on the bentonite underlayment. Any aqueous waste leakage through the primary liner flows through the geonet and gravel to a leachate collection system. The leachate flows to a sump at the northwest corner of each basin, where the leachate is pumped up the side slope and back into the basin above the primary liner. Each liner is constructed of high-density polyethylene. A floating cover made of very low-density polyethylene is stretched over each basin above the primary liner. These covers serve to keep unwanted material from entering the basins, and to minimize evaporation of the liquid contents.

C.2 Effluent Treatment Facility Process Description

The 200 Area ETF is designed as a flexible treatment system that provides treatment for contaminants anticipated in process condensate and other onsite aqueous waste. The design influent flow rate into the 200 Area ETF is approximately 570 liters per minute, with planned outages for activities such as maintenance on the 200 Area ETF systems. Maintenance outages typically are scheduled between treating a batch of aqueous waste, referred to as treatment campaigns. The effluent flow (or volume) is equivalent to the influent flow (or volume).

The 200 Area ETF generally receives aqueous waste directly from the LERF. However, aqueous waste also can be transferred from tanker trucks at the Load-In Station to the 200 Area ETF and from containers (e.g., carboys, drums) directly to ETF. Aqueous waste is treated and stored in the 200 Area ETF process areas in a series of tank systems, referred to as process units. Within the ETF, waste also is managed in containers through treatment and/or storage. Figure C.2 provides the relative locations of the process and container storage areas within the ETF.

The process units are grouped in either the primary or the secondary treatment train. The primary treatment train provides for the removal or destruction of contaminants. Typically, the secondary treatment train processes the waste by-products from the primary treatment train by reducing the volume of waste. In the secondary treatment train, contaminants are concentrated and dried to a powder. The liquid fraction is routed to the primary treatment train. Figure C.3 provides an overview of the layout of the ETF, 2025E Building). Figure C.4 presents the 200 Area ETF floor plan, the relative locations of the individual process units and associated tanks within the ETF, and the location of the Load-In Station.

The dry powder waste and maintenance and operations waste are containerized and stored or treated in the container storage areas or in collection or treatment areas within the Process Area. Secondary containment is provided for all containers and tank systems (including ancillary equipment) housed within the ETF. The trenches and floor of the 200 Area ETF comprise the secondary containment system. The floor includes approximately a 15.2-centimeter rise (berm) along the containing walls of the process and container storage areas. Any spilled or leaked material from within the process area or container storage area is collected into trenches that feed into either sump tank 1 or sump tank 2.

From these sump tanks, the spilled or leaked material (i.e., waste) is fed to either the surge tank and processed in the primary treatment train or the secondary waste receiving tanks and processed in the secondary treatment train. All tank systems outside of the 200 Area ETF are provided with a secondary containment system.

In the following sections, several figures are provided that present general illustrations of the treatment units and the relation to the process.

C.2.1 Load-In Station

The 200 Area ETF receives aqueous waste from LERF or the Load-In Station. The 200 Area ETF Load-In Station, located due east of the surge tank and outside of the perimeter fence (Figure C.4), was designed and constructed to provide the capability to unload, store, and transfer aqueous waste to the LERF or 200 Area ETF from tanker trucks and other containers (such as drums). The Load-In Station consists of two truck bays equipped with load-in tanks, transfer pumps, filtration system, level instrumentation for tanker trucks, leak detection capabilities for the containment basin and transfer line, and an underground transfer line that connects to lines in the surge tank berm, allowing transfers to either the 200 Area ETF surge tank or LERF. The Load-In Station is covered with a steel building for weather protection. Tanker trucks and other containers are used to unload aqueous waste at the Load-In Station. To perform unloading, the tanker truck is positioned on a truck pad, a 'load-in' transfer line is connected to the truck, and the tanker contents are pumped into one of the Load-In Station tanks, the surge tank, or directly to the LERF. For container unloading, the container is placed on the truck pad and the container contents are pumped into one of the Load-In Station tanks, the surge tank, or directly to the LERF.

During unloading operations, solids may be removed from the waste by pumping the contents of the tanker truck or container through a filtration system. If solids removal is not needed, the filtration system is not used and the solution is transferred directly to the Load-In Station tanks, surge tank, or to LERF.

Any leaks at the Load-In Station drain to the sump. A leak detector in the sump alarms locally and in the 200 Area ETF control room. Alarms are monitored continuously in the control room during Load-in Station transfers and at least daily at times when waste is not being received at the Load-In Station. Alternatively, leaks can be visually detected.

C.2.2 Effluent Treatment Facility Operating Configuration

Because the operating configuration of the 200 Area ETF can be adjusted or modified, most aqueous waste streams can be effectively treated to below Delisting and Discharge Permit limits. The operating configuration of the 200 Area ETF depends on the unique chemistry of an aqueous waste stream(s). Before an aqueous waste stream is accepted for treatment, the waste is characterized and evaluated. Information from the characterization is used to adjust the treatment process or change the configuration of the 200 Area ETF process units, as necessary, to optimize the treatment process for a particular aqueous waste stream.

Typically, an aqueous waste is processed first in the primary treatment train, where the 200 Area ETF is configured to process an aqueous waste through the UV/OX unit first, followed by the RO unit. However, under an alternate configuration, an aqueous waste could be processed in the RO unit first. For example, high concentrations of nitrates in an aqueous waste might interfere with the performance of the UV/OX. In this case, the 200 Area ETF could be configured to process the waste in the RO unit before the UV/OX unit.

The flexibility of the 200 Area ETF also allows some aqueous waste to be processed in the secondary treatment train first. For example, for small volume aqueous waste with high concentrations of some anions and metals, the approach could be to first process the waste stream in the secondary treatment train. This approach would prevent premature fouling or scaling of the RO unit. The liquid portion (i.e., untreated overheads from the 200 Area ETF evaporator and thin film dryer) would be sent to the primary treatment train.

Figure C.5 and Figure C.6 provide example process flow diagrams for two different operating configurations.

C.2.3 Primary Treatment Train

The primary treatment train consists of the following processes:

- Influent Receipt/Surge tank - inlet, surge capacity.
- Filtration - for suspended solids removal.

- UV/OX - organic destruction.
- pH adjustment - waste neutralization.
- Hydrogen peroxide decomposition - removal of excess hydrogen peroxide.
- Degasification - removal of carbon dioxide.
- RO - removal of dissolved solids.
- IX - removal of dissolved solids.
- Verification - holding tanks during verification.

Influent Receipt/Surge Tank. Depending on the configuration of the ETF, the surge tank is one inlet used to feed an aqueous waste into the 200 Area ETF for treatment. In Configuration 1 (Figure C.5), the surge tank is the first component downstream of the LERF. The surge tank provides a storage/surge volume for chemical pretreatment and controls feed flow rates from the LERF to the 200 Area ETF. However, in Configuration 2 (Figure C.6), aqueous waste from LERF is fed directly into the treatment units. In this configuration, the surge tank receives aqueous waste, which has been processed in the RO units, and provides the feed stream to the remaining downstream process units. In yet another configuration, some small volume aqueous waste could be received into the secondary treatment train first for processing. In this case, the aqueous waste would be received directly into the secondary waste receiving tanks. Finally, the surge tank also receives waste extracted from various systems within the primary and secondary treatment train while in operation.

The surge tank is located outside the 200 Area ETF on the south side. In the surge tank (Figure C.7), the pH of an aqueous waste is adjusted using the metered addition of sulfuric acid and sodium hydroxide, as necessary, to prepare the waste for treatment in downstream processes. In addition, hydrogen peroxide or biocides could be added to control biological growth in the surge tank. A pump recirculates the contents in the surge tank, mixing the chemical reagents with the waste to a uniform pH.

Filtration. Two primary filter systems remove suspended particles in an aqueous waste: a rough filter removes the larger particulates, while a fine filter removes the smaller particulates. The location of these filters depends on the configuration of the primary treatment train. However, the filters normally are located upstream of the RO units.

The solids accumulating on these filter elements are backwashed to the secondary waste receiving tanks with pulses of compressed air and water, forcing water back through the filter. The backwash operation is initiated either automatically by a rise in differential pressure across the filter or manually by an operator. The filters are cleaned chemically when the backwashing process does not facilitate acceptable filter performance.

Auxiliary fine and rough filters (e.g., disposable filters) have been installed to provide additional filtration capabilities. Depending on the configuration of the ETF, the auxiliary filters are operated either in series with the primary filters to provide additional filtration or in parallel, instead of the primary fine and rough filters, to allow cleaning/maintenance of the primary fine and rough filters while the primary treatment train is in operation.

Ultraviolet Light/Oxidation. Organic compounds contained in an aqueous waste stream are destroyed in the UV/OX system (Figure C.8). Hydrogen peroxide is mixed with the waste.

The UV/OX system uses the photochemical reaction of UV light on hydrogen peroxide to form hydroxyl radicals and other reactive species that oxidize the organic compounds. The final products of the complete reaction are carbon dioxide, water, and inorganic ions.

Organic destruction is accomplished in two UV/OX units operating in parallel. During the UV/OX process, the aqueous waste passes through reaction chambers where hydrogen peroxide is added. While in the UV/OX system, the temperature of an aqueous waste is monitored. Heat exchangers are used to reduce the temperature of the waste should the temperature of the waste approach the upper limits for the UV/OX or RO systems.

pH Adjustment. The pH of a waste stream is monitored and controlled at different points throughout the treatment process. Within the primary treatment train, the pH of a waste can be adjusted with sulfuric acid or sodium hydroxide to optimize operation of downstream treatment processes or adjusted before final discharge. For example, the pH of an aqueous waste would be adjusted in the pH adjustment tank after the UV/OX process and before the RO process. In this example, pH is adjusted to cause certain chemical species such as ammonia to form ammonium sulfate, thereby increasing the rejection rate of the RO.

Hydrogen Peroxide Decomposition. Typically, hydrogen peroxide added into the UV/OX system is not consumed completely by the system. Because hydrogen peroxide is a strong oxidizer, the residual hydrogen peroxide from the UV/OX system is removed to protect the downstream equipment. The hydrogen peroxide decomposer uses a catalyst to break down the hydrogen peroxide that is not consumed completely in the process of organic destruction. The aqueous waste is sent through a column that breaks down the hydrogen peroxide into water and oxygen. The gas generated by the decomposition of the hydrogen peroxide is vented to the vessel off gas system.

Degasification. The degasification column is used to purge dissolved carbon dioxide from the aqueous waste to reduce the carbonate loading to downstream dissolved solids removal processes within the 200 Area ETF primary treatment train. The purged carbon dioxide is vented to the vessel off gas system.

Reverse Osmosis. The RO system (Figure C.9) uses pressure to force clean water molecules through semi-permeable membranes while keeping the larger molecule contaminants, such as dissolved solids, and large molecular weight organic materials, in the membrane. The RO process uses a staged configuration to maximize water recovery. The process produces two separate streams, including a clean 'permeate' and a concentrate (or retentate), which are concentrated as much as possible to minimize the amount of secondary waste produced.

The RO process is divided into first and second stages. Aqueous waste is fed to the first RO stage from the RO feed tank. The secondary waste receiving tanks of the secondary treatment train receive the retentate removed from the first RO stage, while the second RO stage receives the permeate (i.e., 'treated' aqueous waste from the first RO stage). In the second RO stage, the retentate is sent to the first stage RO feed tank while the permeate is sent to the IX system or to the surge tank, depending on the configuration of the ETF.

Two support systems facilitate this process. An anti-scale system injects scale inhibitors as needed into the feed waste to prevent scale from forming on the membrane surface. A clean-in-place system using cleaning agents, such as descalants and surfactants, cleans the membrane pores of surface and subsurface deposits that have fouled the membranes.

Ion Exchange. Because the RO process removes most of the dissolved solids in an aqueous waste, the IX process (Figure C.10) acts as a polishing unit. The IX system consists of three columns containing beds of cation and/or anion resins. This system is designed to allow for regeneration of resins and maintenance of one column while the other two are in operation. Though the two columns generally are operated in series, the two columns also can be operated in parallel or individually.

Typically, the two columns in operation are arranged in a primary/secondary (lead/lag) configuration, and the third (regenerated) column is maintained in standby.

When dissolved solids breakthrough the first IX column and are detected by a conductivity sensor, this column is removed from service for regeneration, and the second column replaces the first column and the third column is placed into service. The column normally is regenerated using sulfuric acid and sodium hydroxide. The resulting regeneration waste is collected in the secondary waste receiving tanks.

Spent resins are transferred into a disposal container should regeneration of the IX resins become inefficient Free water is removed from the container and returned to the surge tank. Dewatered resins are transferred to a final storage/disposal point.

Verification. The three verification tanks (Figure C.11) are used to hold the treated effluent while a determination is made that the effluent meets discharge limits. The effluent can be returned to the primary treatment train for additional treatment, or to the LERF, should a treated effluent not meet Discharge Permit or Final Delisting requirements.

The three verification tanks alternate between three operating modes: receiving treated effluent, holding treated effluent during laboratory analysis and verification, or discharging verified effluent. Treated effluent may also be returned to the 200 Area ETF to provide 'clean' service water for operational and maintenance functions, e.g., for boiler water and for backwashing the filters. This recycling keeps the quantity of fresh water used to a minimum.

C.2.4 Secondary Treatment Train

The secondary treatment system typically receives and processes the following by-products generated from the primary treatment train: concentrate from the first RO stage, filter backwash, regeneration waste from the ion exchange system, and spillage or overflow received into the process sumps. Depending on the operating configuration, however, some aqueous waste could be processed in the secondary treatment train before the primary treatment train (refer to Figure C.5 and Figure C.6 for example operating configurations).

The secondary treatment train provides the following processes:

- Secondary waste receiving - tank receiving and chemical addition.
- Evaporation - concentrates secondary waste streams.
- Concentrate staging - concentrate receipt, pH adjustment, and chemical addition.
- Thin film drying - dewatering of secondary waste streams.
- Container handling - packaging of dewatered secondary waste.

Secondary Waste Receiving. Waste to be processed in the secondary treatment train is received into two secondary waste receiving tanks, where the pH can be adjusted with sulfuric acid or sodium hydroxide for optimum evaporator performance. Chemicals, such as reducing agents, may be added to waste in the secondary waste receiving tanks to reduce the toxicity or mobility of constituents in the powder.

Evaporation. The 200 Area ETF evaporator is fed alternately by the two secondary waste receiving tanks. One tank serves as a waste receiver while the other tank is operated as the feed tank. The 200 Area ETF evaporator vessel (also referred to as the vapor body) is the principal component of the evaporation process (Figure C.12).

Feed from the secondary waste receiving tanks is pumped through a heater to the recirculation loop of the 200 Area ETF evaporator. In this loop, concentrated waste is recirculated from the 200 Area ETF evaporator, to a heater, and back into the evaporator where vaporization occurs. As water leaves the evaporator system in the vapor phase, the concentration of the waste in the evaporator increases. When the concentration of the waste reaches the appropriate density, a portion of the concentrate is pumped to one of the concentrate tanks.

The vapor that is released from the 200 Area ETF evaporator is routed to the entrainment separator, where water droplets and/or particulates are separated from the vapor. The 'cleaned' vapor is routed to the vapor compressor and converted to steam.

The steam from the vapor compressor is sent to the heater (reboiler) and used to heat the recirculating concentrate in the 200 Area ETF evaporator. From the heater, the steam is condensed and fed to the distillate flash tank, where the saturated condensate received from the heater drops to atmospheric pressure and cools to the normal boiling point through partial flashing (rapid vaporization caused by a pressure reduction). The resulting distillate is routed to the surge tank. The non-condensable vapors, such as air, are vented through a vent gas cooler to the vessel off gas system.

Concentrate Staging. The concentrate tanks make up the head end of the thin film drying process. From the 200 Area ETF evaporator, concentrate is pumped into two concentrate tanks, and pH adjusted chemicals, such as reducing agents, may be added to reduce the toxicity or mobility of constituents when converted to powder. Waste is transferred from the concentrate tanks to the thin film dryer for conversion to a powder. The concentrate tanks function alternately between concentrate receiver and feed tank for the thin film dryer. However, one tank may serve as both concentrate receiver and feed tank.

Because low solubility solids (i.e., calcium and magnesium sulfate) tend to settle in the concentrate tanks, these solids must be removed to prevent fouling and to protect the thin film dryer, and to maintain concentrate tank capacity.

Thin Film Drying. From the concentrate tanks, feed is pumped to the thin film dryer (Figure C.13) that is heated by steam. As the concentrated waste flows down the length of the dryer, the waste is dried. The dried film, or powder, is scraped off the dryer cylinder by blades attached to a rotating shaft. The powder is funneled through a cone-shaped powder hopper at the bottom of the dryer and into the Container Handling System.

Overhead vapor released by the drying of the concentrate is condensed in the distillate condenser. Excess heat is removed from the distillate by a water-cooled heat exchanger. Part of the distillate is circulated back to the condenser spray nozzles. The remaining distillate is pumped to the surge tank. Any noncondensible vapors and particulates from the spray condenser are exhausted to the vessel off gas system.

Container Handling. Before an empty container is moved into the Container Handling System (Figure C.14), the lid is removed and the container is placed on a conveyor. The containers are moved into the container filling area after passing through an air lock. The empty container is located under the thin film dryer, and raised into position. The container is sealed to the thin film dryer and a rotary valve begins the transfer of powder to the empty container. Air displaced from the container is vented to the distillate condenser attached to the 200 Area ETF evaporator that exhausts to the vessel off gas system.

The container is filled to a predetermined level, then lowered from the thin film dryer and moved along a conveyor. The filled container is manually recapped, and moved along the conveyor to the airlock. At the airlock, the container is moved onto the conveyor by remote control. The airlock is opened, the smear sample (surface wipe) is taken, and the contamination level counted. A 'C' ring is installed to secure the container lid. If the container has contaminated material on the outside, the container is wiped down and retested. Filled containers that pass the smear test are labeled, placed on pallets, and moved by forklift to the filled container storage area. Section C.3 provides a more detailed discussion of container handling.

C.2.5 Other Effluent Treatment Facility Systems

The 200 Area ETF is provided with support systems that facilitate treatment in the primary and secondary treatment trains and that provide for worker safety and environmental protection. An overview of the following systems is provided:

- Monitor and control system
- Vessel off gas system
- Sump collection system
- Chemical injection feed system
- Verification tank recycle system
- Utilities

C.2.5.1 Monitor and Control System

The operation of the 200 Area ETF is monitored and controlled by a centralized computer system (i.e., monitor and control system or MCS).

The MCS continuously monitors data from various field indicators, such as pH, flow, tank level, temperature, pressure, conductivity, alarm status, and valve switch positions. Data gathered by the MCS enable operations and engineering personnel to document and adjust the operation of the ETF.

Emergency communications equipment and warning systems (e.g. fire alarms and evacuation alarms) are included in Addendum J, Contingency Plan. These emergency response notification alarms are monitored continuously at central Hanford Facility locations (e.g. Hanford Fire Station) and do not rely on staff being present in the ETF control room for notification and response.

C.2.5.2 Vessel Off gas System

Ventilation for various tanks and vessels is provided through the vessel off gas system. The system includes a moisture separator, duct heater, pre-filter, high-efficiency particulate air filters, carbon absorber (when required to reduce organic emissions), exhaust fans, and ductwork. Gasses ventilated from the tanks and vessels enter the exhaust system through the connected ductwork. The vessel off gas system draws vapors and gasses off the following tanks and treatment systems:

- Surge tank
- Vent gas cooler (off the ETF evaporator/distillate flash tank)
- pH adjustment tank
- Concentrate tanks
- Degasification system
- First and second RO stages
- Dry powder hopper
- Effluent pH adjustment tank
- Drum capping station
- Secondary waste receiving tanks
- Distillate condenser (off the thin film dryer)
- Sump tanks 1 and 2

The vessel off gas system maintains a negative pressure with respect to the atmosphere, which produces a slight vacuum within tanks, vessels, and ancillary equipment for the containment of gas vapor. This system also provides for the collection, monitoring, and treatment of confined airborne in-vessel contaminants to preclude over-pressurization. The high-efficiency particulate air filters remove particulates and condensate from the air stream before these are discharged to the heating, ventilation, and air conditioning system.

C.2.5.3 Sump Collection System

Sump tanks 1 and 2 compose the sump collection system that provides containment of waste streams and liquid overflow associated with the 200 Area ETF processes. The process area floor is sloped to two separate trenches that each drain to a sump tank located under the floor of the 200 Area ETF (Figure C.15). One trench runs the length of the primary treatment train and drains to Sump Tank 2, located underneath the verification tank pump floor. The second trench collects spillage primarily from the secondary treatment train and flows to Sump Tank 1, located near the 200 Area ETF evaporator. Sump tanks 1 and 2 are located below floor level (Figure C.15). An eductor in these tanks prevents sludge from accumulating.

C.2.5.4 Chemical Injection Feed System

At several points within the primary and secondary treatment trains, sulfuric acid and sodium hydroxide (or dilute solutions of these reagents) are metered into specific process units to adjust the pH. For example, a dilute solution of 4 percent sulfuric acid and 4 percent sodium hydroxide could be added to the secondary waste receiving tanks to optimize the evaporation process.

C.2.5.5 Verification Tank Recycle System

To reduce the amount of water added to the process, verification tank water (i.e., verified effluent) is recycled throughout the 200 Area ETF process. Tanks and ancillary equipment that use verification tank water include:

- 4 percent H_2SO4 solution tank and ancillary equipment.
- 4 percent NaOH solution tank and ancillary equipment.
- Clean-in-place tank and ancillary equipment.
- IX columns (during resin regeneration).
- 200 Area ETF evaporator boiler and ancillary equipment.
- Thin film dryer boiler and ancillary equipment
- Seal water system.

In addition, verification tank water is used extensively during maintenance activities. For example, it may be used to flush piping systems or to confirm the integrity of piping, a process tank or tank truck.

C.2.5.6 Utilities

The 200 Area ETF maintains the following utility supply systems required for the operation of the ETF:

- Cooling water system - removes heat from process water via heat exchangers and a cooling tower.
- Compressed air system - provides air to process equipment and instrumentation.
- Seal water system - provides cool, clean, pressurized water to process equipment for pump seal cooling and pump seal lubrication, and provides protection against failure and fluid leakage.
- Demineralized water system - removes solids from raw water system to produce high quality, low ion-content, water for steam boilers, and for the hydrogen peroxide feed system.
- Heating, ventilation, and air conditioning system - provides continuous heating, cooling, and air humidity control throughout the ETF.

The following utilities support 200 Area ETF activities:

- Electrical power
- Sanitary water
- Communication systems
- Raw water

C.3 Containers

This section provides specific information on container storage and treatment operations at the 200 Area ETF, including descriptions of containers, labeling, and secondary containment structures.

A list of dangerous and/or mixed waste managed in containers at the 200 Area ETF is presented in Addendum A. The types of dangerous and/or mixed waste managed in containers in the 200 Area ETF could include:

- Secondary waste powder generated from the treatment process.
- Aqueous waste received from other Hanford site sources awaiting treatment.
- Miscellaneous waste generated by operations and maintenance activities.

The secondary treatment train processes the waste by-products from the primary treatment train, which are concentrated and dried into a powder. Containers are filled with dry powder waste from the thin film dryer via a remotely controlled system. Containers of aqueous waste received from other Hanford site sources are stored at 200 Area ETF until their contents can be transferred to the process for treatment.

The waste is usually transferred to the secondary waste receiving or concentration tanks. Miscellaneous waste generated from maintenance and operations activities are stored at the ETF. The waste could include process waste, such as used filter elements; spent RO membranes; damaged equipment, and decontamination and maintenance waste, such as contaminated rags, gloves, and other personal protective equipment. Containers of miscellaneous waste which have free liquids generally are packaged with absorbents.

Several container collection areas could be located within the 200 Area ETF process and container handling areas. These collection areas are used only to accumulate waste in containers. Once a container is filled, the container is transferred to a container storage area (Figure C.3 and Figure C.4), to another TSD unit, or to a less-than-90-day storage pad. Containers stored in the additional storage area (Figure C.4) are elevated or otherwise protected from contact with accumulated liquids. The container storage area within 200 Area ETF is a 22.9 x 8.5-meter room located adjacent to the 200 Area ETF process areas. The containers within the container storage area are clearly labeled, and access to these containers is limited by barriers and by administrative controls. The 200 Area ETF floor provides secondary containment, and the 200 Area ETF roof and walls protects all containers from exposure to the elements.

Waste also could be placed in containers for treatment as indicated in Addendum A. For example, sludge that accumulates in the bottoms of the process tanks is removed periodically and placed into containers. In this example, the waste is solidified by decanting the supernate from the container and the remainder of the waste is allowed to evaporate, or absorbents are added, as necessary, to address remaining liquids. Following treatment, this waste either is stored at the 200 Area ETF or transferred to another treatment, storage, and disposal (TSD) unit.

C.3.1 Description of Containers

The containers used to collect and store dry powder waste are 208-liter steel containers. Most of the aqueous waste received at 200 Area ETF, and maintenance and operation waste generated, are stored in 208-liter steel or plastic containers; however, in a few cases, the size of the container could vary to accommodate the size of a particular waste. For example, some process waste, such as spent filters, might not fit into a 208-liter container. In the case of spent resin from the IX columns, the resin is dewatered, and could be packaged in a special disposal container. In these few cases, specially sized containers could be required. In all cases, however, only approved containers are used and are compatible with the associated waste. Typically, 208-liter containers are used for treatment.

Current operating practices indicate the use of new 208-liter containers that have either a polyethylene liner or a protective coating. Any reused or reconditioned container is inspected for container integrity before use. Overpack containers are available for use with damaged containers. Overpack containers typically are unlined steel or polyethylene.

Per Addendum A, a maximum of 147,630 liters of dangerous and/or mixed waste could be stored in containers in the 200 Area ETF.

C.3.2 Container Management Practices

Before use, each container is checked for signs of damage such as dents, distortion, corrosion, or scratched coating.

For dry powder loading, empty containers on pallets are raised by a forklift and manually placed on the conveyor that transports the containers to the automatic filling station in the container handling room (Figure C.14). The container lids are removed and replaced manually following the filling sequence. After filling, containers exit the container handling room via the filled drum conveyor. Locking rings are installed, the container label is affixed, and the container is moved by dolly or forklift to the container storage area.

Before receipt at 200 Area ETF, each container from other Hanford site sources is inspected for leaks, signs of damage, and a loose lid. The identification number on each container is checked to ensure the proper container is received. The containers are typically placed on pallets and moved by dolly or forklift to the container storage area. These containers are later moved to the process area and the contents transferred to the process for treatment.

Containers used for storing maintenance and operations secondary waste are labeled before being placed in the container storage area or in a collection area. Lids are secured on these containers when not being filled. When the containers in a collection area are full, the containers are transferred by dolly or forklift to the container storage area or to an appropriate TSD unit. Containers used for treating waste also are labeled. The lids on these containers are removed as required to allow for treatment. During treatment, access to these containers is controlled through physical barriers and/or administrative controls.

The filled containers in the container storage area are inventoried, checked for proper labeling, and placed on pallets or in a separate containment device as necessary. Each pallet is moved by forklift. Within the container storage area, palletized containers are stacked no more than three pallets high and in rows no more than two containers wide. Unobstructed aisles with a minimum of 76-centimeter aisle space separate rows.

C.3.3 Container Labeling

Labels are affixed on containers used to store dry powder when the containers leave the container handling room. Labels are affixed on other waste containers before use. Every container is labeled with the date that the container was filled. Appropriate major risk labels, such as "corrosive", "toxic", or "F-listed", also are added. Each container also has a label with an identification number for tracking purposes.

C.3.4 Containment Requirements for Managing Containers

Secondary containment is provided in the container management areas within the ETF. The secondary containment provided for tank systems also serves the container management areas. This section describes the design and operation of the secondary containment structure for these areas.

C.3.4.1 Secondary Containment System Design

For the container management areas, the reinforced concrete floor and a 15.2-centimeter rise (berm) along the walls of the container storage area of the 200 Area ETF provides secondary containment. The engineering assessment required for tanks (Mausshardt 1995) also describes the design and construction of the secondary containment provided for the 200 Area ETF container management areas. All systems were designed to national codes and standards (e.g., American Society for Testing Materials, American Concrete Institute standards).

The floor is composed of cast-in-place, pre-formed concrete slabs, and has a minimum thickness of 15.2 centimeters. All slab joints and floor and wall joints have water stops installed at the mid-depth of the slab. In addition, filler was applied to each joint. The floor and berms are coated with a chemically resistant; high-solids epoxy coating system consisting of primer and top coating. This coating material is compatible with the waste managed in containers and is an integral part of the secondary containment system for containers.

The floor is sloped to drain any solution in the container storage area to floor drains along the west wall. Each floor drain consists of a grating over a 20.3-centimeter diameter drain port connected to a 4-inch polyvinyl chloride transfer pipe. The pipe passes under this wall and connects to a trench running along the east wall of the adjacent process area. This trench drains solution to sump tank 1.

The container storage area is separated from the process area by a common wall and a door for access to the two areas (Figure C.3). These two areas also share a common floor and trenches that, with the 15.2-centimeter rise of the containing walls, form the secondary containment system for the process area and the container storage area.

C.3.4.2 Structural Integrity of Base

Engineering calculations were performed showing the floor of the container storage area is capable of supporting the weight of containers. These calculations were reviewed and certified by a professional engineer (Mausshardt 1995). The concrete was inspected for damage during construction. Cracks were identified and repaired to the satisfaction of the professional engineer. Documentation of these certifications is included in the engineering assessment (Mausshardt 1995).

C.3.4.3 Containment System Capacity

The container storage area is primarily used to store dry powder, aqueous waste awaiting treatment, and maintenance and operation waste. Where appropriate, absorbents are added to fix any trace liquids present. Large volumes of liquid are not stored in the container storage area. However, liquids might be present in those containers that are in the treatment process. The maximum volume of waste that can be stored in containers in the container storage area is 147,630 liters.

Because they are interconnected by floor drains, both the process area and the container storage area are considered in the containment system capacity. The volume available for secondary containment in the process area is approximately 68,000 liters, as discussed in the engineering assessment (Mausshardt 1995). Using the dimensions of the container storage area (23.6 by 8.5 by 0.2 meters), and assuming that 50 percent of the floor area is occupied by containers, the volume of the container storage area is 15,300 liters. The Truck Bay loading areas (see Figure C.4) also provides 10,500 liters of containment as it is connected to the other two areas. The combined volume of the Truck Bay loading areas and process area, and container handling area available for secondary containment, is 93,800 liters. This volume is greater than 10 percent of the maximum total volume of containers allowed for storage in the ETF, as discussed previously.

C.3.4.4 Control of Run-on

The container management areas are located within the ETF, which serves to prevent run-on of precipitation.

C.3.4.5 Removal of Liquids from Containment Systems

The container storage area is equipped with drains that route solution to a trench in the process area, which drains to sump tank 1. The sump tanks are equipped with alarms that notify operating personnel that a leak is occurring. The sump tanks also are equipped with pumps to transfer waste to the surge tank or the secondary treatment train.

C.3.4.6 Prevention of Ignitable, Reactive, and Incompatible Wastes in Containers

Individual waste types (i.e., ignitable, corrosive, and reactive) are stored in separate containers. A waste that could be incompatible with other wastes is separated and protected from the incompatible waste. Incompatible wastes are evaluated using the methodology documented in 40 CFR 264, Appendix V. For example, acidic and caustic wastes are stored in separate containers. Free liquids are absorbed in miscellaneous waste containers that hold incompatible waste. Additionally, ETF-specific packaging requirements for these types of waste provide extra containment with each individual container. For example, each item of acidic waste is individually bagged and sealed within a lined container.

C.4 Tank Systems

This section provides specific information on tank systems and process units. This section also includes a discussion on the types of waste to be managed in the tanks, tank design information, integrity assessments, and additional information on the 200 Area ETF tanks that treat and store dangerous and/or mixed waste. The 200 Area ETF dangerous waste tanks are identified in Section C.4.1.1, and the relative locations of the tanks and process units in the 200 Area ETF are presented in Figure C.3.

C.4.1 Design Requirements

The following sections provide an overview of the design specifications for the tanks within the ETF. A separate discussion on the design of the process units also is provided. In accordance with the new tank system requirements of WAC 173-303-640(3), the following tank components and specifications were assessed:

- Dimensions, capacities, wall thicknesses, and pipe connections.
- Materials of construction and linings and compatibility of materials with the waste being processed.
- Materials of construction of foundations and structural supports.
- Review of design codes and standards used in construction.
- Review of structural design calculations, including seismic design basis.
- Waste characteristics and the effects of waste on corrosion.

This assessment was documented in the *Final RCRA Information Needs Report* (Mausshardt 1995; the engineering assessment performed for the 200 Area ETF tank systems by an independent professional engineer. A similar assessment of design requirements was performed for Load-in tanks 59A-TK-109 and -117 and is documented in *200 Area Effluent BAT/AKART Implementation, ETF Truck Load-in Facility, Project W-291H Integrity Assessment Report* (KEH 1994). An assessment was also performed when Load-in tank 59A-TK-1 was placed into service for receipt of dangerous and mixed wastes. The assessment is documented in *200 Area ETF Purgewater Unloading Facility Tank System Integrity Assessment* (HNF 2009a).

The specifications for the preparation, design, and construction of the tank systems at the 200 Area ETF are documented in the *Design Construction Specification, Project C-018H, 242-A Evaporator/PUREX Plant Process Condensate Treatment Facility* (WHC 1992a). The preparation, design, and construction of Load-in tanks 59A-TK-109 and -117 are provided in the construction specifications in *Project W-291, 200 Area Effluent BAT/AKART Implementation ETF Truck Load-in Facility* (KEH 1994). The preparation, design and construction of Load-in 59A-TK-1 are documented in *Purgewater Unloading Facility Project Documentation* (HNF 2009b).

Most of the tanks in the 200 Area ETF are constructed of stainless steel. According to the design of the ETF, it was determined stainless steel would provide adequate corrosion protection for these tanks. Exceptions include Load-in tank 59A-TK-1, which is constructed of fiberglass-reinforced plastic and the verification tanks, which are constructed of carbon steel with an epoxy coating. The 200 Area ETF evaporator/vapor body (and the internal surfaces of the thin film dryer) is constructed of a corrosion resistant alloy, known as alloy 625, to address the specific corrosion concerns in the secondary treatment train. Finally, the hydrogen peroxide decomposer vessels are constructed of carbon steel and coated with a vinyl ester lining.

The shell thicknesses of the tanks identified in Table C.5 represent a nominal thickness of a new tank when placed into operation. The tank capacities identified in this table represent the maximum volumes. Nominal tank volumes discussed below represent the maximum volume in a tank unit during normal operations.

C.4.1.1 Codes and Standards for Tank System Construction

Specific standards for the manufacture of tanks and process systems installed in the 200 Area ETF are briefly discussed in the following sections. In addition to these codes and industrial standards, a seismic analysis for each tank and process system is required [WAC 173-303-806(4)(a)(xi)]. The seismic analysis was performed in accordance with UCRL-15910 *Design and Evaluation Guidelines for Department of Energy Facilities Subjected to Natural Phenomena Hazards*, Section 4 (UCRL 1987). The results of the seismic analyses are summarized in the engineering assessment of the 200 Area ETF tank systems (Mausshardt 1995).

Storage and Treatment Tanks. The following tanks store and/or treat dangerous waste at the ETF.

Tank name	Tank number
Surge tank	2025E-60A-TK-1
pH adjustment tank	2025E-60C-TK-1
Effluent pH adjustment tank	2025E-60C-TK-2
First RO feed tank	2025E-60F-TK-1
Second RO feed tank	2025E-60F-TK-2
Verification tanks (three)	2025E-60H-TK-1A/1B/1C
Secondary waste receiving tanks (two)	2025E-60I-TK-1A/1B
Evaporator (vapor body)	2025E-601-EV-1
Concentrate tanks (two)	2025E-60J-TK-1A/1B
Sump tanks (two)	2025E-20B-TK-1/2
Distillate flash tank	2025E-60I-TK-2
Load-in tanks	2025ED-59A-TK-1/109/117

The relative location of these tanks is presented in Figure C.3. These tanks are maintained at or near atmospheric pressure. The codes and standards applicable to the design, construction, and testing of the above tanks and ancillary piping systems are as follows:

ASME - B31.3	Chemical Plant and Petroleum Refinery Piping (ASME 1990)
ASME Sect. VIII, Division I	Pressure Vessels (ASME 1992a)
AWS - D1.1	Structural Welding Code - Steel (AWS 1992)
ANSI - B16.5	Pipe Flanges and Flanged Fittings (ANSI 1992)
ASME Sect. IX	Welding and Brazing Qualifications (ASME 1992b)
API 620	Design and Construction of Large Welded Low Pressure Storage Tanks (API 1990)
AWWA - D100	Welded Steel Tanks for Water Storage (AWWA 1989)
AWWA - D103	Factory-Coated Bolted Steel Tanks for Water Storage (AWWA 1987)
AWWA - D120	Thermosetting Fiberglass-Reinforced Plastic Tanks (AWWA 1984)
ASTM-D3299	Filament Wound Glass-Fiber-Reinforced Thermoset Resin Corrosion Resistant Tanks.

The application of these standards to the construction of 200 Area ETF tanks and independent verification of completed systems ensured that the tank and tank supports had sufficient structural strength and that seams and connections were adequate to ensure tank integrity. In addition, each tank met strict quality assurance requirements. Each tank constructed offsite was tested for integrity and leak tightness before shipment to the Hanford Facility. Following installation, the systems were inspected for damage to ensure against leakage and to verify proper operation. If a tank was damaged during shipment or installation, leak tightness testing was repeated onsite.

C.4.1.2 Design Information for Tanks Located Outside of Effluent Treatment Facility

The load-in tanks, surge tank, and verification tanks are located outside the ETF. These tanks are located within concrete structures that provide secondary containment.

Load-In Tanks and Ancillary Equipment. The load-in tanks 59A-TK-109 and -117are heated and constructed of stainless steel, and have a nominal capacity of 31,000 liters. Load-in tank 59A-TK-1 is heated and constructed of fiberglass reinforced plastic and has a nominal capacity of 24,500 liters. Load-in tanks 59A-TK-109 and -117 are located outside of the metal building while Load-in tank 59A-TK-1 is located inside the building. Ancillary equipment includes transfer pumps, filtration systems, a double encased, fiberglass transfer pipeline, level instruments for tanker trucks, and leak detection equipment. From the Load-In Station, aqueous waste can be routed to the surge tank or to the LERF through a double-encased line. The load-in tanks, sump, pumps, and truck pad are all provided with secondary containment.

Surge Tank and Ancillary Equipment. The surge tank is constructed of stainless steel and has a nominal capacity of 421,000 liters. Ancillary equipment to the surge tank includes two underground double encased (i.e., pipe-within-a-pipe) transfer lines connecting to LERF and three pumps for transferring aqueous waste to the primary treatment train. The surge tank is located at the south end of the ETF. The surge tank is insulated and the contents heated to prevent freezing. Eductors in the tank provide mixing.

Verification Tanks and Ancillary Equipment. The verification tanks are located north of the ETF. The verification tanks have a nominal capacity of 2,760,000 liters each. For support, the tanks have a center post with a webbing of beams that extend from the center post to the sides of the tank. The roof is constructed of epoxy covered carbon steel that is attached to the cross beams of the webbing. The tank floor also is constructed of epoxy covered carbon steel and is sloped. Eductors are installed in each tank to provide mixing.

Ancillary equipment includes a return pump that provides circulation of treated effluent through the eductors. The return pump also recycles effluent back to the 200 Area ETF for retreatment and can provide service water for 200 Area ETF functions. Two transfer pumps are used to discharge treated effluent to SALDS or back to the LERF.

C.4.1.3 Design Information for Tanks Located Inside the Effluent Treatment Facility Building

Most of the 200 Area ETF tanks and ancillary equipment that store or treat dangerous and/or mixed waste are located within the ETF. The structure serves as secondary containment for the tank systems.

pH Adjustment Tank and Ancillary Equipment. The pH adjustment tank has a nominal capacity of 16,000 liters. Ancillary equipment for this tank includes overflow lines to a sump tank and pumps to transfer waste to other units in the main treatment train.

Effluent pH Adjustment Tank and Ancillary Equipment. The effluent pH adjustment tank has a nominal capacity of 13,700 liters. Ancillary equipment includes overflow lines to a sump tank and pumps to transfer waste to the verification tanks.

First and Second Reverse Osmosis Feed Tanks and Ancillary Equipment. The first RO feed tank is a vertical, stainless steel tank with a round bottom and has a nominal capacity of 19,700 liters. Conversely, the second RO feed tank is a rectangular vessel with the bottom of the tank sloping sharply to a single outlet in the bottom center. The second RO feed tank has a nominal capacity of 7,800 liters. Each RO tank has a pump to transfer waste to the RO arrays. Overflow lines are routed to a sump tank.

Secondary Waste Receiving Tanks and Ancillary Equipment. Two nominal 69,000-liter secondary waste receiving tanks collect waste from the units in the main treatment train, such as concentrate solution (retentate) from the RO units and regeneration solution from the IX columns.

These are vertical, cylindrical tanks with a semi-elliptical bottom and a flat top. Ancillary equipment includes overflow lines to a sump tank and pumps to transfer aqueous waste to the 200 Area ETF evaporator.

Effluent Treatment Facility Evaporator and Ancillary Equipment. The 200 Area ETF evaporator, the principal component of the evaporation process, is a cylindrical pressure vessel with a conical bottom. Aqueous waste is fed into the lower portion of the vessel. The top of the vessel is domed and the vapor outlet is configured to prevent carryover of liquid during the foaming or bumping (violent boiling) at the liquid surface. The 200 Area ETF evaporator has a nominal operating capacity of approximately 18,500 liters.

The 200 Area ETF evaporator includes the following ancillary equipment:

- Preheater
- Recirculation pump
- Waste heater with steam level control tank
- Concentrate transfer pump
- Entrainment separator
- Vapor compressor with silencers
- Silencer drain pump

Distillate Flash Tank and Ancillary Equipment. The distillate flash tank is a horizontal tank that has a nominal operating capacity of 780 liters. Ancillary equipment includes a pump to transfer the distillate to the surge tank for reprocessing.

Concentrate Tanks and Ancillary Equipment. Each of the two concentrate tanks has an approximate nominal capacity of 22,700 liters. Ancillary equipment includes overflow lines to a sump tank and pumps for recirculation and transfer.

Sump Tanks. Sump tanks 1 and 2 are located below floor level. Both sump tanks are double-walled, rectangular tanks, placed inside concrete vaults. Both tanks have a working volume of 4,000 liters each. The sump tanks are located in pits below grade to allow gravity drain of solutions to the tanks. Each sump tank has two vertical pumps for transfer of waste to the secondary waste receiving tanks or to the surge tank for reprocessing.

C.4.1.4 Design Information for Effluent Treatment Facility Process Units

As with the 200 Area ETF tanks, process units that treat and/or store dangerous and/or mixed waste are maintained at or near atmospheric pressure. These units were constructed to meet a series of design standards, as discussed in the following sections. Table C.6 presents the materials of construction and the ancillary equipment associated with these process units. All piping systems are designed to withstand the effects of internal pressure, weight, thermal expansion and contraction, and any pulsating flow. The design and integrity of these units are presented in the engineering assessment (Mausshardt 1995).

Filters. The load-in fine and rough filter vessels (including the influent and auxiliary filters) are designed to comply with the ASME Section VIII, Division I, Pressure Vessels (ASME 1992a). The application of these standards to the construction of the 200 Area ETF filter system and independent inspection ensure that the filter and filter supports have sufficient structural strength and that the seams and connections are adequate to ensure the integrity of the filter vessels.

Ultraviolet Oxidation System. The UV/OX reaction chamber is designed to comply with manufacturers standards.

Degasification System. The codes and standards applicable to the design, fabrication, and testing of the degasification column are identified as follows:

- ASME - B31.3, Chemical Plant and Petroleum Refinery Piping (ASME 1990)
- AWS - D1.1, Structural Welding Code - Steel (AWS 1992)
- ANSI - B16.5, Pipe Flanges and Flanged Fittings (ANSI 1992)

Reverse Osmosis System. The pressure vessels in the RO unit are designed to comply with ASME Section VIII, Division I, Pressure Vessels (ASME 1992a), and applicable codes and standards.

Ion Exchange (Polishers). The IX columns are designed in accordance with ASME Section VIII, Division I, Pressure Vessels (ASME 1992a), and applicable codes and standards. Polisher piping is fabricated of type 304 stainless steel or polyvinyl chloride (PVC) and meets the requirements of ASME B31.3, Chemical Plant and Petroleum Refinery Piping (ASME 1990).

Effluent Treatment Facility Evaporator. The 200 Area ETF evaporator is designed to meet the requirements of ASME Section VIII, Division I, Pressure Vessels (ASME 1992a), and applicable codes and standards. The 200 Area ETF evaporator piping meets the requirements of ASME B31.3, Chemical Plant and Petroleum Refinery Piping (ASME 1990).

Thin Film Dryer System. The thin film dryer is designed to meet the requirements of ASME Section VIII, Division I, Pressure Vessels (ASME 1992a), and applicable codes and standards. The piping meets the requirements of ASME - B31.3, Chemical Plant and Petroleum Refinery Piping (ASME 1990).

C.4.1.5 Integrity Assessments

The integrity assessment for 200 Area ETF (Mausshardt 1995) attests to the adequacy of design and integrity of the tanks and ancillary equipment to ensure that the tanks and ancillary equipment will not collapse, rupture, or fail over the intended life considering intended uses. For the load-in tanks, a similar integrity assessment was performed (KEH 1995 and HNF 2009a). Specifically, the assessment documents the following considerations:

- Adequacy of the standards used during design and construction of the facility.
- Characteristics of the solution in each tank.
- Adequacy of the materials of construction to provide corrosion protection from the solution in each tank.
- Results of the leak tests and visual inspections.

The results of these assessments demonstrate that tanks and ancillary equipment have sufficient structural integrity and are acceptable for storing and treating dangerous and/or mixed waste. The assessments also state that the tanks and building were designed and constructed to withstand a design-basis earthquake. Independent, qualified registered professional engineers certified these tank assessments.

The scope of the 200 Area ETF tank integrity assessment was based on characterization data from process condensate. To assess the effect that other aqueous waste might have on the integrity of the 200 Area ETF tanks, the chemistry of an aqueous waste will be evaluated for its potential to corrode a tank (e.g., chloride concentrations will be evaluated). The tank integrity assessment for the load-in tanks was based on characterization data from several aqueous waste streams. The chemistry of an aqueous waste stream not considered in the load-in tank integrity assessment also will be evaluated for the potential to corrode a load-in tank.

Consistent with the recommendations of the integrity assessment, a corrosion inspection program was developed. Periodic integrity assessments are scheduled for those tanks predicted to have the highest potential for corrosion. These inspections are scheduled annually or longer, based on age of the tank system, materials of construction, characteristics of the waste, operating experience, and recommendations of the initial integrity assessment. These 'indicator tanks' include the concentrate tanks, secondary waste receiving tanks, and verification tanks. One of each of these tanks will be inspected yearly to determine if corrosion or coating failure has occurred. Should significant corrosion or coating failure be found, an additional tank of the same type would be inspected during the same year.

In the case of the verification tanks, if corrosion or coating failure is found in the second tank, the third tank also will be inspected. If significant corrosion were observed in all three sets of tanks, the balance of the 200 Area ETF tanks would be considered for inspection. For tanks predicted to have lower potential for corrosion, inspections also are performed nonroutinely as part of the corrective maintenance program.

C.4.2 Additional Requirements for New Tanks

Procedures for proper installation of tanks, tank supports, piping, concrete, etc., are included in *Construction Specification, Project C-018H, 242-A Evaporator/PUREX Plant Process Condensate Treatment Facility* (WHC 1992a). For the load-in tanks, procedures are included in the construction specifications in *Project W-291, 200 Area Effluent BAT/AKART Implementation ETF Truck Load-in Facility* (KEH 1994) and Purgewater Unloading Facility Project Documentation (HNF 2009b). Following installation, an independent, qualified, registered professional engineer inspected the tanks and secondary containment. Deficiencies identified included damage to the surge tank, damage to the verification tank liners, and 200 Area ETF secondary containment concrete surface cracking. All deficiencies were repaired to the satisfaction of the engineer. The tanks and ancillary equipment were leak tested as part of acceptance of the system from the construction contractor. Information on the inspections and leak tests are included in the engineering assessment (Mausshardt 1995). No deficiencies were identified during installation of the load-in tanks and ancillary equipment.

C.4.3 Secondary Containment and Release Detection for Tank Systems

This section describes the design and operation of secondary containment and leak detection systems at the ETF.

C.4.3.1 Secondary Containment Requirements for All Tank Systems

The specifications for the preparation, design, and construction of the secondary containment systems at the 200 Area ETF are documented (WHC 1992a). The preparation, design, and construction of the secondary containment for the load-in tanks are provided in the construction specifications (KEH 1994 and HNF 2009b). All systems were designed to national codes and standards. Constructing the 200 Area ETF per these specifications ensured that foundations are capable of supporting tank and secondary containment systems and that uneven settling and failures from pressure gradients should not occur.

C.4.3.1.1 Common Elements

The following text describes elements of secondary containment that are common to all 200 Area ETF tank systems. Details on the secondary containment for specific tanks, including leak detection systems and liquids removal, are provided in Section C.4.3.1,2.

Foundation and Construction. For the tanks within the ETF, except for the sump tanks, secondary containment is provided by a coated concrete floor and a 15.2-centimeter rise (berm) along the containing walls. The double-wall construction of the sump tanks provides secondary containment. Additionally, trenches are provided in the floor that also provides containment and drainage of any liquid to a sump pit. For tanks outside the ETF, secondary containment also is provided with coated concrete floors in a containment pit (load-in tanks) or surrounded by concrete dikes (the surge and verification tanks).

The transfer piping that carries aqueous waste into the 200 Area ETF is pipe-within-a-pipe construction, and is buried approximately 1.2 meters below ground surface. The pipes between the verification tanks and the verification tank pumps within the 200 Area ETF are located in a concrete pipe trench.

For this discussion, there are five discrete secondary containment systems associated with the following tanks and ancillary equipment that treat or store dangerous waste:

- Load-in tanks
- Surge tank
- Process area (including sump tanks)
- Verification tanks

- Transfer piping and pipe trenches

All of the secondary containment systems are designed with reinforcing steel and base and berm thickness to minimize failure caused by pressure gradients, physical contact with the waste, and climatic conditions. Classical theories of structural analysis, soil mechanics, and concrete and structural steel design were used in the design calculations for the foundations and structures. These calculations are maintained at the ETF. In each of the analyses, the major design criteria from the following documents were included:

V-C018HC1-001	Design Construction Specification, Project C-018H, 242A Evaporator/PUREX Plant Process Condensate Treatment Facility (WHC 1992a)
DOE Order 6430.1A	General Design Criteria
SDC-4.1	Standard Architectural-Civil Design Criteria, Design Loads for Facilities (DOE-RL 1988)
UCRL-15910	Design and Evaluation Guidelines for Department of Energy Facilities Subjected to Natural Phenomena Hazards (UCRL 1987)
UBC-91 UBC-97	Uniform Building Code, 1991 Edition (ICBO 1991) Uniform Building Code, 1997 Edition (ICC 1997, for Load-in tank 59A-TK-1)

The design and structural analysis calculations substantiate the structural designs in the referenced drawings. The conclusions drawn from these calculations indicate that the designs are sound and that the specified structural design criteria were met. This conclusion is verified in the independent design review that was part of the engineering assessment (Mausshardt 1995, KEH 1994, and HNF 2009a).

Containment Materials. The concrete floor consists of cast-in-place and preformed concrete slabs. All slab joints and floor and wall joints have water stops installed at the mid-depth of the slab. In addition, filler was applied to each joint.

Except for the sump tank vaults, all of the concrete surfaces in the secondary containment system, including berms, trenches, and pits, are coated with a chemical-resistant, high-solids, epoxy coating that consists of a primer and a top coating. This coating material is compatible with the waste being treated, and with the sulfuric acid, sodium hydroxide, and hydrogen peroxide additives to the process. The coating protects the concrete from contact with any chemical materials that might be harmful to concrete and prevents the concrete from being in contact with waste material. Table C.8 summarizes the specific types of primer and top coats specified for the concrete and masonry surfaces in the ETF. The epoxy coating is considered integral to the secondary containment system for the tanks and ancillary equipment.

The concrete containment systems are maintained such that any cracks, gaps, holes, and other imperfections are repaired in a timely manner. Thus, the concrete containment systems do not allow spilled liquid to reach soil or groundwater. There are a number of personnel doorways and vehicle access points into the 200 Area ETF process areas. Releases of any spilled or leaked material to the environment from these access points are prevented by 15.2-centimeter concrete curbs, sloped areas of the floor (e.g., truck ramp), or trenches.

Containment Capacity and Maintenance. Each of these containment areas is designed to contain more than 100 percent of the volume of the largest tank in each respective system. Secondary containment systems for the surge tank, and the verification tanks, which are outside the ETF, also are large enough to include the additional volume from a 25-year, 24-hour storm event; i.e., 5.3 centimeters of precipitation.

Sprinkler System. The sprinkler system within the 200 Area ETF supplies firewater protection to the process area and the container storage area. This system is connected to a site wide water supply system and has the capacity to supply sufficient water to suppress a fire at the ETF. However, in the event of failure, the sprinkler system can be hooked up to another water source (e.g., tanker truck).

C.4.3.1.2 Specific Containment Systems

The following discussion presents a description of the individual containment systems associated with specific tank systems.

Load-In Tank Secondary Containment. The load-in tanks 59A-TK-109 and -117 are mounted on a 46-centimeter-thick reinforced concrete slab (Drawing H-2-817970). Secondary containment is provided by a pit with 30.5-centimeter-thick walls and a floor constructed of reinforced concrete. The load-in tank pit is sloped to drain solution to a sump. The depth of the pit varies with the slope of the floor, with an average thickness of about 1.1 meters. The volume of the secondary containment is about 73,000 liters, which is capable of containing the volume of at least one load-in tank (i.e., 34,200 liters). Leaks are detected by a leak detector that alarms locally, in the 200 Area ETF control room, and by visual inspection of the secondary containment. Alarms are monitored continuously in the ETF control room during Load-in Station transfers and at least daily when there are no Load-in Station transfers occurring.

Adjacent to the pit is a 25.4-centimeter-thick reinforced concrete pad that serves as secondary containment for the load-in tanker trucks, containers, transfer pumps, and filter system that serve as the first tanker truck unloading bay. The pad is inside the metal Load-in building and is 15.2 centimeters below grade with north and south walls gently sloped to allow truck access. The pad has a 3-inch drain pipe to route waste solution to the adjacent load-in tank pit. The pad does not have protective coating because it would experience excessive wear from the vehicle traffic.

Load-in tank 59A-TK-1 is located on a 25.4-centimeter-thick reinforced concrete slab (Drawing H-2-817970) inside the metal Load-in building. The tank has a flat bottom which sits on a concrete slab in the secondary containment. Secondary containment for the tank, filter system, and truck unloading piping is provided by an epoxy coated catch basin with a capacity of about 3,400 liters. The catch basin is sloped to route solution from the catch basin through a 15.2-centimeter-wide by 14.3-centimeter-deep trench to the adjacent truck unloading pad. This pad drains to the Load-in tank pit discussed above. The volume of the combined secondary containment of these two systems is greater than 76,400 liters, which is capable of holding the volume of tank 59A-TK-1 (i.e., 26,000 liters).

Adjacent to tank 59A-TK-1 catch basin is a 25.4-centimeter-thick reinforced concrete pad that serves as the second tanker truck unloading bay. The pad is inside the metal Load-in building and has a 2.4-meter by 4.0-meter shallow, sloping pit to catch leaks during tanker truck unloading. The pit has a maximum depth of 6.0 centimeters and a 15.2-centimeter-wide by 6.0-centimeter-deep trench to route leaks to the adjacent tank 59A-TK-1 catch basin. The pad does not have protective coating because it would experience excessive wear from the vehicle traffic.

Surge Tank Secondary Containment. The surge tank is mounted on a reinforced concrete ringwall. Inside the ringwall, the flat-bottomed tank is supported by a bed of compacted sand and gravel with a high-density polyethylene liner bonded to the ringwall. The liner prevents galvanic corrosion between the soil and the tank. The secondary containment is reinforced concrete with a 15.2-centimeter thick floor and a 20.3-centimeter thick dike. The secondary containment area shares part of the southern wall of the main process area. The dike extends up 2.9 meters to provide a containment volume of 856,000 liters for the 462,000-liter surge tank.

The floor of the secondary containment slopes to a sump in the northwest corner of the containment area. Leaks into the secondary containment are detected by level instrumentation in the sump which alarms in the ETF control room and/or by routine visual inspections. Sump alarms are monitored continuously in the ETF control room during ETF processing operations and at least daily when ETF is not processing waste. A sump pump is used to transfer solution in the secondary containment to a sump tank.

Process Area Secondary Containment. The process area contains the tanks and ancillary equipment of the primary and secondary treatment trains, and has a jointed, reinforced concrete slab floor. The concrete floor of the process area provides the secondary containment. This floor is a minimum of 15.2 centimeters thick.

With doorsills 15.2 centimeter high, the process area has a containment volume of approximately 93,800 liters. The largest tanks in the process area are the secondary waste receiving tanks, which each have a maximum capacity of 73,800 liters.

The floor of the process area is sloped to drain liquids to two trenches that drain to a sump. Each trench is approximately 38.1 centimeters wide with a sloped trough varying from 39.4 to 76.2 centimeters deep. Leaks into the secondary containment are detected by routine visual inspections of the floor area near the tanks, ancillary equipment, and in the trenches.

A small dam was placed in the trench that comes from the thin film dryer room to contain minor liquid spills originating in the dryer room to minimize the spread of contamination into the process area. The dryer room is inspected for leaks in accordance with the inspection schedule in Addendum I. Operators clean up these minor spills by removing the liquid waste and decontaminating the spill area.

A small dam was also placed in the trench adjacent to the chemical feed skid when the chemical berm area was expanded to accommodate acid and caustic pumps, which were moved indoors from the top of the surge tank to resolve a safety concern. This dam was designed to contain minor spills originating in the chemical berm area and prevent them from entering the process sump.

The northwest corner of the process area consists of a pump pit containing the pumps and piping for transferring treated effluent from the verification tanks to SALDS. The pit is built 1.37 meters below the process area floor level and is sloped to drain to a trench built along its north wall that routes liquid to sump tank 1. Leaks into the secondary containment of the pump pit are detected by routine visual inspections.

Sump Tanks. The sump tanks support the secondary containment system, and collect waste from several sources, including:

- Process area drain trenches.
- Tank overflows and drains.
- Container washing water.
- Resin dewatering solution.
- Steam boiler blow down.
- Sampler system drains.

These double-contained tanks are located within unlined, concrete vaults. The sump tank levels are monitored by remote level indicators or through visual inspections from the sump covers. These indicators are connected to high- and low-level alarms that are monitored in the ETF control room during ETF processing operations and at least daily when ETF is not processing liquid waste. When a high-level alarm is activated, a pump is activated and the sump tank contents usually are routed to the secondary treatment train for processing. The contents also could be routed to the surge tank for treatment in the primary treatment train. In the event of an abnormally high inflow rate, a second sump pump is initiated automatically.

Verification Tank Secondary Containment. The three verification tanks are each mounted on ringwalls with high-density polyethylene liners similar to the surge tank. The secondary containment for the three tanks is reinforced concrete with a 15.2-centimeter thick floor and a 20.3-centimeter thick dike. The dike extends up 2.6 meters to provide a containment of approximately 3,390,000 liters exceeding the capacity of a single verification tank (See Table C.5).

The floor of the secondary containment slopes to a sump along the southern wall of the dike. Leaks into the secondary containment are detected by level instrumentation in the sump and/or by routine visual inspections. Sump alarms are monitored continuously in the ETF control room during ETF processing operations and at least daily when ETF is not processing waste. A sump pump is used to transfer solution in the secondary containment to a sump tank.

C.4.3.2 Additional Requirements for Specific Types of Systems

This section addresses additional requirements in WAC 173-303-640 for double-walled tanks like the sump tanks and secondary containment for ancillary equipment and piping associated with the tank systems.

C.4.3.2.1 Double-Walled Tanks

The sump tanks are the only tanks in the 200 Area ETF classified as 'double-walled' tanks. These tanks are located in unlined concrete vaults and support the secondary containment system for the process area. The sump tanks are equipped with a leak detector between the walls of the tanks that provide continuous monitoring for leaks. The leak detector alarms are monitored in the ETF control room. These sump tank alarms are monitored continuously during ETF processing operations and at least daily when ETF is not processing waste. The inner tanks are contained completely within the outer shells. The tanks are contained completely within the concrete structure of the 200 Area ETF so corrosion protection from external galvanic corrosion is not necessary.

C.4.3.2.2 Ancillary Equipment

The secondary containment provided for the tanks and process systems also serves as secondary containment for the ancillary equipment associated with these systems.

Ancillary Equipment. Section C.4.3.1.2 describes the secondary containment systems that also serve most of the ancillary equipment within the 200 Area ETF. Between the 200 Area ETF and the verification tanks, a pipeline trench provides secondary containment for four pipelines connecting the transfer pumps (i.e., discharge and return pumps) in the 200 Area ETF with the verification tanks (Figure C.2). This concrete trench crosses under the road and extends from the verification tank pumps to the verification tanks. Treated effluent flows through these pipelines from the verification tank pumps to the verification tanks. The return pump is used to return effluent to the 200 Area ETF for use as service water or for reprocessing.

For all of the ancillary equipment housed within the ETF, the concrete floor, trenches, and berms form the secondary containment system. For the ancillary equipment of the surge tank and the verification tanks, secondary containment is provided by the concrete floors and dikes associated with these tanks. The concrete floor and pit provide secondary containment for the ancillary equipment of the load-in tanks.

Transfer Piping and Pipe Trenches. The two buried transfer lines between LERF and the surge tank have secondary containment in a pipe-within-a-pipe arrangement. The 4-inch transfer line has an 8-inch outer pipe, while the 3-inch transfer, line has a 6-inch outer pipe. The pipes are fiberglass and are sloped towards the surge tank. The outer piping ends with a drain valve in the surge tank secondary containment.

These pipelines are equipped with leak detection located in the annulus between the inner and outer pipes; the leak detection equipment can continuously 'inspect' the pipelines during aqueous waste transfers. The alarms on the leak detection system are monitored in the ETF control room. The control room alarms are monitored continuously during aqueous waste transfers between LERF and the ETF surge tank and at least daily when no transfers are occurring. A low-volume air purge of the annulus is provided to prevent condensation buildup and minimize false alarms by the leak detection system. In the event that these leak detectors are not in service, the pipelines are inspected during transfers by opening a drain valve to check for solution in the annular space between the inner and outer pipe.

The 3-inch transfer line between the load-in tanks and the surge tank has a 6-inch outer pipe in a pipe-within-a-pipe arrangement. The piping is made of fiberglass-reinforced plastic and slopes towards the load-in tank secondary containment pit. The drain valve and leak detection system for the load-in tank pipelines are operated similarly to the leak detection system for the LERF to 200 Area ETF pipelines.

As previously indicated, a reinforced concrete pipe trench provides secondary containment for piping under the roadway between the 200 Area ETF and the verification tanks.

Three 15.2 centimeter thick reinforced concrete partitions divide the trench into four portions and support metal gratings over the trench. Each portion of the trench is 1.2 meters wide, 0.76 meter deep, and slopes to route any solution present to 4-inch drain lines through the north wall of the ETF building. These drain lines route solution to sump tank 2 in ETF. The floor of the pipe trench is 30.5 centimeters thick and the sides are 15.2 centimeters thick. The concrete trenches are coated with water sealant and covered with metal gratings at ground level to allow vehicle traffic on the roadway.

C.4.4 Tank Management Practices

When an aqueous waste stream is identified for treatment or storage at 200 Area ETF, the generating unit is required to characterize the waste. Based on characterization data, the waste stream is evaluated to determine if the stream is acceptable for treatment or storage. Specific tank management practices are discussed in the following sections.

C.4.4.1.1 Rupture, Leakage, Corrosion Prevention

Most aqueous waste streams can be managed such that corrosion would not be a concern. For example, an aqueous waste stream with high concentrations of chloride might cause corrosion problems when concentrated in the secondary treatment train. One approach is to adjust the corrosion control measures in the secondary treatment train. An alternative might be to blend this aqueous waste in a LERF basin with another aqueous waste that has sufficient dissolved solids, such that the concentration of the chlorides in the secondary treatment train would not pose a corrosion concern.

Additionally, the materials of construction used in the tanks systems (Table C.5) make it unlikely that an aqueous waste would corrode a tank. For more information on corrosion prevention, refer to Addendum B, Waste Analysis Plan.

If operating experience suggests that most aqueous waste streams can be managed such that corrosion would not be a concern, operating practices and integrity assessment schedules and requirements will be reviewed and modified as appropriate.

When a leak in a tank system is discovered, the leak is immediately contained or stopped by isolating the leaking component. Following containment, the requirements of WAC 173-303-640(7), incorporated by reference, are followed. These requirements include repair or closure of the tank/tank system component, and certification of any major repairs.

C.4.4.2 Overfilling Prevention

Operating practices and administrative controls used at the 200 Area ETF to prevent overfilling a tank are discussed in the following paragraphs. The 200 Area ETF process is controlled by the MCS. The MCS monitors liquid levels in the 200 Area ETF tanks and has alarms that annunciate on high-liquid level to notify operators that actions must be taken to prevent overfilling of these vessels. As an additional precaution to prevent spills, many tanks are equipped with overflow lines that route solutions to sump tanks 1 and 2. These tanks include the pH adjustment tank; RO feed tanks, effluent pH adjustment tank, secondary waste receiving tanks, and concentrate tanks.

The following section discusses feed systems, safety cutoff devices, bypass systems, and pressure controls for specific tanks and process systems.

Tanks. All tanks are equipped with liquid level sensors that give a reading of the tank liquid volume. All of the tanks are equipped further with liquid level alarms that are actuated if the liquid volume is near the tank overflow capacity. In the actuation of the surge tank alarm, a liquid level switch trips, sending a signal to the valve actuator on the tank influent lines, and causing the influent valves to close.

The operating mode for each verification tank, i.e., receiving, holding, or discharging, can be designated through the MCS; modes also switch automatically. When the high-level set point on the receiving verification tank is reached, the flow to this tank is diverted and another tank becomes the receiver. The full tank is switched into verification mode. The third tank is reserved for discharge mode.

The liquid levels in the pH adjustment, first and second RO feed, and effluent pH adjustment tanks are maintained within predetermined operating ranges. Should any of these tanks overflow, the excess waste is piped along with any leakage from the feed pumps to a sump tank.

When waste in a secondary waste-receiving tank reaches the high-level set point, the influent flow of waste is redirected to the second tank. In a similar fashion, the concentrate tanks switch receipt modes when the high-level set point of one tank is reached.

Filter Systems. All filters at 200 Area ETF (i.e., the Load-In Station, rough, fine, and auxiliary filter systems) are in leak-tight steel casings. For the rough and fine filters, a high differential pressure, which could damage the filter element, activates a valve that shuts off liquid flow to protect the filter element from possible damage. To prevent a high-pressure situation, the filters are cleaned routinely with pulses of compressed air that force water back through the filter. Cleaning is terminated automatically by shutting off the compressed air supply if high pressure develops. The differential pressure across the auxiliary filters also is monitored. A high differential pressure in these filters would result in a system shutdown to allow the filters to be changed out.

The Load-In Station filtration system has pressure gauges for monitoring the differential pressure across each filter. A high differential pressure would result in discontinuing filter operation until the filter is replaced.

Ultraviolet Light/Oxidation System and Decomposers. A rupture disk on the inlet piping to each of the UV/OX reaction vessels relieves to the pH adjustment tank in the event of excessive pressure developing in the piping system. Should the rupture disk fail, the aqueous waste would trip the moisture sensor, shut down the UV lamps, and close the surge tank feed valve. Also provided is a level sensor to protect UV lamps against the risk of exposure to air. Should those sensors be actuated, the UV lamps would be shut down immediately.

The piping and valving for the hydrogen peroxide decomposers are configured to split the waste flow: half flows to one decomposer and half flows to the other decomposer. Alternatively, the total flow of waste can be treated in one decomposer or both decomposers can be bypassed. A safety relief valve on each decomposer vessel can relieve excess system pressure to a sump tank.

Degasification System. The degasification column is typically supplied aqueous waste feed by the pH adjustment tank feed pump. This pump transfers waste solution through the hydrogen peroxide decomposer, the fine filter, and the degasification column to the first RO feed tank.

The degasification column is designed for operation at a partial vacuum. A pressure sensor in the outlet of the column detects the column pressure. The vacuum in the degasification column is maintained by a blower connected to the vessel off gas system. The column is protected from extremely low pressure developed by the column blower by the use of an intake vent that is maintained in the open position during operation. The column liquid level is regulated by a flow control system with a high- and low-level alarm. Plate-type heat exchanger cools the waste solution fed to the degasification column.

Reverse Osmosis System. The flow through the first and second RO stages is controlled to maintain constant liquid levels in the first and second stage RO feed tanks.

Polisher. Typically, two of the three columns are in operation (lead/lag) and the third (regenerated) column is in standby. When the capacity of the resin in the first column is exceeded, as detected by an increase in the conductivity of the column effluent, the third column, containing freshly regenerated IX resin, is brought online. The first column is taken offline, and the waste is rerouted to the second column, and to the third. Liquid level instrumentation and automatically operated valves are provided in the IX system to prevent overfilling.

Effluent Treatment Facility Evaporator. Liquid level instrumentation in the secondary waste receiving tanks is designed to preclude a tank overflow. A liquid level switch actuated by a high-tank liquid level causes the valves to reposition, closing off flow to the secondary waste receiving tanks. Secondary containment for these tanks routes liquids to a sump tank.

Valves in the 200 Area ETF evaporator feed line can be positioned to bypass the secondary waste around the 200 Area ETF evaporator and to transfer the secondary waste to the concentrate tanks.

Thin Film Dryer. The two concentrate tanks alternately feed the thin film dryer. Typically, one tank serves as a concentrate waste receiver while the other tank serves as the dryer feed tank. One tank may serve as both concentrate waste receiver and dryer feed tank. Liquid level instrumentation prevents tank overflow by diverting the concentrate flow from the full concentrate tank to the other concentrate tank. Secondary containment for these tanks routes liquids to a sump tank.

An alternate route is provided from the concentrate receiver tank to the secondary waste receiving tanks. Dilute concentrate in the concentrate receiver tank can be reprocessed through the 200 Area ETF evaporator by transferring the concentrate back to a secondary waste-receiving tank.

C.4.5 Labels or Signs

Each tank or process unit in the 200 Area ETF is identified by a nameplate attached in a readily visible location. Included on the nameplate are the equipment number and the equipment title. Those tanks that store or treat dangerous waste at the 200 Area ETF (Section C.4.1.1) are identified with a label, which reads *PROCESS WATER/WASTE.* The labels are legible at a distance of at least fifty feet or as appropriate for legibility within the ETF. Additionally, these tanks bear a legend that identifies the waste in a manner, which adequately warns employees, emergency personnel, and the public of the major risk(s) associated with the waste being stored or treated in the tank system(s).

Caution plates are used to show possible hazards and warn that precautions are necessary. Caution signs have a yellow background and black panel with yellow letters and bear the word *CAUTION.* Danger signs show immediate danger and signify that special precautions are necessary. These signs are red, black, and white and bear the word *DANGER.*

Tanks and vessels containing corrosive chemicals are posted with black and white signs bearing the word *CORROSIVE. DANGER - UNAUTHORIZED PERSONNEL KEEP OUT* signs are posted on all exterior doors of the ETF, and on each interior door leading into the process area. Tank ancillary piping is also labeled *PROCESS WATER* or *PROCESS LIQUID* to alert personnel which pipes in the process area contains dangerous and/or mixed waste.

All tank systems holding dangerous waste are marked with labels or signs to identify the waste contained in the tanks. The labels or signs are legible at a distance of at least 50-feet and bear a legend that identifies the waste in a manner that adequately warns employees, emergency response personnel, and the public, of the major risk(s) associated with the waste being stored or treated in the tank system(s).

C.4.6 Air Emissions

Tank systems that contain extremely hazardous waste that is acutely toxic by inhalation must be designed to prevent the escape of such vapors. To date, no extremely hazardous waste has been managed in 200 Area ETF tanks and is not anticipated. However, the 200 Area ETF tanks have forced ventilation that draws air from the tank vapor spaces to prevent exposure of operating personnel to any toxic vapors that might be present. The vapor passes through a charcoal filter and two sets of high-efficiency particulate air filters before discharge to the environment. The Load-in tanks and verification tanks are vented to the atmosphere.

C.4.7 Management of Ignitable or Reactive Wastes in Tanks Systems

Although the 200 Area ETF is permitted to accept waste that is designated ignitable or reactive, such waste would be treated or blended immediately after placement in the tank system so that the resulting waste mixture is no longer ignitable or reactive. Aqueous waste received does not meet the definition of a combustible or flammable liquid given in National Fire Protection Association (NFPA) code number 30 (NFPA 1996).

The buffer zone requirements in NFPA-30, which require tanks containing combustible or flammable solutions be a safe distance from each other and from public way, are not applicable.

C.4.8 Management of Incompatible Wastes in Tanks Systems

The 200 Area ETF manages dilute solutions that can be mixed without compatibility issues. The 200 Area ETF is equipped with several systems that can adjust the pH of the waste for treatment activities. Sulfuric acid and sodium hydroxide are added to the process through the MCS for pH adjustment to ensure there will be no large pH fluctuations and adverse reactions in the tank systems.

C.5 Surface Impoundments

This section provides specific information on surface impoundment operations at the LERF, including descriptions of the liners and secondary containment structures, as required by WAC 173-303-650 and WAC 173-303-806(4)(d).

The LERF consists of three lined surface impoundments (basins) with a design operating capacity of 29.5 million liters each. The maximum capacity of each basin is 34 million liters. The dimensions of each basin at the anchor wall are approximately 103 meters by 85 meters. The typical top dimensions of the wetted area are approximately 89 meters by 71 meters, while the bottom dimensions are approximately 57 by 38 meters. Total depth from the top of the dike to the bottom of the basin is approximately 7 meters. The typical finished basin bottoms lie at about 4 meters below the initial grade and 175 meters above sea level. The dikes separating the basins have a typical height of 3 meters and typical top width of 11.6 meters around the perimeter of the impoundments.

C.5.1 List of Dangerous Waste

A list of dangerous and/or mixed aqueous waste that can be stored in LERF is presented in Addendum A. Addendum B, Waste Analysis Plan also provides a discussion of the types of waste that are managed in the LERF.

C.5.2 Construction, Operation, and Maintenance of Liner System

General information concerning the liner system is presented in the following sections. Information regarding loads on the liner, liner coverage, UV light exposure prevention, and location relative to the water table are discussed.

C.5.2.1 Liner Construction Materials

The LERF employs a double-composite liner system with a leachate detection, collection, and removal system between the primary and secondary liners. Each basin is constructed with an upper or primary liner consisting of a high-density polyethylene geomembrane laid over a bentonite carpet liner. The lower or secondary liner in each basin is a composite of a geomembrane laid over a layer of soil/bentonite admixture with a hydraulic conductivity less than 10^{-7} centimeters per second. The synthetic liners extend up the dike wall to a concrete anchor wall that surrounds the basin at the top of the dike. A batten system bolts the layers in place to the anchor wall (Figure C.16).

Figure C.17 is a schematic cross-section of the liner system. The liner components, listed from the top to the bottom of the liner system, are the following:

- Primary 1.5-millimeter high-density polyethylene geomembrane
- Bentonite carpet liner
- Geotextile
- Drainage gravel (bottom) and geonet (sides)
- Geotextile
- Secondary 1.5-millimeter high-density polyethylene geomembrane
- Soil/bentonite admixture (91 centimeters on the bottom, 107 centimeters on the sides)

- Geotextile

The primary geomembrane, made of 1.5-millimeter high-density polyethylene, forms the basin surface that holds the aqueous waste. The secondary geomembrane, also 1.5-millimeter high-density polyethylene, forms a barrier surface for leachate that might penetrate the primary liner. The high-density polyethylene chemically is resistant to constituents in the aqueous waste and has a relatively high strength compared to other lining materials. The high-density polyethylene resin specified for the LERF contains carbon black, antioxidants, and heat stabilizers to enhance its resistance to the degrading effects of UV light. The approach to ensuring the compatibility of aqueous waste streams with the LERF liner materials and piping is discussed in Addendum B, Waste Analysis Plan.

Three geotextile layers are used in the LERF liner system. The layers are thin, nonwoven polypropylene fabric that chemically is resistant, highly permeable, and resistant to microbiological growth. The first two layers prevent fine soil particles from infiltrating and clogging the drainage layer. The second geotextile also provides limited protection for the secondary geomembrane from the drainage rock. The third geotextile layer prevents the mixing of the soil/bentonite admixture with the much more porous and granular foundation material.

A 30.5-centimeters-thick gravel drainage layer on the bottom of the basins between the primary and secondary liners provides a flow path for liquid to the leachate detection, collection, and removal system. A geonet (or drainage net) is located immediately above the secondary geomembrane on the basin sidewalls. The geonet functions as a preferential flow path for liquid between the liners, carrying liquid down to the gravel drainage layer and subsequently to the leachate sump. The geonet is a mesh made of high-density polyethylene, with approximately 13-millimeter openings.

The soil/bentonite layer is 91 centimeters thick on the bottom of the basins and 107 centimeters thick on the basin sidewalls; its permeability is less than 10^{-7} centimeters per second. This composite liner design, consisting of a geomembrane laid over essentially impermeable soil/bentonite, is considered best available technology for solid waste landfills and surface impoundments. The combination of synthetic and clay liners is reported in the literature to provide the maximum protection from waste migration (Forseth and Kmet 1983).

A number of laboratory tests were conducted to measure the engineering properties of the soil/bentonite admixture, in addition to extensive field tests performed on three test fills constructed near the LERF site. For establishing an optimum ratio of bentonite to soil for the soil/bentonite admixture, mixtures of various ratios were tested to determine permeability and shear strength. A mixture of 12 percent bentonite was selected for the soil/bentonite liner and tests described in the following paragraphs demonstrated that the admixture meets the desired permeability of less than 10-7 centimeters per second. Detailed discussion of test procedures and results is provided in Report of Geotechnical Investigation, 242-A Evaporation and PUREX Interim Storage Basins (Chen-Northern 1990).

Direct shear tests were performed according to ASTM D3080 test procedures (ASTM 1990) on soil/bentonite samples of various ratios. Based on these results, the conservative minimum Mohr-Coulomb shear strength value of 30 degrees was estimated for a soil/bentonite admixture containing 12 percent bentonite.

The high degree of compaction of the soil/bentonite layer [92 percent per ASTM D1557 (ASTM 1991)] was expected to maximize the bonding forces between the clay particles, thereby minimizing moisture transport through the liner. With respect to particle movement ('piping'), estimated fluid velocities in this low-permeability material are too low to move the soil particles. Therefore, piping is not considered a problem.

For the soil/bentonite layer, three test fills were constructed to demonstrate that materials, methods, and procedures used would produce a soil/bentonite liner that meets the EPA permeability requirement of less than 10^{-7} centimeters per second. All test fills met the EPA requirements.

A thorough discussion of construction procedures, testing, and results is provided in *Report of Permeability Testing, Soil-bentonite Test Fill* (Chen-Northern 1991a).

The aqueous waste stored in the LERF is typically a dilute mixture of organic and inorganic constituents. Though isolated instances of soil liner incompatibility have been documented in the literature (Forseth and Kmet 1983), these instances have occurred with concentrated solutions that were incompatible with the geomembrane liners in which the solutions were contained. Considering the dilute nature of the aqueous waste that is and will be stored in LERF and the moderate pH, and test results demonstrating the compatibility of the high-density polyethylene liners with the aqueous waste [9090 Test Results (WHC 1991)], gross failure of the soil/bentonite layer is not probable.

Each basin also is equipped with a floating very low-density polyethylene cover. The cover is anchored and tensioned at the concrete wall at the top of the dikes, using a patented mechanical tensioning system. Figure C.16 depict the tension mechanism and the anchor wall at the perimeter of each basin. Additional information on the cover system is provided in Section C.5.2.5.

C.5.2.1.1 Material Specifications

Material specifications for the liner system and leachate collection system, including liners, drainage gravel, and drainage net are discussed in the following sections. Material specifications are documented in the *Final Specifications 242-A Evaporator and PUREX Interim Retention Basins* (KEH 1990a) and *Construction Specifications for 242-A Evaporator and PUREX Interim Retention Basins* (KEH 1990b).

Geomembrane Liners. The high-density polyethylene resin for geomembranes for the LERF meets the material specifications listed in Table C.9. Key physical properties include thickness (1.5 millimeters [60 mil]) and impermeability (hydrostatic resistance of over 360,000 kilogram per square meter). Physical properties meet National Sanitation Foundation Standard 54 (NSF 1985). Testing to determine if the liner material is compatible with typical dilute waste solutions was performed and documented in *9090 Test Results* (WHC 1991).

Soil/Bentonite Liner. The soil/bentonite admixture consists of 11.5 to 14.5 percent bentonite mixed into well-graded silty sand with a maximum particle size of 4.75 millimeters (No. 4 sieve). Test fills were performed to confirm the soil/bentonite admixture applied at LERF has hydraulic conductivity less than 10^{-7} centimeters per second, as required by WAC 173-303-650(2)(j) for new surface impoundments.

Bentonite Carpet Liner. The bentonite carpet liner consists of bentonite (90 percent sodium montmorillonite clay) in a primary backing of woven polypropylene with nylon filler fiber, and a cover fabric of open weave spunlace polyester. The montmorillonite is anticipated to retard migration of solution through the liner, exhibiting a favorable cation exchange for adsorption of some constituents (such as ammonium). Based on composition of the bentonite carpet and of the type of aqueous waste stored at LERF, no chemical attack, dissolution, or degradation of the bentonite carpet liner is anticipated.

Geotextile. The nonwoven geotextile layers consist of long-chain polypropylene polymers containing stabilizers and inhibitors to make the filaments resistant to deterioration from UV light and heat exposure. The geotextile layers consist of continuous geotextile sheets held together by needle punching. Edges of the fabric are sealed or otherwise finished to prevent outer material from pulling away from the fabric or raveling.

Drainage Gravel. The drainage layer consists of thoroughly washed and screened, naturally occurring rock meeting the size specifications for Grading Number 5 in Washington State Department of Transportation construction specifications (WSDOT 1988). The specifications for the drainage layer are given in Table C.10. Hydraulic conductivity tests (Chen-Northern 1992a, 1992b, 1992c) showed the drainage rock used at LERF met the sieve requirements and had a hydraulic conductivity of at least 1 centimeter per second, which exceeded the minimum of at least 0.1 centimeters per second required by WAC 173-303-650(2)(j) for new surface impoundments.

Geonet. The geonet is fabricated from two sets of parallel high-density polyethylene strands, spaced 1.3 centimeters center-to-center maximum to form a mesh with minimum two strands per 2.54 centimeter in each direction. The geonet is located between the liners on the sloping sidewalls to provide a preferential flow path for leachate to the drainage gravel and subsequently to the leachate sump.

Leachate Collection Sump. Materials used to line the 3.0-meter by 1.8-meter by 0.30-meter-deep leachate sump, at the bottom of each basin in the northwest corner, include [from top to bottom (Figure C.18)]:

- 25 millimeter high-density polyethylene flat stock (supporting the leachate riser pipe)
- Geotextile
- 1.5-millimeter high-density polyethylene rub sheet
- Secondary composite liner:
  - 1.5-millimeter high-density polyethylene geomembrane
  - 91 centimeters of soil/bentonite admixture
  - Geotextile

Specifications for these materials are identical to those discussed previously.

Leachate System Risers. Risers for the leachate system consist of 10-inch and 4-inch pipes from the leachate collection sump to the catch basin northwest of each basin (Figure C.18). The risers lay below the primary liner in a gravel-filled trench that also extends from the sump to the concrete catch basin.

The risers are high-density polyethylene pipes fabricated to meet the requirements in ASTM D1248 (ASTM 1989). The 10-inch riser is perforated every 20.3 centimeters with 1.3-centimeter holes around the diameter. Level sensors and leachate pump are inserted in the 10-inch riser to monitor and remove leachate from the sump. To prevent clogging of the pump and piping with fine particulate, the end of the riser is encased in a gravel-filled box constructed of high-density polyethylene geonet and wrapped in geotextile. The 4-inch riser is perforated every 10.2 centimeters with 0.64-centimeter holes around the diameter. A level detector is inserted in the 4-inch riser.

Leachate Pump. A deep-well submersible pump, designed to deliver approximately 110 liters per minute, is installed in the 10-inch leachate riser in each basin. Wetted parts of the leachate pump are made of 316L stainless steel, providing both corrosion resistance and durability.

C.5.2.1.2 Loads on Liner System

The LERF liner system is subjected to the following types of stresses.

Stresses from Installation or Construction Operations. Contractors were required to submit construction quality control plans that included procedures, techniques, tools, and equipment used for the construction and care of liner and leachate system. Methods for installation of all components were screened to ensure that the stresses on the liner system were kept to a minimum.

Calculations were performed to estimate the risk of damage to the secondary high-density polyethylene liner during construction (*Calculations for LERF Part B Permit Application* [HNF 1997]). The greatest risk expected was from spreading the gravel layer over the geotextile layer and secondary geomembrane. The results of the calculations show that the strength of the geotextile was sufficiently high to withstand the stress of a small gravel spreader driving on a minimum of 15 centimeters of gravel over the geotextile and geomembrane. The likelihood of damage to the geomembrane lying under the geotextile was considered low.

To avoid driving heavy machinery directly on the secondary liner, a 28-meter conveyer was used to deliver the drainage gravel into the basins. The gravel was spread and consolidated by hand tools and a bulldozer. The bulldozer traveled on a minimum thickness of 30.5 centimeters of gravel.

Where the conveyer assembly was placed on top of the liner, cribbing was placed to distribute the conveyer weight. No heavy equipment was allowed for use directly in contact with the geomembranes.

Additional calculations were performed to estimate the ability of the leachate riser pipe to withstand the static and dynamic loading imposed by lightweight construction equipment riding on the gravel layer (HNF 1997). Those calculations demonstrated that the pipe could buckle under the dynamic loading of small construction equipment; therefore, the pipe was avoided by equipment during spreading of the drainage gravel.

Installation of synthetic lining materials proceeded only when winds were less than 24 kilometers per hour, and not during precipitation. The minimum ambient air temperature for unfolding or unrolling the high-density polyethylene sheets was -10 C, and a minimum temperature of 0 C was required for seaming the high-density polyethylene sheets. Between shifts, geomembranes and geotextile were anchored with sandbags to prevent lifting by wind. Calculations were performed to determine the appropriate spacing of sandbags on the geomembrane to resist lifting caused by 130 kilometer per hour winds (HNF 1997). All of the synthetic components contain UV light inhibitors and no impairment of performance is anticipated from the short-term UV light exposure during construction. Section C.5.2.4 provides further detail on exposure prevention.

During the laying of the soil/bentonite layer and the overlying geomembrane, moisture content of the admixture was monitored and adjusted to ensure optimum compaction and to avoid development of cracks.

C.5.2.1.3 Static and Dynamic Loads and Stresses from the Maximum Quantity of Waste

When a LERF basin is full, liquid depth is approximately 6.4 meters. Static load on the primary liner is roughly 6,400 kilograms per square meter. Load on the secondary liner is slightly higher because of the weight of the gravel drainage layer. Assuming a density of 805 kilograms per square meter for the drainage gravel [conservative estimate based on specific gravity of 2.65 (Ambrose 1988)], the secondary high-density polyethylene liner carries approximately 7,200 kilograms per square meter when a basin is full.

Side slope liner stresses were calculated for each of the layers in the basin sidewalls and for the pipe trench on the northwest corner of each basin (HNF 1997). Results of these calculations indicate factors of safety against shear were 1.5 or greater for the primary geomembrane, geotextile, geonet, and secondary geomembrane.

Because the LERF is not located in an area of seismic concern, as identified in Appendix VI of 40 CFR 264 and WAC 173-303-282(6)(a)(I), discussion and calculation of potential seismic events are not required.

C.5.2.1.4 Stresses Resulting from Settlement, Subsidence, or Uplift

Uplift stresses from natural sources are expected to have negligible impact on the liner. Groundwater lies approximately 62 meters below the LERF, average annual precipitation is only 16 centimeters, and the average unsaturated permeability of the soils near the basin bottoms is high, ranging from about 5.5×10^4 centimeters per second to about 1 centimeter per second (Chen-Northern 1991b). Therefore, no hydrostatic uplift forces are expected to develop in the soil underneath the basins. In addition, the soil under the basins consists primarily of gravel and sand, and contains few or no organic constituents. Therefore, uplift caused by gas production from organic degradation is not anticipated.

Based on the design of the soil-bentonite liner, no structural uplift stresses are present within the lining system (Chen-Northern 1991b).

Regional subsidence is not anticipated because neither petroleum nor extractable economic minerals are present in the strata underlying the LERF basins, nor is karst (erosive limestone) topography present.

Dike soils and soil/bentonite layers were compacted thoroughly and proof-rolled during construction. Calculation of settlement potential showed that combined settlement for the foundation and soil/bentonite layer is expected to be about 2.7 centimeters. Settlement impact on the liner and basin stability is expected to be minimal (Chen-Northern 1991b).

C.5.2.1.5 Internal and External Pressure Gradients

Pressure gradients across the liner system from groundwater are anticipated to be negligible. The LERF is about 62 meters above the seasonal high water table, which prevents buildup of water pressure below the liner. The native gravel foundation materials of the LERF are relatively permeable and free draining. The 2 percent slope of the secondary liner prevents the pooling of liquids on top of the secondary liner. Finally, the fill rate of the basins is slow enough (average 190 liters per minute) that the load of the liquid waste on the primary liner is gradually and evenly distributed.

To prevent the buildup of gas between the liners, each basin is equipped with 21 vents in the primary geomembrane located above the maximum water level that allow the reduction of any excess gas pressure. Gas passing through these vents exit through a single pipe that penetrates the anchor wall into a carbon adsorption filter. This filter extracts nearly all of the organic compounds, ensuring that emissions to the air from the basins are not toxic.

C.5.2.2 Liner System Location Relative to High-Water Table

The lowest point of each LERF basin is the northwest corner of the sump, where the typical subgrade elevation is 175 meters above mean sea level. Based on data collected from the groundwater monitoring wells at the LERF site, the seasonal high-water table is located approximately 62 meters or more below the lowest point of the basins. This substantial thickness of unsaturated strata beneath the LERF provides ample protection to the liner from hydrostatic pressure because of groundwater intrusion into the soil/bentonite layer. Further discussion of the unsaturated zone and site hydrogeology is provided in Addendum D, Groundwater Monitoring Plan.

C.5.2.3 Liner System Foundation

Foundation materials are primarily gravels and cobbles with some sand and silt. The native soils onsite are derived from unconsolidated Holocene sediments. These sediments are fluvial and glaciofluvial sands and gravels deposited during the most recent glacial and postglacial event. Grain-size distributions and shape analyses of the sediments indicate that deposition occurred in a high-energy environment (Chen-Northern 1990).

Analysis of five soil borings from the LERF site was conducted to characterize the natural foundation materials and to determine the suitability of onsite soils for construction of the impoundment dikes and determine optimal design factors. Well-graded gravel containing varying amounts of silt, sand, and cobbles comprises the layer in which the basins were excavated. This gravel layer extends to depths of 10 to 11 meters below land surface (Chen-Northern 1990). The basins are constructed directly on the subgrade. Excavated soils were screened to remove oversize cobbles (greater than 15 centimeters in the largest dimension) and used to construct the dikes.

Settlement potential of the foundation material and soil/bentonite layer was found to be low. The foundation is comprised of undisturbed native soils. The bottom of the basin excavation lies within the well-graded gravel layer, and is dense to very dense. Below the gravel is a layer of dense to very dense poorly graded and well-graded sand. Settlement was calculated for the gravel foundation soils and for the soil/bentonite layer, under the condition of hydrostatic loading from 6.4 meters of fluid depth. The combined settlement for the soils and the soil/bentonite layer is estimated to be about 2.7 centimeters. This amount of settlement is expected to have minimal impact on overall liner or basin stability (Chen-Northern 1991b). Settlement calculations are provided in *Calculations for Liquid Effluent Retention Facility Part B Permit Application* (HNF 1997).

The load bearing capacity of the foundation material, based on the soil analysis discussed previously, is estimated at about 48,800 kilograms per square meter [maximum advisable presumptive bearing capacity (Hough 1969)]. Anticipated static and dynamic loading from a full basin is estimated to be less than 9,000 kilograms per square meter (Section C.5.2.1.3), which provides an ample factor of safety.

When the basins are empty, excess hydrostatic pressure in the foundation materials under the liner system theoretically could result in uplift and damage. However, because the native soil forming the foundations is unsaturated and relatively permeable, and because the water table is located at a considerable depth beneath the basins, any infiltration of surface water at the edge of the basin is expected to travel predominantly downward and away from the basins, rather than collecting under the excavation itself. No gas is expected in the foundation because gas-generating organic materials are not present.

Subsidence of undisturbed foundation materials is generally the result of fluid extraction (water or petroleum), mining, or karst topography. Neither petroleum, mineral resources, nor karst are believed to be present in the sediments overlying the Columbia River basalts. Potential groundwater resources do exist below the LERF. Even if these sediments were to consolidate from fluid withdrawal, their depth most likely would produce a broad, gently sloping area of subsidence that would not cause significant strains in the LERF liner system. Consequently, the potential for subsidence related failures are expected to be negligible.

Borings at the LERF site, and extensive additional borings in the 200 East Area, have not identified any significant quantities of soluble materials in the foundation soil or underlying sediments (Last et al. 1989). Consequently, the potential for sinkholes is considered negligible.

C.5.2.4 Liner System Exposure Prevention

Both primary and secondary geomembranes and the floating cover are stabilized with carbon black to prevent degradation from UV light. Furthermore, none of the liner layers experience long-term exposure to the elements. During construction, thin polyethylene sheeting was used to maintain optimum moisture content and provide protection from the wind for the soil/bentonite layer until the secondary geomembrane was laid in place. The secondary geomembrane was covered by the geonet and geotextile as soon as quality control testing was complete. Once the geotextile layer was completed, drainage material immediately was placed over the geotextile. The final (upper) geotextile layer was placed over the drainage gravel and immediately covered by the bentonite carpet liner. This was covered immediately, in turn, by the primary high-density polyethylene liner.

Both high-density polyethylene liners, geotextile layers, and geonet are anchored permanently to a concrete wall at the top of the basin berm. During construction, liners were held in place with many sandbags on both the basin bottoms and side slopes to prevent wind from lifting and damaging the materials. Calculations were performed to determine the amount of fluid needed in a basin to prevent wind lift damage to the primary geomembrane. Approximately 15 to 20 centimeters of solution are kept in each basin to minimize the potential for uplifting the primary liner (HNF 1997).

The entire lining system is covered by a very low-density polyethylene floating cover that is bolted to the concrete anchor wall. The floating cover prevents evaporation and intrusion from dust, precipitation, vegetation, animals, and birds. A patented tensioning system is employed to prevent wind from lifting the cover and automatically accommodate changes in liquid level in the basins. The cover tension mechanism consists of a cable running from the flexible geosynthetic cover over a pulley on the tension tower (located on the concrete anchor wall) to a dead man anchor. These anchors (blocks) simply hang from the cables on the exterior side of the tension towers. The anchor wall also provides for solid attachment of the liner layers and the cover, using a 6.4-millimeter batten and neoprene gasket to bolt the layers to the concrete wall, effectively sealing the basin from the intrusion of light, precipitation, and airborne dust (Figure C.16).

The floating cover, made of very low-density polyethylene with UV light inhibitors, is not anticipated to experience unacceptable degradation during the service life of the LERF. The very low-density polyethylene material contains carbon black for UV light protection, anti-oxidants to prevent heat degradation, and seaming enhancers to improve its ability to be welded. A typical manufacturer's limited warranty for weathering of very low-density polyethylene products is 20 years (Poly America, undated). This provides a margin of safety for the anticipated medium-term use of the LERF for aqueous waste storage.

The upper 3.4 to 4.6 meters of the sidewall liner also could experience stresses in response to temperature changes. Accommodation of thermal influences for the LERF geosynthetic layers is affected by inclusion of sufficient slack as the liners were installed. Calculations demonstrate that approximately 67 centimeters of slack is required in the long basin bottom dimension, 46 centimeters across the basin, and 34 centimeters from the bottom of the basin to the top of the basin wall (HNF 1997).

Thermal stresses also are experienced by the floating cover. As with the geomembranes, sufficient slack was included in the design to accommodate thermal contraction and expansion.

C.5.2.4.1 Liner Repairs During Operations

Should repair of a basin liner be required while the basin is in operation, the basin contents will be transferred to the 200 Area ETF or another available basin. After the liner around the leaking section is cleaned, repairs to the geomembrane will be made by the application of a piece of high-density polyethylene sheeting, sufficient in size to extend approximately 8 to 15 centimeters beyond the damaged area, or as recommended by the vendor. A round or oval patch will be installed using the same type of equipment and criteria used for the initial field installations.

C.5.2.4.2 Control of Air Emissions

The floating covers limit evaporation of aqueous waste and releases of volatile organic compounds into the atmosphere. To accommodate volumetric changes in the air between the fluid in the basin and the cover, and to avoid problems related to 'sealing' the basins too tightly, each basin is equipped with a carbon filter breather vent system. Any air escaping from the basins must pass through this vent, consisting of a pipe that penetrates the anchor wall and extends into a carbon adsorption filter unit.

C.5.2.5 Liner Coverage

The liner system covers the entire ground surface that underlies the retention basins. The primary liner extends up the side slopes to a concrete anchor wall at the top of the dike encircling the entire basin (Figure C.16).

C.5.3 Prevention of Overtopping

Overtopping prevention is accomplished through administrative controls and liquid-level instrumentation installed in each basin. The instrumentation includes local liquid-level indication as well as remote indication at the ETF. Before an aqueous waste is transferred into a basin, administrative controls are implemented to ensure overtopping will not occur during the transfer. The volume of feed to be transferred is compared to the available volume in the receiving basin. The transfer is not initiated unless there is sufficient volume available in the receiving basin or a cut-off level is established. The transfer into the basin would be stopped when this cut-off level is reached.

In the event of a 25-year, 24-hour storm event, precipitation would accumulate on the basin covers. Through the self-tensioning design of the basin covers and maintenance of adequate freeboard, all accumulated precipitation would be contained on the covers and none would flow over the dikes or anchor walls. The 25-year, 24-hour storm is expected to deliver 5.3 centimeters of rain or approximately 61 centimeters of snow. Cover specifications include the requirement that the covers be able to withstand the load from this amount of precipitation. Because the cover floats on the surface of the fluid in the basin, the fluid itself provides the primary support for the weight of the accumulated precipitation.

Through the cover self-tensioning mechanism, there is ample 'give' to accommodate the overlying load without overstressing the anchor and attachment points.

Rainwater and snow evaporate readily from the cover, particularly in the arid Hanford Facility climate, where evaporation rates exceed precipitation rates for most months of the year. The black color of the cover further enhances evaporation. Thus, the floating cover prevents the intrusion of precipitation into the basin and provides for evaporation of accumulated rain or snow.

C.5.3.1 Freeboard

Under current operating conditions, 0.61 meter of freeboard is maintained at each LERF basin, which corresponds to an operating level of 6.8 meters, or 29.5 million liters.

C.5.3.2 Immediate Flow Shutoff

The mechanism for transferring aqueous waste is either through pump transfers with on/off switches or through gravity transfers with isolation valves. These methods provide positive ability to shut off transfers immediately in the event of overtopping. Overtopping a basin during a transfer is very unlikely because the low flow rate into the basin provides long response times. At a flow rate of 284 liters per minute, approximately 11 days would be required to fill a LERF basin from the 6.8-meter operating level (i.e., 0.61 meter of freeboard) to maximum capacity of 34 million liters (i.e., the 7.4-meter level).

C.5.3.3 Outflow Destination

Aqueous waste in the LERF is transferred routinely to 200 Area ETF for treatment. However, should it be necessary to immediately empty a basin, the aqueous waste either would be transferred to the 200 Area ETF for treatment or transferred to another basin (or basins), whichever is faster. If necessary a temporary pumping system may be installed to increase the transfer rate.

C.5.4 Structural Integrity of Dikes

The structural integrity of the dikes was certified attesting to the structural integrity of the dikes, signed by a qualified, registered professional engineer.

C.5.4.1 Dike Design, Construction, and Maintenance

The dikes of the LERF are constructed of onsite native soils, generally consisting of cobbles and gravels. Well-graded mixtures were specified, with cobbles up to 15 centimeters in the largest dimension, but not constituting more than 20 percent of the volume of the fill. The dikes are designed with a 3:1 (3 units horizontal to 1 unit vertical) slope on the basin side, and 2.25:1 on the exterior side. The dikes are approximately 8.2 meters high from the bottom of the basin, and 3 meters above grade.

Calculations were performed to verify the structural integrity of the dikes (HNF 1997). The calculations demonstrate that the structural strength of the dikes is such that, without dependence on any lining system, the sides of the basins can withstand the pressure exerted by the maximum allowable quantity of fluid in the impoundment. The dikes have a factor of safety greater than 2.5 against failure by sliding.

C.5.4.2 Dike Stability and Protection

In the following paragraphs, various aspects of stability for the LERF dikes and the concrete anchor wall are presented, including slope failure, hydrostatic pressure, and protection from the environment.

Failure in Dike/Impoundment Cut Slopes. A slope stability analysis was performed to determine the factor of safety against slope failure. The computer program 'PCSTABL5' from Purdue University, using the modified Janbu Method, was employed to evaluate slope stability under both static and seismic loading cases. One hundred surfaces per run were generated and analyzed. The assumptions used were as follows (Chen-Northern 1991b):

- Weight of gravel: 2,160 kilograms per cubic meter.
- Maximum dry density of gravel: 2,315 kilograms per cubic meter.

- Mohr-Coulomb shear strength angle for gravel: minimum 33 degrees.
- Weight of soil/bentonite: 1,600 kilograms per cubic meter.
- Mohr-Coulomb shear strength angle for soil/bentonite: minimum 30 degrees.
- Slope: 3 horizontal: 1 vertical.
- No fluid in impoundment (worst case for stability).
- Soils at in-place moisture (not saturated conditions).

Results of the static stability analysis showed that the dike slopes were stable with a minimum factor of safety of 1.77 (Chen-Northern 1991b).

The standard horizontal acceleration required in the *Hanford Plant Standards*, "Standard Architectural-Civil Design Criteria, Design Loads for Facilities" (DOE-RL 1988), for structures on the Hanford Site is 0.12 g. Adequate factors of safety for cut slopes in units of this type generally are considered 1.5 for static conditions and 1.1 for dynamic stability (Golder 1989). Results of the stability analysis showed that the LERF basin slopes were stable under horizontal accelerations of 0.10 and 0.15 g, with minimum factors of safety of 1.32 and 1.17, respectively (Chen-Northern 1991b). Printouts from the PCSTABL5 program are provided in *Calculations for Liquid Effluent Retention Facility Part B Permit Application* (HNF 1997).

Hydrostatic Pressure. Failure of the dikes due to buildup of hydrostatic pressure, caused by failure of the leachate system or liners, is very unlikely. The liner system is constructed with two essentially impermeable layers consisting of a synthetic layer overlying a soil layer with low-hydraulic conductivity. It would require a catastrophic failure of both liners to cause hydrostatic pressures that could endanger dike integrity. Routine inspections of the leachate detection system, indicating quantities of leachate removed from the basins, provide an early warning of leakage or operational problems that could lead to excessive hydrostatic pressure. A significant precipitation event (e.g., a 25-year, 24-hour storm) will not create a hydrostatic problem because the interior sidewalls of the basins are covered completely by the liners. The covers can accommodate this volume of precipitation without overtopping the dike (Section C.5.3), and the coarse nature of the dike and foundation materials on the exterior walls provides for rapid drainage of precipitation away from the basins.

Protection from Root Systems. Risk to structural integrity of the dikes because of penetrating root systems is minimal. Excavation and construction removed all vegetation on and around the impoundments, and native plants (such as sagebrush) grow very slowly. The large grain size of the cobbles and gravel used as dike construction material do not provide an advantageous germination medium for native plants. Should plants with extending roots become apparent on the dike walls, the plants will be controlled with appropriate herbicide application.

Protection from Burrowing Mammals. The cobble size materials that make up the dike construction material and the exposed nature of the dike sidewalls do not offer an advantageous habitat for burrowing mammals. Lack of vegetation on the LERF site discourages foraging. The risk to structural integrity of the dikes from burrowing mammals is therefore minimal. Periodic visual inspections of the dikes provide observations of any animals present. Should burrowing mammals be noted onsite, appropriate pest control methods such as trapping or application of rodenticides will be employed.

Protective Cover. Approximately 7.6 centimeters of crushed gravel serve as the cover of the exterior dike walls. This coarse material is inherently resistant to the effect of wind because of its large grain size. Total annual precipitation is low (16 centimeters) and a significant storm event (e.g., a 25-year, 24-hour storm) could result in about 5.3 centimeters of precipitation in a 24-hour period. The absorbent capacity of the soil exceeds this precipitation rate; therefore, the impact of wind and precipitation run-on to the exterior dike walls will be minimal.

C.5.5 Piping Systems

Aqueous waste from the 242-A Evaporator is transferred to the LERF using a pump located in the 242-A Evaporator and approximately 1,500 meters of pipe, consisting of a 3-inch carrier pipe within a 6-inch outer containment pipeline. Flow through the pump is controlled through a valve at flow rates from 150 to 300 liters per minute. The pipeline exits the 242-A Evaporator below grade and remains below grade at a minimum 1.2 meter depth for freeze protection, until the pipeline emerges at the LERF catch basin, at the corner of each basin. All piping at the catch basin that is less than 1.2 meters below grade is wrapped with electric heat tracing tape and insulated for protection from freezing.

The transfer line from the 242-A Evaporator is centrifugally cast, fiberglass-reinforced epoxy thermoset resin pressure pipe fabricated to meet the requirements of ASME D2997 (ASME 1984). The 3-inch carrier piping is centered and supported within 6-inch containment piping. Pipe supports are fabricated of the same material as the pipe, and meet the strength requirements of ANSI B31.3 (ANSI 1987) for dead weight, thermal, and seismic loads. A catch basin is provided at the northwest corner of each basin where piping extends from the basin to allow for basin-to-basin and basin-to-200 Area ETF liquid transfers. Drawing H-2-88766, Sheets 1 through 4, provide schematic diagrams of the piping system at LERF. Drawing H-2-79604 provides details of the piping from the 242-A Evaporator to LERF.

C.5.5.1 Secondary Containment System for Piping

The 6-inch containment piping encases the 3-inch carrier pipe from the 242-A Evaporator to the LERF. All of the piping and fittings that are not directly over a catch basin or a basin liner are of this pipe-within-a-pipe construction. A catch basin is provided at the northwest corner of each basin where the inlet pipes, leachate risers, and transfer pipe risers emerge from the basin. The catch basin consists of a 20-centimeter-thick concrete pad at the top of the dike. The perimeter of the catch basin has a 20-centimeter-high curb, and the concrete is coated with a chemical resistant epoxy sealant. The concrete pad is sloped so that any leaks or spills from the piping or pipe connections will drain into the basin. The catch basin provides an access point for inspecting, servicing, and operating various systems such as transfer valving, leachate level instrumentation and leachate pump. Drawing H-2-79593 provides a schematic diagram of the catch basins.

C.5.5.2 Leak Detection System

During operation, the 242-A Evaporator receives dilute tank waste directly from the Tank Farms, treats waste by evaporation, and returns the concentrated waste to Tank Farms. The process condensate which is generated is transferred to LERF. Single-point electronic leak detection elements are installed along the transfer line at 305-meter intervals. The leak detection elements are located in the bottom of specially designed test risers. Each sensor element employs a conductivity sensor, which is connected to a cable leading back to the 242-A Evaporator control room. If a leak develops in the carrier pipe, fluid will travel down the exterior surface of the carrier pipe or the interior of the containment pipe. As moisture contacts a sensor unit, an alarm sounds in the ETF control room, which is monitored continuously when the 242-A evaporator is transferring liquids to LERF. If the alarm sounds, ETF Operations staff troubleshoots the alarm and, upon verification of a leak, requests that the pump located in the 242-A Evaporator be shut down to stop the flow of process condensate through the transfer line. The 242-A Evaporator has limited surge capacity, and its operation is closely tied to supporting Tank Farm operations. The flow of process condensate to LERF is not stopped automatically by indication of a possible leak in the primary transfer line. A low-volume air purge of the annulus between the carrier pipe and the containment pipe is provided to prevent condensation buildup and minimize false alarms by the leak detection elements.

The catch basins have conductivity leak detectors that alarm in the ETF control room. Leak detector alarms are monitored in the control room continuously during aqueous waste transfers and at least daily when no transfers are occurring. Leaks into the catch basins drain back to the basin through a 5.1-centimeter drain on the floor of the catch basin.

C.5.5.3 Certification

Although an integrity assessment is not required for piping associated with surface impoundments, an assessment of the transfer liner was performed, including a hydrostatic leak/pressure test at 10.5 kilograms per square centimeter gauge. A statement by an independent, qualified, registered professional engineer attesting to the integrity of the piping system is included in *Integrity Assessment Report for the 242-A Evaporator/LERF Waste Transfer Piping, Project W105* (WHC 1993), along with the results of the leak/pressure test.

C.5.6 Double Liner and Leak Detection, Collection, and Removal System

The double-liner system for LERF is discussed in Section C.5.2. The leachate detection, collection, and removal system (Figure C.18) as designed and constructed to remove leachate that might permeate the primary liner. System components for each basin include:

- 30.5-centimeter layer of drainage gravel below the primary liner at the bottom of the basin
- Geonet below the primary liner on the sidewalls to direct leachate to the gravel layer
- 3.0-meter by 1.8-meter by 0.30-meter-deep leachate collection sump consisting of a 25 millimeter high-density polyethylene flat stock, geotextile to trap large particles in the leachate, and 1.5-millimeter high-density polyethylene rub sheet set on the secondary liner
- 10-inch and 4-inch perforated leachate high-density polyethylene riser pipes from the leachate collection sump to the catch basin northwest of the basin
- Leachate collection sump level instrumentation installed in the 4-inch riser
- Level sensors, submersible leachate pump, and 1.5-inch fiberglass-reinforced epoxy thermoset resin pressure piping installed in the 10-inch riser
- Piping at the catch basin to route the leachate through 1.5-inch high-density polyethylene pipe back to the basins

The bottom of the basins has a two percent slope to allow gravity flow of leachate to the leachate collection sump. This exceeds the minimum of 1 percent slope required by WAC 173-303-650(j) for new surface impoundments. Material specifications for the leachate collection system are given in Section C.5.2.1.1.

Calculations demonstrate that fluid from a small hole (2 millimeter) (EPA 1989, p. 122) at the furthest end of the basin, under a low head situation, would travel to the sump in less than 24 hours (HNF 1997). Additional calculations indicate the capacity of the pump to remove leachate is sufficient to allow time to readily identify a leak and activate emergency procedures (HNF 1997).

The fluid level in each leachate sump is required to be maintained below 33 centimeters to prevent significant liquid backup into the drainage layer. The leachate pump is activated when the liquid level in the sump reaches about 28 centimeters, and is shut off when the sump liquid level reaches about 18 centimeters. This operation may be done either manually or automatically. Liquid level control is accomplished with conductivity probes that trigger relays selected specifically for application to submersible pumps and leachate fluids. A flow meter/totalizer on the leachate return pipe measures fluid volumes pumped and pumping rate from the leachate collection sumps, and indicates volume and flow rate on local readouts. In addition, a timer on the leachate pump tracks the cumulative pump operating time. Other instrumentation provided is real-time continuous level monitoring with readout at the catch basin. Leachate levels are monitored at least weekly. A sampling port is provided in the leachate piping system at the catch basin. The leak rate through the primary liner can be calculated using two methods: 1) measured as the leachate flow meter/totalizer readings (flow meters/totalizers are located on the outflow line from the collection sumps in the bottom of the LERF basins), and 2) calculated using the pump operating time readings multiplied by the pump flow rate (the pump runs at a constant flow rate). Calculations using either method are sufficient for compliance. For more information on inspections, refer to Addendum I.

The stainless steel leachate pump delivers 5 gallons per minute (19 liters per minute). The leachate pump returns draw liquid from the sump via 1.5-inch pipe and discharges into the basin through 1.5-inch high-density polyethylene pipe.

C.5.7 Construction Quality Assurance

The construction quality assurance plan and complete report of construction quality assurance inspection and testing results are provided in *242-A Evaporator Interim Retention Basin Construction Quality Assurance Plan* (KEH 1991). A general description of construction quality assurance procedures is outlined in the following paragraphs.

For excavation of the basins and construction of the dikes, regular inspections were conducted to ensure compliance with procedures and drawings, and compaction tests were performed on the dike soils.

For the soil/bentonite layer, test fills were first conducted in accordance with EPA guidance to demonstrate compaction procedures and to confirm compaction and permeability requirements can be met. The ratio of bentonite to soil and moisture content was monitored; lifts did not exceed 15 centimeters before compaction, and specific compaction procedures were followed. Laboratory and field tests of soil properties were performed for each lift and for the completed test fill. The same suite of tests was conducted for each lift during the laying of the soil/bentonite admixture in the basins.

Geotextiles and geomembranes were laid in accordance with detailed procedures and quality assurance programs provided by the manufacturers and installers. These included destructive and nondestructive tests on the geomembrane seams, and documentation of field test results and repairs.

C.5.8 Proposed Action Leakage Rate and Response Action Plan

An action leakage rate limit is established where action must be taken due to excessive leakage from the primary liner. The action leak rate is based on the maximum design flow rate the leak detection system can remove without the fluid head on the bottom liner exceeding 30 centimeters. The limiting factor in the leachate removal rate is the hydraulic conductivity of the drainage gravel. An action leakage rate (also called the rapid or large leak rate) of 20,000 liters per hectare per day (2,100 gallons/acre/day) was calculated for each basin (WHC 1992b).

When it is determined that the action leakage rate has been exceeded, the response action plan will follow the actions in WAC 173-303-650(11)(b) and (c), which includes notification of Ecology in writing within 7 days, assessing possible causes of the leak, and determining whether waste receipt should be curtailed and/or the basin emptied.

C.5.9 Dike Structural Integrity Engineering Certification

The structural integrity of the dikes was certified attesting to the structural integrity of the dikes, signed by a qualified, registered professional engineer.

C.5.10 Management of Ignitable, Reactive, or Incompatible Wastes

Although ignitable or reactive aqueous waste might be received in small quantities at LERF, such aqueous waste is mixed with dilute solutions in the basins, removing the ignitable or reactive characteristics. For compatibility requirements with the LERF liner, refer to Addendum B, Waste Analysis Plan.

C.6 Air Emissions Control

This section addresses the 200 Area ETF requirements of Air Emission Standards for Process Vents, under 40 CFR 264, Subpart AA (WAC 173-303-690 incorporated by reference) and Subpart CC. The requirements of 40 CFR 264, Subpart BB (WAC 173-303-691) is not applicable because aqueous waste with 10 percent or greater organic concentration would not be acceptable for processing at the ETF.

C.6.1 Applicability of Subpart AA Standards

The 200 Area ETF evaporator and thin film dryer perform operations that specifically require evaluation for applicability of WAC 173-303-690. Aqueous waste in these units routinely contains greater than 10 parts per million concentrations of organic compounds and are, therefore, subject to air emission requirements under WAC 173-303-690. Organic emissions from all affected process vents on the Hanford Facility must be less than 1.4 kilograms per hour and 2.8 mega grams per year, or control devices must be installed to reduce organic emissions by 95 percent.

The vessel off gas system provides a process vent system. This system provides a slight vacuum on the 200 Area ETF process vessels and tanks (refer to Section C.2.5.2). Two vessel vent header pipes combine and enter the vessel off gas system filter unit consisting of a demister, electric heater, prefilter, high-efficiency particulate air filters, activated carbon absorber, and two exhaust fans (one fan in service while the other is backup). The vessel off gas system filter unit is located in the high-efficiency particulate air filter room west of the process area. The vessel off gas system exhaust discharges into the larger building ventilation system, with the exhaust fans and stack located outside and immediately west of the ETF. The exhaust stack discharge point is 15.5 meters above ground level.

The annual average flow rate for the 200 Area ETF stack (which is the combined vessel off gas and building exhaust flow rates) is 1600 cubic meters per minute with a total annual flow of approximately 8.4 E+08 cubic meters. During waste processing, the airflow through just the vessel off gas system is about 23 standard cubic meters per minute.

Organic emissions occur during waste processing, which occurs less than 310 days each year (i.e., 85 percent operating efficiency). This operating efficiency represents the maximum annual operating time for the ETF, as shutdowns are required during the year for planned maintenance outages and for reconfiguring the 200 Area ETF to accommodate different aqueous waste.

C.6.2 Process Vents - Demonstrating Compliance

This section outlines how the 200 Area ETF complies with the requirements and includes a discussion of the basis for meeting the organic emissions limits, calculations demonstrating compliance, and conditions for reevaluation.

C.6.2.1 Basis for Meeting Limits/Reductions

The 242-A Evaporator and the 200 Area ETF are currently the only operating TSD units that contribute to the Hanford Facility volatile organic emissions under 40 CFR 264, Subpart AA. The combined release rate is currently well below the threshold of 1.4 kilograms per hour or 2,800 kilograms per year of volatile organic compounds. As a result, the 200 Area ETF meets these standards without the use of air pollution control devices.

The amount of organic emissions could change as waste streams are changed, or TSD units are brought online or are deactivated. The organic air emissions summation will be re-evaluated periodically as condition warrants. Operations of the TSD units operating under 40 CFR 264, Subpart AA, will be controlled to maintain Hanford Facility emissions below the threshold limits or pollution control device(s) will be added, as necessary, to achieve the reduction standards specified under 40 CFR 264, Subpart AA.

C.6.2.2 Demonstrating Compliance

Calculations to determine organic emissions are performed using the following assumptions:

- Maximum flow rate from LERF to 200 Area ETF is 568 liters per minute.
- Emissions of organics from tanks and vessels upstream of the UV/OX process are determined from flow and transfer rates given in *Clean Air Act Requirements, WAC 173-400, As-built Documentation, Project C-018H, 242-A Evaporator/PUREX Plant Process Condensate Treatment Facility* (Adtechs 1995).

- UV/OX reaction rate constants and residence times are used to determine the amount of organics, which are destroyed in the UV/OX process. These constants are given in *200 Area Effluent Treatment Facility Delisting Petition* (DOE/RL 1992).
- All organic compounds that are not destroyed in the UV/OX process are assumed to be emitted from the tanks and vessels into the vessel off gas system.
- No credit for removal of organic compounds in the vessel off gas system carbon absorber unit is taken. The activated carbon absorbers are used if required to reduce organic emissions.

The calculation to determine organic emissions consists of the following steps:

1. Determine the quantity of organics emitted from the tanks or vessels upstream of the UV/OX process, using transfer rate values
2. Determine the concentration of organics in the waste after the UV/OX process using UV/OX reaction rates and residence times. If the 200 Area ETF is configured such that the UV/OX process is not used, a residence time of zero is used in the calculations (i.e., none of the organics are destroyed)
3. Assuming all the remaining organics are emitted, determine the rate which the organics are emitted using the feed flow rate and the concentrations of organics after the UV/OX process
4. The amount of organics emitted from the vessel off gas system is the sum of the amount calculated in steps 1 and 3.

The organic emission rates and quantity of organics emitted during processing are determined using these calculations and are included in the Hanford Facility Operating Record, LERF and 200 Area ETF file.

C.6.2.3 Reevaluating Compliance with Subpart AA Standards

Calculations to determine compliance with Subpart AA will be reviewed when any of the following conditions occur at the 200 Area ETF:

- Changes in the maximum feed rate to the 200 Area ETF (i.e., greater than the 568 liters per minute flow rate)
- Changes in the configuration or operation of the 200 Area ETF that would modify the assumptions given in Section C.6.2.2 (e.g., taking credit for the carbon absorbers as a control device)
- Annual operating time exceeds 310 days.

C.6.3 Applicability of Subpart CC Standards

The air emission standards of 40 CFR 264, Subpart CC apply to tank, surface impoundment, and container storage units that manage wastes with average volatile organic concentrations equal to or exceeding 500 parts per million by weight, based on the hazardous waste composition at the point of origination (61 FR 59972). However, TSD units that are used solely for management of mixed waste are exempt. Mixed waste is managed at the LERF and 200 Area ETF and dangerous waste could be treated and stored at these TSD units.

TSD owner/operators are not required to determine the concentration of volatile organic compounds in a hazardous waste if the wastes are placed in waste management units that employ air emission controls that comply with the Subpart CC standards. Therefore, the approach to Subpart CC compliance at the LERF and 200 Area ETF is to demonstrate that the LERF and 200 Area ETF meet the Subpart CC control standards (40 CFR 264.1084 – 40 CFR 264.1086).

C.6.3.1 Demonstrating Compliance with Subpart CC for Tanks

Since the 200 Area ETF tanks already have process vents regulated under 40 CFR 264, Subpart AA (WAC 173-303-690), they are exempt from Subpart CC [40 CFR 264.1080(b)(8)].

C.6.3.2 Demonstrating Compliance with Subpart CC for Containers

Container Level 1 and Level 2 standards are met at the 200 Area ETF by managing all dangerous and/or mixed wastes in U.S. Department of Transportation containers [40 CFR 264.1086(f)]. Level 1 containers are those that store more than 0.1 cubic meters and less than or equal to 0.46 cubic meters. Level 2 containers are used to store more than 0.46 cubic meters of waste, which are in 'light material service'. Light material service is defined where a waste in the container has one or more organic constituents with a vapor pressure greater than 0.3 kilopascals at 20 C, and the total concentration of such constituents is greater than or equal to 20 percent by weight.

The monitoring requirements for Level 1 and Level 2 containers include a visual inspection when the container is received at the 200 Area ETF and when the waste is initially placed in the container. Additionally, at least once every 12 months when stored onsite for 1 year or more, these containers must be inspected.

If compliant containers are not used at the 200 Area ETF, alternate container management practices are used that comply with the Level 1 standards. Specifically, the Level 1 standards allow for a "container equipped with a cover and closure devices that form a continuous barrier over the container openings such that when the cover and closure devices are secured in the closed position there are no visible holes, gaps, or other open spaces into the interior of the container. The cover may be a separate cover installed on the container...or may be an integral part of the container structural design..." [40 CFR 264.1086(c)(1)(ii)]. An organic-vapor-suppressing barrier, such as foam, may also be used [40 CFR 264.1086(c)(1)(iii)]. Section C.3 provides detail on container management practices at the 200 Area ETF.

Container Level 3 standards apply when a container is used for the "treatment of a hazardous waste by a waste stabilization process" [40 CFR 264.1086(2)]. Because treatment in containers using the stabilization process is not provided at the 200 Area ETF, these standards do not apply.

C.6.3.3 Demonstrating Compliance with Subpart CC for Surface Impoundments

The Subpart CC emission standards are met at LERF using a floating membrane cover that is constructed of very-low-density polyethylene that forms a continuous barrier over the entire surface area [40 CFR 264.1085(c)]. This membrane has both organic permeability properties equivalent to a high-density polyethylene cover and chemical/physical properties that maintain the material integrity for the intended service life of the material. The additional requirements for the floating cover at the LERF have been met (Section C.5.2.4).

C.7 Engineering Drawings

C.7.1 Liquid Effluent Retention Facility

Drawings of the containment systems at the LERF are summarized in Table C.1. Because the failure of these containment systems at LERF could lead to the release of dangerous waste into the environment, modifications that affect these containment systems will be submitted to the Washington State Department of Ecology, as a Class 1, 2, or 3 Permit modification, as required by WAC 173-303-830.

Table C.1. Liquid Effluent Retention Facility Containment System

LERF System	Drawing Number	Drawing Title
Bottom Liner	H-2-79590, Sheet 1	Civil Plan, Sections and Details; Cell Basin Bottom Liner
Top Liner	H-2-79591, Sheet 1	Civil Plan, Sections and Details; Cell Basin Bottom Liner
Catch Basin	H-2-79593, Sheet 1, 3-5	Civil Plan, Section and Details; Catch Basin

The drawings identified in Table C.2 illustrate the piping and instrumentation configuration within LERF, and of the transfer piping systems between the LERF and the 242-A Evaporator. These drawings are provided for general information and to demonstrate the adequacy of the design of the LERF as a surface impoundment.

Table C.2. Liquid Effluent Retention Facility Piping and Instrumentation

LERF System	Drawing Number	Drawing Title
Transfer Piping to 242-A Evaporator	H-2-79604, Sheet 1	Piping Plot and Key Plans; 242-A Evaporator Condensate Stream
LERF Piping and Instrumentation	H-2-88766, Sheet 1	P&ID; LERF Basin and ETF Influent
	H-2-88766, Sheet 2	P&ID; LERF Basin and ETF Influent
	H-2-88766, Sheet 3	P&ID; LERF Basin and ETF Influent
	H-2-88766, Sheet 4	P&ID; LERF Basin and ETF Influent
Legend	H-2-89351, Sheet 1	Piping & Instrumentation Diagram - Legend

C.7.2 200 Area Effluent Treatment Facility

Drawings of the secondary containment systems for the 200 Area ETF containers, and tanks and process units, and for the Load-In Tanks are summarized in Table C.3. Because the failure of the secondary containment systems could lead to the release of dangerous waste into the environment, modifications, which affect the secondary containment systems, will be submitted to the Washington State Department of Ecology, as a Class 1, 2, or 3 Permit modification, as required by WAC 173-303-830.

Table C.3. Effluent Treatment Facility and Load-In Station Secondary Containment Systems

200 Area ETF Process Unit	Drawing Number	Drawing Title
Surge Tank, Process/ Container Storage Areas and Trenches - Foundation and Containment	H-2-89063, Sheet 1	Architectural/structural – Foundation and Grade Beam Plan
Sump Tank Containment	H-2-89065, Sheet 1	Architectural/structural – Foundation, Sections and Detail
Verification Tank Foundation and Containment	H-2-89068, Sheet 1	Architectural/structural – Verification Tank Foundation
Load-In Facility Foundation and Containment	H-2-817970, Sheet 1	Structural – ETF Truck Load-in Facility Plans and Sections
Load-In Facility Foundation and Containment	H-2-817970, Sheet 2	Structural – ETF Truck Load-in Facility Sections and Details

The drawings identified in Table C.4 provide an illustration of the piping and instrumentation configuration for the major process units and tanks at the ETF, and the Load-In Tanks. Drawings of the transfer piping systems between the LERF and 200 Area ETF, and between the Load-In Station and the 200 Area ETF also are presented in this table. These drawings are provided for general information and to demonstrate the adequacy of the design of the tank systems.

Table C.4. Major Process Units and Tanks at the Effluent Treatment Facility and Load-In Station

Load-In Facility	H-2-817974, Sheet 1	P&ID – ETF Truck Load-In Facility
Load-In Facility	H-2-817974, Sheet 2	P&ID – ETF Truck Load-In Facility
Surge Tank	H-2-89337, Sheet 1	P&ID – Surge Tank System
UV/Oxidation	H-2-88976, Sheet 1	P&ID – UV Oxidizer Part 1

UV/Oxidation	H-2-89342, Sheet 1	P&ID – UV Oxidizer Part 2
Reverse Osmosis	H-2-88980, Sheet 1	P&ID – 1st RO Stage
Reverse Osmosis	H-2-88982, Sheet 1	P&ID – 2nd RO Stage
IX/Polishers	H-2-88983, Sheet 1	P&ID – Polisher
Verification Tanks	H-2-88985, Sheet 1	P&ID – Verification Tank System
ETF Evaporator	H-2-89335, Sheet 1	P&ID – Evaporator
Thin Film Dryer	H-2-88989, Sheet 1	P&ID – Thin Film Dryer
Transfer Piping from LERF to ETF	H-2-88768, Sheet 1	Piping Plan/Profile 4"– 60M-002-M17 and 3"-60M-001-M17
Transfer Piping from Load-In Facility to ETF	H-2-817969, Sheet 1	Civil – ETF Truck Load-In Facility Site Plan

1

Table C.5. 200 Area Effluent Treatment Facility Tank Systems Information

Load-in tanks 59A-TK-109/-117 (2)	304 SS	34,350	3.6	4.7	0.64	Type 304 SS
Load-in tank 59A-TK-1	FRP	26,000	3.0	3.8	0.48 (dome) 0.63 (walls & bottom)	FRP
Surge tank	304 SS	462,000	7.9	9.2	0.48	Type 304 SS
pH adjustment tank	304 SS	16,700	3.0	2.5	0.64	Type 304 SS
First RO feed tank	304 SS	20,600	3.0	3.2	0.64	Type 304 SS
Second RO feed tank	304 SS	9,000	Nonround tank 3.0 m x 1.5 m	1.5	0.48 w/rib stiffeners	Type 304 SS
Effluent pH adjustment tank	304 SS	14,400	2.4	3.6	0.64	Type 304 SS
Verification tanks (3)	Carbon steel with epoxy lining	3,000,000	18.3	11.4	0.79	epoxy coating
Secondary waste receiving tanks (2)	304 SS	73,800	4.3	5.7	0.64	Type 304 SS
Concentrate tanks (2)	316L SS	24,900	3.0	3.8	0.64	Type 316 SS
ETF evaporator (Vapor Body)	Alloy 625	20,000	2.4	6.8	variable	Alloy 625
Distillate flash tank	304 SS	950	Horizontal tank 0.76	Length 2.2	0.7	304 SS
Sump tank 1	304 SS	6,900	1.5 x 1.5	3.4	0.48	304 SS
Sump tank 2	304 SS	6,700	1.5 x 1.5	3.4	0.48	304 SS

[1] The maximum operating volume of the tanks is identified.
[2] The nominal thickness of ETF tanks is represented.
[3] Type 304 SS, 304L, 316 SS and alloy 625 provide corrosion protection.
304 SS = stainless steel type 304 or 304L.
316L SS = stainless steel type 316L
FRP = Fiberglass-reinforced plastic.

Table C.6. 200 Area Effluent Treatment Facility Additional Tank System Information

Load-in tanks 59A-TK-109/-117 (2)	None	vent to atmosphere	concrete slab	SS skirt bolted to concrete	welded	flanged
Load-in tank 59A-TK-1	None	vent to atmosphere	concrete slab	bolted to concrete	none	flanged
Surge tank	None	vacuum breaker valve/vent to VOG	reinforced concrete ring plus concrete slab	structural steel on concrete base	welded	flanged
pH adjustment tank	None	vent to VOG	concrete slab	carbon steel skirt	welded	flanged
First RO feed tank	None	vent to VOG	concrete slab	carbon steel skirt	welded	flanged
Second RO feed tank	None	vent to VOG	concrete slab	carbon steel frame	welded	flanged
Effluent pH adjustment tank	None	vent to VOG	concrete slab	carbon steel skirt	welded	flanged
Verification tanks (3)	Epoxy	filtered vent to atmosphere	reinforced concrete ring plus concrete slab	structural steel on concrete base	welded	flanged
Secondary waste receiving tanks (2)	None	vent to VOG	concrete slab	carbon steel skirt	welded	flanged
Concentrate tanks (2)	None	vent to VOG	concrete slab	carbon steel skirt	welded	flanged
ETF evaporator (vapor body)	None	pressure indicator/pressure relief valve vapor vent to DFT/VOG	concrete slab	carbon steel frame	welded	flanged
Distillate flash tank	None	Pressure relief valve/vent to vent gas cooler/VOG	concrete slab	carbon steel I-beam and cradle	welded	flanged
Sump tank 1	None	vent to VOG	concrete containment	reinforced concrete containment basin	welded	flanged
Sump tank 2	None	vent to VOG	concrete containment	reinforced concrete containment basin	welded	flanged

DFT = distillate flash tank

VOG = vessel off gas system

1 **Table C.7. Ancillary Equipment and Material Data**

Load-in tanks	Load-in/transfer pumps (2)	2025ED-P-103A/-103B	316 SS
		2025ED-P-001A/-001B	Cast iron
	Load-in filters (6)	59A-FL-001/-002/-003/ -004/-005/-006	304 SS
Surge tank	Surge tank pumps (3)	2025E-60A-P-1A/-1B/-1C	304 SS
Rough filter	Rough filter	2025E-60B-FL-1	304 SS
UV/OX	UV oxidation inlet cooler	2025E-60B-E-1	316 SS
	UV oxidizers (4)	2025E-60D-UV-1A/-1B/-2A/-2B	316 SS
pH adjustment	pH adjustment pumps (2)	2025E-60C-P-1A/-1B	304 SS
Peroxide decomposer	H2O2 decomposers (2)	2025E-60D-CO-1A/-1B	CS with epoxy coating
Fine filter	Fine filter	2025E-60B-FL-2	304 SS
Degasification	Degasification column inlet cooler	2025E-60E-E-1	316 SS
	Degasification column	2025E-60E-CO-1	FRP
	Degasification pumps (2)	2025E-60E-P-1A/-1B	316 SS
RO	Feed/booster pumps (6)	2025E-60F-P-1A/-1B/-2A/-2B/-3A/-3B	304 SS
	Reverse osmosis arrays (21)	2025E-60F-RO-01 through -21	Membranes: polyamide Outer piping: 304 SS
IX/Polishers	Polishers (3)	2025E-60G-IX-1A/-1B-1C	CS with epoxy coating
	Resins strainers (3)	2025E-60G-S-1A/-1B/-1C	304 SS
Effluent pH adjustment	Recirculation/transfer pumps (2)	2025E-60C-P-2A/-2B	304 SS/PVC
Verification tanks	Return pump	2025E-60H-P-1	304 SS
	Transfer pumps (2)	2025E-60H-P-2A/-2B	
Secondary waste receiving tanks	Secondary waste feed pumps (2)	2025E-60I-P-1A/-1B	304 SS
ETF evaporator system	Feed/distillate heat exchanger	2025E-60I-E-02	Tubes: 316 SS Shell: 304 SS
	Heater (reboiler)	2025E-60I-E-01	Tubes: alloy 625 Shell: 304 SS
	Recirculation pump	2025E-60I-P-02	316 SS
	Concentrate transfer pump	2025E-60I-P-04	316 SS
	Entrainment separator	2025E-60I-DE-01	Top section: 316 SS Bottom section: alloy 625
	Vapor compressor (incl. silencers)	2025E-60I-C-01	304 SS
	Silencer drain pump	2025E-60I-P-06	316 SS
	Level control tank	2025E-60I-TK-5	304 SS
	Distillate flash tank pump	2025E-60I-P-03	316 SS
Concentrate tanks	Concentrate circulation pumps (2)	2025E-60J-P-1A/-1B	316 SS
Thin film dryer	Concentrate feed pump	2025E-60J-P-2	316 SS
	Thin film dryer	2025E-60J-D-1	Interior surfaces: alloy 625 Rotor and blades: 316 SS
	Powder hopper	2025E-60J-H-1	316 SS
	Spray condenser	2025E-60J-DE-01	316 SS
	Distillate condenser	2025E-60J-CND-01	Tubes: 304 SS Shell: CS
	Dryer distillate pump	2025E-60J-P-3	316 SS
Resin dewatering	Dewatering pump	2025E-80E-P-1	

Table C.8. Concrete and Masonary Coatings

Location	Product Name	Applied Film Thickness, Estimated
ETF Process and Container Storage Areas		
Floor: Topcoat	Steelcote Floor-Nu Finish[1]	2 coats at 10-12 mils
Floor: Primer	Steelcote Monomid Hi-Build[1]	2.0 mils
Walls to 7 feet, Doors & Jambs	Chemproof PermaCoat 4000 Vertical[2]	2 coats at 12-16 mils
Load-in Station Tank Pit		
Floor and Walls	Ameron Amercoat 351[3]	2 coats at 8.0-12 mils
Surge Tank and Verification Tank Berms		
Floors (and Walls at Surge Tank): Topcoat	KCC Corrosion Control Elasti-Liner I[4]	80 mils
Floors (and Walls at Surge Tank): Primer	KCC Corrosion Control Techni-Plus E3[4]	5.0-7.0 mils

[1]Floor-Nu Finish and Monomid Hi-Build are trademarks of Steelcote Manufacturing, Incorporated
[2]PermaCoat is a trademark of Chemproof Polymers, Incorporated
[3]Amercoat is a trademark of Ameron International, Incorporated
[4]Elasti-Liner and Techni-Plus are trademarks of KCC Corrosion Control, Incorporated

Table C.9. Geomembrane Material Specifications

Property	Value
Specific gravity	0.932 to 0.950
Melt flow index	1.0 g/10 min., maximum
Thickness (thickness of flow marks shall not exceed 200% of the nominal liner thickness)	60 mil 310% (1.5 mm 3 10%)
Carbon black content	1.8 to 3%, bottom liner 2 to 3% top liner
Tensile properties (each direction)	
Tensile strength at yield	21.5 kgf/cm width, minimum
Tensile strength at break	32.2 kgf/cm width, minimum
Elongation at yield	10%, minimum
Elongation at break	500%, minimum
Tear resistance	13.6 kgf, minimum
Puncture resistance	31.3 kgf, minimum
Low temperature/brittleness	-400 C, maximum
Dimensional (%change each direction)	32%, maximum
Environmental stress crack	750 h, minimum
Water absorption	0.1 maximum and weight change
Hydrostatic resistance	316,000 kgf/m^2
Oxidation induction time (200 C/l atm. O_2)	90 minutes

Reference: Construction Specifications (KEH 1990b). Format uses NSF 54 table for high-density polyethylene as a guide (NSF 1985). However, RCRA values for dimensional stability and environmental stress crack have been added.

% = percent
g = gram
min = minute
h = hour
max = maximum
kgf = kilograms force
m = meters
mm = millimeters

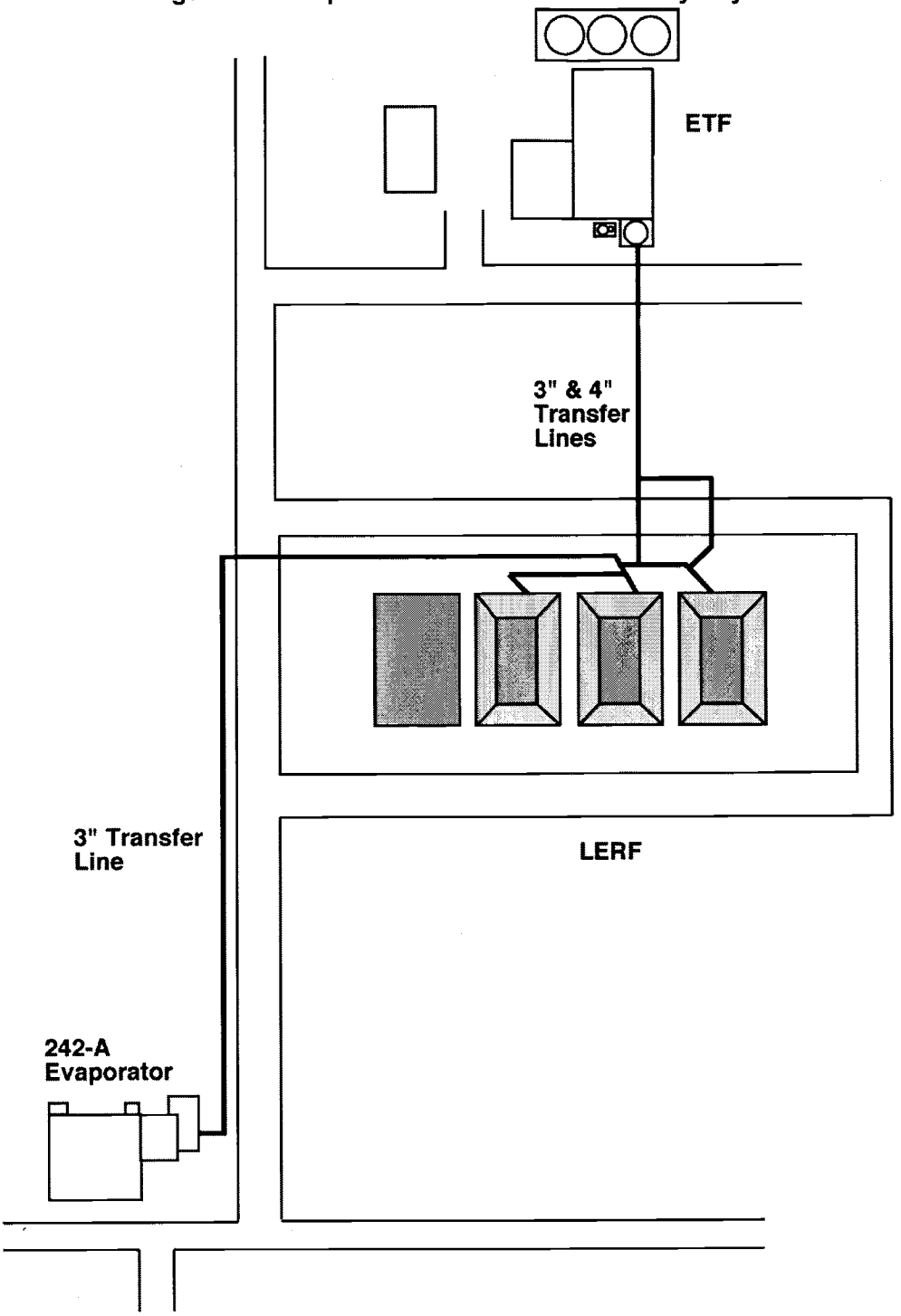
1 **Table C.10. Drainage Gravel Specifications**

Property	Value
Sieve size	
25 millimeters	100 wt% passing
19 millimeters	80 – 100 wt% passing
9.5 millimeters	10 – 40 wt% passing
4.75 millimeters	0 – 4 wt% passing
Permeability	0.1 cm/sec, minimum

2 Reference: Sieve size is from WSDOT M41-10-88, Section 9.03.1(3)C for Grading No. 5 (WSDOT 1988).
3 Permeability requirement is from WAC 173-303-650(2)(j) for new surface impoundments.
4

1

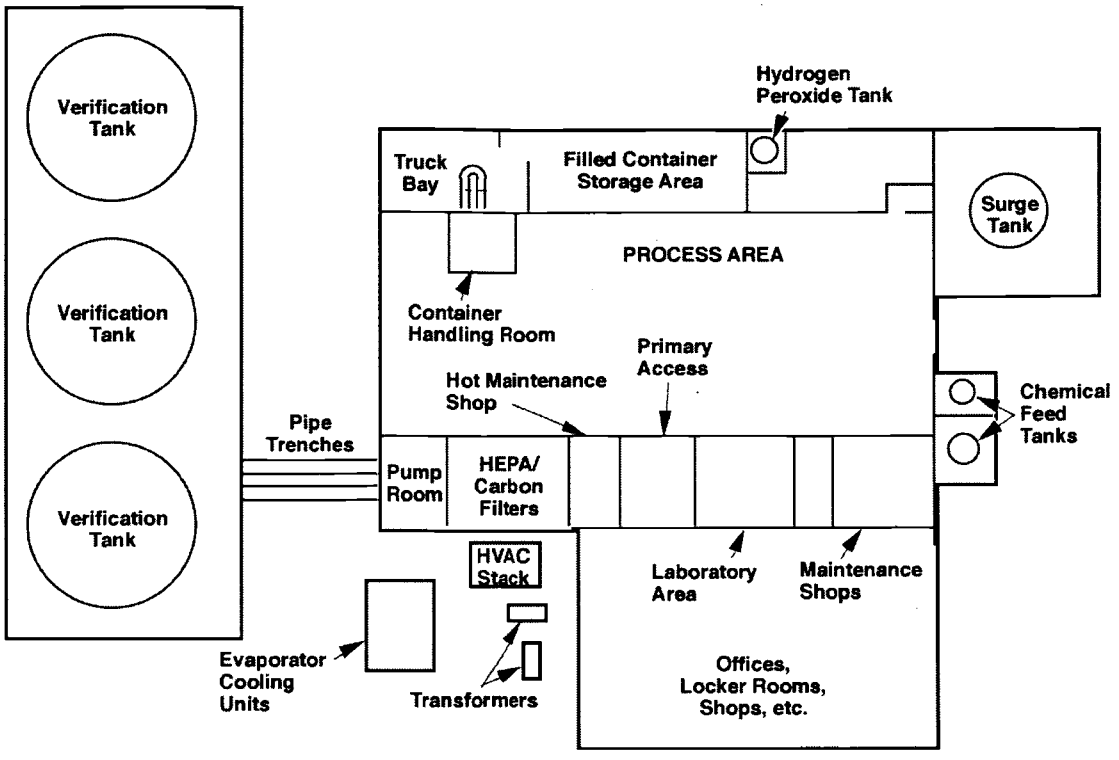
Figure C.1. Liquid Effluent Retention Facility Layout



ETF = Effluent Treatment Facility
LERF = Liquid Effluent Retention Facility

M0704-3.5
4-21-07

2



HEPA = High-efficiency particulate air
HVAC = Heating, ventilation, and air conditioning

M0704-3.6
4-24-07

Figure C.2. Plan View of the 200 Area Effluent Treatment Facility

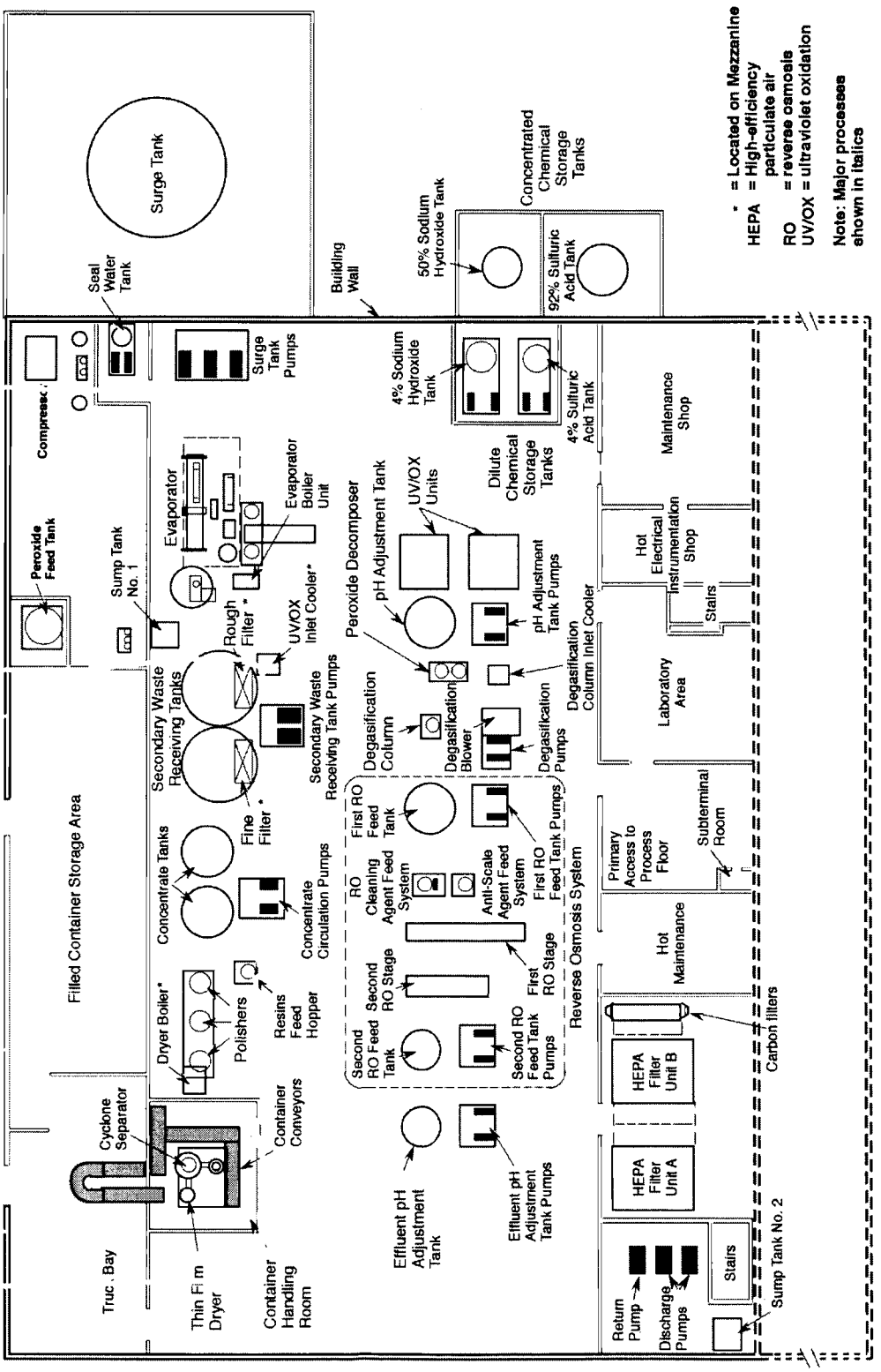
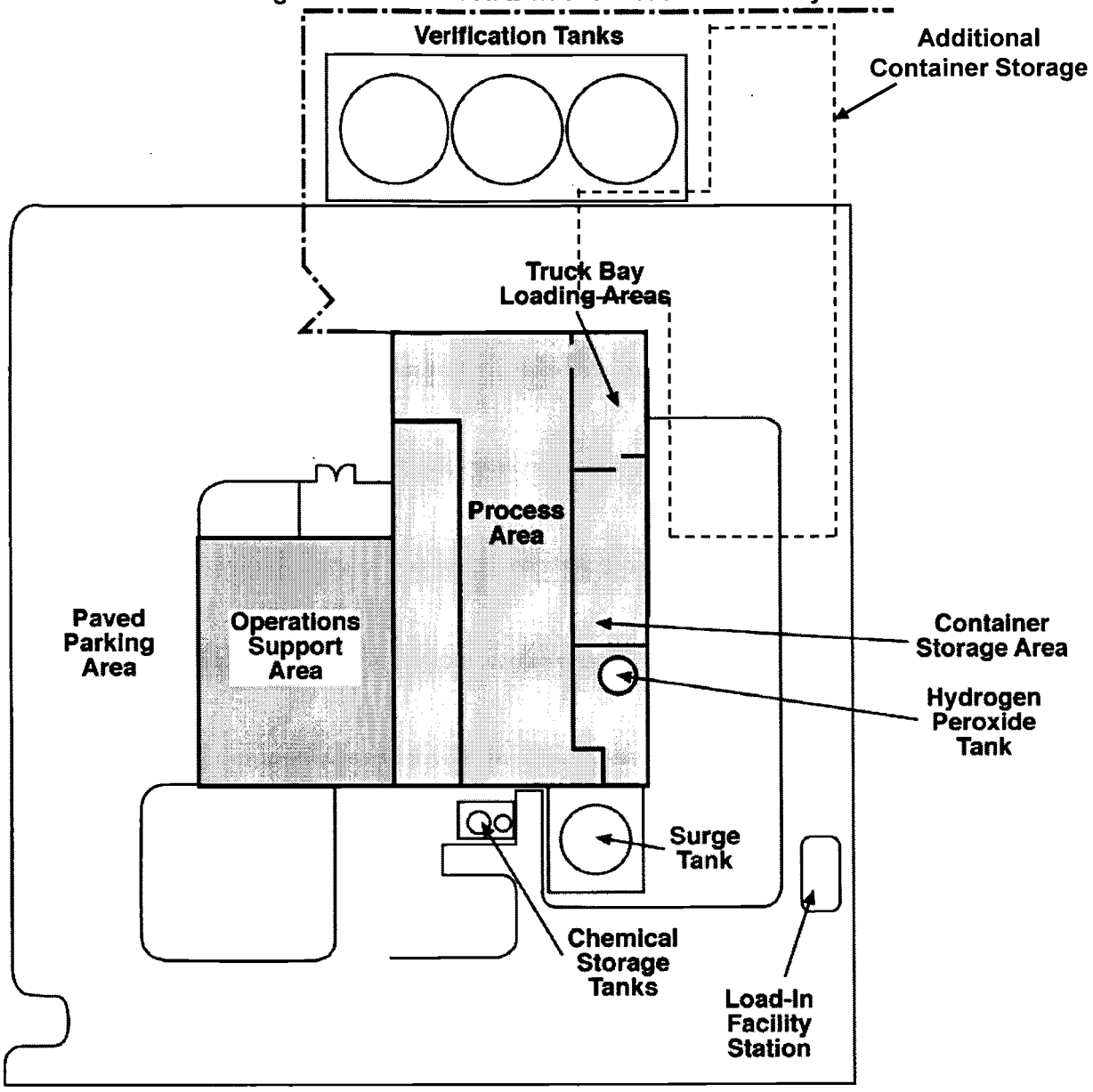


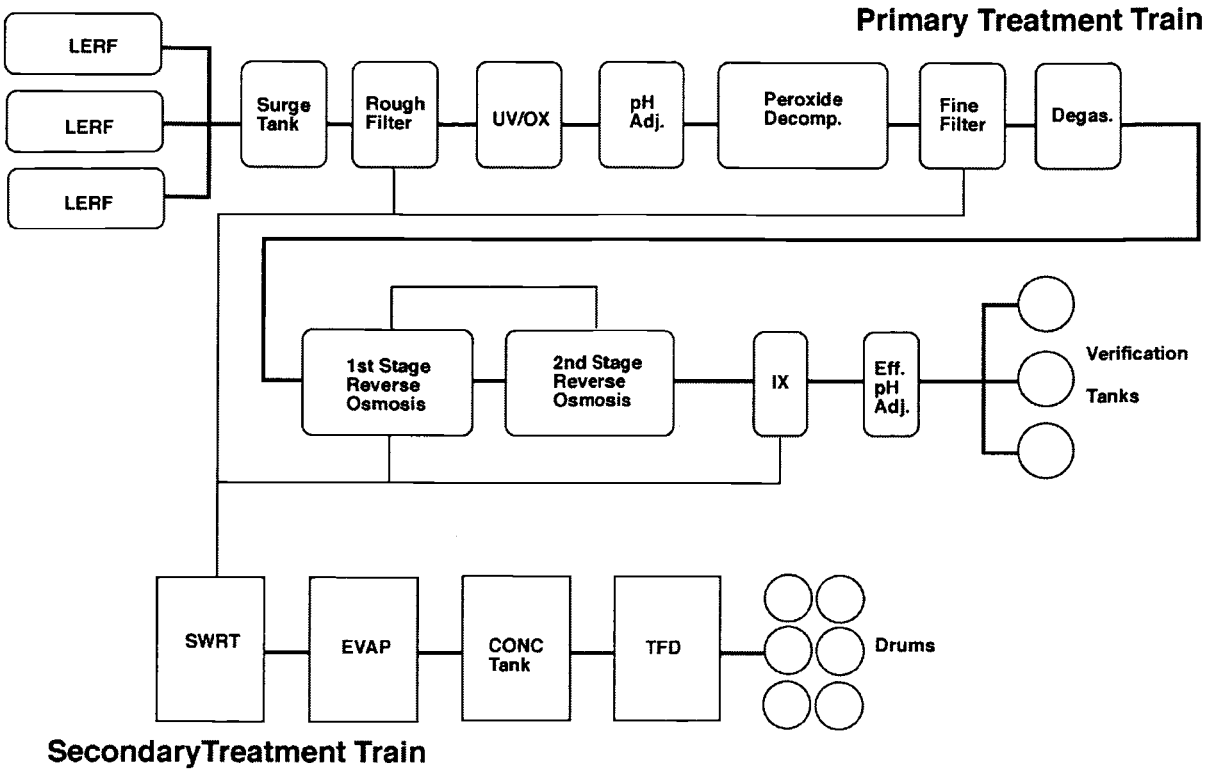
Figure C.3. 200 Area Effluent Treatment Facility Layout

1 **Figure C.4. 200 Area Effluent Treatment Facility**



2

3



CONC Tank = Concentrate tank
Degas. = Degasification column
Eff. pH Adj. = Effluent pH adjustment tank
EVAP = Evaporator
IX = Ion Exchange
LERF = Liquid Effluent Retention Facility
pH Adj. = pH adjustment tank
SWRT = Secondary waste receiving tank
TFD = Thin film dryer
UV/OX = Ultraviolet Oxidation

M0704-3.8
4-21-07

Figure C.5. Example - 200 Area Effluent Treatment Facility Configuration 1

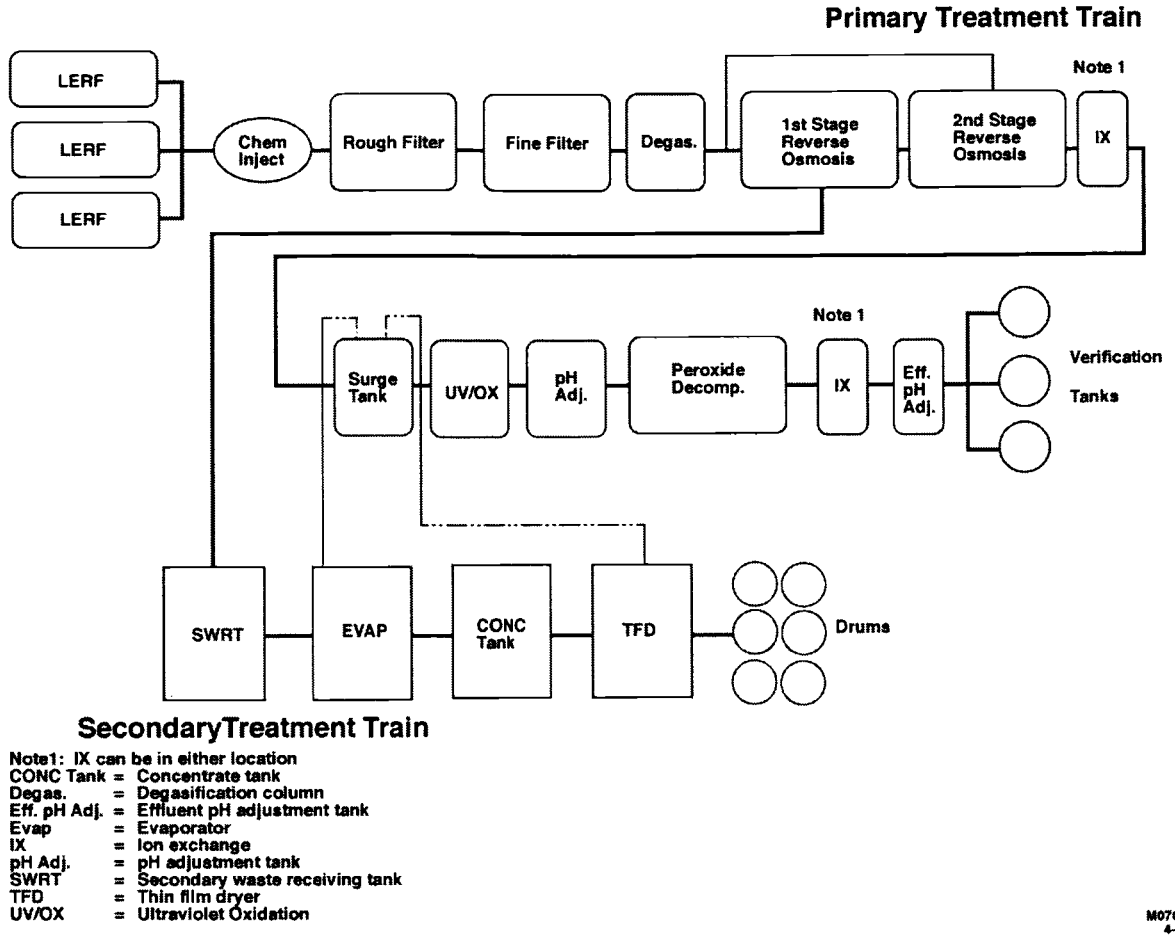


Figure C.6. Example - 200 Area Effluent Treatment Facility Configuration 2

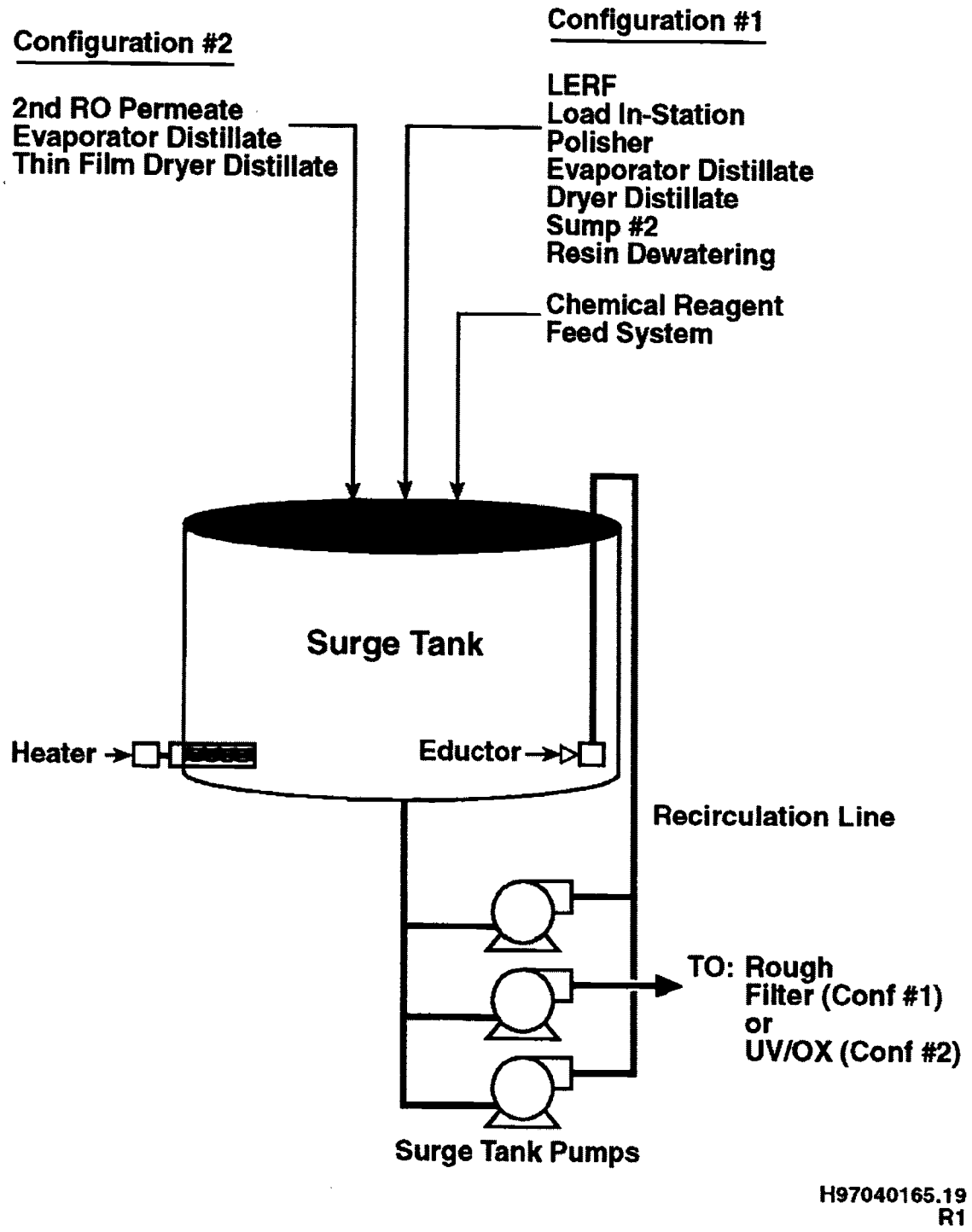


Figure C.7. Surge Tank

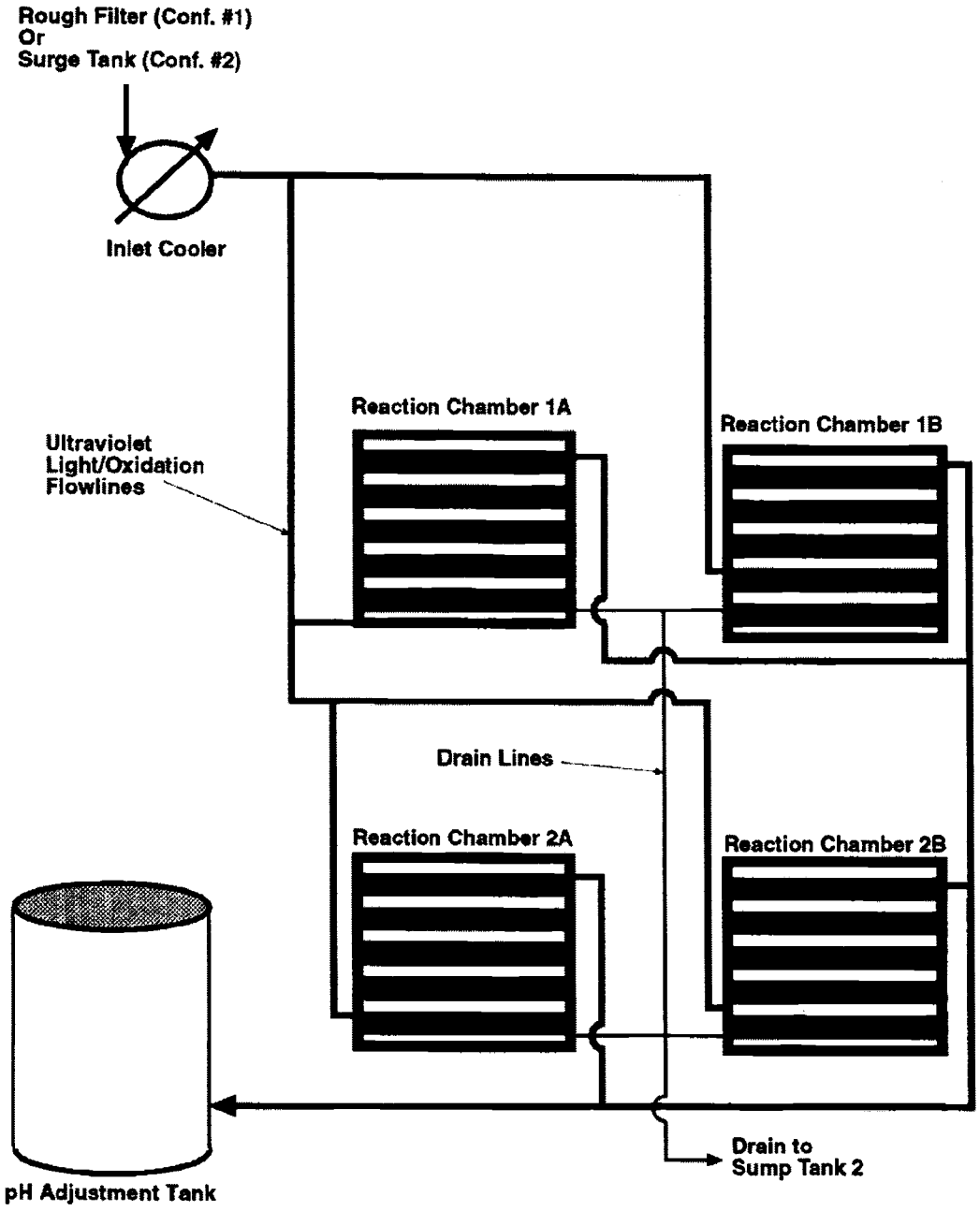
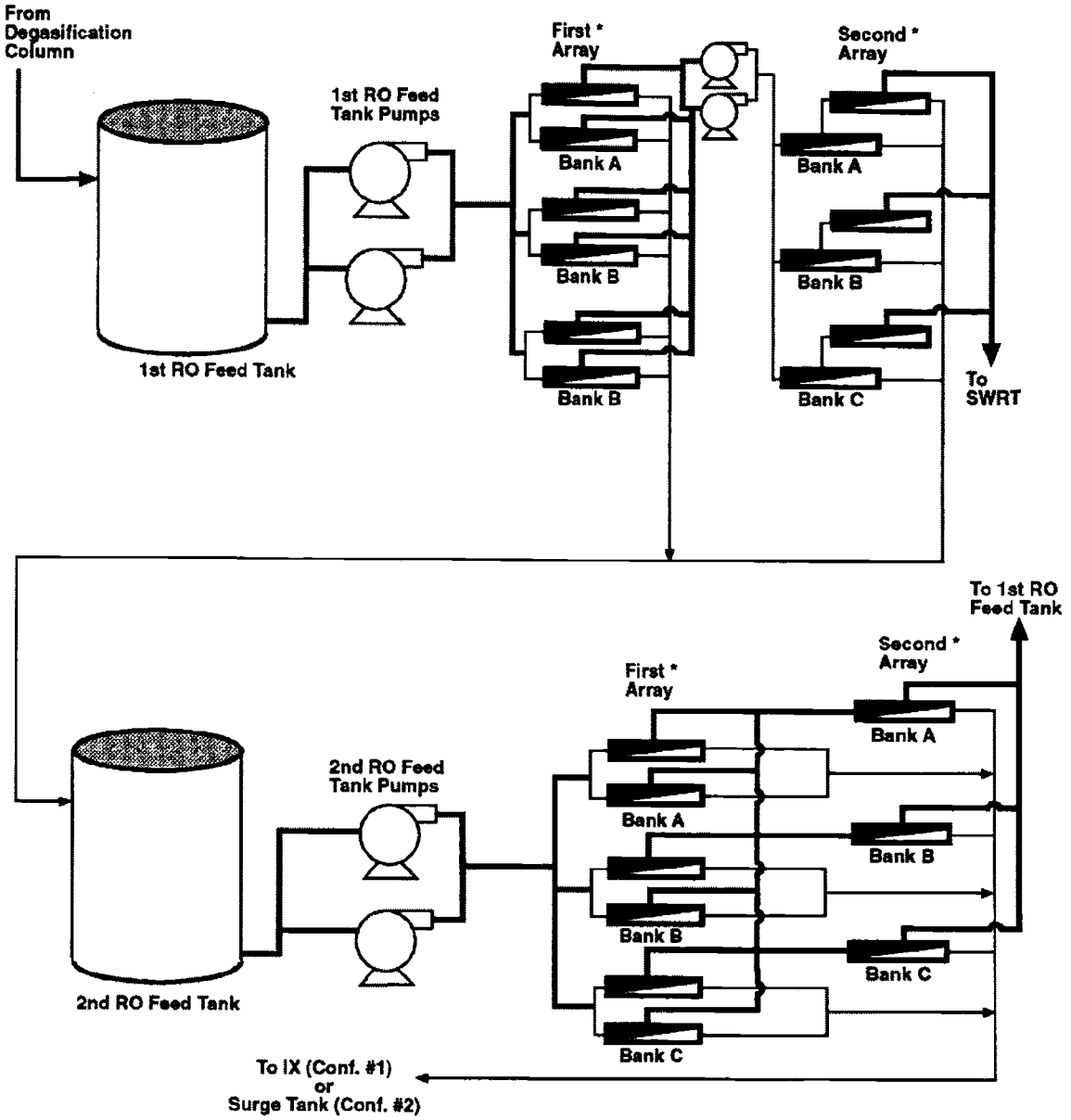


Figure C.8. Ultraviolet Light/Oxidation Unit

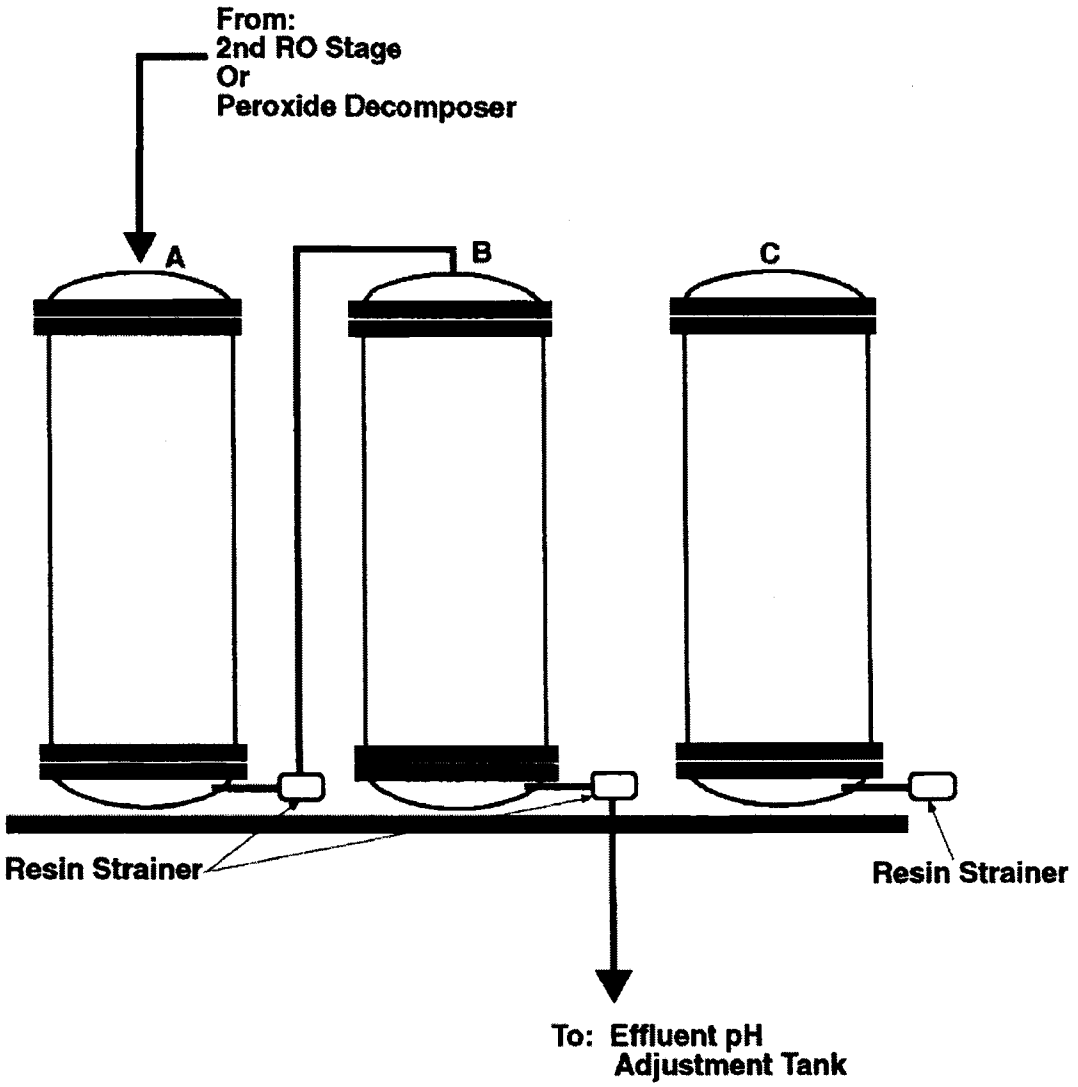
1

Figure C.9. Reverse Osmosis Unit



2

3



NOTE: Example Configuration- Column A and B in Operation, Column C in Standby Mode

H97040165.18

Figure C.10. Ion Exchange Unit

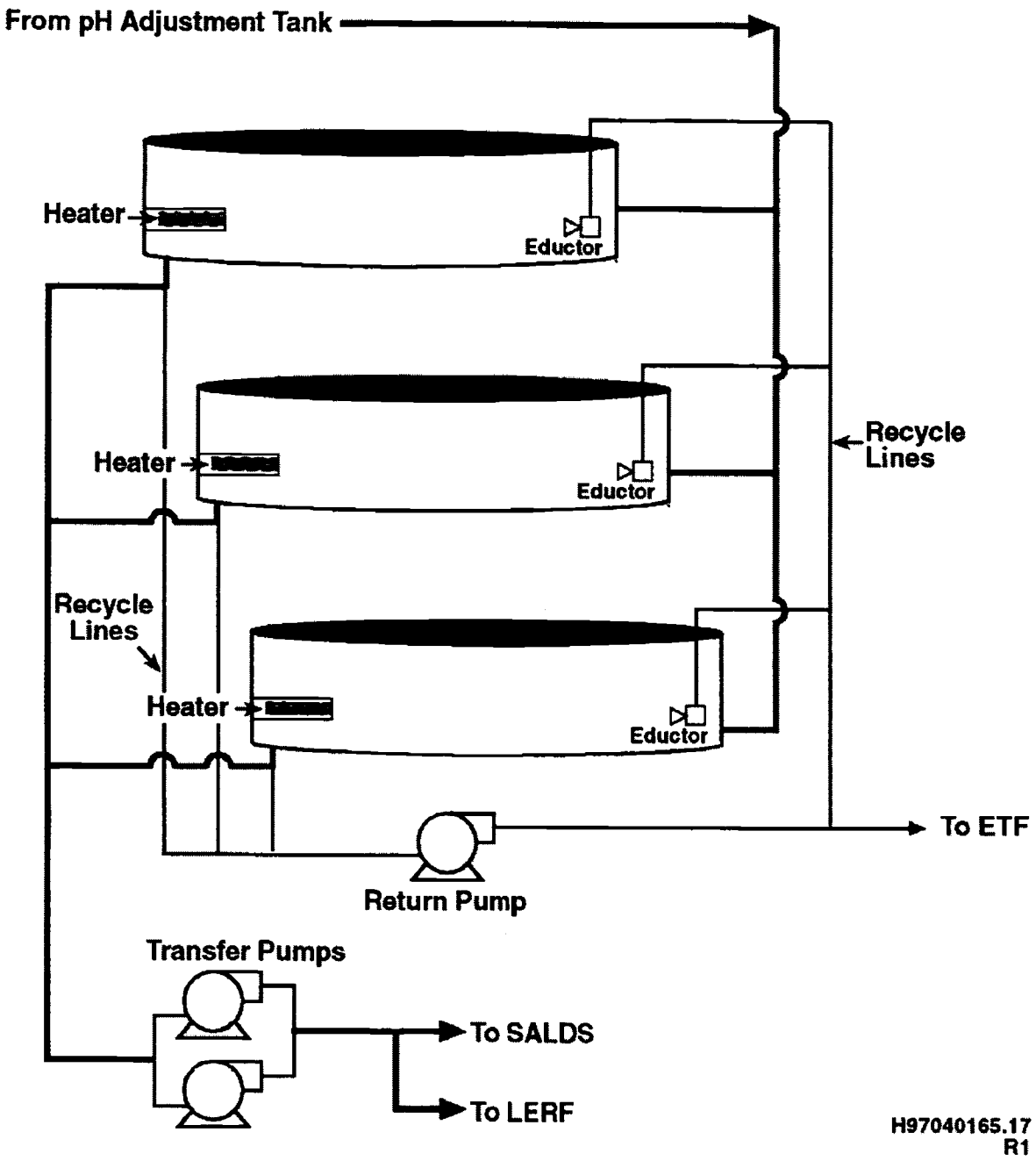


Figure C.11. Verification Tanks

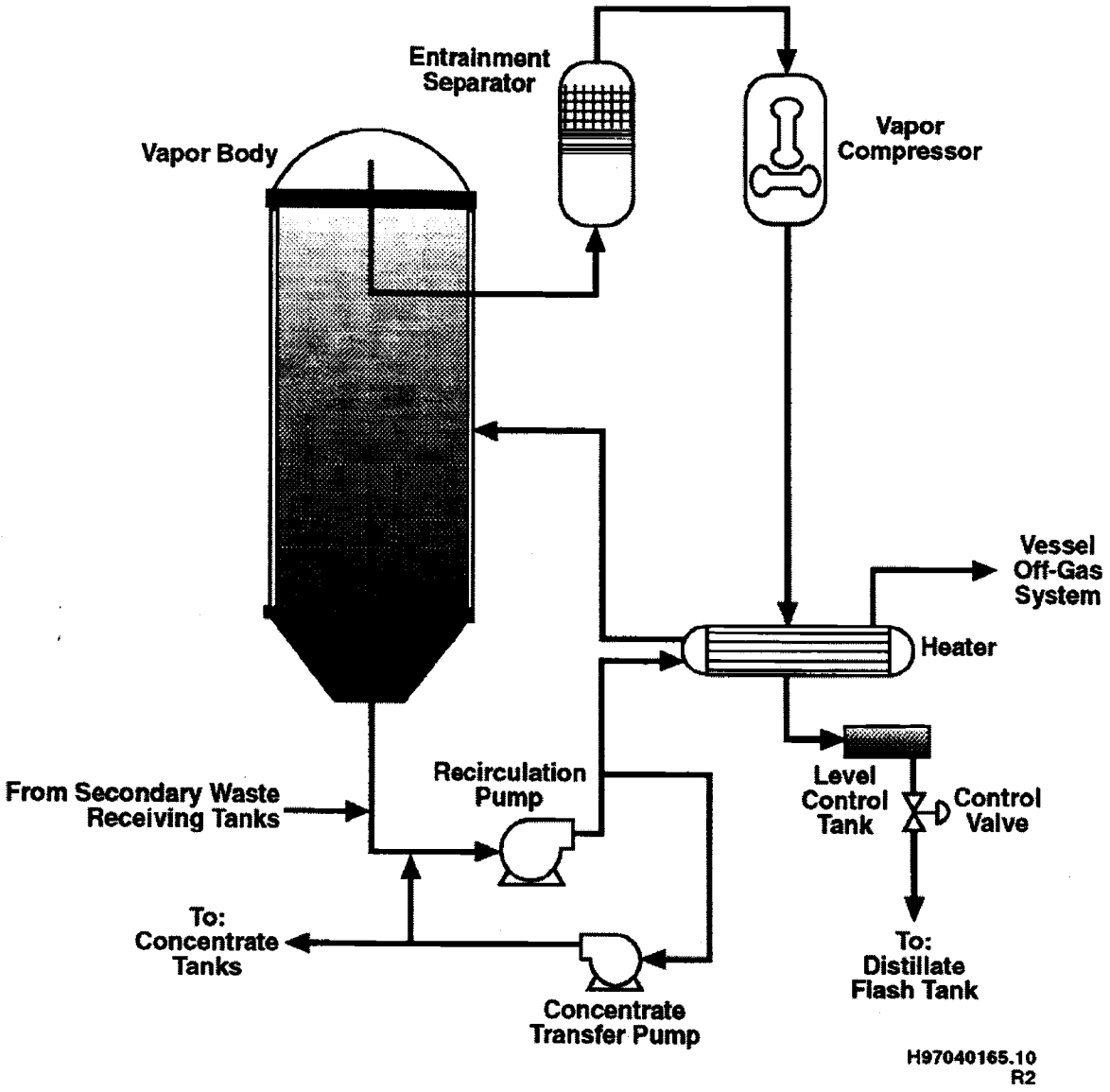


Figure C.12. Effluent Treatment Facility Evaporator

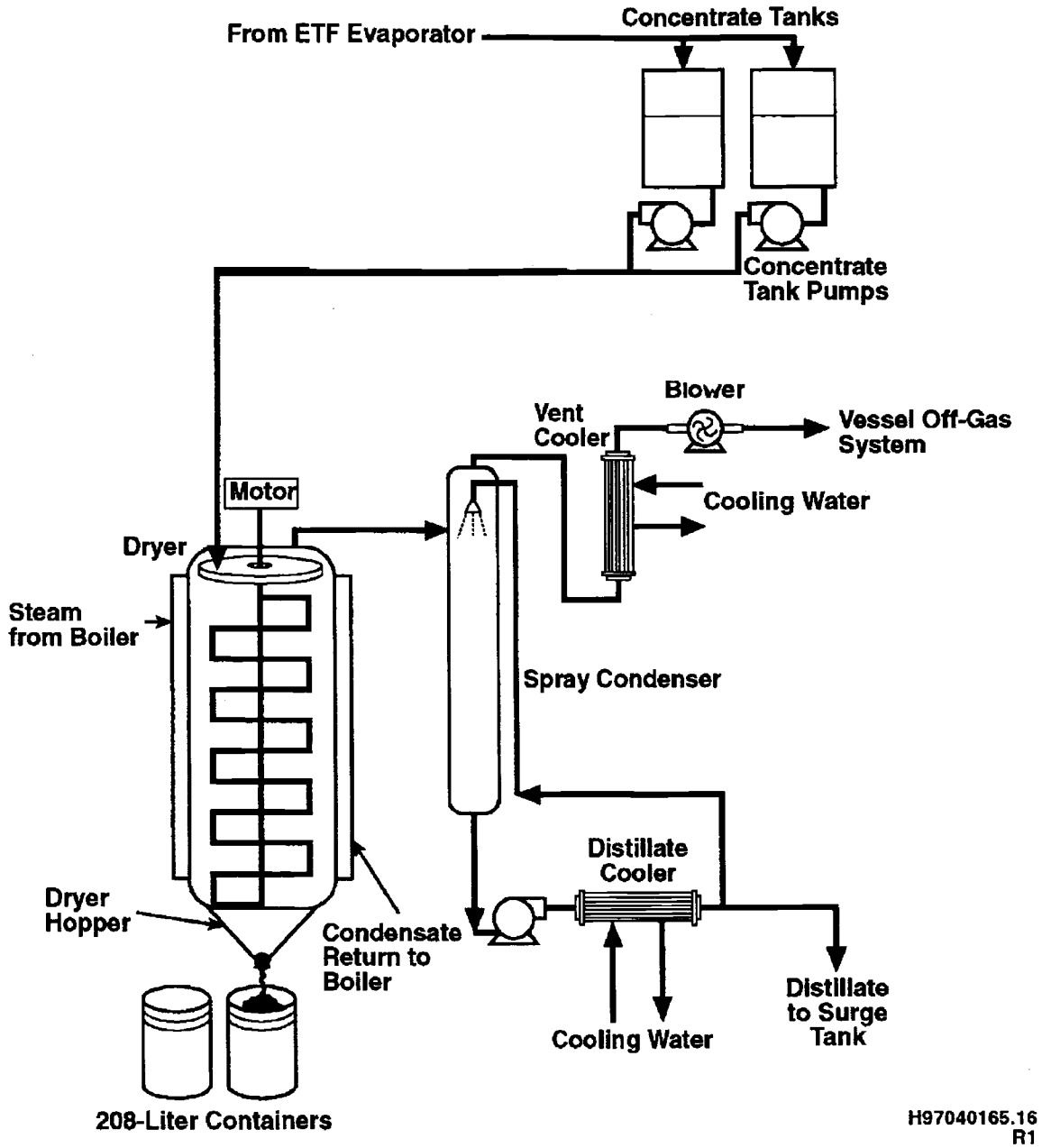


Figure C.13. Thin Film Dryer

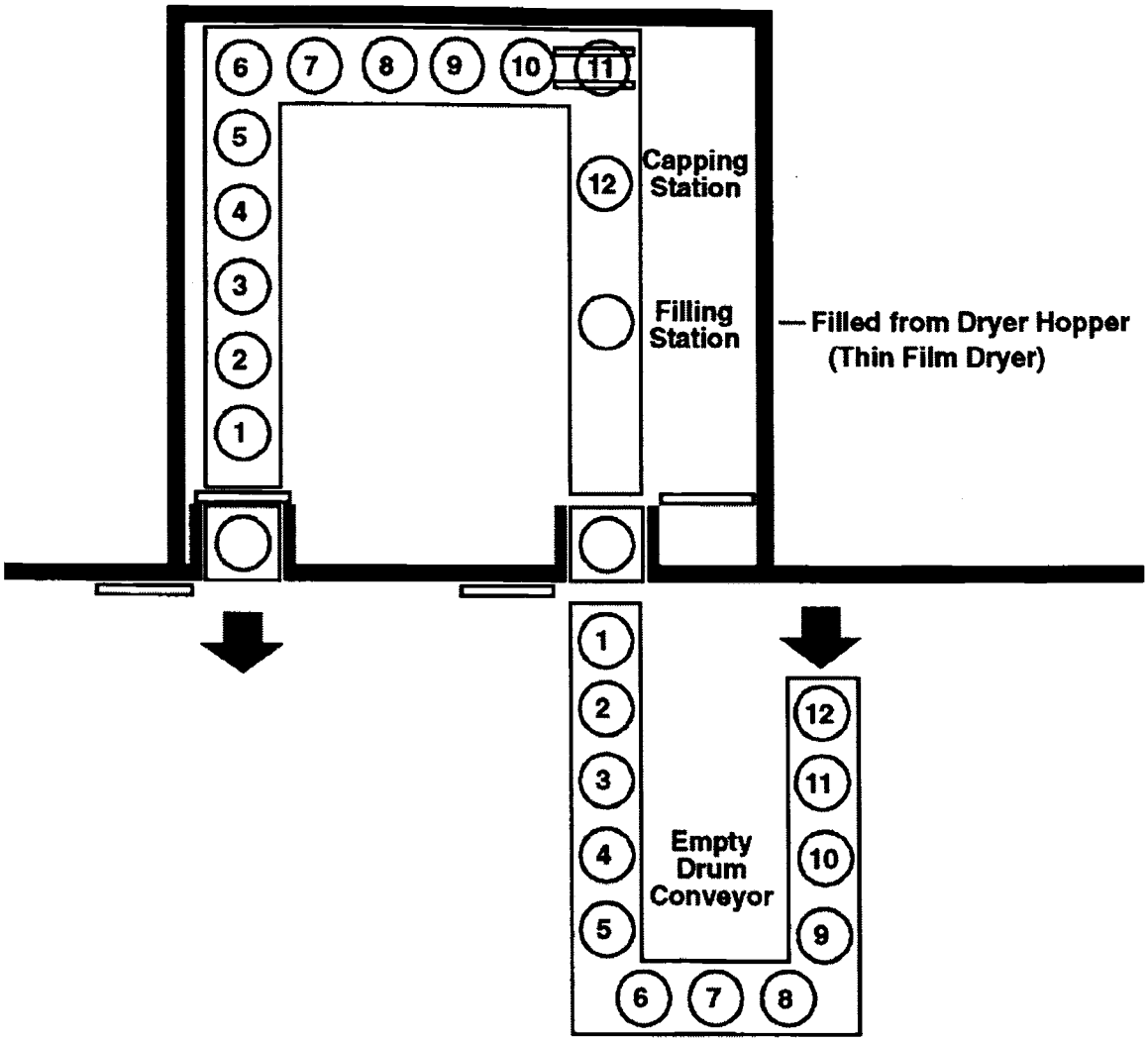


Figure C.14. Container Handling System

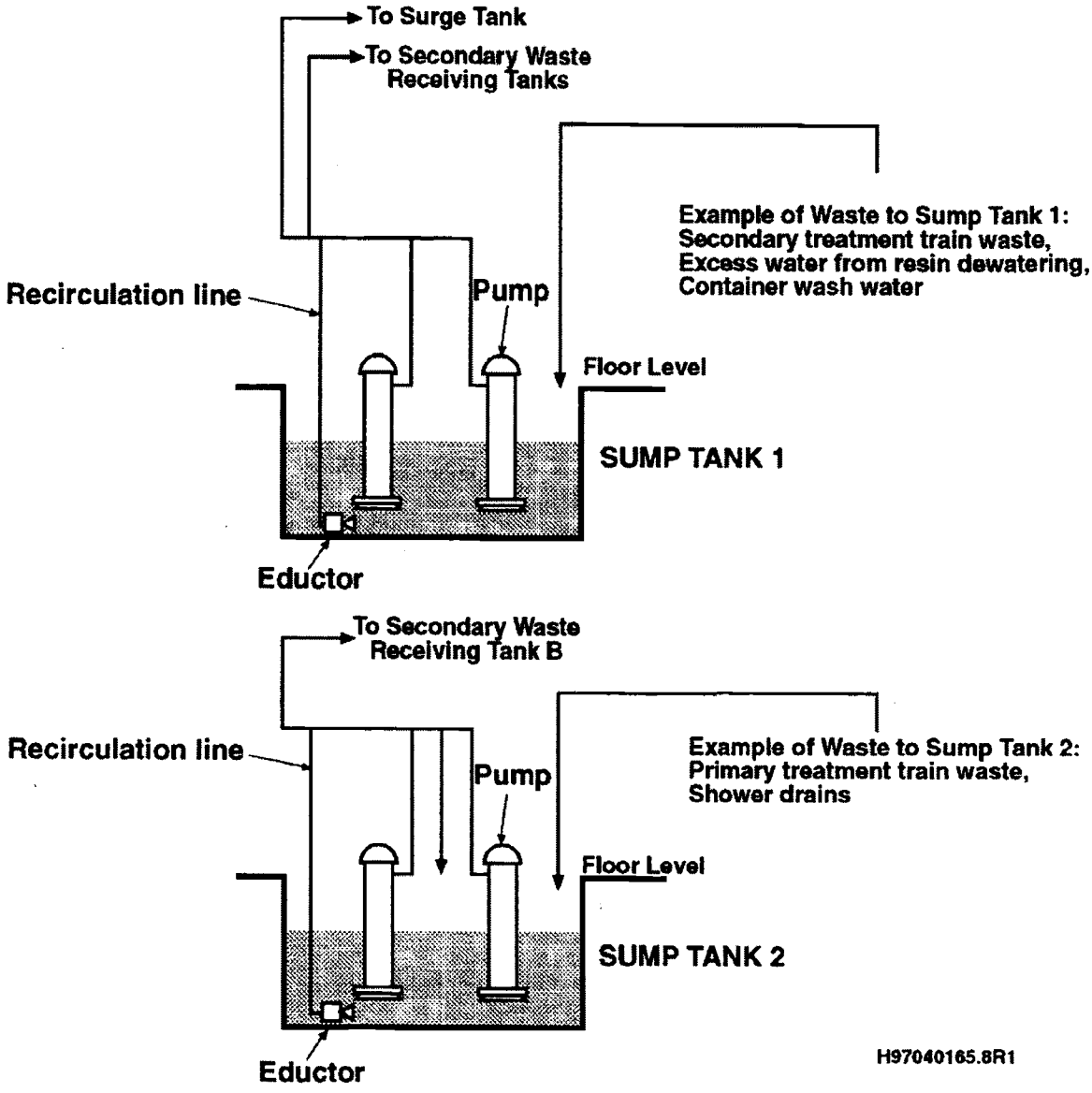


Figure C.15. Effluent Treatment Facility Sump Tanks

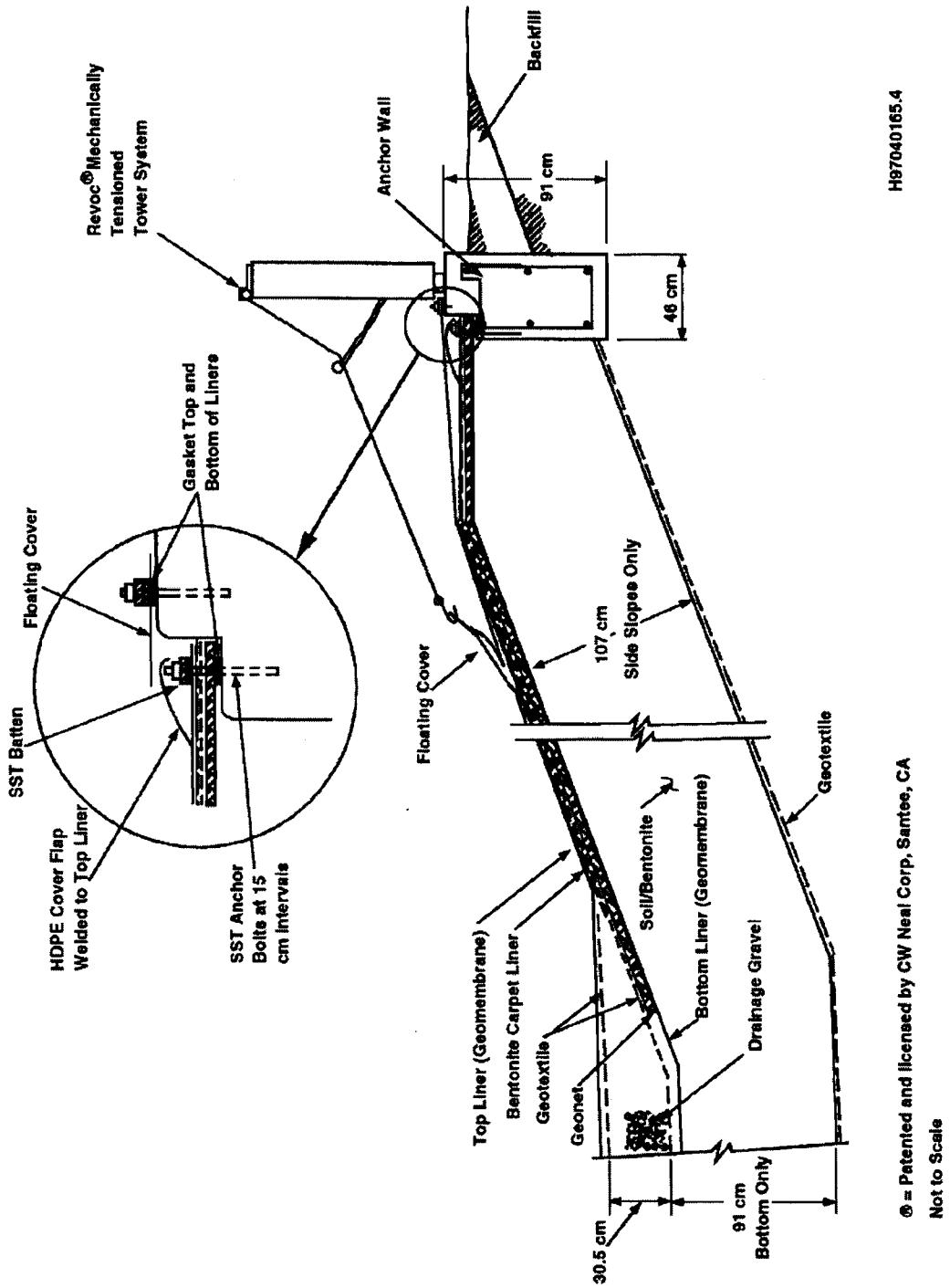


Figure C.16. Liner Anchor Wall and Cover Tension System

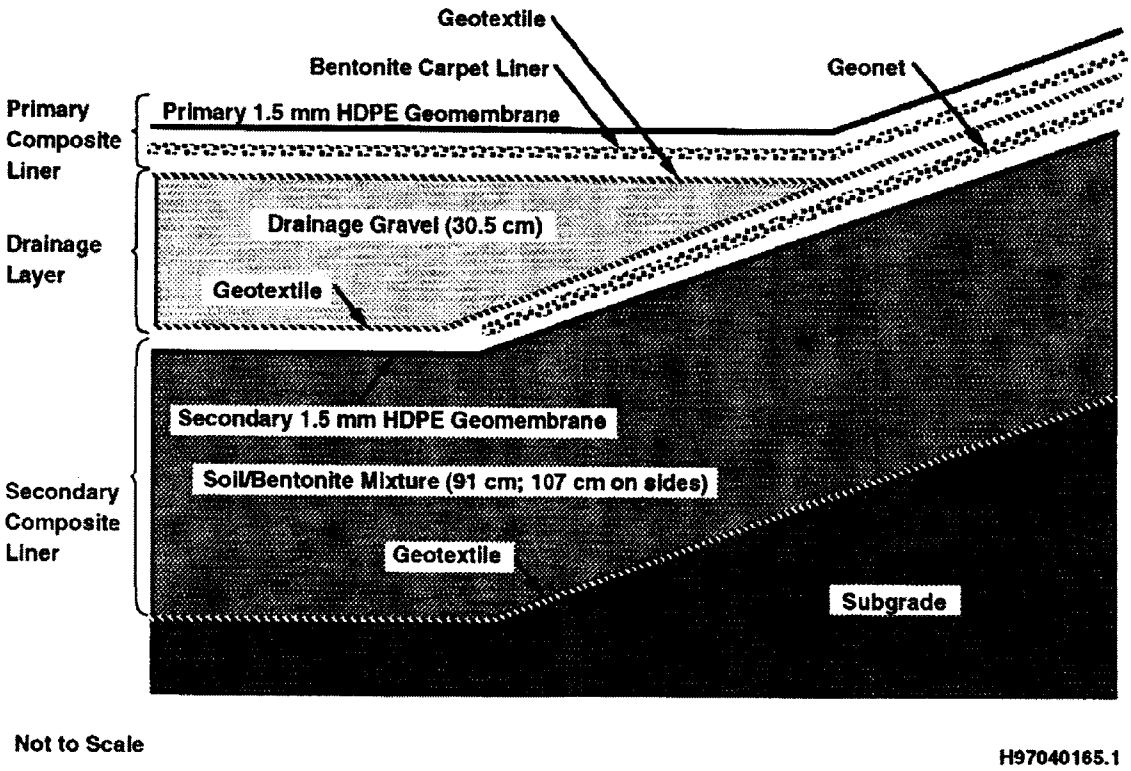


Figure C.17. Liner System Schematic

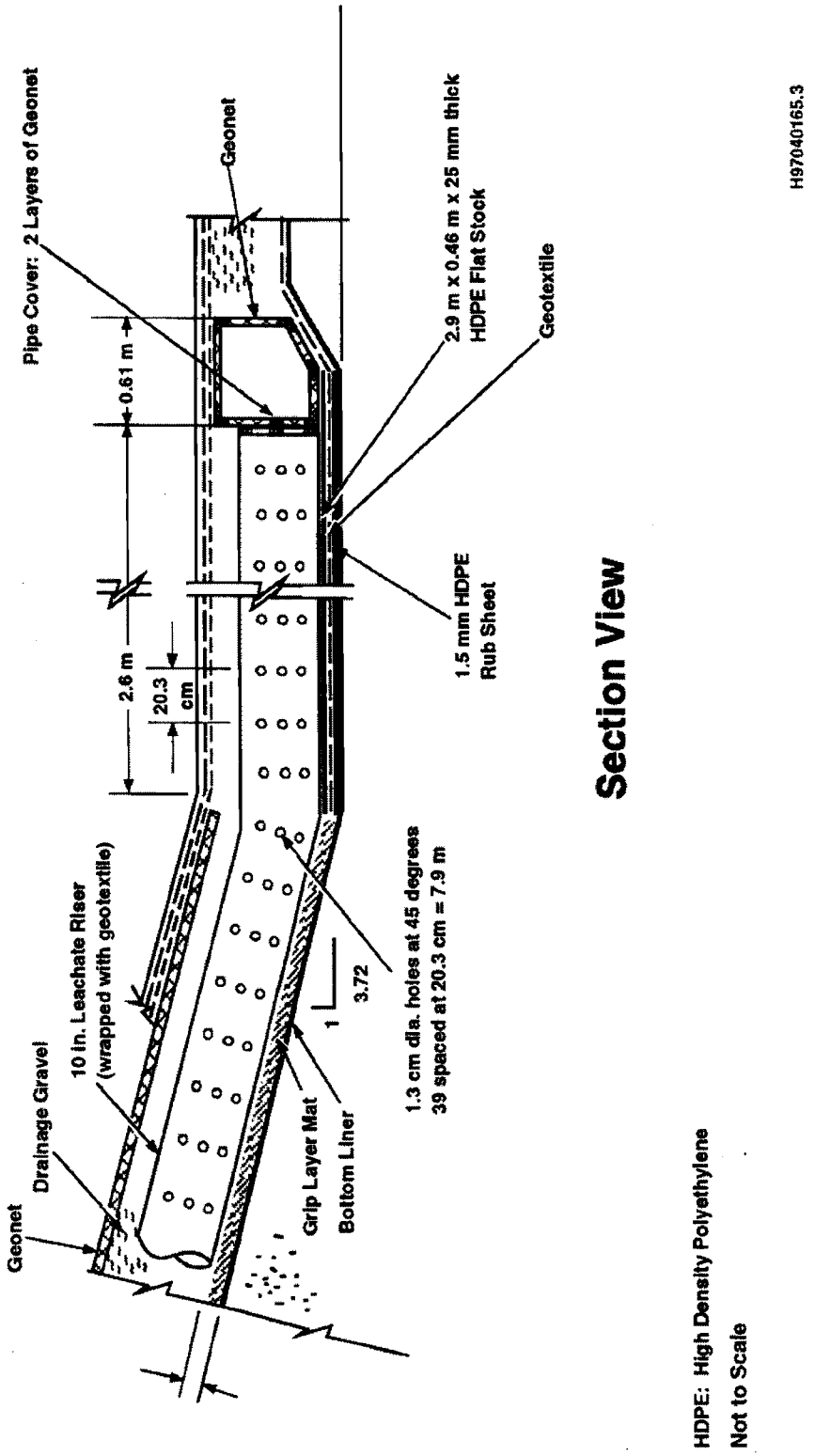
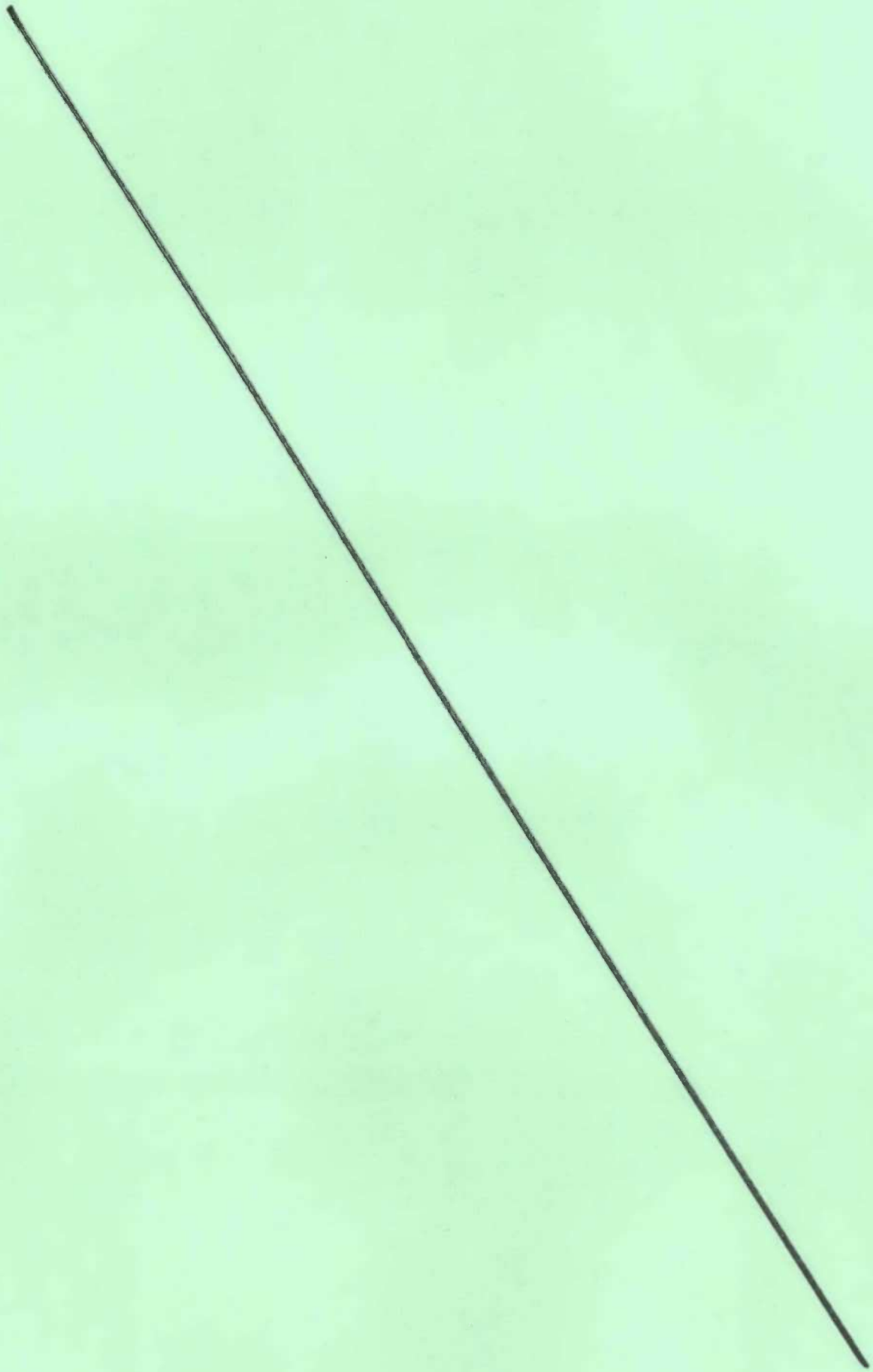


Figure C.18. Detail of Leachate Collection Sump



CHPRC ADMINISTRATIVE DOCUMENT PROCESSING AND APPROVAL

DOCUMENT TITLE:	**OWNING ORGANIZATION/FACILITY:**
Groundwater Monitoring Plan for the Liquid Effluent Retention Facility	S&GRP
Document Number: DOE/RL-2013-46	**Revision/Change Number:** 0

DOCUMENT TYPE *(Check Applicable)* [X] Plan [ ] Report [ ] Study [ ] Description Document [ ] Other

DOCUMENT ACTION *(Check One)* [X] New [ ] Revision [ ] Cancellation

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Prepared for the U.S. Department of Energy
Assistant Secretary for Environmental Management

Contractor for the U.S. Department of Energy
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Addendum D

Groundwater Monitoring Plan

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Terms

bgs	below ground surface
Cr(VI)	hexavalent chromium
CRDL	contract-required detection limit
DQO	data quality objective
EB	equipment blank
Ecology	Washington State Department of Ecology
EMB	Elephant Mountain Member of the Saddle Mountains Basalt
ERDF	Environmental Restoration Disposal Facility
ETF	Effluent Treatment Facility
FTB	full trip blank
FXR	field transfer blank
GC	gas chromatography
HDPE	high-density polyethylene
ICP	inductively coupled plasma
HEIS	Hanford Environmental Information System
KGS	Kansas Geological Survey
LCS	laboratory control sample
LERF	Liquid Effluent Retention Facility
MDL	method detection limit
MS	matrix spike
MSD	matrix spike duplicate
NA	not applicable
OU	operable unit
QC	quality control
RCRA	*Resource Conservation and Recovery Act of 1976*
RPD	relative percent difference
TBD	to be determined
TOC	total organic carbon
TOX	total organic halides
UPR	unplanned release
XRF	X-ray fluorescence

D Groundwater Monitoring Plan, Liquid Effluent Retention Facility

This document describes a groundwater monitoring program for the Liquid Effluent Retention Facility (LERF) (Figure D-1). LERF is a regulated unit under RCW 70.105, "Hazardous Waste Management," and is subject to groundwater monitoring requirements pursuant to WAC 173-303-645, "Dangerous Waste Regulations," "Releases from Regulated Units."

D1 Introduction

This plan describes the LERF groundwater monitoring program, including the monitoring network, constituent list, sampling schedule, sampling and analysis protocols, and data evaluation and reporting methods for LERF groundwater monitoring. Four monitoring wells at LERF (299-E26-10, 299-E26-14, 299-E26-77, and 299-E26-79) provide a monitoring network for establishing the groundwater gradient, and two monitoring wells (299-E26-14 and 299-E26-79) provide upgradient-downgradient comparisons for detection monitoring, respectively (Figure D-2).

D1.1 History of Groundwater Monitoring at the Liquid Effluent Retention Facility

A four-well groundwater monitoring program was established at LERF in 1990 before final construction of the regulated unit. One well (299-E26-11) was completed to the east of LERF as an upgradient monitoring well, and three wells (299-E26-9, 299-E26-10, and 299-E35-2) were completed west of LERF as downgradient monitoring wells. Well 299-E26-77, a replacement well for well 299-E26-9, was located approximately 5 m (15 ft) to the southeast of well 299-E26-9 and because of the scale for Figure D-2, only well 299-E26-77 is identified. Samples were collected quarterly from the four monitoring wells, and evaluation of indicator parameters began before waste was transferred to the basins. Analytes listed in Appendix III, "EPA Interim Primary Drinking Water Standards," of 40 CFR 265, "Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities," Subpart F, "Ground-Water Monitoring," were sampled to establish the suitability of the groundwater as a drinking water supply, as well as several other site-specific constituents the first year of sample collection. Total organic carbon (TOC), total organic halides (TOX), pH, and specific conductivity (indicator parameters) also were analyzed during the first year to derive upgradient/downgradient comparison values for these parameters based on requirements of 40 CFR 265, Subpart F. Detection monitoring continued on a semiannual schedule. Two wells, 299-E26-9 and 299-E35-2, could no longer yield representative samples of groundwater in 1999 and 2001, respectively, due to declining water levels. As a result, inter-well statistical evaluation of LERF groundwater monitoring data has not been performed since 2001. Sampling continued at former downgradient well 299-E26-10 and former upgradient well 299-E26-11. Wells 299-E26-77 and 299-E26-79 were drilled and completed in 2008 to define the aquifer flow rate, flow direction, and hydrogeologic conditions (SGW-41072, *Liquid Effluent Retention Facility Characterization Report*) (Figure D-2). These wells are located west and south of LERF, respectively, and were sampled concurrently with existing wells beginning in January 2009. Water level measurements, after incorporation of the two new wells, demonstrated two different flow conditions, westerly when incorporating well 299-E26-11 and more southerly when data for well 299-E26-11 are not incorporated (SGW-41072). Because of the uncertainty in flow direction, another well, 299-E26-14, was installed north of LERF to clarify current groundwater flow direction. A geophysical investigation was employed to target the best hydraulic location for well 299-E26-14 as discussed further in Section D2.1.1 (Figure D-3). In September 2011, well 299-E26-14 was installed. After two years of water level measurements using well 299-E26-14 and the three other wells (299-E26-10, 299-E26-77, and 299-E26-79), the flow direction was considered southward (discussed further in Section 2.2.2). Various chemical analyses were completed over the past two years to provide an upgradient baseline for dangerous waste constituents specified in this permit.

D1.2 Facility Description

The following subsections provide an overview of the physical structures, operational history, and waste characteristics of LERF. Additional details are provided in Addendum B (Waste Analysis Plan) and Addendum C (Process Information).

D1.3 Physical Structure

LERF is located in the central portion of the Hanford Site on the eastern boundary of the 200 East Area (Figure D-1). Construction of LERF was completed in 1991. The LERF basins consist of three dangerous waste management units classified as surface impoundments: Basins 42, 43, and 44 (Figure D-2). The LERF design uses a dual confinement barrier concept (i.e., dual basin liners and pipe-in-a-pipe transfer piping system) to minimize human exposure and potential for accidental releases to the environment. A leachate detection, collection, and removal system and basin covers are designed to reduce possible environmental or personnel exposures. The leachate detection system is monitored, as required, by the LERF-Effluent Treatment Facility (ETF) permit conditions and Addendum I.

LERF is a 15.8 ha (39 ac) site with three 2.9×10^7 L (7.8 million gal) capacity basins (Figure D-2). The basins are arranged side by side with 18.2 m (60 ft) separations between each basin. The dimensions of each basin (cell) are 100.5 by 82.2 m (330 by 270 ft), with a maximum fluid depth of 6.7 m (22 ft). The side slopes of the basin have a slope ratio of 3:1.

The primary liner for each basin is a 60 mil, high-density polyethylene (HDPE) geomembrane laid directly over a manufactured geotextile/bentonite carpet layer. The secondary liner is also a 60 mil HDPE geomembrane laid directly on 0.9 m (36 in.) of a soil/bentonite mixture. The liners are separated by a synthetic drainage geonet laid on the sides of the basins, with 0.3 m (12 in.) of drainage gravel at the bottom. The sides slope to a sump, which is pumped when the liquid level reaches approximately 28 cm (11 in.) and shuts off when it drops to 18 cm (7 in.). Each basin has a mechanically tensioned cover of very low density polyethylene construction, which is anchored to the perimeter concrete ring wall of the basins with batten plates.

D1.4 Operational History

LERF was constructed for interim storage and treatment for aqueous waste streams prior to final treatment in the 200 Area ETF. Treatment at LERF consists of flow and pH equalization. The flow equalization allows for several smaller waste streams that are intermittently received at the LERF basins to accumulate for continuous higher volume campaign processing at ETF. The pH equalization allows for a uniform wastewater to optimize ETF process campaigns.

LERF began receiving process condensate from the 242-A Evaporator in 1994. In 1995, several new liquid waste feeds were identified for treatment at LERF. These waste streams included Environmental Restoration Disposal Facility (ERDF) leachate, purge water from groundwater monitoring, B Plant waste, and 200-UP-1 groundwater remediation. Between 2000 and 2013, the majority of the liquid waste received at LERF was associated with the following in descending order: 200-UP-1/200-ZP-1 groundwater (181.4 million gal), ERDF leachate (16 million gal), process condensate from the 242-A Evaporator (7.3 million gal), Mixed Waste Burial Trenches leachate (2.9 million gal), K Basins (1.9 million gal), and purge water (1.8 million gal).

Projected ETF influent waste streams for 2010 through 2028 are presented in HNF-23142, *Engineering Study for the 200 Area Effluent Treatment Facility Secondary Waste Treatment of Projected Future Waste Feeds*.

D1.5 Waste Characteristics

As a unit of LERF, the 200 Area ETF was designed to treat a variety of aqueous wastes containing both chemical and radiological contaminants. This aqueous waste is collected in the three LERF basins before transfer to ETF for efficient operations. Before a liquid waste can be transferred to ETF or LERF by a waste generator, a waste profile of the subject waste must be developed. This waste profile is compared against the ETF/LERF acceptance criteria, as explained in Addendum B, "Waste Analysis Plan." Waste streams that have been approved are also periodically re-evaluated for waste characteristics. The results of these periodic re-evaluations (provided in this subsection) help identify reliable chemical contaminants that can be used as or for additional indicator parameters for detection monitoring (as described in WAC 173-303-645(9)(a)). Waste characteristics for liquid effluents that have been historically stored in the three LERF basins (Basins 42, 43, and 44) are provided in the following subsections.

D1.6 Basin 42

Various aqueous waste streams feed Basin 42; however, the 242-A Evaporator waste stream has been the largest volume waste stream associated with Basin 42. Over the past 13 years (1999 through 2012), the liquid volume associated with the 242-A Evaporator waste was 10 times that of any other waste streams sent to Basin 42. Maximum concentration limits for the 242-A Evaporator waste stream during initial startup were provided in WHC-SD-W105-SAR-001, *Final Safety Analysis Report 242-A Evaporator Liquid Effluent Retention Facility*. When the maximum concentrations for the 242-A Evaporator waste stream (Table 9.6 of WHC-SD-W105-SAR-001) were compared with the average contaminant concentration levels (2009 through 2010 weighted average liquid concentrations) in Basin 42 (Table D-1), nearly all of the average Basin 42 concentrations were lower. Constituents with greater concentrations were limited to two anions (chloride and sulfate), one cation (calcium), and four trace metals (barium, manganese, uranium, and zinc). These constituents appear to be associated with other waste streams such as the Mixed Waste Trenches 31 and 34 leachate and Hanford Site purge water which had the second and third largest waste streams by volume. The other 17 waste streams associated with Basin 42 make up approximately 2 percent of the volume.

The makeup of Basin 42 is similar to the groundwater wells upgradient of the Hanford Site or regional background groundwater concentrations, except for alkalinity, nitrogen, and sulfate. A comparison between Basin 42 wastewater and upgradient Hanford Site wells can be seen in the appropriate Table D-1 columns (e.g., 2009 Basin 42 Characterization Results and Basin 42 Average versus Regional Background Concentration of Table D-1). In general, regional groundwater background concentrations are similar to groundwater concentrations beneath LERF, except for anions. Although Basin 42 and groundwater beneath Basin 42 share a common suite of elevated constituents (anions), the source of the elevated anions in the groundwater is from a crossgradient/upgradient groundwater location. The crossgradient/upgradient groundwater location is shown by historical groundwater results at well 299-E34-7 prior to the start of LERF and more recently at the LERF upgradient well 299-E27-14 (Figure D-4). By comparison, the average concentration[1] of sulfate in Basin 42 (55.6 mg/L) is much less than the historical sulfate concentration at well 299-E34-7 of 671 mg/L (sample date 4/3/2003). Even characterization results from Basins 42, 43, and 44 (Tables D-1, D-2, and D-3) do not compare with the maximum groundwater results at well 299-E34-7. The same is true for nitrate in Basin 42 as compared with nitrate at the crossgradient/upgradient well 299-E34-7. Only the contributions of the 200-BP-5 perched water waste streams from Basin 43 and ERDF leachate exceed the groundwater results at well 299-E34-7. However, because of the nature of the elevated groundwater results at well 299-E34-7, including elevated TOC, and the relationship to past unplanned releases (UPRs) near well 299-E34-7

[1] All concentrations are reported as a weighted average.

(e.g., UPR-200-E-32 associated with the 216-B-2-1 Ditch and UPR-200-E-138 associated with the 216-B-2-2 Ditch [Figure D-1]), the elevated groundwater results at well 299-E34-7 appear to be from a source other than LERF (Figure D-5). The UPRs (e.g., UPR-200-E-32 and UPR-200-E-138) were associated with B Plant fractionation waste that had significant levels of nitrate, sulfate, and organic carbon. The nature of these UPRs appears more characteristic of the levels reported at well 299-E34-7. Well 299-E26-10, located to the west of LERF, appears to mimic the historical results at well 299-E34-7 (Figure D-4). As the nitrate and sulfate concentrations decrease over time, if concentrations follow the earlier trends at well 299-E34-7, these constituents may become more appropriate as indicator parameters at LERF. However, the concentration of these constituents in LERF Basins would not be distinguishable from current groundwater conditions beneath LERF. Because nitrate and sulfate may become more appropriate indicator parameters in the future, they will serve currently as groundwater quality parameters at the LERF monitoring wells. Because specific conductance is an indicator of nitrate and sulfate changes, specific conductance will be added as an indicator parameter for documentation of local changes and comparison between the upgradient and downgradient monitoring wells.

None of the toxicity characteristic of dangerous waste constituents received by Basin 42 exceed toxicity characteristics list threshold values (WAC 173-303-090(8)(c), "Dangerous Waste Regulations," "Dangerous Waste Characteristics"). Six of the potentially dangerous waste metal constituent results in the basin were above groundwater background levels (Table D-1): chromium, copper, lead, mercury, nickel, and thallium. Although the results are above the groundwater background levels, the results would not be detectable at groundwater compliance points should there be a potential release into the upper aquifer because of the low waste stream concentrations and dispersive effect associated with infiltrating waste into the groundwater.

There were low levels of organics found in Basin 42 with 1-butanol (288 µg/L) having the highest weighted average. The chemical nature of 1-butanol (e.g., rapidly degrades in water and has a relatively high detection level 100 µg/L) makes this constituent an unlikely indicator parameter.

In conclusion, no reliable waste constituent indicator parameters are presently available for Basin 42 groundwater detection.

D1.7 Basin 43

The largest volume of waste waters received by Basin 43 was the contaminated groundwater from the 200-UP-1/200-ZP-1 operable units (OUs) groundwater pumping systems (Table D-2).
The 200-UP-1/200-ZP-1 OUs waste stream had 20 times more volume sent to LERF than the next closest waste stream (ERDF leachate) over the past decade and a half. The 200-UP-1/200-ZP-1 OUs groundwater effluent waste characteristics are contained in Table D-2. Tables D-2 and D-3 provides characteristics of the ERDF leachate. Table D-2 provides characteristics of ERDF leachate in Basin 43 in 2012 after receipt of the 200-UP-1/200-ZP-1 OUs groundwater effluent waste was terminated, and Table D-3 provides the average ERDF leachate characterization results for Basin 44 from 2000 through September 2011. Overall, the waste characteristics in Basin 43 are most comparable to the waste streams from 200-UP-1/200-ZP-1 OUs groundwater pumping systems because of its significant volume compared with the other waste streams.

The 200-UP-1/200-ZP-1 OUs waste streams have a makeup similar to the groundwater well results near sources of B Plant liquid effluent disposal sites. These sites received and disposed of metal waste, uranium recovery waste, and cesium and strontium scavenging waste which have infiltrated into the aquifer. The highest ionic results are associated with nitrogen. The Basin 43 weighted average concentration was 101 mg/L (nitrogen in nitrate) compared to 10 mg/L in the groundwater beneath LERF. Some of the other waste streams (e.g., ERDF leachate and 200-BP-5 perched water) received at Basin 43

also exceeded regional background groundwater results for chloride, nitrogen, and sulfate, with concentrations as great as 224 mg/L, 220 mg/L, and 597 mg/L, respectively (Table D-2). However, these constituents are not likely to be distinguishable from current groundwater conditions beneath LERF, mainly because of the concentration of these constituents in the groundwater at crossgradient/upgradient locations to LERF, as discussed in Section D1.5.1. As also discussed in Section D1.5.1, as the groundwater concentrations from the crossgradient/upgradient direction decrease, these constituents may become more appropriate as indicator parameters at LERF. Because nitrate and sulfate may become more appropriate indicator parameters in time, they will serve currently as groundwater quality parameters at LERF monitoring wells. Because specific conductance is an indicator of nitrate and sulfate changes, specific conductance will be added as an indicator parameter for documentation of the expected local changes and comparison between the upgradient and downgradient monitoring wells.

None of the toxicity characteristic dangerous waste constituents received by Basin 43 exceed the toxicity characteristics list threshold values (WAC 173-303-090(8)(c)). However, several of the potentially dangerous waste metal constituent results for the basin were above groundwater background levels. Even so, the results appear too low to determine should a potential release reach the aquifer because of the scattering effect associated with infiltrating liquid waste effluents through the vadose zone into the groundwater. However, it may be possible to differentiate hexavalent chromium (Cr(VI)). This constituent will need to be monitored semiannually for two years to develop a local background basis before potentially adding it as an indicator parameter. Total chromium is not a reliable indicator parameter because of the potential concentrations associated with casing corrosion.

Of the 49 volatile and semivolatile constituents, analyzed at various frequencies from 2008 to 2011 for liquid wastes sent to Basin 43, only 3 (carbon tetrachloride, chloroform, and trichloroethene) were detectable. The most significant constituent was carbon tetrachloride, with concentrations ranging between 190 and 800 µg/L. The other two constituents had concentrations less than 10 µg/L. Since carbon tetrachloride is not normally occurring in the groundwater, it should be an excellent indicator parameter. TOC ranged between 0.3 and 2.45 mg/L for liquid waste in Basin 43. The concentrations do not appear to be significant enough to differentiate a groundwater quality impact should a release occur. TOC analyses are subject to a wide range of variability and can lead to a false positive error. A more valid indicator of carbon tetrachloride is TOX (Figures D-6 and D-7). Although not analyzed for in Table D-2, this indicator parameter has a lower level of detection than TOC, and, as shown in Figures D-6 and D-7, mimics the carbon tetrachloride level better than TOC. Thus, detection of both indicators (carbon tetrachloride and TOX) would be conclusive of a dangerous waste constituent impact. As a result, TOX and carbon tetrachloride will be added as indicator parameters for the LERF monitoring network.

D1.8 Basin 44

Basin 44 has received liquid waste dominated by ERDF leachate (7 million gal or 60 percent by volume). Other liquid waste streams include K Basin waste (1.9 million gal or 16 percent by volume), leachate from double-lined burial trenches, Mixed Waste Trenches 31 and 34 located in the 218-W-5 Burial Ground (1.2 million gal or 10 percent by volume), and purge water from well development (1.1 million gal or 10 percent by volume). The purge water and Mixed Waste Trenches 31 and 34 waste streams are lower in all constituents as compared with ERDF leachate. Therefore, waste in Basin 44 is most similar to the ERDF leachate because of volume and concentration.

ERDF waste streams are similar to groundwater well results downgradient from B Plant liquid effluent disposal sites. The most comparable results are associated with chloride, nitrate, and sulfate. The average concentrations were 250 mg/L, 327 mg/L, and 474 mg/L, respectively (Table D-3). However, these constituents are not likely to be distinguishable from current groundwater conditions beneath LERF, mainly because the concentration of these constituents are already present in the groundwater at similar

concentrations both crossgradient and upgradient of LERF, as discussed in Section D1.5.1. As also discussed in Section D1.5.1, as the groundwater concentrations from the crossgradient/upgradient direction decrease, these constituents may become more appropriate as indicator parameters at LERF. Because nitrate and sulfate may become more appropriate indicator parameters in time they will serve currently as groundwater quality parameters at the LERF monitoring wells. Because specific conductance is an indicator of nitrate and sulfate changes, specific conductance will be added as an indicator parameter for documentation of the expected local changes and comparison between the upgradient and downgradient monitoring wells.

None of the toxicity characteristic dangerous waste constituents received by LERF exceed the toxicity characteristics list threshold values (WAC 173-303-090(8)(c)). Several of the potentially dangerous waste metal constituents received at LERF were above groundwater background levels (Table D-3). Nevertheless, the results would not show a measurable difference should a potential release to the aquifer occur because of the low waste stream concentrations and the scattering effect associated with infiltrating of liquid waste effluents through the vadose zone into the groundwater.

The organic chemical analytical results associated with Basin 44 were at very low levels (<5 µg/L) and were only periodically detected. Therefore, the ability to detect a potential release in the aquifer for organic chemicals is not practicable for the same reason as discussed for the metals and anions. TOC averaged 13.2 mg/L in Basin 44. TOC concentrations seem to be correlated with the elevated oil and grease results. Because oil and grease are viscous, TOC does not appear to be a good indicator parameter. The concentrations do not appear significant enough to be detectable in groundwater should a release occur. As noted previously, TOC analyses are subject to a wide range of variability and can potentially lead to a false positive error.

In conclusion, no reliable waste constituent indicator parameters are presently available for Basin 44 groundwater detection.

D1.9 Potential Contaminate Indicator Parameters in Groundwater

Based on the projected LERF influent waste streams and concentration levels from 2010 through 2028 as presented in HNF-23142, there does not appear a significant change in waste streams expected. Thus, the indicator parameters identified above appear to be sufficient for future detection monitoring at LERF. From review of the waste stream characterization data for Basins 42, 43, and 44, one additional indicator parameter (carbon tetrachloride) has been identified as a reliable indication of the presence of a potential dangerous waste constituent release into the groundwater. Another potential indicator parameter may be Cr(VI), based on local background results collected during 2014 through 2016.

D2 Hydrogeology and Groundwater-Chemistry

This section describes the geology, hydrogeology, and groundwater chemistry beneath the LERF area. To date, seven wells have been installed for monitoring the groundwater quality beneath the LERF basins. Table D-4 provides the well attributes for reference when reviewing this section.

D2.1 Geology

The geology near LERF consists of Columbia River Basalt overlain by a series of sedimentary units of the Ringold Formation and Hanford formation. The interpretations are based on information from the following sources:

- *Miocene- to Pliocene-Aged Suprabasalt Sediments of the Hanford Site, South-Central Washington* (BHI-00184)

- *Revised Hydrogeology for the Suprabasalt Aquifer System, 200-East Area and Vicinity, Hanford Site, Washington* (PNNL-12261)
- *Hydrogeologic Model for the Gable Gap Area, Hanford Site* (PNNL-19702)
- *Borehole Summary Report for the Installation of RCRA Wells 299-E26-77 (C6455), 299-E26-79 (C6826), 299-E25-236 (C6542) and 199-N-165 (C6693), FY 2008* (SGW-39344)
- *Liquid Effluent Retention Facility Characterization Report* (SGW-41072)
- *Landstreamer/Gimbaled GeoPhone Acquisition of High Resolution Seismic Reflection Data North of the 200 Area – Hanford Site* (SGW-43746)
- *Borehole Summary Report for the Installation of Two RCRA Groundwater Monitoring Wells in the 200 Areas, FY2011* (SGW-51467)
- *Seismic Reflection Investigation at the Liquid Effluent Retention Facility, 200 East Area, Hanford Site Richland, Washington* (SGW-52162)
- *Integrated Surface Geophysical Investigation Results at Liquid Effluent Retention Facility, 200 East Area, Hanford, Washington* (SGW-52467)
- *Site Characterization Report for the Liquid Effluent Retention Facility* (WHC-SD-EN-EV-024)
- *Borehole Completion Data Package for the Liquid Effluent Retention Facility* (WHC-MR-0235)

LERF lies in the Pasco Basin, between the axis of the Umtanum-Gable Mountain anticlinal ridge and the axis of the Cold Creek syncline. The terrain surrounding the LERF basins is flat to slightly undulating, and the average elevation is approximately 182 to 184 m (597 to 604 ft) above mean sea level.

The stratigraphy beneath LERF was interpreted from geologic observations during the drilling of seven boreholes, select analyses of sediment samples, aquifer tests, and geophysical investigations over the past two decades. The three principal stratigraphic units present near LERF, in ascending order, are the Elephant Mountain Member of the Saddle Mountains Basalt (EMB), the Ringold Formation, and the Hanford formation. The thickness of the suprabasalt sediments near the LERF basins ranges from 60 to 69 m (198 to 225 ft).

D2.1.1 Elephant Mountain Member

The nature and extent of the EMB, one of the youngest members of the Saddle Mountains Basalt and the uppermost basalt in this area, is based on result of observations and documentation of archive samples collected during drilling, X-ray fluorescence (XRF) analysis, seismic analyses, and hydraulic tests performed within the upper basalt flow top. The EMB in this area was characterized in WHC-SD-EN-EV-024 as consisting of only the oldest EMB flow (Elephant Mountain I). This flow is generally continuous throughout the area, with a thickness ranging from approximately 12 m (39 ft) where partially eroded, to greater than 35.1 m (115 ft) north of the 200 East Area. The EMB I flow contains three intraflow structures: colonnade, entablature, and flow top. The colonnade makes up the bottom third of the flow. The upper part of the colonnade grades from moderate- to well-developed columns into a platy cross-fractured colonnade and then into a hackly entablature. The entablature has numerous, irregular cross-fractures, vertical fractures, and small scattered vesicles near its top. The flow top is characterized by abundant vesicles and is brecciated and/or palagonitic (WHC-SD-EN-EV-024).

Observations during drilling near the LERF basins, when initially encountering the EMB surface, were described in WHC-MR-0235 as reddish weathered basalt with vesicles partially filled, except in

wells 299-E26-9 and 299-E26-10, located to the west. However, well 299-E26-77, located next to well 299-E26-9, was reported with heavy weathering and the presences of vesicles (SGW-41072). The drilling rate was moderate through the upper EMB to a depth of 2 to 3 m (6.5 to 9.8 ft) when drilling wells 299-E26-77 and 299-E26-79, respectively (SGW-39344). It was concluded in SGW-41072 that hydraulic communication of the uppermost aquifer (e.g., unconfined) extends from the suprabasalt sediments into the basalt, at least in the western half of LERF, because there was no impediment associated with the overlying Hanford formation sediments. The thickness of the flow top was interpreted to range from 2 m (6.5 ft) at well 299-E26-77 (west of LERF) to 3.2 m (10.5 ft) at well 299-E26-79 (south of LERF), and 1.5 m (5 ft) at well 299-E26-11 (east of LERF).

The EMB surface expression in the immediate vicinity of the LERF basins forms a depression centered at the newest well 299-E26-14 (Figure D-3). The contours presented in Figure D-3 are based on a combination of basalt contact during drilling and various geophysical investigations (e.g., seismic reflection and refraction, electrical resistivity, and time-domain electromagnetic sounding). Seismic results to the east and west of well 299-E26-14 portray limited aquifer conditions above the basalt (Figure D-8). Paleochannels are interpreted to the north and northwest of well 299-E26-14 and continued to the south-southeast, as displayed in Figures D-3 and D-9. Seismic reflection results suggest an even deeper depression to the east of well 299-E26-79, centered almost directly south of Basin 43, with as much as 8 m (26 ft) of aquifer thickness (Figure D-10, black line in figure provides the interpreted top of basalt). Continuing east of this depression to the south of LERF, the basalt surface is interpreted to rise to the current water table level. The apparent contact with the water table is estimated to be just south of the west boundary of Basin 44. Further east, the basalt is interpreted to plateau beyond well 299-E26-11. West of well 299-E26-79, the basalt surface is interpreted to increase in elevation linearly to the elevation of 121.3 m (398 ft) at well 299-E26-10. Finally, Figure D-11 provides an angle view of the well casing extensions from ground surface to basalt in the LERF area and to the west/northwest, including remnant Ringold Unit A sediments and groundwater extent above basalt. The depiction of the groundwater implies flow through the basalt flow top as discussed further in Section D2.2. Figure D-12 provides an interpretation of the basalt surface and Ringold sediments without the groundwater overlay.

D2.1.2 Ringold Formation

The Ringold Formation represents ancient fluvial and lacustrine deposits associated with the ancestral Columbia River, and the formation exhibits consolidation and weathering. Where present, this Formation overlies the EMB (Figure D-12). According to WHC-SD-EN-EV-024, remnant muds associated with the Ringold period exist to the east and northwest of the LERF site at wells 299-E26-11 and 299-E35-2, respectively. *200 East Groundwater Aggregate Area Management Study Report* (DOE/RL-92-19) reported approximately 2.74 m (9 ft) of the Ringold Lower Mud Unit in well 299-E26-11 and mapped the Lower Mud Unit extending to this location from the east. BHI-00184 identified the Ringold muds east of the 200 East Area as paleosol-overbank deposits. WHC-SD-EN-EV-024 concluded that the sediment layer was a paleosol based on XRF analysis. BHI-00184 states that pedogenically altered silt- and clay-rich overbank-paleosol (facies association III) deposits of the Ringold Formation are easily distinguished from the basalt-rich sand and gravel of the Hanford formation. In 2000, PNNL-12261 defined the sediments near well 299-E26-11 hydraulically as the Ringold Formation Unit A and, more specifically, the hydrogeologic unit 9C (Figure D-12).

The Ringold sediment at well 299-E26-11, as described in WHC-MR-0235, consists of a slightly gravelly sandy mud (5 percent gravel, 30 percent sand, and 65 percent mud). The color was reported as very dark grayish brown (10YR3/2). The gravel content was described as 90 percent mafic, and the sand content was 50 percent mafic. The sediments had no reaction to hydrochloric acid.

During drilling of the new well 299-E26-14, low permeability sediments were encountered at 65.5 to 66.1 m (215.5 to 217 ft) below ground surface (bgs). The sediments were described as 95 percent silt and 5 percent gravel. Photographic review of this sediment layer, presented in SGW-51467, showed a distinct texture and color change from the overlying Hanford sandy gravels. The reddish brown hue and yellow tints associated with this layer correlate well with the distal overbank description provided in BHI-00184. Other characteristics associated with this layer included no reaction to hydrochloric, similar to Ringold sediments described at well 299-E26-11. An alternative explanation may be that the apparent Ringold sediments are rework, removed from one location and deposited at this location, possibly associated with cataclysmic glacial fluvial floods.

Most of the area beneath LERF is considered devoid of Ringold sediments because of the high energy scouring associated glacial fluvial flooding in the Pleistocene and the lack of reflectors in the suprabasalt section during 2011 seismic data reviews. PNNL-19702 presents a conceptual model of various paleochannels originating to the northwest (Figure D-9). Some of these paleochannels may have been formed during Ringold times, and isolated remnants of Ringold sediments are sometimes found within these older paleochannels.

D2.1.3 Hanford Formation

The Hanford formation near LERF ranges in thickness from approximately 59 to 66 m (193 to 215 ft) or more (Figures D-13 and D-14). The texture of the Hanford formation is loose to weakly cemented, sandy, pebble-cobble gravels to gravelly sand, with occasional layers of sand and/or muddy sand. Regionally, the Hanford formation is subdivided into an upper gravel sequence (H1), a sandy sequence (H2), and a lower gravel sequence (H3). The sandy sequence is present locally and, where it is missing, a single sequence of gravel-dominated facies exists, which is undifferentiated in cross-sections.

LERF is located along the southern flank of a major west-northwest/east-southeast trending cataclysmic flood channel. Because of multiple flood events and the turbulence and extremely high energy associated with these floods, it is difficult to correlate individual strata within flood sequences. In outcrops of the Hanford formation elsewhere in the Pasco Basin, for example, it is common to see changes from gravel-dominated sediments to sand and silt-dominated sediments over a distance of a few tens of meters.

In general, more silt or mud was present to the west and east than north or south of the LERF basins based on geologic logs for the seven wells drilled within the LERF vicinity. However, high silt and clay content to the north and south of LERF is present near the contact with the EMB within the aquifer. These silt and clay layers ranged in thickness between 0.3 to 1.5 m (1 to 5 ft) and appear to be of Ringold age as discussed in D.2.1.2. The basalt content in layers above the silt and clay indicates Hanford origin. Above these initial layers, the gravel content was generally about 60 percent, consisting of 40 to 70 percent mafics. Significantly more cobbles were described in the north and south boreholes than to the east and west throughout the borehole log descriptions. The grayish brown to very dark grayish brown color description of the sediments was consistent throughout the area. Calcium carbonate levels are low to within 21 m (70 ft) of ground surface, based on little to no reaction to hydrochloric acid. The upper zone with increased calcium carbonate levels correlates with low modeled velocities during refraction and resistivity modeling, as stated in SGW-52467, and may be a distinctive feature to differentiate the H1 and H3 in this area. Moisture observations ranged from dry to wet; however, the damp and wet descriptions in the vadose zone pertained to zones where water was added during drilling. In conclusion, based on the larger gravel content and size to the north and south of the LERF basins, the dominant flow during deposition appears to be from the northwest, aligning with the conceptual model in PNNL-19702 (Figure D-9). In addition, there were no significant zones of silt or clay above the aquifer indicating no perching horizons in the suprabasalt sediments beneath the LERF vicinity.

D2.2 Groundwater Hydrology

The vadose zone beneath LERF consists of the Hanford formation and portions of the EMB above the water table, as well as potentially some of the Ringold Formation near well 299-E26-11. There have been no observations indicating perched water table conditions near the LERF basins; however, perched conditions could be present west and northwest of the westernmost LERF monitoring wells.

The uppermost aquifer directly beneath LERF is thin to moderate in thickness (e.g., ranging from possibly not present to greater than 8 m (26.25 ft) and exists in the Hanford and EMB flow top (Figures D-13 and D-14). This aquifer is unconfined, except to the east where barometric analyses within well 299-E26-11 indicate semiconfined conditions. This is consistent with the rise in groundwater elevation when drilling advanced through the lower Ringold sediments, present at this well, causing the groundwater elevation to rise nearly 3.1 m (10 ft) in the temporary casing (WHC-MR-0235). The westward extent of the Ringold sediments is uncertain; however, it has been portrayed to pinch out west of well 299-E26-11 (Figure D-12). Although well 299-E26-11 is still capable of yielding representative samples from the same hydrostratigraphic unit as the other LERF wells, the chemical nature of the samples is different and has been more characteristic of groundwater to the east of LERF.

Well construction details are discussed in Section D2.4 and presented in Table D-4. To date, seven wells have been installed for detection monitoring since 1990. Three of the wells (299-E26-11, 299-E26-77, and 299-E26-79) were screened either entirely or primarily within the EMB flow top. The wells produce at a minimum 22.7 L/min (6 gal/min), which is sufficient for groundwater sampling, and the flow top is sufficiently permeable for adequate hydraulic connection with the overlying sediments.

Basalt flow top fracturing, brecciation, and/or weathering provide localized zones of higher permeability. Where these conditions exist and are in hydraulic communication with overlying saturated sediments, the basalt flow top is part of the overlying unconfined aquifer system. Based on evaluations of drill cuttings, drilling rates, and water production noted during drilling wells 299-E26-77 and 299-E26-79, the EMB flow top functions as a component of the unconfined aquifer and forms a laterally continuous aquifer beneath LERF.

The uppermost aquifer is thickest north of Basin 42 and appears to thicken south of Basin 43 (Figures D-8, D-10, D-13, and D-14) due to paleochannel development. The flow interior of the EMB represents the lower boundary of the uppermost aquifer. This was verified by observations during drilling at wells 299-E26-77 and 299-E26-79, as discussed in Section D2.1.1.

D2.2.1 Aquifer Properties

Hydraulic tests were conducted in 1990, 2003, 2008, and 2011 to derive representative hydraulic parameters for the various saturated formations beneath the LERF general vicinity. Slug tests were completed for each of the seven wells with a derived hydraulic conductivity value. The 1990 slug tests were completed in wells 299-E26-9, 299-E26-10, 299-E26-11, and 299-E35-2, which were constructed with a 10.2 cm (4 in.) diameter wire wrapped screen and 0.25 mm (0.010 in.) slot width. A 20-40 silica sand filter pack encases the screen interval. The following paragraphs summarize the results for each well, and WHC-SD-EN-EV-024 provides further detailed discussion. The 2003 hydraulic tests were completed at wells 299-E26-10 and 299-E26-11 and consisted of slug tests at each well and the following additional tests at well 299-E26-10: tracer test, tracer-pumpback test, and constant-rate pumping test. This subsection summarizes the results for each well, and PNNL-14804, *Results of Detailed Hydrologic Characterization Tests Fiscal Year 2003*, provides further discussion. The 2008 hydraulic slug tests were completed at wells 299-E26-77 and 299-E26-79, constructed with a 10.2 cm (4 in.) diameter wire wrapped screens and 0.5 mm (0.020 in.) slot width. A 10-20 silica sand filter pack encases the screen interval. A slug test at well 299-E26-11 also was included in 2008. This subsection summarizes the 2008

results for each well, and SGW-41072 provides further discussion. Finally, a 2011 constant rate pumping test was completed at well 299-E26-14, which was constructed with 10.2 cm (4 in.) diameter wire wrapped screens and 0.5 mm (0.020 in.) slot width. A 10-20 silica sand filter pack encases the screen interval. Because several of the well screens cross various formations, a summary of the screen interval is provided in the following text and in Table D-4. When heterogeneous conditions exist, the hydraulic results are an arithmetic average of the individual formational layers based on a weighted-thickness (PNNL-14804).

Well 299-E26-9 (now sample dry) was screened only in the Hanford formation. The 1990 slug test derived transmissivity values for well 299-E26-9 ranged from 11 to 230 m^2/day (118 to 2,476 ft^2/day). The derived hydraulic conductivity ranged between approximately 6 to 120 m/day (20 to 394 ft/day), assuming an aquifer thickness of 2 m (6.6 ft).

Well 299-E26-10 is screened primarily across the Hanford formation with a small section across the EMB flow top (0.5 m [1.6 ft]). Transmissivity values for well 299-E26-10 were not derived for the 1990 tests because of the fast recovery response (e.g., less than 3 seconds). In 2003, four hydraulic slug tests, two low and two high stress, were performed at well 299-E26-10. The results produced a hydraulic conductivity range, based on the Kansas Geological Survey (KGS) type-curve method, of 36.7 to 42.8 m/day for both stress-level tests (KGS, 1991, *Seismic-Reflection Processing Demonstration Using Eavesdropper*). The KGS type-curve method was used to derive the hydraulic conductivity as explained in PNNL-14804. The 2003 screened thickness across the saturated Hanford formation was 1.48 m (4.85 ft). Four additional hydraulic tests were completed at this well in 2003. The tracer-dilution test provided qualitative evidence that the overlying Hanford formation sediments had a considerably higher hydraulic conductivity than the EMB flow top. The tracer-pumpback test was used to derive the effective porosity; however, due to test complexities, the calculation did not appear representative of the aquifer conditions. The constant-rate pumping test provided another means of deriving the hydraulic conductivity, which was reported at 36.2 m/d with a transmissivity of 71.6 m^2/day. Based on the consistency of the 2003 results, the hydraulic conductivity ranges between 36.2 and 42.8 m/day.

Well 299-E26-11 is screened only across the EMB flow top. The 1990-derived transmissivity value for well 299-E26-11 was 6.1 m^2/d (20 ft^2/d) with a hydraulic conductivity of 11.2 m/day (120 ft/day). Five additional hydraulic slug tests were completed at well 299-E26-11 in 2003, which derived a range of hydraulic conductivity values from 5.85 to 6.8 m/day. Four additional slug tests were performed in 2008 producing a reported hydraulic conductivity value of 10 m/day. The hydraulic conductivity values for the three times range from 5.85 to 11.2 m/day. Because of the analysis methods used by PNNL-14804, the most representative value appears to be 6.3 m/day.

Well 299-E26-14 was completed in 2011 with 5.5 m (18 ft) of screen across the Ringold and Hanford sediments. Only a small portion (0.27 m or 0.9 ft) of the Ringold sediments are adjacent to the bottom of the well screen. A 27.3 gal/min constant pump test was completed on November 26, 2011. A transducer was installed to collect changing water table elevations during the 75 minute pumping test. In total, 2,048 gal were pumped during the test, as described in the field activity log. Because no hydraulic parameters were calculated from the field activity records, type-curve matching methods were used to derive transmissivity and hydraulic conductivity results for this well. The computer program AQTESOLV was used for curve matching. AQTESOLV uses a nonlinear least squares procedure to match a type-curve or straight-line solution for the data provided. Through a sequence of iterations, the procedure systematically adjusts the values of hydraulic properties to achieve the best statistical match between a solution (type-curve) and the test data. Each iteration seeks to minimize the sum of squared residuals. AQTESOLV provides five different solution methods for unconfined aquifer pumping tests. Initially, the Theis and Cooper-Jacob methods were evaluated against the field data, but the curve matching associated with these solution methods did not align (Theis, 1935, "The Relation Between the Lowering of the

Piezometric Surface and the Rate and Duration of Discharge of a Well Using Ground-Water Storage;" Cooper and Jacob, 1946, "A Generalized Graphical Method of Evaluating Formation Constants and Summarizing Well-Field History"). The Moench method provides independent parameters for wellbore storage, wellbore skin, and delayed gravity response in anisotropic unconfined aquifers (Moench, 1997, "Flow to a well of finite diameter in a homogeneous, anisotropic water table aquifer"). After manual manipulation of the independent parameter for the wellbore skin factor and delayed drainage parameter, the Moench derived curve nearly matched the field results as provided in Figure D-15. The derived hydraulic conductivity from this curve matching solution was 27.3 m/d. Another solution method, Neuman, with less independent parameters for manipulation, produced the type-curve in Figure D-16 (Neuman, 1974, "Effect of Partial Penetration on Flow in Unconfined Aquifers Considering Delayed Gravity Response"). The derived hydraulic conductivity from this curve matching solution was 24.4 m/day. These results agree with the slug results derived for the other wells in the LERF vicinity. The best estimate is considered 27.3 m/day.

Well 299-E26-77 was completed in 2008 with 6.1 m (20.1 ft) of screen across the EMB flow top and 0.71 m (2.3 ft) across the overlying silty sandy gravel Hanford formation. The 2008 derived hydraulic conductivity was reported in SGW-41072 at several tens of meters/day. Because there were no specific values presented in this report, the data from the two slug withdraw tests were retrieved and reanalyzed with type-curve methods, as discussed in PNNL-14804. Briefly, the type-curve method is useful for analyzing unconfined aquifer conditions because it uses all or any part of the slug test response. The computer program AQTESOLV was used for curve matching, as discussed previously. The automated matching option with default setting was applied to the KGS Model, KGS model with skin effects, and the Springer-Gelhar inertial effects method (Water-Resources Investigation Report 91-4034, *U.S. Geological Survey Toxic Substances Hydrology Program—Proceedings of the technical meeting, Monterey, California, March 11-15, 1991*). The most comparable slug test derived curve was the Springer-Gelhar ~~critically~~ dampened method. This method nearly matched the second slug withdraw results, as shown in Figure D-17. One of the assumptions for this type-curve is a quasi steady-state of the aquifer. A quasi steady-state flow neglects specific storage, unlike the Cooper-Bredehoeft-Papadopulos method (Cooper et al., 1967, "Response of a Finite-Diameter Well to an Instantaneous Charge of Water"). When the Cooper-Bredehoeft-Papadopulos method was run, it did not converge with the test data, indicating the aquifer conditions are more suitable for the Springer-Gelhar method. In addition, the Barker-Black fractured aquifer solution method failed to converge (Barker and Black, 1983, "Slug Test in Fissured Aquifers"). The Springer-Gelhar results derived a hydraulic conductivity of 134 m/d. For comparison, three additional methods (Bouwer-Rice, Hvorslev, and Barker-Black double porosity fractured aquifer method) also were analyzed; however, the curve-type matching alignment with the data was either significantly different and did not converge or only visually applied to the later recovering slug test results using line-matching, which produced much greater hydraulic conductivity results (Bouwer and Rice, 1976, "A Slug Test for Determining Hydraulic Conductivity of Unconfined Aquifers With Completely or Partially Penetrating Wells;" Hvorslev, 1951, *Time Lag and Soil Permeability in Ground-Water Observations*). As discussed in PNNL-14804, the semi-empirical nature of the Bouwer and Rice method for complex well/aquifer conditions can lead to declining levels of accuracy beyond 30 percent. Thus, the best estimate of the hydraulic conductivity for well 299-E26-77 is 134 m/d using the Springer-Gelhar solution. Because hydraulic conductivity results from other tests in the area produce much lower results for the Hanford formation, the fractured flow top appears to be the dominant flow regime at this well. If the fractured flow top is thinner and the borehole diameter within the basalt is smaller, the hydraulic conductivity value would be even higher. Conversely, if the flow top is thicker and the borehole diameter is larger, the hydraulic conductivity value would be smaller.

Well 299-E26-79 was completed in 2008 with 4 m (13.2 ft) of screen across the EMB flow top and 2.7 m (8.9 ft) across the overlying Ringold and Hanford sediments. The 2008 derived hydraulic conductivity was reported in SGW-41072 at several tens of meters/day. Because there were no specific values presented in this report, the two slug withdraw test data were retrieved and reanalyzed with type-curve methods, as discussed previously for well 299-E26-77. The early slug test data could not be fit by any of the AQTESOLV solution methods. Fitting the remaining portion of the data produced significantly larger hydraulic conductivity results by one to two orders of magnitude than at well 299-E26-77. Because the results are not consistent with the other LERF well results, the data do not appear to be useable; therefore, no hydraulic conductivity results were generated for this well.

Well 299-E35-2 (now sample dry) was screened mainly across the sediments above the EMB flow top (1.9 m [6.2 ft]) with a portion of the screen across the EMB flow top (0.4 m [1.3 ft]). The 1990 derived transmissivity value for well 299-E35-2 was 6 m^2/day (20 ft^2/day), with a hydraulic conductivity of 39.7 m/day (130 ft/day).

In summary, the multiple slug test results at six of the seven wells described in this subsection adequately define the hydraulic conductivity for the basalt flow top and Hanford sediments. The basalt flow top slug test data produced varying results of hydraulic conductivity. To the east, the results were low, while results to the south and west of LERF were significantly greater than the overlying Ringold and Hanford sediment results. The range of hydraulic conductivity beneath and west of LERF appears to exceed 100 m/day. A best estimate is 134 m/day. The overlying suprabasalt sediments were consistent with a hydraulic conductivity range of 24.4 to 42.8 m/day, with a best estimate of 39.5 m/day. These values will be used to derive the rate of flow for LERF. Although effective porosity was not derived from tests completed at LERF, the effective porosity to be used for flow rate calculations at LERF is 0.1. This value was chosen because of the evaluation process discussed in SGW-54508, *WMA C September 2012 Quarterly Groundwater Monitoring Report.*

D2.2.2 Flow Dynamics

Regional groundwater flow was initially from west to east but was impacted by groundwater mounding resulting from wastewater discharges primarily to the east. These impacts have diminished significantly and do not appear to contribute to the flow regime beneath LERF; however, they still appear to affect the groundwater quality at well 299-E26-11.

Recently, statistical methods have been applied to deriving the flow direction. Table D-5 provides the gradient and magnitude associated with the statistically corrected calculations since the installation and water level collection at well 299-E26-14. Although the p-value indicates the derived values have a moderate amount of uncertainty (e.g., 20 to 30 percent), the direction has been very constant ranging between 186 and 198 degrees from north (e.g., southwest of south flow). The average direction is 190 degrees from north. In addition, the gradient magnitude has been constant, ranging between 2.39E-04 and 2.98E-04, with an average of 2.7E-4. If these average values are applied to the following formula $V=(K*G)/n_e$ (Driscoll, 1986, *Groundwater and Wells*); where V is the flow rate, K is the hydraulic conductivity, G is the gradient, and n_e is the effective porosity, then the average flow rate in the suprabasalt sediments could be 0.11 m/day or 38.9 m/year. This value correlates with the movement of a sulfate plume originating to the northwest and west of LERF, as explained in Section 2.10.3.6 and displayed in Figure 2.10-42 of DOE/RL-2008-01, *Hanford Site Groundwater Monitoring for Fiscal Year 2007.*

Although the derived gradient magnitude and associated flow rate beneath LERF is consistent with other observances of migrating plume rates, as discussed in the previous paragraph, the 190 degree flow direction and increasing anion and cation concentrations at well 299-E26-14 (Figure D-18) do not correlate with the perceived source of anion and cation increases. One of the most distinguishable

constituents associated with the anion and cation increases is sulfate. Sulfate was initially observed increasing at significant levels at well 299-E34-7, located northwest of LERF, in the mid-1990s, as shown in Figure D-19. This well became sample dry in 2005, two years after concentrations had peaked at 671 mg/L. The extent and source of the sulfate is uncertain; however, various conceptual models have been discussed to a limited degree. Movement of the sulfate also has been discussed in several of the Hanford Site Groundwater Monitoring Reports over the past two decades, and an interpreted snapshot of the sulfate plume is provided for 2008 and 2013 (Figure D-20). Historically, sulfate increases along the west side of the LERF monitoring network appear to be associated with transverse dispersivity because of the lack of vadose zone moisture during drilling well 299-E26-77, the derived southward flow direction from monitoring network, and smaller slope of sulfate increase at well 299-E26-10 than at well 299-E34-7 (Figure D-19).

More recently, the rate and direction of groundwater flow appear to be in a state of change regionally. Sulfate concentrations leveled off between 2009 and 2011 near LERF, when the regional flow conditions were considered to be at a minimum compared to previous years (Figure D-18). Since the middle of 2011, when the Columbia River elevation began to exceed the 200 East groundwater elevations, concentrations have been increasing at a greater rate in wells farther east than at well 299-E26-10 (Figure D-18). Three explanations are provided for what may be occurring:

- The northwest sulfate source may be diminishing and because well 299-E26-10 is spatially closer to the source, it is beginning to decrease with the decreasing front, while wells 299-E26-14 and 299-E26-79, farther spatially from the proposed northwest source, are still within the increasing front of the sulfate plume.
- The flow direction has shifted to a southeast of east to an easterly flow direction, causing greater concentrations to migrate preferentially toward wells 299-E26-14 and 299-E26-79.
- The source of sulfate increases may be from a more regional source as sulfate increases have also been seen at well 299-E26-11, but to a smaller degree, and delayed compared to well 299-E26-10 (Figure D-21).

Because of the consistent flow direction derived by the current monitoring network and the larger influence of sulfate increases seen across the LERF monitoring network, the sulfate increases are considered to be from a larger regional source. As such, well 299-E26-14 provides a sufficient representation of the groundwater quality migrating into the area from the north. However, to ensure that conditions continue to reflect this conceptual flow model, well 299-E26-77 will be monitored but considered a crossgradient monitoring well and not included in upgradient statistical measurements. Water levels will continue to be collected at wells 299-E26-10 and 299-E26-77 to maintain statistical analyses of the flow direction. Should conditions change in the statistically derived flow direction or groundwater quality parameters at well 299-E26-77 suggest a change in sulfate migration, then the information will be relayed in the *Resource Conservation and Recovery Act of 1976* (RCRA) quarterly report with a proposed action.

Based on the accepted southward flow direction, well 299-E26-79 is the only downgradient well currently at LERF. Because the basalt flow top appears to be connected hydraulically to the suprabasalt sediments and provides a potentially more transmissive pathway, well 299-E26-79 does not appear to be sufficiently located to monitor the easternmost basin.

D2.3 Groundwater Chemistry

Groundwater chemistry in the uppermost aquifer beneath LERF was affected by several years of diluted liquid waste discharge to the 216-B-3 Pond System, which ceased in 1997. Figure D-22 provides an

illustration of the groundwater chemical facies at various LERF wells from the early 1990s to 2013 using Stiff Diagrams. As can be seen in the figure during the early to mid-1990s, the groundwater chemical facies was calcium-bicarbonate, except to the west of LERF where a calcium-bicarbonate-sulfate facies was present at well 299-E34-7. By 1999, the groundwater at well 299-E34-7 was a strong calcium-sulfate facies, and well 299-E26-10, located to the southeast of well 299-E34-7, was changing to a calcium-sulfate facies. In 2001, well 299-E26-10 was a calcium-sulfate facies.

In 2006, well 299-E26-11, located to the east of LERF, was beginning to show signs of changing from a calcium-bicarbonate facies. By 2011, well 299-E26-11 also had changed to a calcium-sulfate facies. Although well 299-E26-11 saw a change in the chemical nature of the groundwater after wells to the west, it has had a greater water elevation than the wells to the west. This indicates the water facies change must be from north of well 299-E26-11. Further east at well 699-45-42, located east of well 299-E26-11, a calcium-bicarbonate chemical facies was still present in 2012.

The wells to the north and south of LERF portray an intermediate chemical facies, which is between the strong calcium-sulfate facies to the west and the more dilute calcium-sulfate facies to the east. For comparison, the calcium and sulfate milliequivalents in well 299-E26-10 in January of 2013 were 6.4 to 5.5, respectively. The milliequivalents at well 299-E26-11 in January 2013 were 2.6 for both calcium and sulfate. The January 2013 milliequivalent results for wells 299-E26-14 and 299-E26-79 were 3.9 to 3.3 and 3.7 to 3.0, respectively. Thus, the chemical facies is slightly stronger to the north of LERF than south, which is downgradient of LERF.

The fact that all the wells near LERF are showing chemical facies changes to a calcium-sulfate indicates that the wells are hydraulically connected and that there is a sulfate source to the north of LERF. Water quality parameters will continue to be collected semiannually for purposes of further evaluation, as shown in Table D-6.

D2.4 Well Completions and Conditions

The basic well information is summarized in Table D-4 and in Figures D-23 through D-27. Five wells are provided for discussions related to the geology and hydrogeology; however, only four of the wells (299-E26-10, 299-E26-14, 299-E26-77, and 299-E26-79) are being used for monitoring the groundwater near LERF. The four wells allow use of statistical measures to derive a groundwater gradient and direction.

All four wells extend beyond 61 m (200 ft) in depth. Although the new wells extend 5.5 to 6.1 m (18 to 20 ft) into the EMB, the screened intervals in all four wells intercept the unconfined aquifer as discussed in Section D2.2.1.

The initial LERF groundwater monitoring wells were installed in 1990 and included three downgradient wells on the west end of the facility boundary and one upgradient well at the east end of the facility. This configuration was based on the east-to-west groundwater flow direction, caused by the recharge mound created by years of liquid effluent disposal to B Pond. Wells 299-E26-9, 299-E26-10, and 299-E35-2 were originally installed as downgradient wells and well 299-E26-11 as an upgradient well. Wells 299-E26-77 and 299-E26-79 were installed in 2008. Well 299-E26-77 is adjacent to the location of well 299-E26-9, and well 299-E26-79 is south of LERF between Basins 42 and 43 (Figure D-2). Well 299-E26-10 (Figure D-23) has a 4.5 m (15 ft) screen, screening across the entire saturated suprabasalt sediments. The well screen in 299-E26-10 penetrates approximately 0.5 m (1.8 ft) into the basalt. Well 299-E26-11 (Figure D-3 and D.24) was completed with a 1.5 m (5 ft) long channel-pack screen placed completely within the basalt flow top and includes a sand pack that extends 1.3 m (4.4 ft) above the screen top.

Well 299-E26-14 encountered groundwater at approximately 60.5 m (198.4 ft) bgs and was drilled to a total depth of 73.3 m (240.6 ft) bgs (Figure D-25). The well is constructed with 6.1 m (20 ft) total length of screen installed across approximately 5.5 m (18 ft) of Ringold and Hanford sediments. Only a small portion (0.27 m or 0.9 ft) of the Ringold sediments are adjacent the bottom of the well screen. The screen is 10 cm (4 in.) in diameter, 20 slot, stainless-steel wire-wrap. The well has a 1 m (3 ft) blank sump below the screen. The casing from the top of the screen to land surface is 10 cm (4 in.) diameter stainless steel.

Well 299-E26-77 encountered groundwater at approximately 63.4 m (208 ft) bgs and was drilled to a total depth of 71 m (232.8 ft) bgs (Figure D-26). The well is constructed with 7.6 m (25 ft) total length of screen installed across approximately 1.4 m (4.6 ft) of sediments and 6.2 m (21.4 ft) of basalt flow top. Well 299-E26-79 encountered groundwater at 61.5 m (201.7 ft) bgs and was drilled to a total depth of 68.5 m (224.8 ft) bgs (Figure D-27). The well is constructed with 7.6 m (25 ft) total length of screen installed across approximately 3.7 m (12 ft) of sediments and 3.9 m (13 ft) of basalt flow top. The screens are 10 cm (4 in.) in diameter, 20-slot, stainless-steel wire-wrap. Both wells have a 1 m (3 ft) blank sump below the screens. The casing from the top of the screen to land surface is 10 cm (4 in.) diameter stainless steel.

The longevity of the operable monitoring lifetime for the remaining LERF wells is not a concern as water levels are only being collected from well 299-E26-10 and, based on recent water level declines, should be useable for decades. The other three wells have significant water for sample collection and should not go dry, based on pre-Hanford groundwater elevations.

D3 Groundwater-Monitoring Program

Groundwater monitoring at LERF is in detection monitoring and the indicator parameters are discussed further in Section D3.6.1. The indicator parameters were derived as summarized in Section D3.2 and discussed in further detail in Section D1.5. The detection monitoring sample frequency is semiannual as discussed in Section D3.6.2. Sampling procedures and required documentation is provided in Sections D3.6.3 and D3.6.4, respectively. The analytical procedures, analytical quality control (QC), data management are discussed in Sections D3.9, D3.9.1, and D3.9.2, respectively.

Statistical methods are employed to determine local background conditions for the upgradient well 299-E26-14 as provided in Section D3.9.3. Detection monitoring at LERF is discussed in Section D3.4. Should indicator parameter results exceed local background levels then resampling will be implemented for determining if a false positive result has occurred or if assessment monitoring must be undertaken as discussed in Sections D3.3, D3.9.3, and D3.11.

Reporting will be annually through the Hanford Site Annual Groundwater Monitoring Report unless a significant exceedance of the background values determined for the upgradient well 299-E26-14 is verified. If an exceedance is verified then the notification process discussed in Section D3.11 will be followed.

As discussed in Sections D1 and D2 and their subsections, the following characteristics describe the hydrogeology in the LERF area:

- Representative groundwater samples can be collected from the uppermost aquifer.
- Upgradient background samples at well 299-E26-14 are representative of unaffected groundwater from LERF.
- Groundwater samples collected at well 299-E26-79 are representative of the quality of groundwater passing the LERF point of compliances.

Hydraulic characterization tests conducted over the past two decades and the groundwater chemical facies changes indicate the hydrostratigraphic units underlying the LERF basins constitute an aquifer unit that is continuous beneath the LERF basins and is capable of yielding representative groundwater samples.

D3.1 Objectives of Dangerous Waste Groundwater Monitoring and Past Monitoring Results

A groundwater monitoring program, in accordance with the requirements of WAC 173-303-645, is designed to determine whether there is statistically significant evidence of contamination in the uppermost aquifer attributable to the LERF basins. The statistical parametric *t*-test approach at LERF compares two distinct statistical populations for true differences in population means as discussed further in Section D3.9.3.

By the date of this permit, the action leakage rate has not been exceeded during operations, and results of the LERF groundwater monitoring program indicate the LERF basins have not impacted groundwater quality beneath the site. Past monitoring results from former downgradient wells 299-E26-10 and 299-E26-11, and more recent results from newer wells 299-E26-77 and 299-E26-79, have not indicated dangerous constituents above background levels, with the exception of one positive carbon tetrachloride result at each well. Because the detections were followed by a series of non-detect values and the results were associated with out-of-limit QC samples, the reported concentrations appeared to be associated with a laboratory error and were flagged as suspect. As a result, a detection monitoring program in accordance with WAC 173-303-645(9) is appropriate for the site to provide compliance with the requirements of WAC 173-303-645.

D3.2 Dangerous Constituents

A list of dangerous and/or mixed aqueous waste that can be accepted in LERF is defined by the requirements of Addendum B (Waste Analysis Plan).

Dangerous constituents and suitable indicator parameters that provide a reliable indication of the presence of dangerous constituents in groundwater for purposes of groundwater monitoring were based on target parameter constituents from Addendum B (Waste Analysis Plan), and results of LERF basin water samples collected between July 1999 and June 2013. Several target parameters in the Waste Analysis Plan (Addendum B) occur in the LERF basin influent data and were evaluated relative to the dangerous waste characteristics (groundwater monitoring list in WAC 173-303-090, "Dangerous Waste Characteristics," and Ecology Publication 97-407, *Chemical Testing Methods for Designating Dangerous Waste: WAC 173-303-090 & -100*, Appendix 5. As discussed in Section D1.5, dangerous waste constituents measured as part of routine liquid sampling in the LERF basins were included as indicator parameters.

Tables D-1 through D-3 present a list of dangerous constituents measured as part of routine liquid sampling in the LERF basins from as early as February of 2000 through 2011. The results were further evaluated to identify reliable parameters for the indication or identification of dangerous waste constituents in groundwater, as discussed in Section D1.5. The full list of groundwater monitoring indicator parameters is provided in Section D3.6.1.

D3.3 Concentration Limits

A series of events that triggers the shift from detection monitoring to compliance monitoring is prescribed in WAC 173-303-645. If there is statistically significant evidence of contamination, as required in WAC 173-303-645(9)(f), groundwater protection standards and concentration limits will be established subsequently in accordance with WAC 173-303-645(9)(g)(iv)(D). Section D3.11, Evaluation and Notification, provides the process and schedule for actions, notification, and permit modification, if necessary.

If a tolerance limit is exceeded at a statistically significant level, additional measurements will be conducted to verify that a detection event has occurred. If the detection of a dangerous constituent is verified, as discussed in Section D3.11, compliance monitoring will be implemented in accordance with WAC 173-303-645(10).

D3.4 Groundwater Monitoring System and Point of Compliance

The groundwater monitoring system for LERF uses existing wells, 299-E26-14 and 299-E26-79. Well 299-E26-14 is an upgradient well and well 299-E26-79 is a downgradient well based on the flow direction presented in Section D2.2.2. A third detection monitoring well will need to be installed just south of the open interval between Basin 43 and 44 in order to compare the groundwater quality downgradient of LERF Basin 44 (Figure D-28). All three of these wells will be monitored in accordance with the requirements provided in this permit. The additional well to be installed, 299-E26-15, will be planned through Tri-Party Agreement (Ecology et al., 1989, *Hanford Federal Facility Agreement and Consent Order*) Milestone M-024, which is updated on a yearly basis. The well is tentatively planned to be installed prior before fiscal year 2016.

D3.5 Compliance Period

The compliance period will be the number of years equal to the active life of the waste management area including any additional years required for corrective actions, if necessary. Any additional years associated with corrective actions will be completed after three consecutive years in which the groundwater protective standard for any specific dangerous waste constituent has not been exceeded in accordance with WAC 173-303-645(7).

D3.6 Sampling and Analysis

This section describes the groundwater detection sampling and analysis program for the three LERF regulated units (Basins 42, 43, and 44), including monitoring parameters, analytical methods, monitoring frequency, and sampling protocols.

D3.6.1 Monitoring Parameters

Monitoring parameters include the indicator and geochemical parameters. The monitoring of these two parameters is similar, and sampling and analysis frequencies are the same and will be done concurrently on a semiannual basis.

As identified in Section D1.5, carbon tetrachloride and TOX are reliable indicator parameters for the presence of dangerous constituents associated with LERF. In addition, the standard parameters of pH, specific conductance, and TOC provide the requirements of detection monitoring in accordance with WAC 173-303-645(9)(a). Table D-7 provides a list of these constituents and the frequency of sampling.

Samples will also be collected semiannually and analyzed for major anions, cations, and alkalinity to evaluate groundwater geochemistry, as discussed in Table D-6.

Samples also will be collected for Cr(VI) for evaluation as an additional indicator parameter, as discussed in Section D1.5.

D3.6.2 Sampling Frequency

Samples will be collected semiannually from wells 299-E26-14 and 299-E26-79 to determine whether there is statistically significant evidence of contamination for the indicator parameters established in Section D3.6.1.

Samples will be collected semiannually and analyzed for major anions, cations, and alkalinity to evaluate groundwater geochemistry, as discussed in Section D2.3 and shown in Table D-6.

Finally, samples will be collected semiannually for Cr(VI) evaluation as an indicator parameter, as discussed in Section D1.5 and Table D-7.

D3.6.3 Sampling Procedures

Groundwater sampling procedures, sample collection documentation, sample preservation and shipment, and chain-of-custody requirements are described in this subsection. The Permittees will develop, maintain, and conduct work according to procedures consistent with, and no less stringent than, those described to be conducted. The Permittees will maintain current copies of these procedures in the Hanford Facility Operating Record, LERF, and 200 Area ETF file, as required by Permit Condition II.I.1.

Samplers fill out groundwater sample report forms as they purge and sample each well. Field personnel measure water levels in each well before sampling and then purge stagnant water from the well. Field personnel also record time of sampling, which allows correlation with barometric pressure measurements at the Hanford Meteorological Station. Water levels are typically measured with laminated-steel electrical sounding tapes with a precision of 2 mm. Procedures require sample collection after three casing volumes of water have been purged from the well and after field parameters (pH, temperature, specific conductance, and turbidity) have stabilized. Field parameters are measured in a flow-through chamber. Both filtered and unfiltered samples are collected for metals analyses. Filtering is performed in the field with 0.45-micron, in-line, disposable filters to ensure that results represent dissolved metals and do not include particulates. Dissolved trace metals analysis (from filtered samples) will be used for statistical analyses of trace metal arsenic.

Sample preservation techniques will follow generally accepted practices (e.g., U.S. Environmental Protection Agency [EPA]-approved guidelines such as SW-846, *Test Methods for Evaluating Solid Waste: Physical/Chemical Methods, Third Edition; Final Update IV-B*, Table 11-1, or equivalent) and will be documented in sample authorization forms generated by the Sample and Data Management organization. Chemical preservatives are added to collection bottles before use in the field. A chemical preservative label is affixed to the sample container listing the specific preservative. The preservative's brand name, lot number, concentration, and date opened are recorded. As part of sample preservation, samples may be refrigerated or stored on ice as necessary prior to delivery to the analyzing laboratory.

D3.6.4 Sample Chain-of-Custody

Groundwater samplers use chain-of-custody forms to document the integrity of groundwater samples from the time of collection through data reporting. The forms are generated during scheduling and are managed through a documented procedure. Required information recorded on the forms includes the following:

- Sampler's name
- Method of shipment and destination
- Collection date and time
- Sample identification numbers
- Analysis methods
- Preservation methods

Samples are labeled and sealed with evidence tape, wrapped with bubble wrap, and placed in a U.S. Department of Transportation-approved container with ice, as appropriate. The packaging parameters for samples are determined by associated hazards. Samples for offsite laboratories are shipped according to U.S. Department of Transportation regulations. A chain-of-custody form accompanies all samples.

When samples are transferred from one custodian to another (e.g., from sampler to shipper, or from shipper to analytical laboratory), the receiving custodian inspects the form and the samples, noting any deficiencies. Each transfer of custody is documented by the printed names and signatures of the custodian relinquishing the samples and the custodian receiving the samples, as well as the time and date of transfer. Commercial shippers do not sign chain-of-custody forms, but the forms are signed by the receiving laboratory, and sample integrity is verified by inspecting the bottle seals.

D3.7 Decontamination of Drilling and Sampling Equipment

The following information is included relative to well drilling equipment for new wells installed at LERF for this Permit. Well drilling equipment is decontaminated using high temperature and pressure washing. The equipment then is rinsed with clean water.

Equipment for collecting soil samples during drilling for later chemical analysis is decontaminated. Equipment is washed with phosphate-free detergent, rinsed three times with de-ionized water, rinsed once with nitric acid (glass or stainless-steel equipment only), rinsed three more times with de-ionized water, and then finally rinsed with hexane. After heat drying, equipment is wrapped in unused aluminum foil and sealed with tape until needed. The tape shall not come into contact with the equipment to avoid any contamination from the materials in the tape.

Monitoring wells for LERF shall be equipped with dedicated sampling pumps. Sample pumps are placed at approximately mid-depth within the screen interval. Water-level measuring tapes are cleaned with potable or deionized water and a clean towel. Sample manifolds used at the well head require decontamination as follows: wash with a phosphate-free detergent, rinse three times in high-purity water, rinse in a 1 M solution of nitric acid, rinse three more times in high-purity water, then rinse in hexane, and finally dry in drying chamber. These are done in accordance with established procedures.

D3.8 Quality Objectives and Criteria

The QC program is designed to assess and assure the reliability and validity of groundwater data, and to document whether the resulting data are of the quantity and quality necessary for the intended decision-making purpose. In groundwater detection monitoring, the primary decision-making purpose is to determine whether a statistically significant increase in a dangerous constituent concentration is observed in groundwater downgradient from the permitted site. Consequently, data quality is monitored by evaluating the results of QC samples, conducting audits, validating groundwater data, and comparing these results to data quality requirements established in this groundwater monitoring plan. Accuracy, precision, and detection are the primary parameters used to assess data quality. Data for these parameters are obtained from two categories of QC samples: field QC samples that provide checks on field and laboratory activities, and laboratory QC samples that monitor laboratory performance. Table D-8 summarizes the types of samples in each category and the sample frequencies and characteristics evaluated.

D3.9 Analytical Procedures

All field and laboratory instrumentation are calibrated using approved procedures, and analytical measurements are generated according to approved procedures. These procedures include quality checks to ensure the resulting analytical values are of known quality.

Instruments for field measurements (e.g., pH, specific conductance, temperature, and turbidity) are verified using standard solutions before use. These include, for pH, 4, 7, and 10 buffer/standard solutions; for specific conductance, 445 µS/cm and 1,413 µS/cm solutions; and for turbidity, Gelex standards 0-10, 0-100, and 0-1,000 nephelometric turbidity units. Instruments are operated in accordance with the

manufacturer's instructions. Each instrument is assigned a unique number that is tracked via calibration documentation and field logbooks and sampling reports.

Laboratory analytical methods are specified in Table D-9 and are generally specified in contracts with the laboratories. Laboratory methods for chemical parameters are typically standard methods from SW-846; EPA-600/4-79-020, *Methods for Chemical Analysis of Water and Wastes*; or APHA/AWWA/WEF, 2012, *Standard Methods for the Examination of Water and Wastewater*. Analytes, analytical methods, and required maximum practical quantitation limits are shown in Table D-9.

D3.9.1 Quality Control

QC data are evaluated based on acceptance criteria for each QC sample type, as summarized by constituent in Table D-10. These criteria limits are intended to provide confidence that the analytical and field methods are in control and provide data of known quality. For field and method blanks, the acceptance limit is two times the instrument detection limit (metals) or method detection limit (other chemical parameters), except for the common laboratory contaminants 2-butanone, acetone, methylene chloride, toluene, and phthalate esters where the limit is five times the method detection limit. Groundwater samples that are associated (i.e., collected on the same date and analyzed by the same method) with out-of-limit field blanks are given a review qualifier of "Q" in the Hanford Environmental Information System (HEIS) database to indicate a potential problem, and then recorded in the Hanford Facility Operating Record, LERF, and 200 Area ETF file pursuant to Permit Condition III.3.D.1.b.

Field duplicates must agree within 20 percent (as measured by relative percent difference) to be acceptable. Only those field duplicates with at least one result greater than five times the appropriate detection limit shall be evaluated. In the case where one result is a non-detect, the detection limit is used to calculate the relative percent difference. Unacceptable field duplicate results are given a review qualifier of "Q" in the database and recorded in the Hanford Facility Operating Record, LERF, and 200 Area ETF file.

The specified frequency for laboratory duplicates, matrix spikes, matrix spike duplicates, surrogates, and laboratory control samples are defined in Table D-10 in accordance with SW-846. The acceptance criteria for the associated parameter data shall be analyzed and recorded in accordance with Section D3.10.2.

Sample holding times depend on the analyte and are specified in the Environmental Quality Assurance Program Plan. Data associated with exceeded holding times are given a review qualifier of "H" in the HEIS database and noted in the Hanford Facility Operating Record, LERF, and 200 Area ETF file. Data exceeding holding times shall be maintained but potentially may not be used in statistical analyses.

Table D-11 lists the acceptable accuracy for the blind standards for carbon tetrachloride and TOX. These samples are prepared by spiking Hanford background well water (currently, wells 699-19-88 and 699-49-100C) with known concentrations of constituents of interest. Spiking concentrations range from the detection limit to the upper limit of concentration determined in groundwater on the Hanford Site. Investigations shall be conducted for blind standards that are outside of acceptance limits. The results from these standards shall be used to determine acceptability of the associated parameter data.

Additional QC measures include laboratory audits and participation in nationally based performance evaluation studies. Audit results are used to improve performance. Summaries of audit results and performance evaluation studies shall be incorporated into the Hanford Facility Operating Record, LERF, and 200 Area ETF file as appropriate to substantiate data quality objectives (DQOs) and data acceptance criteria.

D3.9.2 Data Management

This section describes data management practices.

Loading Data

The contract laboratories report analytical results electronically and in hardcopy. The electronic results shall be loaded into the HEIS database as they are received from the laboratories. The appropriate sections of the HEIS shall be incorporated by reference into the Hanford Facility Operating Record, LERF, and 200 Area ETF file to satisfy Permit Condition III.3D.1.b. Field data (e.g., specific conductance, pH, temperature, turbidity, and depth to water) are recorded on field records. Data management staff enter field data into the HEIS database manually through data-entry screens and verify each value against the hardcopy. An electronic field data collection system may be implemented soon, which would replace the manual field data collection and the manual data entry process when it is implemented.

Data not available electronically may include well logbooks, borehole videos, geologic descriptions, field screening data, or other information.

Data Review, Verification, Validation, and Usability

The final data review shall determine whether data meet the criteria specified in this subsection. The work activities shall follow documented procedures and processes for data validation and verification. Validation of groundwater data involves assessing whether the data collected and measured meet contractual quality requirements. Verification involves assessing data accuracy, completeness, consistency, availability, and internal control practices to determine overall reliability of the data collected. Other DQOs that shall be met include the proper chain-of-custody, sample handling, use of proper analytical techniques for each constituent, and the quality and acceptability of the laboratory analyses conducted.

Groundwater monitoring staff performs checks on laboratory electronic data files for formatting, allowed values, data flagging (qualifiers), and completeness. A percentage of hardcopy results are verified to check for completeness; notes on condition of samples upon receipt by the laboratory; notes on problems that arose during the analysis of the samples; and correct reporting of results. If data are incomplete or deficient, staff will work with the laboratory to correct the problem discovered during the analysis.

The data validation process provides the requirements and guidance for validating groundwater data that are routinely collected. Validation is a systematic process of reviewing verified data against a set of criteria (listed in Table D-10) to determine whether the data are acceptable for their intended use.

Results of laboratory and field QC evaluations, blind sample results, laboratory performance evaluation samples, and holding-time criteria are considered when determining data usability. Staff review the data to identify whether observed changes reflect changes in groundwater quality or potential data errors, and they may request data reviews of laboratory, field, or water-level data for usability purposes. The laboratory may be requested to check calculations or reanalyze the sample, or the well may be resampled. Results of the data reviews are used to determine what appropriate review qualifier should be applied to the analytical results in the HEIS database (e.g., "R" for reject, "Y" for suspect, or "G" for good) and/or to add comments.

Upon final data acceptance, both the raw data and the accepted/validated data shall be incorporated into the Hanford Facility Operating Record, LERF, and 200 Area ETF file.

Data Review Corrective Actions

The responses to data quality defects are identified through the verification/validation process. Corrective actions are shown in Table D-8.

D3.9.3 Statistical Analysis of Groundwater Monitoring Data

Groundwater monitoring constituents have been identified for the LERF basins and are listed in Table D-10. The dangerous constituents and indicator parameters used to indicate the presence of contamination (WAC 173-303-645(9)(a)) and subject to statistical evaluation are listed in Table D-7 and include carbon tetrachloride, pH, specific conductance, TOC, and TOX.

To establish background conditions, the previous data collected over the past two years will be used. Every year, background results will be evaluated for updating the critical mean for each indicator parameter identified in Table D-8. Sample collection and analysis will continue on a semiannual basis.

The statistical method for comparing baseline (background) groundwater quality with compliance-point groundwater quality is the Welch's *t*-test in accordance with WAC 173-303-645(8)(h)(i), and it is recommended for detection monitoring when population variances might differ between two groups, as stated in EPA 530/R-09-007, *Statistical Analysis of Groundwater Monitoring Data at RCRA Facilities Unified Guidance*. Applying this parametric *t*-test provides a reasonably robust statistical procedure and assurance when background data are at a minimum and the underlying populations may not meet normality. However, normality can usually be met by log transforming the data. As more background data is generated for well 299-E26-14, additional tests may be applied if spatial variability becomes an issue, such as the Wilcoxon Rank-sum Test.

The Welch's *t*-test procedure can be implemented as follows:

I. Compute the sample mean $\bar{x}$, standard deviation s, and variance s^2, in each of the background and compliance well data sets.

II. Compute Welch's *t*-statistic using the following equation $t = (\overline{X_c} - \overline{X_{BG}})/\sqrt{\frac{s^2_{BG}}{n_{BG}} + \frac{s^2_c}{n_c}}$

III. Compute the approximate degrees of freedom using the following equation

$$df = \left[\left[\frac{s^2_{BG}}{n_{BG}} + \frac{s^2_c}{n_c}\right] / \left[\frac{(s^2_{BG}/n_{BG})^2}{n_{BG}-1} + \frac{(s^2_c/n_c)^2}{n_c - 1}\right]\right]$$

IV. Use Table 16-1 of Appendix D in EPA 530/R-09-007 to assign the upper 95 percent critical mean based on the degrees of freedom.

V. Compare the t-statistic against the critical point, t_{cp}. When the condition $t \leq t_{cp}$, conclude there is no statistically significant difference between the background and compliance point population means. If, however, $t > t_{cp}$, conclude that the compliance point population mean is significantly greater than the background mean at the α level of significance.

As monitoring continues and the process is shown to be in control (i.e., there is no statistically significant evidence of facility impact to groundwater), the baseline mean and standard deviation should be updated periodically (e.g., every 1 or 2 years) to incorporate the new data (EPA 530/R-09-007). This reduces uncertainty in the background and helps adjust for groundwater influences from outside sources. This updating process should continue for the lifetime of the monitoring program.

If an exceedance occurs, resampling will be undertaken to verify or refute the original exceedance. The analytical result from the resample is substituted into the previous formulas in place of the original value obtained, and the Welch's *t*-test statistic is updated. If resampling does not confirm the exceedance, and *if the exceedance can be shown to be a measurement in error or a confirmed outlier, it should be excluded from the revised background. Otherwise, any disconfirmed exceedances (including any*

resamples that exceed the background limit but are disconfirmed by other resamples) should probably be included when updating the background. The reason is that background limits designed to incorporate retesting are computed as low as possible to ensure adequate statistical power (EPA 530/R-09-007).

D3.10 Reporting and Recordkeeping

Reporting of monitoring evaluations for LERF will be carried out through the Hanford Site Annual Groundwater Monitoring Report.

Pertinent information for groundwater monitoring and electronic files for groundwater data shall be maintained in the Hanford Facility Operating Record, LERF, and 200 Area ETF file required by Permit Condition II.I.1. Records may be stored in either electronic or hardcopy format.

The Hanford Facility Operating Record, LERF, and 200 Area ETF file will also include, consistent with Permit Condition III.3.D.1.b, the following items:

- Groundwater sample reports
- Chain-of-custody forms
- Sample receipt records

D3.11 Evaluation and Notification

Groundwater flow rate and direction in the uppermost aquifer will be evaluated and reported annually. Groundwater indicator parameter data collected under this permit will be reviewed semiannually to determine if there is statistically significant evidence of contamination (in accordance with WAC 173-303-645(9)(f)) using the statistical method provided in Section D3.9.3. The results of the statistical evaluation and associated information will be submitted to the Washington State Department of Ecology (Ecology) annually through the Hanford Site Annual Groundwater Monitoring Report (WAC 173-303-645(9)(c)).

If statistically significant evidence of contamination is determined for one or more of the dangerous constituents or indicator parameters, at any monitoring well at the compliance point, the owner or operator may resample within one month and repeat the analysis for the detected compounds in accordance with WAC 173-303-645(9)(g)(ii). The resample data will be compared with the control limit. If resampling confirms statistically significant evidence of contamination, the following actions will be performed in accordance with WAC 173-303-645(9)(g):

- Notify Ecology in writing within seven days of the finding, indicating which chemical parameters have shown statistically significant evidence of contamination.
- Sample the groundwater in all monitoring wells and determine if constituents included in Ecology Publication 97-407, *Chemical Testing Methods for Designating Dangerous Waste: WAC 173-303-090 & -100*, Appendix 5, are present, and if so, in what concentration. For any of these compounds detected, the owner or operator may resample within one month of receiving the results and repeat the analysis for those compounds detected. If the constituents are detected in the second analysis, they will form the basis for compliance monitoring.
- If dangerous constituent(s) are detected, submit an application for a Permit modification to Ecology within 90 days to establish a compliance monitoring program in accordance with WAC 173-303-645(9)(g)(iv).
- If dangerous constituents are not detected, continue to monitor in accordance with the detection monitoring program.

If a source other than LERF caused the contamination or the detection is an artifact caused by an error in sampling, analysis, or statistical evaluation or natural variation in groundwater (as allowed by WAC 173-303-645(9)(g)(vi), the following guidelines will apply:

- Notify Ecology in writing within seven days of the finding (i.e., exceedance) and indicate the intent to make a demonstration to this effect.
- Submit a report to Ecology within 90 days. The report should demonstrate that a source other than the regulated unit caused the contamination, or that the contamination resulted from an error in sampling, analysis, evaluation, or natural variation in groundwater chemistry.
- Continue monitoring in accordance with the detection monitoring program.

If it is determined, in accordance with WAC 173-303-645(9)(h), that the detection monitoring program no longer satisfies the requirements of WAC 173-303-645(9), submit an application to Ecology for a Permit modification within 90 days to make any appropriate changes to the program.

D4 Compliance-Monitoring Program

Reserved.

D5 Corrective-Action Program

Reserved.

D6 References

40 CFR 265, "Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities," Subpart F, "Ground-Water Monitoring," *Code of Federal Regulations*. Available at: http://www.gpo.gov/fdsys/pkg/CFR-2010-title40-vol25/xml/CFR-2010-title40-vol25-part265-subpartF.xml.

APHA/AWWA/WEF, 2012, *Standard Methods For the Examination of Water and Wastewater*, 22nd Edition, American Public Health Association, American Water Works Association, and Water Environment Federation, Washington, D.C.

Barker, John A. and John H. Black, 1983, "Slug Tests in Fissured Aquifers," *Water Resources Research* 19(6):1558-1564. Available at: http://onlinelibrary.wiley.com/doi/10.1029/WR019i006p01558/pdf.

BHI-00184, 1995, *Miocene- to Pliocene-Aged Suprabasalt Sediments of the Hanford Site, South-Central Washington*, Rev. 00, Bechtel Hanford, Inc., Richland, Washington.

Bouwer, Herman and R.C. Rice, 1976, "A Slug Test for Determining Hydraulic Conductivity of Unconfined Aquifers With Completely or Partially Penetrating Wells," *Water Resources Research* 12(3):423-428. Available at: http://www.cof.orst.edu/cof/fe/watershd/fe538/Watershed_10_HillslopeModule/Bouwer_Rice_slug_test_hydraulic_conductivity_WRR1976.pdf.

Cooper, H.H. and C.E. Jacob, 1946, "A Generalized Graphical Method of Evaluating Formation Constants and Summarizing Well-Field History," *Trans. Am. Geophys. Union* 27(4):526-534. Available at: http://core.ecu.edu/geology/spruill/spruill/Groundwater%20Notes%20No.%207.pdf.

Cooper, Hilton H., Jr., John D. Bredehoeft, and Istavros S. Papadopulos, 1967, "Response of a Finite-Diameter Well to an Instantaneous Charge of Water," *Water Resources Research* 3(1):263-269. Available at: http://onlinelibrary.wiley.com/doi/10.1029/WR003i001p00263/pdf.

DOE/RL-92-19, 1993, *200 East Groundwater Aggregate Area Management Study Report*, Rev. 0, U.S. Department of Energy, Richland Operations Office, Richland, Washington. Available at: http://pdw.hanford.gov/arpir/index.cfm/viewDoc?accession=D196136305.

DOE/RL-96-61, 1997, *Hanford Site Background: Part 3, Groundwater Background*, Rev. 0, U.S. Department of Energy, Richland Operations Office, Richland, Washington. Available at: http://pdw.hanford.gov/arpir/index.cfm/viewDoc?accession=D197226378.

DOE/RL-2008-01, 2008, *Hanford Site Groundwater Monitoring for Fiscal Year 2007*, Rev. 0, U.S. Department of Energy, Richland Operations Office, Richland, Washington. Available at: http://pdw.hanford.gov/arpir/index.cfm/viewDoc?accession=00098824.

Driscoll, Fletcher G., 1986, *Groundwater and Wells*, Second Edition, Johnson Division, St. Paul, Minnesota.

Ecology, EPA, and DOE, 1989, *Hanford Federal Facility Agreement and Consent Order*, 2 vols., as amended, Washington State Department of Ecology, U.S. Environmental Protection Agency, and U.S. Department of Energy, Olympia, Washington. Available at: http://www.hanford.gov/?page=81.

Ecology Publication 97-407, 2009, *Chemical Testing Methods for Designating Dangerous Waste: WAC 173-303-090 & -100*, Appendix 5, "Appendix IX of 40 CFR 264," Washington State Department of Ecology, Olympia, Washington. Available at: http://www.ecy.wa.gov/pubs/97407.pdf.

EPA 530/R-09-007, 2009, *Statistical Analysis of Groundwater Monitoring Data at RCRA Facilities Unified Guidance*, Office of Resource Conservation and Recovery, U.S. Environmental Protection Agency, Washington, D.C. Available at: http://www.epa.gov/osw/hazard/correctiveaction/resources/guidance/sitechar/gwstats/unified-guid.pdf.

EPA-600/4-79-020, 1983, *Methods for Chemical Analysis of Water and Wastes*, Environmental Monitoring and Support Laboratory, U.S. Environmental Protection Agency, Cincinnati, Ohio. Available at: http://pdw.hanford.gov/arpir/index.cfm/viewDoc?accession=D196019611.

Hanford Environmental Information System, Hanford Site database, Richland, Washington.

HNF-23142, 2004, *Engineering Study for the 200 Area Effluent Treatment Facility Secondary Waste Treatment of Projected Future Waste Feeds*, Rev. 0, Fluor Hanford, Inc., Richland, Washington.

Hvorslev, M. Juul, 1951, *Time Lag and Soil Permeability in Ground-Water Observations*, Bulletin No. 36, Waterways Experiment Station, U.S. Army Corps of Engineers, Vicksburg, Mississippi. Available at: http://www.csus.edu/indiv/h/hornert/Geol_210_Summer_2012/Week%203%20readings/Hvorslev%201951.pdf.

KGS, 1991, *Seismic-Reflection Processing Demonstration Using Eavesdropper*, Open-File Report 91-27, Kansas Geological Survey, Lawrence, Kansas. Available at: http://www.kgs.ku.edu/Geophysics/Reports2/EavesTutorial.pdf.

Moench, Allen F., 1997, "Flow to a well of finite diameter in a homogeneous, anisotropic water table aquifer," *Water Resources Research* 33(6):1397-1407. Available at: http://onlinelibrary.wiley.com/doi/10.1029/97WR00651/pdf.

Neuman, Shlomo P., 1974, "Effect of Partial Penetration on Flow in Unconfined Aquifers Considering Delayed Gravity Response," *Water Resources Research* 10(2):303-312. Available at: http://onlinelibrary.wiley.com/doi/10.1029/WR010i002p00303/abstract.

PNNL-12261, 2000, *Revised Hydrogeology for the Suprabasalt Aquifer System, 200-East Area and Vicinity, Hanford Site, Washington*, Pacific Northwest National Laboratory, Richland, Washington. Available at: http://pdw.hanford.gov/arpir/index.cfm/viewDoc?accession=0906180659.

PNNL-14804, 2004, *Results of Detailed Hydrologic Characterization Tests Fiscal Year 2003*, Pacific Northwest National Laboratory, Richland, Washington. Available at: http://www.pnl.gov/main/publications/external/technical_reports/PNNL-14804.pdf.

PNNL-19702, 2010, *Hydrogeologic Model for the Gable Gap Area, Hanford Site*, Pacific Northwest National Laboratory, Richland, Washington. Available at: www.pnl.gov/main/publications/external/technical_reports/PNNL-19702.pdf.

RCW 70.105, "Hazardous Waste Management," *Revised Code of Washington*, Olympia, Washington. Available at: http://apps.leg.wa.gov/RCW/default.aspx?cite=70.105.

Resource Conservation and Recovery Act of 1976, 42 USC 6901, et seq. Available at: http://www.epa.gov/epawaste/inforesources/online/index.htm.

SGW-39344, 2008, *Borehole Summary Report for the Installation of RCRA Wells 299-E26-77 (C6455), 299-E26-79 (C6826), 299-E25-236 (C6542) and 199-N-165 (C6693), FY 2008*, Rev. 0, CH2M HILL Plateau Remediation Company, Richland, Washington. Available at: http://pdw.hanford.gov/arpir/index.cfm/viewDoc?accession=0906180658.

SGW-41072, 2009, *Liquid Effluent Retention Facility Characterization Report*, Rev. 0, CH2M HILL Plateau Remediation Company, Richland, Washington. Available at: http://pdw.hanford.gov/arpir/index.cfm/viewDoc?accession=0906160165.

SGW-43746, 2009, *Landstreamer/Gimbaled GeoPhone Acquisition of High Resolution Seismic Reflection Data North of the 200 Areas – Hanford Site*, Rev. 0, CH2M HILL Plateau Remediation Company, Richland, Washington. Available at: http://www.osti.gov/scitech/servlets/purl/968021.

SGW-51467, 2012, *Borehole Summary Report for the Installation of Two RCRA Groundwater Monitoring Wells in the 200 Areas, FY2011*, Rev. 0, CH2M HILL Plateau Remediation Company, Richland, Washington.

SGW-52162, 2012, *Seismic Reflection Investigation at the Liquid Effluent Retention Facility, 200 East Area, Hanford Site Richland, Washington*, Rev. 0, CH2M HILL Plateau Remediation Company, Richland, Washington.

SGW-52467, 2012, *Integrated Surface Geophysical Investigation Results at Liquid Effluent Retention Facility, 200 East Area, Hanford, Washington*, Rev. 0, CH2M HILL Plateau Remediation Company, Richland, Washington.

SGW-54508, *WMA C September 2012 Quarterly Groundwater Monitoring Report*, Rev. 0, CH2M HILL Plateau Remediation Company, Richland, Washington. Available at: http://pdw.hanford.gov/arpir/index.cfm/viewDoc?accession=0089297.

SW-846, 2007, *Test Methods for Evaluating Solid Waste: Physical/Chemical Methods, Third Edition; Final Update IV-B*, as amended, Office of Solid Waste and Emergency Response, U.S. Environmental Protection Agency, Washington, D.C. Available at: http://www.epa.gov/epawaste/hazard/testmethods/sw846/online/index.htm.

Theis, C. V., 1935, "The Relation Between the Lowering of the Piezometric Surface and the Rate and Duration of Discharge of a Well Using Ground-Water Storage," *American Geophysical Union Transactions* 16(2):519-524.

WAC 173-303-090, "Dangerous Waste Regulations," "Dangerous Waste Characteristics," *Washington Administrative Code*, Olympia, Washington. Available at: http://apps.leg.wa.gov/WAC/default.aspx?cite=173-303-090.

WAC 173-303-645, "Dangerous Waste Regulations," "Releases from Regulated Units," *Washington Administrative Code*, Olympia, Washington. Available at: http://apps.leg.wa.gov/WAC/default.aspx?cite=173-303-645.

Water-Resources Investigation Report 91-4034, 1991, *U.S. Geological Survey Toxic Substances Hydrology Program—Proceedings of the technical meeting, Monterey, California, March 11-15, 1991*, U.S. Geological Survey, Reston, Virginia. Available at: http://pubs.usgs.gov/wri/1991/4034/report.pdf.

WHC-MR-0235, 1990, *Borehole Completion Data Package for the Liquid Effluent Retention Facility*, Westinghouse Hanford Company, Richland, Washington.

WHC-SD-EN-EV-024, 1994, *Site Characterization Report for the Liquid Effluent Retention Facility*, Rev. 0, Westinghouse Hanford Company, Richland, Washington. Available at: http://pdw.hanford.gov/arpir/index.cfm/viewDoc?accession=D196090383.

WHC-SD-W105-SAR-001, 1991, *Final Safety Analysis Report 242-A Evaporator Liquid Effluent Retention Facility*, Rev. 0, Westinghouse Hanford Company, Richland, Washington.

Table D-1. Constituent Results for 242-A Process Condensate Characterization Results and Basin 42 Characterization Results

Sample Location	2009 Basin 42 Characterization Results					Basin 42 Average[b]		Process Condensate to LERF[c]		Process Condensate Cold Run[c]	Process Condensate to LERF[c]		Process Condensate to LERF[c]		Regional Groundwater Background Concentration[d]		Constituents
Sample Date	Units	Wtd Avg[a]		Max[b]		2009 - 2010		6/16/2009			8/31/2010		10/5/2010			Units	
Volume	gal	6.76E+06		6.87E+06		5.19E+06		5.36E+05		3.14E+05	3.61E+05		3.61E+05				
Nitrogen in ammonium	mg/L	111.41		140.00		127		140		0.02	26.3		22.1		NL	mg/L	Nitrogen in ammonium
Bromide	mg/L	0.07	U	0.09	U	0.08	U	0.05	U	0.05	0.05	U	0.05	U	0.151	mg/L	Bromide
Chloride	mg/L	5.37		7.75		6.91		0.04	U	1.30	0.04	U	0.04	U	19.58	mg/L	Chloride
Fluoride	mg/L	0.05		0.06		0.05	U	0.03	U	0.03	0.06		0.04	U	1.298	mg/L	Fluoride
Nitrogen in Nitrate	mg/L	0.08		0.10		0.09		0.03	U	0.12	0.0097	U	0.01		9.42	mg/L	Nitrogen in Nitrate
Nitrogen in Nitrite	mg/L	0.03	U	0.04	U	0.03	U	0.02	U	0.02	0.02	U	0.02	U	0.045	mg/L	Nitrogen in Nitrite
Phosphorus in phosphate	mg/L	0.19		0.27		0.22		0.07	U	0.07	0.07	U	0.07	U	0.072	mg/L	Phosphorus in phosphate
Sulfate	mg/L	55.36		80.20		71.50		0.07	U	10.10	0.08	U	0.08	U	54.95	mg/L	Sulfate
Aluminum	µg/L	17.78	U	34.00	U	17.00	U	34	U	0.15	19	U	19	U	170	µg/L	Aluminum
Antimony	µg/L	0.29	U	0.30	U	0.30	U	0.3	U		0.3	U	0.3	U	69.8	µg/L	Antimony
Arsenic	µg/L	3.59		5.20		4.55		0.4	U	0.60	0.4	U	0.4	U	11.8	µg/L	Arsenic
Barium	µg/L	9.43		12.30		10.90		8	U		4	U	4	U	149	µg/L	Barium
Beryllium	µg/L	0.05	U	0.05	U	0.05	U	0.05	U		0.05	U	0.05	U	3.38	µg/L	Beryllium
Cadmium	µg/L	0.10	U	0.10	U	0.10	U	0.1	U	0.10	0.1	U	0.1	U	1.29	µg/L	Cadmium
Calcium	µg/L	10,691.93		14,400.00		12830.00		78	U	18000.00	27	U	27	U	58389	µg/L	Calcium
Chromium	µg/L	5.52		7.90		7.06		0.5	U	0.09	0.5	U	0.5	U	3.17	µg/L	Chromium
Cobalt	µg/L	4.13	U	8.00	U	4.00	U	8	U		4	U	4	U	1.29	µg/L	Cobalt
Copper	µg/L	4.60		6.96		5.30		4.52		0.74	0.469		2.04		1.04	µg/L	Copper
Cyanide	µg/L	3.81	U	4.00	U	4.00	U	4	U		4	U	4	U	9.52	µg/L	Cyanide
Iron	µg/L	51.87		58.10		49.50		36	U	150.00	38	U	38	U	1104	µg/L	Iron
Lead	µg/L	1.33		9.01		0.30		9.01		1.60	3.52		2.32		1.3	µg/L	Lead

Table D-1. Constituent Results for 242-A Process Condensate Characterization Results and Basin 42 Characterization Results

Sample Location	2009 Basin 42 Characterization Results					Basin 42 Average[b]		Process Condensate to LERF[c]		Process Condensate Cold Run[c]	Process Condensate to LERF[c]		Process Condensate to LERF[c]		Regional Groundwater Background Concentration[d]		Constituents
Sample Date	Units	Wtd Avg[a]		Max[b]		2009 - 2010		6/16/2009			8/31/2010		10/5/2010			Units	
Magnesium	µg/L	2,533.13		3,380.0		2986.67		32	U	5100.0	14	U	14	U	31051	µg/L	Magnesium
Manganese	µg/L	5.69		8.00		5.27		8	U	8.00	6	U	6	U	86.4	µg/L	Manganese
Mercury	µg/L	0.09		0.12		0.11		0.1		0.05	0.05	U	0.05	U	0.006	µg/L	Mercury
Nickel	µg/L	7.53		10.60		8.40		8	U	0.55	4	U	4	U	1.98	µg/L	Nickel
Potassium	µg/L	1,498.00		2,060.0		1826.67		586		900.00	73	U	73	U	11089	µg/L	Potassium
Selenium	µg/L	0.60		0.87		0.69		0.3	U	0.32	0.3	U	0.3	U	20.7	µg/L	Selenium
Silicon	µg/L	3,453.02		4,610.0		4120.00		275		5300.0	43		388		43904	µg/L	Silicon
Silver	µg/L	5.38	U	10.00	U	5.00	U	10	U		7	U	7	U	5.98	µg/L	Silver
Sodium	µg/L	18,276.24		26,700.00		23633.33		260		2500.0	11	U	11	U	32919	µg/L	Sodium
Thallium	µg/L	43.83		148.00		35.00	U	148		0.02	49	U	49	U	1.87	µg/L	Thallium
Titanium	µg/L	4.13	U	8.00	U	4.00	U	8	U		4	U	4	U	30	µg/L	Titanium
Uranium	µg/L	8.54		13.40		11.12		0.05	U		0.05	U	0.05	U	14.4	µg/L	Uranium
Vanadium	µg/L	2.93	U	24.00	U	12.00	U	24	U		17	U	17	U	19.3	µg/L	Vanadium
Zinc	µg/L	12.93		17.60		14.97		12	U	1.40	4	U	4	U	48.9	µg/L	Zinc
Specific Conductance	µS/cm	430.52		583.00		533.00		113		168.00	45.1		42.6		TBD	µS/cm	Specific Conductance
pH Measurement	unitless	9.65		10.40		9.65		10.4		8.20	9.87		9.54		TBD	unitless	pH Measurement
Alkalinity	mg/L	490.00		500.00		473.33		ND		71.20	ND		ND		156367	mg/L	Alkalinity
Total dissolved solids	mg/L	113.17		162.00		136.33		10	U	100.00	31		27		277190	mg/L	Total dissolved solids
Total suspended solids	mg/L	2.49		10.00		2.20		2	U		10	U	2	U		mg/L	Total suspended solids
Total organic carbon	mg/L	7.10		9.59		7.69		9.59			3.78		4.39		TBD	mg/L	Total organic carbon
1-Butanol	µg/L	287.66		1,700.0		163.33		680			1700		330		0	µg/L	1-Butanol
2-Butanone	µg/L	6.17		10.00		6.83		8.0			4.4		1	U	0	µg/L	2-Butanone
2-Pentanone	µg/L	3.34		5.70		3.70		1	U		5.7		2.1		0	µg/L	2-Pentanone
Acetone	µg/L	220.09		1,700.0		83.33		1700			260		140		0	µg/L	Acetone

Table D-1. Constituent Results for 242-A Process Condensate Characterization Results and Basin 42 Characterization Results

Sample Location	2009 Basin 42 Characterization Results					Basin 42 Average[b]		Process Condensate to LERF[c]		Process Condensate Cold Run[c]	Process Condensate to LERF[c]		Process Condensate to LERF[c]		Regional Groundwater Background Concentration[d]		Constituents
Sample Date	Units	Wtd Avg[a]		Max[b]		2009 - 2010		6/16/2009			8/31/2010		10/5/2010			Units	
Benzene	µg/L	0.95	U	1.0	U	1.0	U	1.0	U		1.0	U	1.0	U	0	µg/L	Benzene
Carbon tetrachloride	µg/L	0.95	U	1.0	U	1.0	U	1.0	U		1.0	U	1.0	U	0	µg/L	Carbon tetrachloride
Chloroform	µg/L	0.95	U	1.0	U	1.0	U	1.0	U		1.0	U	1.0	U	0	µg/L	Chloroform
Methylene chloride	µg/L	1.16		1.60		1.27		1	U		1	U	1	U	0	µg/L	Methylene chloride
Tetrahydrofuran	µg/L	36.89		84.00		30.33		74			84		61		0	µg/L	Tetrahydrofuran
2-Butoxyethanol	µg/L	50.95		330.00		18.33		220			34		330		0	µg/L	2-Butoxyethanol
2-Methylphenol (cresol, o-)	µg/L	1.26		4.30		1.00		0.8	U		3.8		4.3		0	µg/L	2-Methylphenol (cresol, o-)
Benzyl alcohol	µg/L	3.06		23.00		0.70		23			6.7		6.4		0	µg/L	Benzyl alcohol
n-Nitrosodimethylamine	µg/L	176.61		290.00		190.00		290			67		79		0	µg/L	n-Nitrosodimethylamine
Total cresols	µg/L	0.95		4.30		0.60	U	0.5	U		4.3		4.2		0	µg/L	Total cresols
Tributyl phosphate	µg/L	47.73		72.00		62.00		0.5	U		1	U	1	U	0	µg/L	Tributyl phosphate
Formate	pCi/L	0.00	U	0.01		ND	U	0.00467	U		0.00467	U	0.00629		0	pCi/L	Formate
Gross alpha	pCi/L	136.24		190.00		176.67		4.7		0.60	2.3	U	2.3	U	0	pCi/L	Gross alpha
Gross beta	pCi/L	23,218.16		34,000.00		30000.00		930		2.80	2100		140		4.15	pCi/L	Gross beta

Note: Spreadsheet data were provided by Effluent Treatment Facility personnel.

a. Weighted average for Basin 42 based on samples collected in Risers 2, 4, and 7 from June 2009, August 2010, and October 2010, respectively.

b. Maximum and average results are derived from the three sample dates June 2009, August 2010, and October 2010.

c. Process condensate results are associated with characterization results collected at 242-A Evaporator.

d. Results based on Hanford Site Background: Part 3, Groundwater Background (DOE/RL-96-61).

gal = gallons
mg/L = milligrams per liter
µg/L = micrograms per liter
µS/cm = microSiemens per centimeter
u = less than detection
Wtd Ave = weighted average

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Table D-2. Basin 43 Constituent Characterization Results for Past 200-UP-1 and 200-ZP-1 Groundwater Waste Streams and Recent Waste Characterization Results For Basin 43

Constituent	Units	200-UP-1 and 200-ZP-1 Groundwater[a]		Basin 43 Characterization Results[c]		ERDF Leachate[d]		200-BP-5 Perched Water[e]	
Volume	gal	7.03E+7[b]		9.13E+05					
Added Vol.	gal			1.26E+06		5.62E+05		2.36E+04	
Ammonium (N)	mg/L	0.064		0.1		0.1			
Bromide	mg/L	0.4		1.2		1.5		1.2	
Chloride	mg/L	22.1		176.9		224.0		83.7	
Fluoride	mg/L	2.7		1.2		0.2		22	
Nitrate (N)	mg/L	101		63.8		64.6		219.7	
Nitrite (N)	mg/L	0.036	U	3.4	U	7.6	U	0.2	U
Phosphate (P)	mg/L	0.12	U	0.2	U	0.3	U	0.3	
Sulfate	mg/L	57.2		404.4		597.0		556.4	
Aluminum	µg/L	44		17.5		19.7	U	125	
Antimony	µg/L	0.3	U	3.3	U	6.0	U	31	U
Arsenic	µg/L	5.5		6.9		7.7		7.5	
Barium	µg/L	71.1		96.7		129.1		62.1	
Beryllium	µg/L	0.05	U	0.8	U	1.3	U	3	U
Cadmium	µg/L	0.1	U	0.5		0.3		4.4	
Calcium	µg/L	56861.5		181161.2		248000.0		167000	
Chromium	µg/L	121.1		36.1		29.2		143.9	
Cobalt	µg/L	4	U	67.7		145.0		9	
Copper	µg/L	0.15		121.1		145.0		21.7	
Hexavalent Chromium	µg/L	113							
Iron	µg/L	18	U	21.2		14.3		130.4	
Lead	µg/L	0.1	U	5.1		10.9			
Magnesium	µg/L	18361.5		44035.4		53750.0		71300	
Manganese	µg/L	4	U	7.1		6.9		129.7	
Mercury	µg/L	0.05	U	0.1		0.2	U		
Nickel	µg/L	4	U	6.7		6.3		19.9	U
Potassium	µg/L	5536.2		13579.6		17138.0		10100	
Selenium	µg/L	4.8		5.7		8.0			
Silicon	µg/L	21300		17465.4		21750.0			
Silver	µg/L	5	U	5.5		5.0		33	U
Sodium	µg/L	161846.2		187496.6		191250.0		391000	
Thallium	µg/L	36	U	27.7		5.0	U		
Titanium	µg/L	4	U	4.4		4.0			
Uranium	µg/L	25.6		2249.2		1100.6		43500	
Vanadium	µg/L	38.5		32.9		45.1		17.4	
Zinc	µg/L	37.5		25.5		27.2		92.2	
Specific Conductance	µS/cm	1206.2		2041.7		2483.8		2592	
pH Measurement	unitless	7.95		6.9		7.8		7.7	
Alkalinity	mg/L	151		224.9		296.0		231	
Total dissolved solids	mg/L	906.8		1351.1		1688.8			
Total suspended solids	mg/L	1.62		9.7		19.4			
Total organic carbon	mg/L	0.64		6.0		6.4			
Carbon tetrachloride	µg/L	490.7		12.1		5.0	U	1.0	U
Chloroform	µg/L	8.5		0.6	U			1.0	U
Tetrahydrofuran	µg/L	2	U	1.1	U			1.0	U
Gross alpha	pCi/L	29.4		1510.2		587.6		38800	
Gross beta	pCi/L	2830.8		8065.1		394.8		34600	

a. 200-UP-1 and 200-ZP-1 Average Groundwater Characterization Results Based on up to 14 samples collected between 2008 and 2011.

b. Total volume of 200-UP-1 and 200-ZP-1 groundwater received between 2008 and 2011.

c. Characterization results of the Basin 43 after removal of the 200-UP-1 and 200-ZP-1 groundwater from the basin.

d. Represents the dominant waste stream for Basin 43 since 200-UP-1 and 200-ZP-1 groundwater transfers have ceased (2012 total gallons to Basin 43 were 2,770,000).

e. Represents the third most dominant waste stream received at Basin 43 (2012 total gallons to Basin 43 were 130,000).

Notes: The second and fourth dominant waste streams were MODU-Tanks and Hanford purge water (2012 total gallons to Basin 43 were 291,500), which are the same streams and contain significantly less contaminant concentrations than the ERDF leachate and 200-BP-5 perched water. Spreadsheet data were provided by Effluent Treatment Facility Personnel.

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Table D-3. ERDF Leachate

CAS #	Constituent	Average Concentration between February 2000 and September 2012	Units	Regional Groundwater Background Concentration*	Units
7429-90-5	Aluminum	31	µg/L	170	µg/L
7440-36-0	Antimony	1	µg/L	69.8	µg/L
7440-38-2	Arsenic	9	µg/L	11.8	µg/L
7440-39-3	Barium	97	µg/L	149	µg/L
	Beryllium	0	µg/L	3.38	µg/L
7440-70-2	Calcium	213,735	µg/L	58,389	µg/L
7440-47-3	Chromium	27	µg/L	3.17	µg/L
7440-50-8	Copper	20	µg/L	1.04	µg/L
7439-89-6	Iron	35	µg/L	1,104	µg/L
7439-92-1	Lead	2.8	µg/L	1.3	µg/L
7439-95-4	Magnesium	69,580	µg/L	31,051	µg/L
7440-02-0	Nickel	13	µg/L	1.98	µg/L
7440-09-7	Potassium	20,573	µg/L	11,089	µg/L
7782-49-2	Selenium	5	µg/L	20.7	µg/L
7440-21-3	Silicon	20,063	µg/L	43,904	µg/L
7440-23-5	Sodium	254,237	µg/L	32,919	µg/L
7440-31-5	Tin	1	µg/L	23.6	µg/L
	Thallium	0	µg/L	1.87	µg/L
7440-62-2	Vanadium	26	µg/L	19.3	µg/L
7440-66-6	Zinc	14	µg/L	48.9	µg/L
56-23-5	Carbon Tetrachloride	0	µg/L	0	µg/L
67-56-1	Trichloroethene	0	µg/L	0	µg/L
75-69-4	Methyl Alcohol	0	µg/L	0	µg/L
	Trichlorofluoromethane	3.2	µg/L	0	µg/L
pH	pH	7 to 8	unitless	TBD	unitless
CONDUCT	Specific Conductance	2509	µmS/cm	TBD	µmS/cm
24959-67-9	Bromide	1242	µg/L	151	µg/L

Table D-3. ERDF Leachate

CAS #	Constituent	Average Concentration between February 2000 and September 2012	Units	Regional Groundwater Background Concentration*	Units
16887-00-6	Chloride	249,638	µg/L	19,580	µg/L
16984-48-8	Fluoride	521	µg/L	1,298	µg/L
14797-55-8	Nitrate	327,241	µg/L	41,723	µg/L
14797-65-0	Nitrite	500U	µg/L	130	µg/L
14808-79-8	Sulfate	473,776	µg/L	54,950	µg/L
TOC	Total Organic Carbon	13,148	µg/L	TBD	µg/L
OIL/GREASE	Oil and Grease	3,213	µg/L	0	µg/L
TDS	Total Dissolved Solids	1,926,897	µg/L	TBD	µg/L
TSS	Total Suspended Solids	15,686	µg/L	TBD	µg/L
12587-46-1	Alkalinity	264,813	µg/L	156,367	µg/L
12587-47-2	Gross alpha	965	pCi/L	0	pCi/L
14762-75-5	Gross beta	643	pCi/L	4.15	pCi/L

* Results are based on *Hanford Site Background: Part 3, Groundwater Background* (DOE/RL-96-61).

Note: Spreadsheet data were provided by Effluent Treatment Facility personnel.

TBD = to be determined

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Table D-4. LERF Well Attributes

Wells	299-E26-9	299-E26-10	299-E26-11	299-E26-14	299-E26-77	299-E26-79	299-E35-2
Date Drilled	August, 1990	August, 1990	August, 1990	September, 2011	October, 2008	September, 2008	August, 1990
Top of Casing Elevation (m/ft)	184.854/606.48	184.418/605.05	183.88/603.281	183.224/601.129	184.782/606.24	183.115/600.771	184.611/605.679
Ground Surface Elevation (m/ft)	183.941/603.48	183.512/602.07	182.979/600.325	182.494/598.734	184.011/603.371	182.356/598.281	183.712/602.73
Total Depth Drilled (m/ft)	61.722/202.5	62.972/206.6	61.417/201.5	73.334/240.6	70.957/232.8	68.507/224.76	61.661/202.3
Elevation of Total Depth Drilled (m/ft)	122.219/400.98	120.54/395.47	121.562/398.825	109.16/358.134	113.054/370.571	113.849/373.521	122.051/400.43
Depth to Top of Basalt (m/ft)	61.271/201.02	62.271/204.3	60.35/198	67.361/221	62.636/205.5	63.094/207	60.991/200.1
Top of Basalt Elevation (m/ft)	122.67/402.46	121.241/397.77	122.629/402.325	115.133/377.734	121.375/397.871	119.262/391.281	122.722/402.63
Bottom of Sump Elevation (m/ft)	None	None	None	115.773/379.834	114.334/374.771	114.325/375.081	None
Fill Below Bottom of Sump/Screen	20-40 Sand	20-40 Sand	20-40 Sand	Bentonite Pellets/ 10-20 Sand[d]	10-20 Sand	10-20 Sand	20-40 Sand
Bottom of Screen Elevation (m/ft)	122.707/402.58	120.693/395.97	120.251/394.525	116.688/382.834	115.248/377.771	115.239/378.081	122.295/401.23
Top of Screen Elevation (m/ft)	125.937/413.18	125.448/411.57	121.958/400.125	122.784/402.834	122.792/402.521	122.859/403.081	125.526/411.83
Sand Pack	20-40 Sand	20-40 Sand	20-40 Sand	10-20 Sand	10-20 Sand	10-20 Sand	20-40 Sand
Water Table Elevation After Drilling (m/ft)	124.444/408.28[a]	124.594/408.773[b]	125.097/410.425[c]	121.922/400[c]	121.987/400.22[f]	121.976/400.184[g]	124.611/408.83[h]
Water Table Elevation 4/10/2013 (m/ft)	Dry	121.81/399.639	122.602/402.238	121.878/399.862	121.825/399.688	121.808/399.633	Dry
Water Height Across Screen 4/10/2013 (m/ft)	Dry	1.117/3.67	2.351/7.713	5.19/17.028	6.577/21.917	6.569/24.552	Dry
Water above Basalt 4/10/2013 (m/ft)	Dry	0.569/1.87	None	6.745/22.128	0.45/1.817	2.546/8.352	Dry
Ringold Present	No	No	Yes	Yes	No	Yes	Yes[i]
Depth to Top of Ringold (m/ft)	NA	NA	57.607/189	65.532/215	NA	62.789/206	Not Provided
Top of Ringold Elevation (m/ft)	NA	NA	125.372/411.325	116.962/383.734	NA	119.567/392.281	Uncertain
Thickness of Ringold Across Screen 4/10/2013 (m/ft)	NA	NA	NA	0.274/0.9	NA	0.3048/1	Uncertain
Best Estimate of Hydraulic Conductivity (m/day)	6 to 120	36.2 to 42.8	6.3	27.3	134		39.7

a. Date 8/1/1990

b. Date 9/4/1990

c. Date 8/13/1990

d. Bentonite pellets to 0.7 ft below bottom of sump

e. Date 12/29/2011

f. Date 11/26/2008

g. Date 11/26/2008

h. Date 8/2/1990

i. Basis: WHC-SD-EN-EV-024, *Site Characterization Report for the Liquid Effluent Retention Facility*

NA = not applicable

Table D-5. LERF Groundwater Gradient and Magnitude Calculations

Well Name	December, 2011	February, 2012	April, 2012	August, 2012	September, 2012	November, 2012	December, 2012	January, 2013
	New RP, Gyro Correction & BP Adjustment	New RP, Gyro Correction & BP Adjustment	New RP, Gyro Correction & BP Adjustment	New RP, Gyro Correction & BP Adjustment	New RP, Gyro Correction & BP Adjustment	New RP, Gyro Correction & BP Adjustment	New RP, Gyro Correction & BP Adjustment	New RP, Gyro Correction & BP Adjustment
299-E26-10	121.838	121.834	121.816	121.808	121.809	121.828	121.830	121.825
299-E26-14	121.911	121.895	121.890	121.878	121.878	121.898	121.899	121.891
299-E26-77	121.844	121.843	121.824	121.820	121.822	121.839	121.841	121.835
299-E26-79	121.847	121.835	121.818	121.813	121.816	121.830	121.837	121.839
Range (m):	0.073	0.061	0.074	0.070	0.069	0.070	0.069	0.066
Gradient Magnitude (m/m):	2.75E-04	2.49E-04	2.98E-04	2.77E-04	2.69E-04	2.85E-04	2.67E-04	2.39E-04
Direction (azimuth):	195	186	188	188	189	186	190	198
R^2:	0.92	0.93	0.91	0.94	0.95	0.93	0.94	0.96
p-Value:	0.2858	0.2670	0.2928	0.2348	0.2132	0.2569	0.2355	0.2071
Statistically Significant?:	No	No	No	No	No	No	No	No

RP = Reference Point

BP = Barometric Pressure

In order to minimize the measurement error associated with the monitoring network, steps were taken that included deviation correction from vertical of the well bore, barometric corrections completed within each well and correlated with barometric conditions measured at weather station in 200 Areas; precision geodetic surveys to a common benchmark; dedicated e-tape measurements for depth-to-groundwater from the dedicated point associated with the precision survey at each well; dedicated person for measuring water levels all groundwater level measurements completed on same day per month. Further discussion on these corrections is provided in *Calculations in Support of the Low Hydraulic Gradient Evaluation Study for the 200 East Area Unconfined Aquifer* (ECF-200EAST-12-0086).

After these corrections have been applied, a least square regression of the plane to elevations associated with the well locations is completed to derive the gradient. The gradient, azimuth, R2 ratio, the p-value are provided in the lower part of the table. Briefly, the R2 ratio is the goodness of fit coefficient. Basically, it is the ratio of the sum of squares because of the regression to the total sum of squares. If the ratio of the two sums is close to 1, indicating unity, then the fit is considered good. Likewise, the p-value is the probability that the degree of an apparent spatially dependent trend observed in the data (or a trend of even greater degree) would occur solely by random chance. Thus, if the p-value is less than 0.05, the fitted trend surface is deemed statistically significant. Again, further details of this process calculation are provided in ECF-200EAST-12-0086.

Table D-6. Geochemical Constituent Evaluation for Groundwater

Constituent	Sample Frequency	Comment
• Major anions • Major cations • Alkalinity	Semiannually	Aid geochemical evaluation

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Table D-7. Dangerous Constituents and Indicators to be Analyzed as Indicators of Groundwater Contamination at the LERF Basins

Constituent	Sample Frequency	Comment
• Carbon tetrachloride • pH • Specific conductance • Total organic carbon • Total organic halides	Samples collected semiannually for detection monitoring.	Subject to statistical evaluation, based on the standard sampling plan outlined in WAC 173-303-645(8)(g)(i) and WAC 173-303-645(8)(h)(i). Analyze hexavalent chromium as potential indicator parameter.

Source: WAC 173-303-645, "Dangerous Waste Regulations," "Releases from Regulated Units."

Notes: Wells 299-E26-14 and 299-E26-79 are sampled as upgradient and downgradient compliance wells semiannually.

Well 299-E26-77 will be sampled semiannually for geochemical and field parameters, but not used for background calculations.

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Table D-8. Quality Control Samples

Sample Type	Primary Characteristics Evaluated	Frequency
Field Quality Control		
Full trip blank	Contamination from containers or transportation	One per 20 well trips
Field transfer blank	Airborne contamination from the sampling site	One each day volatile organic compound samples are collected
Equipment blank	Contamination from non-dedicated sampling equipment	As needed[a]
Duplicate samples	Reproducibility	One per 20 well trips
Laboratory Quality Control		
Method blank	Laboratory contamination	One per batch
Laboratory duplicates	Laboratory reproducibility	[b]
Matrix spike	Matrix effects and laboratory accuracy	[b]
Matrix spike duplicate	Laboratory reproducibility and	[b]

Table D-8. Quality Control Samples

Sample Type	Primary Characteristics Evaluated	Frequency
	accuracy	
Surrogates	Recovery/yield	[b]
Laboratory control sample	Method accuracy	One per batch

a. For portable Grundfos pumps, equipment blanks are collected 1 per 10 well trips. Whenever a new type of non-dedicated equipment is used, an equipment blank is collected every time sampling occurs until it can be shown that less frequent collection of equipment blanks is adequate to monitor the decontamination procedure for the non-dedicated equipment.

b. As defined in the laboratory contract or quality assurance plan and/or analysis procedures.

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Table D-9. Preservation Techniques, Analytical Methods Used, and Current Required Quantitation Limits for Chemical Constituents

Constituent	Collection & Preservation[a,b]	Analysis Methods[c]	Method Quantitation Limit (µg/L)[d]
Metals			
Calcium	P, HNO_3 to pH<2	SW-846[e] Method 6010,or EPA/600 Method 200.8	1,000
Magnesium			750
Potassium			4,000
Sodium			500
Anions by Ion Chromatography			
Nitrate	P, none	EPA/600 Method 300.0[f]	250
Sulfate			500
Chloride			200
Nitrite			250
Volatile Organics			
Carbon Tetrachloride	G, no headspace	SW-846 8260	2
Total Organic Halides			
Total Organic Carbon	G, no headspace	SW-846 9060A	140
Total Organic Halides			
Total Organic Halides	G, no headspace	SW-846 9020B	20
Hexavalent Chromium			
Hexavalent Chromium	P, none	EPA/7196A	10
Alkalinity			

Table D-9. Preservation Techniques, Analytical Methods Used, and Current Required Quantitation Limits for Chemical Constituents

Constituent	Collection & Preservation[a,b]	Analysis Methods[c]	Method Quantitation Limit (µg/L)[d]
Alkalinity	G/P, none	EPA Standard Method[g] 2320 EPA/600 Method 310.1 EPA/600 Method 310.2	5,000

a. P = plastic; G = glass.

b. All samples will be cooled to 4°C upon collection.

c. Constituents grouped together are analyzed by the same method, unless otherwise indicated.

d. Detection limit units, except where indicated.

e. SW-846, *Methods for Evaluation of Solid Waste: Physical/Chemical Methods, Third Edition; Final Update IV-B.*

f. Analytical method adapted from Method 300.0, *Test Methods for Determination of Inorganic Anions in Water by Ion Chromatography* (EPA-600/4-84-017).

g. *Standard Methods for the Examination of Water and Wastewater* (APHA /AWWA/WEF, 2012).

EPA = U.S. Environmental Protection Agency

N/A = not applicable

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Table D-10. Field and Laboratory Quality Control Elements and Acceptance Criteria

Constituent[a]	QC Element	Acceptance Criteria	Corrective Action
General Chemical Parameters			
Alkalinity	MB	<MDL	Flagged with "C"
	LCS	80-120% recovery[c]	Data reviewed[d]
	DUP	<20% RPD[c]	Data reviewed[d]
	MS[e]	75-125% recovery[c]	Flagged with "N"
	EB, FTB	<2 times MDL	Flagged with "Q"
	Field duplicate	<20% RPD[f]	Flagged with "Q"
Anions			
Anions by IC	MB	<MDL	Flagged with "C"
	LCS	80-120% recovery[c]	Data reviewed[d]
	DUP	<20% RPD[c]	Data reviewed[d]
	MS	75-125% recovery[c]	Flagged with "N"
	EB, FTB	<2 times MDL	Flagged with "Q"
	Field duplicate	<20% RPD[f]	Flagged with "Q"

Table D-10. Field and Laboratory Quality Control Elements and Acceptance Criteria

Constituent[a]	QC Element	Acceptance Criteria	Corrective Action
Metals			
ICP metals	MB	<CRDL	Flagged with "C"
	LCS	80-120% recovery[c]	Data reviewed[d]
	MS	75-125% recovery[c]	Flagged with "N"
	MSD	<20% RPD[c]	Data reviewed[d]
	EB, FTB	<2 times MDL	Flagged with "Q"
	Field duplicate	<20% RPD[f]	Flagged with "Q"
Volatile Organic Compounds			
Carbon tetrachloride	MB	<MDL	Flagged with "B"
	LCS	Statistically derived[g]	Data reviewed
	MS	Statistically derived[g]	Flagged with "N"
	MSD	Statistically derived[g]	Data reviewed[d]
	SUR	Statistically derived[g]	Data reviewed[d]
	EB, FTB, FXR	<2 times MDL	Flagged with "Q"
	Field duplicate	<20% RPD[f]	Flagged with "Q"
Hexavalent Chromium			
Hexavalent Chromium	MB	<2 times MDL	Flagged with "B"
	LCS	Statistically derived[g]	Data reviewed
	MS	Statistically derived[g]	Flagged with "N"
	MSD	Statistically derived[g]	Data reviewed[d]
	SUR	Statistically derived[g]	Data reviewed[d]
	EB, FTB	<2 times MDL	Flagged with "Q"
	Field duplicate	<20% RPD[f]	Flagged with "Q"

Table D-10. Field and Laboratory Quality Control Elements and Acceptance Criteria

Constituent[a]	QC Element	Acceptance Criteria	Corrective Action

a. Refer to Table D-7 for specific analytical methods.

c. Laboratory-determined, statistically derived control limits may also be used. Such limits are reported with the data.

d. After review, corrective actions are determined on a case-by-case basis. Corrective actions may include a laboratory recheck or flagging the data as suspect ("Y" flag) or rejected ("R" flag).

e. Applies to total organic carbon and total organic halides only.

f. Applies only in cases where one or both results are greater than 5 times the detection limit.

g. Determined by the laboratory based on historical data. Control limits are reported with the data.

Data flags:

B, C = possible laboratory contamination (analyte was detected in the associated method blank)

N = result may be biased (associated matrix spike result was outside the acceptance limits)

Q = problem with associated field QC sample (blank and/or duplicate results were out of limits)

Abbreviations:

CRDL = contract-required detection limit
DUP = laboratory matrix duplicate
EB = equipment blank
FTB = full trip blank
FXR = field transfer blank
GC = gas chromatography
ICP = inductively coupled plasma
LCS = laboratory control sample
MB = method blank
MDL = method detection limit
MS = matrix spike
MSD = matrix spike duplicate
QC = quality control
RPD = relative percent difference
SUR = surrogate

1

Table D-11. Blind-Standard Constituents and Schedule

Constituents	Frequency	Accuracy (%)[a]	Precision (% RSD)[a]
Carbon Tetrachloride	Quarterly	±25%	<25%
Total Organic Halides[b]	Quarterly	±25%	<25%

a. If the results are less than 5 times the required detection limit, then the criterion is that the difference of the results of the replicates is less than the required detection limit.

b. Two sets of spikes for total organic halides will be used. The spiking compound for one set should be 2,4, 5-trichlorophenol. The spiking compound for the second set should include the constituents used for the volatile organic compounds sample (carbon tetrachloride).

RSD = relative standard deviation

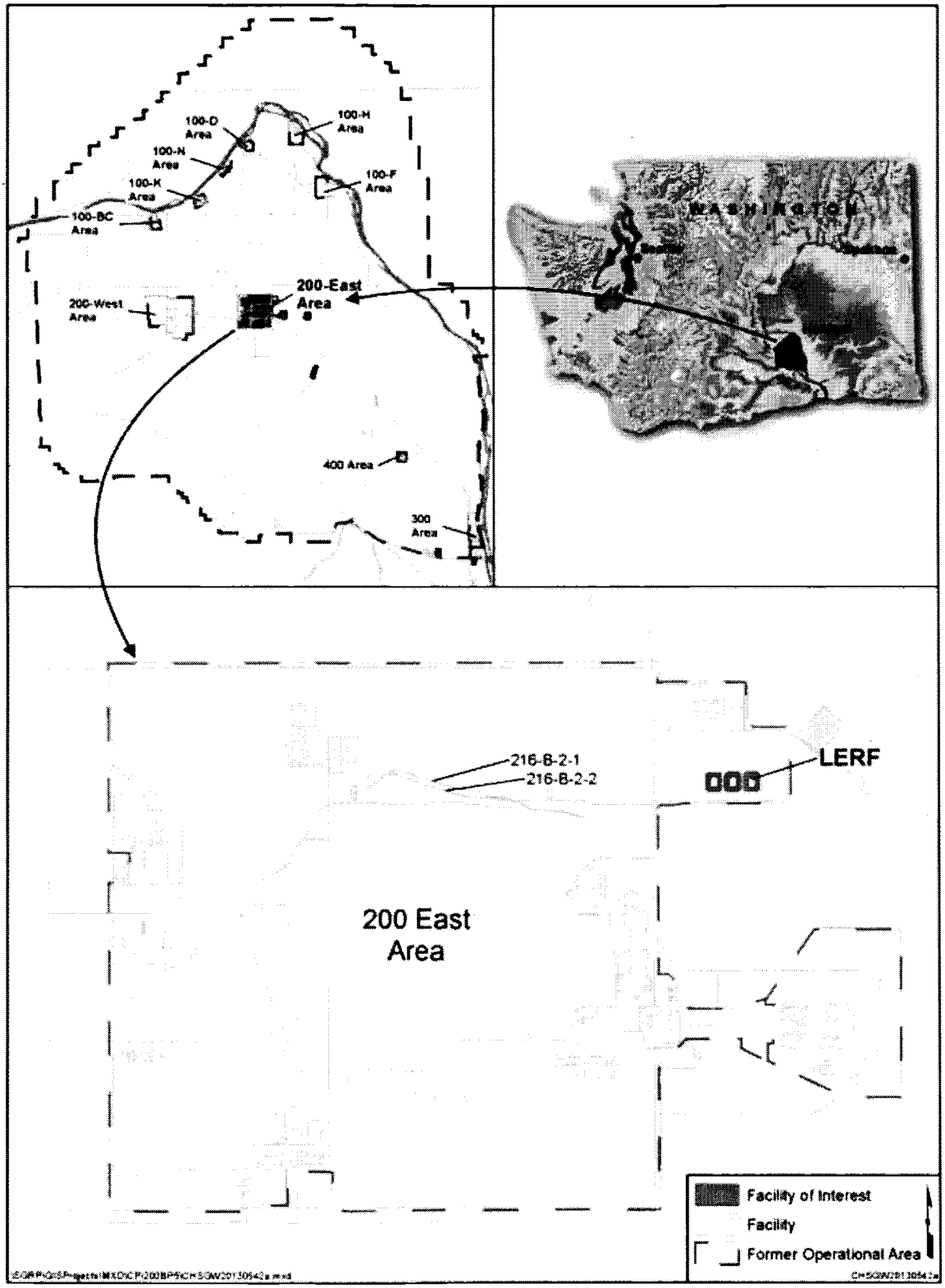

Figure D-1. LERF Location Map

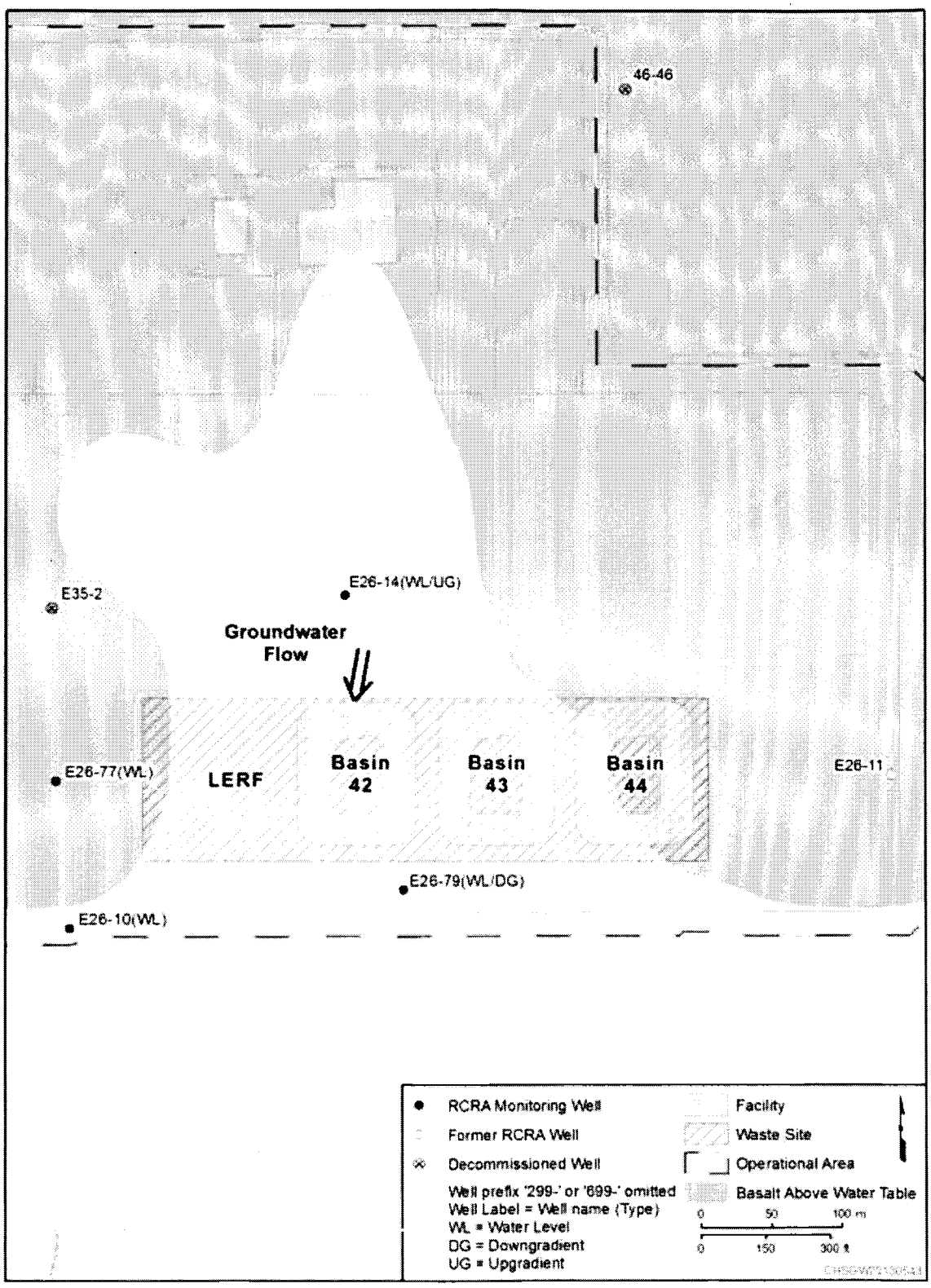

Figure D-2. LERF Well and Facility Description Map

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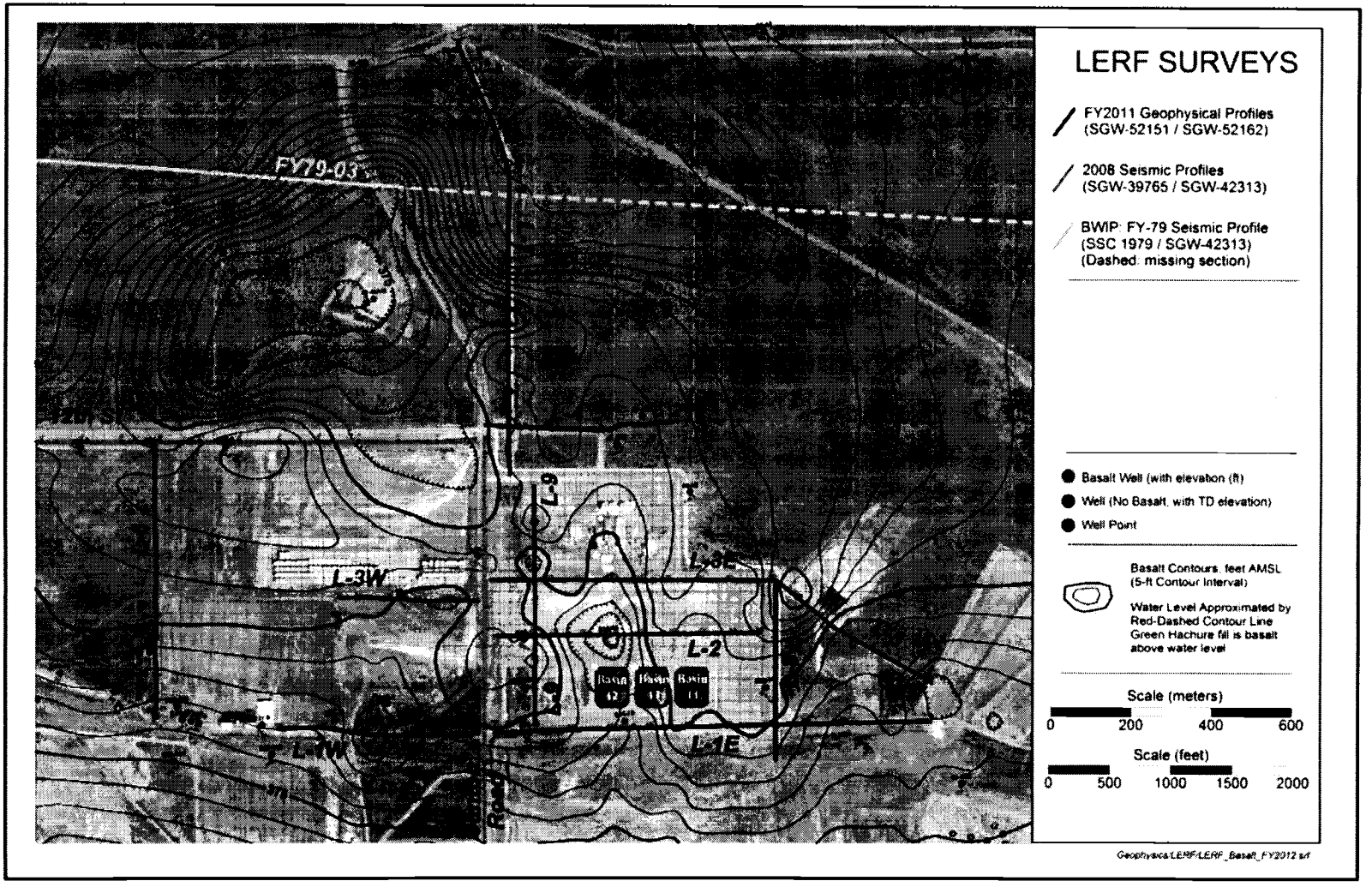

Figure D-3. LERF Location Map Showing Revised Unconfined Aquifer Thickness Based on Geophysical and Previous Borehole Data

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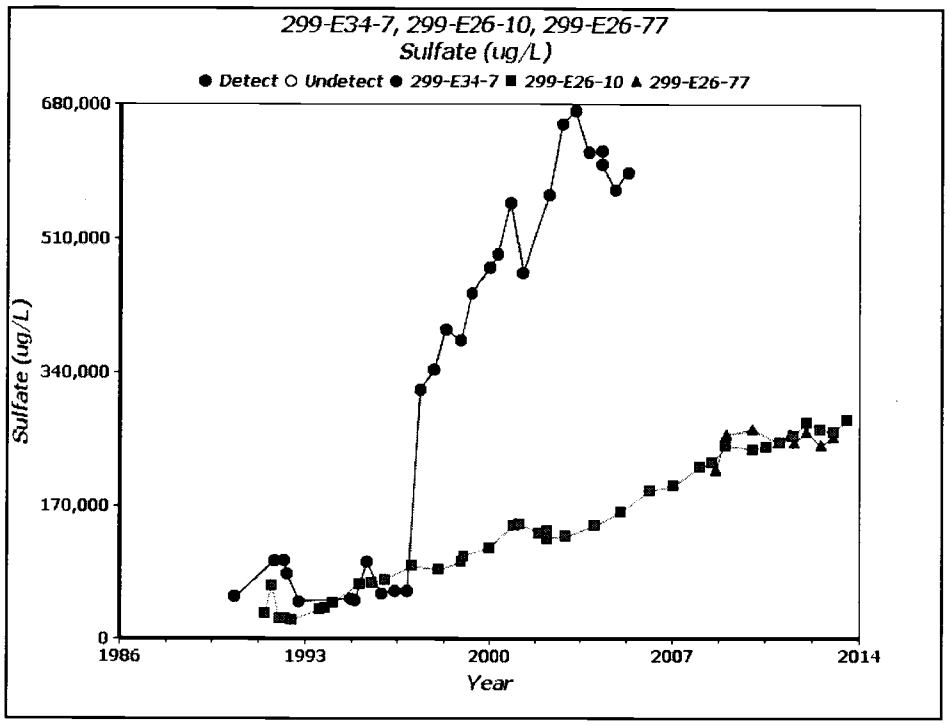

Figure D-4. Comparison of Sulfate at Wells 299-E34-7, 299-E26-14, and 299-E26-10

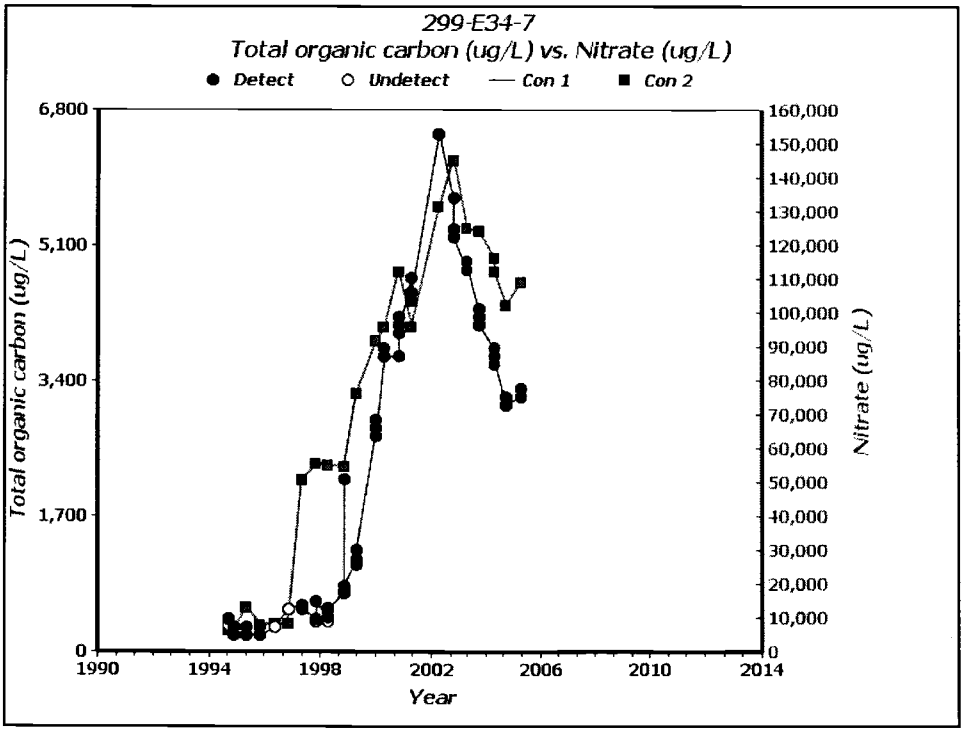

Figure D-5. History of Total Organic Carbon and Nitrate at Well 299-E34-7

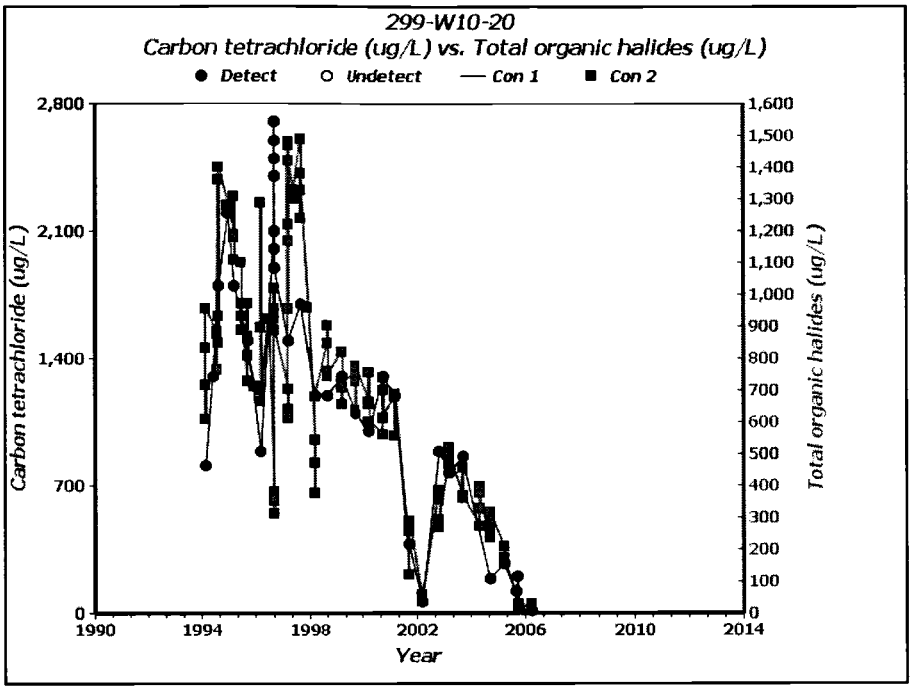

Figure D-6. Comparison of Carbon Tetrachloride to Total Organic Halides

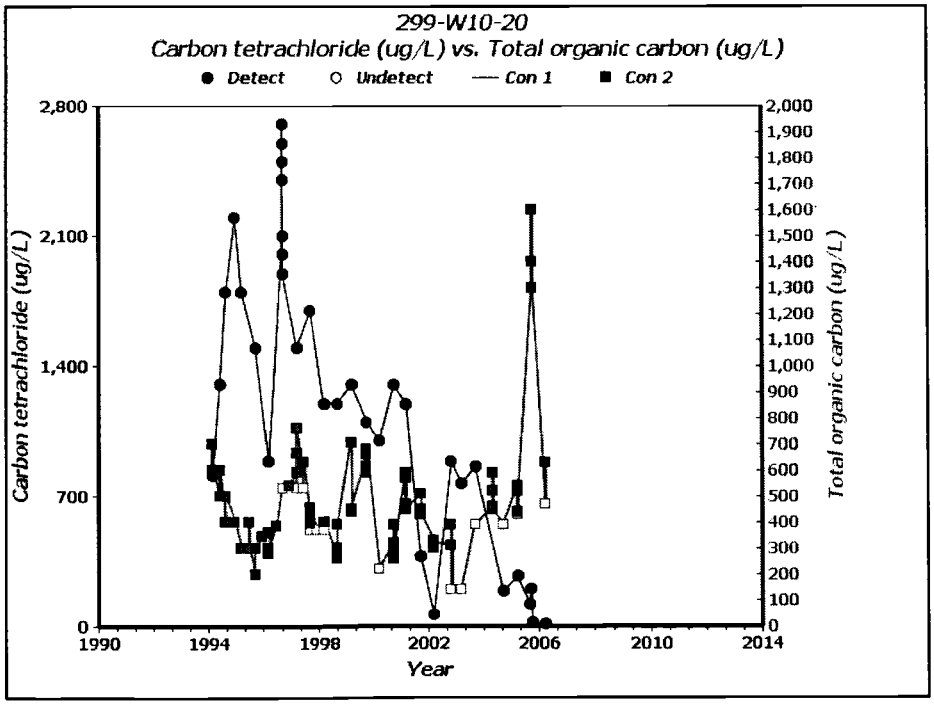

Figure D-7. Comparison of Carbon Tetrachloride to Total Organic Carbon

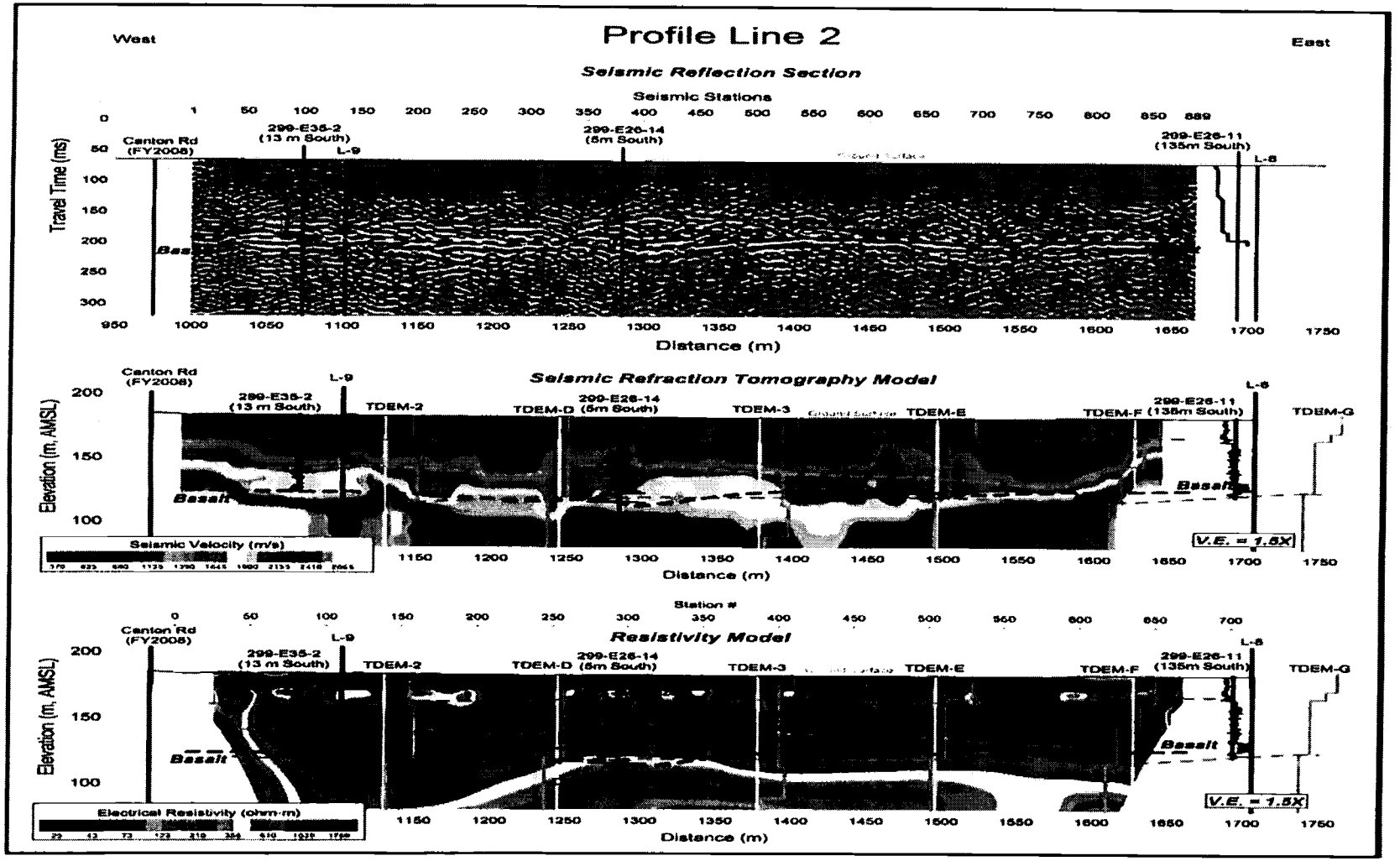

Figure D-8. Seismic Profile North of LERF extending from Well 299-E35-2 to Well 299-E26-11 and including Well 299-E26-11

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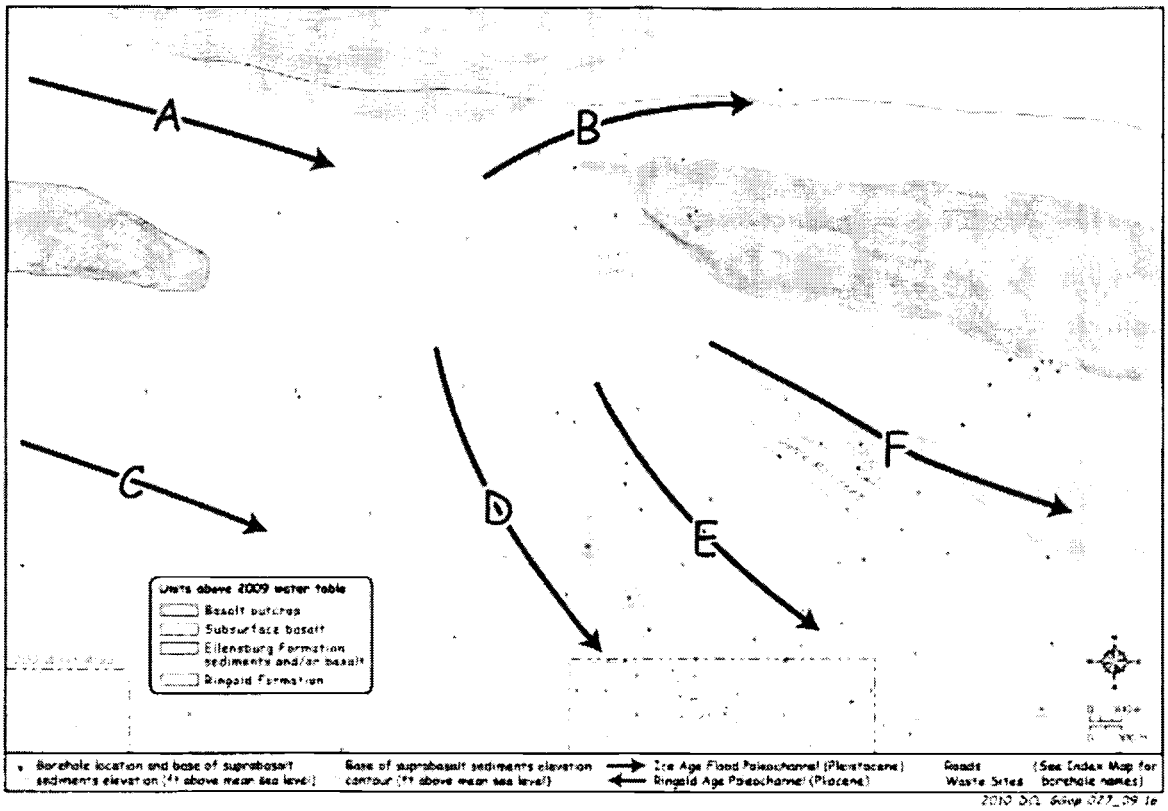

Figure D-9. Conceptual Model of Buried Paleochannels Extending Through Gable Gap

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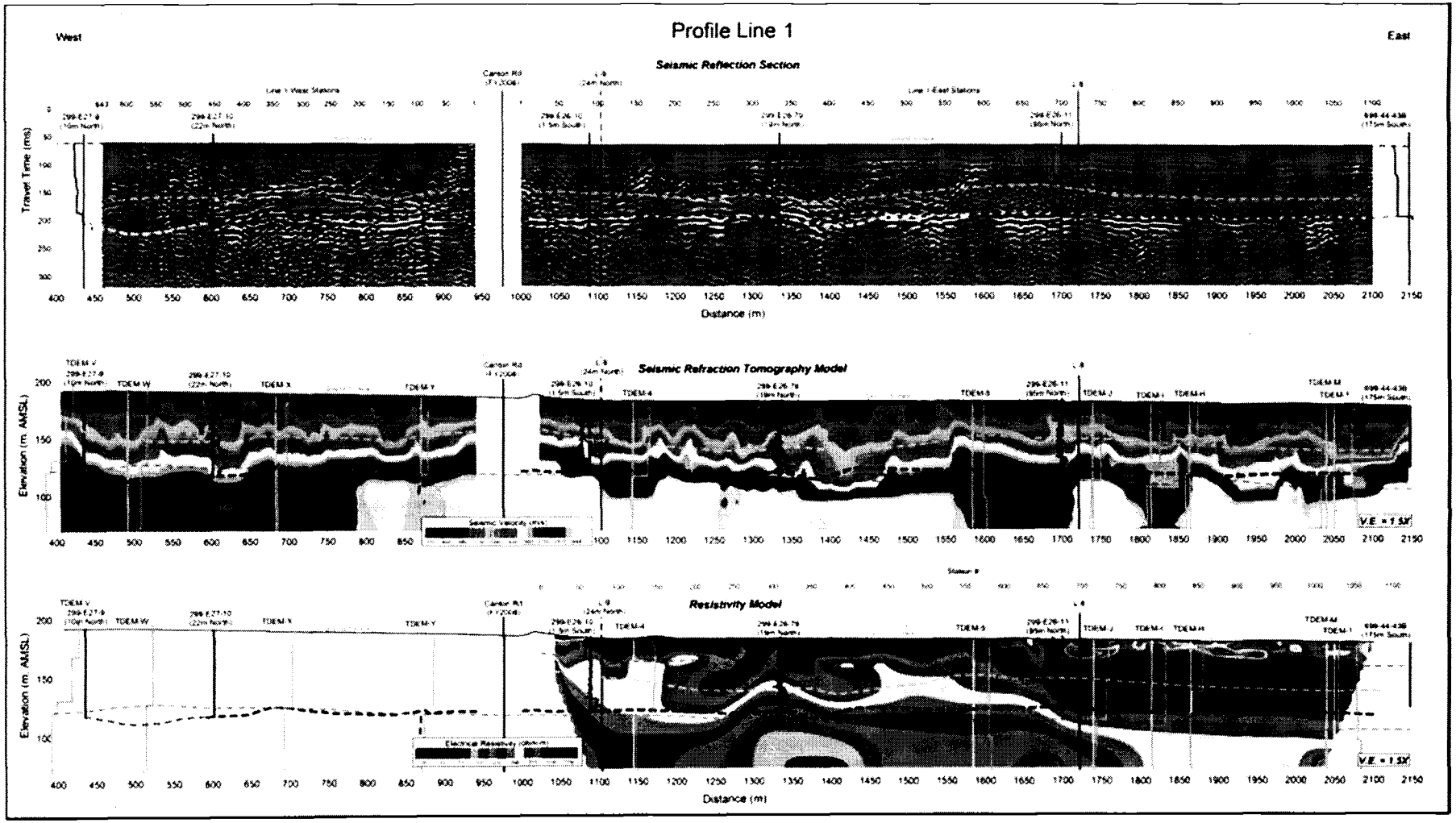

Figure D-10. Seismic Profile South of LERF with Focus on Area between Wells 299-E26-10 and 299-E26-11

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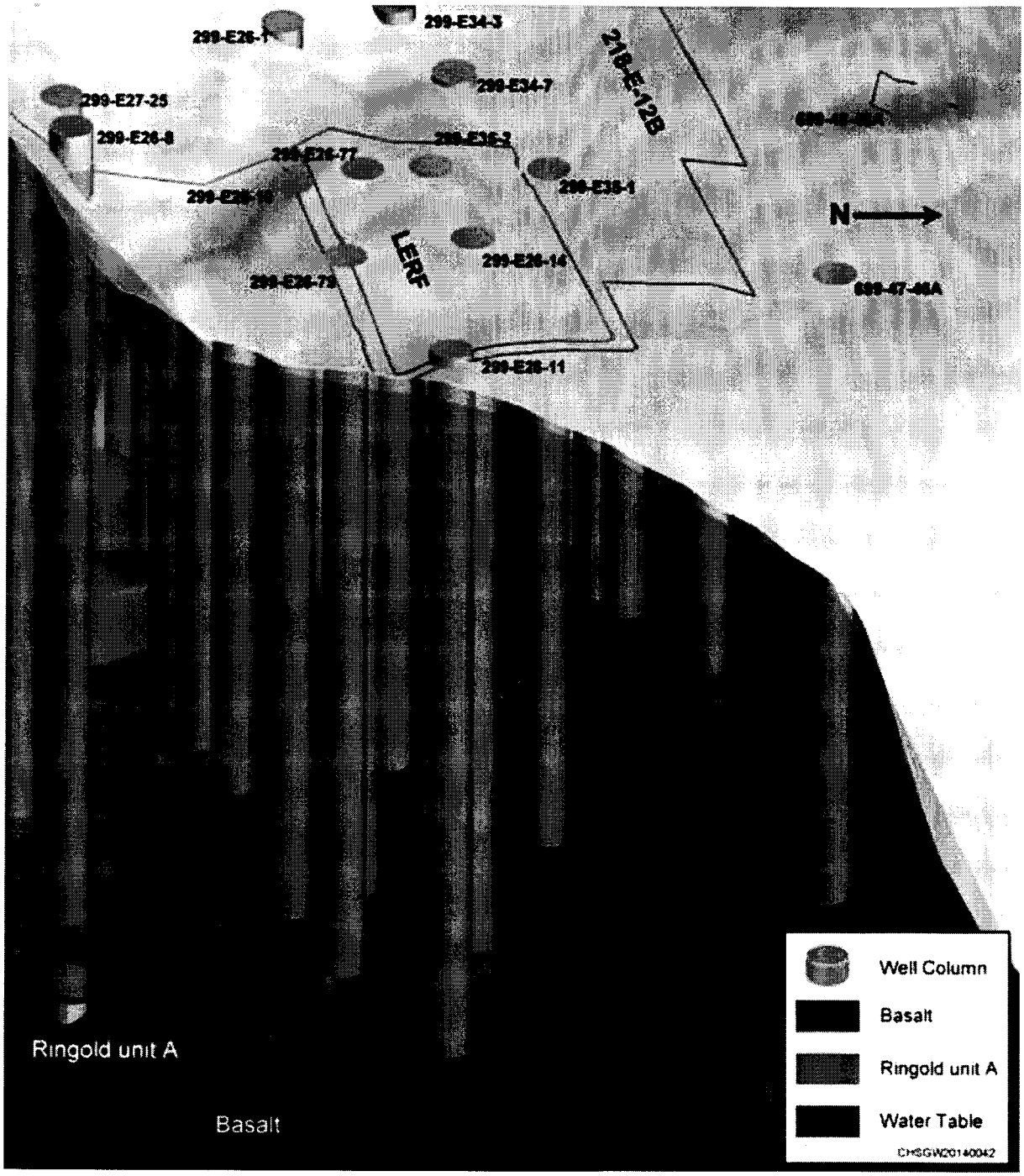
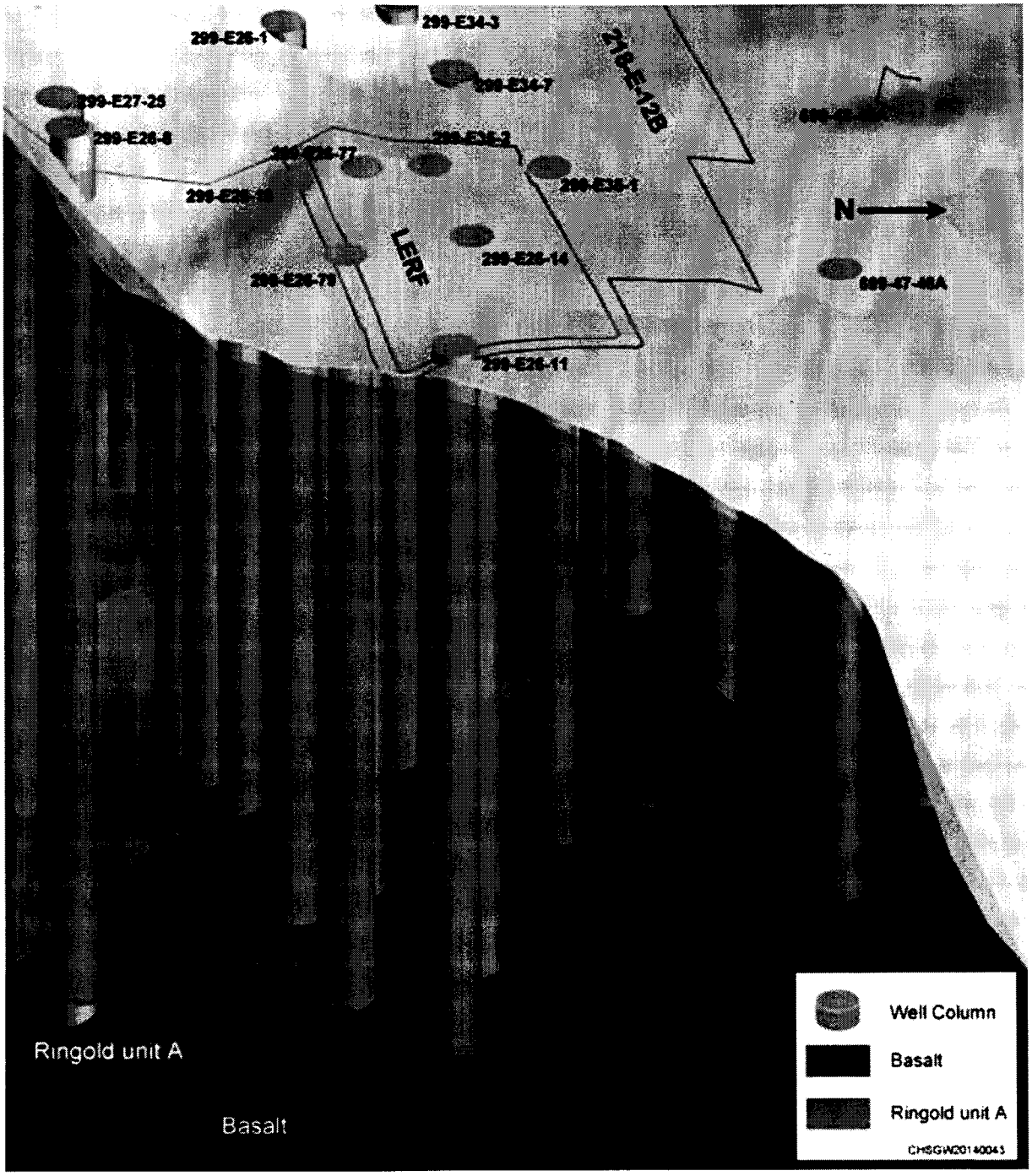


Note: The Hanford sediments have been removed to portray the basalt surface, remnant Ringold Formation, and groundwater elevation.

Figure D-11. Three Dimensional View from East of LERF Looking West-Northwest into the Soil Column



Note: The Hanford sediments and groundwater have been removed to portray the basalt surface and remnant Ringold Formation sediments.

Figure D-12. Three Dimensional View from East of LERF Looking West-Northwest into the Soil Column

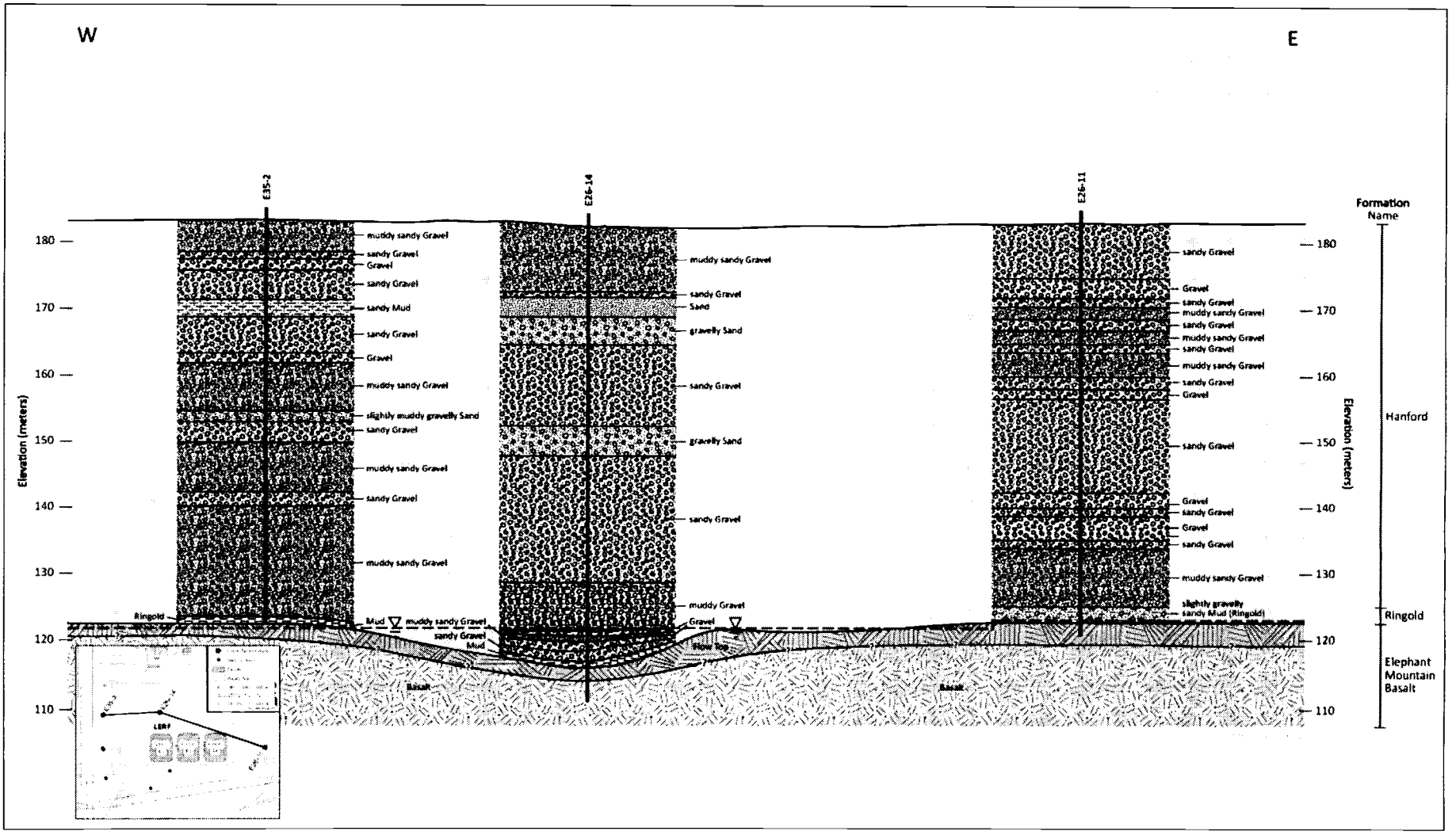


Figure D-13. West to East Geologic Cross Section Just North of LERF

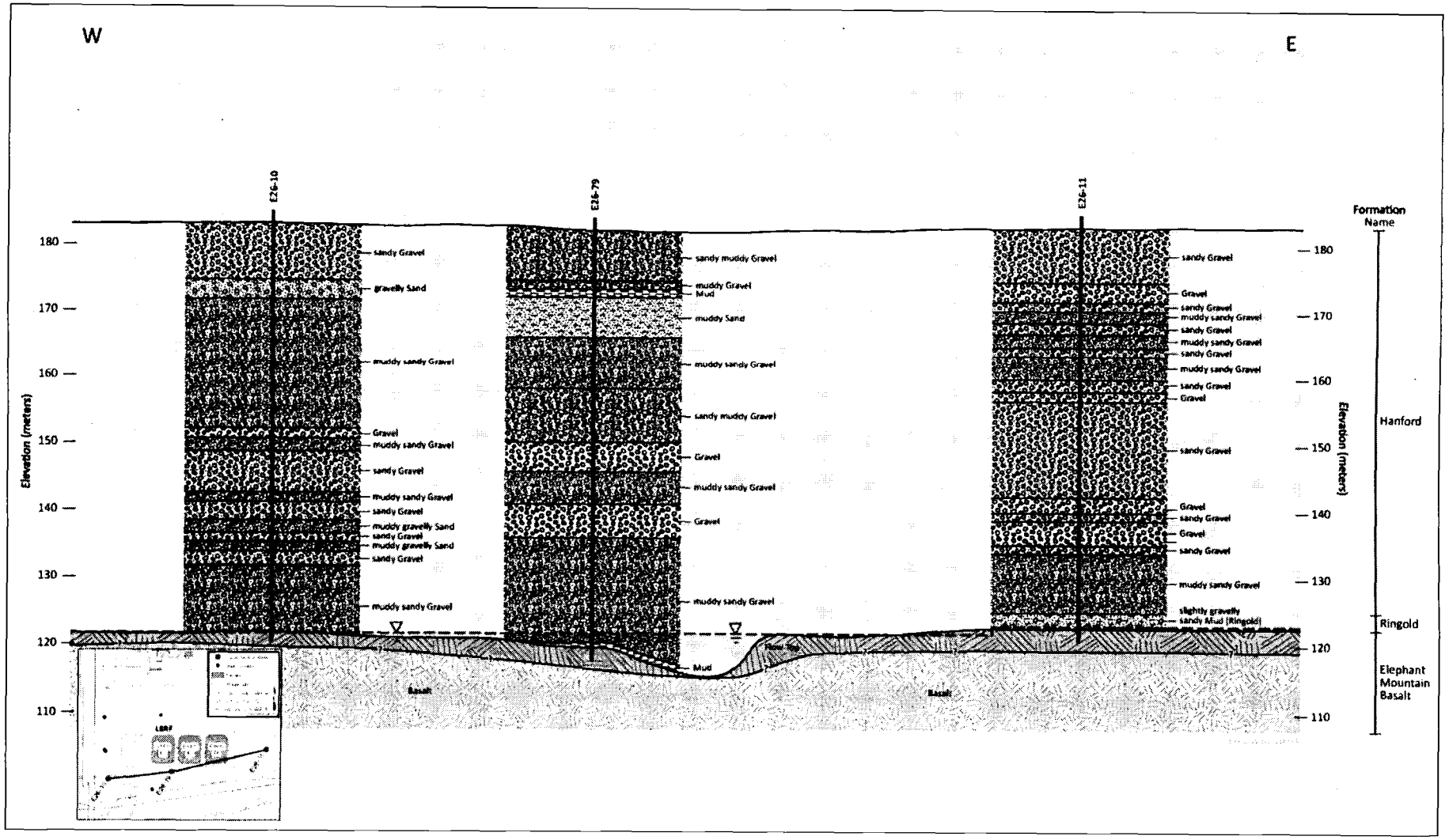
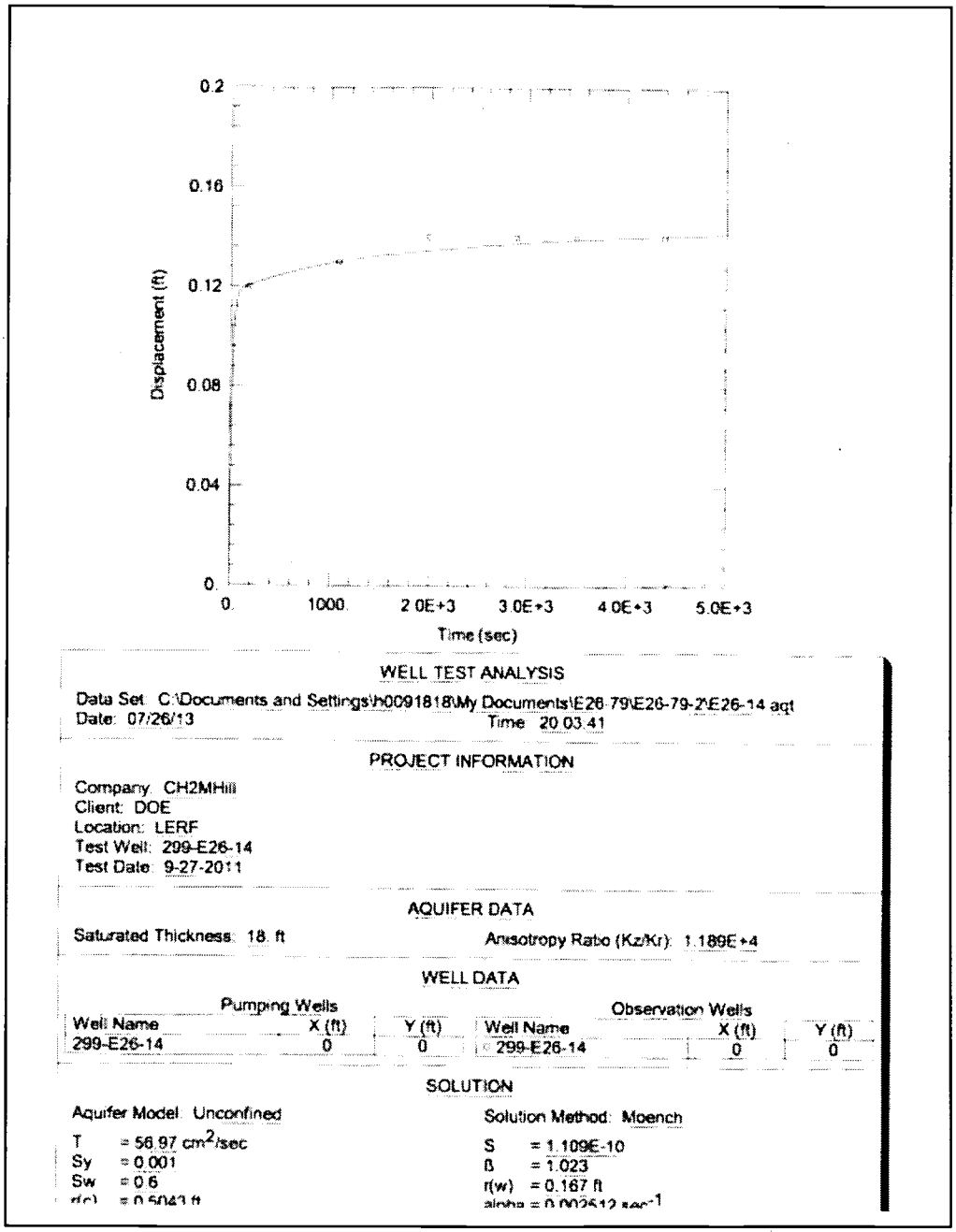


Figure D-14. West to East Geologic Cross Section Just South of LERF



α = empirical constant (fitting parameter, early part of curve matching) for noninstantaneous drainage

β = empirical constant (fitting parameter, late part of curve matching)

Kz = vertical hydraulic conductivity

Kr = horizontal hydraulic conductivity

T = transmissivity

S = storativity

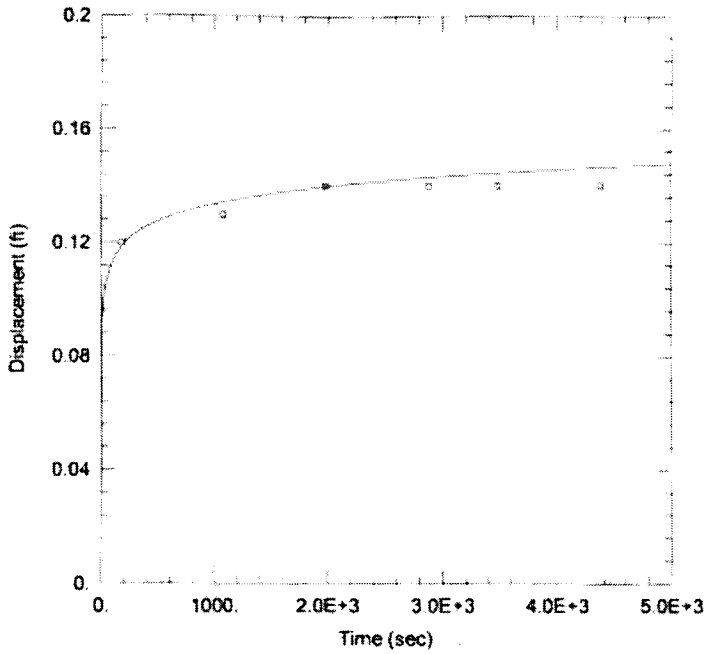
Sy = specific conductance

Sw = wellbore skin factor

r(c) = casing radius

r(w) = well radius

Figure D-15. AQTESOLV Moench Unconfined Aquifer Pumping/Recovery Test for Type-Curve Match to Well 299-E26-14 with Wellbore Skin Affects and Delayed Gravity Response



WELL TEST ANALYSIS

Data Set: C:\Documents and Settings\h0091818\My Documents\E26-79\E26-79-2\E26-14.aqt
Date: 07/26/13 Time: 22:12:52

PROJECT INFORMATION

Company: CH2MHill
Client: DOE
Location: LERF
Test Well: 299-E26-14
Test Date: 9-27-2011

AQUIFER DATA

Saturated Thickness: 18. ft

WELL DATA

Pumping Wells			Observation Wells		
Well Name	X (ft)	Y (ft)	Well Name	X (ft)	Y (ft)
299-E26-14	0	0	299-E26-14	0	0

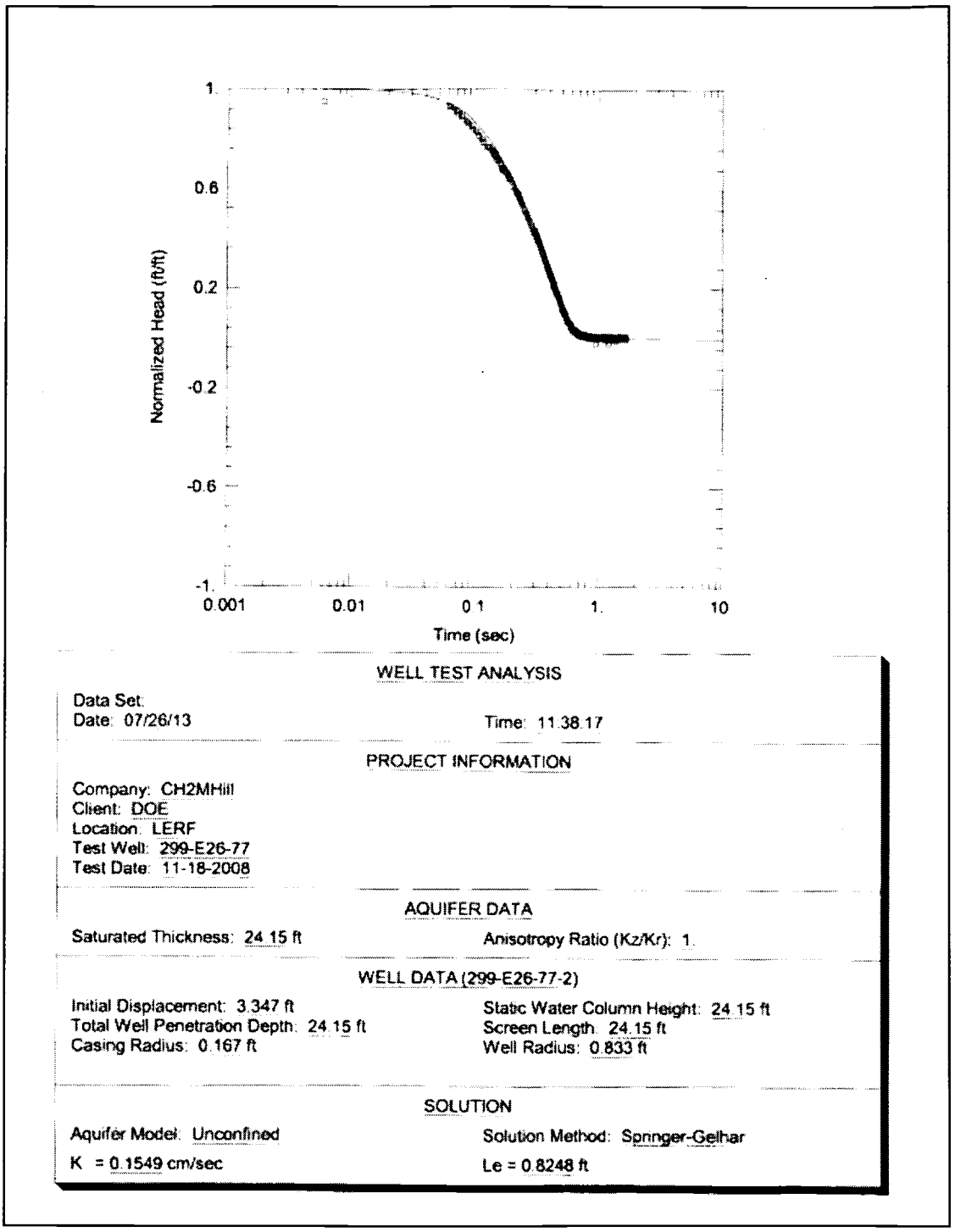
SOLUTION

Aquifer Model: Unconfined Solution Method: Neuman

T = 50.78 cm^2/sec S = 1.109E-10
Sy = 0.001 ß = 1.023

T = transmissivity S = storativity
β = empirical constant (fitting parameter, late part of curve matching)
Sy = specific conductance

Figure D-16. AQTESOLV Neuman Unconfined Aquifer Pumping/Recovery Test for Type-Curve Match to Well 299-E26-14 with Delayed Gravity Response



K = Hydraulic conductivity

Le = Effective water column length

Figure D-17. AQTESOLV Springer-Gelhar Critically Dampened Type-Curve Match to Well 299-E26-77 Second Slug Withdraw Test November 18, 2008

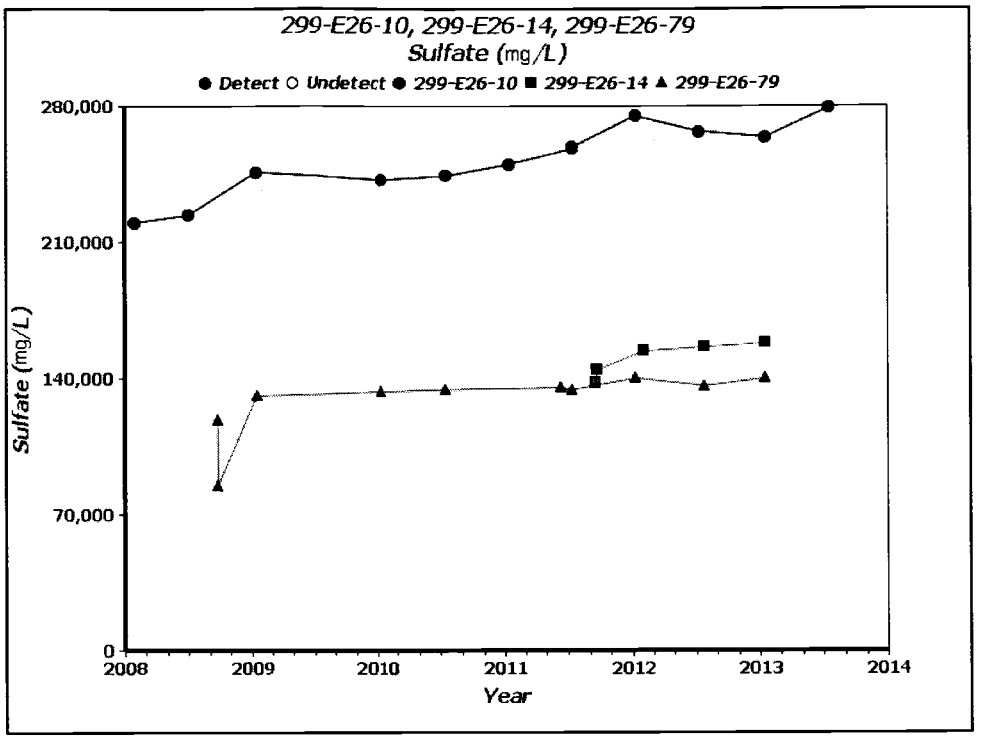


Figure D-18. Comparison of Sulfate Concentration Results in Wells 299-E26-10, 299-E26-14, and 299-E26-79

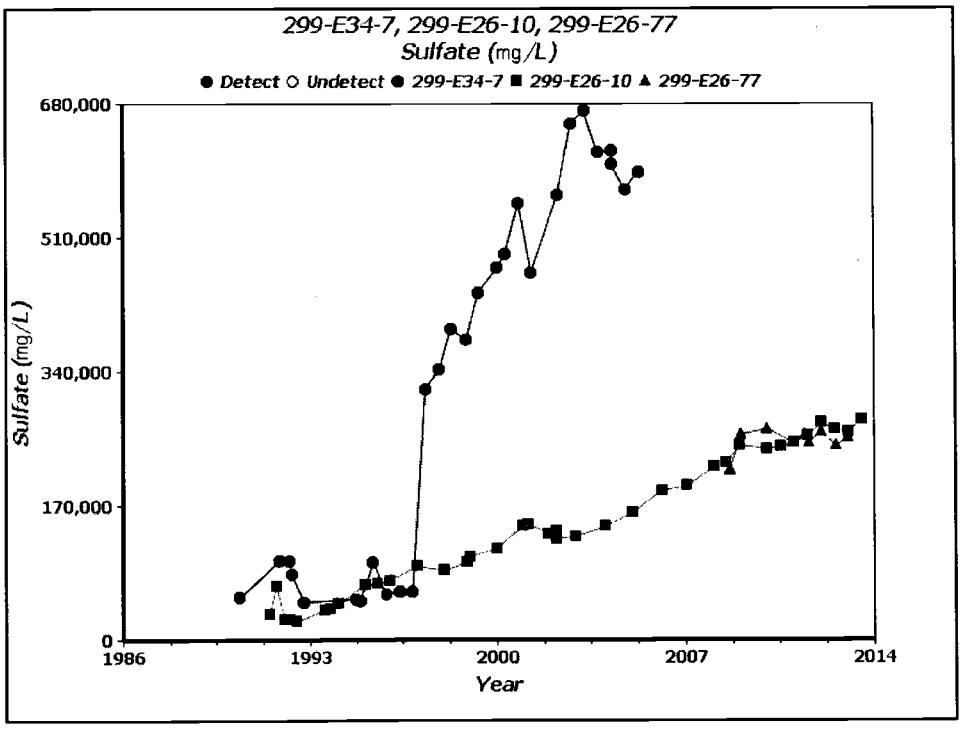


Figure D-19. History of Sulfate Increases in Wells 299-E26-10, 299-E26-77, and 299-E34-7

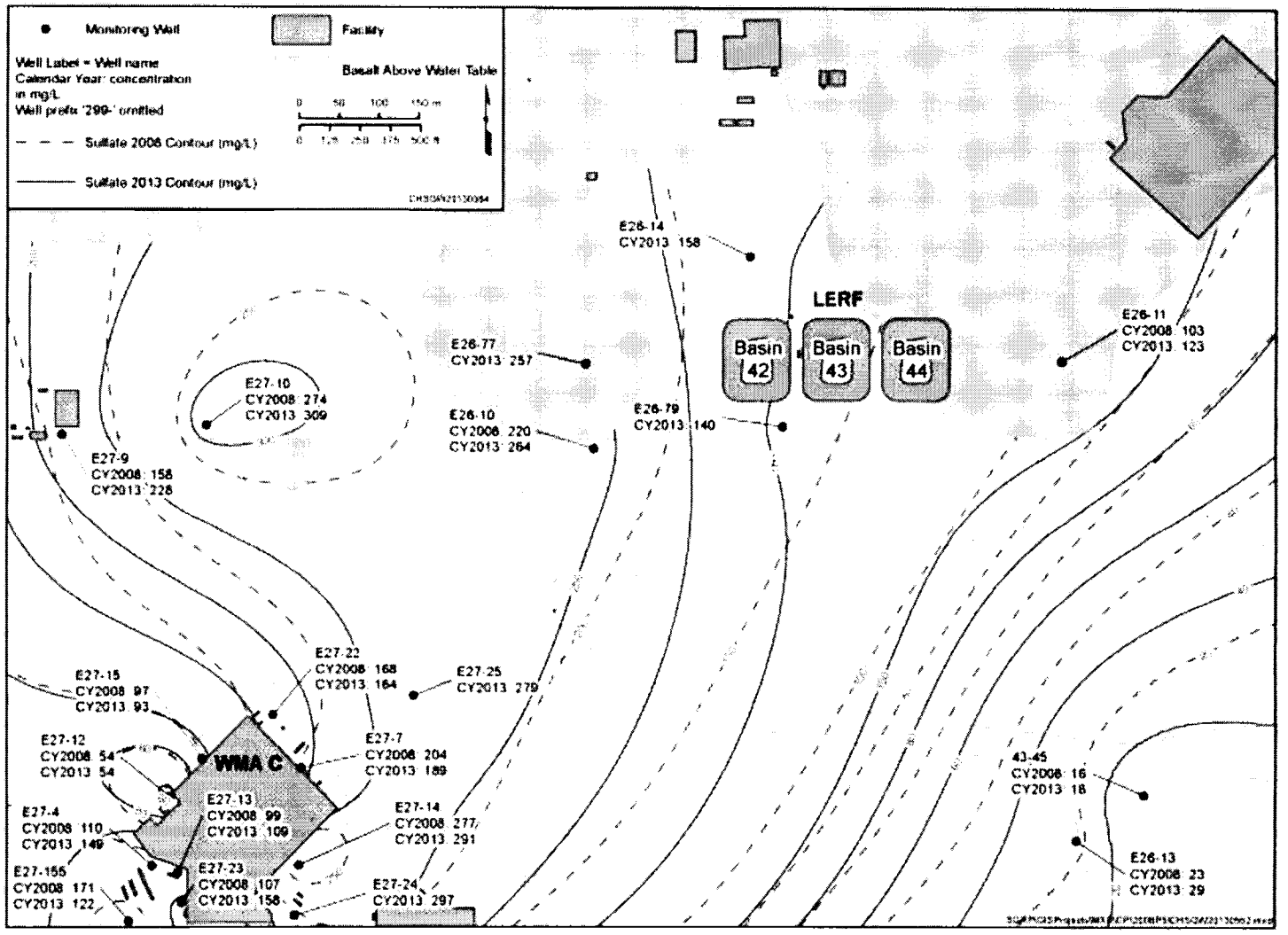


Figure D-20. 2008 and 2013 Regional Sulfate Plume

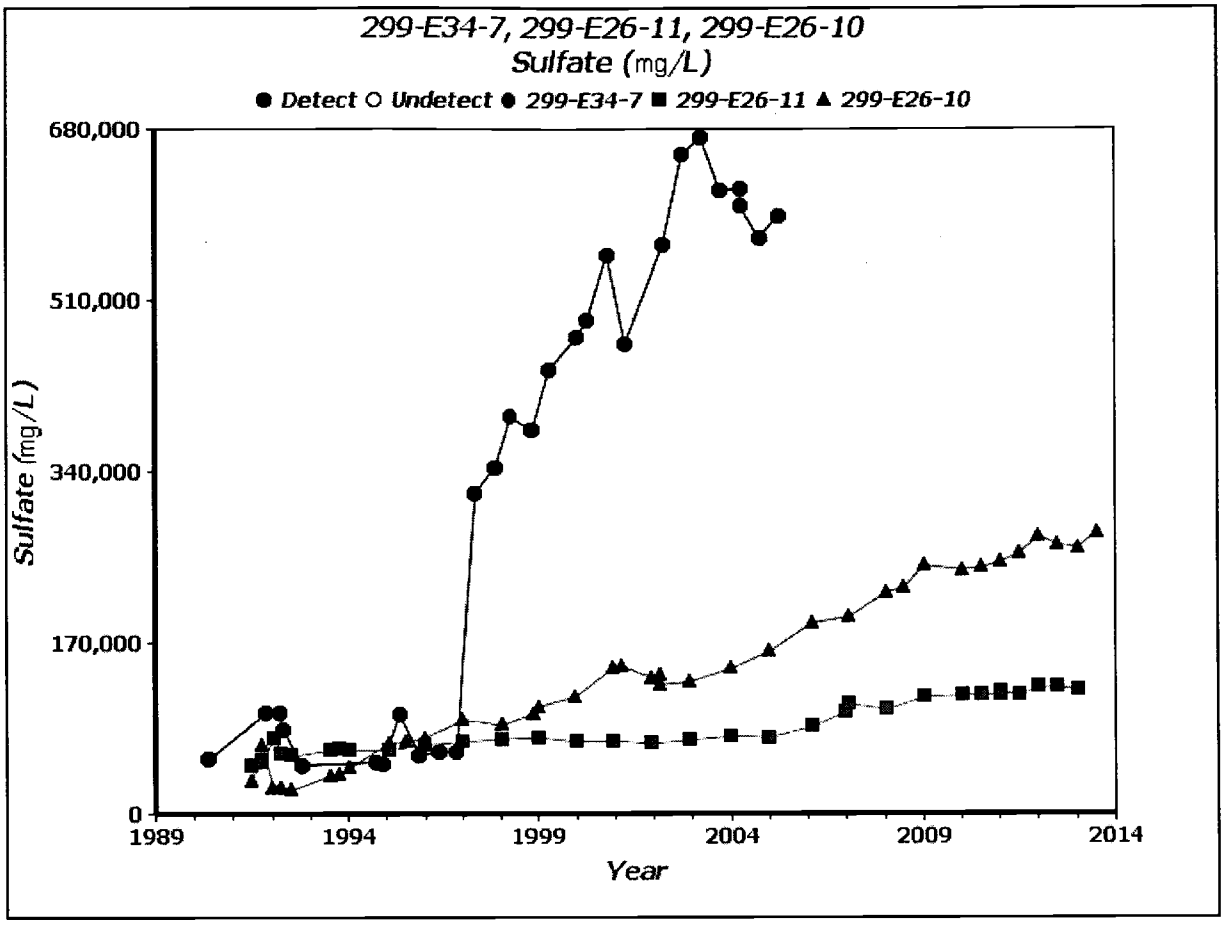


Figure D-21. Comparison of Sulfate Increases in Wells 299-E26-10, 299-E26-11, and 299-E34-7

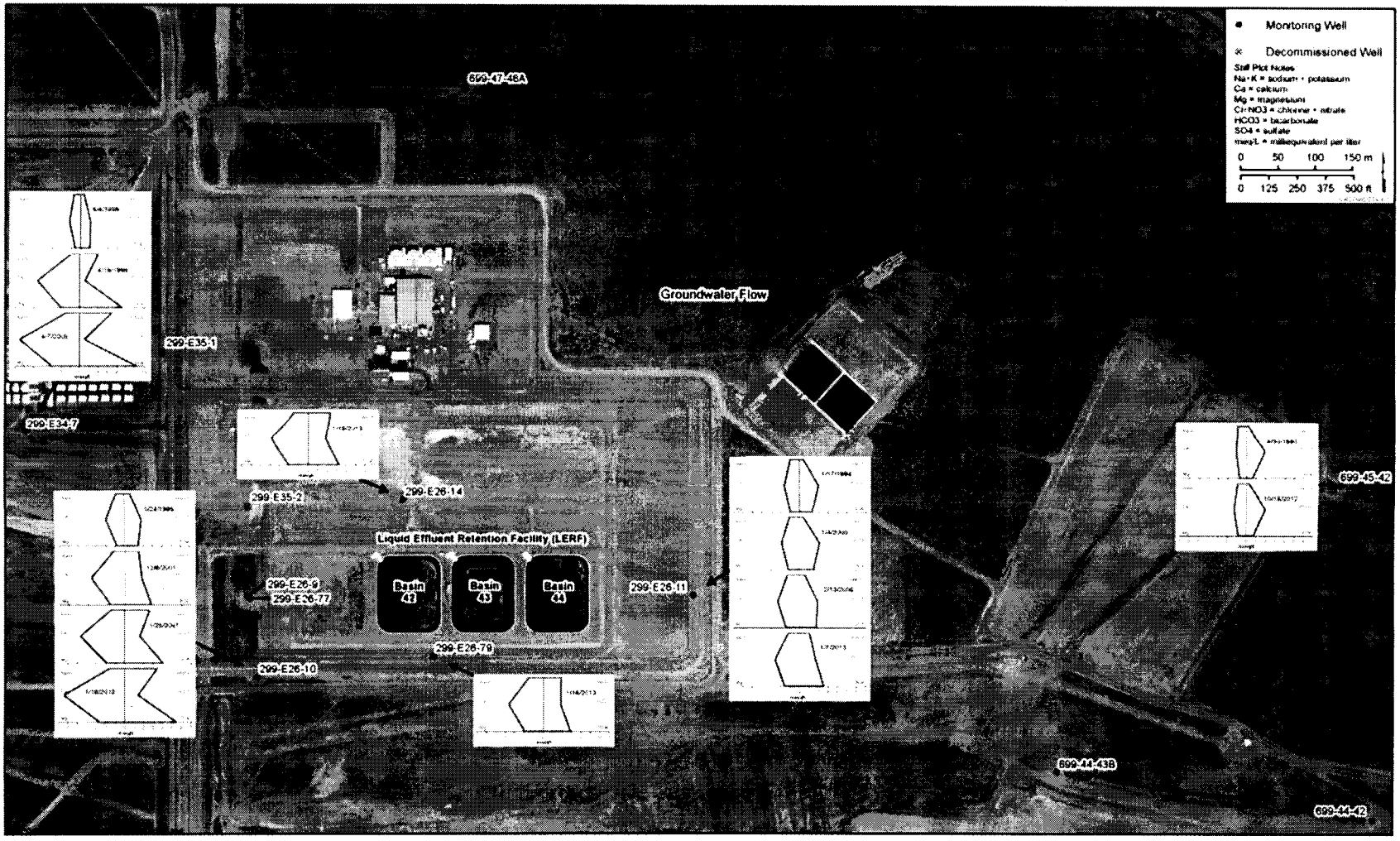


Figure D-22. Historical View of Changing Groundwater Conditions Beneath the Vicinity of LERF

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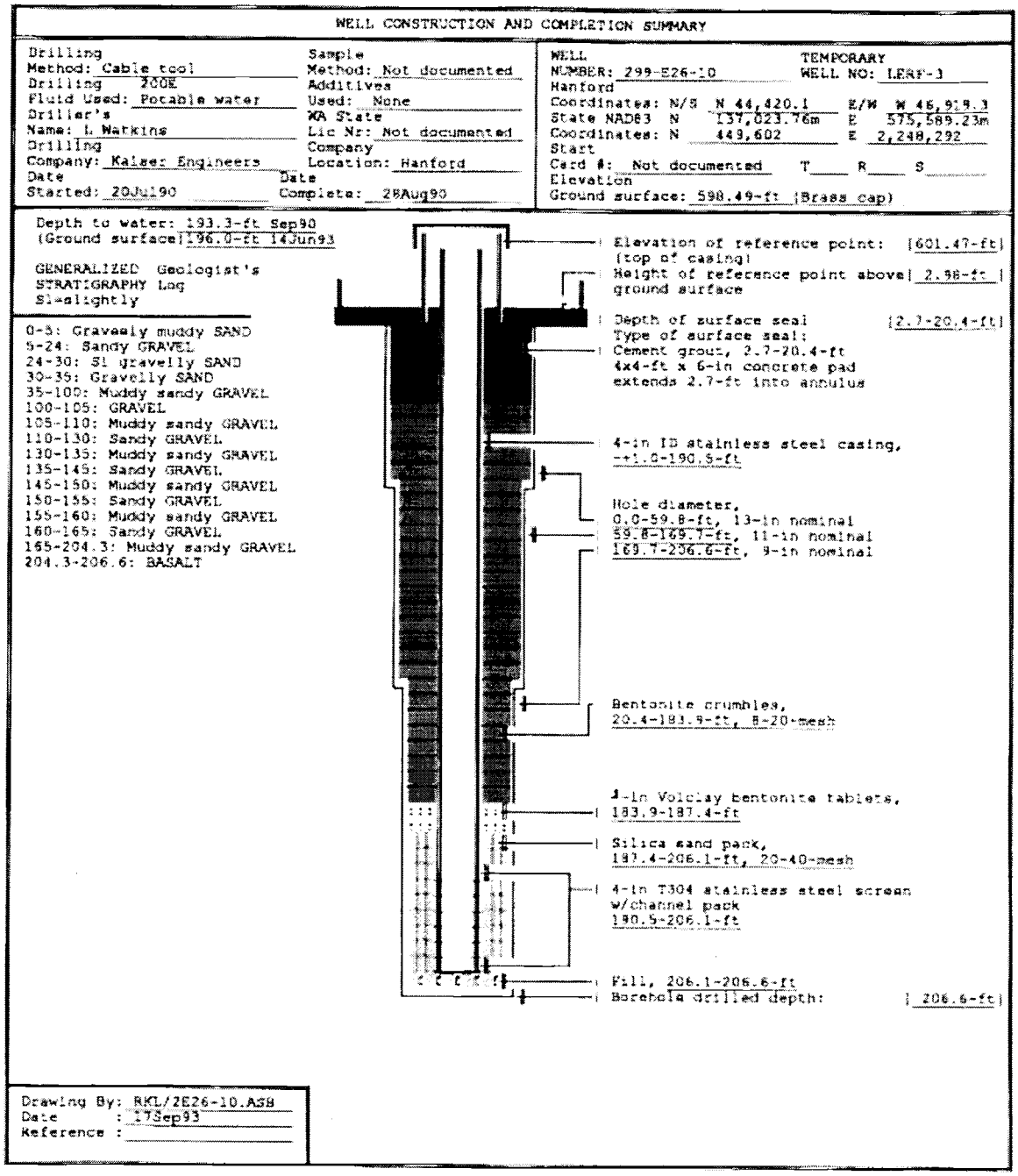


Figure D-23. Well Construction Diagram for Well 299-E26-10 in LERF Groundwater Monitoring Network

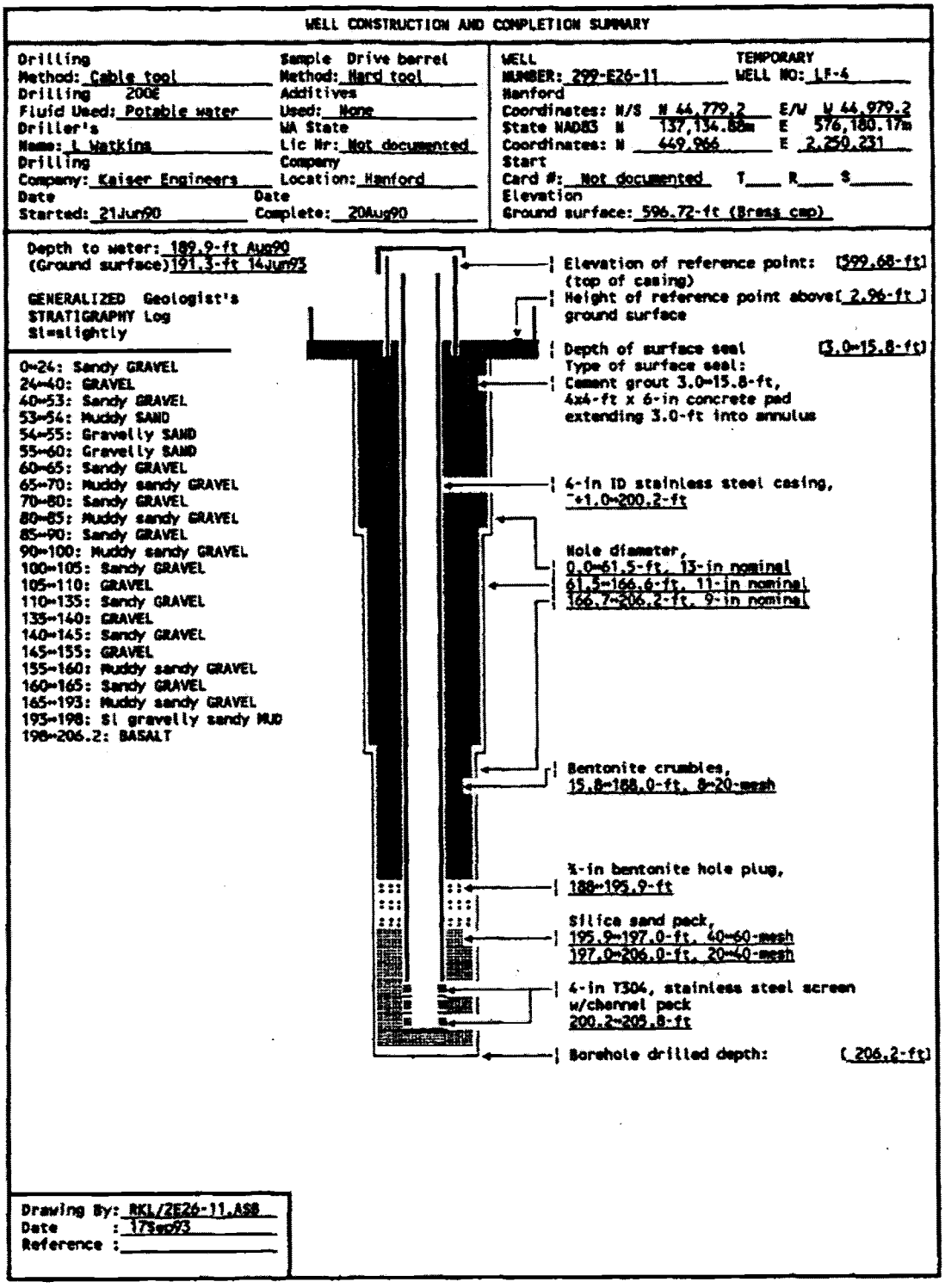


Figure D-24. Well Construction Diagram for Well 299-E26-11 in the Past LERF Groundwater Monitoring Network

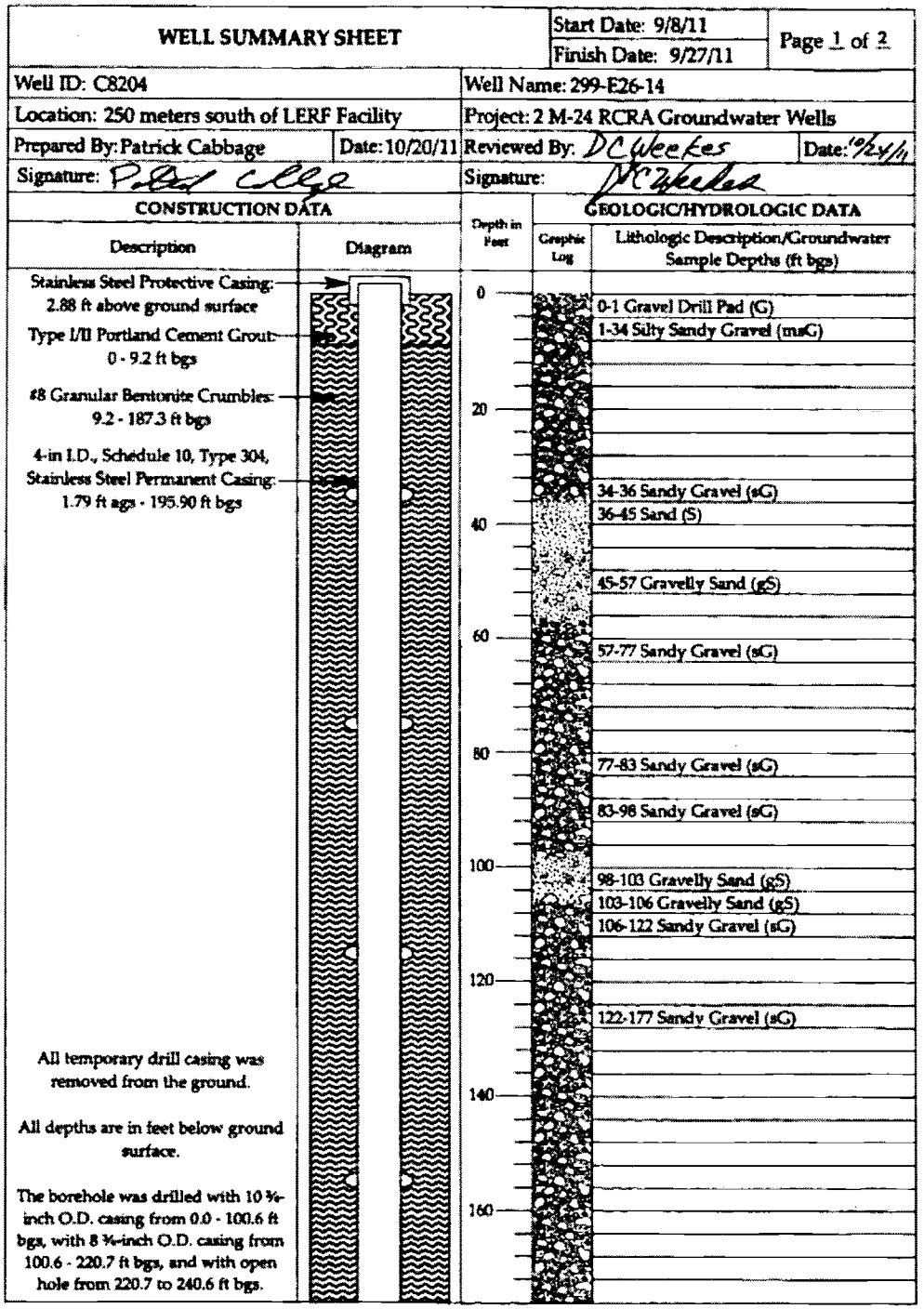


Figure D-25. Well Construction Diagram for Well 299-E26-14 in LERF Groundwater Monitoring Network

WELL SUMMARY SHEET | Start Date: 9/8/11 | Finish Date: 9/27/11 | Page 2 of 2

Well ID: C8204		Well Name: 299-E26-14	
Location: 250 meters south of LERF Facility		Project: 2 M-24 RCRA Groundwater Wells	
Prepared By: Patrick Cabbage	Date: 10/21/11	Reviewed By: DC Weekes	Date: 10/24/11
Signature: Patrick Cabbage		Signature: DC Weekes	

CONSTRUCTION DATA

Description	Diagram
#8 Granular Bentonite Crumbles: 9.2 - 187.3 ft bgs	
4-in I.D., Schedule 10, Type 304, Stainless Steel Permanent Casing: 1.79 ft ags - 195.90 ft bgs	
3/8-in Bentonite Pellets: 187.3 - 190.0 ft bgs	
Static Water Level: 198.4 ft bgs (9/27/11)	
4-in I.D., Schedule 10, Type 304, Stainless Steel 20-slot Screen: 195.90 - 215.90 ft bgs	
Primary Filter Pack 10-20 Mesh Colorado Silica Sand: 190.0 - 219.6 ft bgs	
4-in I.D., Schedule 10, Type 304, Stainless Steel Sump: 215.90 - 218.90 ft bgs	
3/8-in Bentonite Pellets: 219.6 - 240.6 ft bgs	

All temporary drill casing was removed from the ground.

All depths are in feet below ground surface.

The borehole was drilled with 10 ¾-inch O.D. casing from 0.0 - 100.6 ft bgs, with 8 ¾-inch O.D. casing from 100.6 - 220.7 ft bgs, and with open hole from 220.7 to 240.6 ft bgs.

GEOLOGIC/HYDROLOGIC DATA

Depth in Feet	Graphic Log	Lithologic Description/Groundwater Sample Depths (ft bgs)
180		177-180 Silty Gravel (mG)
		180-200 Silty Sandy Gravel (msG)
200		200-202.5 Gravel (G)
		202.5-205 Silty Sandy Gravel (msG)
		205-210 Gravel (G)
		210-215.5 Sandy Gravel (sG)
		215.5-217 Silt (M)
220		217-220 Gravelly Silt (gM)
		220-221 Silty Gravel (mG)
		221-240.6 Basalt
240		TD = 240.6 ft bgs (09-21-2011)
260		

Figure D-25. Well Summary Sheet for Well 299-E26-14 in the LERF Groundwater Monitoring Network (Cont.)

WELL SUMMARY SHEET

Start Date: 09/12/2008 Finish Date: 02/05/2009 Page 1 of 2

Well ID: C16455 Well Name: 299-E26-77

Location: Inside fence, LERF 200E Project: M-24 RCRA Wells (LERF)

Prepared By: S. Sexton Date: 9/08/08 Reviewed By: L.D. Walker Date: 10/12/08

Signature: Signature:

CONSTRUCTION DATA

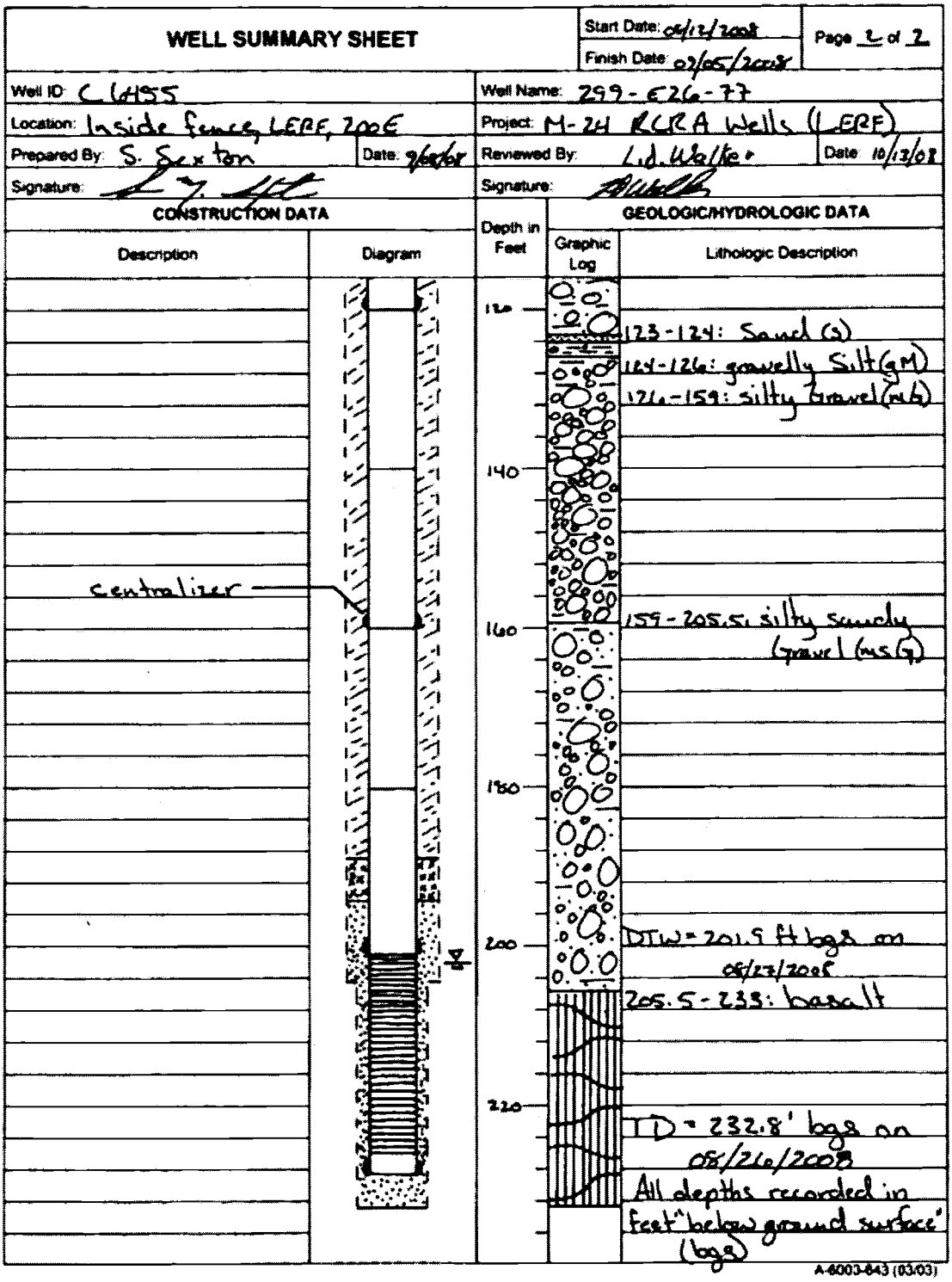
Description	Diagram
6" x 4' x 4' concrete pad	
6" stainless steel protective monument:	
4" ID Sch. 10 stainless steel casing: 1.96 ft ags to 200.85 ft bgs	
4" stainless steel wire wrap screen, 0.020": 200.85 to 225.6 ft bgs	
4" Sch. 10 stainless steel sump: 225.6 to 228.6 ft bgs	
Portland cement grout: 0 to 11.05 ft bgs	
Bentonite granules: 11.05 to 189.0 ft bgs	
Bentonite pellets: 189.0 to 194.6 ft bgs	
10-20 silica sand: 194.6 to 232.8 ft bgs	
casing joint	

GEOLOGIC/HYDROLOGIC DATA

Depth in Feet	Graphic Log	Lithologic Description
0		0-0.5: gravel drill pad
		0.5-5: gravelly Silt (gM)
		5-34: silty Gravel (gM)
20		
		34-38: gravelly silty Sand (gsM)
		38-43: silty sandy Gravel (gsM)
40		43-45.5: gravelly Sand (gS)
		45.5-52: silty Sand (sM)
		52-75: silty sandy Gravel (gsM)
60		
		75-123: sandy silty Gravel (gsM)
80		
100		

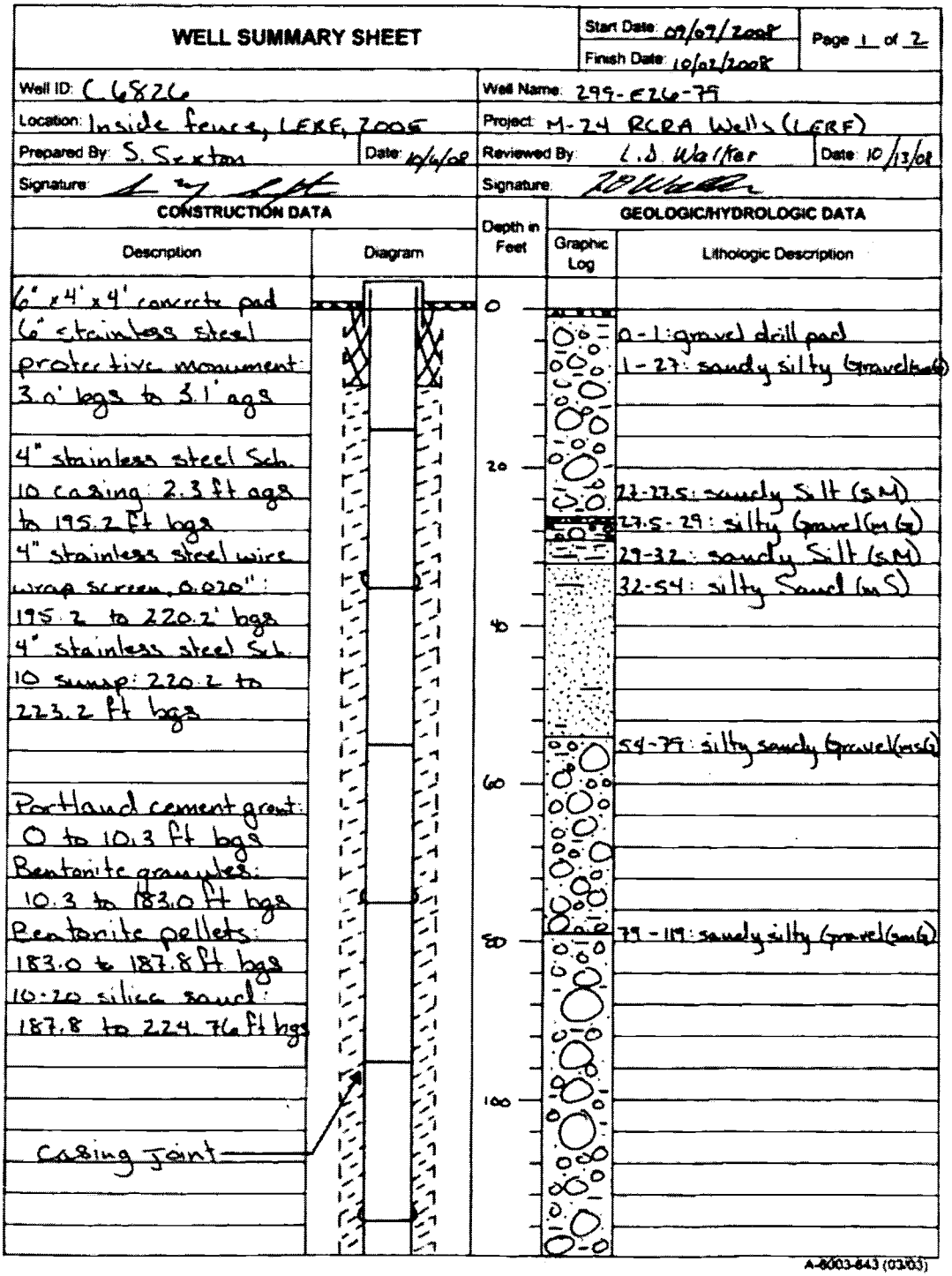
A-6003-643 (03/03)

Figure D-26. Well Summary Sheet for Well 299-E26-77 in LERF Groundwater Monitoring Network



1 **Figure D-26. Well Summary Sheet for Well 299-E26-77 in LERF Groundwater Monitoring Network (Cont.)**

2



WELL SUMMARY SHEET

Start Date: 09/09/2008 Finish Date: 10/02/2008 Page 1 of 2

Well ID: C 6826 Well Name: 299-E26-79

Location: Inside fence, LERF, 200E Project: M-24 RCRA Wells (LERF)

Prepared By: S. Sexton Date: 10/6/08 Reviewed By: L.D. Walker Date: 10/13/08

Signature: Signature:

CONSTRUCTION DATA

Description	Diagram
6" x 4' x 4' concrete pad	
6" stainless steel protective monument: 3.0' bgs to 3.1' ags	
4" stainless steel Sch. 10 casing: 2.3 ft ags to 195.2 Ft bgs	
4" stainless steel wire wrap screen, 0.020": 195.2 to 220.2' bgs	
4" stainless steel Sch. 10 sump: 220.2 to 223.2 ft bgs	
Portland cement grout: 0 to 10.3 ft bgs	
Bentonite granules: 10.3 to 183.0 ft bgs	
Bentonite pellets: 183.0 to 187.8 ft bgs	
10-20 silica sand: 187.8 to 224.76 ft bgs	
Casing joint	

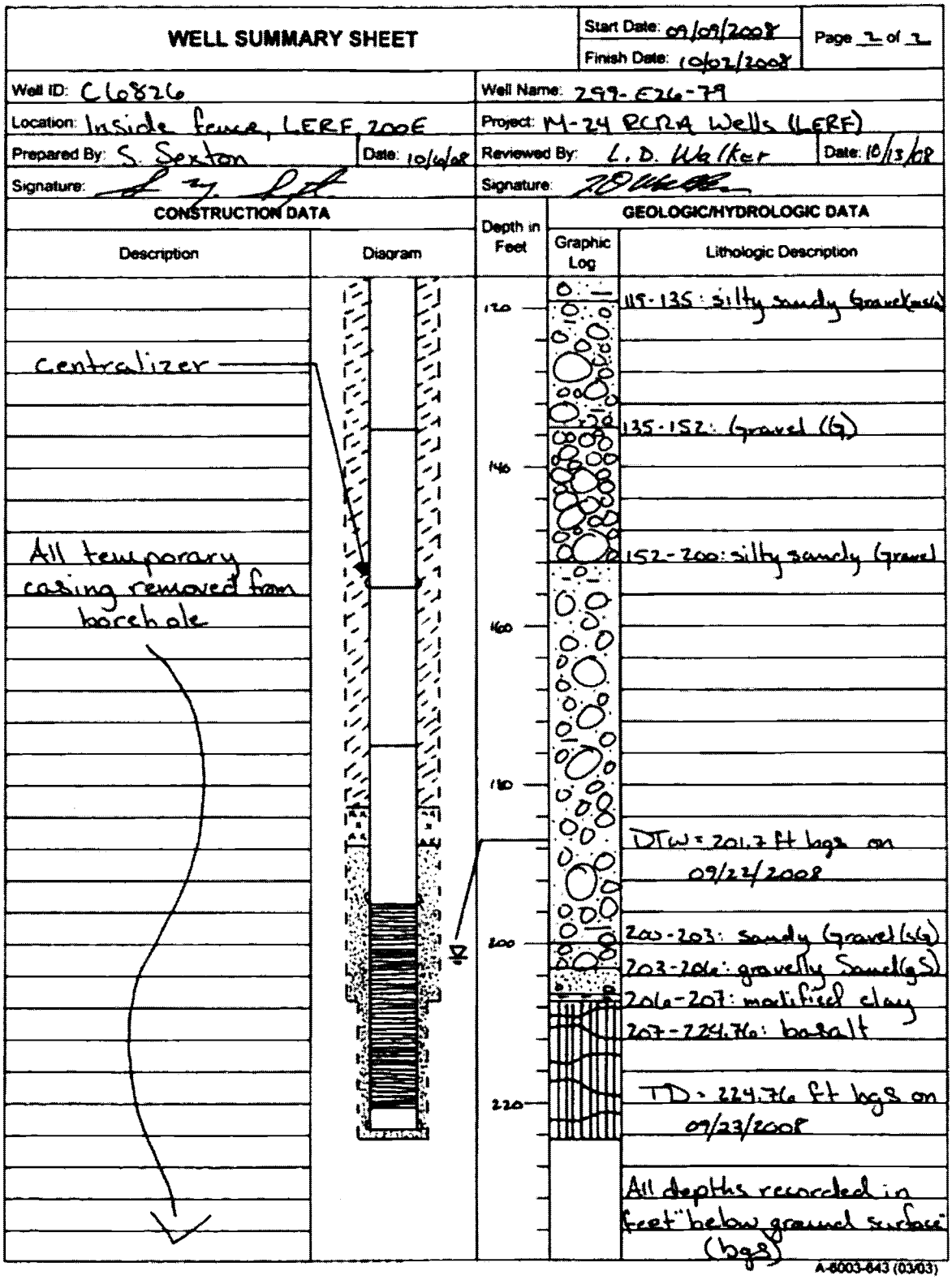
GEOLOGIC/HYDROLOGIC DATA

Depth in Feet	Graphic Log	Lithologic Description
0		0-1: gravel drill pad
		1-27: sandy silty (gravel)
20		22-27.5: sandy Silt (SM)
		27.5-29: silty (gravel) (mG)
		29-32: sandy Silt (SM)
		32-54: silty Sand (mS)
40		
		54-79: silty sandy (gravel) (msG)
60		
80		79-119: sandy silty (gravel) (smG)
100		

A-8003-843 (03/03)

1 **Figure D-27. Well Summary Sheet for Well 299-E26-79 in LERF Groundwater Monitoring Network**

2



1 **Figure D-27. Well Summary Sheet for Well 299-E26-79 in LERF Groundwater Monitoring Network (Cont.)**

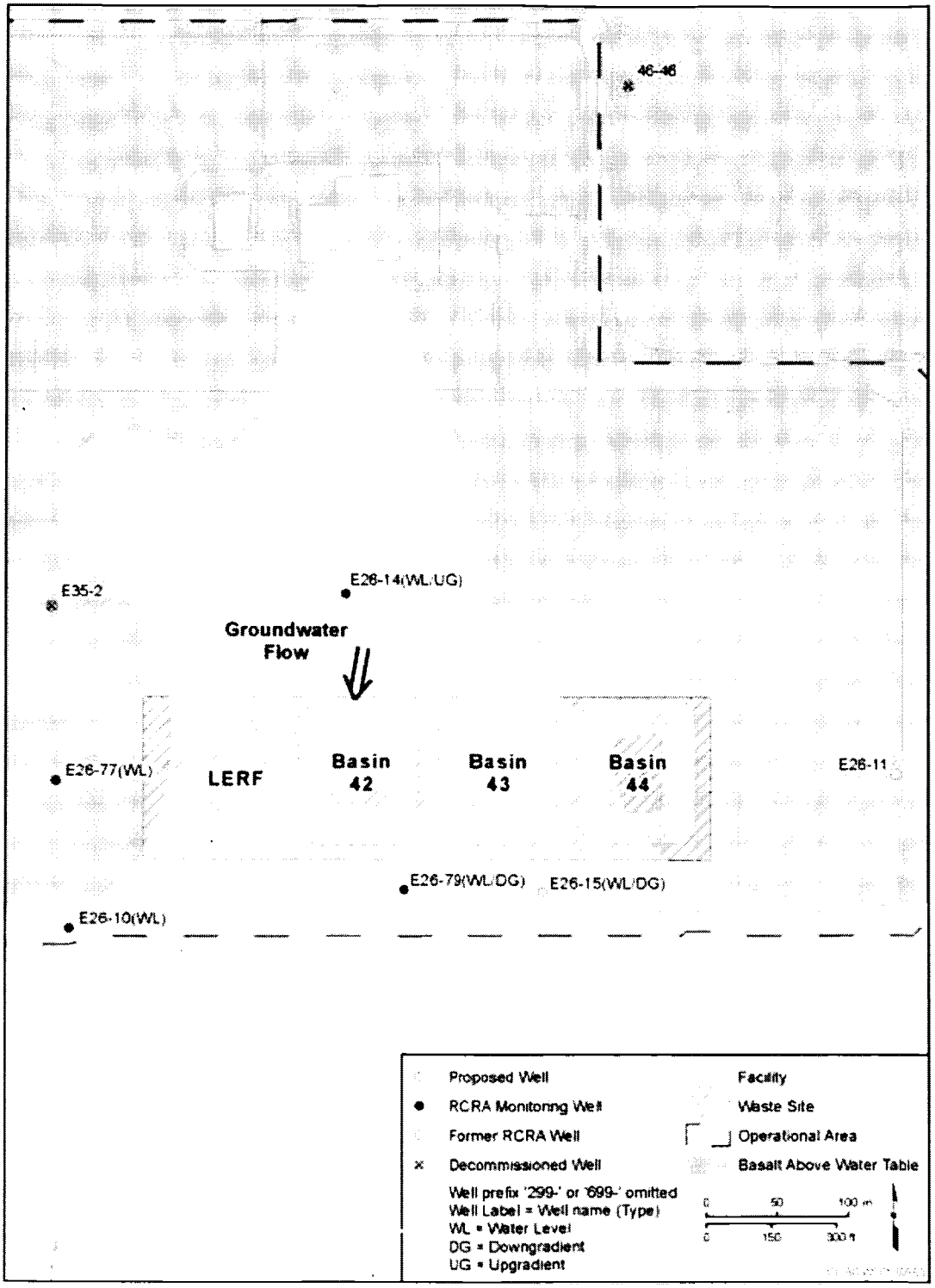


Figure D-28. Proposed LERF Well and Facility Description Map

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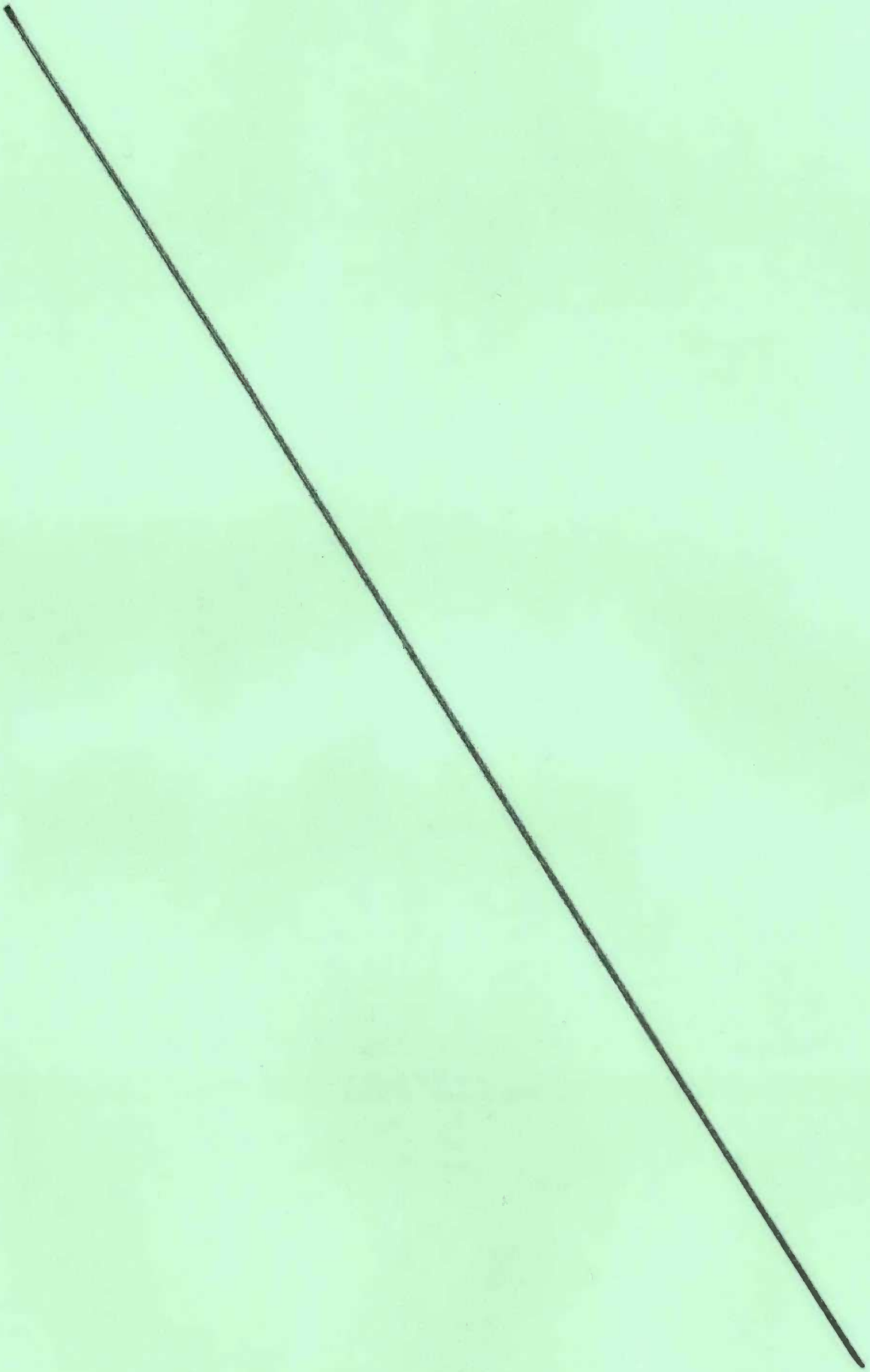

ADDENDUM E
SECURITY REQUIREMENTS

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ADDENDUM E
SECURITY REQUIREMENTS

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E. SECURITY REQUIREMENTS

The Liquid Effluent Retention Facility (LERF) and 200 Area Effluent Treatment Facility (200 Area ETF) are operated to minimize exposure of the general public and operating personnel to dangerous waste.

E.1 Security

Refer to Permit Attachment 3, Security. The LERF and 200 Area ETF are located within the 200 Area of the Hanford Facility and access is controlled by physical barriers, which complies with WAC 173-303-310(2)(c). Signs stating *Danger-Unauthorized Personnel Keep Out*, or equivalent language, legible at 25 feet or more, are posted near the entrance of each storage/processing area.

E.1.1 Waiver

Waiver of the security procedures and equipment requirements for LERF and 200 Area ETF are not requested, therefore; WAC 173-303-310(1)(a) and (b) are not applicable to LERF and 200 Area ETF.

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ADDENDUM F
PREPAREDNESS AND PREVENTION

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ADDENDUM F
PREPAREDNESS AND PREVENTION

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F.2 Preventive Procedures, Structures, and Equipment 6
F.2.1 Unloading Operations, Spill Prevention, and Control 6
F.2.2 Runoff 6
F.2.3 Water Supplies 7
F.2.4 Equipment and Power Failure 7
F.2.5 Personnel Exposure 8
F.3 Prevention of Reaction of Ignitable, Reactive, and Incompatible Waste 8

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F. PREPAREDNESS AND PREVENTION

F.1 Preparedness and Prevention Requirements

The following sections document the preparedness and prevention measures taken at the Liquid Effluent Retention Facility (LERF) and 200 Area Effluent Treatment Facility (ETF).

F.1.1 Equipment Requirements

The following sections describe the internal and external communications systems and the emergency equipment required that could be activated by the LERF and 200 Area ETF Building Emergency Director (BED).

F.1.1.1 Internal Communications

When operators are present at the LERF, the operators carry two-way radios to maintain contact with 200 Area ETF personnel. The operators at LERF are informed of emergencies (e.g., building and/or area evacuations, take-cover events, high airborne contamination, fire, and/or explosion), and are provided with emergency instructions by several systems. These systems include the mobile two-way radios, and the telephone in the LERF instrument building.

The 200 Area ETF is equipped with an internal communication system to provide immediate emergency instruction to personnel. The onsite communication system at the 200 Area ETF includes telephones, mobile two-way radios, a public address system, and alarm systems. The telephone and radio systems provide for internal and external communication. Alarm systems exist to allow personnel to respond appropriately to various emergencies, including building evacuations, take cover events, and fire and/or explosion. Addendum J provides additional information on the response activities.

F.1.1.2 External Communications

The LERF and its operators are equipped with devices for summoning emergency assistance from the Hanford Fire Department, the Hazardous Materials Response Team, and/or Hanford patrol, as necessary. External communication to summon emergency assistance is made by a normal telephone system or mobile two-way radios. The LERF telephone is available in the instrumentation building. The 200 Area ETF uses fire alarm pull boxes and telephones for external communication and are located at numerous locations throughout the 200 Area ETF.

F.1.1.3 Emergency Equipment

The LERF and 200 Area ETF rely primarily on the Hanford Fire Department to respond to fires and other emergencies as described in Permit Attachment 4, *Hanford Emergency Management Plan*, (DOE/RL-94-02). All LERF and 200 Area ETF operators are familiar with the LERF and 200 Area ETF contingency plans (Addendum J) and are trained in the use of emergency pumping of LERF and 200 Area ETF systems, fire, and communications equipment.

Portable fire extinguishers, fire control equipment, spill control equipment, and decontamination equipment is available at various locations in the 200 Area ETF.

The 200 Area ETF has fire extinguishers, automatic fire suppression systems (200 Area ETF control room and electrical room), fire alarm pull boxes, and a water spray system (200 Area ETF operating and administrative portions).

Respirators, hazardous material protective gear, and special work procedure clothing for 200 Area ETF personnel are kept in the change room at the 200 Area ETF. Safety showers are located in convenient locations in the 200 Area ETF, and emergency eyewashes are available for use. Water for these devices is supplied from the 200 Area ETF sanitary water system.

F.1.1.4 Water for Fire Control

A water main is not provided to the LERF. The Hanford Fire Department is equipped with fire engines for fire control for fires requiring high water volume and pressure. The 200 Area ETF is serviced by two 12-inch raw water lines that are tied into the 200 East Area raw water distribution grids. These lines provide a looped configuration that supplies two independent sources of raw water for fire protection and raw water uses. Connections from the 200 Area ETF raw water system supply fire hydrants and the wet pipe sprinkler system. In the event that water pressure is lost, the Hanford Fire Department is equipped with fire engines to provide needed water.

F.1.2 Aisle Space Requirement

The operation of the LERF does not involve aisle space. Nevertheless, the LERF and the individual basins are easily accessible to emergency response personnel and vehicles. A 6.1-meter-wide service road runs along the base of the basin area on the east, south, and west sides within the operational security fence.

Aisle spacing at 200 Area ETF is sufficient to allow the movement of personnel and fire protection equipment in and around the containers. This storage arrangement also meets the requirements of the National Fire Protection Association (NFPA 1996) for the protection of personnel and the environment. A minimum 30-inch aisle space is maintained between rows of containers as required by WAC 173-303-630(5)(c).

F.2 Preventive Procedures, Structures, and Equipment

The following sections describe preventive procedures, structures, and equipment.

F.2.1 Unloading Operations, Spill Prevention, and Control

Underground pipelines that transfer aqueous waste to and from the LERF are encased in a secondary pipe. If a leak is detected in a pipeline, flow in the pipeline will be stopped and the cause of the leak investigated and remediated.

If it is required to transfer aqueous waste from one LERF basin to another, existing transfer pumps are used as described in Addendum C.

The 200 Area ETF Load-in Station is monitored continuously during tank-filling operations and filling is stopped immediately if leaks occur. Care is taken to ensure that even minor leaks are cleaned up immediately and disposed of in accordance with approved management procedures. Any spill that is determined to be a dangerous waste will be managed according to the requirements of WAC 173-303.

F.2.2 Runoff

The LERF is constructed and operated to ensure that all aqueous waste is contained within the basins. The basins are designed and operated to prevent overtopping. Furthermore, the basins are provided with very low-density polyethylene floating covers to prevent the introduction of precipitation into the basins. The basins also are graded to ensure that all precipitation outside the basins is directed away from the surface impoundments.

The basins are constructed so that the top of the basin dikes are approximately 3 meters above grade. The exterior side slopes of the basins have a 2.25 (horizontal) to 1 (vertical) slope. Run-on of precipitation to the basins from the surrounding area is not possible because the surrounding area slopes away from the LERF.

Dangerous waste and hazardous chemical handling areas at the 200 Area ETF are designed to contain spills, leaks, and wash water, thereby preventing run-off and subsequent releases. All dangerous and/or mixed waste loading and unloading areas are provided with secondary containment structures as described in Addendum C, Process Information.

F.2.3 Water Supplies

The LERF uses operating practices, structures, and equipment to prevent the contamination of natural water supplies (i.e., groundwater and surface water). The LERF is monitored continuously in the ETF control room during liquid waste transfers and at least daily at other times to detect abnormal conditions (e.g., leaks), and regularly inspected to detect equipment and structural deteriorations that could allow possible water supply contamination. The basins are provided with a leachate collection system that is designed to contain any leachate generated. These systems, in conjunction with the double-composite liner system and underlying low permeable clay liner, ensure that should a release occur, the release will be fully contained within the basin configuration and, therefore, water supplies will be protected. Addendum J, Contingency Plan, provides information on procedures that are implemented if a release is detected at the LERF.

There are no drinking water wells near the 200 Area ETF. Therefore, a release would not immediately contaminate drinking water supplies. The 200 Area ETF uses operating practices, structures, and equipment to prevent the contamination of natural water supplies (i.e., groundwater and surface water). The 200 Area ETF is continuously monitored in the ETF control room during liquid waste processing operations and/or Load-In Station transfer to detect abnormal conditions and at least daily at other times , and is inspected regularly to detect equipment and structural deteriorations that could allow spills to the environment. Areas in contact with dangerous and/or mixed waste are monitored continuously in the ETF Control room during Load-in Station and/or ETF processing operations through a series of level and pressure indicators, leak detection alarms, equipment failure alarms, and control panel readouts. In addition, the 200 Area ETF is inspected regularly for the presence of leaks or other off normal conditions wherever possible (in all areas that can be safely entered).

In addition to detailed operating practices, structures and equipment are used at the 200 Area ETF to prevent contamination of water supplies. The structures and equipment designed to prevent contamination of water supplies are the same as the structures and equipment used to prevent run-off from dangerous and/or mixed waste handling areas.

F.2.4 Equipment and Power Failure

The storage function of the LERF is not affected by loss of power and a temporary loss of power would not pose a threat to the environment. Loss of electrical power would not cause the storage of the waste to be jeopardized. For process condensate transferred from the 242-A Evaporator, appropriate valving procedures are followed to ensure a smooth restart of the flow to the LERF in the event of a power failure at the 242-A Evaporator.

The 200 Area ETF does not have a standby power source. Power to selected lighting, computers, and process controls is configured with an uninterruptible power supply. During partial loss of normal power, the effected pumps and subsystems will be shut down. Complete loss of power to the 200 Area ETF shuts down the entire 200 Area ETF except for the instruments in the control room connected to the uninterruptible power supply. Redundant pumps allow the process to continue to operate when only one component is out of service.

When power at the 200 Area ETF is lost, the valves assume a fail-safe position to allow the process to remain in a safe shutdown mode until restoration of power. This action allows the operators to perform equipment surveys during shutdown and to confirm that there are no safety issues because the 200 Area ETF is shut down. Because a power failure would also shutoff flow into the 200 Area ETF, there will not be any increase in volume in any of the holdup basins, tanks, or other systems.

A combination of reliability, redundancy, maintenance, and repair features are used in the 200 Area ETF equipment and systems to minimize random failure of equipment. For crucial systems such as ventilation filters, redundant trains are provided to mitigate equipment and system failure. Spare parts are maintained for essential production and safety equipment.

F.2.5 Personnel Exposure

At the LERF and 200 Area ETF, operating practices, structures, and equipment are used to prevent undue exposure of personnel to dangerous and/or mixed waste. All personnel handling waste use protective clothing and equipment. All operations are conducted so that exposure to dangerous and/or mixed waste and hazardous materials are maintained as low as reasonably achievable (ALARA).

Protective clothing and equipment are prescribed for personnel handling chemicals or dangerous waste. Before the start of any operation that could expose personnel to the risk of injury or illness, a review of the operation is performed to ensure that the nature of hazards that might be encountered is considered and appropriate protective gear is selected. Personnel are instructed to wear personal protective equipment in accordance with training, posting, and instructions.

A change trailer at LERF is located between Basins 42 and 43. In addition, the change trailer has an operations office for working with procedures. Exits within the change trailer are clearly marked. A storage building is located within the perimeter fence, northwest of the basins. The LERF storage building also is provided with separate storage areas for clean and contaminated equipment. A decontamination shower and decontamination building is located at the 272-AW Building, approximately 1.6 kilometers from the LERF or at the 200 Area ETF.

The 200 Area ETF has eyewash stations and safety showers in convenient locations for use by personnel. The following structures and equipment were incorporated into the 200 Area ETF design to minimize personnel exposure.

- Offices, control room, clean- and soiled-clothes storage areas, change rooms, and the lunchroom are situated to minimize casual exposure of personnel.
- Building exit pathways are located to provide rapid egress in emergency evacuations.
- Emergency lighting devices are located strategically throughout the 200 Area ETF.
- Audio and/or visual alarms are provided for all room air samplers, area alarms, and liquid monitors. Visual readouts for these alarm systems are located in less contaminated areas to minimize exposure to personnel.
- Areas for decontaminating and maintaining equipment are provided in contaminated areas to limit the spread of contamination to uncontaminated areas such as the control room.
- Instrument interlock systems automatically return process operations to a safe condition if an unsafe condition should occur.
- The 200 Area ETF ventilation systems are designed to provide airflow from uncontaminated zones to progressively more contaminated zones.

Whenever possible, exposures to hazards are controlled by accepted engineering and/or administrative controls. Protective gear is used where effective engineering or administrative controls are not feasible.

F.3 Prevention of Reaction of Ignitable, Reactive, and Incompatible Waste

Typically, aqueous waste managed at the LERF or 200 Area ETF does not display the characteristics of reactivity or ignitability. Any aqueous waste streams exhibiting these characteristics are blended or mixed at LERF to a concentration where the waste no longer exhibits reactive or ignitable characteristics.

Incompatible aqueous waste is not expected to be stored or treated at the LERF or 200 Area ETF (Addendum B, Waste Analysis Plan). Therefore, the requirements of WAC 173-303-806(4)(a) are not applicable.

ADDENDUM G
PERSONNEL TRAINING

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ADDENDUM G
PERSONNEL TRAINING

Specific requirements for the Hanford Facility Personnel Training program are described in Permit Attachment 5. The Permittees will comply with the training matrix below which provides training requirements for Hanford Facility personnel associated with the Liquid Effluent Retention Facility (LERF) and the 200 Area Effluent Treatment Facility (ETF). Refer to the LERF & 200 Area ETF Dangerous Waste Training Plan (DWTP) for a complete description of the personnel training requirements. As required by Permit Condition II.I.1, a copy of the LERF & 200 Area ETF DWTP will be placed in the Hanford Facility Operating Record, LERF & 200 Area ETF file, and will be updated by the Permittee as unit-specific conditions change. Training received by facility personnel will be commensurate with the duties they perform. Individuals are not required to receive training for work/duties they do not perform.

LERF and 200 Area ETF Training Matrix

	Training Category						
Permit Attachment 5, Training Category	**General Hanford Facility Training**	**Contin-gency Plan Training**	**Emergency Coordi-nator Training**	**Operations Training**			
LERF & 200 Area ETF DWTP implementing category	Orient-ation Program	Emergency Response (contingency plan)	Emergency Coordinator Training	General Waste Manage-ment	Contain er Manage-ment	Tank System Manage-ment	Surface Impound-ment
JOB TITLE/POSITION							
Nuclear Chemical Operator (NCO)	X	X		X	X	X	X
Operations supervisor Shift Operations Manager (SOM)	X	X	X				
Engineer/scientist	X			X			
Environmental Field Representative	X			X			
Waste Service Provider	X			X	X		
Sampler	X			X			X

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ADDENDUM H
CLOSURE PLAN

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ADDENDUM H
CLOSURE PLAN

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H. CLOSURE PLAN

This addendum describes the planned activities and performance standards for closing Liquid Effluent Retention Facility (LERF) and 200 Area Effluent Treatment Facility (ETF).

H.1 Closure Plan

The LERF and 200 Area ETF will be closed by removal or decontamination with respect to dangerous waste contamination that resulted from operation asTtreatment, Storage, and Disposal (TSD) units, with closure of LERF occurring first. To facilitate closure, the LERF retention basins are being viewed as consisting of seven components: the covers and primary liner, drainage layer system/bentonite carpet liner, secondary liner, soil/bentonite, internal and/or external piping, ancillary equipment, and concrete basins. To facilitate closure of 200 Area ETF, the 200 Area ETF is being viewed as consisting of six components: tanks, internal and/or external piping, ancillary equipment, concrete floors/dikes/ encasements, structures, and soil directly beneath the structure. It is anticipated that closure of LERF and 200 Area ETF will begin after the projected 30-year active life of LERF and 200 Area ETF. If it is determined that closure by removal or decontamination is not possible, the closure plan will be modified to address required post closure activities.

Uncontaminated structures will be left for future use or disassembled, dismantled, and removed for disposal. Uncontaminated equipment and structures could include aqueous makeup, HVAC and piping, steam condensate and cooling water piping, and the control room and office areas.

Closure by removal or decontamination requires decontamination or removal and disposal of all dangerous waste, waste residues, contaminated equipment, soil, or other material established in accordance with the removal or decontamination closure performance standards of WAC 173-303-610(2). This and future closure plan revisions will provide for compliance with these performance standards.

H.2 Closure Performance Standard

Closure by removal or decontamination, as provided for in this plan based on the requirements of WAC 173-303-610(2), will eliminate future maintenance and will be protective of human health and the environment by removing or reducing chemical contamination at LERF and 200 Area ETF to levels that are below concern with respect to human health and the environment.

This plan proposes to leave clean structures and equipment in place after closure for potential use in future operations. This need will be evaluated at the time of closure.

H.2.1 Closure Standards for Metal Surfaces, Rubber, Tanks, and Concrete

This closure plan proposes use of a 'clean debris surface' (defined in the following paragraph) as the clean closure performance standard for the metal surfaces, rubber (i.e., basin covers, liners, etc.), tanks, and concrete that will remain after closure. This approach is consistent with the Washington State Department of Ecology (Ecology) guidance (Publication #94-111, Ecology 2005) for achievement of clean closure. Additionally, adherence to this guidance ensures that all residues have been removed as required by WAC 173-303-640 for closure of the 200 Area ETF tank systems.

The clean debris surface standard is verified visually.

> *A clean debris surface means the surface, when viewed without magnification, shall be free of all visible contaminated soil and hazardous waste except residual staining from soil and waste consisting of light shadows, slight streaks, or minor discolorations and soil and waste in cracks, crevices, and pits may be present provided that such staining and waste and soil in cracks, crevices, and pits shall be limited to no more than 5% of each square inch of surface area* (40 CFR 268.45).

When a physical extraction method is used on concrete, the performance standard is based on removal of the contaminated layer of debris. The physical extraction performance standard for concrete is removal of 0.6 centimeter of the surface layer and treatment to a clean debris surface. Inspections to verify achievement of a clean debris surface will be performed and documented.

H.2.2 Closure Standards for Piping and Ancillary Equipment

The internal and external piping of both LERF and 200 Area ETF that has contacted dangerous waste will be flushed and drained as part of closure. When practical, ancillary equipment, which has contacted dangerous waste will also be flushed and drained. For piping and ancillary equipment where the contaminated surfaces can be inspected, an inspection will be performed to see if the surfaces meets the clean debris surface standard in 40 CFR 268.45, incorporated by reference by WAC 173-303-140, and can be declared non-dangerous in accordance with WAC 173-303-071(3)(qq). If it is not possible to inspect the contaminated surfaces or meet the clean debris surface performance standard, the particular piping or ancillary equipment of concern will be removed, designated, and disposed of accordingly.

Dangerous and/or mixed-waste materials generated during closure activities will be managed in accordance with WAC 173-303-610(5). Removal of any dangerous wastes or dangerous constituents during partial or final closure will be handled in accordance with applicable requirements of WAC 173-303-610(5).

H.2.3 Closure Standards for Underlying Soils

The LERF retention basins have a leachate collection system that channels the leachate to sumps at the bottom of the basins. The collected liquid is pumped back into the basins, thereby limiting fluid head on the secondary liner. The secondary liner is comprised of several protective layers, including a high-density polyethylene geomembrane and a soil/bentonite admixture. The soil below the LERF only could be contaminated if the layers of the secondary liner had failed. The primary liner and the drainage gravel, geotextile, and geonet between the primary and secondary liners cannot easily be decontaminated. The high-density polyethylene layer of the secondary liner also cannot be decontaminated. These materials will be removed and disposed according to the requirements of WAC 173-303-170. The soil/bentonite admixture will be sampled and analyzed for constituents of concerns according to the sampling and analysis plan developed prior to the time of closure. If the analytical results determine that the constituents of concern are at or below the levels in WAC 173-303-610(2)(b)(i), or background levels for Hanford soil if background is greater, the soil/bentonite admixture and the soil below LERF will be considered clean closed.

Clean closure of soil under the 200 Area ETF will be accomplished by demonstrating that the coated concrete floor kept contaminants from reaching the soil. The coated concrete floor provided secondary containment for all the tanks and process piping. Unless inspections identify potential through-thickness cracks indicating containment failure and a subsequent potential for soil contamination from TSD unit operations, the soil will be considered clean closed. However, if inspections identify such cracks and there have been documented spills in the vicinity, potential soil contamination will be investigated. Soils will be sampled and analyzed for constituents of concern according to the sampling and analysis plan. The sampling and analysis plan will be prepared following the completion of a data quality objectives process in accordance with EPA/600/R-96/055 (QA/G-4), *Data Quality Objectives Process*, as amended. The data quality objectives process will be initiated prior to closure on a schedule to ensure timely closure of LERF. The sampling and analysis plan will be submitted to Ecology as part of a permit modification request meeting the requirements of WAC 173-303-830. The sampling and analysis plan will be prepared consistent with EPA/240/B-01/003 (EPA/QA R-5), *EPA Requirements for Quality Assurance Project Plans*, as amended.

If the soil analytical results determine that the constituents of concern are at or below the levels in WAC 173-303-610(2)(b)(i), or background levels in the Hanford soil if background is greater, the soil will be considered clean closed. If the constituents of concern exceed background levels, the soil will be closed per the standards of WAC 173-303-610(2)(b).

H.3 Closure Activities

The LERF and 200 Area ETF were designed for a 30-year active life.

At the time of closure, the closure plan will be modified as necessary to reflect current regulation or informational revisions in accordance with WAC 173-303-610(3)(b). If it is determined that clean closure is not possible, the closure plan will be modified to address required post closure activities.

H.3.1 General Closure Activities

The approach to LERF closure is to dispose of accumulated basin aqueous waste by processing the waste through 200 Area ETF. Primary basin liners, covers, drainage gravel, geonets, and secondary High Density Polyethylene (HDPE) liners will be removed, designated, and disposed of as described in Section G.3.4.1. Any remaining solids (residue) within the basins will also be removed, designated, and disposed of accordingly. Piping associated with LERF closure is intended to be decontaminated, drained, and inspected. Piping that meets the closure standard in Section G.2.2 will be left in place. Piping that does not meet the closure standard, or cannot be inspected, will be disposed of accordingly. Rinsate generated during decontamination also will be disposed of through 200 Area ETF. Sampling will assess whether contamination beneath the secondary HDPE liner has occurred. Contamination above background levels, if present, will be removed or decontaminated to meet the regulatory requirements of WAC 173-303-610(2)(b).

The approach to 200 Area ETF closure is to process any aqueous waste through the effluent treatment system. Any waste, which cannot be treated at 200 Area ETF as the facility is being closed, will be transferred to other TSD units or off-site TSD facility. Piping will be rerouted and temporary piping installed to allow the isolation of tanks and ancillary equipment for draining, decontamination, and closure. Rerouted and temporary piping will be closed in the same manner as process piping. All structures and equipment will be decontaminated to the closure standards in Section G.2.2 or disposed. Piping associated with 200 Area ETF closure is intended to be decontaminated, drained, and inspected. Piping that meets the closure standard in Section G.2.2 will be left in place. Piping that does not meet the closure standard, or cannot be inspected, will be disposed of accordingly. Contamination, if present, will be managed in compliance with regulatory requirements.

Equipment or materials used in performing closure activities will be decontaminated or disposed at a permitted facility.

H.3.2 Constituents of Concern for Closure for the Liquid Effluent Retention Facility and 200 Area Effluent Treatment Facility

Using the list of dangerous waste numbers in the Addendum A, Part A Form, constituents in the final delisting in 40 CFR 261 Appendix IX, sample results from wastes added to LERF and 200 Area ETF, process knowledge and the risk to human health and the environment, the constituents of concern for closure will be determined through the data quality objective process. Based on constituents in wastewater received at LERF from 2000 to 2006 which are present at five percent of their delisting levels or higher, the constituents of concern are:

- Acetone
- Ammonia
- Barium
- Chromium
- Carbon tetrachloride
- Fluoride
- Lead
- Mercury
- Methyl ethyl ketone
- n-Butyl alcohol
- Total cresols
- Tributyl phosphate
- Vanadium

Arsenic and beryllium are excluded because they are present in Hanford soils and may therefore give a false positive sample result. Constituents of concern vary in each basin. For example, ammonia may be present only in LERF Basin 42. The constituents of concern for each basin will be determined by process knowledge as part of the Data Quality Objectives process for the Sampling and Analysis Plan.

H.3.3 Removing Dangerous Waste

At the start of LERF closure, aqueous waste will be transferred sequentially from each basin to another LERF basin or to 200 Area ETF for treatment.

At a pump rate of about 284 liters per minute, it will take approximately 60 days to empty a full basin. Basin covers will remain in place to prevent possible wind dispersion of waste until all basin waste has been removed.

All of the aqueous waste inventory at the 200 Area ETF will be processed before closure. Any residue remaining in piping, equipment, or the LERF liner will be removed to an appropriate disposal unit. All containerized waste will be dispositioned. All secondary waste in containers will be transferred to an appropriate TSD unit.

H.3.4 Decontaminating Structures, Equipment, and Soils

This section discusses the activities necessary to implement a clean closure strategy for the LERF and 200 Area ETF.

H.3.4.1 Covers and Primary Liner

The following steps will be performed to close each LERF basin cover and primary liner:

- Wastewater will be removed from the basins and transferred to another LERF basin or to 200 Area ETF. Additional pumps and piping may be installed to empty the basin as low as possible.
- The basin cover will be cut into pieces and disposed in containers.
- As much as practical of the remaining residue within the basins will be removed and transferred to containers, another LERF basin, or 200 Area ETF. Rinsing may be performed to facilitate removal.
- The pipe risers, transfer pump, HDPE primary liner and bentonite carpet liner will be cut into pieces and disposed in containers.

H.3.4.2 Drainage Layer and Secondary Liner

The following steps will be performed to close each LERF basin drainage layer and secondary liner:

- The drainage gravel, geotextile, and geonet will be cut into pieces, and disposed in containers.
- As much as practical of the remaining residue on the secondary liner will be removed and transferred to containers, another LERF basin or 200 Area ETF. Rinsing may be performed to facilitate removal of residue.
- The HDPE liner portion of the secondary liner will be visually inspected for physical damage. This will provide potential sampling locations to determine if the soil/bentonite below the HDPE liner may be clean closed.
- The leachate pump, pump riser, and HDPE liner portion of the secondary liner will be removed, cut into pieces, and disposed in containers.
- The soil/bentonite portion of the secondary liner will be visually inspected for signs of contamination. This will provide potential sampling locations to determine if the soil/bentonite may be clean closed.

Assessment of contamination beneath the LERF's secondary liner will be performed within each basin by sampling the top surface of the 97-centimeter thick layer of soil/bentonite. Biased and random location selection will be used to increase the probability of detecting leachate contamination. Some sampling points will be chosen randomly, while others will be chosen where physical damage was noted during the inspection of the secondary HDPE liner and soil/bentonite layer, and in areas where the underlying material porosity and permeability and the hydraulic head would most likely drive any leachate. The leakage rate through the liner would increase toward the bottom of the liner as hydraulic head increases. Any leakage that did occur in the sloped sides could be expected to travel down slope through the geotextile between the primary and secondary liner until reaching the bottom of the liner.

Therefore, the most likely area of contamination would be the soil/bentonite in the leachate sump and at the bottom of the basin. Sampling and disposal objectives will be determined at the time prior to closure activities through the data quality objectives process. The sampling and analysis plan will be prepared following the completion of a data quality objectives process in accordance with EPA/600/R-96/055 (QA/G-4) *Data Quality Objectives Process*, as amended.

The data quality objectives process will be initiated prior to closure on a schedule to ensure timely closure of LERF. The sampling and analysis plan will be submitted to Ecology as part of a permit modification request meeting the requirements of WAC 173-303-830. The sampling and analysis plan will be prepared consistent with EPA/240/B-01/003 (EPA/QA R-5), *EPA Requirements for Quality Assurance Project Plans*, as amended.

Sampling of the soil/bentonite will be performed in accordance with the sampling methods allowed for in WAC 173-303-110(2). Special care will be needed in sampling for volatiles. To aid in ensuring sample integrity, the initial sampling of the soil/bentonite may proceed while the secondary HDPE liner is in the process of being removed.

If no constituents of concern are found above soil closure performance standards (Section G.2.3), no further analysis will be done. If the initial sample analysis indicates liner leakage, additional samples from different depths and locations will be taken to determine the spatial extent of contamination. The soil/bentonite will be removed in the area around the contamination and placed in containers. If contamination is found to extend through the entire depth of the soil/bentonite layer, soil beneath the basin that is contaminated above closure performance standards will also be removed and placed in containers.

H.3.4.3 Tanks

The following general steps will be performed to close, each 200 Area ETF tank and ancillary equipment:

- Wastewater and chemical additions to the tank will be isolated or rerouted to a downstream tank.
- Piping and ancillary equipment associated with the tank will be flushed with water and drained to the tank being closed, to another tank, or to containers.
- Wastewater will be removed from the tank and transferred to another tank. Additional pumps and piping may be installed to empty the tank as low as possible.
- All remaining residue at the bottom of the tank will be removed and transferred to another tank or containers. Rinsing may be performed to facilitate removal of residue.
- An initial visual inspection of the tank's interior and exterior surfaces will be performed to determine the type of flushing that will allow the tank to be clean closed, or whether the tank cannot be clean closed.
- The tank's surfaces, piping and ancillary equipment will be cleaned by chemical or physical extraction techniques described in 40 CFR 268.45. Flush solution will be transferred to another tank or containers. All flush solution at the bottom of the tank will be removed before visual inspection.
- The tank, piping, and ancillary equipment will be inspected visually for compliance with the performance standard in Sections H.2.1 and H.2.2.

Closure will begin with the Load-In Station tanks, surge tank, and other tanks of the main treatment train. The secondary treatment train will operate as long as possible to reduce the volume of flush water requiring disposal. Condensate from the secondary treatment train will be routed to the main treatment train or the verification tanks for storage or treatment.

After rinsing, the tanks will be inspected visually for compliance with the performance standard. Visual inspection might be made remotely using a camera or other device that allows verification of meeting the performance standard.

If any areas are found not meeting the clean debris surface performance standard, these areas will be decontaminated in-place, or the contaminated portions will be removed, designated, and disposed accordingly. Per 40 CFR 268.45, Table 1 incorporated by reference at WAC 173-303-140, only removal of contaminants from the surface layer is necessary for metal surfaces.

The outside of the tanks also will be inspected for compliance to the performance standard. Any areas found not to meet this performance standard will be decontaminated in-place, or the contaminated portions will be removed, designated, and disposed accordingly.

Before using decontamination solutions on the outside of the tanks, the floor will be inspected for cracks or other openings that could provide a pathway to soil. This inspection will be performed as described in Section G.2.3 in conjunction with mapping of potential through-thickness cracks. Any such cracks will be mapped. The cracks will be sealed before beginning treatment or other engineered containment devices (e.g., portable catch basins, liners) will be used to collect and contain solutions.

Decontamination residues will be collected, designated, and managed as appropriate. If it is not possible to meet the clean closure performance standard, contaminated portions of the tanks could be removed, designated, and disposed of accordingly. The inspections for a clean debris surface will be documented on an inspection record.

H.3.4.4 Internal and External Piping and Ancillary Equipment

The internal piping and ancillary equipment for both LERF and 200 Area ETF, which have contacted dangerous waste will be flushed and drained as part of closure. Any treatment media, such as filters, reverse osmosis membranes, ion exchange resins, will be removed from the ancillary equipment, and disposed of accordingly. Where the contaminated surfaces can be inspected, an inspection will be performed to see if the piping and ancillary equipment meet the clean debris surface standard in 40 CFR 268.45 and can be declared non-dangerous. If it is not possible to meet the clean debris surface standard or the piping or ancillary equipment cannot be inspected, those portions of the piping and ancillary equipment will be removed, designated, and disposed of accordingly.

External piping (transfer lines) associated with LERF and 200 Area ETF consist of below grade and above grade piping. Below grade, piping will be dispositioned at closure consistent with the practices for below grade piping in the 200 Areas at the time of closure consistent with the 200-IS-1 operable unit decisions. Above grade piping will be dispositioned consistent with the provisions for internal piping.

Rinsate from the LERF and 200 Area ETF external piping and LERF internal piping will be processed through 200 Area ETF. Dangerous and/or mixed-waste solutions and materials generated during closure activities, which cannot be treated at 200 Area ETF will be managed in accordance with WAC 173-303-610(5).

H.3.4.5 Concrete

At LERF, the concrete catch basins are located at the northeast corner of each retention basin, where inlet pipes, leachate risers, and transfer pipe risers emerge for the basin. The concrete catch basin is curbed, and coated with a chemical resistant epoxy sealant. The concrete catch basin is sloped so that any leaks or spills from the piping or connections will drain into the basin. At the 200 Area ETF, the coated concrete floor and berm provides secondary containment for all the tanks and process piping.

Closure of concrete at LERF and 200 Area ETF will be performed after the associated tanks, piping, ancillary equipment, and structures have been closed. All concrete will be inspected visually and surveyed before any decontamination. The purpose of the inspection will be twofold: to identify and map any cracks in the concrete that might have allowed contaminants a pathway to the soil below (Section G.2.3.), and to identify areas that potentially are contaminated with dangerous waste or dangerous waste residues. The inspection standard will be a clean debris surface as defined in Section G.2.1. The inspection of the concrete for a clean debris surface will be documented on an inspection record. Those areas already meeting the standard can be clean closed as is.

Those potentially contaminated areas will undergo decontamination to meet the clean closure standard of a clean debris surface. The concrete will be washed down; the rinsate collected, designated, and disposed of accordingly. The concrete will be reinspected for a clean debris surface. Concrete surfaces indicated by visual examination, as still being potentially contaminated will have the surface layer removed to a depth of 0.6 centimeter by scabbing or other approved methods. This will not threaten the environment, even if potential through-thickness cracks had been found during the inspection, because concrete decontamination (scabbing) will not employ liquid solutions that could enter cracks and because scabbing residues will be vacuumed away from cracks as, any residue is generated.

Achievement of a clean debris surface will be documented on an inspection record. Decontamination residues will be collected, designated, and managed as appropriate.

H.3.4.6 Structures

If contaminated with either dangerous or mixed waste constituents, the 200 Area ETF structures will be decontaminated and/or disassembled, if necessary, packaged, and disposed of in accordance with existing land disposal restrictions (WAC 173-303-140).

Closure steps could include the following activities.

- Containerize (as necessary and practicable) and remove any remaining waste.
- Review operating records for spillage incidents and visually inspect storage area surfaces for evidence of contamination or for cracks that could harbor contamination or allow the escape of decontamination solutions. Inspect storage area surfaces for visible evidence of contamination (e.g., discoloration, material degradation, wetness, and odor). If contamination is evident, the affected area(s) will be decontaminated.
- Decontaminate 200 Area ETF walls and floors to minimize the potential for loose contamination and facilitate any required surveys and/or chemical field screening. The structures could be cleaned by water rinse or high-pressure, low-volume steam cleaning coupled with a detergent wash. After decontamination, the walls and floors will be compared to closure performance standards.
- Collect rinsate and manage as dangerous waste for appropriate disposal.
- Secure (lock) personnel entries into building and post doors with appropriate warning signs.

H.3.4.7 Underlying Soils

Clean closure of soil under LERF's secondary liner will be accomplished by demonstrating that the liners and leak detection system kept contaminants from reaching the soil. The secondary liner provided secondary containment for the LERF basins. Unless inspections identify potential leaks, punctures, cracks, or tears indicating containment failure and a subsequent potential for soil contamination from TSD unit operations, the soil will be considered clean closed. However, if inspections identify such leaks, punctures, etc., potential soil contamination will be investigated.

Clean closure of soil under 200 Area ETF will be accomplished by demonstrating that the coated concrete floor kept contaminants from reaching the soil. The coated concrete floor and bermed area provided secondary containment for all the tanks and process piping. Unless inspections identify potential through-thickness cracks indicating containment failure and a subsequent potential for soil contamination from TSD unit operations, the soil will be considered clean closed. However, if inspections identify such cracks and there have been documented spills in the vicinity, potential soil contamination will be investigated.

Where it is possible visually to inspect directly beneath the tanks, a visual inspection will be performed. Where it is not possible visually to inspect beneath the tanks, an evaluation of the tank integrity will be made. The condition of the tank will be evaluated to determine if there was any potential for leakage. If no cracks, severe corrosion, or evidence of leaks is observed, it will be reasoned that mixed or dangerous waste solutions could not have penetrated to the soil directly below the tank.

External piping (transfer lines) between the 242-A Evaporator and LERF and 200 Area ETF are double lined with a leak detection system. If records indicate that no leaks from the primary piping occurred, the soil will be considered clean with respect to RCRA closure.

Where there is evidence that contamination may have leaked into the soil below tanks, concrete, or the soil/bentonite layer at LERF, the contaminated tank, concrete, or soil/bentonite layer will be removed to allow the underlying soil to be sampled to determine the depth of the contamination. Soil that is contaminated above the closure performance standards in Section G.2.3 will be removed, placed in containers, and disposed accordingly.

H.4 Maximum Waste Inventory

The maximum waste inventory for LERF and 200 Area ETF is in Addendum A.

H.5 Closure of Containers, Tanks, and Surface Impoundments

The following sections cover closure of containers, closure of tanks, and closure of surface impoundments.

H.5.1 Closure of Containers

Containers at 200 Area ETF will be used to contain dangerous waste in the event of a spill, unexpected release, or equipment failure. Containers will be used to accumulate nonradioactive dangerous waste and/or mixed wastes. All containers will be emptied and treated prior to closure of 200 Area ETF. Any containers used to contain dangerous and/or mixed waste at the 200 Area ETF that is generated during the closure process and therefore cannot be treated at 200 Area ETF will be designated and shipped to an onsite TSD unit or off-site TSD facility. Containers of dangerous and/or mixed waste will not be left in the 200 Area ETF after closure.

H.5.2 Closure of Tanks

Clean closure of 200 Area ETF will consist of the removal and disposal of all dangerous waste and the decontamination and/or removal and disposal of equipment which does not meet the performance standards in Section G.2, including tanks. The 200 Area ETF was designed to incorporate removable components. This design facilitates closure by allowing complete removal of equipment, which does not meet the performance standards.

H.5.3 Closure of Surface Impoundments

At closure, all of LERF that received regulated waste will be closed in accordance with the requirements of this approved closure plan, which are intended to ensure compliance with the requirements of WAC 173-303-650(6)(a)(i). All equipment, structures, and other material associated with closure of LERF will be decontaminated or removed in accordance with WAC 173-303-610(2). All basin waste and decontamination rinsate will be transferred to 200 Area ETF. Sampling and testing will be conducted as described in Section G.3.4.2.

H.6 Schedule for Closure

Closure of LERF and 200 Area ETF is not anticipated to occur within the next 30 years. The actual year of closure will depend on the time required for current waste to be processed and what role the LERF and 200 Area ETF will play in processing additional waste generated during future activities in the 200 Areas. Other factors affecting the year of closure include changes in operational requirements, lifetime extension upgrades, and unforeseen factors. When a definite closure date is established, notification of closure will be provided in accordance with Permit Condition II.J.1.

The activities required to complete closure are planned to be accomplished within 180 days in accordance with WAC 173-303-610(4)(b). Should a modified schedule be necessary, a revised schedule will be proposed through the permit modification procedure in accordance with WAC 173-303-610(4)(b).

ADDENDUM I
INSPECTION REQUIREMENTS

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ADDENDUM I
INSPECTION REQUIREMENTS

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I. INSPECTION REQUIREMENTS

I.1 Inspection Plan

This addendum describes the method and schedule for inspections of the Liquid Effluent Retention Facility (LERF) and 200 Area Effluent Treatment Facility (ETF). The purpose of inspections is to help ensure that situations do not exist that might cause or lead to the release of dangerous and/or mixed waste that could pose a threat to human health and the environment. Abnormal conditions identified by an inspection will be corrected on a schedule that prevents hazards to workers, the public, and the environment.

I.1.1 General Inspection Requirements

The content and frequency of inspections are described in this section. Inspection records are retained in the Hanford Facility Operating Record, LERF and 200 Area ETF file, or other approved locations, in accordance with Permit Condition II.I.1.

In certain areas of the 200 Area ETF, many inspections are performed remotely to maintain as low as reasonably achievable (ALARA) exposure. Monitoring instruments are connected to audible alarms and visual indicators track alarm status. The monitoring system provides trending of selected monitoring data, graphics, and equipment summary displays.

A preventive maintenance recall system is employed to direct preventive maintenance activities at the LERF and 200 Area ETF. Equipment requiring maintenance is checked as indicated by the maintenance history and the manufacturer's recommendations. The preventive maintenance of certain equipment might not be possible if the LERF or the 200 Area ETF is in an operational mode. Thus, the preventive maintenance could be performed slightly earlier or later than planned to minimize impact on operations.

Instrumentation at 200 Area ETF is calibrated regularly to ensure accuracy and reliability. All process control instrumentation is calibrated on a schedule depending on previous calibration experience. An instrument calibration and recall system is employed to manage calibrations.

I.1.1.1 Types of Problems

Key components of the LERF inspection program include the following areas:

- Structural integrity of the basins.
- Catch basin secondary containment system integrity.
- Evidence of release from basins.
- Safety, communications, and emergency equipment.

Key components of the 200 Area ETF inspection program include the following areas:

- Condition of tanks and ancillary piping.
- Condition of containers.
- Condition of the process control equipment.
- Condition of emergency equipment.
- Condition of secondary containment.

Table I.1 and Table I.2 provide a description of LERF and 200 Area ETF items to be inspected.

I.1.1.2 Frequency of Inspections

The frequency of inspections is based on the rate of possible deterioration of equipment and the probability of a threat to human health or the environment.

The LERF and 200 Area ETF is inspected as indicated in Table I.1 and Table I.2.

I.1.2 Specific Process Inspection Requirements

The following sections describe the specific process inspections performed at LERF and 200 Area ETF.

I.1.2.1 Container Inspections

Containers are used at the 200 Area ETF to store solidified secondary waste, such as the powder waste from the thin film dryer and maintenance and operations waste. When containers are being held in container storage areas, the following inspection schedule is maintained:

- Daily visual inspection of container storage area for leaks, spills, accumulated liquids, and open or improperly sealed containers.
- Weekly visual inspection of container labels to ensure labels are not obscured, removed, or otherwise unreadable.
- Weekly visual inspection for deterioration of containers, containment systems, or cracks in protective coating or foundations caused by corrosion, mishandling, or other factors.

Following the inspections, an inspection datasheet is signed and dated by the inspector and supervisor.

I.1.2.2 Tank Inspections

A description of the tank systems and ancillary equipment at the 200 Area ETF is given in Addendum C. Inspections and frequencies are given in Table I.1 and Table I.2. This section includes a brief discussion of the inspections.

I.1.2.2.1 Overfill Protection

Tanks that have the possibility of being overfilled have level instrumentation that alarms before the tanks reach overflow. High tank level alarms annunciate in the control room, allowing operating personnel to take immediate action to stop the vessels from overfilling. These alarms are monitored continuously in the control room during solution transfers.

I.1.2.2.2 Visual Inspections

Visual inspections of tanks and secondary containments are performed to check for leaks, signs of corrosion or damage, and malfunctioning equipment. Inspections are performed on tanks, secondary containment within the 200 Area ETF, surge tank, and verification tank, and associated secondary containment.

I.1.2.2.3 Secondary Containment Leak Detectors

The surge tank and verification tank secondary containment systems have sloped floors that drain solutions to sumps equipped with leak detectors that alarm in the control room. These alarms are monitored continuously in the control room during ETF processing operations and at least daily at other times. If an alarm is activated, further investigation is performed to determine if the source is a tank leak or other solution (i.e., precipitation).

I.1.2.2.4 Integrity Assessments

The initial integrity assessment was issued in 1995 (Addendum C). Consistent with the recommendations of the integrity assessment, a periodic integrity assessment program was developed for the 200 Area ETF tanks and is discussed in detail in Addendum C, Section C.4.2.

I.1.2.2.5 Effluent Treatment Facility Piping

The 200 Area ETF employs an extensive piping system. During inspections at the 200 Area ETF, any aboveground piping is inspected visually for signs of leakage and for general structural integrity. During the visual inspection, particular attention is paid to valves and fittings for signs of cracking, deformation, and leakage.

I.1.2.3 Surface Impoundments and Condition Assessment

The following describes the surface impoundment inspections performed at LERF.

I.1.2.3.1 Overtopping Control

Under current operating conditions, 1.34 meters of freeboard is maintained at each LERF basin, which corresponds to a normal operating level of 6.1 meters, or 24.6 million liters. Level indicators at each basin are monitored to confirm that this level is not exceeded.

Before an aqueous waste is transferred into a basin, administrative controls are implemented to ensure overtopping will not occur during the transfer. The volume of feed to be transferred is compared to the available volume in the receiving basin. The transfer is not initiated unless there is sufficient volume available in the receiving basin or a cut-off level is established. The transfer into the basin would be stopped when this cut-off level is reached.

The LERF basins also are provided with floating very low-density polyethylene covers that are designed and constructed to prevent overtopping by the introduction of precipitation and dust into the basins. Overtopping and flow control also are discussed in Addendum C.

I.1.2.3.2 Impoundment Contents

The LERF basins are inspected weekly to assess whether the contents are escaping from a basin. Level indicators are inspected weekly to check for unaccountable change in the level of the basins.

I.1.2.3.3 Leak Detection

The leachate detection, collection, and removal system is described in Addendum C. The leachate collection sump pump is activated when the liquid level in the leachate sump reaches a preset level. A flow meter/totalizer measures the amount of leachate removed. In addition, the timer on the leachate pump tracks the cumulative pump run time. The leak rate through the primary liner can be determined using one of two methods: 1) measured as the leachate flow meter/totalizer readings or pump operating time readings (flow meters/totalizers are located on the outflow line from the collection sumps in the bottom of the LERF basins) or 2) calculated using the pump operating time readings multiplied by the pump flow rate (the pump runs at a constant flow rate). Calculations using either method are sufficient for compliance. If either the flow meter/ totalizer or pump operating time system is not functioning, this is identified as an abnormal condition (see I.1).The LERF employs a double walled transfer piping between 242-A Evaporator and LERF and between LERF and 200 Area ETF. The WAC 173-303-650 regulations do not require a discussion of piping for surface impoundments. However, for the purposes of comprehensive coverage of the LERF, inspections and integrity assessments are performed on the piping system. Aqueous waste (e.g., process condensate) is transferred from the 242-A Evaporator to the LERF via a buried pipeline. Likewise, aqueous waste is transferred to the 200 Area ETF via buried pipelines. At the LERF dikes, aboveground piping serves to transfer waste from one basin to another.

The buried pipelines normally are continuously monitored during transfers by a leak detection system (Addendum C). The alarms on the leak detection system signal to the 200 Area ETF control room, which is monitored continuously during waste transfers and daily when no waste is transferring. . As an alternative to continuous leak detection, the transfer lines can be inspected daily during transfers by opening the secondary containment drain lines at the LERF catch basins (for 242-A Evaporator transfers to LERF) and the surge tank (for LERF transfers to 200 Area ETF) to inspect for leakage. During the routine inspections at LERF, the aboveground piping system is inspected for signs of leakage and for general structural integrity. During the visual inspection, particular attention is paid to valves and fittings for signs of cracking, deformation, and leakage.

I.1.2.3.4 Dike Erosion

The LERF basins and dikes are visually inspected weekly and after significant precipitation events for run-on, run-off, cover integrity, erosion problems, or other signs of deterioration in the dikes from precipitation, wind, burrowing mammals, or vegetation.

I.1.2.3.5 Structural Integrity

A written certification attesting to the structural integrity of the basin dikes, signed by a qualified, registered professional engineer, is provided in Addendum C.

I.1.2.3.6 Container Inspection

Normal operation of the LERF does not involve the storage of dangerous waste in containers. Therefore, the inspection requirements of this section normally are not applicable to the LERF. Any containerized dangerous waste generated at LERF will be brought to the 200 Area ETF and managed in accordance with WAC 173-303-630 and is discussed in Section I.1.3.

I.1.3 Inspection Log

Observations made and deficiencies noted during an inspection are recorded on inspection log sheets (also called turnover sheets). On completion, the log sheet includes the inspector's printed name, signature, date, and time; the log sheet is submitted for review and approval by LERF and200 Area ETF management or their designee, as required by operating procedures. Once approved, the log sheet is kept in the Hanford Facility Operating Record, LERF and 200 Area ETF files. Inspection records are retained in the Hanford Facility Operating Record, LERF and 200 Area ETF files, or other approved locations, in accordance with Permit Condition II.I.1. The inspection records are used to help determine any necessary corrective actions. Problems identified during the inspections are prioritized and addressed in a timely fashion to mitigate health risks to workers, maintain integrity of the TSD units, and prevent hazards to public health and the environment.

If while performing an inspection, a leak or spill is discovered, facility operations responds per the emergency response procedures action is taken to stop the leak and determine the cause. The waste is removed from the secondary containment in a timely manner that prevents harm to human health and the environment.

I.1.4 Storage of Ignitable or Reactive Wastes

The LERF could receive an aqueous waste that is designated reactive or ignitable. Any aqueous waste exhibiting these characteristics is managed (e.g., through blending in LERF) such that the waste no longer exhibits the reactive or ignitable characteristics.

Though unlikely, the 200 Area ETF secondary wastes might have the characteristics of being reactive or ignitable. A qualified inspector performs annual fire inspections of the 200 Area ETF using a checklist developed specifically for facilities that handle dangerous and/or mixed waste.

1 **Table I.1. Visual Inspection Schedule for the LERF and 200 Area ETF**

Item	Inspection	Frequency
Load-In Facility		
Load-In tank system	Inspect area for leaks. Note any unusual noises or vibration from the system pumps. Inspect secondary containment system for signs of deterioration.	Daily
Main Treatment Train		
Surge tank system	Inspect area for leaks. Note any unusual noises or vibration from the system pumps. Inspect secondary containment system for signs of deterioration.	Daily
Rough filter	Inspect for leaks.	Daily
Ultraviolet oxidation system	Inspect module for leaks Inspect peroxide storage tank, ancillary equipment for leaks.	Daily
pH adjustment tank	Inspect tank and ancillary equipment for leaks	Daily
H_2O_2 decomposer	Inspect tank and ancillary equipment for leaks	Daily
Fine filter	Inspect module for leaks	Daily
Degasification system	Inspect module for leaks. Note any unusual noises or vibration from the degasification blower.	Daily
Reverse osmosis system	Inspect tanks and ancillary equipment for leaks. Note any unusual noises or vibration from the system pumps.	Daily
Polishers	Inspect tanks and ancillary equipment for leaks.	Daily
Effluent pH adjustment tank	Inspect tank and ancillary equipment for leaks.	Daily
Verification tanks	Inspect tanks and ancillary equipment for leaks. Note any unusual noises or vibration from the system pumps. Inspect secondary containment system for signs of deterioration.	Daily
Secondary Treatment Train		
Secondary waste receiving tank	Inspect tank and ancillary equipment for leaks	Daily
ETF evaporator	Inspect tank and equipment for leaks. Note any unusual noises or vibration from the system pumps or compressor.	Daily
Concentrate tank	Inspect tank and ancillary equipment for leaks.	Daily
Thin film dryer	Inspect tanks and ancillary equipment for leaks (viewed through camera). Note any unusual noises or vibration from the system pumps or blower.	Daily
Container handling	Inspect area for spills, leaks, accumulated liquids.	Daily
Container handling	Inspect for deterioration of containers and secondary containment, including corrosion and cracks in secondary containment foundation and coating. Inspect container labels to ensure that they are readable.	Weekly
Support Systems		
Vessel ventilation system	Inspect filters (HEPA and pre-filters), check vessel off gas pressures, system flow, and discharge temperatures.	Daily
Sump tank system	Inspect sump trenches for unexpected liquids, which indicate spills or leaks from process equipment.	Daily
Safety Systems		
Eye wash stations	Check status; check for adequate pressure	Monthly
Safety showers	Check status; check for adequate pressure	Monthly

Emergency Systems		
Fire extinguishers	Check for adequate charge.	Monthly
Emergency lighting	Test operability.	Monthly
Processing Area		
Uninterruptible power supply	Check output voltage and visually inspect battery pack for corrosion and leakage. Check indicator lights for fault conditions.	Annually
LERF (Surface Impoundment)		
LERF basins and dikes	Check to overtopping controls and integrity of the basins and dikes	Weekly
LERF contents	Check basin level indicators for unaccountable changes in the level of the basins	Weekly
Leak Detections	Determine the leak rate per wetted surface area	Weekly
LERF basins and dikes	Check for run-on, run-off, cover integrity, erosion problems, and other signs of deterioration	Weekly & After significant precipitation events
Ignitable and Reactive		
Ignitable and reactive waste	Storage in compliance with Hanford Site fire protection standards and WAC 173-303-630(8)	Annually†
Container Storage Areas Other Than Secondary Treatment Train		
Container Storage	Container labels to ensure labels are not obscured, removed, or otherwise unreadable	Weekly
	Deterioration of containers, containment systems, or cracks in protective coating or foundations caused by corrosion, mishandling, or other factors	Weekly
	Leaks, spills, and accumulated liquids	Daily
HEPA – High efficiency particulate air † When waste management activities occur		

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1 **Table I.2. Inspection Plan for Instrumentation Monitoring**

Item	Inspection	Frequency*
Load-In Facility		
Level alarm LAHH-59A-109/-117	Monitor liquid level in load-in tanks TK-109 and TK-117 to prevent overflow	Continuously
Level alarm LSH-59A-003	Monitor liquid level in load-in tanks TK-1 to prevent overflow	Continuously
Leak detector	Monitor for leakage in the load-in tank pit sump	Continuously
Main Treatment Train		
Leak detector LAH-20B009	Monitor for leakage in the surge tank drainage sump	Continuously
Level alarm LAH-60A013	Monitor surge tank level to prevent overflow	Continuously
Level alarm LAHL-60C-111	Monitor liquid levels in the pH adjustment tank to prevent overflow	Continuously
Level alarm LAHL-60F-101	Monitor liquid levels in the first RO feed tank to prevent overflow	Continuously
Level alarm LAHL-60F-201	Monitor liquid levels in the second RO feed tank to prevent overflow	Continuously
Level alarms LAHL-60F-211	Monitor liquid levels in the effluent pH adjustment tank to prevent overflow	Continuously
Level transmitter LAHX-60H001A/B/C	Monitor liquid level in verification tanks to prevent overflow	Continuously
Leak detector LAH-20B010	Monitor for leakage in the verification tank drainage sump	Continuously
Secondary Treatment Train		
Level alarm LAHL-60I-001A/B	Monitor liquid levels in secondary waste receiver tanks A and B to prevent overflow.	Continuously
Level alarm LAHL-60J-001A/B	Monitor liquid levels in concentrate tanks A and B to prevent overflow.	Continuously
Level alarm LAHL-60I-107	Monitor liquid levels in the evaporator tank to prevent overflow.	Continuously
Level alarm LAHL-60J-036	Monitor liquid levels in the spray condenser tank to prevent overflow.	Continuously
Level alarm LAHL-60I-108	Monitor liquid levels in the distillate flash tank to prevent overflow.	Continuously
Level alarm LAH-60I-119	Monitor liquid levels in the entrainment separator tank to prevent overflow.	Continuously
Level transmitter LAH-20B001	Monitor liquid level in sump tank No. 1 to prevent overflow.	Continuously
Level transmitter LAH-20B002	Monitor liquid level in sump tank No. 2 to prevent overflow.	Continuously
Leak detector LAH-20B003	Monitor for leakage to sump No. 1.	Continuously**

Leak detector LAH-20B005	Monitor for leakage to sump No. 2.	Continuously**
Leak detector	Monitor for leakage from pipeline between 200 Area ETF and load in station.	Continuously**
Leak detector	Monitor for leakage from pipeline between 200 Area ETF and LERF.	Continuously**
Leak detector	Monitor for leakage from pipeline between LERF and the 242-A Evaporator.	Continuously**

* Frequency of "continuously" applies during ETF processing operations and/or Load-in Station transfers. Data from alarms, leak detectors, and level transmitters is monitored in the control room at least daily at other times, even though many of these instruments record continuously (see C.2.5.1). Emergency communications equipment and warning systems (e.g. fire alarms, take cover alarms, and evacuation alarms) are included in addendum J, Contingency Plan. These alarms are monitored continuously and the response to these alarms do not rely on personnel being present the ETF control room.

** In the event of a malfunction of one of the electronic leak detectors, daily visual inspections will be performed while the facilities are in operation.

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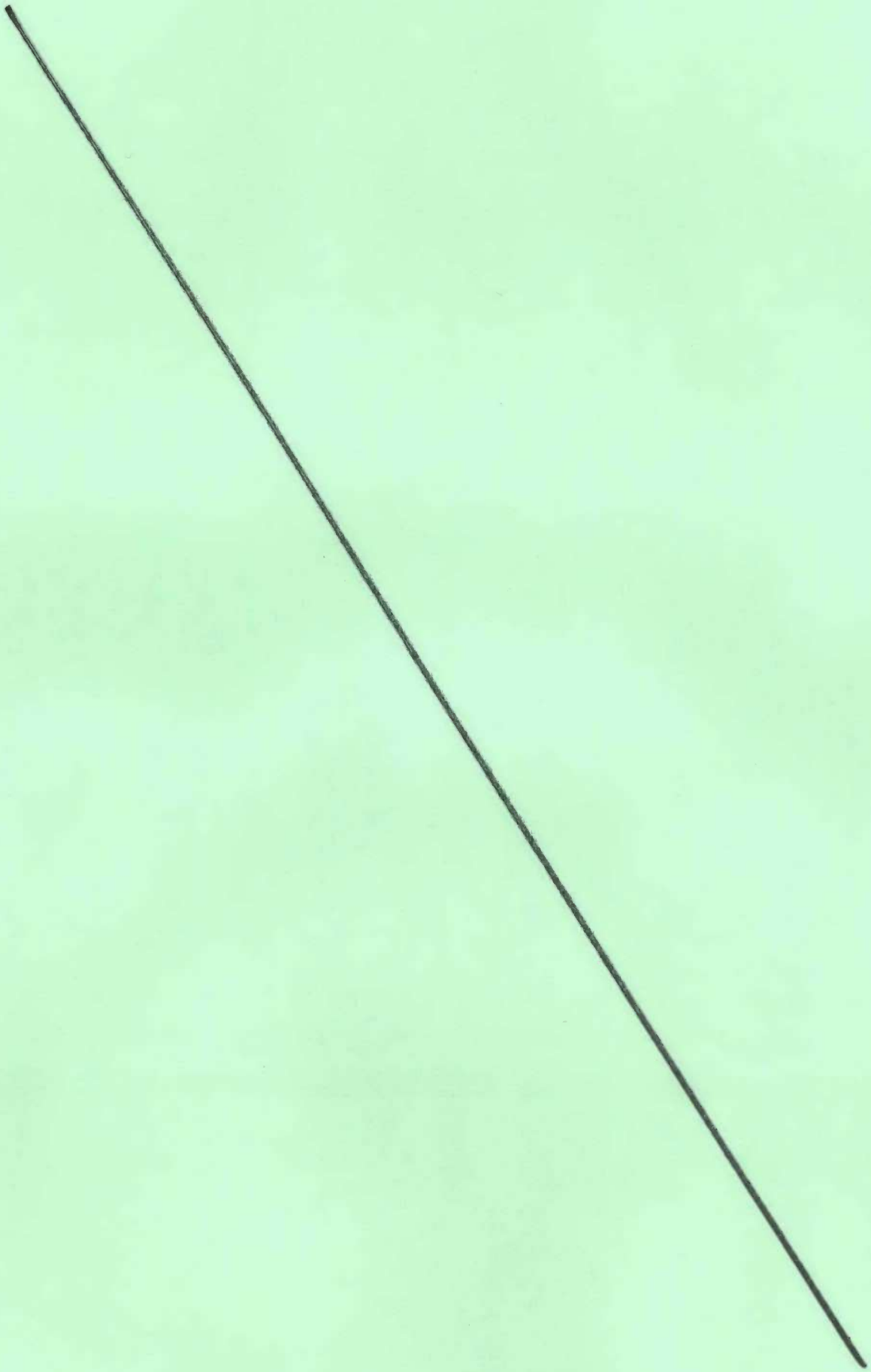

ADDENDUM J
CONTINGENCY PLAN

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ADDENDUM J
CONTINGENCY PLAN

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TABLE

Table J.1. Hanford Facility Documents Containing Contingency Plan Requirements of WAC 173-303-350(3) 5

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J. CONTINGENCY PLAN

The requirements for a contingency plan at the Liquid Effluent Retention Facility (LERF)/200 Area Effluent Treatment Facility (ETF) are satisfied in the following documents: portions of Hanford Facility Permit (Permit) Attachment 4, *Hanford Emergency Management Plan* (DOE/RL-94-02) and this Addendum.

The unit specific building emergency plan also serves to satisfy a broad range of other requirements [e.g., Occupational Safety and Health Administration standards (29 CFR 191029 Code of Federal Regulations [CFR] 1910), *Toxic Substance Control Act of 1976* (40 CFR 761) and U.S. Department of Energy Orders]. Therefore, revisions made to portions of this unit specific building emergency plan that are not governed by the requirements of Washington Administrative Code (WAC) 173-303 will not be considered as a modification subject to WAC 173-303-830 or Permit Condition I.C.3.

Table J.1 identifies the sections of the unit specific building emergency plan written to meet WAC 173-303-350(3) contingency plan requirements. In addition, Section 12.0 of the unit specific building emergency plan is written to meet WAC 173-303 requirements identifying where copies of Permit Attachment 4, *Hanford Emergency Management Plan* (DOE/RL-94-02) and the building emergency plan are located and maintained on the Hanford Facility. Therefore, revisions to Addendum J require a modification subject to WAC 173-303-830 and/or Permit Condition I.C.3.

Table J.1. Hanford Facility Documents Containing Contingency Plan Requirements of WAC 173-303-350(3)

Requirement	Permit Attachment 4, *Hanford Emergency Management Plan* (DOE/RL-94-02)	Building Emergency Plan[1] (HNF-IP-0263-ETF)	Part III, OU-3, LERF & 200 Area ETF, Addendum J
-350(3)(a) - A description of the actions, which facility personnel must take to comply with this section and WAC 173-303-360.	X[2] Section 1.3.4	X[2] Sections 7.1, 7.2 through 7.2.5, and 7.3[3] Sections 4.0 (1st paragraph), 8.2, 8.3, 8.4, 11.0	X[2] Sections J.3.1, J.3.2, through J.3.2.5, and J.3.3[3] Sections J.3, J.3.4, J.3.5, J.3.6, and J.5
-350(3)(b) - A description of the actions which shall be taken in the event that a dangerous waste shipment, which is damaged or otherwise presents a hazard to the public health and the environment, arrives at the facility, and is not acceptable to the owner or operator, but cannot be transported pursuant to the requirements of WAC 173-303-370(5), Manifest system, reasons for not accepting dangerous waste shipments.	X[2] Section 1.3.4	X[2, 4] Section 7.2.5.1	X[2,4] Section J.3.2.5.1
-350(3)(c) - A description of the arrangements agreed to by local police departments, fire departments, hospitals, contractors, and state and local emergency response teams to coordinate emergency services as required in WAC 173-303-340(4).	X Sections 3.2.3, 3.3.1, 3.3.2, 3.4, 3.4.1.1, 3.4.1.2, 3.4.1.3, 3.7, and Table 3-1		

Table J.1. Hanford Facility Documents Containing Contingency Plan Requirements of WAC 173-303-350(3)

Requirement	Permit Attachment 4, *Hanford Emergency Management Plan* (DOE/RL-94-02)	Building Emergency Plan[1] (HNF-IP-0263-ETF)	Part III, OU-3, LERF & 200 Area ETF, Addendum J
-350(3)(d) - A current list of names, addresses, and phone numbers (office and home) of all persons qualified to act as the emergency coordinator required under WAC 173-303-360(1). Where more than one person is listed, one must be named as primary emergency coordinator, and others must be listed in the order in which they will assume responsibility as alternates. For new facilities only, this list may be provided to the department at the time of facility certification (as required by WAC 173-303-810(14)(a)(I)), rather than as part of the permit application.		X[5] Section 3.1, 13.0	X[5] Sections J.2 and J.7
-350(3)(e) - A list of all emergency equipment at the facility (such as fire extinguishing systems, spill control equipment, communications and alarm systems, and decontamination equipment), where this equipment is required. This list must be kept up to date. In addition, the plan must include the location and a physical description of each item on the list, and a brief outline of its capabilities.		X Section 9.0	X Section J.4
-350(3)(f) - An evacuation plan for facility personnel where there is a possibility that evacuation could be necessary. This plan must describe the signal(s) to be used to begin evacuation, evacuation routes, and alternate evacuation routes.	X[6] Figure 7-3 and Table 5-1	X[7] Section 1.5	X[7] Section J.1

An "X" indicates requirement applies.

[1] Portions of Permit Attachment 4, *Hanford Emergency Management Plan* (DOE/RL-94-02) not enforceable through Appendix A of that document are not made enforceable by reference in the building emergency plan.

[2]Permit Attachment 4, *Hanford Emergency Management Plan* (DOE/RL-94-02) contains descriptions of actions relating to the Hanford Site Emergency Preparedness System. No additional description of actions are required if at the site level. If other credible scenarios exist or if emergency procedures at the unit are different, the description of actions contained in the building emergency plan will be used during an event by a building emergency director.

[3]Sections J.1, J.2 through J.2.5, and J.3 of the building emergency plan are those sections subject to the Class 2 "Changes in emergency procedures (i.e., spill or release response procedures)" described in WAC 173-303-830, Appendix I, Section B.6.a.

[4]This requirement only applies to TSD units, which receive shipment of dangerous or mixed waste defined as off-site shipments in accordance with WAC 173-303.

[5]Emergency Coordinator names and home telephone numbers are maintained separate from any contingency plan document, on file in accordance with Permit Condition II.A.4 and are updated, at a minimum, monthly.

[6]The Hanford Facility (site wide) signals are provided in this document. No unit/building signal information is required unless unique devices are used at the unit/building.

[7]An evacuation route for the TSD unit must be provided. Evacuation routes for occupied buildings surrounding the TSD unit are provided through information boards posted within buildings.

J.1 Building Evacuation Routing

Figures J.1 and J.2 provide identification of the primary and secondary staging areas and a general layout of the 2025E and ETF/LERF. Alternate evacuation routes will be used on a case-by-case basis based on meteorological conditions at the time of the event.

J.2 Building Emergency Director

Emergency response will be directed by the Building Emergency Director (BED) until the Incident Commander (IC) arrives. The Incident Command System and staff with supporting on-call personnel fulfill the responsibilities of the Emergency Coordinator as discussed in WAC 173-303-360.

During events, ETF/LERF personnel perform response duties under the direction of the BED. The Incident Command Post (ICP) is managed by the senior Hanford Fire Department official, unless the event is determined to be primarily a security event, in which case the Hanford Fire Department and Hanford Patrol will operate under a unified command system with Hanford Patrol making all decisions pertaining to security. These individuals are designated as the IC and as such, have the authority to request and obtain any resources necessary for protecting people and the environment. The BED becomes a member of the ICP and functions under the direction of the IC. In this role, the BED continues to manage and direct LERF/ETF operations.

A listing of BEDs by title, work location, and work telephone numbers is contained in Section J.7 of this plan. The BED is on the premises or is available through an "on-call" list 24 hours a day. Names and home telephone numbers of the BEDs are available from the Patrol Operations Center (POC) in accordance with Permit Condition II.A.4.

J.3 Implementation of the Plan

In accordance with WAC 173-303-360(2)(b) the BED ensures that trained personnel identify the character, source, amount, and areal extent of the release, fire, or explosion to the extent possible. Identification of waste can be made by activities that can include, but are not limited to, visual inspection of involved containers, sampling activities in the field, reference to inventory records, or by consulting with facility personnel. Samples of materials involved in an emergency might be taken by qualified personnel and analyzed as appropriate. These activities must be performed with a sense of immediacy and shall include available information.

The BED shall use the following guidelines to determine if an event has met the requirements of WAC 173-303-360(2)(d):

1. The event involved an unplanned spill, release, fire, or explosion,

AND

2.a The unplanned spill or release involved a dangerous waste, or the material involved became a dangerous waste as a result of the event (e.g., product that is not recoverable.), or

2.b The unplanned fire or explosion occurred at the ETF/LERF or transportation activity subject to RCRA contingency planning requirements,

AND

3. Time urgent response from an emergency services organization was required to mitigate the event or a threat to human health or the environment exists.

As soon as possible, after stabilizing event conditions, the BED shall determine, in consultation with the site contractor environmental single point-of-contact, if notification to the Washington State Department of Ecology (Ecology) is needed to meet WAC 173-303-360(2)(d) reporting requirements. If all of the conditions under 1, 2, and 3 are met, notifications are to be made to Ecology. Additional information is found in Permit Attachment 4, *Hanford Emergency Management Plan*, (DOE/RL-94-02), Section 4.2.

If review of all available information does not yield a definitive assessment of the danger posed by the incident, a worst-case condition will be presumed and appropriate protective actions and notifications will be initiated. The BED is responsible for initiating any protective actions based on their best judgment of the incident.

The BED must assess each incident to determine the response necessary to protect the personnel, facility, and the environment. If assistance from Hanford Patrol, Hanford Fire Department, or ambulance units is required, the Hanford Emergency Response Number (911 from site office phones/373-0911 from cellular phones) must be used to contact the POC and request the desired assistance. To request other resources or assistance from outside the ETF/LERF, the POC business number is 373-3800.

J.3.1 Protective Actions Responses

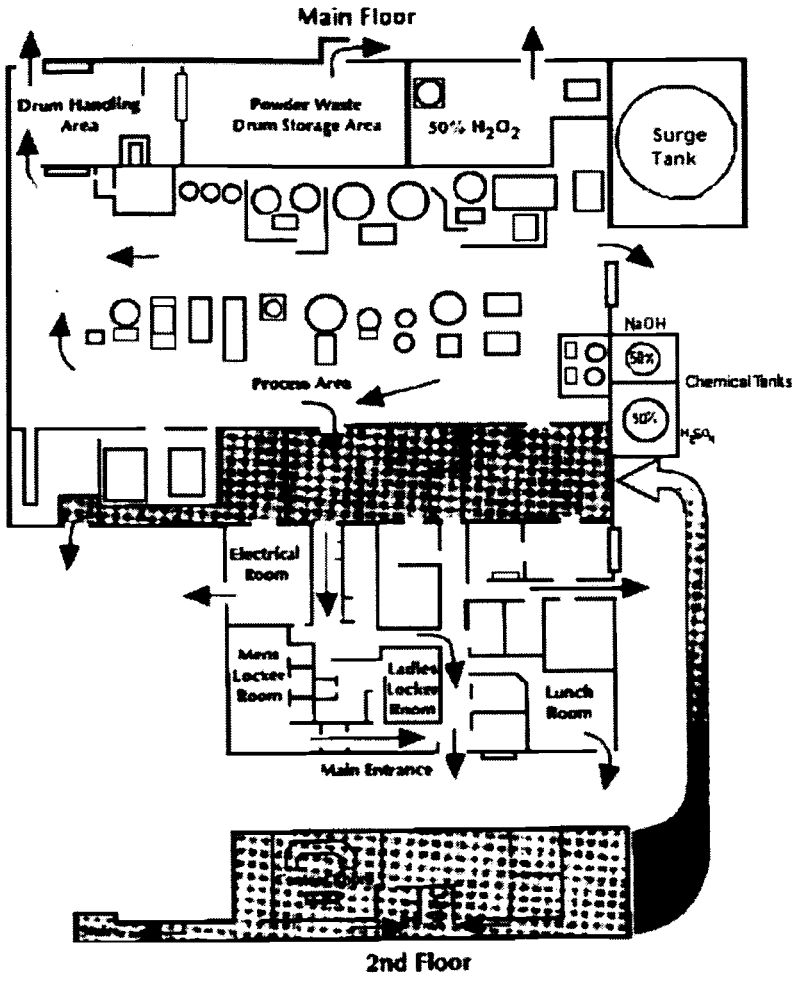
Protective action responses are discussed in the following sections. The steps identified in the following description of actions do not have to be performed in sequence because of the unanticipated sequence of incident events.

J.3.1.1 Evacuation

The objective of a facility evacuation order is to limit personnel exposure to hazardous materials or dangerous/mixed waste by increasing the distance between personnel and the hazard. The scope of the evacuation includes evacuation of the facility because of an event at the facility as well as evacuation of the facility in response to a site evacuation order. Evacuation will be directed by the BED when conditions warrant and will apply to all personnel not actively involved in the event response or emergency plan related activities.

The BED will initiate the evacuation by directing an announcement be made to evacuate along with the evacuation location over a public address system, facility radios, and, as conditions warrant, by activating the 200 Area site evacuation alarms by calling the POC using 911 from site office phones/373-0911 from cellular phones. Personnel proceed to a predetermined staging area (shown in Figure J.2), or other safe upwind location, as determined by the BED. The BED will determine the operating configuration of the facility and identify any additional protective actions to limit personnel exposure to the hazard.

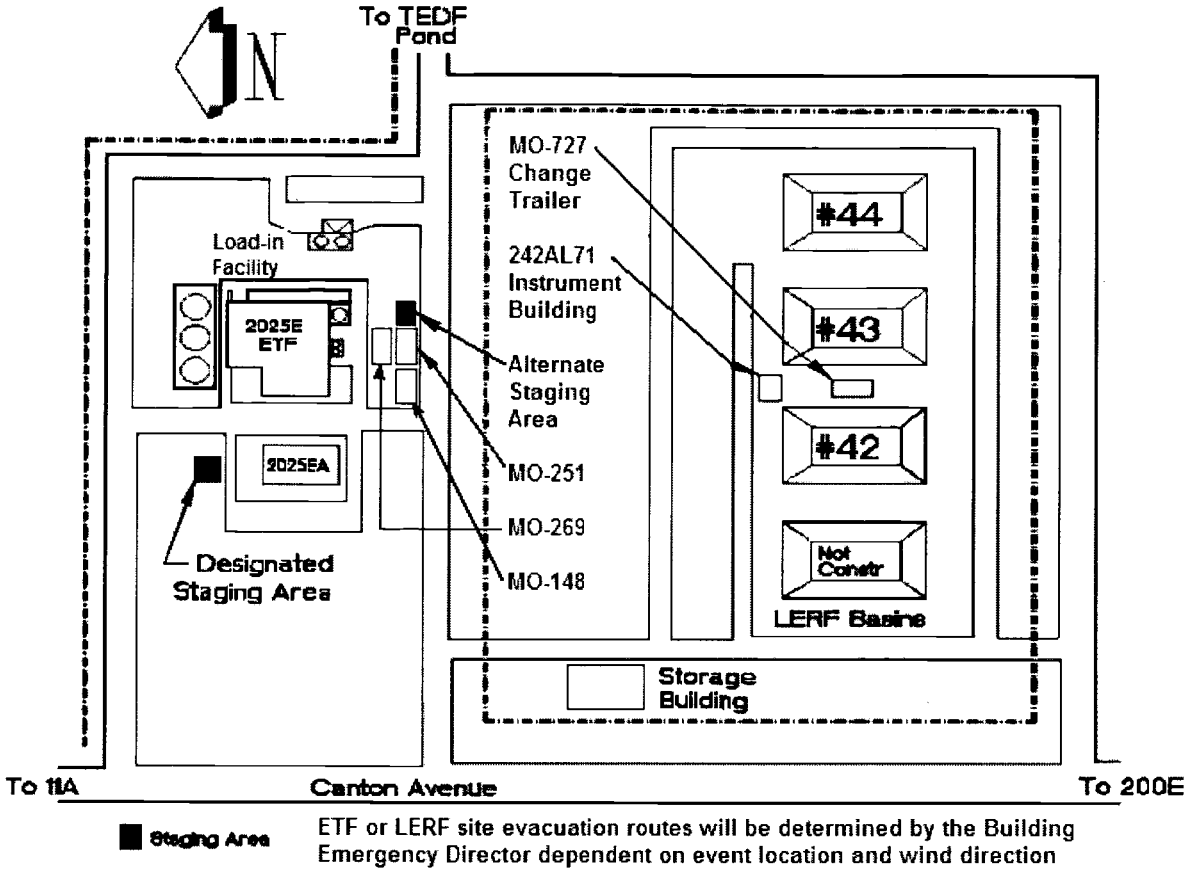
Emergency organization personnel or assigned operations personnel will conduct a sweep of occupied buildings to ensure that all non-essential personnel and visitors have evacuated. For an immediate evacuation, accountability will be performed at the staging area. The BED will assign personnel as accountability aides and staging managers with the responsibility to ensure that evacuation actions are taken at all occupied buildings at the ETF/LERF. All implementing actions executed by the aides/managers are directed by the emergency response procedures. When evacuation actions are complete, the aides/managers will provide a status report to the BED. The BED will provide status to the IC.



1 **Figure J.1 Evacuation Routes from 2025E**

2

Figure J.2. LERF and 200 Area ETF Site Plan



J.3.1.2 Take Cover

The objective of the take cover order is to limit personnel exposure to hazardous materials, or dangerous/mixed waste when evacuation is inappropriate or not practical. Evacuation might not be practical or appropriate because of extreme weather conditions or the material release might limit the ability to evacuate safely personnel.

The BED will initiate the take cover by directing an announcement be made over the public address system, facility radios, and, as conditions warrant, by activating the 200 Area site take cover alarms by calling the POC using 911 from site office phones/373-0911 from cellular phones). Actions to complete a facility take-cover will be directed by the emergency response procedure. Protective actions associated with operations include configuring, or shutting down, the ventilation systems. Determination of additional take cover response is based on plant operating configuration, weather conditions, amount and duration of release, and other conditions, as applicable to the event and associated hazard. As a minimum, personnel exposure to the hazard will be minimized. The BED will assign personnel as accountability aides with responsibility to ensure that take-cover actions are taken at all occupied buildings at the ETF complex. All implementing actions executed by the aides/managers are directed by the emergency response procedure. When take cover actions are complete, the aides/manager will provide the BED with a status report.

J.3.2 Response to Facility Operations Emergencies

Depending on the severity of the following events, the BED reviews the site wide procedures and ETF/LERF emergency response procedure(s) and, as required, categorizes and/or classifies the event. If necessary, the BED initiates area protective actions and Hanford Site Emergency Response Organization activation. The steps identified in the following description of actions do not have to be performed in sequence because of the unanticipated sequence of incident events.

J.3.2.1 Loss of Utilities

A case-by-case evaluation is required for each event to determine loss of utility impacts. When a BED determines a loss of utility impact, actions are taken to ensure dangerous and/or mixed waste is being properly managed, to the extent possible given event circumstances. As necessary, the BED will stop operations and take appropriate actions until the utility is restored.

J.3.2.2 Major Process Disruption/Loss of Plant Control

The hazards assessment has determined that this occurrence does not pose significant risk to human health or the environment.

J.3.2.3 Pressure Release

The hazards assessment has determined that a pressure release does not pose significant risk to human health or the environment. Hazardous material release and dangerous/mixed waste releases are addressed in Section J.2.5.

J.3.2.4 Fire and/or Explosion

In the event, of a fire, the discoverer activates a fire alarm (pull box); calls 911 from site office phones/373-0911 from cellular phones or verifies that the Hanford Emergency Response Number has been called. Automatic initiation of a fire alarm (through the smoke detectors, and sprinkler systems) is also possible.

- Unless otherwise instructed, personnel shall evacuate the area/building by the nearest safe exit and proceed to the designated staging area for accountability.
- On actuation of the fire alarm, ONLY if time permits, personnel should shut down equipment, secure waste, and lock up classified materials (or hand carry them out). The alarm automatically signals the Hanford Fire Department.

- The BED proceeds directly to the ICP, obtains all necessary information pertaining to the incident, and sends a representative to meet Hanford Fire Department.
- The BED provides a formal turnover to the IC when the IC arrives at the ICP.
- The BED informs the Hanford Site Emergency Response Organization as to the extent of the emergency (including estimates of dangerous waste and mixed waste quantities released to the environment).
- If operations are stopped in response to the fire, the BED ensures that systems are monitored for leaks, pressure buildup, gas generation, and ruptures.
- Hanford Fire Department firefighters extinguish the fire as necessary.

Note: Following a fire and/or explosion, WAC 173-303-640(7) will be addressed for the ETF regarding fitness for use.

J.3.2.5 Hazardous Material, Dangerous and/or Mixed Waste Spill

Spills can result from many sources including process leaks, container spills or leaks, damaged packages or shipments, or personnel error. Spills of mixed waste are complicated by the need to deal with the extra hazards posed by the presence of Atomic Energy Act materials. These controls include containment berms, dedicated spill control sumps, remote gauges, and level indicators as well as spray shields on chemical pipe flanges. WRPS procedures provide alarm response and maintenance actions for leak detection equipment, surveillance of possible leak locations, and response actions for detected spills.

- The discoverer notifies BED and initiates SWIMS response:

  Stops work

  Warns others in the vicinity

  Isolates the area

  Minimizes the exposure to the hazards

  Requests the BED **S**ecure ventilation
- If Operations are stopped, the BED ensures that the plant is put in a safe shutdown configuration.
- The BED determines if emergency conditions exist requiring response from the Hanford Fire Department based on classification of the spill and injured personnel, and evaluates need to perform additional protective actions.
- If the Hanford Fire Department resources are not needed, the spill is mitigated with resources identified in Section J.4 of this plan and proper notifications are made.
- If the Hanford Fire Department resources are needed, the BED calls 911 from site office phones/373-0911 from cellular phones.
- The BED sends a representative to meet the Hanford Fire Department.
- The BED provides a formal turnover to the IC when the IC arrives at the ICP.
- The BED informs the Hanford Site Emergency Response Organization as to the extent of the emergency (including estimates of dangerous waste and mixed waste quantities released to the environment).
- If operations are stopped in response to the spill, the BED ensures that systems are monitored for leaks, pressure buildup, gas generation, and ruptures.
- Hanford Fire Department stabilizes the spill.

Note: For response to leaks or spills and disposition of leaking or unfit-for-use tank systems, refer to WAC 173-303-640(7).

J.3.2.5.1 Damaged, or Unacceptable Shipments

During the course of receiving an onsite transfer of dangerous and/or mixed waste at ETF/LERF an unanticipated event could be discovered resulting in a conformance issue concerning the waste.

Damaged or unacceptable shipments resulting from onsite transfers are not subject to WAC 173-303-370 however conformance issues must be resolved in order to maintain proper records.

The following actions are taken to resolve the conformance issue:

- Operations management is notified of the damaged or unacceptable waste to be received.
- If the conformance issue results in a spill or release, actions described in Section J.3.2.5 are taken.
- The generating organization is notified of the conformance issue.

An operations representative, in conjunction with the generating organization, determines the course of action to resolve the conformance issue.

J.3.3 Prevention of Recurrence or Spread of Fires, Explosions, or Releases

The BED, as part of the ICP, takes the steps necessary to ensure that a secondary release, fire, or explosion does not occur. The BED will take measures, where applicable, to stop processes and operations, collect and contain released waste, and remove or isolate containers. The BED also monitors for leaks, pressure buildups, gas generation, or ruptures in valves, pipes, or other equipment, whenever this is appropriate.

J.3.4 Incident Recovery and Restart of Operations

A recovery plan is developed when necessary in accordance with Permit Attachment 4, *Hanford Emergency Management Plan*, (DOE/RL-94-02), Section 9.2. A recovery plan is needed following an event where further risk could be introduced to personnel, the ETF/LERF, or the environment through recovery action and/or to maximize the preservation of evidence.

If this plan was implemented according to Section J.3 of this plan, Ecology is notified before operations can resume. The Permit Attachment 4, *Hanford Emergency Management Plan*, (DOE/RL-94-02), Section 5.1 discusses different reports to outside agencies. This notification is in addition to those required reports and includes the following statements:

- There are no incompatibility issues with the waste and released materials from the incident.
- All the equipment has been cleaned, fit for its intended use, and placed back into service.

The notification required by WAC 173-303-360(2)(j) may be made via telephone conference. Additional information that Ecology requests regarding these restart conditions will be included in the required 15-day report identified in Section J.5 of this plan.

For emergencies not involving activation of the Hanford EOC, the BED ensures that conditions are restored to normal before operations are resumed. If the Hanford Site Emergency Response Organization was activated and the emergency phase is complete, a special recovery organization could be appointed at the discretion of RL to restore conditions to normal. This process is detailed in RL and contractor emergency procedures. The makeup of this organization depends on the extent of the damage and the effects. The onsite recovery organization will be appointed by the appropriate contractor's management.

J.3.5 Incompatible Waste

After an event, the BED or the onsite recovery organization ensures that no waste that might be incompatible with the released material is treated, stored, and/or disposed of until cleanup is completed. Cleanup actions are taken by ETF/LERF personnel or other assigned personnel. Permit Attachment 4, *Hanford Emergency Management Plan*, (DOE/RL-94-02), Section 9.2.3, describes actions to be taken.

Waste from cleanup activities is designated and managed as newly generated waste. A field check for compatibility before storage is performed as necessary. Incompatible wastes are not placed in the same container. Containers of waste are placed in storage areas appropriate for their compatibility class.

If incompatibility of wastes was a factor in the incident, the BED or the onsite recovery organization ensures that the cause is corrected.

J.3.6 Post Emergency Equipment Maintenance and Decontamination

All equipment used during an incident is decontaminated (if practicable) or disposed of as spill debris. Decontaminated equipment is checked for proper operation before storage for subsequent use. Consumable and disposed materials are restocked. Fire extinguishers are replaced.

The BED ensures that all equipment is cleaned and fit for its intended use before operations are resumed. Depleted stocks of neutralizing and absorbing materials are replenished; protective clothing is cleaned or disposed of and restocked, etc.

J.4 Emergency Equipment

Emergency resources and equipment for the ETF/LERF are presented in this section.

J.4.1 Fixed Emergency Equipment

TYPE	LOCATION	CAPABILITY
Safety shower/ eye wash stations (ETF only)	1 - 2025E Rm 122 Decon Station 1 - 2025E South Wall of Process Area 1- 2025E Rm 131 1 - 2025E Rm 134 1 - Outside south 2025E near acid/ caustic tanks 1 - Outside at Load-in station 1 - 2025E Rm 112 Laboratory	Assist in flushing chemicals/ materials from the body and/ or eyes and face of personnel.
Wet pipe sprinkler (ETF only)	Throughout the ETF except those areas protected by preactive sprinklers	Assist in the control of a fire.
Preactive sprinkler (ETF only)	Control room, communications room, electrical equipment room	Assist in the control of a fire. Maintained dry to prevent accidental damage to equipment
Fire alarm pull boxes (ETF only)	All high traffic areas in operations administration and support areas, truck bay, and process area	Activate the local fire alarm
E-lights	Throughout ETF	1 hour temporary lighting

J.4.2 Portable Emergency Equipment

TYPE	LOCATION	CAPABILITY
Fire extinguisher ABC type	Throughout ETF (Administrative/Support areas), LERF, and TEDF	Fire suppression for Class A, B, and C fires
Fire extinguisher BC type	Throughout ETF (process area and electrical room)	Fire suppression for Class B and C fires
Portable safety showers and Eye Wash Stations	As needed for special evolutions and maintenance	Assist in flushing chemicals/ materials from the body and/or eyes and face of personnel.

J.4.3 Communications Equipment/Warning Systems

TYPE	LOCATION	CAPABILITY
Fire alarms (ETF only)	Corridors, locker rooms, process area, drum storage, and truck bay	Audible throughout ETF
Take cover/evacuation	Throughout the ETF	Audible outside buildings and inside administrative buildings
Public address system (ETF Only)	Throughout the ETF	Audible throughout ETF
Portable radios	Operations and maintenance personnel	Communication to control room
Telephone	**ETF**– control room, 2025E, 2025EA offices, MO-148, MO-269, MO-251, 2025EC71. **LERF**– MO-727 and 242AL71 instrument building, LERF Garage 242AL11 **TEDF**– 225E(pump house 1), 225W (pump house 2), 6653 (sample building), 6653A (pump house 3)	Internal and external communications. Allows notification of outside resources (POC, HFD, Hanford Patrol, etc.

Note: Sitewide communications and warning systems are identified in Permit Attachment 4, *Hanford Emergency Management Plan*, (DOE/RL-94-02), Table 5.1.

J.4.4 Personal Protective Equipment

TYPE	LOCATION	CAPABILITY
Acid suits	In the spill response cabinets in 2025E	Chemical protection for personnel during containment and isolation
Respirators	2025E, 1st Floor	Filtered air for recovery of known hazards

J.4.5 Spill Control and Containment Supplies

SPILL KITS AND SPILL CONTROL EQUIPMENT		
TYPE	**LOCATION**	**CAPABILITY**
Spill bags, drums, carts, etc.	4 – 2025E in process area 1 – TEDF 6653 Disposal Building 1 – 2025E upper level process area 1 – 2025E Rm 125A 1 – 2025ED Load-In Station CONEX	Support containment and cleanup of hazardous material spills
Spill response cabinet	1 – 2025E Rm 122 2 – container storage CONEX East of 2025E building within the TSD unit boundary 1 – TEDF 6653 Disposal Building 1 – MO-727 Change Trailer 1 – outside southeast side of 2025E	Support equipment for spill response

J.4.6 Incident Command Post

The ICPs for the ETF/LERF are in ETF control room or 2025 EA. Emergency resource materials are stored at each location. The IC could activate the Hanford Fire Department Mobile Command Unit if necessary.

J.5 Required Reports

Post incident, written reports are required for certain incidents on the Hanford Site. The reports are described in Permit Attachment 4, *Hanford Emergency Management Plan*, (DOE/RL-94-02), Section 5.1.

Facility management must note in the Hanford Facility Operating Record, LERF & 200 Area ETF File, the time, date and details of any incident that requires implementation of the contingency plan (refer to Section J.3). Within fifteen (15) days after the incident, a written report must be submitted to Ecology. The report must include the elements specified in WAC 173-303-360(2)(k).

J.6 Plan Location and Amendments

Copies of this plan are maintained at the following locations:

- ETF control room
- Building 2025EA ICP

This plan will be reviewed and immediately amended as necessary, in accordance with Permit Attachment 4, *Hanford Emergency Management Plan*, (DOE/RL-94-02), Section 14.3.1.1.

J.7 Facility/Building Emergency Response Organization

ETF/LERF Building Emergency Directors		
TITLE	**WORK LOCATION**	**WORK PHONE**
Shift Operation Manager (SOM)	2025E Building	373-9000 or 373-9500

Names and home telephone numbers of the BEDs are available from the POC (373-3800) in accordance with Permit Condition II.A.4.

PUREX

Change Control Log

Change Control Logs ensure that changes to this unit are performed in a methodical, controlled, coordinated and transparent manner. Each unit addendum will have a "**Last Modification Date**" which represents the last date the portion of the unit has been modified. The "**Modification Number**" represents Ecology's method for tracking the different versions of the permit. This log will serve as an up to date record of modifications and version history of the unit.

Last modification to PUREX **June 30, 2012**

Chapters	Last Modification Date	Modification Number
Conditions	06/30/2012	
1.0 Part A Form	10/01/2008	
2.0 Reserved		
3.0 Waste Analysis Plan	10/2006	
4.0 Process Information	10/2006	
5.0 Reserved		
6.0 Reserved		
7.0 Reserved		
8.0 Reserved		
9.0 Reserved		
10.0 Reserved		
11.0 Closure and Financial Assurance	10/2006	

Addenda	Last Modification Date	Modification Number
A. Reserved		
B. Reserved		
C. Reserved		
D. Reserved		
E. Security	09/30/2010	
F. Preparedness and Prevention	09/30/2010	
G. Personnel Training	09/30/2012	
H. Reserved		
I. Inspection Requirements	09/30/2010	
J. Contingency Plan (OUO)	09/30/2010	

PART III, OPERATING UNIT 2 PERMIT CONDITIONS
PUREX STORAGE TUNNELS

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PART III, OPERATING UNIT 2 PERMIT CONDITIONS
PUREX STORAGE TUNNELS

UNIT DESCRIPTION

The Plutonium Uranium Extraction Facility (PUREX) Storage Tunnels are mixed waste storage units consisting of two underground railroad tunnels: Tunnel Number 1, designated 218-E-14, and Tunnel Number 2, designated 218-E-15. This Chapter sets forth the operating Conditions for this treatment, storage, and disposal (TSD) unit.

OPERATING UNIT 2

Chapter 1.0	Part A Form, dated October 1, 2008
Chapter 3.0	Waste Analysis Plan, dated October 2006
Chapter 4.0	Process Information, dated October 2006
Addendum E	Security, dated September 30, 2010
Addendum F	Preparedness and Prevention, dated September 30, 2010
Addendum G	Personnel Training, dated June 30, 2012
Addendum I	Inspection Requirements, dated September 30, 2010
Addendum J	Contingency Plan, dated September 30, 2010
Chapter 11.0	Closure and Financial Assurance, dated October 2006

III.2.A COMPLIANCE WITH UNIT SPECIFIC PERMIT CONDITIONS

III.2.A.1 The Permittees will comply with all conditions in this Chapter and its addenda and chapters with respect to storage of waste in the miscellaneous units, (PUREX Storage Tunnels), in addition to applicable requirements in Permit Parts I and II.

III.2.A.2 In the event that the Part III, Unit Specific Conditions for Operating Unit 2, PUREX Storage Tunnels conflict with the Part I, Standard Conditions and/or Part II, General Facility Conditions of the Permit, the unit specific conditions for Operating Unit 2, PUREX Storage Tunnels prevail.

III.2.B UNIT SPECIFIC CONDITIONS

III.2.B.1 Portions of Permit Attachment 4, *Hanford Emergency Management Plan*, (DOE/RL-94-02) that are not made enforceable by inclusion in the applicability matrix for that document are not made enforceable by reference in this document.

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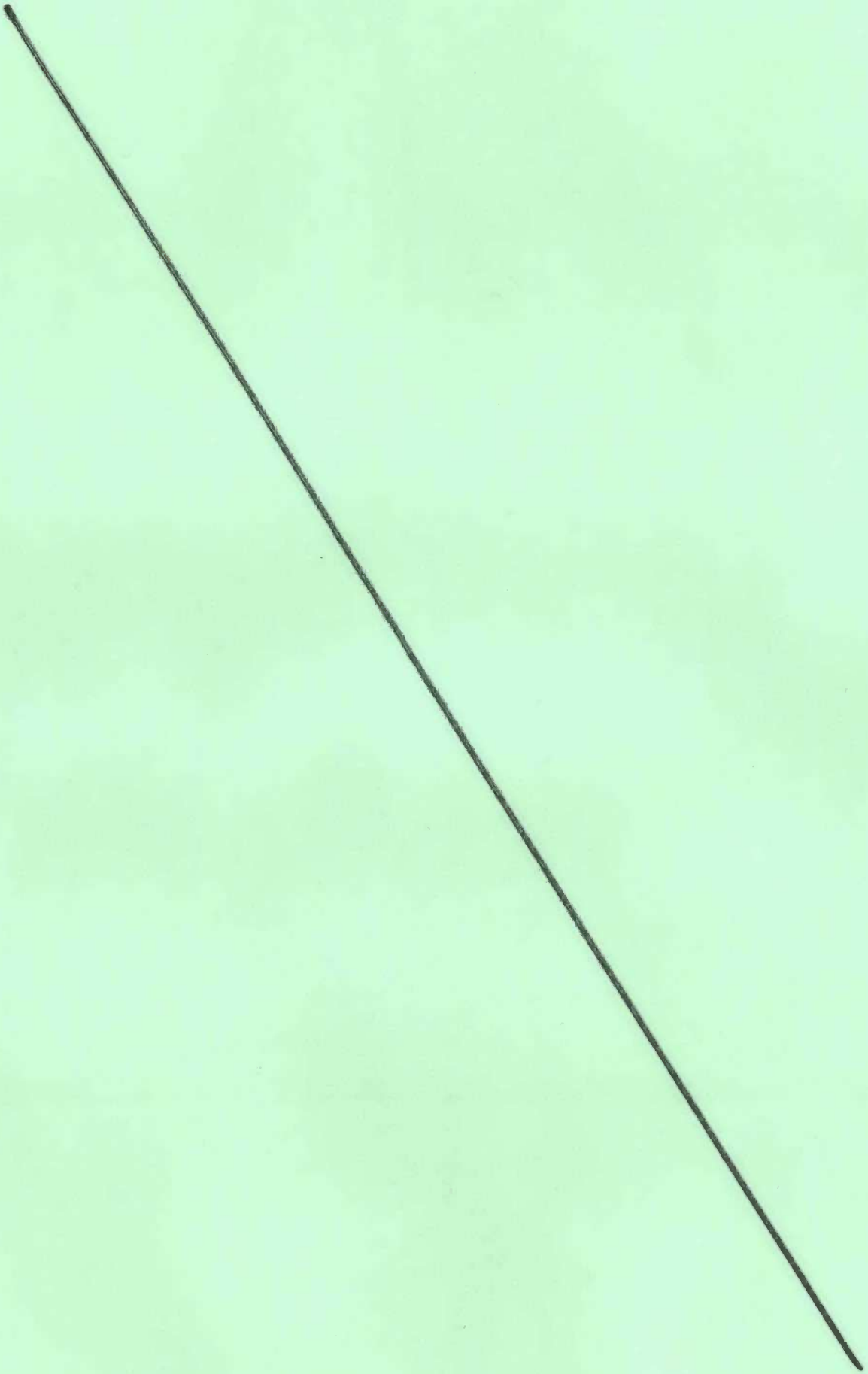

CHAPTER 1
PART A FORM

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1
2

WASHINGTON STATE DEPARTMENT OF ECOLOGY

Dangerous Waste Permit Application
Part A Form

Date Received	Reviewed	Date:	0	9	2	2	2	0	0	8
Month Day Year	Approved	Date:	0	9	2	2	2	0	0	8
0 9 1 9 2 0 0 8										

I. This form is submitted to: (place an "X" in the appropriate box)

☒	**Request modification to a final status permit (commonly called a "Part B" permit)**
☐	**Request a change under interim status**
☐	**Apply for a final status permit. This includes the application for the initial final status permit for a site or for a permit renewal (i.e., a new permit to replace an expiring permit).**
☐	**Establish interim status because of the wastes newly regulated on:** (Date)
	List waste codes:

II. EPA/State ID Number

W A 7 8 9 0 0 0 8 9 6 7

III. Name of Facility

US Department of Energy - Hanford Facility

IV. Facility Location (Physical address not P.O. Box or Route Number)
A. Street

825 Jadwin

City or Town	State	ZIP Code
Richland	WA	99352

County Code (if			County Name
0	0	5	Benton

B. Land Type	C. Geographic Location Latitude (degrees, mins, secs)	Longitude (degrees, mins, secs)	D. Facility Existence Date Month Day Year
F	Refer to TOPO Map (Section XV.)		0 3 0 2 1 9 4 3

V. Facility Mailing Address

Street or P.O. Box

P.O. Box 550

City or Town	State	ZIP Code
Richland	WA	99352

VI. Facility contact (Person to be contacted regarding waste activities at facility)		
Name (last)	**(first)**	
Brockman	David	
Job Title	**Phone Number (area code and number)**	
Manager	(509) 376-7395	
Contact Address		
Street or P.O. Box		
P.O. Box 550		
City or Town	**State**	**ZIP Code**
Richland	WA	99352

VII. Facility Operator Information	
A. Name	**Phone Number**
Department of Energy * Owner/Operator CH2M HILL Plateau Remediation Company Co-Operator for PUREX Storage Tunnels*	(509) 376-7395 (509) 376-0556*
Street or P.O. Box	
P.O. Box 550 P.O. Box 1600 *	

City or Town	State	ZIP Code
Richland	WA	99352

B. Operator Type F

C. Does the name in VII.A reflect a proposed change in operator? ☐ Yes ☒ No Co-Operator* change

If yes, provide the scheduled date for the change:

Month	Day	Year
10	01	2008

D. Is the name listed in VII.A. also the owner? If yes, skip to Section VIII.C. ☐ Yes ☒ No

VIII. Facility Owner Information	
A. Name	**Phone Number (area code and number)**
David A. Brockman, Operator/Facility-Property Owner	(509) 376-7395
Street or P.O. Box	
P.O. Box 550	

City or Town	State	ZIP Code
Richland	WA	99352

B. Owner Type F

C. Does the name in VIII.A reflect a proposed change in owner? ☐ Yes ☒ No

If yes, provide the scheduled date for the change:

Month	Day	Year

IX. NAICS Codes (5/6 digit codes)

A. First		B. Second	
562211	Waste Treatment & Disposal	924110	Administration of Air & Water Resource & Solid Waste Management Programs
C. Third		**D. Fourth**	
541713	Research & Development in the Physical, Engineering, & Life Sciences		

X. Other Environmental Permits (see instructions)

A. Permit Type	B. Permit Number	C. Description
E	AIR02-1221	WAC-246-247, NOC

XI. Nature of Business (provide a brief description that includes both dangerous waste and non-dangerous waste areas and activities)

SectionXIV

*The estimated annual quantity of waste listed represents the maximum quantity of waste placed in either tunnel in a given year.

Process Code X99 is being used to designate the PUREX Storage Tunnels as a *Miscellaneous Unit* per WAC 173-303-680. The PUREX Storage Tunnels, a miscellaneous unit (X99), are used for storage of mixed waste subject to the requirements of WAC 173-303-680. The two tunnels store waste from the PUREX Plant and other onsite sources. Since being placed into service, mixed waste has been stored in the tunnels on railcars. Not all material stored in the tunnels contains mixed waste.

The construction of Tunnel Number 1 was completed in 1956. The tunnel is approximately 5.8 meters (19 feet) wide by 6.7 meters (22 feet) high by 109 meters (358 feet) long and provides storage space for eight railcars. Between June 1960 and January 1965, all eight railcar positions were filled and the tunnel subsequently was sealed. The combined volume of the equipment stored on the eight railcars presently in Tunnel Number 1 is approximately 596 cubic meters (780 cubic yards). The maximum process design capacity for storage in Tunnel Number 1 is approximately 4,129 cubic meters (5,400 cubic yards).

The construction of Tunnel Number 2 was completed in 1964. Tunnel Number 2 is approximately 5.8 meters (19 feet) wide by 6.7 meters (22 feet) high by 514 meters (1,686 feet) long and provides storage space for 40 railcars. The first railcar was placed in Tunnel Number 2 in December 1967 and as of August 2000, 28 railcars have been placed in the tunnel. The combined volume of equipment stored on the 28 railcars presently in Tunnel Number 2 is approximately 2,204 cubic meters (2,883 cubic yards). The maximum process design capacity for storage in Tunnel Number 2 is approximately 19,878 cubic meters (26,000 cubic yards).

The waste stored in the tunnels could include barium(D005), cadmium (D006), chromium (D007), lead (D008), mercury (D009), selenium (D010), silver (D011), and light mineral oil (WT02, state-only, toxic, dangerous waste) contained in oil absorption material. The silver is predominately in the form of salts and is considered ignitable (D001) because of the presence of silver nitrate ($AgNO_3$). Cadmium also could be considered state-only, toxic, dangerous waste (WT02).

EXAMPLE FOR COMPLETING ITEMS XII and XIII (shown in lines numbered X-1, X-2, and X-3 below): A facility has two storage tanks that hold 1200 gallons and 400 gallons respectively. There is also treatment in tanks at 20 gallons/hr. Finally, a one-quarter acre area that is two meters deep will undergo *in situ vitrification.*

Section XII. Process Codes and Design Capacities								Section XIII. Other Process Codes							
Line Number		A. Process Codes (enter code)			B. Process Design Capacity		C. Process Total Number of Units	Line Number		A. Process Codes (enter code)			B. Process Design Capacity		C. Process Total Number of Units	D. Process Description
					1. Amount	2. Unit of Measure (enter code)							1. Amount	2. Unit of Measure (enter code)		
X	1	S	0	2	1,600	G	002	X	1	T	0	4	700	C	001	In situ vitrification
X	2	T	0	3	20	E	001									
X	3	T	0	4	700	C	001									
	1	X	9	9	24,007	C	002		1							
	2								2							
	3								3							
	4								4							
	5								5							
	6								6							
	7								7							
	8								8							
	9								9							
1	0							1	0							
1	1							1	1							
1	2							1	2							
1	3							1	3							
1	4							1	4							
1	5							1	5							
1	6							1	6							
1	7							1	7							
1	8							1	8							
1	9							1	9							
2	0							2	0							
2	1							2	1							
2	2							2	2							
2	3							2	3							
2	4							2	4							
2	5							2	5							

XIV. Description of Dangerous Wastes

Example for completing this section: A facility will receive three non-listed wastes, then store and treat them on-site. Two wastes are corrosive only, with the facility receiving and storing the wastes in containers. There will be about 200 pounds per year of each of these two wastes, which will be neutralized in a tank. The other waste is corrosive and ignitable and will be neutralized then blended into hazardous waste fuel. There will be about 100 pounds per year of that waste, which will be received in bulk and put into tanks.

Line Number	A. Dangerous Waste No. (enter code)	B. Estimated Annual Quantity of Waste	C. Unit of Measure (enter code)	D. Processes (1) Process Codes (enter)	D. Processes (2) Process Description [If a code is not entered in D (1)]
X 1	D 0 0 2	400	P	S 0 1 T 0 1	
X 2	D 0 0 1	100	P	S 0 2 T 0 1	
X 3	D 0 0 2				Included with above
1	D 0 0 5	454*	K	X 9 9	Includes Debris
2	D 0 0 6	454*	K	X 9 9	Includes Debris
3	W T 0 2		K	X 9 9	Includes Debris
4	D 0 0 7	454*	K	X 9 9	Includes Debris
5	D 0 0 8	8,000*	K	X 9 9	Includes Debris
6	D 0 0 9	45*	K	X 9 9	Includes Debris
7	D 0 1 0	454*	K	X 9 9	Includes Debris
8	D 0 1 1	680*	K	X 9 9	Includes Debris
9	D 0 0 1		K	X 9 9	Includes Debris
10	W T 0 2	454	K	X 9 9	Includes Debris
11					
12					
13					
14					
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16					
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19					
20					
21					
22					
23					
24					
25					

XV. Map

Attach to this application a topographic map of the area extending to at least one (1) mile beyond property boundaries. The map must show the outline of the facility; the location of each of its existing and proposed intake and discharge structures; each of its dangerous waste treatment, storage, recycling, or disposal units; and each well where fluids are injected underground. Include all springs, rivers, and other surface water bodies in this map area, plus drinking water wells listed in public records or otherwise known to the applicant within ¼ mile of the facility property boundary. The instructions provide additional information on meeting these requirements.

Topographic map is located in the Ecology Library

XVI. Facility Drawing

All existing facilities must include a scale drawing of the facility (refer to instructions for more detail).

XVII. Photographs

All existing facilities must include photographs (aerial or ground-level) that clearly delineate all existing structures; existing storage, treatment, recycling, and disposal areas; and sites of future storage, treatment, recycling, or disposal areas (refer to instructions for more detail).

XVIII. Certifications

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

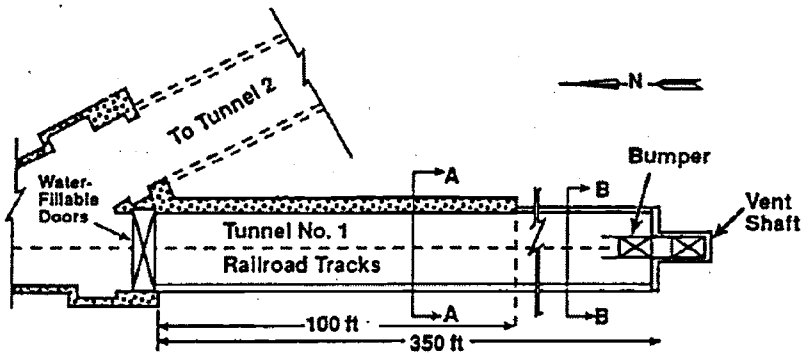
Operator Name and Official Title (type or print) David A. Brockman, Manager U.S. Department of Energy Richland Operations Office	**Signature** Doug A. Shoop for	**Date Signed** 9/19/08
Co-Operator* Name and Official Title (type or print) John G. Lehew, III President and Chief Executive Officer CH2M HILL Plateau Remediation Company	**Signature**	**Date Signed** 9/2/08
Co-Operator – Address and Telephone Number* P.O. Box 1600 Richland, WA 99352 (509) 376-0556		
Facility-Property Owner Name and Official Title (type or print) David A. Brockman, Manager U.S. Department of Energy Richland Operations Office	**Signature** Doug A. Shoop for	**Date Signed** 9/19/08

1

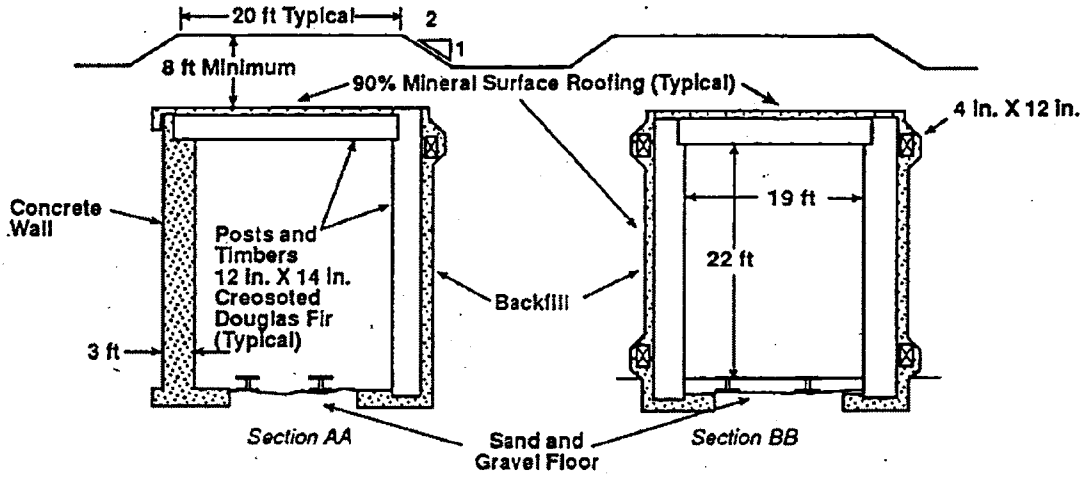
Comments

In Section VII. Facility Operator Information, there is no change to DOE as the Facility Owner/Operator; only a change in Co-Operator*. The change in Co-Operator* will be effective October 1, 2008.

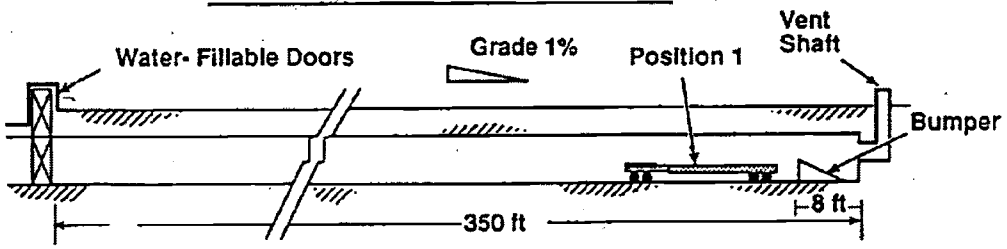
PUREX Tunnel No. 1 - Details



PUREX Tunnel No.1 - Plan View



PUREX Tunnel No.1 - Section View

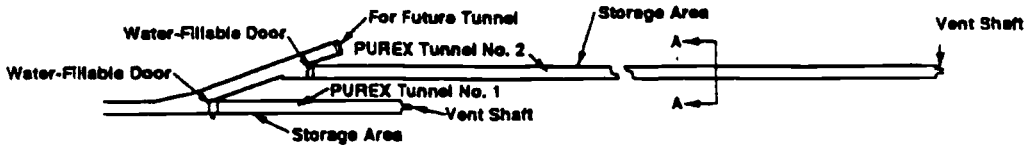


PUREX Tunnel No.1 - Elevation View

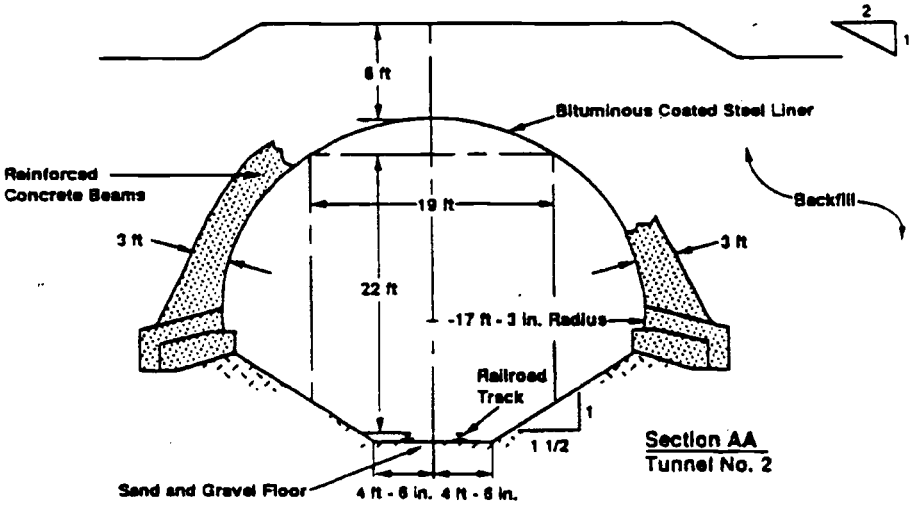
For conversion to meters, multiply feet by 0.3048.
For conversion to centimeters, multiply inches by 2.54.

H96030106.2

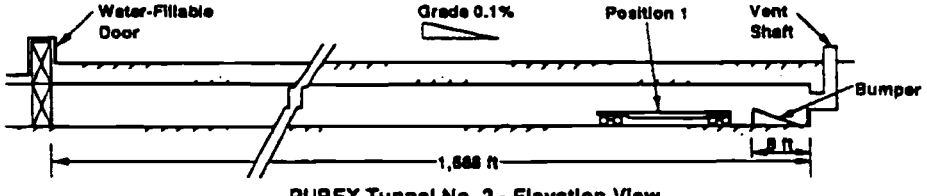
PUREX Tunnel No. 2 – Details



PUREX Tunnels - Plan View



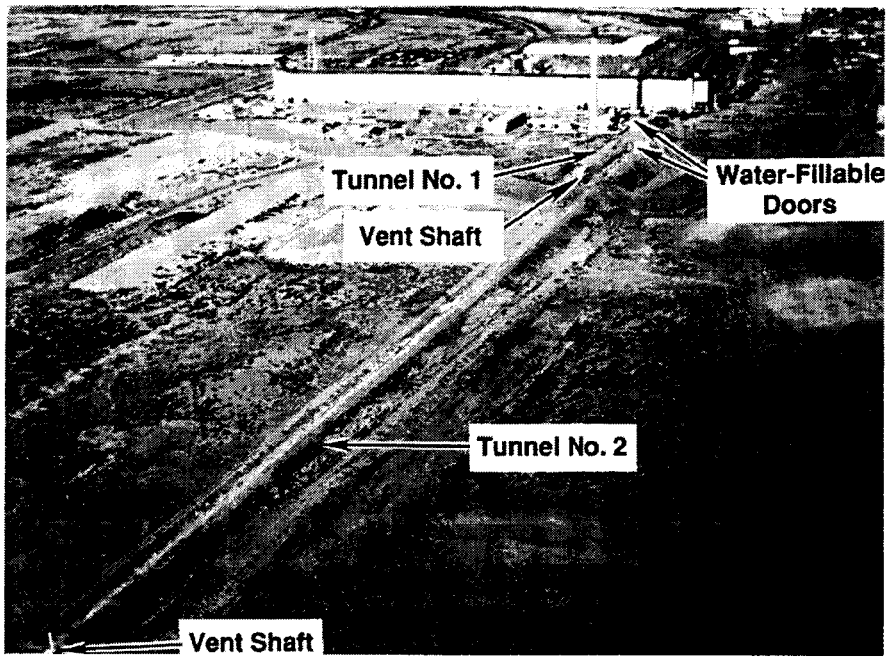
Section AA
Tunnel No. 2



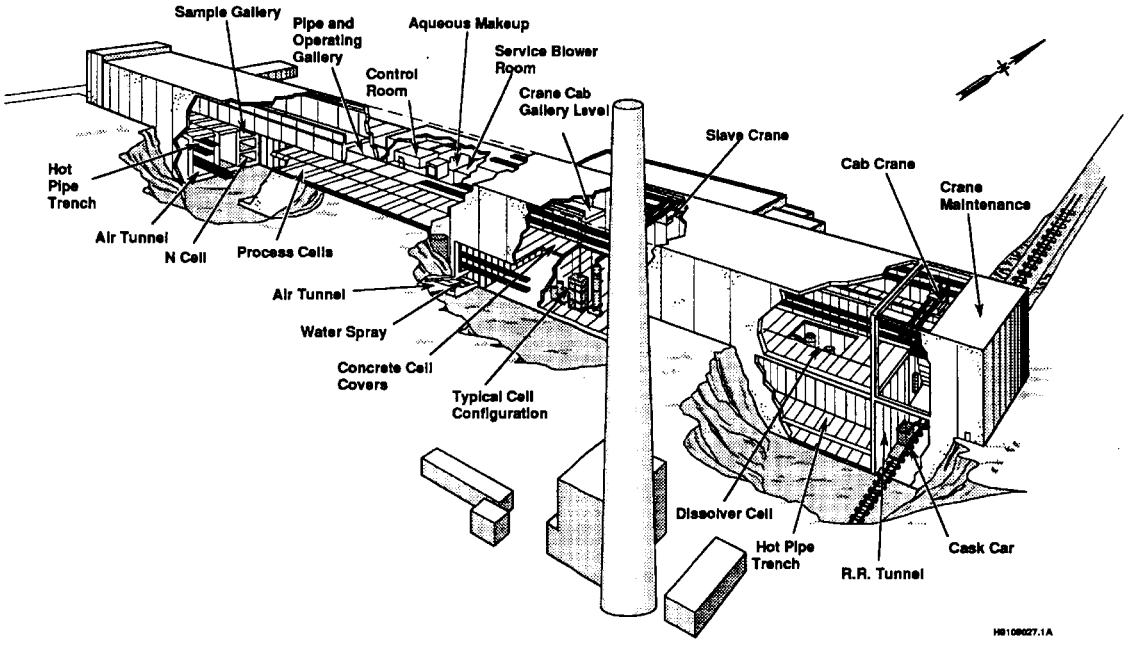
PUREX Tunnel No. 2 - Elevation View

78910032.1

PUREX Storage Tunnels

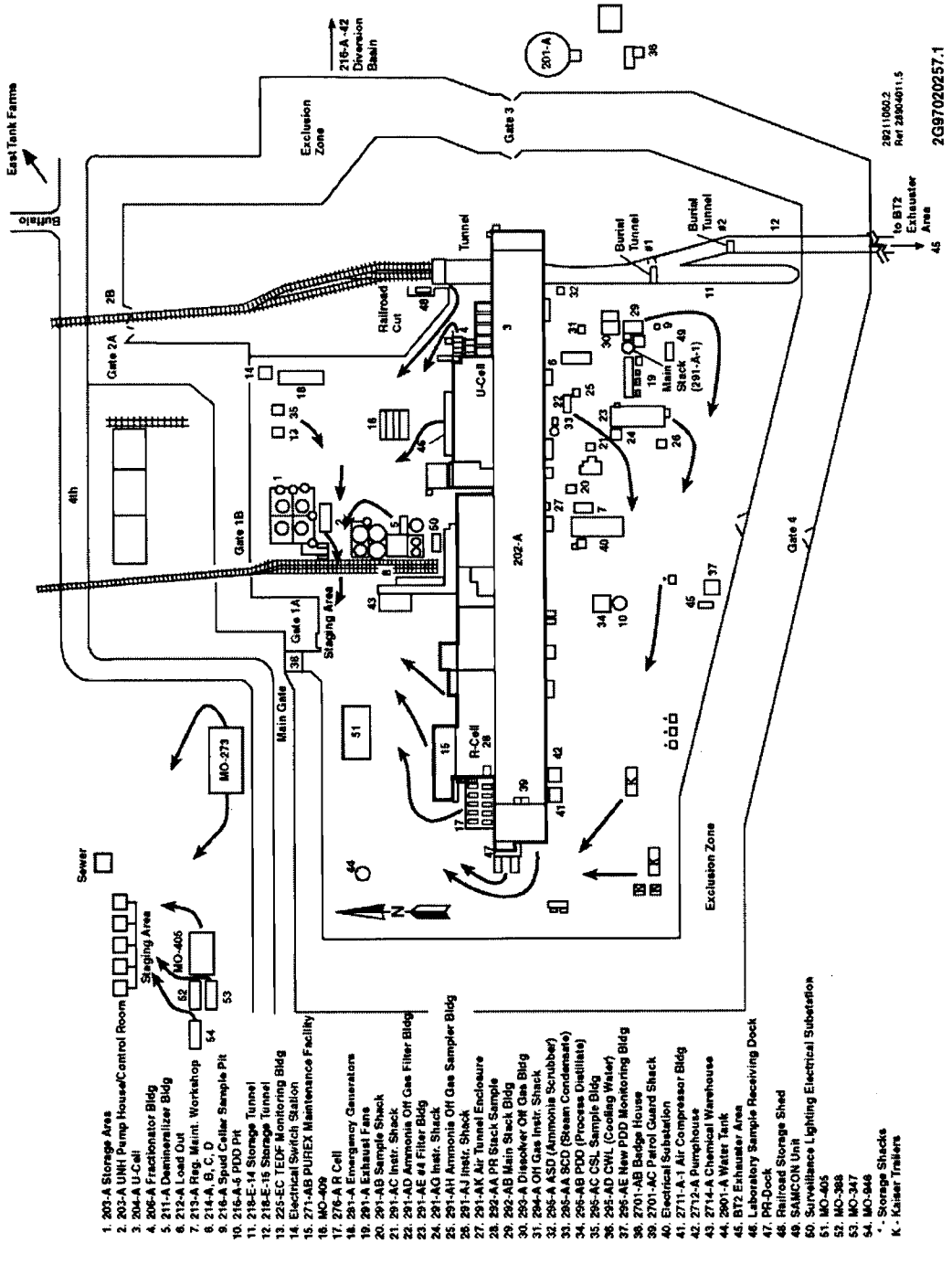


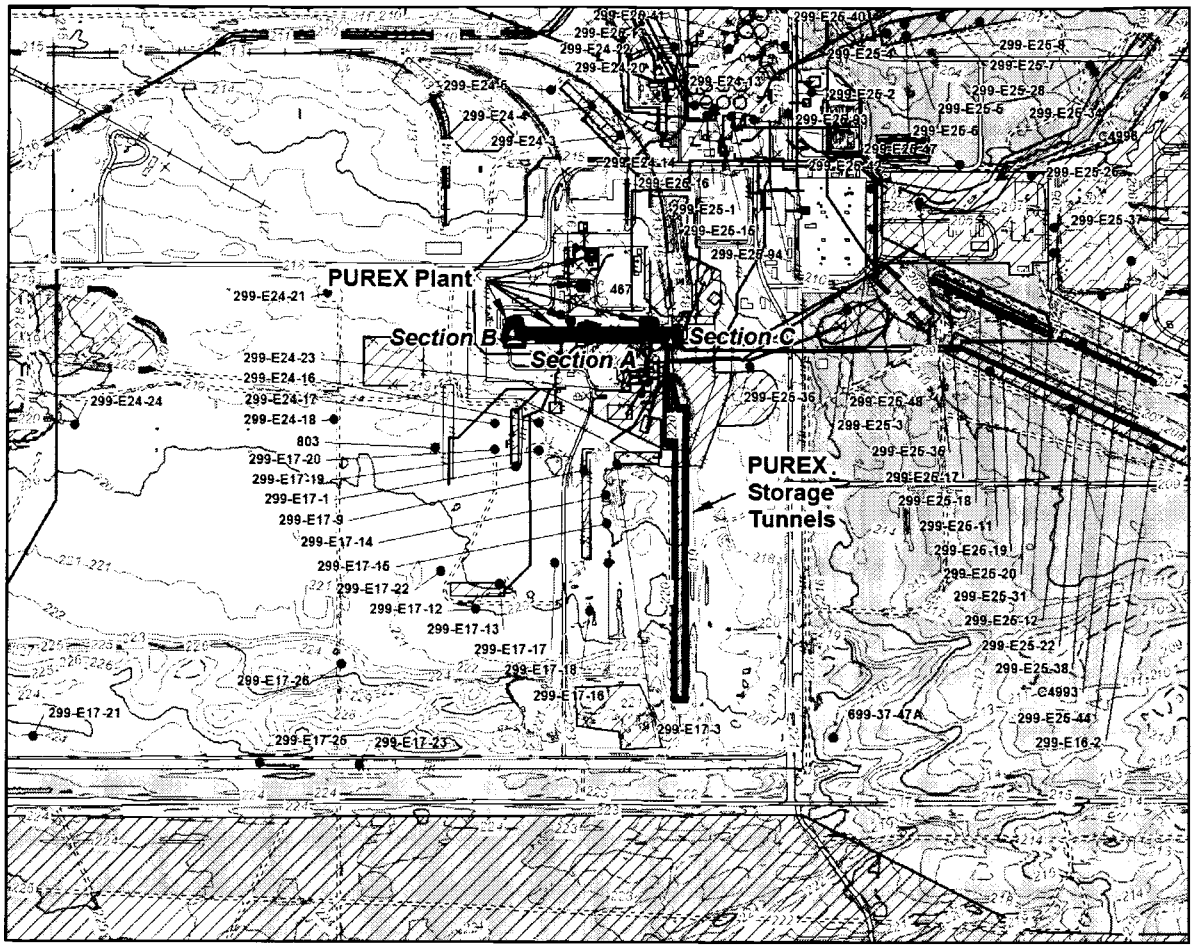
89100252-3CN Photo Taken 1989



PUREX Storage Tunnels

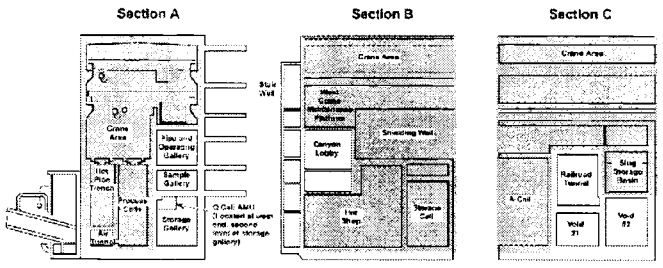
§





Sections
(Not to scale)

Note:
Shaded portions denote areas that are within the TSD boundary.



PUREX

Prepared for:
US DEPARTMENT OF ENERGY
RICHLAND OPERATIONS OFFICE
Created and Published by:
Central Mapping Services
Fluor Hanford, Richland, WA
(509) 373-9076
Intended Use: REFERENCE ONLY
Topographic Data:
1996, Bechtel Hanford, Inc.

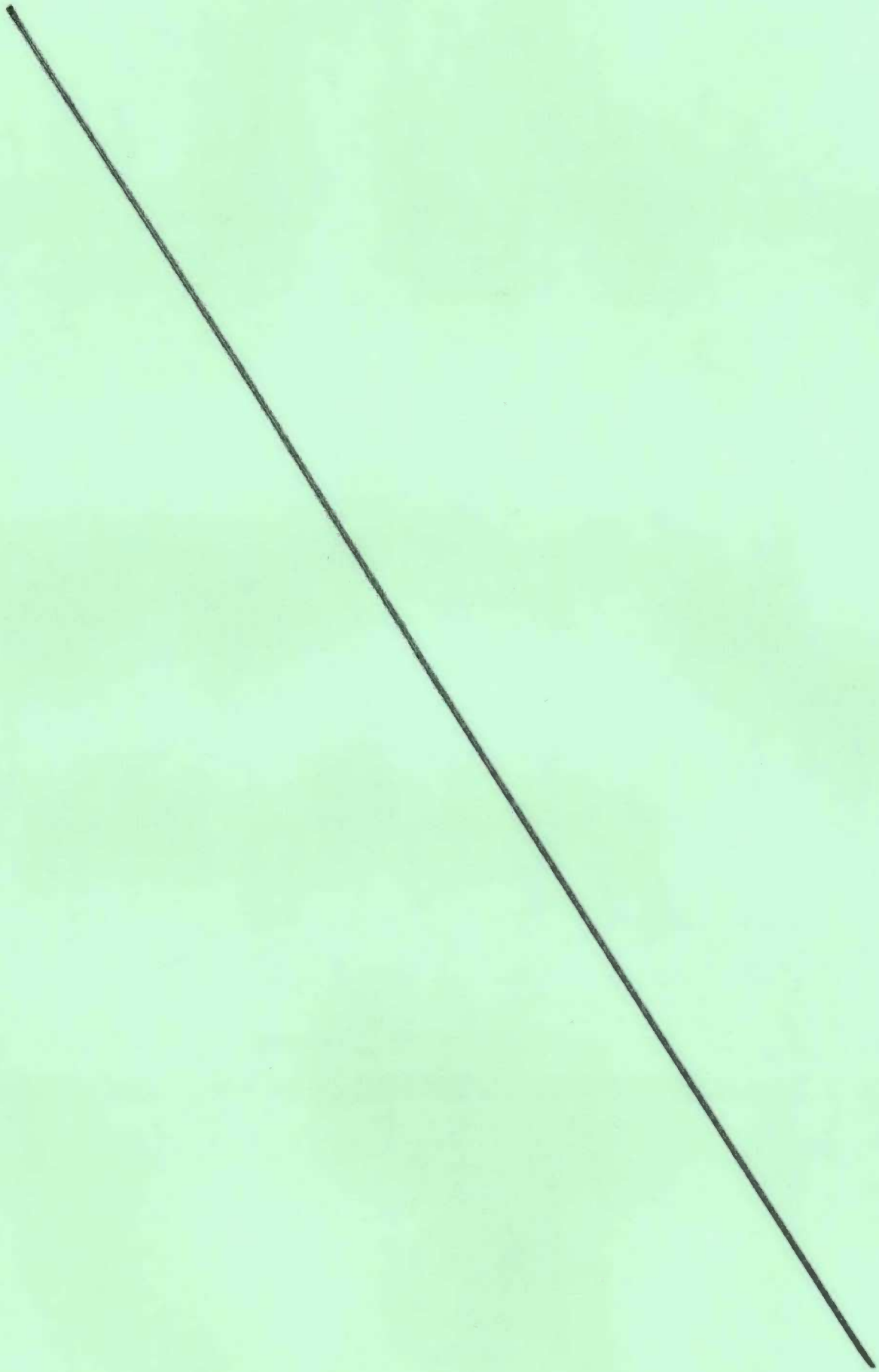
Hanford Site

- TSD Unit Boundary
- DOE Operating Areas
- Hanford Facility
- Injection and Withdrawal Wells
- Contours at 1 Meter Intervals
- Depression Contours
- SWMUs and Known Releases
- Linear SWMUs and Known Releases
- Spot SWMUs and Known Releases
- Buildings and Mobiles
- Structures
- Concrete
- Major Roads
- Service Roads
- Railroads
- Fences

Meters: 0 100 200 300 400
Feet: 0 300 600 900 1,200

CMS

O:\Projects\2005\RCRA_TSD\050614_2ndPriorityFacilityTopos2005_Thompson\Maps\080723_PUREX_LineDwg_85x11_Rev1.mxd - 8/5/2008 @ 2:50:03 PM



CHAPTER 3.0
WASTE ANALYSIS PLAN

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CHAPTER 3.0
WASTE ANALYSIS PLAN

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TABLE

Table 3.1. PUREX Storage Tunnels Inventory 19

GLOSSARY

ALARA	as low as reasonably achievable
ECOLOGY	Washington State Department of Ecology
EHW	extremely hazardous waste
EPA	U.S. Environmental Protection Agency
pH	negative logarithm of the hydrogen-ion concentration
PUREX	plutonium-uranium extraction
QA/QC	quality assurance and quality control
TSD	treatment, storage, and/or disposal
WAC	Washington Administrative Code
WAP	waste analysis plan

METRIC CONVERSION CHART

The following conversion chart provides the reader an aid in conversion.

Into metric units			Out of metric units		
If you know	Multiply by	To get	If you know	Multiply by	To get
Length			Length		
inches	25.40	millimeters	millimeters	0.0393	inches
inches	2.54	centimeters	centimeters	0.393	inches
feet	0.3048	meters	meters	3.2808	feet
yards	0.914	meters	meters	1.09	yards
miles	1.609	kilometers	kilometers	0.62	miles
Area			Area		
square inches	6.4516	square centimeters	square centimeters	0.155	square inches
square feet	0.092	square meters	square meters	10.7639	square feet
square yards	0.836	square meters	square meters	1.20	square yards
square miles	2.59	square kilometers	square kilometers	0.39	square miles
acres	0.404	hectares	hectares	2.471	acres
Mass (weight)			Mass (weight)		
ounces	28.35	grams	grams	0.0352	ounces
pounds	0.453	kilograms	kilograms	2.2046	pounds
short ton	0.907	metric ton	metric ton	1.10	short ton
Volume			Volume		
fluid ounces	29.57	milliliters	milliliters	0.03	fluid ounces
quarts	0.95	liters	liters	1.057	quarts
gallons	3.79	liters	liters	0.26	gallons
cubic feet	0.03	cubic meters	cubic meters	35.3147	cubic feet
cubic yards	0.76456	cubic meters	cubic meters	1.308	cubic yards
Temperature			Temperature		
Fahrenheit	subtract 32 then multiply by 5/9ths	Celsius	Celsius	multiply by 9/5ths, then add 32	Fahrenheit

Source: *Engineering Unit Conversions*, M. R. Lindeburg, PE, Second Ed., 1990, Professional Publications, Inc., Belmont, California.

3.0 WASTE ANALYSIS PLAN

This chapter provides information on the chemical, biological, and physical characteristics of the dangerous waste stored in the PUREX Storage Tunnels. Waste in the tunnels is stored and managed as mixed waste. The PUREX Storage Tunnels provide the necessary shielding for the protection of employees and the environment from mixed waste.

3.1 Chemical, Biological, and Physical Analyses

Regulated material presently stored in the PUREX Storage Tunnels contains the following dangerous waste:

- Lead
- Mercury
- Silver and silver salts
- Chromium
- Cadmium
- Barium
- Mineral oil

In general, dangerous waste is either attached to, contained within, or actually is material removed from the PUREX Plant and other onsite sources. Changes in dangerous waste stored is updated annually in the annual dangerous waste report submitted to Ecology. Future storage of barium and selenium may occur in Tunnel Number 2. The PUREX Storage Tunnels are permitted as a miscellaneous unit under WAC 173-303-680 because the tunnels are not a typical containerized storage unit. That is, the bulk of the material stored in the tunnels is not placed in a container; rather, this material is placed on a portable device (railcar) used as a storage platform. The mixed waste stored in the PUREX Storage Tunnels is encased or contained within carbon or stainless steel plate, pipe, or vessels that meet the WAC 173-303-040 definition of container. Therefore, the mixed waste normally is not exposed to the tunnel environment.

The only free-liquid dangerous waste stored in the tunnels is elemental mercury. The mercury is contained within thick-walled (0.8-centimeter) thermowells. The amount of mercury per thermowell is less than 1.7 liters.

Other liquid containers, such as large discarded process tanks, are stored in the PUREX Storage Tunnels. These containers are 'empty' [per WAC 173-303-160(2)(a)]. In the future, containers will be flushed and the final rinsate sampled and analyzed to verify that the residual heel is not a dangerous waste.

The only stored mixed waste that is designated as either reactive or ignitable (D001) is silver nitrate in the silver reactors [WAC 173-303-090(5)]. There is no mixed waste designated as reactive (D003). The potential for ignition from this source is considered negligible because this material is dispersed on ceramic packing and is physically isolated from contact with any combustible material or ignition source.

3.2 Tracking System

Specific waste tracking forms for the movement of waste destined for the PUREX Storage Tunnels are used. These waste tracking forms effectively track waste inventories from generation through storage.

The waste tracking forms and other supporting documentation will be maintained at the Hanford Facility for a minimum of 5 years following closure of the PUREX Storage Tunnels.

3.3 Facility Description

This waste analysis plan (WAP) has been prepared for the PUREX Storage Tunnels, located on the Hanford Facility, Richland, Washington. This WAP applies to all mixed waste (containing dangerous components) regulated by WAC 173-303 that is transferred to and/or contained in the PUREX Storage Tunnels.

The PUREX Storage Tunnels are permitted as a miscellaneous unit under WAC 173-303-680. The bulk of the waste stored in the PUREX Storage Tunnels is not placed in a typical container; rather, this waste is placed on a portable device (railcar) that is used as a storage platform. In general, the mixed waste stored in the PUREX Storage Tunnels is encased or contained within carbon or stainless steel plate, pipe, or vessels. Therefore, the mixed waste normally is not exposed to the tunnel environment.

The PUREX Facility, located in the 200 East Area, consists of two separate treatment, storage, and/or disposal (TSD) units, the PUREX Plant (202-A Building) and the PUREX Storage Tunnels. Access to the PUREX Storage Tunnels is by means of the railroad tunnel.

The PUREX Storage Tunnels branch off from the railroad tunnel and extend southward from the east end of the PUREX Plant. The tunnels are used for storage of mixed waste from the PUREX Plant and from other onsite sources. Each storage tunnel is isolated from the railroad tunnel by a water-fillable shielding door. There are no electrical utilities, water lines, drains, fire detection or suppression systems, or communication systems provided inside the PUREX Storage Tunnels.

Material selected for storage is loaded on railcars modified to serve as both transport and storage platforms. Normally, a remote-controlled, battery-powered locomotive was used to position the railcar in the storage tunnel. In the past and possibly in the future, other remote movers, e.g., standard locomotive with a string of railcar spacers, power winch, etc., have or could be used to position a railcar in the tunnel or to withdraw a car from the tunnel. The railcar storage positions are numbered sequentially, commencing with Position 1 that abuts the rail stop bumper at the south end of each tunnel. Position 2 is the location of the railcar that abuts the railcar in Position 1 and so forth. The railcars and material remain in the storage tunnel until final disposition is determined. Each railcar is retrievable; however, because the railcars are stored on a single, dead-end railroad track, the railcars can be removed only in reverse order (i.e., last in, first out).

Construction of Tunnel Number 1 was completed in 1956. The Tunnel has three areas, the water-fillable door, the storage area, and the vent shaft. The water-fillable door is located at the north end of Tunnel Number 1 and separates the storage tunnel from the railroad tunnel. The door is 7.5 meters high, 6.6 meters wide, and 2.1 meters thick, and is constructed of 1.3-centimeter steel plate. The door is hollow so that the door can be filled with water to act as a shield when the door is in the down (closed) position. If the door is filled with water, the water must be pumped from the door before the door can be raised. Above the door is a reinforced concrete structure in which the door is raised to open the tunnel. Electric hoists used for opening and closing the door are located on the top of this concrete structure.

The storage area is that portion of the tunnel that extends southward from the water-fillable door. Inside dimensions of Tunnel Number 1 are 109.1 meters long, 6.7 meters high, and 5.9 meters wide. Ceiling and walls are 35.6-centimeters thick and constructed of 30.5- by 35.6-centimeter creosote pressure-treated Douglas fir timbers arranged side by side. The first 30.5 meters of the east wall are constructed of 0.9-meter-thick reinforced concrete. A 40.8-kilogram mineral-surface roofing material was used to cover the exterior surface of the timbers before placement of 2.4 meters of earth fill. The earth cover serves as protection from the elements and as shielding. The timbers that form the walls rest on reinforced concrete footings 0.9 meter wide by 0.3 meter thick. The floor consists of a railroad track laid on a gravel bed. The space between the ties is filled to top-of-tie with gravel ballast. The tracks are on a 1.0 percent downward slope to the south to ensure that the railcars remain in their storage position. A railcar bumper is located 2.4 meters from the south end of the tracks to act as a stop. The capacity of the storage area is eight, 12.8-meter-long railcars.

In June 1960, the first two railcars were loaded with a single, approximately 12.5-meter-long, failed separation column and placed in Tunnel Number 1. Between June 1960 and January 1965, six more railcars were placed in Tunnel Number 1, filling the tunnel. After the last car was placed in the northern-most storage position (Position 8), the water-fillable door was closed, filled with water, and deactivated electrically.

Construction of Tunnel Number 2 was started and completed in 1964. Like Tunnel Number 1, Tunnel Number 2 consists of three functional areas: the water-fillable door, the storage area, and the vent shaft. Construction of Tunnel Number 2 differs from that of Tunnel Number 1 as follows.

- A combination of steel and reinforced concrete was used in the construction of the storage area for Tunnel Number 2 rather than wood timbers, as used in Tunnel Number 1.
- Tunnel Number 2 is longer, having a storage capacity of five times that of Tunnel Number 1.
- The floor of Tunnel Number 2, outboard of the railroad ties, slopes upward to a height of approximately 1.8 meters above the railroad bed, whereas the floor in Tunnel Number 1 remains flat all the way out to the sidewalls.
- The railroad tunnel approach to Tunnel Number 2 angles eastward then angles southward to parallel Tunnel Number 1. The approach to Tunnel Number 1 is a straight extension southward from the PUREX Plant. Center-line to center-line distance between the two tunnels is approximately 18.3 meters.

The physical structure of the water-fillable door at the north end of Tunnel Number 2 essentially is identical to the water-fillable door for Tunnel Number 1. The water-fillable door for Tunnel Number 2 is approximately 57.9 meters south and 18.3 meters east of the water-fillable door for Tunnel Number 1. As of March 1997, the door is empty and there is no plan to fill the door.

The storage area of Tunnel Number 2 is that portion of the tunnel extending southward from the water-fillable door. Construction of this portion of Tunnel Number 2 consists of a 10.4-meter diameter, steel (0.5 centimeter plate), and semicircular-shaped roof, supported by internal I-beam wales attached to external, reinforced concrete arches. The concrete arches are 0.4 meter thick and vary in width from 0.4 to 1.8 meters. The arches are spaced on 4.8-meter centers. This semicircular structure is supported on reinforced concrete grade beams approximately 1.8 meters wide by 1.2 meters thick (one on each side) that run the full length of Tunnel Number 2. The interior and exterior surfaces of the steel roof are coated with a bituminous coating compound to inhibit corrosion. The entire storage area is covered with 2.4 meters of earth fill to serve as shielding.

The nominal inside dimensions of Tunnel Number 2 are 514.5 meters long, 7.9 meters high, and 10.4 meters wide. However, because of the arch-shaped cross-section of Tunnel Number 2 and entry clearance at the water-fillable door, the usable storage area (width and height above top-of-rail) is 6.7 meters high and 5.8 meters wide, the same dimensions as for Tunnel Number 1. The floor consists of a railroad track laid on a gravel bed. The space between ties is filled to top-of-tie with gravel ballast. Commencing at the ends of the 2.4-meter-long ties, the earth floor is sloped upward on a 1 (vertical) to 1 1/2 (horizontal) grade. The tracks are on a 1/10 of 1 percent downgrade slope to the south to ensure the railcars remain in their storage position. A railcar bumper is located 2.4 meters from the south end of the tracks to act as a stop. The capacity of the storage area is 40, 12.8-meter-long railcars.

The first railcar was placed in storage in December 1967. Table 3.1 contains an approximate inventory of waste stored in the PUREX Storage Tunnels.

3.3.1 Process and Activities

The function of the PUREX Tunnels is to store mixed waste until the waste can be processed for final disposal. When waste is to be placed in the storage tunnels, a work plan, describing the overall transfer activities, and a storage tunnel checklist are prepared. The work plan and storage tunnel checklist are routed for review and concurrence by key personnel and forwarded to management for approval.

3.3.2 Physical Characterization of Material to be Stored

Physical characterization of waste includes an evaluation of the following physical properties:

- Length, width, and height
- Gross weight and volume
- Preferred orientation for transport and storage
- Presence of dangerous waste constituents

Information sources used in physical characterization include equipment fabrication and installation drawings, operational records, and process knowledge. Physical characterization provides information necessary to describe the waste material. Such information also is used to design and fabricate, if required, supports on the railcar.

Before removal from service, the equipment could be flushed to minimize loss of products, to reduce contamination, and to reduce dangerous waste constituents present in a residual heel to nonregulated levels. When equipment is flushed, analysis of the rinsate is used to determine when these goals have been achieved.

3.4 Identification/Classification and Quantities Of Dangerous Waste Managed Within The PUREX Storage Tunnels

Table 3.1 contains an approximation of the total amount of waste stored within the PUREX Storage Tunnels.

3.5 Waste Analysis Parameters

Analytical requirements were selected based on knowledge required for the safe handling and storage of the waste within the PUREX Storage Tunnels, including any operational compliance issues.

3.5.1 Waste Identification

A prerequisite step in proper waste management is to address whether waste being considered for management within the PUREX Storage Tunnels falls within the scope of this unit's permit. This includes identifying any dangerous waste in accordance with regulatory and permit requirements and applicability of any land disposal restrictions.

This section provides information on how the chemical and physical characteristics of the mixed waste currently stored in the PUREX Storage Tunnels were determined so that the waste is stored and managed properly.

Regulated material presently stored in the PUREX Storage Tunnels contains the following dangerous waste:

- Lead
- Mercury
- Silver and silver salts
- Chromium
- Cadmium
- Barium
- Mineral oil

Storage of non-PUREX Plant waste is reviewed on a case-by-case basis. Sampling, chemical analysis, process knowledge (as discussed in the following section), and/or inventory information from waste tracking forms provided from other onsite sources are required to confirm the characteristics and quantities of mixed waste to be stored. Future waste and dangerous constituents might not be in the same configuration or form as described in the following sections.

3.5.1.1 Lead

Lead stored was used in various capacities during past Hanford Facility operations. Primary functions of lead included use as weights, counterweights, and shielding. Often the lead is encased in steel (carbon or stainless) to facilitate its attachment to various types of equipment.

Lead exhibits the characteristic of toxicity as determined by the toxicity characteristics leaching procedure and is designated D008 [WAC 173-303-090(8)]. The quantity of lead present could produce an extract greater than 500 milligrams per liter should the lead be exposed to a leachate. However, because the bulk of the lead is encased in steel, is stored inside a weather-tight structure, and is elevated above floor level on railcars that isolate the lead from other materials stored, the potential for exposure of bare lead to a leachate is considered negligible.

Sampling and chemical analysis is not performed on lead associated with the material placed in the PUREX Storage Tunnels. Therefore, the accuracy of the estimate on the amount of lead presently stored in each tunnel is limited to the data available from process knowledge. Counterweights on equipment dunnage and lead used for shielding cannot be quantified by existing historical records and are not included for lead listed on Table 3.1. However, if removed from the tunnels, the material will be examined and any suspect attachments will be removed, evaluated, and disposed of in accordance with established methods.

3.5.1.2 Mercury

Mercury is contained within thermowells that are an integral part of spent reactor fuel dissolvers used at the PUREX Plant. The dissolvers are large 304L stainless steel process vessels that are approximately 2.7 meters in diameter, 7.3 meters tall, and weigh approximately 26,309 kilograms. The outer shell is constructed of a 1-centimeter-thick plate. The dissolvers were used in decladding and dissolving spent reactor fuel in the PUREX Plant.

Depending on the specific dissolver in question, 19.1 or 45.4 kilograms of mercury (1.4 or 1.77 liters) were poured in each of the two thermowells per dissolver (38.2 or 90.8 kilograms total per dissolver) following vertical installation of the dissolvers inside the PUREX canyon and before the dissolver was installed in a process cell. The mercury served to transfer heat from the dissolver interior to the thermohm temperature sensor mounted within the thermowell. This mercury remains within the thermowells of discarded dissolvers. In preparation for storage, the thermohms were removed and the upper end of each thermowell was plugged with a 304L stainless steel nozzle plug. In storage, the discarded dissolver rests in an inclined position in a cradle on the railcar. The mercury contained in the thermowells remains in the lower portion of each thermowell and, under normal conditions, is never in contact with the mechanical closure on the nozzle end of the thermowell.

Mercury exhibits the characteristic of toxicity as determined by the toxicity characteristics leaching procedure and is designated D009 [WAC 173-303-090(8)].

The potential for mercury to become exposed to leachate is considered negligible. The PUREX Storage Tunnels are designed and constructed as weather-tight structures. Further, the mercury is encased in a stainless steel pipe within a stainless steel vessel that is stored on a railcar above the floor level of the tunnels. Therefore, exposure of the mercury stored in the tunnels to leachate is not considered a credible occurrence.

Sampling and chemical analysis is not performed on mercury associated with the dissolvers stored in Tunnel Number 2. The quantity of mercury present in each thermowell is documented on Table 3.1.

3.5.1.3 Silver

Silver, mostly in the form of silver salts deposited on unglazed ceramic packing, is contained within the discarded silver reactors stored in Tunnel Number 2. The silver reactors were used to remove iodine from the offgas streams of the spent reactor fuel dissolvers. The reactor vessel is approximately 1.4 meters in diameter by 4.1 meters tall and is constructed of 1-centimeter 304L stainless steel. The vessel contains two 1.2-meter-deep beds of packing. Each bed consists of a 30.5-centimeter depth of 2.5-centimeter unglazed ceramic saddles topped with a 0.6-meter depth of 1.3-centimeter unglazed ceramic saddles. The two beds are separated vertically by a distance of about 0.6 meter, and each bed rests on a support made of stainless steel angles and coarse screen. The packing was coated initially with 113.4 kilograms of silver nitrate used for iodine retention. Nozzles on the top of the reactor were provided to allow flushing and/or regeneration of the packing with silver nitrate solution as the need arose.

Because of competing reactions, which include conversion of silver nitrate to silver iodide, reduction of silver nitrate to metallic silver, and formation of silver chloride, the packing of a stored silver reactor contains a mixture of silver nitrate, silver halides, and silver fines.

Silver salts exhibit the characteristics of toxicity as determined by the toxicity characteristics leaching procedure and are designated D011 [WAC 173-303-090(8)]. Silver salts exhibit the characteristic of ignitability and are designated as D001 [WAC 173-303-090(5)].

The potential of silver, including silver salts, stored in the PUREX Storage Tunnels to become exposed to leachate is considered negligible. Silver is contained within a stainless steel vessel, stored inside a weather-tight structure, and elevated above floor level on a railcar. Therefore, exposure of the silver stored in the tunnels to leachate is not considered a credible occurrence. In addition, the contained silver is isolated from contact with any combustibles; therefore, the possibility of ignition is considered extremely remote.

Provisions for taking samples of the packing were not provided in the design of the vessels. Therefore, sampling and chemical analysis are not performed for silver salts before placing a silver reactor in storage. However, for accountability, the total silver content (Table 3.1) is considered silver nitrate, the salt that exhibits the characteristics of both ignitability and toxicity.

The quantity of silver salts contained within a discarded silver reactor is a function of silver nitrate regeneration history. Operating records (process knowledge) of regenerations and flushes are used to estimate the total accumulation of silver within each reactor.

3.5.1.4 Chromium

Presently, chromium stored in Tunnel Number 2 is contained within a failed concentrator removed from the PUREX Plant, and within stainless steel containers received from the 324 Building. The concentrator is a vertical tube structure that was used to concentrate aqueous streams from the final uranium cycle, final plutonium cycles, final neptunium cycles, and condensate from the acid recovery system for recycle. Following service, the concentrator was inspected and found to contain silicate solids with high levels of chromium from the corrosion of stainless steel. The existence of chromium within the 324 Building waste was determined through process knowledge. Chromium exhibits the characteristic of toxicity as determined by the toxicity characteristics leaching procedure and is designated D007 [WAC 173-303-090(8)]. The potential for the chromium stored in Tunnel Number 2 to become exposed to leachate is considered negligible. Tunnel Number 2, is designed and constructed to be weather-tight. Further, the chromium is encased within stainless steel vessels and containers that are stored on railcars above the floor level of the tunnel. Therefore, exposure of the chromium stored in the tunnel to leachate is not considered a credible occurrence.

The quantity of chromium within the concentrator was estimated by calculating the volume of silicate solids and the percentage of chromium within the silicate solids. The quantity of chromium in the 324 Building waste was based on process knowledge.

3.5.1.5 Cadmium

Presently, cadmium stored in the PUREX Storage Tunnel Number 2 is associated with shielding and with a dissolver moderator removed from the PUREX Plant, and within stainless steel containers received from the 324 Building. The cadmium was used to shield equipment and consists of sheets of the metal attached to lead, both of which could be encased in steel. The cadmium received from the 324 Building was used in waste technology research and development programs.

The dissolvers are annular vessels that are geometrically favorable for criticality safety. The dissolvers were placed over cadmium lined (neutron absorbers) moderators for additional criticality safety. The moderator is a centrally located, cylindrical, cadmium-jacketed 0.08-centimeter-thick concrete 15.2-centimeter-thick neutron absorber. The moderators are approximately 4.4 meters tall by approximately 1.5 meters outer diameter.

Cadmium exhibits the characteristic of toxicity as determined by the toxicity characteristics leaching procedure and is designated D006 [WAC 173-303-090(8)]. If exposed to a leachate, the quantity of cadmium present could produce an extract having a concentration of greater than or equal to 1 milligram per liter, but less than 100 milligrams per liter; therefore, the mixed waste is managed as a WT02 [WAC 173-303-100(5)].

The potential for the cadmium stored in Tunnel Number 2 to become exposed to leachate is considered negligible. Tunnel Number 2 is designed and constructed to be weather-tight. Further, the cadmium is stored on railcars above the floor level of the tunnel. Therefore, exposure of the cadmium stored in the tunnel to leachate is not considered a credible occurrence.

3.5.1.6 Barium

Presently, barium is stored in Tunnel Number 2 in stainless steel containers received from the 324 Building. The waste was generated during numerous research and development programs conducted in B-Cell of the Waste Technology Engineering Laboratory (324 Building). The existence of barium within the 324 Building waste was determined through process knowledge.

Barium exhibits the characteristic of toxicity as determined by the toxicity characteristics leaching procedure and is designated D005 [WAC 173-303-090(8)].

The potential for barium stored in Tunnel Number 2 to become exposed to leachate is considered negligible. Tunnel Number 2 is designed and constructed to be weather-tight. Further, the barium is encased in steel containers stored on a railcar above the floor level of the tunnel. Therefore, exposure of the barium stored in the tunnel to leachate is not considered a credible occurrence.

3.5.1.7 Mineral Oil

Presently, mineral oil is stored in Tunnel Number 2 in stainless steel containers received from the 324 Building. The mineral oil was used in the B-Cell viewing windows in the 324 Building. Oil leaking from the windows was absorbed on rags and clay absorbent material.

The material safety data sheet for the mineral oil lists a lethal dose (LD50) of 2 grams per kilogram (dermal rabbit). Therefore, the oil designates as a Toxic Category A WT02 [WAC 173-303-100(5)].

The potential for the absorbed mineral oil stored in Tunnel Number 2 to become exposed to leachate is considered negligible. Tunnel Number 2 is designed and constructed to be weather-tight. Further, the mineral oil is encased in steel containers stored on a railcar above the floor level of the tunnel. Therefore, exposure of the mineral oil stored in the tunnel to leachate is not considered a credible occurrence.

3.5.1.8 Identification of Incompatible Waste

The next step is to ensure that sufficient information concerning the waste has been provided so the waste can be managed properly. This includes identifying incompatible waste. These safety issues primarily are related to prevention of unwanted chemical reactions that could create a catastrophic situation, such as a fire, an explosion, or a large chemical release.

3.5.1.9 Operational Considerations

Sufficient information must be available to ensure that incoming waste meets operational acceptance limits, e.g., physical size, ALARA concerns, and WAC 173-303 requirements. These operating specifications are limits and controls imposed on a process or operation that, if violated, could jeopardize the safety of personnel, and could damage equipment, facilities, or the environment. Operating specifications have been established from operating experience, process knowledge, and calculations.

3.5.2 Parameter and Rationale Selection Process

This WAP describes the process to ensure that the dangerous waste components of the material stored in the tunnels are properly characterized and designated so that dangerous and mixed waste is managed properly.

The parameters considered for waste designation under WAC 173-303-070(3) and the rationale for there application is discussed in the following sections.

3.5.2.1 Discarded Chemical Products

The first category of dangerous waste designation is "Discarded Chemical Products" (WAC 173-303-081). The waste stored in the tunnels does not fit the definitions in WAC 173-303-081 for a discarded chemical product. Therefore, the waste stored in the PUREX Storage Tunnels is not designated as a discarded chemical product.

3.5.2.2 Dangerous Waste Sources

The second category of dangerous waste designation is "Dangerous Waste Sources" (WAC 173-303-082). The waste stored in the tunnels is not listed on the "Dangerous Waste Sources List" (WAC 173-303-9904). Therefore, the waste stored in the PUREX Storage Tunnels is not designated as a dangerous waste source.

3.5.2.3 Dangerous Waste Characteristics

The third category of dangerous waste designation is "Dangerous Waste Characteristics" (WAC 173-303-090). The characteristics are as follows.

- Characteristic of Ignitability – Although the solid silver nitrate has not been tested in accordance with Appendix F of 49 CFR 173, the waste is assumed to be an oxidizer as specified in 49 CFR 173.127(a). Therefore, the silver nitrate waste is assumed to exhibit the characteristic of ignitability under WAC 173-303-090(5) and is designated as D001.
- Characteristic of Corrosivity – Some of the material stored within the tunnels either has contained or has been in contact with corrosive liquids. The standard operating procedure has been to flush vessels with water to recover as much special nuclear material as practical. In addition, flushing removes much of the mixed waste contamination, minimizing the spread of contamination during handling. Currently, the final aqueous rinse is sampled and analyzed to confirm that the pH is greater than 2 and less than 12.5. Therefore, the waste stored in the PUREX Storage Tunnels is not designated as corrosive waste.

- Characteristic of Reactivity – The waste stored in the tunnels does not meet any of the definitions of reactivity as defined in WAC 173-303-090(7). The waste material is not unstable, does not react violently with water, does not form explosive mixtures, or does not generate toxic gases. Therefore, the waste stored in the PUREX Storage Tunnels is not designated as reactive waste.
- Characteristic of Toxicity – Lead, mercury, silver, chromium, barium, and cadmium are identified on the Toxicity Characteristics list. The quantity of these materials stored in the tunnels is sufficient that, should the substances come in contact with a leachate (an event considered unlikely), the concentration of the extract could be above the limits identified in the list. Therefore, this waste is designated D005, D006, D007, D008, D009, and D011.

The PUREX Storage Tunnels also are permitted for selenium (D010). Currently, there is no waste stored in the tunnels that is designated for D010; however, there is a potential for waste with this waste number to be stored within the tunnels.

3.5.2.4 Dangerous Waste Criteria

The fourth category of dangerous waste designation is "Dangerous Waste Criteria" (WAC 173-303-100). The criteria are as follows:

- Toxicity Criteria – Cadmium meets the toxicity criteria in WAC 173-303-100(5) when performing a book designation. Because of the concentrations present, the waste containing these constituents is designated as dangerous waste (DW) and is assigned the dangerous waste number of WT02.
- Persistence Criteria – Currently, no waste stored in the tunnels has been designated as persistent per WAC 173-303-100(6).

3.5.2.5 Waste Designation Summary

The mixed waste currently stored in the PUREX Storage Tunnels is designated as follows:

- Lead – D008; EHW
- Mercury – D009; EHW
- Silver and silver salts – D001, D011; EHW
- Chromium – D007; EHW
- Cadmium – D006, WT02; DW
- Barium – D005; EHW
- Mineral Oil – WT02; DW

3.5.3 Rationale for Parameter Selection

Refer to Section 3.5.2.

3.5.4 Special Parameter Selection

Refer to Section 3.5.2.

3.5.5 Selection of Sampling Procedures

The following sections discuss the sampling methods and procedures that will be used. Sampling usually will be in accordance with requirements contained in the pertinent sampling analysis plan, procedures, and/or other documents that specify sampling and analysis parameters.

3.5.6 Sampling Strategies

The only analysis presently used in support of the PUREX Storage Tunnels operation is a corrosivity check on the final in-place aqueous rinse of discarded vessels before the vessels are released for storage.

The pH is determined by a pH meter using U.S. Environmental Protection Agency (EPA) Test Method 9040 or 9041 in Test Methods for the Evaluation of Solid Waste: Physical/Chemical Methods (EPA 1986). The Resource Conservation and Recovery Act (RCRA) sampling will not be performed on any waste currently stored in the PUREX Storage Tunnels.

Waste received that is not generated at the PUREX Plant could require sampling strategies associated with this waste that will be developed on a case-by-case basis.

Sampling Methods

Process knowledge of the characteristics and the quantities of the dangerous waste to be stored in the PUREX Storage Tunnels is considered sufficient to properly designate and manage the stored waste.

The waste currently stored in the tunnels is lead, mercury, chromium, cadmium, barium, mineral oil, silver, and silver salts. Sampling and chemical analysis of the lead, mercury, cadmium, barium, mineral oil, or chromium to confirm their presence would not provide additional data beneficial to proper management of the waste and would not comply with ALARA principles. The silver salts are dispersed over a large area on ceramic packing contained within a large stainless steel reactor vessel. Representative sampling of the ceramic packing is not considered to be practical and therefore was not performed.

If RCRA sampling is required for operation of the PUREX Storage Tunnels, representative sampling methods referenced in WAC 173-303-110 or some other method approved by the Washington State Department of Ecology (Ecology) will be used. For waste received from other Hanford Facility activities, existing sampling, chemical analysis, and/or process knowledge documentation is used to confirm the characteristics and quantities of mixed waste to be stored. Storage of non-PUREX Facility waste is reviewed on a case-by-case basis.

3.5.6.1 Frequency of Analyses

Because the dangerous waste components of mixed waste stored in the PUREX Storage Tunnels are stable and will remain undisturbed for a long time, the waste designations and quantities present will remain the same as assigned at the time of storage. Therefore, repeated analysis is not considered necessary to ensure that waste designation data are representative.

3.5.7 Selection of Sampling Equipment

The only analysis presently used in support of the PUREX Storage Tunnels operation is for corrosivity on the final in-place aqueous rinse of discarded vessels before the vessels are released for storage. The pH is determined by Method 9040 or 9041 (SW-846). The RCRA sampling methods, as referenced in WAC 173-303-110, will not be performed on any waste currently stored in the PUREX Storage Tunnels.

3.5.8 Maintaining and Decontaminating Field Equipment

All RCRA sampling equipment used to collect and transport samples must be free of contamination that could alter test results. Equipment used to obtain and contain samples must be clean. Acceptable cleaning procedures for sample bottles and equipment include, but are not limited to, washing with soap or solvent, and steam cleaning. After cleaning, cleaning residues must be removed from all equipment that could come in contact with the waste. One method to remove these residues would be a solvent (acetone or other suitable solvent) rinse followed by a final rinse with deionized water. Equipment must be cleaned before use for another sampling event.

After completion of sampling, equipment should be cleaned as indicated previously. If decontamination of the equipment is not feasible, the sampling equipment should be disposed of properly.

3.5.9 Sample Preservation and Storage

Following RCRA sampling, sample preservation follows methods set forth for the specific analysis identified. Preservation is in accordance with the methods stated in SW-846 or any of the test methods adopted by the Hanford Facility that meet WAC 173-303 requirements. No preservation method will be used when there are ALARA concerns.

3.5.10 Quality Assurance and Quality Control Procedures

The only test method presently used in support of the PUREX Storage Tunnels operation is a corrosivity check on the final in-place aqueous rinse of discarded vessels before the vessels are released for storage. The RCRA sampling will not be performed on any waste currently stored in the PUREX Storage Tunnels. Field duplicates, field blanks, trip blanks, and equipment blanks will not be taken. Split samples could be taken at the request of Ecology.

Generally, quality assurance and quality control (QA/QC) requirements for sampling will be divided between paperwork requirements, such as chain-of-custody, and sampling and analysis activities. This section addresses sampling QA/QC requirements. Analytical QA/QC is discussed in Section 3.6.

A chain-of-custody procedure is required for all sampling identified by this WAP. At a minimum, the chain of custody must include the following: (1) description of waste collected, (2) names and signatures of samplers, (3) date and time of collection and number of containers in the sample, and (4) names and signatures of persons involved in transferring the samples.

3.5.11 Health and Safety Protocols

The safety and health protocol requirements established for the Hanford Site must be followed for all RCRA sampling activities required by this WAP.

3.6 Laboratory Selection and Testing and Analytical Methods

This section discusses laboratory selection and the types of acceptable analytical methods for RCRA samples.

3.6.1 Laboratory Selection

Laboratory selection is limited as only a few laboratories are equipped to handle mixed waste because of the special equipment and procedures that must be used to minimize personnel exposure to mixed waste. Laboratory selection depends on laboratory capability, nature of the sample, timing requirements, and cost. At a minimum, the selected laboratory must have the following:

- A comprehensive QA/QC program (both qualitative and quantitative).
- Technical analytical expertise.
- An effective information management system.

These requirements will be met if the selected laboratory follows the pertinent requirements contained in the Hanford Federal Facility Agreement and Consent Order Action Plan, Section 6.5. The selected laboratory also can meet these requirements by having some other type of QA/QC program as long as equivalent data quality is achieved.

3.6.2 Testing and Analytical Methods

The testing and analytical methods for corrosivity used by the various onsite analytical laboratories are outlined in SW-846. These methods in some cases deviate from SW-846 and American Society for Testing and Materials-accepted specifications for holding times, sample preservation, and other specific analytical procedures. These deviations are discussed in Analytical Methods for Mixed Waste Analyses at the Hanford Site (DOE/RL-94-97).

3.7 Waste Re-Evaluation Frequencies

Re-evaluation of waste within the PUREX Storage Tunnels will not occur because of the personnel and environmental exposure to mixed waste and the way the railcars are positioned in the tunnels. The waste is expected to remain stable.

3.8 Special Procedural Requirements

The following sections describe special procedural requirements associated with waste in the PUREX Storage Tunnels.

3.8.1 Procedures for Receiving Wastes Generated Offsite

The PUREX Storage Tunnels do not accept waste generated off the Hanford Site.

3.8.2 Procedures for Ignitable, Reactive, and Incompatible Waste

Presently, the only ignitable, reactive, or incompatible dangerous waste stored in the PUREX Storage Tunnels is the silver nitrate coating on the ceramic packing inside the silver reactors. This material is confined to the interior of a large stainless steel vessel (Section 3.5.1.1) that separates this material from all other waste material stored in the tunnel. The requirements in WAC 173-303-395(1)(a) require 'No Smoking' signs be conspicuously placed wherever there is a hazard present from ignitable or dangerous waste. 'No Smoking' signs are not considered appropriate at the PUREX Storage Tunnels because of ALARA principles. Smoking is not allowed in any area with ALARA concerns and rules prohibiting smoking are strictly enforced. This policy serves to achieve the no smoking intent of WAC 173-303-395(1)(a), posting and maintaining 'No Smoking' signs are not considered appropriate.

Isolated areas within the PUREX Storage Tunnels make periodic inspections inconsistent with ALARA guidelines[e.g., an annual fire inspection as required by WAC 173-303-395(1)(d) for storage areas containing ignitable waste]. Therefore, such inspections are not performed.

3.8.3 Provisions for Complying with Land Disposal Restriction Requirements

Operation of the PUREX Storage Tunnels does not involve land disposal or treatment of dangerous waste. The information provided by the generating unit regarding land disposal restrictions of dangerous waste is sufficient to operate the PUREX Storage Tunnels in compliance with land disposal restriction requirements. When final disposition of the waste occurs, this information will be passed on for final treatment or disposal of the waste.

3.8.4 Deviations from the Requirements of this Plan

Management may approve deviations from this plan if special circumstances arise that make this prudent. These deviations must be documented in writing with a copy to be retained by the management.

3.9 Recordkeeping

Records associated with this waste analysis plan and waste verification program are maintained on the Hanford Facility. These records will be maintained until closure of the PUREX Storage Tunnels. Records associated with the waste inventory will be maintained for 5 years.

3.10 References

Ecology, EPA, and DOE, 2003, Hanford Federal Facility Agreement and Consent Order, Washington State Department of Ecology, U.S. Environmental Protection Agency, and U.S Department of Energy, Richland, Washington, amended periodically.

DOE/RL-94-97, Analytical Methods for Mixed Waste Analyses at the Hanford Site, Rev. 0, U.S. Department of Energy, Richland Operations Office, Richland, Washington.

EPA, 1986, Test Methods for the Evaluation of Solid Waste: Physical/Chemical Methods, SW-846, 3rd ed., U.S. Environmental Protection Agency, Washington, D.C.

Table 3.1. PUREX Storage Tunnels Inventory

PUREX #1 Storage Tunnel (218-E-14)
Tunnel #1 is at its Capacity as of 1/22/65

PUREX #1 Storage Tunnel is located at the southeast end of the PUREX Plant and is an extension of the railroad tunnel. The storage area is approximately 109 meters long, 6.9 meters high, and 5.8 meters wide. The tracks have a one percent downgrade toward the south end of the tunnel. The capacity of the Storage Tunnel is eight modified railroad cars, 12.8 meters long.

Position	PUREX #1 Storage Tunnel (218-E-14)
1. & 2.	HA column and miscellaneous jumpers in box placed in Tunnel #1 on 6/60 HA 4,700 Cu. Ft. Jumpers 2,190 Cu. Ft., Pb-~115 Kg
3.	E-F11 #1 (1WW Waste) Concentrator failed 7/24/60. Placed in Tunnel #1 on 7/29/60, 1,900 Cu. Ft.
4.	G-E2 Centrifuge, miscellaneous jumpers in box and two tube bundles. Placed in Tunnel #1 on 12/24/60. (FUG SER# 762) 2,465 Cu. Ft., Pb-~115 Kg.,
5.	E-H4 (3WB) Concentrator failed 1/4/61. Placed in Tunnel #1 on 1/4/61, 2,336 Cu. Ft.
6.	E-F6 (2\NW Waste) Original Concentrator failed 4/21/61. Placed in Tunnel #1 on 4/21/61, 2,336 Cu. Ft.
7.	E-F11 (1WW Waste) #2 Concentrator failed 2/1/62. Placed in Tunnel #1 on 2/8/62, 2,336 Cu. Ft.
8.	E-F6 (2WW Waste) #3 Spare Concentrator failed 5/23/64. Placed in Tunnel #1 on 1/22/65 Flat Car 3621, 2400 Cu. Ft.

Table 3.1. PUREX Storage Tunnels Inventory (con't)

The storage area is approximately 514.5 meters long, 7.9 meters high, and 10.4 meters wide. The tracks have a one percent downgrade toward the south end of the tunnel. The capacity of the Storage Tunnel is 38-40 modified railroad cars, 12.8 meters long. The Tunnel contains 21 cars as of 2/95.

Position	PUREX #2 Storage Tunnel (218-E-15)
1.	E-F6 # (2WW Waste) Concentrator, TK F 15-2, One tube bundle and agitator motors, placed in Tunnel on 12/12/67 on Car 61439. 2,400 Cu. Ft.
2.	E-F6 #5 (E-H4 3WB) Concentrator, two tube bundles placed in Tunnel on 3/26/69. On Car MILW 60883, 2,400 Cu. Ft.
3.	E-F6 #6 (2WW Waste) Concentrator, two tube bundles failed placed in Tunnel on 3/19/70. On Car 3612, 2,400 Cu. Ft.
4.	L Cell Package in a sealed steel box (H2-66012) placed in Tunnel on 12/30/70 on Car MILW 60033, 2,400 Cu. Ft.
5.	F2 Silver Reactor, F6 Demister, Vessel Vent Line, Steel Catwalk and Guard Rails, placed in Tunnel on 2/26/71. On Gondola Car 4610, 2,400 Cu. Ft., Ag-~625 Kg
6.	Modified A3-1 tower, scrubber, liquid, and vapor line placed in Tunnel on 12/12/71. On Gondola Car 4611, 2,400 Cu. Ft.
7.	A3 Dissolver placed in Tunnel on 12/22/71. On 9 Ft. shortened Car B58, 2,400 Cu. Ft.,

Position	PUREX #2 Storage Tunnel (218-E-15)
	Hg--~45 Kg.
8.	A1W1 Fuel ends in steel liner box and NPR fuel handling equipment. Used with the suspected canisters, on Car 19808. Placed in Tunnel on 8/29/72, 800 Cu. Ft.
9.	C3 Dissolver placed in Tunnel on 9/30/72, on Car 19811, 1590 Cu. Ft., Hg--~45 Kg.
10.	E-H4 (3WB) Concentrator, #61 tube bundle, prototype cooling coil, and F-Fl Filter Tank, placed in Tunnel 8/30/83, on Car CDX-1, 2,400 Cu. Ft.
11.	A3 Dissolver (Vessel #10 and Heater Vessel #6), placed in Tunnel on 1/18/86, on Car 3613, 3960 Cu. Ft. Hg--~40 Kg., Cd--~43 Kg
12.	White box (H2-58456) containing eight tube bundles #S 57.
13.	J5 Tank (Vessel #30), FL condenser (Vessel #13), and F12-B Cell Block, old four-way dumper, disc yoke, and flange plate placed in Tunnel on 1/21/86, on Car 19806, 2,500 Cu. Ft.
14.	L-l Pulser, 2-column cartridges, 1-jumper cutter, 3-jumper alignment tools, 9-exterior dumping trunnions, 10-pumps, 3-agitators, 4-tube bundles, 2-vent jumpers and 7-yokes placed in Tunnel on 11/18/87, on Car PX-10 (10A-19380) & Rack H2-96629.50. 50 tons, 3,600 Cu. Ft., Pb--~2540 Kg.
15.	Silver Reactor, E-F2 steam heater, and storage liner (H2-65095), full of cut up jumpers placed in Tunnel on 5/13/88, on Car PX-9 (10A-19809) & S/R Cradle SK-GLR-11-2-87. 20 tons, 2,775 Cu. Ft., Cd--~13 Kg., Ag--~115 Kg., Pb--~230 Kg.
16.	E-J8-1 Unitized Concentrator Vessel #1 H2-52477, failed 3/11/89. Placed on storage Car H2-99608, Px-6 (10A-19028) and in #2 Tunnel 4/6/89 graveyards. Estimated 42 tons, 6,000 Cu. Ft.
17.	North storage liner H2-65095 containing six pumps, one agitator, and cut up jumper (14 tons). South storage liner H2-65095 containing one pump, one #15 yoke and cut up jumpers (11.5 tons). Placed on storage Car PX-19 (10A-19030) and in #2 Tunnel on days 8/5/89. Estimated 25.5 tons, 2,574 Cu. Ft.
18.	T-F5 Acid absorber, ID#1-T-F5/F-168713, H2-52535 and H2-52487/488. Placed on storage Car PX-2 and in #2 Tunnel on 4/8/94. Estimated 22 tons, 835 Cu. Ft.
19.	Four metal liner storage boxes H-2-65095-3/H-2-100187-0 containing failed jumpers and miscellaneous obsolete canyon equipment items. Placed on storage Car PX-23 and in #2 Tunnel 9/16/94. Estimated 60 tons, 4032 Cu. Ft.
20.	E-H4-1 unitized concentrator (H-2-52477/56213)/(E-H4-1) Placed in Tunnel on 1/27/95, on Car Px-28. Estimated 40 tons, 5,760 Cu. Ft., Cr--~8 Kg.
21.	Tank E-5 (H-2-52453)/(F-166955), lead storage box assembly (H-2-131629)/ (H-2-131629-1), H4 concentrator tower (H-2-58102)/(F-223017-CBT-4), hot shop cover plate (H-2-52222)/("Q"), tube bundle wash capsule (H-2-58647), dissolver charging insert (H-2-75875)/ (H-2-75875-1), lifting yoke #7A (H-2-96837), lifting yoke #9 (H-2-52458). Placed in tunnel on 2/8/95 on Car PX-3609. Estimated 44 tons, 3,457 Cu. Ft., Pb--~1830 Kg
22.	Metal liner box (H-2-65096) containing jumpers and failed/obsolete canyon equipment. F7 neutron monitor (H-2-75825), lead storage box (H-2-131629) containing jumper counterweights and miscellaneous lead items, scrap hopper (H-2-57347) containing

Position	PUREX #2 Storage Tunnel (218-E-15)
	miscellaneous canyon equipment, canister capping station (H-2-821831), test canister containing various lengths of carbon steel pipe. Placed in Tunnel on 3-11-06, on Car #3616. Estimated weight 22 tons, 1,712 Cu. Ft., Pb-~3232 Kg., Cd-~2 Kg.
23.	Two burial boxes (H-2-100187) containing jumpers and failed/obsolete canyon equipment, lifting yoke (H-2-99652). Placed in Tunnel 3-11-96 on Car #PX-31. Estimated weight 21 tons, 2,116 Cu. Ft.
24.	Concrete burial box (H-1-44980) storing 8 containers of 324 Building, B-Cell waste. For additional details, see PUREX Work Plan WP-P-95-60. Placed in Tunnel on Car #PX-29, on April 26, 1996. Estimated weight 36 tons, 1,890 Cu. Ft. Cd-~10.5 kg., absorbed oil-~8.5 kg., Cr-~1 kg., Ba-~ 3 kg
25.	Concrete burial box (H-1-44980) storing 9 containers of 324 and 325 Building waste. For additional details, see PUREX Work Plan WP-P-96-015. Placed in tunnel on Car #10A-3619, on June 12, 1996. Estimated weight 46.5 tons, 1,890 Cu. Ft. Ba-~4g., Cd-<1g., Cr-~2g., Pb- <1g
26.	20,000-gallon liquid waste tank Car HO-10H-18580, empty per RCRA, placed in Tunnel on June 19, 1996, approximately 30 tons.
27.	20,000 gallon liquid waste tank Car HO-10H-18579, empty per RCRA, placed in Tunnel on June 19, 1996, approximately 30 Tons
28.	20,000-gallon liquid waste tank Car HO-10H-18582, empty per RCRA, placed in Tunnel on June 19, 1996, approximately 30 tons.

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CHAPTER 4.0
PROCESS INFORMATION

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CHAPTER 4.0
PROCESS INFORMATION

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4.0 PROCESS INFORMATION

This chapter discusses the processes involved in the operation of the Plutonium Uranium Extraction Facility (PUREX) Storage Tunnels. The PUREX Storage Tunnels are used for the storage of mixed waste from the PUREX Plant and other onsite sources.

The PUREX Storage Tunnels were designed and constructed to provide a means of protecting personnel and the environment from exposure to mixed waste associated with stored material. This design also serves to protect personnel and the environment from the dangerous waste component of the mixed waste stored inside the tunnels.

The PUREX Storage Tunnels are being permitted as a miscellaneous unit under Washington Administrative Code (WAC) 173-303-680. The WAC regulations require that miscellaneous unit permit terms and provisions address appropriate requirements provided for other treatment, storage, and disposal (TSD) units. Because the operation and construction of the PUREX Storage Tunnels most closely resemble that of a container storage unit, the appropriate requirements prescribed for a container storage unit are addressed in this chapter.

4.1 Operation of the PUREX Storage Tunnels

This section describes the selection, characterization, preparation, placement, and removal activities associated with storage of mixed waste in the PUREX Storage Tunnels.

4.1.1 Preparation for Tunnel Activities

Management, with the concurrence of an appropriate cognizant engineer, determines when material is to be removed and transported to the PUREX Storage Tunnels. A job specific work plan describing the overall transfer activities is prepared.

4.1.1.1 Storage/Removal Equipment Preparation

A remotely controlled, battery-powered locomotive normally was used to move railcars into and out of the PUREX Storage Tunnels. Other mechanical means such as a standard locomotive or a winch also can be used independently or in combination with the remote locomotive should the need arise. Methods for use of the remote locomotive are described in this chapter as this represents the normal placement and removal of railcars at the PUREX Storage Tunnels. Should storage activities require the use of a mechanical means other than the remote locomotive to place or withdraw a railcar, methods for that application will be developed.

Preparatory activities associated with the remote-controlled locomotive included the following:

- Charging the batteries for both the locomotive and the radio transmitter.
- Performing operational checks.
- Installing a plastic shroud over the locomotive to facilitate decontamination.
- Installing an anticoupling device on the south coupler of the locomotive (storage only).
- Performing physical inspections of the railroad track within the railroad tunnel to ensure that the track switches are positioned properly and the track is clear of obstructions.

4.1.1.2 Water-Fillable Door Preparation

Each PUREX Storage Tunnel has a water-fillable door that isolates the storage area from the PUREX railroad tunnel.

Currently, the water-fillable door to Tunnel Number 2 is empty and is not expected to be filled. Operational checks are performed on the door hoists. Before performing operational checks on the water-fillable door, the operator confirms with a dispatcher that the railroad tunnel area is clear of personnel.

4.1.1.3 Other Preparation Tasks

Before material storage, the following preparatory tasks are completed.

- The storage tunnel exhaust fan is verified to be operating.
- Labels will be attached to the railcar in accordance with WAC 173-303-395(6) and 173-303-630(3) if the material contains dangerous waste components.

4.1.2 Tunnel Storage Activities

This section describes the placement of material within the PUREX Storage Tunnels.

4.1.2.1 Physical Characterization of Material to be Stored

Physical characterization includes an evaluation of the following physical properties:

- Length, width, and height.
- Gross weight and volume.
- Preferred orientation for transport and storage.
- Presence of mixed waste.

Information sources used in physical characterization include equipment fabrication and installation drawings, operational records, and process knowledge. Physical characterization provides information necessary to appropriately describe the mixed waste materials. Such information also is used to design and fabricate, if required, supports on the railcar.

Specific material known to contain constituents that would cause the equipment to be designated as mixed waste is discussed in the waste analysis plan (Chapter 3.0). The material includes but is not limited to dissolvers that contain elemental mercury; silver reactors that contain silver salts; jumpers and other equipment that have elemental lead counterweights; a concentrator that contains chromium; neutron absorbing equipment containing cadmium. Characteristics of these materials when stored as mixed waste are described in Chapter 3.0. Waste transferred to the PUREX Storage Tunnels from other than PUREX Plant also would be physically characterized.

4.1.2.2 Material Flushing

Before removal from service, the material from the PUREX Plant was flushed to minimize loss of products, to reduce contamination, and to reduce to nonregulatory levels the concentration of any dangerous chemicals present in a residual heel. In the future the analysis of the rinsate will be used to determine when these goals have been achieved. The analysis of the final flush will be retained as part of the PUREX Storage Tunnel records. Material removed from other onsite units will be prepared for transfer to the tunnels in accordance with this dangerous waste permit.

4.1.2.3 Railcar Preparation

Railcars are modified to serve as dedicated storage platforms and transporters for material placed in the PUREX Storage Tunnels. The wooden decking on the railcars is removed to minimize the amount of combustible material placed in the PUREX Storage Tunnels. The south coupler is disabled or removed to prevent the railcar from coupling to the railcar stored ahead. Brakes are disabled to ensure freewheeling of the railcar. Steel decking, catch pans filled with absorbent, and equipment cradles are provided as needed to modify the railcar for its specific task.

4.1.2.4 Placement of Material into Storage Position

With all preparations complete and with the approval of cognizant management, transferring material to the PUREX Storage Tunnels proceeds as follows.

- The water-fillable door to the storage tunnel is opened.
- The railcar is loaded as specified in the storage tunnel checklist.
- An inventory of items loaded on the railcar and a record of their location on the railcar are recorded in the storage tunnel checklist.
- A survey is obtained of the loaded railcar at a distance commensurate with ALARA practices.
- The railcar is pushed into the storage tunnel to its storage position.
- Once the railcar is in position, the water-fillable door is closed.

4.1.3 Removal of Stored Material

Removal of material stored within the PUREX Storage Tunnels is not conducted routinely. It is planned that the material will remain in storage until a means to accommodate processing and repackaging of the material for disposal or further storage or until another final disposition option becomes available. Removal of material from storage within the PUREX Storage Tunnels would proceed after the preparation activities identified in Section 4.1.1.

With all preparations complete and approval of management, removal of material from the storage area of the PUREX Storage Tunnels would proceed as follows.

- The equipment that will be used to remove material is positioned in the PUREX railroad tunnel.
- Verification is made that the PUREX railroad tunnel is configured properly to proceed with entrance into the PUREX Storage Tunnels (i.e., tunnel ventilation system is operating, the overhead door is closed and a survey of the area is performed for as low as reasonably achievable [ALARA] concerns).
- The water-fillable door is opened.
- The equipment that will be used to remove material is moved into the storage tunnel and connected to the railcar.
- Verification is made that the railcar is connected to the removal equipment and the railcar is extracted from the storage tunnel and positioned within the PUREX railroad tunnel.
- The water-fillable door is closed.

The loaded railcar retrieved from the tunnel would be remotely viewed and measurements may be obtained to determine the possibility of mixed waste containment failure during storage in the PUREX Storage Tunnels. If evidence of containment failure is detected, the specific details (i.e., material, location on railcar, storage position) would be documented and attached to the waste tracking form. This information would be maintained in the files and would be used to establish sampling locations within the tunnels at closure. After remote viewing and surveying, the railcar and associated material may be prepared as required for transfer to an appropriate onsite TSD unit for treatment or further storage.

4.1.4 Filling the Water-Fillable Door (Tunnel Number 2)

If shielding beyond that provided by the empty water-fillable door becomes necessary, the door can be filled with water. In the past, this was accomplished by connecting a fire hose from the water hydrant to the wall stub on the exterior of the door housing (Figure 4.1). Once the fire hose was in place, the hydrant valve was opened and the door was filled with water.

The hydrant was closed by personnel when a high-level indicator light illuminated. Although attendance by an operator is required at all times during filling operations, should the door overfill, excess water is channeled through a vent/spill pipe to the door sump. A 15.2-centimeter drain is provided in each door sump. Water accumulated in the door sump was pumped out to the Double-Shell Tank System, and the sump and drain were made inoperable during PUREX Facility deactivation activities.

The drain was sealed during PUREX Facility deactivation. In the future, a temporary source of water could be provided for filling the water-fillable door.

4.1.5 Post Storage Activities

The following post storage activities would conclude the tunnel storage task.

- Decontamination activities, if required, are performed.
- Management is notified of any unusual conditions observed during the storage/retrieval activities.

4.1.6 Operation of the Tunnel Ventilation System

The ventilation systems for Tunnel Number 1 and Tunnel Number 2 were designed to ventilate air from within the tunnels so the airborne contamination is vented through a High Efficiency Particulate Air (HEPA) filtered exhaust system.

4.1.6.1 Tunnel Number 1 Ventilation

Active ventilation of Tunnel Number 1 presently is not provided. After placement of the last railcar into Tunnel Number 1, the tunnel was sealed (Chapter 2.0). As part of the sealing activities, the ventilation fan was deactivated electrically and the exhaust stack and filter were isolated from the system by installing blanks upstream and downstream of both the exhaust fan and filter and the stack was removed. In the event railcar removal activities are initiated, it is planned that the ventilation system would be reactivated. Operation of the ventilation system would be similar to that for Tunnel Number 2.

4.1.6.2 Tunnel Number 2 Ventilation

The Tunnel Number 2 ventilation system presently is inactive. As part of PUREX Facility deactivation, the water-fillable door and outer PUREX railroad tunnel door were sealed. The seal may be temporary or permanent depending on the future need for storing waste in the tunnel. The ventilation system may be operated continuously, or de-energized and reactivated during waste placement activities. During deactivation, a blank was installed on the downstream side of the filter and the stack was capped. When the determination has been made that Tunnel Number 2 will no longer receive waste, the ventilation system will be blanked and deactivated electrically similar to the Tunnel Number 1 ventilation system. While the Tunnel Number 2 ventilation system is operating and the water-fillable door is closed, the exhaust system, which discharges approximately 100 cubic meters per minute, maintains a slightly negative pressure in the tunnel. The exhaust air is replaced by infiltration around the water-fillable door and through the porosity of the tunnel structure (e.g., the rail-bed ballast). When the water-fillable door is open (during transfer activities), inward airflow is maintained through the open doorway. This inward airflow channels airborne radioactive contamination away from both the railroad tunnel and personnel following railcars (if allowed) into the storage tunnel. A HEPA filter provides filtration of all exhaust air before release to the atmosphere. When the ventilation system is operating, the HEPA filter is tested in place at least annually to ensure radioactive particulate removal efficiency. Exhausted air is sampled periodically and analyzed for airborne radionuclides.

4.2 Containers

This section describes the various types of containment used to isolate mixed waste stored in the PUREX Storage Tunnels. The PUREX Storage Tunnels are considered to be a miscellaneous unit most closely resembling that of a container storage unit. The mixed waste stored in the PUREX Storage Tunnels is contained and is not considered a risk to human health or to the environment.

4.2.1 Containers with Free Liquids

The only mixed waste stored as a free liquid is elemental mercury. A small quantity, less than 1.7 liters, of mercury is contained in each of the two thermowells attached to and contained within each dissolver (Chapter 3.0).

Primary containment of the mercury is provided by the all-welded construction of the thermowell itself, which is fabricated from 7.6-centimeter, Schedule 80, 304L stainless steel pipe. The open upper end of the thermowell was plugged with a 304L stainless steel nozzle plug in preparation for storage. The dissolver rests on a cradle on its railcar in an inclined position. This ensures that the mercury remains in the lower portion of the thermowell and is not in contact with the mechanical closure on the nozzle end of the thermowell.

A secondary containment barrier for mercury, should it leak from the thermowell, is provided by the dissolver itself. The dissolver is a 304L stainless steel process vessel constructed from 1-centimeter-thick plate and is approximately 2.7 meters in diameter. The dissolver is of all-welded construction and contains no drains or nozzle outlets in the bottom several feet of its lower section, which contains both thermowells.

The 304L stainless steel used to contain the elemental mercury is both compatible with the waste itself and the storage environment. The potential for significant deterioration of either the primary or secondary containment barrier material before closure is considered to be negligible.

The dissolvers stored within the PUREX Storage Tunnels are not labeled as containing characteristic toxic mercury (D009) [WAC 173-303-090(8)(c)]. Procedures for labeling were not in place at the time of storage. Personnel access into the storage area for purposes such as labeling is not feasible and cannot be justified under ALARA guidelines. Based on ALARA, mixed waste presently within the PUREX Storage Tunnels will remain unlabeled. However, during future transfers of mixed waste into the PUREX Storage Tunnels the railcars will be labeled as specified by WAC 173-303-395(6) and WAC 173-303-630(3).

4.2.2 Containers without Free Liquids that do not Exhibit Ignitability or Reactivity

Most lead is fully contained in all-welded encasements of either carbon steel or 304L stainless steel (refer to Chapter 3.0, Table 1). The encasement serves as support, protection against mechanical damage, and protection of the lead from exposure to the environment. Also, lead has been placed in burial boxes of appropriate size. The boxes provide secondary containment for the lead in the unlikely event the primary encasement should fail. Although boxes may be open on the top, the PUREX Storage Tunnels are enclosed; therefore, the containers are protected from the elements.

Both carbon steel and 304L stainless steel used to encase the lead are compatible with the waste and the storage environment. Significant deterioration of either the primary or secondary containment barrier materials before closure is not considered to be credible.

In the past, material that contains lead or that has encased lead attached was not labeled as containing characteristic toxic lead (D008) [WAC 173-303-090(8)], because the requirements were not yet on line. As stated in Section 4.2.1, personnel entry into the tunnel storage area for purposes of labeling would be inconsistent with ALARA guidelines. However, during future storage of material containing lead the railcars will be labeled in accordance with WAC 173-303-395(6) and WAC 173-303-630(3).

4.2.3 Protection of Extremely Hazardous Waste in Containers

The present amount of mixed waste stored in the PUREX Storage Tunnels is sufficient to characterize this material as extremely hazardous waste. Because the PUREX Storage Tunnels are enclosed totally, protective covering from the elements and from run-on is provided for the storage of extremely hazardous waste. Periodic inspection of the equipment stored in the PUREX Storage Tunnels is not feasible and cannot be justified under ALARA guidelines. Safe management of this waste is based on the following considerations.

- The operation of the PUREX Storage Tunnels is passive, i.e., once a storage position is filled, the storage position remains undisturbed until closure.

- The extremely hazardous waste is compatible with its storage container and the storage environment.

4.2.4 Prevention of Reaction of Ignitable, Reactive, and Incompatible Waste in Containers

There is no reactive or incompatible waste known to be stored in the PUREX Storage Tunnels. The only mixed waste stored in the PUREX Storage Tunnels considered an ignitable waste is the silver nitrate in Tunnel Number 2. The silver nitrate fraction of the silver salts, within the silver reactors, exhibits the characteristic of ignitability as defined in 49 CFR 173.127(a). Therefore, the silver salts are managed as an ignitable dangerous waste in accordance with WAC 173-303-395.

- The risk of fire associated with the storage of silver nitrate in the PUREX Storage Tunnels is considered to be extremely low. This conclusion is based on the following considerations.
- The operation of the PUREX Storage Tunnels is passive; i.e., once a storage position is filled, the storage position remains undisturbed until closure.
- The silver nitrate is contained within large, heavy-walled stainless steel vessels that isolate the silver nitrate from contact with any combustibles that might be in the tunnels.
- The silver nitrate is dispersed over a large surface area on a ceramic packing substraight and is not conducive to build-up of heat that could lead to spontaneous combustion.
- Personnel access to the occupied areas of the tunnels is not permitted, thereby precluding activities that could present a fire hazard (e.g., smoking, flame cutting, welding, grinding, and other electrical activities).

Although ignitable waste storage units are required by WAC 173-303-395(1)(d) to have inspections conducted at least yearly by a fire marshall or professional fire inspector familiar with the requirements of the uniform fire code, the ALARA concerns within the PUREX Storage Tunnels make such inspections impractical. These inspections are not considered appropriate or necessary for the safe operation of the unit because of the nature of the ignitable waste, the means of storage, and ALARA concerns (Chapter 6.0, §6.2).

4.3 Engineering Drawings

As-built drawings for the PUREX Storage Tunnels:

H-2-55587	218-E-14 Structural Floor Plan and Section
H-2-55588	Structural Sections and Details: Disposal Facility for Failed Equipment
H-2-55589	Structural Sections and Details: Disposal Facility for Failed Equipment
H-2-55590	Door and Hoist Details
H-2-55591	Door and Hoist Details
H-2-55592	Door and Hoist Details
H-2-55593	Electrical Details
H-2-55594	Shielding Door Fill and Drain Lines Arrangement: Disposal Facility for Failed Equipment
H-2-55599	Electrical Door Control Plan, Elementary Diagram and Miscellaneous Details: Disposal Facility for Failed PUREX Equipment
H-2-58134	Ventilation Details; Sheet 1, Sheet 2, Sheet 3, Sheet 4
H-2-58175	PUREX Tunnel

H-2-58193	Sump Details
H-2-58194	Sump Details
H-2-58195	Structural Sections and Details: Equipment Disposal - PUREX
H-2-58206	Sump Details
H-2-58208	Fan Details; Sheet 1, Sheet 2, Sheet 3
H-2-94756	Filter Details; Sheet 1, Sheet 2

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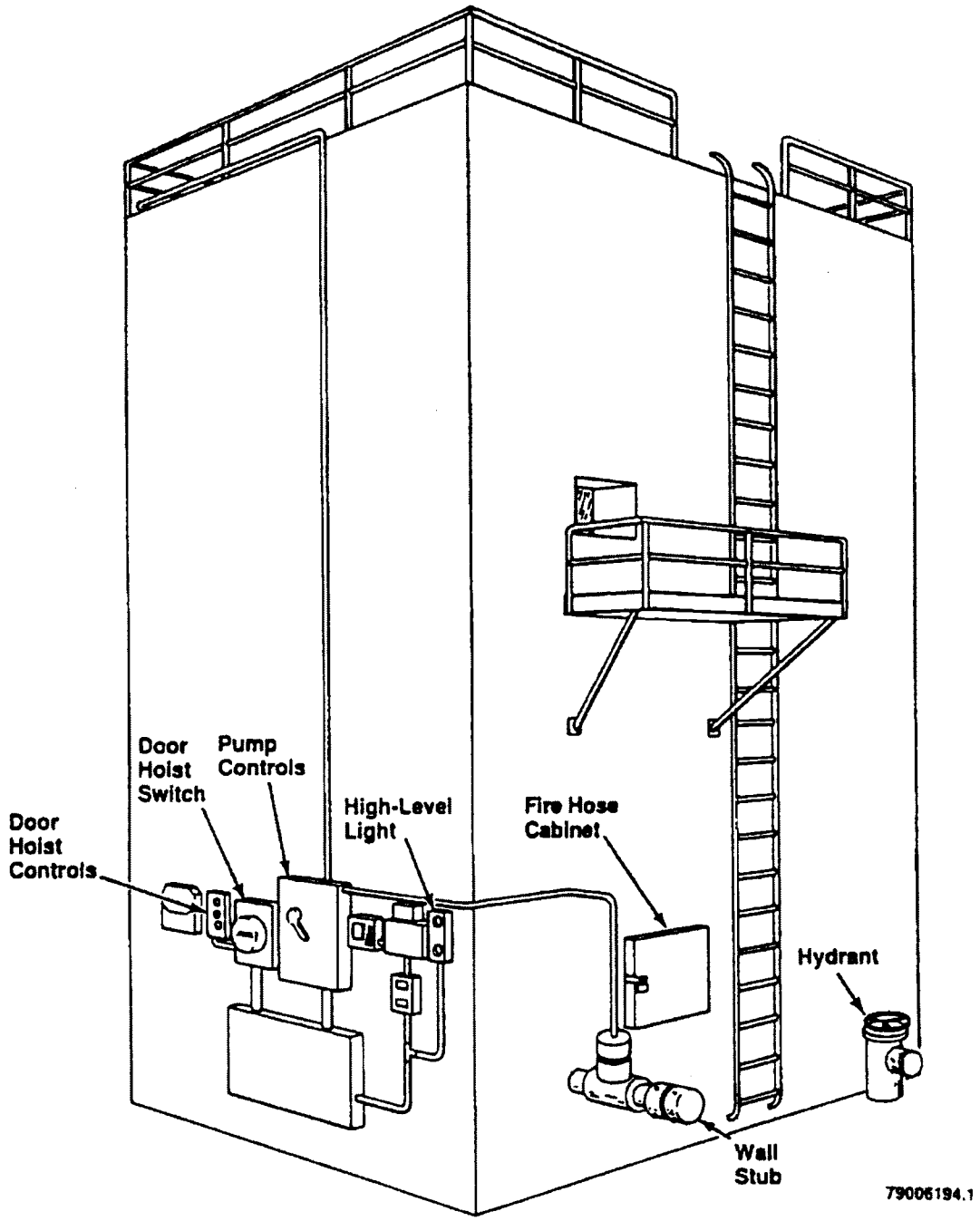
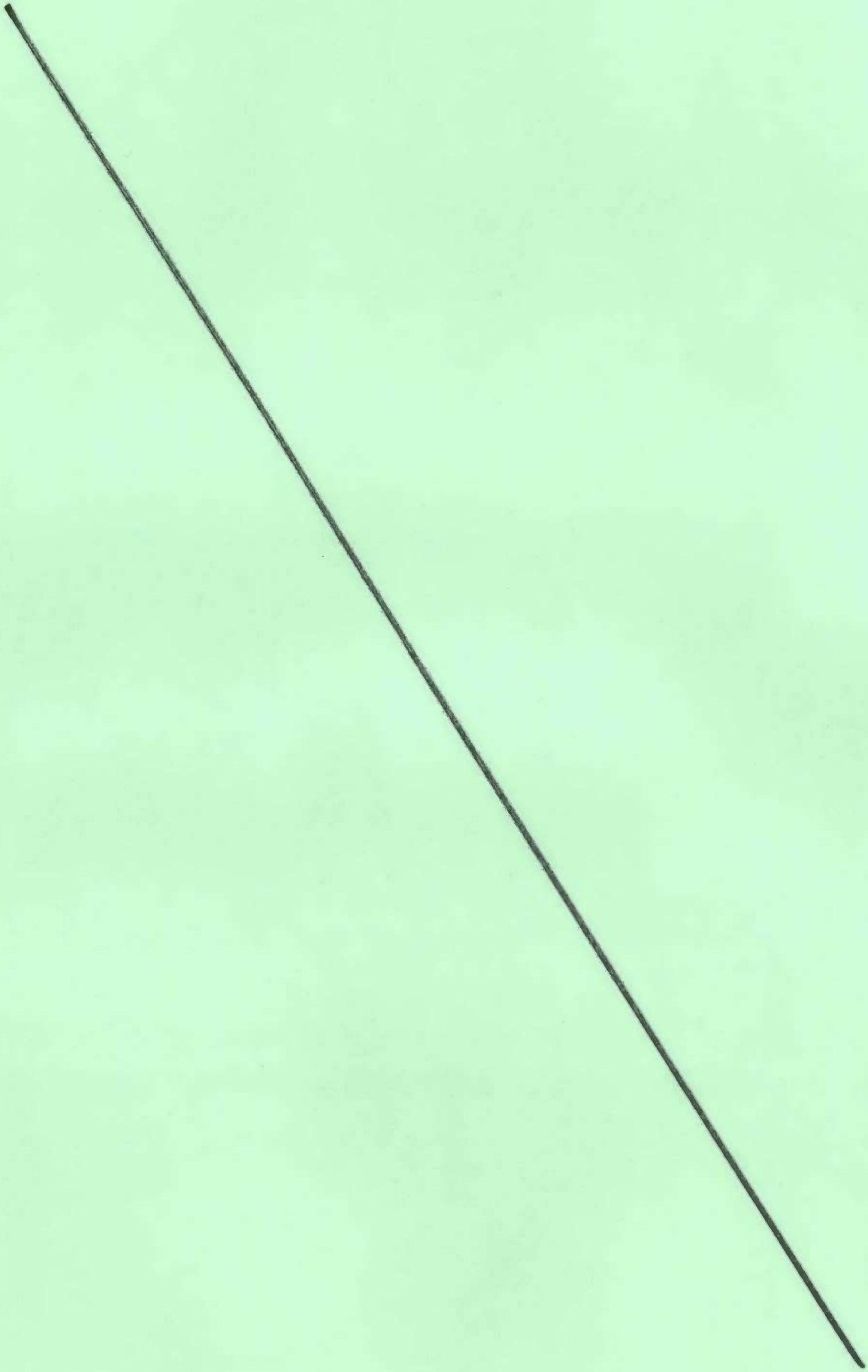


Figure 4.1. Water Fillable Door Exterior (Tunnel Number 2)



CHAPTER 11.0
CLOSURE AND FINANCIAL ASSURANCE

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CLOSURE AND FINANCIAL ASSURANCE

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11.0 CLOSURE AND FINANCIAL ASSURANCE

Closure of the Plutonium Uranium Extraction Facility (PUREX) Storage Tunnels requires coordination with closure of the PUREX Plant to ensure a cost effective closure for both units. In addition, the nature of the mixed waste located within the PUREX Plant and PUREX Storage Tunnels precludes the determination of the type of treatment and/or disposition of the waste at this time.

The PUREX Storage Tunnels will be managed as a Resource Conservation and Recovery Act (RCRA) storage unit until closure can be coordinated with the final closure plan for the PUREX Plant. The PUREX Storage Tunnels closure plan will be submitted after any required National Environmental Policy Act of 1969 documentation and land usage agreements, which initiate disposition and aid in identifying or developing necessary disposition activities, have been adopted. The PUREX Storage Tunnels closure plan will be submitted for the Washington State Department of Ecology (Ecology) approval with the PUREX Plant closure plan.

The PUREX Storage Tunnels closure plan will be written to meet the requirements of Washington Administrative Code (WAC) 173-303-140 and WAC 173-303-610. This closure plan might consider but will not be limited to the following options for either in situ disposal or retrieval/clean closure of this unit.

Federal facilities are not required to comply with WAC 173-303-620 as is stated in the regulations and as described in Hanford Facility RCRA Permit, WA7890008967 (Permit) Condition II.H.3.

11.1 In Situ Disposal Options

This closure plan might consider but will not be limited to the following options for in situ disposal of waste in this unit.

11.1.1 Backfilling the PUREX Storage Tunnels with Gravel

This option could involve backfilling the tunnels with gravel to eliminate void space and prevent ground subsidence. A modified commercially available centrifugal rock-throwing device could be placed in newly constructed risers evenly spaced along each tunnel roof. Fill material could be supplied and dispersed into the tunnels by automated controls. Following the fill process, all equipment could be removed from the tunnel roofs and all means of access to the tunnels could be permanently sealed. Final activities could involve the construction of a final surface barrier that meets RCRA landfill cover requirements to prevent water from leaching mixed waste contained in the tunnels.

11.1.2 Injecting the PUREX Storage Tunnels with Grout

This option could involve the injection of grout material into each tunnel to stabilize and immobilize contained materials and prevent ground subsidence. A grout injector could be alternately placed in newly constructed risers evenly spaced along each tunnel roof. Grout material could be supplied and dispersed into the tunnels by automated controls. The grout material could be injected in lifts to accommodate curing and heat dissipation normally associated with the use of this type of material. Final activities could involve the construction of a final surface barrier that meets RCRA landfill cover requirements to prevent water from leaching mixed waste contained in the tunnels.

11.1.3 Combination of Grout Injection and Backfilling

This option combines grout injection with gravel backfilling similar to the processes discussed previously. Grout could be injected first to fill void spaces under the railcars and provide a basal structure. Gravel could be dispersed to fill remaining void space and prevent ground subsidence. Final activities could involve the construction of a final surface barrier that meets RCRA landfill cover requirements to prevent water from leaching mixed waste contained in the tunnels.

11.2 Retrieval/Clean Closure Options

This closure plan might consider but will not be limited to the following options for retrieval/clean closure of this unit.

11.2.1 Retrieval and Disposal in the PUREX Plant

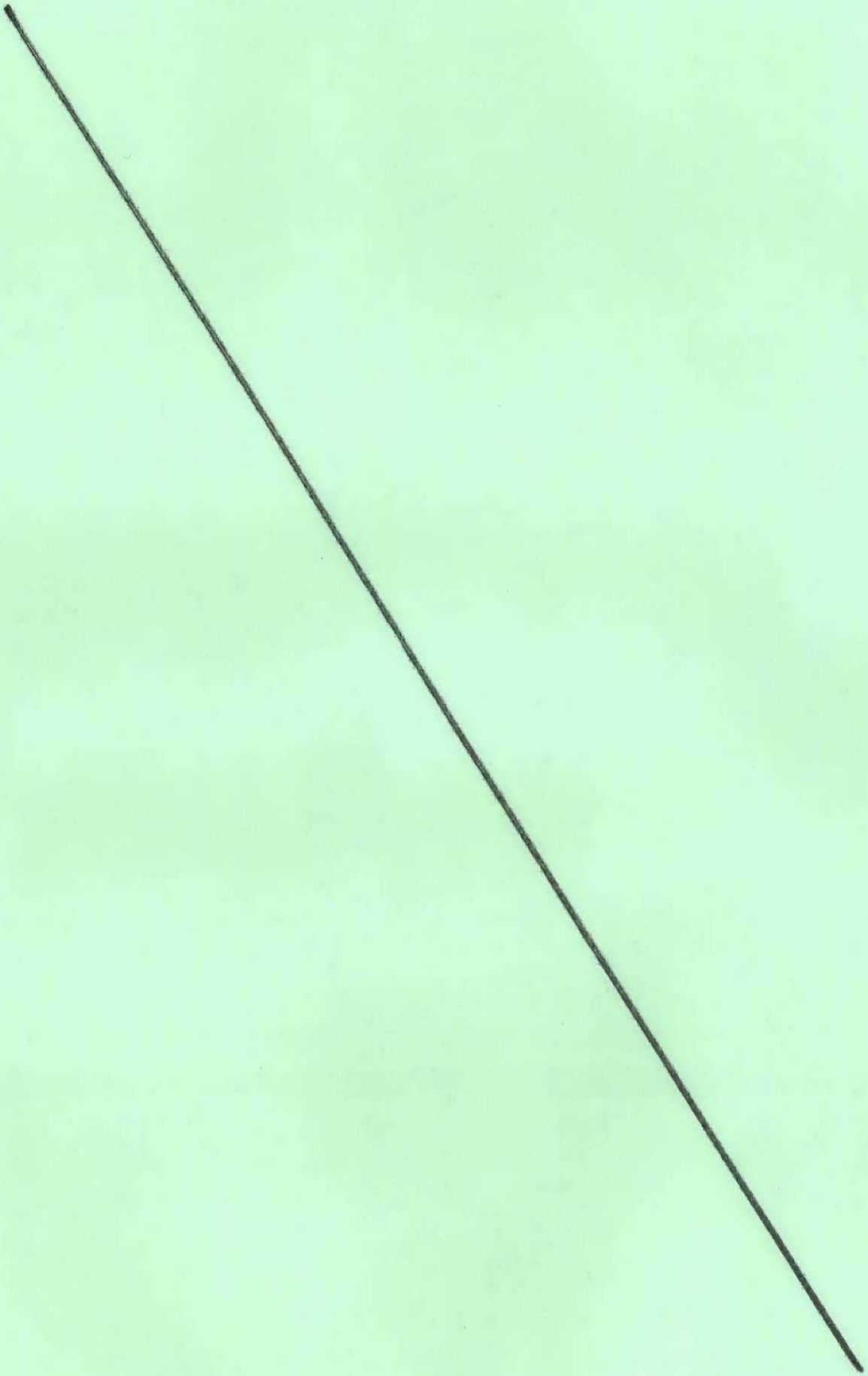
Railcars stored in both tunnels could be remotely retrieved one at a time and moved beneath the horizontal door of the railroad tunnel extension for remote viewing, and if possible, characterization. Transfer procedures could be initiated to move waste material from the railcars to the PUREX Plant canyon deck area. Following transfer of the waste material, the railcars could be decontaminated and removed for final disposition at other onsite units. Final disposition of the waste transferred to the canyon deck area could be in accordance with PUREX Plant closure documentation. The PUREX Storage Tunnels could be closed after submittal and implementation of a PUREX Storage Tunnels closure plan in conjunction with PUREX Plant closure documentation. The PUREX Storage Tunnels closure plan will detail verification sampling and analysis to be performed as a part of closure activities.

11.2.2 Retrieval and Physical Processing (size reduction) in the PUREX Plant and Subsequent Disposal

Retrieval of waste material stored in the tunnels could be similar to that described in the previous section. Once the waste material was transferred to the PUREX Plant canyon deck area, characterization and size reduction of waste material could proceed. An area located on the canyon deck or in a process cell could be modified to include all necessary equipment to perform characterization and size reduction activities. Size reduction could be performed through various technologies that include, but are not limited to, flame cutting, water jet cutting, sawing, or other technologies. Final disposition of the processed waste material either onsite or offsite could be in accordance with regulations and procedures in place at that time. The PUREX Storage Tunnels could be closed after submittal and implementation of a PUREX Storage Tunnels closure plan in conjunction with PUREX Plant closure documentation. The PUREX Storage Tunnels closure plan will detail verification sampling and analysis to be performed as a part of closure activities.

11.2.3 Construction of a New Facility for Retrieval, Processing, and Treatment of Equipment for Disposal

This option involves the construction of a new unit that is either mobile or stationary to excavate, retrieve, and treat waste material stored in the tunnels. The unit could be constructed in a manner consistent with the retrieval and handling requirements for large, contaminated waste material. Following retrieval, the waste material could be treated in accordance with final onsite or offsite disposition requirements identified at such time. The excavated tunnels could have a temporary surface barrier placed in position until verification and sampling analysis could be performed as a part of closure activities to be performed in conjunction with PUREX Plant closure.



ADDENDUM E
SECURITY

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SECURITY

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E. SECURITY

This addendum discusses security for the Plutonium Uranium Extraction Facility (PUREX) Storage Tunnels.

E.1 Security

The Permittees will post signs at access points to the PUREX Storage Tunnels stating, DANGER-UNAUTHORIZED PERSONNEL KEEP OUT, or an equivalent legend. These signs will be written in English, legible from a distance of 7.6 meters, and visible from any approach to the active portion. [WAC 173-303-310(2)(a)].

E.1.1 Waiver

A waiver of the security procedures and equipment requirements for the PUREX Storage Tunnels is not requested. Therefore, the requirements of WAC 173-303-310(1)(a) and (b) are not applicable to the PUREX Storage Tunnels.

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ADDENDUM F
PREPAREDNESS AND PREVENTION

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PREPAREDNESS AND PREVENTION

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F. PREPARDENESS AND PREVENTION

F.1 Preparedness and Prevention Requirements

This addendum discusses preparedness and prevention measures for the Plutonium Uranium Extraction Facility (PUREX) Storage Tunnels.

F.1.1 Equipment Requirements

The following sections describe the internal and external communications systems and emergency equipment required.

F.1.1.1 Internal Communications

The PUREX Storage Tunnels are not occupied and personnel entry is allowed only on a very limited basis and under close supervision. Normal and emergency communications equipment (portable two-way radios) is available for use.

F.1.1.2 External Communications

External communications equipment for summoning emergency assistance from the Hanford Fire Department and/or emergency response teams are provided by two-way portable radios or other devices.

F.1.1.3 Emergency Equipment

Equipment included in the emergency plan for the PUREX Storage Tunnels is provided in Addendum J.

F.1.1.4 Water for Fire Control

The fire hazard associated with the operation of the PUREX Storage Tunnels is considered to be very low because of the minimal amount of combustibles stored within the tunnels and the lack of an ignition source. In the event it is determined there is a fire in the storage area of the tunnels, the contingency plan will be activated. Because of the potential of the mixed waste stored within the tunnels to leach, the use of water for fire control will be avoided if possible. Reductions of the air supply to the storage area by isolation of the tunnel exhaust system, if operating, should permit the fire to self-extinguish. Should the fire continue to propagate, heavy equipment and cranes will be called to the scene to cover areas of the tunnels that might collapse. Heavy equipment and cranes are readily available on the Hanford Facility at all times and generally are available for deployment to the scene of an emergency within 1 hour. In the event that a fire resulted in the collapse of the tunnels, a recovery plan will be developed in accordance with emergency response procedures included in Addendum J. The recovery plan will take into consideration plans, if any, for retrieval of the waste stored within the tunnel(s).

F.1.2 Aisle Space Requirement

Requirements for aisle space are not considered appropriate for the safe operation of the PUREX Storage Tunnels and were not included in design documents.

F.2 Preventive Procedures, Structures, and Equipment

The following sections describe preventive procedures, structures, and equipment.

F.2.1 Unloading Operations

Operation of the PUREX Storage Tunnels does not involve the loading or unloading of dangerous waste. All loading and unloading operations are conducted at the PUREX Facility or other onsite units. Therefore, the requirements of WAC 173-303-806(4)(a)(viii)(A) are not applicable to the PUREX Storage Tunnels.

F.2.2 Runoff

The design of the PUREX Storage Tunnels included consideration and provisions for the control of runoff and run-on. Construction of both tunnels included the application of a moisture barrier before placement of the soil overburden. On Tunnel Number 1, 40.8-kilogram mineral surface roofing was applied to the external surfaces of the structural timbers (top and sides). The roofing material was nailed in place with an overlap of approximately 10 centimeters at all joints and seams. All interior and exterior steel surfaces of Tunnel Number 2 were coated with at least a 0.9-millimeter bituminous, solvent coal tar base, coating compound. The coating was applied using a two coat system, with each coat not less than 0.45 millimeters, ensuring a total dry film thickness of not less than 0.9 millimeter.

The soil overburden covering the PUREX Storage Tunnels also is contoured to provide a side slope of 2 (horizontal) to 1 (vertical). This construction serves to divert any seasonal or unanticipated run-on away from the storage area of the PUREX Storage Tunnels. For potential situations where a natural catastrophic event occurs, inspections of the tunnel side slopes are conducted to ensure the contours remain in a condition that ensures proper runoff and continues to divert run-on away from the tunnel storage areas.

Run-on at the PUREX Storage Tunnels is controlled by the design features of the exterior of the tunnels that serve to divert run-on away from the interior of the tunnels. Additionally, all waste within the tunnels is stored well above the floor level on railcars. The control of run-on combined with the storage of all waste above the floor elevation provides adequate assurance that runoff will not occur at the PUREX Storage Tunnels.

F.2.3 Water Supplies

Water was supplied to the PUREX Storage Tunnels from the PUREX Plant. This water was used for the sole purpose of filling the water-fillable doors should it have been determined necessary. There are no other sources or uses of water at the PUREX Storage Tunnels. The line that supplied water to the PUREX Storage Tunnels was blanked and emptied during deactivation activities. In the future, a temporary source of water would be provided for filling the water fillable door.

F.2.4 Equipment and Power Failures

The procedures, structures, and equipment used to mitigate the effects of equipment failure and power outage are described in the following sections.

F.2.4.1 Mitigation of the Effects of Equipment Failure

Maintaining safe storage of materials in the PUREX Storage Tunnels is not contingent on continued operation of equipment. The operable equipment associated with the PUREX Storage Tunnels were the remote controlled locomotive or waste placement and removal equipment, the railcars, and the water-fillable door and ventilation system for both tunnels. No operable equipment is associated with either tunnel, as these tunnels have been sealed and may no longer receive dangerous waste. Backup or redundant systems are not provided for either tunnel, as failure of the equipment would not have the potential to result in a release of dangerous waste to the environment. There are no hazards associated with tunnel equipment failure.

F.2.4.2 Mitigation of the Effects of Power Failure

Maintaining safe storage of materials in the PUREX Storage Tunnels is not contingent on continued supply of electrical power. Electrical power is required to operate the water-fillable door and the ventilation fan in both tunnels. Backup or redundant ventilation systems are not provided as the system is operated only to maintain air balance and provide secondary control of mixed waste airborne particulate. Power failure to either tunnel would not have the potential to result in the release of dangerous waste or mixed waste to the environment. There are no hazards associated with the shutdown of the tunnel ventilation systems due to loss of electrical power.

F.2.5 Personnel Protection Equipment

Personnel entering the PUREX Storage Tunnels are required to wear special protective clothing and respiratory protection at all times because of the material stored in the PUREX Storage Tunnels. Protective clothing and full-face respirators with filters are considered to be sufficient protection from the dangerous waste stored within the PUREX Storage Tunnels. Personnel are trained and qualified in using the protective equipment and are checked routinely for mask fit.

F.3 Prevention of Reaction of Ignitable, Reactive, and/or Incompatible Waste

There is no reactive or incompatible waste stored in the PUREX Storage Tunnels. The only ignitable waste stored within the tunnels is silver nitrate. The silver nitrate is present within the silver reactors (deposited on unglazed ceramic packing) stored in Tunnel Number 2.

Although silver nitrate exhibits the characteristic of ignitability, it is contained within stainless steel vessels, stored on railcars above the floor level, and isolated from combustible materials and other dangerous waste. Additional measures to prevent reaction of the ignitable waste are not considered necessary.

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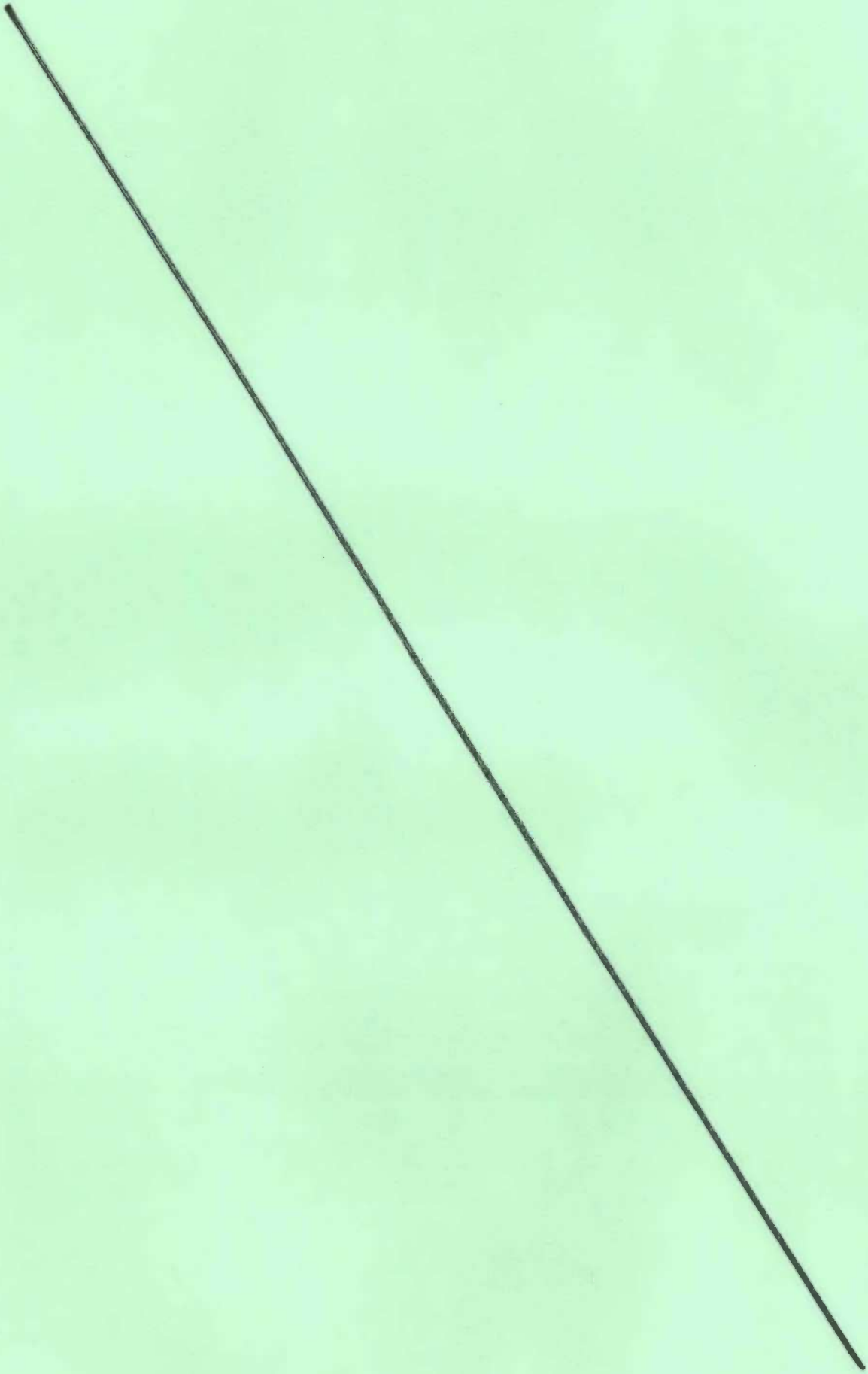

ADDENDUM G
PERSONNEL TRAINING

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ADDENDUM G
PERSONNEL TRAINING

PUREX Storage Tunnels Training Matrix

Permit (WA7890008967) Training Category	Training Category[1]					
	General Hanford Facility Training	Contingency Plan Training	Emergency Coordinator Training	Operations Training		
PUREX Storage Tunnels DWTP Implementing Plan	Orientation Program	Emergency Response (Contingency Plan)	Emergency Coordinator Training	General Waste Management		Misc. Unit
Job title/position						
Nuclear Chemical Operator	X	X		X		X
D&D worker	X	X				
Operations Manager	X	X				
Field Work Supervisor	X	X				
Environmental Compliance Officer	X			X		
Building Emergency Director	X		X			

[1] Refer to the Dangerous Waste Training Plan (DWTP) prepared for PUREX Storage Tunnels for a complete description

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ADDENDUM I
INSPECTION REQUIREMENTS

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INSPECTION REQUIREMENTS

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I. INSPECTION REQUIREMENTS

This Addendum discusses the inspection requirements for dangerous waste management units within the Plutonium Uranium Extraction Facility (PUREX) Storage Tunnels.

I.1 PUREX Storage Tunnels

Because waste within the PUREX Storage Tunnels is inaccessible and exposure to the mixed waste would be extremely hazardous to personnel, inspection of the tunnel interior will not be performed. External inspections of the tunnels will be performed annually. The inspection schedule, log, and records will be maintained in the Hanford Facility Operating Record, PUREX Storage Tunnels File for a minimum of 5 years.

Information from inspections will be recorded on inspection logs. The log forms are used to initiate corrective action if necessary. The elements inspected at the PUREX Storage Tunnels are identified in Table I.1. Abnormal conditions are recorded, evaluated, and corrective action initiated as necessary.

Table I.1. Inspection Schedule

Requirement Description	Inspection Frequency	Types of Problems
Perform external surveillance of PUREX Storage Tunnels	Annual	External surfaces of the PUREX Storage Tunnels are observed for evidence of structural deterioration. Tunnel subsidence, erosion of the earth cover, and vent stack damage are of primary concern. The points of access to the PUREX Storage Tunnels are inspected to ensure warning signs are in place, visible, and legible.

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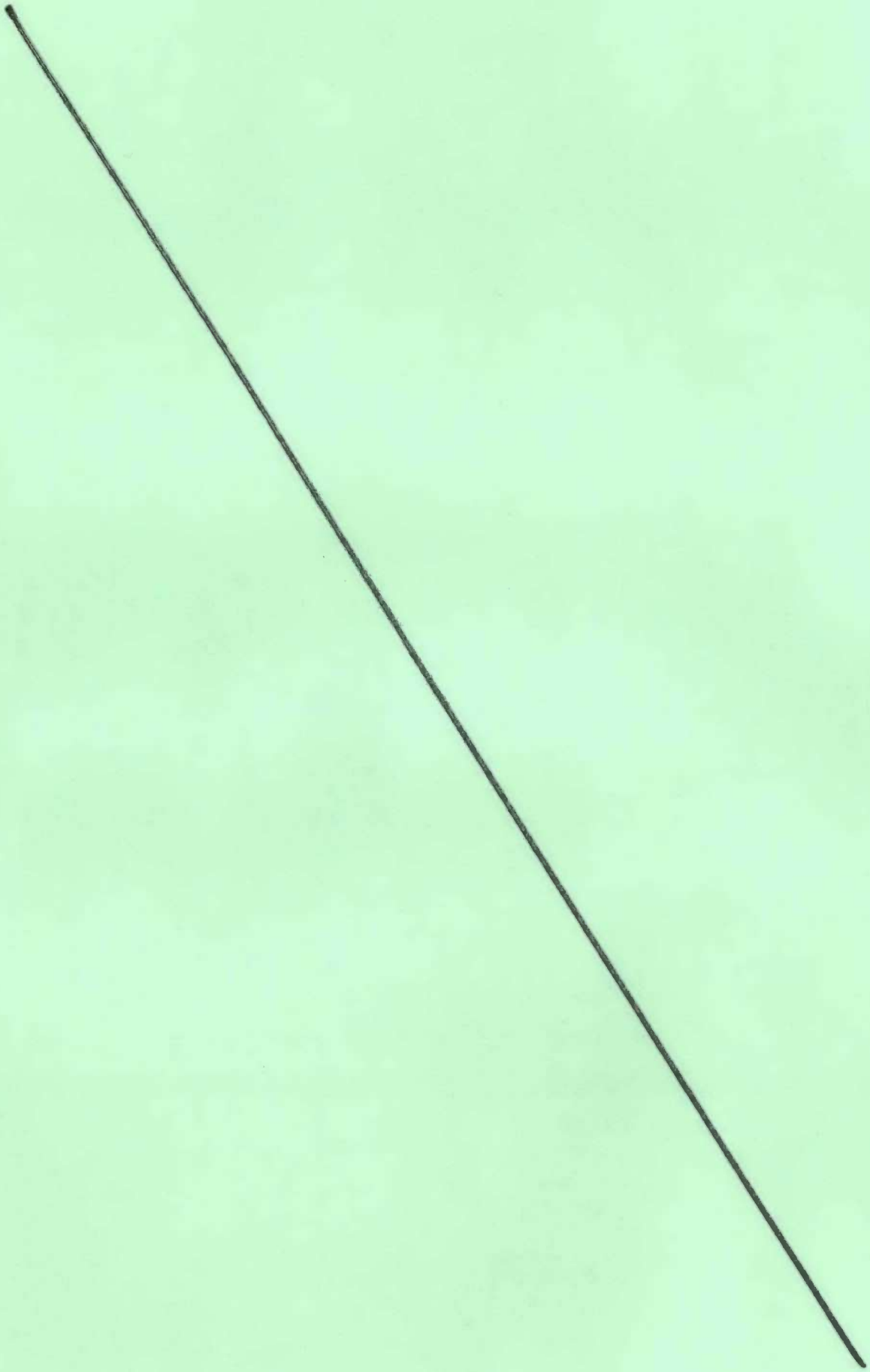

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Title: Addendum J: Contingency Plan

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